

Volume 2

MATHEMATICS FOR JEE (MAIN & ADVANCED)

TRIGONOMETRY | VECTOR ALGEBRA | PROBABILITY

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Dr. U M SWAMY



Mathematics for IIT-JEE

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Dedication

*Dedicated to
my father*

Ganti Chinavenkata Chainulu (Chiranjeevi Babu)
who expired when I was three years old.

Dr. G. S. N. Murthi

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Features and Benefits at a Glance

Feature	Benefit to student
Chapter Opener	Peaks the student's interest with the chapter opening vignette, definitions of the topic, and contents of the chapter.
Clear, Concise, and Inviting Writing Style, Tone and Layout	Students are able to Read this book, which reduces math anxiety and encourages student success.
Theory and Applications	Unlike other books that provide very less or no theory, here theory is well matched with solved examples.
Theorems	Relevant theorems are provided along with proofs to emphasize conceptual understanding.
Solved Examples	Topics are followed by solved examples for students to practice and understand the concept learned.
Examples	Wherever required, examples are provided to aid understanding of definitions and theorems.
Quick Look	Formulae/concepts that do not require extensive thought but can be looked at the last moment.
Try It Out	Practice problems for students in between the chapter.
Worked Out Problems	Based on IIT-JEE pattern problems are presented in the form of Single Correct Choice Type Questions Multiple Correct Choice Type Questions Matrix-Match Type Questions Comprehension-Type Questions Assertion–Reasoning Type Questions Integer Answer Type Questions In-depth solutions are provided to all problems for students to understand the logic behind.
Summary	Key formulae, ideas and theorems are presented in this section in each chapter.
Exercises	Offer self-assessment. The questions are divided into subsections as per requirements of IIT-JEE.
Answers	Answers are provided for all exercise questions for student's to validate their solution.

Note to the Students

The IIT-JEE is one of the hardest exams to crack for students, for a very simple reason – concepts cannot be learned by rote, they have to be absorbed, and IIT believes in strong concepts. Each question in the IIT-JEE entrance exam is meant to push the analytical ability of the student to its limit. That is why the questions are called brainteasers!

Students find Mathematics the most difficult part of IIT-JEE. We understand that it is difficult to get students to love mathematics, but one can get students to love succeeding at mathematics. In order to accomplish this goal, the book has been written in clear, concise, and inviting writing style. It can be used as a self-study text as theory is well supplemented with examples and solved examples. Whenever required, figures have been provided for clear understanding.

If you take full advantage of the unique features and elements of this textbook, we believe that your experience will be fulfilling and enjoyable. Let's walk through some of the special book features that will help you in your efforts to crack IIT-JEE.

To crack mathematics paper for IIT-JEE the five things to remember are:

- 1. Understanding the concepts**
- 2. Proper applications of concepts**
- 3. Practice**
- 4. Speed**
- 5. Accuracy**

About the Cover Picture

Medieval mathematician and businessman Fibonacci (*Leonardo Pisano*) posed the following problem in his treatise *Liber Abaci* (pub. 1202):

How many pairs of rabbits will be produced in a year, beginning with a single pair, if in every month each pair bears a new pair which becomes productive from the second month on?

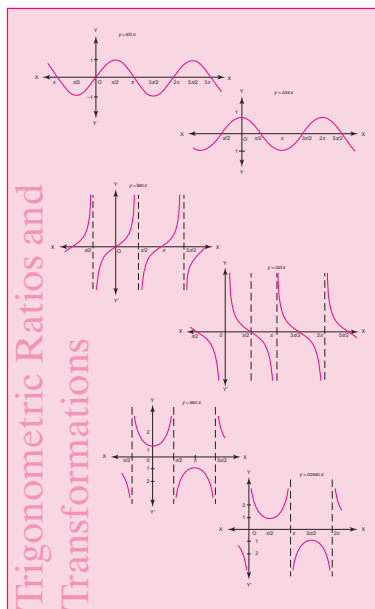
The solution to this problem leads to the recursive sequence, obeying the simple rule that to calculate the next term one simply sums the preceding two. The Fibonacci numbers are recursively defined by $F_{n+1} = F_n + F_{n-1}$, where $F_0 = 0$, $F_1 = 1$. The first few are

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, ...

The image on the cover is that of a Fibonacci spiral.

Trigonometric Ratios and Transformations

1



Contents

- 1.1 Angles and Their Measures
- 1.2 Trigonometric Ratios
- 1.3 Periodicity and Variance
- 1.4 Trigonometric Ratios of Compound Angles
- 1.5 Trigonometric Ratios of Multiple and Submultiple Angles
- 1.6 Sum and Product Transformations

Worked-Out Problems
Summary
Exercises
Answers

Trigonometric Ratios: The relationships between the angles and the sides of a right triangle are expressed in terms of trigonometric ratios.

CHAPTER OPENER

Each chapter starts with an opening vignette, definition of the topic, and contents of the chapter that give you an overview of the chapter to help you see the big picture.

CLEAR, CONCISE, AND INVITING WRITING

Special attention has been paid to present an engaging, clear, precise narrative in the layout that is easy to use and designed to reduce math anxiety students may have.

DEFINITIONS

Every new topic or concept starts with defining the concept for students. Related examples to aid the understanding follow the definition.

DEFINITION 7.1 Random Experiment A random experiment is an experiment in which

1. The experiment can be repeated any number of times under identical conditions.
2. All possible outcomes of the experiment are known in advance.
3. The actual outcome in a particular experiment is not known in advance.

Examples

(1) Rolling of an Unbiased Die

Rolling of an unbiased die is a random experiment in which all the possible outcomes are 1, 2, 3, 4, 5 and 6, if we denote the six faces of the die with the numbers 1, 2, 3, 4, 5 and 6. The faces of the die may also contain dots in numbers 1, 2, 3, 4, 5 and 6. In any case, we identify the faces of a die with the numbers 1, 2, 3, 4, 5 and 6. The actual outcome in a particular experiment (rolling of the die) is the number that appears on the uppermost face of the die and this is not known in advance.

This experiment can be performed any number of times under identical conditions.

(2) Tossing of an Unbiased Coin

Tossing of an unbiased coin is a random experiment in which there are only two possible outcomes, namely, Head (H) and Tail (T). In a particular experiment (tossing of the coin), the outcome is not known in advance. This experiment can also be performed any number of times under identical conditions and therefore this is a random experiment.

“Tossing of an unbiased coin till tail appears” is also a random experiment. However, experiments such as “measuring the acceleration due to gravity using a compound pendulum” is not a random experiment, since the experiment cannot be repeated under identical conditions and the possible outcomes are not known in advance.

Throughout this chapter, by a coin or die we always mean an unbiased coin (or a fair coin) or unbiased die (or a fair die) unless otherwise mentioned.

DEFINITION 7.2 Some of the important definitions are as follows:

1. A set of possible outcomes of an experiment is called an *event*.
2. Two or more events are said to be *exhaustive* if the performance of the experiment always results in the occurrence of atleast one of them.
3. Two or more events are said to be *mutually exclusive* if the occurrence of one of the events prevents the occurrence of any of the remaining events.
4. Two or more events are said to be *equally likely* (or *equiprobable*) if there is no reason to expect one of them in preference to others.

EXAMPLES

Example 7.3

Suppose that an integer is picked from among 1 to 20 (both inclusive). What is the probability of picking a prime?

Solution: There are 20 outcomes of the experiment of picking an integer. The primes between 1 and 20 are

2, 3, 5, 7, 11, 13, 17 and 19 and these are 8 in number. Therefore, 8 are favourable to the event of picking a prime and hence the probability of picking a prime is

$$\frac{8}{20} = \frac{2}{5}$$

Example 7.4

Suppose that 8 dice are thrown. Find the probability that none of the dice shows 3 (on the upper most face).

Solution: Any outcome of “throwing 8 dice” can be expressed as an 8-tuple of integers from 1 to 6 and hence the total number of possible outcomes is 6^8 .

An outcome that none of the dice shows 3 can be expressed as 8-tuple of integers from the 5-element set $\{1, 2, 4, 5, 6\}$ and there are 5^8 such outcomes. Thus, the probability that none of the dice shows 3 is

$$\frac{5^8}{6^8} = \left(\frac{5}{6}\right)^8$$

Example 7.5

Suppose that a bag contains 6 red, 5 black and 4 blue balls. Find the probability that three balls drawn simultaneously are one blue, one black and one red.

Solution: The total number of balls is

$$6 + 5 + 4 = 15$$

Out of these 15 balls, 3 balls can be drawn in ${}^{15}C_3$ ways. Therefore, 3 balls can be drawn simultaneously in

$$\frac{15 \cdot 14 \cdot 13}{1 \cdot 2 \cdot 3} = 455 \text{ ways}$$

Drawing one blue, one black and one red ball simultaneously can be expressed as a triple (a_1, a_2, a_3) , where $1 \leq a_1 \leq 4$, $1 \leq a_2 \leq 5$ and $1 \leq a_3 \leq 6$. The number of such tuples is

$$4 \times 5 \times 6 = 120$$

Thus, the probability that 3 balls drawn simultaneously are one blue, one black and one red is

$$\frac{120}{455} = \frac{24}{91}$$

Examples pose a specific problem using concepts already presented and then work through the solution. These serve to enhance the students' understanding of the subject matter.

THEOREMS

Relevant theorems are provided along with proofs to emphasize conceptual understanding rather than rote learning.

DEFINITION 1.8 Complementary Angles Two angles are said to be *complementary* if their sum is a right angle. Therefore, for $0 \leq \theta < \pi/2$, θ and $\pi/2 - \theta$ are complementary.

THEOREM 1.6 Let θ and ψ be complementary angles. Then

1. $\sin \theta = \cos \psi$ and $\cos \theta = \sin \psi$
2. $\tan \theta = \cot \psi$ and $\cot \theta = \tan \psi$
3. $\sec \theta = \operatorname{cosec} \psi$ and $\operatorname{cosec} \theta = \sec \psi$

PROOF

Draw a right-angled triangle OAP with $\widehat{OAP} = 90^\circ$ and $\widehat{AOP} = \theta$ (Figure 1.12). Since the sum of the angles of any triangle is 180° and since $\theta + \psi = 90^\circ$, it follows that $\widehat{OPA} = \psi$.

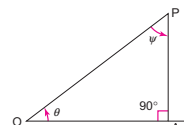


FIGURE 1.12 Theorem 1.6.



QUICK LOOK 4

Unlike the trigonometric functions sine and cosine, the trigonometric functions secant, tangent, cosecant and cotangent are not defined on the whole real line. The functions $\sec x$ and $\tan x$ are not defined at $x = (2n + 1)\pi/2$, $n \in \mathbb{Z}$, since $\cos x = 0$ at these x . Similarly cosec

and $\cot x$ are not defined at $x = n\pi$, $n \in \mathbb{Z}$, since $\sin x = 0$ at these x . The domains of all the trigonometric functions are collected in Table 1.1 for convenience and for quick reference.

QUICK LOOK

Some important formulae and concepts that do not require exhaustive explanation, but their mention is important, are presented in this section. These are marked with a **magnifying glass**.

TRY IT OUT

Within each chapter the students would find problems to reinforce and check their understanding. This would help build confidence as one progresses in the chapter. These are marked with a **pointed finger**.

Example 7.8

Let $S = \{1, 2, 3, 4, 5, 6\}$ be the sample space of a random experiment. Define $P(E) = \sum_{s \in E} P(s)$ for any $E \subseteq S$, where

$$P(1) = 0.1$$

$$P(2) = 0.2 = P(3)$$

$$P(4) = 0.4$$

$$P(5) = 0.05 = P(6)$$

Show that P is a probability function.

Solution: We have

(1) Positive axiom:

$$P(E) = \sum_{s \in E} P(s) > 0$$

since $P(s) > 0$ for all $s \in S$.

(2) Completeness axiom:

$$\begin{aligned} P(S) &= \sum_{s \in S} P(s) = P(1) + P(2) + P(3) + P(4) + P(5) + P(6) \\ &= 0.1 + 0.2 + 0.2 + 0.4 + 0.05 + 0.05 \\ &= 1 \end{aligned}$$

(3) Union axiom: For any E_1 and $E_2 \in \wp(S)$ with $E_1 \cap E_2 = \emptyset$, we have

$$P(E_1 \cup E_2) = \sum_{s \in E_1 \cup E_2} P(s) = \sum_{s \in E_1} P(s) + \sum_{s \in E_2} P(s) = P(E_1) + P(E_2)$$

since $s \in E_1 \cup E_2$ if and only if $s \in E_1$ or $s \in E_2$ but not both.

Thus P is a probability function.

Try it out Let $S = \{H, T\}$ be the sample space of a random experiment ξ and define

$$P(H) = \frac{1}{2} = P(T)$$

$$P(\emptyset) = 0$$

and

$$P(S) = 1$$

Then verify that P is a probability function.

SUMMARY

7.1 Random experiment: An experiment is called random experiment if the following conditions are satisfied.

- (1) The experiment can be repeated any number of times under similar conditions.
- (2) All possible outcomes of the experiment are known in advance
- (3) The actual outcome in a particular experiment cannot be exactly predicted.

7.2 Sample space and event: The set of all possible outcomes of a random experiment is called *sample space* of the experiment and any subset of the sample space is called an *event*.

7.3 Exhaustive events: Two or more events are called *exhaustive events* if the performance of the experiment results in the occurrence of at least one of these events.

7.4 Mutually exclusive events: Two or more events are said to be *mutually exclusive* if the occurrence of one of the events prevents the occurrence of any one of the other events.

7.5 Equally likely events: Two or more events are said to be *equally likely* (or *equiprobable*) if there is no reason to expect one of them in preference to the others.

7.6 Probability (classical definition): Suppose in a random experiment there are n exhaustive, mutually exclusive, equally likely outcomes. If m of them are favourable to an event E , then the probability $P(E)$ of E is defined as

7.8 Usual probability: The classical probability is also called usual probability.

7.9 Sample points and sample space: Any possible outcome of a random experiment is called a sample point and the set of all sample points is called the sample space of the random experiment. An elementary event means a sample point. Generally sample space is denoted by S .

7.10 Finite sample space: A set A is called finite if either A is an empty set or it is bijective with the set $\{1, 2, 3, \dots, n\}$ for some positive integer n . If a sample space is finite, then it is called a finite sample space.

7.11 Countably infinite sample space: A set A is called countably infinite set if it is bijective with the set $\mathbb{Z}^+$ of all positive integers. If the sample space of a random experiment is countably infinite set, then the sample space is called countably infinite. For example, tossing a fair coin till head appears has a countably infinite sample space.

7.12 Definition: Here afterwards events mean subsets of the sample space. If A and B are two events, then $A \cup B$ means at least one of A or B and $A \cap B$ means both A and B . Impossible event is denoted by empty set $\emptyset$ and a certain event means the entire sample space.

7.13 Various events in set theoretical form:

- (1) Events $E_1, E_2, \dots, E_n$ are said to be *mutually exclusive* if $E_i \cap E_j = \emptyset$ for $i \neq j$.
- (2) $E_1, E_2, \dots, E_n$ are called *exhaustive events* if $E_1 \cup E_2 \cup \dots \cup E_n = S$ (sample space).

SUMMARY

At the end of every chapter, a summary is presented that organizes the key formulae and theorems in an easy to use layout. The related topics are indicated so that one can quickly summarize a chapter.

B.**WORKED-OUT PROBLEMS AND ASSESSMENT – AS PER IIT-JEE PATTERN**

Mere theory is not enough. It is also important to practice and test what has been proved theoretically. The worked-out problems and exercise at the end of each chapter are in resonance with the IIT-JEE paper pattern. Keeping the IIT-JEE pattern in mind, the worked-out problems and exercises have been divided into:

1. Single Correct Choice Type Questions
2. Multiple Correct Choice Type Questions
3. Matrix-Match Type Questions
4. Comprehension-Type Questions
5. Assertion–Reasoning Type Questions
6. Integer Answer Type Questions

WORKED-OUT PROBLEMS

In-depth solutions are provided to all worked-out problems for students to understand the logic behind and formula used.

WORKED-OUT PROBLEMS**Single Correct Choice Type Questions**

1. If $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = \tan^2 \theta + \cot^2 \theta + k$, then k equals
(A) 9 (B) 7 (C) 5 (D) 8

Solution:

$$\begin{aligned}\text{LHS} &= \sin^2 \theta + 2 + \operatorname{cosec}^2 \theta + \cos^2 \theta + 2 + \sec^2 \theta \\ &= 5 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) \\ &= \tan^2 \theta + \cot^2 \theta + 7\end{aligned}$$

Answer: (B)

2. If $\sin \theta + \sin^2 \theta = 1$, then $\cos^{12} \theta + 3\cos^{10} \theta + 3\cos^8 \theta + \cos^6 \theta$ is equal to
(A) 0 (B) 2 (C) 1 (D) 4

Solution: Given that $\sin \theta + \sin^2 \theta = 1$. This implies $\sin \theta = \cos^2 \theta$.

$$\begin{aligned}1 &= \cos^2 \theta + \sin^2 \theta \\ &= m^4 n^2 + m^2 n^4 \\ &= m^2 n^2 (m^2 + n^2)\end{aligned}$$

Answer: (D)

4. If $a \sec \theta = 1 - b \tan \theta$ and $a^2 \sec^2 \theta = 5 + b^2 \tan^2 \theta$, then $a^2 b^2 + 4a^2 = kb^2$ where value of k is
(A) 3 (B) 4 (C) 5 (D) 9

Solution: $(1 - b \tan \theta)^2 = a^2 \sec^2 \theta = 5 + b^2 \tan^2 \theta$
Therefore

$$\begin{aligned}1 - 2b \tan \theta + b^2 \tan^2 \theta &= 5 + b^2 \tan^2 \theta \\ \tan \theta &= \frac{-2}{b}\end{aligned}$$

SINGLE CORRECT CHOICE TYPE QUESTIONS

These are the regular multiple choice questions with four choices provided. Only one among the four choices will be the correct answer.

MULTIPLE CORRECT CHOICE TYPE QUESTIONS

Multiple correct choice type questions have four choices provided, but one or more of the choices provided may be correct.

Multiple Correct Choice Type Questions

1. If $\cos A = -7/25$ and A lies between 450° and 540° then

- (A) $\sin \frac{A}{2} = \frac{-4}{5}$ (B) $\cos \frac{A}{2} = \frac{-3}{5}$
(C) $\sin \frac{A}{2} = \frac{-3}{5}$ (D) $\cos \frac{A}{2} = \frac{-4}{5}$

Solution: By hypothesis $450^\circ < A < 540^\circ$. This implies $225^\circ < A/2 < 270^\circ$. So

$$\left| \sin \frac{A}{2} \right| = \sqrt{\frac{1 - \cos A}{2}} = \sqrt{\frac{1 + (7/25)}{2}} = \frac{4}{5}$$

Now $A/2$ lies in the third quadrant. This means

$$\sin \frac{A}{2} = \frac{-4}{5}$$

Again

$$\left| \cos \frac{A}{2} \right| = \sqrt{\frac{1 + \cos A}{2}} = \sqrt{\frac{1 - (7/25)}{2}} = \frac{3}{5}$$

$$\begin{aligned}\sin 22\frac{1}{2}^\circ &= +\sqrt{1 - \cos^2 22\frac{1}{2}^\circ} \\ &= +\sqrt{1 - \frac{2 + \sqrt{2}}{4}} = \frac{\sqrt{2 - \sqrt{2}}}{2}\end{aligned}$$

Answers: (A), (B), (C), (D)

3. Let $\cos x + \cos y = a$, $\cos 2x + \cos 2y = b$ and $\cos 3x + \cos 3y = c$. Which of the following is (are) true?

- (A) $\cos^2 x + \cos^2 y = 1 + \frac{b}{2}$
(B) $\frac{1}{4}(2a^2 - b - 2) = \cos x \cos y$
(C) $2a^3 + c = 3a(1 + b)$
(D) $abc = 0$ for all real x and y

Solution:

$$\cos^2 x + \cos^2 y + 2 \cos x \cos y = a^2 \quad (1.11)$$

MATRIX-MATCH TYPE QUESTIONS

These questions are the regular “Match the Following” variety. Two columns each containing 4 subdivisions or first column with four subdivisions and second column with more subdivisions are given and the student should match elements of column I to that of column II. There can be one or more matches.

Matrix-Match Type Questions

1. Match the items of Column I with those of Column II.

Column I	Column II
(A) $\left(\frac{\sin 2\theta}{1 + \cos 2\theta}\right)\left(\frac{\cos \theta}{1 + \cos \theta}\right)$ is equal to	(p) $\cot \frac{\theta}{2}$
(B) $\left(\frac{\sin 2\theta}{1 - \cos 2\theta}\right)\left(\frac{1 - \cos \theta}{\cos \theta}\right)$ equals	(q) $\tan \frac{\theta}{2}$
(C) $\frac{\cot \theta - 1}{\cot \theta + 1} =$	(r) $\frac{1 - \sin 2\theta}{\cos 2\theta}$
(D) $\sec \theta + \tan \theta =$	(s) $\frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}}$

Solution:

$$\begin{aligned} \text{(A)} \quad & \left(\frac{\sin 2\theta}{1 + \cos 2\theta}\right)\left(\frac{\cos \theta}{1 + \cos \theta}\right) = \left(\frac{2\sin \theta \cos \theta}{2\cos^2 \theta}\right)\left(\frac{1}{2\cos^2(\theta/2)}\right) \\ & = \frac{2\sin(\theta/2)\cos(\theta/2)}{2\cos^2(\theta/2)} \\ & = \tan \frac{\theta}{2} \end{aligned}$$

$$\begin{aligned} \text{(D)} \quad \sec \theta + \tan \theta &= \frac{1 + \sin \theta}{\cos \theta} = \frac{\left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2}\right)^2}{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}} \\ &= \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} = \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} \end{aligned}$$

Answer: (D) → (s)

2. Match the items of Column I with those of Column II.

Column I	Column II
(A) The value of $\cos^2 \theta + \cos^2(60^\circ + \theta) + \cos^2(60^\circ - \theta)$ is	(p) $\frac{3}{2}$
(B) $\cos 20^\circ \cos 40^\circ \cos 80^\circ$	(q) $\frac{1}{8}$
(C) $\sin^2 \theta + \sin^2(120^\circ + \theta) + \sin^2(120^\circ - \theta)$ equals	(r) $\frac{\sqrt{3}}{8}$
(D) $\sin 20^\circ \sin 40^\circ \sin 80^\circ$ is equal to	(s) $\frac{3}{8}$

COMPREHENSION-TYPE QUESTIONS

Comprehension-Type Questions

1. **Passage:** $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$. Based on this information, answer the following questions.

(i) $\sin A + 2\sin 3A + \sin 5A$ is equal to

- (A) $4\sin 3A \cos^2 A$ (B) $4\cos 3A \sin^2 A$
(C) $4\sin 3A \sin^2 A$ (D) $4\cos 3A \cos^2 A$

- (A) $\cot 2A$ (B) $\tan 2A$
(C) $2\cot 2A$ (D) $2\tan 2A$

Solution: From the given information, we have

$$2\sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2\cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2\cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2\sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$\begin{aligned} \text{(i)} \quad \sin A + 2\sin 3A + \sin 5A &= (\sin A + \sin 5A) + 2\sin 3A \\ &= 2\sin 3A \cos 2A + 2\sin 3A \\ &= 2\sin 3A(\cos 2A + 1) \\ &= 4\sin 3A \cos^2 A \end{aligned}$$

Answer: (A)

$$\begin{aligned} \text{(ii)} \quad \text{Numerator} &= (\sin A + \sin 5A) + (\sin 2A + \sin 4A) \\ &= 2\sin 3A \cos 2A + 2\sin 3A \cos A \\ &= 2\sin 3A(\cos 2A + \cos A) \\ \text{Denominator} &= (\cos A + \cos 5A) + (\cos 2A + \cos 4A) \\ &= 2\cos 3A \cos 2A + 2\cos 3A \cos A \\ &= 2\cos 3A(\cos 2A + \cos A) \end{aligned}$$

$$\text{(ii)} \quad \frac{\sin A + \sin 2A + \sin 4A + \sin 5A}{\cos A + \cos 2A + \cos 4A + \cos 5A} =$$

- (A) $\cot 3A$ (B) $\tan 3A$
(C) $2\cot 3A$ (D) $2\tan 3A$

$$\text{(iii)} \quad \frac{\cos 7A + \cos 3A - \cos 5A - \cos A}{\sin 7A - \sin 3A - \sin 5A + \sin A} =$$

(ii) $(\cot B + \cot C)(\cot C + \cot A)(\cot A + \cot B)$ equals

- (A) $\sin A \sin B \sin C$
(B) $\sec A \sec B \sec C$
(C) $\operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C$
(D) $\cos A \cos B \cos C$

$$\text{(iii)} \quad \frac{1 + \cos A - \cos B + \cos C}{1 + \cos A + \cos B - \cos C} =$$

- (A) $\cot \frac{B}{2} \tan \frac{C}{2}$ (B) $\tan \frac{B}{2} \cot \frac{C}{2}$
(C) $\tan \frac{A}{2} \cot \frac{B}{2}$ (D) $\tan \frac{C}{2} \cot \frac{A}{2}$

Solution:

$$\begin{aligned} \text{(i)} \quad 1 - 2\sin B \sin C \cos A + \cos^2 A &= 1 + \cos^2 A - [\cos(B - C) - \cos(B + C)]\cos A \\ &= 1 + \cos^2 A - \cos(B - C)\cos A + \cos(B + C)\cos A \\ &= 1 + \cos^2 A + \cos(B - C)\cos(B + C) - \cos^2 A \\ &= 1 + \cos^2 B - \sin^2 C = \cos^2 B + \cos^2 C \end{aligned}$$

Answer: (A)

$$\text{(ii)} \quad (\cot B + \cot C)(\cot C + \cot A)(\cot A + \cot B)$$

Comprehension-type questions consist of a small passage, followed by three multiple choice questions. The questions are of single correct answer type.

ASSERTION-REASONING TYPE QUESTIONS

These questions check the analytical and reasoning skills of the students. Two statements are provided – Statement I and Statement II. The student is expected to verify if (a) both statements are true and if both are true, verify if statement I follows from statement II; (b) both statements are true and if both are true, verify if statement II is not the correct reasoning for statement I; (c), (d) which of the statements is untrue.

Assertion-Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I.
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I.
 (C) Statement I is true and Statement II is false.
 (D) Statement I is false and Statement II is true.

1. Statement I: If $0 < \theta < \pi/2$ and $\cos 2\theta = \sqrt{2}(\cos \theta - \sin \theta)$, then $\tan \theta$ is equal to 1.

Statement II: If $0 < \alpha, \beta < \pi/2$ and $\sin(\alpha + \beta) = 1$, then $\alpha + \beta = \pi/2$.

Solution: Statement II is clear. Now

$$\begin{aligned}\cos 2\theta &= \sqrt{2}(\cos \theta - \sin \theta) \Rightarrow \cos^2 \theta - \sin^2 \theta \\ &= \sqrt{2}(\cos \theta - \sin \theta)\end{aligned}$$

This is true when $\cos \theta = \sin \theta$ (i.e., $\theta = \pi/4$) in which case $\tan \theta = 1$. Suppose $\cos \theta \neq \sin \theta$. Therefore

$$\begin{aligned}\cos \theta + \sin \theta &= \sqrt{2} \\ \sin\left(\frac{\pi}{4} + \theta\right) &= 1 \\ \frac{\pi}{4} + \theta &= \frac{\pi}{2}\end{aligned}$$

This implies that either

$$\sin \frac{\alpha + \beta}{2} = 0$$

$$\text{or} \quad \cos\left(\frac{\alpha - \beta}{2}\right) = -\sqrt{3} \sin \frac{\alpha - \beta}{2}$$

Therefore

$$\alpha + \beta = 2n\pi \quad \text{or} \quad \tan\left(\frac{\alpha - \beta}{2}\right) = \frac{-1}{\sqrt{3}} = \tan\left(\frac{-\pi}{6}\right)$$

$$\alpha + \beta = 2n\pi \quad \text{or} \quad \frac{\alpha - \beta}{2} = n\pi - \frac{\pi}{6}$$

$$\alpha + \beta = 2n\pi \quad \text{or} \quad \alpha = \beta + 2n\pi - \frac{\pi}{3}$$

Now $\alpha + \beta = 2n\pi$ implies

$$\sin 3\alpha = \sin(6n\pi - 3\beta) \quad \text{or} \quad \sin 3\alpha = \sin(6n\pi + 3\beta - \pi)$$

In any case $\sin 3\alpha = -\sin 3\beta$ so that $\sin 3\alpha + \sin 3\beta = 0$.

Answer: (A)

3. Statement I: $\sin(\pi/18)$ is a root of $8x^3 - 6x + 1 = 0$.

Statement II: For any real α , $\sin 3\alpha = 3\sin \alpha - 4\sin^3 \alpha$.

Solution: Statement II is a standard formula. Put $\alpha = \pi/18$. Therefore

INTEGER-TYPE QUESTIONS

The questions in this section are numerical problems for which no choices are provided. The students are required to find the exact answers to numerical problems and enter the same in OMR sheets. Answers can be one-digit or two-digit numerals.

Integer Answer Type Questions

1. In $\triangle ABC$, $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = k \sin A \sin B \sin C$ where k equals ____.

Solution: $A + B + C = \pi \Rightarrow B + C - A = \pi - 2A$, etc. Therefore

$$\begin{aligned}\text{LHS} &= \sin(\pi - 2A) + \sin(\pi - 2B) + \sin(\pi - 2C) \\ &= \sin 2A + \sin 2B + \sin 2C \\ &= 4 \sin A \sin B \sin C\end{aligned}$$

Answer: 4

2. If $A + B + C = \pi$, then

$$\sin^2 \frac{A}{2} - \sin^2 \frac{C}{2} + \sin^2 \frac{B}{2} = p - q \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

where $p + q$ is ____

Solution:

$$\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2}$$

$$= 1 - 2 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

Therefore $p = 1, q = 2$.

Answer: 3

3. If $\frac{1}{\tan 3\theta + \tan \theta} - \frac{1}{\cot 3\theta + \cot \theta} = \cot k\theta$ then the value of k is ____

Solution:

$$\begin{aligned}\frac{1}{\tan 3\theta + \tan \theta} - \frac{1}{\cot 3\theta + \cot \theta} &= \frac{1}{\tan 3\theta + \tan \theta} - \frac{\tan \theta \tan 3\theta}{\tan 3\theta + \tan \theta} \\ &= \frac{1 - \tan 3\theta \tan \theta}{\tan 3\theta + \tan \theta} = \frac{1}{\tan 4\theta} = \cot 4\theta\end{aligned}$$

Answer: 4

EXERCISES

EXERCISES

Single Correct Choice Type Questions

- The value of $\sin 36^\circ =$
 (A) $\frac{(\sqrt{5}+1)\sqrt{10+2\sqrt{5}}}{8}$ (B) $\frac{\sqrt{5}+1}{4}$
 (C) $\frac{\sqrt{10+2\sqrt{5}}}{8}$ (D) $\frac{(\sqrt{5}-1)\sqrt{10+2\sqrt{5}}}{8}$
- If $a = \frac{2\sin\theta}{1+\cos\theta+\sin\theta}$, then $\frac{1-\cos\theta+\sin\theta}{1+\sin\theta}$ is equal to
 (A) $1-a$ (B) $1+a$ (C) a (D) $1/a$
- Suppose that $\sin^3 x \sin 3x = \sum_{m=0}^n A_m \cos mx$, where $A_0, A_1, \dots, A_n$ are constants and $A_n \neq 0$, then the value of n is
 (A) 8 (B) 6 (C) 4 (D) 9
- If a, b, c and d are smallest positive angles in the ascending order such that the sine of each angle is equal to a positive constant λ , then $4\sin(a/2) + 3\sin(b/2) + 2\sin(c/2) + \sin(d/2)$ is equal to
 (A) $2\sqrt{1+\lambda}$ (B) $\sqrt{1+\lambda}$

Multiple Correct Choice Type Questions

- Let $x = \left(\frac{\cos A + \cos B}{\sin A - \sin B} \right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B} \right)^n$
 Then
 (A) $x = 0$ if n is an odd positive integer
 (B) $x = \tan^n(A-B)/2$ if n is an even positive integer
 (C) $x = 2 \cot^n(A-B)/2$ if n is an even positive integer
 (D) $x = 0$ if n is an even positive integer
- Which of the following statements are true?
 (A) If $\sin \theta = (m^2 - n^2)/(m^2 + n^2)$, then
 (A) $\tan \theta = \frac{m^2 - n^2}{2mn}$ (B) $\operatorname{cosec} \theta = \frac{m^2 + n^2}{m^2 - n^2}$
 (C) $\sec \theta = \frac{m^2 + n^2}{2mn}$ (D) $\cot \theta = \frac{2mn}{m^2 + n^2}$
- Which of the following are true?
 (A) $\tan 20^\circ + \tan 72^\circ + \tan 88^\circ = \tan 20^\circ \cdot \tan 72^\circ \cdot \tan 88^\circ$
 (B) $\tan 51^\circ + \tan 62^\circ + \tan 67^\circ = \tan 51^\circ \cdot \tan 62^\circ \cdot \tan 67^\circ$
 (C) $\cot 27^\circ + \cot 32^\circ + \cot 31^\circ = \cot 27^\circ \cdot \cot 32^\circ \cdot \cot 31^\circ$
 (D) $\sqrt{3} + \tan 40^\circ + \tan 80^\circ = \sqrt{3} \tan 40^\circ \tan 80^\circ$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with one or more statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are (A) $\rightarrow$ (p),(s); (B) $\rightarrow$ (q),(s),(t); (C) $\rightarrow$ (r); (D) $\rightarrow$ (r),(t); that is if the

matches are (A) $\rightarrow$ (p) and (s); (B) $\rightarrow$ (q),(s) and (t); (C) $\rightarrow$ (r); and (D) $\rightarrow$ (r),(t) then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

Comprehension-Type Questions

- Passage:** The equation $a \cos \theta + b \sin \theta = c$ can be solved when $|c| \leq \sqrt{a^2 + b^2}$. Based on this answer the following questions (i), (ii) and (iii).
 (i) If $0 \leq x \leq \pi$ and $\sin x + \cos x = 1$, then the number of values of x is
 (A) 1 (B) 2 (C) 4 (D) 0
 (ii) The maximum and minimum values of $3 \cos \theta + 4 \sin \theta - 5$ are respectively (θ is real)
 (A) 5, -5 (B) 4, 3 (C) 0, -10 (D) 4, -3
 (iii) The maximum and minimum values of $5 \sin \theta + 12 \cos \theta + 13$ are respectively (θ is real)
 (A) 13
- Passage:** To eliminate a parameter, we need two equations involving the parameter. For example, if $x = r \cos \theta$ and $y = r \sin \theta$, then by squaring and adding, we have $x^2 + y^2 = r^2$. This shows that θ is eliminated from the given equations.
 Answer the following questions (i), (ii) and (iii).
 (i) If $\tan \theta + \sin \theta = a$ and $\tan \theta - \sin \theta = b$, then after eliminating θ , $(a^2 - b^2)^2$ is equal to
 (A) $4\sqrt{ab}$ (B) $4ab$ (C) $16\sqrt{ab}$ (D) $16ab$
 (ii) Eliminating θ from the equations $x = \cot \theta + \tan \theta$ and $y = \sec \theta - \cos \theta$, we have

Assertion-Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- Both Statements I and II are true and Statement II is a correct explanation for Statement I
- Both Statements I and II are true but Statement II is not a correct explanation for Statement I
- Statement I is true and Statement II is false
- Statement I is false and Statement II is true

- Statement I:** If $x \cos \alpha + y \sin \alpha = x \cos \beta + y \sin \beta = 2a$ and $2 \sin \alpha / 2 \sin \beta / 2 = 1$, then $y^2 = 4a(a-x)$

Statement II: $\sin^2 \alpha + \sin^2 \beta = 1$

- Statement I:** For all values of θ , $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) = -1$

Statement II: $x^3 + y^3 = (x+y)(x^2 - xy + y^2)$

- Statement I:** If $\tan(\theta/2) = m$, then

$$\frac{1 - 2 \sin^2(\theta/2)}{1 + \sin \theta} = \frac{1 + m}{1 - m}$$

Statement II: $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$ and $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

- Statement I:** If

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
☐	☐	☐	☐
☐	☐	☐	☐
☐	☒	☐	☐
☐	☐	☐	☐
☐	☒	☐	☐
☐	☐	☐	☐

- If $2 \tan(A+B) = 3 \tan A$, then $\sin(2A+B) = k \sin B$, where k is equal to ____.
- If $3 \cos x = 2 \cos(x-2y)$, then $\cot(x-y) \cot y$ is equal to ____.
- If $\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \dots + \sin 178^\circ = n \cot 1^\circ$, then n is equal to ____.
- Let $0 < \alpha_j < \pi/2$ for $j = 1, 2, 3, \dots, 8$ and $\cot \alpha_1 \cdot \cot \alpha_2 \cdot \dots \cdot \cot \alpha_8 = 1$. If M is the maximum value of $\cot \alpha_1 \cot \alpha_2 \cdot \dots \cdot \cot \alpha_8$, then the value of $(32)M$ is ____.
- The value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$ is ____.

For self-assessment, each chapter has adequate number of exercise problems where the questions have been subdivided into the same categories as asked in IIT-JEE.

ANSWERS

The Answer key at the end of each chapter contains answers to all exercise problems.

ANSWERS

Single Correct Choice Type Questions

- | | |
|---------|---------|
| 1. (D) | 29. (A) |
| 2. (C) | 30. (C) |
| 3. (A) | 31. (A) |
| 4. (B) | 32. (A) |
| 5. (D) | 33. (A) |
| 6. (B) | 34. (B) |
| 7. (A) | 35. (C) |
| 8. (C) | 36. (A) |
| 9. (D) | 37. (B) |
| 10. (C) | 38. (D) |
| 11. (A) | 39. (A) |
| 12. (C) | 40. (A) |
| 13. (B) | 41. (B) |
| 14. (A) | 42. (B) |
| 15. (D) | 43. (A) |
| 16. (A) | 44. (C) |
| 17. (C) | 45. (D) |
| 18. (C) | 46. (B) |
| 19. (C) | 47. (A) |
| 20. (A) | 48. (B) |
| 21. (A) | 49. (A) |
| 22. (A) | 50. (B) |
| 23. (B) | 51. (B) |
| 24. (B) | 52. (C) |
| 25. (D) | 53. (A) |
| 26. (A) | 54. (D) |
| 27. (B) | 55. (A) |
| 28. (B) | 56. (D) |

Multiple Correct Choice Type Questions

- | | |
|-----------------------|------------------------|
| 1. (A), (C) | 9. (B), (C) |
| 2. (A), (B), (C) | 10. (A), (C), (D) |
| 3. (A), (B), (C) | 11. (A), (B), (C) |
| 4. (A), (B), (C), (D) | 12. (B), (C) |
| 5. (A), (B), (C), (D) | 13. (A), (D) |
| 6. (A), (B), (C) | 14. (A), (C), (D) |
| 7. (A), (B), (D) | 15. (A), (B), (C), (D) |
| 8. (A), (D) | 16. (A), (B), (C), (D) |

Matrix-Match Type Questions

- | | |
|--|---|
| 1. (A) $\rightarrow$ (p), (s), (B) $\rightarrow$ (q), (C) $\rightarrow$ (r), (D) $\rightarrow$ (t) | 4. (A) $\rightarrow$ (s), (B) $\rightarrow$ (r), (C) $\rightarrow$ (q), (D) $\rightarrow$ (p) |
| 2. (A) $\rightarrow$ (p), (B) $\rightarrow$ (p), (C) $\rightarrow$ (p), (D) $\rightarrow$ (s) | 5. (A) $\rightarrow$ (t), (B) $\rightarrow$ (p), (C) $\rightarrow$ (r), (D) $\rightarrow$ (q) |
| 3. (A) $\rightarrow$ (r), (B) $\rightarrow$ (s), (C) $\rightarrow$ (q), (D) $\rightarrow$ (p) | 6. (A) $\rightarrow$ (p), (B) $\rightarrow$ (p), (C) $\rightarrow$ (q), (D) $\rightarrow$ (q) |

Comprehension-Type Questions

- | | |
|---------------------------------|---------------------------------|
| 1. (i) (B); (ii) (C); (iii) (C) | 3. (i) (D); (ii) (B); (iii) (A) |
| 2. (i) (A); (ii) (B); (iii) (D) | |

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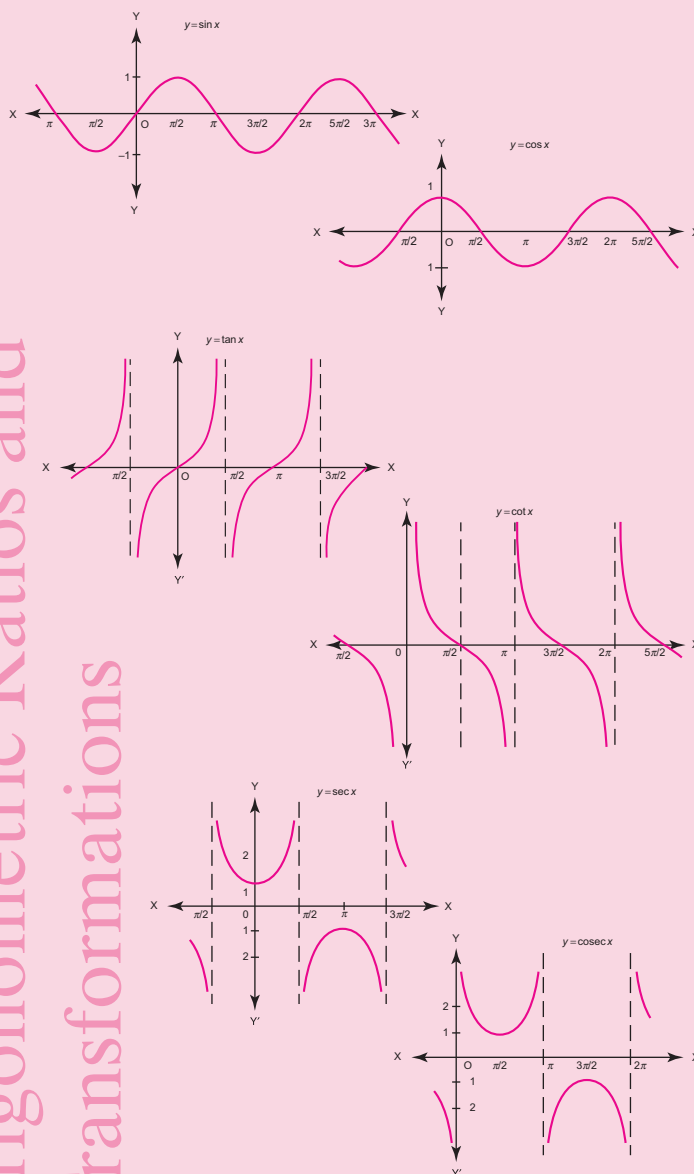
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Trigonometric Ratios and Transformations

1

Trigonometric Ratios and Transformations



Contents

- 1.1 Angles and Their Measures
- 1.2 Trigonometric Ratios
- 1.3 Periodicity and Variance
- 1.4 Trigonometric Ratios of Compound Angles
- 1.5 Trigonometric Ratios of Multiple and Submultiple Angles
- 1.6 Sum and Product Transformations

Worked-Out Problems
Summary
Exercises
Answers

Trigonometric Ratios: The relationships between the angles and the sides of a right triangle are expressed in terms of *trigonometric ratios*.

Trigonometry is the study of the relation between angles and sides of a triangle. This is an essential branch of mathematics useful for the measurement of areas, distances and heights.

1.1 | Angles and Their Measures

Consider a straight line extending indefinitely in both the directions and let O be a point on this straight line. Then the point O divides the line into two parts and each of these two parts is called a *ray* and O is called the *vertex* of the two rays. Usually rays are denoted by OA , where O is the vertex and A is any point on the ray (ray is a part of the straight line). In Figure 1.1, OA and OB are rays. The vertex of a ray is called the *origin* of the ray or the *initial point* of the ray.



FIGURE 1.1 Rays OA and OB .

DEFINITION 1.1 The figure formed by two rays with the common vertex is called an *angle* and is denoted by $\angle AOB$ or $\widehat{AOB}$, where O is the common vertex, A is a point on one ray and B is a point on the other ray as shown in Figure 1.2.

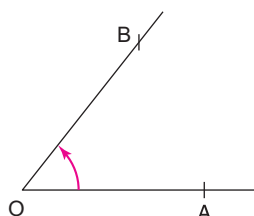


FIGURE 1.2 Angle.

A real number is associated with each angle and this number is called the *measure of the angle*. There are two important systems of measurement of an angle: the *sexagesimal system* (or British system) and the *radian system* (or *circular measure system*).

DEFINITION 1.2 Sexagesimal System An angle $\widehat{AOB}$ is called a right angle if OB is perpendicular to OA , geometrically. In sexagesimal system a right angle is divided into 90 equal parts called *degrees*. Each degree is divided to 60 equal parts called *minutes* and each minute is divided into 60 equal parts called *seconds*. One degree, one minute and one second are denoted by 1° , $1'$ and $1''$, respectively. Therefore,

$$1 \text{ right angle} = 90^\circ$$

$$1^\circ = 60'$$

and

$$1' = 60''$$

Figure 1.3 illustrates the sexagesimal system.

DEFINITION 1.3 Radian System The angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle is called a *radian* and is denoted by 1^c .

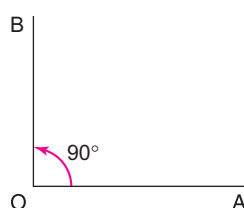


FIGURE 1.3 Sexagesimal system.

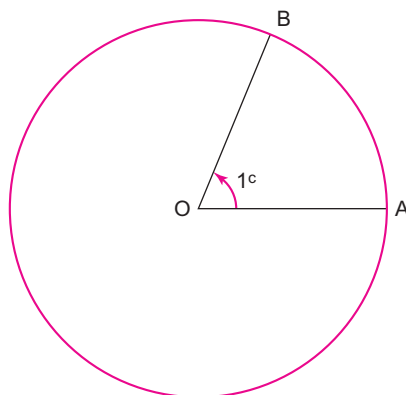


FIGURE 1.4 Radian system.

Figure 1.4 shows the radian system. In the following theorem, we prove that this angle is independent of the radius of the circle, we have considered.

THEOREM 1.1 The radian is constant, in the sense that it is independent of the radius of the circle.

PROOF Consider a circle with centre O and radius r . Let A and B be points on the circle such that the length of arc AB is equal to the radius r of the circle (Figure 1.5). Then by definition,

$$\widehat{AOB} = 1 \text{ radian} = 1^\circ$$

Produce AO to meet the circle at C . Then AC is a diameter of the circle and the length of the arc ABC is equal to half of the circumference of the circle. Therefore,

$$\widehat{AOC} = 2 \text{ right angles} = 180^\circ$$

We know that the angles subtended at the centre of a circle are proportional to the lengths of the arcs which subtend them. Therefore

$$\frac{\widehat{AOC}}{\widehat{AOB}} = \frac{\text{arc } ABC}{\text{arc } AB}$$

Also, we know that the ratio of the circumference and diameter of a circle is always a constant and is denoted by π . Therefore, the circumference of any circle is equal to $\pi d (= 2\pi r)$, where d is the length of any diameter of the circle. From these, we have

$$\frac{180^\circ}{\widehat{AOB}} = \frac{(2\pi r)/2}{r} = \pi$$

and hence $\widehat{AOB} = 180^\circ/\pi$, which is independent of the radius of the circle.

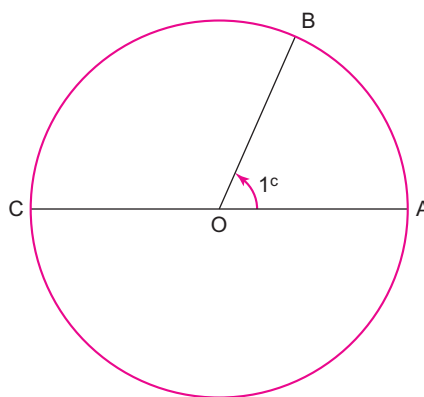


FIGURE 1.5 Theorem 1.1.

QUICK LOOK 1

1. The value of the constant π mentioned in Theorem 1.1 is approximately $22/7$ ($= 3.1415\dots$). In fact π is not a rational number; nor it is an algebraic number in the sense that π is not a root of any nonzero polynomial with rational coefficients. However, $22/7$ or $3.1415\dots$ are only approximate values of π .

2. From Theorem 1.1, it follows that

$$1^\circ = \frac{180^\circ}{\pi}$$

or π radians $= 180^\circ = 2$ right angles

$$\pi^\circ = 180^\circ$$

and therefore

$$\text{A right angle} = \left(\frac{\pi}{2}\right) \text{ radians} = 90^\circ$$

Also,

$$1^\circ = \frac{\pi}{180} \text{ radians}$$

These provide us methods to convert the measurement of an angle in sexagesimal system into that in radian system and vice-versa.

3. Usually angles are measured in radians or degrees. For an angle having degree measure θ , we write θ° . When we do not mention the superscript $^\circ$, the measure is considered to be in radians.

Example 1.1

Express $40^\circ 36' 20''$ in radian measure.

Therefore

Solution:

$$20'' = \frac{20}{60} \text{ minutes}$$

$$= \frac{20}{60 \times 60} \text{ degree} = \left(\frac{1}{180}\right)^\circ$$

$$36' = \left(\frac{36}{60}\right)^\circ = \left(\frac{6}{10}\right)^\circ$$

$$\begin{aligned} 40^\circ 36' 20'' &= \left(40 + \frac{6}{10} + \frac{1}{180}\right)^\circ \\ &= \left(40 + \frac{6}{10} + \frac{1}{180}\right) \left(\frac{\pi}{180}\right) \text{ radians} \\ &= \frac{7200 + 108 + 1}{180 \times 180} \pi \text{ radians} \\ &= \left(\frac{7309}{32400} \pi\right) \end{aligned}$$

Example 1.2

Express $(5\pi/6)^\circ$ in degrees.

$$\left(\frac{5\pi}{6}\right)^\circ = \left(\frac{5}{6} \cdot 180\right)^\circ = 150^\circ$$

Solution:

The given radian measure in degrees is

THEOREM 1.2

Let θ be the angle in radians subtended by an arc of length a at the centre of a circle of radius r . Then

$$\theta = \frac{a}{r} \text{ radians}$$

PROOF

We have from the definition of a radian,

1 radian = The angle subtended by an arc of length r at the centre of circle

Therefore

$\frac{1}{r}$ radian = The angle subtended by an arc of length 1 at the centre

$\frac{a}{r}$ radian = The angle subtended by an arc of length a at the centre

Let us consider a unit circle (a circle of unit radius) having centre at C . Let O be a point on the circle. Let CO be the initial side of an angle. Since the radius of the circle is one unit, the length of an arc of this circle will be the radian measure of the angle subtended by it at the centre C . Consider the line AOB , which is the tangent to the circle at O . We can consider AOB as the real line with O representing the real number zero, the points towards OB representing positive real numbers and those towards OA representing negative real numbers. Now, if the line OB is roped in anticlockwise direction along the circle, then every positive real number will correspond to a radian measure and conversely. Similarly, if the line OA is roped in clockwise direction along the circle, then each negative real number will correspond to a radian measure and conversely. Thus, the radian measures and real numbers are same. However, regarding angles, note that two different real numbers may represent the same angle; for example, $\pi/2$ and $5\pi/2$ represent the same angle. In fact for any angle θ , in radians, $\theta + 2n\pi$ represents the angle θ only for any integer n .

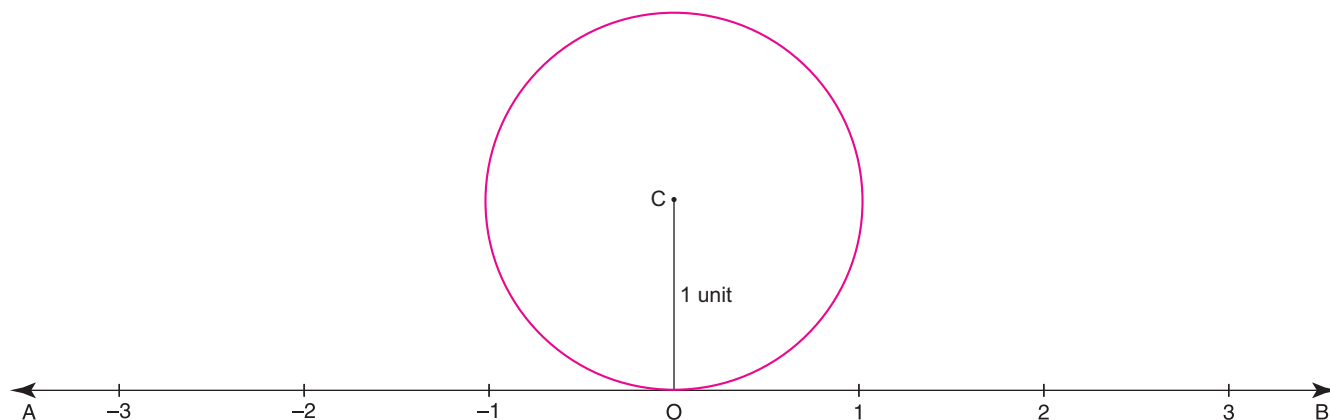


FIGURE 1.6 Circle of unit radius.

THEOREM 1.3 For any real number x , there exist unique integer n and a real number θ such that

$$x = 2n\pi + \theta \quad \text{and} \quad 0 \leq \theta < 2\pi$$

PROOF Let n be the integral part of $x/2\pi$; that is, n is the largest integer which is less than or equal to $x/2\pi$. Then

$$n \leq \frac{x}{2\pi} < n + 1$$

Therefore

$$n(2\pi) \leq x < n(2\pi) + 2\pi$$

$$0 \leq x - 2n\pi < 2\pi$$

Now, put $\theta = x - 2n\pi$. Then

$$x = 2n\pi + \theta \quad \text{and} \quad 0 \leq \theta < 2\pi$$

To prove the uniqueness of n and θ , let n and m be integers and θ and ϕ be real numbers such that

$$2n\pi + \theta = 2m\pi + \phi, \quad 0 \leq \theta < 2\pi \quad \text{and} \quad 0 \leq \phi < 2\pi$$

Then

$$|\theta - \phi| = 2|m - n|\pi \geq 2\pi \quad \text{if } m \neq n$$

and

$$|\theta - \phi| < 2\pi$$

Therefore

$$m = n \quad \text{and} \quad \theta = \phi.$$

Note: The integer n above can be considered as the number of completed revolutions of the circle in Figure 1.6 to reach x .

Example 1.3

Suppose that a clock shows half past 3. Find the angle between the hours hand and the minutes hand in degrees and radians.

Solution: From the hypothesis, the hours hand is exactly in middle of 3 and 4 and minutes hand is at 6 (Figure 1.7). The 12 divisions in the clock totally represent 360° and hence each division represents 30° . Therefore, the angle between the hours hand and the minutes hand is $30^\circ + 30^\circ + 30^\circ/2 = 75^\circ$, which is equal to $75 \times \pi/180 = 5\pi/12$ radians.

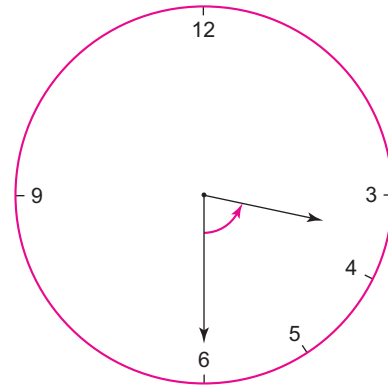


FIGURE 1.7 Example 1.3.

Example 1.4

Determine the radius of the circle in which an arc of length 16 cm subtends an angle of 80° at the centre of the circle.

Solution: If l is the length of the arc and θ is the angle subtended at the centre of the circle of radius r , we know, from Theorem 1.2, that

$$\theta = \frac{l}{r} \text{ radians}$$

We are given here that $\theta = 80^\circ$ and $l = 16$ and we have to find r . First we have to convert the measure of θ into radians. We have

$$\theta = 80^\circ = 80 \times \frac{\pi}{180} = \frac{4\pi}{9}$$

Therefore

$$\begin{aligned} \frac{4\pi}{9} &= \frac{l}{r} = \frac{16}{r} \\ r &= \frac{9 \times 16}{4\pi} = \frac{36}{\pi} \quad \left(= 36 \times \frac{7}{22} \text{ approx.} \right) \end{aligned}$$

Example 1.5

Let a triangle be given such that its angles are in arithmetic progression and that the ratio of the greatest to the smallest angles, in degrees, is $\pi:60$. Find all the angles of the triangle.

Solution: Let the angles be $a - d$, a and $a + d$, where $d > 0$. The sum of the angles in a triangle is 180° . Therefore

$$3a = 180^\circ$$

$$a = 60^\circ$$

Since $d > 0$, $a - d < a < a + d$, from the hypothesis, we have

$$\frac{a + d}{a - d} = \frac{\pi}{60} \quad \text{or} \quad \frac{60 + d}{60 - d} = \frac{180}{60} = 3$$

Therefore $d = 30^\circ$ and so 30° , 60° and 90° are all the angles of the given triangle.

1.2 | Trigonometric Ratios

In this section we will define the trigonometric ratios sine and cosine as real-valued functions in the interval $[0, 2\pi]$ first and later extend the domain of these to the entire real line. Using the definitions of these, we define the other trigonometric functions such as cosecant, secant, tangent and cotangent on appropriate domains.

DEFINITION 1.4 Sine and Cosine Ratios Let θ be a real number such that $0 \leq \theta \leq 2\pi$. Consider a rectangular coordinate system OXY . Draw a circle with centre at O and radius $r > 0$ in the coordinate plane. Choose a point P on this circle such that OP makes an angle of θ radians with OX , measured in the anticlockwise sense (see Figure 1.8).

With reference to the coordinate system OXY , let (x, y) be the coordinates of the point P . Then, we define the *sine ratio* of θ as

$$\sin \theta = \frac{y}{r}$$

and the *cosine ratio* of θ as

$$\cos \theta = \frac{x}{r}$$

First note that these ratios $\sin \theta$ and $\cos \theta$ are independent of the circle and its radius r . Consider Figure 1.9 in which two circles C_1 and C_2 of radius r_1 and r_2 are given and the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on these circles, respectively, such that O, P and Q are collinear.

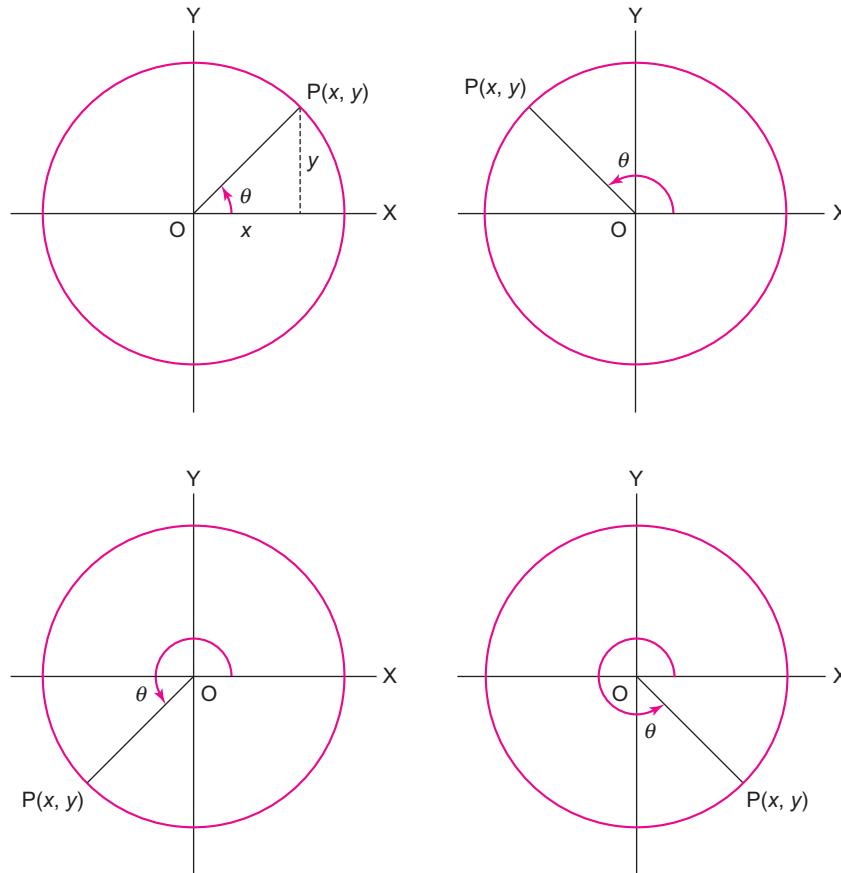


FIGURE 1.8

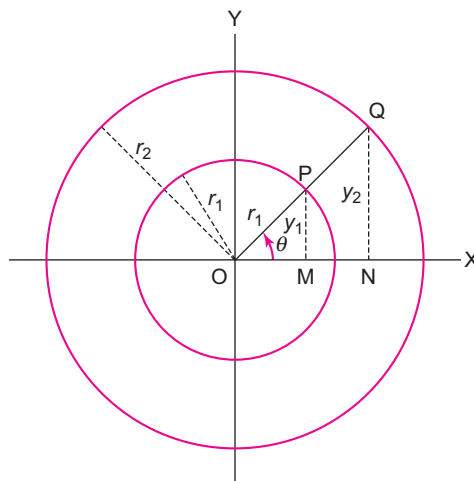


FIGURE 1.9

Let M and N be feet of the perpendiculars drawn from P and Q, respectively, on OX. Then, $PM = y_1$, $QN = y_2$. The triangles OPM and OQN are similar and hence

$$\frac{PM}{OP} = \frac{QN}{OQ}$$

Therefore

$$\frac{y_1}{r_1} = \frac{y_2}{r_2}$$

Thus, $\sin \theta$ is independent of the circle and its radius. Similarly, $\cos \theta$ is also independent of the circle and its radius.



QUICK LOOK 2

1. If $P(x, y)$ is a point on the circle of radius r with centre at the origin O such that OP makes an angle of θ radians with the ray OX in the anticlockwise sense, then we have

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta$$

2. For any $0 \leq \theta \leq 2\pi$, we have

$$\sin^2 \theta + \cos^2 \theta = 1$$

and hence

$$\sin^2 \theta = 1 - \cos^2 \theta$$

and

$$\cos^2 \theta = 1 - \sin^2 \theta$$

where $\sin^2 \theta = \sin \theta \cdot \sin \theta$ and $\cos^2 \theta = \cos \theta \cdot \cos \theta$.

3. If $P(x, y)$ is a point on the unit circle (circle of radius 1) with centre at the origin O and OP makes an angle θ radians with the X-axis, then

$$x = \cos \theta \quad \text{and} \quad y = \sin \theta$$

4. For any $0 \leq \theta \leq 2\pi$,

$$\sin \theta = 0 \Leftrightarrow y = 0 \quad (\text{since } r > 0)$$

$$\Leftrightarrow P(x, y) \text{ lies on the X-axis}$$

$$\Leftrightarrow \theta = 0 \quad \text{or} \quad \pi \text{ radians}$$

and

$$\cos \theta = 0 \Leftrightarrow x = 0$$

$$\Leftrightarrow P(x, y) \text{ lies on the Y-axis}$$

$$\Leftrightarrow \theta = \frac{\pi}{2} \quad \text{or} \quad \frac{3\pi}{2}$$

DEFINITION 1.5 Let x be a real number. Then there exist unique integer n and a real number θ such that

$$x = 2n\pi + \theta \quad \text{and} \quad 0 \leq \theta < 2\pi$$

(see Theorem 1.3). Now, we define

$$\sin x = \sin \theta$$

and

$$\cos x = \cos \theta$$

In other words, we are defining

$$\sin(2n\pi + \theta) = \sin \theta$$

and

$$\cos(2n\pi + \theta) = \cos \theta$$

for any integer n and real number θ such that $0 \leq \theta < 2\pi$.

QUICK LOOK 3

1. Sine ($\sin$) and cosine ($\cos$) are real-valued functions defined on the entire real line and both are of period 2π , that is,

$$\sin(\theta + 2\pi) = \sin \theta$$

and

$$\cos(\theta + 2\pi) = \cos \theta$$

for all real numbers.

2. $\sin x = 0 \Leftrightarrow x = n\pi$ for some integer n and $\cos x = 0 \Leftrightarrow x = (2n + 1)\pi/2$ for some integer n .

DEFINITION 1.6 Secant and Tangent For any real number $x \neq (2n + 1)\pi/2$, $n \in \mathbb{Z}$, we define

$$\secant x = \frac{1}{\cos x} \quad \text{and} \quad \text{tangent } x = \frac{\sin x}{\cos x}$$

and denote secant x and tangent x simply by $\sec x$ and $\tan x$; that is,

$$\sec x = \frac{1}{\cos x} \quad \text{and} \quad \tan x = \frac{\sin x}{\cos x}$$

for any real number x which is not of the form $(2n + 1)\pi/2$, where n is an integer.

DEFINITION 1.7 Cosecant and Cotangent For any real number $x \neq n\pi$, $n \in \mathbb{Z}$, we define

$$\text{cosecant } x = \frac{1}{\sin x} \quad \text{and} \quad \text{cotangent } x = \frac{\cos x}{\sin x}$$

and denote cosecant x and cotangent x simply by $\text{cosec } x$ and $\cot x$; that is,

$$\text{cosec } x = \frac{1}{\sin x} \quad \text{and} \quad \cot x = \frac{\cos x}{\sin x}$$

for any real number x which is not of the form $n\pi$, where n is an integer.

QUICK LOOK 4

Unlike the trigonometric functions sine and cosine, the trigonometric functions secant, tangent, cosecant and cotangent are not defined on the whole real line. The functions $\sec x$ and $\tan x$ are not defined at $x = (2n + 1)\pi/2$, $n \in \mathbb{Z}$, since $\cos x = 0$ at these x . Similarly cosec x

and $\cot x$ are not defined at $x = n\pi$, $n \in \mathbb{Z}$, since $\sin x = 0$ at these x . The domains of all the trigonometric functions are collected in Table 1.1 for convenience and for quick reference.

Table 1.1 Domains of all the trigonometric functions

Function	Domain
$\sin x$	$\mathbb{R}$
$\cos x$	$\mathbb{R}$
$\operatorname{cosec} x$	$\mathbb{R} - \{n\pi n \in \mathbb{Z}\}$
$\sec x$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2} n \in \mathbb{Z}\right\}$
$\tan x$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2} n \in \mathbb{Z}\right\}$
$\cot x$	$\mathbb{R} - \{n\pi n \in \mathbb{Z}\}$

We have discussed earlier about all the values of x for which the values of each of the trigonometric ratios $\sin x$, $\cos x$, $\tan x$, $\operatorname{cosec} x$ and $\cot x$ are defined. In the following paragraphs, we discuss the values of x for which these are positive and those values of x for which they are negative.

QUICK LOOK 5

Let θ be a real number such that $0 < \theta < 2\pi$, $\theta \neq \pi/2$, $\theta \neq \pi$ and $\theta \neq 3\pi/2$. Consider a rectangular coordinate system OXY, as in Definition 1.4, with O as origin. Let r be a positive real number. Choose a point P in the coordinate plane such that $OP = r$ and OP makes an angle of θ radians with OX measured in the anticlockwise sense. Let (x, y) be the coordinates of the point P. We have the following important observations. Recall that both x and y are nonzero since $\theta \notin \{0, \pi/2, \pi, 3\pi/2\}$.

1. If $0 < \theta < \pi/2$, then P lies in the first quadrant of the plane and hence $x > 0$ and $y > 0$, so that

$$\sin \theta = \frac{y}{r} > 0; \quad \cos \theta = \frac{x}{r} > 0$$

$$\tan \theta = \frac{y}{x} > 0; \quad \sec \theta = \frac{r}{x} > 0$$

$$\cot \theta = \frac{x}{y} > 0; \quad \operatorname{cosec} \theta = \frac{r}{y} > 0$$

Therefore the values of all the trigonometric functions are positive at θ when $0 \leq \theta \leq \pi/2$. In this case we say that θ lies in the first quadrant.

2. If $\pi/2 < \theta < \pi$, then P lies in the second quadrant of the plane and hence $x < 0$ and $y > 0$, so that $\sin \theta > 0$ and $\operatorname{cosec} \theta > 0$ and all others are negative. In this case, we say that θ lies in the second quadrant.

3. If $\pi < \theta < 3\pi/2$, then P lies in the third quadrant of the plane and hence $x < 0$ and $y < 0$, so that $\tan \theta > 0$ and $\cot \theta > 0$ and all others are negative. In this case, θ is said to be in the third quadrant.
4. If $3\pi/2 < \theta \leq 2\pi$, then P lies in the fourth quadrant of the plane and hence $x > 0$ and $y < 0$, so that $\cos \theta$ and $\sec \theta$ are positive and all the others are negative. In this case, θ is said to be in the fourth quadrant. The angles $0, \pi/2, \pi, 3\pi/2$ and 2π are called the *quadrant angles*.
5. The signs (positive or negative) of the trigonometric ratios in the four quadrants can be remembered as shown below.

I	II	III	IV
All	Silver	Tea	Cups

- In quadrant I, all are positive.
- In quadrant II, only sine (and its reciprocal cosecant) is positive.
- In quadrant III, only tangent (and its reciprocal cotangent) are positive.
- In quadrant IV, only cosine (and its reciprocal secant) is positive.

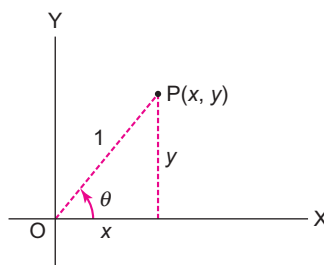
Now, we shall prove certain basic identities satisfied by the trigonometric functions. For convenience, we write $f^2(x)$ for $(f(x))^2$, where f is any function.

THEOREM 1.4

1. $\sin^2 x + \cos^2 x = 1$ for all $x \in \mathbb{R}$
2. $1 + \tan^2 x = \sec^2 x$ for all $x \in \mathbb{R} - \{(2n+1)\pi/2 | n \in \mathbb{Z}\}$
3. $1 + \cot^2 x = \operatorname{cosec}^2 x$ for all $x \in \mathbb{R} - \{n\pi | n \in \mathbb{Z}\}$

PROOF

Let $x \in \mathbb{R}$ and $x = 2n\pi + \theta$, where n is an integer and θ is a real number such that $0 \leq \theta < 2\pi$. Consider a rectangular coordinate system OXY with O as origin. Choose a point $P(x, y)$ in the coordinate plane such that $OP = 1$ and OP makes an angle of θ radians with OX measured in anticlockwise sense (Figure 1.10).

**FIGURE 1.10** Theorem 1.4.

1. By Definition 1.4,

$$\sin \theta = \frac{y}{OP} = y$$

and

$$\cos \theta = \frac{x}{OP} = x$$

Since $x^2 + y^2 = (OP)^2 = 1$, it follows that

$$\sin^2 \theta + \cos^2 \theta = 1$$

Now,

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= \sin^2(2n\pi + \theta) + \cos^2(2n\pi + \theta) \\ &= \sin^2 \theta + \cos^2 \theta \\ &= 1 \end{aligned}$$

2. Let $x \in \mathbb{R} - \{(2n+1)\pi/2 | n \in \mathbb{Z}\}$. Then $\cos x \neq 0$ and hence $\tan x$ and $\sec x$ are defined

$$\begin{aligned} 1 + \tan^2 x &= 1 + \frac{\sin^2 x}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x \end{aligned}$$

3. Let $x \in \mathbb{R} - \{n\pi | n \in \mathbb{Z}\}$. Then $\sin x \neq 0$ and $\operatorname{cosec} x$ and $\cot x$ are defined and

$$\begin{aligned} 1 + \cot^2 x &= 1 + \frac{\cos^2 x}{\sin^2 x} \\ &= \frac{\sin^2 x + \cos^2 x}{\sin^2 x} \\ &= \frac{1}{\sin^2 x} = \operatorname{cosec}^2 x \end{aligned}$$

■

COROLLARY 1.1

1. For any $x \in \mathbb{R} - \{(2n+1)\pi/2 | n \in \mathbb{Z}\}$,

$$(\sec x - \tan x)(\sec x + \tan x) = \sec^2 x - \tan^2 x = 1$$

that is, $\sec x - \tan x$ and $\sec x + \tan x$ are reciprocal to each other.

2. For any $x \in \mathbb{R} - \{n\pi | n \in \mathbb{Z}\}$,

$$(\operatorname{cosec} x - \cot x)(\operatorname{cosec} x + \tan x) = \operatorname{cosec}^2 x - \cot^2 x = 1$$

that is, $\operatorname{cosec} x - \cot x$ and $\operatorname{cosec} x + \cot x$ are reciprocal to each other.

COROLLARY 1.2

1. $|\sin x| \leq 1$ and $|\cos x| \leq 1$ for all $x \in \mathbb{R}$

2. $|\operatorname{cosec} x| \geq 1$ for all $x \in \mathbb{R} - \{n\pi | n \in \mathbb{Z}\}$ and $|\sec x| \geq 1$ for all $x \in \mathbb{R} - \{(2n+1)\pi/2 | n \in \mathbb{Z}\}$

PROOF

This is an immediate consequence from the identity $\sin^2 x + \cos^2 x = 1$. ■



Try it out Express all the trigonometric ratios of an angle in terms of parts (1) or (2) of Corollary 1.2.

In the following examples, we have expressed all the trigonometric ratios of an angle in terms of sine and in terms of cotangent.

Example 1.6

Express each trigonometric ratios of an angle θ in terms of $\sin \theta$.

Solution: Since $\sin^2 \theta + \cos^2 \theta = 1$, we have

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\sqrt{1 - \sin^2 \theta}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta}$$

All the above are valid for approximate values of θ .

Example 1.7

Express all trigonometric ratios of an angle in terms of $\cot \theta$.

Solution: From part (3) of Theorem 1.4, we have, for all approximate θ ,

$$\operatorname{cosec} \theta = \sqrt{1 + \cot^2 \theta}$$

$$\sin \theta = \frac{1}{\operatorname{cosec} \theta} = \frac{1}{\sqrt{1 + \cot^2 \theta}}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{1}{1 + \cot^2 \theta}} = \frac{\cot \theta}{\sqrt{1 + \cot^2 \theta}}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{\sqrt{1 + \cot^2 \theta}}{\cot \theta}$$

Example 1.8

If $\cos \theta = 2\sqrt{2}/3$, find the numerical magnitude of all trigonometric ratios of θ .

Solution: Using $\cos \theta = 2\sqrt{2}/3$, we get

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \left(\frac{2\sqrt{2}}{3}\right)^2} = \frac{1}{3}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = 3$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{3}{2\sqrt{2}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{1}{2\sqrt{2}}$$

$$\cot \theta = \frac{1}{\tan \theta} = 2\sqrt{2}$$

Inter-Relationships Among Trigonometric Ratios

In Table 1.2, we display formulae expressing each of the trigonometric ratios in terms of any other ratio.

Table 1.2 Relationship among trigonometric ratios

Function	$\sin \theta$	$\cos \theta$	$\operatorname{cosec} \theta$	$\sec \theta$	$\tan \theta$	$\cot \theta$
$\sin \theta$	$\sin \theta$	$\sqrt{1 - \sin^2 \theta}$	$\frac{1}{\sin \theta}$	$\frac{1}{\sqrt{1 - \sin^2 \theta}}$	$\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$	$\frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta}$
$\cos \theta$	$\sqrt{1 - \cos^2 \theta}$	$\cos \theta$	$\frac{1}{\sqrt{1 - \cos^2 \theta}}$	$\frac{1}{\cos \theta}$	$\frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta}$	$\frac{\cos \theta}{\sqrt{1 - \cos^2 \theta}}$
$\operatorname{cosec} \theta$	$\frac{1}{\operatorname{cosec} \theta}$	$\frac{\sqrt{\operatorname{cosec}^2 \theta - 1}}{\operatorname{cosec} \theta}$	$\operatorname{cosec} \theta$	$\frac{\operatorname{cosec} \theta}{\sqrt{\operatorname{cosec}^2 \theta - 1}}$	$\frac{1}{\sqrt{\operatorname{cosec}^2 \theta - 1}}$	$\sqrt{\operatorname{cosec}^2 \theta - 1}$
$\sec \theta$	$\frac{\sqrt{\sec^2 \theta - 1}}{\sec \theta}$	$\frac{1}{\sec \theta}$	$\frac{\sec \theta}{\sqrt{\sec^2 \theta - 1}}$	$\sec \theta$	$\sqrt{\sec^2 \theta - 1}$	$\frac{1}{\sqrt{\sec^2 \theta - 1}}$
$\tan \theta$	$\frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$	$\frac{1}{\sqrt{1 + \tan^2 \theta}}$	$\frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta}$	$\sqrt{1 + \tan^2 \theta}$	$\tan \theta$	$\frac{1}{\tan \theta}$
$\cot \theta$	$\frac{1}{\sqrt{1 + \cot^2 \theta}}$	$\frac{\cot \theta}{\sqrt{1 + \cot^2 \theta}}$	$\sqrt{1 + \cot^2 \theta}$	$\frac{\sqrt{1 + \cot^2 \theta}}{\cot \theta}$	$\frac{1}{\cot \theta}$	$\cot \theta$

Example 1.9

If $\tan \theta + \sec \theta = 3/2$, evaluate the value of $\sec \theta$.

Solution: First, express $\tan \theta$ and $\sec \theta$ in terms of $\sin \theta$.

We have

$$\tan \theta = \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} \quad \text{and} \quad \sec \theta = \frac{1}{\sqrt{1 - \sin^2 \theta}}$$

Put $\sin \theta = x$. We are given that

$$\tan \theta + \sec \theta = \frac{3}{2}$$

Therefore

$$\frac{x}{\sqrt{1 - x^2}} + \frac{1}{\sqrt{1 - x^2}} = \frac{3}{2}$$

$$4(x + 1)^2 = 9(1 - x^2)$$

$$13x^2 + 8x - 5 = 0$$

$$(13x - 5)(x + 1) = 0$$

$$x = -1 \quad \text{or} \quad \frac{5}{13}$$

If $x = -1$, then $\cos \theta = 0$ (since $\cos^2 \theta + \sin^2 \theta = 1$) and hence $\tan \theta$ and $\sec \theta$ are not defined. Therefore $x = 5/13$; that is, $\sin \theta = 5/13$.

Now, let us evaluate the trigonometric ratios of certain angles and prepare a table useful for ready reckoning. First recall the following in degrees and radians.

Radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	2π
Degrees	30°	45°	60°	90°	120°	135°	150°	180°	360°

THEOREM 1.5

1. $\sin 30^\circ = \sin \frac{\pi}{6} = \frac{1}{2}$
2. $\sin 45^\circ = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$
3. $\sin 60^\circ = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$
4. $\sin 90^\circ = \sin \frac{\pi}{2} = 1$
5. $\sin 0^\circ = \sin 0 = 0$

PROOF

1. Consider a rectangular coordinate plane OXY with origin O. Let P(a, b) be a point such that OP makes an angle of 30° with OX in anticlockwise sense. Draw a perpendicular through P to OX to meet at M. Then PM = b and OM = a. Produce PM to Q such that PM = MQ. Now, by the properties of triangles, OPQ is an equilateral triangle and PQ = 2b and hence OP = 2b = OQ. Therefore

$$\sin 30^\circ = \frac{b}{2b} = \frac{1}{2}$$

This is graphically illustrated in Figure 1.11.

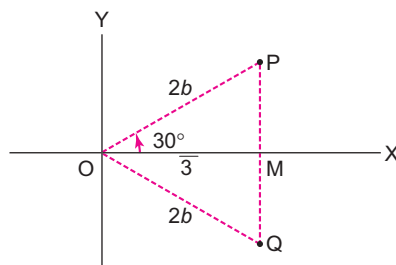


FIGURE 1.11 Theorem 1.5 part (1).

By similar techniques, we can prove (2) through (5) also. ■

Using the values of sine given in Theorem 1.5 and Table 1.2, we can evaluate the values of other trigonometric ratios also at these angles.

DEFINITION 1.8 Complementary Angles Two angles are said to be *complementary* if their sum is a right angle. Therefore, for $0 \leq \theta < \pi/2$, θ and $\pi/2 - \theta$ are complementary.

THEOREM 1.6

Let θ and ψ be complementary angles. Then

1. $\sin \theta = \cos \psi$ and $\cos \theta = \sin \psi$
2. $\tan \theta = \cot \psi$ and $\cot \theta = \tan \psi$
3. $\sec \theta = \operatorname{cosec} \psi$ and $\operatorname{cosec} \theta = \sec \psi$

PROOF

Draw a right-angled triangle OAP with $\widehat{OAP} = 90^\circ$ and $\widehat{AOP} = \theta$ (Figure 1.12). Since the sum of the angles of any triangle is 180° and since $\theta + \psi = 90^\circ$, it follows that $\widehat{OPA} = \psi$.

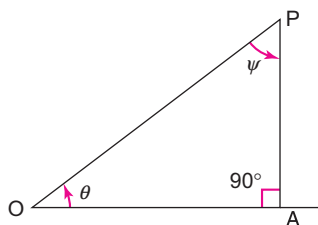


FIGURE 1.12 Theorem 1.6.

If O is treated as origin and the line OA as the X-axis, then the x-coordinate of P is OA and the y-coordinate of P is PA. By the definition of the ratio $\sin \theta$, we have

$$\sin \theta = \frac{PA}{OP} \quad \text{and} \quad \cos \theta = \frac{OA}{OP}$$

Also, when the angle ψ is considered, PA becomes the base and OA becomes the perpendicular and hence

$$\sin \psi = \frac{OA}{OP} \quad \text{and} \quad \cos \psi = \frac{PA}{OP}$$

And therefore (1) is proved. Similarly (2) and (3) can be proved. ■

COROLLARY 1.3 For any $0 \leq \theta < \pi/2$,

1. $\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$ and $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$
2. $\tan \theta = \cot\left(\frac{\pi}{2} - \theta\right)$ and $\cot \theta = \tan\left(\frac{\pi}{2} - \theta\right)$
3. $\sec \theta = \operatorname{cosec}\left(\frac{\pi}{2} - \theta\right)$ and $\operatorname{cosec} \theta = \sec\left(\frac{\pi}{2} - \theta\right)$

 **Try it out** Prove parts (1)–(3) of Corollary 1.3.

Thus, we have that

1. The *sine* of any angle is the *cosine* of its complement.
2. The *tangent* of any angle is the *cotangent* of its complement.
3. The *secant* of any angle is the *cosecant* of its complement.

For these reasons, sine and cosine, tangent and cotangent, and secant and cosecant can be remembered as complementary.

From these observations, we can construct Table 1.3 containing the trigonometric ratios of certain important angles. The reader is advised before proceeding any further to table that if he/she can memorize accurately the portion of the table included between thick lines, then one should be able to reproduce the rest easily.

Table 1.3 Trigonometric ratios of certain angles

Ratios	Degree	0°	30°	45°	60°	90°	180°	270°	360°
	Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin		0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	−1	0
cos		1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	−1	0	1
cosec		Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	Not defined	−1	Not defined
sec		1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined	−1	Not defined	1
tan		0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined	0	Not defined	0
cot		Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	Not defined	0	Not defined

Next, we consider variations and periodicity of the trigonometrical ratios. In fact, we have defined $\sin x$ and $\cos x$ for all real numbers x , $\sec x$ and $\tan x$ for all $x \in \mathbb{R} - \{(2n+1)\pi/2, n \in \mathbb{Z}\}$, and $\operatorname{cosec} x$ and $\cot x$ for all $x \in \mathbb{R} - \{n\pi | n \in \mathbb{Z}\}$. By the very definition of these functions, it follows that

$$f(2\pi + x) = f(x)$$

for all trigonometric functions f . Now, we have the following theorem.

THEOREM 1.7 For all real numbers x ,

$$\sin(-x) = -\sin x \quad \text{and} \quad \cos(-x) = \cos x$$

PROOF The theorem is evidently true when $x = 0$ or $\pi/2$ or π or $3\pi/2$ or 2π . Let x be any real number. Then we can write $x = 2n\pi + \theta$, where $0 \leq \theta < 2\pi$ and $n \in \mathbb{Z}$.

In view of the above remark, we can assume that $0 \leq \theta < 2\pi$ and $\theta \neq \pi/2, \pi, 3\pi/2$. Consider a rectangular coordinate system OXY with O as origin. Draw a unit circle in the coordinate plane with the origin as the centre. Choose a point P on the unit circle such that $\widehat{XOP}$ measured in anti-clockwise sense is θ radians. Figure 1.13 illustrates the cases for each of the four quadrants. Draw a perpendicular from P to OX to meet the circle at Q .

In each of the four cases, the X -coordinates of P and Q are same and their Y -coordinates are same in absolute value but with different sign. Now, we have

$$\begin{aligned} \sin(-\theta) &= \sin(\theta + 2\pi) \\ &= \text{Y-coordinate of } Q \end{aligned}$$

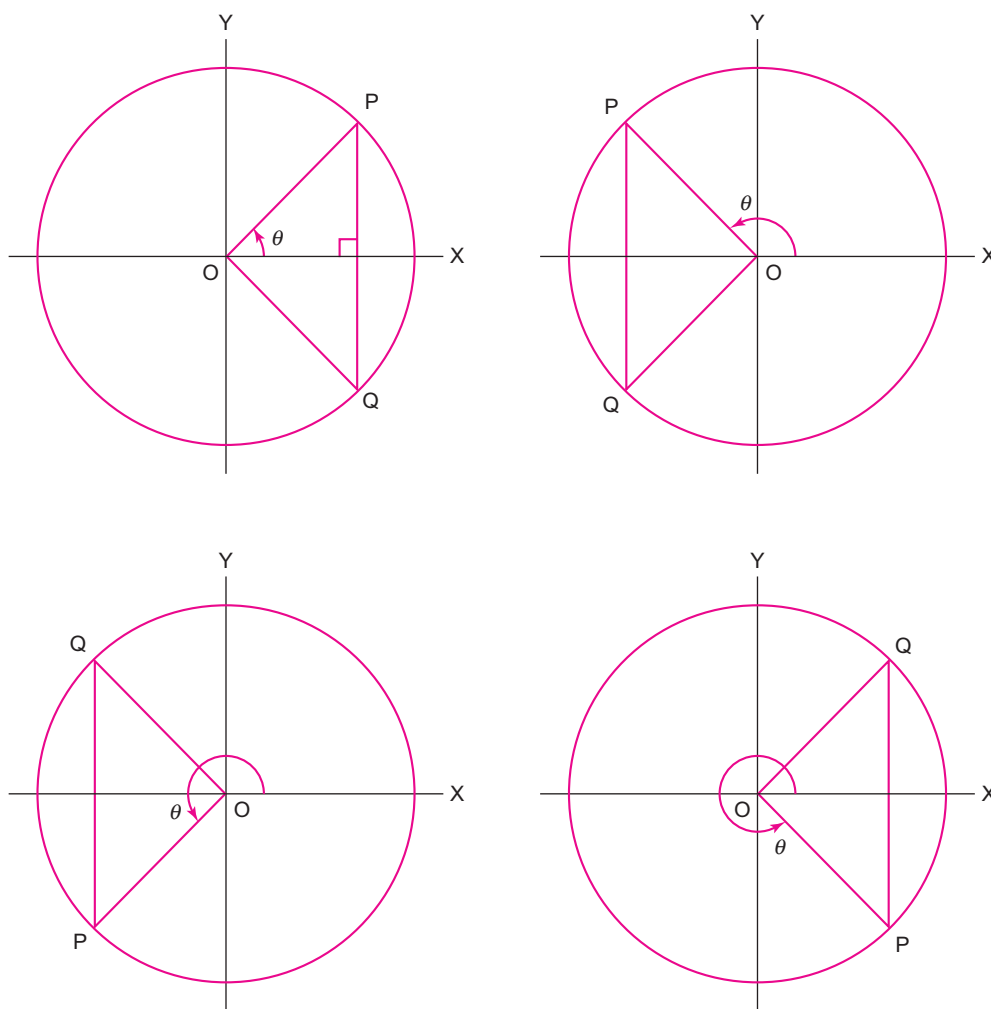


FIGURE 1.13 Theorem 1.7.

$$\begin{aligned}
 &= -(\text{Y-coordinate of P}) \\
 &= -\sin \theta \\
 \text{and} \quad &\cos(-\theta) = \cos(\theta + 2\pi) \\
 &= \text{X-coordinate of Q} \\
 &= \text{X-coordinate of P} \\
 &= \cos \theta
 \end{aligned}$$

Next, we have $x = 2n\pi + \theta$ and $2n\pi \leq x < 2(n+1)\pi$ (recall that n is the integral part of $x/2\pi$). Put $\phi = 2(n+1)\pi - x$. Then $0 < \phi \leq 2\pi$. From the above argument, we have

$$\sin(-\phi) = -\sin \phi \quad \text{and} \quad \cos(-\phi) = \cos \phi$$

Therefore

$$\begin{aligned}
 \sin(-x) &= \sin[2(n+1)\pi - x] \\
 &= \sin \phi \\
 &= -\sin(-\phi) \\
 &= -\sin[x + 2(-n-1)\pi] \\
 &= -\sin x
 \end{aligned}$$

and

$$\begin{aligned}
 \cos(-x) &= \cos[2(n+1)\pi - x] \\
 &= \cos \phi \\
 &= \cos(-\phi) \\
 &= \cos[x + 2(-n-1)\pi] \\
 &= \cos x
 \end{aligned}$$

Argument similar to the one given above can be used to prove the following also.

THEOREM 1.8 The following hold for any real number x .

1. $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x$
2. $\sin(\pi + x) = -\sin x$ and $\cos(\pi + x) = -\cos x$
3. $\cos\left(\frac{\pi}{2} - x\right) = \sin x$ and $\sin\left(\frac{\pi}{2} - x\right) = \cos x$
4. $\cos\left(\frac{\pi}{2} + x\right) = -\sin x$ and $\sin\left(\frac{\pi}{2} + x\right) = \cos x$

 **Try it out** Prove Theorem 1.8.

DEFINITION 1.9 Supplementary Two angles θ and ϕ are said to be *supplementary* if $\theta + \phi = \pi$.

Examples

(1) $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ are supplementary angles.

(2) 45° and 135° are supplementary angles, as $\sin \theta = \sin \phi$ and $\cos \theta = -\cos \phi$.

The following are consequences of Theorems 1.7 and 1.8. They can be obtained by using the “Principle of Mathematics Induction”.

COROLLARY 1.4 For any real number x and for any integer n , the following are true.

1. $\sin(n\pi - x) = (-1)^{n+1} \sin x$ and $\cos(n\pi - x) = (-1)^n \cos x$
2. $\sin(n\pi + x) = (-1)^n \sin x$ and $\cos(n\pi + x) = (-1)^n \cos x$
3. $\sin\left((2n+1)\frac{\pi}{2} + x\right) = (-1)^n \cos x$ and $\cos\left((2n+1)\frac{\pi}{2} + x\right) = (-1)^{n+1} \sin x$
4. $\sin\left((2n+1)\frac{\pi}{2} - x\right) = (-1)^n \cos x$ and $\cos\left((2n+1)\frac{\pi}{2} - x\right) = (-1)^n \sin x$

 **Try it out** Prove parts (1)–(4) of Corollary 1.4.

Table 1.4 contains values of other trigonometric functions also. Here n stands for an integer and x for any real number in the domain of corresponding functions.

Table 1.4 Values of other trigonometric functions

x	$\sin x$	$\cos x$	$\tan x$	$\cot x$	$\operatorname{cosec} x$	$\sec x$
$n\pi - \theta$	$(-1)^{n+1} \sin \theta$	$(-1)^n \cos \theta$	$-\tan \theta$	$-\cot \theta$	$(-1)^{n+1} \operatorname{cosec} \theta$	$(-1)^n \sec \theta$
$n\pi + \theta$	$(-1)^n \sin \theta$	$(-1)^n \cos \theta$	$\tan \theta$	$\cot \theta$	$(-1)^n \operatorname{cosec} \theta$	$(-1)^n \sec \theta$
$(2n+1)\frac{\pi}{2} - \theta$	$(-1)^n \cos \theta$	$(-1)^n \sin \theta$	$\cot \theta$	$\tan \theta$	$(-1)^n \sec \theta$	$(-1)^n \operatorname{cosec} \theta$
$(2n+1)\frac{\pi}{2} + \theta$	$(-1)^n \cos \theta$	$(-1)^{n+1} \sin \theta$	$\cot \theta$	$\tan \theta$	$(-1)^n \sec \theta$	$(-1)^{n+1} \operatorname{cosec} \theta$

DEFINITION 1.10 **Coterminal** Two angles θ and ϕ are said to be *coterminal* if the difference between them is an integral multiple of 2π or 360° according as the angles are measured in radians or degrees.

Example

The following are pairs of coterminal angles:

(1) $\frac{\pi}{2}$ and $\frac{5\pi}{2}$

(2) 40° and 400°

(3) 50° and -310°

If θ and ϕ are coterminal angles, then $\theta = 2n\pi + \phi$ for some integer n and hence, for any trigonometric function f

$$f(\theta) = f(2n\pi + \phi) = f(\phi)$$

Example 1.10

If $\sin \theta = -5/13$ and θ is in the third quadrant, evaluate the value of

$$5\cot^2 \theta + 12\tan \theta + 13\operatorname{cosec} \theta$$

Solution: Consider a rectangular coordinate plane OXY with O as origin. Let P(x , y) be a point in the plane

such that OP makes of θ in the anticlockwise sense, with OX. Since θ is given in the third quadrant, $x < 0$ and $y < 0$ (Figure 1.14). Now, since $\sin \theta = -5/13$, we get that $y = -5$, $OP = 13$ and hence $x = -12 (= \pm \sqrt{13^2 - y^2})$. Therefore

$$\tan \theta = \frac{y}{x} = \frac{-5}{-12} = \frac{5}{12}$$

$$\cot \theta = \frac{x}{y} = \frac{12}{5}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{-13}{5}$$

Hence

$$\begin{aligned} & 5 \cot^2 \theta + 12 \tan \theta + 13 \operatorname{cosec} \theta \\ &= 5 \left(\frac{12}{5} \right)^2 + 12 \left(\frac{5}{12} \right) + 13 \left(\frac{-13}{5} \right) \\ &= \frac{144}{5} + 5 - \frac{169}{5} = 0 \end{aligned}$$

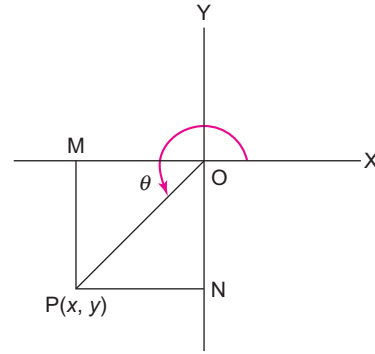


FIGURE 1.14 Example 1.10.

Example 1.11

Find the value of

$$\sin 600^\circ \cos 330^\circ + \cos 120^\circ \sin 150^\circ$$

Solution:

$$\begin{aligned} \sin 600^\circ \cos 330^\circ + \cos 120^\circ \sin 150^\circ &= \sin(3\pi + 60^\circ) \\ &\times \cos\left(\frac{3\pi}{2} + 60^\circ\right) + \cos\left(\frac{\pi}{2} + 30^\circ\right) \sin(\pi - 30^\circ) \end{aligned}$$

$$= -\sin 60^\circ \cdot \sin 60^\circ + (-\sin 30^\circ) \sin 30^\circ \quad (\text{by Table 1.4})$$

$$= -\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \left(-\frac{1}{2} \cdot \frac{1}{2}\right) = -1$$

Example 1.12

If $\sec \theta + \tan \theta = 1/5$ and $0 < \theta < 2\pi$, find the quadrant in which θ lies and the value of $\sin \theta$. Using this, find all the other trigonometric ratios of θ .

Solution: Since $\sec \theta + \tan \theta$ and $\sec \theta - \tan \theta$ are reciprocal to each other, we have

$$\sec \theta - \tan \theta = 5$$

By adding this to the given equation, we get

$$2 \sec \theta = \frac{1}{5} + 5 \quad \text{or} \quad \sec \theta = \frac{13}{5}$$

Therefore

$$\cos \theta = \frac{1}{\sec \theta} = \frac{5}{13} > 0$$

$$\tan \theta = \frac{1}{5} - \sec \theta = \frac{1}{5} - \frac{13}{5} = \frac{-12}{5}$$

$$\sin \theta = \tan \theta \cdot \cos \theta = \frac{-12}{5} \cdot \frac{5}{13} = \frac{-12}{13} < 0$$

Since $\sin \theta < 0$ and $\cos \theta > 0$, θ lies in the fourth quadrant.

1.3 | Periodicity and Variance

Let us recall that a function f defined on a subset E of the real number system $\mathbb{R}$ is called *periodic* if there exists a positive real number p such that $f(x+p) = f(x)$ for all $x \in E$ whenever $x+p$ is also in E . In this case, p is called a *period* of f . Note that there may be several periods of the same function. For example, if p is a period of f , then np is also a period of f for any positive integer n .

DEFINITION 1.11 Let f be a function defined on a subset E of $\mathbb{R}$. If the set of all periods of f has minimum, then that minimum period is called *the period* of f .

A function may possess a period, but still it may not possess the minimum period. For example, consider a constant function f defined on any $E \subseteq \mathbb{R}$. Then every positive real number is a period of f and each period is a positive real

number and, evidently, there is no minimum positive real number. That is, a function may be periodic without having the period (i.e., the minimum period).

We know that each trigonometric function is periodic and 2π is a period of each of these. In the following, we prove that each trigonometric function has minimum period.

THEOREM 1.9 The sine function is periodic and its minimum period is 2π .

PROOF By Definition 1.5 of the sine function, $\sin(2\pi + \theta) = \sin \theta$ for all real numbers θ and hence 2π is period of the sine function. We shall prove that any positive real number less than 2π is not a period of sine. Suppose, on the contrary, that $0 < p < 2\pi$ and p is a period of the sine function. Put $x_0 = 2\pi - p$. Then since p is a period

$$\sin(x_0 + p) = \sin x_0$$

Therefore

$$\sin 2\pi = \sin(2\pi - p) = \sin(-p) = -\sin p$$

Therefore

$$\sin p = 0$$

and hence $p = \pi$ (since $0 < p < 2\pi$). But π is not a period of the sine function, since

$$\sin\left(\frac{\pi}{4} + \pi\right) = -\frac{1}{\sqrt{2}} \neq \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}$$

This implies that there is no p such that $0 < p < 2\pi$ and p is a period of the sine function. Thus, 2π is the minimum period of the sine function. ■

Similar arguments as in Theorem 1.9 yield the following theorem.

THEOREM 1.10

1. The sine, cosine, cosecant and secant functions are all periodic with minimum period 2π .
2. The tangent and cotangent functions are periodic with minimum period π .

Example 1.13

Determine the value of the following

(1) $\tan 780^\circ$

(2) $\sec 840^\circ$

(3) $\operatorname{cosec} 420^\circ$

(4) $\cot 240^\circ$

(2) $\sec 840^\circ = \sec(4\pi + 120^\circ) = \sec 120^\circ = \sec\left(\frac{\pi}{2} + 30^\circ\right)$

$= -\operatorname{cosec} 30^\circ = -2$

(3) $\operatorname{cosec} 420^\circ = \operatorname{cosec}(2\pi + 60^\circ) = \operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}$

(4) $\cot 240^\circ = \cot(\pi + 60^\circ) = \cot 60^\circ = \frac{1}{\sqrt{3}}$

Solution:

(1) $\tan 780^\circ = \tan(4\pi + 60^\circ) = \tan 60^\circ = \sqrt{3}$

Next, we discuss the variance of different trigonometric functions; that is, how does the change in value of x affects each trigonometric ratio at x .

Recall that the domain of each of the sine and cosine is the whole real number system $\mathbb{R}$, while that of secant and tangent is $\mathbb{R} - \{(2n+1)\pi/2, n \in \mathbb{Z}\}$ and that of cosecant and cotangent is $\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$. In Table 1.5 we display these together with the ranges (or images) of these functions.

Table 1.5 Domains and ranges of trigonometric functions

	Domain	Range
Sine	$\mathbb{R}$	$[-1, 1]$
Cosine	$\mathbb{R}$	$[-1, 1]$
Tangent	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2} \mid n \in \mathbb{Z}\right\}$	$\mathbb{R}$
Cotangent	$\mathbb{R} - \{n\pi \mid n \in \mathbb{Z}\}$	$\mathbb{R}$
Secant	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2} \mid n \in \mathbb{Z}\right\}$	$(-\infty, -1] \cup [1, +\infty)$
Cosecant	$\mathbb{R} - \{n\pi \mid n \in \mathbb{Z}\}$	$(-\infty, -1] \cup [1, \infty)$

DEFINITION 1.12 **Increasing and Decreasing Functions** Let f be a real-valued function defined on a subset of $\mathbb{R}$ and $E \subseteq \text{Dom } f$. Then f is said to be *increasing on E* if

$$x, y \in E \quad \text{and} \quad x < y \Rightarrow f(x) \leq f(y)$$

f is said to *decreasing on E* if

$$x, y \in E \quad \text{and} \quad x < y \Rightarrow f(x) \geq f(y)$$

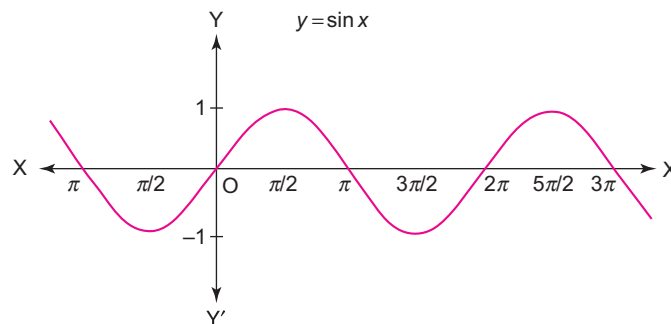
The following can be easily proved.

Variance and Graph of $\sin x$

1. The function $\sin x$ is increasing on $[0, \pi/2]$. As x increases from 0 to $\pi/2$, $\sin x$ increases from 0 to 1.
2. The function $\sin x$ is decreasing on $[\pi/2, 3\pi/2]$. As x increases from $\pi/2$ and $3\pi/2$ $\sin x$ decreases from 1 to -1 .
3. The function $\sin x$ is increasing on $[3\pi/2, 2\pi]$. As x increases from $3\pi/2$ to 2π , $\sin x$ increases from -1 to 0.

We can plot the graph of $\sin x$ in the coordinate plane by taking angles x in radian measure on X-axis and the values of $y = \sin x$ on Y-axis. The graph of $\sin x$ is given in Figure 1.15. The sine curve passes through the origin and the values of $\sin x$ vary between -1 and $+1$ which are, respectively, the minimum and maximum. This curve is in the shape of a wave whose wavelength is 2π , which is nothing but the period of $\sin x$.

X	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$	3π
Y	0	-1	0	1	0	-1	0	1	0

**FIGURE 1.15** Graph of $\sin x$.

Variance and Graph of $\cos x$

1. The function $\cos x$ is decreasing on $[0, \pi]$. As x increases from 0 to π , $\cos x$ decreases from 1 to -1 .
2. The function $\cos x$ is increasing on $[\pi, 2\pi]$. As x increases from π to 2π , $\cos x$ increases from -1 to 1.

By choosing a suitable scale, plot the points $(x, \cos x)$ given in the table below and join these points with a smooth curve to get the graph of $\cos x$. This cosine curve does not pass through the origin. The maximum and minimum values of $\cos x$ are 1 and -1 , respectively. Each real number in $[-1, 1]$ is a value of $\cos x$. The curve also looks like a wave with wavelength 2π , which is same as the period of $\cos x$. The graph of $\cos x$ is given in Figure 1.16.

X	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$
Y	-1	0	1	0	-1	0	1	0

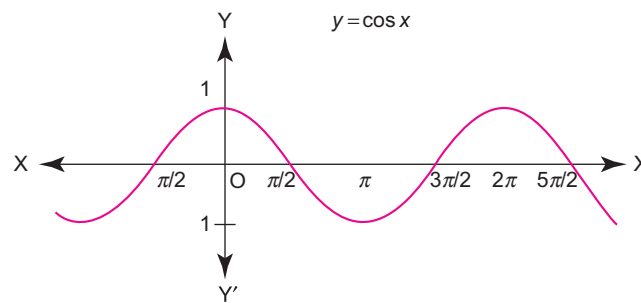


FIGURE 1.16 Graph of $\cos x$.

Variance and Graph of $\tan x$

1. The function $\tan x$ is of period π , the minimum period being π .
2. The function $\tan x$ is not defined at $x = [(2n + 1)/2]\pi$, for any integer n .
3. The function $\tan x$ is increasing in each of the intervals $[0, \pi/2)$ and $(\pi/2, \pi]$. As x increases from 0 to $\pi/2$, $\tan x$ increases from 0 to $+\infty$, and as x increases from $\pi/2$ to π , $\tan x$ increases from $-\infty$ to 0.
4. The $\tan x$ curve passes through the origin and does not intersect the vertical lines at $x = [(2n + 1)/2]\pi$, for any integer n rather it looks like touching these lines.
5. The function $\tan x$ has no minimum and maximum values.

Figure 1.17 illustrates the graph of $\tan x$.

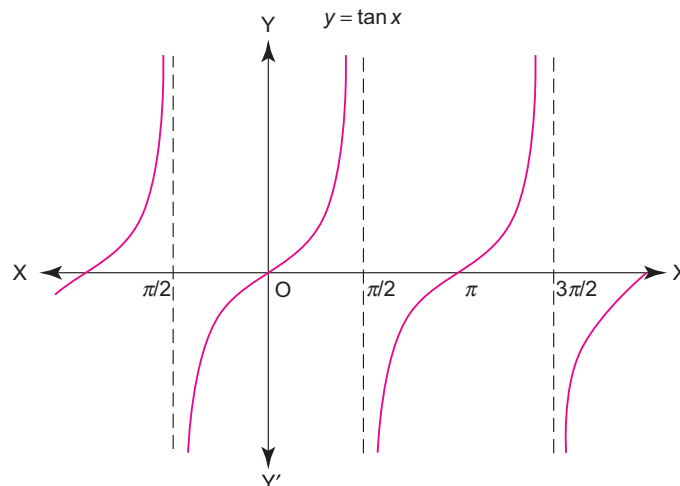


FIGURE 1.17 Graph of $\tan x$.

Variance and Graph of $\cot x$

1. The function $\cot x$ is of period π , the minimum period being π .
2. The function $\cot x$ is not defined at $x = n\pi$, for any integer n .
3. The function $\cot x$ is decreasing on the interval $(0, \pi)$ and on each of the intervals $(n\pi, (n+1)\pi)$, $n \in \mathbb{Z}$. As x increases from 0 to π , $\cot x$ decreases from $+\infty$ to $-\infty$.
4. The $\cot x$ curve does not pass through the origin and does not intersect the vertical lines $x = n\pi$, for any integer n , but looks like touching these lines.
5. The function $\cot x$ has no minimum and maximum values.

Figure 1.18 illustrates the graph of $\cot x$.

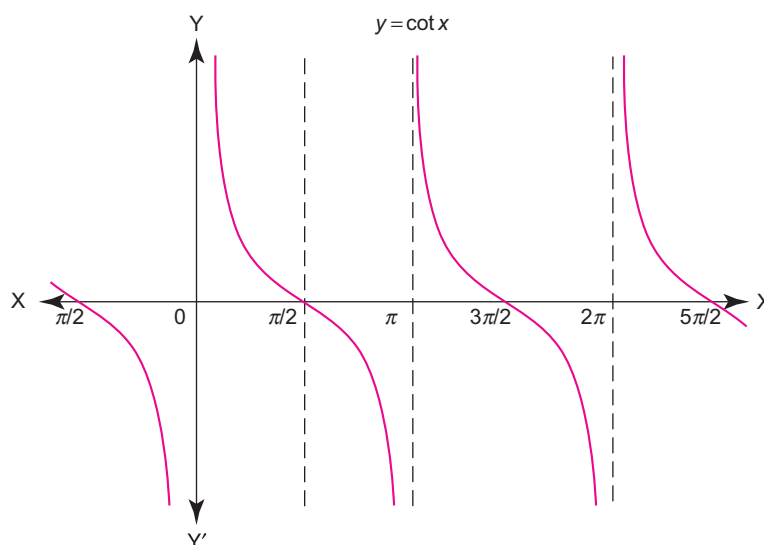


FIGURE 1.18 Graph of $\cot x$.

Variance and Graph of $\sec x$

1. The function $\sec x$ is periodic and 2π is its minimum period.
2. The function $\sec x$ is not defined at $n\pi/2$ for any integer n .
3. The function $\sec x$ is increasing on each of the intervals

$$\left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right), \left(2n\pi\left(2n + \frac{1}{2}\right)\pi\right) \quad \text{and} \quad \left(\left(2n + \frac{1}{2}\right)\pi, (2n+1)\pi\right)$$

for any integer n . Also, $\sec x$ is decreasing on each of the intervals

$$\left((2n+1)\pi, \left(2n + \frac{3}{2}\right)\pi\right) \quad \text{and} \quad \left(\left(2n + \frac{3}{2}\right)\pi, (2n+2)\pi\right)$$

for any integer n .

4. The $\sec x$ curve does not intersect the X -axis and any line $x = n\pi/2$, $n \in \mathbb{Z}$ but nearly touches these lines.
5. The function $\sec x$ has no minimum and maximum values and for no x , $-1 < \sec x < 1$.

Figure 1.19 illustrates the graph of $\sec x$.

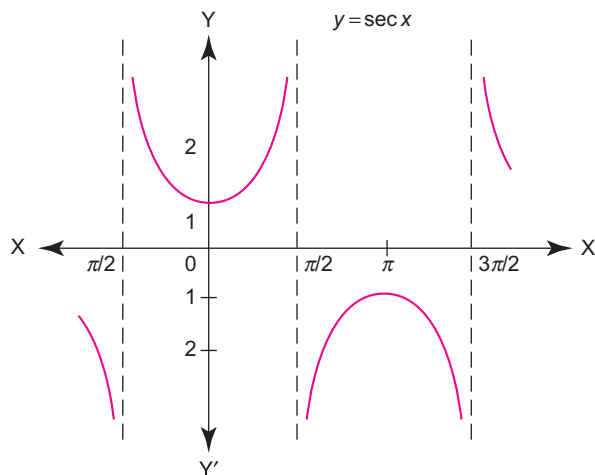


FIGURE 1.19 Graph of $\sec x$.

Variance and Graph of Cosec x

1. The function $\operatorname{cosec} x$ is periodic and 2π is its minimum period.
2. The function $\operatorname{cosec} x$ is not defined at $n\pi$ for any integer n .
3. The function $\operatorname{cosec} x$ is decreasing on each of the intervals

$$\left(n\pi, \left(n + \frac{1}{2}\right)\pi\right) \quad \text{and} \quad \left(\left(n + \frac{3}{2}\right)\pi, (n + 2)\pi\right)$$

and increasing on each of the intervals

$$\left(\left(n + \frac{1}{2}\right)\pi, (n + 1)\pi\right) \quad \text{and} \quad \left((n + 1)\pi, \left(n + \frac{3}{2}\right)\pi\right)$$

for any integer n .

4. The $\operatorname{cosec} x$ curve does not intersect the X-axis and the lines $x = n\pi$ for any integer n . But it nearly touches these lines $x = n\pi$.
5. The function $\operatorname{cosec} x$ has neither minimum nor maximum values and there is no x such that $-1 < \operatorname{cosec} x < 1$.

Figure 1.20 illustrates the graph of $\operatorname{cosec} x$.

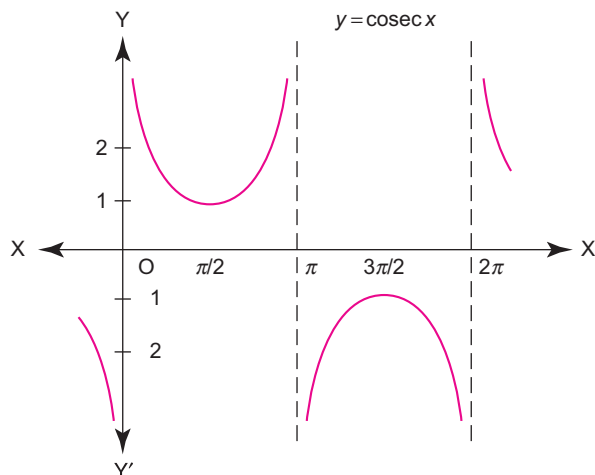


FIGURE 1.20 Graph of $\operatorname{cosec} x$.

Next we consider graphs of functions of the form $\sin(ax + b)$, $\cos(ax + b)$, $a \sin bx$, etc. First of all note that if p is a period of a function $f(x)$, then $p/|a|$ is a period of $f(ax + b)$ for any nonzero real number a and real number b ; for, if $p > 0$ and $f(p + x) = f(x)$ for all $x \in \mathbb{R}$, then

$$\frac{p}{|a|} > 0$$

and

$$f\left(a\left(\frac{p}{|a|} + x\right) + b\right) = f(\pm p + ax + b) = f(ax + b)$$

for all $x \in \mathbb{R}$ and hence $p/|a|$ is a period of $f(ax + b)$.

Further, if p is a period of $f(x)$, then p is also a period of $af(x)$ for any nonzero real number a .

Examples

- | | |
|---|---|
| <p>(1) $\sin(2x + 3)$ is a periodic function of period $2\pi/ 2 = \pi$, since $\sin x$ is of period 2π.</p> <p>(2) $3 \cos(6x - 9)$ is a periodic function of period $2\pi/ 6 = \pi/3$, since $\cos x$ is of a period 2π.</p> | <p>(3) $7 \tan(-3x)$ is periodic and $\pi/ -3 = \pi/3$ is its period, since $\tan x$ is of period π.</p> <p>(4) $6 \cot(4 - 5x)$ is of period $\pi/ -5 = \pi/5$, since $\cot x$ is of period π.</p> |
|---|---|

Graph of $3 \sin 2x$

The function $\sin x$ is a periodic function with period 2π and hence $\sin 2x$ is also a periodic function with period $2\pi/|2| = \pi$. Therefore, $3 \sin 2x$ is of period π . Also, since $-1 \leq \sin 2x \leq 1$, hence $-3 \leq 3 \sin 2x \leq 3$ for all $x \in \mathbb{R}$.

To draw the graph of $y = 3 \sin 2x$, consider the graph of $y = \sin x$. On the X -axis change k as $k/2$ (e.g., $\pi/4$ for $\pi/2$, $\pi/2$ for π , etc.) and on the Y -axis change k as $3k$ (e.g., write 3 in place of 1, -3 in place of -1, $3/2$ in place of $1/2$, etc.). The graph of $y = 3 \sin 2x$ is as given in Figure 1.21.

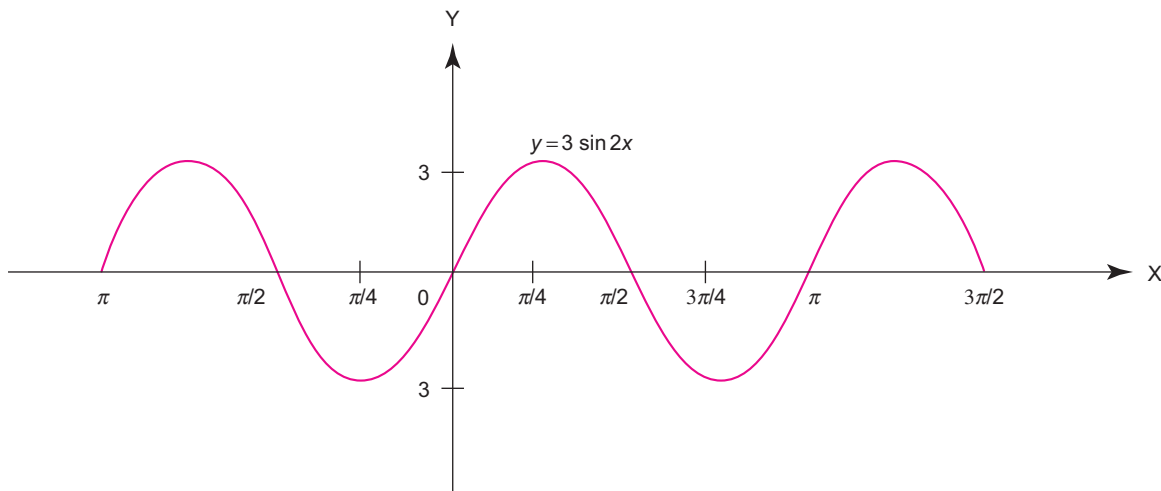


FIGURE 1.21 Graph of $y = 3 \sin 2x$.

Graph of $\cos(x - \pi/4)$

To draw the graph of $y = \cos(x - \pi/4)$, draw the graph of $y = \cos x$ and on X -axis write $k + \pi/4$ in place of k (e.g., write $\pi/2$ in place of $\pi/4$, $3\pi/4$ in place of $\pi/2$, etc.). The graph of $\cos(x - \pi/4)$ is shown in Figure 1.22.

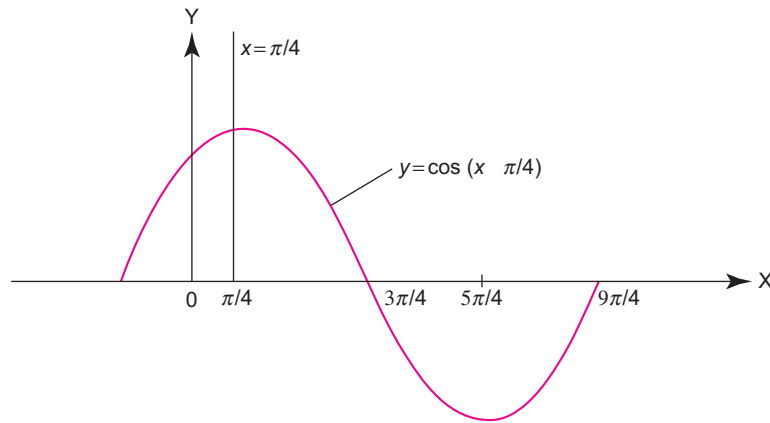


FIGURE 1.22 Graph of $\cos(x - \pi/4)$.

Another way to draw the graph of $y = \cos(x - \pi/4)$ is as follows: Draw the graph of $y = \cos x$ and shift it to the right side through a distance of $\pi/4$ units. The resultant graph is shown in Figure 1.23.

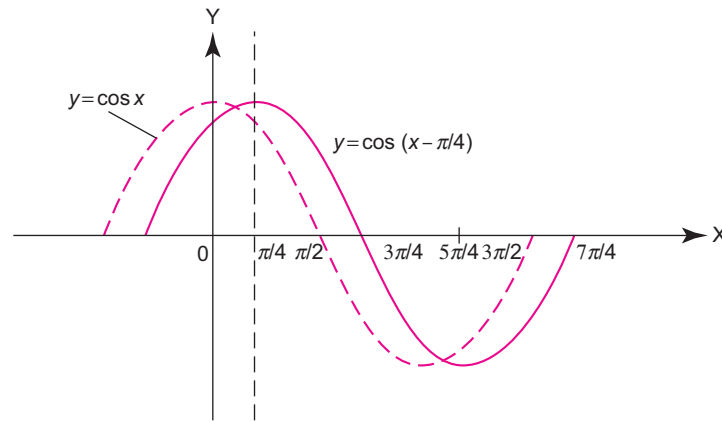


FIGURE 1.23 Alternate graph of $y = \cos(x - \pi/4)$.

Graphs of $y = \cos 2x$ and $y = \cos x$

Period of $\cos x$ is 2π and hence the period of $\cos 2x$ is π . Also, $-1 \leq \cos x \leq 1$ and $-1 \leq \cos 2x \leq 1$ for all x . The graphs of $y = \cos x$ and $y = \cos 2x$ are given in Figure 1.24.

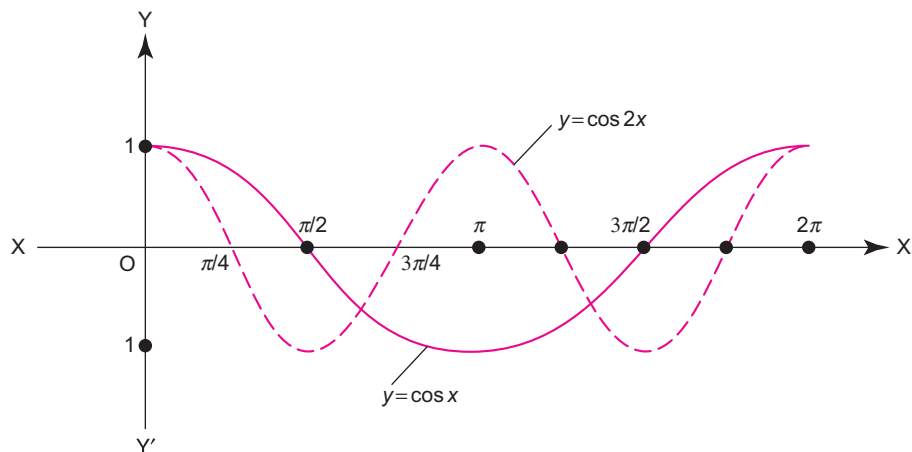


FIGURE 1.24 Graphs of $y = \cos x$ and $y = \cos 2x$.

Example 1.14

Draw the graph of $y = \sin(x/2)$.

Solution: The graph of the given function is as shown in Figure 1.25.

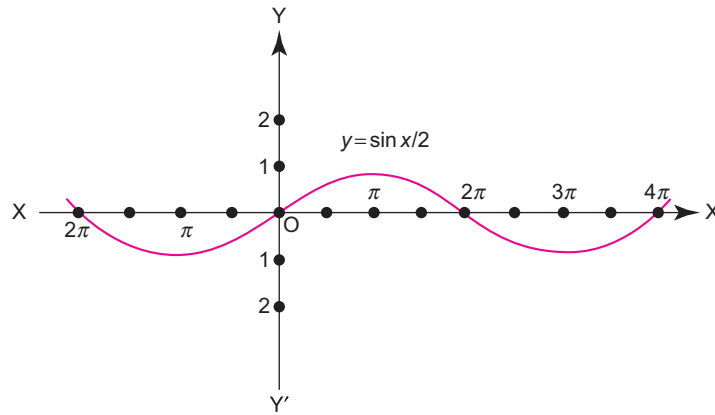


FIGURE 1.25 Solution of Example 1.14.

Example 1.15

Draw the graph of $y = 4 \cos 2x$.

Solution: The graph of the given function is as shown in Figure 1.26.

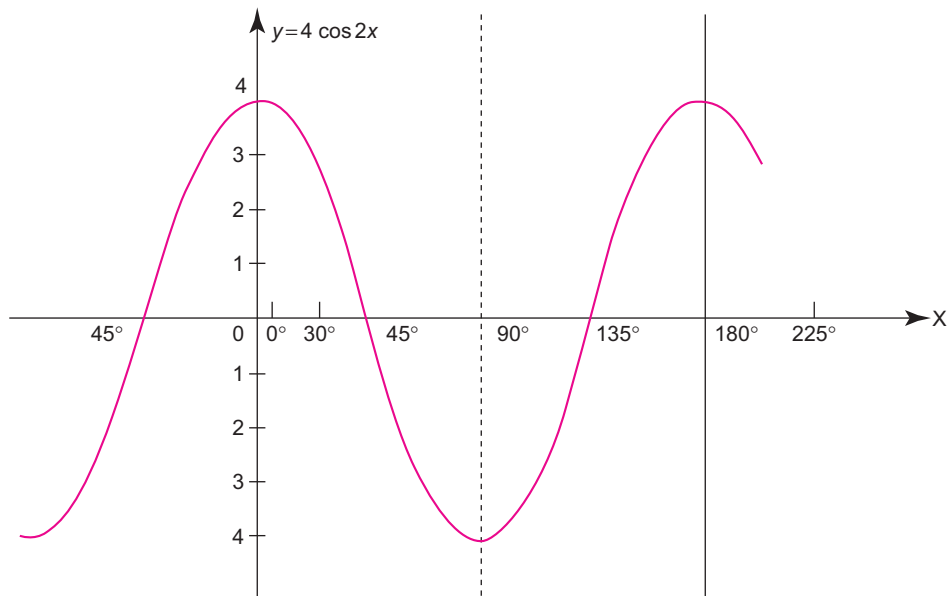


FIGURE 1.26 Solution of Example 1.15.

Example 1.16

Draw the graph of $y = 3 \sin x$.

Solution: The graph of the given function is as shown in Figure 1.27.

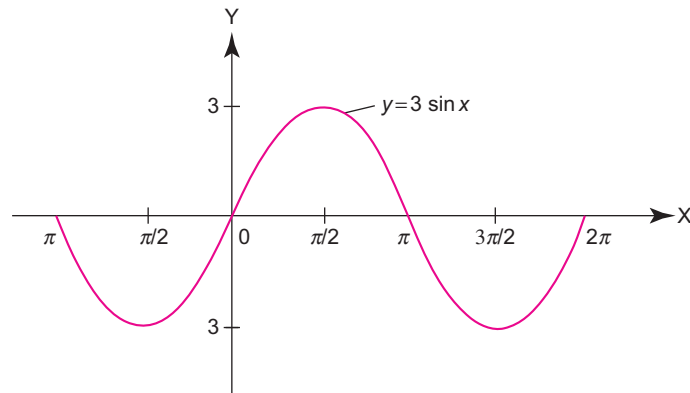


FIGURE 1.27 Solution of Example 1.16.

Example 1.17

Draw the graph of $y = 2 \tan x$.

Solution: The graph of the given function is as shown in Figure 1.28.

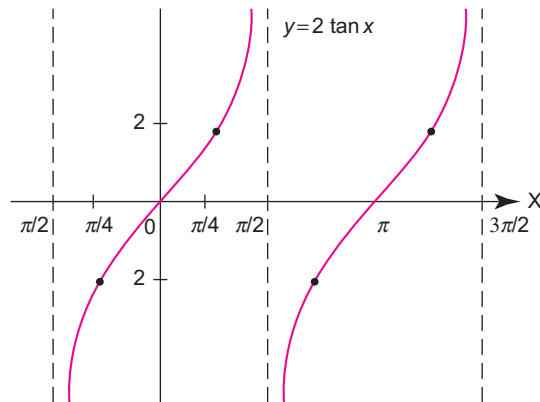


FIGURE 1.28 Solution of Example 1.17.

Example 1.18

Draw the graph of $y = \sin[x - (\pi/4)]$.

Solution: The graph of the given function is as shown in Figure 1.29.

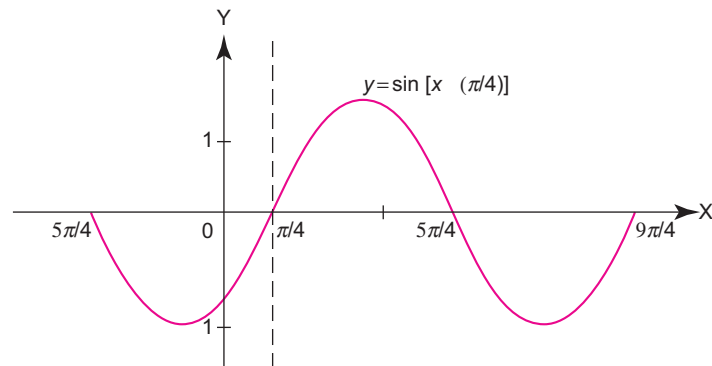


FIGURE 1.29 Solution of Example 1.18.

Example 1.19

Draw the graphs of $y = \sin x$, $y = \sin 2x$.

Solution: The graphs of the given functions are as shown in Figure 1.30.

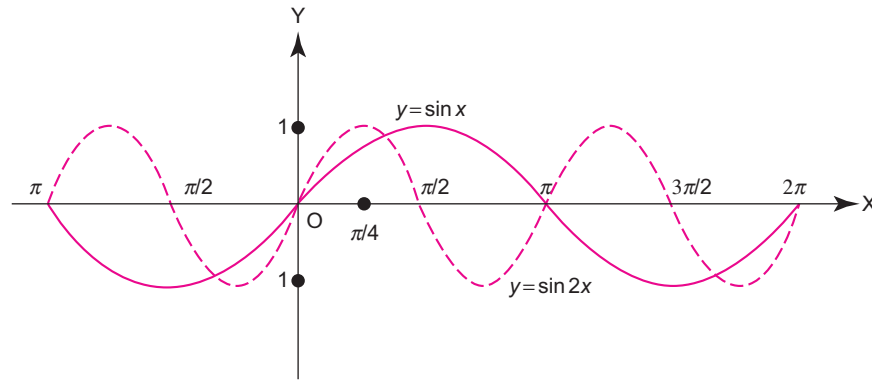


FIGURE 1.30 Solution of Example 1.19.

Example 1.20

Draw the graphs of $y = \sin x$, $y = \cos x$.

Solution: The graphs of the given functions are as shown in Figure 1.31.

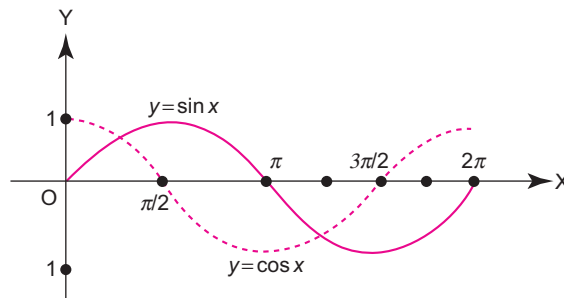


FIGURE 1.31 Solution of Example 1.20.

Variance of all the Trigonometric Ratios in the Interval $[0, 2\pi]$

In Table 1.6, we have summarized the behavior of the curves representing the trigonometric ratios of angles in the four quadrants. This can be used as a ready reference.

Table 1.6 Behavior of the curves representing the trigonometric ratios of angles in the four quadrants

<i>Second quadrant</i> $\left(\frac{\pi}{2}, \pi\right)$	<i>First quadrant</i> $\left(0, \frac{\pi}{2}\right)$
sine decreases from 1 to 0	sine increases from 0 to 1
cosine decreases from 0 to -1	cosine decreases from 1 to 0
cosecant increases from 1 to $+\infty$	cosecant decreases from $+\infty$ to 1
secant increases from $-\infty$ to -1	secant increases from 1 to $+\infty$
tangent increases from $-\infty$ to 0	tangent increases from 0 to $+\infty$
cotangent decreases from 0 to $-\infty$	cotangent decreases from $+\infty$ to 0

(Continued)

Table 1.6 Continued

Third quadrant $\left(\pi, \frac{3\pi}{2}\right)$	Fourth quadrant $\left(\frac{3\pi}{2}, 2\pi\right)$
sine decreases from 0 to -1	sine increases from -1 to 0
cosine increases from -1 to 0	cosine increases from 0 to 1
cosecant increases from $-\infty$ to -1	cosecant decreases from -1 to $-\infty$
secant decreases from -1 to $-\infty$	secant decreases from $+\infty$ to 1
tangent increases from 0 to $+\infty$	tangent increases from $-\infty$ to 0
cotangent decreases from $+\infty$ to 0	cotangent decreases from 0 to $-\infty$

1.4 | Trigonometric Ratios of Compound Angles

The algebraic sum of two or more angles is called a *compound angle*. In this section we derive formulae for the trigonometric ratios of compound angles.

THEOREM 1.11 Let x and y be any real numbers. Then

$$\cos(x + y) = \cos x \cdot \cos y - \sin x \sin y \quad (1.1)$$

PROOF Choose integers m and n and real numbers θ and ϕ such that $0 \leq \theta < 2\pi$, $0 \leq \phi < 2\pi$, $x = 2m\pi + \theta$ and $y = 2n\pi + \phi$. We shall prove that

$$\cos(\theta + \phi) = \cos \theta \cdot \cos \phi - \sin \theta \sin \phi \quad (1.2)$$

Since $\cos 0 = 1$ and $\sin 0 = 0$, Eq. (1.1) becomes trivial if $\theta = 0$ or $\phi = 0$. Suppose that $\theta > 0$ and $\phi > 0$. We have to distinguish several cases.

Case I: Suppose that $\theta + \phi < 2\pi$

Consider a rectangular coordinate system OXY with O as origin. Consider the unit circle in the coordinate plane with centre at the origin. Let P be the point with coordinates (1, 0). Choose points Q, R and S on the unit circle such that $\widehat{POQ}$, $\widehat{POR}$ and $\widehat{POS}$, measured in the anticlockwise sense, are θ , $\theta + \phi$ and $2\pi - \phi$, respectively. Then the coordinates of Q, R and S are $(\cos \theta, \sin \theta)$, $(\cos(\theta + \phi), \sin(\theta + \phi))$ and $(\cos(2\pi - \phi), \sin(2\pi - \phi))$, respectively. Since $\cos(2\pi - \phi) = \cos \phi$ and $\sin(2\pi - \phi) = -\sin \phi$, the coordinates of S are $(\cos \phi, -\sin \phi)$. We will prove that $PR = QS$ in each of the following three subcases under case I.

1. Suppose that $\theta + \phi < \pi$: In the triangle POR are QOS

$$OR = 1 = OS$$

$$OP = 1 = OQ$$

and

$$\widehat{POR} = \theta + \phi = \widehat{QOS}$$

therefore $PR = QS$ (Figure 1.32).

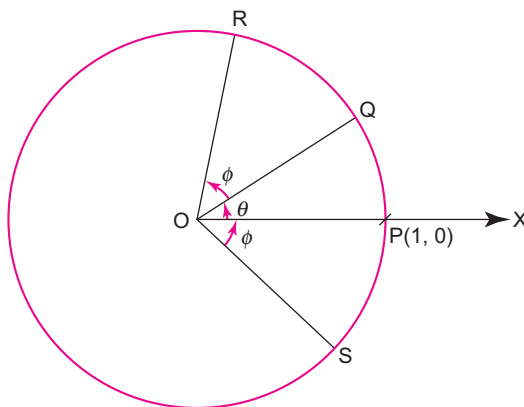


FIGURE 1.32 Case when $\theta + \phi < \pi$.

2. See Figure 1.33. Suppose that $\theta + \phi = \pi$. Then

$$PR = \text{Diameter} = 2 = QS$$

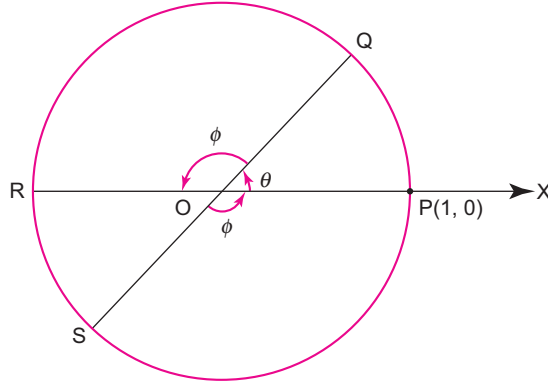


FIGURE 1.33 Case when $\theta + \phi = \pi$.

3. See Figure 1.34. Suppose that $\theta + \phi > \pi$. Then

$$2\pi - (\theta + \phi) < \pi$$

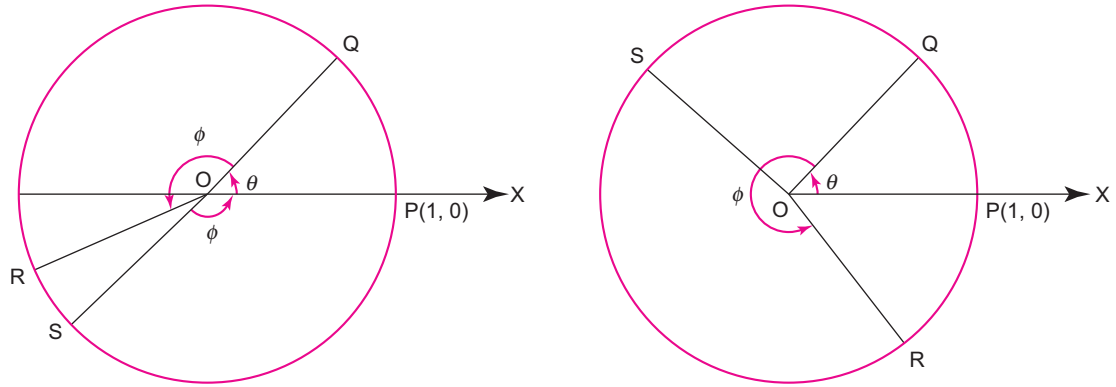


FIGURE 1.34 Case when $\theta + \phi > \pi$.

In triangles POR and QOS, we have

$$\angle POR = 2\pi - (\theta + \phi) = \angle QOS$$

and $OP = 1 = OQ$ and $OR = 1 = OS$. Therefore

$$PR = QS$$

Thus, in each of the three subcases under case I, we have proved that $PR = QS$ and hence $(PR)^2 = (QS)^2$.

Since $P = (1, 0)$, $Q = (\cos \theta, \sin \theta)$, $R = (\cos(\theta + \phi), \sin(\theta + \phi))$ and $S = (\cos(2\pi - \phi), \sin(2\pi - \phi)) = (\cos \phi, -\sin \phi)$, it follows that

$$(\cos(\theta + \phi) - 1)^2 + (\sin(\theta + \phi) - 0)^2 = (\cos \phi - \cos \theta)^2 + (-\sin \phi - \sin \theta)^2$$

Therefore

$$\begin{aligned} \cos^2(\theta + \phi) + \sin^2(\theta + \phi) + 1 - 2\cos(\theta + \phi) &= \cos^2 \phi + \cos^2 \theta - 2\cos \phi \cos \theta \\ &\quad + \sin^2 \phi + \sin^2 \theta + 2\sin \phi \sin \theta \end{aligned}$$

Since $\sin^2 x + \cos^2 x = 1$ we get

$$\begin{aligned} 2 - 2\cos(\theta + \phi) &= 2 - 2\cos\phi\cos\theta + 2\sin\phi\sin\theta \\ \Rightarrow \cos(\theta + \phi) &= \cos\theta\cos\phi - \sin\theta\sin\phi \end{aligned}$$

Case II: Suppose that $\theta + \phi = 2\pi$, then $\phi = 2\pi - \theta$. In this case

$$\begin{aligned} \cos(\theta + \phi) &= \cos 2\pi = 1 = \cos\theta\cos(2\pi - \theta) - \sin\theta\sin(2\pi - \theta) \\ &= \cos\theta\cos\phi - \sin\theta\sin\phi \end{aligned}$$

Case III: Suppose that $\theta + \phi > 2\pi$. If $\theta \leq \pi$ and $\phi \leq \pi$, then $\theta + \phi \leq 2\pi$ and the required identity is proved as in case I and case II. Therefore, we can assume that $\theta > \pi$ or $\phi > \pi$; that is, $\pi < \theta < 2\pi$ or $\pi < \phi < 2\pi$.

1. Suppose that $\pi < \theta < 2\pi$ and $0 < \phi \leq \pi$. Put $\psi = \theta - \pi$. Then $0 < \psi \leq \pi$ and $\psi + \phi \leq 2\pi$. Hence

$$\begin{aligned} \cos(\theta + \phi) &= \cos(\pi + \psi + \phi) \\ &= -\cos(\psi + \phi) \\ &= -(\cos\psi\cos\phi - \sin\psi\sin\phi) \\ &= \cos(\pi + \psi)\cos\phi - \sin(\pi + \psi)\sin\phi \\ &= \cos\theta\cos\phi - \sin\theta\sin\phi \end{aligned}$$

2. Suppose that $0 < \theta \leq 2\pi$ and $\pi < \phi \leq \pi$. The proof in this case is similar to the one in (1) above.

3. Suppose that $\pi < \theta < 2\pi$ and $\pi < \phi < 2\pi$. Put $\alpha = \theta - \pi$ and $\beta = \phi - \pi$. Then $0 < \alpha < \pi$ and $0 < \beta < \pi$. By case I,

$$\begin{aligned} \cos(\theta + \phi) &= \cos(\pi + \alpha + \pi + \beta) \\ &= \cos(2\pi + \alpha + \beta) \\ &= \cos(\alpha + \beta) \\ &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ &= \cos(\pi + \alpha)\cos(\pi + \beta) - \sin(\pi + \alpha)\sin(\pi + \beta) \\ &= \cos\theta\cos\phi - \sin\theta\sin\phi \end{aligned}$$

Thus, Eq. (1.2) is proved in all cases. Finally

$$\begin{aligned} \cos(x + y) &= \cos(2m\pi + \theta + 2n\pi + \phi) \\ &= \cos(2(m + n)\pi + \theta + \phi) \\ &= \cos(\theta + \phi) \\ &= \cos\theta\cos\phi - \sin\theta\sin\phi \\ &= \cos(2m\pi + \theta)\cos(2n\pi + \phi) - \sin(2m\pi + \theta)\sin(2n\pi + \phi) \\ &= \cos x\cos y - \sin x\sin y \end{aligned}$$

This completes the proof of the theorem. ■

THEOREM 1.12 For any real numbers x and y ,

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

PROOF Let $x, y \in \mathbb{R}$. From Theorem 1.11, we have

$$\cos\left(\frac{\pi}{2} + x + y\right) = \cos\left(\frac{\pi}{2} + x\right) \cos y - \sin\left(\frac{\pi}{2} + x\right) \sin y$$

Therefore

$$-\sin(x + y) = -\sin x \cos y - \cos x \sin y$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

■

COROLLARY 1.5 For any real numbers x and y ,

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

and

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

PROOF This follows from Theorems 1.11 and 1.12 by substituting $-y$ for y .

■

THEOREM 1.13 If none of x , y and $x + y$ is an odd multiple of $\pi/2$, then

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

PROOF Suppose that none of x , y and $x + y$ is an odd multiple of $\pi/2$. Then $\cos x$, $\cos y$ and $\cos(x + y)$ are nonzero. Now,

$$\begin{aligned} \tan(x + y) &= \frac{\sin(x + y)}{\cos(x + y)} \\ &= \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y} \quad (\text{by Theorem 1.11 and Quick Look 3}) \\ &= \frac{(\sin x \cos y / \cos x \cos y) + (\cos x \sin y / \cos x \cos y)}{(\cos x \cos y / \cos x \cos y) - (\sin x \sin y / \cos x \cos y)} \\ &= \frac{\tan x + \tan y}{1 - \tan x \tan y} \end{aligned}$$

■

THEOREM 1.14 Suppose that none of x , y and $x + y$ is an integral multiple of π . Then

$$\cot(x + y) = \frac{\cot x \cot y - 1}{\cot x + \cot y}$$

PROOF Since x , y and $x + y$ are all not of the form $n\pi$, $n \in \mathbb{Z}$, we have $\sin x \neq 0$, $\sin y \neq 0$ and $\sin(x + y) \neq 0$ and therefore $\cot x$, $\cot y$ and $\cot(x + y)$ are defined. Now, consider

$$\begin{aligned} \frac{\cot x \cot y - 1}{\cot x + \cot y} &= \frac{(\cos x / \sin x)(\cos y / \sin x) - 1}{(\cos x / \sin x) + (\cos y / \sin y)} \\ &= \frac{\cos x \cos y - \sin x \sin y}{\cos x \sin y + \cos y \sin x} \\ &= \frac{\cos(x + y)}{\sin(x + y)} = \cot(x + y) \end{aligned}$$

■

Note:

1. The formula for $\tan(x + y)$ in terms $\tan x$ and $\tan y$ given in Theorem 1.13 is valid if and only if $\tan x$ and $\tan y$ are defined and $\tan x \tan y \neq 1$, which are equivalent to saying that none of x , y and $x + y$ is an odd multiple of $\pi/2$.
2. The formula for $\cot(x + y)$ in terms of $\cot x$ and $\cot y$ given in Theorem 1.14 is valid if and only if $\cot x$ and $\cot y$ are defined and $\cot x + \cot y \neq 1$, which are equivalent to saying that none of x , y and $x + y$ is an integral multiple of π .

COROLLARY 1.6 1. If none of x , y and $x - y$ is an odd multiple of $\pi/2$, then

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

2. If none of x , y and $x - y$ is an integral multiple of π , then

$$\cot(x - y) = \frac{\cot x \cot y + 1}{\cot y - \cot x}$$

PROOF These follow from Theorems 1.13 and 1.14 by substituting $-y$ for y . ■

THEOREM 1.15 The following hold good for any real numbers x and y .

1. $\sin(x + y)\sin(x - y) = \sin^2 x - \sin^2 y = \cos^2 y - \cos^2 x$

2. $\cos(x + y)\cos(x - y) = \cos^2 x - \sin^2 y = \cos^2 y - \sin^2 x$

PROOF Let x and y be any real numbers.

1. $\sin(x + y)\sin(x - y) = (\sin x \cos y + \cos x \sin y) \cdot (\sin x \cos y - \cos x \sin y)$

$$= \sin^2 x \cos^2 y - \cos^2 x \sin^2 y$$

$$= \sin^2 x(1 - \sin^2 y) - (1 - \sin^2 x)\sin^2 y$$

$$= \sin^2 x - \sin^2 y$$

$$= (1 - \cos^2 x) - (1 - \cos^2 y)$$

$$= \cos^2 y - \cos^2 x$$

2. $\cos(x + y)\cos(x - y) = (\cos x \cos y - \sin x \sin y) \cdot (\cos x \cos y + \sin x \sin y)$

$$= \cos^2 x \cos^2 y - \sin^2 x \sin^2 y$$

$$= \cos^2 x(1 - \sin^2 y) - (1 - \cos^2 x)\sin^2 y$$

$$= \cos^2 x - \sin^2 y$$

$$= (1 - \sin^2 x) - (1 - \cos^2 y)$$

$$= \cos^2 y - \sin^2 x$$
■

THEOREM 1.16 1. If none of x , y , $x + y$ and $x - y$ is an odd multiple of $\pi/2$,

$$\tan(x + y)\tan(x - y) = \frac{\tan^2 x - \tan^2 y}{1 - \tan^2 x \tan^2 y}$$

2. If none of x , y , $x + y$ and $x - y$ is an integral multiple of π , then

$$\cot(x + y)\cot(x - y) = \frac{\cot^2 y \cot^2 x + 1}{\cot^2 y - \cot^2 x}$$

PROOF This is similar to the one given for Theorem 1.15. ■

THEOREM 1.17 For any real numbers x, y and z ,

$$\sin(x + y + z) = \sin x \cos y \cos z + \cos x \sin y \cos z + \cos x \cos y \sin z - \sin x \sin y \sin z$$

and

$$\cos(x + y + z) = \cos x \cos y \cos z - \cos x \sin y \sin z - \sin x \cos y \sin z - \sin x \sin y \cos z$$

PROOF Consider

$$\begin{aligned}\sin(x + y + z) &= \sin x \cos(y + z) + \cos x \sin(y + z) \\ &= \sin x(\cos y \cos z - \sin y \sin z) + \cos x(\sin y \cos z + \cos y \sin z) \\ &= \sin x \cos y \cos z + \cos x \sin y \cos z + \cos x \cos y \sin z - \sin x \sin y \sin z\end{aligned}$$

The expansion for $\cos(x + y + z)$ can be similarly proved. ■

THEOREM 1.18 Let none of x, y, z and $x + y + z$ be an odd multiple of $\pi/2$ and at least one of $x + y, y + z$ and $z + x$ be not an odd multiple of $\pi/2$. Then

$$\tan(x + y + z) = \frac{\Sigma \tan x - \pi \tan x}{1 - \Sigma \tan x \tan y}$$

where

$$\Sigma \tan x = \tan x + \tan y + \tan z$$

$$\Sigma \tan x \tan y = \tan x \tan y + \tan y \tan z + \tan z \tan x$$

and

$$\pi \tan x = \tan x \tan y \tan z$$

PROOF Suppose that none of $x, y, z, x + y$ and $x + y + z$ is an odd multiple of $\pi/2$. Then

$$\begin{aligned}\tan(x + y + z) &= \tan[(x + y) + z] \\ &= \frac{\tan(x + y) + \tan z}{1 - \tan(x + y)\tan z} \\ &= \frac{[(\tan x + \tan y)/(1 - \tan x \tan y)] + \tan z}{1 - [(\tan x + \tan y)/(1 - \tan x \tan y)]\tan z} \\ &= \frac{\tan x + \tan y + \tan z - \tan x \tan y \tan z}{1 - \tan x \tan y - \tan y \tan z - \tan z \tan x} \\ &= \frac{\Sigma \tan x - \pi \tan x}{1 - \Sigma \tan x \tan y}\end{aligned}$$
■

The following can be proved similarly.

THEOREM 1.19 If none of x, y, z and $x + y + z$ is an integral multiple of π and at least one of $x + y, y + z$ and $z + x$ is not an integral multiple of π , then

$$\cot(x + y + z) = \frac{\Sigma \cot x - \pi \cot x}{1 - \Sigma \cot x \cot y}$$

Example 1.21

Evaluate $\sin 15^\circ$, $\cos 15^\circ$ and $\tan 15^\circ$.

Solution:

$$\begin{aligned}\sin 15^\circ &= \sin(60^\circ - 45^\circ) \\ &= \sin 60^\circ \cos 45^\circ - \cos 60^\circ \sin 45^\circ \\ &= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{3} - 1}{2\sqrt{2}} \\ \cos 15^\circ &= \cos(60^\circ - 45^\circ) \\ &= \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

$$\begin{aligned}\tan 15^\circ &= \tan(60^\circ - 45^\circ) \\ &= \frac{\tan 60^\circ - \tan 45^\circ}{1 + \tan 60^\circ \tan 45^\circ} \\ &= \frac{\sqrt{3} - 1}{1 + \sqrt{3}} \\ &= \frac{(\sqrt{3} - 1)^2}{(1 + \sqrt{3})(\sqrt{3} - 1)} \\ &= \frac{4 - 2\sqrt{3}}{2} \\ &= 2 - \sqrt{3}\end{aligned}$$

Example 1.22

Prove that $\cos 42^\circ + \cos 78^\circ + \cos 162^\circ = 0$

Solution:

$$\begin{aligned}\cos 42^\circ + \cos 78^\circ + \cos 162^\circ \\ = \cos(60^\circ - 18^\circ) + \cos(60^\circ + 18^\circ) + \cos(180^\circ - 18^\circ)\end{aligned}$$

$$\begin{aligned}&= \cos 60^\circ \cos 18^\circ + \sin 60^\circ \sin 18^\circ \\ &\quad + \cos 60^\circ \cos 18^\circ - \sin 60^\circ \sin 18^\circ - \cos 18^\circ \\ &= 2\left(\frac{1}{2}\right)\cos 18^\circ - \cos 18^\circ = 0\end{aligned}$$

Example 1.23

Prove that $\cot\left(\frac{\pi}{4} + \theta\right)\cot\left(\frac{\pi}{4} - \theta\right) = 1$

Solution:

$$\cot\left(\frac{\pi}{4} + \theta\right)\cot\left(\frac{\pi}{4} - \theta\right)$$

$$\begin{aligned}&= \frac{\cot(\pi/4)\cot\theta - 1}{\cot(\pi/4) + \cot\theta} \cdot \frac{\cot(\pi/4)\cot\theta + 1}{\cot\theta - \cot(\pi/4)} \\ &= \frac{\cot\theta - 1}{\cot\theta + 1} \cdot \frac{\cot\theta + 1}{\cot\theta - 1} = 1\end{aligned}$$

1.5 | Trigonometric Ratios of Multiple and Submultiple Angles

In the formulae for $\cos(x + y)$ and $\sin(x + y)$ if we substitute x for y , we get formulae for $\cos 2x$ and $\sin 2y$. If x is an angle, then $2x, 3x, 4x, \dots$ are called *multiple angles* of x and $x/2, x/3, 2x/3, 2x/5, \dots$ are called *submultiple angles* of x . In this section, we derive formulae for the trigonometric ratios of multiple and submultiple angles.

THEOREM 1.20

For any real number x ,

$$\sin 2x = 2 \sin x \cos x$$

and

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

PROOF

Let x be any real number. Then

$$\sin 2x = \sin(x + x) = \sin x \cos x + \cos x \sin x = 2 \sin x \cos x$$

and

$$\begin{aligned}\cos 2x &= \cos(x+x) = \cos x \cos x - \sin x \sin x \\ &= \cos^2 x - \sin^2 x \\ &= \cos^2 x - (1 - \cos^2 x) = 2\cos^2 x - 1 \\ &= (1 - \sin^2 x) - \sin^2 x = 1 - 2\sin^2 x\end{aligned}$$

By substituting $x/2$ for x in the above, we get the following.

COROLLARY 1.7

For any real number x ,

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

and

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} = 2 \cos^2 \frac{x}{2} - 1 = 1 - 2 \sin^2 \frac{x}{2}$$

THEOREM 1.21

1. If x and $2x$ are not odd multiples of $\pi/2$, then

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

2. If $2x$ is not an integral multiple of π , then

$$\cot 2x = \frac{\cot^2 x - 1}{2 \cot x}$$

or

$$2 \cot 2x = \cot x - \tan x$$

PROOF

$$1. \tan(2x) = \tan(x+x) = \frac{\tan x + \tan x}{1 - \tan x \cdot \tan x}$$

$$= \frac{2 \tan x}{1 - \tan^2 x}$$

$$2. \cot(2x) = \cot(x+x) = \frac{\cot x \cot x - 1}{\cot x + \cot x} = \frac{\cot^2 x - 1}{2 \cot x}$$

Therefore

$$2 \cot 2x = \frac{\cot^2 x - 1}{\cot x} = \cot x - \tan x$$

COROLLARY 1.8

1. If x and $x/2$ are not odd multiples of $\pi/2$, then

$$\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$$

2. If x is not an integral multiple of π , then

$$\cot x = \frac{\cot^2(x/2) - 1}{2 \cot(x/2)}$$

or

$$2 \cot x = \cot \frac{x}{2} - \tan \frac{x}{2}$$

THEOREM 1.22 Let x be any real number. Then

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

and

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

PROOF We have

$$\begin{aligned}\sin 3x &= \sin(2x + x) \\ &= \sin 2x \cdot \cos x + \cos 2x \cdot \sin x \\ &= 2 \sin x \cos x \cos x + (1 - 2 \sin^2 x) \sin x \\ &= 2 \sin x (1 - \sin^2 x) + (1 - 2 \sin^2 x) \sin x \\ &= 3 \sin x - 4 \sin^3 x \\ \cos 3x &= \cos(2x + x) \\ &= \cos 2x \cos x - \sin 2x \sin x \\ &= (2 \cos^2 x - 1) \cos x - (2 \sin x \cos x) \sin x \\ &= 2 \cos^3 x - \cos x - 2 \sin^2 x \cos x \\ &= 2 \cos^3 x - \cos x - 2(1 - \cos^2 x) \cos x \\ &= 4 \cos^3 x - 3 \cos x\end{aligned}$$

THEOREM 1.23 1. If $3x$ is not an odd multiple of $\pi/2$, then

$$\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$

2. If $3x$ is not an integral multiple of π , then

$$\cot 3x = \frac{3 \cot x - \cot^3 x}{1 - 3 \cot^2 x}$$

PROOF 1. By hypothesis none of x and $3x$ is an odd multiple of $\pi/2$. Then

$$\begin{aligned}\tan 3x &= \tan(x + 2x) \\ &= \frac{\tan x + \tan 2x}{1 - \tan x \tan 2x} \\ &= \frac{\tan x + [2 \tan x / (1 - \tan^2 x)]}{1 - \tan x [2 \tan x / (1 - \tan^2 x)]} \\ &= \frac{(1 - \tan^2 x) \tan x + 2 \tan x}{(1 - \tan^2 x) - 2 \tan^2 x} \\ &= \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}\end{aligned}$$

Note that the above formula for $\tan 3x$ is valid even if $2x$ is an odd multiple of $\pi/2$. Similarly, formula (2) for $\cot 3x$ can also be proved. Note that (2) remains valid even if $2x$ is an integral multiple of π . If $3x$ is not integral multiple of π , then so is x .

THEOREM 1.24 1. For any real number x ,

$$\sin x = \pm \sqrt{\frac{1 - \cos 2x}{2}}$$

and

$$\cos x = \pm \sqrt{\frac{1 + \cos 2x}{2}}$$

2. If x is not an odd multiple of $\pi/2$, then

$$\tan x = \pm \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$$

The signs $\pm$ for $\sin x$, $\cos x$ and $\tan x$ in the above formulae can be determined depending on the quadrants in which the angle x lies.

PROOF

From Theorem 1.20,

$$1 - \cos 2x = 2 \sin^2 x$$

and therefore

$$\sin x = \pm \sqrt{\frac{1 - \cos 2x}{2}}$$

Similarly, the other equalities can also be proved. ■

By substituting $x/2$ for x in the above theorem, we get the following corollary.

COROLLARY 1.9

1. For all real numbers x ,

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}$$

and

$$\cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}$$

2. If $x/2$ is not an odd multiple of $\pi/2$, then

$$\tan \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}}$$

The signs $\pm$ in the above can be determined depending on the quadrant in which the angle $x/2$ lies.

THEOREM 1.25

1. If x is not an odd multiple of $\pi/2$, then

$$\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$$

and

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

2. If neither x nor $2x$ is an odd multiple of $\pi/2$, then

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

PROOF

1. We have

$$\begin{aligned} \sin 2x &= 2 \sin x \cos x \\ &= \frac{2(\sin x / \cos x)}{1/\cos^2 x} \\ &= \frac{2 \tan x}{\sec^2 x} = \frac{2 \tan x}{1 + \tan^2 x} \end{aligned}$$

and

$$\begin{aligned}
 \cos 2x &= \cos^2 x - \sin^2 x \\
 &= \cos^2 x \cdot \left(1 - \frac{\sin^2 x}{\cos^2 x}\right) \\
 &= \frac{1}{\sec^2 x} (1 - \tan^2 x) \\
 &= \frac{1 - \tan^2 x}{1 + \tan^2 x}
 \end{aligned}$$

2. From (1) it follows that

$$\tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{2 \tan x}{1 - \tan^2 x}$$

On substituting $\frac{x}{2}$ for x in the above, we get the following corollary.

COROLLARY 1.10 1. If $x/2$ is not an odd multiple of $\pi/2$, then

$$\sin x = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \quad \text{and} \quad \cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$$

2. If neither x nor $x/2$ is an odd multiple of $\pi/2$, then

$$\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$$

Example 1.24

Determine the values of all the trigonometric ratios at 18° , 36° , 54° and 72° .

Solution: Put $x = 18^\circ$. Then $2x + 3x = 90^\circ$. Therefore

$$\begin{aligned}
 \sin 2x &= \sin(90^\circ - 3x) = \cos 3x \\
 2 \sin x \cos x &= 4 \cos^3 x - 3 \cos x \\
 \cos x(2 \sin x - 4 \cos^2 x + 3) &= 0
 \end{aligned}$$

Since $x = 18^\circ$, $\cos x \neq 0$. Therefore

$$\begin{aligned}
 2 \sin x - 4 \cos^2 x + 3 &= 0 \\
 2 \sin x - 4(1 - \sin^2 x) + 3 &= 0 \\
 4 \sin^2 x + 2 \sin x - 1 &= 0
 \end{aligned}$$

Solving this quadratic equation in $\sin x$, we get

$$\sin x = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{\pm \sqrt{5} - 1}{4}$$

Since $\sin 18^\circ > 0$, we get that

$$\sin 18^\circ = \frac{\sqrt{5} - 1}{4}$$

$$\begin{aligned}
 \cos 18^\circ &= \sqrt{1 - \sin^2 18^\circ} \\
 &= \sqrt{1 - \left(\frac{\sqrt{5} - 1}{4}\right)^2} \\
 &= \sqrt{\frac{16 - (5 + 1 - 2\sqrt{5})}{16}} \\
 &= \frac{\sqrt{10 + 2\sqrt{5}}}{4}
 \end{aligned}$$

$$\operatorname{cosec} 18^\circ = \frac{1}{\sin 18^\circ} = \frac{4}{\sqrt{5} - 1}$$

$$\sec 18^\circ = \frac{1}{\cos 18^\circ} = \frac{4}{\sqrt{10 + 2\sqrt{5}}}$$

$$\tan 18^\circ = \frac{\sin 18^\circ}{\cos 18^\circ} = \frac{\sqrt{5} - 1}{\sqrt{10 + 2\sqrt{5}}}$$

$$\cot 18^\circ = \frac{\cos 18^\circ}{\sin 18^\circ} = \frac{\sqrt{10 + 2\sqrt{5}}}{\sqrt{5} - 1}$$

$$\cos 36^\circ = 2 \cos^2 18^\circ - 1 = 2 \left[\frac{10 + 2\sqrt{5}}{16} \right] - 1$$

$$\begin{aligned}
&= \frac{2+2\sqrt{5}}{8} = \frac{\sqrt{5}+1}{4} \\
\sin 36^\circ &= \sqrt{1-\cos^2 36^\circ} = \sqrt{1-\left(\frac{\sqrt{5}+1}{4}\right)^2} \\
&= \frac{\sqrt{16-(6+2\sqrt{5})}}{4} = \frac{\sqrt{10-2\sqrt{5}}}{4}
\end{aligned}$$

$$\sin 54^\circ = \sin(90^\circ - 36^\circ) = \cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

$$\sin 72^\circ = \cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

$$\cos 54^\circ = \sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}, \text{ etc.}$$

By using these, we get

$$\cos 72^\circ = \cos(90^\circ - 18^\circ) = \sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

Example 1.25

Determine the value of

$$\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8}$$

Solution: We have

$$\sin \frac{3\pi}{8} = \sin\left(\frac{\pi}{2} - \frac{\pi}{8}\right) = \cos \frac{\pi}{8}$$

$$\sin \frac{5\pi}{8} = \sin\left(\pi - \frac{3\pi}{8}\right) = \sin \frac{3\pi}{8} = \cos \frac{\pi}{8}$$

$$\sin \frac{7\pi}{8} = \sin\left(\pi - \frac{\pi}{8}\right) = \sin \frac{\pi}{8}$$

Therefore

$$\begin{aligned}
&\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} \\
&= \sin^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{8}\right) \\
&= 2(\sin^4 x + \cos^4 x), \quad \text{where } x = \frac{\pi}{8} \\
&= 2(\sin^2 x + \cos^2 x)^2 - 4\sin^2 x \cos^2 x \\
&= 2 - (2\sin x \cos x)^2 \\
&= 2 - (\sin 2x)^2 \\
&= 2 - \left(\sin \frac{\pi}{4}\right)^2 \\
&= 2 - \left(\frac{1}{\sqrt{2}}\right)^2 = 2 - \frac{1}{2} = \frac{3}{2}
\end{aligned}$$

Let us recall that the sine and cosine functions are periodic with minimum period 2π . Also, we have $-1 \leq \sin x \leq 1$ for all $x \in \mathbb{R}$ and

$$\sin \frac{\pi}{2} = 1 \quad \text{and} \quad \sin \frac{3\pi}{2} = -1$$

Therefore, 1 and -1 are the extreme values of $\sin x$, 1 being the maximum value and -1 being the minimum value. Similarly, 1 and -1 are the extreme values of $\cos x$; note that $\cos 0 = 1$ and $\cos \pi = -1$.

The trigonometric functions other than cosine and sine are not bounded and hence they do not have extreme values. In this context, we have the following.

THEOREM 1.26 Let a, b and c be any real numbers and

$$f(x) = a \cos x + b \sin x + c$$

for all $x \in \mathbb{R}$. Then $f(x)$ has extreme values and $c + \sqrt{a^2 + b^2}$ and $c - \sqrt{a^2 + b^2}$ are its maximum and minimum values, respectively.

PROOF Put $a = r \cos \theta$ and $b = r \sin \theta$, so that $\sqrt{a^2 + b^2} = r$. Now, consider

$$\begin{aligned} f(x) &= a \cos x + b \sin x + c \\ &= r(\cos \theta \cos x + \sin \theta \sin x) + c \quad \left(\text{where } r = \sqrt{a^2 + b^2}, \cos \theta = \frac{a}{r}, \sin \theta = \frac{b}{r} \right) \\ &= r \cos(x - \theta) + c \end{aligned}$$

Since $-1 \leq \cos(x - \theta) \leq 1$, we have

$$-r \leq r \cos(x - \theta) \leq r$$

Therefore

$$-r + c \leq r \cos(x - \theta) + c \leq r + c$$

That is,

$$c - \sqrt{a^2 + b^2} \leq f(x) \leq c + \sqrt{a^2 + b^2}$$

Also,

$$f(\theta) = c + \sqrt{a^2 + b^2} \quad \text{and} \quad f(\pi + \theta) = c - \sqrt{a^2 + b^2}$$

Thus $c + \sqrt{a^2 + b^2}$ and $c - \sqrt{a^2 + b^2}$ are the extreme values of $a \cos x + b \sin x + c$, $x \in \mathbb{R}$. ■

Example 1.26

Find the extreme values of

$$5 \cos x + 3 \cos\left(x + \frac{\pi}{3}\right) + 8, \quad x \in \mathbb{R}$$

Solution: Let

$$\begin{aligned} f(x) &= 5 \cos x + 3 \cos\left(x + \frac{\pi}{3}\right) + 8 \\ &= 5 \cos x + 3 \left(\cos x \cos \frac{\pi}{3} - \sin x \sin \frac{\pi}{3} \right) + 8 \\ &= 5 \cos x + 3 \left(\cos x \cdot \frac{1}{2} - \sin x \cdot \frac{\sqrt{3}}{2} \right) + 8 \\ &= \frac{13}{2} \cos x - \frac{3\sqrt{3}}{2} \sin x + 8 \end{aligned}$$

This is of the form $a \cos x + b \sin x + c$, where $a = 13/2$, $b = -3\sqrt{3}/2$ and $c = 8$. We have

$$\sqrt{a^2 + b^2} = \sqrt{\left(\frac{13}{2}\right)^2 + \left(\frac{-3\sqrt{3}}{2}\right)^2} = 7$$

From Theorem 1.26, $8 + 7 (= 15)$ and $8 - 7 (= 1)$ are the maximum and minimum values of $f(x)$, respectively.

Example 1.27

Find the period of $\sin 5x/3$.

Solution: Since the period p of $\sin x$ is 2π , the period of $\sin 5x/3$ is

$$\frac{p}{5/3} = 2\pi \cdot \frac{3}{5} = \frac{6\pi}{5} = 216^\circ$$

Example 1.28

Find the period of the function $\sin^4 x + \cos^4 x$.

Solution: We have

$$\begin{aligned}\sin^4 x + \cos^4 x &= (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x \\ &= 1 - \frac{1}{2}(\sin 2x)^2\end{aligned}$$

$$\begin{aligned}&= 1 - \frac{1}{2} \left(\frac{1 - \cos 4x}{2} \right) \\ &= \frac{3}{4} + \frac{1}{4} \cos 4x\end{aligned}$$

Therefore the required period is

$$\frac{1}{4}(\text{period of } \cos x) = \frac{2\pi}{4} = \frac{\pi}{2}$$

Example 1.29

For any real number x , express $\cos 5x$ in terms of $\cos x$.

Solution: We have

$$\begin{aligned}\cos 5x &= \cos(3x + 2x) \\ &= \cos 3x \cos 2x - \sin 3x \sin 2x \\ &= (4\cos^3 x - 3\cos x)(2\cos^2 x - 1) \\ &\quad - (3\sin x - 4\sin^3 x)(2\sin x \cos x) \\ &= (8\cos^5 x - 10\cos^3 x + 3\cos x) \\ &\quad - 2\cos x \sin^2 x(3 - 4\sin^2 x)\end{aligned}$$

$$\begin{aligned}&= (8\cos^5 x - 10\cos^3 x + 3\cos x) \\ &\quad - 2\cos x(1 - \cos^2 x)(3 - 4(1 - \cos^2 x)) \\ &= (8\cos^5 x - 10\cos^3 x + 3\cos x) \\ &\quad - 2\cos x(5\cos^2 x - 4\cos^4 x - 1) \\ &= 16\cos^5 x - 20\cos^3 x + 5\cos x\end{aligned}$$

Recall that, in Corollary 1.9, we have expressed $\sin x/2$ and $\cos x/2$ in terms of $\cos x$. In the following we express these in terms of $\sin x$.

THEOREM 1.27

For any real number x ,

$$2\sin \frac{x}{2} = \pm \sqrt{1 + \sin x} \pm \sqrt{1 - \sin x}$$

and

$$2\cos \frac{x}{2} = \pm \sqrt{1 + \sin x} \mp \sqrt{1 - \sin x}$$

PROOF

From Theorem 1.20, we have

$$2\sin \frac{x}{2} \cos \frac{x}{2} = \sin x \quad (1.3)$$

Also, we have

$$\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} = 1 \quad (1.4)$$

Adding Eqs. (1.3) and (1.4), we get

$$\sin^2 \frac{x}{2} + 2\sin \frac{x}{2} \cos \frac{x}{2} + \cos^2 \frac{x}{2} = 1 + \sin x$$

$$\left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)^2 = 1 + \sin x$$

$$\text{or} \quad \sin \frac{x}{2} + \cos \frac{x}{2} = \pm \sqrt{1 + \sin x} \quad (1.5)$$

Again subtracting Eq. (1.3) from Eq. (1.4), we get

$$\sin^2 \frac{x}{2} - 2 \sin \frac{x}{2} \cos \frac{x}{2} + \cos^2 \frac{x}{2} = 1 - \sin x$$

Therefore

$$\left(\sin \frac{x}{2} - \cos \frac{x}{2} \right)^2 = 1 - \sin x$$

$$\text{or} \quad \sin \frac{x}{2} - \cos \frac{x}{2} = \pm \sqrt{1 - \sin x} \quad (1.6)$$

Adding Eqs. (1.5) and (1.6), we get

$$2 \sin \frac{x}{2} = \pm \sqrt{1 + \sin x} \pm \sqrt{1 - \sin x}$$

Subtracting Eq. (1.6) from Eq. (1.5), we get

$$2 \cos \frac{x}{2} = \pm \sqrt{1 + \sin x} \mp \sqrt{1 - \sin x}$$

In the formulae for $\sin x/2$ and $\cos x/2$ in terms of $\sin x$ given in the above theorem, there are two ambiguous signs. In the following two examples, we have demonstrated how to determine the ambiguity in any particular case.

Example 1.30

Find the values of $\sin 15^\circ$ and $\cos 15^\circ$.

Solution: By taking $x = 30^\circ$, we get from the proof of Theorem 1.27 that

$$\begin{aligned} \sin 15^\circ + \cos 15^\circ &= \pm \sqrt{1 + \sin 30^\circ} \\ &= \pm \sqrt{1 + \frac{1}{2}} = \pm \frac{\sqrt{3}}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{and} \quad \sin 15^\circ - \cos 15^\circ &= \pm \sqrt{1 - \sin 30^\circ} \\ &= \pm \frac{1}{\sqrt{2}} \end{aligned}$$

Since $\sin 15^\circ$ and $\cos 15^\circ$ are both positive and since $\cos 15^\circ > \sin 15^\circ$, we get that $\sin 15^\circ + \cos 15^\circ$ is positive and $\sin 15^\circ - \cos 15^\circ$ is negative. Therefore, we have

$$\sin 15^\circ + \cos 15^\circ = + \frac{\sqrt{3}}{\sqrt{2}}$$

and

$$\sin 15^\circ - \cos 15^\circ = - \frac{1}{\sqrt{2}}$$

Therefore

$$\sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}} \quad \text{and} \quad \cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

Example 1.31

Find the value of $\sin(19\pi/6)$ and deduce the values of $\sin(19\pi/12)$ and $\cos(19\pi/12)$. Therefore

Solution: Note that

$$\frac{19\pi}{6} = 2\pi + \left(\pi + \frac{\pi}{6} \right)$$

$$\begin{aligned} \sin\left(\frac{19\pi}{6}\right) &= \sin(\pi + 30^\circ) \\ &= -\sin 30^\circ = -\frac{1}{2} \end{aligned}$$

Also,

$$\frac{19\pi}{6} = 570^\circ \quad \text{and} \quad \frac{19\pi}{12} = 285^\circ$$

As in Example 1.30, we get

$$\begin{aligned} \sin \frac{19\pi}{12} + \cos \frac{19\pi}{12} &= \pm \sqrt{1 + \sin \frac{19\pi}{6}} \\ &= \pm \sqrt{1 - \frac{1}{2}} = \pm \frac{1}{\sqrt{2}} \end{aligned}$$

and

$$\begin{aligned} \sin \frac{19\pi}{12} - \cos \frac{19\pi}{12} &= \pm \sqrt{1 - \sin \frac{19\pi}{6}} \\ &= \pm \frac{\sqrt{3}}{\sqrt{2}} \end{aligned}$$

Since $19\pi/12 > 3\pi/2$, $19\pi/12$ is in the fourth quadrant and hence its cosine is positive and its sine is negative. Therefore, $\sin 19\pi/12 < 0$ and $\cos 19\pi/12 > 0$. Also,

$$-\sin \frac{19\pi}{12} = \left| \sin \frac{19\pi}{12} \right| > \cos \frac{19\pi}{12}$$

Therefore

(a) $\sin \frac{19\pi}{12} + \cos \frac{19\pi}{12}$ is negative

(b) $\sin \frac{19\pi}{12} - \cos \frac{19\pi}{12}$ is also negative

Hence

$$\sin \frac{19\pi}{12} + \cos \frac{19\pi}{12} = -\frac{1}{\sqrt{2}}$$

and

$$\sin \frac{19\pi}{12} - \cos \frac{19\pi}{12} = -\frac{\sqrt{3}}{\sqrt{2}}$$

Therefore

$$\sin \frac{19\pi}{12} = -\frac{1 + \sqrt{3}}{2\sqrt{2}}$$

and

$$\cos \frac{19\pi}{12} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

Note: In general, the ambiguity of the signs $\pm$ in the formulae for $\sin x/2$ and $\cos x/2$ in the terms $\sin x$ can be cleared as follows.

We have

$$\begin{aligned} \sin \frac{x}{2} + \cos \frac{x}{2} &= \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin \frac{x}{2} + \frac{1}{\sqrt{2}} \cos \frac{x}{2} \right) \\ &= \sqrt{2} \left(\sin \frac{x}{2} \cos \frac{\pi}{4} + \cos \frac{x}{2} \sin \frac{\pi}{4} \right) \\ &= \sqrt{2} \sin \left(\frac{\pi}{4} + \frac{x}{2} \right) \\ \sin \left(\frac{\pi}{4} + \frac{x}{2} \right) &> 0 \end{aligned}$$

if $(\pi/4) + (x/2)$ lies between $2n\pi$ and $2n\pi + \pi$. That is if

$$2n\pi - \frac{\pi}{4} < \frac{x}{2} < 2n\pi + \frac{3\pi}{4}$$

Therefore

(a) $\sin \frac{x}{2} + \cos \frac{x}{2}$ is positive if $\frac{x}{2}$ lies between $2n\pi - \frac{\pi}{4}$ and $2n\pi + \frac{3\pi}{4}$

(b) It is negative otherwise.

Similarly we can prove that

$$\sin \frac{x}{2} - \cos \frac{x}{2} = \sqrt{2} \sin \left(\frac{x}{2} - \frac{\pi}{4} \right)$$

and therefore

(a) $\sin x/2 - \cos x/2$ is positive if $x/2 - \pi/4$ lies between $2n\pi$ and $2n\pi + \pi$ (i.e., if $x/2$ lies between $2n\pi + \pi/4$ and $2n\pi + 5\pi/4$).

(b) It is negative otherwise.

These results are displayed in Figure 1.35 in which OXY is the rectangular coordinate system and OP, OQ, OR and OS bisect the angles in the first, second, third and fourth quadrants, respectively.

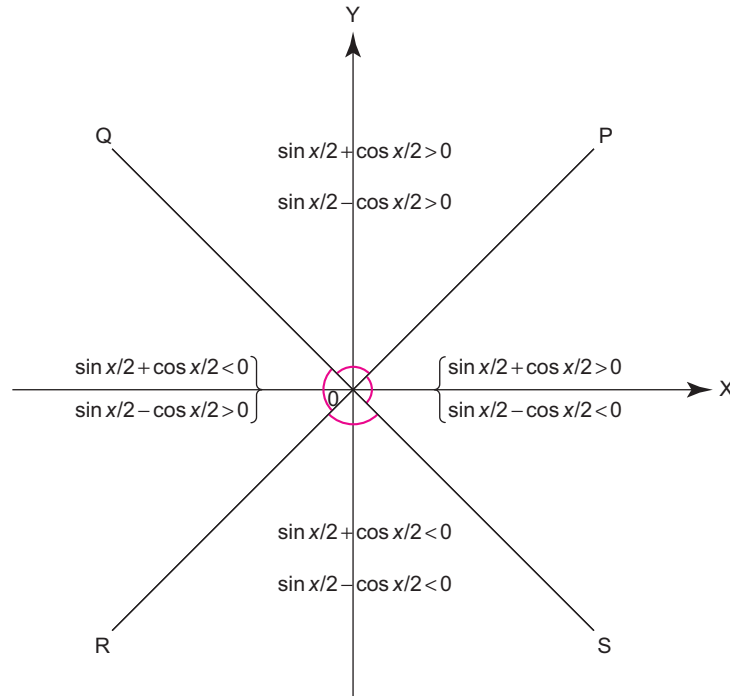


FIGURE 1.35

Example 1.32

If $2\sin(x/2) = -\sqrt{1 + \sin x} - \sqrt{1 - \sin x}$, determine the interval in which $x/2$ lies.

Solution: From Theorem 1.27 and from our hypothesis, we should have

$$\sin \frac{x}{2} + \cos \frac{x}{2} = -\sqrt{1 + \sin x}$$

and
$$\sin \frac{x}{2} - \cos \frac{x}{2} = -\sqrt{1 - \sin x}$$

From Figure 1.35 it follows that $x/2$ lies between $2n\pi - 3\pi/4$ and $2n\pi - \pi/4$ (between OR and OS).

THEOREM 1.28 If neither x nor $x/2$ is an odd multiple of $\pi/2$, then

$$\tan \frac{x}{2} = \frac{\pm \sqrt{1 + \tan^2 x} - 1}{\tan x}$$

PROOF From part (2) of Corollary 1.10, we have

$$\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$$

$$1 - \tan^2\left(\frac{x}{2}\right) = \frac{2 \tan(x/2)}{\tan x}$$

$$\begin{aligned}
\tan^2\left(\frac{x}{2}\right) + 2\frac{1}{\tan x} \cdot \tan\left(\frac{x}{2}\right) + \frac{1}{\tan^2 x} &= 1 + \frac{1}{\tan^2 x} = \frac{1 + \tan^2 x}{\tan^2 x} \\
\left(\tan\frac{x}{2} + \frac{1}{\tan x}\right)^2 &= \frac{1 + \tan^2 x}{\tan^2 x} \\
\tan\frac{x}{2} + \frac{1}{\tan x} &= \frac{\pm\sqrt{1 + \tan^2 x}}{\tan x} \\
\tan\frac{x}{2} &= \frac{\pm\sqrt{1 + \tan^2 x} - 1}{\tan x}
\end{aligned}$$

The ambiguity of $\pm$ in the above formula can only be determined when we know about the magnitude of x . Consider the following example.

Example 1.33

It is given that $\sec 15^\circ = 2 - \sqrt{3}$ (see Example 1.31). Find $\tan 7(1/2)^\circ$ and $\cot 7(1/2)^\circ$.

Solution: Taking $x = 15^\circ$ in Theorem 1.28, we have

$$\begin{aligned}
\tan 7\frac{1^\circ}{2} &= \frac{\pm\sqrt{1 + (2 - \sqrt{3})^2} - 1}{2 - \sqrt{3}} \\
&= \frac{\pm\sqrt{8 - 4\sqrt{3}} - 1}{2 - \sqrt{3}} = \frac{\pm(\sqrt{6} - \sqrt{2})}{2 - \sqrt{3}}
\end{aligned}$$

Since $\tan 7(1/2)^\circ$ is positive, we must have

$$\begin{aligned}
\tan 7\frac{1^\circ}{2} &= \frac{+\sqrt{8 - 4\sqrt{3}} - 1}{2 - \sqrt{3}} \\
&= \frac{(\sqrt{6} - \sqrt{2}) - 1}{2 - \sqrt{3}} \\
&= (\sqrt{6} - \sqrt{2} - 1)(2 + \sqrt{3}) \\
&= \sqrt{6} - \sqrt{3} + \sqrt{2} - 2 \\
&= (\sqrt{3} - \sqrt{2})(\sqrt{2} - 1)
\end{aligned}$$

Since $\tan 15^\circ = \tan(\pi + 15^\circ) = \tan(195^\circ)$, we get from the above

$$\tan \frac{195^\circ}{2} = \frac{\pm(\sqrt{6} - \sqrt{2}) - 1}{2 - \sqrt{3}}$$

But we know that

$$\tan \frac{195^\circ}{2} = \tan\left(\frac{\pi}{2} + 7\frac{1^\circ}{2}\right) < 0$$

Therefore

$$\begin{aligned}
\tan \frac{195^\circ}{2} &= \frac{-(\sqrt{6} - \sqrt{2}) - 1}{2 - \sqrt{3}} \\
&= (-\sqrt{6} + \sqrt{2} - 1)(2 + \sqrt{3}) \\
&= -(\sqrt{2} + \sqrt{3})(\sqrt{2} + 1) \\
\cot 7\frac{1^\circ}{2} &= -\tan\left(\frac{\pi}{2} + 7\frac{1^\circ}{2}\right) \\
&= -\tan \frac{195^\circ}{2} = (\sqrt{2} + \sqrt{3})(\sqrt{2} + 1)
\end{aligned}$$

1.6 | Sum and Product Transformations

In this section, we derive certain formulae through which we can transform the sum or difference of two trigonometric ratios into products and vice versa. First we take up transformation of sum into product of trigonometric ratios.

THEOREM 1.29 The following hold good for any real numbers x and y .

1. $\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$
2. $\sin(x + y) - \sin(x - y) = 2 \cos x \sin y$
3. $\cos(x + y) + \cos(x - y) = 2 \cos x \cos y$
4. $\cos(x + y) - \cos(x - y) = -2 \sin x \sin y$

PROOF This is a simple consequence of the following identities:

1. $\sin(x + y) = \sin x \cos y + \cos x \sin y$
2. $\sin(x - y) = \sin x \cos y - \cos x \sin y$
3. $\cos(x + y) = \cos x \cos y - \sin x \sin y$
4. $\cos(x - y) = \cos x \cos y + \sin x \sin y$



Try it out Prove Theorem 1.29.

By taking

$$x = \frac{a+b}{2} \quad \text{and} \quad y = \frac{a-b}{2}$$

in the above, we get the following theorem (note that $x + y = a$ and $x - y = b$).

THEOREM 1.30

1. $\sin a + \sin b = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$
2. $\sin a - \sin b = 2 \cos\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$
3. $\cos a + \cos b = 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$
4. $\cos a - \cos b = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$
5. $\cos b - \cos a = 2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$

Try it out Prove Theorem 1.30.

QUICK LOOK 6

Regarding formulae given in Theorem 1.29 for transforming a sum into a product, the following may be noted and used as tips to remember the formulae.

$$1. \quad 2 \sin x \cos y = \sin(x + y) + \sin(x - y)$$

$$= \sin(\text{sum}) + \sin(\text{difference})$$

$$2. \quad 2 \cos x \sin y = \sin(x + y) - \sin(x - y)$$

$$= \sin(\text{sum}) - \sin(\text{difference})$$

$$3. \quad 2 \cos x \cos y = \cos(x + y) + \cos(x - y)$$

$$= \cos(\text{sum}) + \cos(\text{difference})$$

$$4. \quad -2 \sin x \sin y = \cos(x + y) - \cos(x - y)$$

$$= \cos(\text{sum}) - \cos(\text{difference})$$

Therefore

$$2 \sin x \sin y = \cos(\text{difference}) - \cos(\text{sum})$$

Example 1.34

Find the value of $4(\cos 66^\circ + \sin 84^\circ)$.

Solution:

$$\begin{aligned} 4(\cos 66^\circ + \sin 84^\circ) &= 4\left(\cos 66^\circ + \sin\left(\frac{\pi}{2} - 6^\circ\right)\right) \\ &= 4(\cos 66^\circ + \cos 6^\circ) \end{aligned}$$

$$\begin{aligned} &= 4\left(2 \cos\left(\frac{66^\circ + 6^\circ}{2}\right) \cos\left(\frac{66^\circ - 6^\circ}{2}\right)\right) \\ &= 8 \cos 36^\circ \cos 30^\circ \\ &= 8\left(\frac{\sqrt{5}+1}{4}\right)\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3} + \sqrt{15} \end{aligned}$$

(See Example 1.24 for $\cos 36^\circ$.)

Example 1.35

Prove that for any real numbers x, y and z ,

$$\begin{aligned} \cos(y+z-x) - \cos(z+x-y) + \cos(x+y-z) \\ - \cos(x+y+z) = 4 \sin x \cos y \sin z \end{aligned}$$

Solution: We have

$$\begin{aligned} \cos(y+z-x) - \cos(z+x-y) \\ = 2 \sin\left(\frac{(z+x-y) + (y+z-x)}{2}\right) \\ \sin\left(\frac{(z+x-y) - (y+z-x)}{2}\right) \\ = 2 \sin z \sin(x-y) \end{aligned}$$

and

$$\begin{aligned} \cos(x+y-z) - \cos(x+y+z) \\ = 2 \sin\left(\frac{(x+y+z) + (x+y-z)}{2}\right) \\ \sin\left(\frac{(x+y+z) - (x+y-z)}{2}\right) \\ = 2 \sin(x+y) \sin z \end{aligned}$$

Therefore

$$\begin{aligned} \cos(y+z-x) - \cos(z+x-y) + \cos(x+y+z) \\ = 2 \sin z \sin(x-y) + 2 \sin(x+y) \sin z \\ = 2 \sin z [\sin(x+y) + \sin(x-y)] \\ = 2 \sin z (2 \sin x \sin y) \\ = 4 \sin x \cos y \sin z \end{aligned}$$

Example 1.36

Prove that $\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ = 1/16$.

Solution: Consider

$$\begin{aligned} \cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ \\ = \frac{1}{2} (2 \cos 80^\circ \cdot \cos 40^\circ) \cdot \cos 20^\circ \left(\frac{1}{2}\right) \left(\text{since } \cos 60^\circ = \frac{1}{2}\right) \\ = \frac{1}{4} \cos 20^\circ (\cos(80^\circ + 40^\circ) + \cos(80^\circ - 40^\circ)) \\ = \frac{1}{4} \cos 20^\circ (\cos 120^\circ + \cos 40^\circ) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{4} \cos 20^\circ \left(\frac{-1}{2} + \cos 40^\circ\right) \\ &= \frac{-1}{8} \cos 20^\circ + \frac{1}{8} (2 \cos 40^\circ \cos 20^\circ) \\ &\quad \left(\text{since } \cos 120^\circ = \frac{-1}{2}\right) \\ &= \frac{-1}{8} \cos 20^\circ + \frac{1}{8} (\cos 60^\circ + \cos 20^\circ) \\ &= \frac{1}{8} \cos 60^\circ = \frac{1}{8} \cdot \frac{1}{2} = \frac{1}{16} \end{aligned}$$

Hence proved.

Example 1.37

If A, B, C are angles of a triangle, prove that

$$\sin 2A - \sin 2B + \sin 2C = 4 \cos A \sin B \cos C$$

Solution: Note that

$$A + B + C = 180^\circ = \pi$$

Now

$$\begin{aligned} (\sin 2A - \sin 2B) + \sin 2C \\ = 2 \cos(A+B) \sin(A-B) + 2 \sin C \cos C \\ = 2 \cos(\pi - C) \sin(A-B) + 2 \sin C \cos C \end{aligned}$$

$$\begin{aligned} &= -2 \cos C \sin(A-B) + 2 \sin C \cos C \\ &= 2 \cos C [\sin C - \sin(A-B)] \\ &= 2 \cos C [\sin\{\pi - (A+B)\} - \sin(A-B)] \\ &= 2 \cos C (2 \cos A \sin B) \\ &= 4 \cos A \sin B \cos C \end{aligned}$$

Example 1.38

If $A + B + C = \pi$, prove that

$$\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

Solution: We have

$$\begin{aligned} (\sin A + \sin B) + \sin C &= 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2} \\ &\quad + 2 \sin \frac{C}{2} \cos \frac{C}{2} \end{aligned}$$

$$\begin{aligned} &= 2 \cos \frac{C}{2} \cos \frac{A-B}{2} + 2 \cos \frac{C}{2} \cos \frac{A+B}{2} \\ &= 2 \cos \frac{C}{2} \left(\cos \frac{A-B}{2} + \cos \frac{A+B}{2} \right) \\ &= 2 \cos \frac{C}{2} \left(2 \cos \frac{A}{2} \cos \frac{B}{2} \right) \\ &= 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \end{aligned}$$

Hence proved.

Example 1.39

If $A + B + C = \pi$, then prove that

- (i) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
 (ii) $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$
 (iii) When $\tan A, \tan B, \tan C$ are not equal to zero, then

$$\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

and

$$\cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2} = \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}$$

Solution:

- (i) If $\tan(A+B) = \tan(\pi - C) = -\tan C$, then

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

- (ii) Consider

$$\frac{A+B}{2} = \frac{\pi}{2} - C$$

It gives

$$\tan \left(\frac{A+B}{2} \right) = \cot \frac{C}{2}$$

$$\frac{\tan(A/2) + \tan(B/2)}{1 - \tan(A/2) \tan(B/2)} = \frac{1}{\tan(C/2)}$$

$$\sum \tan \frac{B}{2} \tan \frac{C}{2} = 1$$

- (iii) follows from (i) and (ii).

**QUICK LOOK 7**

Values of trigonometrical ratios of some important angles.

1. $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$

2. $\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$

3. $\tan 15^\circ = \cot 75^\circ = 2 - \sqrt{3}$

4. $\cot 15^\circ = \tan 75^\circ = 2 + \sqrt{3}$

5. $\sin 22\frac{1}{2}^\circ = \frac{1}{2} \sqrt{2 - \sqrt{2}}$

6. $\cos 22\frac{1}{2}^\circ = \frac{1}{2} \sqrt{2 + \sqrt{2}}$

7. $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$

8. $\sin 18^\circ = \cos 72^\circ = \frac{1}{4}(\sqrt{5} - 1)$

9. $\cos 18^\circ = \sin 72^\circ = \frac{1}{4}\sqrt{10 + 2\sqrt{5}}$

10. $\sin 36^\circ = \cos 54^\circ = \frac{1}{4}\sqrt{10 - 2\sqrt{5}}$

11. $\cos 36^\circ = \sin 54^\circ = \frac{1}{4}(\sqrt{5} + 1)$

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. If $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = \tan^2 \theta + \cot^2 \theta + k$, then k equals
(A) 9 (B) 7 (C) 5 (D) 8

Solution:

$$\begin{aligned}\text{LHS} &= \sin^2 \theta + 2 + \operatorname{cosec}^2 \theta + \cos^2 \theta + 2 + \sec^2 \theta \\ &= 5 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) \\ &= \tan^2 \theta + \cot^2 \theta + 7\end{aligned}$$

Answer: (B)

2. If $\sin \theta + \sin^2 \theta = 1$, then $\cos^{12} \theta + 3\cos^{10} \theta + 3\cos^8 \theta + \cos^6 \theta$ is equal to
(A) 0 (B) 2 (C) 1 (D) 4

Solution: Given that $\sin \theta + \sin^2 \theta = 1$. This implies $\sin \theta = \cos^2 \theta$.

$$\begin{aligned}\cos^{12} \theta + 3\cos^{10} \theta + 3\cos^8 \theta + \cos^6 \theta \\ &= \cos^6 \theta [\cos^6 \theta + 3\cos^4 \theta + 3\cos^2 \theta + 1] \\ &= \cos^6 \theta (1 + \cos^2 \theta)^3 \\ &= \sin^3 (1 + \sin \theta)^3 \\ &= (\sin \theta + \sin^2 \theta)^3 \\ &= 1\end{aligned}$$

Answer: (C)

3. If $\operatorname{cosec} \theta - \sin \theta = m^3$ and $\sec \theta - \cos \theta = n^3$, then $m^2 n^2 (m^2 + n^2)$ is equal to
(A) 4 (B) $2\sqrt{2}$ (C) $\sqrt[3]{2}$ (D) 1

Solution:

$$\frac{1}{\sin \theta} - \sin \theta = m^3 \quad \text{and} \quad \frac{1}{\cos \theta} - \cos \theta = n^3$$

Therefore

$$\cos^2 \theta = m^3 \sin \theta \quad \text{and} \quad \sin^2 \theta = n^3 \cos \theta$$

Now

$$\cos^4 \theta = m^6 \sin^2 \theta = m^6 n^3 \cos \theta$$

Therefore

$$\cos^3 \theta = m^6 n^3 \quad \text{and} \quad \sin^3 \theta = m^3 n^6$$

Now

$$\begin{aligned}1 &= \cos^2 \theta + \sin^2 \theta \\ &= m^4 n^2 + m^2 n^4 \\ &= m^2 n^2 (m^2 + n^2)\end{aligned}$$

Answer: (D)

4. If $a \sec \theta = 1 - b \tan \theta$ and $a^2 \sec^2 \theta = 5 + b^2 \tan^2 \theta$, then $a^2 b^2 + 4a^2 = kb^2$ where value of k is
(A) 3 (B) 4 (C) 5 (D) 9

Solution: $(1 - b \tan \theta)^2 = a^2 \sec^2 \theta = 5 + b^2 \tan^2 \theta$
Therefore

$$\begin{aligned}1 - 2b \tan \theta + b^2 \tan^2 \theta &= 5 + b^2 \tan^2 \theta \\ \tan \theta &= \frac{-2}{b}\end{aligned}$$

Now

$$\begin{aligned}a^2 (1 + \tan^2 \theta) &= a^2 \sec^2 \theta = 5 + b^2 \tan^2 \theta \\ a^2 \left(1 + \frac{4}{b^2}\right) &= 5 + b^2 \left(\frac{4}{b^2}\right) \\ a^2 (b^2 + 4) &= 9b^2 \\ k &= 9\end{aligned}$$

Answer: (D)

5. If $\tan^2 \theta = 1 - a^2$, then the value of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ is
(A) $(2 - a^2)^{1/2}$ (B) $(2 - a^2)^{3/2}$
(C) $(2 + a^2)^{1/2}$ (D) $(2 + a^2)^{3/2}$

Solution:

$$\begin{aligned}\sec \theta + \tan^3 \theta \operatorname{cosec} \theta &= \sec \theta (1 + \tan^3 \theta \cot \theta) \\ &= \sec \theta (1 + \tan^2 \theta) \\ &= (1 + \tan^2 \theta)^{3/2} \\ &= (2 - a^2)^{3/2}\end{aligned}$$

Answer: (B)

6. For positive integer n , let $f(n) = \sin^n \theta + \cos^n \theta$. Then

$$\frac{f(3) - f(5)}{f(5) - f(7)} =$$

(A) $\frac{f(1)}{f(3)}$ (B) $\frac{f(3)}{f(1)}$ (C) $f(2)$ (D) $\frac{f(2)}{f(3)}$

Solution: We have

$$\begin{aligned} f(3) - f(5) &= (\sin^3 \theta + \cos^3 \theta) - (\sin^5 \theta + \cos^5 \theta) \\ &= \sin^3 \theta \cos^2 \theta + \cos^3 \theta \sin^2 \theta \\ &= \sin^2 \theta \cos^2 \theta (\sin \theta + \cos \theta) \\ &= (\sin^2 \theta \cos^2 \theta) f(1) \end{aligned}$$

Similarly

$$f(5) - f(7) = (\sin^2 \theta \cos^2 \theta) f(3)$$

Therefore

$$\frac{f(3) - f(5)}{f(5) - f(7)} = \frac{f(1)}{f(3)}$$

Answer: (A)

7. Let $f(x) = \sin x(\sin x + \sin 3x)$. Then $f(x)$

- (A) ≥ 0 only when $x \leq 0$
- (B) ≤ 0 for all real x
- (C) ≥ 0 for all real x
- (D) ≤ 0 only when $x \leq 0$

Solution:

$$\begin{aligned} f(x) &= \sin^2 x + \sin x \sin 3x \\ &= \frac{1 - \cos 2x}{2} + \frac{\cos 2x - \cos 4x}{2} \\ &= \frac{1}{2}(1 - \cos 4x) \\ &= \sin^2 2x \geq 0 \text{ for all real } x \end{aligned}$$

Answer: (C)

8. Let

$$f_n(x) = \frac{1}{n}(\sin^n x + \cos^n x)$$

for $n = 1, 2, 3, \dots$. Then $f_4(x) - f_6(x)$ is equal to

- (A) $\frac{1}{10}$
- (B) 10
- (C) 12
- (D) $\frac{1}{12}$

Solution:

$$\begin{aligned} f_4(x) - f_6(x) &= \frac{1}{4}(\sin^4 x + \cos^4 x) - \frac{1}{6}(\sin^6 x + \cos^6 x) \\ &= \frac{1}{12}[3\sin^4 x + 3\cos^4 x - 2\sin^6 x - 2\cos^6 x] \\ &= \frac{1}{12}[3\sin^4 x + 3\cos^4 x - 2\sin^4 x(1 - \cos^2 x) \\ &\quad - 2\cos^4 x(1 - \sin^2 x)] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{12}[\sin^4 x + \cos^4 x + 2\sin^2 x \cos^2 x] \\ &= \frac{1}{12}(\sin^2 x + \cos^2 x)^2 \\ &= \frac{1}{12} \end{aligned}$$

Answer: (D)

9. $(1 - \cos 22^\circ)(1 - \cos 23^\circ)$ equals

- (A) 1
- (B) 2
- (C) 4
- (D) -2

Solution: It is known that

$$\cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$$

Take $A = 22^\circ$, $B = 23^\circ$. Therefore

$$\begin{aligned} \cot 22^\circ + \cot 23^\circ &= \cot 22^\circ \cot 23^\circ - 1 \\ (1 - \cot 22^\circ)(1 - \cot 23^\circ) &= 2 \end{aligned}$$

Answer: (B)

10. $\left(4\cos^2 \frac{\pi}{20} - 3\right)\left(4\cos^2 \frac{3\pi}{20} - 3\right) =$

- (A) $\sec \frac{\pi}{20}$
- (B) $\tan \frac{\pi}{20}$
- (C) $\cot \frac{\pi}{20}$
- (D) $\operatorname{cosec} \frac{\pi}{20}$

Solution: It is known that

$$\cos 3x = 4\cos^3 x - 3\cos x$$

Therefore

$$4\cos^2 x - 3 = \frac{\cos 3x}{\cos x}$$

for all $x \neq (2k+1)(\pi/2)$ where k is an integer. Hence

$$\begin{aligned} \left(4\cos^2 \frac{\pi}{20} - 3\right)\left(4\cos^2 \frac{3\pi}{20} - 3\right) &= \frac{\cos(3\pi/20)}{\cos(\pi/20)} \times \frac{\cos(9\pi/20)}{\cos(3\pi/20)} \\ &= \frac{\cos(9\pi/20)}{\cos(\pi/20)} \\ &= \frac{\cos[(\pi/2) - (\pi/20)]}{\cos(\pi/20)} \\ &= \tan \frac{\pi}{20} \end{aligned}$$

Answer: (B)

11. If

$$\frac{\sin^4 \alpha}{a} + \frac{\cos^4 \alpha}{b} = \frac{1}{a+b}$$

then

$$\frac{\sin^8 \alpha}{a^3} + \frac{\cos^8 \alpha}{b^3} = \frac{1}{(a+b)^k}$$

where the value of k is

- (A) 3 (B) 2 (C) -3 (D) -2

Solution: From the hypothesis

$$(a+b) \left(\frac{\sin^4 \alpha}{a} + \frac{\cos^4 \alpha}{b} \right) = 1 = (\sin^2 \alpha + \cos^2 \alpha)^2$$

Therefore

$$\begin{aligned} \cos^4 \alpha + \sin^4 \alpha + \frac{b}{a} \sin^4 \alpha + \frac{a}{b} \cos^4 \alpha \\ = \sin^4 \alpha + 2 \sin^2 \alpha \cos^2 \alpha + \cos^4 \alpha \\ \frac{b}{a} \sin^4 \alpha + \frac{a}{b} \cos^4 \alpha - 2 \sin^2 \alpha \cos^2 \alpha = 0 \\ \left(\sqrt{\frac{b}{a}} \sin^2 \alpha - \sqrt{\frac{a}{b}} \cos^2 \alpha \right)^2 = 0 \end{aligned}$$

Let

$$\frac{b}{a} \sin^4 \alpha = \frac{a}{b} \cos^4 \alpha = \lambda \text{ (say)}$$

Then

$$\frac{1}{a+b} = \frac{\sin^4 \alpha}{a} + \frac{\cos^4 \alpha}{b} = \frac{\lambda}{b} + \frac{\lambda}{a} = \lambda \frac{(a+b)}{ab}$$

This gives

$$\lambda = \frac{ab}{(a+b)^2}$$

Hence

$$\begin{aligned} \frac{\sin^8 \alpha}{a^3} + \frac{\cos^8 \alpha}{b^3} &= \lambda^2 \left[\frac{a^2}{a^3 b^2} + \frac{b^2}{a^2 b^3} \right] \\ &= \frac{a^2 b^2}{(a+b)^4} \left[\frac{a+b}{a^2 b^2} \right] \\ &= \frac{1}{(a+b)^3} \end{aligned}$$

Answer: (A)12. $\tan \alpha \tan \left(\frac{\pi}{3} + \alpha \right) \tan \left(\frac{\pi}{3} - \alpha \right)$ is equal to

- (A)
- $\cot 3\alpha$
- (B)
- $\cot 2\alpha$
-
- (C)
- $\tan 3\alpha$
- (D)
- $\tan 2\alpha$

Solution:

$$\begin{aligned} \tan \alpha \tan \left(\frac{\pi}{3} + \alpha \right) \tan \left(\frac{\pi}{3} - \alpha \right) \\ = \tan \alpha \left[\frac{\sqrt{3} + \tan \alpha}{1 - \sqrt{3} \tan \alpha} \right] \left[\frac{\sqrt{3} - \tan \alpha}{1 + \sqrt{3} \tan \alpha} \right] \\ = \tan \alpha \left[\frac{3 - \tan^2 \alpha}{1 - 3 \tan^2 \alpha} \right] \\ = \frac{3 \tan \alpha - \tan^3 \alpha}{1 - 3 \tan^2 \alpha} \\ = \tan 3\alpha \end{aligned}$$

Answer: (C)13. If $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \cdots (1 + \tan 45^\circ) = 2^n$ then n is equal to

- (A) 22 (B) 23 (C) 21 (D) 24

Solution:

$$\begin{aligned} (1 + \tan k^\circ)[1 + \tan(45^\circ - k^\circ)] \\ = (1 + \tan k^\circ) \left[1 + \frac{1 - \tan k^\circ}{1 + \tan k^\circ} \right] = 2 \end{aligned}$$

Therefore

$$\begin{aligned} 2^n &= [(1 + \tan 1^\circ)(1 + \tan 44^\circ)][(1 + \tan 2^\circ) \\ &\quad (1 + \tan 43^\circ)] \cdots [(1 + \tan 22^\circ)(1 + \tan 23^\circ)] \times 2 \\ &= (2 \times 2 \times \cdots \times 22 \text{ times}) \times 2 = 2^{23} \end{aligned}$$

Hence

$$n = 23$$

Answer: (B)14. $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)$ is equal to

- (A) 0 (B) 1 (C) -1 (D) 2

Solution:

$$\begin{aligned} 2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) \\ = 2[\sin^2 \theta + \cos^2 \theta][\sin^4 \theta - \sin^2 \theta \cos^2 \theta + \cos^4 \theta] \\ - 3[(\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta] \end{aligned}$$

$$\begin{aligned}
&= 2[(\sin^2 \theta + \cos^2 \theta)^2 - 3\sin^2 \theta \cos^2 \theta] \\
&\quad - 3[1 - 2\sin^2 \theta \cos^2 \theta] \\
&= 2(1 - 3\sin^2 \theta \cos^2 \theta) - 3(1 - 2\sin^2 \theta \cos^2 \theta) \\
&= 2 - 3 = -1
\end{aligned}$$

Answer: (C)**15.** The value of the expression

$$\frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ}$$

is

(A) $\frac{2\sin 20^\circ}{\cos 40^\circ}$

(B) $\frac{2\sin 20^\circ}{\sin 40^\circ}$

(C) 2

(D) 4

Solution:

$$\begin{aligned}
\frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ} &= \frac{\sqrt{3} \cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cos 40^\circ} \\
&= \frac{2[(\sqrt{3}/2)\cos 20^\circ - 1/2 \sin 20^\circ]}{\sin 20^\circ \cos 20^\circ} \\
&= \frac{2[\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin 20^\circ]}{\sin 20^\circ \cos 20^\circ} \\
&= \frac{2 \sin(60^\circ - 20^\circ)}{\sin 20^\circ \cos 20^\circ} \\
&= \frac{2 \sin 40^\circ}{\sin 20^\circ \cos 20^\circ} \\
&= \frac{2(2 \sin 20^\circ \cos 20^\circ)}{\sin 20^\circ \cos 20^\circ} \\
&= 4
\end{aligned}$$

Answer: (D)**16.** The value of

$$\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14}$$

is

(A) $\frac{1}{32}$

(B) $\frac{1}{64}$

(C) $\frac{1}{128}$

(D) $\frac{1}{256}$

Solution: Since

$$\frac{13\pi}{14} = \pi - \frac{\pi}{14}, \frac{11\pi}{14} = \pi - \frac{3\pi}{14}, \frac{9\pi}{14} = \pi - \frac{5\pi}{14} \text{ and } \frac{7\pi}{14} = \frac{\pi}{2}$$

the given expression can be written as

$$\sin^2 \frac{\pi}{14} \sin^2 \frac{3\pi}{14} \sin^2 \frac{5\pi}{14}$$

Let

$$s = \sin^2 \frac{\pi}{14} \sin^2 \frac{3\pi}{14} \sin^2 \frac{5\pi}{14}$$

and

$$t = \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14}$$

Now

$$\begin{aligned}
\sin \left(\frac{\pi}{14} \right) &= \sin \left(\frac{\pi}{2} - \frac{6\pi}{14} \right) = \cos \frac{6\pi}{14} \\
&= \cos \left(\pi - \frac{8\pi}{14} \right) = -\cos \left(\frac{8\pi}{14} \right)
\end{aligned}$$

$$\sin \left(\frac{3\pi}{14} \right) = \sin \left(\frac{\pi}{2} - \frac{4\pi}{14} \right) = \cos \frac{4\pi}{14}$$

$$\sin \left(\frac{5\pi}{14} \right) = \sin \left(\frac{\pi}{2} - \frac{2\pi}{14} \right) = \cos \frac{2\pi}{14}$$

Therefore

$$\begin{aligned}
t &= -\cos \frac{8\pi}{14} \cdot \cos \frac{4\pi}{14} \cos \frac{2\pi}{14} \\
&= -\cos \frac{2\pi}{14} \cos \frac{4\pi}{14} \cdot \cos \frac{8\pi}{14} \\
&= \frac{-[2 \sin(2\pi/14) \cos(2\pi/14)]}{2 \sin(2\pi/14)} \cdot \cos \frac{4\pi}{14} \cdot \cos \frac{8\pi}{14} \\
&= \frac{-1}{2 \sin(2\pi/14)} \left[\sin \frac{4\pi}{14} \cos \frac{4\pi}{14} \right] \cos \frac{8\pi}{14} \\
&= \frac{-1}{2^2 \sin(2\pi/14)} \left(\sin \frac{8\pi}{14} \right) \cos \frac{8\pi}{14} \\
&= \frac{-\sin(16\pi/14)}{8 \sin(2\pi/14)} = \frac{-\sin(8\pi/7)}{8 \sin(\pi/7)} = \frac{1}{8}
\end{aligned}$$

Hence

$$s = t^2 = \left(\frac{1}{8} \right)^2 = \frac{1}{64}$$

Answer: (B)**17.** If $\sec(\theta + \alpha) + \sec(\theta - \alpha) = 2\sec \theta$ and α is not an even multiple of π , then $(1 + \cos \alpha)$ is equal to

(A) $\sec^2 \theta$

(B) $\operatorname{cosec}^2 \theta$

(C) $\sin^2 \theta$

(D) $\cos^2 \theta$

Solution: Given that $\sec(\theta + \alpha) + \sec(\theta - \alpha) = 2\sec \theta$

$$\frac{1}{\cos(\theta + \alpha)} + \frac{1}{\cos(\theta - \alpha)} = \frac{2}{\cos \theta}$$

Therefore

$$\begin{aligned}\frac{\cos(\theta - \alpha) + \cos(\theta + \alpha)}{\cos(\theta + \alpha)\cos(\theta - \alpha)} &= \frac{2}{\cos \theta} \\ (2 \cos \theta \cos \alpha) \cos \theta &= 2 \cos(\theta + \alpha) \cos(\theta - \alpha) \\ \cos^2 \theta \cos \alpha &= \cos^2 \theta - \sin^2 \alpha \\ \sin^2 \alpha &= \cos^2 \theta (1 - \cos \alpha) \\ 1 - \cos^2 \alpha &= \cos^2 \theta (1 - \cos \alpha) \\ 1 + \cos \alpha &= \cos^2 \theta \quad (\because \cos \alpha \neq 1)\end{aligned}$$

Answer: (D)

18. If $\tan \alpha = \frac{1}{7}$ and $\sin \beta = \frac{1}{\sqrt{10}}$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\alpha + 2\beta$ is equal to

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Solution:

$$\begin{aligned}\tan(\alpha + 2\beta) &= \frac{\tan \alpha + \tan 2\beta}{1 - \tan \alpha \tan 2\beta} \\ &= \frac{(1/7) + \tan 2\beta}{1 - (1/7)\tan 2\beta} \\ &= \frac{(1/7) + [2 \tan \beta / (1 - \tan^2 \beta)]}{1 - (1/7)[2 \tan \beta / (1 - \tan^2 \beta)]} \\ &\quad \left(\because \sin \beta = \frac{1}{\sqrt{10}} \Rightarrow \beta \neq \frac{\pi}{4} \right) \\ &= \frac{\frac{1}{7} + \left[\left(2 \times \frac{1}{3} \right) / \left(1 - \frac{1}{9} \right) \right]}{1 - \frac{1}{7} \left[\left(2 \times \frac{1}{3} \right) / \left(1 - \frac{1}{9} \right) \right]} \\ &= \frac{(1/7) + (3/4)}{1 - (3/4) \cdot (1/7)} \\ &= \frac{4 + 21}{28 - 3} = 1\end{aligned}$$

$0 < \alpha, \beta < \frac{\pi}{2}$ and $\tan 2\beta > 0 \Rightarrow 0 < \alpha + 2\beta < \pi$. Therefore

$$\alpha + 2\beta = \frac{\pi}{4}$$

Answer: (A)

19. If $A - B = \pi/4$ and $x = (1 + \tan A)(1 - \tan B)$, then $(1 + x)^x$ is
- (A) 27 (B) 9 (C) 18 (D) 81

Solution:

$$\tan(A - B) = \tan \frac{\pi}{4} = 1$$

Therefore

$$\begin{aligned}\tan A - \tan B &= 1 + \tan A \tan B \\ \tan A - \tan B - \tan A \tan B &= 1 \\ 1 + \tan A - \tan B - \tan A \tan B &= 2 \\ (1 + \tan A)(1 - \tan B) &= 2\end{aligned}$$

Now

$$x = 2 \Rightarrow (1 + x)^x = 3^2 = 9$$

Answer: (B)

20. The value of $\tan 20^\circ \tan 80^\circ \cot 50^\circ$ is equal to

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\frac{1}{2\sqrt{3}}$ (D) $\frac{1}{\sqrt{3}}$

Solution:

$$\begin{aligned}\tan 20^\circ \tan 80^\circ \cot 50^\circ &= (\tan 20^\circ \tan 40^\circ \tan 80^\circ) \cot 40^\circ \cot 50^\circ \\ &= \left[\tan(\theta) \tan\left(\frac{\pi}{3} - \theta\right) \tan\left(\frac{\pi}{3} + \theta\right) \right] \times 1 \\ &\quad \text{(where } \theta = 20^\circ) \\ &= \tan 3\theta = \tan 60^\circ = \sqrt{3}\end{aligned}$$

Answer: (A)

21. If α and β are acute angles and such that

$$\tan \alpha = \frac{m}{m+1} \text{ and } \tan \beta = \frac{1}{2m+1}$$

then $\alpha + \beta$ equals

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{3\pi}{4}$

Solution:

$$\begin{aligned}\tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{[m/(m+1)] + [1/(2m+1)]}{1 - [m/(m+1)](1/(2m+1))} \\ &= \frac{2m^2 + 2m + 1}{(m+1)(2m+1) - m} \\ &= \frac{2m^2 + 2m + 1}{2m^2 + 3m + 1 - m} \\ &= \frac{2m^2 + 2m + 1}{2m^2 + 2m + 1} \\ &= 1\end{aligned}$$

since $0 < \alpha, \beta < \pi/2 \Rightarrow 0 < \alpha + \beta < \pi$ and $\tan(\alpha + \beta) = 1$ we have

$$\alpha + \beta = \frac{\pi}{4}$$

Answer: (B)

22. If $0 \leq \theta \leq \pi$ and $81^{\sin^2 \theta} + 81^{\cos^2 \theta} = 30$, then the number of values of θ is

- (A) 4 (B) 3 (C) 2 (D) 8

Solution: Put $t = 81^{\sin^2 \theta}$ so that

$$81^{\cos^2 \theta} = \frac{81}{t}$$

Therefore

$$\begin{aligned} t + \frac{81}{t} &= 30 \\ \Rightarrow t^2 - 30t + 81 &= 0 \\ \Rightarrow (t - 3)(t - 27) &= 0 \end{aligned}$$

Hence

$$\begin{aligned} 81^{\sin^2 \theta} &= 3 \text{ or } 27 \Rightarrow \sin^2 \theta = \frac{1}{4} \text{ or } \frac{3}{4} \\ 0 \leq \theta \leq \pi &\Rightarrow \theta = \frac{\pi}{6} \text{ or } \frac{5\pi}{6} \text{ or } \frac{\pi}{3} \text{ or } \frac{2\pi}{3} \end{aligned}$$

Answer: (A)

23. Let $0 < \theta < \pi/4$ and

$$\begin{aligned} t_1 &= (\tan \theta)^{\tan \theta}, \quad t_2 = (\tan \theta)^{\cot \theta} \\ t_3 &= (\cot \theta)^{\tan \theta}, \quad t_4 = (\cot \theta)^{\cot \theta} \end{aligned}$$

then

- (A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$
(C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

Solution: Since $0 < \theta < \pi/4$, we can take

$$\tan \theta = 1 - x \quad \text{and} \quad \cot \theta = 1 + y$$

where x and y are positive and $x < 1$. Observe that $0 < \tan \theta < 1$ and $\cot \theta > 1$. Now

$$\begin{aligned} t_1 &= (1 - x)^{1-x}, \quad t_2 = (1 - x)^{1+y} \\ t_3 &= (1 + y)^{1-x}, \quad t_4 = (1 + y)^{1+y} \end{aligned}$$

Clearly

$$t_4 > t_3 > t_1 > t_2 \quad (\because 1 - x < 1 + y)$$

Answer: (B)

$$\mathbf{24.} \quad \frac{\sec 8\theta - 1}{\sec 4\theta - 1} =$$

- (A) $\cot 2\theta \tan 8\theta$ (B) $\cot 8\theta \tan 2\theta$
(C) $\tan 8\theta \tan 2\theta$ (D) $\cot 8\theta \cot 2\theta$

Solution:

$$\begin{aligned} \frac{\sec 8\theta - 1}{\sec 4\theta - 1} &= \left(\frac{1 - \cos 8\theta}{1 - \cos 4\theta} \right) \left(\frac{\cos 4\theta}{\cos 8\theta} \right) \\ &= \left(\frac{2 \sin^2 4\theta}{2 \sin^2 2\theta} \right) \left(\frac{\cos 4\theta}{\cos 8\theta} \right) \\ &= \frac{\sin 4\theta (2 \sin 4\theta \cos 4\theta)}{(2 \sin^2 2\theta) \cos 8\theta} \\ &= \frac{2 \sin 2\theta \cos 2\theta (\sin 8\theta)}{2 \sin^2 2\theta \cos 8\theta} \\ &= \cot 2\theta \tan 8\theta \end{aligned}$$

Answer: (A)

25. The value $\cos 12^\circ + \cos 84^\circ + \cos 132^\circ + \cos 156^\circ$ is

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{4}$

Solution: Given expression is

$$\begin{aligned} &(\cos 12^\circ + \cos 132^\circ) + (\cos 84^\circ + \cos 156^\circ) \\ &= (2 \cos 72^\circ \cos 60^\circ) + (2 \cos 120^\circ \cos 36^\circ) \\ &= 2 \frac{(\sqrt{5} - 1)}{4} \times \frac{1}{2} + 2 \left(-\frac{1}{2} \right) \frac{(\sqrt{5} + 1)}{4} \\ &= \frac{\sqrt{5} - 1 - (\sqrt{5} + 1)}{4} \\ &= -\frac{1}{2} \end{aligned}$$

Answer: (B)

26. The value of $4(\sin 24^\circ + \cos 6^\circ)$ is

- (A) $\sqrt{3} + \sqrt{2}$ (B) $\sqrt{15} - \sqrt{3}$
(C) $\frac{\sqrt{15} + \sqrt{3}}{2}$ (D) $\sqrt{15} + \sqrt{3}$

Solution:

$$\begin{aligned} 4(\sin 24^\circ + \cos 6^\circ) &= 4(\sin 24^\circ + \sin 84^\circ) \\ &= 4(2 \sin 54^\circ \cos 30^\circ) \\ &= 8 \frac{\sqrt{5} + 1}{4} \times \frac{\sqrt{3}}{2} \\ &= \sqrt{15} + \sqrt{3} \end{aligned}$$

Answer: (D)

27. If $\sin \theta$, $\cos \theta$ and $\tan \theta$ are in GP, then $\cos^9 \theta + \cos^6 \theta + 3 \cos^5 \theta$ is equal to
 (A) 1 (B) 0 (C) 2 (D) -1

Solution: By hypothesis

$$\cos^2 \theta = (\sin \theta) \tan \theta = \frac{\sin^2 \theta}{\cos \theta}$$

Therefore

$$\cos^3 \theta = \sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^3 \theta + \cos^2 \theta = 1$$

$$(\cos^3 \theta + \cos^2 \theta)^3 = 1$$

$$\cos^9 \theta + 3 \cos^8 \theta + 3 \cos^7 \theta + \cos^6 \theta = 1$$

$$\cos^9 \theta + 3 \cos^5 \theta (\cos^3 \theta + \cos^2 \theta) + \cos^6 \theta = 1$$

$$\cos^9 \theta + 3 \cos^5 \theta + \cos^6 \theta = 1$$

$$(\because \cos^3 \theta + \cos^2 \theta = 1)$$

Answer: (A)

28. The least period of the function $f(x) = |\sin x| + |\cos x|$ is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{2}$ (D) 2π

Solution:

$$\begin{aligned} f\left(x + \frac{\pi}{2}\right) &= \left| \sin\left(x + \frac{\pi}{2}\right) \right| + \left| \cos\left(x + \frac{\pi}{2}\right) \right| \\ &= |\cos x| + |-\sin x| \\ &= |\sin x| + |\cos x| \\ &= f(x) \end{aligned}$$

Answer: (A)

29. If $0 < \theta < \pi/2$ and

$$\sin \theta = \sin \frac{\pi}{12} + \sin \frac{\pi}{4}$$

then θ equals

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{8}$

Solution:

$$\sin \theta = \frac{\sqrt{3}-1}{2\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ = \sin \frac{5\pi}{12}$$

Now $0 < \theta < \pi/2$ and $\sin \theta = \sin 75^\circ$. This implies

$$\theta = \frac{5\pi}{12}$$

Answer: (C)

30. If

$$f(x) = \sin^2\left(\frac{\pi}{8} + \frac{x}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{x}{2}\right)$$

then the period of f is

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) 2π

Solution:

$$\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$$

This implies

$$f(x) = \sin\left(\frac{\pi}{4}\right) \sin x = \frac{1}{\sqrt{2}} \sin x$$

Therefore period of $f(x)$ is 2π .

Answer: (D)

31. If $a = \tan 15^\circ$, $b = \operatorname{cosec} 75^\circ$ and $c = 4 \sin 18^\circ$, then

(A) $c > b > a$ (B) $a > c > b$
 (C) $b > a > c$ (D) $b > c > a$

Solution:

$$a = \tan 15^\circ = 2 - \sqrt{3} = \frac{1}{2 + \sqrt{3}} = \frac{4}{8 + 4\sqrt{3}}$$

$$b = \operatorname{cosec} 75^\circ = \sqrt{6} - \sqrt{2} = \frac{4}{\sqrt{6} + \sqrt{2}}$$

$$c = 4 \sin 18^\circ = \frac{4(\sqrt{5}-1)}{4} = \sqrt{5} - 1 = \frac{4}{\sqrt{5} + 1}$$

Now

$$8 + 4\sqrt{3} > \sqrt{6} + \sqrt{2} > \sqrt{5} + 1$$

Therefore

$$\frac{4}{\sqrt{5} + 1} > \frac{4}{\sqrt{6} + \sqrt{2}} > \frac{4}{8 + 4\sqrt{3}}$$

Hence

$$c > b > a$$

Answer: (A)

32. If $\cos(x-y)$, $\cos x$, $\cos(x+y)$ are in HP, then

$$\left| \sec x \cos \frac{y}{2} \right| =$$

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 1 (D) $\sqrt{3}$

Solution: By hypothesis

$$\begin{aligned}\cos x &= \frac{2\cos(x-y)\cos(x+y)}{\cos(x-y) + \cos(x+y)} \\ &= \frac{2(\cos^2 x - \sin^2 y)}{2\cos x \cos y}\end{aligned}$$

This implies

$$\begin{aligned}\cos^2 x \cos y &= \cos^2 x - \sin^2 y \\ \sin^2 y &= \cos^2 x (1 - \cos y) \\ 4\sin^2 \frac{y}{2} \cos^2 \frac{y}{2} &= \cos^2 x \left(2\sin^2 \frac{y}{2} \right) \\ 2\cos^2 \frac{y}{2} &= \cos^2 x \\ \cos^2 \frac{y}{2} \sec^2 x &= \frac{1}{2}\end{aligned}$$

Therefore

$$\left| \cos \frac{y}{2} \sec x \right| = \frac{1}{\sqrt{2}}$$

Answer: (A)

33. If $\cos x = \cos y$, $\sin x = -\sin y$, then

$$\begin{aligned}\sin(2010)x + \sin(2010)y &= \\ (A) 1 \quad (B) 2010 \quad (C) -2010 \quad (D) 0\end{aligned}$$

Solution:

$$\begin{aligned}\sin x + \sin y = 0 &\Rightarrow 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = 0 \\ \cos x - \cos y = 0 &\Rightarrow 2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right) = 0\end{aligned}$$

Both $\cos\left(\frac{x-y}{2}\right)$ and $\sin\left(\frac{x-y}{2}\right)$ cannot be simultaneously zero. So this implies

$$\begin{aligned}\sin\left(\frac{x+y}{2}\right) &= 0 \\ \Rightarrow x+y &= 2n\pi\end{aligned}$$

where n is an integer. Now

$$\begin{aligned}\sin(2010)x + \sin(2010)y &= \sin(2010)x + \sin 2010(2n\pi - x) \\ &= \sin(2010)x - \sin(2010)x \\ &= 0\end{aligned}$$

Answer: (D)

34. If

$$\frac{\cos x}{\cos \theta} + \frac{\sin x}{\sin \theta} = 1 = \frac{\cos y}{\cos \theta} + \frac{\sin y}{\sin \theta}$$

Then

$$\begin{aligned}\frac{\cos x \cos y}{\cos^2 \theta} + \frac{\sin x \sin y}{\sin^2 \theta} &= \\ (A) 2 \quad (B) 0 \quad (C) 1 \quad (D) -1\end{aligned}$$

Solution: From the hypothesis

$$\sin(x + \theta) = \sin(y + \theta)$$

Therefore

$$2\cos\left(\frac{x+y+2\theta}{2}\right)\sin\left(\frac{x-y}{2}\right) = 0$$

Now

$$x - y \neq 2n\pi \Rightarrow \sin\left(\frac{x-y}{2}\right) \neq 0$$

Hence

$$\begin{aligned}\frac{x+y+2\theta}{2} &= (2n+1)\frac{\pi}{2} \\ x+y &= (2n+1)\pi - 2\theta\end{aligned} \quad (1.7)$$

Again

$$\left(\frac{\cos x}{\cos \theta} + \frac{\sin x}{\sin \theta}\right)\left(\frac{\cos y}{\cos \theta} + \frac{\sin y}{\sin \theta}\right) = 1$$

Therefore

$$\begin{aligned}\left(\frac{\cos x \cos y}{\cos^2 \theta} + \frac{\sin x \sin y}{\sin^2 \theta}\right) &+ \frac{\sin x \cos y + \cos x \sin y}{\sin \theta \cos \theta} = 1 \\ \frac{\cos x \cos y}{\cos^2 \theta} + \frac{\sin x \sin y}{\sin^2 \theta} + \frac{\sin(x+y)}{\sin \theta \cos \theta} &= 1 \\ \frac{\cos x \cos y}{\cos^2 \theta} + \frac{\sin x \sin y}{\sin^2 \theta} + \frac{\sin 2\theta}{\sin \theta \cos \theta} &= 1 \\ \frac{\cos x \cos y}{\cos^2 \theta} + \frac{\sin x \sin y}{\sin^2 \theta} + 2 &= 1\end{aligned}$$

In the third step we have used Eq. (1.7).

Answer: (D)

35. The function $f(x) = \sin(x/3) + \cos(x/2)$ has period equal to

$$(A) \pi \quad (B) 2\pi \quad (C) 6\pi \quad (D) 12\pi$$

Solution: Period of $\sin(x/3)$ is 6π and period of $\cos(x/2)$ is 4π . Therefore period of $f(x)$ is 12π .

Answer: (D)

36. If $A + B + C = \pi$, then the minimum value of

$$\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2}$$

is

- (A) 1 (B) 3 (C) 9 (D) 27

Solution:

$$\begin{aligned} \frac{A}{2} + \frac{B}{2} + \frac{C}{2} &= \frac{\pi}{2} \Rightarrow \tan \frac{A}{2} = \cot \left(\frac{B+C}{2} \right) \\ &= \frac{1 - \tan(B/2) \tan(C/2)}{\tan(B/2) + \tan(C/2)} \end{aligned}$$

Therefore

$$\Sigma \tan \frac{A}{2} \tan \frac{B}{2} = 1$$

Now,

$$\begin{aligned} &\left(\tan \frac{A}{2} - \tan \frac{B}{2} \right)^2 + \left(\tan \frac{B}{2} - \tan \frac{C}{2} \right)^2 \\ &\quad + \left(\tan \frac{C}{2} - \tan \frac{A}{2} \right)^2 \geq 0 \\ \Rightarrow &2 \left(\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \right) - 2 \Sigma \tan \frac{A}{2} \tan \frac{B}{2} \geq 0 \\ \Rightarrow &\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} - 1 \geq 0 \\ \Rightarrow &\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1 \end{aligned}$$

and the equality holds when

$$\tan \frac{A}{2} = \tan \frac{B}{2} = \tan \frac{C}{2}$$

so that when $A = B = C = \pi/3$, the minimum value of the given expression is 1.

Answer: (A)

37. $\cos \theta \cos 2\theta \cos 4\theta \cos 8\theta \cdots \cos 2^{n-1} \theta$ is equal to

- (A) $\frac{2^n \sin \theta}{\sin 2^n \theta}$ (B) $\frac{\sin 2^n \theta}{2^n \sin \theta}$
(C) $\frac{2^n \sin 2^n \theta}{\sin \theta}$ (D) $\frac{\sin \theta}{2^n \sin 2^n \theta}$

Solution: Let $P = \cos \theta \cos 2\theta \cos 4\theta \cos 2^{n-1} \theta$.

$$\begin{aligned} (2^n \sin \theta)P &= (2 \sin \theta \cos \theta)(2 \cos 2\theta) \\ &\quad (2 \cos 4\theta) \cdots (2 \cos 2^{n-1} \theta) \\ &= [(\sin 2\theta)(2 \cos 2\theta)](2 \cos 4\theta) \\ &\quad (2 \cos 8\theta) \cdots (2 \cos 2^{n-1} \theta) \\ &= [(\sin 4\theta)(2 \cos 4\theta)] \\ &\quad 2 \cos 8\theta \cdots (2 \cos 2^{n-1} \theta) \\ &\quad \vdots \quad \vdots \quad \vdots \\ &= (\sin 2^{n-1} \theta)(2 \cos 2^{n-1} \theta) \\ &= \sin 2^n \theta \end{aligned}$$

Hence

$$P = \frac{\sin 2^n \theta}{2^n \sin \theta}$$

Answer: (B)

38. $\cos 36^\circ \cos 72^\circ \cos(108^\circ) \cos(144^\circ)$ is equal to

- (A) $\frac{1}{16}$ (B) $\frac{1}{32}$ (C) $\frac{1}{8}$ (D) $\frac{1}{64}$

Solution: The given product is

$$\begin{aligned} &\cos 36^\circ \sin 18^\circ (-\sin 18^\circ) (-\cos 36^\circ) = \sin^2 18^\circ \cos^2 36^\circ \\ &= \left(\frac{\sqrt{5}-1}{4} \right)^2 \left(\frac{\sqrt{5}+1}{4} \right)^2 \\ &= \frac{(5-1)^2}{4^4} \\ &= \frac{1}{16} \end{aligned}$$

Answer: (A)

39. If $A + B + C = \pi$, then

$$\sin(B+C-A) + \sin(C+A-B) + \sin(A+B-C) =$$

- (A) $4 \cos A \cos B \cos C$ (B) $4 \cos A \sin B \sin C$
(C) $4 \sin B \cos C \cos A$ (D) $4 \sin A \sin B \sin C$

Solution: The given sum is

$$\begin{aligned} &\sin(\pi - 2A) + \sin(\pi - 2B) + \sin(\pi - 2C) \\ &= \sin 2A + \sin 2B + \sin 2C \\ &= 4 \sin A \sin B \sin C \end{aligned}$$

Answer: (D)

40. $\tan \alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha =$

- (A) $\tan 2\alpha$ (B) $\cot 2\alpha$
(C) $\cot \alpha$ (D) $\tan 4\alpha$

Solution:

$$\cot \alpha - \tan \alpha = \frac{\cos^2 \alpha - \sin^2 \alpha}{\sin \alpha \cos \alpha} = 2 \cot 2\alpha$$

Therefore

$$\cot 2\alpha - \tan 2\alpha = 2 \cot 4\alpha$$

$$\cot 4\alpha - \tan 4\alpha = 2 \cot 8\alpha$$

$$\cot \alpha - 2 \cot 2\alpha = \tan \alpha$$

$$2 \cot 2\alpha - 4 \cot 4\alpha = 2 \tan 2\alpha$$

$$4 \cot 4\alpha - 8 \cot 8\alpha = 4 \tan 4\alpha$$

On adding, we have

$$\cot \alpha - 8 \cot 8\alpha = \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha$$

Hence

$$\cot \alpha = \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha$$

Answer: (C)

41. If x is any real number, then the value of $\tan[(\pi/4)\sin^2 x]$ lies between

- (A) -1 and 1 (B) 0 and ∞
(C) 0 and 1 (D) $-\infty$ and ∞

Solution:

$$0 \leq \sin^2 x \leq 1 \Rightarrow 0 \leq \frac{\pi}{4} \sin^2 x \leq \frac{\pi}{4}$$

This implies $\tan[(\pi/4)\sin^2 x]$ lies between 0 and 1 .

Answer: (C)

42. If

$$\frac{\cos x}{\cos y} = \frac{a}{b}$$

then $a \tan x + b \tan y$ equals

- (A) $(a+b) \tan\left(\frac{x+y}{2}\right)$
(B) $(a+b) \left(\tan \frac{x}{2} + \tan \frac{y}{2}\right)$
(C) $(a+b) \left(\cot \frac{x}{2} + \cot \frac{y}{2}\right)$
(D) $(a+b) \cot\left(\frac{x+y}{2}\right)$

Solution: Given that

$$\frac{\cos x}{\cos y} = \frac{a}{b}$$

Let

$$\frac{a}{\cos x} = \frac{b}{\cos y} = k \text{ (suppose)}$$

Therefore

$$\frac{a+b}{\cos x + \cos y} = k \quad (1.8)$$

Now

$$\begin{aligned} a \tan x + b \tan y &= \frac{a \sin x}{\cos x} + \frac{b \sin y}{\cos y} \\ &= k[\sin x + \sin y] \\ &= (a+b) \left[\frac{\sin x + \sin y}{\cos x + \cos y} \right] \\ &\quad \text{[from Eq. (1.8)]} \\ &= (a+b) \tan\left(\frac{x+y}{2}\right) \end{aligned}$$

Answer: (A)

43. If $90^\circ < \theta < 180^\circ$ and $\tan \theta = -2/3$, then

$$\frac{\tan(90^\circ + \theta) + \cos(180^\circ - \theta)}{\sin(270^\circ - \theta) - \cot(-\theta)} =$$

- (A) $\frac{2 + \sqrt{13}}{2 - \sqrt{13}}$ (B) $\frac{2 - \sqrt{13}}{2 + \sqrt{13}}$
(C) $\frac{2 + \sqrt{39}}{2 - \sqrt{39}}$ (D) $2 + \sqrt{39}$

Solution: $90^\circ < \theta < 180^\circ$ implies

$$\sin \theta = \frac{2}{\sqrt{13}} \quad \text{and} \quad \cos \theta = \frac{-3}{\sqrt{13}}$$

Therefore

$$\begin{aligned} \frac{\tan(90^\circ + \theta) + \cos(180^\circ - \theta)}{\sin(270^\circ - \theta) - \cot(-\theta)} &= \frac{-\cot \theta - \cos \theta}{-\cos \theta + \cot \theta} \\ &= \frac{-(-3/2) - (-3/\sqrt{13})}{-(-3/\sqrt{13}) - (3/2)} \\ &= \frac{(3/2) + (3/\sqrt{13})}{(3/\sqrt{13}) - (3/2)} \\ &= \frac{2 + \sqrt{13}}{2 - \sqrt{13}} \end{aligned}$$

Answer: (A)

44. For all angles $A \neq (2n+1)\pi$,

$$\frac{\sin 2A \cos A}{(1 + \cos 2A)(1 + \cos A)} =$$

- (A) $\sin \frac{A}{2}$ (B) $\cos \frac{A}{2}$
 (C) $\tan \frac{A}{2}$ (D) $\sin A$

Solution:

$$\begin{aligned} \frac{\sin 2A \cos A}{(1 + \cos 2A)(1 + \cos A)} &= \frac{2 \sin A \cos^2 A}{(2 \cos^2 A)[2 \cos^2(A/2)]} \\ &= \frac{\sin(A/2) \cos(A/2)}{\cos^2(A/2)} \\ &= \tan \frac{A}{2} \end{aligned}$$

Answer: (C)

45. If $x \cos \alpha + y \sin \alpha = x \cos \beta + y \sin \beta = 2a$ and $(2 \sin \alpha/2) \times \sin \beta/2 = 1$, then

- (A) $y^2 = 4a(x - a)$ (B) $y^2 = 4a(a - x)$
 (C) $x^2 = 4a(y - a)$ (D) $x^2 = 4a(a - y)$

Solution: Consider the equation

$$(2a - x \cos \theta)^2 = y^2 \sin^2 \theta$$

which can be written as

$$(x^2 + y^2) \cos^2 \theta - (4ax) \cos \theta + 4a^2 - y^2 = 0 \quad (1.9)$$

From the hypothesis, $\cos \alpha$ and $\cos \beta$ are roots of Eq. (1.9). Therefore

$$\cos \alpha + \cos \beta = \frac{4ax}{x^2 + y^2} \quad \text{and} \quad \cos \alpha \cos \beta = \frac{4a^2 - y^2}{x^2 + y^2}$$

Also

$$2 \sin \frac{\alpha}{2} \sin \frac{\beta}{2} = 1 \Rightarrow 4 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2} = 1$$

Therefore

$$\begin{aligned} 1 &= 4 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2} = (1 - \cos \alpha)(1 - \cos \beta) \\ &= 1 - (\cos \alpha + \cos \beta) + \cos \alpha \cos \beta \\ &= 1 - \frac{4ax}{x^2 + y^2} + \frac{4a^2 - y^2}{x^2 + y^2} \end{aligned}$$

Hence

$$\begin{aligned} -4ax + 4a^2 - y^2 &= 0 \\ y^2 &= 4a(a - x) \end{aligned}$$

Answer: (B)

46. If

$$\frac{\tan(A - B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1$$

then $\tan A \tan B$ is equal to

- (A) $\tan^2 C$ (B) $2 \tan^2 C$
 (C) $\cot^2 C$ (D) $2 \cot^2 C$

Solution: From the hypothesis

$$\frac{\sin^2 C}{\sin^2 A} = 1 - \frac{\tan(A - B)}{\tan A} = \frac{\tan B(1 + \tan^2 A)}{\tan A(1 + \tan A \tan B)}$$

Therefore

$$\begin{aligned} \sin^2 C &= \frac{\tan B}{\tan A} \times \frac{\sin^2 A}{\cos^2 A} \times \frac{1}{1 + \tan A \tan B} \\ &= \frac{\tan B \tan A}{1 + \tan A \tan B} \end{aligned}$$

This gives

$$\sin^2 C + \sin^2 C \tan A \tan B = \tan B \tan A$$

$$\sin^2 C = (1 - \sin^2 C) \tan A \tan B$$

$$\tan^2 C = \tan A \tan B$$

Answer: (A)

47. If

$$\frac{\cos x - \cos \alpha}{\cos x - \cos \beta} = \frac{\sin^2 \alpha \cos \beta}{\sin^2 \beta \cos \alpha}$$

then

$$\tan^2 \frac{\alpha}{2} \tan^2 \frac{\beta}{2} =$$

- (A) $\cot^2 x$ (B) $\cot^2 \frac{x}{2}$
 (C) $\tan^2 x$ (D) $\tan^2 \frac{x}{2}$

Solution: From the hypothesis

$$(\cos x - \cos \alpha) \sin^2 \beta \cos \alpha = (\cos x - \cos \beta) \sin^2 \alpha \cos \beta$$

$$\cos x [\sin^2 \beta \cos \alpha - \sin^2 \alpha \cos \beta]$$

$$= \sin^2 \beta \cos^2 \alpha - \cos^2 \beta \sin^2 \alpha$$

$$\cos x [(1 - \cos^2 \beta) \cos \alpha - (1 - \cos^2 \alpha) \cos \beta]$$

$$= (1 - \cos^2 \beta) \cos^2 \alpha - \cos^2 \beta (1 - \cos^2 \alpha)$$

$$\cos x [(\cos \alpha - \cos \beta) + \cos \alpha \cos \beta (\cos \alpha - \cos \beta)]$$

$$= \cos^2 \alpha - \cos^2 \beta$$

$$\cos x(1 + \cos \alpha \cos \beta) = \cos \alpha + \cos \beta$$

$$\cos x = \frac{\cos \alpha + \cos \beta}{1 + \cos \alpha \cos \beta}$$

Therefore

$$\begin{aligned}\tan^2 \frac{x}{2} &= \frac{1 - \cos x}{1 + \cos x} \\&= \frac{1 - [(\cos \alpha + \cos \beta) / (1 + \cos \alpha \cos \beta)]}{1 + [(\cos \alpha + \cos \beta) / (1 + \cos \alpha \cos \beta)]} \\&= \frac{1 + \cos \alpha \cos \beta - (\cos \alpha + \cos \beta)}{1 + \cos \alpha \cos \beta + \cos \alpha + \cos \beta} \\&= \frac{(1 - \cos \alpha)(1 - \cos \beta)}{(1 + \cos \alpha)(1 + \cos \beta)} \\&= \frac{[2 \sin^2(\alpha/2)][2 \sin^2(\beta/2)]}{[2 \cos^2(\alpha/2)][2 \cos^2(\beta/2)]} \\&= \tan^2 \frac{\alpha}{2} \tan^2 \frac{\beta}{2}\end{aligned}$$

Answer: (D)

48. If

$$\tan \theta = \frac{\sin \alpha \sin \beta}{\cos \alpha + \cos \beta}$$

then

$$\tan^2 \frac{\alpha}{2} \tan^2 \frac{\beta}{2} =$$

- (A) $\cot^2 \theta$ (B) $\cot^2 \frac{\theta}{2}$
(C) $\tan^2 \frac{\theta}{2}$ (D) $\tan^2 \theta$

Solution: We have

$$\begin{aligned}\sec^2 \theta &= 1 + \tan^2 \theta = 1 + \frac{\sin^2 \alpha \sin^2 \beta}{(\cos \alpha + \cos \beta)^2} \\&= \frac{(\cos \alpha + \cos \beta)^2 + \sin^2 \alpha \sin^2 \beta}{(\cos \alpha + \cos \beta)^2} \\&= \frac{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta + (1 - \cos^2 \alpha)(1 - \cos^2 \beta)}{(\cos \alpha + \cos \beta)^2} \\&= \frac{(1 + \cos \alpha \cos \beta)^2}{(\cos \alpha + \cos \beta)^2}\end{aligned}$$

Therefore

$$\cos \theta = \frac{\cos \alpha + \cos \beta}{1 + \cos \alpha \cos \beta}$$

Hence

$$\begin{aligned}\tan^2 \frac{\theta}{2} &= \frac{1 - \cos \theta}{1 + \cos \theta} = \frac{1 - [(\cos \alpha + \cos \beta) / (1 + \cos \alpha \cos \beta)]}{1 + [(\cos \alpha + \cos \beta) / (1 + \cos \alpha \cos \beta)]} \\&= \frac{(1 - \cos \alpha)(1 - \cos \beta)}{(1 + \cos \alpha)(1 + \cos \beta)} \\&= \tan^2 \frac{\alpha}{2} \tan^2 \frac{\beta}{2}\end{aligned}$$

Answer: (C)

49. If α, β lie between 0 and $\pi/4$ and $\cos(\alpha + \beta) = 4/5$, $\sin(\alpha - \beta) = 5/13$, then value of $\tan 2\alpha$ is

- (A) $\frac{11}{12}$ (B) $\frac{46}{33}$ (C) $\frac{56}{33}$ (D) $\frac{13}{11}$

Solution:

$$\begin{aligned}\tan 2\alpha &= \tan(\alpha + \beta + \alpha - \beta) \\&= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)} \\&= \frac{(3/4) + (5/12)}{1 - (3/4) \times (5/12)} \\&= \frac{56}{33}\end{aligned}$$

Answer: (C)

50. If $\alpha + \beta = \pi/2$ and $\beta + \gamma = \alpha$, then $\tan \alpha$ equals

- (A) $2(\tan \beta + \tan \gamma)$ (B) $\tan \beta + \tan \gamma$
(C) $\tan \beta + 2 \tan \gamma$ (D) $2 \tan \beta + \tan \gamma$

Solution:

$$\begin{aligned}\alpha + \beta &= \frac{\pi}{2} \\ \Rightarrow \tan \alpha &= \tan\left(\frac{\pi}{2} - \beta\right) \\ \Rightarrow \tan \alpha \tan \beta &= 1\end{aligned}$$

Again,

$$\begin{aligned}\beta + \gamma &= \alpha \\ \Rightarrow \gamma &= \alpha - \beta \\ \Rightarrow \tan \gamma &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{\tan \alpha - \tan \beta}{1 + 1} \\ \Rightarrow 2 \tan \gamma + \tan \beta &= \tan \alpha\end{aligned}$$

Answer: (C)

51. Let n be a positive integer such that

$$\sin \frac{\pi}{2n} + \cos \frac{\pi}{2n} = \frac{\sqrt{n}}{2}$$

Then

(A) $6 \leq n \leq 8$

(B) $4 < n \leq 8$

(C) $4 \leq n \leq 8$

(D) $4 < n < 8$

Solution: $n = 1, 2$ cannot satisfy the given equation.

$$\begin{aligned}\sin \frac{\pi}{2n} + \cos \frac{\pi}{2n} &= \frac{\sqrt{n}}{2} \Rightarrow \left(\sin \frac{\pi}{2n} + \cos \frac{\pi}{2n} \right)^2 = \frac{n}{4} \\ &\Rightarrow 1 + \sin \frac{\pi}{n} = \frac{n}{4} \\ &\Rightarrow \sin \frac{\pi}{n} = \frac{n-4}{4}\end{aligned}$$

Therefore

$$0 < \frac{n-4}{4} < 1$$

$$0 < n-4 < 4$$

$$4 < n < 8$$

Answer: (D)

52. Let $0 < x < \pi/4$. Then $(\sec 2x - \tan 2x)$ equals

(A) $\tan\left(x - \frac{\pi}{4}\right)$

(B) $\tan\left(\frac{\pi}{4} - x\right)$

(C) $\tan\left(x + \frac{\pi}{4}\right)$

(D) $\tan^2\left(x + \frac{\pi}{4}\right)$

Solution:

$$\sec 2x - \tan 2x = \frac{1 - \sin 2x}{\cos 2x} = \frac{(\cos x - \sin x)^2}{\cos^2 x - \sin^2 x}$$

Now $0 < x < \pi/4 \Rightarrow \cos x - \sin x > 0$. Therefore

$$\sec 2x - \tan 2x = \frac{\cos x - \sin x}{\cos x + \sin x} = \tan\left(\frac{\pi}{4} - x\right)$$

Answer: (B)

53. The value of

$$\sin \frac{2\pi}{7} \sin \frac{4\pi}{7} + \sin \frac{4\pi}{7} \sin \frac{8\pi}{7} + \sin \frac{8\pi}{7} \sin \frac{2\pi}{7}$$

is

(A) 0

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

(D) 1

Solution: The given expression can be written as

$$\begin{aligned}\frac{1}{2} \left[\left(\cos \frac{2\pi}{7} - \cos \frac{6\pi}{7} \right) + \left(\cos \frac{4\pi}{7} - \cos \frac{12\pi}{7} \right) \right. \\ \left. + \left(\cos \frac{6\pi}{7} - \cos \frac{10\pi}{7} \right) \right]\end{aligned}$$

$$\begin{aligned}&= \frac{1}{2} \left[\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \frac{12\pi}{7} - \cos \frac{10\pi}{7} \right] \\ &= \frac{1}{2} \left[\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \left(2\pi - \frac{2\pi}{7} \right) - \cos \left(2\pi - \frac{4\pi}{7} \right) \right] \\ &= \frac{1}{2} \left[\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \frac{2\pi}{7} - \cos \frac{4\pi}{7} \right] \\ &= 0\end{aligned}$$

Answer: (A)

54. $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} =$

(A) 0

(B) 1

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

Solution: Let

$$s = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}$$

Therefore

$$\begin{aligned}\left(2 \sin \frac{\pi}{7} \right) s &= 2 \sin \frac{\pi}{7} \cos \frac{2\pi}{7} + 2 \sin \frac{\pi}{7} \cos \frac{4\pi}{7} \\ &\quad + 2 \sin \frac{\pi}{7} \cos \frac{6\pi}{7} \\ &= \left[\sin \frac{3\pi}{7} - \sin \frac{\pi}{7} \right] + \left[\sin \frac{5\pi}{7} - \sin \frac{3\pi}{7} \right] \\ &\quad + \left[\sin \frac{7\pi}{7} - \sin \frac{5\pi}{7} \right] \\ &= \sin \frac{7\pi}{7} - \sin \frac{\pi}{7} \\ &= 0 - \sin \frac{\pi}{7}\end{aligned}$$

This gives

$$s = -\frac{1}{2}$$

Answer: (D)

55. The value of

$$\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$$

is

(A) $\frac{\sqrt{5}}{2}$

(B) $\frac{5}{2}$

(C) $\frac{\sqrt{7}}{2}$

(D) $\frac{7}{2}$

Solution: Let

$$s = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$$

Therefore

$$\begin{aligned} s^2 &= \sin^2 \frac{2\pi}{7} + \sin^2 \frac{4\pi}{7} + \sin^2 \frac{8\pi}{7} + 2\sum \sin \frac{2\pi}{7} \sin \frac{4\pi}{7} \\ &= \sin^2 \frac{2\pi}{7} + \sin^2 \frac{4\pi}{7} + \sin^2 \frac{8\pi}{7} + 0 \quad (\text{see Q.53}) \\ &= \frac{1 - \cos(4\pi/7)}{2} + \frac{1 - \cos(8\pi/7)}{2} + \frac{1 - \cos(16\pi/7)}{2} \\ &= \frac{3}{2} - \frac{1}{2} \left[\cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{16\pi}{7} \right] \\ &= \frac{3}{2} - \frac{1}{2} \left[\cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{2\pi}{7} \right] \\ &= \frac{3}{2} - \frac{1}{2} \left(-\frac{1}{2} \right) \quad (\text{see Problem 54}) \\ &= \frac{7}{4} \end{aligned}$$

Therefore

$$s = \frac{\sqrt{7}}{2}$$

Answer: (C)

56. If

$$\begin{aligned} f(x) &= \sin^2 x + \sin^2 \left(x + \frac{\pi}{3} \right) \\ &\quad + \cos x \cos \left(x + \frac{\pi}{3} \right) \text{ and } g\left(\frac{5}{4}\right) = 1 \end{aligned}$$

then

$$(g \circ f)\left(\frac{\pi}{8}\right) =$$

$$(A) \frac{5}{4} \quad (B) 1 \quad (C) 2 \quad (D) \pi$$

Solution:

$$\begin{aligned} &\sin^2 x + \sin^2 \left(x + \frac{\pi}{3} \right) + \cos x \cos \left(x + \frac{\pi}{3} \right) \\ &= \frac{1}{2} \left[2\sin^2 x + 1 - \cos \left(2x + \frac{2\pi}{3} \right) + \cos \left(2x + \frac{\pi}{3} \right) + \cos \frac{\pi}{3} \right] \\ &= \frac{1}{2} \left[2\sin^2 x - \cos \left(2x + \frac{2\pi}{3} \right) + \cos \left(2x + \frac{\pi}{3} \right) + \frac{3}{2} \right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left[2\sin^2 x + \cos 2x + \frac{3}{2} \right] \\ &= \frac{1}{2} \left(1 + \frac{3}{2} \right) = \frac{5}{4} \end{aligned}$$

Therefore $f(x) = 5/4$ for all x . Hence

$$(g \circ f)\left(\frac{\pi}{8}\right) = g\left(\frac{5}{4}\right) = 1$$

Answer: (B)

57. Period of the function $f(x) = \sin 3x + |\cos 6x|$ is

$$(A) \frac{\pi}{3} \quad (B) \frac{2\pi}{3} \quad (C) \frac{\pi}{6} \quad (D) \pi$$

Solution: Period of $\sin 3x = 2\pi/3$

$$\text{Period of } |\cos 6x| = \frac{\pi}{6}$$

Therefore

$$1 \cdot \left(\frac{2\pi}{3} \right) = 4 \cdot \left(\frac{\pi}{6} \right) = \frac{2\pi}{3}$$

Hence period of $f(x) = 2\pi/3$.

Answer: (B)

58. Period of $f(x) = 3\sin(\pi x/3) + 4\cos(\pi x/4)$ is

$$(A) 6 \quad (B) 12 \quad (C) 16 \quad (D) 24$$

Solution:

$$\text{Period of } \sin \frac{\pi x}{3} = \frac{2\pi}{\pi/3} = 6$$

$$\text{Period of } \cos \frac{\pi x}{4} = \frac{2\pi}{\pi/4} = 8$$

Now $4(6) = 3(8) = 24$. Therefore period of $f(x) = 24$

Answer: (D)

59. Period of $f(x) = \sin(\pi x/5!) + \cos(\pi x/6!)$ is

$$(A) 5! \quad (B) 2(5!) \quad (C) 2(6!) \quad (D) 6!$$

Solution:

$$\text{Period of } \sin \frac{\pi x}{5!} = \frac{2\pi}{(\pi/5!)} = 2(5!)$$

$$\text{Period of } \cos \frac{\pi x}{6!} = \frac{2\pi}{(\pi/6!)} = 2(6!)$$

Now $6(2 \cdot 5!) = 2(6!)$. Therefore, period of $f(x) = 2(6!)$

Answer: (C)

60. If period of $f(x) = \sum_{r=0}^{n-1} \sin(x/2^r) + \sum_{r=1}^n \tan(x/2^r)$ is $k\pi$, then k equals
 (A) 1 (B) 2 (C) 2^n (D) $\frac{1}{2^n}$

Solution: Periods of $\sin x, \sin(x/2), \sin(x/2)^2, \dots, \sin(x/2^{n-1})$ are $2\pi, 2^2\pi, 2^3\pi, \dots, 2^n\pi$ and those of $\tan(x/2), \tan(x/2^2), \tan(x/2^3), \dots, \tan(x/2^n)$ are $2\pi, 2^2\pi, 2^3\pi, \dots, 2^n\pi$. Therefore Period of $f(x)$ is $2^n\pi$ and hence $k = 2^n$.

Answer: (C)

61. If the function $f(x) = \cos x + \cos(bx)$ is periodic where $b \neq 0$, then b must be
 (A) an even integer (B) odd integer
 (C) irrational number (D) rational number

Solution: Period of $\cos x = 2\pi$

$$\text{Period of } \cos bx = \frac{2\pi}{|b|}$$

Since f is periodic, there exist positive integers m and n such that

$$m(2\pi) = n\left(\frac{2\pi}{|b|}\right)$$

Therefore

$$|b| = \frac{n}{m}$$

which implies b is rational.

Answer: (D)

62. Let $f(x) = x^2 - 1$ and $g(x) = \sin x + \cos x$. Then $f \circ g$ is invertible if x belongs to the interval

- (A) $\left[0, \frac{3\pi}{2}\right]$ (B) $\left[0, \frac{\pi}{2}\right]$
 (C) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (D) $\left[-\frac{\pi}{2}, 0\right]$

Solution:

$$(f \circ g)(x) = f(g(x)) = (\sin x + \cos x)^2 - 1 = \sin 2x$$

$\sin 2x$ is invertible, if $-\pi/2 \leq 2x \leq \pi/2$ that is, $-\pi/4 \leq x \leq \pi/4$.

Answer: (C)

63. $\cos 20^\circ + \cos 40^\circ + \cos 60^\circ - 4\cos 10^\circ \cos 20^\circ \cos 30^\circ$ is equal to

- (A) 0 (B) 1
 (C) -1 (D) $\frac{\sqrt{5}+1}{4}$

Solution: The given expression is equivalent to

$$\begin{aligned} & \cos 20^\circ + \cos 40^\circ + \frac{1}{2} - 2[\cos 30^\circ + \cos 10^\circ] \frac{\sqrt{3}}{2} \\ &= 2\cos 30^\circ \cos 10^\circ + \frac{1}{2} - \frac{3}{2} - \sqrt{3} \cos 10^\circ \\ &= 2\left(\frac{\sqrt{3}}{2}\right) \cos 10^\circ - 1 - \sqrt{3} \cos 10^\circ \\ &= -1 \end{aligned}$$

Answer: (C)

64. $\frac{\cos 8x - \cos 7x}{1 + 2\cos 5x} =$

- (A) $\cos 3x + \cos 2x$ (B) $\cos 3x - \cos 2x$
 (C) $\sin 3x - \sin 2x$ (D) $\sin 3x + \sin 2x$

Solution:

$$\begin{aligned} \frac{\cos 8x - \cos 7x}{1 + 2\cos 5x} &= \frac{\sin 5x(\cos 8x - \cos 7x)}{\sin 5x + \sin 10x} \\ &= \frac{-\sin 5x[2\sin(15x/2)\sin(x/2)]}{2\sin(15x/2)\cos(5x/2)} \\ &= \frac{-2\sin(5x/2)\cos(5x/2)\sin(x/2)}{\cos(5x/2)} \\ &= -(\cos 2x - \cos 3x) \\ &= \cos 3x - \cos 2x \end{aligned}$$

Answer: (B)

65. $\cos^2 x + \cos^2(\theta + x) - 2\cos \theta \cos x \cos(\theta + x)$ is equal to
 (A) $\cos^2 x$ (B) $\sin^2 x$ (C) $\cos^2 \theta$ (D) $\sin^2 \theta$

Solution: The given expression is equivalent to

$$\begin{aligned} & \cos^2 x + \cos(\theta + x)[\cos(\theta + x) - 2\cos \theta \cos x] \\ &= \cos^2 x - \cos(\theta + x)\cos(\theta - x) \\ &= \cos^2 x - (\cos^2 x - \sin^2 \theta) \\ &= \sin^2 \theta \end{aligned}$$

Answer: (D)

66. If A, B and C are positive such that $A + B + C = \pi$, then the minimum value of

$$\frac{\cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A+B}{2}\right)} + \frac{\cos\left(\frac{B-C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} + \frac{\cos\left(\frac{C-A}{2}\right)}{\cos\left(\frac{C+A}{2}\right)}$$

is

- (A) 3 (B) $3/2$ (C) 6 (D) $\frac{3}{16}$

Solution:

$$\begin{aligned}
\frac{\cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A+B}{2}\right)} &= \frac{\cos\left(\frac{A-B}{2}\right)\sin\left(\frac{A+B}{2}\right)}{\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A+B}{2}\right)} \\
&= \frac{\sin A + \sin B}{\sin(A+B)} \\
&= \frac{\sin A + \sin B}{\sin C} \\
&= \frac{\sin A}{\sin C} + \frac{\sin B}{\sin C}
\end{aligned}$$

Similarly

$$\frac{\cos\left(\frac{B-C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} = \frac{\sin B}{\sin A} + \frac{\sin C}{\sin A}$$

and

$$\frac{\cos\left(\frac{C-A}{2}\right)}{\cos\left(\frac{C+A}{2}\right)} = \frac{\sin C}{\sin B} + \frac{\sin A}{\sin B}$$

Therefore

$$\begin{aligned}
&\frac{\cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A+B}{2}\right)} + \frac{\cos\left(\frac{B-C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} + \frac{\cos\left(\frac{C-A}{2}\right)}{\cos\left(\frac{C+A}{2}\right)} \\
&= \left(\frac{\sin A}{\sin C} + \frac{\sin B}{\sin C}\right) + \left(\frac{\sin B}{\sin A} + \frac{\sin C}{\sin A}\right) \\
&\quad + \left(\frac{\sin C}{\sin B} + \frac{\sin A}{\sin B}\right) \\
&= \left(\frac{\sin A}{\sin C} + \frac{\sin C}{\sin A}\right) + \left(\frac{\sin B}{\sin C} + \frac{\sin C}{\sin B}\right) \\
&\quad + \left(\frac{\sin A}{\sin B} + \frac{\sin B}{\sin A}\right) \\
&\geq 2 + 2 + 2 \quad (\because \text{AM} \geq \text{GM}) \\
&= 6
\end{aligned}$$

Note that equality occurs when $A = B = C = \pi/3$.**Answer: (C)**

67. $\tan 20^\circ + 4 \sin 20^\circ =$

- (A) 3 (B) 2 (C) $\sqrt{2}$ (D) $\sqrt{3}$

Solution:

$$\begin{aligned}
\tan 20^\circ + 4 \sin 20^\circ &= \frac{\sin 20^\circ + 2 \sin 40^\circ}{\cos 20^\circ} \\
&= \frac{(\sin 20^\circ + \sin 40^\circ) + \sin 40^\circ}{\cos 20^\circ} \\
&= \frac{\cos 10^\circ + \cos 50^\circ}{\cos 20^\circ} \\
&= \frac{2 \cos 30^\circ \cos 20^\circ}{\cos 20^\circ} \\
&= \sqrt{3}
\end{aligned}$$

Answer: (D)

68. In $\triangle PQR$, $\angle R = \pi/2$. If $\tan(P/2)$ and $\tan(Q/2)$ are roots of the quadratic equation $ax^2 + bx + c = 0$ then

- (A) $b + c = a$ (B) $c + a = b$
 (C) $a + b = c$ (D) $b = c$

Solution:

$$\angle R = \frac{\pi}{2} \Rightarrow \angle P + \angle Q = \pi/2$$

Therefore

$$\begin{aligned}
\frac{P}{2} + \frac{Q}{2} &= \frac{\pi}{4} \\
1 &= \tan \frac{\pi}{4} = \frac{\tan(P/2) + \tan(Q/2)}{1 - \tan(P/2)\tan(Q/2)} \\
&= \frac{(-b/a)}{1 - (c/a)} = \frac{-b}{a - c} \\
a - c &= -b \quad \text{or} \quad a + b = c
\end{aligned}$$

Answer: (C)

69. $\frac{3 + \cot 76^\circ \cot 16^\circ}{\cot 76^\circ + \cot 16^\circ} =$

- (A) $\tan 46^\circ$ (B) $\cot 76^\circ$
 (C) $\tan 16^\circ$ (D) $\cot 44^\circ$

Solution:

$$\begin{aligned}
\frac{3 + \cot 76^\circ \cot 16^\circ}{\cot 76^\circ + \cot 16^\circ} &= \frac{3 \sin 76^\circ \sin 16^\circ + \cos 76^\circ \cos 16^\circ}{\sin 16^\circ \cos 76^\circ + \cos 16^\circ \sin 76^\circ} \\
&= \frac{2 \sin 76^\circ \sin 16^\circ + \cos 60^\circ}{\sin 92^\circ} \\
&= \frac{\cos 60^\circ - \cos 92^\circ + \cos 60^\circ}{\sin 92^\circ} \\
&= \frac{1 - \cos 92^\circ}{\sin 92^\circ} = \tan 46^\circ
\end{aligned}$$

Answer: (A)

70. In $\triangle ABC$, if $\cot \theta = \cot A + \cot B + \cot C$, then

$$\frac{\sin(A - \theta)\sin(B - \theta)\sin(C - \theta)}{\sin^3 \theta} =$$

- (A) 1 (B) -1 (C) $-\sqrt{2}$ (D) $\sqrt{2}$

Solution: We have $\cot \theta - \cot A = \cot B + \cot C$. Therefore

$$\frac{\sin(A - \theta)}{\sin A \sin \theta} = \frac{\sin(B + C)}{\sin B \sin C} = \frac{\sin A}{\sin B \sin C}$$

$$\frac{\sin(A - \theta)}{\sin \theta} = \frac{\sin^2 A}{\sin B \sin C}$$

Similarly

$$\frac{\sin(B - \theta)}{\sin \theta} = \frac{\sin^2 B}{\sin C \sin A}, \quad \frac{\sin(C - \theta)}{\sin \theta} = \frac{\sin^2 C}{\sin A \sin B}$$

$$\frac{\sin(A - \theta)\sin(B - \theta)\sin(C - \theta)}{\sin^3 \theta} = 1$$

Answer: (A)

71. Which one of the following numbers is rational?

- (A) $\sin 15^\circ$ (B) $\cos 15^\circ$
(C) $\sin 15^\circ \cos 15^\circ$ (D) $\sin 15^\circ \cos 75^\circ$

Solution:

$$\begin{aligned} \sin 15^\circ &= \sin(45^\circ - 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}-1}{2\sqrt{2}} \text{ is irrational} \end{aligned}$$

$$\cos 15^\circ = \cos(45^\circ - 30^\circ) = \frac{\sqrt{3}+1}{2\sqrt{2}} \text{ is irrational}$$

$$\sin 15^\circ \cos 15^\circ = \frac{3-1}{8} = \frac{1}{4} \text{ is rational}$$

$$\begin{aligned} \sin 15^\circ \cos 75^\circ &= \sin^2 15^\circ = \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right)^2 \\ &= \frac{4-2\sqrt{3}}{8} = \frac{2-\sqrt{3}}{4} \text{ is irrational} \end{aligned}$$

Answer: (C)

72. $\sec^2 \theta = 4xy/(x+y)^2$ is true if and only if

- (A) $x+y=0$ (B) $x=y, x \neq 0$
(C) $x=y$ (D) $x \neq 0, y \neq 0$

Solution:

$$\frac{4xy}{(x+y)^2} = \frac{(x+y)^2 - (x-y)^2}{(x+y)^2} = 1 - \left(\frac{x-y}{x+y} \right)^2 \leq 1$$

But $\sec^2 \theta \geq 1$ for all θ . Therefore

$$1 = 1 - \left(\frac{x-y}{x+y} \right)^2 \Leftrightarrow x = y, x \neq 0$$

Answer: (B)

73. If $\tan A = (1 - \cos B)/\sin B$, then $\tan 3A$ is equal to

- (A) $\tan\left(\frac{3B}{2}\right)$ (B) $\tan 2B$
(C) $\tan B$ (D) $-\tan 2B$

Solution:

$$\tan A = \frac{1 - \cos B}{\sin B} = \tan \frac{B}{2}$$

Therefore

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} = \frac{2 \tan(B/2)}{1 - \tan^2(B/2)} = \tan B$$

$$\tan 3A = \frac{\tan A + \tan 2A}{1 - \tan A \tan 2A} = \frac{\tan(B/2) + \tan B}{1 - \tan(B/2) \tan B} = \tan \frac{3B}{2}$$

Answer: (A)

74. Consider the following two statements.

S_1 : All cyclic quadrilaterals $ABCD$ satisfy

$$\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2} \tan \frac{D}{2} = 1$$

S_2 : All trapeziums $ABCD$ satisfy

$$\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2} \tan \frac{D}{2} = 1$$

Then

- (A) both S_1 and S_2 are true
(B) S_1 is true but S_2 is not true
(C) S_1 is not true and S_2 is true
(D) neither S_1 nor S_2 is true

Solution: In a cyclic quadrilateral $ABCD$, it is known that

$$A + C = 180^\circ = B + D$$

Therefore

$$\tan \frac{A}{2} = \cot \frac{C}{2} \quad \text{and} \quad \tan \frac{B}{2} = \cot \frac{D}{2}$$

$$\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2} \tan \frac{D}{2} = 1$$

This implies S_1 is true. Let $ABCD$ be a trapezium in which AB and DC are parallel sides (see Figure 1.36). Therefore

$$\angle DCA = \angle CAB \quad (1.10)$$

Now in $\triangle ADC$,

$$\angle CAD + D + \angle DCA = 180^\circ$$

Therefore

$$(A - \angle CAB) + D + \angle DCA = 180^\circ$$

$$A + D = 180^\circ \quad [\text{From Eq. (1.10)}]$$

Similarly $B + C = 180^\circ$. Therefore

$$\tan \frac{A}{2} = \cot \frac{D}{2}, \quad \tan \frac{B}{2} = \cot \frac{C}{2}$$

$$\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2} \tan \frac{D}{2} = 1$$

Hence S_2 is also true. This implies both S_1 and S_2 are true.

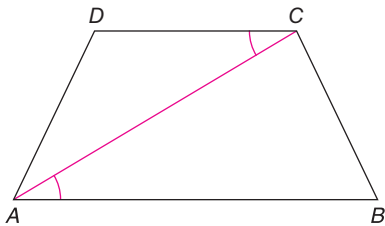


FIGURE 1.36 Single correct choice type question 74.

Answer: (A)

75. Let $A_1, A_2, \dots, A_n$ be the interior angles of an n -sided convex polygon. Then the value of

$$\frac{\cos(A_1 + A_2 + \dots + A_k)}{\cos(A_{k+1} + A_{k+2} + \dots + A_n)}$$

where

$$\cos\left(\sum_{i=1}^k A_i\right) \neq 0 \quad \text{for } k = 1, 2, 3, \dots, n-1$$

- (A) is independent of both k and n
- (B) is independent of k but depends on n
- (C) is independent of n but depends on k
- (D) depends on both k and n

Solution: It is known that the sum of the interior angles of an n -sided convex polygon is $(2n-4)$ right angles. Therefore

$$\begin{aligned} \cos(A_{k+1} + A_{k+2} + \dots + A_n) &= \cos[(n-2)\pi \\ &\quad - (A_1 + A_2 + \dots + A_k)] \\ &= \pm \cos(A_1 + A_2 + \dots + A_k) \end{aligned}$$

Now

$$\frac{\cos(A_1 + A_2 + \dots + A_k)}{\cos(A_{k+1} + A_{k+2} + \dots + A_n)} = \pm 1$$

according as n is even or odd.

Answer: (B)

76. If $\cos x = \tan y$, $\cos y = \tan z$, $\cos z = \tan x$, then $\sin x = \sin y = \sin z = 2 \sin \theta$ where θ is

- (A) 15°
- (B) 18°
- (C) $22\frac{1}{2}^\circ$
- (D) 75°

Solution:

$$\cos^2 x = \tan^2 y = \sec^2 z - 1 = \cot^2 z - 1$$

Therefore

$$\begin{aligned} 1 + \cos^2 x &= \cot^2 z = \frac{\cos^2 z}{\sin^2 z} = \frac{\tan^2 x}{1 - \tan^2 x} \\ &= \frac{\sin^2 x}{\cos^2 x - \sin^2 x} = \frac{\sin^2 x}{1 - 2\sin^2 x} \end{aligned}$$

$$(1 + \cos^2 x)(1 - 2\sin^2 x) = \sin^2 x$$

$$(2 - \sin^2 x)(1 - 2\sin^2 x) = \sin^2 x$$

$$2\sin^4 x - 6\sin^2 x + 2 = 0$$

$$\sin^4 x - 3\sin^2 x + 1 = 0$$

$$\sin^2 x = \frac{3 \pm \sqrt{5}}{2}$$

Since $(3 + \sqrt{5}/2) > 1$, it follows that

$$\sin^2 x = \frac{3 - \sqrt{5}}{2} = \left(\frac{\sqrt{5} - 1}{2}\right)^2$$

Hence

$$\sin x = \frac{\sqrt{5} - 1}{2} = 2 \left(\frac{\sqrt{5} - 1}{4}\right) = 2 \sin 18^\circ$$

Answer: (B)

Multiple Correct Choice Type Questions

1. If $\cos A = -7/25$ and A lies between 450° and 540° then

(A) $\sin \frac{A}{2} = \frac{-4}{5}$

(B) $\cos \frac{A}{2} = \frac{-3}{5}$

(C) $\sin \frac{A}{2} = \frac{-3}{5}$

(D) $\cos \frac{A}{2} = \frac{-4}{5}$

Solution: By hypothesis $450^\circ < A < 540^\circ$. This implies $225^\circ < A/2 < 270^\circ$. So

$$\left| \sin \frac{A}{2} \right| = \sqrt{\frac{1 - \cos A}{2}} = \sqrt{\frac{1 + (7/25)}{2}} = \frac{4}{5}$$

Now $A/2$ lies in the third quadrant. This means

$$\sin \frac{A}{2} = \frac{-4}{5}$$

Again

$$\left| \cos \frac{A}{2} \right| = \sqrt{\frac{1 + \cos A}{2}} = \sqrt{\frac{1 - (7/25)}{2}} = \frac{3}{5}$$

Again $A/2$ lies in the third quadrant implies

$$\cos \frac{A}{2} = \frac{-3}{5}$$

Answers: (A), (B)

2. Which of the following are true?

(A) $\sin 22\frac{1}{2}^\circ = \frac{\sqrt{2} - \sqrt{2}}{2}$

(B) $\cos 22\frac{1}{2}^\circ = \frac{\sqrt{2} + \sqrt{2}}{2}$

(C) $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$

(D) $\cot 22\frac{1}{2}^\circ = \sqrt{2} + 1$

Solution:

$$\frac{1}{\sqrt{2}} = \cos 45^\circ = 2 \cos^2 22\frac{1}{2}^\circ - 1$$

Therefore

$$\cos^2 22\frac{1}{2}^\circ = \frac{1}{2} \left(\frac{1}{\sqrt{2}} + 1 \right) = \frac{1}{2} \left(\frac{\sqrt{2} + 2}{2} \right) = \frac{\sqrt{2} + 2}{4}$$

$$\cos 22\frac{1}{2}^\circ = \frac{\sqrt{\sqrt{2} + 2}}{2} \left(\because \cos 22\frac{1}{2}^\circ > 0 \right)$$

$$\begin{aligned} \sin 22\frac{1}{2}^\circ &= +\sqrt{1 - \cos^2 22\frac{1}{2}^\circ} \\ &= +\sqrt{1 - \frac{2 + \sqrt{2}}{4}} = \frac{\sqrt{2} - \sqrt{2}}{2} \end{aligned}$$

Answers: (A), (B), (C), (D)

3. Let $\cos x + \cos y = a$, $\cos 2x + \cos 2y = b$ and $\cos 3x + \cos 3y = c$. Which of the following is (are) true?

(A) $\cos^2 x + \cos^2 y = 1 + \frac{b}{2}$

(B) $\frac{1}{4}(2a^2 - b - 2) = \cos x \cos y$

(C) $2a^3 + c = 3a(1 + b)$

(D) $abc = 0$ for all real x and y

Solution:

$$\cos^2 x + \cos^2 y + 2 \cos x \cos y = a^2 \quad (1.11)$$

$$b = \cos 2x + \cos 2y = 2(\cos^2 x + \cos^2 y) - 2$$

Therefore

$$\cos^2 x + \cos^2 y = \frac{b + 2}{2} = 1 + \frac{b}{2} \quad (1.12)$$

Hence (A) is true. From Eqs. (1.11) and (1.12)

$$\cos x \cos y = \frac{1}{4}(2a^2 - b - 2)$$

Hence (B) is true. Now

$$c = \cos 3x + \cos 3y = 4(\cos^3 x + \cos^3 y) - 3(\cos x + \cos y)$$

$$= 4[(\cos x + \cos y)(\cos^2 x + \cos^2 y - \cos x \cos y)]$$

$$- 3(\cos x + \cos y)$$

$$= 4 \left[a \left(\frac{b + 2}{2} - \frac{1}{4}(2a^2 - b - 2) \right) \right] - 3a$$

$$= 2a(b + 2) - 2a^3 + ab + 2a - 3a$$

$$= -2a^3 + 3ab + 3a$$

Therefore

$$2a^3 + c = 3a(b + 1)$$

Hence (C) is true. $abc = 0$ for all x, y is not true. Hence (D) is not true.

Answers: (A), (B), (C)

4. Which of the following are true?

- (A) $\cos 2\theta \cos \theta - \sin 4\theta \sin \theta = \cos 3\theta \cos 2\theta$
 (B) $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ = 0$
 (C) $\sin 5\theta + \sin 2\theta - \sin \theta = \sin 2\theta(2 \cos 3\theta + 1)$
 (D) $\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} = \tan 2\theta + \sec 2\theta$

Solution:

(A) $\cos 2\theta \cos \theta - \sin 4\theta \sin \theta$

$$\begin{aligned} &= \frac{1}{2}(\cos 3\theta + \cos \theta) - \frac{1}{2}(\cos 3\theta - \cos 5\theta) \\ &= \frac{1}{2}(\cos 5\theta + \cos \theta) \\ &= \cos 3\theta \cos 2\theta \end{aligned}$$

Therefore (A) is true.

(B) $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$
 $= \cos 20^\circ + 2 \cos 120^\circ \cos 20^\circ$
 $= \cos 20^\circ - \cos 20^\circ$
 $= 0$

Hence (B) is true.

(C) $\sin 5\theta + \sin 2\theta - \sin \theta = (\sin 5\theta - \sin \theta) + \sin 2\theta$
 $= 2 \cos 3\theta \sin 2\theta + \sin 2\theta$
 $= \sin 2\theta(2 \cos 3\theta + 1)$

So (C) is true.

(D) $\tan 2\theta + \sec 2\theta = \frac{\sin 2\theta + 1}{\cos 2\theta} = \frac{(\cos \theta + \sin \theta)^2}{\cos^2 \theta - \sin^2 \theta}$
 $= \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$

This implies that (D) is true.

Answers: (A), (B), (C), (D)

5. Which of the following are true?

- (A) $\frac{1}{\tan 3A - \tan A} - \frac{1}{\cot 3A - \cot A} = \cot 2A$
 (B) $\frac{\cos \alpha - \sin 3\alpha}{\cos \alpha + \sin \alpha} = 1 - 2 \sin 2\alpha$
 (C) $\frac{\cot \theta}{\cot \theta - \cot 3\theta} + \frac{\tan \theta}{\tan \theta - \tan 3\theta} = 1$
 (D) $4(\cos^6 \theta + \sin^6 \theta) = 1 + 3 \cos^2 2\theta$

Solution:

(A) $\text{LHS} = \frac{1}{(\sin 3A / \cos 3A) - (\sin A / \cos A)}$
 $= \frac{1}{(\cos 3A / \sin 3A) - (\cos A / \sin A)}$
 $= \frac{\cos 3A \cos A}{\sin 3A \cos A - \cos 3A \sin A}$
 $= \frac{\sin 3A \sin A}{\sin A \cos 3A - \cos A \sin 3A}$
 $= \frac{\cos 3A \cos A + \sin 3A \sin A}{\sin 3A \cos A - \cos 3A \sin A}$
 $= \frac{\cos(3A - A)}{\sin(3A - A)} = \frac{\cos 2A}{\sin 2A}$
 $= \cot 2A$

Therefore (A) is true.

(B) $\frac{\cos 3\alpha - \sin 3\alpha}{\cos \alpha + \sin \alpha}$
 $= \frac{(4 \cos^3 \alpha - 3 \cos \alpha) - (3 \sin \alpha - 4 \sin^3 \alpha)}{\cos \alpha + \sin \alpha}$
 $= \frac{4(\cos^3 \alpha + \sin^3 \alpha) - 3(\cos \alpha + \sin \alpha)}{\cos \alpha + \sin \alpha}$
 $= \frac{4[(\cos \alpha + \sin \alpha)(1 - \sin \alpha \cos \alpha)] - 3(\cos \alpha + \sin \alpha)}{\cos \alpha + \sin \alpha}$
 $= 4 - 4 \sin \alpha \cos \alpha - 3$
 $= 1 - 2 \sin 2\alpha$

Therefore (B) is true.

(C) $\frac{\cot \theta}{\cot \theta - \cot 3\theta} + \frac{\tan \theta}{\tan \theta - \tan 3\theta}$
 $= \frac{\left(\frac{\cos \theta}{\sin \theta}\right)}{\left(\frac{\cos \theta}{\sin \theta}\right) - \left(\frac{\cos 3\theta}{\sin 3\theta}\right)} + \frac{\left(\frac{\sin \theta}{\cos \theta}\right)}{\left(\frac{\sin \theta}{\cos \theta}\right) - \left(\frac{\sin 3\theta}{\cos 3\theta}\right)}$
 $= \frac{\cos \theta \sin 3\theta}{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta} + \frac{\sin \theta \cos 3\theta}{\sin \theta \cos 3\theta - \cos \theta \sin 3\theta}$
 $= \frac{\cos \theta \sin 3\theta}{\sin 2\theta} - \frac{\sin \theta \cos 3\theta}{\sin 2\theta}$

$$= \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin 2\theta}$$

$$= \frac{\sin 2\theta}{\sin 2\theta} = 1$$

Therefore (C) is true.

$$\begin{aligned} \text{(D)} \quad & 4(\cos^6 \theta + \sin^6 \theta) \\ &= 4[(\cos^2 \theta)^3 + (\sin^2 \theta)^3] \\ &= 4[\cos^2 \theta + \sin^2 \theta][\cos^4 \theta - \cos^2 \theta \sin^2 \theta + \sin^4 \theta] \\ &= 4[(\cos^2 \theta + \sin^2 \theta)^2 - 3\cos^2 \theta \sin^2 \theta] \\ &= 4[1 - 3\cos^2 \theta \sin^2 \theta] \\ &= 4\left[1 - \frac{3}{4}\sin^2 2\theta\right] \\ &= 4 - 3\sin^2 2\theta \\ &= 1 + 3(1 - \sin^2 2\theta) \\ &= 1 + 3\cos^2 2\theta \end{aligned}$$

Hence (D) is true.

Answers: (A), (B), (C), (D)

6. Which of the following are true?

- (A) $\cos^3 \alpha \cos 3\alpha + \sin^3 \alpha \sin 3\alpha = \cos^3 2\alpha$
 (B) $4\sin^3 \alpha \cos 3\alpha + 4\cos^3 \alpha \sin 3\alpha = 3\sin 4\alpha$
 (C) $2\sin 75^\circ \sin 15^\circ = 1/2$
 (D) $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) - \sin(A + B - C) = 4\sin A \sin B \sin C$

Solution:

$$\begin{aligned} \text{(A)} \quad \text{LHS} &= \cos^3 \alpha (4\cos^3 \alpha - 3\cos \alpha) \\ &\quad + \sin^3 \alpha (3\sin \alpha - 4\sin^3 \alpha) \\ &= 4(\cos^6 \alpha - \sin^6 \alpha) - 3(\cos^4 \alpha - \sin^4 \alpha) \\ &= 4(\cos^2 \alpha - \sin^2 \alpha)[\cos^4 \alpha + \sin^2 \alpha \cos^2 \alpha + \sin^4 \alpha] \\ &\quad - 3(\cos^2 \alpha - \sin^2 \alpha) \\ &= \cos 2\alpha [4\cos^4 \alpha + 4\sin^2 \alpha \cos^2 \alpha + 4\sin^4 \alpha - 3] \\ &= \cos 2\alpha [4\cos^2 \alpha (1 - \sin^2 \alpha) + 4\sin^2 \alpha \cos^2 \alpha \\ &\quad + 4\sin^2 \alpha (1 - \cos^2 \alpha) - 3] \\ &= \cos 2\alpha [4 - 4\sin^2 \alpha \cos^2 \alpha - 3] \\ &= \cos 2\alpha [1 - \sin^2 2\alpha] \\ &= \cos^3 2\alpha \end{aligned}$$

So (A) is true.

$$\begin{aligned} \text{(B)} \quad \text{LHS} &= (3\sin \alpha - \sin 3\alpha)\cos 3\alpha \\ &\quad + (3\cos \alpha + \cos 3\alpha)\sin 3\alpha \\ &= 3[\sin \alpha \cos 3\alpha + \sin 3\alpha \cos \alpha] \\ &= 3\sin 4\alpha \end{aligned}$$

Hence (B) is true.

$$\begin{aligned} \text{(C)} \quad 2\sin 75^\circ \sin 15^\circ &= \cos(75^\circ - 15^\circ) - \cos(75^\circ + 15^\circ) \\ &= \cos 60^\circ - \cos 90^\circ \\ &= \frac{1}{2} - 0 \\ &= \frac{1}{2} \end{aligned}$$

(C) is also true.

$$\begin{aligned} \text{(D)} \quad \text{LHS} &= 2\sin C \cos(B - A) + 2\cos(A + B)\sin(-C) \\ &= 2\sin C [\cos(B - A) - \cos(A + B)] \\ &= 2\sin C (2\sin A \sin B) \\ &= 4\sin A \sin B \sin C \end{aligned}$$

Therefore (D) is true.

Answers: (A), (B), (C), (D)

7. Which of the following are true?

- (A) $\cos(30^\circ - A) + \cos(30^\circ + A) = \sqrt{3} \cos A$
 (B) $\sin(60^\circ + A) - \sin(60^\circ - A) = \sin A$
 (C) $\frac{\cos 2\theta - \cos 3\theta}{\sin 2\theta + \sin 3\theta} = \tan \frac{\theta}{2}$
 (D) $\sin^2 5\theta - \sin^2 3\theta = \sin 8\theta \sin 2\theta$

Solution:

$$\begin{aligned} \text{(A)} \quad & \cos(30^\circ - A) + \cos(30^\circ + A) \\ &= 2\cos 30^\circ \cos A \\ &= 2\left(\frac{\sqrt{3}}{2}\right)\cos A = \sqrt{3} \cos A \end{aligned}$$

Therefore (A) is true.

$$\begin{aligned} \text{(B)} \quad & \sin(60^\circ + A) - \sin(60^\circ - A) \\ &= 2\cos 60^\circ \sin A = 2\left(\frac{1}{2}\right)\sin A = \sin A \end{aligned}$$

Therefore (B) is true.

$$\text{(C)} \quad \frac{\cos 2\theta - \cos 3\theta}{\sin 2\theta + \sin 3\theta} = \frac{2\sin(5\theta/2)\sin(\theta/2)}{2\sin(5\theta/2)\cos(\theta/2)} = \tan \frac{\theta}{2}$$

Therefore (C) is true.

$$\begin{aligned}
 \text{(D)} \quad \sin^2 A - \sin^2 B &= \sin(A+B)\sin(A-B) \\
 \sin^2 5\theta - \sin^2 3\theta &= \sin(5\theta+3\theta)\sin(5\theta-3\theta) \\
 &= \sin 8\theta \sin 2\theta
 \end{aligned}$$

Therefore (D) is true.

Answers: (A), (B), (C), (D)

8. Which of the following are true?

$$\text{(A)} \quad 4(\cos^3 10^\circ + \sin^3 20^\circ) = 3(\cos 10^\circ + \sin 20^\circ)$$

$$\text{(B)} \quad \frac{1}{\tan 3\theta + \tan \theta} - \frac{1}{\cot 3\theta + \cot \theta} = \cot 4\theta$$

$$\text{(C)} \quad \frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \cot \theta$$

$$\text{(D)} \quad \frac{\cot \theta - 1}{\cot \theta + 1} = \frac{1 - \sin 2\theta}{\cos 2\theta}$$

Solution:

$$\begin{aligned}
 \text{(A)} \quad 4(\cos^3 10^\circ + \sin^3 20^\circ) &= (\cos 30^\circ + 3\cos 10^\circ) \\
 &\quad + (3\sin 20^\circ - \sin 60^\circ) \\
 &= \left(\frac{\sqrt{3}}{2} + 3\cos 10^\circ \right) \\
 &\quad + \left(3\sin 20^\circ - \frac{\sqrt{3}}{2} \right) \\
 &= 3(\cos 10^\circ + \sin 20^\circ)
 \end{aligned}$$

Therefore (A) is true.

$$\begin{aligned}
 \text{(B)} \quad \text{LHS} &= \frac{\cos \theta \cos 3\theta}{\sin 3\theta \cos \theta + \cos 3\theta \sin \theta} \\
 &\quad - \frac{\sin 3\theta \sin \theta}{\sin \theta \cos 3\theta + \cos \theta \sin 3\theta} \\
 &= \frac{\cos 3\theta \cos \theta - \sin 3\theta \sin \theta}{\sin(3\theta + \theta)} \\
 &= \frac{\cos 4\theta}{\sin 4\theta} \\
 &= \cot 4\theta
 \end{aligned}$$

Hence (B) is true.

$$\begin{aligned}
 \text{(C)} \quad \frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} &= \frac{2\sin^2 \theta + 2\sin \theta \cos \theta}{2\cos^2 \theta + 2\sin \theta \cos \theta} \\
 &= \frac{\sin \theta (\sin \theta + \cos \theta)}{\cos \theta (\cos \theta + \sin \theta)} \\
 &= \tan \theta
 \end{aligned}$$

Hence (C) is not true.

$$\begin{aligned}
 \text{(D)} \quad \frac{\cot \theta - 1}{\cot \theta + 1} &= \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \\
 &= \frac{(\cos \theta - \sin \theta)^2}{\cos^2 \theta - \sin^2 \theta} \\
 &= \frac{1 - 2\sin \theta \cos \theta}{\cos 2\theta} \\
 &= \frac{1 - \sin 2\theta}{\cos 2\theta}
 \end{aligned}$$

Therefore (D) is true.

Answers: (A), (B), (D)

9. Which of the following are true?

$$\text{(A)} \quad \text{If } \cos \theta - \sin \theta = \sqrt{2} \sin \theta, \text{ then } \cos \theta + \sin \theta = \sqrt{2} \cos \theta$$

$$\text{(B)} \quad \text{Period of the function } f(x) = 2\sin \frac{\pi x}{4} + 3\cos \frac{\pi x}{3} \text{ is } 24$$

$$\text{(C)} \quad \text{Period of } \tan(x + 2x + 3x) \text{ is } \frac{\pi}{3}$$

$$\text{(D)} \quad \tan 855^\circ + \tan 45^\circ = -1$$

Solution:

$$\begin{aligned}
 \text{(A)} \quad \cos \theta - \sin \theta &= \sqrt{2} \sin \theta \\
 \cos \theta &= (\sqrt{2} + 1) \sin \theta \\
 (\sqrt{2} - 1) \cos \theta &= \sin \theta \\
 \sqrt{2} \cos \theta &= \cos \theta + \sin \theta
 \end{aligned}$$

Therefore (A) is true.

$$\text{(B)} \quad \text{Period of } \sin\left(\frac{\pi x}{4}\right) = (2\pi) \frac{4}{\pi} = 8$$

$$\text{Period of } \cos\left(\frac{\pi x}{3}\right) = (2\pi) \frac{3}{\pi} = 6$$

$3(8) = 4(6) = 24 \Rightarrow 24$ is a period of $f(x)$
This implies (B) is also true.

$$\text{(C)} \quad \text{Period of } \tan 6x = \frac{\pi}{6}$$

So (C) is not true.

$$\begin{aligned}
 \text{(D)} \quad \tan 855^\circ + \tan 45^\circ &= \tan(720^\circ + 135^\circ) + 1 \\
 &= \tan 135^\circ + 1 \\
 &= -1 + 1 \\
 &= 0
 \end{aligned}$$

So (D) is not true.

Answers: (A), (B)

10. A, B and C are angles of triangle such that

$$\cos A \cos B + \sin A \sin B \sin C = 1$$

then $\triangle ABC$ is

- (A) isosceles triangle
(B) right-angled triangle
(C) isosceles and right angled
(D) equilateral triangle

Solution:

$$\cos A \cos B + \sin A \sin B \sin C = 1$$

$$\cos A \cos B + \sin A \sin B$$

$$- \sin A \sin B + \sin A \sin B \sin C = 1$$

$$\cos(A - B) + \sin A \sin B(\sin C - 1) = 1$$

$$\sin A \sin B(\sin C - 1) = 1 - \cos(A - B)$$

Now $\text{LHS} \leq 0$ and $\text{RHS} \geq 0$. Therefore $\text{LHS} = \text{RHS} = 0$.

Hence

$$\sin C - 1 = 0 \quad \text{and} \quad 1 - \cos(A - B) = 0$$

$$\sin C = 1 \quad \text{and} \quad \cos(A - B) = 1$$

$$C = 90^\circ \quad \text{and} \quad A = B$$

Answers: (A), (B), (C)

11. Which of the following are true?

- (A) Minimum value of $27 \tan^2 \theta + 3 \cot^2 \theta$ is 18
(B) $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 179^\circ = 0$
(C) $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \dots + \sin^2 85^\circ = \frac{17}{2}$
(D) Period of $\sin^2\left(\frac{\pi}{8} + \frac{x}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{x}{2}\right)$ is 2π

Answers: (A), (B), (C), (D)

Solution:

- (A) If a, b are positive, then

$$\frac{a+b}{2} \geq \sqrt{ab}$$

and equality holds if and only if $a = b$. Therefore

$$\begin{aligned} 27 \tan^2 \theta + 3 \cot^2 \theta &\geq 2\sqrt{(27 \tan^2 \theta)(3 \cot^2 \theta)} \\ &= 2 \times 9 = 18 \end{aligned}$$

Hence the minimum value of $27 \tan^2 \theta + 3 \cot^2 \theta$ is 18 and it will occur when $27 \tan^2 \theta = 3 \cot^2 \theta$. So (A) is true.

- (B) It is known that $\cos(180^\circ - \theta) = -\cos \theta$. Therefore the given expression is

$$(\cos 1^\circ - \cos 1^\circ) + (\cos 2^\circ - \cos 2^\circ) + \dots$$

$$+ (\cos 89^\circ - \cos 89^\circ) + \cos 90^\circ = 0$$

Hence (B) is true.

- (C) $\sin^2 85^\circ = \sin^2(90^\circ - 5^\circ) = \cos^2 5^\circ$, $\sin^2 80^\circ = \cos^2 10^\circ$, etc.

The given expression is

$$8 + \sin^2 45^\circ = 8 + \frac{1}{2} = \frac{17}{2}$$

Therefore (C) is true.

$$\begin{aligned} \text{(D)} \quad \sin^2\left(\frac{\pi}{8} + \frac{x}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{x}{2}\right) \\ &= \sin\left(\frac{\pi}{8} + \frac{x}{2} + \frac{\pi}{8} - \frac{x}{2}\right) \sin\left(\frac{\pi}{8} + \frac{x}{2} - \frac{\pi}{8} + \frac{x}{2}\right) \\ &= \left(\sin \frac{\pi}{4}\right) \sin x \\ &= \frac{1}{\sqrt{2}} \sin x \end{aligned}$$

Therefore period = 2π . Hence (D) is also true.

Answers: (A), (B), (C), (D)

12. Which of the following are true?

- (A) If $\sin A + \sin B = \sqrt{3}(\cos B - \cos A)$, then $\sin 3A + \sin 3B$ is equal to 0
(B) The value of $\cos^2 \frac{\pi}{16} + \cos^2 \frac{3\pi}{16} + \cos^2 \frac{5\pi}{16} + \cos^2 \frac{7\pi}{16}$ is $\frac{3}{2}$
(C) If $A + B + C = 180^\circ$ and $\cos A = \cos B \cos C$, then $\tan A - \tan B - \tan C = 0$
(D) For all values of θ , $3 - \cos \theta + \cos((\pi/3) + \theta)$ lies in the interval $[2, 4]$

Solution:

- (A) Given $\sin A + \sin B = \sqrt{3}(\cos B - \cos A)$. Now two cases arise

Case 1: $\cos B - \cos A = 0$

Therefore

$$\sin A + \sin B = 0$$

$$\sin A = -\sin B$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$= -3 \sin B + 4 \sin^3 B$$

$$= -\sin 3B$$

$$\sin 3A + \sin 3B = 0$$

Case 2: $\cos B - \cos A \neq 0$

Therefore

$$\frac{\sin A + \sin B}{\cos B - \cos A} = \sqrt{3}$$

$$\frac{2\sin[(A+B)/2]\cos[(A-B)/2]}{2\sin[(A+B)/2]\sin[(A-B)/2]} = \sqrt{3}$$

$$\tan\left(\frac{A-B}{2}\right) = \frac{1}{\sqrt{3}} = \tan\frac{\pi}{6}$$

$$\frac{A-B}{2} = n\pi + \frac{\pi}{6}$$

$$A = B + 2n\pi + \frac{\pi}{3}$$

$$\sin 3A = \sin(3B + 6n\pi + \pi)$$

$$= \sin(\pi + 3B)$$

$$= -\sin 3B$$

$$\sin 3A + \sin 3B = 0$$

Therefore (A) is true.

$$\begin{aligned} \text{(B)} \quad & \cos^2 \frac{\pi}{16} + \cos^2 \frac{3\pi}{16} + \cos^2 \frac{5\pi}{16} + \cos^2 \frac{7\pi}{16} \\ &= \frac{1 + \cos(\pi/8)}{2} + \frac{1 + \cos(3\pi/8)}{2} \\ &\quad + \frac{1 + \cos(5\pi/8)}{2} + \frac{1 + \cos(7\pi/8)}{2} \\ &= 2 + \frac{1}{2} \left[\cos \frac{\pi}{8} + \cos \frac{3\pi}{8} + \cos \frac{5\pi}{8} + \cos \frac{7\pi}{8} \right] \\ &= 2 + \frac{1}{2} \left[\cos \frac{\pi}{8} + \cos \frac{3\pi}{8} - \cos \frac{3\pi}{8} - \cos \frac{\pi}{8} \right] \\ &= 2 + 0 = 2 \end{aligned}$$

Hence (B) is not true.

(C) Given $A + B + C = 180^\circ$ and $\cos A = \cos B \cos C$.
Now,

$$\begin{aligned} A + B + C = 180^\circ &\Rightarrow \tan A + \tan B + \tan C \\ &= \tan A \tan B \tan C \end{aligned}$$

Also

$$\begin{aligned} \cos A = \cos B \cos C &\Rightarrow -\cos(B+C) = \cos B \cos C \\ &\Rightarrow -\cos B \cos C + \sin B \sin C = \cos B \cos C \\ &\Rightarrow \sin B \sin C = 2 \cos B \cos C \\ &\Rightarrow \tan B \tan C = 2 \end{aligned}$$

Therefore

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C = 2 \tan A$$

$$\tan A - \tan B - \tan C = 0$$

Hence (C) is true.

$$\begin{aligned} \text{(D)} \quad & 3 - \cos \theta + \cos\left(\frac{\pi}{3} + \theta\right) \\ &= 3 - \cos \theta + \frac{1}{2} \cos \theta - \frac{\sqrt{3}}{2} \sin \theta \\ &= 3 - \frac{1}{2} [\cos \theta + \sqrt{3} \sin \theta] \end{aligned}$$

The maximum and minimum values of $\cos \theta + \sqrt{3} \sin \theta$ are $\pm\sqrt{1+3} = \pm 2$. Therefore, the given expression lies in $[2, 4]$. So, (D) is true.

Answers: (A), (C), (D)

13. Which of the following are true?

(A) If $\cos \alpha + \cos \beta = 0 = \sin \alpha + \sin \beta$, then $\cos 2\alpha + \cos 2\beta$ is equal to $2 \cos(\alpha + \beta)$

(B) $\sin^2 18^\circ$ and $\cos^2 36^\circ$ are roots of the quadratic equation $16x^2 - 12x + 1 = 0$

(C) If $\sin \theta = \lambda \sin(\theta + 2\alpha)$, then $\tan(\theta + \alpha)$ equals $\left(\frac{1+\lambda}{1-\lambda}\right) \tan \alpha$

(D) If $\sqrt{\frac{1+\cos A}{1-\cos A}} = \frac{m}{n}$, then $\tan A = \frac{2mn}{m^2 + n^2}$

Solution:

(A) $(\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 = 0 \Rightarrow 1 + \cos(\alpha - \beta) = 0$

Therefore

$$\cos(\alpha - \beta) = -1$$

$$\cos 2\alpha + \cos 2\beta = 2 \cos(\alpha + \beta) \cos(\alpha - \beta)$$

Hence (A) is not true.

$$\text{(B)} \quad \sin 18^\circ = \frac{\sqrt{5}-1}{4} \text{ and } \cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

Therefore $\sin^2 18^\circ$ and $\cos^2 36^\circ$ are roots of

$$x^2 - \left[\left(\frac{\sqrt{5}-1}{4} \right)^2 + \left(\frac{\sqrt{5}+1}{4} \right)^2 \right] x + \left(\frac{\sqrt{5}-1}{4} \right)^2 \left(\frac{\sqrt{5}+1}{4} \right)^2 = 0$$

$$x^2 - \frac{12x}{16} + \frac{1}{16} = 0$$

$$16x^2 - 12x + 1 = 0$$

Hence (B) is true.

$$(C) \quad \frac{1}{\lambda} = \frac{\sin(\theta + 2\alpha)}{\sin \theta}$$

Therefore

$$\begin{aligned} \frac{\lambda + 1}{1 - \lambda} &= \frac{\sin(\theta + 2\alpha) + \sin \theta}{\sin(\theta + 2\alpha) - \sin \theta} \\ &= \frac{2 \sin(\theta + \alpha) \cos \alpha}{2 \cos(\theta + \alpha) \sin \alpha} \\ &= \tan(\theta + \alpha) \cot \alpha \\ \tan(\theta + \alpha) &= \left(\frac{\lambda + 1}{1 - \lambda} \right) \tan \alpha \end{aligned}$$

Hence (C) is true.

$$(D) \quad \frac{m}{n} = \sqrt{\frac{2 \cos^2(A/2)}{\sin^2(A/2)}} = \cot \frac{A}{2}$$

Therefore

$$\tan \frac{A}{2} = \frac{n}{m} \quad \text{and} \quad \tan A = \frac{2 \tan(A/2)}{1 - \tan^2(A/2)} = \frac{2mn}{m^2 - n^2}$$

So (D) is not true.

Answers: (B), (C)

- 14.** If $A + B + C = 180^\circ$ where A, B, C are positive angles and

$$\begin{aligned} \cos A \cos B + \cos B \cos C + \cos C \cos A \\ = 1 - 2 \cos A \cos B \cos C \end{aligned}$$

then

$$(A) \quad A = B$$

$$(B) \quad B = C$$

$$(C) \quad C = A$$

$$(D) \quad A = B = C$$

Solution:

$$\Sigma \cos B \cos C$$

$$= 1 - 2 \cos A \cos B \cos C$$

$$= 1 - \cos A [\cos(B + C) + \cos(B - C)]$$

$$= 1 + \cos^2 A + \cos(B + C) \cos(B - C)$$

$$(\because A + B + C = 180^\circ)$$

$$= 1 + \cos^2 A + \cos^2 B - \sin^2 C$$

$$= \cos^2 A + \cos^2 B + \cos^2 C$$

Therefore

$$\begin{aligned} \cos^2 A + \cos^2 B + \cos^2 C - \cos B \cos C - \cos C \cos A \\ - \cos A \cos B = 0 \end{aligned}$$

$$\begin{aligned} \frac{1}{2} [(\cos A - \cos B)^2 + (\cos B - \cos C)^2 \\ + (\cos C - \cos A)^2] = 0 \end{aligned}$$

$$\cos A = \cos B = \cos C$$

$$A = B = C \quad (\because A + B + C = 180^\circ)$$

Answers: (A), (B), (C), (D)

- 15.** For a positive integer n , let

$$\begin{aligned} f_n(\theta) &= \left(\tan \frac{\theta}{2} \right) (1 + \sec \theta) (1 + \sec 2\theta) \\ &\quad (1 + \sec 4\theta) \cdots (1 + \sec 2^n \theta) \end{aligned}$$

Then

$$(A) \quad f_2\left(\frac{\pi}{16}\right) = 1$$

$$(B) \quad f_3\left(\frac{\pi}{32}\right) = 1$$

$$(C) \quad f_4\left(\frac{\pi}{64}\right) = 1$$

$$(D) \quad f_5\left(\frac{\pi}{128}\right) = 1$$

Solution:

$$\tan \frac{\theta}{2} (1 + \sec \theta) = \frac{\sin(\theta/2)}{\cos(\theta/2)} \left(\frac{1 + \cos \theta}{\cos \theta} \right) = \tan \theta$$

Therefore

$$\begin{aligned} f_n(\theta) &= \tan \theta (1 + \sec 2\theta) (1 + \sec 4\theta) \cdots (1 + \sec 2^n \theta) \\ &= \tan 2\theta (1 + \sec 4\theta) (1 + \sec 8\theta) \cdots (1 + \sec 2^n \theta) \end{aligned}$$

Finally

$$f_n(\theta) = \tan 2^{n-1} \theta (1 + \sec 2^n \theta) = \tan(2^n \theta)$$

Therefore

$$f_2\left(\frac{\pi}{16}\right) = \tan\left(2^2 \cdot \frac{\pi}{16}\right) = \tan \frac{\pi}{4} = 1$$

Hence (A) is true.

$$f_3\left(\frac{\pi}{32}\right) = \tan\left(2^3 \cdot \frac{\pi}{32}\right) = \tan \frac{\pi}{4} = 1$$

So (B) is true.

$$f_4\left(\frac{\pi}{64}\right) = \tan\left(2^4 \cdot \frac{\pi}{64}\right) = \tan \frac{\pi}{4} = 1$$

So (C) is true.

$$f_5\left(\frac{\pi}{128}\right) = \tan\left(2^5 \cdot \frac{\pi}{128}\right) = \tan \frac{\pi}{4} = 1$$

Hence (D) is true.

Answers: (A), (B), (C), (D)

16. Which of the following are true

(A) If $90^\circ < \theta < 180^\circ$ and $\tan \theta = \frac{-2}{3}$, then

$$\frac{\tan(90^\circ + \theta) + \cos(180^\circ + \theta)}{\sin(270^\circ - \theta) - \cot(-\theta)} = \frac{2 + \sqrt{13}}{2 - \sqrt{13}}$$

(B) For any angle θ ,

$$\frac{\sin 2\theta \cos \theta}{(1 + \cos 2\theta)(1 + \cos \theta)} = \tan \frac{\theta}{2}$$

(C) If $\frac{-\pi}{4} < \theta < \frac{\pi}{4}$, then $\cos \theta - \sin \theta$ is always positive

(D) If $\frac{\cos x}{\cos y} = \frac{a}{b}$, then $a \tan x + b \tan y$ is equal to $(a + b) \tan \left(\frac{x + y}{2} \right)$

Solution: In the second quadrant $\sin \theta > 0$, $\cos \theta < 0$, $\tan \theta < 0$ and $\cot \theta < 0$.

$$\begin{aligned} \text{(A)} \quad & \frac{\tan(90^\circ + \theta) + \cos(180^\circ + \theta)}{\sin(270^\circ - \theta) - \cot(-\theta)} \\ &= \frac{-\cot \theta - \cos \theta}{-\cos \theta + \cot \theta} \\ &= \frac{\cot \theta + \cos \theta}{\cos \theta - \cot \theta} = \frac{(-3/2) + (-3/\sqrt{13})}{-(3/\sqrt{13}) + (3/2)} \\ &= \frac{2 + \sqrt{13}}{2 - \sqrt{13}} \end{aligned}$$

Therefore (A) is true.

$$\begin{aligned} \text{(B)} \quad & \frac{\sin 2\theta \cos \theta}{(1 + \cos 2\theta)(1 + \cos \theta)} = \frac{2 \sin \theta \cos^2 \theta}{(2 \cos^2 \theta)(2 \cos^2(\theta/2))} \\ &= \frac{2 \sin(\theta/2) \cos(\theta/2)}{2 \cos^2(\theta/2)} \\ &= \tan \frac{\theta}{2} \end{aligned}$$

Therefore (B) is true.

(C) $\frac{-\pi}{4} < \theta < 0 \Rightarrow \cos \theta > 0$ and $\sin \theta < 0$

$$\Rightarrow \cos \theta - \sin \theta > 0$$

$$0 < \theta < \frac{\pi}{4} \Rightarrow \cos \theta > \sin \theta$$

$$\Rightarrow \cos \theta - \sin \theta > 0$$

$$\theta = 0 \Rightarrow \cos \theta - \sin \theta = 1$$

Therefore

$$\frac{-\pi}{4} < \theta < \frac{\pi}{4} \Rightarrow \cos \theta - \sin \theta > 0$$

Hence (C) is true.

(D) $\frac{a}{\cos x} = \frac{b}{\cos y} = k$ (say)

so that

$$k = \frac{a + b}{\cos x + \cos y}$$

Now

$$\begin{aligned} a \tan x + b \tan y &= (k \cos x) \tan x + (k \cos y) \tan y \\ &= k(\sin x + \sin y) \\ &= \frac{(a + b)(\sin x + \sin y)}{\cos x + \cos y} \\ &= (a + b) \tan \left(\frac{x + y}{2} \right) \end{aligned}$$

Therefore (D) is true.

Answers: (A), (B), (C), (D)

17. Which of the following are true?

(A) If x is any real number, then $\tan \left(\frac{\pi}{4} \sin^2 x \right)$ lies between 0 and 1

(B) The value of $\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8}$ is $2/3$

(C) If α and β are two different solutions lying between $-\pi/2$ and $\pi/2$ of the equation $2 \tan \theta + \sec \theta = 2$, then the value of $\tan \alpha + \tan \beta$ is $4/3$

(D) The value of $2 \sin \left(\frac{\theta}{2} \right) \cos \left(\frac{3\theta}{2} \right) + 4 \sin \theta \sin^2 \left(\frac{\theta}{2} \right)$ equals $\sin \theta$.

Solution:

(A) $0 \leq \sin^2 x \leq 1$ for all $x \Rightarrow 0 \leq \tan \left(\frac{\pi}{4} \sin^2 x \right) \leq 1$

Left equality holds when x is a multiple of π and right equality holds when x is an odd multiple of $\pi/2$.

Therefore (A) is true.

(B) The given expression is equivalent to

$$\begin{aligned} 2 \cos^4 \frac{\pi}{8} + 2 \cos^4 \frac{3\pi}{8} &= \frac{1}{2} \left(1 + \cos \frac{\pi}{4} \right)^2 + \frac{1}{2} \left(1 + \cos \frac{3\pi}{4} \right)^2 \\ &= \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right)^2 + \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right)^2 \end{aligned}$$

$$= \left(1 + \frac{1}{2}\right) = \frac{3}{2}$$

Hence (B) is not true.

(C) $\tan \alpha, \tan \beta$ are roots of the equation

$$\sec^2 \theta = (2 - 2 \tan \theta)^2$$

$$3 \tan^2 \theta - 8 \tan \theta + 3 = 0$$

Therefore

$$\tan \alpha + \tan \beta = \frac{8}{3}$$

Hence (C) is not true.

$$\begin{aligned} \text{(D)} \quad & 2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{3\theta}{2}\right) + 4 \sin \theta \sin^2\left(\frac{\theta}{2}\right) \\ &= (\sin 2\theta - \sin \theta) + 2 \sin \theta (1 - \cos \theta) \\ &= (\sin 2\theta - \sin \theta) + (2 \sin \theta - \sin 2\theta) \\ &= \sin \theta \end{aligned}$$

Hence (D) is true.

Answers: (A), (D)

18. Which of the following are true?

- (A) If a, b are positive numbers, then the values of r and θ ($0 \leq \theta \leq \pi$) for which $a \sin x + b \cos x = r \sin(x + \theta)$ are $r = \sqrt{a^2 + b^2}$ and $\tan \theta = b/a$
- (B) $\cos(\alpha - \beta) = -1$, if and only if $\sin \alpha + \sin \beta = 0 = \cos \alpha + \cos \beta$
- (C) If $A + B + C = 180^\circ$, then the value of $1 - (\sin^2 A + \sin^2 B + \sin^2 C)$ is $2 \sin A \sin B \sin C$
- (D) If $\tan \alpha$ and $\tan \beta$ are roots of the equation $x^2 - bx + c = 0$, then the value of $\sin^2(\alpha + \beta)$ is $b^2/(b^2 + c^2)$

Solution:

(A) $a \sin x + b \cos x$

$$\begin{aligned} &= \sqrt{a^2 + b^2} \left[\frac{a}{\sqrt{a^2 + b^2}} \sin x + \frac{b}{\sqrt{a^2 + b^2}} \cos x \right] \\ &= \sqrt{a^2 + b^2} [\cos \theta \sin x + \sin \theta \cos x] \\ &= \sqrt{a^2 + b^2} \sin(x + \theta) \end{aligned}$$

where

$$\cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \sin \theta = \frac{b}{\sqrt{a^2 + b^2}}$$

Therefore

$$r = \sqrt{a^2 + b^2} \quad \text{and} \quad \tan \theta = b/a$$

(A) is true.

(B) $\cos(\alpha - \beta) = -1 \Rightarrow \alpha - \beta = (2k + 1)\pi$ (where k is an integer)

$$\begin{aligned} \sin \alpha + \sin \beta &= 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \\ &= 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos(2k + 1)\frac{\pi}{2} = 0 \end{aligned}$$

$$\begin{aligned} \cos \alpha + \cos \beta &= 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \\ &= 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos(2k + 1)\frac{\pi}{2} = 0 \end{aligned}$$

The converse is clear. Hence (B) is true.

$$\begin{aligned} \text{(C)} \quad & 1 - (\sin^2 A + \sin^2 B + \sin^2 C) \\ &= 1 - \left[\frac{1 - \cos 2A}{2} + \frac{1 - \cos 2B}{2} + \frac{1 - \cos 2C}{2} \right] \\ &= \frac{-1}{2} + \frac{1}{2} [\cos 2A + \cos 2B + \cos 2C] \\ &= \frac{-1}{2} + \frac{1}{2} [-1 - 4 \cos A \cos B \cos C] \\ &= -1 - 2 \cos A \cos B \cos C \end{aligned}$$

Hence (C) is not true.

(D) $\tan \alpha + \tan \beta = b, \tan \alpha \tan \beta = c$
Now

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{b}{1 - c} \\ \cos 2(\alpha + \beta) &= \frac{1 - \tan^2(\alpha + \beta)}{1 + \tan^2(\alpha + \beta)} \\ &= \frac{1 - [b^2/(1 - c)^2]}{1 + [b^2/(1 - c)^2]} \\ &= \frac{(1 - c)^2 - b^2}{(1 - c)^2 + b^2} \end{aligned}$$

Therefore

$$\begin{aligned} \sin^2(\alpha + \beta) &= \frac{1 - \cos 2(\alpha + \beta)}{2} \\ &= \frac{1}{2} \left[1 - \frac{(1 - c)^2 - b^2}{(1 - c)^2 + b^2} \right] \\ &= \frac{b^2}{(1 - c)^2 + b^2} \end{aligned}$$

So (D) is not true.

Answers: (A), (B)

Matrix-Match Type Questions

1. Match the items of Column I with those of Column II.

Column I	Column II
(A) $\left(\frac{\sin 2\theta}{1 + \cos 2\theta}\right)\left(\frac{\cos \theta}{1 + \cos \theta}\right)$ is equal to	(p) $\cot \frac{\theta}{2}$
(B) $\left(\frac{\sin 2\theta}{1 - \cos 2\theta}\right)\left(\frac{1 - \cos \theta}{\cos \theta}\right)$ equals	(q) $\tan \frac{\theta}{2}$
(C) $\frac{\cot \theta - 1}{\cot \theta + 1} =$	(r) $\frac{1 - \sin 2\theta}{\cos 2\theta}$
(D) $\sec \theta + \tan \theta =$	(s) $\frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}}$

Solution:

$$\begin{aligned}
 \text{(A)} \quad \left(\frac{\sin 2\theta}{1 + \cos 2\theta}\right)\left(\frac{\cos \theta}{1 + \cos \theta}\right) &= \left(\frac{2 \sin \theta \cos^2 \theta}{2 \cos^2 \theta}\right)\left(\frac{1}{2 \cos^2(\theta/2)}\right) \\
 &= \frac{2 \sin(\theta/2) \cos(\theta/2)}{2 \cos^2(\theta/2)} \\
 &= \tan \frac{\theta}{2}
 \end{aligned}$$

Answer: (A) → (q)

$$\begin{aligned}
 \text{(B)} \quad \left(\frac{\sin 2\theta}{1 - \cos 2\theta}\right)\left(\frac{1 - \cos \theta}{\cos \theta}\right) &= \frac{(2 \sin \theta \cos \theta) 2 \sin^2(\theta/2)}{(2 \sin^2 \theta)(\cos \theta)} \\
 &= \frac{2 \sin^2(\theta/2)}{\sin \theta} \\
 &= \frac{2 \sin^2(\theta/2)}{2 \sin(\theta/2) \cos(\theta/2)} \\
 &= \tan\left(\frac{\theta}{2}\right)
 \end{aligned}$$

Answer: (B) → (q)

$$\begin{aligned}
 \text{(C)} \quad \frac{1 - \sin 2\theta}{\cos 2\theta} &= \frac{(\cos \theta - \sin \theta)^2}{\cos^2 \theta - \sin^2 \theta} \\
 &= \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \\
 &= \frac{\cot \theta - 1}{\cot \theta + 1}
 \end{aligned}$$

Answer: (C) → (r)

$$\begin{aligned}
 \text{(D)} \quad \sec \theta + \tan \theta &= \frac{1 + \sin \theta}{\cos \theta} = \frac{\left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2}\right)^2}{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}} \\
 &= \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} = \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}}
 \end{aligned}$$

Answer: (D) → (s)

2. Match the items of Column I with those of Column II.

Column I	Column II
(A) The value of $\cos^2 \theta + \cos^2(60^\circ + \theta) + \cos^2(60^\circ - \theta)$ is	(p) $\frac{3}{2}$
(B) $\cos 20^\circ \cos 40^\circ \cos 80^\circ$	(q) $\frac{1}{8}$
(C) $\sin^2 \theta + \sin^2(120^\circ + \theta) + \sin^2(120^\circ - \theta)$ equals	(r) $\frac{\sqrt{3}}{8}$
(D) $\sin 20^\circ \sin 40^\circ \sin 80^\circ$ is equal to	(s) $\frac{3}{8}$

Solution:

$$\begin{aligned}
 \text{(A)} \quad \text{Given expression is equivalent to} \\
 \frac{1}{2} [3 + \cos 2\theta + \cos(120^\circ + 2\theta) + \cos(120^\circ - 2\theta)] \\
 = \frac{1}{2} \left[3 + \cos 2\theta - \frac{1}{2} \cos 2\theta - \frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta \right. \\
 \left. + \frac{\sqrt{3}}{2} \sin 2\theta \right] \\
 = \frac{3}{2}
 \end{aligned}$$

Answer: (A) → (p)

$$\begin{aligned}
 \text{(B)} \quad \cos 20^\circ \cos 40^\circ \cos 80^\circ \\
 = \cos \theta \cos(60^\circ - \theta) \cos(60^\circ + \theta) \\
 = \frac{\cos 3\theta}{4} \\
 = \frac{\cos 60^\circ}{4} = \frac{1}{8} \\
 \text{where } \theta = 20^\circ.
 \end{aligned}$$

Answer: (B) → (q)

(C) Given expression is

$$\begin{aligned} \frac{3}{2} - \frac{1}{2} [\cos 2\theta + \cos(240^\circ + 2\theta) + \cos(240^\circ - 2\theta)] \\ = \frac{3}{2} - 0 = \frac{3}{2} \end{aligned}$$

Answer: (C) → (p)

(D) Given that $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \sin \theta \sin(60^\circ - \theta) \sin(60^\circ + \theta)$

$$\begin{aligned} &= \frac{\sin 3\theta}{4} \\ &= \frac{\sin 60^\circ}{4} = \frac{\sqrt{3}}{8} \end{aligned}$$

where $\theta = 20^\circ$.

Answer: (D) → (r)

3. Match the items of Column I with those of Column II.

Column I	Column II
(A) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to	(p) 0
(B) If $\tan A + \tan B + \tan C = 6$ and $\tan A \tan B = 2$ where $A + B + C = 180^\circ$, then $\tan C$ value is	(q) $\sqrt{3}$
(C) The value of $\cot 70^\circ + 4 \cos 70^\circ$ is	(r) 4
(D) If $\tan^2 \theta = 2 \tan^2 \alpha + 1$, then the value of $\cos 2\theta + \sin^2 \alpha$ is	(s) 3

Solution:

$$\begin{aligned} \text{(A)} \quad & \tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ \\ &= (\tan 9^\circ + \tan 81^\circ) - (\tan 27^\circ + \tan 63^\circ) \\ &= (\tan 9^\circ + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ) \\ &= \left(\frac{\sin 9^\circ}{\cos 9^\circ} + \frac{\cos 9^\circ}{\sin 9^\circ} \right) - \left(\frac{\sin 27^\circ}{\cos 27^\circ} + \frac{\cos 27^\circ}{\sin 27^\circ} \right) \\ &= \frac{1}{\sin 9^\circ \cos 9^\circ} - \frac{1}{\sin 27^\circ \cos 27^\circ} \\ &= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} \\ &= \frac{2[\sin 54^\circ - \sin 18^\circ]}{\sin 18^\circ \sin 54^\circ} \\ &= \frac{4 \cos 36^\circ \sin 18^\circ}{\sin 18^\circ \sin 54^\circ} \\ &= 4 \quad (\because \cos 36^\circ = \sin 54^\circ) \end{aligned}$$

Answer: (A) → (r)

(B) $A + B + C = 180^\circ$

$$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\Rightarrow 6 = 2 \tan C$$

$$\Rightarrow \tan C = 3$$

Answer: (B) → (s)

$$\begin{aligned} \text{(C)} \quad \cot 70^\circ + 4 \cos 70^\circ &= \frac{\cos 70^\circ + 4 \cos 70^\circ \sin 70^\circ}{\sin 70^\circ} \\ &= \frac{\cos 70^\circ + 2 \sin 140^\circ}{\sin 70^\circ} \\ &= \frac{\cos 70^\circ + 2 \sin 40^\circ}{\sin 70^\circ} \\ &= \frac{\sin 20^\circ + \sin 40^\circ + \sin 40^\circ}{\sin 70^\circ} \\ &= \frac{2 \cos 10^\circ \sin 30^\circ + \sin 40^\circ}{\sin 70^\circ} \\ &= \frac{\cos 10^\circ + \cos 50^\circ}{\sin 70^\circ} \\ &= \frac{2 \cos 30^\circ \cos 20^\circ}{\sin 70^\circ} \\ &= \frac{\sqrt{3} \cos 20^\circ}{\cos 20^\circ} = \sqrt{3} \end{aligned}$$

Answer: (C) → (q)

$$\begin{aligned} \text{(D)} \quad \cos 2\theta + \sin^2 \alpha &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} + \sin^2 \alpha \\ &= \frac{1 - (2 \tan^2 \alpha + 1)}{1 + (2 \tan^2 \alpha + 1)} + \sin^2 \alpha \\ &= \frac{-\tan^2 \alpha}{1 + \tan^2 \alpha} + \sin^2 \alpha \\ &= -\sin^2 \alpha + \sin^2 \alpha = 0 \end{aligned}$$

Answer: (D) → (p)

4. Match the items of Column I with those of Column II.

Column I	Column II
(A) If $\tan \theta = \frac{4}{5}$, then the value of $\frac{5 \sin \theta - 3 \cos \theta}{5 \sin \theta + 2 \cos \theta}$ is	(p) 2
(B) The minimum value of $\cos^2 \theta + \sec^2 \theta$ is	(q) 2^n
(C) The minimum value of $3 \sin^2 \theta + 2 \cos^2 \theta$ is	(r) 2^{n+1}
(D) If $\sin x + \operatorname{cosec} x = 2$, then for any positive integer n , $\sin^n x + \operatorname{cosec}^n x$ is equal to	(s) $1/6$

Solution:

$$(A) \frac{5 \sin \theta - 3 \cos \theta}{5 \sin \theta + 2 \cos \theta} = \frac{5 \tan \theta - 3}{5 \tan \theta + 2} = \frac{5(4/5) - 3}{5(4/5) + 2} = \frac{1}{6}$$

Answer: (A) → (s)

$$(B) \cos^2 \theta + \sec^2 \theta \geq 2\sqrt{\cos^2 \theta \cdot \sec^2 \theta} = 2$$

Therefore minimum value of $\cos^2 \theta + \sec^2 \theta$ is 2.**Answer: (B) → (p)**

$$(C) 3 \sin^2 \theta + 2 \cos^2 \theta = 2(\sin^2 \theta + \cos^2 \theta) + \sin^2 \theta \\ = 2 + \sin^2 \theta \geq 2$$

Answer: (C) → (p)

$$(D) \sin x + \operatorname{cosec} x = 2 \Rightarrow (\sin x - 1)^2 \equiv 0 \\ \Rightarrow \sin x = 1$$

Therefore

$$\sin^n x + \operatorname{cosec}^n x = 1 + 1 = 2$$

Answer: (D) → (p)**5. Match the items of Column I with those of Column II.**

Column I	Column II
(A) If $\cos(A - B) = 3/5$ and $\tan A \tan B = 2$, the value of $\cos A \cos B$ is	(p) $\frac{-1}{3}$
(B) If $\theta = \pi/4n$ (n is a positive integer), then the value of $\tan \theta \tan 2\theta \tan 3\theta \cdots \tan 2(n-1)\theta \tan(2n-1)\theta$ is	(q) 1
(C) If $\cos \alpha = 2 \cos \beta$, then the value of $\tan\left(\frac{\alpha - \beta}{2}\right) \tan\left(\frac{\alpha + \beta}{2}\right)$	(r) $\frac{1}{5}$
(D) If $A + B + C = 180^\circ$, $\sin A \cos B = 1/4$ and $3 \tan A = \tan B$, then $\cot^2 A$ is equal to	(s) 3

Solution:

$$(A) \cos A \cos B + \sin A \sin B = \frac{3}{5} \quad (1.13)$$

$$\tan A \tan B = 2 \Rightarrow \sin A \sin B = 2 \cos A \cos B$$

Therefore from Eq. (1.13) we have

$$3 \cos A \cos B = \frac{3}{5}$$

$$\cos A \cos B = \frac{1}{5}$$

Answer: (A) → (r)

$$(B) \theta = \frac{\pi}{4n} \Rightarrow 2n\theta = \frac{\pi}{2}$$

Therefore

$$\tan(2n-1)\theta = \tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

$$\tan 2(n-1)\theta = \tan\left(\frac{\pi}{2} - 2\theta\right) = \cot 2\theta$$

$$\tan(2n-3)\theta = \tan\left(\frac{\pi}{2} - 3\theta\right) = \cot 3\theta$$

$$\vdots \quad \vdots \quad \vdots$$

$$\tan n\theta = \tan \frac{\pi}{4} = 1$$

The given expression is equal to

$$(\tan \theta \cot \theta)(\tan 2\theta \cot 2\theta)(\tan 3\theta \cot 3\theta) \cdots$$

$$(\tan(n-1)\theta \cot(n-1)\theta) \tan n\theta$$

$$= (1)(1)(1) \cdots (1) 1 \quad \left(\because \tan n\theta = \tan \frac{\pi}{4} = 1 \right)$$

$$= 1$$

Answer: (B) → (q)

$$(C) \tan\left(\frac{\alpha - \beta}{2}\right) \tan\left(\frac{\alpha + \beta}{2}\right) = \frac{\sin\left(\frac{\alpha - \beta}{2}\right) \sin\left(\frac{\alpha + \beta}{2}\right)}{\cos\left(\frac{\alpha - \beta}{2}\right) \cos\left(\frac{\alpha + \beta}{2}\right)} \\ = \frac{\sin^2(\alpha/2) - \sin^2(\beta/2)}{\cos^2(\alpha/2) - \sin^2(\beta/2)} \\ = \frac{(1 - \cos \alpha) - (1 - \cos \beta)}{(1 + \cos \alpha) - (1 - \cos \beta)} \\ = \frac{-\cos \alpha + \cos \beta}{\cos \alpha + \cos \beta} \\ = \frac{-2 \cos \beta + \cos \beta}{2 \cos \beta + \cos \beta} = \frac{-1}{3}$$

Answer: (C) → (p)

$$(D) 3 \tan A = \tan B \Rightarrow 3 \sin A \cos B = \sin B \cos A$$

$$\Rightarrow \frac{3}{4} = \cos A \sin B$$

Therefore

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$= \frac{1}{4} + \frac{3}{4}$$

$$= 1$$

This implies

$$A + B = 90^\circ$$

$$C = 90^\circ$$

$$3 \tan A = \tan B = \tan(90^\circ - A) = \cot A$$

$$\cot^2 A = 3$$

Answer: (D) → (s)

6. Match the items of Column I with those of Column II.

Column I	Column II
(A) If $0 < \alpha < \pi/16$ and $(1 + \tan \alpha)(1 + \tan 4\alpha) = 2$ then α is equal to	(p) $\frac{\pi}{3}$
(B) If $0 < x < \pi/2$ and $1 + \sin x + \sin^2 x + \sin^3 x + \dots + \infty$ is equal to $4 + 2\sqrt{3}$, then x equals	(q) $\frac{\pi}{20}$
(C) The value of $\frac{3 + \cot 80^\circ \cot 20^\circ}{\cot 80^\circ + \cot 20^\circ}$ is equal to $\tan \theta$, then θ can be	(r) $\frac{5\pi}{18}$
(D) If $2 \sec 2\theta = \tan \alpha + \cot \alpha$, then one of the values of $\alpha + \theta$ is equal to	(s) $\frac{2\pi}{3}$
	(t) $\frac{\pi}{4}$

Solution:

(A) We have

$$1 + \tan \alpha + \tan 4\alpha + \tan \alpha \tan 4\alpha = 2$$

Therefore

$$\tan(\alpha + 4\alpha) = 1$$

$$5\alpha = \frac{\pi}{4} \quad \left(\because 0 < \alpha < \frac{\pi}{16} \right)$$

$$\alpha = \frac{\pi}{20}$$

Answer: (A) → (q)

$$(B) \frac{1}{1 - \sin x} = 4 + 2\sqrt{3}$$

$$\sin x = 1 - \frac{1}{4 + 2\sqrt{3}}$$

$$= \frac{3 + 2\sqrt{3}}{4 + 2\sqrt{3}}$$

$$= \frac{\sqrt{3}}{2} \frac{(2 + \sqrt{3})}{(2 + \sqrt{3})}$$

$$= \frac{\sqrt{3}}{2}$$

This implies

$$x = \frac{\pi}{3} \quad \text{or} \quad \frac{2\pi}{3}$$

Answer: (B) → (p)

$$\begin{aligned} (C) \quad & \frac{3 + \cot 80^\circ \cot 20^\circ}{\cot 80^\circ + \cot 20^\circ} \\ &= \frac{3 \sin 80^\circ \sin 20^\circ + \cos 80^\circ \cos 20^\circ}{\sin 20^\circ \cos 80^\circ + \cos 20^\circ \sin 80^\circ} \\ &= \frac{2 \sin 80^\circ \sin 20^\circ + \cos 80^\circ \cos 20^\circ + \sin 80^\circ \sin 20^\circ}{\sin 100^\circ} \\ &= \frac{(\cos 60^\circ - \cos 100^\circ) + \cos 60^\circ}{\sin 100^\circ} \\ &= \frac{1 - \cos 100^\circ}{\sin 100^\circ} \\ &= \tan 50^\circ \end{aligned}$$

Therefore

$$\theta = \frac{5\pi}{18}$$

Answer: (C) → (r)

$$(D) \frac{2}{\cos 2\theta} = \frac{1}{\sin \alpha \cos \alpha} = \frac{2}{\sin 2\alpha}$$

Therefore

$$\cos 2\theta = \sin 2\alpha$$

$$\text{Hence one of the values of } 2\theta = \frac{\pi}{2} - 2\alpha$$

$$\text{So one of the values of } \alpha + \theta \text{ is } \frac{\pi}{4}$$

Answer: (D) → (t)

Comprehension-Type Questions

1. **Passage:** $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$. Based on this information, answer the following questions.

(i) $\sin A + 2 \sin 3A + \sin 5A$ is equal to

- (A) $4 \sin 3A \cos^2 A$ (B) $4 \cos 3A \sin^2 A$
(C) $4 \sin 3A \sin^2 A$ (D) $4 \cos 3A \cos^2 A$

$$(ii) \frac{\sin A + \sin 2A + \sin 4A + \sin 5A}{\cos A + \cos 2A + \cos 4A + \cos 5A} =$$

- (A) $\cot 3A$ (B) $\tan 3A$
(C) $2 \cot 3A$ (D) $2 \tan 3A$

$$(iii) \frac{\cos 7A + \cos 3A - \cos 5A - \cos A}{\sin 7A - \sin 3A - \sin 5A + \sin A} =$$

- (A) $\cot 2A$ (B) $\tan 2A$
 (C) $2 \cot 2A$ (D) $2 \tan 2A$

Solution: From the given information, we have

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$\begin{aligned} \text{(i)} \quad \sin A + 2 \sin 3A + \sin 5A &= (\sin A + \sin 5A) + 2 \sin 3A \\ &= 2 \sin 3A \cos 2A + 2 \sin 3A \\ &= 2 \sin 3A (\cos 2A + 1) \\ &= 4 \sin 3A \cos^2 A \end{aligned}$$

Answer: (A)

$$\begin{aligned} \text{(ii)} \quad \text{Numerator} &= (\sin A + \sin 5A) + (\sin 2A + \sin 4A) \\ &= 2 \sin 3A \cos 2A + 2 \sin 3A \cos A \\ &= 2 \sin 3A (\cos 2A + \cos A) \end{aligned}$$

$$\begin{aligned} \text{Denominator} &= (\cos A + \cos 5A) + (\cos 2A + \cos 4A) \\ &= 2 \cos 3A \cos 2A + 2 \cos 3A \cos A \\ &= 2 \cos 3A (\cos 2A + \cos A) \end{aligned}$$

Therefore the given quotient is

$$\frac{\sin 3A}{\cos 3A} = \tan 3A$$

Answer: (B)

$$\begin{aligned} \text{(iii)} \quad \text{Numerator} &= (\cos 7A - \cos A) + (\cos 3A - \cos 5A) \\ &= -2 \sin 4A \sin 3A + 2 \sin 4A \sin A \\ &= -2 \sin 4A (\sin 3A - \sin A) \\ &= -2 \sin 4A (2 \cos 2A \sin A) \\ &= -4 \sin 4A \cos 2A \sin A \end{aligned}$$

$$\begin{aligned} \text{Denominator} &= (\sin 7A + \sin A) - (\sin 3A + \sin 5A) \\ &= 2 \sin 4A \cos 3A - 2 \sin 4A \cos A \\ &= 2 \sin 4A (\cos 3A - \cos A) \\ &= 2 \sin 4A (-2 \sin 2A \sin A) \\ &= -4 \sin 4A \sin 2A \sin A \end{aligned}$$

Therefore the given quotient is $\cot 2A$.

Answer: (A)

2. Passage: It is given that A , B and C are angles such that $A + B + C = \pi$. Answer the following questions.

- (i) $1 - 2 \sin B \sin C \cos A + \cos^2 A$ equals
 (A) $\cos^2 B + \cos^2 C$ (B) $\cos^2 B - \cos^2 C$
 (C) $\sin^2 B + \sin^2 C$ (D) $\sin^2 B - \sin^2 C$

(ii) $(\cot B + \cot C)(\cot C + \cot A)(\cot A + \cot B)$ equals

- (A) $\sin A \sin B \sin C$
 (B) $\sec A \sec B \sec C$
 (C) $\operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C$
 (D) $\cos A \cos B \cos C$

(iii) $\frac{1 + \cos A - \cos B + \cos C}{1 + \cos A + \cos B - \cos C} =$

- (A) $\cot \frac{B}{2} \tan \frac{C}{2}$ (B) $\tan \frac{B}{2} \cot \frac{C}{2}$
 (C) $\tan \frac{A}{2} \cot \frac{B}{2}$ (D) $\tan \frac{C}{2} \cot \frac{A}{2}$

Solution:

$$\begin{aligned} \text{(i)} \quad 1 - 2 \sin B \sin C \cos A + \cos^2 A &= 1 + \cos^2 A - [\cos(B - C) - \cos(B + C)] \cos A \\ &= 1 + \cos^2 A - \cos(B - C) \cos A + \cos(B + C) \cos A \\ &= 1 + \cos^2 A + \cos(B - C) \cos(B + C) - \cos^2 A \\ &= 1 + \cos^2 B - \sin^2 C = \cos^2 B + \cos^2 C \end{aligned}$$

Answer: (A)

$$\begin{aligned} \text{(ii)} \quad (\cot B + \cot C)(\cot C + \cot A)(\cot A + \cot B) &= \frac{\sin(B + C)}{\sin B \sin C} \cdot \frac{\sin(C + A)}{\sin C \sin A} \cdot \frac{\sin(A + B)}{\sin A \sin B} \\ &= \frac{1}{\sin A \sin B \sin C} = \operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C \end{aligned}$$

Answer: (C)

$$\begin{aligned} \text{(iii)} \quad \text{Numerator} &= 1 - 2 \sin \frac{A+B}{2} \sin \frac{A-B}{2} + 2 \cos^2 \frac{C}{2} - 1 \\ &= 2 \cos \frac{C}{2} \left[\cos \frac{C}{2} - \sin \left(\frac{A-B}{2} \right) \right] \\ &= 2 \cos \frac{C}{2} \left[\sin \frac{A+B}{2} - \sin \frac{A-B}{2} \right] \\ &= 2 \cos \frac{C}{2} \left(2 \cos \frac{A}{2} \sin \frac{B}{2} \right) \\ &= 4 \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} \end{aligned}$$

$$\text{Similarly, the denominator} = 4 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

Therefore the given quotient is

$$\tan \frac{B}{2} \cot \frac{C}{2}$$

Answer: (B)

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I.
- (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I.
- (C) Statement I is true and Statement II is false.
- (D) Statement I is false and Statement II is true.

1. Statement I: If $0 < \theta < \pi/2$ and $\cos 2\theta = \sqrt{2}(\cos \theta - \sin \theta)$, then $\tan \theta$ is equal to 1.

Statement II: If $0 < \alpha, \beta < \pi/2$ and $\sin(\alpha + \beta) = 1$, then $\alpha + \beta = \pi/2$.

Solution: Statement II is clear. Now

$$\begin{aligned}\cos 2\theta &= \sqrt{2}(\cos \theta - \sin \theta) \Rightarrow \cos^2 \theta - \sin^2 \theta \\ &= \sqrt{2}(\cos \theta - \sin \theta)\end{aligned}$$

This is true when $\cos \theta = \sin \theta$ (i.e., $\theta = \pi/4$) in which case $\tan \theta = 1$. Suppose $\cos \theta \neq \sin \theta$. Therefore

$$\begin{aligned}\cos \theta + \sin \theta &= \sqrt{2} \\ \sin\left(\frac{\pi}{4} + \theta\right) &= 1 \\ \frac{\pi}{4} + \theta &= \frac{\pi}{2} \\ \theta &= \frac{\pi}{4} \\ \tan \theta &= 1\end{aligned}$$

Answer: (A)

2. Statement I: If $\sin \alpha + \sin \beta = \sqrt{3}(\cos \alpha - \cos \beta)$, then $\sin 3\alpha + \sin 3\beta = 0$.

Statement II: $\sin \theta = 0 \Leftrightarrow \theta = n\pi$ and $\tan \theta = \tan \phi \Leftrightarrow \theta = n\pi + \phi$, where n is an integer.

Solution:

$$\sin \alpha + \sin \beta = \sqrt{3}(\cos \alpha - \cos \beta)$$

If $\cos \alpha = \cos \beta$, then $\sin \alpha = -\sin \beta$. Therefore

$$\begin{aligned}3\sin \alpha - 4\sin^3 \alpha &= -3\sin \beta + 4\sin^3 \beta \\ \sin 3\alpha &= -\sin 3\beta \\ \sin 3\alpha + \sin 3\beta &= 0\end{aligned}$$

Suppose $\cos \alpha \neq \cos \beta$. Then

$$2\sin \frac{\alpha + \beta}{2} \cos \left(\frac{\alpha - \beta}{2}\right) = -\sqrt{3} \times 2\sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

This implies that either

$$\sin \frac{\alpha + \beta}{2} = 0$$

$$\text{or} \quad \cos \left(\frac{\alpha - \beta}{2}\right) = -\sqrt{3} \sin \frac{\alpha - \beta}{2}$$

Therefore

$$\alpha + \beta = 2n\pi \quad \text{or} \quad \tan \left(\frac{\alpha - \beta}{2}\right) = \frac{-1}{\sqrt{3}} = \tan \left(\frac{-\pi}{6}\right)$$

$$\alpha + \beta = 2n\pi \quad \text{or} \quad \frac{\alpha - \beta}{2} = n\pi - \frac{\pi}{6}$$

$$\alpha + \beta = 2n\pi \quad \text{or} \quad \alpha = \beta + 2n\pi - \frac{\pi}{3}$$

Now $\alpha + \beta = 2n\pi$ implies

$$\sin 3\alpha = \sin(6n\pi - 3\beta) \quad \text{or} \quad \sin 3\alpha = \sin(6n\pi + 3\beta - \pi)$$

In any case $\sin 3\alpha = -\sin 3\beta$ so that $\sin 3\alpha + \sin 3\beta = 0$.

Answer: (A)

3. Statement I: $\sin(\pi/18)$ is a root of $8x^3 - 6x + 1 = 0$.

Statement II: For any real α , $\sin 3\alpha = 3\sin \alpha - 4\sin^3 \alpha$.

Solution: Statement II is a standard formula. Put $\alpha = \pi/18$. Therefore

$$\begin{aligned}3\alpha &= \frac{\pi}{6} \quad \text{and hence} \quad \sin 3\alpha = \frac{1}{2} \\ 3\sin \alpha - 4\sin^3 \alpha &= \frac{1}{2} \\ 8\sin^3 \alpha - 6\sin \alpha + 1 &= 0\end{aligned}$$

That is $\sin \alpha = \sin(\pi/18)$ is a root of $8x^3 - 6x + 1 = 0$.

Answer: (A)

4. Statement I:

$$\begin{aligned}\frac{1}{\cos 0^\circ \cos 1^\circ} + \frac{1}{\cos 1^\circ \cos 2^\circ} + \frac{1}{\cos 2^\circ \cos 3^\circ} + \dots \\ + \frac{1}{\cos 88^\circ \cos 89^\circ} = \frac{\cos 1^\circ}{\sin^2 1^\circ}\end{aligned}$$

Statement II: $\frac{\sin(A - B)}{\cos A \cos B} = \tan A - \tan B$.

Solution:

$$\begin{aligned}\frac{\sin(A - B)}{\cos A \cos B} &= \frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B} \\ &= \tan A - \tan B\end{aligned}$$

Statement II is true.

Now

$$\begin{aligned} & \frac{\sin 1^\circ}{\cos 0^\circ \cos 1^\circ} + \frac{\sin 1^\circ}{\cos 1^\circ \cos 2^\circ} + \frac{\sin 1^\circ}{\cos 2^\circ \cos 3^\circ} \\ & \quad + \cdots + \frac{\sin 1^\circ}{\cos 88^\circ \cos 89^\circ} \\ &= \frac{\sin(1^\circ - 0^\circ)}{\cos 0^\circ \cos 1^\circ} + \frac{\sin(2^\circ - 1^\circ)}{\cos 1^\circ \cos 2^\circ} + \frac{\sin(3^\circ - 2^\circ)}{\cos 2^\circ \cos 3^\circ} \\ & \quad + \cdots + \frac{\sin(89^\circ - 88^\circ)}{\cos 88^\circ \cos 89^\circ} \\ &= (\tan 1^\circ - \tan 0^\circ) + (\tan 2^\circ - \tan 1^\circ) + (\tan 3^\circ - \tan 2^\circ) \\ & \quad + \cdots + (\tan 89^\circ - \tan 88^\circ) \\ &= \tan 89^\circ - \tan 0^\circ = \cot 1^\circ \end{aligned}$$

Therefore

$$\sum_{k=1}^{89} \frac{1}{\cos(k-1) \cos k} = \frac{\cot 1^\circ}{\sin 1^\circ} = \frac{\cos 1^\circ}{\sin^2 1^\circ}$$

Statement I is true.

Answer: (A)

5. Statement I: Let n be a positive integer and let

$$S_n = \sin \theta + \sin 2\theta + \sin 3\theta + \cdots + \sin n\theta$$

and

$$C_n = \cos \theta + \cos 2\theta + \cos 3\theta + \cdots + \cos n\theta$$

Then

$$S_n = \frac{\left(\sin \frac{n\theta}{2}\right) \left(\sin \left(\frac{n+1}{2}\theta\right)\right)}{\sin(\theta/2)}$$

and

$$C_n = \frac{\left(\sin \frac{n\theta}{2}\right) \left(\cos \left(\frac{n+1}{2}\theta\right)\right)}{\sin(\theta/2)}$$

Statement II: If $z = \cos \theta + i \sin \theta$, then $z^k = \cos k\theta + i \sin k\theta$, where k is an integer.

Solution: Statement II is true (by De'Moivre's Theorem). Now, by Statement II, we have

$$\begin{aligned} C_n + iS_n &= z + z^2 + z^3 + \cdots + z^n = \frac{z(z^n - 1)}{z - 1} \\ &= z \frac{[\cos n\theta + i \sin n\theta - 1]}{\cos \theta + i \sin \theta - 1} \\ &= z \frac{[-2 \sin^2(n\theta/2) + 2i \sin(n\theta/2) \cos(n\theta/2)]}{-2 \sin^2(\theta/2) + 2i \sin(\theta/2) \cos(\theta/2)} \end{aligned}$$

$$\begin{aligned} &= \frac{z[2i \sin(n\theta/2)][\cos(n\theta/2) + i \sin(n\theta/2)]}{2i \sin(\theta/2)(\cos(\theta/2) + i \sin(\theta/2))} \\ &= \frac{\sin(n\theta/2)(\cos \theta + i \sin \theta)[\cos(n\theta/2) + i \sin(n\theta/2)]}{\sin(\theta/2)[\cos(\theta/2) + i \sin(\theta/2)]} \\ &= \frac{\sin(n\theta/2)}{\sin(\theta/2)} \left[\cos\left(\frac{n}{2} + 1 - \frac{1}{2}\right)\theta + i \sin\left(\frac{n}{2} + 1 - \frac{1}{2}\right)\theta \right] \\ &= \frac{\sin(n\theta/2)}{\sin(\theta/2)} \left[\cos\left(\frac{n+1}{2}\theta\right) + i \sin\left(\frac{n+1}{2}\theta\right) \right] \end{aligned}$$

Equating real and imaginary parts both sides, we have

$$C_n = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \left(\cos\left(\frac{n+1}{2}\theta\right) \right)$$

and

$$S_n = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \left(\sin\left(\frac{n+1}{2}\theta\right) \right)$$

Answer: (A)

6. Statement I: $(4 \cos^2 9^\circ - 3)(4 \cos^2 27^\circ - 3) = \tan 9^\circ$.

Statement II: If $x \neq (2n+1)(\pi/2)$ where n is an integer, then

$$4 \cos^2 x - 3 = \frac{\cos 3x}{\cos x}$$

Solution: We know that

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

Therefore

$$x \neq (2n+1)\frac{\pi}{2} \Rightarrow \frac{\cos 3x}{\cos x} = 4 \cos^2 x - 3$$

Hence Statement II is true. Now

$$\begin{aligned} (4 \cos^2 9^\circ - 3)(4 \cos^2 27^\circ - 3) &= \frac{\cos 27^\circ}{\cos 9^\circ} \cdot \frac{\cos 81^\circ}{\cos 27^\circ} \\ &= \frac{\cos 81^\circ}{\cos 9^\circ} = \frac{\sin 9^\circ}{\cos 9^\circ} = \tan 9^\circ \end{aligned}$$

Answer: (A)

7. Statement I: $\sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x} = \cos 2x$

Statement II: $\cos^2 x + \sin^2 x = 1$

Solution: Statement II is obvious.

$$\begin{aligned} & \sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x} \\ &= \sqrt{\sin^4 x + 4(1 - \sin^2 x)} - \sqrt{\cos^4 x + 4(1 - \cos^2 x)} \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{(2 - \sin^2 x)^2} - \sqrt{(2 - \cos^2 x)^2} \\
 &= (2 - \sin^2 x) - (2 - \cos^2 x) \\
 &= \cos^2 x - \sin^2 x = \cos 2x
 \end{aligned}$$

Statement I is true.

Answer: (A)

8. Statement I: $(1 - \cot 23^\circ)(1 - \cot 22^\circ) = 2$

Statement II: If a and b are non-zero real numbers, then $a \sin x \pm b \cos x = \sqrt{a^2 + b^2} \sin(x \pm \alpha)$ where α satisfies the relations

$$\cos \alpha = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \sin \alpha = \frac{b}{\sqrt{a^2 + b^2}}$$

Solution:

$$a \sin x \pm b \cos x = \sqrt{a^2 + b^2} \left[\frac{a}{\sqrt{a^2 + b^2}} \sin x \pm \frac{b}{\sqrt{a^2 + b^2}} \cos x \right]$$

$$\begin{aligned}
 &= \sqrt{a^2 + b^2} [\cos \alpha \sin x \pm \sin \alpha \cos x] \\
 &= \sqrt{a^2 + b^2} \sin(x \pm \alpha)
 \end{aligned}$$

Statement II is true. Now

$$\begin{aligned}
 &(1 - \cot 23^\circ)(1 - \cot 22^\circ) \\
 &= \frac{(\sin 23^\circ - \cos 23^\circ)(\sin 22^\circ - \cos 22^\circ)}{\sin 23^\circ \sin 22^\circ} \\
 &= \frac{\sqrt{2} \sin(23^\circ - 45^\circ) \times \sqrt{2} \sin(22^\circ - 45^\circ)}{\sin 23^\circ \sin 22^\circ} \\
 &= \frac{(-\sqrt{2} \sin 22^\circ)(-\sqrt{2} \sin 23^\circ)}{\sin 23^\circ \sin 22^\circ} = 2
 \end{aligned}$$

Statement I is also true.

Answer: (A)

Integer Answer Type Questions

1. In $\triangle ABC$, $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = k \sin A \sin B \sin C$ where k equals ____.

Solution: $A + B + C = \pi \Rightarrow B + C - A = \pi - 2A$, etc. Therefore

$$\begin{aligned}
 \text{LHS} &= \sin(\pi - 2A) + \sin(\pi - 2B) + \sin(\pi - 2C) \\
 &= \sin 2A + \sin 2B + \sin 2C \\
 &= 4 \sin A \sin B \sin C
 \end{aligned}$$

Answer: 4

2. If $A + B + C = \pi$, then

$$\sin^2 \frac{A}{2} - \sin^2 \frac{C}{2} + \sin^2 \frac{B}{2} = p - q \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

where $p + q$ is ____.

Solution:

$$\begin{aligned}
 &\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} \\
 &= \frac{1 - \cos A}{2} + \frac{1 - \cos B}{2} - \frac{1 - \cos C}{2} \\
 &= \frac{1}{2} - \frac{1}{2}(\cos A + \cos B - \cos C) \\
 &= \frac{1}{2} - \frac{1}{2} \left(4 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} - 1 \right)
 \end{aligned}$$

$$= 1 - 2 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

Therefore $p = 1, q = 2$.

Answer: 3

3. If $\frac{1}{\tan 3\theta + \tan \theta} - \frac{1}{\cot 3\theta + \cot \theta} = \cot k\theta$ then the value of k is ____.

Solution:

$$\begin{aligned}
 &\frac{1}{\tan 3\theta + \tan \theta} - \frac{1}{\cot 3\theta + \cot \theta} \\
 &= \frac{1}{\tan 3\theta + \tan \theta} - \frac{\tan \theta \tan 3\theta}{\tan 3\theta + \tan \theta} \\
 &= \frac{1 - \tan 3\theta \tan \theta}{\tan 3\theta + \tan \theta} = \frac{1}{\tan 4\theta} = \cot 4\theta
 \end{aligned}$$

Answer: 4

4. $(2 \cos A + 1)(2 \cos A - 1) - 2 \cos 2A$ is equal to ____.

Solution: $(2 \cos A + 1)(2 \cos A - 1) - 2 \cos 2A = 4 \cos^2 A - 1 - 2(2 \cos^2 A - 1) = 1$

Answer: 1

5. Let $0 < \alpha, \beta < \pi/2$. If $\tan \alpha = 1/7$ and $\sin \beta = 1/\sqrt{10}$, then value of $\tan(\alpha + 2\beta)$ is ____.

Solution:

$$\sin \beta = \frac{1}{\sqrt{10}} \quad \text{and} \quad 0 < \beta < \frac{\pi}{2} \Rightarrow \cos \beta = \frac{3}{\sqrt{10}}$$

Therefore

$$\tan \beta = \frac{1}{3}$$

$$\tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta} = \frac{2(1/3)}{1 - (1/9)} = \frac{3}{4}$$

Hence

$$\tan(\alpha + 2\beta) = \frac{\tan \alpha + \tan 2\beta}{1 - \tan \alpha \tan 2\beta} = \frac{(1/7) + (3/4)}{1 - (3/28)} = \frac{25}{25} = 1$$

Answer: 1

6. The value of $2 \left[\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8} \right]$ is _____.

Solution:

$$\frac{7\pi}{8} = \pi - \frac{\pi}{8}, \quad \frac{5\pi}{8} = \pi - \frac{3\pi}{8}$$

The given expression is equal to

$$\begin{aligned} & 4 \left[\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} \right] \\ &= 4 \left[\left(\frac{1 + \cos(\pi/4)}{2} \right)^2 + \left(\frac{1 + \cos(3\pi/4)}{2} \right)^2 \right] \\ &= \left(1 + \cos \frac{\pi}{4} \right)^2 + \left(1 - \cos \frac{\pi}{4} \right)^2 \\ &= \left(1 + \frac{1}{\sqrt{2}} \right)^2 + \left(1 - \frac{1}{\sqrt{2}} \right)^2 \\ &= 2 \left(1 + \frac{1}{2} \right) \\ &= 3 \end{aligned}$$

Answer: 3

7. If $\operatorname{cosec} 10^\circ - \sqrt{3} \sec 10^\circ = k$, then k equals _____.

Solution:

$$k = \frac{\cos 10^\circ - \sqrt{3} \sin 10^\circ}{\sin 10^\circ \cos 10^\circ}$$

$$\begin{aligned} &= 2 \frac{[(1/2)\cos 10^\circ - (\sqrt{3}/2)\sin 10^\circ]}{\sin 10^\circ \cos 10^\circ} \\ &= \frac{2 \sin(30^\circ - 10^\circ)}{\sin 10^\circ \cos 10^\circ} \\ &= \frac{4 \sin 20^\circ}{\sin 20^\circ} \\ &= 4 \end{aligned}$$

Answer: 4

8. $\tan \theta + 2 \tan 2\theta + 4 \tan 4\theta - \cot \theta = -k \cot 8\theta$, where the value of k is _____.

Solution:

$$\cot \theta - \tan \theta = \frac{1 - \tan^2 \theta}{\tan \theta} = 2 \frac{(1 - \tan^2 \theta)}{2 \tan \theta} = 2 \cot 2\theta$$

Therefore

$$\cot \theta - \tan \theta = 2 \cot 2\theta$$

$$\cot \theta - 2 \cot 2\theta = \tan \theta$$

$$2 \cot 2\theta - 4 \cot 4\theta = 2 \tan 2\theta$$

$$4 \cot 4\theta - 8 \cot 8\theta = 4 \tan 4\theta$$

Adding we get $\cot \theta - 8 \cot 8\theta = \tan \theta + 2 \tan 2\theta + 4 \tan 4\theta$
Therefore

$$\tan \theta + 2 \tan 2\theta + 4 \tan 4\theta - \cot \theta = -8 \cot 8\theta$$

Answer: 8

9. The value of $\frac{(\cos 3A - \cos 5A)(\sin 9A + \sin 3A)}{(\sin 7A + \sin A)(\cos 5A - \cos 7A)}$ is _____.

Solution: Numerator = $(2 \sin 4A \sin A)(2 \sin 6A \cos 3A)$
Denominator = $(2 \sin 4A \cos 3A)(2 \sin 6A \sin A)$
Therefore quotient = 1.

Answer: 1

10. If $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 = 3$, then $\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3 =$ _____

Solution: It is known that maximum value of $\sin \theta$ is 1.
Therefore

$$\sin \theta_1 + \sin \theta_2 + \sin \theta_3 = 3$$

$$\Rightarrow \sin \theta_1 = \sin \theta_2 = \sin \theta_3 = 1$$

$$\cos \theta_1 = \cos \theta_2 = \cos \theta_3 = 0$$

Answer: 0

SUMMARY

1.1 Domain and ranges of trigonometric functions (Table 1).

Table 1

Function	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1, 1]$
$\cos x$	$\mathbb{R}$	$[-1, 1]$
$\tan x$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$	$\mathbb{R}$
$\operatorname{cosec} x$	$\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$	$\mathbb{R} - (-1, 1)$
$\sec x$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$	$\mathbb{R} - (-1, 1)$
$\cot x$	$\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$	$\mathbb{R}$

1.2 Increasing/decreasing nature of trigonometric functions in different quadrants. Upward $\uparrow$ denotes increasing nature and downward $\downarrow$ denotes decreasing nature (Table 2).

Table 2

Function	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
$\sin x$	$\uparrow$ from 0 to 1	$\downarrow$ from 1 to 0	$\downarrow$ from 0 to -1	$\uparrow$ from -1 to 0
$\cos x$	$\downarrow$ from 1 to 0	$\downarrow$ from 0 to -1	$\uparrow$ from -1 to 0	$\uparrow$ from 0 to 1
$\tan x$	$\uparrow$ from 0 to $+\infty$	$\uparrow$ from $-\infty$ to 0	$\uparrow$ from 0 to $+\infty$	$\uparrow$ from $-\infty$ to 0
$\operatorname{cosec} x$	$\downarrow$ from $+\infty$ to 1	$\uparrow$ from 1 to $+\infty$	$\uparrow$ from $-\infty$ to -1	$\downarrow$ from -1 to $-\infty$
$\sec x$	$\uparrow$ from 1 to $+\infty$	$\uparrow$ from $-\infty$ to -1	$\downarrow$ from -1 to $-\infty$	$\downarrow$ from $+\infty$ to 1
$\cot x$	$\downarrow$ from $+\infty$ to 0	$\downarrow$ from 0 to $-\infty$	$\downarrow$ from $+\infty$ to 0	$\downarrow$ from 0 to $-\infty$

1.3 Ratios of some of the standard angles (Table 3).

Table 3

Angle	Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
	Degree	0°	30°	45°	60°	90°	120°	135°	150°	180°
Functions										
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$\frac{-1}{2}$	$\frac{-1}{\sqrt{2}}$	$\frac{-\sqrt{3}}{2}$	-1	
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined	$-\sqrt{3}$	-1	$\frac{-1}{\sqrt{3}}$	0	
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined	
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined	-2	$-\sqrt{2}$	$\frac{-2}{\sqrt{3}}$	-1	
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$\frac{-1}{\sqrt{3}}$	-1	$-\sqrt{3}$	Not defined	

1.4 Period: Let f be a real valued function defined on a subset E of the real number set $\mathbb{R}$. If there exists positive real number p such that for x and $x + p$ belonging to E , $f(x + p) = f(x)$, then f is said to be a periodic function and p is called a period of f . Among the periods the least one (if it exists) is called the least period of f . If p is a period of f , and n is any positive integer then np is also a period.

1.5 Least periods of trigonometric functions:

- (1) $\sin x: 2\pi$ (2) $\cos x: 2\pi$ (3) $\tan x: \pi$
 (4) $\operatorname{cosec} x: 2\pi$ (5) $\sec x: 2\pi$ (6) $\cot x: \pi$

1.6 Trigonometrical ratios of sum and difference of two and three angles:

- (1) $\sin(x + y) = \sin x \cos y + \cos x \sin y$, $\sin(x - y) = \sin x \cos y - \cos x \sin y$
 (2) $\cos(x + y) = \cos x \cos y - \sin x \sin y$, $\cos(x - y) = \cos x \cos y + \sin x \sin y$
 (3) $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$ (provided $\tan x \tan y \neq 1$)
 $\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$ (provided $\tan x \tan y \neq -1$)

- 1.7** (1) $\sin(x + y + z) = \sin x \cos y \cos z + \cos x \sin y \cos z + \cos x \cos y \sin z - \sin x \sin y \sin z$.
 (2) $\cos(x + y + z) = \cos x \cos y \cos z - \cos x \sin y \sin z - \sin x \cos y \sin z - \sin x \sin y \cos z$.
 (3) $\tan(x + y + z) = \frac{\tan x + \tan y + \tan z - \tan x \tan y \tan z}{1 - 3 \tan x \tan y}$

1.8 Useful formulae:

- (1) $\sin(x + y) \sin(x - y) = \sin^2 x - \sin^2 y$
 (2) $\cos(x + y) \cos(x - y) = \cos^2 x - \sin^2 y$

1.9 Trigonometrical ratios of multiple and submultiple angles:

- (1) $\sin(2x) = 2 \sin x \cos x$, $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$ and
 $\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$

$$(2) \cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x,$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}, \cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$(3) \sin^2 x = \frac{1 - \cos 2x}{2} \text{ and } \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$(4) \tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}, \tan x = \frac{2 \tan x/2}{1 - \tan^2 x/2} \text{ and}$$

$$\tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

$$(5) \sin(3x) = 3 \sin x - 4 \sin^3 x$$

$$(6) \cos(3x) = 4 \cos^3 x - 3 \cos x$$

$$(7) \tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$

1.10 Useful identities:

- (1) $\sin x \sin(60^\circ - x) \sin(60^\circ + x) = \frac{1}{4} \sin(3x)$
 (2) $\cos x \cos(60^\circ - x) \cos(60^\circ + x) = \frac{1}{4} \cos(3x)$
 (3) $\tan x \tan(60^\circ - x) \tan(60^\circ + x) = \tan(3x)$

1.11 Trigonometrical ratios of $7\frac{1}{2}^\circ$, 15° , 18° , $22\frac{1}{2}^\circ$, 36° , 54° , 72° , 75° (Table 4):

1.12 Transformations:

- (1) $\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$
 $\sin C + \sin D = 2 \sin\left(\frac{C + D}{2}\right) \cos\left(\frac{C - D}{2}\right)$
 (2) $\sin(x + y) - \sin(x - y) = 2 \cos x \sin y$
 $\sin C - \sin D = 2 \cos\left(\frac{C + D}{2}\right) \sin\left(\frac{C - D}{2}\right)$
 (3) $\cos(x + y) + \cos(x - y) = 2 \cos x \cos y$
 $\cos C + \cos D = 2 \cos\left(\frac{C + D}{2}\right) \cos\left(\frac{C - D}{2}\right)$

Table 4

Function	$7\frac{1}{2}^\circ$	15°	18°	$22\frac{1}{2}^\circ$	36°	54°	72°	75°
$\sin \theta$	$\frac{\sqrt{8 - 2\sqrt{6} - 2\sqrt{2}}}{4}$	$\frac{\sqrt{3} - 1}{2\sqrt{2}}$	$\frac{\sqrt{5} - 1}{4}$	$\frac{\sqrt{2} - \sqrt{2}}{2}$	$\frac{\sqrt{10 - 2\sqrt{5}}}{4}$	$\frac{\sqrt{5} + 1}{4}$	$\frac{\sqrt{10 + 2\sqrt{5}}}{4}$	$\frac{\sqrt{3} + 1}{2\sqrt{2}}$
$\cos \theta$	$\frac{\sqrt{8 + 2\sqrt{6} + 2\sqrt{2}}}{4}$	$\frac{\sqrt{3} + 1}{2\sqrt{2}}$	$\frac{\sqrt{10 + 2\sqrt{5}}}{4}$	$\frac{\sqrt{2} + \sqrt{2}}{2}$	$\frac{\sqrt{5} + 1}{4}$	$\frac{\sqrt{10 - 2\sqrt{5}}}{4}$	$\frac{\sqrt{5} - 1}{4}$	$\frac{\sqrt{3} - 1}{2\sqrt{2}}$
$\tan \theta$	$(\sqrt{3} - \sqrt{2})(\sqrt{2} - 1)$	$2 - \sqrt{3}$	—	$\sqrt{2} - 1$	—	—	—	$2 + \sqrt{3}$

$$(4) \cos(x+y) - \cos(x-y) = -2 \sin x \sin y$$

$$\cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

1.13 Some of the useful formulae:

It is given that $A + B + C = 180^\circ$ (or π). Then

$$(1) \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$$

$$(2) \sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$(3) \tan A + \tan B + \tan C = \tan A \tan B \tan C \text{ and}$$

$$\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$$

1.14 Sum to n terms of sine and cosine series where the angles are in AP:

$$(1) \sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \cdots + \sin(\alpha + (n-1)\beta) =$$

$$\frac{\sin \left[\frac{2\alpha + (n-1)\beta}{2} \right] \sin \left(\frac{n\beta}{2} \right)}{\sin \left(\frac{\beta}{2} \right)}$$

$$(2) \cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \cdots + \cos(\alpha + (n-1)\beta) =$$

$$\frac{\cos \left[\frac{2\alpha + (n-1)\beta}{2} \right] \sin \left(\frac{n\beta}{2} \right)}{\sin \left(\frac{\beta}{2} \right)}$$

EXERCISES

Single Correct Choice Type Questions

1. The value of $\sin 36^\circ =$

- (A) $\frac{(\sqrt{5}+1)\sqrt{10+2\sqrt{5}}}{8}$ (B) $\frac{\sqrt{5}+1}{4}$
 (C) $\frac{\sqrt{10+2\sqrt{5}}}{8}$ (D) $\frac{(\sqrt{5}-1)\sqrt{10+2\sqrt{5}}}{8}$

2. If $a = \frac{2 \sin \theta}{1 + \cos \theta + \sin \theta}$, then $\frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta}$ is equal to

- (A) $1 - a$ (B) $1 + a$ (C) a (D) $1/a$

3. $\frac{\sec 8\theta - 1}{\sec 4\theta - 1} =$

- (A) $\frac{\tan 8\theta}{\tan 2\theta}$ (B) $\frac{\cot 8\theta}{\cot 2\theta}$
 (C) $\frac{\tan 8\theta}{\tan 4\theta}$ (D) $\frac{\cot 8\theta}{\cot 4\theta}$

4. If $0 < \theta < \pi/2$ and $\sin \theta = 3/5$, then $\tan 2\theta =$

- (A) $\frac{25}{24}$ (B) $\frac{24}{7}$ (C) $\frac{25}{7}$ (D) $\frac{23}{7}$

5. $-\sin^2 48^\circ + \cos^2 12^\circ =$

- (A) $\frac{\sqrt{5}-1}{8}$ (B) $\frac{\sqrt{5}-1}{4}$
 (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{5}+1}{8}$

6. Suppose that $\sin^3 x \sin 3x = \sum_{m=0}^n A_m \cos mx$, where $A_0, A_1, \dots, A_m$ are constants and $A_n \neq 0$, then the value of n is
 (A) 8 (B) 6 (C) 4 (D) 9

7. If a, b, c and d are smallest positive angles in the ascending order such that the sine of each angle is equal to a positive constant λ , then $4\sin(a/2) + 3\sin(b/2) + 2\sin(c/2) + \sin(d/2)$ is equal to

- (A) $2\sqrt{1+\lambda}$ (B) $\sqrt{1+\lambda}$
 (C) $1+\lambda$ (D) $3\sqrt{1+\lambda}$

8. If $\cos(\alpha + \beta) = 4/5$, $\sin(\alpha - \beta) = 5/13$ and $0 < \alpha, \beta < \pi/4$ then $\tan 2\alpha$ is equal to

- (A) $\frac{46}{33}$ (B) $\frac{56}{23}$ (C) $\frac{56}{33}$ (D) $\frac{65}{33}$

9. If $\tan A = (1 - \cos B)/\sin B$, then $\tan 2A$

- (A) $\cot B$ (B) $\tan \frac{B}{2}$ (C) $\cot \frac{B}{2}$ (D) $\tan B$

10. $\cos \frac{\pi}{11} + \cos \frac{3\pi}{11} + \cos \frac{5\pi}{11} + \cos \frac{7\pi}{11} + \cos \frac{9\pi}{11} =$

- (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) -1

11. If $\cos 2B = \cos(A+C)/\cos(A-C)$, then $\tan A, \tan B, \tan C$ are in

- (A) GP (B) HP (C) AP (D) AGP

12. If α is a root of the equation $25\cos^2\theta + 5\cos\theta - 12 = 0$ and $\pi/2 < \alpha < \pi$, then $\sin 2\alpha =$
 (A) $\frac{8}{13}$ (B) $\frac{24}{25}$ (C) $\frac{-24}{25}$ (D) $\frac{-8}{13}$
13. $\sin^2 A + \sin^2(A - B) - 2\sin A \cos B \sin(A - B) =$
 (A) $\cos^2 B$ (B) $\sin^2 B$ (C) $\cos^2 A$ (D) $\sin^2 A$
14. If $\tan \theta = n \sin \alpha \cos \alpha / (1 - n \cos^2 \alpha)$, then $\tan(\theta + \alpha) =$
 (A) $\frac{1}{1-n} \tan \alpha$ (B) $\frac{1}{1-n} \cot \alpha$
 (C) $(1+n) \tan \alpha$ (D) $(1+n) \cot \alpha$
15. Let $z = \cos \theta + i \sin \theta$. Then, at $\theta = 2^\circ$, the value of $\sum_{n=1}^{15} \operatorname{Re}(z^{2n-1})$ is
 (A) $\frac{\sqrt{3}}{2} \operatorname{cosec} 2^\circ$ (B) $\frac{\sqrt{3}}{4} \sec 2^\circ$
 (C) $\frac{\sqrt{3}}{2} \sec 2^\circ$ (D) $\frac{\sqrt{3}}{4} \operatorname{cosec} 2^\circ$
16. If α and β are solutions of the equation $a \tan \theta + b \sec \theta = c$, then $\tan(\alpha + \beta) =$
 (A) $\frac{2ca}{a^2 - c^2}$ (B) $\frac{2ab}{a^2 - b^2}$ (C) $\frac{2bc}{b^2 - c^2}$ (D) $\frac{a^2 + c^2}{a^2 - c^2}$
17. In a $\triangle ABC$, if $p = \sin A \sin B \sin C$ and $q = \cos A \cos B \cos C$, then $\sum \tan B \tan C =$
 (A) $\frac{1+p}{p}$ (B) $\frac{1+p}{q}$ (C) $\frac{1+q}{q}$ (D) $\frac{1+q}{p}$
18. If A and B are acute angles ($A > B$) such that $A + B$ and $A - B$ satisfy the equation $\tan^2 \theta - 4 \tan^2 \theta + 1 = 0$, then
 (A) $A = \frac{\pi}{3}, B = \frac{\pi}{4}$ (B) $A = \frac{\pi}{3}, B = \frac{\pi}{6}$
 (C) $A = \frac{\pi}{4}, B = \frac{\pi}{6}$ (D) $A = \frac{\pi}{3}, B = \frac{\pi}{12}$
19. If $\tan \alpha \tan \beta = a$ and $\alpha + \beta = \pi/6$, then $\tan \alpha$ and $\tan \beta$ are roots of the equation
 (A) $x^2 + \sqrt{3}(1+a)x - a = 0$
 (B) $x^2 - \sqrt{3}(1-a)x + a = 0$
 (C) $\sqrt{3}x^2 - (1-a)x + a\sqrt{3} = 0$
 (D) $\sqrt{3}x^2 + (1+a)x - a\sqrt{3} = 0$
20. If $\tan \alpha = m/(m-1)$ and $\tan \beta = 1/(2m-1)$, then $\alpha - \beta$ is
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$
21. If $\tan \alpha$ and $\tan \beta$ are roots of the equation $x^2 + ax + b = 0$ ($a \neq 0$), then $\tan(\alpha + \beta) =$
 (A) $\frac{a}{b-1}$ (B) $\frac{a}{b+1}$
 (C) $\frac{b}{a-1}$ (D) $\frac{b}{a+1}$
22. If n is a positive integer and

$$\sin \frac{\pi}{2n} + \cos \frac{\pi}{2n} = \frac{\sqrt{n}}{2}$$
 then
 (A) $1 \leq n \leq 8$ (B) $1 \leq n \leq 4$ (C) $n = 6$ (D) $n > 8$
23. If $\cos(\theta - \alpha) = a$ and $\sin(\theta - \beta) = b$, then the value of $\cos^2(\alpha - \beta) + 2ab \sin(\alpha - \beta)$ is
 (A) $4a^2b^2$ (B) $a^2 + b^2$
 (C) $a^2 - b^2$ (D) $a(a+b)$
24. If $\tan \alpha = 3$, then

$$\frac{2 \sin 2\alpha - 3 \cos 2\alpha}{4 \sin 2\alpha + 5 \cos 2\alpha} =$$
 (A) $\frac{9}{4}$ (B) $\frac{-9}{4}$ (C) $\frac{13}{4}$ (D) $\frac{-13}{4}$
25. $\frac{1 + \sin 2\alpha}{\sin \alpha + \cos \alpha} - \frac{1 - \tan^2 \alpha/2}{1 + \tan^2 \alpha/2} =$
 (A) $-\cos \alpha$ (B) $\cos \alpha$ (C) $-\sin \alpha$ (D) $\sin \alpha$
26. $\sin^2 70^\circ \sin^2 50^\circ \sin^2 10^\circ =$
 (A) $\frac{1}{64}$ (B) $\frac{1}{32}$ (C) $\frac{1}{128}$ (D) $\frac{1}{256}$
27. If $0 < \alpha, \beta < \pi/2$, $\sin(\alpha + \beta) = 1$ and $\sin(\alpha - \beta) = 1/2$, then $\tan(\alpha + 2\beta) =$
 (A) $\sqrt{3}$ (B) $-\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{-1}{\sqrt{3}}$
28. If $\tan \alpha/2 = m$, then

$$\frac{1 - 2 \sin^2 \alpha/2}{1 + \sin \alpha} =$$
 (A) $\frac{1+m}{1-m}$ (B) $\frac{1-m}{1+m}$ (C) $\frac{m}{1+m}$ (D) $\frac{-m}{1+m}$
29. If $\tan \alpha + \cot \alpha = a$, then $\tan^4 \alpha + \cot^4 \alpha =$
 (A) $(a^2 - 2)^2 - 2$ (B) $(a^2 + 2)^2 - 2$
 (C) $(a^2 - 2)^2 + 2$ (D) $(a^2 - 2)^2 + 4$

30. If $0 < \alpha < \pi$ such that $\sin \alpha + \cos \alpha = 1/5$, then $\tan \alpha/2 =$
 (A) $-\frac{1}{3}$ (B) $\frac{1}{3}$ (C) 2 (D) -2
31. If $\cos(\alpha + \beta) = 0$, then $\sin(\alpha + 2\beta) =$
 (A) $\sin \alpha$ (B) $\cos \alpha$ (C) $\sin \beta$ (D) $\cos \beta$
32. If $\frac{\sin \beta}{\sin(2\alpha + \beta)} = \frac{n}{m}$ and $|n| < |m|$, then
 (A) $\frac{1 + (\tan \beta / \tan \alpha)}{m + n} = \frac{1 - \tan \alpha \tan \beta}{m - n}$
 (B) $\frac{1 - (\tan \beta / \tan \alpha)}{m + n} = \frac{1 - \tan \alpha \tan \beta}{m - n}$
 (C) $\frac{1 + \tan \alpha \tan \beta}{m + n} = \frac{1 - (\tan \alpha / \tan \beta)}{m - n}$
 (D) $\frac{1 - \tan \alpha \tan \beta}{m + n} = \frac{1 + (\tan \beta / \tan \alpha)}{m - n}$
33. If α, β, γ are in AP, then

$$\frac{\sin \alpha - \sin \gamma}{\cos \gamma - \cos \alpha} =$$

 (A) $\cot \beta$ (B) $\tan \beta$
 (C) $\tan \alpha \cot \beta$ (D) $\tan \gamma \cot \beta$
34. $\frac{\sin 3\alpha \cdot \cos^3 \alpha + \cos 3\alpha \cdot \sin^3 \alpha}{3} =$
 (A) $\frac{1}{2} \sin 4\alpha$ (B) $\frac{1}{4} \sin 4\alpha$
 (C) $\frac{1}{4} \cos 4\alpha$ (D) $4 \cos 4\alpha$
35. $\frac{\sin 2\alpha - \sin 3\alpha + \sin 4\alpha}{\cos 2\alpha - \cos 3\alpha + \cos 4\alpha} =$
 (A) $\tan 2\alpha$ (B) $\cot 2\alpha$
 (C) $\tan 3\alpha$ (D) $\cot 3\alpha$
36. If $\cot x, \cot y, \cot z$ are in AP, then $\cot(y - x), \cot y, \cot(y - z)$ are in
 (A) AP (B) GP (C) HP (D) AGP
37. If $\sin \theta + \cos \theta = \sqrt{7}/2$ and $0 < \theta < \pi/6$, then $\tan(\theta/2) =$
 (A) $\frac{1}{3}(3 - \sqrt{7})$ (B) $\frac{1}{3}(\sqrt{7} - 2)$
 (C) $\sqrt{7} - 2$ (D) $\sqrt{7} + 2$
38. If $\theta = \pi/(2^n + 1)$, the $\cos \theta \cos 2\theta \cos 2^2\theta \cdots \cos 2^{n-1}\theta =$
 (A) 2^n (B) 2^{n+1} (C) $\frac{1}{2^{n+1}}$ (D) $\frac{1}{2^n}$
39. The value of $\tan \theta \cdot \tan(\theta + 60^\circ) + \tan \theta \tan(\theta - 60^\circ) + \tan(\theta - 60^\circ) \tan(\theta + 60^\circ) + 3$ is
 (A) 0 (B) 1 (C) 6 (D) $\frac{1}{3}$
40. The value of $\sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ$ is
 (A) $-\frac{3}{8}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{5}$ (D) $\frac{2}{5}$
41. If

$$\sin \frac{A+B}{2} \cos \frac{A-B}{2} + \sin \frac{C+D}{2} \cos \frac{C-D}{2} = 2$$

 then $\sum \cos(A/2) \cos(B/2) =$
 (A) 0 (B) 3 (C) 2 (D) 4
42. If $450^\circ < \theta < 540^\circ$ and $\sin \theta = 336/625$, then $\sin(\theta/4) =$
 (A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{7}{8}$ (D) $\frac{1}{5\sqrt{2}}$
43. If $A + B + C = \pi$, then

$$\frac{\sin^2 A + \sin^2 B + \sin^2 C}{\cos A \cos B \cos C} =$$

 (A) 2 (B) 4 (C) -2 (D) -4
44. If $A + B + C = \pi$, then

$$\sin \frac{A}{2} + \sin \frac{B}{2} + \sin \frac{C}{2}$$

$$= 1 + k \sin\left(\frac{\pi - A}{4}\right) \sin\left(\frac{\pi - B}{4}\right) \sin\left(\frac{\pi - C}{4}\right)$$

 where k is equal to
 (A) 2 (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{4}$
45. If $A + B + C = 2s$, then $\sin(s - A) \sin(s - B) + \sin s \cdot \sin(s - C) =$
 (A) $\sin A \cos B$ (B) $\sin B \cos A$
 (C) $\cos A \cos B$ (D) $\sin A \sin B$
46. In $\triangle ABC$, if $\cos 3A + \cos 3B + \cos 3C = 1$, then one angle of the triangle must be
 (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) π (D) $\frac{4\pi}{3}$
47. If $\cos \theta = (a \cos \alpha + b)/(a + b \cos \alpha)$, then $\tan^2(\theta/2) =$
 (A) $\frac{a-b}{a+b} \tan^2 \frac{\alpha}{2}$ (B) $\frac{a+b}{a-b} \cos^2 \frac{\alpha}{2}$
 (C) $\frac{a-b}{a+b} \sin^2 \frac{\alpha}{2}$ (D) $\frac{a-b}{a+b} \cot^2 \frac{\alpha}{2}$

48. Minimum value of $4x^2 - 4x|\sin \theta| - \cos^2 \theta$ is
 (A) -2 (B) -1 (C) 0 (D) $-\frac{1}{2}$

49. If $3\theta, 9\theta$ and 27θ are not odd multiples of $\pi/2$, then

$$2\left(\frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta}\right) =$$

- (A) $\tan 27\theta - \tan \theta$ (B) $\tan \theta - \tan 27\theta$
 (C) $\cot 27\theta - \cot \theta$ (D) $\cot \theta - \cot 27\theta$

50. If

$$\frac{\cos x}{a} = \frac{\cos(x+\theta)}{b} = \frac{\cos(x+2\theta)}{c} = \frac{\cos(x+3\theta)}{d}$$

then $(a+c)/(b+d)$ is equal to

- (A) $\frac{c}{d}$ (B) $\frac{b}{c}$ (C) $\frac{d}{a}$ (D) $\frac{a}{d}$

51. If $\sin \alpha = -3/5$ and $\pi < \alpha < 3\pi/2$, then $\cos(\alpha/2) =$

- (A) $\frac{1}{5}$ (B) $-\frac{1}{\sqrt{10}}$ (C) $-\frac{1}{5}$ (D) $\frac{1}{\sqrt{10}}$

52. If $a_n = \sin(n\theta) \cdot \sec^n \theta$ and $b_n = \cos(n\theta) \sec^n \theta \neq 1$, then

$$\frac{b_n - b_{n-1}}{a_{n-1}} + \frac{a_n}{nb_n} =$$

- (A) 0 (B) $\tan \theta$
 (C) $-\tan \theta + \frac{1}{n} \tan(n\theta)$ (D) $\tan \theta + \frac{1}{n} \tan(n\theta)$

53. If $\sin(y+z-x), \sin(z+x-y), \sin(x+y-z)$ are in AP, then $\tan x, \tan y, \tan z$ are in

- (A) AP (B) GP (C) HP (D) AGP

54. If $\sec(\theta - \alpha), \sec \theta, \sec(\theta + \alpha)$ are in AP, then the ratio $\cos \theta : \cos(\alpha/2)$ can be

- (A) 2:1 (B) 1:2 (C) $1:\sqrt{2}$ (D) $\sqrt{2}:1$

55. If

$$\tan B = \frac{2 \sin A \sin C}{\sin(A+C)}$$

then $\cot A, \cot B, \cot C$ are in

- (A) AP (B) GP (C) HP (D) AGP

56. If

$$\tan x = \frac{(1+m)\tan \alpha}{1-m\tan^2 \alpha} \quad \text{and} \quad \tan y = \frac{(1-m)\tan \alpha}{1+m\tan^2 \alpha}$$

then

$$\tan\left(\frac{x-y}{2}\right) \cot\left(\frac{x+y}{2}\right) =$$

- (A) $2m$ (B) $1+m^2$ (C) m^2-1 (D) m

Multiple Correct Choice Type Questions

1. Let

$$x = \left(\frac{\cos A + \cos B}{\sin A - \sin B}\right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B}\right)^n$$

Then

- (A) $x = 0$ if n is an odd positive integer
 (B) $x = \tan^n(A-B)/2$ if n is an even positive integer
 (C) $x = 2 \cot^n(A-B)/2$ if n is an even positive integer
 (D) $x = 0$ if n is an even positive integer

2. Which of the following statements are true?

- (A) If $\cos x + \sin x = \sqrt{2} \cos x$, then $\cos x - \sin x = \sqrt{2} \sin x$
 (B) If $3 \sin \theta + 5 \cos \theta = 5$, then $5 \sin \theta - 3 \cos \theta = \pm 3$
 (C) If $A = \sin^2 \theta + \cos^4 \theta$, then $\frac{3}{4} \leq A \leq 1$
 (D) If $(1 + \sin A)(1 + \sin B)(1 + \sin C) = (1 - \sin A)(1 - \sin B)(1 - \sin C)$ then each is equal to ± 1

3. If $\sin \theta = (m^2 - n^2)/(m^2 + n^2)$, then

- (A) $\tan \theta = \frac{m^2 - n^2}{2mn}$ (B) $\operatorname{cosec} \theta = \frac{m^2 + n^2}{m^2 - n^2}$
 (C) $\sec \theta = \frac{m^2 + n^2}{2mn}$ (D) $\cot \theta = \frac{2mn}{m^2 + n^2}$

4. Which of the following are true?

- (A) $\tan 20^\circ + \tan 72^\circ + \tan 88^\circ = \tan 20^\circ \cdot \tan 72^\circ \cdot \tan 88^\circ$
 (B) $\tan 51^\circ + \tan 62^\circ + \tan 67^\circ = \tan 51^\circ \cdot \tan 62^\circ \cdot \tan 67^\circ$
 (C) $\cot 27^\circ + \cot 32^\circ + \cot 31^\circ = \cot 27^\circ \cdot \cot 32^\circ \cdot \cot 31^\circ$
 (D) $\sqrt{3} + \tan 40^\circ + \tan 80^\circ = \sqrt{3} \tan 40^\circ \tan 80^\circ$

5. Which of the following are true?

- (A) $\cos 10^\circ \cdot \cos 30^\circ \cdot \cos 50^\circ \cdot \cos 70^\circ = \frac{3}{16}$
 (B) $\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ = \frac{1}{16}$
 (C) $\tan 20^\circ \cdot \tan 40^\circ \cdot \tan 80^\circ = \sqrt{3}$
 (D) $\cos 55^\circ + \cos 65^\circ + \cos 175^\circ = 0$

6. Which of the following are true?
- (A) If $\tan(\alpha + \theta) = n \tan(\alpha - \theta)$, then $\sin 2\theta / \sin 2\alpha = (n-1)/(n+1)$
 (B) If $\sin \theta = n \sin(\theta + 2\alpha)$, then $(1-n)\tan(\theta + \alpha) = (1+n)\tan \alpha$
 (C) If $A + B + C = \pi$ and $\sin[A + (C/2)] = n \sin(C/2)$, then $\tan(A/2) \cdot \tan(B/2) = (n-1)/(n+1)$
 (D) If $A + B + C = \pi$, then $\tan(A/2) + \tan(B/2) + \tan(C/2) = 1$
7. Which of the following are (is) irrational?
- (A) $\sin 15^\circ$ (B) $\cos 15^\circ$
 (C) $\sin 15^\circ \cdot \cos 15^\circ$ (D) $\sin 15^\circ \cdot \cos 75^\circ$
8. If $p = \sin(\alpha - \beta)\sin(\gamma - \delta)$, $q = \sin(\beta - \gamma)\sin(\alpha - \delta)$ and $r = \sin(\gamma - \alpha)\sin(\beta - \delta)$, then
- (A) $p + q + r = 0$ (B) $p + q - r = 0$
 (C) $q + r - p = 0$ (D) $p^3 + q^3 + r^3 - 3pqr = 0$
9. If $0 \leq \theta \leq \pi/2$, $x = \alpha \cos \theta + \beta \sin \theta$, $y = \alpha \sin \theta - \beta \cos \theta$ and $x^2 + 4xy + y^2 = a\alpha^2 + b\beta^2$, then
- (A) $a = -1, b = 3$ (B) $\theta = \frac{\pi}{4}$
 (C) $a = 3, b = -1$ (D) $\theta = \frac{\pi}{3}$
10. If $\cos^3 x \cdot \sin 2x = \sum_{m=1}^n a_m \sin(mx)$ is an identity in x , then
- (A) $a_3 = 3/8, a_2 = 0$ (B) $n = 6, a_1 = 1/2$
 (C) $n = 5, a_1 = 1/4$ (D) $\sum_{m=1}^n a_m = 3/4$
11. If $7 \cos x - 24 \sin x = \lambda \cos(x + \theta)$ and $0 < \theta < \pi/2$, then
- (A) $\lambda = 25$ (B) $\cos \theta = 7/25$
 (C) $\sin \theta = 24/25$ (D) $\lambda = 15$
12. If $1 + \cos(x - y) = 0$, then
- (A) $\cos x - \cos y = 0$ (B) $\cos x + \cos y = 0$
 (C) $\sin x + \sin y = 0$ (D) $\cos x + \sin y = 1$
13. If $y = \sin x \cdot \sin(60^\circ - x) \sin(60^\circ + x)$, then
- (A) Minimum value of y is $-1/4$
 (B) Maximum value of y is 1
 (C) Minimum value of y is -1
 (D) $y < 1$
14. If $(\cos^2 x + \sec^2 x)(1 + \tan^2 2y)(3 + \sin 3z) = 4$, then
- (A) x can be a multiple of π
 (B) $x \neq 2n\pi, n \in \mathbb{Z}$
 (C) $z = (4n-1)\frac{\pi}{6}, n \in \mathbb{Z}$
 (D) $y = \frac{n\pi}{2}$
15. If $A + B + C = \pi$, which of the following are true?
- (A) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$
 (B) $\sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C$
 (C) $\cos A + \cos B + \cos C = 1 + 4 \sin(A/2) \sin(B/2) \sin(C/2)$
 (D) $\cot B \cot C + \cot C \cot A + \cot A \cot B = 1$
16. In a triangle ABC , which of the following are true?
- (A) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
 (B) $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1$
 (C) $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$
 (D) $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with *one or more* statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are (A) $\rightarrow$ (p), (s); (B) $\rightarrow$ (q), (s), (t); (C) $\rightarrow$ (r); (D) $\rightarrow$ (r), (t); that is if the

matches are (A) $\rightarrow$ (p) and (s); (B) $\rightarrow$ (q), (s) and (t); (C) $\rightarrow$ (r); and (D) $\rightarrow$ (r), (t) then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

1. Match the items in Column I with those in Column II.

Column I	Column II
(A) $\tan\left(45^\circ + \frac{A}{2}\right)$	(p) $\sec A + \tan A$
(B) $\sin^2\left(\frac{\pi}{8} + \frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{A}{2}\right)$	(q) $\frac{1}{\sqrt{2}} \sin A$
(C) $(\cos A + \cos B)^2 + (\sin A + \sin B)^2$	(r) $4\cos^2\left(\frac{A-B}{2}\right)$
(D) $(\cos A - \cos B)^2 + (\sin A - \sin B)^2$	(s) $\sqrt{\frac{1+\sin A}{1-\sin A}}$
	(t) $4\sin^2\left(\frac{A-B}{2}\right)$

2. Match the items in Column I with those in Column II.

Column I	Column II
(A) $\cos^2\theta + \cos^2(\theta + 120^\circ) + \cos^2(\theta - 120^\circ)$	(p) $\frac{3}{2}$
(B) $\cos^4\frac{\pi}{8} + \cos^4\frac{3\pi}{8} + \cos^4\frac{5\pi}{8} + \cos^4\frac{7\pi}{8}$	(q) 2
(C) $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8}$	(r) $\frac{1}{16}$
(D) $\sin\frac{\pi}{10} \cdot \sin\frac{13\pi}{10}$	(s) $-\frac{1}{4}$
	(t) $\frac{1}{4}$

3. Assuming that $A + B + C = \pi$, match the items in Column I with those in Column II.

Column I	Column II
(A) $\cos^2 A + \cos^2 B + \cos^2 C$	(p) $4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
(B) $\cos A + \cos B + \cos C - 1$	(q) $4\sin\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$
(C) $\sin A - \sin B + \sin C$	(r) $1 - 2\cos A \cos B \cos C$
(D) $\sin A + \sin B + \sin C$	(s) $4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

4. Let $\theta = 22(1/2)^\circ$. Match the items in Column I with those in Column II.

Column I	Column II
(A) $\sin \theta$	(p) $\sqrt{2} + 1$
(B) $\cos \theta$	(q) $\sqrt{2} - 1$
(C) $\tan \theta$	(r) $\frac{1}{2}\sqrt{2+\sqrt{2}}$
(D) $\cot \theta$	(s) $\frac{1}{2}\sqrt{2-\sqrt{2}}$
	(t) $\frac{1}{2}(\sqrt{2} + 1)$

5. Match the items in Column I with those in Column II.

Column I	Column II
(A) If $\tan(x-y)/2, \tan z, \tan(x+y)/2$ are in GP, then $\cos x$ is equal to	(p) $\cos 2\theta$
(B) If $\sin x, \sin \theta, \cos x$ are in GP, then $2\cos^2[(\pi/4) + x]$ is equal to	(q) $\frac{\sqrt{3}\sin\theta}{2+\cos\theta}$
(C) If $\sec(\theta-x), \sec \theta, \sec(\theta+x)$ are in AP, then $\cos \theta$ is equal to	(r) $\sqrt{2}\cos\frac{x}{2}$
(D) If $\cos \theta = (2\cos x - 1)/(2 - \cos x)$, then $\sin x$ is equal to	(s) $\frac{\sqrt{2}}{2}\sin x$
	(t) $\cos y \cos 2z$

6. Match the items in Column I with those in Column II.

Column I	Column II
(A) $\frac{\tan(A-B) + \tan B}{1 - \tan(A-B)\tan B}$	(p) $\tan A$
(B) $\frac{\tan 4A - \tan 3A}{1 - \tan 3A \cdot \tan 4A}$	(q) $\cot A$
(C) $\sin 2A(1 + \cot 2A \cdot \cot A)$	(r) $\tan 2A$
(D) $\cos 2A(\tan 2A + \cos A \cdot \operatorname{cosec} A)$	(s) $\cot 2A$

Comprehension-Type Questions

1. Passage: The equation $a \cos \theta + b \sin \theta = c$ can be solved when $|c| \leq \sqrt{a^2 + b^2}$. Based on this answer the following questions (i), (ii) and (iii).

- (i) If $0 \leq x \leq \pi$ and $\sin x + \cos x = 1$, then the number of values of x is
(A) 1 (B) 2 (C) 4 (D) 0
- (ii) The maximum and minimum values of $3 \cos \theta + 4 \sin \theta - 5$ are respectively (θ is real)
(A) 5, -5 (B) 4, 3 (C) 0, -10 (D) 4, -3
- (iii) The maximum and minimum values of $5 \sin \theta + 12 \cos \theta + 13$ are respectively (θ is real)
(A) 13, 0 (B) 1, -1 (C) 26, 0 (D) 26, 13

2. Passage: If α , β and γ are roots of the equation $ax^3 + bx^2 + cx + d = 0$ and $a \neq 0$, then $\alpha + \beta + \gamma = -b/a$, $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$ and $\alpha\beta\gamma = -d/a$. It is given that $\cos \pi/7$, $\cos 3\pi/7$ and $\cos 5\pi/7$ are roots of $8x^3 - 4x^2 - 4x + 1 = 0$.

Answer questions (i), (ii) and (iii).

- (i) The value of $\sec(\pi/7) + \sec(3\pi/7) + \sec(5\pi/7)$ is
(A) 4 (B) 2 (C) 8 (D) 6
- (ii) The value of $\sin(\pi/14) \sin(3\pi/14) \sin(5\pi/14)$ is
(A) $\frac{\sqrt{7}}{8}$ (B) $\frac{1}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{1}{4}$
- (iii) The value of $\cos(\pi/14) \cos(3\pi/14) \cos(5\pi/14)$ is
(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{7}}{8}$

3. Passage: To eliminate a parameter, we need two equations involving the parameter. For example, if $x = r \cos \theta$ and $y = r \sin \theta$, then by squaring and adding, we have $x^2 + y^2 = r^2$. This shows that θ is eliminated from the given equations.

Answer the following questions (i), (ii) and (iii).

- (i) If $\tan \theta + \sin \theta = a$ and $\tan \theta - \sin \theta = b$, then after eliminating θ , $(a^2 - b^2)^2$ is equal to
(A) $4\sqrt{ab}$ (B) $4ab$ (C) $16\sqrt{ab}$ (D) $16ab$
- (ii) Eliminating θ from the equations $x = \cot \theta + \tan \theta$ and $y = \sec \theta - \cos \theta$, we have
(A) $xy(x+y) = 1$ (B) $(x^2y)^{2/3} - (xy^2)^{2/3} = 1$
(C) $x^{2/3} - y^{2/3} = 1$ (D) $xy = 1$
- (iii) Eliminating θ from the equations $x/a = \cos \theta + \cos 2\theta$ and $y/b = \sin \theta + \sin 2\theta$, which of the following relations between x and y can be obtained?

- (A) $\frac{2x}{a} = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 3 \right)$
(B) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2x}{a} + 3$
(C) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{x}{a} + 3$
(D) $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 3 \right) = 2 \left(\frac{x}{a} + \frac{y}{b} \right)$

Assertion-Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
(B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
(C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true

1. Statement I: If $x \cos \alpha + y \sin \alpha = x \cos \beta + y \sin \beta = 2a$ and $2 \sin \alpha/2 \sin \beta/2 = 1$, then $y^2 = 4a(a - x)$

Statement II: If x_1 and x_2 are roots of the quadratic equation $ax^2 + bx + c = 0$, then $x_1 + x_2 = -b/a$ and $x_1x_2 = c/a$

2. Statement I: If $x = \cot \theta + \tan \theta$ and $y = \csc \theta - \sin \theta$, then $(x^2y)^{2/3} - (xy^2)^{2/3} = 1$

Statement II: The conclusion of Statement I is that $\csc^2 \theta - \cot^2 \theta = 1$

3. Statement I: For all values of θ , $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) = -1$

Statement II: $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

4. Statement I: If $\tan(\theta/2) = m$, then

$$\frac{1 - 2 \sin^2(\theta/2)}{1 + \sin \theta} = \frac{1 + m}{1 - m}$$

Statement II: $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$ and $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

5. Statement I: If

$$\frac{\sin 3\alpha}{\sin \alpha} = \frac{11}{25}$$

then $\tan(\alpha/2)$ has four values.

Statement II: $\sin \theta = \frac{2 \tan(\theta/2)}{1 + \tan^2(\theta/2)}$

6. Statement I: If $0 < \alpha < \pi/2$ and $\cot \alpha = 3$, then the value of $\tan(\alpha/2)$ is $1/(3 + \sqrt{10})$.

Statement II: $\tan \frac{\alpha}{2} = \frac{\sin \alpha}{1 + \cos \alpha}$

7. Statement I: If

$$\tan A = \frac{\sqrt{3}}{4 - \sqrt{3}} \quad \text{and} \quad \tan B = \frac{\sqrt{3}}{4 + \sqrt{3}}$$

then $\tan(A - B) = 0.355$.

Statement II: $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

8. Statement I: If $\alpha + \beta + \gamma = \pi/2$, then $\sum \tan \beta \tan \gamma = 1$.

Statement II: If $A + B + C = \pi$, then $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

9. Statement I: If $\alpha + \beta + \gamma = \pi/2$, then

$$\frac{\sin 2\alpha + \sin 2\beta + \sin 2\gamma}{\sin 2\alpha + \sin 2\beta - \sin 2\gamma} = \cot \alpha \cot \beta$$

Statement II: If $A + B + C = \pi$, then $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$.

10. Statement I: If $A + B + C = \pi$, then

$$\begin{aligned} \cos \frac{A}{2} + \cos \frac{B}{2} + \cos \frac{C}{2} \\ = 4 \cos \left(\frac{\pi - A}{4} \right) \cos \left(\frac{\pi - B}{4} \right) \cos \left(\frac{\pi - C}{4} \right) \end{aligned}$$

Statement II: $\cos(\alpha + \beta) \cos(\alpha - \beta) = \cos^2 \alpha - \cos^2 \beta$ for all α and β .

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. $\tan 12^\circ \tan 24^\circ \tan 48^\circ \tan 84^\circ$ is equal to ____.

2. If $\sin 18^\circ$ and $\cos 36^\circ$ are roots of the equation $ax^2 - 2\sqrt{5}x + c = 0$, then the value of $a + c$ is ____.

3. $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ$ is equal to ____.

4. $\frac{\sin 3\alpha}{\sin \alpha} + \frac{\cos 3\alpha}{\cos \alpha} = k \cos 2\alpha$, where k is equal to ____.

5. $(1 - 2 \sin \theta)(1 + 2 \sin \theta)(1 + 2 \cos 2\theta) = 1 + k \cos 4\theta$, where k is equal to ____.

6. If $2 \tan(A + B) = 3 \tan A$, then $\sin(2A + B) = k \sin B$, where k is equal to ____.

7. If $3 \cos x = 2 \cos(x - 2y)$, then $\cot(x - y) \cot y$ is equal to ____.

8. If $\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \dots + \sin 178^\circ = n \cot 1^\circ$, then n is equal to ____.

9. Let $0 < \alpha_j < \pi/2$ for $j = 1, 2, 3, \dots, 8$ and $\cot \alpha_1 \cdot \cot \alpha_2 \cdots \cot \alpha_8 = 1$. If M is the maximum value of $\cot \alpha_1 \cot \alpha_2 \cdots \cot \alpha_8$, then the value of $(32)M$ is ____.

10. The value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$ is ____.

11. If $\sin x + \sin y + \sin z = 3$, then $|\cos 2x + \cos 2y + \cos 2z|$ is equal to ____.

12. $(\sin \alpha + \operatorname{cosec} \alpha)^2 + (\cos \alpha + \sec \alpha)^2 - (\tan^2 \alpha + \cot^2 \alpha)$ is equal to ____.

13. If $\tan^2 \theta = 1 - e^2$, then $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta = (k - e^2)^{3/2}$, where k is ____.

14. If θ_j ($j = 1, 2, 3, 4$) satisfy the equation $3 \tan 3\theta = \tan(45^\circ + \theta)$, then the value of $\tan \theta_1 + \tan \theta_2 + \tan \theta_3 + \tan \theta_4$ is ____.

15. For $n > 2$, the value of

$$\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots + \sin(2n - 1) \frac{\pi}{n}$$

is ____.

ANSWERS

Single Correct Choice Type Questions

- | | |
|---------|---------|
| 1. (D) | 29. (A) |
| 2. (C) | 30. (C) |
| 3. (A) | 31. (A) |
| 4. (B) | 32. (A) |
| 5. (D) | 33. (A) |
| 6. (B) | 34. (B) |
| 7. (A) | 35. (C) |
| 8. (C) | 36. (A) |
| 9. (D) | 37. (B) |
| 10. (C) | 38. (D) |
| 11. (A) | 39. (A) |
| 12. (C) | 40. (A) |
| 13. (B) | 41. (B) |
| 14. (A) | 42. (B) |
| 15. (D) | 43. (A) |
| 16. (A) | 44. (C) |
| 17. (C) | 45. (D) |
| 18. (C) | 46. (B) |
| 19. (C) | 47. (A) |
| 20. (A) | 48. (B) |
| 21. (A) | 49. (A) |
| 22. (A) | 50. (B) |
| 23. (B) | 51. (B) |
| 24. (B) | 52. (C) |
| 25. (D) | 53. (A) |
| 26. (A) | 54. (D) |
| 27. (B) | 55. (A) |
| 28. (B) | 56. (D) |

Multiple Correct Choice Type Questions

- | | |
|-----------------------|------------------------|
| 1. (A), (C) | 9. (B), (C) |
| 2. (A), (B), (C) | 10. (A), (C), (D) |
| 3. (A), (B), (C) | 11. (A), (B), (C) |
| 4. (A), (B), (C), (D) | 12. (B), (C) |
| 5. (A), (B), (C), (D) | 13. (A), (D) |
| 6. (A), (B), (C) | 14. (A), (C), (D) |
| 7. (A), (B), (D) | 15. (A), (B), (C), (D) |
| 8. (A), (D) | 16. (A), (B), (C), (D) |

Matrix-Match Type Questions

- | | |
|--|---|
| 1. $(A) \rightarrow (p), (s), (B) \rightarrow (q), (C) \rightarrow (r), (D) \rightarrow (t)$ | 4. $(A) \rightarrow (s), (B) \rightarrow (r), (C) \rightarrow (q), (D) \rightarrow (p)$ |
| 2. $(A) \rightarrow (p), (B) \rightarrow (p), (C) \rightarrow (p), (D) \rightarrow (s)$ | 5. $(A) \rightarrow (t), (B) \rightarrow (p), (C) \rightarrow (r), (D) \rightarrow (q)$ |
| 3. $(A) \rightarrow (r), (B) \rightarrow (s), (C) \rightarrow (q), (D) \rightarrow (p)$ | 6. $(A) \rightarrow (p), (B) \rightarrow (p), (C) \rightarrow (q), (D) \rightarrow (q)$ |

Comprehension-Type Questions

- | | |
|---------------------------------|---------------------------------|
| 1. (i) (B); (ii) (C); (iii) (C) | 3. (i) (D); (ii) (B); (iii) (A) |
| 2. (i) (A); (ii) (B); (iii) (D) | |

Assertion–Reasoning Type Questions

- | | |
|--------|---------|
| 1. (A) | 6. (A) |
| 2. (A) | 7. (D) |
| 3. (A) | 8. (B) |
| 4. (D) | 9. (B) |
| 5. (A) | 10. (C) |

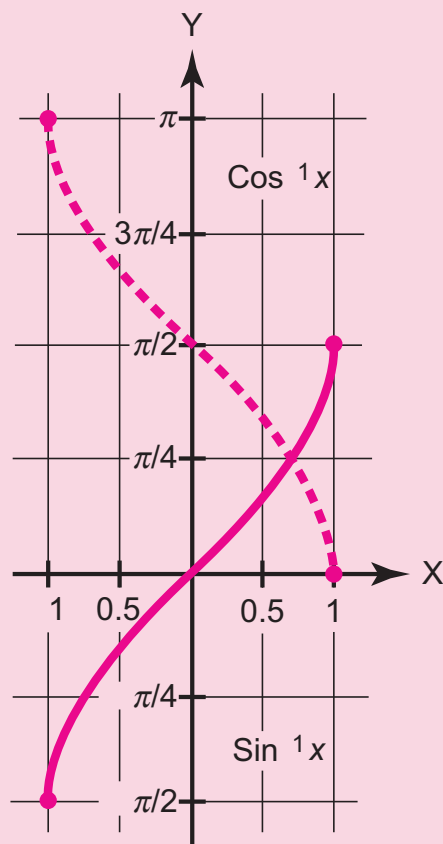
Integer Answer Type Questions

- | | |
|------|-------|
| 1. 1 | 9. 2 |
| 2. 5 | 10. 2 |
| 3. 1 | 11. 3 |
| 4. 4 | 12. 7 |
| 5. 2 | 13. 2 |
| 6. 5 | 14. 0 |
| 7. 5 | 15. 0 |
| 8. 1 | |

Inverse Trigonometric Functions

2

Inverse Trigonometric Functions



Contents

- 2.1 Domains and Ranges of Inverse Trigonometric Functions
- 2.2 Properties of Inverse Trigonometric Functions

Worked-Out Problems
Summary
Exercises
Answers

The **inverse trigonometric functions** (or **cyclometric functions**) are the inverse functions of the trigonometric functions, though they do not meet the official definition for inverse functions as their ranges are subsets of the domains of the original functions. Since none of the six trigonometric functions are one-to-one (by failing the horizontal line test), they must be restricted in order to have inverse functions.

Let us recall that a function f is a bijection (injection and surjection) if and only if there is a function g such that $f \circ g$ and $g \circ f$ are both identity functions. In this case, g is unique and is denoted by f^{-1} . The trigonometric functions are not bijections, since each of them is of period 2π and hence not injective. For example, $\sin 30^\circ = \sin(2\pi + 30^\circ)$ and therefore the sine function is not an injection. Neither, it is surjective, since we cannot find a real number x such that $\sin x = 2$. However, by restricting the natural domain and codomain of the sine function, we can arrive at a bijection.

In this chapter, we discuss various properties of the inverses of certain trigonometric functions by first determining the appropriate domains and codomains of these functions.

2.1 | Domains and Ranges of Inverse Trigonometric Functions

We have first defined the sine function on the interval $[0, 2\pi)$ and later extended it to all real numbers x by defining

$$\sin x = \sin \theta$$

where $x = 2n\pi + \theta$, $0 \leq \theta < 2\pi$ (here n is the integral part $x/2\pi$). However, the sine function is not an injection on $[0, 2\pi)$, since $\sin 0 = \sin \pi$ and $\sin \pi/4 = \sin 3\pi/4$. Instead of interval $[0, 2\pi)$, if we consider the interval $[-\pi/2, \pi/2]$, then the sine function is one-to-one on this interval. Also, the range of sine function is the interval $[-1, 1]$. Therefore, the function

$$f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

defined by $f(x) = \sin x$ for all $x \in [-\pi/2, \pi/2]$, is a bijection and hence there exists a unique function

$$g: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

such that $f \circ g$ is the identity on $[-1, 1]$ and $g \circ f$ is the identity on $[-\pi/2, \pi/2]$; that is,

$$\sin g(y) = y$$

for all $-1 \leq y \leq 1$ and

$$g(\sin x) = x \text{ for all } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

DEFINITION 2.1 The unique function $g: [-1, 1] \rightarrow [-\pi/2, \pi/2]$ such that

$$g(\sin x) = x \text{ and } \sin(g(y)) = y$$

for all $-\pi/2 \leq x \leq \pi/2$ and $-1 \leq y \leq 1$ is called the *inverse sine function* and is denoted by Sin^{-1} . Therefore, Sin^{-1} is a function whose domain is $[-1, 1]$ and codomain is $[-\pi/2, \pi/2]$ and it satisfies the relations

$$\text{Sin}^{-1}(\sin x) = x \text{ and } \sin(\text{Sin}^{-1}y) = y$$

for all $-\pi/2 \leq x \leq \pi/2$ and $-1 \leq y \leq 1$.

Note that, in place of $[-\pi/2, \pi/2]$ if we take $[\pi/2, 3\pi/2]$ (or $[(2n-1)\pi/2, (2n+1)\pi/2]$ for any integer n), then the function

$$\sin: \left[\frac{(2n-1)\pi}{2}, \frac{(2n+1)\pi}{2}\right] \rightarrow [-1, 1]$$

is a bijection and hence has inverse whose domain is $[-1, 1]$ and range is $[(2n-1)\pi/2, (2n+1)\pi/2]$. This can also be called the *inverse of the sine function* (restricted to $[(2n-1)\pi/2, (2n+1)\pi/2]$). However, if we consider the sine function restricted to $[-\pi/2, \pi/2]$ then its inverse

$$\text{Sin}^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

will be usually considered the inverse of the sine function. Also, for any $-1 \leq y \leq 1$, $\text{Sin}^{-1}y$ in $[-\pi/2, \pi/2]$ is called the *principal value* of $\text{Sin}^{-1}y$ and is often denoted by $\arcsin y$. The principal value of $\text{Sin}^{-1}y$ will be denoted by $\text{Sin}^{-1}y$ (with a capital letter S).

QUICK LOOK 1

1. $\frac{-\pi}{2} \leq \sin^{-1} y \leq \frac{\pi}{2}$ for all $-1 \leq y \leq 1$
2. $-1 \leq \sin x \leq 1$ for all $\frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$
3. $\sin(\sin^{-1} y) = y$ for all $-1 \leq y \leq 1$
4. $\sin^{-1}(\sin x) = x$ for all $\frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$
5. $\sin^{-1} y = 0 \Leftrightarrow y = 0$
6. $\sin^{-1} y \in \left[\frac{-\pi}{2}, 0 \right) \Leftrightarrow y \in [-1, 0)$
7. $\sin^{-1} y \in \left(0, \frac{\pi}{2} \right] \Leftrightarrow y \in (0, 1]$
8. $\sin^{-1} y = x \Leftrightarrow \sin x = y$

The discussion made in Quick Look 1 can be extended to other trigonometric functions also. The restrictions of the functions cosine, tangent, cosecant, secant and cotangent restricted to $[0, \pi]$, $(-\pi/2, \pi/2)$, $[-\pi/2, \pi/2] - \{0\}$, $[0, \pi] - \{\pi/2\}$ and $(0, \pi)$, respectively, are injections and hence can be considered as bijections onto their respective ranges. Therefore, these functions have inverses which are denoted by $\cos^{-1}$, $\tan^{-1}$, $\csc^{-1}$, $\sec^{-1}$ and $\cot^{-1}$, respectively. The inverse trigonometric functions are also called *inverse circular functions*. The domains and ranges of the inverse trigonometric functions are given in Table 2.1.

Table 2.1 Domains and ranges of the inverse trigonometric functions

Function	Domain	Range
$\sin^{-1}$	$[-1, 1]$	$\left[\frac{-\pi}{2}, \frac{\pi}{2} \right]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$	$\mathbb{R}$	$\left(\frac{-\pi}{2}, \frac{\pi}{2} \right)$
$\cot^{-1}$	$\mathbb{R}$	$(0, \pi)$
$\csc^{-1}$	$(-\infty, -1] \cup [1, \infty)$	$\left[\frac{-\pi}{2}, 0 \right) \cup \left(0, \frac{\pi}{2} \right]$
$\sec^{-1}$	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{-\pi}{2} \right) \cup \left(\frac{\pi}{2}, \pi \right]$

The graphs of inverse trigonometric functions are illustrated in Figure 2.1 by taking the domains on the X-axis and the ranges on the Y-axis.

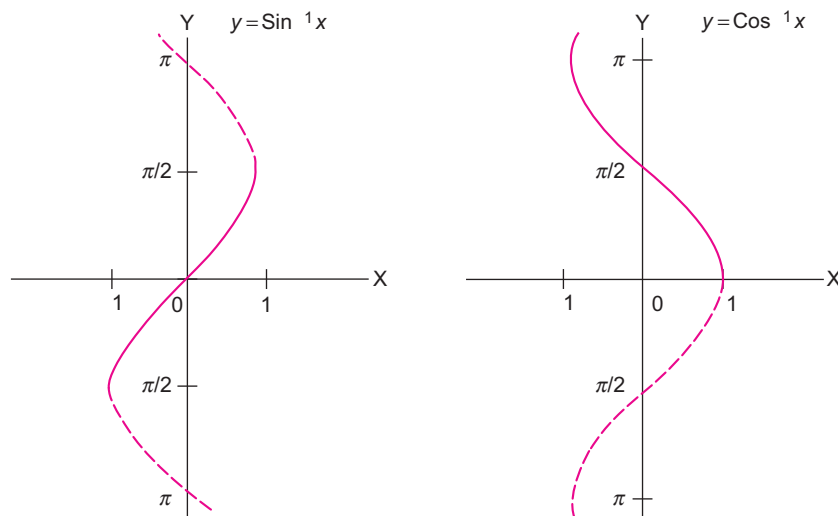


FIGURE 2.1 Graphs of inverse trigonometric functions.

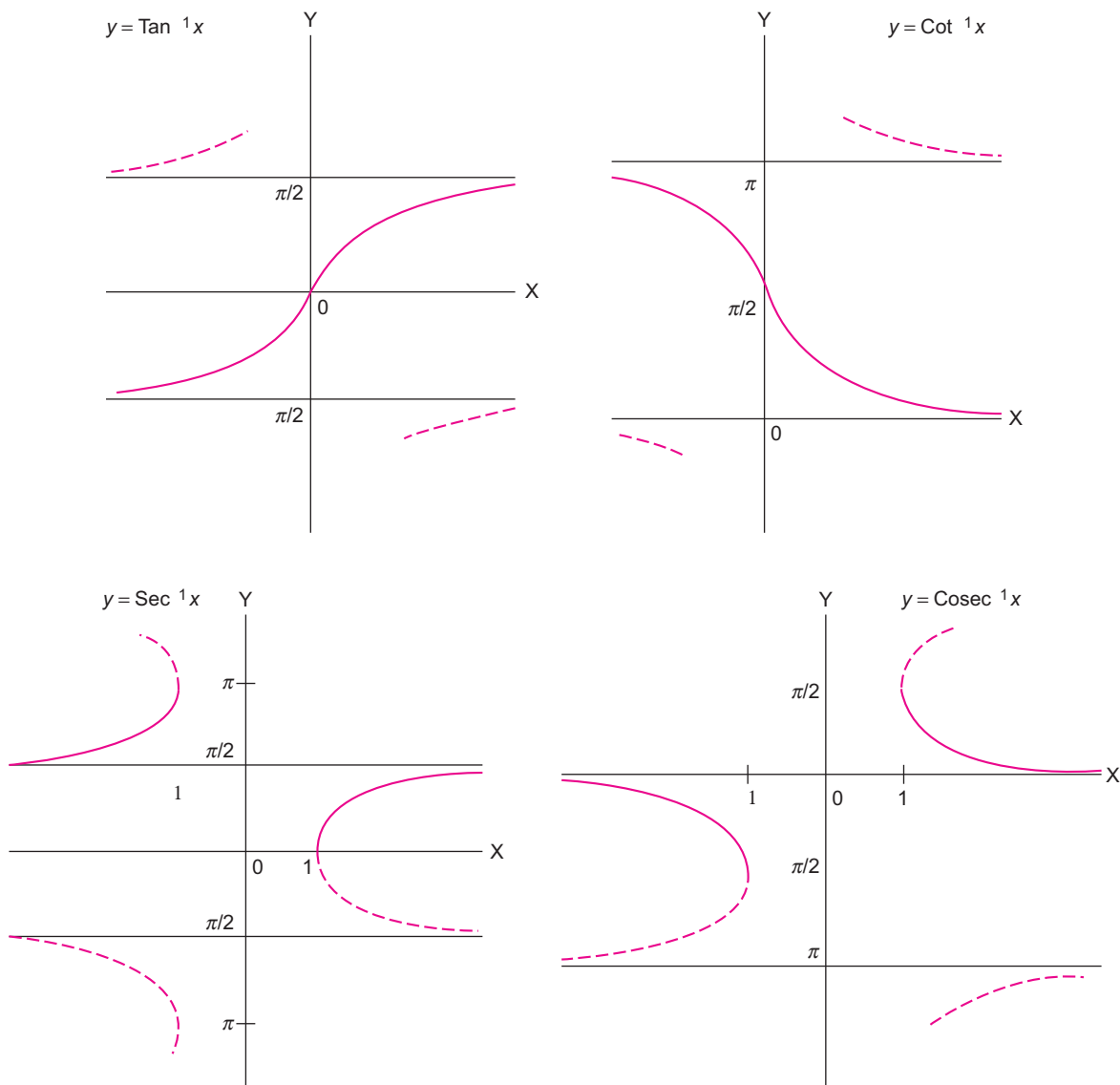


FIGURE 2.1 Continued.

2.2 | Properties of Inverse Trigonometric Functions

In this section we collect certain important properties of inverse trigonometric functions.

THEOREM 2.1 For any $-1 \leq y \leq 1$,

1. $\sin^{-1}(-y) = -\sin^{-1}y$
2. $\cos^{-1}(-y) = \pi - \cos^{-1}y$

PROOF Let $-1 \leq y \leq 1$. Then y and $-y \in [-1, 1]$.

1. Put $\sin^{-1}(-y) = x$. Then

$$-y = \sin x \quad \text{and} \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

Therefore

$$y = -\sin x = \sin(-x) \quad \text{and} \quad -x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

and hence

$$\sin^{-1} y = \sin^{-1}(\sin(-x)) = -x = -\sin^{-1}(-y)$$

Therefore

$$\sin^{-1}(-y) = -\sin^{-1} y$$

2. Put $\cos^{-1}(-y) = z$. Then $-y = \cos z$ and $z \in [0, \pi]$. Therefore $y = -\cos z = \cos(\pi - z)$ and $\pi - z \in [0, \pi]$. Hence

$$\cos^{-1} y = \pi - z = \pi - \cos^{-1}(-y)$$

or

$$\cos^{-1}(-y) = \pi - \cos^{-1} y$$

The following can be similarly proved.

THEOREM 2.2 For any real number y ,

and

$$\tan^{-1}(-y) = -\tan^{-1} y$$

$$\cot^{-1}(-y) = \pi - \cot^{-1} y$$

THEOREM 2.3 For any $y \in \mathbb{R} - (-1, 1)$,

and

$$\sec^{-1}(-y) = \pi - \sec^{-1} y$$

$$\operatorname{cosec}^{-1}(-y) = -\operatorname{cosec}^{-1} y$$

Examples

$$(1) \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = -\sin^{-1}(\sin 60^\circ) = -60^\circ$$

$$(2) \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\cos \frac{\pi}{4}\right) \\ = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$(3) \tan^{-1}(-\sqrt{3}) = -\tan^{-1}(\sqrt{3}) = -\tan^{-1}\left(\tan \frac{\pi}{3}\right) = -\frac{\pi}{3}$$

$$(4) \cot^{-1}(-1) = \pi - \cot^{-1}(1) = \pi - \cot^{-1}\left(\cot \frac{\pi}{4}\right) \\ = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$(5) \sec^{-1}(-\sqrt{2}) = \pi - \sec^{-1}(\sqrt{2}) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$(6) \operatorname{cosec}^{-1}(-\sqrt{2}) = -\operatorname{cosec}^{-1}(\sqrt{2}) \\ = -\operatorname{cosec}^{-1}\left(\operatorname{cosec} \frac{\pi}{6}\right) = -\frac{\pi}{6}$$

Example 2.1

Find the values of the following:

$$(1) \sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$$

$$(2) \cos^{-1}\left(\cos\frac{2\pi}{3}\right)$$

$$(3) \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$$

Solution:

$$(1) \sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right) = \sin^{-1}\left(\sin\left(\pi - \frac{\pi}{6}\right)\right)$$

$$= \sin^{-1}\left(\sin \frac{\pi}{6}\right)$$

$$= \frac{\pi}{6}, \quad \text{since } \frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Here, note that

$$\frac{5\pi}{6} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

and therefore $\sin^{-1}(\sin(5\pi/6))$ is not equal to $5\pi/6$.

$$(2) \quad \cos^{-1}\left(\cos \frac{2\pi}{3}\right) = \frac{2\pi}{3}, \quad \text{since } \frac{2\pi}{3} \in [0, \pi]$$

$$(3) \quad \tan^{-1}\left(\tan \frac{3\pi}{4}\right) = \tan^{-1}\left(\tan\left(\pi - \frac{\pi}{4}\right)\right)$$

$$= \tan^{-1}\left(-\tan \frac{\pi}{4}\right)$$

$$= -\tan^{-1}\left(\tan \frac{\pi}{4}\right)$$

$$= -\frac{\pi}{4}, \quad \text{since } \frac{-\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

THEOREM 2.4 For any $0 \neq y \in [-1, 1]$

$$1. \quad \sin^{-1}\left(\frac{1}{y}\right) = \operatorname{Cosec}^{-1} y$$

$$2. \quad \cos^{-1}\left(\frac{1}{y}\right) = \operatorname{Sec}^{-1} y$$

PROOF 1. Put $\sin^{-1}(1/y) = x$. Then $\sin x = 1/y$ and hence $y = 1/\sin x = \operatorname{cosec} x$.
Therefore

$$\sin^{-1}\left(\frac{1}{y}\right) = x = \operatorname{Cosec}^{-1}(\operatorname{cosec} x) = \operatorname{Cosec}^{-1} y$$

Similarly (2) and the following theorem can be proved. ■

THEOREM 2.5 1. For any $y > 0$,

$$\tan^{-1}\left(\frac{1}{y}\right) = \cot^{-1} y$$

2. For any $y < 0$,

$$\tan^{-1}\left(\frac{1}{y}\right) = \cot^{-1} y - \pi$$

THEOREM 2.6 1. For any $0 \leq x \leq \pi$, $\sin^{-1}(\cos x) = \pi/2 - x$.

2. For any $-\pi/2 \leq x \leq \pi/2$, $\cos^{-1}(\sin x) = \pi/2 - x$.

PROOF 1. Let $0 \leq x \leq \pi$. Then $-1 \leq \cos x \leq 1$. Put $\sin^{-1}(\cos x) = z$. Then

$$\frac{-\pi}{2} \leq z \leq \frac{\pi}{2} \quad \text{and} \quad \sin z = \cos x = \sin\left(\frac{\pi}{2} - x\right)$$

Therefore, $z = (\pi/2) - x$ (since both z and $\pi/2 - x$ belong to $[-\pi/2, \pi/2]$). Hence

$$\sin^{-1}(\cos x) = \pi/2 - x$$

Similarly (2) and the following theorem can be proved. ■

- THEOREM 2.7**
1. For any $0 < x < \pi$, $\tan^{-1}(\cot x) = (\pi/2) - x$
 2. For any $-\pi/2 < x < \pi/2$, $\cot^{-1}(\tan x) = (\pi/2) - x$
 3. For any $0 \neq x \in [-\pi/2, \pi/2]$, $\sec^{-1}(\operatorname{cosec} x) = (\pi/2) - x$
 4. For any $\pi/2 \neq x \in [0, \pi]$, $\operatorname{Cosec}^{-1}(\sec x) = (\pi/2) - x$

THEOREM 2.8 For any $-1 \leq y \leq 1$,

$$\sin^{-1} y = \begin{cases} \cos^{-1} \sqrt{1-y^2} & \text{if } 0 \leq y \leq 1 \\ -\cos^{-1} \sqrt{1-y^2} & \text{if } -1 \leq y < 0 \end{cases}$$

PROOF Let $0 \leq y \leq 1$. Then $0 \leq \sin^{-1} y \leq \pi/2$. Put $\sin^{-1} y = x$. Then

$$\sin x = y \quad \text{and} \quad 0 \leq x \leq \pi/2$$

Therefore

$$\cos x = \sqrt{1-y^2} \quad \text{and} \quad x = \cos^{-1} \sqrt{1-y^2}$$

That is

$$\sin^{-1} y = \cos^{-1} \sqrt{1-y^2}$$

Next, let $-1 \leq y < 0$. Then $-\pi/2 \leq \sin^{-1} y < 0$. Put $\sin^{-1} y = -z$. Then $0 < z \leq \pi/2$ and $y = \sin(-z) = -\sin z$. Therefore

$$\sin z = -y \quad \text{and} \quad \cos z = \sqrt{1-\sin^2 z} = \sqrt{1-y^2}$$

This implies

$$z = \cos^{-1} \sqrt{1-y^2}$$

So

$$\sin^{-1} y = -z = -\cos^{-1} \sqrt{1-y^2}$$



The following can be proved on the lines of the above proof.

THEOREM 2.9 1. If $-1 < y < 1$,

$$\sin^{-1} y = \tan^{-1} \left(\frac{y}{\sqrt{1-y^2}} \right)$$

2. If $0 < y \leq 1$,

$$\cos^{-1} y = \sin^{-1} (\sqrt{1-y^2}) = \tan^{-1} \left(\frac{\sqrt{1-y^2}}{y} \right)$$

3. If $-1 \leq y < 0$,

$$\cos^{-1} y = \pi + \tan^{-1} \left(\frac{\sqrt{1-y^2}}{y} \right)$$

4. If $y > 0$,

$$\tan^{-1} y = \sin^{-1} \left(\frac{y}{\sqrt{1+y^2}} \right) = \cos^{-1} \left(\frac{1}{\sqrt{1+y^2}} \right)$$

Example 2.2

Find the value of

(1) $\sin\left(\cos^{-1}\frac{3}{5}\right)$

(2) $\cos\left(\tan^{-1}\frac{5}{12}\right)$

$$\begin{aligned}
 \text{(2) } \cos\left(\tan^{-1}\frac{5}{12}\right) &= \cos\left(\cos^{-1}\left(\frac{1}{\sqrt{1+(5/12)^2}}\right)\right) \\
 &= \cos\left(\cos^{-1}\left(\frac{12}{13}\right)\right) = \frac{12}{13}
 \end{aligned}$$

Solution:

$$\begin{aligned}
 \text{(1) } \sin\left(\cos^{-1}\frac{3}{5}\right) &= \sin\left(\sin^{-1}\sqrt{1-\left(\frac{3}{5}\right)^2}\right) \\
 &\quad \text{[by part (2) of Theorem 2.9]} \\
 &= \sin\left(\sin^{-1}\frac{4}{5}\right) = \frac{4}{5}
 \end{aligned}$$

THEOREM 2.10 For any $-1 \leq y \leq 1$, $\cos^{-1}y + \sin^{-1}y = \pi/2$.**PROOF** If $y = 0$, then

$$\cos^{-1}y + \sin^{-1}y = \cos^{-1}0 + \sin^{-1}0 = \frac{\pi}{2} + 0 = \frac{\pi}{2}$$

Let $0 < y \leq 1$. Then

$$0 \leq \cos^{-1}y < \frac{\pi}{2} \quad \text{and} \quad 0 < \sin^{-1}y \leq \frac{\pi}{2}$$

Put $\sin^{-1}y = x$. Then $0 < x \leq \pi/2$ and $\sin x = y$. Therefore, $0 \leq \pi/2 - x < \pi/2$ and $\cos(\pi/2 - x) = \sin x = y$ and hence

$$\cos^{-1}y = \frac{\pi}{2} - x$$

$$\cos^{-1}y + \sin^{-1}y = \left(\frac{\pi}{2} - x\right) + x = \frac{\pi}{2}$$

Next, let $-1 \leq y < 0$. Then

$$-\frac{\pi}{2} \leq \sin^{-1}y < 0 \quad \text{and} \quad \frac{\pi}{2} < \cos^{-1}y \leq \pi$$

Put $\sin^{-1}y = z$. Then $\sin z = y$ and $-\pi/2 \leq z < 0$ and hence $\pi/2 \geq -z > 0$ so that $\pi/2 < (\pi/2 - z) \leq \pi$. Now, $\cos(\pi/2 - z) = \sin z = y$. Therefore

$$\cos^{-1}y = \frac{\pi}{2} - z$$

$$\cos^{-1}y + \sin^{-1}y = \left(\frac{\pi}{2} - z\right) + z = \frac{\pi}{2}$$

THEOREM 2.11

For any real number y ,

$$\tan^{-1} y + \cot^{-1} y = \frac{\pi}{2}$$

PROOF

Let y be a real number and $x = \tan^{-1} y$. Then $\tan x = y$ and $-\pi/2 < x < \pi/2$. Also,

$$\cot(\pi/2 - x) = \tan x = y$$

We shall distinguish two cases.

Case I: Suppose that $y \geq 0$. Then $0 \leq x < \pi/2$ and therefore $0 < (\pi/2 - x) \leq \pi/2$. Since $\cot(\pi/2 - x) = y$, we get

$$\cot^{-1} y = \frac{\pi}{2} - x$$

Therefore

$$\tan^{-1} y + \cot^{-1} y = x + \left(\frac{\pi}{2} - x\right) = \frac{\pi}{2}$$

Case II: Suppose that $y < 0$. In such case $-\pi/2 < x < 0$ and therefore $\pi/2 < (\pi/2 - x) < \pi$. Again, since $\cot(\pi/2 - x) = y$, we get that

$$\cot^{-1} y = \frac{\pi}{2} - x$$

Therefore

$$\tan^{-1} y + \cot^{-1} y = x + \left(\frac{\pi}{2} - x\right) = \frac{\pi}{2}$$

COROLLARY 2.1

For any nonzero real number y ,

$$\tan^{-1} y + \tan^{-1} \left(\frac{1}{y}\right) = \begin{cases} \frac{\pi}{2} & \text{if } y > 0 \\ -\frac{\pi}{2} & \text{if } y < 0 \end{cases}$$

PROOF

If $y > 0$, then

$$\begin{aligned} \tan^{-1} y + \tan^{-1} \left(\frac{1}{y}\right) &= \tan^{-1} y + \cot^{-1} y \quad [\text{by part (1) of Theorem 2.5}] \\ &= \frac{\pi}{2} \quad (\text{by Theorem 2.10}) \end{aligned}$$

If $y < 0$, then

$$\begin{aligned} \tan^{-1} y + \tan^{-1} \left(\frac{1}{y}\right) &= \tan^{-1} y + (\cot^{-1} y - \pi) \quad [\text{by part (2) of Theorem 2.5}] \\ &= \frac{\pi}{2} - \pi \quad (\text{by Theorem 2.10}) \\ &= -\frac{\pi}{2} \end{aligned}$$

THEOREM 2.12 Let x and y be positive real numbers. Then

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \left(\frac{x+y}{1-xy} \right) & \text{if } xy < 1 \\ \tan^{-1} \left(\frac{x+y}{1-xy} \right) + \pi & \text{if } xy > 1 \\ \frac{\pi}{2} & \text{if } xy = 1 \end{cases}$$

PROOF If $xy = 1$, then $x = 1/y$ and $y > 0$ and hence, by Corollary 2.1,

$$\tan^{-1} x + \tan^{-1} y = \frac{\pi}{2}$$

Suppose that $xy \neq 1$ (note that both x and y are given to be positive). Put $\tan^{-1} x = \theta$ and $\tan^{-1} y = \phi$. Then $\tan \theta = x$ and $\tan \phi = y$. Since x and y are positive, both θ and ϕ belong to $(0, \pi/2)$ and hence

$$0 < \theta + \phi < \pi$$

Since $xy \neq 1$, $\tan \theta \cdot \tan \phi \neq 1$ and hence

$$\theta + \phi \neq \frac{\pi}{2}$$

$$\text{and} \quad \tan(\theta + \phi) = \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi} = \frac{x + y}{1 - xy} \quad (2.1)$$

Now, suppose that $xy < 1$. Then, by Eq. (2.1)

$$\tan(\theta + \phi) > 0$$

Since $0 < \theta + \phi < \pi$ and $\tan(\theta + \phi) > 0$, we have $0 < \theta + \phi < \pi/2$. Therefore, by Eq. (2.1),

$$\tan^{-1} x + \tan^{-1} y = \theta + \phi = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$$

Next, suppose that $xy > 1$. Then, by Eq. (2.1),

$$\tan(\theta + \phi) < 0$$

Since $0 < \theta + \phi < \pi$ and $\tan(\theta + \phi) < 0$, we have

$$\frac{\pi}{2} < \theta + \phi < \pi$$

and hence

$$\frac{-\pi}{2} < \theta + \phi - \pi < 0$$

Also,

$$\tan(\theta + \phi - \pi) = -\tan(\pi - (\theta + \phi)) = \tan(\theta + \phi) = \frac{x+y}{1-xy}$$

Therefore

$$\tan^{-1} \left(\frac{x+y}{1-xy} \right) = \theta + \phi - \pi$$

$$\tan^{-1} x + \tan^{-1} y = \theta + \phi = \tan^{-1} \left(\frac{x+y}{1-xy} \right) + \pi$$

The following can be proved on the lines of the above proof.

THEOREM 2.13

1. For any negative real numbers x and y ,

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \frac{x+y}{1-xy} & \text{if } xy < 1 \\ -\pi + \tan^{-1} \frac{x+y}{1-xy} & \text{if } xy > 1 \\ \frac{-\pi}{2} & \text{if } xy = 1 \end{cases}$$

2. If $xy + yz + zx < 1$ and x, y, z have the same sign (that is, either all the three are positive or all the three are negative), then

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x+y+z-xyz}{1-(xy+yz+zx)} \right]$$

THEOREM 2.14

If x and y are positive real numbers, then

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

PROOF

Let $x > 0, y > 0, \tan^{-1} x = \theta$ and $\tan^{-1} y = \phi$. Then $\tan \theta = x$ and $\tan \phi = y$. Since x and y are positive, both θ and ϕ belong to $(0, \pi/2)$ and hence

$$\frac{-\pi}{2} < \theta - \phi < \frac{\pi}{2}$$

Also,

$$\tan(\theta - \phi) = \frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi} = \frac{x - y}{1 + xy}$$

and hence

$$\theta - \phi = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

Therefore

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

The following can be proved similarly.

THEOREM 2.15

Let x and y be real numbers such that $xy \neq -1$. Then

$$\tan^{-1} x - \tan^{-1} y = \begin{cases} \tan^{-1} \left(\frac{x-y}{1+xy} \right) & \text{if } xy > -1 \\ \pi + \tan^{-1} \left(\frac{x-y}{1+xy} \right) & \text{if } xy < -1, x > 0 \text{ and } y < 0 \\ -\pi + \tan^{-1} \left(\frac{x-y}{1+xy} \right) & \text{if } xy < -1, x < 0 \text{ and } y > 0 \end{cases}$$

THEOREM 2.16 Let x and y be non-negative real numbers such that $x^2 + y^2 \leq 1$. Then

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$$

PROOF Since $\sin^{-1} 0 = 0$, the theorem is trivial if $x = 0$ or $y = 0$. Therefore, we can assume that both x and y are positive and belong to the interval $(0, 1)$. Put $\sin^{-1} x = \theta$ and $\sin^{-1} y = \phi$. Then $\sin \theta = x$, $\sin \phi = y$ and θ and $\phi \in (0, \pi/2)$. Therefore

$$\cos \theta = \sqrt{1-x^2} \quad \text{and} \quad \cos \phi = \sqrt{1-y^2}$$

Also, $0 < \theta + \phi < \pi$ and we have

$$\begin{aligned} \cos(\theta + \phi) &= \cos \theta \cos \phi - \sin \theta \sin \phi \\ &= \sqrt{1-x^2} \sqrt{1-y^2} - xy \\ &= \sqrt{1-x^2y^2 - (x^2 + y^2) - xy} \geq 0 \quad (\text{since } x^2 + y^2 \leq 1) \end{aligned}$$

Therefore

$$\theta + \phi \in (0, \pi) \text{ and hence } 0 < \theta + \phi \leq \pi/2$$

Now,

$$\begin{aligned} \sin(\theta + \phi) &= \sin \theta \cos \phi + \cos \theta \sin \phi \\ &= x\sqrt{1-y^2} + y\sqrt{1-x^2} \end{aligned}$$

therefore

$$\begin{aligned} \theta + \phi &= \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) \\ \sin^{-1} x + \sin^{-1} y &= \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) \end{aligned}$$

The following can be proved similarly.

THEOREM 2.17 1. If $0 < x < 1$, $0 < y < 1$ and $x^2 + y^2 > 1$, then

$$\sin^{-1} x + \sin^{-1} y = \pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$$

2. For any x and $y \in [0, 1]$,

$$\sin^{-1} x + \sin^{-1} y = \cos^{-1}[\sqrt{(1-x^2)(1-y^2)} - xy]$$

3. For any x and $y \in [0, 1]$,

$$\sin^{-1} x - \sin^{-1} y = \sin^{-1}[x\sqrt{1-y^2} - y\sqrt{1-x^2}]$$

4. If $0 \leq y \leq x \leq 1$,

$$\sin^{-1} x - \sin^{-1} y = \cos^{-1}[\sqrt{(1-x^2)(1-y^2)} + xy]$$

THEOREM 2.18 Let $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

1. $\cos^{-1} x + \cos^{-1} y = \cos^{-1}[xy - \sqrt{(1-x^2)(1-y^2)}]$

2. If $x^2 + y^2 \geq 1$, $\cos^{-1} x + \cos^{-1} y = \sin^{-1}[x\sqrt{1-y^2} + y\sqrt{1-x^2}]$

$$3. \cos^{-1} x - \cos^{-1} y = \sin^{-1} [y\sqrt{1-x^2} - x\sqrt{1-y^2}]$$

$$4. \text{ If } x \leq y, \cos^{-1} x - \cos^{-1} y = \cos^{-1} [xy + \sqrt{(1-x^2)(1-y^2)}]$$

Example 2.3

Evaluate the following:

$$(1) \sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{8}{17}\right)$$

$$(2) \cos^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right)$$

$$(3) \tan^{-1} 2 + \tan^{-1} 3$$

Solution:

(1) Using Theorem 2.16 we have

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1} [x\sqrt{1-y^2} + y\sqrt{1-x^2}]$$

if $x^2 + y^2 \leq 1$. Here

$$\left(\frac{3}{5}\right)^2 + \left(\frac{8}{17}\right)^2 = \frac{9}{25} + \frac{64}{289} = \frac{4201}{7225} < 1$$

Therefore

$$\begin{aligned} \sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{8}{17}\right) &= \sin^{-1}\left[\frac{3}{5}\sqrt{1-\left(\frac{8}{17}\right)^2} + \frac{8}{17}\sqrt{1-\left(\frac{3}{5}\right)^2}\right] \end{aligned}$$

$$= \sin^{-1}\left[\frac{3}{5} \cdot \frac{15}{17} + \frac{8}{17} \cdot \frac{4}{5}\right]$$

$$= \sin^{-1}\left(\frac{77}{85}\right)$$

(2) Using part (1) of Theorem 2.18,

$$\cos^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right)$$

$$= \cos^{-1}\left[\frac{3}{5} \cdot \frac{12}{13} - \sqrt{1-\left(\frac{3}{5}\right)^2} \sqrt{1-\left(\frac{12}{13}\right)^2}\right]$$

$$= \cos^{-1}\left[\frac{36}{65} - \frac{4}{5} \cdot \frac{5}{13}\right]$$

$$= \cos^{-1}\left(\frac{16}{65}\right)$$

(3) By Theorem 2.12,

$$\tan^{-1} 2 + \tan^{-1} 3 = \pi + \tan^{-1}\left(\frac{2+3}{1-2 \cdot 3}\right)$$

$$= \pi + \tan^{-1}(-1)$$

$$= \pi - \tan^{-1}(1)$$

$$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Example 2.4

Prove that for any $-1 < x < 1$,

$$\tan^{-1} x = \frac{1}{2} \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

Solution: Put $\theta = \tan^{-1} x$. Then

$$\tan \theta = x \quad \text{and} \quad -\pi/2 < \theta < \pi/2$$

Since $-1 < x < 1$, it follows that

$$\frac{-\pi}{4} < \theta < \frac{\pi}{4}$$

or

$$\frac{-\pi}{2} < 2\theta < \frac{\pi}{2}$$

Now, we have

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right)$$

$$= \tan^{-1}(\tan 2\theta)$$

$$= 2\theta \quad \left(\text{since } \frac{-\pi}{2} < 2\theta < \frac{\pi}{2}\right)$$

$$= 2 \tan^{-1} x$$

Therefore

$$\tan^{-1} x = \frac{1}{2} \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. $\tan^{-1}2 + \tan^{-1}3 =$

- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{2\pi}{3}$

Solution: Here $x=2$, $y=3$ so that $xy > 1$. From Theorem 2.12 we have

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

Therefore substituting the values we get

$$\begin{aligned}\tan^{-1}2 + \tan^{-1}3 &= \pi + \tan^{-1}\left(\frac{2+3}{1-2\cdot3}\right) \\ &= \pi + \tan^{-1}(-1) \\ &= \pi - \frac{\pi}{4} \\ &= \frac{3\pi}{4}\end{aligned}$$

Answer: (B)

2. $\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8}$ is equal to

- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

Solution:

$$\begin{aligned}&\left(\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7}\right) + \left(\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8}\right) \\ &= \tan^{-1}\left(\frac{\frac{1}{5} + \frac{1}{7}}{1 - \frac{1}{5} \cdot \frac{1}{7}}\right) + \tan^{-1}\left(\frac{\frac{1}{3} + \frac{1}{8}}{1 - \frac{1}{3} \cdot \frac{1}{8}}\right) \quad (\because xy < 1) \\ &= \tan^{-1}\left(\frac{6}{17}\right) + \tan^{-1}\left(\frac{11}{23}\right) \\ &= \tan^{-1}\left(\frac{\frac{6}{17} + \frac{11}{23}}{1 - \frac{6}{17} \cdot \frac{11}{23}}\right) \quad (\because xy < 1) \\ &= \tan^{-1}\left(\frac{325}{325}\right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4}\end{aligned}$$

Answer: (A)

3. If $\sum_{i=1}^n \cos^{-1}(x_i) = 0$, then $\sum_{i=1}^n x_i$ is

- (A) 0 (B) $-n$ (C) n (D) $\frac{n\pi}{2}$

Solution: Since $0 \leq \cos^{-1}x \leq \pi$ and $\sum_{i=1}^n \cos^{-1}x_i = 0$ we have

$$\cos^{-1}x_i = 0 \quad \text{and} \quad x_i = 1 \quad \text{for } i = 1, 2, 3, \dots, n$$

Therefore

$$\sum_{i=1}^n x_i = n$$

Answer: (C)

4. The value of

$$\tan\left(\cos^{-1}\frac{4}{5} + \tan^{-1}\left(\frac{2}{3}\right)\right)$$

is

- (A) $\frac{6}{17}$ (B) $\frac{17}{6}$ (C) $\frac{16}{7}$ (D) $\frac{18}{7}$

Solution: Put $\cos^{-1}(4/5) = \alpha$ and $\tan^{-1}(2/3) = \beta$. Therefore $0 < \alpha, \beta < \pi/2$ and so

$$\cos\alpha = \frac{4}{5} \quad \text{and} \quad \tan\beta = \frac{2}{3}$$

This gives

$$\tan\alpha = \frac{3}{4} \quad \text{and} \quad \tan\beta = \frac{2}{3}$$

Therefore

$$\begin{aligned}n(\alpha + \beta) &= \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \tan\beta} \\ &= \frac{(3/4) + (2/3)}{1 - (3/4) \cdot (2/3)} \\ &= \frac{17}{6}\end{aligned}$$

Answer: (B)

5. $\tan\left(\cos^{-1}\frac{1}{5\sqrt{2}} - \sin^{-1}\frac{4}{\sqrt{17}}\right)$ is

- (A) $\frac{\sqrt{29}}{3}$ (B) $\frac{29}{3}$ (C) $\frac{\sqrt{3}}{29}$ (D) $\frac{3}{29}$

Solution: Put

$$\cos^{-1} \frac{1}{5\sqrt{2}} = \alpha \quad \text{and} \quad \sin^{-1} \frac{4}{\sqrt{17}} = \beta$$

Therefore $0 < \alpha, \beta < \pi/2$ and so

$$\cos \alpha = \frac{1}{5\sqrt{2}} \quad \text{and} \quad \sin \beta = \frac{4}{\sqrt{17}}$$

From these values we can find out $\sin \alpha$ and $\cos \beta$ as

$$\sin \alpha = \frac{7}{5\sqrt{2}} \quad \text{and} \quad \cos \beta = \frac{1}{\sqrt{17}}$$

Therefore

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = 7$$

and

$$\tan \beta = \frac{\sin \beta}{\cos \beta} = 4$$

Now

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{7 - 4}{1 + 7 \cdot 4} = \frac{3}{29}$$

Answer: (D)

6. The number of real values of x satisfying the equation

$$\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \pi/2 \text{ is}$$

- (A) 0 (B) 1 (C) 2 (D) infinite

Solution: Observe that $x(x+1) \geq 0$ and $0 < x^2 + x + 1 \leq 1$. Therefore

$$x(x+1) + 1 \geq 1 \quad \text{and} \quad x^2 + x + 1 \leq 1$$

This implies

$$x^2 + x + 1 = 1$$

or

$$x = 0, -1$$

Answer: (C)

7. The value of x for which $\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1}x)$ is

- (A) $\frac{1}{2}$ (B) 1 (C) 0 (D) $-\frac{1}{2}$

Solution: Let $\cot^{-1}(1+x) = \alpha$ and $\tan^{-1}x = \beta$. Therefore

$$\cot \alpha = 1+x \quad \text{and} \quad \tan \beta = x$$

This gives

$$0 < \alpha < \pi \quad \text{and} \quad -\frac{\pi}{2} < \beta < \frac{\pi}{2} \quad \text{and} \quad \sin \alpha = \cos \beta$$

We have that

$$\frac{1}{\sqrt{(1+x)^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}}$$

Therefore

$$(1+x)^2 + 1 = x^2 + 1$$

$$2x + 1 = 0$$

$$x = -\frac{1}{2}$$

Answer: (D)

8. If $0 < x < 1$, then

$$\sqrt{1+x^2} [\{x \cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1]^{1/2} =$$

(A) $\frac{x}{\sqrt{1+x^2}}$ (B) x

(C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$

Solution: Put $\cot^{-1}x = \alpha$. Then

$$0 < x < 1 \Rightarrow 0 < \alpha < \frac{\pi}{2}$$

Therefore the given expression equals

$$\begin{aligned} & \sqrt{1+x^2} \left[\left(\frac{x^2}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \right)^2 - 1 \right]^{1/2} \\ &= \sqrt{1+x^2} [(\sqrt{1+x^2})^2 - 1]^{1/2} \\ &= \sqrt{1+x^2} \sqrt{x^2} \\ &= x\sqrt{1+x^2} \quad (\because x > 0) \end{aligned}$$

Answer: (C)

9. The numerical value of $\tan(2 \tan^{-1}(1/5) - (\pi/4))$ is

- (A) $\frac{7}{17}$ (B) $\frac{17}{7}$ (C) $-\frac{17}{7}$ (D) $-\frac{7}{17}$

Solution: Put $\tan^{-1}(1/5) = \alpha$. Since $1/5 < 1$ we have $0 < \alpha < \pi/4$. Therefore

$$0 < 2\alpha < \frac{\pi}{2}$$

Hence

$$\begin{aligned} \tan\left(2\alpha - \frac{\pi}{4}\right) &= \frac{\tan 2\alpha - 1}{\tan 2\alpha + 1} = \frac{[2 \tan \alpha / (1 - \tan^2 \alpha)] - 1}{[2 \tan \alpha / (1 - \tan^2 \alpha)] + 1} \\ &= \frac{2 \tan \alpha + \tan^2 \alpha - 1}{2 \tan \alpha + 1 - \tan^2 \alpha} \end{aligned}$$

$$\begin{aligned}
 &= \frac{(2/5) + (1/25) - 1}{(2/5) + 1 - (1/25)} \\
 &= \frac{-14}{34} = \frac{-7}{17}
 \end{aligned}$$

Answer: (D)**10.** $\cos(\tan^{-1}(\sin(\cot^{-1}x)))$ is equal to

$$\begin{array}{ll}
 \text{(A)} \sqrt{\frac{x^2+1}{x^2+2}} & \text{(B)} \sqrt{\frac{x^2-1}{x^2+2}} \\
 \text{(C)} \frac{x^2-1}{x^2+2} & \text{(D)} \frac{x^2+1}{x^2+2}
 \end{array}$$

Solution: Put $\cot^{-1}x = \alpha$ so that $\cot \alpha = x$. Now

$$\sin \alpha = \frac{1}{\operatorname{cosec} \alpha} = \frac{1}{\sqrt{x^2+1}}$$

Again put

$$\tan^{-1}(\sin \alpha) = \tan^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right) = \beta$$

so that

$$\tan \beta = \frac{1}{\sqrt{x^2+1}}$$

Now

$$\begin{aligned}
 \cos \beta &= \frac{1}{\sec \beta} = \frac{1}{\sqrt{1+\tan^2 \beta}} \\
 &= \frac{1}{\sqrt{1+[1/(x^2+1)]}} = \sqrt{\frac{x^2+1}{x^2+2}}
 \end{aligned}$$

Answer: (A)**11.** If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$, then the value of $xy + yz + zx$ is

$$\begin{array}{llll}
 \text{(A)} 0 & \text{(B)} -3 & \text{(C)} \frac{1}{3} & \text{(D)} 3
 \end{array}$$

Solution: Since $0 \leq \cos^{-1}x, \cos^{-1}y, \cos^{-1}z \leq \pi$, we have

$$\cos^{-1}x = \pi, \quad \cos^{-1}y = \pi \quad \text{and} \quad \cos^{-1}z = \pi$$

Therefore

$$x = y = z = -1$$

Substituting we get

$$xy + yz + zx = 3$$

Answer: (D)**12.** The number of real values of x satisfying the equation

$$\sin[2 \cos^{-1}\{\cot(2 \tan^{-1}x)\}] = 0$$

is

$$\text{(A)} 2 \quad \text{(B)} 4 \quad \text{(C)} 6 \quad \text{(D)} 8$$

Solution: Clearly $x = 0$ cannot be a solution and $x = \pm 1$ are solutions. We suppose $x \neq 1$ and -1 . Put $\tan^{-1}x = \alpha$. Therefore

$$\tan \alpha = x \neq 1, -1$$

Now

$$\cot 2\alpha = \frac{1}{\tan 2\alpha} = \frac{1 - \tan^2 \alpha}{2 \tan \alpha} = \frac{1 - x^2}{2x}$$

Now put

$$\cos^{-1}\left(\frac{1 - x^2}{2x}\right) = \beta$$

so that

$$\cos \beta = \frac{1 - x^2}{2x}$$

Therefore

$$\begin{aligned}
 \text{LHS} &= \sin 2\beta = 2 \sin \beta \cos \beta \\
 &= 2 \sqrt{1 - \left(\frac{1 - x^2}{2x}\right)^2} \cdot \frac{1 - x^2}{2x}
 \end{aligned}$$

Now

$$\begin{aligned}
 \sin 2\beta = 0 &\Rightarrow 1 - \left(\frac{1 - x^2}{2x}\right)^2 = 0 \quad (\because x \neq 1, -1) \\
 &\Rightarrow (1 - x^2)^2 = 4x^2 \\
 &\Rightarrow (1 + 2x - x^2)(1 - 2x - x^2) = 0
 \end{aligned}$$

This implies

$$x = -1 \pm \sqrt{2} \quad \text{or} \quad x = 1 \pm \sqrt{2}$$

which gives

$$x = \pm 1, \quad -1 \pm \sqrt{2} \quad \text{and} \quad 1 \pm \sqrt{2}$$

Answer: (C)**13.** If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\pi/2$, then the value of

$$x^{2010} + y^{2011} + z^{2012} + \frac{9}{x^{2010} + y^{2011} + z^{2012}}$$

is

$$\text{(A)} 0 \quad \text{(B)} 3 \quad \text{(C)} 6 \quad \text{(D)} 9$$

Solution: We have $-\pi/2 \leq \sin^{-1}x, \sin^{-1}y, \sin^{-1}z \leq \pi/2$ and $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\pi/2$. This implies

$$\sin^{-1}x = \sin^{-1}y = \sin^{-1}z = \frac{\pi}{2}$$

Therefore $x = y = z = 1$ and hence

$$x^{2010} + y^{2011} + z^{2012} + \frac{9}{x^{2010} + y^{2011} + z^{2012}} = 3 + \frac{9}{3} = 6$$

Answer: (C)

14. If $0 \leq p, q < 1$ and

$$\sin^{-1}\left(\frac{2p}{1+p^2}\right) - \cos^{-1}\left(\frac{1-q^2}{1+q^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

then x is equal to

- (A) $\frac{p-q}{1+pq}$ (B) $\frac{p+q}{1+pq}$
(C) $\frac{p-q}{p+q}$ (D) $\frac{p+q}{p-q}$

Solution: Put $p = \tan \theta$ so that $0 \leq \theta < \pi/4$ ($\because 0 \leq p < 1$). Therefore

$$\begin{aligned} \sin^{-1}\left(\frac{2p}{1+p^2}\right) &= \sin^{-1}(\sin 2\theta) = 2\theta \quad \left(\because 0 \leq 2\theta < \frac{\pi}{2}\right) \\ &= 2 \tan^{-1} p \end{aligned}$$

Similarly

$$\cos^{-1}\left(\frac{1-q^2}{1+q^2}\right) = 2 \tan^{-1} q$$

Therefore

$$\begin{aligned} \sin^{-1}\left(\frac{2p}{1+p^2}\right) - \cos^{-1}\left(\frac{1-q^2}{1+q^2}\right) &= 2(\tan^{-1} p - \tan^{-1} q) \\ &= 2 \tan^{-1}\left(\frac{p-q}{1+pq}\right) \\ &= \tan^{-1}\left(\frac{2x}{1-x^2}\right) \quad (\text{Given}) \\ &= 2 \tan^{-1} x \end{aligned}$$

This implies

$$x = \frac{p-q}{1+pq}$$

Answer: (A)

15. $3 \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \sin^{-1}\left(\frac{142}{65\sqrt{5}}\right)$ is equal to

- (A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{4}$ (D) $-\pi$

Solution:

$$\sin^{-1}\left(\frac{142}{65\sqrt{5}}\right) = \tan^{-1}\left(\frac{142}{\sqrt{(65\sqrt{5})^2 - (142)^2}}\right) = \tan^{-1}\left(\frac{142}{31}\right)$$

$$2 \tan^{-1}\left(\frac{1}{2}\right) = \tan^{-1}\left(\frac{2/5}{1-(1/25)}\right) = \tan^{-1}\left(\frac{5}{12}\right)$$

$$3 \tan^{-1}\left(\frac{1}{2}\right) = \tan^{-1}\left(\frac{(3/2)-(1/8)}{1-(3/4)}\right) = \tan^{-1}\left(\frac{11}{2}\right)$$

Therefore the given sum equals

$$\begin{aligned} &\tan^{-1}\left(\frac{11}{2}\right) + \tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{142}{31}\right) \\ &= \pi + \tan^{-1}\left(\frac{(11/2)+(5/12)}{1-(55/24)}\right) + \tan^{-1}\left(\frac{142}{31}\right) \\ &= \pi - \tan^{-1}\left(\frac{142}{31}\right) + \tan^{-1}\left(\frac{142}{31}\right) \\ &= \pi \end{aligned}$$

Answer: (B)

16. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, then the value of $x^2 + y^2 + z^2 + 2xyz$ is

- (A) 1 (B) 0 (C) $x+y+z$ (D) $x^2y^2z^2$

Solution: Let $\cos^{-1}x = \alpha$, $\cos^{-1}y = \beta$ and $\cos^{-1}z = \gamma$. Therefore $\cos \alpha = x$, $\cos \beta = y$ and $\cos \gamma = z$ and at least one of x, y and z is non-negative. Now

$$\alpha + \beta + \gamma = \pi \Rightarrow \alpha + \beta = \pi - \gamma$$

Therefore

$$\begin{aligned} \cos(\alpha + \beta) &= \cos(\pi - \gamma) = -\cos \gamma \\ \cos \alpha \cos \beta - \sin \alpha \sin \beta &= -\cos \gamma \end{aligned}$$

This gives

$$\begin{aligned} xy - \sqrt{1-x^2}\sqrt{1-y^2} &= -z \\ (xy+z)^2 &= (1-x^2)(1-y^2) \\ x^2y^2 + z^2 + 2xyz &= 1 - x^2 - y^2 + x^2y^2 \\ x^2 + y^2 + z^2 + 2xyz &= 1 \end{aligned}$$

Answer: (A)

17. If $a^2 + b^2 = c^2$, $c \neq 0$, then the number of values of x such that $\sin^{-1}(ax/c) + \sin^{-1}(bx/c) = \sin^{-1}x$ is

- (A) 2 (B) 1 (C) 0 (D) 3

Solution: Clearly $x = 0$ is a solution. Suppose $x \neq 0$. Put

$$\sin^{-1} \frac{ax}{c} = \alpha \quad \text{and} \quad \sin^{-1} \frac{bx}{c} = \beta$$

Therefore

$$\sin \alpha = \frac{ax}{c} \quad \text{and} \quad \sin \beta = \frac{bx}{c}$$

Now

$$\begin{aligned} \alpha + \beta = \sin^{-1}x &\Rightarrow \sin(\alpha + \beta) = x \\ &\Rightarrow \sin \alpha \cos \beta + \cos \alpha \sin \beta = x \\ &\Rightarrow \frac{ax}{c} \sqrt{1 - \frac{b^2x^2}{c^2}} + \left(\sqrt{1 - \frac{a^2x^2}{c^2}} \right) \frac{bx}{c} = x \\ &\Rightarrow \frac{a}{c^2} \sqrt{c^2 - b^2x^2} + \frac{b}{c^2} \sqrt{c^2 - a^2x^2} = 1 \\ &\Rightarrow a^2(c^2 - b^2x^2) + b^2(c^2 - a^2x^2) + \\ &\quad 2ab\sqrt{c^2 - b^2x^2} \cdot \sqrt{c^2 - a^2x^2} = c^4 \end{aligned}$$

Therefore

$$\begin{aligned} c^2(a^2 + b^2) - 2a^2b^2x^2 + 2ab\sqrt{c^2 - b^2x^2} \cdot \sqrt{c^2 - a^2x^2} &= c^4 \\ a^2b^2x^2 &= ab\sqrt{c^2 - b^2x^2} \cdot \sqrt{c^2 - a^2x^2} \quad (\because a^2 + b^2 = c^2) \\ a^2b^2x^4 &= (c^2 - b^2x^2)(c^2 - a^2x^2) \\ &= c^4 - (a^2 + b^2)x^2 + a^2b^2x^4 \\ (a^2 + b^2)x^2 &= c^4 \end{aligned}$$

This gives $x^2 = 1$ or $x = \pm 1$.

Answer: (D)

18. If $0 \leq x \leq 1/2$, then the value of

$$\tan \left[\sin^{-1} \left\{ \frac{x}{\sqrt{2}} + \frac{\sqrt{1-x^2}}{\sqrt{2}} \right\} - \sin^{-1}x \right]$$

is

- (A) -1 (B) 1 (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

Solution: Put $x = \sin \theta$ where $0 \leq \theta \leq \pi/6$. Now

$$\begin{aligned} \sin^{-1} \left\{ \frac{x}{\sqrt{2}} + \frac{\sqrt{1-x^2}}{\sqrt{2}} \right\} &= \sin^{-1} \left(\frac{\sin \theta + \cos \theta}{\sqrt{2}} \right) \\ &= \sin^{-1} \left(\sin \left(\theta + \frac{\pi}{4} \right) \right) = \theta + \frac{\pi}{4} \quad \left(\because 0 \leq \theta + \frac{\pi}{4} \leq \frac{\pi}{2} \right) \end{aligned}$$

Therefore

$$\begin{aligned} \tan \left[\sin^{-1} \left\{ \frac{x}{\sqrt{2}} + \frac{\sqrt{1-x^2}}{\sqrt{2}} \right\} - \sin^{-1}x \right] &= \tan \left(\theta + \frac{\pi}{4} - \theta \right) \\ &= \tan \frac{\pi}{4} = 1 \end{aligned}$$

Answer: (B)

19. $\sin \left(2 \tan^{-1} \frac{1}{3} \right) + \cos(\tan^{-1} 2\sqrt{2}) =$

- (A) $\frac{1}{15}$ (B) $\frac{3}{15}$ (C) $\frac{14}{15}$ (D) $\frac{11}{15}$

Solution: Put $\tan^{-1}(1/3) = \alpha$ and $\tan^{-1} 2\sqrt{2} = \beta$.

Therefore

$$\tan \alpha = \frac{1}{3} \quad \text{and} \quad \tan \beta = 2\sqrt{2}$$

This implies

$$\sin \alpha = \frac{1}{\sqrt{10}}, \quad \cos \alpha = \frac{3}{\sqrt{10}} \quad \text{and} \quad \cos \beta = \frac{1}{3}$$

Therefore

$$\begin{aligned} \text{LHS} &= \sin 2\alpha + \cos \beta = 2 \sin \alpha \cos \alpha + \cos \beta \\ &= \frac{2}{\sqrt{10}} \times \frac{3}{\sqrt{10}} + \frac{1}{3} \\ &= \frac{6}{10} + \frac{1}{3} = \frac{14}{15} \end{aligned}$$

Answer: (C)

20. If

$$\sin \left(\sin^{-1} \frac{1}{5} + \cos^{-1}x \right) = 1$$

then the value of x is

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$

Solution: Put $\sin^{-1}(1/5) = \alpha$ and $\cos^{-1}x = \beta$.

Therefore

$$\sin \alpha = \frac{1}{5} \quad \text{and} \quad \cos \beta = x$$

Now

$$\begin{aligned} 1 &= \sin(\alpha + \beta) \\ 1 &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ 1 &= \frac{1}{5}(x) + \frac{\sqrt{24}}{5} \sqrt{1-x^2} \end{aligned}$$

$$\begin{aligned}
 5 - x &= \sqrt{24}\sqrt{1-x^2} \\
 (5-x)^2 &= 24(1-x^2) \\
 25x^2 - 10x + 1 &= 0 \\
 (5x-1)^2 &= 0 \\
 x &= \frac{1}{5}
 \end{aligned}$$

Answer: (D)

21. If

$$\tan^{-1} \frac{1}{2x+1} + \tan^{-1} \frac{1}{4x+1} = \tan^{-1} \frac{2}{x^2}$$

and $x \neq 0$, then x equals

(A) 3 (B) $\frac{-2}{3}$ (C) -3 (D) $\frac{2}{3}$

Solution:

$$\begin{aligned}
 \tan^{-1} \left(\frac{\{1/(2x+1)\} + \{1/(4x+1)\}}{1 - \{1/(2x+1)\} \cdot \{1/(4x+1)\}} \right) &= \tan^{-1} \frac{2}{x^2} \\
 \tan^{-1} \left(\frac{6x+2}{8x^2+6x} \right) &= \tan^{-1} \frac{2}{x^2}
 \end{aligned}$$

Therefore

$$\begin{aligned}
 \frac{3x+1}{4x^2+3x} &= \frac{2}{x^2} \\
 x^2(3x+1) &= 2(4x^2+3x) \\
 3x^3 - 7x^2 - 6x &= 0 \\
 x(3x^2 - 7x - 6) &= 0 \\
 x(x-3)(3x+2) &= 0
 \end{aligned}$$

We take $x=3$ because when $x=-2/3$, LHS will be negative whereas $\tan^{-1}(2/x^2)$ is positive.

Answer: (A)

22. If $\sin^{-1}(1/\sqrt{5})$ and $\cos^{-1}(3/\sqrt{10})$ are angles in $[0, \pi/2]$, then their sum is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\sin^{-1} \frac{1}{\sqrt{50}}$

Solution: Let $\sin^{-1}(1/\sqrt{5}) = \alpha$ and $\cos^{-1}(3/\sqrt{10}) = \beta$. Then

$$\sin \alpha = \frac{1}{\sqrt{5}}, \cos \beta = \frac{3}{\sqrt{10}} \quad \text{and} \quad 0 < \alpha, \beta < \frac{\pi}{2}$$

From these we get

$$\cos \alpha = \frac{2}{\sqrt{5}} \quad \text{and} \quad \sin \beta = \frac{1}{\sqrt{10}}$$

Therefore

$$\tan \alpha = \frac{1}{2} \quad \text{and} \quad \tan \beta = \frac{1}{3}$$

Hence

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{(1/2) + (1/3)}{1 - (1/6)} = \frac{5}{5} = 1$$

$$\alpha + \beta = \tan^{-1}(1) = \frac{\pi}{4}$$

Answer: (B)

23. If $\cot[\sin^{-1}(\sqrt{13/17})] = \sin(\tan^{-1} \alpha)$, then α is

(A) $\frac{4}{17}$ (B) $\sqrt{\frac{17^2 - 13^2}{17 \times 13}}$
 (C) $\sqrt{\frac{17^2 - 13^2}{17^2 + 13^2}}$ (D) $\frac{2}{3}$

Solution: Put $\sin^{-1}(\sqrt{13/17}) = \theta$. Therefore

$$\sin \theta = \sqrt{\frac{13}{17}}$$

Also

$$\begin{aligned}
 \cot \theta &= \sqrt{\operatorname{cosec}^2 \theta - 1} \\
 &= \sqrt{\frac{1}{\sin^2 \theta} - 1} \\
 &= \sqrt{\frac{17}{13} - 1} = \frac{2}{\sqrt{13}}
 \end{aligned}$$

Again put $\tan^{-1} \alpha = \phi$ so that $\tan \phi = \alpha$. Now

$$\begin{aligned}
 \sin \phi &= \frac{1}{\operatorname{cosec} \phi} = \frac{1}{\sqrt{\cot^2 \phi + 1}} \\
 &= \frac{1}{\sqrt{(1/\alpha^2) + 1}} = \frac{\alpha}{\sqrt{\alpha^2 + 1}}
 \end{aligned}$$

Therefore

$$\begin{aligned}
 \cot \theta = \sin \phi &\Rightarrow \frac{2}{\sqrt{13}} = \frac{\alpha}{\sqrt{1 + \alpha^2}} \\
 &\Rightarrow \frac{4}{13} = \frac{\alpha^2}{1 + \alpha^2} \\
 &\Rightarrow \frac{13}{4} = \frac{1 + \alpha^2}{\alpha^2} = \frac{1}{\alpha^2} + 1
 \end{aligned}$$

$$\begin{aligned}\Rightarrow \frac{13}{4} - 1 &= \frac{1}{\alpha^2} \\ \Rightarrow \frac{9}{4} &= \frac{1}{\alpha^2} \\ \Rightarrow \alpha &= \frac{2}{3}\end{aligned}$$

Answer: (D)

24. If x, y, z are in GP and $\tan^{-1}x, \tan^{-1}y, \tan^{-1}z$ are in AP, then

(A) $x = y = z$ or $y = \pm 1$

(B) $z = \frac{1}{x}$

(C) $x = y = z$, but their common value may not necessarily be 0

(D) $x = y = z = 0$

Solution: We have

$$y^2 = zx \text{ and } \tan^{-1}x + \tan^{-1}z = 2\tan^{-1}y$$

$$y^2 = zx \text{ and } \tan^{-1}\left(\frac{x+z}{1-xz}\right) = \tan^{-1}\left(\frac{2y}{1-y^2}\right) \text{ if } y \neq \pm 1$$

$$y^2 = zx \text{ and } \tan^{-1}\left(\frac{x+z}{1-y^2}\right) = \tan^{-1}\left(\frac{2y}{1-y^2}\right)$$

$$x + z = 2y$$

This implies x, y, z are in AP. That is x, y, z are in G.P. as well as in AP. Therefore

$$x = y = z \text{ or } y = \pm 1$$

Answer: (A)

25. If $\sin^{-1}(1/\sqrt{5})$ and $\sin^{-1}(1/\sqrt{10})$ are two angles of a triangle, then the third angle is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{2\pi}{3}$

Solution: We have

$$\sin^{-1}\frac{1}{\sqrt{5}} = \tan^{-1}\left(\frac{1}{2}\right) \text{ and } \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) = \tan^{-1}\left(\frac{1}{3}\right)$$

(by Section 2.2)

Put $\tan^{-1}(1/2) = \alpha$ and $\tan^{-1}(1/3) = \beta$. Therefore

$$\tan \alpha = \frac{1}{2} \text{ and } \tan \beta = \frac{1}{3}$$

This implies that $0 < \alpha$ and $\beta < \pi/4$. Now

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{(1/2) + (1/3)}{1 - (1/2) \cdot (1/3)} = \frac{5}{5} = 1$$

Therefore

$$\alpha + \beta = \frac{\pi}{4}$$

and hence the third angle is $3\pi/4$.

Answer: (C)

26. Consider the following two statements P and Q .

$$P: \cos^{-1}\left(\cos \frac{4\pi}{3}\right) = \frac{2\pi}{3}$$

$$Q: \sec^2\left(\cot^{-1}\frac{1}{2}\right) + \operatorname{cosec}^2\left(\tan^{-1}\frac{1}{3}\right) = 15$$

Which of the following is true?

(A) Both P and Q are true

(B) P is true, but Q is false

(C) P is false, but Q is true

(D) Both P and Q are false

Solution: We have

$$\cos\left(\frac{4\pi}{3}\right) = -\cos\frac{\pi}{3} = \frac{-1}{2}$$

Therefore

$$\cos^{-1}\left(\frac{-1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

This implies P is true. Now,

$$\begin{aligned}\sec^2\left(\cot^{-1}\frac{1}{2}\right) + \operatorname{cosec}^2\left(\tan^{-1}\frac{1}{3}\right) &= [1 + \tan^2(\tan^{-1}2)] \\ &\quad + [1 + \cot^2(\cot^{-1}3)] \\ &= 1 + 4 + 1 + 9 = 15\end{aligned}$$

Therefore Q is also true.

Answer: (A)

27. The value of x satisfying the equation

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$$

is

(A) -1 (B) $\frac{1}{6}$ (C) 1 (D) $\frac{-1}{6}$

Solution: Observe that $x = \pm 1$ is not a solution. Let $\tan^{-1}(2x) = \alpha$ and $\tan^{-1}(3x) = \beta$. Therefore $\tan \alpha = 2x$, $\tan \beta = 3x$. Now

$$\begin{aligned}
 \alpha + \beta = \frac{\pi}{4} &\Rightarrow \tan(\alpha + \beta) = 1 \\
 &\Rightarrow \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = 1 \\
 &\Rightarrow \frac{2x + 3x}{1 - 6x^2} = 1 \\
 &\Rightarrow 6x^2 + 5x - 1 = 0 \\
 &\Rightarrow (6x - 1)(x + 1) = 0 \\
 &\Rightarrow x = \frac{1}{6} \quad (\because x \neq \pm 1)
 \end{aligned}$$

Answer: (B)

28. The value of $\cos(2\cos^{-1}x + \sin^{-1}x)$ at $x = 1/5$ where $\cos^{-1}x$ and $\sin^{-1}x$ are principal values is

- (A) $\frac{\sqrt{24}}{5}$ (B) $\frac{-4}{5}$ (C) $\frac{4}{5}$ (D) $-\frac{\sqrt{24}}{5}$

Solution:

$$\begin{aligned}
 \cos(2\cos^{-1}x + \sin^{-1}x) &= \cos\left(\cos^{-1}x + \frac{\pi}{2}\right) \\
 &= -\sin(\cos^{-1}x) = -\sin(\sin^{-1}\sqrt{1-x^2}) \\
 &= -\sqrt{1-x^2}
 \end{aligned}$$

Therefore at $x = 1/5$ the required value is

$$\begin{aligned}
 \cos(2\cos^{-1}x + \sin^{-1}x) &= -\sqrt{1 - \frac{1}{25}} \\
 &= -\frac{\sqrt{24}}{5}
 \end{aligned}$$

Answer: (D)

29. For any $0 \leq x \leq 1$,

$$\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right]$$

is equal to

- (A) $\frac{1-x^2}{1+x^2}$ (B) $\frac{2x}{1-x^2}$
 (C) $\frac{2x}{1+x^2}$ (D) 0

Solution: Put $\tan^{-1}x = \theta$. Then $\tan \theta = x$ and $-\pi/2 < \theta < \pi/2$. Since $0 \leq x < 1$, we get that $0 \leq \theta < \pi/4$. Hence $0 \leq 2\theta < \pi/2$. Now

$$\begin{aligned}
 \sin^{-1}\left(\frac{2x}{1+x^2}\right) &= \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) \\
 &= \sin^{-1}(\sin 2\theta) = 2\theta
 \end{aligned}$$

Again

$$\begin{aligned}
 \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) &= \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right) \\
 &= \cos^{-1}(\cos 2\theta) \\
 &= 2\theta
 \end{aligned}$$

Therefore, the given expression equals

$$\begin{aligned}
 \tan\left(\frac{1}{2}(2\theta) + \frac{1}{2}(2\theta)\right) &= \tan 2\theta \\
 &= \frac{2\tan\theta}{1-\tan^2\theta} \\
 &= \frac{2x}{1-x^2}
 \end{aligned}$$

Answer: (B)

30. If

$$\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$$

then x is equal to

- (A) 17 (B) 15 (C) 13 (D) 19

Solution: $\sin^{-1}(5/x)$ and $\sin^{-1}(12/x)$ are both defined when $x > 12$. Therefore by Theorem 2.7,

$$\begin{aligned}
 \cos^{-1}\left(\sqrt{1-\frac{25}{x^2}}\right) + \sin^{-1}\left(\frac{12}{x}\right) &= \frac{\pi}{2} \\
 &= \cos^{-1}\left(\frac{12}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right)
 \end{aligned}$$

Comparing we get

$$\frac{12}{x} = \sqrt{1-\frac{25}{x^2}}$$

Therefore

$$144 = x^2 - 25 \quad \text{or} \quad x = 13$$

Answer: (C)

31. If x, y, z are positive real numbers, then

$$\begin{aligned}
 \tan^{-1}\sqrt{\frac{x(x+y+z)}{yz}} + \tan^{-1}\sqrt{\frac{y(x+y+z)}{zx}} \\
 + \tan^{-1}\sqrt{\frac{z(x+y+z)}{xy}} =
 \end{aligned}$$

- (A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) π

Solution: Put

$$a = \sqrt{\frac{x(x+y+z)}{yz}}$$

$$b = \sqrt{\frac{y(x+y+z)}{zx}}$$

$$c = \sqrt{\frac{z(x+y+z)}{xy}}$$

Therefore

$$\begin{aligned} ab &= (x+y+z) \sqrt{\frac{(xy)}{xyz^2}} \\ &= \frac{x+y+z}{z} = 1 + \frac{x+y}{z} > 1 \end{aligned}$$

Now

$$\begin{aligned} \tan^{-1} a + \tan^{-1} b &= \pi + \tan^{-1} \left(\frac{a+b}{1-ab} \right) \\ &\quad \text{(by Theorem 2.11)} \end{aligned}$$

$$\begin{aligned} &= \pi + \tan^{-1} \left[\frac{\sqrt{\frac{x+y+z}{z}} \left(\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} \right)}{\frac{-(x+y)}{z}} \right] \\ &= \pi + \tan^{-1} \left[-\sqrt{\frac{x+y+z}{z}} \left(\frac{x+y}{\sqrt{xy}} \right) \left(\frac{z}{x+y} \right) \right] \\ &= \pi - \tan^{-1} \left[\sqrt{\frac{z(x+y+z)}{xy}} \right] \\ &= \pi - \tan^{-1} c \end{aligned}$$

So

$$\tan^{-1} a + \tan^{-1} b + \tan^{-1} c = \pi$$

Answer: (D)

32. $\cos^{-1} \left(\frac{4}{5} \right) + \sin^{-1} \left(\frac{3}{\sqrt{34}} \right) =$

- (A) $\tan^{-1} \left(\frac{27}{13} \right)$ (B) $\tan^{-1} \left(\frac{27}{11} \right)$
 (C) $\tan^{-1} \left(\frac{23}{11} \right)$ (D) $\tan^{-1} \left(\frac{23}{12} \right)$

Solution: We have

$$\cos^{-1} \left(\frac{4}{5} \right) = \tan^{-1} \left(\frac{3}{4} \right) \quad \text{and} \quad \sin^{-1} \left(\frac{3}{\sqrt{34}} \right) = \tan^{-1} \left(\frac{3}{5} \right)$$

(by Theorem 2.9)

Therefore

$$\begin{aligned} &\cos^{-1} \left(\frac{4}{5} \right) + \sin^{-1} \left(\frac{3}{\sqrt{34}} \right) \\ &= \tan^{-1} \left(\frac{3}{4} \right) + \tan^{-1} \left(\frac{3}{5} \right) \\ &= \tan^{-1} \left(\frac{(3/4) + (3/5)}{1 - (3/4) \cdot (3/5)} \right) \quad \left(\because \frac{3}{4} \cdot \frac{3}{5} < 1 \right) \\ &= \tan^{-1} \left(\frac{27}{11} \right) \end{aligned}$$

Answer: (B)

33. If

$$\cot^{-1} \frac{1}{x} + \cot^{-1} \frac{1}{y} + \cot^{-1} \frac{1}{z} = \frac{\pi}{2}$$

then $xy + yz + zx$ is equal to

- (A) 1 (B) -1
 (C) $x + y + z$ (D) $-(x + y + z)$

Solution: Put $\cot^{-1}(1/x) = \alpha$, $\cot^{-1}(1/y) = \beta$ and $\cot^{-1}(1/z) = \gamma$. Therefore

$$\begin{aligned} \cot(\alpha + \beta) &= \cot \left(\frac{\pi}{2} - \gamma \right) = \tan \gamma \\ \frac{\cot \alpha \cot \beta - 1}{\cot \beta + \cot \alpha} &= \tan \gamma \\ \frac{(1/x) \cdot (1/y) - 1}{(1/y) + (1/x)} &= z \\ 1 - xy &= z(x + y) \\ xy + yz + zx &= 1 \end{aligned}$$

Answer: (A)

34. If $\cos^{-1}(3/5) + \cos^{-1}(12/13) = \cos^{-1}(k)$, then the value of k is

- (A) $\frac{16}{65}$ (B) $\frac{12}{65}$ (C) $\frac{11}{65}$ (D) $\frac{19}{65}$

Solution: It is known that

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1} (xy - \sqrt{1-x^2} \sqrt{1-y^2})$$

when $x + y \geq 0$.

Therefore

$$\begin{aligned}\cos^{-1}(k) &= \cos^{-1}\left(\frac{3}{5} \cdot \frac{12}{13} - \sqrt{1 - \frac{9}{25}} \cdot \sqrt{1 - \frac{144}{169}}\right) \\ &= \cos^{-1}\left(\frac{36}{65} - \frac{20}{65}\right) \\ &= \cos^{-1}\left(\frac{16}{65}\right)\end{aligned}$$

This implies

$$k = \frac{16}{65}$$

Answer: (A)

35. $\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \tan^{-1}\left(\frac{1}{1+3 \cdot 4}\right) + \dots$
upto n terms is equal to

(A) $\tan^{-1}\left(\frac{n-1}{n+1}\right)$ (B) $\tan^{-1}\left(\frac{n+1}{n+2}\right)$

(C) $\tan^{-1}\left(\frac{n}{n+2}\right)$ (D) $\tan^{-1}\left(\frac{n}{n+1}\right)$

Solution: Let

$$\begin{aligned}u_k &= \tan^{-1}\left(\frac{1}{1+k(k+1)}\right) = \tan^{-1}\left(\frac{k+1-k}{1+k(k+1)}\right) \\ &= \tan^{-1}(k+1) - \tan^{-1}k\end{aligned}$$

Put $k = 1, 2, 3, \dots, n$ and add so that the given sum equals

$$\begin{aligned}\tan^{-1}(n+1) - \tan^{-1}(1) &= \tan^{-1}\left(\frac{n+1-1}{1+1 \cdot (n+1)}\right) \\ &= \tan^{-1}\left(\frac{n}{n+2}\right)\end{aligned}$$

Answer: (C)

Multiple Correct Answer Type Questions

1. Which of the following are true?

(A) $\tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{13} = \tan^{-1}\frac{2}{9}$

(B) $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$

(C) $\tan^{-1}(3) + \tan^{-1}(4) = \frac{3\pi}{4}$

(D) $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ if $xy > 1$

Solution:

(A) We know that

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \quad \text{if } xy < 1$$

Therefore

$$\begin{aligned}\tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) &= \tan^{-1}\left(\frac{(1/7) + (1/13)}{1 - (1/7) \cdot (1/13)}\right) \\ &= \tan^{-1}\left(\frac{13+7}{90}\right) \\ &= \tan^{-1}\left(\frac{2}{9}\right)\end{aligned}$$

Therefore (A) is true.

$$\begin{aligned}\text{(B)} \quad \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) &= \tan^{-1}\left(\frac{(1/2) + (1/3)}{1 - (1/2) \cdot (1/3)}\right) \\ &= \tan^{-1}\left(\frac{5}{5}\right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4}\end{aligned}$$

Therefore (B) is true.

(C) Here $x > 0, y > 0$ and $xy > 1$ so that

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

Therefore

$$\begin{aligned}\tan^{-1}3 + \tan^{-1}4 &= \pi + \tan^{-1}\left(\frac{4+3}{1-12}\right) \\ &= \pi + \tan^{-1}\left(\frac{-7}{11}\right) \neq \frac{3\pi}{4}\end{aligned}$$

Hence (C) is not true.

(D) According to Theorem 2.12, (D) is not true.

Answers: (A), (B)

2. The values of x satisfying the equation

$$\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(x - \frac{2}{x}\right) - \tan^{-1}\left(\frac{4}{x}\right) = 0$$

are

(A) $\sqrt{3}$ (B) $-\sqrt{3}$ (C) $-\sqrt{2}$ (D) $\sqrt{2}$

Solution: The given equation can be written as

$$\tan^{-1}\left(\frac{x + (2/x) - x + (2/x)}{1 + [x^2 - (4/x^2)]}\right) - \tan^{-1}\left(\frac{4}{x}\right) = 0$$

$$\tan^{-1}\left(\frac{4x}{x^4 + x^2 - 4}\right) - \tan^{-1}\left(\frac{4}{x}\right) = 0$$

$$\tan^{-1}\left(\frac{[4x/(x^4 + x^2 - 4)] - (4/x)}{1 + [16/(x^4 + x^2 - 4)]}\right) = 0$$

$$\frac{4x}{x^4 + x^2 - 4} - \frac{4}{x} = 0$$

$$x^2 - (x^4 + x^2 - 4) = 0$$

Therefore

$$x^4 = 4 \quad \text{or} \quad x = \pm\sqrt{2}$$

Answers: (C), (D)

3. When $0 \leq x < 1$, then $\cos^{-1}x$ is equal to

(A) $\sin^{-1}\sqrt{1-x^2}$ (B) $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

(C) $\operatorname{cosec}^{-1}\frac{1}{\sqrt{1-x^2}}$ (D) $\frac{\pi}{2} - \sin^{-1}x$

Solution: Let $\cos^{-1}x = \theta$. Since $0 \leq \cos^{-1}x \leq \pi$ and $x > 0$, it follows that $0 < \theta < \pi/2$.

(A) $\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - x^2}$

Therefore

$$\theta = \sin^{-1}\sqrt{1-x^2}$$

Hence (A) is true.

(B) $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{1-x^2}}{x}$ and hence (B) is true.

(C) $\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{1}{\sqrt{1-x^2}}$

Therefore

$$\theta = \operatorname{cosec}^{-1}\frac{1}{\sqrt{1-x^2}}$$

Hence (C) is also true. By Theorem 2.10 (D) is also true.

Answers: (A), (B), (C), (D)

4. Identify the correct statements.

(A) The number of solutions of the equation

$$\tan^{-1}\left(\frac{x}{1-x^2}\right) + \tan^{-1}\left(\frac{1}{x^3}\right) = \frac{3\pi}{4}$$

belonging to the interval $(0, 1)$ is two.

(B) $\tan^{-1}(1/2) + \tan^{-1}(1/8) + \tan^{-1}(1/18) + \dots$ upto n terms is $(\pi/4) - \tan^{-1}[1/(2n+1)]$.

(C) If $0 \leq x \leq 1$ and $\theta = \sin^{-1}x + \cos^{-1}x - \tan^{-1}x$, then $\pi/4 \leq \theta \leq \pi/2$.

(D) The value of $\sin^{-1}[\sin(4\pi/3)] + \cos^{-1}[\cos(7\pi/6)]$ is $\pi/2$.

Solution:

(A) $\left(\frac{x}{1-x^2}\right)\left(\frac{1}{x^3}\right) = \frac{1}{(1-x^2)x^2} > 1$ because $0 < x < 1$

Therefore

$$\tan^{-1}\left(\frac{x}{1-x^2}\right) + \tan^{-1}\left(\frac{1}{x^3}\right) = \frac{3\pi}{4}$$

$$\Rightarrow \pi + \tan^{-1}\left(\frac{\{x/(1-x^2)\} + (1/x^3)}{1 - \{1/x^2(1-x^2)\}}\right) = \frac{3\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{x^4 - x^2 + 1}{x(-x^4 + x^2 - 1)}\right) = \frac{-\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{-1}{x}\right) = \frac{-\pi}{4}$$

$\Rightarrow x = 1$. But $0 < x < 1$. Therefore (A) is false.

(B) Let

$$u_k = \tan^{-1}\left(\frac{1}{2k^2}\right) \quad \text{for } k = 1, 2, 3, \dots$$

$$= \tan^{-1}\left(\frac{2}{4k^2}\right)$$

$$= \tan^{-1}\left(\frac{2}{4k^2 - 1 + 1}\right)$$

$$= \tan^{-1}\left(\frac{2}{1 + (2k+1)(2k-1)}\right)$$

$$= \tan^{-1}\left(\frac{2k+1 - (2k-1)}{1 + (2k+1)(2k-1)}\right)$$

$$= \tan^{-1}\left(\frac{[1/(2k-1)] - [1/(2k+1)]}{1 + [1/(2k+1)][1/(2k-1)]}\right)$$

$$= \tan^{-1}\left(\frac{1}{2k-1}\right) - \tan^{-1}\left(\frac{1}{2k+1}\right)$$

Therefore

$$u_1 + u_2 + \cdots + u_n = \tan^{-1}(1) - \tan^{-1}\left(\frac{1}{2n+1}\right)$$

Hence (B) is true.

$$(C) \sin^{-1}x + \cos^{-1}x - \tan^{-1}x = \frac{\pi}{2} - \tan^{-1}x$$

Therefore

$$\theta = \frac{\pi}{2} - \tan^{-1}x$$

Now

$$0 \leq x \leq 1 \Rightarrow 0 \leq \tan^{-1}x \leq \frac{\pi}{4}$$

$$\Rightarrow \frac{-\pi}{4} \leq -\tan^{-1}x \leq 0$$

$$\Rightarrow \frac{\pi}{4} \leq \frac{\pi}{2} - \tan^{-1}x \leq \frac{\pi}{2}$$

Therefore

$$\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$$

Hence (C) is true.

$$(D) \sin^{-1}\left(\sin\frac{4\pi}{3}\right) + \cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right) \\ = \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) = \frac{\pi}{2}$$

Hence (D) is also true.

Answers: (B), (C), (D)

5. Which of the following are true?

$$(A) \text{ If } 0 < x < \frac{\pi}{2}, \text{ then}$$

$$\cot^{-1}\left[\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}\right] = \pi - \frac{x}{2}$$

$$(B) \cos^{-1}(x) = 2\sin^{-1}\sqrt{\frac{1-x}{2}}$$

$$(C) \tan^{-1}x + \tan^{-1}\frac{1}{x} = \frac{\pi}{2} \quad \text{if } x > 0$$

$$(D) \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{1}{2}\tan^{-1}(x)$$

Solution:

$$(A) \sqrt{1-\sin x} = \cos\frac{x}{2} - \sin\frac{x}{2} \quad \left(\because \cos\frac{x}{2} > \sin\frac{x}{2}\right)$$

$$\sqrt{1+\sin x} = \cos\frac{x}{2} + \sin\frac{x}{2}$$

Therefore

$$\begin{aligned} \cot^{-1}\left[\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}\right] &= \cot^{-1}\left(-\cot\frac{x}{2}\right) \\ &= \cot^{-1}\left(\cot\left(\pi - \frac{x}{2}\right)\right) \\ &= \pi - \frac{x}{2} \end{aligned}$$

Hence (A) is true.

(B) Let $\cos^{-1}x = \theta$ so that $\cos\theta = x$ where $-1 \leq x \leq 1$ and $0 \leq \theta \leq \pi$. Now

$$2\sin^2\frac{\theta}{2} = 1 - \cos\theta = 1 - x$$

Therefore

$$\sin\frac{\theta}{2} = \sqrt{\frac{1-x}{2}} \quad \left(\because 0 \leq \theta \leq \frac{\pi}{2}\right)$$

$$\frac{\theta}{2} = \sin^{-1}\sqrt{\frac{1-x}{2}}$$

$$\theta = 2\sin^{-1}\sqrt{\frac{1-x}{2}}$$

Hence (B) is true.

(C) When $x > 0$, then we know that $\tan^{-1}(1/x) = \cot^{-1}x$. Therefore

$$\tan^{-1}x + \tan^{-1}\frac{1}{x} = \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

So (C) is true.

(D) Put $x = \tan\theta$ where $-\pi/2 < \theta < \pi/2$. Therefore

$$\begin{aligned} \tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right] &= \tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right) \\ &= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) \\ &= \tan^{-1}\left(\tan\frac{\theta}{2}\right) \\ &= \frac{\theta}{2} \quad \left(\because -\frac{\pi}{4} < \frac{\theta}{2} < \frac{\pi}{4}\right) \\ &= \frac{1}{2}\tan^{-1}x \end{aligned}$$

Hence (D) is also true.

Answers: (A), (B), (C), (D)

Matrix-Match Type Questions

1. Match the items of Column I with those of Column II.

Column I	Column II
(A) If $\sin^{-1}x + \sin^{-1}y = 2\pi/3$, then the value of $\cos^{-1}x + \cos^{-1}y$ is	(p) $\frac{\pi}{4}$
(B) $\tan^{-1}(\sec x + \tan x)$ at $x = \pi/3$ is equal to	(q) $\frac{3\pi}{10}$
(C) For $-1 \leq x \leq 1$, if $\sin^{-1}x = \pi/5$, then $\cos^{-1}x$ equals	(r) $\frac{\pi}{3}$
(D) $2\cot^{-1}5 + \cot^{-1}7 + 2\cot^{-1}8$ equals	(s) $\frac{5\pi}{12}$

Solution:

$$(A) \sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$$

Suppose $\cos^{-1}x + \cos^{-1}y = \alpha$. Adding we get

$$\frac{\pi}{2} + \frac{\pi}{2} = \frac{2\pi}{3} + \alpha$$

Therefore

$$\alpha = \frac{\pi}{3}$$

Answer: (A) $\rightarrow$ (r)

$$\begin{aligned} (B) \tan^{-1}(\sec x + \tan x) &= \tan^{-1}\left(\frac{1 + \sin x}{\cos x}\right) \\ &= \tan^{-1}\left(\frac{1 - \cos\{(\pi/2) + x\}}{\sin\{(\pi/2) + x\}}\right) \\ &= \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right) = \frac{\pi}{4} + \frac{x}{2} \end{aligned}$$

Therefore at $x = \pi/3$,

$$\tan^{-1}(\sec x + \tan x) = \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$$

Answer: (B) $\rightarrow$ (s)

(C) We have

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$$

$$\cos^{-1}x = \frac{\pi}{2} - \frac{\pi}{5} = \frac{3\pi}{10}$$

Answer: (C) $\rightarrow$ (q)

$$\begin{aligned} (D) 2\cot^{-1}5 + \cot^{-1}7 + 2\cot^{-1}8 \\ = 2(\cot^{-1}5 + \cot^{-1}8) + \tan^{-1}\frac{1}{7} \end{aligned}$$

$$\begin{aligned} &= 2\left(\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right) + \tan^{-1}\frac{1}{7} \\ &= 2\tan^{-1}\left(\frac{(1/5) + (1/8)}{1 - (1/40)}\right) + \tan^{-1}\frac{1}{7} \\ &= 2\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\frac{1}{7} \\ &= \tan^{-1}\left(\frac{(2/3)}{1 - (1/9)}\right) + \tan^{-1}\frac{1}{7} \\ &= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right) \\ &= \tan^{-1}\left(\frac{(3/4) + (1/7)}{1 - (3/28)}\right) \\ &= \tan^{-1}\left(\frac{25}{25}\right) = \frac{\pi}{4} \end{aligned}$$

Answer: (D) $\rightarrow$ (p)

2. Match the items of Column I with those of Column II.

Column I	Column II
(A) In $\triangle ABC$, if $\angle A = 90^\circ$, then $\tan^{-1}[b/(c+a)] + \tan^{-1}[c/(a+b)] =$	(p) $\frac{\pi}{4}$
(B) The sides of a triangle are 5, 12 and 13. Then $\sin^{-1}(12/13) + \cos^{-1}(5/13) =$	(q) $\frac{\pi}{2}$
(C) $\sin^{-1}\left(\frac{x-1}{x+1}\right) + \sec^{-1}\left(\frac{x+1}{x-1}\right) =$	(r) $\frac{\pi}{6}$
(D) $\cos^{-1}\left(\frac{3x+2}{2x+3}\right) + \operatorname{cosec}^{-1}\left(\frac{2x+3}{3x+2}\right) =$	(s) π

Solution:

(A) It is given that $a^2 = b^2 + c^2$. Therefore

$$\begin{aligned} &\tan^{-1}\left(\frac{b}{c+a}\right) + \tan^{-1}\left(\frac{c}{a+b}\right) \\ &= \tan^{-1}\left(\frac{\{b/(c+a)\} + \{c/(a+b)\}}{1 - \{bc/(c+a)(a+b)\}}\right) \\ &= \tan^{-1}\left(\frac{b^2 + c^2 + ab + ca}{a^2 + ab + ca}\right) \\ &= \tan^{-1}(1) \quad (\because b^2 + c^2 = a^2) \\ &= \frac{\pi}{4} \end{aligned}$$

Answer: (A) $\rightarrow$ (p)

- (B) The given triangle is a right-angled triangle with hypotenuse 13. Hence the other two angles are $\sin^{-1}(12/13)$ and $\cos^{-1}(5/13)$ whose sum must be equal to $\pi/2$.

Answer: (B) → (q)

- (C) We have

$$\sec^{-1} x = \cos^{-1} \left(\frac{1}{x} \right)$$

Therefore

$$\begin{aligned} & \sin^{-1} \left(\frac{x-1}{x+1} \right) + \sec^{-1} \left(\frac{x+1}{x-1} \right) \\ &= \sin^{-1} \left(\frac{x-1}{x+1} \right) + \cos^{-1} \left(\frac{x-1}{x+1} \right) \\ &= \frac{\pi}{2} \end{aligned}$$

Answer: (C) → (q)

- (D) We have

$$\operatorname{Cosec}^{-1}(x) = \sin^{-1} \left(\frac{1}{x} \right)$$

Therefore

$$\begin{aligned} & \cos^{-1} \left(\frac{3x+2}{2x+3} \right) + \operatorname{Cosec}^{-1} \left(\frac{2x+3}{3x+2} \right) \\ &= \cos^{-1} \left(\frac{3x+2}{2x+3} \right) + \sin^{-1} \left(\frac{3x+2}{2x+3} \right) \\ &= \frac{\pi}{2} \end{aligned}$$

Answer: (D) → (q)

3. Match the items of Column I with the items of Column II.

Column I	Column II
(A) Value of $\sin[\cos^{-1}(3/5)]$ is	(p) $\frac{1}{2}$
(B) $\sin\left(\frac{\pi}{2} - \sin^{-1} \frac{\sqrt{3}}{2}\right) =$	(q) $\frac{-3}{5}$
(C) $\cos\left(\tan^{-1} \frac{3}{4}\right) =$	(r) $\frac{-1}{2}$
(D) $\cos\left(\frac{\pi}{2} - \sin^{-1} \left(\frac{-1}{2}\right)\right) =$	(s) $\frac{4}{5}$

Solution:

$$(A) \sin\left(\cos^{-1} \frac{3}{5}\right) = \sin \theta \left(\text{where } \cos \theta = \frac{3}{5}, 0 < \theta < \pi \right)$$

$$\begin{aligned} &= \sqrt{1 - \cos^2 \theta} \\ &= \sqrt{1 - \frac{9}{25}} \\ &= \frac{4}{5} \end{aligned}$$

Answer: (A) → (s)

$$\begin{aligned} (B) \sin\left(\frac{\pi}{2} - \sin^{-1} \frac{\sqrt{3}}{2}\right) &= \sin\left(\frac{\pi}{2} - \frac{\pi}{3}\right) \\ &= \sin \frac{\pi}{6} \\ &= \frac{1}{2} \end{aligned}$$

Answer: (B) → (p)

$$(C) \cos\left(\tan^{-1} \frac{3}{4}\right) = \cos \theta \left(\text{where } 0 < \theta < \frac{\pi}{2} \text{ and } \tan \theta = \frac{3}{4} \right)$$

$$\begin{aligned} &= \frac{1}{\sqrt{1 + \tan^2 \theta}} \\ &= \frac{1}{\sqrt{1 + \frac{9}{16}}} \\ &= \frac{4}{5} \end{aligned}$$

Answer: (C) → (s)

$$\begin{aligned} (D) \cos\left(\frac{\pi}{2} - \sin^{-1} \left(\frac{-1}{2}\right)\right) &= \cos\left(\frac{\pi}{2} - \left(\frac{-\pi}{6}\right)\right) \\ &= \cos\left(\frac{2\pi}{3}\right) \\ &= \frac{-1}{2} \end{aligned}$$

Answer: (D) → (r)

4. Match the items of Column I with those of Column II.

Column I	Column II
(A) If $1/\sqrt{2} \leq x \leq 1$, then $2\sin^{-1}x$ is equal to	(p) $\pi - \sin^{-1}(3x - 4x^3)$
(B) If $1/2 < x \leq 1$, then $3\sin^{-1}x$ is equal to	(q) $\pi - \sin^{-1}(2x\sqrt{1-x^2})$
(C) If $-1 \leq x \leq -1/\sqrt{2}$, then $2\sin^{-1}x$ equals	(r) $-\pi - \sin^{-1}(3x - 4x^3)$
(D) $-1 < x \leq -1/2$, then $3\sin^{-1}x$ is	(s) $-\pi - \sin^{-1}(2x\sqrt{1-x^2})$

Solution:

- (A) It is given that $1/\sqrt{2} \leq x \leq 1$. Put $\sin^{-1}x = \theta$ so that $\pi/4 \leq \theta \leq \pi/2$ and $\sin \theta = x$. Therefore

$$\begin{aligned}\frac{\pi}{2} \leq 2\theta \leq \pi &\Rightarrow -\pi \leq -2\theta \leq \frac{-\pi}{2} \\ &\Rightarrow 0 \leq \pi - 2\theta \leq \frac{\pi}{2}\end{aligned}$$

Also

$$\sin(\pi - 2\theta) = \sin 2\theta = 2x\sqrt{1-x^2} \quad (\because 0 \leq 2x\sqrt{1-x^2} \leq 1)$$

Therefore

$$\begin{aligned}\pi - 2\theta &= \sin^{-1}(2x\sqrt{1-x^2}) \\ 2\sin^{-1}x &= \pi - \sin^{-1}(2x\sqrt{1-x^2})\end{aligned}$$

Answer: (A) $\rightarrow$ (q)

- (B) It is given that $1/2 < x \leq 1$. Put $\sin^{-1}x = \theta$ so that $\pi/6 < \theta < \pi/2$ and $\sin \theta = x$. We know that

$$\sin 3\theta = 3\sin \theta - 4\sin^3 \theta = 3x - 4x^3$$

Therefore

$$\begin{aligned}\frac{\pi}{6} < \theta \leq \frac{\pi}{2} &\Rightarrow \frac{\pi}{2} < 3\theta \leq \frac{3\pi}{2} \\ &\Rightarrow \frac{-3\pi}{2} \leq -3\theta < \frac{-\pi}{2} \\ &\Rightarrow \frac{-\pi}{2} \leq \pi - 3\theta < \frac{\pi}{2}\end{aligned}$$

This implies

$$\begin{aligned}\sin(\pi - 3\theta) &= \sin 3\theta = 3x - 4x^3 \\ \pi - 3\theta &= \sin^{-1}(3x - 4x^3) \\ 3\theta &= \pi - \sin^{-1}(3x - 4x^3)\end{aligned}$$

Hence

$$3\sin^{-1}x = \pi - \sin^{-1}(3x - 4x^3)$$

Answer: (B) $\rightarrow$ (p)

- (C) It is given that $-1 \leq x \leq -1/\sqrt{2}$. Put $\sin^{-1}x = \theta$ so that $-\pi/2 \leq \theta \leq -\pi/4$ and $\sin \theta = x$. Therefore

$$-\pi \leq 2\theta \leq \frac{-\pi}{2}$$

This implies

$$0 \leq \pi + 2\theta \leq \frac{\pi}{2}$$

and so

$$\sin(\pi + 2\theta) = -\sin 2\theta$$

$$= -2\sin \theta \cos \theta = -2x\sqrt{1-x^2}$$

Now

$$\begin{aligned}\pi + 2\theta &= -\sin^{-1}(2x\sqrt{1-x^2}) \\ 2\theta &= -\pi - \sin^{-1}(2x\sqrt{1-x^2})\end{aligned}$$

Therefore

$$2\sin^{-1}x = -\pi - \sin^{-1}(2x\sqrt{1-x^2})$$

Answer: (C) $\rightarrow$ (s)

- (D) It is given that $-1 \leq x \leq -1/2$. Put $\sin^{-1}x = \theta$. Therefore $x = \sin \theta$ and

$$\begin{aligned}\frac{-\pi}{2} \leq \theta < \frac{-\pi}{6} \\ \frac{-3\pi}{2} \leq 3\theta < \frac{-\pi}{2} \\ -\frac{\pi}{2} \leq \pi + 3\theta < \frac{\pi}{2}\end{aligned}$$

Now

$$\begin{aligned}\sin 3\theta &= 3\sin \theta - 4\sin^3 \theta = 3x - 4x^3 \\ \sin(\pi + 3\theta) &= -\sin 3\theta = -(3x - 4x^3)\end{aligned}$$

Therefore

$$\begin{aligned}\pi + 3\theta &= -\sin^{-1}(3x - 4x^3) \\ 3\theta &= -\pi - \sin^{-1}(3x - 4x^3) \\ 3\sin^{-1}x &= -\pi - \sin^{-1}(3x - 4x^3)\end{aligned}$$

Answer: (D) $\rightarrow$ (r)

5. In Column I, some equations are given and in Column II their corresponding solutions are given. Match them.

Column I	Column II
(A) $2\tan^{-1}(2x+1) = \cos^{-1}x$	(p) $\sqrt{3}$
(B) $3(\tan^{-1}x)^2 - 4\pi(\tan^{-1}x) + \pi^2 = 0$	(q) 1
(C) $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(3x-2)$	(r) 0
(D) $\tan^{-1}\left(x + \frac{1}{2}\right) + \tan^{-1}\left(x - \frac{1}{2}\right) = \frac{\pi}{4}$	(s) $\frac{1}{2}$

Solution:

- (A) $\cos^{-1}x$ is defined when $-1 \leq x \leq 1$. Put $\tan^{-1}(2x+1) = \theta$ so that $\tan \theta = 2x+1$. Therefore

$$\cos^{-1}x = 2\theta$$

$$\Rightarrow x = \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - (2x+1)^2}{1 + (2x+1)^2}$$

$$\begin{aligned} \Rightarrow x + x(2x + 1)^2 &= 1 - (2x + 1)^2 \\ \Rightarrow x + 4x^3 + 4x^2 + x &= 1 - 4x^2 - 4x - 1 \\ \Rightarrow 4x^3 + 8x^2 + 6x &= 0 \\ \Rightarrow x(2x^2 + 4x + 3) &= 0 \end{aligned}$$

The equation $2x^2 + 4x + 3 = 0$ has no real solutions. Therefore $x = 0$ satisfies the original equation.

Answer: (A) → (r)

(B) Put $\tan^{-1} x = y$. Then the equation transforms to

$$\begin{aligned} 3y^2 - 4\pi y + \pi^2 &= 0 \\ (3y - \pi)(y - \pi) &= 0 \end{aligned}$$

Therefore

$$y = \frac{\pi}{3} \quad \text{or} \quad y = \pi$$

This implies

$$\tan^{-1} x = \frac{\pi}{3} \quad \text{or} \quad \tan^{-1} x = \pi$$

Since $-\pi/2 < \tan^{-1} x < \pi/2$, it follows that

$$\tan^{-1} x = \frac{\pi}{3}$$

Therefore

$$x = \tan \frac{\pi}{3} = \sqrt{3}$$

Answer: (B) → (p)

(C) The given equation can be written as

$$\sin^{-1} x - \left(\frac{\pi}{2} - \sin^{-1} x \right) = \sin^{-1}(3x - 2)$$

Therefore

$$2\sin^{-1} x - \frac{\pi}{2} = \sin^{-1}(3x - 2)$$

$$3x - 2 = \sin \left(2\theta - \frac{\pi}{2} \right) \quad (\text{where } \theta = \sin^{-1} x)$$

$$= -\cos 2\theta$$

$$= -(1 - 2\sin^2 \theta)$$

$$= 2x^2 - 1$$

Therefore

$$2x^2 - 3x + 1 = 0$$

$$(2x - 1)(x - 1) = 0$$

Hence $x = 1/2, 1$ and these values satisfy the original equation.

Answer: (C) → (q), (s)

(D) The given equation can be transformed to

$$\tan^{-1} \left(\frac{x + (1/2) + x - (1/2)}{1 - [x^2 - (1/4)]} \right) = \frac{\pi}{4}$$

$$\frac{2x}{(5/4) - x^2} = 1$$

$$4x^2 + 8x - 5 = 0$$

$$(2x - 1)(2x + 5) = 0$$

Therefore

$$x = \frac{1}{2} \quad \text{or} \quad x = \frac{-5}{2}$$

$x = 1/2$ satisfies the equation and $x = -5/2$ does not.

Answer: (D) → (s)

Comprehension-Type Questions

1. Passage: To solve an equation of the form $af(x) + bg(x) = ch(x)$, where $f(x)$, $g(x)$ and $h(x)$ are some inverse trigonometric functions reduce the equation to an algebraic equation by calculating a certain trigonometric function of both sides and isolate the extraneous roots by verification. Answer the following three questions.

(i) If $\sin^{-1} 6x + \sin^{-1}(6\sqrt{3}x) = -\pi/2$, then the value of x is

(A) $\frac{1}{12}$ (B) $\frac{-1}{12}$ (C) $\frac{1}{13}$ (D) $\frac{-1}{13}$

(ii) The number of solutions of the equation

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{3\pi}{4}$$

is

(A) 0 (B) 1 (C) 2 (D) 4

(iii) $\sin^{-1}(2x) = 3\sin^{-1} x$ has solutions whose number is

(A) 0 (B) 1 (C) 2 (D) 3

Solution:

- (i) Observe that the equation has meaning if $-1/6\sqrt{3} \leq x \leq 1/6\sqrt{3}$. This gives

$$\begin{aligned}\sin^{-1} 6x &= \frac{-\pi}{2} - \sin^{-1}(6\sqrt{3}x) \\ \Rightarrow 6x &= -\sin\left(\frac{\pi}{2} + \sin^{-1}(6\sqrt{3}x)\right) \\ &= -\cos(\sin^{-1}(6\sqrt{3}x)) \\ &= -\sqrt{1 - 108x^2} \\ \Rightarrow 36x^2 &= 1 - 108x^2\end{aligned}$$

Hence

$$144x^2 = 1 \quad \text{or} \quad x = \pm \frac{1}{12}$$

So $x = -1/12$ is a solution and $x = 1/12$ is not a solution because $x = 1/12$ makes LHS positive and RHS negative.

Answer: (B)

- (ii) Put $\tan^{-1}(2x) = \alpha$, $\tan^{-1}(3x) = \beta$. Therefore $\tan \alpha = 2x$ and $\tan \beta = 3x$. Now

$$\begin{aligned}-1 &= \tan \frac{3\pi}{4} = \tan(\alpha + \beta) \\ &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{5x}{1 - 6x^2}\end{aligned}$$

Therefore

$$\begin{aligned}6x^2 - 5x - 1 &= 0 \\ (6x + 1)(x - 1) &= 0\end{aligned}$$

The solution is $x = 1, -1/6$. We can verify that $x = -1/6$ is not a solution. Therefore $x = 1$ is the only solution and it satisfies the given equation.

Answer: (B)

- (iii) Put $\sin^{-1} x = \theta$ so that $-1 \leq x \leq 1$ and $\sin \theta = x$. Therefore

$$\begin{aligned}2x &= \sin 3\theta = 3\sin \theta - 4\sin^3 \theta = 3x - 4x^3 \\ \Rightarrow 4x^3 - x &= 0 \\ \Rightarrow x &= 0, \pm \frac{1}{2}\end{aligned}$$

Answer: (D)

2. Passage: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) + \lambda\pi$ where

$$\lambda = \begin{cases} 0 & \text{if } xy < 1 \\ -1 & \text{if } xy > 1 \text{ and } x < 0 \\ 1 & \text{if } xy > 1 \text{ and } x > 0 \end{cases}$$

$$\text{and } \tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right) + \mu\pi$$

where

$$\mu = \begin{cases} 0 & \text{if } xy > -1 \\ 1 & \text{if } x > 0, y < 0 \text{ and } xy < -1 \\ -1 & \text{if } x < 0, y > 0 \text{ and } xy < -1 \end{cases}$$

Answer the following questions:

- (i) $\tan^{-1}(5) + \tan^{-1}(3) - \cot^{-1}\left(\frac{4}{7}\right)$ is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{4}$ (D) $\frac{-\pi}{2}$

- (ii) $3\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{1}{20}\right) + \tan^{-1}\left(\frac{1}{1985}\right)$ equals

(A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

- (iii) The number of solutions of the equation

$$\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{8}{31}\right)$$

when $-\sqrt{2} < x < \sqrt{2}$ is

(A) 1 (B) 2 (C) 3 (D) 0

Solution:

- (i) $x = 5, y = 3$ and $xy = 15 > 1$. Therefore

$$\begin{aligned}\tan^{-1}(5) + \tan^{-1}(3) &= \pi + \tan^{-1}\left(\frac{5+3}{1-15}\right) \\ &= \pi - \tan^{-1}\left(\frac{4}{7}\right)\end{aligned}$$

$$\tan^{-1}(5) + \tan^{-1}(3) - \cot^{-1}\left(\frac{4}{7}\right)$$

$$= \pi - \left(\tan^{-1} \frac{4}{7} + \cot^{-1} \frac{4}{7} \right)$$

$$= \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

Answer: (A)

- (ii) $3\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{1}{20}\right) + \tan^{-1}\left(\frac{1}{1985}\right)$

$$= \tan^{-1}\left(\frac{(3/4) - (1/64)}{1 - (3/16)}\right) + \tan^{-1} \frac{1}{20}$$

$$+ \tan^{-1}\left(\frac{1}{1985}\right) \left(\because \frac{1}{4} < \frac{1}{\sqrt{3}} \right)$$

$$\begin{aligned}
 &= \tan^{-1}\left(\frac{47}{52}\right) + \tan^{-1}\frac{1}{20} + \tan^{-1}\left(\frac{1}{1985}\right) \\
 &= \tan^{-1}\left(\frac{(47/52) + \frac{1}{20}}{1 - (47/1040)}\right) + \tan^{-1}\left(\frac{1}{1985}\right) \\
 &= \tan^{-1}\left(\frac{992}{993}\right) + \tan^{-1}\left(\frac{1}{1985}\right) \\
 &= \tan^{-1}\left(\frac{1 - (1/1985)}{1 + (1/1985)}\right) + \tan^{-1}\left(\frac{1}{1985}\right) \\
 &= \tan^{-1}(1) - \tan^{-1}\left(\frac{1}{1985}\right) + \tan^{-1}\left(\frac{1}{1985}\right) \\
 &= \frac{\pi}{4}
 \end{aligned}$$

Answer: (D)

(iii) $-\sqrt{2} < x < \sqrt{2} \Rightarrow x^2 - 1 < 1$. Therefore

$$\begin{aligned}
 \tan^{-1}(x+1) + \tan^{-1}(x-1) &= \tan^{-1}\left(\frac{8}{31}\right) \\
 \Rightarrow \tan^{-1}\left(\frac{x+1+x-1}{1-(x^2-1)}\right) &= \tan^{-1}\left(\frac{8}{31}\right) \\
 \Rightarrow \tan^{-1}\left(\frac{2x}{2-x^2}\right) &= \tan^{-1}\left(\frac{8}{31}\right) \\
 \Rightarrow \frac{2x}{2-x^2} &= \frac{8}{31} \\
 \Rightarrow 8x^2 + 62x - 16 &= 0 \\
 \Rightarrow 4x^2 + 31x - 8 &= 0 \\
 \Rightarrow 4x^2 + 32x - x - 8 &= 0 \\
 \Rightarrow (x+8)(4x-1) &= 0
 \end{aligned}$$

Therefore $x = 1/4$ [since $-8 \notin (-\sqrt{2}, \sqrt{2})$] and one can check that $x = 1/4$ satisfies the initial equation.

Answer: (A)

3. Passage: Consider the equation

$$\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$$

Answer the following questions.

(i) If $a = 1$ and $b = 0$, then (x, y) satisfies the equation

(A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$

(C) $x^2 + y^2 = \frac{1}{2}$ (D) $y = x$

(ii) If $a = 1, b = 1$, then (x, y) is a solution of the equation

(A) $(x^2 + 1)(y^2 - 1) = 0$ (B) $(x^2 - 1)(y^2 + 1) = 0$

(C) $(x^2 - 1)(y^2 - 1) = 0$ (D) $x^2 + y^2 = \sqrt{2}$

(iii) If $a = 2, b = 2$, then (x, y) satisfies the equation

(A) $(x^2 - 1)(4y^2 - 1) = 0$

(B) $(4x^2 - 1)(y^2 - 1) = 0$

(C) $(4x^2 - 1)(y^2 - 2) = 0$

(D) $(x^2 - 2)(4y^2 - 1) = 0$

Solution:

(i) When $a = 1$ and $b = 0$, the given equation reduces to

$$\sin^{-1}x + \cos^{-1}y = 0 \quad \left(\because \cos^{-1}(0) = \frac{\pi}{2} \right)$$

Therefore

$$\sin^{-1}x = \cos^{-1}y = -\sin^{-1}\sqrt{1-y^2}$$

This gives

$$x = -\sqrt{1-y^2} \quad \text{or} \quad x^2 + y^2 = 1$$

Answer: (A)

(ii) When $a = 1, b = 1$, the equation is

$$\sin^{-1}x + \cos^{-1}y + \cos^{-1}xy = \frac{\pi}{2}$$

Therefore

$$\sin^{-1}x + \cos^{-1}y = \frac{\pi}{2} - \cos^{-1}(xy) = \sin^{-1}xy$$

$$xy = \sin(\sin^{-1}x + \cos^{-1}y)$$

$$= xy - \sqrt{1-x^2}\sqrt{1-y^2}$$

$$(x^2 - 1)(y^2 - 1) = 0$$

Answer: (C)

(iii) When $a = 2, b = 2$, the equation becomes

$$\sin^{-1}(2x) + \cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2}$$

Therefore

$$\sin^{-1}2x + \cos^{-1}y = \frac{\pi}{2} - \cos^{-1}(2xy) = \sin^{-1}(2xy)$$

$$2xy = \sin(\sin^{-1}2x + \cos^{-1}y)$$

$$= 2xy - \sqrt{1-4x^2}\sqrt{1-y^2}$$

$$(1-4x^2)(1-y^2) = 0$$

That is

$$(4x^2 - 1)(y^2 - 1) = 0$$

Answer: (B)

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true

1. Statement I: $\tan\left(\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right) = \frac{17}{6}$

Statement II: If $-1 \leq x \leq 1$, then $\cos^{-1}x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

Solution: Statement II is false, because

$$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

whereas

$$\tan^{-1}\left(\frac{\sqrt{1-(1/4)}}{(-1/2)}\right) = \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$$

But,

$$\begin{aligned} \tan\left(\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right) \\ = \tan\left(\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3}\right) \\ = \tan(\alpha + \beta) \end{aligned}$$

where

$$\alpha = \tan^{-1}\frac{3}{4} \quad \text{and} \quad \beta = \tan^{-1}\frac{2}{3}$$

Now

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{(3/4) + (2/3)}{1 - (3/4) \cdot (2/3)} = \frac{17}{6}$$

Answer: (C)

2. Statement I: $\cos^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{63}{65}\right) = \frac{\pi}{2}$

Statement II: If $0 < x, y \leq 1$, then $\cos^{-1}x + \cos^{-1}y = \cos^{-1}(xy - \sqrt{1-x^2} \cdot \sqrt{1-y^2})$

Solution: Put $\cos^{-1}x = \alpha$ and $\cos^{-1}y = \beta$ so that $0 \leq \alpha < \pi/2$ and $0 \leq \beta < \pi/2$ and hence $0 \leq \alpha + \beta < \pi$. Also

$$\begin{aligned} \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= xy - \sqrt{1-x^2} \sqrt{1-y^2} \\ \Rightarrow \alpha + \beta &= \cos^{-1}(xy - \sqrt{1-x^2} \sqrt{1-y^2}) \end{aligned}$$

Therefore Statement II is true. Now

$$\begin{aligned} \cos^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{63}{65}\right) \\ = \cos^{-1}\left(\frac{3}{5} \cdot \frac{12}{13} - \sqrt{1-\frac{9}{25}} \cdot \sqrt{1-\frac{144}{169}}\right) + \cos^{-1}\left(\frac{63}{65}\right) \\ = \cos^{-1}\left(\frac{36}{65} - \frac{4}{5} \cdot \frac{5}{13}\right) + \cos^{-1}\left(\frac{63}{65}\right) \\ = \cos^{-1}\left(\frac{16}{65}\right) + \cos^{-1}\left(\frac{63}{65}\right) \\ = \sin^{-1}\sqrt{1-\frac{16^2}{65^2}} + \cos^{-1}\left(\frac{63}{65}\right) \\ = \sin^{-1}\left(\frac{63}{65}\right) + \cos^{-1}\left(\frac{63}{65}\right) \\ = \frac{\pi}{2} \end{aligned}$$

Answer: (A)

3. Statement I: $\sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) + \sin^{-1}\left(\frac{16}{65}\right) = \frac{\pi}{4}$

Statement II: If $x > 0, y > 0$ and $x^2 + y^2 < 1$, then $\sin^{-1}x + \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$

Solution: Put $\sin^{-1}x = \alpha$ and $\sin^{-1}y = \beta$. Therefore

$$0 < \alpha < \frac{\pi}{2}, \quad 0 < \beta < \frac{\pi}{2}$$

So

$$\begin{aligned} x^2 + y^2 < 1 &\Rightarrow 0 < \alpha + \beta < \frac{\pi}{2} \\ \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= x\sqrt{1-y^2} + y\sqrt{1-x^2} \\ 0 < \alpha + \beta < \frac{\pi}{2} &\Rightarrow \alpha + \beta \\ &= \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) \end{aligned}$$

Therefore Statement II is true.

Now

$$\begin{aligned}
 & \sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) + \sin^{-1}\left(\frac{16}{65}\right) \\
 &= \sin^{-1}\left(\frac{4}{5} \cdot \sqrt{1 - \left(\frac{5}{13}\right)^2} + \frac{5}{13} \sqrt{1 - \left(\frac{4}{5}\right)^2}\right) + \sin^{-1}\left(\frac{16}{65}\right) \\
 &= \sin^{-1}\left(\frac{4}{5} \cdot \frac{12}{13} + \frac{5}{13} \cdot \frac{3}{5}\right) + \sin^{-1}\left(\frac{16}{65}\right) \\
 &= \sin^{-1}\left(\frac{63}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right) \\
 &= \cos^{-1}\left(\frac{16}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

Hence Statement I is false.

Answer: (D)

4. Statement I: If $x > y > z > 0$, then

$$\cot^{-1}\left(\frac{xy+1}{x-y}\right) + \cot^{-1}\left(\frac{yz+1}{y-z}\right) + \cot^{-1}\left(\frac{zx+1}{z-x}\right) = \pi$$

$$\text{Statement II: } \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1}x, & \text{if } x > 0 \\ -\pi + \cot^{-1}x, & \text{if } x < 0 \end{cases}$$

Solution: Let $\cot^{-1}x = \theta$ so that $0 < \theta < \pi$ and $\cot \theta = x$.
Therefore

$$\begin{aligned}
 \tan \theta &= \frac{1}{x} \\
 \tan^{-1}\left(\frac{1}{x}\right) &= \begin{cases} \theta, & \text{if } 0 < \theta < \frac{\pi}{2} \\ -\pi + \theta & \text{if } \frac{\pi}{2} < \theta < \pi \end{cases} \\
 &= \begin{cases} \cot^{-1}x, & \text{if } 0 < \cot^{-1}x < \frac{\pi}{2} \\ -\pi + \cot^{-1}x, & \text{if } \frac{\pi}{2} < \cot^{-1}x < \pi \end{cases} \\
 &= \begin{cases} \cot^{-1}x, & \text{if } x > 0 \\ -\pi + \cot^{-1}x, & \text{if } x < 0 \end{cases}
 \end{aligned}$$

Therefore Statement II is true.

Integer Answer Type Questions

1. If $\cos(2\sin^{-1}x) = 1/9$, then the value of $3|x|$ is ____.

Solution: Put $\sin^{-1}x = \theta$ so that $\sin \theta = x$.

Now $x > y > z > 0 \Rightarrow x - y > 0$, $y - z > 0$ and $z - x < 0$.
Therefore

$$\begin{aligned}
 & \cot^{-1}\left(\frac{xy+1}{x-y}\right) + \cot^{-1}\left(\frac{yz+1}{y-z}\right) + \cot^{-1}\left(\frac{zx+1}{z-x}\right) \\
 &= \tan^{-1}\left(\frac{x-y}{xy+1}\right) + \tan^{-1}\left(\frac{y-z}{yz+1}\right) + \pi + \tan^{-1}\left(\frac{z-x}{zx+1}\right) \\
 &= (\tan^{-1}x - \tan^{-1}y) + (\tan^{-1}y - \tan^{-1}z) + \\
 &\quad (\tan^{-1}z - \tan^{-1}x) + \pi \\
 &= \pi
 \end{aligned}$$

Therefore Statement I is also true and Statement II is a correct explanation of Statement I.

Answer: (A)

5. Statement I: For $x > 14$, if $\sin^{-1}(14/x) + \sin^{-1}(2\sqrt{15}/x) = \pi/2$, then $x = 16$

Statement II: If $0 < x \leq 1$, then $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2}$

Solution: Statement II is obviously true.

$$\begin{aligned}
 \sin^{-1}\left(\frac{14}{x}\right) &= \frac{\pi}{2} - \sin^{-1}\left(\frac{2\sqrt{15}}{x}\right) \\
 &= \cos^{-1}\left(\frac{2\sqrt{15}}{x}\right) \\
 &= \sin^{-1}\sqrt{1 - \frac{60}{x^2}}
 \end{aligned}$$

Therefore

$$\begin{aligned}
 \frac{14}{x} &= \frac{\sqrt{x^2 - 60}}{x} \\
 \Rightarrow x^2 - 60 &= 196 \\
 \Rightarrow x^2 &= 256 \\
 \Rightarrow x &= 16
 \end{aligned}$$

Therefore Statements I and II are true and Statement II is a correct explanation of Statement I.

Answer: (A)

Now,

$$\frac{1}{9} = \cos 2\theta = 1 - 2\sin^2 \theta = 1 - 2x^2$$

Therefore

$$2x^2 = 1 - \frac{1}{9} = \frac{8}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \pm \frac{2}{3}$$

or

$$3|x| = 2$$

Answer: 2

2. $1 + \cos[\cos^{-1}(-\sqrt{3}/2) + (\pi/6)]$ is equal to ____.

Solution:

$$\begin{aligned} 1 + \cos\left(\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \frac{\pi}{6}\right) &= 1 + \cos\left(\frac{5\pi}{6} + \frac{\pi}{6}\right) \\ &= 1 + \cos(\pi) = 0 \end{aligned}$$

Answer: 0

3. Number of values of x satisfying the equation $\sec^{-1}x + \sin^{-1}x = \pi/2$ is ____.

Solution: $\sec^{-1}x$ is defined for $|x| \geq 1$ whereas $\sin^{-1}x$ is defined for $|x| \leq 1$. Therefore $\sec^{-1}x + \sin^{-1}x$ is defined when $|x| = 1$ in which case their sum is $\pi/2$.

Answer: 2

4. If $x > 1$, then the value of

$$-\sin\left(\frac{1}{2}\tan^{-1}\left(\frac{2x}{1-x^2}\right) - \tan^{-1}x\right)$$

is ____.

Solution:

$$\begin{aligned} &-\sin\left(\frac{1}{2}\tan^{-1}\left(\frac{2x}{1-x^2}\right) - \tan^{-1}x\right) \\ &= -\sin\left(\frac{1}{2}(-\pi + 2\tan^{-1}x) - \tan^{-1}x\right) \\ &= -\sin\left(\frac{-\pi}{2}\right) = 1 \end{aligned}$$

Answer: 1

5. $2\pi + \tan^{-1}(\tan 5)$ is ____.

Solution: 5 does not lie between $-\pi/2$ and $\pi/2$. But

$$5 - 2\pi = 5 - \left(\frac{44}{7}\right) = 5 - (6.28) = -(1.28)$$

lies between $-\pi/2$ and $\pi/2$. Also

$$\tan^{-1}(\tan 5) = \tan^{-1}(\tan(5 - 2\pi)) = 5 - 2\pi$$

Therefore

$$2\pi + \tan^{-1}(\tan 5) = 5$$

Answer: 5

SUMMARY

Inverse Trigonometric Functions

2.1 Definition of $\sin^{-1}x$ or $\arcsin x$: The function f :

$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$ defined by $f(x) = \sin x$ is a bijection and hence it has unique inverse function $g: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin(f(y)) = y$ for all $-1 \leq y \leq 1$

and $g(\sin x) = x$ for all $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. This function

$g: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is called the inverse of sine function and is noted by $\sin^{-1}x$ or $\arcsin x$.



QUICK LOOK

1. For $-1 \leq x \leq 1$, and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, $\sin^{-1}x = y \Leftrightarrow \sin y = x$.

2. Domain of $\sin^{-1}x$ is the closed interval $[-1, 1]$ and range is the closed interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

3. $\sin(\sin^{-1}x) = x$ for all $x \in [-1, 1]$ and $\sin^{-1}(\sin y) = y$ for all $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

4. $\sin^{-1}x = 0 \Leftrightarrow x = 0$.

5. $\sin^{-1}x \in \left[-\frac{\pi}{2}, 0\right] \Leftrightarrow y \in [-1, 0]$ and $\sin^{-1}x \in \left[0, \frac{\pi}{2}\right] \Leftrightarrow x \in [0, 1]$.

2.2 Table showing the domains and ranges of inverse trigonometric functions

Function	Domain	Range
$\sin^{-1}x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1}x$	$\mathbb{R}$	$(0, \pi)$
$\operatorname{cosec}^{-1}x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
$\sec^{-1}x$	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

2.3 Fundamental Identities:

- (1) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ for all $-1 \leq x \leq 1$
- (2) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$ for all real x
- (3) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$ for all $x \in (-\infty, -1] \cup [1, \infty)$
- (4) $\sin(\cos^{-1}x) = \sqrt{1-x^2}$ and $\cos(\sin^{-1}x) = \sqrt{1-x^2}$
- (5) $\sin(\tan^{-1}x) = \frac{x}{\sqrt{1+x^2}}$ and $\cos(\tan^{-1}x) = \frac{1}{\sqrt{1+x^2}}$
- (6) $\tan(\sin^{-1}x) = \frac{x}{\sqrt{1-x^2}}$ for $-1 < x < 1$
- (7) $\tan(\cos^{-1}x) = \frac{\sqrt{1-x^2}}{x}$ for $0 \neq x \in [-1, 1]$
- (8) $\tan(\cot^{-1}x) = \frac{1}{x}$ for $x \neq 0$
- (9) $\sin(\cot^{-1}x) = \frac{1}{\sqrt{1+x^2}}$
- (10) $\cos(\cot^{-1}x) = \frac{x}{\sqrt{1+x^2}}$
- (11) $\sin^{-1}x = \cos^{-1}(\sqrt{1-x^2})$ for $0 \leq x \leq 1$
- (12) $\sin^{-1}\frac{x}{\sqrt{1+x^2}} = \cos^{-1}\frac{1}{\sqrt{1+x^2}} = \cot^{-1}\frac{1}{x} = \tan^{-1}x$
for all $x > 0$
- (13) $\sin^{-1}(-x) = -\sin^{-1}x$ and $\cos^{-1}(-x) = \pi - \cos^{-1}x$
for $-1 \leq x \leq 1$
- (14) $\tan^{-1}(-x) = -\tan^{-1}x$

2.4 Useful Formulae:

$$(1) \tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \frac{\pi}{2} & \text{if } x > 0 \\ -\frac{\pi}{2} & \text{if } x < 0 \end{cases}$$

(2) (a) If x and y are positive real numbers, then

$$\tan^{-1}x + \tan^{-1}y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy < 1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy > 1 \\ \frac{\pi}{2} & \text{if } xy = 1 \end{cases}$$

(b) If x and y are negative real numbers, then

$$\tan^{-1}x + \tan^{-1}y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy < 1 \\ -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy > 1 \\ -\frac{\pi}{2} & \text{if } xy = 1 \end{cases}$$

2.5 (a) If x and y are positive reals, then

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$$

(b) Let x, y be reals and $xy \neq -1$. Then

$$\begin{aligned} &\tan^{-1}x - \tan^{-1}y \\ &= \begin{cases} \tan^{-1}\left(\frac{x-y}{1+xy}\right) & \text{if } xy > -1 \\ \pi + \tan^{-1}\left(\frac{x-y}{1+xy}\right) & \text{if } x > 0, y < 0 \text{ and } xy < -1 \\ -\pi + \tan^{-1}\left(\frac{x-y}{1+xy}\right) & \text{if } x < 0, y > 0 \text{ and } xy < -1 \end{cases} \end{aligned}$$

2.6 If $x \geq 0, y \geq 0$ and $x^2 + y^2 \leq 1$, then

$$\sin^{-1}x + \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$$

2.7 If $-1 \leq x, y \leq 1, xy < 0$ and $x^2 + y^2 > 1$, then

$$\sin^{-1}x + \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$$

2.8 If $-1 \leq x, y < 0$ and $x^2 + y^2 > 1$, then

$$\sin^{-1}x + \sin^{-1}y = -\pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$$

2.9 If $-1 \leq x, y \leq 1$ and $x^2 + y^2 \leq 1$, then

$$\sin^{-1}x - \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$$

2.10 If $0 < x, y \leq 1$ and $x^2 + y^2 > 1$, then

$$\sin^{-1}x - \sin^{-1}y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$$

2.11 If $0 < x \leq 1, -1 \leq y < 0$ and $x^2 + y^2 > 1$, then

$$\sin^{-1}x - \sin^{-1}y = \pi - \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$$

2.12 If $-1 \leq x < 0, 0 < y \leq 1$ and $x^2 + y^2 > 1$, then

$$\sin^{-1}x - \sin^{-1}y = -\pi - \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$$

2.13 If $-1 \leq x, y \leq 1$ and $x + y \geq 0$, then

$$\cos^{-1}x + \cos^{-1}y = \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$$

2.14 If $-1 \leq x, y \leq 1$ and $x + y < 0$, then

$$\cos^{-1}x + \cos^{-1}y = 2\pi - \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$$

2.15 If $-1 \leq x, y \leq 1$ and $x \leq y$, then

$$\cos^{-1}x - \cos^{-1}y = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$$

2.16 If $-1 \leq y \leq 0, 0 < x \leq 1$, then

$$\cos^{-1}x - \cos^{-1}y = -\cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$$

$$\mathbf{2.17} \quad 2\sin^{-1}x = \begin{cases} \sin^{-1}(2x\sqrt{1-x^2}), & \text{if } -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ \pi - \sin^{-1}(2x\sqrt{1-x^2}), & \text{if } \frac{1}{\sqrt{2}} \leq x \leq 1 \\ -\pi - \sin^{-1}(2x\sqrt{1-x^2}), & \text{if } -1 \leq x \leq -\frac{1}{\sqrt{2}} \end{cases}$$

$$\mathbf{2.18} \quad 3\sin^{-1}x = \begin{cases} \sin^{-1}(3x - 4x^3), & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - \sin^{-1}(3x - 4x^3), & \text{if } \frac{1}{2} < x \leq 1 \\ -\pi - \sin^{-1}(3x - 4x^3), & \text{if } -1 \leq x < -\frac{1}{2} \end{cases}$$

$$\mathbf{2.19} \quad 2\cos^{-1}x = \begin{cases} \cos^{-1}(2x^2 - 1), & \text{if } 0 \leq x \leq 1 \\ 2\pi - \cos^{-1}(2x^2 - 1), & \text{if } -1 \leq x \leq 0 \end{cases}$$

$$\mathbf{2.20} \quad 3\cos^{-1}x = \begin{cases} \cos^{-1}(4x^3 - 3x), & \text{if } \frac{1}{2} \leq x \leq 1 \\ 2\pi - \cos^{-1}(4x^3 - 3x), & \text{if } -\frac{1}{2} < x \leq \frac{1}{2} \\ 2\pi + \cos^{-1}(4x^3 - 3x), & \text{if } -1 \leq x \leq -\frac{1}{2} \end{cases}$$

$$\mathbf{2.21} \quad 2\tan^{-1}x = \begin{cases} \tan^{-1}\frac{2x}{1-x^2}, & \text{if } -1 < x < 1 \\ \pi + \tan^{-1}\frac{2x}{1-x^2}, & \text{if } x > 1 \\ -\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right), & \text{if } x < -1 \end{cases}$$

$$\mathbf{2.22} \quad 2\tan^{-1}x = \begin{cases} \sin^{-1}\frac{2x}{1+x^2}, & \text{if } -1 \leq x \leq 1 \\ \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } x > 1 \\ -\pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } x < -1 \end{cases}$$

$$\mathbf{2.23} \quad 2\tan^{-1}x = \begin{cases} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), & \text{if } x \geq 0 \\ -\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), & \text{if } x \leq 0 \end{cases}$$

$$\mathbf{2.24} \quad 3\tan^{-1}x = \begin{cases} \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), & \text{if } \frac{-1}{\sqrt{3}} \leq x \leq \frac{1}{\sqrt{3}} \\ \pi + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), & \text{if } x \geq \frac{1}{\sqrt{3}} \\ -\pi + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), & \text{if } x \leq -\frac{1}{\sqrt{3}} \end{cases}$$

EXERCISES

Single Correct Choice Type Questions

- $\sin^{-1}[\sin(33\pi/7)]$ is equal to
(A) $\frac{33\pi}{7}$ (B) $\frac{\pi}{7}$ (C) $\frac{-\pi}{7}$ (D) $\frac{2\pi}{7}$
- $\tan^{-1}[\tan(2\pi/3)]$ is equal to
(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{-\pi}{3}$ (D) $\frac{-2\pi}{3}$
- If $p \leq \tan^{-1}[(1-x)/(1+x)] \leq q$ for all $0 \leq x \leq 1$, then
(A) $p=0, q=\frac{\pi}{4}$ (B) $p=\frac{-\pi}{4}, q=0$
(C) $p=\frac{\pi}{4}, q=\frac{\pi}{2}$ (D) $p=\frac{-\pi}{2}, q=0$
- The value of $\cos^{-1}(\cos 10)$ is
(A) $4\pi - 10$ (B) $2\pi - 10$ (C) $2\pi + 10$ (D) 10
- $\tan^{-1}(1/2) + \tan^{-1}(1/3)$ is equal to
(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$
- $2\tan^{-1}(1/5) + \sec^{-1}(5\sqrt{2}/7) + 2\tan^{-1}(1/8) =$
(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{-\pi}{2}$ (D) $\frac{-\pi}{4}$
- $\tan^{-1}(1/4) + \tan^{-1}(2/9) =$
(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$
- If $x = (\sqrt{2}-\sqrt{3})/2$, $y = \sqrt{12}/4$ and $z = \sqrt{2}$, then the value of $\sin^{-1}\{\cot(\sin^{-1}x + \cos^{-1}y + \sec^{-1}z)\}$ is
(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
- $\tan^{-1}(1/21) + \tan^{-1}(1/13) + \tan^{-1}(-1/8) =$
(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) 0
- $\sum_{k=1}^n \tan^{-1}\left(\frac{1}{1+k+k^2}\right)$ is equal to
(A) $\tan^{-1}(n+1) + \frac{\pi}{4}$ (B) $\tan^{-1}(n+1) - \frac{\pi}{4}$
(C) $\tan^{-1}(n+2) + \frac{\pi}{4}$ (D) $\tan^{-1}\left(\frac{1}{n+1}\right) + \frac{\pi}{4}$
- $\tan^{-1}(1/7) + \tan^{-1}(1/3)$ is equal to
(A) $\frac{\pi}{4} + \tan^{-1}\left(\frac{1}{3}\right)$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)$ (D) $\tan^{-1}\left(\frac{2}{3}\right)$
- If $\tan^{-1}x = 2\tan^{-1}(2/3)$, then x is equal to
(A) $\frac{7}{5}$ (B) $\frac{9}{5}$ (C) $\frac{11}{5}$ (D) $\frac{12}{5}$
- If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\pi/2$, then the value of $\sin[(\pi/2)(x+y+z)]$ is
(A) 0 (B) -1 (C) 1 (D) $\frac{1}{\sqrt{2}}$
- Suppose that $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$ and $\alpha = \sin[\pi(x+y+z)] + i\cos[\pi(x+y+z)]$, then $\text{Arg } \alpha$ is
(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) $\frac{3\pi}{2}$ (D) $\frac{-\pi}{2}$
- $\sin^{-1}(\sin 5)$ is equal to
(A) 5 (B) $5 - 2\pi$ (C) $2\pi - 5$ (D) $\pi - 5$
- If $u_k = \text{Cosec}^{-1}\sqrt{(k^2+1)(k^2+2k+2)}$, then $\sum_{k=1}^n u_k =$
(A) $\tan^{-1}(n+1) - \frac{\pi}{4}$ (B) $\tan^{-1}(n+1) + \frac{\pi}{4}$
(C) $\cot^{-1}(n+1) - \frac{\pi}{4}$ (D) $\cot^{-1}(n+1) + \frac{\pi}{4}$
- If
$$\frac{(\sin^{-1}x)^3 + (\cos^{-1}x)^3}{(\tan^{-1}x + \cot^{-1}x)^3} = 7$$
then the value (s) of x is (are)
(A) -1 (B) 1 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$
- The value of x satisfying the equation
$$\tan^{-1}\left(\frac{x+1}{x-1}\right) + \tan^{-1}\left(\frac{x-1}{x}\right) = \pi + \tan^{-1}(-7)$$
is
(A) -2 (B) $\frac{-1}{2}$ (C) 2 (D) $\frac{1}{2}$

19. Domain of the function

$$f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$$

is

- (A) $\left[\frac{-1}{4}, \frac{1}{2}\right]$ (B) $\left[\frac{-1}{2}, \frac{1}{4}\right]$ (C) $\left[\frac{-1}{2}, \frac{1}{2}\right]$ (D) $\left[\frac{-1}{4}, \frac{1}{4}\right]$

20. If

$$\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \frac{x^4}{8} + \cdots + \infty\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \frac{x^8}{8} + \cdots + \infty\right) = \frac{\pi}{2}$$

for $0 < |x| < \sqrt{2}$, then x is equal to

- (A) $\frac{1}{2}$ (B) 1 (C) $-\frac{1}{2}$ (D) -1

21. The equation
- $2\tan^{-1}x = \sin^{-1}[4x/(1+4x^2)]$
- is valid if
- x
- belongs to the interval

- (A) $\left(\frac{1}{2}, \infty\right)$ (B) $\left(-\infty, \frac{1}{2}\right)$
(C) $\left[\frac{-1}{2}, \frac{1}{2}\right]$ (D) $[-1, 1]$

22. If
- $0 < x < 1$
- , then

$$\tan^{-1}\left(\frac{1-x^2}{2x}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) =$$

- (A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{\pi}{2}$ (D) π

23. If

$$\alpha = 2\tan^{-1}\left(\frac{1+x}{1-x}\right) \quad \text{and} \quad \beta = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

where $0 < x < 1$, then $\alpha + \beta$ is equal to

- (A) 2π (B) π (C) 0 (D) $\frac{\pi}{2}$

- 24.
- $\sum_{k=1}^n \sin^{-1}\left(\frac{\sqrt{k} - \sqrt{k-1}}{\sqrt{k(k+1)}}\right) =$

- (A) $\tan^{-1}\sqrt{n} + \frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{n+1} + \frac{\pi}{4}$
(C) $\tan^{-1}\sqrt{n} - \frac{\pi}{4}$ (D) $\tan^{-1}\sqrt{n}$

Multiple Correct Choice Type Questions

1. Which of the following are integer pair solutions of the equation
- $\tan^{-1}x + \tan^{-1}(1/y) = \tan^{-1}(3)$
- ?

- (A) (1, 2) (B) (4, -13) (C) (5, -8) (D) (2, 7)

2. Let
- $y = (\pi/2) + \cos^{-1}(\cos x)$
- . Then the values of
- y
- such that
- $y = |\tan x|$
- (
- $0 \leq x \leq 2\pi$
-) are

- (A) $x + \frac{\pi}{4}$ (B) $x + \frac{\pi}{2}$ (C) $-x + \frac{5\pi}{2}$ (D) $-x + \frac{3\pi}{2}$

3. If
- x
- and
- y
- are real numbers such that
- $\tan^{-1}y = 4\tan^{-1}x$
- , then

- (A) $y = \frac{4x - 4x^3}{1 - 6x^2 + x^4}$ (B) $y = \frac{4x + x^3}{1 - 6x^2 + x^4}$
(C) $x \neq \tan 22\frac{1}{2}^\circ$ (D) $y = \frac{4x - x^3}{1 + 6x^2 - x^4}$

4. If
- $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\pi/2$
- , then

- (A) $x + y + z = xyz$
(B) $x + y + z = 3xyz$
(C) $1 + w^{x+y+z} = 2$, where $w \neq 1$ is a cube root of unity
(D) $w^x + w^{2y} + w^{3z} = 0$, where w is a nonreal root of $x^3 - 1 = 0$

5. If
- $f(x) = \sin^{-1}x$
- and
- $g(x) = \cos^{-1}x$
- , then

- (A) $f(x) > g(x)$ if $\frac{1}{\sqrt{2}} < x \leq 1$
(B) $f(x) \leq g(x)$ if $-1 \leq x \leq \frac{1}{\sqrt{2}}$
(C) $g(x) \geq f(x)$ if $\frac{1}{\sqrt{2}} \leq x \leq 1$
(D) $g(x) \leq f(x)$ if $-1 \leq x \leq \frac{1}{\sqrt{2}}$

6. Which of the following statements are true?

- (A) $x = 1$ is a solution of the equation $\cos^{-1}(1-x) + \sin^{-1}x = \pi$
(B) If $\cos^{-1}x + \cos^{-1}y = \pi/3$, then $x^2 - xy + y^2 = 3/4$
(C) $\tan^{-1}(4/3) = \tan^{-1}(1/7) + (\pi/4)$
(D) For $0 < x < a$, $2\tan^{-1}\sqrt{\frac{x}{a}} = \cos^{-1}\left(\frac{a-x}{a+x}\right)$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with *one or more* statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are (A) $\rightarrow$ (p), (s); (B) $\rightarrow$ (q), (s), (t); (C) $\rightarrow$ (r); (D) $\rightarrow$ (r), (t); that is if the matches are (A) $\rightarrow$ (p) and (s); (B) $\rightarrow$ (q), (s) and (t); (C) $\rightarrow$ (r); and (D) $\rightarrow$ (r), (t) then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

1. Match the items in Column I with those in Column II.

Column I	Column II
(A) $\sin^{-1}(1/2) + \cos^{-1}(-1/2)$ is equal to	(p) $\frac{\pi}{6}$
(B) $\tan^{-1}\sqrt{3} + \cot^{-1}(-\sqrt{3})$ is equal to	(q) $\frac{5\pi}{6}$
(C) $\sin^{-1}(-1/2) + \cos^{-1}(1/2)$ is equal to	(r) $\frac{\pi}{3}$
(D) $\cos^{-1}[\cos(7\pi/6)] + \sin^{-1}[\sin(2\pi/3)] =$	(s) $\frac{7\pi}{6}$
	(t) $\frac{2\pi}{3}$

2. Match the items in Column I with those in Column II

Column I	Column II
(A) If $\sin^{-1}x - \cos^{-1}x = \frac{\pi}{6}$, then $x =$	(p) 6
(B) If $\tan^{-1}x + \tan^{-1}3 = \tan^{-1}8$, then $x =$	(q) $\frac{\sqrt{5}}{3}$
(C) If $\tan(\sec^{-1}x) = \sin(\tan^{-1}2)$ and $x > 0$, then x is equal to	(r) $\frac{1}{5}$
(D) If $\tan^{-1}(4/x) + \tan^{-1}(9/x) = \pi/2$, then $x =$	(s) $\frac{\sqrt{3}}{2}$
	(t) $\frac{1}{\sqrt{3}}$

3. Match the items in Column I with those in Column II

Column I	Column II
(A) $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(\frac{1}{2}\right)\right) =$	(p) $\frac{3 - \sqrt{5}}{2}$
(B) $\cos\left(\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \frac{\pi}{6}\right) =$	(q) $\frac{-24}{7}$
(C) $\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{2}\right) =$	(r) $\frac{1}{2}$
(D) $-\tan\left(2\sin^{-1}\frac{4}{5}\right) =$	(s) $\frac{24}{7}$
	(t) $-\frac{1}{2}$

4. Match the items in Column I with those in Column II.

Column I	Column II
(A) If $\cot^{-1}x + \cot^{-1}2x = 3\pi/4$, then $x =$	(p) $\frac{-3 \pm \sqrt{17}}{4}$
(B) If $\tan^{-1}x = \cot^{-1}x$, then $x =$	(q) 0
(C) If $\cos^{-1}x - \sin^{-1}x = \cos^{-1}(\sqrt{3}x)$, then $x =$	(r) 1
(D) If $\tan^{-1}x + \tan^{-1}(1-x) = 2\tan^{-1}\sqrt{x-x^2}$, then $x =$	(s) $\frac{1}{2}$
	(t) $-\frac{1}{2}$

5. Match the items in Column I with those in Column II.

Column I	Column II
(A) If $f_1(x) = \cos^{-1}x + \sin^{-1}x + \cot^{-1}x$, then $f_1(x)$ belongs to	(p) $\left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$
(B) If $f_2(x) = \tan^{-1}x + \cot^{-1}x + \operatorname{cosec}^{-1}x$, then $f_2(x)$ belongs to	(q) $[0, \pi] - \{\pi/2\}$
(C) Range of the function $f_3(x) = \cot^{-1}x + \tan^{-1}x + \cos^{-1}x$ lies in the interval	(r) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$
(D) The range of $f_4(x) = \cot^{-1}x + \tan^{-1}x + \sin^{-1}x$ lies in	(s) $\left[0, \frac{\pi}{2}\right]$
	(t) $[0, \pi]$

Comprehension-Type Questions

1. Passage: If $x > 0$ and $y > 0$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

Answer the following questions.

(i) $\tan^{-1} 2 - \tan^{-1} x = \pi/4$ implies that the value of x is

- (A) 2 (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) $\frac{1}{3}$

(ii) If x and y are positive, then

$$\tan^{-1} \left(\frac{x}{y} \right) + \tan^{-1} \left(\frac{y-x}{y+x} \right)$$

is equal to

- (A) $\frac{\pi}{4}$ (B) $\frac{-\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{-\pi}{3}$

(iii) If

$$3 \tan^{-1} \left(\frac{1}{2+\sqrt{3}} \right) - \tan^{-1} \left(\frac{1}{x} \right) = \tan^{-1} \left(\frac{1}{3} \right)$$

then x is equal to

- (A) 2 (B) $\frac{1}{2}$ (C) 1 (D) -1

2. Passage: If $x > 0$ and $y > 0$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

Answer the following questions.

(i) Sum to n terms of the series

$$\tan^{-1} \frac{x}{1+2x^2} + \tan^{-1} \frac{x}{1+6x^2} + \tan^{-1} \frac{x}{1+12x^2} + \dots$$

is

- (A) $\tan^{-1}(n+1) - \tan^{-1} x$
 (B) $\tan^{-1} nx - \tan^{-1} x$
 (C) $\tan^{-1} \left(\frac{1}{n+1} \right) - \tan^{-1} x$
 (D) $\tan^{-1}(n(n+1)x) - \tan^{-1} x$

(ii) Sum of the first 2010 terms of the series

$$\tan^{-1} x + \tan^{-1} \frac{x}{1+1 \cdot 2x^2} + \tan^{-1} \frac{x}{1+2 \cdot 3x^2} + \tan^{-1} \frac{x}{1+3 \cdot 4x^2} + \dots$$

is

- (A) $\tan^{-1}(2011x) - \tan^{-1} x$
 (B) $\tan^{-1}(2010x) - \tan^{-1} x$
 (C) $\tan^{-1}(2010x)$
 (D) $\tan^{-1}(2009x)$

(iii) If the sum of the first n terms of the series

$$\tan^{-1} \frac{4}{13} + \tan^{-1} \frac{6}{73} + \tan^{-1} \frac{8}{241} + \dots$$

is $\tan^{-1}((n+1)(n+2)) - \tan^{-1} x$, then x is equal to

- (A) 1 (B) -2 (C) -1 (D) 2

Assertion-Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true

1. Statement I: If $\alpha = \tan^{-1}(1/7)$ and $\beta = \tan^{-1}(1/3)$, then $\cos 2\alpha = \sin 4\beta$.

Statement II: $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ and

$$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

2. Statement I: If $\beta = \cot^{-1} \sqrt{\cos \alpha} - \tan^{-1} \sqrt{\cos \alpha}$, then $\sin \beta = \tan^2(\alpha/2)$

Statement II: $\cot^{-1} x = \tan^{-1}(1/x)$ for any $x > 0$

3. Statement I: If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi/2$ and $x + y + z = \sqrt{3}$, then $x = y = z$

Statement II: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$, if $xy < 1$

4. Statement I: $\tan^{-1}x + \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \pi$ for $x > 1$

Statement II: For $-1 \leq x \leq 1$, $-\pi/2 \leq \sin^{-1}x \leq \pi/2$

5. Statement I: $\sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{8}{17}\right) = \sin^{-1}\left(\frac{77}{85}\right)$

Statement II: For $0 \leq x \leq 1$, $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$ and $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2}$

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. $4\tan^{-1}(1/5) - \tan^{-1}(1/239) = \pi/K$, where k is equal to _____.

2. $\sin^{-1}(1/2) + \cos^{-1}(1/\sqrt{2}) = k\pi/12$, where k is equal to _____.

3. If $2\tan^{-1}(1/2) + \cos^{-1}(4/5) = \pi/K$, then k is equal to _____.

4. If $2\tan^{-1}(1/5) + \tan^{-1}(1/4) = \tan^{-1}k$, then the integral part of k is _____.

5. $\cot^{-1}(\cot 4) = k - \pi$, where k is equal to _____.

6. If $\sin^{-1}(4/5) + \sin^{-1}(3/5) = 2\tan^{-1}k$, then k is equal to _____.

7. $\frac{15}{2} \left[\sin \left(2\tan^{-1} \left(\frac{1}{3} \right) \right) + \cos(\tan^{-1}(2\sqrt{2})) \right] = \text{_____}$.

8. $\frac{1}{6} + \tan \left[\cos^{-1} \left(\frac{4}{5} \right) + \tan^{-1} \left(\frac{2}{3} \right) \right] = \text{_____}$.

9. If x satisfies the equation $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$, then $|x|$ is _____.

10. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$, then $x^4 + y^4 + z^4 + 4x^2y^2z^2 = k(x^2y^2 + y^2z^2 + z^2x^2)$, where k is equal to _____.

ANSWERS

Single Correct Choice Type Questions

- (D)
- (C)
- (A)
- (A)
- (B)
- (B)
- (D)
- (A)
- (D)
- (B)
- (C)
- (D)

- (B)
- (D)
- (B)
- (A)
- (A)
- (C)
- (A)
- (B)
- (D)
- (C)
- (B)
- (D)

Multiple Correct Choice Type Questions

1. (A), (B), (C), (D)
2. (B), (C)
3. (A), (C)
4. (B), (C), (D)
5. (A), (B)
6. (A), (B), (C), (D)

Matrix-Match Type Questions

1. $(A) \rightarrow (q), (B) \rightarrow (s), (C) \rightarrow (p), (D) \rightarrow (s)$
2. $(A) \rightarrow (s), (B) \rightarrow (r), (C) \rightarrow (q), (D) \rightarrow (p)$
3. $(A) \rightarrow (r), (B) \rightarrow (r), (C) \rightarrow (p), (D) \rightarrow (q)$
4. $(A) \rightarrow (p), (B) \rightarrow (r), (C) \rightarrow (q), (s), (t), (D) \rightarrow (s)$
5. $(A) \rightarrow (p), (B) \rightarrow (q), (t), (C) \rightarrow (r), (D) \rightarrow (t)$

Comprehension-Type Questions

1. (i) (D); (ii) (A); (iii) (A)
2. (i) (A); (ii) (C); (iii) (D)

Assertion–Reasoning Type Questions

1. (A)
2. (A)
3. (A)
4. (A)
5. (A)

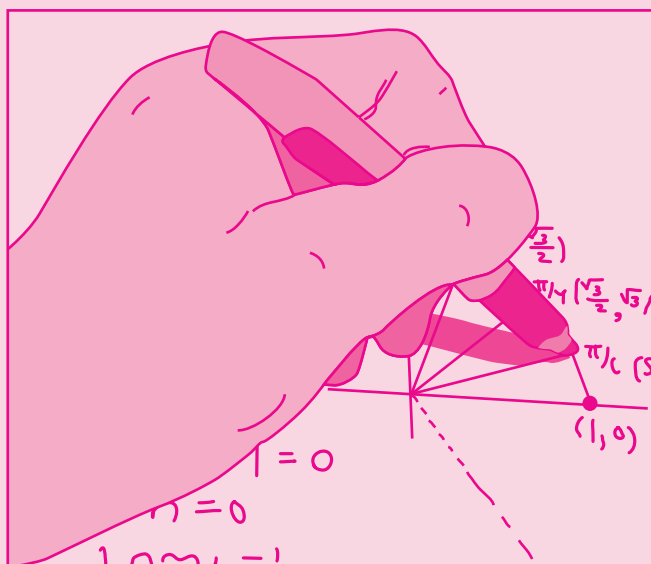
Integer Answer Type Questions

1. 4
2. 5
3. 2
4. 0
5. 4
6. 1
7. 7
8. 3
9. 4
10. 2

Trigonometric Equations

3

Trigonometric Equations



Contents

- 3.1 Least Positive Angles with a Given Trigonometric Ratio
- 3.2 General Solution of Equations of the Form $\sin x = a$
- 3.3 Solutions of Simple Trigonometric Equations

Worked-Out Problems
Summary
Exercises
Answers

A **trigonometric equation** is one that involves one or more of the six functions sine, cosine, tangent, cotangent, secant, and cosecant.

Consider a polynomial equation of the form $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$, where $a_0, a_1, \dots, a_n$ are real numbers. If a is a real number such that

$$a_n a^n + a_{n-1} a^{n-1} + \cdots + a_1 a + a_0 = 0$$

then a is called a **root** or a **solution** of the equation $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$. If at least one a_i is nonzero, then the equation has at most n roots. Note that if all the a_i 's are zero, then every real number is a root of this equation. If we consider a quadratic equation, then we know from Chapter 4 of Vol. 1 that the equation has at most two roots. In trigonometry, we come across equations such as

$$\sin \theta = 1/2$$

$$a \cos \theta + b \sin \theta = 1$$

$$a \cos^2 \theta + b \cos \theta + c = 0, \text{ etc}$$

where a, b, c are given real numbers. Equations of this type are called **trigonometric equations** and an angle θ satisfying a trigonometric equation is called a **root** or **solution** of this equation.

Unlike polynomial equations, any trigonometric equation has infinite number of solutions, if at all there is a solution. The trivial reason is that each trigonometric function is periodic and 2π is a period. Therefore, if θ is a solution of a given trigonometric equation, then $\theta + 2n\pi$ is also a solution of the same equation for every integer n . In this chapter we discuss trigonometric equations and their solutions.

DEFINITION 3.1 The solutions of a trigonometric equation lying in the interval $[0, 2\pi)$ are called **principal solutions**.

Example

$\pi/6$ and $5\pi/6$ are the principal solutions of the equation $\sin x = 1/2$.

3.1 | The Least Positive Angles with a Given Trigonometric Ratio

For any trigonometric function f and for any real number a , we explore the possibility of the existence of a real number x such that $f(x) = a$. If f is a sine function or a cosine function and a is a real number such that $|a| > 1$, then clearly there is no real number x for which $f(x) = a$, since the range of sine or cosine function is the interval $[-1, 1]$ and hence any real number outside this interval is neither $\sin x$ nor $\cos x$ for any $x \in \mathbb{R}$. However, if $|a| \leq 1$, we have infinite number of real numbers x for which $\sin x = a$. Among these, there must be least positive x (recall that each trigonometric function is having a minimum period). In the following, we construct such least positive real numbers, represented by angles, between 0 and 2π .

3.1.1 Construction of Least Positive Angle whose Sine is Equal to a , $|a| \leq 1$

Let a be a given real number with $|a| \leq 1$. If $a = 0$, then clearly π is the least positive real number x satisfying the equation $\sin x = 0$. If $a = 1$ or $a = -1$, then $\pi/2$ or $3\pi/2$ are, respectively, the least positive reals x for which $\sin x = 1$ or $\sin x = -1$. Therefore, we can assume that $0 < |a| < 1$.

Consider a rectangular coordinate system OXY. Let A be a point on OY such that $OA = a$ (if a is negative, the point A has to be chosen on YO produced, as shown in Figure 3.1).

Draw a circle of unit radius with O as the centre. Draw a line parallel to the X-axis through A meeting the circle at P. Draw line PB perpendicular to the X-axis through P meeting the X-axis at B. Let θ be the angle $\angle BOP$. Then $BP = a$ and $OP = 1$ and hence

$$\sin \theta = \sin \angle BOP = \frac{BP}{OP} = \frac{a}{1} = a$$

Therefore θ is the required angle.

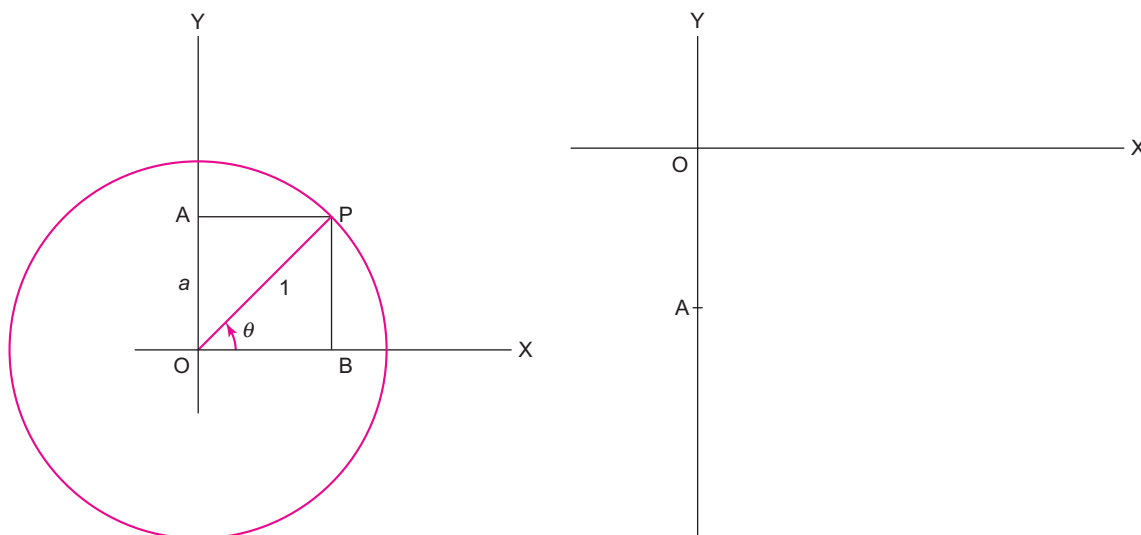


FIGURE 3.1

3.1.2 Construction of Least Positive Angle whose Cosine is Equal to a , $|a| \leq 1$

Let a be a real number with $|a| \leq 1$. Now $\pi/2$ is the least positive real x such that $\cos x = 0$. We can suppose that $a \neq 0$. Along the X-axis, let A be a point such that $OA = a$. (If a is negative, then A has to be chosen on the left part of the X-axis). Draw a circle of unit radius with the origin O as the centre. Draw a perpendicular to X-axis at A meeting the circle at P and P' , with P on the semicircle above the x-axis (Figure 3.2). Then $\theta = \angle AOP$ is the required angle, since

$$\cos \theta = \cos \angle AOP = \frac{OA}{OP} = \frac{a}{1} = a$$

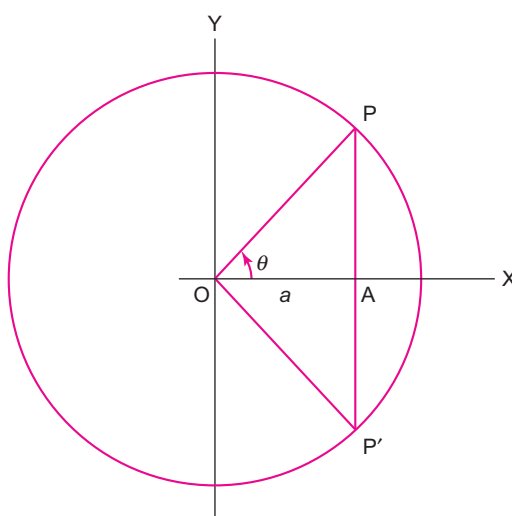


FIGURE 3.2

3.1.3 Construction of Least Positive Angle whose Tangent is a Given Real Number a

Let a be a nonzero real number. Let A be a point on the X-axis with OA as unit length. Draw a perpendicular to X-axis at A and choose a point P on the perpendicular such that $AP = a$. If a is negative, then we have to consider P' in the lower part of the plane (Figure 3.3). Then $\theta = \angle AOP$ is the required angle, since

$$\tan \theta = \tan \angle AOP = \frac{AP}{OA} = \frac{a}{1} = a$$

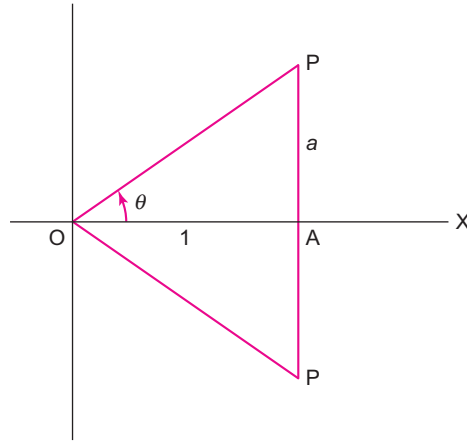


FIGURE 3.3

3.2 | General Solution of Equations of the Form $\sin x = a$

As we have discussed earlier, for any real number a with $|a| > 1$, there is no real number x satisfying $\sin x = a$ or $\cos x = a$. However, if $|a| \leq 1$, there are infinite number of solutions of the equation $\sin x = a$ or the equation $\cos x = a$. In the following, general forms of solutions of such type of equations are discussed.

3.2.1 General Solution of the Equation $\sin x = 0$

For $x \in [0, 2\pi)$, we know that

$$\sin x = 0 \Leftrightarrow x = 0 \quad \text{or} \quad \pi$$

Now, for any $x \in \mathbb{R}$, we have $\sin x = \sin(x - 2k\pi)$, where $k = [x/2\pi]$, the integral part of $x/2\pi$. Also, $x - 2k\pi \in [0, 2\pi)$. Therefore

$$\begin{aligned} \sin x &= 0 \\ \Leftrightarrow \sin(x - 2k\pi) &= 0 \\ \Leftrightarrow x - 2k\pi &= 0 \quad \text{or} \quad \pi \\ \Leftrightarrow x &= 2k\pi \quad \text{or} \quad (2k + 1)\pi \end{aligned}$$

Since k is an integer, it follows that $\sin x = 0 \Leftrightarrow x = n\pi$ for any integer n . Therefore, the general solution of the equation $\sin x = 0$ is $x = n\pi$, $n \in \mathbb{Z}$, or the solution set of the equation $\sin x = 0$ is $\{n\pi | n \in \mathbb{Z}\}$.

3.2.2 General Solution of Equations $\cos x = 0$ and $\tan x = 0$

For any real number x , we have

$$\begin{aligned} \cos x = 0 &\Leftrightarrow \sin\left(x - \frac{\pi}{2}\right) = 0 \\ \Leftrightarrow x - \frac{\pi}{2} &= n\pi \quad \text{for some integer } n \\ \Leftrightarrow x &= \frac{2n + 1}{2} \pi \quad \text{for some integer } n \end{aligned}$$

Therefore, the solution set of the equation $\cos x = 0$ is

$$\left\{ \left(\frac{2n + 1}{2} \right) \pi \mid n \in \mathbb{Z} \right\}$$

Also

$$\tan x = 0 \Leftrightarrow \sin x = 0 \Leftrightarrow x = n\pi, n \in \mathbb{Z}$$

Note that $\sin x = 0$ implies that $\cos x \neq 0$ and hence $\tan x$ is defined. On the other hand, $\tan x$ is defined; also $\tan x = 0$ implies that $\sin x = 0$ and $\cos x \neq 0$. Thus, for any real number x , $\tan x = 0 \Leftrightarrow x = n\pi$ for some integer n .

3.2.3 General Solution of the Equation $\sin x = a$

We can assume that $|a| \leq 1$; for, otherwise $\sin x = a$ has no solution. Then $-1 \leq a \leq 1$. Let θ be a solution of $\sin x = a$; that is, $\sin \theta = a$. Now, for any x ,

$$\begin{aligned} \sin x = a &\Leftrightarrow \sin x = \sin \theta \\ &\Leftrightarrow \sin x - \sin \theta = 0 \\ &\Leftrightarrow 2 \cos\left(\frac{x+\theta}{2}\right) \sin\left(\frac{x-\theta}{2}\right) = 0 \\ &\Leftrightarrow \cos \frac{x+\theta}{2} = 0 \quad \text{or} \quad \sin \frac{x-\theta}{2} = 0 \end{aligned}$$

Now two cases arise:

$$\text{Case 1: } \frac{x+\theta}{2} = (2n+1)\frac{\pi}{2} \text{ (by Section 3.2.2)}$$

$$\text{Case 2: } \frac{x-\theta}{2} = n\pi \text{ (by Section 3.2.1)}$$

For Case 1:

$$\begin{aligned} x &= (2n+1)\pi - \theta, n \in \mathbb{Z} \\ &\Leftrightarrow x = (2n+1)\pi + (-1)^{2n+1}\theta, n \in \mathbb{Z} \end{aligned}$$

For Case 2:

$$\begin{aligned} x &= 2n\pi + \theta, n \in \mathbb{Z} \\ &\Leftrightarrow x = 2n\pi + (-1)^{2n}\theta, n \in \mathbb{Z} \end{aligned}$$

Combining both conclusions for Cases 1 and 2 we get

$$x = m\pi + (-1)^m \theta, m \in \mathbb{Z}$$

Thus the general solution of the equation $\sin x = a$ is $m\pi + (-1)^m \theta, m \in \mathbb{Z}$, where θ is any real number such that $\sin \theta = a$; for example, using Section 3.1.1 we can take the least positive angle θ such that $\sin \theta = a$ or we can take $\sin^{-1} a$ for θ .

Note: For any real number a such that $|a| \geq 1$ (i.e., $a \in (-\infty, -1] \cup [1, \infty)$), general solution of the equation $\operatorname{cosec} x = a$ is $m\pi + (-1)^m \theta$, where θ is such that $\operatorname{cosec} \theta = a$.

Example 3.1

Find the general solution of the equation

$$\sin x = \frac{\sqrt{3}}{2}$$

Solution: The smallest positive angle whose sine is $\sqrt{3}/2$ is 60° or $\pi/3$. Therefore, the general solution of the equation $\sin x = \sqrt{3}/2$ is $m\pi + (-1)^m \pi/3, m \in \mathbb{Z}$.

3.2.4 General Solution of the Equation $\cos x = a$

If $|a| > 1$, then the equation $\cos x = a$ has no solution. Suppose that $|a| \leq 1$ and θ is a solution of $\cos x = a$; that is, $\cos \theta = a$. Now, for any real number x ,

$$\begin{aligned}\cos x = a &\Leftrightarrow \cos x = \cos \theta \\ &\Leftrightarrow \cos x - \cos \theta = 0 \\ &\Leftrightarrow -2 \sin\left(\frac{x+\theta}{2}\right) \sin\left(\frac{x-\theta}{2}\right) = 0 \\ &\Leftrightarrow \sin\left(\frac{x+\theta}{2}\right) = 0 \quad \text{or} \quad \sin\left(\frac{x-\theta}{2}\right) = 0 \\ &\Leftrightarrow \frac{x+\theta}{2} = n\pi, n \in \mathbb{Z} \quad \text{or} \quad \frac{x-\theta}{2} = n\pi, n \in \mathbb{Z} \\ &\Leftrightarrow x = 2n\pi \pm \theta, n \in \mathbb{Z}\end{aligned}$$

Thus, if θ is any angle such that $\cos \theta = a$, then the solution set of $\cos x = a$ is $\{2n\pi \pm \theta | n \in \mathbb{Z}\}$. We can take $\theta = \cos^{-1} a$.

Note: If $|a| \geq 1$, the general solution of the equation $\sec x = a$ is $2n\pi \pm \theta, n \in \mathbb{Z}$, where θ is any solution of $\sec x = a$.

Example

The general solution of the equation $\cos x = -1/2$ is $2n\pi \pm (2\pi/3), n \in \mathbb{Z}$ since $120^\circ (=2\pi/3)$ is a solution of $\cos x = -1/2$.

Example 3.2

Find the general solution of the equation $\sin^2 x = 1/4$.

Solution: The given equation can be written as $\sin x = \pm 1/2$. Since $\sin \pi/6 = 1/2$, the general solution of $\sin x = 1/2$ is $n\pi + (-1)^n(\pi/6), n \in \mathbb{Z}$.

Also, since $\sin(-\pi/6) = -1/2$, the general solution of $\sin x = -1/2$ is $n\pi + (-1)^n(-\pi/6)$. Thus the general solution of $\sin^2 x = 1/4$ is

$$n\pi \pm (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$$

3.2.5 General Solution of the Equation $\tan x = a$

Let $a \in \mathbb{R}$ be fixed and θ be any solution of $\tan x = a$; that is, θ is a real number such that $\tan \theta = a$. Then, for any real number x ,

$$\begin{aligned}\tan x = a &\Leftrightarrow \tan x = \tan \theta \\ &\Leftrightarrow \frac{\sin x}{\cos x} = \frac{\sin \theta}{\cos \theta} \\ &\Leftrightarrow \sin x \cos \theta - \cos x \sin \theta = 0 \\ &\Leftrightarrow \sin(x - \theta) = 0 \\ &\Leftrightarrow x - \theta = n\pi, n \in \mathbb{Z} \\ &\Leftrightarrow x = n\pi + \theta, n \in \mathbb{Z}\end{aligned}$$

Therefore, the general solution $\tan x = a$ is $n\pi + \theta, n \in \mathbb{Z}$, where θ is a solution of $\tan x = a$; here also, we can take $\theta = \tan^{-1} a$.

Note: For any $a \in \mathbb{R}$, the general solution of the equation $\cot x = a$ is $n\pi + \theta, n \in \mathbb{Z}$, where θ is a solution of $\cot x = a$.

Example

The general solution of the equation $\tan x = 1/\sqrt{3}$ is $n\pi + (\pi/6)$, $n \in \mathbb{Z}$, since $\tan \pi/6 = 1/\sqrt{3}$; that is, $\pi/6$ is a solution of $\tan x = 1/\sqrt{3}$.

3.2.6 General Solution of the Equation $\sin^2 x = \sin^2 \theta$

For any given real number θ , let us find the general solution of the equation $\sin^2 x = \sin^2 \theta$. Note that, for any real number x ,

$$\begin{aligned}\sin^2 x = \sin^2 \theta &\Leftrightarrow \frac{1 - \cos 2x}{2} = \frac{1 - \cos 2\theta}{2} \\ &\Leftrightarrow \cos 2x = \cos 2\theta \\ &\Leftrightarrow 2x = 2n\pi \pm 2\theta, n \in \mathbb{Z} \\ &\Leftrightarrow x = n\pi \pm \theta, n \in \mathbb{Z}\end{aligned}$$

If x is a solution $\sin^2 x = \sin^2 \theta$, then $-x$ is also a solution of the same equation. Therefore, the general solution of the equation $\sin^2 x = \sin^2 \theta$ is $n\pi \pm \theta$, $n \in \mathbb{Z}$.

Note: The general solution of $\cos^2 x = \cos^2 \theta$ and $\tan^2 x = \tan^2 \theta$ are $n\pi \pm \theta$, $n \in \mathbb{Z}$

Example

The general solution of the equation $\tan^2 x = 1/3$ is $n\pi \pm \pi/6$, $n \in \mathbb{Z}$; for, $\tan(\pi/6) = 1/\sqrt{3}$ and hence $\tan^2(\pi/6) = 1/3$.

Example 3.3

Find the general solution of the equation $\sec^2 x = 4/3$.

$$\Leftrightarrow \cos^2 x = \cos^2(30^\circ) = \cos^2 \frac{\pi}{6}$$

Solution: We have

$$\Leftrightarrow x = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$$

$$\sec^2 x = \frac{4}{3} \Leftrightarrow \cos^2 x = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2$$

Example 3.4

Find the general solution satisfying both the equations $\sin x = -1/2$ and $\tan x = 1/\sqrt{3}$.

Solution: Consider the equation $\sin x = -1/2$. The only angles between 0 and 2π that satisfy $\sin x = -1/2$ are $x = 330^\circ = 11\pi/6$ and $x = 210^\circ = 7\pi/6$. Also, the only angles

between 0 and 2π that satisfy the equation $\tan x = 1/\sqrt{3}$ are $x = 30^\circ = \pi/6$ and $x = 210^\circ = 7\pi/6$. Therefore, the only angle between 0 and 2π that satisfies both the equations $\sin x = -1/2$ and $\tan x = 1/\sqrt{3}$ is $210^\circ = 7\pi/6$. Hence, the general solution of these two equations is $2n\pi + (7\pi/6)$, $n \in \mathbb{Z}$.

Example 3.5

Find the general solution of the combined equations $\cot x = -\sqrt{3}$ and $\operatorname{cosec} x = -2$.

$$\cot\left(\frac{-\pi}{6}\right) = -\sqrt{3}$$

Solution: The general solution of the combined equations $\cot x = -\sqrt{3}$ and $\operatorname{cosec} x = -2$ is $2n\pi - (\pi/6)$, $n \in \mathbb{Z}$ because

and

$$\operatorname{cosec}\left(\frac{-\pi}{6}\right) = -2$$

3.3 | Solutions of Simple Trigonometric Equations

DEFINITION 3.2 Equations involving trigonometric ratios of an unknown angle are called *trigonometric equations*.

An equation is not completely solved unless we obtain a general expression for all angles which satisfy it. In this section, we solve certain elementary types of trigonometric equations.

3.3.1 General Solution of an Equation of the Form $a \sin x + b \cos x = c$

Let a , b and c be any given real numbers such that $a \neq 0$ or $b \neq 0$. Consider the equation

$$a \sin x + b \cos x = c$$

Put $r = \sqrt{a^2 + b^2}$. Then $r > 0$ (since $a \neq 0$ or $b \neq 0$). Suppose that $|c/r| \leq 1$. Choose a real number θ such that

$$\sin \theta = \frac{a}{r}$$

and

$$\cos \theta = \frac{b}{r}$$

Then the given equation can be written as

$$r \sin \theta \sin x + r \cos \theta \cos x = c$$

that is

$$\cos(x - \theta) = \frac{c}{r}$$

This can be solved for x , since $|c/r| \leq 1$.

We can choose $\alpha \in [0, \pi]$ such that $\cos \alpha = c/r$. Then $x = 2n\pi \pm \alpha + \theta$, $n \in \mathbb{Z}$ is the general solution of the given equation $a \sin x + b \cos x = c$.

Alternate Form

The alternate form of the general solution of the above equation can be obtained as follows: Let $r = \sqrt{a^2 + b^2}$ and suppose that $|c/r| \leq 1$. Then we can choose $\theta \in \mathbb{R}$ such that

$$\cos \theta = a/r \quad \text{and} \quad \sin \theta = b/r$$

Then the given equation can be written as

$$\cos \theta \sin x + \sin \theta \cos x = \frac{c}{r}$$

that is

$$\sin(x + \theta) = \frac{c}{r}$$

This can be solved for x . Since $|c/r| \leq 1$, there exists $\alpha \in [-\pi/2, \pi/2]$ such that $\sin \alpha = c/r$ this α is precisely $\sin^{-1}(c/r)$. Then $x = n\pi + (-1)^n \alpha - \theta$, $n \in \mathbb{Z}$ is the general solution of the given equation.

Example 3.6

Solve $\sin x + \sqrt{3} \cos x = \sqrt{2}$.

Therefore we have

$$r = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

Solution: The given equation can be written in the form $a \sin x + b \cos x = c$. Now r is given by

$$r = \sqrt{a^2 + b^2}$$

Dividing both sides of the given equation by 2, we get

$$\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x = \frac{1}{\sqrt{2}}$$

Since $\sin \pi/6 = 1/2$ and $\cos \pi/6 = \sqrt{3}/2$, the equation can be written as

$$\sin \frac{\pi}{6} \sin x + \cos \frac{\pi}{6} \cos x = \frac{1}{\sqrt{2}}$$

that is

$$\cos\left(x - \frac{\pi}{6}\right) = \frac{1}{\sqrt{2}} = \cos \frac{\pi}{4}$$

Therefore

$$x - \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{4}, \quad n \in \mathbb{Z} \quad (\text{by Section 3.2.4})$$

$$x = 2n\pi + \frac{\pi}{6} \pm \frac{\pi}{4}, \quad n \in \mathbb{Z}$$

This gives

$$x = 2n\pi + \frac{5\pi}{12}, \quad n \in \mathbb{Z}$$

or

$$x = 2n\pi - \frac{\pi}{12}, \quad n \in \mathbb{Z}$$

Alternate Solution

As mentioned in the note “Alternate Form”, the given equation can be solved as follows also. Since $\sin \pi/3 = \sqrt{3}/2$ and $\cos \pi/3 = 1/2$, the given equation can be written as

$$\cos \frac{\pi}{3} \sin x + \sin \frac{\pi}{3} \cos x = \frac{1}{\sqrt{2}}$$

that is

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}$$

Therefore

$$x + \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}, \quad n \in \mathbb{Z} \quad (\text{by Section 3.2.3})$$

$$x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}, \quad n \in \mathbb{Z}$$

It can be seen that both the solution sets obtained above are equal. That is,

$$\left\{2n\pi + \frac{\pi}{6} \pm \frac{\pi}{4} \mid n \in \mathbb{Z}\right\} = \left\{n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3} \mid n \in \mathbb{Z}\right\}$$

Example 3.7

Solve $2\sin^2 x - \cos 2x = 0$.

Solution: Since $\cos 2x = 1 - 2\sin^2 x$, we have

$$2\sin^2 x - \cos 2x = 0$$

$$\Leftrightarrow 4\sin^2 x = 1$$

$$\Leftrightarrow \sin x = \pm \frac{1}{2}$$

$$\Leftrightarrow x = n\pi \pm \frac{\pi}{6} \left(\text{since } \sin \frac{\pi}{6} = \frac{1}{2} \right)$$

Therefore $n\pi \pm \pi/6$, $n \in \mathbb{Z}$ is the general solution of the given equation.

Example 3.8

Solve $\sin 5x = \cos 2x$.

Solution: We have

$$\sin 5x = \cos 2x$$

$$\Leftrightarrow \sin 5x = \sin\left(\frac{\pi}{2} - 2x\right)$$

$$\Leftrightarrow 5x = n\pi + (-1)^n \left(\frac{\pi}{2} - 2x\right), \quad n \in \mathbb{Z}$$

$$\Leftrightarrow 5x + (-1)^n 2x = n\pi + (-1)^n \frac{\pi}{2}, \quad n \in \mathbb{Z}$$

$$\Leftrightarrow x = \left(\frac{2n + (-1)^n}{5 + 2(-1)^n}\right) \frac{\pi}{2}, \quad n \in \mathbb{Z}$$

Therefore

$$\left(\frac{2n + (-1)^n}{5 + 2(-1)^n}\right) \frac{\pi}{2}, \quad n \in \mathbb{Z}$$

is the general solution of the equation $\sin 5x = \cos 2x$.

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. The general solution of the equation $11\sin x + 2\cos^2 x = 7$ is

- (A) $2n\pi \pm (\pi/3)$
 (B) $2n\pi + (-1)^n(\pi/6), n \in \mathbb{Z}$
 (C) $(2n+1)\pi + (-1)^n(\pi/6), n \in \mathbb{Z}$
 (D) $n\pi + (-1)^n(\pi/6), n \in \mathbb{Z}$

Solution:

$$\begin{aligned} 11\sin x + 2\cos^2 x &= 7 \\ \Leftrightarrow 2(1 - \sin^2 x) + 11\sin x &= 7 \\ \Leftrightarrow 2\sin^2 x - 11\sin x + 5 &= 0 \\ \Leftrightarrow (\sin x - 5)(2\sin x - 1) &= 0 \\ \Leftrightarrow \sin x = 5 \quad \text{or} \quad \sin x &= 1/2 \\ \Leftrightarrow \sin x = \frac{1}{2} \quad (\text{since } \sin x = 5 \text{ is impossible}) \\ \Leftrightarrow x = n\pi + (-1)^n(\pi/6), n \in \mathbb{Z} \end{aligned}$$

Answer: (D)

2. The values of x and y , when $x + y = 2\pi/3$ and $\cos x + \cos y = \sqrt{3}/2$, are respectively

- (A) $2n\pi + \frac{\pi}{3}$ and $\frac{\pi}{3} - 2n\pi, n \in \mathbb{Z}$
 (B) $2n\pi + \frac{\pi}{6}$ and $\frac{\pi}{2} - 2n\pi, n \in \mathbb{Z}$
 (C) $n\pi + \frac{\pi}{6}$ and $\frac{\pi}{2} - n\pi, n \in \mathbb{Z}$
 (D) $n\pi + \frac{\pi}{3}$ and $\frac{\pi}{3} - 2n\pi, n \in \mathbb{Z}$

Solution:

$$\begin{aligned} x + y &= \frac{2\pi}{3} \quad \text{and} \quad \cos x + \cos y = \frac{\sqrt{3}}{2} \\ \Leftrightarrow x + y &= \frac{2\pi}{3} \quad \text{and} \quad 2\cos \frac{x+y}{2} \cdot \cos \frac{x-y}{2} = \frac{\sqrt{3}}{2} \\ \Leftrightarrow x + y &= \frac{2\pi}{3} \quad \text{and} \quad 2\cos \frac{\pi}{3} \cos \frac{x-y}{2} = \frac{\sqrt{3}}{2} \\ \Leftrightarrow x + y &= \frac{2\pi}{3} \quad \text{and} \quad 2 \cdot \frac{1}{2} \cos \frac{x-y}{2} = \frac{\sqrt{3}}{2} \\ \Leftrightarrow x + y &= \frac{2\pi}{3} \quad \text{and} \quad \cos \frac{x-y}{2} = \cos \frac{\pi}{6} \\ \Leftrightarrow x + y &= \frac{2\pi}{3} \quad \text{and} \quad \frac{x-y}{2} = 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z} \end{aligned}$$

$$\Leftrightarrow x + y = \frac{2\pi}{3} \quad \text{and} \quad x - y = 4n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\Leftrightarrow x = 2n\pi + \frac{\pi}{6} \quad \text{and} \quad y = \frac{\pi}{2} - 2n\pi, n \in \mathbb{Z} \text{ or vice-versa}$$

Answer: (B)

3. If $0 < x < \pi/2$ and $\sin(x + 28^\circ) = \cos(3x - 78^\circ)$, then the possible value(s) of x is (are)

- (A) 8° or 35° (B) 9° or 36°
 (C) 18° or 45° (D) 28° or 55°

Solution: Suppose that $\sin(x + 28^\circ) = \cos(3x - 78^\circ)$. Then

$$\sin(x + 28^\circ) = \sin\left(\frac{\pi}{2} - (3x - 78^\circ)\right) = \sin(168^\circ - 3x)$$

Therefore

$$x + 28^\circ = 2n\pi + 168^\circ - 3x, n \in \mathbb{Z}$$

$$\text{or} \quad x = \pi - (168^\circ - 3x) + 2n\pi, n \in \mathbb{Z}$$

Hence

$$4x = 2n\pi + 140^\circ, n \in \mathbb{Z}$$

$$\text{or} \quad 2x = 16^\circ - 2n\pi, n \in \mathbb{Z}$$

This gives

$$x = 35^\circ + n(90^\circ)$$

$$\text{or} \quad x = 8^\circ - n(180^\circ), n \in \mathbb{Z}$$

$$\text{Now } 0 < x < 90^\circ \Rightarrow x = 35^\circ \text{ or } 8^\circ.$$

Answer: (A)

4. Suppose that $\cos 3x \cos 2x \cos x = 1/4$. If $0 < x < \pi/4$, then the value of x is

- (A) $\frac{\pi}{5}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{7}$

Solution: It is given that $\cos 3x \cos 2x \cos x = 1/4$. Therefore

$$4\cos x \cos 2x \cos 3x = 1$$

$$(2\cos x \cos 3x)2\cos 2x = 1$$

$$(\cos 4x + \cos 2x)2\cos 2x = 1$$

$$2\cos^2 2x + 2\cos 2x \cos 4x - 1 = 0$$

$$\cos 4x + 2\cos 2x \cos 4x = 0$$

$$\cos 4x(1 + 2\cos 2x) = 0$$

This gives

$$\cos 4x = 0$$

or $1 + 2 \cos 2x = 0$

Therefore

$$4x = \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$$

or $\cos 2x = \frac{-1}{2}$

Hence

$$x = \left(\frac{2n+1}{8}\right)\pi, n \in \mathbb{Z}$$

or $2x = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$

or $x = n\pi \pm \frac{2\pi}{6}, n \in \mathbb{Z}$

If $0 < x < \pi/4$, then $x = \pi/8$.

Answer: (C)

5. General solution of the equation $\cos 5x = -1/2$ is

(A) $n\pi + (-1)^n \frac{2\pi}{3}, n \in \mathbb{Z}$

(B) $2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$

(C) $\frac{2n\pi}{5} \pm \frac{\pi}{15}, n \in \mathbb{Z}$

(D) $\frac{2n\pi}{5} \pm \frac{2\pi}{15}, n \in \mathbb{Z}$

Solution:

$$\cos 5x = \frac{-1}{2}$$

$$\Leftrightarrow \cos 5x = \cos \frac{2\pi}{3}$$

$$\Leftrightarrow 5x = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$$

$$\Leftrightarrow x = \frac{2n\pi}{5} \pm \frac{2\pi}{15}, n \in \mathbb{Z}$$

Answer: (D)

6. A principal solution of the equation $\tan x + \cot x = 2 \operatorname{cosec} x$ is

(A) $\frac{7\pi}{8}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{3}$ (D) $\frac{2\pi}{3}$

Solution:

$$\tan x + \cot x = 2 \operatorname{cosec} x$$

$$\Rightarrow \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{2}{\sin x}$$

$$\Rightarrow \sin^2 x + \cos^2 x = 2 \cos x$$

$$\Rightarrow \cos x = \frac{1}{2} = \cos \frac{\pi}{3}$$

$$\Rightarrow x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

Hence $\pi/3, 2\pi - (\pi/3) (= 5\pi/3)$ are the principal solutions.

Answer: (C)

7. General solution of the equation $\tan x + \tan(\pi/3 + x) + \tan(2\pi/3 + x) = 3$ is

(A) $n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$ (B) $n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$

(C) $\frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{Z}$ (D) $\frac{n\pi}{3} + \frac{\pi}{4}, n \in \mathbb{Z}$

Solution:

$$\tan x + \tan\left(\frac{\pi}{3} + x\right) + \tan\left(\frac{2\pi}{3} + x\right) = 3$$

$$\Rightarrow \tan x + \frac{\tan(\pi/3) + \tan x}{1 - \tan(\pi/3)\tan x} + \frac{\tan(2\pi/3) + \tan x}{1 - \tan(2\pi/3)\tan x} = 3$$

$$\Rightarrow \tan x + \frac{\sqrt{3} + \tan x}{1 - \sqrt{3}\tan x} + \frac{-\sqrt{3} + \tan x}{1 + \sqrt{3}\tan x} = 3$$

$$\Rightarrow \tan x + \frac{8\tan x}{1 - 3\tan^2 x} = 3$$

$$\Rightarrow \tan x - 3\tan^3 x + 8\tan x = 3(1 - 3\tan^2 x)$$

$$\Rightarrow \frac{3(3\tan x - \tan^3 x)}{1 - 3\tan^2 x} = 3$$

$$\Rightarrow \tan 3x = 1 = \tan \frac{\pi}{4}$$

$$\Rightarrow 3x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\Rightarrow x = \frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{Z}$$

Answer: (C)

8. The general solution of the equation $\sin x + \sin 3x + \sin 5x = 0$ is

(A) $\frac{2n\pi}{3}, n \in \mathbb{Z}$ (B) $(3n \pm 1)\frac{\pi}{3}, n \in \mathbb{Z}$

(C) $\frac{4n\pi}{3}, n \in \mathbb{Z}$ (D) $\frac{n\pi}{3}, n \in \mathbb{Z}$

Solution:

$$\begin{aligned}
&\sin x + \sin 3x + \sin 5x = 0 \\
&\Rightarrow (\sin 5x + \sin x) + \sin 3x = 0 \\
&\Rightarrow 2 \sin 3x \cos 2x + \sin 3x = 0 \\
&\Rightarrow (\sin 3x)(2 \cos 2x + 1) = 0
\end{aligned}$$

This implies that either

$$\sin 3x = 0$$

or

$$2 \cos 2x + 1 = 0$$

Therefore

$$3x = n\pi, n \in \mathbb{Z}$$

or

$$\cos 2x = \frac{-1}{2} = \cos \frac{2\pi}{3}$$

From these we get

$$x = \frac{n\pi}{3}, n \in \mathbb{Z}$$

or

$$2x = 2n\pi \pm \left(\frac{2\pi}{3}\right), n \in \mathbb{Z}$$

which further implies that

$$x = \frac{n\pi}{3}, n \in \mathbb{Z}$$

or

$$x = (3n \pm 1)\frac{\pi}{3}, n \in \mathbb{Z}$$

Hence

$$x = \frac{n\pi}{3}, n \in \mathbb{Z} \quad (\text{since } 3n \pm 1 \in \mathbb{Z})$$

Answer: (D)

9. The general values of θ satisfying the equation $\tan^2 \theta + \sec 2\theta = 1$ are

(A) $(2n+1)\pi, n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

(B) $n\pi, n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

(C) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

(D) $n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$

Solution:

$$\begin{aligned}
&\tan^2 \theta + \sec 2\theta = 1 \\
&\Rightarrow \tan^2 \theta + \frac{1 + \tan^2 \theta}{1 - \tan^2 \theta} = 1
\end{aligned}$$

$$\Rightarrow \tan^2 \theta - \tan^4 \theta + 1 + \tan^2 \theta = 1 - \tan^2 \theta$$

$$\Rightarrow \tan^4 \theta = 3 \tan^2 \theta$$

$$\Rightarrow \tan^2 \theta (\tan^2 \theta - 3) = 0$$

This implies that either

$$\tan^2 \theta = 0$$

or

$$\tan^2 \theta = 3$$

which further gives

$$\tan \theta = 0$$

or

$$\tan \theta = \pm \sqrt{3} = \tan \left(\pm \frac{\pi}{3} \right)$$

Hence

$$\theta = n\pi, n \in \mathbb{Z}$$

or

$$\theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

Note that for $\tan \theta$ to be defined, we must have $\theta \neq (2n+1)\pi/2$ and for $\sec 2\theta$ to be defined, we must have $2\theta \neq (2n+1)\pi/2$ or $\theta \neq (2n+1)\pi/4$. Therefore, the general values of θ satisfying the given equation are

$$n\pi, n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

Answer: (B)

10. General value of x for which the sets $P = \{\sin x, \sin 2x, \sin 3x\}$ and $Q = \{\cos x, \cos 2x, \cos 3x\}$ are equal is

(A) $\frac{\pi}{4} + \frac{n\pi}{2}, n \in \mathbb{Z}$

(B) $\frac{\pi}{8} + \frac{n\pi}{2}, n \in \mathbb{Z}$

(C) $\frac{\pi}{12} + \frac{n\pi}{2}, n \in \mathbb{Z}$

(D) $\frac{\pi}{15} + \frac{n\pi}{2}, n \in \mathbb{Z}$

Solution: Since $P = Q$, the sum of elements in P and sum of the elements in Q are equal. Therefore

$$\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$$

$$(\sin x + \sin 3x) + \sin 2x = (\cos x + \cos 3x) + \cos 2x$$

$$2 \sin 2x \cos x + \sin 2x = 2 \cos 2x \cos x + \cos 2x$$

$$\sin 2x(2 \cos x + 1) = \cos 2x(2 \cos x + 1)$$

If $\cos x = -1/2$, then $x = 2n\pi \pm (2\pi/3)$ in which case $P \neq Q$. Hence $\sin 2x = \cos 2x$ which gives that $\tan 2x = 1$ and hence $2x = n\pi + (\pi/4)$ or $x = (n\pi/2) + (\pi/8)$ and these values of x satisfy the condition that $P = Q$ (check this).

Answer: (B)

11. If the angles A and B of $\triangle ABC$ satisfy the relations $3\sin A + 4\cos B = 6$ and $4\sin B + 3\cos A = 1$, then the third angle C is

- (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$

Solution: Squaring and adding the two given equations we get

$$24\sin(A+B) + 25 = 37$$

Therefore

$$\sin(A+B) = 1/2$$

$$A+B = 30^\circ \text{ or } 150^\circ$$

This gives that either $C = 30^\circ$ or 150° . If $C = 150^\circ$, then $A < 30^\circ$ so that $6 = 3\sin A + 4\cos B < \frac{3}{2} + 4 = \frac{11}{2} = 5\frac{1}{2}$ which is impossible. Therefore

$$C = 30^\circ$$

Answer: (C)

12. General solution of the equation $\sin^6 x = 1 + \cos^4 3x$ is

- (A) $(2K+1)\frac{\pi}{6}, K \in \mathbb{Z}$ (B) $\frac{K\pi}{3}, K \in \mathbb{Z}$
(C) $(2K+1)\frac{\pi}{2}, K \in \mathbb{Z}$ (D) $\frac{K\pi}{2}, K \in \mathbb{Z}$

Solution:

$$\sin^6 x \leq 1 \text{ and } 1 + \cos^4 3x \geq 1$$

$$\Rightarrow \sin^6 x = 1 \text{ and } \cos^4 3x = 0$$

$$\Rightarrow \sin^2 x = 1 \text{ and } \cos 3x = 0$$

Therefore

$$\cos x = 0 \text{ and } \cos 3x = 0$$

which gives

$$x = (2K+1)\pi/2 \text{ and } 3x = (2K+1)\pi/2$$

Hence

$$x = (2K+1)\pi/2 \text{ and } x = (2K+1)\pi/6$$

All solutions of the form $x = (2K+1)\pi/2$ are included in the solution set

$$\left\{ (2K+1)\frac{\pi}{6} \mid K \in \mathbb{Z} \right\}$$

which is the solution set.

Answer: (A)

13. General solution of the equation $\tan x + \tan 2x + \tan x \tan 2x = 1$ is

- (A) $\frac{k\pi}{3} + \frac{\pi}{12}, k \in \mathbb{Z}$ (B) $(2k+1)\frac{\pi}{12}, k \in \mathbb{Z}$
(C) $\frac{k\pi}{12} + \frac{\pi}{3}, k \in \mathbb{Z}$ (D) $\frac{k\pi}{6} + \frac{\pi}{3}, k \in \mathbb{Z}$

Solution: The given equation can be written as

$$\frac{\tan x + \tan 2x}{1 - \tan x \tan 2x} = 1$$

that is

$$\tan 3x = 1$$

Therefore

$$3x = k\pi + \frac{\pi}{4}, k \in \mathbb{Z}$$

$$x = \frac{k\pi}{3} + \frac{\pi}{12}, k \in \mathbb{Z}$$

Answer: (A)

14. The number of solutions in the interval $[0, \pi]$ of the equation $\sin^3 x \cos 3x + \sin 3x \cos^3 x = 0$ is

- (A) 7 (B) 6 (C) 5 (D) 4

Solution: The given equation can be written as

$$\left(\frac{3\sin x - \sin 3x}{4} \right) \cos 3x + \sin 3x \left(\frac{3\cos x + \cos 3x}{4} \right) = 0$$

which is equivalent to

$$\sin x \cos 3x + \sin 3x \cos x = 0$$

Therefore

$$\sin 4x = 0$$

or

$$x = \frac{n\pi}{4}, n \in \mathbb{Z}$$

Answer: (C)

15. The general solution of the equation $2\sin x \cos x + 5\cos^2 x = 4$ is

- (A) $n\pi + \tan^{-1}\left(\frac{\sqrt{5}+1}{4}\right), n\pi - \tan^{-1}\left(\frac{\sqrt{5}-1}{4}\right), n \in \mathbb{Z}$
(B) $\frac{n\pi}{2} + \tan^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right), \frac{n\pi}{2} - \tan^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right), n \in \mathbb{Z}$
(C) $(2n+1)\pi + \tan^{-1}\left(\frac{\sqrt{6}+\sqrt{2}}{2\sqrt{2}}\right),$
 $(2n+1)\pi - \tan^{-1}\left(\frac{\sqrt{6}-\sqrt{2}}{2\sqrt{2}}\right), n \in \mathbb{Z}$
(D) $(2n+1)\frac{\pi}{2} + \tan^{-1}\left(\frac{\sqrt{5}+1}{4}\right),$
 $(2n+1)\frac{\pi}{2} + \tan^{-1}\left(\frac{\sqrt{5}-1}{4}\right)$

Solution:

$$\begin{aligned}
 2 \sin x \cos x + 5 \cos^2 x &= 4(\sin^2 x + \cos^2 x) \\
 4 \sin^2 x - 2 \sin x \cos x - \cos^2 x &= 0 \\
 4 \tan^2 x - 2 \tan x - 1 &= 0
 \end{aligned}$$

(since $\cos x = 0$ is not a solution of the original equation.)
Therefore

$$\begin{aligned}
 \tan x &= \frac{2 \pm \sqrt{20}}{8} = \frac{1 \pm \sqrt{5}}{4} \\
 x &= n\pi + \tan^{-1}\left(\frac{\sqrt{5}+1}{4}\right), n\pi - \tan^{-1}\left(\frac{\sqrt{5}-1}{4}\right), n \in \mathbb{Z}
 \end{aligned}$$

Answer: (A)

16. The number of solutions of the equation $\sqrt{1 + \sin 2x} = \sqrt{2} \cos 3x$ belonging to the interval $[\pi, 3\pi/2]$ is

- (A) 5 (B) 4 (C) 3 (D) 2

Solution: Squaring both sides of the equation we get

$$1 + \sin 2x = 2 \cos^2 3x = 1 + \cos 6x$$

Therefore

$$\begin{aligned}
 \cos 6x &= \sin 2x = \cos\left(\frac{\pi}{2} - 2x\right) \\
 6x &= (2n\pi) \pm \left(\frac{\pi}{2} - 2x\right) \\
 x &= \frac{n\pi}{4} + \frac{\pi}{16}, x = \frac{n\pi}{2} - \frac{\pi}{8}, n \in \mathbb{Z}
 \end{aligned}$$

(i) $n = 4 \Rightarrow x = \frac{17\pi}{16} \in \left[\pi, \frac{3\pi}{2}\right]$

When $x = 17\pi/16$, the left-hand side of the original equation is positive while the right-hand side is negative. Hence $x = 17\pi/16$ is *not a solution*.

(ii) $n = 5 \Rightarrow x = \frac{21\pi}{16} \in \left[\pi, \frac{3\pi}{2}\right]$

(iii) $n = 3 \Rightarrow x = \frac{3\pi}{2} - \frac{\pi}{8} = \frac{11\pi}{8} \in \left[\pi, \frac{3\pi}{2}\right]$

Answer: (D)

17. General solution of the equation $\sin 10x + \cos 10x = \sqrt{2} \sin 15x$ is

- (A) $x = \frac{n\pi}{10} + \frac{\pi}{20}, x = \frac{n\pi}{25} - \frac{\pi}{100}$
 (B) $x = \frac{n\pi}{20} + \frac{\pi}{10}, x = \frac{n\pi}{50} - \frac{\pi}{50}$

(C) $x = \frac{n\pi}{15} + \frac{\pi}{20}, x = \frac{n\pi}{20} + \frac{\pi}{100}$

(D) $x = \frac{2n\pi}{15} + \frac{\pi}{20}, x = \frac{2n\pi}{25} - \frac{\pi}{50}$

Solution: Squaring both sides of the given equation we get

$$(\sin 10x + \cos 10x)^2 = 2 \sin^2 15x$$

Therefore

$$\begin{aligned}
 1 + \sin 20x &= 1 - \cos 30x \\
 \cos 30x &= -\sin 20x = \cos\left(\frac{\pi}{2} + 20x\right) \\
 30x &= (2n\pi) \pm \left(\frac{\pi}{2} + 20x\right) \\
 10x &= 2n\pi + \frac{\pi}{2}, 50x = 2n\pi - \frac{\pi}{2} \\
 x &= \frac{n\pi}{5} + \frac{\pi}{20}, x = \frac{n\pi}{25} - \frac{\pi}{100}
 \end{aligned}$$

Answer: (A)

18. General solution of the equation $\sin^2 x - 5 \sin x \cos x + 6 \cos^2 x = 0$ is

- (A) $(2n+1)\pi + \tan^{-1}2, 2n\pi + \tan^{-1}(1/2)$
 (B) $n\pi + \tan^{-1}2, n\pi + \tan^{-1}3$
 (C) $n\pi + \tan^{-1}(1/2), n\pi + \tan^{-1}(1/3)$
 (D) $2n\pi + \tan^{-1}\sqrt{2}, 2n\pi + \tan^{-1}3$

Solution: Clearly $\cos x = 0$ is not a solution of the equation. Therefore by dividing the given equation with $\cos^2 x$, we get

$$\tan^2 x - 5 \tan x + 6 = 0$$

Solving this we get that $\tan x = 2$ and $\tan x = 3$ are solutions. Therefore the general solution is

$$\begin{aligned}
 x &= n\pi + \tan^{-1}2 \\
 \text{and } x &= n\pi + \tan^{-1}3
 \end{aligned}$$

Answer: (B)**19.** The equation

$$\left(2 \cos^2 \frac{x}{2}\right) \sin^2 x = \left(\frac{x^2 + 1}{x^2}\right), x \leq \pi/9$$

has

- (A) no solution
 (B) one solution
 (C) more than one real solution
 (D) cannot be said

Solution: We have

$$2\cos^2\left(\frac{x}{2}\right)\sin^2 x \leq 2 \quad \text{and} \quad \frac{x^2+1}{x^2} \geq 2$$

and $x \neq 0, x \leq \pi/9$. Therefore

$$2\cos^2\left(\frac{x}{2}\right)\sin^2 x = 2 \quad \text{and} \quad \frac{x^2+1}{x^2} = 2$$

which gives

$$(1 + \cos x)(1 - \cos^2 x) = 2 \quad \text{and} \quad x = \pm 1$$

$$\cos^3 x + \cos^2 x - \cos x + 1 = 0 \quad \text{and} \quad x = \pm 1$$

which is impossible. Therefore the equation has no real solution.

Answer: (A)

20. The number of solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is

- (A) 0 (B) 1 (C) 2 (D) infinitely many

Solution: We have

$$\sin(e^x) \leq 1 \quad \text{and} \quad 5^x + 5^{-x} \geq 2$$

Therefore the equation has no solution.

Answer: (A)

21. The general value of θ satisfying the equation $2\sin^2 \theta - 3\sin \theta - 2 = 0$ is

- (A) $n\pi + (-1)^n \frac{\pi}{6}$ (B) $n\pi + (-1)^n \frac{\pi}{2}$
 (C) $n\pi + (-1)^n \frac{5\pi}{6}$ (D) $n\pi + (-1)^n \frac{7\pi}{6}$

Solution: The given equation can be written as

$$(2\sin \theta + 1)(\sin \theta - 2) = 0$$

Therefore

$$2\sin \theta + 1 = 0$$

$$\sin \theta = \frac{-1}{2} = \sin\left(\frac{7\pi}{6}\right)$$

$$\theta = n\pi + (-1)^n \frac{7\pi}{6}$$

Answer: (D)

22. The number of roots of the equation $\cos^7 x + \sin^4 x = 1$ in the open interval $(-\pi, \pi)$ is

- (A) 4 (B) 3 (C) 2 (D) 1

Solution: The given equation is

$$\cos^7 x + (1 - \cos^2 x)^2 = 1$$

Therefore

$$\cos^7 x + \cos^4 x - 2\cos^2 x = 0$$

$$\cos^2 x(\cos^5 x + \cos^2 x - 2) = 0$$

Now $\cos^2 x = 0$ which means $x = (2n+1)\pi/2$ or $\cos^5 x + \cos^2 x - 2 = 0$ which implies that $x = 2n\pi$. Therefore the roots which lie in $(-\pi, \pi)$ are $-\pi/2, 0, \pi/2$ corresponding to $n = -1, 0$.

Answer: (B)

23. The values of $\theta \in (0, 2\pi)$ for which $2\sin^2 \theta - 5\sin \theta + 2 > 0$ are

- (A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$
 (C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{41\pi}{48}, \pi\right)$

Solution: We have

$$2\sin^2 \theta - 5\sin \theta + 2 > 0$$

$$\Leftrightarrow (2\sin \theta - 1)(\sin \theta - 2) > 0$$

$$\Leftrightarrow 2\sin \theta - 1 < 0 \quad (\because \sin \theta < 2)$$

$$\Leftrightarrow \sin \theta < \frac{1}{2}$$

$$\Leftrightarrow 0 < \theta < \frac{\pi}{6} \quad \text{or} \quad \frac{5\pi}{6} < \theta < 2\pi$$

Answer: (A)

24. Number of roots of the equation $\sin 2x + 2\sin x - \cos x - 1 = 0$ in the range $0 \leq x \leq 2\pi$ is

- (A) 1 (B) 2 (C) 3 (D) 4

Solution: The given equation can be written as

$$2\sin x \cos x + 2\sin x - \cos x - 1 = 0$$

$$(2\sin x - 1)(\cos x + 1) = 0$$

Therefore

$$\sin x = 1/2 \quad \text{or} \quad \cos x = -1$$

which implies

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \pi$$

are the roots of the equation in the range $[0, 2\pi]$.

Answer: (C)

25. In $\triangle ABC$, angle A is greater than angle B . If the measures of the angles satisfies the equation $3\sin x - 4\sin^3 x - k = 0, 0 < k < 1$, then the measure of the angle C is

(A) $\frac{\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{2\pi}{3}$

Solution: By hypothesis

$$\sin 3A = k = \sin 3B$$

Therefore

$$2 \cos \frac{3A+3B}{2} \sin \frac{(3A-3B)}{2} = 0$$

$$\frac{3}{2}(A+B) = \frac{\pi}{2} \quad \text{or} \quad A=B$$

But $A > B$, which implies

$$A+B = \frac{\pi}{3}$$

$$\Rightarrow \pi - C = \frac{\pi}{3}$$

$$\Rightarrow C = \frac{2\pi}{3}$$

Answer: (D)

26. General solution of the equation $\sin x - 3 \sin 2x + \sin 3x = \cos x - 3 \cos 2x + \cos 3x$ is

(A) $n\pi + \frac{\pi}{8}$ (B) $\frac{n\pi}{2} + \frac{\pi}{8}$

(C) $(-1)^n \frac{n\pi}{2} + \frac{\pi}{8}$ (D) $2n\pi + \cos^{-1} \frac{2}{3}$

Solution: We have

$$(\sin x + \sin 3x) - 3 \sin 2x = (\cos x + \cos 3x) - 3 \cos 2x$$

$$2 \sin 2x \cos x - 3 \sin 2x = 2 \cos 2x \cos x - 3 \cos 2x$$

$$\sin 2x(2 \cos x - 3) = \cos 2x(2 \cos x - 3)$$

This gives

$$\sin 2x = \cos 2x \quad (\because 2 \cos x \neq 3)$$

$$2x = n\pi + \frac{\pi}{4}$$

That is

$$x = \frac{n\pi}{2} + \frac{\pi}{8}$$

Answer: (B)

27. The equation $|3 \sin^4 x - 2 \sin^6 x + y - 2 \cos^6 x + 3 \cos^4 x|^2 = 9$ is true

- (A) for any value of x and $y = 2$ or $y = -4$
 (B) only for $x = \pi/4$ or π and $y = -2$ or 4
 (C) only for $x = \pi/2$ or π and $y = 2$ or -4
 (D) only for $x = 0$ or $\pi/2$ and $y = \pm 2$

Solution: We simplify the left-hand side of the given equation:

$$|3(\sin^4 x + \cos^4 x) - 2(\sin^6 x + \cos^6 x) + y|^2$$

$$= |3[1 - 2 \sin^2 x \cos^2 x] - 2[1 - 3 \sin^2 x \cos^2 x] + y|^2$$

$$= |1 + y|^2$$

This is equal to the right-hand side (=9) when $y = 2$ or -4 .

Answer: (A)

28. The number of values of x in the range $-\pi/2 < x < \pi/2$ and satisfying the equation $\sin^3 x + \sin^2 x + \sin x - \sin x \sin 2x - \sin 2x - 2 \cos x = 0$ is

- (A) 0 (B) 1 (C) 2 (D) 3

Solution: The given equation can be written as

$$\sin x(1 + \sin x + \sin^2 x) - (1 + \sin x + \sin^2 x)2 \cos x = 0$$

$$(1 + \sin x + \sin^2 x)(\sin x - 2 \cos x) = 0$$

Therefore

$$\sin x - 2 \cos x = 0 \quad (\because 1 + \sin x + \sin^2 x > 0 \text{ for all real } x)$$

$$\tan x = 2$$

as $\cos x = 0$ is not a root of the initial equation. Therefore $x = \tan^{-1} 2$ is the only solution in $(-\pi/2, \pi/2)$.

Answer: (B)

29. If $2 \sec 2\alpha = \tan \beta + \cot \beta$, then $\alpha + \beta$ may be

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) 0

Solution:

$$2 \sec 2\alpha = \tan \beta + \cot \beta$$

$$= \frac{\sin \beta}{\cos \beta} + \frac{\cos \beta}{\sin \beta}$$

$$= \frac{\sin^2 \beta + \cos^2 \beta}{\sin \beta \cos \beta}$$

$$= \frac{1}{\sin \beta \cos \beta}$$

$$= 2 \operatorname{cosec} 2\beta$$

Therefore

$$\sec 2\alpha = \operatorname{cosec} 2\beta$$

Now $2\alpha = 2\beta = \pi/4$ is possible. Therefore

$$\alpha + \beta = \frac{\pi}{4}$$

is a possible value.

Answer: (B)

30. General value of θ satisfying the equation $\sin 6\theta = \sin 4\theta - \sin 2\theta$ is

- (A) $\frac{n\pi}{4}, n \in \mathbb{Z}$ (B) $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
 (C) $\frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (D) $\frac{n\pi}{2}, n \in \mathbb{Z}$

Solution: The given equation can be written as $\sin 6\theta + \sin 2\theta = \sin 4\theta$. Therefore

$$2\sin 4\theta \cos 2\theta = \sin 4\theta$$

$$\sin 4\theta(2\cos 2\theta - 1) = 0$$

Now either

$$4\theta = n\pi$$

or $2\theta = 2n\pi \pm \frac{\pi}{3}$

Therefore

$$\theta = \frac{n\pi}{4}$$

or $\theta = n\pi \pm \frac{\pi}{6}$

Answer: (C)

31. Which of the following statements (only one) is true with regard to the solutions of the equation $\tan^2 x - \sqrt{5}\tan x + 1 = 0$?

- (A) No solution exists in the range $0 \leq x \leq \pi/2$.
 (B) There exist two solutions α and β in the interval $[0, \pi/2]$ such that $\alpha + \beta = \pi/2$.
 (C) Two solutions α, β exist in $[0, \pi/2]$ such that $\alpha + \beta = \pi/4$.
 (D) Two solutions α, β exist in $[0, \pi/2]$ such that $\alpha - \beta = \pi/4$.

Solution: From the given equation we have

$$\tan x = \frac{\sqrt{5} \pm 1}{2}$$

Let

$$\tan \alpha = \frac{\sqrt{5} + 1}{2} \quad \text{and} \quad \tan \beta = \frac{\sqrt{5} - 1}{2}$$

Clearly $\alpha, \beta \in [0, \pi/2]$ and also

$$\tan \beta = \frac{\sqrt{5} - 1}{2} = \frac{2}{\sqrt{5} + 1} = \cot \alpha = \tan \left(\frac{\pi}{2} - \alpha \right)$$

Therefore

$$\beta = \frac{\pi}{2} - \alpha$$

$$\alpha + \beta = \frac{\pi}{2}$$

Answer: (B)

32. The set of values of θ which satisfy the equation $\cos 2\theta = \sin \theta + \cos \theta$ is

- (A) $\theta = n\pi + (\pi/2), n \in \mathbb{Z}$
 (B) $\theta = 2n\pi$ or $\theta = n\pi/4$ where $n \in \mathbb{Z}$
 (C) $\theta = 0$
 (D) $\theta = 2n\pi$ or $2n\pi - (n\pi/2)$ or $\theta = n\pi - (\pi/4), n \in \mathbb{Z}$

Solution: The given equation, $\cos 2\theta = \sin \theta + \cos \theta$, can be written as

$$(\cos \theta + \sin \theta)(\cos \theta - \sin \theta - 1) = 0$$

Therefore

$$\cos \theta + \sin \theta = 0$$

or $\cos \theta - \sin \theta = 1$

Hence

$$\tan \theta = -1$$

or $\cos \left(\theta + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$

This implies that

$$\theta = n\pi - \frac{\pi}{4}$$

or $\theta + \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$

Hence

$$\theta = n\pi - \frac{\pi}{4}$$

or $\theta = 2n\pi$

or $\theta = 2n\pi - \frac{\pi}{2}$

Answer: (D)

33. General solution of the equation $\sin^2 6x + 8\sin^2 3x = 0$ is

- (A) $\frac{n\pi}{3}, n \in \mathbb{Z}$ (B) $\frac{n\pi}{2}, n \in \mathbb{Z}$
 (C) $\frac{n\pi}{4}, n \in \mathbb{Z}$ (D) $\frac{n\pi}{6}, n \in \mathbb{Z}$

Solution: We have

$$\sin^2 6x + 8\sin^2 3x = 0$$

$$4\sin^2 3x(\cos^2 3x + 2) = 0$$

Therefore

$$\sin^2 3x = 0$$

and hence $\sin 3x = 0$. This gives

$$3x = n\pi, n \in \mathbb{Z}$$

$$x = \frac{n\pi}{3}, n \in \mathbb{Z}$$

Answer: (A)

34. The equation $\sin(\cos x) = \cos(\sin x)$ has

- (A) only one real solution
- (B) infinitely many solutions
- (C) no real solution
- (D) two real solutions

Solution: We have

$$\cos[(\pi/2) - \cos x] = \cos(\sin x)$$

Therefore

$$(\pi/2) - \cos x = 2n\pi \pm \sin x, n \in \mathbb{Z}$$

$$\cos x + \sin x = (\pi/2) - 2n\pi = \cos x - \sin x$$

Hence $\sin x = 0$ and $\cos x = (\pi/2) - 2n\pi$ which is impossible, because $\sin x = 0$ implies $\cos x = \pm 1$.

Answer: (C)

35. The value of x satisfying the equation $2\cos^2(x^3 + x) = 2^x + 2^{-x}$ is

- (A) 0
- (B) -1
- (C) $x = 2n\pi$
- (D) $x = n\pi$

Solution: The given equation can be written as

$$1 + \cos(2x^3 + 2x) = 2^x + 2^{-x}$$

Left-hand side is ≤ 2 and right-hand side is ≥ 2 . Therefore

$$\cos(2x^3 + 2x) = 1$$

and $2^x + 2^{-x} = 2$

Now

$$2^x + 2^{-x} = 2 \Leftrightarrow 2^{2x} = 1$$

$$\Leftrightarrow x = 0$$

$$x = 0 \Rightarrow \cos(2x^3 + 2x) = 1$$

Therefore $x = 0$ is the only solution of the equation.

Answer: (A)

36. The number of solutions of the pair of equations $2\sin^2 \theta - \cos 2\theta = 0$ and $2\cos^2 \theta - 3\sin \theta = 0$ in the interval $[0, 2\pi]$ is

- (A) 0
- (B) 1
- (C) 2
- (D) 4

Solution: The first equation is

$$-1 + 4\sin^2 \theta = 0$$

Therefore

$$\sin \theta = \pm \frac{1}{2}$$

which implies

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

The second equation is

$$2\sin^2 \theta + 3\sin \theta - 2 = 0$$

$$(2\sin \theta - 1)(\sin \theta + 2) = 0$$

which implies

$$\sin \theta = \frac{1}{2}$$

or

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Therefore $\pi/6$ and $5\pi/6$ satisfy both equations.

Answer: (C)

37. The number of solutions of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$ in the range $0 \leq x \leq 2\pi$ is

- (A) 4
- (B) 6
- (C) 8
- (D) 10

Solution: We have

$$16^{\sin^2 x} + 16^{\cos^2 x} = 10$$

$$16^{\sin^2 x} + \frac{16}{16^{\sin^2 x}} = 10$$

Substituting $t = 16^{\sin^2 x}$ we get

$$t^2 - 10t + 16 = 0$$

$$(t - 2)(t - 8) = 0$$

$$\Rightarrow t = 2, 8$$

Case 1: When $t = 2$ we have

$$16^{\sin^2 x} = 2$$

$$2^{4\sin^2 x} = 1$$

$$\Rightarrow 4\sin^2 x = 1$$

$$\Rightarrow \sin x = \pm \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Case 2: When $t = 8 = 2^3$ we have

$$16^{\sin^2 x} = 2^3$$

$$2^{4\sin^2 x} = 2^3$$

$$\begin{aligned}\Rightarrow 4 \sin^2 x &= 3 \\ \Rightarrow \sin x &= \pm \frac{\sqrt{3}}{2} \\ \Rightarrow x &= \frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\end{aligned}$$

Therefore there are total eight solutions in $[0, 2\pi]$.

Answer: (C)

38. General solution of the equation $4 \cot 2\theta + \tan^2 \theta = \cot^2 \theta$ is

$$\begin{aligned}(\text{A}) \quad n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z} & \quad (\text{B}) \quad n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z} \\ (\text{C}) \quad 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z} & \quad (\text{D}) \quad 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}\end{aligned}$$

Solution: The given equation can be written as

$$\begin{aligned}\frac{2(1 - \tan^2 \theta)}{\tan \theta} + \tan^2 \theta &= \frac{1}{\tan^2 \theta} \\ \frac{2(1 - \tan^2 \theta) + \tan^3 \theta}{\tan \theta} &= \frac{1}{\tan^2 \theta}\end{aligned}$$

Therefore

$$\begin{aligned}2(1 - \tan^2 \theta) \tan \theta + \tan^4 \theta - 1 &= 0 \\ (1 - \tan^2 \theta)[2 \tan \theta - (1 + \tan^2 \theta)] &= 0 \\ (1 - \tan^2 \theta)(1 - \tan \theta)^2 &= 0 \\ (1 + \tan \theta)(1 - \tan \theta)(1 - \tan \theta)^2 &= 0 \\ (1 - \tan \theta)^3(1 + \tan \theta) &= 0\end{aligned}$$

This gives

$$\begin{aligned}\tan \theta &= \pm 1 \\ \theta &= n\pi + \frac{\pi}{4}, \theta = n\pi - \frac{\pi}{4} \\ \theta &= n\pi \pm \frac{\pi}{4}\end{aligned}$$

Answer: (A)

39. The number of distinct real roots of the equation

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

in the interval $-\pi/4 \leq x \leq \pi/4$ is

$$(\text{A}) \quad 0 \quad (\text{B}) \quad 2 \quad (\text{C}) \quad 3 \quad (\text{D}) \quad 1$$

Solution: Given equation is

$$(\sin x + 2 \cos x)(\sin x - \cos x) = 0$$

Since $-\pi/4 \leq x \leq \pi/4$, $\sin x + 2 \cos x \neq 0$. Therefore

$$\sin x - \cos x = 0 \quad \text{or} \quad \tan x = 1$$

Therefore $x = \pi/4$ is the only solution.

Answer: (D)

40. The number of values of θ in the range $0 \leq \theta \leq 360^\circ$ satisfying the equation $\cos^4 2\theta + 2 \sin^2 2\theta = 17(\sin \theta + \cos \theta)^8$ is

$$(\text{A}) \quad 4 \quad (\text{B}) \quad 6 \quad (\text{C}) \quad 2 \quad (\text{D}) \quad 8$$

Solution: Given equation is

$$(1 - \sin^2 2\theta)^2 + 2 \sin^2 2\theta = 17(1 + \sin 2\theta)^4$$

Put $x = \sin 2\theta$. Therefore

$$(1 - x^2)^2 + 2x^2 = 17(1 + x)^4$$

$$x^4 + 1 = 17(1 + x)^4 = 17(1 + 2x + x^2)^2 \quad (3.1)$$

Clearly $x = \sin 2\theta = 0$ is not a solution of the given equation. Therefore dividing both sides of Eq. (3.1) by x^2 we get

$$x^2 + \frac{1}{x^2} = 17 \left(x + \frac{1}{x} + 2 \right)^2$$

Putting $x + (1/x) = y$, we have

$$\begin{aligned}y^2 - 2 &= 17(y + 2)^2 \\ 16y^2 + 68y + 70 &= 0 \\ 8y^2 + 34y + 35 &= 0 \\ (4y + 7)(2y + 5) &= 0\end{aligned}$$

From this we get

$$y = \frac{-7}{4} \quad \text{or} \quad \frac{-5}{2}$$

Case 1: If $y = -7/4$, then $4x^2 + 7x + 4 = 0$ has no real roots.

Case 2: If $y = -5/2$, then

$$\begin{aligned}2x^2 + 5x + 2 &= 0 \\ x &= \frac{-1}{2}, -2\end{aligned}$$

Now $\sin 2\theta = x = -2$ is not possible. Therefore

$$\sin 2\theta = x = -1/2$$

$$\Rightarrow \theta = (n\pi/2) + (-1)^n (7\pi/12)$$

Now

$$n = 0 \Rightarrow \theta = 7\pi/12 = 105^\circ$$

$$n = 2 \Rightarrow \theta = \pi + (7\pi/12) = 285^\circ$$

$$n = 3 \Rightarrow \theta = (3\pi/2) - (7\pi/12) = 165^\circ$$

$$n = 5 \Rightarrow \theta = (5\pi/2) - (7\pi/12) = 345^\circ$$

Therefore there are total four roots.

Answer: (A)

41. If x is not a multiple of π and satisfies the equation $\sin 5x \cos 3x = \sin 6x \cos 2x$, then x is

(A) $\frac{\pi}{6} + \frac{n\pi}{3}, n \in \mathbb{Z}$ (B) $\frac{\pi}{3} + \frac{n\pi}{6}, n \in \mathbb{Z}$

(C) $\frac{\pi}{6} + n\pi, n \in \mathbb{Z}$ (D) $n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$

Solution: From the given equation we have

$$\frac{1}{2}(\sin 8x + \sin 2x) = \frac{1}{2}(\sin 8x + \sin 4x)$$

Therefore

$$\sin 2x = \sin 4x$$

$$2 \cos 3x \sin x = 0$$

Now

$$x \neq n\pi, n \in \mathbb{Z} \Rightarrow \cos 3x = 0$$

$$\Rightarrow 3x = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow x = \frac{n\pi}{3} + \frac{\pi}{6}$$

Answer: (A)

42. General solution of the equation

$$\left(\cos \frac{x}{4} - 2 \sin x\right) \sin x + \left(1 + \sin \frac{x}{4} - 2 \cos x\right) \cos x = 0$$

is

(A) $(4n+1)\pi, n \in \mathbb{Z}$ (B) $(4n+1)2\pi, n \in \mathbb{Z}$

(C) $(2n+1)2\pi, n \in \mathbb{Z}$ (D) $(2n+1)\pi, n \in \mathbb{Z}$

Solution: Given equation can be written as

$$\sin x \cos \frac{x}{4} + \cos x \sin \frac{x}{4} - 2(\sin^2 x + \cos^2 x) + \cos x = 0$$

$$\sin \frac{5x}{4} + \cos x - 2 = 0$$

Since maximum values of $\sin 5x/4$ and $\cos x$ are 1, it follows that $\sin 5x/4 = 1$ and $\cos x = 1$. Therefore

$$\frac{5x}{4} = 2m\pi + \frac{\pi}{2}, m \in \mathbb{Z}$$

and

$$x = 2k\pi, k \in \mathbb{Z}$$

Therefore

$$x = \frac{8m\pi}{5} + \frac{2\pi}{5} \quad \text{and} \quad x = 2k\pi$$

which gives

$$2k\pi = \left(\frac{8\pi}{5}\right)m + \frac{2\pi}{5}$$

so that

$$k = \frac{4m+1}{5}$$

Since k is an integer, we can take $m = 5n + 1$. Therefore

$$x = 2\pi + 8\pi n = 2\pi(4n+1), n \in \mathbb{Z}$$

Answer: (B)

43. General solution of the equation $\sin^4 2x + \cos^4 2x = \sin 2x \cos 2x$ is

(A) $(2n+1)\frac{\pi}{8}, n \in \mathbb{Z}$ (B) $(4n+1)\frac{\pi}{8}, n \in \mathbb{Z}$

(C) $(2n+1)\frac{\pi}{6}, n \in \mathbb{Z}$ (D) $(4n+1)\frac{\pi}{3}, n \in \mathbb{Z}$

Solution: We have

$$(\sin^2 2x + \cos^2 2x)^2 - 2\sin^2 2x \cos^2 2x = \sin 2x \cos 2x$$

$$2\sin^2 2x \cos^2 2x + \sin 2x \cos 2x - 1 = 0$$

Put $\sin 2x \cos 2x = t$. Therefore

$$2t^2 + t - 1 = 0$$

$$(2t-1)(t+1) = 0$$

$$t = \frac{1}{2}, -1$$

Case 1: When $t = -1$ we have

$$\sin 2x \cos 2x = -1$$

$$\Rightarrow \sin 4x = -2$$

which is impossible. Therefore $t = -1$ is to be rejected.

Case 2: When $t = 1/2$ we have

$$\sin 4x = 1$$

$$\Rightarrow 4x = 2n\pi + (\pi/2)$$

$$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8} = (4n+1)\frac{\pi}{8}$$

Answer: (B)

44. The number of values of x in the range $0 \leq x \leq 2\pi$ satisfying the equation $\tan x + \tan[(\pi/4) + x] = 2$ is

(A) 2 (B) 4 (C) 6 (D) 1

Solution: We have

$$\tan x + 1 + \frac{\tan x}{1 - \tan x} = 2$$

$$\tan^2 x - 4 \tan x + 1 = 0$$

$$\tan x = 2 \pm \sqrt{3}$$

$$x = 15^\circ, 75^\circ, 195^\circ, 255^\circ$$

Answer: (B)

45. Number of solutions of the equation

$$\sin^3 x \cos x + \sin^2 x \cos^2 x + \cos^3 x \sin x = 1$$

in the interval $[0, 2\pi]$ is

- (A) 0 (B) 2 (C) 4 (D) 8

Solution: The given equation can be written as

$$\sin x \cos x (\sin^2 x + \cos^2 x) + \sin^2 x \cos^2 x = 1$$

Substituting $t = \sin x \cos x$ in this equation, we get

$$t^2 + t + 1 = 0$$

The equation $t^2 + t + 1 = 0$ has no real roots.

Answer: (A)

46. The number of solutions of the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ in $[0, 5\pi]$ is

- (A) 0 (B) 5 (C) 6 (D) 10

Solution:

$$3 \sin^2 x - 6 \sin x - \sin x + 2 = 0$$

$$3 \sin x (\sin x - 2) - 1 (\sin x - 2) = 0$$

$$(3 \sin x - 1)(\sin x - 2) = 0$$

This gives $\sin x = 1/3$ (since $\sin x \neq 2$). Therefore

$$x = n\pi + (-1)^n \sin^{-1}(1/3) \quad \text{for } n = 0, 1, 2, 3, 4, 5.$$

Answer: (C)

47. For the following equation

$$3(\sin \theta + \cos \theta) - 2(\sin^3 \theta + \cos^3 \theta) = 8$$

exactly one of the following statements is true. Which one is it?

- (A) The equation has no real solutions.
 (B) The equation has exactly one solution.
 (C) The equation has exactly two solutions in the interval $[0, \pi]$.
 (D) The equation has infinitely many solutions.

Solution: The equation can be written as

$$3(\sin^2 \theta + \cos^2 \theta)(\sin \theta + \cos \theta) - 2(\sin^3 \theta + \cos^3 \theta) = 8$$

$$\sin^3 \theta + \cos^3 \theta + 3 \sin^2 \theta \cos \theta + 3 \sin \theta \cos^2 \theta = 8$$

Therefore

$$(\sin \theta + \cos \theta)^3 = 8 = 2^3$$

Therefore $\sin \theta + \cos \theta = 2$. Hence $\sin \theta = 1$ and $\cos \theta = 1$, which is impossible and so the initial equation has no real roots.

Answer: (A)

48. If real numbers x and y satisfy the equations $4 \tan x = \tan y$ and $5 \sin x \cos y = 1$, then

(A) $x + y = (2n + (-1)^n)\pi/2$ and $x - y =$

$$k\pi + (-1)^k \sin^{-1}(-3/5)$$

(B) $x + y = (2n + 1)\pi/2$ and $x - y = k\pi + \sin^{-1}(3/5)$

(C) $x + y = (n + 1)\pi/2$, $x - y = n\pi/2$

(D) $x + y = 2n\pi$, $x - y = (k + 1)\sin^{-1}(3/5)$

Solution:

$$\frac{\tan x}{\tan y} = \frac{1}{4}$$

$$\Rightarrow \frac{\sin x \cos y}{\cos x \sin y} = \frac{1}{4}$$

$$\Rightarrow \cos x \sin y = \frac{4}{5} \quad \left(\because \sin x \cos y = \frac{1}{5} \right)$$

Therefore from $\sin x \cos y = 1/5$ and $\cos x \sin y = 4/5$, we have

$$\sin(x + y) = 1 \quad \text{and} \quad \sin(x - y) = -3/5$$

Therefore

$$x + y = n\pi + (-1)^n \frac{\pi}{2}$$

and $x - y = k\pi + (-1)^k \sin^{-1} \left(\frac{-3}{5} \right)$

Answer: (A)

49. The number of solutions of the equation

$$1 + \sqrt{3} \tan^2 \theta = (1 + \sqrt{3}) \tan \theta$$

in $[0, 3\pi/2]$ is

- (A) 2 (B) 3 (C) 1 (D) 4

Solution: The given equation is

$$1 + \sqrt{3} \tan^2 \theta - (1 + \sqrt{3}) \tan \theta = 0$$

$$(\sqrt{3} \tan \theta - 1)(\tan \theta - 1) = 0$$

Therefore

$$\tan \theta = 1, \frac{1}{\sqrt{3}}$$

Hence

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{\pi}{6}, \frac{7\pi}{6}$$

belong to $[0, 3\pi/2]$.

Answer: (D)

50. If $\tan^2(x+y) + \cot^2(x+y) = 1 - 2x - x^2$, then

- (A) $x = 1, y = n\pi \pm \frac{\pi}{4} - 1$
 (B) $x = -1, y = n\pi \pm \frac{\pi}{4} + 1$
 (C) $x = -1, y = \frac{n\pi}{2} \pm \frac{\pi}{4} - 1$
 (D) $x = 1, y = \frac{n\pi}{2} \pm \frac{\pi}{4} - 1$

where $n \in \mathbb{Z}$.

Solution: Left-hand side is greater than or equal to 2. This implies

$$1 - 2x - x^2 \geq 2$$

$$(x+1)^2 \leq 0$$

$$\Rightarrow x = -1$$

Hence the given equation is

$$\tan^2(y-1) + \cot^2(y-1) = 2$$

Substituting $\tan(y-1) = a$, we get

$$a^2 + \frac{1}{a^2} - 2 = 0 \quad \text{where}$$

which gives

$$(a^2 - 1)^2 = 0 \Rightarrow a^2 = 1$$

$$\Rightarrow a = \pm 1$$

$$\Rightarrow \tan(y-1) = \pm 1$$

$$\Rightarrow y-1 = n\pi \pm \pi/4$$

$$\Rightarrow y = \left(n\pi \pm \frac{\pi}{4}\right) + 1$$

Therefore

$$x = -1 \quad \text{and} \quad y = \left(n\pi \pm \frac{\pi}{4}\right) + 1$$

Answer: (B)

Multiple Correct Choice Type Questions

1. If x and y are solutions of the equations

$$\sin x \sin y = \frac{\sqrt{3}}{4} \quad \text{and} \quad \cos x \cos y = \frac{\sqrt{3}}{4}$$

then some of the values of x, y (in degrees) are

- (A) $x = 60^\circ, y = 30^\circ$ (B) $x = 30^\circ, y = 60^\circ$
 (C) $x = -30^\circ, y = -60^\circ$ (D) $x = -60^\circ, y = -30^\circ$

Solution: Adding and subtracting the equations, we have

$$\cos(x+y) = 0$$

and

$$\cos(x-y) = \sqrt{3}/2$$

Therefore

$$x+y = (2n+1)\pi/2$$

and

$$x-y = 2n\pi \pm (\pi/6)$$

Now

$$x+y = 90^\circ, x-y = 30^\circ \Rightarrow x = 60^\circ, y = 30^\circ$$

$$x+y = 90^\circ, x-y = -30^\circ \Rightarrow x = 30^\circ, y = 60^\circ$$

$$x+y = -90^\circ, x-y = 30^\circ \Rightarrow x = -30^\circ, y = -60^\circ$$

$$x+y = -90^\circ, x-y = -30^\circ \Rightarrow x = -60^\circ, y = -30^\circ$$

Answers: (A), (B), (C), (D)

2. Some of the common roots of the equations $\cos 2x + \sin 2x = \cot x$ and $2\cos^2 x + \cos^2 2x = 1$ are

- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{-\pi}{4}$ (D) $\frac{\pi}{2}$

Solution: We have

$$\cos 2x + \sin 2x = \cot x$$

$$\Rightarrow \cos 2x + 2\sin x \cos x - \frac{\cos x}{\sin x} = 0$$

$$\Rightarrow \cos 2x - \cos x \left(\frac{1 - 2\sin^2 x}{\sin x} \right) = 0$$

$$\Rightarrow \cos 2x(1 - \cot x) = 0 \quad (3.2)$$

The second equation is

$$2\cos^2 x + \cos^2 2x = 1$$

$$\Rightarrow \cos^2 2x + \cos 2x = 0$$

$$\Rightarrow \cos 2x(\cos 2x + 1) = 0 \quad (3.3)$$

From Eqs. (3.2) and (3.3), the common roots are given by the equation $\cos 2x = 0$. Therefore

$$2x = (2n+1)\frac{\pi}{2}$$

or
$$x = (2n+1)\frac{\pi}{4}$$

Answers: (A), (B), (C)

3. Let

$$f(x) = \frac{1}{\tan^2 x - 1}$$

$$g(x) = 1 + \cos 2x$$

and
$$h(x) = 2^{x+1} - 8$$

Then

(A) $f\left(\frac{\pi}{3}\right) = g\left(\frac{\pi}{3}\right)$ and $h\left(\frac{\pi}{3}\right) < 0$

(B) $f\left(\frac{-\pi}{3}\right) = g\left(\frac{-\pi}{3}\right)$ and $h\left(\frac{-\pi}{3}\right) < 0$

(C) $f\left(\frac{\pi}{6}\right) \neq g\left(\frac{\pi}{6}\right)$ and $h\left(\frac{\pi}{6}\right) < 0$

(D) $f\left(\frac{4\pi}{3}\right) = g\left(\frac{4\pi}{3}\right)$ and $h\left(\frac{4\pi}{3}\right) > 0$

Solution: Consider the equation $f(x) = g(x)$. First, we have that $\cos x \neq 0$. Domain of $f(x)$ is the set of all real x for which $\tan x \neq \pm 1$.

$$\begin{aligned} f(x) &= g(x) \\ \Rightarrow \frac{1}{\tan^2 x - 1} &= 1 + \cos 2x \\ \Rightarrow \frac{-\cos^2 x}{\cos 2x} &= 1 + \cos 2x \\ \Rightarrow \frac{-(1 + \cos 2x)}{2 \cos 2x} &= 1 + \cos 2x \\ \Rightarrow (1 + \cos 2x) \left(1 + \frac{1}{2 \cos 2x} \right) &= 0 \\ \Rightarrow \cos 2x = -1 \quad \text{or} \quad \cos 2x &= \frac{-1}{2} \\ \Rightarrow 2x = 2n\pi \pm \pi, 2x = 2n\pi \pm \frac{2\pi}{3} \\ \Rightarrow x = n\pi \pm \frac{\pi}{2}, x = n\pi \pm \frac{\pi}{3} \end{aligned}$$

$x = n\pi \pm (\pi/2)$ implies $\cos x = 0$ so that we can reject this value. Therefore $x = n\pi \pm (\pi/3)$ satisfy the equation $f(x) = g(x)$. But

$$h(x) = 2^{x+1} - 8 > 0 \Leftrightarrow x > 2$$

Therefore all (A), (B), (C) and (D) are true.

Answers: (A), (B), (C), (D)

4. The values of $\theta \in (0, \pi/2)$ and satisfying the equation

$$\begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & 1 + \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$$

are

(A) $\frac{7\pi}{24}$ (B) $\frac{5\pi}{24}$ (C) $\frac{11\pi}{24}$ (D) $\frac{\pi}{24}$

Solution: Adding column 2 to column 1 we get

$$\begin{vmatrix} 2 & \cos^2 \theta & 4 \sin 4\theta \\ 2 & 1 + \cos^2 \theta & 4 \sin 4\theta \\ 1 & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$$

Now $R_2 - R_1$, $R_3 - R_1$ implies

$$\begin{vmatrix} 2 & \cos^2 \theta & 4 \sin 4\theta \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 1(2 + 4 \sin 4\theta) = 0$$

$$\Rightarrow \sin 4\theta = \frac{-1}{2} = \sin\left(\frac{-\pi}{6}\right)$$

$$\Rightarrow 4\theta = n\pi + (-1)^n \left(\frac{-\pi}{6}\right), n \in \mathbb{Z}$$

Therefore

$$\theta = \frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$$

is the general value of θ . Therefore

$$\theta = \frac{7\pi}{24} \quad (\text{when } n = 1)$$

and
$$\theta = \frac{11\pi}{24} \quad (\text{when } n = 2)$$

Answers: (A), (C)

5. Solution of the equation $\sin x + \cos x - 2\sqrt{2} \sin x \cos x = 0$ is

(A) $x = n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$

(B) $2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$

(C) $n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$

(D) $n\pi - \frac{\pi}{4} + (-1)^{n+1} \frac{\pi}{6}, n \in \mathbb{Z}$

Solution: Put $\sin x + \cos x = y$. Therefore

$$2 \sin x \cos x = y^2 - 1$$

The given equation is

$$\begin{aligned} y - \sqrt{2}(y^2 - 1) &= 0 \\ \Rightarrow \sqrt{2}y^2 - y - \sqrt{2} &= 0 \\ \Rightarrow (y - \sqrt{2})(\sqrt{2}y + 1) &= 0 \end{aligned}$$

Therefore we get

$$y = \sqrt{2}, \frac{-1}{\sqrt{2}}$$

Case 1: When $y = \sqrt{2}$ we have

$$\sin x + \cos x = \sqrt{2}$$

Therefore

$$\begin{aligned} \sin\left(x + \frac{\pi}{4}\right) &= 1 \\ \Rightarrow x &= n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z} \end{aligned}$$

Case 2: When $y = -1/\sqrt{2}$ we have

$$\sin x + \cos x = \frac{-1}{\sqrt{2}}$$

Therefore

$$\begin{aligned} \sin\left(x + \frac{\pi}{4}\right) &= \frac{-1}{2} \\ \Rightarrow x &= n\pi - \frac{\pi}{4} + (-1)^{n+1} \frac{\pi}{6}, n \in \mathbb{Z} \end{aligned}$$

Answers: (A), (D)

6. If x is a solutions of the equation $\sin^2 x (\tan x + 1) = 3 \sin x (\cos x - \sin x) + 3$, then x is

(A) $n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$ (B) $n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$

(C) $n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$ (D) $n\pi - \frac{\pi}{3}, n \in \mathbb{Z}$

Solution: The given equation can be written as

$$\sin^2 x \tan x + \sin^2 x - 3 \sin x \cos x + 3 \sin^2 x - 3 = 0$$

Dividing by $\cos^2 x$ (clearly $\cos x \neq 0$), we have

$$\tan^3 x + \tan^2 x - 3 \tan x - 3 = 0$$

(using $\sec^2 x = 1 + \tan^2 x$). Clearly $\tan x = -1$ is a root and hence

$$(\tan x + 1)(\tan^2 x - 3) = 0$$

Therefore

$$\tan x = -1, \tan x = \pm \sqrt{3}$$

From this we get

$$x = n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$$

and

$$x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

Answers: (B), (C), (D)

7. Solution of the equation $1 + \sin x + \cos 3x = \cos x + \cos 2x + \sin 2x$ is

(A) $x = n\pi, n \in \mathbb{Z}$

(B) $x = 2n\pi \pm (\pi/3), n \in \mathbb{Z}$

(C) $x = 2n\pi \pm (\pi/8)$

(D) $x = n\pi + (-1)^{n+1}(\pi/6), n \in \mathbb{Z}$

Solution: Given equation is

$$1 + \sin x + \cos 3x = \cos x + \cos 2x + \sin 2x$$

Solving this we get

$$1 + \sin x - \sin 2x = \cos x - \cos 3x + \cos 2x$$

$$1 + \sin x - 2 \sin x \cos x = 2 \sin 2x \sin x + 1 - 2 \sin^2 x$$

$$\sin x(1 - 2 \cos x) = 4 \sin^2 x \cos x - 2 \sin^2 x$$

$$\sin x(1 - 2 \cos x) = -2 \sin^2 x(1 - 2 \cos x)$$

$$\sin x(1 - 2 \cos x)(1 + 2 \sin x) = 0$$

Therefore

$$\sin x = 0$$

or

$$\cos x = 1/2$$

or

$$\sin x = -1/2$$

Hence

$$x = n\pi, n \in \mathbb{Z}$$

or

$$x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

or

$$x = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in \mathbb{Z}$$

Answers: (A), (B), (D)

8. Let α be a solution of $\tan x + \sin 2x = \sec x$. Then

(A) $\alpha \neq 2n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$

(B) $\alpha = n\pi + (-1)^n \sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right), n \in \mathbb{Z}$

(C) $\alpha = 2n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$

(D) $\alpha = n\pi + (-1)^n \sin^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right), n \in \mathbb{Z}$

Solution: The given equation can be written as

$$\sin x + 2 \sin x \cos^2 x = 1$$

$$\sin x + 2 \sin x (1 - \sin^2 x) = 1$$

$$2 \sin^3 x - 3 \sin x + 1 = 0$$

$$(\sin x - 1)(2 \sin^2 x + 2 \sin x - 1) = 0$$

Now

$$\sin x = 1 \Rightarrow \cos x = 0$$

so that $\tan x$ and $\sec x$ are not defined. Therefore

$$x \neq (2n\pi) + \frac{\pi}{2}$$

and hence

$$\sin x = \frac{-1 \pm \sqrt{3}}{2}$$

Since

$$\frac{-1 - \sqrt{3}}{2} < -1$$

we have

$$\sin x = \frac{\sqrt{3}-1}{2}$$

or

$$x = \sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$$

Answers: (A), (B)

9. Let θ be a solution of the equation $(1 - \tan x)(1 + \sin 2x) = 1 + \tan x$. Then

(A) $\theta = (4n+1)\frac{\pi}{4}$

(B) $\theta = (4n-1)\frac{\pi}{4}$

(C) $\theta = n\pi$

(D) $\theta = n\pi + \frac{\pi}{8}, n \in \mathbb{Z}$

Solution: Clearly $\tan x = 1$ is not a solution while $\tan x = -1$ is a solution. Therefore

$$\theta \neq n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$$

and $\theta = n\pi - \frac{\pi}{4} = (4n-1)\frac{\pi}{4}, n \in \mathbb{Z}$

If $\tan x \neq \pm 1$, then the given equation reduces to

$$1 - \tan^2 x = 1 + \tan^2 x$$

Therefore

$$2 \tan^2 x = 0 \Rightarrow x = n\pi$$

Hence

$$\theta = n\pi, n \in \mathbb{Z}$$

Answers: (B), (C)

10. If $\cos(x+1) \sin 2(x+1) = \cos 3(x+1) \sin 4(x+1)$, then

(A) $x+1 = \frac{n\pi}{2}, n \in \mathbb{Z}$

(B) $x+1 = \frac{n\pi}{3}, n \in \mathbb{Z}$

(C) $x+1 = \frac{n\pi}{6}, n \in \mathbb{Z}$

(D) $x+1 = (2n+1)\frac{n\pi}{10}$

Solution: The given equation can be transformed to

$$\sin 3(x+1) + \sin(x+1) = \sin 7(x+1) + \sin(x+1)$$

$$\sin 3(x+1) = \sin 7(x+1)$$

$$\sin 7(x+1) - \sin 3(x+1) = 0$$

$$2 \cos 5(x+1) \sin 2(x+1) = 0$$

This gives

$$5(x+1) = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

or

$$2(x+1) = n\pi, n \in \mathbb{Z}$$

Answers: (A), (D)

11. If

$$\frac{2 \tan(x/2)}{1 + \tan^2(x/2)} = y^2 - 4y + 5$$

then (x, y) can be

(A) $\left(\frac{\pi}{2}, 2\right)$

(B) $\left(\frac{5\pi}{2}, 2\right)$

(C) $\left((4n+1)\frac{\pi}{2}, 2\right), n \in \mathbb{Z}$

(D) $\left(\frac{n\pi}{2}, -2\right), n \in \mathbb{Z}$

Solution: Given equation is

$$\sin x = y^2 - 4y + 5$$

$$y^2 - 4y + 5 = (y-2)^2 + 1 \geq 1$$

(equals to 1 if and only if $y = 2$). Therefore the minimum value of right-hand side is 1, whereas the maximum value of $\sin x$ is 1.

$$\sin x = 1 \Leftrightarrow x = (4n + 1)\frac{\pi}{2}$$

and in such a case $y = 2$.

Answers: (A), (B), (C)

12. If

$$\frac{1}{\cos x \cos 2x} + \frac{1}{\cos 2x \cos 3x} + \frac{1}{\cos 3x \cos 4x} = 0$$

then possible values of x are

- (A) $\frac{\pi}{3}$ (B) $\frac{-\pi}{3}$
 (C) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (D) $\frac{n\pi}{3}, n \in \mathbb{Z}$

Solution: Suppose $\sin x \neq 0$. Then

$$\begin{aligned} T_1 &= \frac{\sin x}{\sin x \cos x \cos 2x} = \frac{\sin(2x - x)}{\sin x \cos x \cos 2x} \\ &= \frac{1}{\sin x} [\tan 2x - \tan x] \\ T_2 &= \frac{1}{\sin x} [\tan 3x - \tan 2x] \\ T_3 &= \frac{1}{\sin x} [\tan 4x - \tan x] \end{aligned}$$

Therefore

$$\begin{aligned} T_1 + T_2 + T_3 &= 0 \\ \Rightarrow \tan 4x - \tan x &= 0 \\ \Rightarrow \sin 3x &= 0 \\ \Rightarrow 3x &= n\pi, n \in \mathbb{Z} \quad \text{and} \quad n \neq 3k, k \in \mathbb{Z} \\ \Rightarrow x &= \frac{n\pi}{3}, n \text{ is an integer} \neq 3k, k \in \mathbb{Z} \\ \Rightarrow x &= k\pi \pm \frac{\pi}{3}, k \in \mathbb{Z} \end{aligned}$$

Answers: (A), (B), (C)

13. If

$$\frac{\cos^2 2x}{\cos x + (1/\sqrt{2})} = \cos x - \frac{1}{\sqrt{2}}$$

then x is equal to

- (A) $2n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$ (B) $2n\pi \pm \frac{\pi}{6}$
 (C) $n\pi \pm \frac{\pi}{2}, n \in \mathbb{Z}$ (D) $n\pi \pm \frac{\pi}{3}$

Solution: Given equation is

$$\cos^2 2x = \cos^2 x - \frac{1}{2}$$

$$\Rightarrow 2\cos^2 2x = \cos 2x$$

$$\Rightarrow \cos 2x(2\cos 2x - 1) = 0$$

Case 1: $\cos 2x = 0$. Therefore

$$\cos x = \pm 1/\sqrt{2}$$

But $\cos x \neq -1/\sqrt{2}$. Therefore $\cos x = +1/\sqrt{2}$. So,

$$x = 2n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$$

Case 2: $2\cos 2x - 1 = 0$. Therefore

$$\cos 2x = 1/2$$

$$\Rightarrow \cos x = \pm \sqrt{3}/2$$

$$\Rightarrow x = 2n\pi \pm \pi/6$$

Answers: (A), (B)

14. The values of θ in $[0, 2\pi]$ satisfying the equation

$$(\sin 2\theta + \sqrt{3} \cos 2\theta)^2 = 5 + \cos(2\theta - \pi/6)$$

are

- (A) $\theta = \frac{7\pi}{12}$ (B) $\theta = \frac{3\pi}{4}$
 (C) $\frac{5\pi}{12}$ (D) $\frac{19\pi}{12}$

Solution: Given equation is

$$4\left(\frac{1}{2}\sin 2\theta + \frac{\sqrt{3}}{2}\cos 2\theta\right)^2 = 5 + \cos\left(2\theta - \frac{\pi}{6}\right)$$

$$4\left[\sin \frac{\pi}{6} \sin 2\theta + \cos \frac{\pi}{6} \cos 2\theta\right]^2 = 5 + \cos\left(2\theta - \frac{\pi}{6}\right)$$

Therefore

$$4\cos^2\left(2\theta - \frac{\pi}{6}\right) = 5 + \cos\left(2\theta - \frac{\pi}{6}\right)$$

Substituting $2\theta - (\pi/6) = \alpha$ we get

$$4\cos^2 \alpha - \cos \alpha - 5 = 0$$

$$(4\cos \alpha - 5)(1 + \cos \alpha) = 0$$

Now, one solution is $\cos \alpha = -1$. Then

$$2\theta - \frac{\pi}{6} = \alpha = 2n\pi \pm \pi$$

$$2\theta = 2n\pi + \frac{7\pi}{6}$$

or
$$\theta = 2n\pi - \frac{5\pi}{6}$$

Therefore

$$\theta = n\pi + \frac{7\pi}{12}$$

or
$$\theta = n\pi - \frac{5\pi}{12}$$

Hence

$$\theta = \frac{7\pi}{12} (n=0) \quad \text{and} \quad \theta = \frac{19\pi}{12} (n=1)$$

Answers: (A), (D)

15. If $\sin^3 x + \cos^3 x + \sin x \cos x = 1$, then x is equal to

- (A) $2n\pi, n \in \mathbb{Z}$ (B) $n\pi, n \in \mathbb{Z}$
 (C) $2n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$ (D) $2n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$

Solution: Given equation can be written as

$$(\sin x + \cos x)(\sin^2 x - \sin x \cos x + \cos^2 x)$$

$$-(1 - \sin x \cos x) = 0$$

$$(\sin x + \cos x)(1 - \sin x \cos x) - (1 - \sin x \cos x) = 0$$

$$(\sin x + \cos x - 1)(1 - \sin x \cos x) = 0$$

Therefore

$$\sin x + \cos x = 1$$

or
$$1 - \sin x \cos x = 0$$

This gives

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

or
$$\sin 2x = 2$$

But

$$\sin 2x \neq 2$$

Therefore

$$x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$$

$$x = 2n\pi + \frac{\pi}{2}$$

or
$$x = 2n\pi, n \in \mathbb{Z}$$

Answers: (A), (C)

16. Let $f(x) = \tan^2 x + \cos 2x - 1$. Then $f(x) = 0$ has

- (A) four solutions in $(0, 5\pi)$
 (B) two solutions in $(0, \pi)$
 (C) two solutions in $(\pi, 3\pi/2)$
 (D) one solution in $(0, \pi/2)$

Solution:

$$f(x) = 0$$

$$\Rightarrow \tan^2 x - 2\sin^2 x = 0$$

$$\Rightarrow \sin^2 x(1 - 2\cos^2 x) = 0$$

$$\Rightarrow \sin x = 0, \cos x = \pm 1/\sqrt{2}$$

$$\Rightarrow x = n\pi \quad \text{or} \quad x = 2n\pi \pm \frac{\pi}{4}, 2n\pi \pm \frac{3\pi}{4}$$

Answers: (A), (B), (C), (D)

17. If $2(\cos x + \cos 2x) + (1 + 2\cos x)\sin 2x = 2\sin x$, then some of the values of x in the interval $[-\pi, \pi]$ are

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{-\pi}{3}$ (D) $\frac{-\pi}{2}$

Solution: Given that

$$2\cos x + 2\cos 2x + \sin 2x + 2\cos x \sin 2x - 2\sin x = 0$$

$$2\cos x + 2\cos 2x + 2\sin x \cos x + 4\sin x \cos^2 x - 2\sin x = 0$$

$$2\cos x + 2\cos 2x + 2\sin x \cos x + 4\sin x(1 - \sin^2 x)$$

$$- 2\sin x = 0$$

$$2\cos x + 2\cos 2x + 2\sin x \cos x + 2\sin x - 4\sin^3 x = 0$$

$$2\cos x + 2\cos 2x + 2\sin x \cos x + 2\sin x$$

$$- (3\sin x - \sin 3x) = 0$$

$$2\cos x + 2\cos 2x + 2\sin x \cos x + \sin 3x - \sin x = 0$$

$$2\cos x + 2\cos 2x + 2\sin x \cos x + 2\cos 2x \sin x = 0$$

$$\begin{aligned}\cos x(1 + \sin x) + \cos 2x(1 + \sin x) &= 0 \\ (1 + \sin x)(\cos x + \cos 2x) &= 0 \\ (1 + \sin x)2\left(\cos \frac{3x}{2} \cos \frac{x}{2}\right) &= 0\end{aligned}$$

Therefore

$$\begin{aligned}1 + \sin x = 0 &\Rightarrow x = n\pi + (-1)^{n+1} \frac{\pi}{2}, n \in \mathbb{Z} \\ \cos \frac{x}{2} = 0 &\Rightarrow x = 2(2n + 1) \frac{\pi}{2} = (2n + 1)\pi, n \in \mathbb{Z} \\ \cos \frac{3x}{2} = 0 &\Rightarrow x = (2n + 1) \frac{\pi}{3}, n \in \mathbb{Z}\end{aligned}$$

The values $\pi/3$ and $-\pi/3$ correspond to $n = 0, -1$. When $n = 0$ in

$$x = n\pi + (-1)^{n+1} \frac{\pi}{2}$$

then $x = -\pi/2$.

Answers: (A), (C), (D)

18. If the equation $\cos 2x + a \sin x = 2a - 7$ has solutions, then

- (A) $0 < a < 2$
- (B) $2 \leq a \leq 6$
- (C) $x = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$, if $a = 5$
- (D) $x = -\pi/6$ is a solution, if $a = 3$

Solution: The given equation is equivalent to

$$(1 - 2\sin^2 x) + a \sin x = 2a - 7$$

$$2\sin^2 x - a \sin x - 8 + 2a = 0$$

Therefore

$$\begin{aligned}\sin x &= \frac{a \pm \sqrt{a^2 - 8(2a - 8)}}{4} \\ &= \frac{a \pm \sqrt{(a - 8)^2}}{4} \\ &= \frac{a + (a - 8)}{4} \quad \text{or} \quad \frac{a - (a - 8)}{4} \\ &= \frac{a - 4}{2} \quad \text{or} \quad 2\end{aligned}$$

But $\sin x \neq 2$. Therefore

$$\sin x = \frac{a - 4}{2}$$

provided

$$-1 \leq \frac{a - 4}{2} \leq 1$$

Therefore equation possess solution if $2 \leq a \leq 6$.

Case 1: When $a = 5$, then $\sin x = 1/2$ so that

$$x = n\pi + (-1)^n \frac{\pi}{6}$$

is the general solution.

Case 2: If $a = 3$, then $\sin x = -1/2$ and hence $x = -\pi/6$ is a solution.

Answers: (B), (C), (D)

19. If $\alpha \neq n\pi, n \in \mathbb{Z}$, then the equation $\sin 3\alpha = 4 \sin \alpha \sin(x + \alpha) \sin(x - \alpha)$ has

- (A) a solution in the first quadrant
- (B) no solution in the second quadrant
- (C) a solution in the third quadrant
- (D) no solution in the fourth quadrant

Solution: $\alpha \neq n\pi \Rightarrow \sin \alpha \neq 0$. Therefore

$$\frac{\sin 3\alpha}{\sin \alpha} = 4(\sin^2 x - \sin^2 \alpha)$$

$$3 - 4\sin^2 \alpha = 4\sin^2 x - 4\sin^2 \alpha$$

$$\sin x = \pm \frac{\sqrt{3}}{2}$$

Therefore, $x = 60^\circ, 120^\circ, 240^\circ$ and 300° are solutions of the equation in each of the quadrants.

Answers: (A), (C)

20. The equation $\cos^7 x + \sin^4 x = 1$ has

- (A) exactly two solutions in $0 \leq x \leq \pi/2$
- (B) exactly one solution in $\pi/2 < x \leq \pi$
- (C) exactly one solution in $\pi < x \leq 3\pi/2$
- (D) no solution in $3\pi/2 < x < 2\pi$

Solution: Clearly $x = 2n\pi, n \in \mathbb{Z}$ is a solution of the given equation. Also $x = (2k + 1)\pi/2, k \in \mathbb{Z}$ is also a solution. Further, the equation can be written as

$$\cos^3 x(1 - \sin^2 x)^2 = (1 - \sin^2 x)(1 + \sin^2 x)$$

$$(1 - \sin^2 x)(\cos^5 x - 1 - \sin^2 x) = 0$$

$$\sin x = \pm 1 \quad (\because \cos^5 x \neq 1 + \sin^2 x)$$

Answers: (A), (C), (D)

Matrix-Match Type Questions

1. Some trigonometric equations are given in Column I and some of their solutions are in Column II. Match them.

Column I	Column II
(A) $\sin x + \cos x = \sqrt{2}$	(p) $\frac{\pi}{6}$
(B) $\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x = 1$	(q) $\frac{9\pi}{4}$
(C) $\cos 3x + \sin 2x - \sin 4x = 0$	(r) $\frac{\pi}{4}$
(D) $\sin 5x \cos 3x = \sin 6x \cos 2x$	(s) $\frac{13\pi}{6}$
	(t) $\frac{\pi}{2}$

Solution:

$$(A) \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = 1$$

$$\sin\left(x + \frac{\pi}{4}\right) = 1 = \sin \frac{\pi}{2}$$

Therefore

$$x + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{2}, n \in \mathbb{Z}$$

Now

$$n = 0 \Rightarrow x = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$n = 2 \Rightarrow x = 2\pi + \frac{\pi}{4} = \frac{9\pi}{4}$$

Answer: (A) $\rightarrow$ (q), (r)

$$(B) 1 = \frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x = \sin\left(x + \frac{\pi}{3}\right)$$

Therefore

$$x + \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{2}, n \in \mathbb{Z}$$

Now

$$n = 0 \Rightarrow x = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$$

$$n = 2 \Rightarrow x = 2\pi + \frac{\pi}{6} = \frac{13\pi}{6}$$

Answer: (B) $\rightarrow$ (p), (s)

$$(C) \cos 3x + \sin 2x - \sin 4x = 0$$

$$\Rightarrow \cos 3x - 2 \cos 3x \sin x = 0$$

$$\Rightarrow \cos 3x(1 - 2 \sin x) = 0$$

$$\Rightarrow \cos 3x = 0 \quad \text{or} \quad \sin x = 1/2$$

Case 1: $\cos 3x = 0$. Then

$$3x = (2n + 1) \frac{\pi}{2}, n \in \mathbb{Z}$$

$$\Rightarrow x = (2n + 1) \frac{\pi}{6}$$

Now $n = 1 \Rightarrow x = \pi/2$.

Case 2: $\sin x = 1/2$. Then

$$x = n\pi + (-1)^n \frac{\pi}{6}$$

Answer: (C) $\rightarrow$ (p), (s), (t)

$$(D) \sin 5x \cos 3x = \sin 6x \cos 2x$$

$$\Rightarrow \sin 8x + \sin 2x = \sin 8x + \sin 4x$$

$$\Rightarrow \sin 4x - \sin 2x = 0$$

$$\Rightarrow 2 \cos 3x \sin x = 0$$

Therefore

$$x = n\pi, x = k\pi/3 + \pi/6$$

Answer: (D) $\rightarrow$ (p), (s), (t)

2. Match the trigonometric equations of Column I with some of their solutions in Column II.

Column I	Column II
(A) $\sin x + \cos x = 1 + \sin x \cos x$	(p) $\frac{\pi}{2}$
(B) $2 \sin \frac{x}{2} \cos^2 x - 2 \sin \frac{x}{2} \sin^2 x = \cos 2x$	(q) $\frac{\pi}{6}$
(C) $\sin^2 x + \sin^2 2x = 1$	(r) $\frac{\pi}{4}, \frac{5\pi}{3}$
	(s) 0
(D) $\sin^4 x - \cos^4 x = \frac{1}{2}$	(t) $\pm \frac{\pi}{3}$

Solution:

$$(A) \sin x + \cos x = 1 + \sin x \cos x$$

Put $\sin x + \cos x = y$. Therefore

$$y = 1 + \frac{y^2 - 1}{2}$$

$$\begin{aligned}y^2 - 2y + 1 &= 0 \\(y - 1)^2 &= 0 \\y &= 1\end{aligned}$$

Hence

$$\begin{aligned}\sin x + \cos x &= 1 \\\cos\left(x - \frac{\pi}{4}\right) &= \frac{1}{\sqrt{2}} \\x &= \frac{\pi}{4} + \left(2n\pi \pm \frac{\pi}{4}\right)\end{aligned}$$

For $n = 0$, we have $x = \pi/2, 0$.

Answer: (A) → (p), (s)

(B) We have

$$\begin{aligned}2\sin\frac{x}{2}\cos 2x &= \cos 2x \\ \left(2\sin\frac{x}{2} - 1\right)\cos 2x &= 0 \\\sin\frac{x}{2} = \frac{1}{2} \quad \text{or} \quad \cos 2x &= 0\end{aligned}$$

Case 1: We have $\sin(x/2) = 1/2$. This implies

$$x = 2n\pi + (-1)^n \frac{\pi}{3}$$

For $n = 1$, $x = 5\pi/3$.

Case 2: We have $\cos 2x = 0$. This implies

$$x = (2n + 1)\frac{\pi}{4}$$

For $n = 0$, $x = \pi/4$.

Answer: (B) → (r)

(C) $\sin^2 x + \sin^2 2x = 1$. Therefore

$$\begin{aligned}\frac{1 - \cos 2x}{2} + \frac{1 - \cos 4x}{2} &= 1 \\\cos 4x + \cos 2x &= 0 \\2\cos 3x \cos x &= 0 \\x &= (2n + 1)\frac{\pi}{6}, \quad x = (2n + 1)\frac{\pi}{2}\end{aligned}$$

But

$$x = \left\{(2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$$

is a subset of

$$x = \left\{(2n + 1)\frac{\pi}{6}, n \in \mathbb{Z}\right\}$$

Therefore, the general solution is

$$x = (2n + 1)\frac{\pi}{6}$$

Now $n = 0, 1 \Rightarrow x = \pi/6$ and $x = \pi/2$.

Answer: (C) → (p), (q)

(D) We have

$$\begin{aligned}\sin^4 x - \cos^4 x &= 1/2 \\\Rightarrow \sin^2 x - \cos^2 x &= \frac{1}{2} \\\Rightarrow \cos 2x &= -\frac{1}{2} \\\Rightarrow 2x &= 2n\pi \pm \frac{2\pi}{3} \\\Rightarrow x &= n\pi \pm \frac{\pi}{3}\end{aligned}$$

Therefore $n = 0 \Rightarrow x = \pm\pi/3$.

Answer: (D) → (t)

3. Column I consists of trigonometric equations and Column II consists of their solutions. Match them.

Column I	Column II
(A) $\cos 3x + \sin 5x = 0$	(p) $\frac{3\pi}{4}$
(B) $\sin x \cos 5x = \sin 9x \cos 3x$	(q) π
(C) $\cos 2x - \cos 8x + \cos 6x = 1$	(r) $\frac{\pi}{3}$
(D) $4\sin x \sin 2x \sin 3x = \sin 4x$	(s) $\frac{\pi}{8}$
	(t) $\frac{3\pi}{16}$

Solution:

(A) We have

$$\begin{aligned}\cos 3x + \sin 5x &= 0 \\\cos 3x + \cos\left(\frac{\pi}{2} - 5x\right) &= 0 \\2\cos\left(\frac{3x + (\pi/2) - 5x}{2}\right)\cos\left(\frac{3x - (\pi/2) + 5x}{2}\right) &= 0 \\\cos\left(\frac{\pi}{4} - x\right)\cos\left(4x - \frac{\pi}{4}\right) &= 0\end{aligned}$$

Therefore

$$x - \frac{\pi}{4} = (2n + 1)\frac{\pi}{2}$$

$$\text{or } 4x - \frac{\pi}{4} = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

which gives

$$x = n\pi + \frac{3\pi}{4} \quad \text{or} \quad x = \frac{n\pi}{4} + \frac{3\pi}{16}, n \in \mathbb{Z}$$

When

$$n=0 \Rightarrow x = 3\pi/4 \quad \text{or} \quad 3\pi/16$$

Answer: (A) $\rightarrow$ (p), (t)

(B) The given equation is $\sin x \cos 5x = \sin 9x \cos 3x$. Solving it we get

$$\sin 6x - \sin 4x = \sin 12x + \sin 6x$$

$$\sin 12x + \sin 4x = 0$$

$$2 \sin 8x \cos 4x = 0$$

Now two cases arise.

Case 1: $\sin 8x = 0$. Then $x = n\pi/8$.

Case 2: $\cos 4x = 0$. Then $x = (2n+1)\pi/8$.

Also

$$n=6 \Rightarrow x = 3\pi/4$$

$$n=8 \Rightarrow x = \pi$$

Answer: (B) $\rightarrow$ (p), (q)

(C) The given equation is $\cos 2x - \cos 8x + \cos 6x = 1$. So

$$2 \sin 5x \sin 3x = 1 - \cos 6x = 2 \sin^2 3x$$

$$\sin 3x(\sin 5x - \sin 3x) = 0$$

$$\sin 3x(-2 \cos 4x \sin x) = 0$$

Three cases arise.

Case 1: $\sin x = 0$. Then $x = n\pi, n \in \mathbb{Z}$.

Case 2: $\sin 3x = 0$. Then $x = n\pi/3, n \in \mathbb{Z}$.

Case 3: $\cos 4x = 0$. Then $x = (2n+1)\pi/8, n \in \mathbb{Z}$.

Therefore $x = \pi, \pi/3, \pi/8$ are solutions.

Answer: (C) $\rightarrow$ (q), (r), (s)

(D) We have

$$4 \sin x \sin 2x \sin 3x = \sin 4x = 2 \sin 2x \cos 2x$$

$$\sin 2x(2 \sin x \sin 3x - \cos 2x) = 0$$

$$\sin 2x(\cos 2x - \cos 4x - \cos 2x) = 0$$

$$\sin 2x \cos 4x = 0$$

Therefore

$$x = \frac{n\pi}{2} \quad \text{or} \quad x = (2n+1)\frac{\pi}{8}$$

Hence $x = \pi$ or $x = \pi/8$ are solutions.

Answer: (D) $\rightarrow$ (q), (s)

4. Match the entries of Column I with the entries of Column II.

Column I

Column II

- | | |
|---|---------------------------|
| (A) If $\sqrt{\tan x + \sin x} + \sqrt{\tan x - \sin x} = \sqrt{3 \tan x}$ then $\sin x$ value is | (p) 0 |
| (B) If $1 + \cos x + \cos^2 x + \cos x ^3 + \dots + \infty$ is 2 ($0 < x < \pi/2$) then $\cos x$ is equal to | (q) $\frac{\sqrt{3}}{2}$ |
| (C) If $\tan\left(\frac{\pi}{4} + x\right) + \tan\left(\frac{\pi}{4} - x\right) = 2$, then $\tan x$ is equal to | (r) $\frac{-\sqrt{3}}{2}$ |
| (D) If $7 \cos^2 \theta + 3 \sin^2 \theta = 4$, then $\sin \theta$ is | (s) $\frac{1}{2}$ |
| | (t) $\frac{-1}{2}$ |

Solution:

(A) Squaring both sides, we have

$$\tan x = 2\sqrt{\tan^2 x - \sin^2 x}$$

$$\tan^2 x = 4(\tan^2 x - \sin^2 x) = \frac{4 \sin^4 x}{\cos^2 x}$$

$$\sin^2 x(1 - 4 \sin^2 x) = 0$$

$$\sin x = 0 \quad \text{or} \quad \pm 1/2$$

Answer: (A) $\rightarrow$ (p), (s), (t)

(B) $0 < x < \frac{\pi}{2} \Rightarrow |\cos x| \neq 1$. Therefore the given sum equals

$$\frac{1}{1 - |\cos x|} = 2$$

$$1 - |\cos x| = 1/2$$

$$|\cos x| = 1/2$$

$$\cos x = \pm 1/2$$

Answer: (B) $\rightarrow$ (s), (t)

(C) We have

$$\tan\left(\frac{\pi}{4} + x\right) + \tan\left(\frac{\pi}{4} - x\right) = 2$$

Therefore

$$\tan\left(\frac{\pi}{4} + x\right) + \cot\left(\frac{\pi}{4} + x\right) = 2$$

The left-hand side is ≥ 2 . Therefore

$$\tan\left(\frac{\pi}{4} + x\right) = 1$$

$$\tan x = 0$$

Answer: (C) $\rightarrow$ (p)

(D) We have

$$7(1 - \sin^2 \theta) + 3\sin^2 \theta = 4$$

Therefore

$$-4\sin^2 \theta = -3$$

$$\sin^2 \theta = \frac{3}{4}$$

Answer: (D) → (q), (r)

Comprehension-Type Questions

1. Passage: Using trigonometric transformations, answer questions (i), (ii) and (iii).

(i) If $\sin 5x \sin 4x + \cos 6x \cos 3x = 0$, then x is equal to

(A) $\left\{ \frac{\pi}{4} + \frac{n\pi}{2}, n \in \mathbb{Z} \right\} \cup \left\{ \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\}$

(B) $\left\{ n\pi + \frac{\pi}{3}, n \in \mathbb{Z} \right\} \cup \left\{ \frac{n\pi}{3} + \frac{\pi}{4}, n \in \mathbb{Z} \right\}$

(C) $\left\{ k\pi + \frac{\pi}{4}, k \in \mathbb{Z} \right\} \cup \left\{ \frac{n\pi}{2} + \frac{\pi}{6}, n \in \mathbb{Z} \right\}$

(D) $\left\{ 2n\pi \pm \frac{\pi}{8}, n \in \mathbb{Z} \right\}$

(ii) If $\sin 2x \sin 6x = \cos x \cos 3x$, then x can be

(A) $\frac{\pi}{8}, \frac{\pi}{4}$ (B) $\frac{\pi}{12}, \frac{\pi}{10}$

(C) $\frac{\pi}{10}, \frac{\pi}{6}$ (D) $\frac{\pi}{12}, \frac{\pi}{4}$

(iii) General solution of the equation $\cos 2x - \cos 8x + \cos 6x = 1$ is

(A) $\left\{ n\pi + \frac{\pi}{4} \right\} \cup \left\{ (2n+1)\frac{\pi}{3} \right\}, n \in \mathbb{Z}$

(B) $\left\{ (2n+1)\frac{\pi}{2} \right\} \cup \left\{ \frac{n\pi}{8} \right\}, n \in \mathbb{Z}$

(C) $\left\{ \frac{n\pi}{3} \right\} \cup \left\{ (2n+1)\frac{\pi}{4} \right\}, n \in \mathbb{Z}$

(D) $\left\{ (2n+1)\frac{\pi}{8} \right\} \cup \left\{ \frac{n\pi}{3} \right\}, n \in \mathbb{Z}$

Solution:

(i) We have

$$\begin{aligned} \sin 5x \sin 4x + \cos 6x \cos 3x &= 0 \\ \frac{\cos x - \cos 9x}{2} + \frac{\cos 9x + \cos 3x}{2} &= 0 \\ \cos x + \cos 3x &= 0 \\ 2 \cos 2x \cos x &= 0 \end{aligned}$$

Two cases arise.

Case 1: $\cos x = 0$. Therefore

$$x = (2k+1)\frac{\pi}{2}, k \in \mathbb{Z}$$

Case 2: $\cos 2x = 0$. Therefore

$$2x = (2n+1)\frac{\pi}{2}$$

From both cases we get

$$x = \left\{ \frac{n\pi}{2} + \frac{\pi}{4}, n \in \mathbb{Z} \right\} \cup \left\{ k\pi + \frac{\pi}{2}, k \in \mathbb{Z} \right\}$$

Answer: (A)

(ii) We have $\sin 2x \sin 6x = \cos x \cos 3x$. Therefore

$$\frac{\cos 4x - \cos 8x}{2} = \frac{\cos 4x + \cos 2x}{2}$$

$$\cos 8x + \cos 2x = 0$$

$$2 \cos 5x \cos 3x = 0$$

Two cases arise.

Case 1: $\cos 3x = 0$. Therefore

$$x = (2n+1)\frac{\pi}{6}$$

Case 2: $\cos 5x = 0$. Therefore

$$x = (2k+1)\frac{\pi}{10}$$

From both cases we get

$$x = \left\{ (2n+1)\frac{\pi}{6}, n \in \mathbb{Z} \right\} \cup \left\{ (2k+1)\frac{\pi}{10}, k \in \mathbb{Z} \right\}$$

If $n = 0$ and $k = 0$, then $x = \pi/6, \pi/10$, respectively.

Answer: (C)

(iii) We have $\cos 2x - \cos 8x + \cos 6x = 1$. Therefore

$$\begin{aligned} 2 \sin 5x \sin 3x &= 1 - \cos 6x = 2 \sin^2 3x \\ \sin 3x(\sin 5x - \sin 3x) &= 0 \\ \sin 3x(2 \cos 4x \sin x) &= 0 \end{aligned}$$

This gives either $\sin x = 0$ or $\sin 3x = 0$ or $\cos 4x = 0$.

Case 1: $\sin x = 0$ implies $\sin 3x = 0$. Therefore

$$x = \frac{n\pi}{3}$$

Case 2: $\cos 4x = 0$. This implies $x = (2K+1)\frac{\pi}{8}$

Hence

$$x = \left\{ (2k+1)\frac{\pi}{8}, k \in \mathbb{Z} \right\} \cup \left\{ \frac{n\pi}{3}, n \in \mathbb{Z} \right\}$$

Answer: (D)

2. Passage: When the equation is of the form $R(\sin x + \cos x, \sin x \cos x) = 0$, where R is a rational function of $\sin x + \cos x$ and $\sin x \cos x$ then substitution $t = \sin x + \cos x$ can be used, so that $\sin x \cos x = (t^2 - 1)/2$. Answer the following questions (i), (ii) and (iii).

(i) If $\sin x + \cos x + \sin x \cos x = 1$, then the general value of x is

(A) $\left\{ (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \right\} \cup \{k\pi, k \in \mathbb{Z}\}$

(B) $\{2k\pi, k \in \mathbb{Z}\} \cup \left\{ 2n\pi + \frac{\pi}{2}, n \in \mathbb{Z} \right\}$

(C) $\left\{ \frac{k\pi}{2}, k \in \mathbb{Z} \right\} \cup \left\{ n\pi + \frac{\pi}{4}, n \in \mathbb{Z} \right\}$

(D) $\left\{ (2k+1)\frac{\pi}{2}, k \in \mathbb{Z} \right\} \cup \left\{ 2n\pi + \frac{\pi}{4}, n \in \mathbb{Z} \right\}$

(ii) Which of the following values of x satisfy the equation $\sin x + \cos x - 2\sqrt{2} \sin x \cos x = 0$.

(A) $\frac{\pi}{2}, \frac{-\pi}{4}$ (B) $\frac{-\pi}{2}, \frac{-\pi}{4}$

(C) $\frac{\pi}{4}, \frac{-5\pi}{12}$ (D) $\frac{\pi}{6}, \frac{\pi}{3}$

(iii) The set of all solutions of the equation $\sin x + \cos x - 2\sin x \cos x = 1$ is

(A)

$$\left\{ k\pi + \frac{3\pi}{4}, k \in \mathbb{Z} \right\} \cup \left\{ (4n-1)\frac{\pi}{4} + (-1)^n \frac{\pi}{4} (n \in \mathbb{Z}) \right\}$$

(B) $\left\{ k\pi + \frac{\pi}{4}, k \in \mathbb{Z} \right\} \cup \left\{ n\pi + \frac{\pi}{6}, n \in \mathbb{Z} \right\}$

(C) $\left\{ (2k+1)\frac{\pi}{3}, k \in \mathbb{Z} \right\} \cup \left\{ n\pi + \frac{\pi}{3}, n \in \mathbb{Z} \right\}$

(D) $\{2k\pi, k \in \mathbb{Z}\} \cup \left\{ n\pi + \frac{\pi}{6}, n \in \mathbb{Z} \right\}$

Solution:

(i) We have $\sin x + \cos x + \sin x \cos x = 1$. Put $\sin x + \cos x = t$. Therefore

$$t + \frac{t^2 - 1}{2} = 1$$

$$t^2 + 2t - 3 = 0$$

$$(t-1)(t+3) = 0$$

$$t = 1, -3$$

Case 1: $t = 1$. Therefore

$$\sin x + \cos x = 1$$

$$\sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}$$

$$x + \frac{\pi}{4} = k\pi + (-1)^k \frac{\pi}{4}$$

1. If k is even, then $x = 2m\pi, m \in \mathbb{Z}$.

2. If k is odd, then so that $x = 2n\pi + (\pi/2)$.

Therefore the solution set is

$$\{2n\pi, n \in \mathbb{Z}\} \cup \left\{ 2k\pi + \frac{\pi}{2}, k \in \mathbb{Z} \right\}$$

Case 2: $t = -3$. That is $\sin x + \cos x = -3$. This equation has no solution, because $3 > \sqrt{2}$. Recall that $a \sin x + b \cos x = c$ is solvable only if

$$\frac{|c|}{\sqrt{a^2 + b^2}} \leq 1$$

Answer: (B)

(ii) We have

$$\sin x + \cos x = 2\sqrt{2} \sin x \cos x = 0$$

Substituting $\sin x + \cos x = t$ we get

$$t - \sqrt{2}(t^2 - 1) = 0$$

$$\sqrt{2}t^2 - t - \sqrt{2} = 0$$

$$(\sqrt{2}t + 1)(t - \sqrt{2}) = 0$$

$$t = \sqrt{2}, -1/\sqrt{2}$$

Case 1: $t = \sqrt{2}$. Therefore

$$\sin x + \cos x = \sqrt{2}$$

$$\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = 1$$

$$\sin\left(x + \frac{\pi}{4}\right) = 1 = \sin \frac{\pi}{2}$$

$$x + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{2}, n \in \mathbb{Z}$$

when $n = 0$, then $x = \pi/4$.

Case 2: $t = -1/\sqrt{2}$. This implies

$$\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{-1}{2}$$

Therefore

$$\sin\left(x + \frac{\pi}{4}\right) = \frac{-1}{2} = \sin\left(\frac{-\pi}{6}\right)$$

$$x + \frac{\pi}{4} = k\pi + (-1)^k \left(\frac{-\pi}{6}\right)$$

When $k = 0$, then $x = -5\pi/12$. Therefore $x = \pi/4, -5\pi/12$ are solutions.

Answer: (C)

(iii) We have

$$\sin x + \cos x - 2 \sin x \cos x = 1$$

Substituting $\sin x + \cos x = t$ we get

$$t - t^2 = 0$$

$$t = 0, 1$$

Case 1: $t = 0$. Therefore

$$\sin x + \cos x = 0$$

$$\sin\left(x + \frac{\pi}{4}\right) = 0$$

$$x + \frac{\pi}{4} = n\pi, n \in \mathbb{Z}$$

$$x = n\pi - \frac{\pi}{4} = (n-1)\pi + \frac{3\pi}{4}$$

$$x = k\pi + \frac{3\pi}{4}$$

Case 2: $t = 1$. Therefore

$$\sin x + \cos x = 1$$

$$\sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}$$

$$x + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$$

$$x = n\pi - \frac{\pi}{4} + (-1)^n \frac{\pi}{4} = (4n-1)\frac{\pi}{4} + (-1)^n \frac{\pi}{4}$$

Therefore the solution set is

$$\left\{k\pi + \frac{3\pi}{4}, k \in \mathbb{Z}\right\} \cup \left\{(4n-1)\frac{\pi}{4} + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}\right\}$$

Answer: (A)

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
- (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
- (C) Statement I is true and Statement II is false
- (D) Statement I is false and Statement II is true

1. Statement I: The solution set of $\cos^{100} x - \sin^{100} x = 1$ is $\{n\pi, n \in \mathbb{Z}\}$.

Statement II: $-1 \leq \cos x \leq 1$ for all real values of x .

Solution: We have

$$\cos^{100} x - \sin^{100} x = 1$$

$$\Rightarrow \cos^{100} x = 1 + \sin^{100} x \geq 1$$

$$\Rightarrow \cos^{100} x = 1$$

$$\Rightarrow \sin x = 0$$

$$\Rightarrow x = n\pi, n \in \mathbb{Z}$$

Answer: (A)

2. Statement I: If $2x$ is not an odd multiple of $\pi/2$ and $(\tan 4x - \tan 2x)/(1 + \tan 4x \tan 2x) = 1$, then $x = (k\pi/2) + (\pi/8), k \in \mathbb{Z}$.

Statement II: General solution of the equation $\tan \theta = \tan \alpha$ is $\theta = n\pi + \alpha, n \in \mathbb{Z}$.

Solution: Statement II is true (see Section 3.2.5). Now

$$\frac{\tan 4x - \tan 2x}{1 + \tan 4x \tan 2x} = 1$$

$$\Rightarrow \tan 2x = 1 = \tan \frac{\pi}{4}$$

$$\Rightarrow 2x = k\pi + \frac{\pi}{4}$$

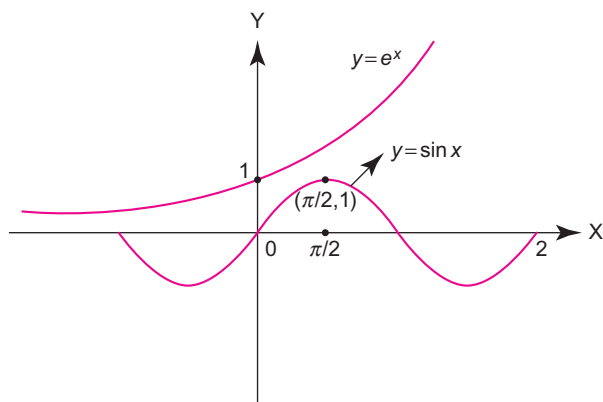
$$\Rightarrow x = k\frac{\pi}{2} + \frac{\pi}{8}$$

Answer: (A)

3. Statement I: The equation $\sin x = e^x$ has a solution for $x \geq 0$.

Statement II: $-1 \leq \sin x \leq 1$ for all real x .

Solution: Since $e^x \geq 1$ for all $x \geq 0$ and $|\sin x| \leq 1$, the equation $\sin x = e^x$ has no solution (see the following figure).



Therefore Statement II is true, but Statement I is false.

Answer: (D)

Note: Since $e^x \rightarrow 0$ as $x \rightarrow -\infty$, the graph of $y = \sin x$ intersects the graph of $y = e^x$ at infinitely many points for $x < 0$ and hence $\sin x = e^x$ has infinitely many solutions when $x < 0$.

4. Statement I: The general solution of the equation $\cos x + \sec x = 2$ when x is not an odd multiple of $\pi/2$ is $x = 2n\pi, n \in \mathbb{Z}$.

Statement II: $\cos x + \sec x \geq 2$ for all real values of x .

Solution: Since the maximum value of $\cos x$ is 1 and $|\sec x| \geq 1$, $\cos x + \sec x = 2$ if $\cos x = 1$ and $\sec x = 1$. Therefore

$$x = 2n\pi, n \in \mathbb{Z}$$

Hence Statement I is true.

Statement II is false because $\cos x + \sec x < 0$ for infinitely many values of x .

Answer: (C)

5. Statement I: General solution of the equation $5 \tan^4 x - \sec^4 x - 29 = 0$ is $x = n\pi \pm (\pi/3), n \in \mathbb{Z}$.

Statement II: The equation $a \sin x + b \cos x = c$ has solutions if $|c| \leq \sqrt{a^2 + b^2}$.

Solution: According to Section 3.3.1, Statement II is true. Now

$$5 \tan^4 x - \sec^4 x - 29 = 5 \tan^4 x - (1 + \tan^2)^2 - 29$$

$$4 \tan^4 x - 2 \tan^2 x - 30 = 0$$

$$2 \tan^4 x - \tan^2 x - 15 = 0$$

$$(2 \tan^2 x + 5)(\tan^2 x - 3) = 0$$

$$\tan^2 x = 3$$

$$\tan x = \pm \sqrt{3} = \tan\left(\pm \frac{\pi}{3}\right)$$

$$x = n\pi \pm \frac{\pi}{3}$$

Statement I is also true, but Statement II is not a correct explanation of Statement I.

Answer: (B)

Integer Answer Type Questions

1. The number of solutions of the equation $\sin 3x + \cos 3x = 0$ in the interval $[0, \pi]$ is ____.

Solution: As $\cos 3x = 0$ is not a solution of the initial equation, the given equation is equivalent to the equation $\tan 3x = -1 = \tan(-\pi/4)$. Therefore

$$3x = \frac{-\pi}{4} + n\pi, n \in \mathbb{Z}$$

$$x = \frac{-\pi}{12} + \frac{n\pi}{3}, n \in \mathbb{Z}$$

When $n = 1, 2, 3$, respectively, then $x = \pi/4, 7\pi/12$ and $11\pi/12$.

Answer: 3

2. If the general solution of the equation $\cos x = 2 \sin^2(x/2)$ is given by $x = 2n\pi \pm (\pi/k), n \in \mathbb{Z}$, then the value of k is ____.

Solution: We have
Therefore

$$\cos x = 2 \sin^2(x/2) = 1 - \cos x$$

$$\Rightarrow 2 \cos x = 1$$

$$\Rightarrow \cos x = 1/2$$

$$\Rightarrow x = 2n\pi \pm (\pi/3), n \in \mathbb{Z}$$

Therefore $k = 3$.

Answer: 3

3. The general solution of the equation

$$\tan 2x \sin x + \sqrt{3}(\sin x - \sqrt{3} \tan 2x) = 3\sqrt{3}$$

is $x = (3n - 1)\pi/k, n \in \mathbb{Z}$ where k is equal to ____.

Solution: The given equation can be written as

$$\tan 2x(\sin x - 3) + \sqrt{3}(\sin x - 3) = 0$$

As $\sin x \neq 3$,

$$\tan 2x = -\sqrt{3} = \tan\left(\frac{-\pi}{3}\right)$$

Therefore

$$2x = n\pi - \frac{\pi}{3} = (3n-1)\frac{\pi}{3}$$

$$x = (3n-1)\frac{\pi}{6}$$

This implies $k = 6$.

Answer: 6

4. The equation $3\sin x + 4\cos x = c$ is solvable if $|c|$ value is less than or equal to ____.

Solution: It is known that (see Section 3.3.1) $a\sin x + b\cos x = c$ is solvable if $|c| \leq \sqrt{a^2 + b^2}$. Therefore

$$|c| \leq \sqrt{3^2 + 4^2} = 5$$

Answer: 5

5. If

$$\tan^3 x - 1 + \frac{1}{\cos^2 x} - 3\cot\left(\frac{\pi}{2} - x\right) = 3$$

then the number of values of $\tan x$ is ____.

Solution: Given equation can be written as

$$\tan^3 x - 1 + \sec^2 x - 3\tan x - 3 = 0$$

$$\tan^3 x + \tan^2 x - 3\tan x - 3 = 0$$

$$(\tan x + 1)(\tan^2 x - 3) = 0$$

$$\tan x = -1, \pm\sqrt{3}$$

Answer: 3

6. The number values of x belonging to interval $[0, 2\pi]$ and satisfying the equation $\sin^4 x + \cos^4 x = \cos 4x$ is ____.

Solution: Given equation can be written as

$$(1 - 2\sin^2 x \cos^2 x) - \cos 4x = 0$$

$$1 - \frac{1}{2}\sin^2 2x - \cos 4x = 0$$

$$1 - \frac{1(1 - \cos 4x)}{2} - \cos 4x = 0$$

$$4 - 1 + \cos 4x - 4\cos 4x = 0$$

$$3\cos 4x = 3$$

$$\cos 4x = 1$$

$$x = \frac{n\pi}{2}, n \in \mathbb{Z}$$

When $n = 0, 1, 2, 3$ and 4 , then $x = 0, \pi/2, \pi, 3\pi/2$ and 2π .

Answer: 5

7. For $0 < x, y < \pi$, the number of ordered pairs (x, y) satisfying the equations $\sin x + \cos y = 0$ and $\sin^2 x + \cos^2 y = 1/2$ is ____.

Solution: We have

$$\sin x + \cos y = 0 \quad (3.4)$$

$$\sin^2 x + \cos^2 y = \frac{1}{2} \quad (3.5)$$

$$0 < x < \pi \quad \text{and} \quad 0 < y < \pi \quad (3.6)$$

Substituting $\cos y = -\sin x$ in Eq. (3.5) we get

$$2\sin^2 x = \frac{1}{2}$$

Therefore

$$\sin x = \pm \frac{1}{2}$$

$$x = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$$

$$x = n\pi + (-1)^n \left(\frac{-\pi}{6}\right), n \in \mathbb{Z}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \quad (\because 0 < x < \pi)$$

Now $\sin x = \pm 1/2 \Rightarrow \cos y = \mp 1/2$. So

$$\sin x = \frac{1}{2} \Rightarrow \cos y = \frac{-1}{2}$$

$$\Rightarrow y = 2n\pi \pm \frac{2\pi}{3}$$

$$\Rightarrow y = \frac{2\pi}{3} \quad (\because 0 < y < \pi)$$

Between 0 to π , $\sin x$ cannot be $-1/2$ so that $\cos y \neq 1/2$. Therefore the ordered pairs (x, y) are

$$\left(\frac{\pi}{6}, \frac{2\pi}{3}\right) \quad \text{and} \quad \left(\frac{5\pi}{6}, \frac{2\pi}{3}\right)$$

Answer: 2

8. The number of roots of the equation $(1 - \cos 2x)\sin 2x = \sqrt{3}\sin^2 x$ which lie in the interval $[-\pi, \pi/3]$ is ____.

Solution: We have

$$(1 - \cos 2x)\sin 2x = \sqrt{3}\sin^2 x = \frac{\sqrt{3}}{2}(1 - \cos 2x)$$

Therefore

$$(1 - \cos 2x)\left(\sin 2x - \frac{\sqrt{3}}{2}\right) = 0$$

Two cases arise.

Case 1: $\cos 2x = 1$. Then $2x = 2n\pi$ or $x = n\pi$. This implies $x = -\pi, 0$ when $n = -1, 0$.

Case 2: $\sin 2x = \sqrt{3}/2$. Then

$$2x = k\pi + (-1)^k \frac{\pi}{3}; n, k \in \mathbb{Z}$$

$\therefore$

or
$$x = \frac{k\pi}{2} + (-1)^k \frac{\pi}{6}$$

When $k = -2, -1, 0, 1$, respectively, the n

$$x = -\frac{5\pi}{6}, -\frac{2\pi}{3}, \frac{\pi}{6}, \frac{\pi}{3}$$

$$x = -\pi, -\frac{5\pi}{6}, -\frac{2\pi}{3}, 0, \frac{\pi}{6}, \frac{\pi}{3}$$

Answer: 6

9. If $x = \tan^{-1}(-k)$ is a solution of the equation $\cos^2 x - 2\cos x = 4\sin x - \sin 2x$, then the value of $2k$ is _____.

Solution: The given equation can be written as

$$\cos x(\cos x - 2) = 2\sin x(2 - \cos x)$$

Therefore

$$(\cos x - 2)(\cos x + 2\sin x) = 0$$

Now $\cos x \neq 2$ implies $\cos x + 2\sin x = 0$. Clearly $\cos x = 0$ is not a solution of the initial equation. Hence $\tan x = -1/2$.

Therefore

$$x = \tan^{-1}\left(-\frac{1}{2}\right)$$

is the principal solution. Hence

$$k = \frac{1}{2}$$

so that $2k = 1$.

Answer: 1

10. The number of values x lying in the interval $(\pi/9, 4\pi/9)$ satisfying the equation $1 + \cos 10x \cos 6x = 2\cos^2 8x + \sin^2 8x$ is _____.

Solution: We have

$$\begin{aligned} 1 + \cos 10x \cos 6x &= 2\cos^2 8x + \sin^2 8x \\ &= \cos^2 8x + 1 \end{aligned}$$

Therefore

$$\cos 10x \cos 6x = \cos^2 8x = \frac{1 + \cos 16x}{2}$$

$$2\cos 10x \cos 6x = 1 + \cos 16x$$

$$\cos 16x + \cos 4x = 1 + \cos 16x$$

$$\cos 4x = 1$$

$$4x = 2n\pi, n \in \mathbb{Z}$$

$$x = n\pi/2, n \in \mathbb{Z}$$

Therefore for any value of n ,

$$x \notin \left(\frac{\pi}{9}, \frac{4\pi}{9}\right)$$

Therefore the number of values of $x \in (\pi/9, 4\pi/9)$ is zero.

Answer: 0

SUMMARY

Trigonometric Equations

3.1 Principal Solution: Solutions of a trigonometric equation belonging to the interval $[0, 2\pi]$ is called principal solution of the equation.

3.2 (a) General solution of the equation $\sin x = K$ where $-1 \leq K \leq 1$ is $x = n\pi + (-1)^n \alpha$ where $\sin \alpha = K$ and $n \in \mathbb{Z}$.

(b) General solution of the equation $\cos x = K$ ($-1 \leq K \leq 1$) is $x = 2n\pi \pm \alpha$ where $\cos \alpha = K$ and $n \in \mathbb{Z}$.

(c) General solution of the equation $\tan x = a$ where 'a' is any real number is $x = n\pi + \theta$ where $\tan \theta = a$.

3.3 If a, b and c are real numbers, then the equation $a\sin x + b\cos x = c$ is solvable only when $|c| \leq \sqrt{a^2 + b^2}$ and the general solution is $x = (2n\pi \pm \alpha) + \theta$

where

$$\sin \theta = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \cos \theta = \frac{b}{\sqrt{a^2 + b^2}}$$

and $\alpha \in [0, \pi]$ such that

$$\cos \alpha = \frac{c}{\sqrt{a^2 + b^2}}, n \in \mathbb{Z}$$

or $x = n\pi + (-1)^n \alpha - \theta$ where

$$\cos \theta = \frac{a}{\sqrt{a^2 + b^2}}$$

$$\sin \theta = \frac{b}{\sqrt{a^2 + b^2}}$$

and $\sin \alpha = \frac{c}{\sqrt{a^2 + b^2}}, n \in \mathbb{Z}, -\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$

Note: Writing

$$\sin x = \frac{2 \tan x/2}{1 + \tan^2 x/2}$$

and $\cos x = \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}$

we can solve the equations.

3.4 Useful substitution: In some trigonometric equation, the substitution $\sin x + \cos x = t$ will be useful.

Example: Solve $\sin x + \cos x = 1 + \sin x \cos x$.

3.5 If a trigonometric equation is a rational function of $\sin mx, \cos nx, \tan px$, etc. and equals zero, where m, n, p , etc. are positive integers, then the formulae

$$\sin x = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)}$$

$$\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$$

$$\tan x = \frac{2 \tan(x/2)}{1 - \tan^2(x/2)}$$

etc. will be useful for substitution.

Example: The equation

$$(\cos x - \sin x)(2 \tan x + \sec x) + 2 = 0$$

can be solved by substituting $\tan(x/2) = t$.

EXERCISES

Single Correct Choice Type Questions

1. The number of roots of the equation

$$\frac{\cot 2x}{\cot x} + \frac{\cot x}{\cot 2x} + 2 = 0$$

in the interval $[0, 2\pi]$ is

- (A) 1 (B) 2 (C) 3 (D) 4

2. General solution of the equation $\sin 6x + 2 = 2 \cos 4x$ is

- (A) $\frac{n\pi}{6}, n \in \mathbb{Z}$ (B) $n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$
(C) $\frac{n\pi}{2}, n \in \mathbb{Z}$ (D) $2n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$

3. The number of solutions of the equation $\tan^2 x \cdot \tan^2 3x \cdot \tan 4x = \tan^2 x - \tan^2 3x + \tan 4x$ in the interval $[0, \pi]$ is

- (A) 0 (B) 2 (C) 3 (D) 4

4. General solution of the equation

$$(\sin 19^\circ)^{\cot x + \cot 3x} = \sin^2(2\pi - x) - \cos(\pi - x) \\ \times \sin\left(\frac{\pi}{2} + x\right)$$

is

- (A) $x = \frac{n\pi}{4}, n \in \mathbb{Z}$
(B) $x = \frac{n\pi}{4}, n \in \mathbb{Z}$ and $n \neq 4k, k \in \mathbb{Z}$
(C) $x = \frac{n\pi}{2}, n \in \mathbb{Z}$
(D) $x = \frac{n\pi}{2}, n \in \mathbb{Z}$ and $n \neq 2k, k \in \mathbb{Z}$

5. General solution of the equation

$$\frac{\sin^3(x/2) - \cos^3(x/2)}{2 + \sin x} = \frac{1}{3} \cos x$$

is

- (A) $x = (4n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$ (B) $x = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$
(C) $x = (2n + 1)\pi, n \in \mathbb{Z}$ (D) $x = \frac{n\pi}{4}, n \in \mathbb{Z}$

6. General solution of the equation $\tan x \sin x - \cos x = (1/2) \sec x$ is

- (A) $(2n \pm 1)\frac{\pi}{3}, n \in \mathbb{Z}$ (B) $(3n \pm 1)\frac{\pi}{3}, n \in \mathbb{Z}$
(C) $(4n + 1)\frac{\pi}{6}, n \in \mathbb{Z}$ (D) $(2n \pm 1)\frac{\pi}{6}, n \in \mathbb{Z}$

7. If $\cos^2 3x + \sin^2[(\pi/2) + 5x] = \cos^2(7x) + \cos^2(9x)$, then $12x$ is equal to

- (A) $4n\pi, n \in \mathbb{Z}$ (B) $3n\pi, n \in \mathbb{Z}$
(C) $2n\pi, n \in \mathbb{Z}$ (D) $n\pi, n \in \mathbb{Z}$

8. The number of solutions of the equation $1 - 2 \sin x - 2 \cos x + \cot x = 0$ ($0 < x < 2\pi$) is

- (A) 4 (B) 6 (C) 2 (D) 1

9. If $\cos 3x + \sin[2x - (7\pi/6)] = -2$, then x is of the form

- (A) $(4m + 1)\frac{\pi}{3}, m \in \mathbb{Z}$ (B) $(6m + 1)\frac{\pi}{3}, m \in \mathbb{Z}$
(C) $(3m + 1)\frac{\pi}{3}, m \in \mathbb{Z}$ (D) $(2m + 1)\frac{\pi}{3}, m \in \mathbb{Z}$

10. General solution of $1 + \sin^3 x + \cos^3 x = (3/2) \sin 2x$ is

- (A) $\{(2n+1)\pi, n \in \mathbb{Z}\} \cup \{(4k-1)\frac{\pi}{2}, k \in \mathbb{Z}\}$
 (B) $\{n\pi, n \in \mathbb{Z}\} \cup \{(2k-1)\frac{\pi}{2}, k \in \mathbb{Z}\}$
 (C) $\{(4n+1)\pi, n \in \mathbb{Z}\} \cup \{2k\pi, k \in \mathbb{Z}\}$
 (D) $\{(3n+1)\pi, n \in \mathbb{Z}\} \cup \{k\pi, k \in \mathbb{Z}\}$

11. If $(\sqrt{3}/2) \sin x = \cos x + \cos^2 x$, then x is equal to

- (A) $n\pi$ or $(2n \pm 1)\frac{\pi}{3}, n \in \mathbb{Z}$
 (B) $(2n \pm 1)\pi$ or $(6n \pm 1)\frac{\pi}{3}, n \in \mathbb{Z}$
 (C) $\frac{n\pi}{2}$ or $(2n-1)\frac{\pi}{3}, n \in \mathbb{Z}$
 (D) $2n\pi$ or $n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$

12. The general value of x satisfying the equation $\sin 2x + 5 \sin x + 5 \cos x + 1 = 0$ is

- (A) $(3n+1)\frac{\pi}{4}, n \in \mathbb{Z}$ (B) $(2n+1)\frac{\pi}{4}, n \in \mathbb{Z}$
 (C) $(4n-1)\frac{\pi}{4}, n \in \mathbb{Z}$ (D) $(6n+1)\frac{\pi}{4}, n \in \mathbb{Z}$

13. The number of solutions of the equation $2 \sin^3 x = \cos x$ in the interval $(-\pi, 5\pi)$ is

- (A) 4 (B) 6 (C) 5 (D) 2

14. General solution of the equation $\sin x + \cos x = \sqrt{2}$ ($\sin^3 x + \cos^3 x$) belongs to

- (A) $\left\{(2n-1)\frac{\pi}{2}, n \in \mathbb{Z}\right\} \cup \left\{n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}\right\}$
 (B) $\left\{\frac{n\pi}{2} + (-1)^n \sin^{-1}\left(\frac{1}{2\sqrt{2}}\right), n \in \mathbb{Z}\right\} \cup \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$
 (C) $\left\{2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right), n \in \mathbb{Z}\right\} \cup \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$
 (D) $\left\{-\frac{\pi}{4} + n\pi, n \in \mathbb{Z}\right\} \cup \left\{\frac{n\pi}{2} + \frac{(-1)^n}{2} \sin^{-1}\left(\frac{2-\sqrt{2}}{2}\right), n \in \mathbb{Z}\right\}$

15. If $a > 0$, the number of solutions of the equation $\sin x = (|x|/2) + a$ in the interval $(-2, 0)$ is

- (A) infinite (B) 4 (C) 2 (D) 0

16. The number of solutions of the equation

$$\cos\left(\frac{\sqrt{2}+1}{2}x\right) \cdot \cos\left(\frac{\sqrt{2}-1}{2}x\right) = 1$$

in the interval $[-\pi, \pi]$ is

- (A) 0 (B) 1 (C) 2 (D) 4

17. The number of solutions of the equation

$$\tan\left(\frac{\pi}{4} + x\right) = \tan\left(\frac{\pi}{4} - x\right)$$

in the interval $(-\infty, \infty)$ is

- (A) 0 (B) 2 (C) finite (D) infinite

18. If the pair (x, y) satisfies both the equations $\sin^2 x + \sin^2 y = 1/2$ and $x - y = 4\pi/3$, then

- (A) $x = n\pi + \frac{7\pi}{6}, y = \frac{-\pi}{6} + n\pi, n \in \mathbb{Z}$
 (B) $x = n\pi + \frac{5\pi}{6}, y = n\pi - \frac{\pi}{3}, n \in \mathbb{Z}$
 (C) $x = n\pi - \frac{7\pi}{6}, y = n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$
 (D) $x = n\pi - \frac{5\pi}{6}, y = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$

19. If (x, y) satisfies the system of equations $\tan x + \tan y = 4$ and $\cot x + \cot y = 5$, then

- (A) $x = n\pi + \tan^{-1}[(10 \pm 4\sqrt{5})/5], y = m\pi + \tan^{-1}[(10 \mp 4\sqrt{5})/5], n$ and $m \in \mathbb{Z}$ (either both have upper signs or both have lower signs)
 (B) $x = (n\pi/2) + \tan^{-1}[2 \pm (0.8)\sqrt{5}], y = (m\pi/2) + \tan^{-1}[2 \mp (0.8)\sqrt{5}], n, m \in \mathbb{Z}$
 (C) $x = 2n\pi + \tan^{-1}(\pm 4/\sqrt{5}), y = 2m\pi + \tan^{-1}(\mp 4/\sqrt{5}), n, m \in \mathbb{Z}$
 (D) $x = [(2n+1)\pi/2] + \tan^{-1}(2 \pm \sqrt{5}), y = [(2m+1)\pi/2] + \tan^{-1}(2 \mp \sqrt{5}), n, m \in \mathbb{Z}$

20. The number of solutions in the interval $[-\pi, \pi]$ of the equation $4 \sin 3x + \sin 5x - 2 \sin x \cos 2x = 0$ is

- (A) 8 (B) 7 (C) 6 (D) 4

21. The number of values of x such that $\tan[(2\pi x)/(1+x+x^2)] = -\sqrt{3}$ is

- (A) 8 (B) 7 (C) 5 (D) 6

22. The general solution of the equation $\cos 2x + \sin 2x = \cos x + \sin x$ belongs to

- (A) $\{2n\pi, n \in \mathbb{Z}\} \cup \{(4m+1)\frac{\pi}{6}, m \in \mathbb{Z}\}$
 (B) $\{n\pi, n \in \mathbb{Z}\} \cup \{(2m+1)\frac{\pi}{6}, m \in \mathbb{Z}\}$

(C) $\{n\pi, n \in \mathbb{Z}\} \cup \left\{(4m+1)\frac{\pi}{3}, m \in \mathbb{Z}\right\}$

(D) $\left\{\frac{n\pi}{2}, n \in \mathbb{Z}\right\} \cup \left\{(2m+1)\frac{\pi}{3}, m \in \mathbb{Z}\right\}$

23. The number of solutions of the equation $\sqrt{2}\sin x + \cot x = 0$ in the interval $(-\pi, 2\pi)$ is
(A) 6 (B) 4 (C) 3 (D) 2

24. The number of solutions of the equation $2(\sin^3 x + \cos^3 x) - 3(\sin x + \cos x) = 8$ in the interval $(-\pi, \pi)$ is
(A) infinite (B) 4 (C) 2 (D) 0

25. The equation $\sin^6 x + \cos^6 x = a$ has a solution if and only if

(A) $1 \leq a \leq 2$ (B) $\frac{1}{6} \leq a \leq \frac{1}{3}$

(C) $\frac{1}{4} \leq a \leq 1$ (D) $\frac{1}{8} \leq a \leq \frac{1}{4}$

26. The number of solutions of the equation $\cos^7 x + \sin^4 x = 1$ in the interval $(-\pi, \pi)$ is
(A) 2 (B) 3 (C) 4 (D) 5

27. The smallest positive value of x (in degrees) for which $\tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)$ is
(A) 30° (B) 45° (C) 55° (D) 60°

28. The smallest positive value of p for which the equation $\cos(p \sin x) = \sin(p \cos x)$ has a solution is
(A) $\frac{\pi}{2\sqrt{3}}$ (B) $\frac{\pi}{\sqrt{3}}$ (C) $\frac{\pi}{\sqrt{2}}$ (D) $\frac{\pi}{2\sqrt{2}}$

29. The number of integral values of k for which the equation $7\cos x + 5\sin x = 2k + 1$ has a solution is
(A) 4 (B) 8 (C) 10 (D) 12

30. The number of solutions of the equation $\sin x + \sin 5x = \sin 3x$ in the interval $[0, \pi]$ is
(A) 6 (B) 5 (C) 4 (D) 3

31. The number of solutions of the equation $\tan^2 x = \cos 2x - 1$ in the interval $[-\pi, \pi]$ is
(A) 6 (B) 5 (C) 4 (D) 3

Multiple Correct Choice Type Questions

1. Solution of the equation $1 + \sin^4 2x = \cos^2 6x$ is

(A) $n\pi/3, n \in \mathbb{Z}$ (B) $n\pi/2, n \in \mathbb{Z}$
(C) $n\pi, n \in \mathbb{Z}$ (D) $(2n+1)\pi/4, n \in \mathbb{Z}$

2. If $7\cos^2 x + \cos x \sin x - 3 = 0$, then x is

(A) $n\pi + \tan^{-1}(4/3), n \in \mathbb{Z}$
(B) $n\pi + (3\pi/4), n \in \mathbb{Z}$
(C) $(n\pi/2) + \tan^{-1}(4/3), n \in \mathbb{Z}$
(D) $(n\pi/2) + (3\pi/4), n \in \mathbb{Z}$

3. If $2^{\cos 2x} + 1 = 3 \cdot 2^{-\sin^2 x}$, then x may be equal to

(A) $n\pi$ (B) $2n\pi$
(C) $n\pi/2$ (D) $(2n \pm 1)\pi/2 (n \in \mathbb{Z})$

4. If $z = (3 + 2i \sin \theta)/(1 - 2i \sin \theta)$, then

(A) z is pure real, if $\theta = n\pi, n \in \mathbb{Z}$
(B) z is pure real, if $\theta = 2n\pi, n \in \mathbb{Z}$
(C) z is pure imaginary, if $\theta = n\pi + (\pi/3), n \in \mathbb{Z}$
(D) z is pure imaginary, if $\theta = n\pi - (\pi/3), n \in \mathbb{Z}$

5. If α and β are smallest positive values of x and y satisfying the equation $2(\sin x + \sin y) - 2\cos(x - y) = 3$, then

(A) $\sin[(\alpha + \beta)/2] = 1$
(B) $\cos[(\alpha - \beta)/2] = 1/2$

(C) the number of pairs (α, β) is 2

(D) the number of pairs (α, β) is 4

6. For the equation $\sin^2 x + (1/4)\sin^2 3x = \sin x \sin^2 3x$

(A) $x = 13\pi/6$ is a solution
(B) $x = 0, \pi$ are the solutions
(C) the number of solutions in the interval $[-\pi, \pi]$ is 5
(D) $x = \pi/2$ is not a solution

7. For the equation $\tan x \cdot \tan 4x = 1$

(A) the number of solutions in $[0, \pi]$ is 5
(B) $x = 3\pi/4$ is a solution
(C) $x = 11\pi/4$ is not a solution
(D) $x = (2n+1)\pi/10$ is the general solution

8. For the equation $\cos x \cos[(\pi/3) - x] \cos[(\pi/3) + x] = 1/4$,

(A) the number of solutions in $[-\pi, \pi]$ is 3
(B) the number of solutions in $[0, 6\pi]$ is 10
(C) the sum of all the solutions in $[0, 6\pi]$ is 30π
(D) the general solution is $x = 2n\pi/3, n \in \mathbb{Z}$

9. Let $f(x) = \sqrt{\sin^4 x + 4(1 - \sin^2 x)} - \sqrt{\cos^4 x + 4(1 - \cos^2 x)}$. Then

- (A) general solution of $f(x) = 0$ is $n\pi, n \in \mathbb{Z}$
 (B) the number of values of x belonging to $[-\pi, \pi]$ satisfying $f(x) = 0$ is 4
 (C) general solution of $f(x) = 0$ is $2n\pi, n \in \mathbb{Z}$
 (D) the sum of all the solutions of $f(x) = 0$ belonging to $[-2\pi, 2\pi]$ is 0
10. The region S consists of all the points (x, y) such that $x^2 + y^2 \leq 100$ and $\sin(x + y) \geq 0$. Then
 (A) S is cut by the lines of the form $x + y = k\pi$ for some $k \in \mathbb{Z}$
 (B) if (x, y) belongs to the region S and lies between two lines of the form $x + y = k\pi$, then either $\sin(x + y) > 0$ or $\sin(x + y) < 0$
 (C) the area of the region S is 50π square units
 (D) the area of the region S is 100π square units

11. Consider the equation

$$\frac{\sqrt{3} - 1}{\sin x} + \frac{\sqrt{3} + 1}{\cos x} = 4\sqrt{2}$$

For this equation

- (A) $x = \pi/12, n \in \mathbb{Z}$, is a solution
 (B) $x = 11\pi/36, n \in \mathbb{Z}$, is a solution
 (C) the number of solutions in $(0, \pi/2)$ is 2
 (D) $2x = n\pi + (-1)^n(x + \pi/12)$
12. In triangle ABC , the angles A and B satisfy the relations $3\sin A + 4\cos B = 6$ and $3\cos A + 4\sin B = 1$. Then

- (A) $A + B = 30^\circ$ (B) $C = 150^\circ$
 (C) $A + B = 150^\circ$ (D) $C = 30^\circ$

13. Consider the following two statements:

I. If $0 < x < \pi/2$ and $\log_{\cos x}^{\sin x} + \log_{\sin x}^{\cos x} = 2$, then $x = \pi/4$.

II. If α and β are any two angles such that $\sin \alpha = \sin \beta$ and $\cos \alpha = \cos \beta$, then $\alpha - \beta = 2n\pi, n \in \mathbb{Z}$.

which of the following are correct?

- (A) I is true (B) II is false
 (C) I is false (D) II is true

14. Consider the following statements

P: The equation $\cos x + \cos 3x = \cos 2x$ has two solutions in the interval $(0, \pi/2)$.

Q: If $1 + \sin x \cdot \sin 2x = \cos x \cos 2x$, then $x = 2n\pi/3, n \in \mathbb{Z}$.

which of the following are correct?

- (A) P is true
 (B) Q is true
 (C) P is true, but Q is false
 (D) P is false, but Q is true

15. Consider the equation $2\sqrt{2}|\cos x| = \operatorname{cosec} x$.

- (A) This has four solutions in $[0, 2\pi]$
 (B) This has two solutions in $[\pi/2, \pi]$
 (C) Solutions of this equation belonging to $[0, \pi]$ form an AP in some order
 (D) This equation has no solutions in $(\pi, 3\pi/2)$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with *one or more* statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are (A) $\rightarrow$ (p), (s); (B) $\rightarrow$ (q), (s), (t); (C) $\rightarrow$ (r); (D) $\rightarrow$ (r), (t); that is if the matches are (A) $\rightarrow$ (p) and (s); (B) $\rightarrow$ (q), (s) and (t); (C) $\rightarrow$ (r); and (D) $\rightarrow$ (r), (t), then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

1. Match the items in Column I with those in Column II.

Column I	Column II
(A) If $\cos 3x - \cos 2x = \sin 3x$, then some of the values of x can be	(p) $-\frac{\pi}{4}, \frac{3\pi}{4}$
(B) If $\sin 3x - 4\sin x \cos 2x = 0$, then x can be	(q) $-\pi, \pi, 2\pi$
(C) If $\sec^2 x - \tan^2 x + \cot(\pi/2 + x) = \cos 2x \sec^2 x$, then x can be	(r) $\frac{\pi}{4}, \frac{5\pi}{4}$
(D) If $\tan x + \sin(\pi + x) = 2\sin^2 x/2$, then x is	(s) $2n\pi, n \in \mathbb{Z}$
	(t) $(4n-1)\frac{\pi}{4}, n \in \mathbb{Z}$

2. Some trigonometric equations are given in Column I and some or whole of their solutions are given in Column II. Match these.

Column I	Column II
(A) If $0 \leq x, y \leq 2\pi$ and α and β are solutions of the equation $\sin x + \sin y = 2$, then $\alpha + \beta$ is	(p) $\frac{\pi}{3}$
(B) If $4\cos^2 x + 4\cos x - 3 = 0$, then the value of x in the interval $[0, 2\pi]$ is	(q) $\frac{5\pi}{3}$
(C) The solution of the equation $2\sin^2 x + (4 - \sqrt{3})\sin x - 2\sqrt{3} = 0$ in the interval $[0, \pi]$ is	(r) $\frac{2\pi}{3}$
(D) Solutions of the equation $\sqrt{3} - \tan x = \tan(\pi/3 - x)$ are	(s) $\frac{\pi}{2}$
	(t) π

3. Certain trigonometric equations are given in Column I and some of their solutions are given in Column II. Match these.

Column I	Column II
(A) $\sin^2 x = \sin 3x + \cos x(\cos x - 1)$	(p) $\frac{2n\pi}{3}, n \in \mathbb{Z}$
(B) $\frac{1 + \tan x}{1 - \tan x} = (\sin x + \cos x)^2$	(q) $n\pi, (4k - 1)\frac{\pi}{4}$
(C) $\tan 3x + \cos 6x = 1$	(r) $(2n + 1)\frac{\pi}{6}$
(D) $\cos 4x + 2\cos^2 x = 1$	(s) $\frac{n\pi}{3}, (4k + 1)\frac{\pi}{12}$
	(t) $(4k + 1)\frac{\pi}{4}$

4. Certain trigonometric equations are given in Column I and their solution sets are given in Column II. Match these.

Column I	Column II
(A) $1 + \sin^4 x = \cos^5 x$	(p) Empty set
(B) $\sin^4 x = 1 + \cos^6 x$	(q) $2n\pi, n \in \mathbb{Z}$
(C) $\sin^2 2x = 1 + \cos^4 x$	(r) $2n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$
(D) $\sin^4 x = 2\sin^2 x + 1$	(s) $(2n + 1)\pi/2, n \in \mathbb{Z}$

5. Match the items in Column I with those in Column II.

Column I	Column II
(A) The number of solutions of the equation $\tan x + \sec x = 2\cos x$ in the interval $[0, 2\pi]$ is	(p) 0
(B) The number of solution of the equation $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1$ in $[0, 2\pi]$ is	(q) 3
(C) The number of values of x in $[0, \pi]$ satisfying the equation $\sin x + \sin 3x + \sin 5x = 0$ is	(r) 4
(D) The number of values of x in $[0, 3\pi/2]$ satisfying the equation $\sin x(2\cos x + 1) = 0$ is	(s) 2
	(t) 6

Comprehension-Type Questions

1. **Passage:** Using trigonometric transformations, answer the following questions.

- (i) The number of solutions of the equation $\sin 2x \cdot \sin 6x = \cos x \cos 3x$ in the interval $[0, \pi/2]$ is
(A) 4 (B) 6 (C) 2 (D) 8
- (ii) The number of solutions of the equation $\tan x + \tan(x + \pi/3) + \tan(x + 2\pi/3) = 3$ in the interval $[-\pi, \pi]$ is
(A) 2 (B) 3 (C) 4 (D) 6
- (iii) The number of solutions of the equation $\sin x + \sin 7x = \sin 4x$ in the interval $[0, \pi/2]$ is
(A) 2 (B) 3 (C) 4 (D) 6

2. **Passage:** Equations of the form $a \cos x + b \sin x = c$ can be solved if $|c| \leq \sqrt{a^2 + b^2}$, by substituting $a = r \cos \theta$ and $b = r \sin \theta$ or $a = r \sin \theta$ and $b = r \cos \theta$. Based on this, answer the following questions.

- (i) General solution set of the equation $\sin 8x - \cos 6x = \sqrt{3}(\sin 6x + \cos 8x)$ is

- (A) $\left\{n\pi + \frac{\pi}{2} : n \in \mathbb{Z}\right\} \cup \left\{2k\pi + \frac{\pi}{6} : k \in \mathbb{Z}\right\}$
 (B) $\left\{n\frac{\pi}{7} + \frac{\pi}{12} : n \in \mathbb{Z}\right\} \cup \left\{(4k + 1)\frac{\pi}{4} : k \in \mathbb{Z}\right\}$
 (C) $\left\{n\pi + \frac{\pi}{6} : n \in \mathbb{Z}\right\} \cup \left\{k\pi + \frac{\pi}{2} : k \in \mathbb{Z}\right\}$
 (D) $\left\{(2n + 1)\frac{\pi}{3} : n \in \mathbb{Z}\right\} \cup \left\{(2k + 1)\frac{\pi}{7} : k \in \mathbb{Z}\right\}$

- (ii) The number of solutions of the equation $\cos 5x + \sin 5x = 0$ in the interval $[0, \pi]$ is

- (A) 4 (B) 5 (C) 6 (D) 8

- (iii) The number of solutions of the equation $\sin x + \cos x = (\sqrt{3} + 1)/2$ in $[0, \pi/2]$ is

- (A) 4 (B) 3 (C) 2 (D) 1

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true

1. **Statement I:** $\cos x + \sin x = 2$ has solutions.

Statement II: $a \cos x + b \sin x = c$ has solutions if $|c| \leq \sqrt{a^2 + b^2}$.

2. **Statement I:** $\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$ has solutions.

Statement II: The sets $\{\sin x, \sin 2x, \sin 3x\}$ and $\{\cos x, \cos 2x, \cos 3x\}$ are equal for all real values of x .

3. **Statement I:** $\cos x = 1/2 \Rightarrow x = 2n\pi \pm (\pi/3), n \in \mathbb{Z}$.

Statement II: $\sin x = 1/2 \Rightarrow x = k\pi + (-1)^k(\pi/6), k \in \mathbb{Z}$.

4. **Statement I:** $\tan x = 1 \Leftrightarrow x = n\pi + (\pi/4), n \in \mathbb{Z}$.

Statement II: $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = \tan x$ for all $x \in \mathbb{Z}$.

5. **Statement I:** If $y > 0$ and $0 \leq x \leq \pi/2$ are such that $\sin x + \cos x = \sqrt{y + 1/y}$, then $x = \pi/4$ and $y = 1$.

Statement II: $|\sin x + \cos x| \leq \sqrt{2}$ and $y + (1/y) \geq 2$.

6. **Statement I:** For $0 \leq x, y \leq 2\pi$, $\sin x + \sin y = 2 \Rightarrow x + y = \pi$.

Statement II: $|\sin \theta| \leq 1$ for all θ .

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. If $\cos 2x + a \sin x = 2a - 7$ has real solutions, then the sum of the minimum and maximum values of a is _____.

2. The number of values of x in the interval $(0, 5\pi)$ satisfying the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ is _____.

3. The number of solutions of the equation $\sin^4 x - 2 \sin^2 x - 1 = 0$ in the interval $[0, 2\pi]$ is _____.

4. Let $f(x) = 2 \sin x [\cos x + \cos 3x + \cos 5x + \cos 7x + \cos 9x]$. Then the number of values of x satisfying the equation $f(x) = 1$ in the interval $[0, \pi]$ is _____.

5. The number of values of x satisfying the equation $2[\cos^2(x/2)] \sin^2 x = x^2 + (1/x^2)$, $x \leq \pi/9$ is _____.

6. The number of solutions of the equation $e^{\sin x} - e^{-\sin x} = 4$ is _____.

7. The number of values of x satisfying the equation $2^{\cos 2x} + 1 = 3 \cdot 2^{-\sin^2 x}$ in the interval $[0, 2\pi]$ is _____.

8. The number of solutions of the equation $\sin x + 2 \sin 2x = 3 + \sin 3x$, $0 < x < \pi$ is _____.

9. Let $0 < x, y < \pi$. Then the number of ordered pairs (x, y) satisfying the system of equations $5^{\csc^2 x - 3 \sec^2 y} = 1$ and $2^{\csc x + \sqrt{3} |\sec y|} = 64$ is _____.

ANSWERS**Single Correct Choice Type Questions**

- | | |
|---------|---------|
| 1. (D) | 17. (D) |
| 2. (C) | 18. (A) |
| 3. (D) | 19. (A) |
| 4. (B) | 20. (B) |
| 5. (A) | 21. (C) |
| 6. (B) | 22. (A) |
| 7. (D) | 23. (C) |
| 8. (A) | 24. (D) |
| 9. (B) | 25. (C) |
| 10. (A) | 26. (B) |
| 11. (B) | 27. (A) |
| 12. (C) | 28. (D) |
| 13. (B) | 29. (B) |
| 14. (D) | 30. (A) |
| 15. (D) | 31. (D) |
| 16. (B) | |

Multiple Correct Choice Type Questions

- | | |
|-----------------------|------------------------|
| 1. (B), (C) | 9. (B), (D) |
| 2. (A), (B) | 10. (A), (B), (C) |
| 3. (A), (B), (D) | 11. (A), (B), (C), (D) |
| 4. (A), (B), (C), (D) | 12. (C), (D) |
| 5. (A), (B), (C) | 13. (A), (D) |
| 6. (A), (B), (C), (D) | 14. (A), (B) |
| 7. (A), (C), (D) | 15. (A), (C) |
| 8. (A), (B), (C), (D) | |

Matrix-Match Type Questions

- $(A) \rightarrow (p), (s), (t), (B) \rightarrow (q), (s), (C) \rightarrow (q), (r), (s), (D) \rightarrow (r), (s)$
- $(A) \rightarrow (t), (B) \rightarrow (p), (q), (C) \rightarrow (p), (r), (D) \rightarrow (p), (t)$
- $(A) \rightarrow (p), (B) \rightarrow (q), (C) \rightarrow (s), (D) \rightarrow (r)$
- $(A) \rightarrow (q), (B) \rightarrow (s), (C) \rightarrow (p), (D) \rightarrow (p)$
- $(A) \rightarrow (q), (B) \rightarrow (p), (C) \rightarrow (q), (D) \rightarrow (r)$

Comprehension-Type Questions

- | | |
|---------------------------------|---------------------------------|
| 1. (i) (A); (ii) (D); (iii) (C) | 2. (i) (B); (ii) (B); (iii) (C) |
|---------------------------------|---------------------------------|

Assertion–Reasoning Type Questions

- | | |
|--------|--------|
| 1. (D) | 4. (C) |
| 2. (C) | 5. (A) |
| 3. (B) | 6. (A) |

Integer Answer Type Questions

- | | |
|------|------|
| 1. 8 | 6. 0 |
| 2. 6 | 7. 5 |
| 3. 0 | 8. 0 |
| 4. 5 | 9. 4 |
| 5. 0 | |

Properties of Triangles

4

Properties of Triangles



Contents

- 4.1 *Relations Between the Sides and Trigonometric Ratios of Angles of a Triangle*
- 4.2 *Solution of Triangles*
- 4.3 *Properties of Triangles*

Worked-Out Problems
Summary
Exercises
Answers

Triangles are one of the most fundamental geometric shapes and have a variety of often studied properties relating to their **Interior angles** (angles on the inside) and **relationship between measurement of the sides and angles in a triangle**.

The three sides and the three angles of a triangle are called the *elements* of the triangle. One important aspect of trigonometry is to find all the elements of triangle when some of its sides and angles are known. The process of finding all the unknown elements of a triangle with the help of known elements is called *solving a triangle* and the values of the unknown elements are called *solution of the triangle*. It is known that when all the three angles are known, then the triangle is not uniquely determined. For example, each angle in an equilateral triangle is $\pi/3$ and an equilateral triangle is determined by its side. In this chapter, we will be proving that a triangle is completely determined (i.e., all its elements are known) if any three of the six elements of the triangle, with at least one side, are known. Further, we discuss various methods and formulae for solving a triangle.

4.1 | Relations Between the Sides and the Trigonometric Ratios of Angles of a Triangle

We derive certain important formulae relating the sides of a triangle and the trigonometric ratios of the triangle which are useful in solving a triangle. First, let us fix certain notations. A triangle is usually denoted by $\triangle ABC$ or simply ABC if A , B and C are vertices of the triangle. The letters A , B and C also denote angles $\angle BAC$, $\angle ABC$ and $\angle BCA$, respectively. The lengths of the sides opposite to A , B and C are denoted by a , b and c , respectively, as shown in Figure 4.1.

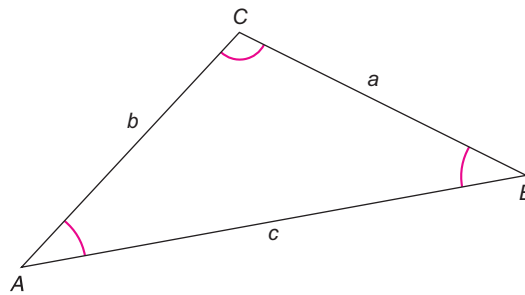


FIGURE 4.1

Since the secondary school level, we are familiar with the following geometrical properties of the sides and angles of a triangle ABC .

1. $A + B + C = \pi = 180^\circ$. That is the sum of the three angles is π radians or 180° .
2. $a < b + c$; $b < c + a$ and $c < a + b$. That is, the sum of any two sides is greater than the third side.
3. a , b and c are positive real numbers.

THEOREM 4.1 (SINE FORMULA)

In any triangle ABC , the sines of the angles are proportional to opposite sides; that is

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \quad (4.1)$$

PROOF

Consider angle C . We shall prove the theorem using the following three cases:

Case I: Suppose ABC is a right-angled triangle (see Figure 4.2); say

$$C = \frac{\pi}{2} = 90^\circ$$

Then

$$\sin B = \frac{b}{c}, \quad \sin C = 1 \quad \text{and} \quad \sin A = \frac{a}{c}$$

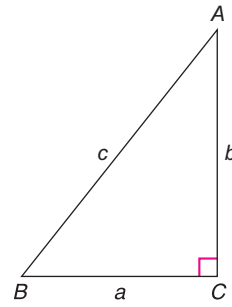


FIGURE 4.2 Case I.

Therefore

$$\frac{\sin A}{a} = \frac{1}{c} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Case II: Suppose that each of the angles A , B , and C is acute. Then draw a perpendicular to BC through A to meet BC at D (see Figure 4.3). Now

$$\sin B = \frac{AD}{AB} = \frac{AD}{c}$$

and hence

$$c \sin B = AD$$

Also,

$$\sin C = \frac{AD}{AC} = \frac{AD}{b}$$

and hence

$$b \sin C = AD = c \sin B$$

Therefore

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

Similarly

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

Therefore

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

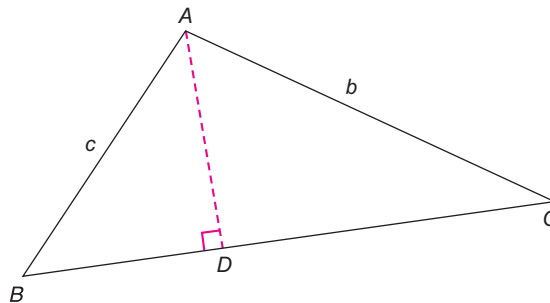


FIGURE 4.3 Case II.

Case III: Suppose that one of the angles is obtuse; say C is obtuse. Then A and B are acute angles (Figure 4.4).

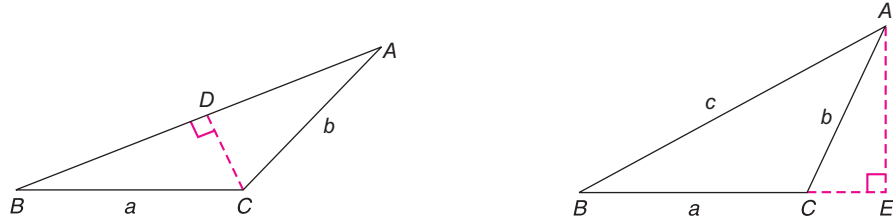


FIGURE 4.4 Case III.

Draw perpendicular from C touching AB at D and from A touching extended BC at E . Then

$$\sin A = \frac{CD}{CA} = \frac{CD}{b} \quad \text{and} \quad \sin B = \frac{CD}{CB} = \frac{CD}{a}$$

Therefore

$$b \sin A = CD = a \sin B$$

and hence

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

Also,

$$\sin B = \frac{AE}{AB} = \frac{AE}{c}$$

and

$$\sin C = \sin(\pi - C) = \sin \angle ECA = \frac{AE}{AC} = \frac{AE}{b}$$

Therefore

$$b \sin C = AE = c \sin B$$

and hence

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

Thus

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

THEOREM 4.2
(**COSINE**
FORMULA)

In any triangle ABC ,

$$1. \cos A = \frac{b^2 + c^2 - a^2}{2bc} \text{ or } a^2 = b^2 + c^2 - 2bc \cos A$$

$$2. \cos B = \frac{a^2 + c^2 - b^2}{2ac} \text{ or } b^2 = a^2 + c^2 - 2ac \cos B$$

$$3. \cos C = \frac{a^2 + b^2 - c^2}{2ab} \text{ or } c^2 = a^2 + b^2 - 2ab \cos C$$

PROOF

Consider angle A . If A is a right angle (Figure 4.5), then $\cos A = 0$. Clearly $a^2 + b^2 = c^2$ implies

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Therefore, we can suppose that $A \neq 90^\circ$. Now draw a perpendicular to CA through B to meet CA at D (extend CA if necessary). Two cases arise:

Case I: If A is acute (as in Figure 4.6), then

$$\cos A = \frac{AD}{BA} = \frac{AD}{c} \quad \text{or} \quad AD = c \cos A$$

and

$$\sin A = \frac{BD}{BA} = \frac{BD}{c} \quad \text{or} \quad BD = c \sin A$$

and therefore

$$CD = AC - AD = b - c \cos A$$

Case II: If A is obtuse (as in Figure 4.7), then

$$\sin A = \sin(\pi - A) = \sin \angle DAB = \frac{BD}{AB} = \frac{BD}{c}$$

or

$$BD = c \sin A$$

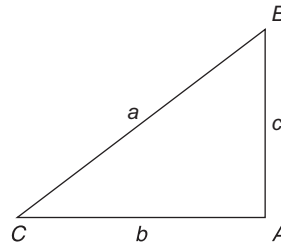


FIGURE 4.5

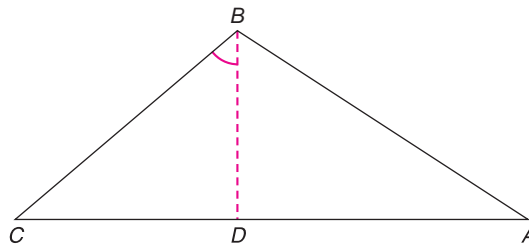


FIGURE 4.6 Case I.

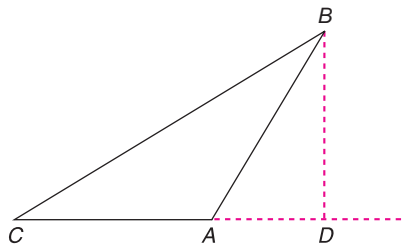


FIGURE 4.7 Case II.

and

$$\cos A = -\cos(\pi - A) = -\cos \angle DAB = -\frac{AD}{AB} = -\frac{AD}{c}$$

or

$$AD = -c \cos A$$

Hence

$$CD = CA + AD = b - c \cos A$$

From both Cases I and II, we have

$$CD = b - c \cos A \quad \text{and} \quad BD = c \sin A$$

From the triangle BCD , we have

$$\begin{aligned} a^2 &= BC^2 \\ &= CD^2 + BD^2 \\ &= (b - c \cos A)^2 + (c \sin A)^2 \\ &= b^2 + c^2 \cos^2 A - 2bc \cos A + c^2 \sin^2 A \\ &= b^2 + c^2 (\cos^2 A + \sin^2 A) - 2bc \cos A \\ &= b^2 + c^2 - 2bc \cos A \end{aligned}$$

This proves the formula (1). Similarly, formulae (2) and (3) also can be proved. ■

Try it out Show that

1. $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$ or $b^2 = a^2 + c^2 - 2ac \cos B$
2. $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ or $c^2 = a^2 + b^2 - 2ab \cos C$

Example 4.1

If ABC is a triangle, such that $a = 36$, $b = 15$ and $c = 39$, then verify the sine formula.

Solution: We have

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{(15)^2 + (39)^2 - (36)^2}{2 \times 15 \times 39} = \frac{450}{1170} = \frac{5}{13}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca} = \frac{(39)^2 + (36)^2 - (15)^2}{2 \times 39 \times 36} = \frac{2592}{2808} = \frac{12}{13}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{(36)^2 + (15)^2 - (39)^2}{2 \times 36 \times 15} = 0$$

That is, C is a right angle and ABC is a right-angled triangle (Figure 4.8). Note that, in this case

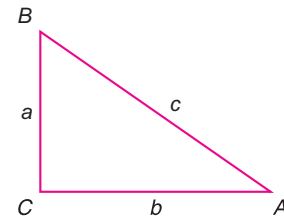


FIGURE 4.8 Example 4.1.

$$\cos A = \frac{CA}{BA} = \frac{b}{c} = \frac{15}{39} = \frac{5}{13}$$

$$\cos B = \frac{CB}{AB} = \frac{a}{c} = \frac{36}{39} = \frac{12}{13}$$

$$\sin A = \frac{CB}{AB} = \frac{a}{c} = \frac{36}{39} = \frac{12}{13}$$

$$\sin B = \frac{b}{c} = \frac{15}{39} = \frac{5}{13}$$

Therefore

$$\begin{aligned}\frac{\sin A}{a} &= \frac{12}{13} = \frac{1}{36} = \frac{1}{39} \\ \frac{\sin B}{b} &= \frac{5}{13} = \frac{1}{15} = \frac{1}{39} \\ \frac{\sin C}{c} &= \frac{1}{39}\end{aligned}$$

which implies

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Hence the sine formula [Eq. (4.1)] is verified.

Example 4.2

Let ABC be a triangle such that $a = 2$, $b = 3$ and $c = 4$. Verify the sine formula.

Solution: From the given data we get

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{3^2 + 4^2 - 2^2}{2 \times 3 \times 4} = \frac{7}{8}\end{aligned}$$

$$\begin{aligned}\cos B &= \frac{c^2 + a^2 - b^2}{2ca} \\ &= \frac{4^2 + 2^2 - 3^2}{2 \times 4 \times 2} = \frac{11}{16}\end{aligned}$$

$$\begin{aligned}\cos C &= \frac{a^2 + b^2 - c^2}{2ab} \\ &= \frac{2^2 + 3^2 - 4^2}{2 \times 2 \times 3} = \frac{-3}{12}\end{aligned}$$

$$\sin A = \sqrt{1 - \cos^2 A} = \sqrt{1 - \left(\frac{7}{8}\right)^2} = \frac{\sqrt{15}}{8}$$

$$\sin B = \sqrt{1 - \cos^2 B} = \sqrt{1 - \left(\frac{11}{16}\right)^2} = \frac{3\sqrt{15}}{16}$$

$$\sin C = \sqrt{1 - \cos^2 C} = \sqrt{1 - \left(\frac{-3}{12}\right)^2} = \frac{3\sqrt{15}}{12}$$

Therefore

$$\frac{\sin A}{a} = \frac{\sqrt{15}}{8} \cdot \frac{1}{2} = \frac{\sqrt{15}}{16}$$

$$\frac{\sin B}{b} = \frac{3\sqrt{15}}{16} \cdot \frac{1}{3} = \frac{\sqrt{15}}{16}$$

$$\frac{\sin C}{c} = \frac{3\sqrt{15}}{12} \cdot \frac{1}{4} = \frac{\sqrt{15}}{16}$$

Therefore the sine formula [Eq. (4.1)] is verified.

Note that A and B are acute angles and C is an obtuse angle, since $\cos C < 0$. Now

THEOREM 4.3 (PROJECTION FORMULAE)

In any triangle ABC ,

1. $a = b \cos C + c \cos B$
2. $b = c \cos A + a \cos C$
3. $c = a \cos B + b \cos A$

PROOF

We shall prove formula (1) in each of the following three cases.

Case I: Suppose that $C = 90^\circ$. See Figure 4.9.

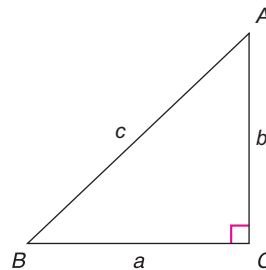


FIGURE 4.9 Case I.

Then,

$$\cos B = \frac{BC}{AB} = \frac{a}{c}$$

and hence $a = c \cos B$. This can be written as

$$a = b \cos C + c \cos B$$

since $\cos C = 0$.

Case II: Suppose that C is an acute angle. Draw a perpendicular to BC through A to meet BC at D (Figure 4.10). Then

$$\cos B = \frac{BD}{AB} = \frac{BD}{c}$$

and therefore $BD = c \cos B$. Also,

$$\cos C = \frac{DC}{AC} = \frac{DC}{b}$$

and hence $DC = b \cos C$.

Therefore

$$a = BC = BD + DC = c \cos B + b \cos C$$

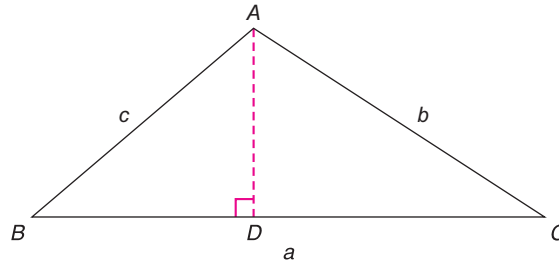


FIGURE 4.10 Case II.

Case III: Suppose that C is an obtuse angle. Here again, draw a perpendicular through A to extended BC meeting it at D (Figure 4.11). Then

$$\cos B = \frac{BD}{BA} = \frac{BD}{c}$$

Therefore

$$BD = c \cos B$$

Also,

$$\cos \angle ACD = \frac{CD}{AC} = \frac{CD}{b}$$

Now,

$$\begin{aligned} \cos C &= \cos(\pi - \angle ACD) \\ &= -\cos \angle ACD \\ &= -\frac{CD}{b} \end{aligned}$$

Therefore $CD = -b \cos C$. Hence

$$\begin{aligned} a &= BC \\ &= BD - CD \\ &= c \cos B - (-b \cos C) \\ &= c \cos B + b \cos C \end{aligned}$$

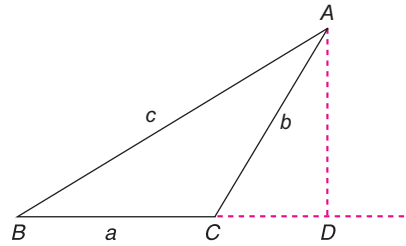


FIGURE 4.11 Case III.

Thus, in each of the above cases, $a = c \cos B + b \cos C$.
Similarly we can prove formulae (2) and (3). ■

Try it out Prove the following:

1. $b = c \cos A + a \cos C$
2. $c = a \cos B + b \cos A$

Remark: The above formulae can be proved by using the cosine formulae given in Theorem 4.2. Assuming the cosine formulae in Theorem 4.2, we get that

$$b \cos C + c \cos B = \frac{a^2 + b^2 - c^2}{2a} + \frac{c^2 + a^2 - b^2}{2a} = \frac{2a^2}{2a} = a$$

Similarly the other formulae in Theorem 4.3 can be proved. Conversely, by assuming the formulae in Theorem 4.3, we get

$$\begin{aligned} a^2 + b^2 - c^2 &= a(b \cos C + c \cos B) + b(c \cos A + a \cos C) - c(a \cos B + b \cos A) \\ &= 2ab \cos C \end{aligned}$$

and similarly the other cosine formulae can be proved.

In the following, we derive formulae which give sine of half the angles of a triangle in terms of the sides. If s denotes half of the perimeter of the triangle then

$$s = \frac{a + b + c}{2} \quad \text{or} \quad 2s = a + b + c$$

We will use the above-mentioned formulae for s in the rest of the chapter.

THEOREM 4.4 In any triangle ABC

1. $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$
2. $\sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$
3. $\sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

PROOF We have

$$\cos A = 1 - 2 \sin^2 \frac{A}{2}$$

and hence

$$\begin{aligned} 2 \sin^2 \frac{A}{2} &= 1 - \cos A \\ &= 1 - \frac{b^2 + c^2 - a^2}{2bc} \quad (\text{by the cosine formula}) \\ &= \frac{2bc - (b^2 + c^2 - a^2)}{2bc} \\ &= \frac{a^2 - (b^2 + c^2 - 2bc)}{2bc} \\ &= \frac{a^2 - (b - c)^2}{2bc} \\ &= \frac{[a + (b - c)][a - (b - c)]}{2bc} \\ &= \frac{(a + b - c)(a + c - b)}{2bc} \\ &= \frac{2(s - c) \cdot 2(s - b)}{2bc} \quad (\text{since } 2s = a + b + c) \end{aligned}$$

Therefore

$$\sin^2 \frac{A}{2} = \frac{(s - b)(s - c)}{bc} \quad \text{or} \quad \sin \frac{A}{2} = \sqrt{\frac{(s - b)(s - c)}{bc}}$$

Similarly,

$$\sin \frac{B}{2} = \sqrt{\frac{(s - c)(s - a)}{ca}} \quad \text{and} \quad \sin \frac{C}{2} = \sqrt{\frac{(s - a)(s - b)}{ab}}$$

Note: The sine of any angle of a triangle is always positive (since $\sin \theta > 0$ for all $0 < \theta < \pi$) and hence, in the above formulae, we have to take positive square roots only.

THEOREM 4.5 In any triangle ABC ,

1. $\cos \frac{A}{2} = \sqrt{\frac{s(s - a)}{bc}}$
2. $\cos \frac{B}{2} = \sqrt{\frac{s(s - b)}{ca}}$
3. $\cos \frac{C}{2} = \sqrt{\frac{s(s - c)}{ab}}$

PROOF Since

$$\cos A = 2 \cos^2 \frac{A}{2} - 1$$

we have

$$\begin{aligned} 2 \cos^2 \frac{A}{2} &= 1 + \cos A \\ &= 1 + \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{2bc + b^2 + c^2 - a^2}{2bc} \\ &= \frac{(b+c)^2 - a^2}{2bc} \\ &= \frac{(b+c-a)(b+c+a)}{2bc} \\ &= \frac{2(s-a) \cdot 2s}{2bc} \quad (\text{since } a+b+c=2s) \end{aligned}$$

Therefore

$$\cos^2 \frac{A}{2} = \frac{s(s-a)}{bc} \quad \text{or} \quad \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

Similarly,

$$\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}} \quad \text{and} \quad \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

THEOREM 4.6

In any triangle ABC ,

1. $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$
2. $\tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$
3. $\tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

PROOF

We know that

$$\begin{aligned} \tan \frac{A}{2} &= \sin \frac{A}{2} \div \cos \frac{A}{2} \\ &= \sqrt{\frac{(s-b)(s-c)}{bc}} \div \sqrt{\frac{s(s-a)}{bc}} \\ &= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \end{aligned}$$

(Using Theorems 4.4 and 4.5.) Similarly, the formulae for $\tan(B/2)$ and $\tan(C/2)$ can be derived. ■

Try it out Show that

$$1. \tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$$

$$2. \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

Example 4.3

In a triangle ABC , suppose $a = 6$, $b = 8$ and $c = 12$. Find the following:

$$\sin \frac{A}{2}; \sin \frac{B}{2}; \sin \frac{C}{2}; \cos \frac{A}{2}; \cos \frac{B}{2}; \cos \frac{C}{2}$$

and hence

$$\tan \frac{A}{2}; \tan \frac{B}{2}; \tan \frac{C}{2}$$

Solution: Given $a = 6$, $b = 8$ and $c = 12$, we have

$$s = \frac{a+b+c}{2} = \frac{6+8+12}{2} = 13$$

Therefore

$$s-a=7, s-b=5 \text{ and } s-c=1$$

Now

$$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}} = \sqrt{\frac{5 \times 1}{8 \times 12}} = \sqrt{\frac{5}{96}}$$

$$\sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}} = \sqrt{\frac{1 \times 7}{12 \times 6}} = \sqrt{\frac{7}{72}}$$

$$\sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}} = \sqrt{\frac{7 \times 5}{6 \times 8}} = \sqrt{\frac{35}{48}}$$

$$\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}} = \sqrt{\frac{13 \times 7}{8 \times 12}} = \sqrt{\frac{91}{96}}$$

$$\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}} = \sqrt{\frac{13 \times 5}{12 \times 6}} = \sqrt{\frac{65}{72}}$$

$$\cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}} = \sqrt{\frac{13 \times 1}{6 \times 8}} = \sqrt{\frac{13}{48}}$$

Also

$$\tan \frac{A}{2} = \sin \frac{A}{2} \div \cos \frac{A}{2} = \sqrt{\frac{5}{91}}$$

$$\tan \frac{B}{2} = \sin \frac{B}{2} \div \cos \frac{B}{2} = \sqrt{\frac{7}{65}}$$

$$\tan \frac{C}{2} = \sin \frac{C}{2} \div \cos \frac{C}{2} = \sqrt{\frac{35}{13}}$$

THEOREM 4.7 In any triangle ABC ,

$$\sin A = \frac{2}{bc} \sqrt{s(s-a)(s-b)(s-c)}$$

$$\sin B = \frac{2}{ca} \sqrt{s(s-a)(s-b)(s-c)}$$

$$\sin C = \frac{2}{ab} \sqrt{s(s-a)(s-b)(s-c)}$$

PROOF Since

$$\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

we have

$$\begin{aligned}\sin A &= 2\sqrt{\frac{(s-b)(s-c)}{bc}}\sqrt{\frac{s(s-a)}{bc}} \\ &= \frac{2}{bc}\sqrt{s(s-a)(s-b)(s-c)}\end{aligned}$$

Similarly the formulae for $\sin B$ and $\sin C$ can be derived.

Note: From the above formulae for $\sin A$, we have

$$\frac{\sin A}{a} = \frac{2}{abc}\sqrt{s(s-a)(s-b)(s-c)}$$

The right-hand side of the above is symmetric in a, b and c and, therefore,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \left(= \frac{2}{abc}\sqrt{s(s-a)(s-b)(s-c)} \right)$$

which is precisely the sine formulae proved in the beginning of this chapter.

COROLLARY 4.1 In any triangle ABC ,

$$\tan A = \frac{4\sqrt{s(s-a)(s-b)(s-c)}}{b^2 + c^2 - a^2}$$

$$\tan B = \frac{4\sqrt{s(s-a)(s-b)(s-c)}}{c^2 + a^2 - b^2}$$

$$\tan C = \frac{4\sqrt{s(s-a)(s-b)(s-c)}}{a^2 + b^2 - c^2}$$

Example 4.4

Suppose that in a triangle ABC , $a = 8$, $b = 9$ and $c = 13$. Then find $\tan A$, $\tan B$ and $\tan C$.

Solution: Given $a = 8$, $b = 9$ and $c = 13$ we get

$$s = \frac{8+9+13}{2} = 15$$

$$s-a=7, s-b=6 \quad \text{and} \quad s-c=2$$

$$\sqrt{s(s-a)(s-b)(s-c)} = \sqrt{15 \times 7 \times 6 \times 2} = 6\sqrt{35}$$

Therefore

$$\sin A = \frac{2}{bc}\sqrt{s(s-a)(s-b)(s-c)}$$

$$= \frac{2}{9 \times 13} \times 6\sqrt{35} = \frac{4\sqrt{35}}{39}$$

$$\sin B = \frac{2}{13 \times 8} \times 6\sqrt{35} = \frac{3\sqrt{35}}{26}$$

$$\sin C = \frac{2}{8 \times 9} \times 6\sqrt{35} = \frac{\sqrt{35}}{6}$$

Hence

$$\frac{\sin A}{a} = \frac{4\sqrt{35}}{39 \times 8} = \frac{\sqrt{35}}{78} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Now

$$\tan A = \frac{4\sqrt{s(s-a)(s-b)(s-c)}}{b^2 + c^2 - a^2} = \frac{4 \times 6\sqrt{35}}{9^2 + 13^2 - 8^2}$$

$$= \frac{4 \times 6\sqrt{35}}{9^2 + 13^2 - 8^2} = \frac{24\sqrt{35}}{186} = \frac{4\sqrt{35}}{31}$$

$$\tan B = \frac{4 \times 6\sqrt{35}}{13^2 + 8^2 - 9^2} = \frac{24\sqrt{35}}{152} = \frac{6\sqrt{35}}{38} = \frac{3\sqrt{35}}{19}$$

$$\tan C = \frac{4 \times 6\sqrt{35}}{8^2 + 9^2 - 13^2} = -\sqrt{35}$$

THEOREM 4.8 In any triangle ABC ,

$$\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\tan \frac{C-A}{2} = \frac{c-a}{c+a} \cot \frac{B}{2}$$

$$\tan \frac{A-B}{2} = \frac{a-b}{a+b} \cot \frac{C}{2}$$

PROOF From the sine formulae, we have

$$\frac{\sin B}{b} = \frac{\sin C}{c} = K \text{ (say)}$$

Now consider

$$\begin{aligned} \frac{b-c}{b+c} &= \frac{(1/K)(\sin B - \sin C)}{(1/K)(\sin B + \sin C)} \\ &= \frac{\sin B - \sin C}{\sin B + \sin C} \\ &= \frac{2 \cos \left(\frac{B+C}{2} \right) \sin \left(\frac{B-C}{2} \right)}{2 \sin \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right)} \\ &= \cot \left(\frac{B+C}{2} \right) \tan \left(\frac{B-C}{2} \right) \\ &= \cot \left(90^\circ - \frac{A}{2} \right) \tan \left(\frac{B-C}{2} \right) \quad (\text{since } A+B+C=\pi) \\ &= \tan \frac{A}{2} \tan \left(\frac{B-C}{2} \right) \end{aligned}$$

Therefore

$$\tan \left(\frac{B-C}{2} \right) = \left(\frac{b-c}{b+c} \right) \cot \frac{A}{2}$$

Similarly, the other formulae can be proved. ■

Note: The formulae given above are called *tangent rules* or *Napier analogy*.

THEOREM 4.9 In any triangle ABC ,

$$\cos \left(\frac{A-B}{2} \right) = \frac{a+b}{c} \sin \frac{C}{2}$$

$$\cos \left(\frac{B-C}{2} \right) = \frac{b+c}{a} \sin \frac{A}{2}$$

$$\cos \left(\frac{C-A}{2} \right) = \frac{c+a}{b} \sin \frac{B}{2}$$

PROOF Consider

$$\begin{aligned}
 \frac{a+b}{c} &= \frac{\sin A + \sin B}{\sin C} \left(\text{since } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \right) \\
 &= \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \sin \frac{C}{2} \cdot \cos \frac{C}{2}} \\
 &= \frac{\sin \left(90^\circ - \frac{C}{2} \right) \cos \left(\frac{A-B}{2} \right)}{\sin \frac{C}{2} \cdot \cos \frac{C}{2}} \\
 &= \frac{\cos \frac{C}{2} \cdot \cos \left(\frac{A-B}{2} \right)}{\sin \frac{C}{2} \cdot \cos \frac{C}{2}} = \frac{\cos \left(\frac{A-B}{2} \right)}{\sin \frac{C}{2}}
 \end{aligned}$$

Therefore

$$\cos \left(\frac{A-B}{2} \right) = \frac{a+b}{c} \cdot \sin \frac{C}{2}$$

Similarly,

$$\cos \left(\frac{C-A}{2} \right) = \frac{c+a}{b} \cdot \sin \frac{B}{2}$$

and

$$\cos \left(\frac{B-C}{2} \right) = \frac{b+c}{a} \cdot \sin \frac{A}{2}$$

The following can be proved similarly.

THEOREM 4.10 In any triangle ABC ,

$$\sin \left(\frac{A-B}{2} \right) = \frac{a-b}{c} \cdot \cos \frac{C}{2}$$

$$\sin \left(\frac{B-C}{2} \right) = \frac{b-c}{a} \cdot \cos \frac{A}{2}$$

$$\sin \left(\frac{C-A}{2} \right) = \frac{c-a}{b} \cdot \cos \frac{B}{2}$$

PROOF Left as an exercise for the readers.

Note: The formulae given in the above two theorems are called *Mollweide rules*.

THEOREM 4.11 In any triangle ABC ,

$$(b^2 - c^2) \cot A + (c^2 - a^2) \cot B + (a^2 - b^2) \cot C = 0$$

PROOF We have by sine formula that

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = K, \text{ say}$$

Now,

$$\cot A = \frac{\cos A}{\sin A} = \frac{\cos A}{aK} = \frac{b^2 + c^2 - a^2}{2abcK}$$

Similarly

$$\cot B = \frac{c^2 + a^2 - b^2}{2abcK} \quad \text{and} \quad \cot C = \frac{a^2 + b^2 - c^2}{2abcK}$$

Therefore

$$\begin{aligned} & (b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C \\ &= \frac{1}{2abcK} [(b^2 - c^2)(b^2 + c^2 - a^2) + (c^2 - a^2)(c^2 + a^2 - b^2) + (a^2 - b^2)(a^2 + b^2 - c^2)] \\ &= \frac{1}{2abcK} [b^4 - c^4 - a^2(b^2 - c^2) + c^4 - a^4 - b^2(c^2 - a^2) + a^4 - b^4 - c^2(a^2 - b^2)] \\ &= 0 \end{aligned}$$

THEOREM 4.12 In any triangle ABC , if a, b, c are in Arithmetical Progression, then so are $\cot(A/2), \cot(B/2), \cot(C/2)$.

PROOF Suppose that a, b, c are in Arithmetical Progression. Then $a + c = 2b$ and hence

$$(s - a) + (s - c) = 2(s - b)$$

Now,

$$\begin{aligned} \cot \frac{A}{2} + \cot \frac{C}{2} - 2\cot \frac{B}{2} &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} - 2\sqrt{\frac{s(s-b)}{(s-c)(s-a)}} \\ &= \frac{\sqrt{s}}{\sqrt{(s-a)(s-b)(s-c)}} [(s-a) + (s-c) - 2(s-b)] \\ &= 0 \end{aligned}$$

Therefore $\cot(A/2), \cot(B/2), \cot(C/2)$ are in Arithmetical Progression.

THEOREM 4.13 In any triangle ABC ,

$$\begin{aligned} (a + b + c) \left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) &= 2c \cot \frac{C}{2} \\ (a + b + c) \left(\tan \frac{B}{2} + \tan \frac{C}{2} \right) &= 2a \cot \frac{A}{2} \\ (a + b + c) \left(\tan \frac{C}{2} + \tan \frac{A}{2} \right) &= 2b \cot \frac{B}{2} \end{aligned}$$

PROOF From Theorem 4.6, we have

$$\begin{aligned}
 (a + b + c) \left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) &= 2s \left(\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} + \sqrt{\frac{(s-c)(s-a)}{s(s-b)}} \right) \\
 &= 2s \sqrt{\frac{s-c}{s}} \left(\sqrt{\frac{s-b}{s-a}} + \sqrt{\frac{s-a}{s-b}} \right) \\
 &= 2\sqrt{s(s-c)} \left(\frac{(s-b) + (s-a)}{\sqrt{(s-a)(s-b)}} \right) \\
 &= 2 \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \cdot c \quad (\text{since } a + b + c = 2s) \\
 &= 2c \cot \frac{A}{2}
 \end{aligned}$$

4.2 | Solution of Triangles

The three sides and the three angles of a triangle are known to be the elements of the triangle. When any three elements with at least one side are given, then the triangle is completely known, in the sense that its other elements can be found. When all the three angles are given, only the ratios of lengths of the sides can be found and therefore only the shape of the triangle can be found and not the size of the triangle. The process of calculating the values of all the elements of the triangle is termed as *solving the triangle* and the set of values of the elements is called the *solution* of the triangle. In the following, we discuss the solutions of triangles in several cases.

4.2.1 Solutions of Right-Angled Triangles

Let ABC be a right-angled triangle with $C = 90^\circ$ (Figure 4.12). Then c is the hypotenuse and A and B are acute angles such that $A + B = 90^\circ$. We need two more elements to solve the triangle ABC , in which one is necessarily a side.

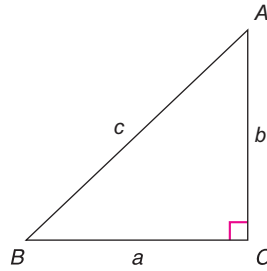


FIGURE 4.12

Case 1: Suppose that the hypotenuse and one side are given

Let b be the given side and c be the given hypotenuse. Since

$$\sin B = \frac{b}{c} < 1 \quad (\because b < c)$$

we have

$$B = \sin^{-1} \frac{b}{c} \in \left(0, \frac{\pi}{2} \right)$$

and

$$A = 180^\circ - (B + C) = 90^\circ - B$$

Also, $a = c \cos B$. Thus, we can calculate all the three sides and the three angles.

Case 2: Suppose that the two sides a and b are given

In this case

$$c = \sqrt{a^2 + b^2}$$

Since

$$\tan B = \frac{b}{a}$$

so

$$B = \tan^{-1} \frac{b}{a}$$

Therefore B is known and

$$A = \pi - (B + C) = \frac{\pi}{2} - B$$

Thus all the elements of the triangle are known. The hypotenuse c can also be calculated using the formula

$$\sin B = \frac{b}{c}$$

Case 3: Suppose that the angle B and the hypotenuse c are given

Then $A = 90^\circ - B$ and a and b can be obtained from

$$\cos B = \frac{a}{c} \quad \text{and} \quad \sin B = \frac{b}{c}$$

Case 4: Suppose that an angle B and one of the sides a are given

Then $A = 90^\circ - B$ and b is obtained from $\tan B = b/a$. Also c can be found from the relation $\cos B = a/c$.

Example 4.5

The length of the perpendicular from one vertex A of a triangle to the opposite side is 3 cm and the lengths of the sides for which A is the common point are 4 cm and 5 cm. Solve the triangle.

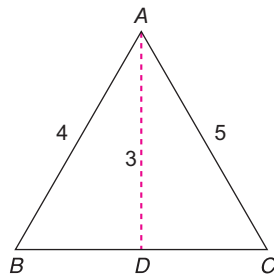


FIGURE 4.13 Example 4.5.

$$\sin B = \frac{AD}{AB} = \frac{3}{4}$$

and

$$\sin C = \frac{AD}{AC} = \frac{3}{5}$$

Therefore

$$B = \sin^{-1} \frac{3}{4}$$

$$C = \sin^{-1} \frac{3}{5}$$

and

$$A = \pi - (B + C)$$

Also

$$a = BD + DC = 4 \cos B + 5 \cos C$$

Solution: Let ABC be the triangle and AD be the perpendicular from A to BC . We are given that $AD = 3$ cm, $AB = 4$ cm and $AC = 5$ cm (Figure 4.13). Then

Thus, all the three angles and three sides are found.

Example 4.6

Solve the right-angled triangle whose hypotenuse is four times as long as the perpendicular drawn to it from the opposite vertex.

Solution: Let ABC be the given right angled triangle with $C = 90^\circ$. Let CD be the perpendicular from C to AB . Then AB is the hypotenuse. Let $CD = x$. Then $AB = 4x$. See Figure 4.14.

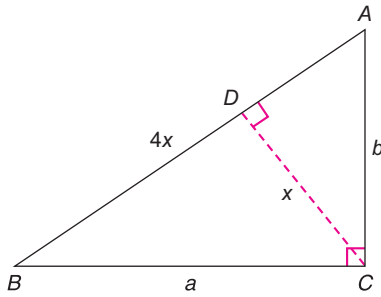


FIGURE 4.14 Example 4.6.

From the triangle ABC , we have

$$\sin B = \frac{b}{4x}$$

From the triangle BCD , we have

$$\sin B = \frac{x}{a}$$

Therefore

$$\frac{b}{4x} = \frac{x}{a} \quad \text{or} \quad 4x^2 = ab$$

Also, we have

$$a^2 + b^2 = (4x)^2 = 4ab \quad (\because 4x^2 = ab)$$

Now

$$\left(\frac{a}{b}\right)^2 + 1 = 4\left(\frac{a}{b}\right)$$

$$\left(\frac{a}{b}\right)^2 - 4\left(\frac{a}{b}\right) + 1 = 0$$

This is a quadratic equation solving which we get

$$\frac{a}{b} = \frac{4 \pm \sqrt{16 - 4}}{2} = 2 \pm \sqrt{3}$$

Therefore

$$\tan A = \frac{a}{b} = 2 \pm \sqrt{3}$$

Hence $A = 75^\circ$ or 15° and $B = 15^\circ$ or 75° .

Without loss of generality, we can suppose that $A = 75^\circ$ and $B = 15^\circ$ and we are given that $C = 90^\circ$. Therefore

$$\frac{a}{b} = \tan A = 2 + \sqrt{3} \Rightarrow a = b(2 + \sqrt{3})$$

$$\text{and} \quad c = \sqrt{a^2 + b^2} = b\sqrt{8 + 4\sqrt{3}} = 2b\sqrt{2 + \sqrt{3}}$$

Therefore knowing the value of b we can determine the values of a and c . Hence, there are infinitely many triangles with angles 90° , 75° and 15° , if the altitude from the right-angled vertex onto the hypotenuse is one-fourth of the hypotenuse.

Note: In the above example, the triangle can be determined provided one of the sides is given.

4.2.2 Solution of a Triangle when Three Sides are Given

Let ABC be the triangle and let the sides a , b and c be given. Then the perimeter $a + b + c = 2s$ and hence s is known. Then using the formulae

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \quad \text{and} \quad \tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$$

the angles A and B can be found and the third angle $C = \pi - (A + B)$. The angles A and B can also be found using the cosine formulae

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad \text{and} \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

Example 4.7

Solve the triangle ABC if $a = 2$, $b = \sqrt{6}$ and $c = \sqrt{3} - 1$. Also

Solution: We have

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{(\sqrt{6})^2 + (\sqrt{3} - 1)^2 - (2)^2}{2 \times \sqrt{6} \times (\sqrt{3} - 1)} \\ &= \frac{6 + 3 + 1 - 2\sqrt{3} - 4}{2\sqrt{2}\sqrt{3}(\sqrt{3} - 1)} \\ &= \frac{2\sqrt{3}(\sqrt{3} - 1)}{2\sqrt{2}\sqrt{3}(\sqrt{3} - 1)} = \frac{1}{\sqrt{2}}\end{aligned}$$

Therefore $A = 45^\circ$.

$$\begin{aligned}\cos B &= \frac{c^2 + a^2 - b^2}{2ca} = \frac{(\sqrt{3} - 1)^2 + 2^2 - (\sqrt{6})^2}{2(\sqrt{3} - 1) \times 2} \\ &= \frac{3 + 1 - 2\sqrt{3} + 4 - 6}{4(\sqrt{3} - 1)} = \frac{2(1 - \sqrt{3})}{4(\sqrt{3} - 1)} = \frac{-1}{2}\end{aligned}$$

Therefore $B = 120^\circ$.

Now

$$C = \pi - (A + B) = 180^\circ - (45^\circ + 120^\circ) = 15^\circ$$

Thus A, B, C are $45^\circ, 120^\circ$ and 15° , respectively.

Example 4.8

Solve the triangle ABC if $a = 2$, $b = \sqrt{6}$ and $c = \sqrt{3} + 1$. Also

Solution: Consider

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{(\sqrt{6})^2 + (\sqrt{3} + 1)^2 - 2^2}{2\sqrt{6}(\sqrt{3} + 1)} \\ &= \frac{6 + 3 + 1 + 2\sqrt{3} - 4}{2\sqrt{2}(3 + \sqrt{3})} = \frac{1}{\sqrt{2}}\end{aligned}$$

Therefore $A = 45^\circ$.

$$\begin{aligned}\cos B &= \frac{c^2 + a^2 - b^2}{2ca} = \frac{(\sqrt{3} + 1)^2 + 2^2 - (\sqrt{6})^2}{2(\sqrt{3} + 1)2} \\ &= \frac{3 + 1 + 2\sqrt{3} + 4 - 6}{4(\sqrt{3} + 1)} = \frac{2(1 + \sqrt{3})}{4(\sqrt{3} + 1)} = \frac{1}{2}\end{aligned}$$

Therefore $B = 60^\circ$. Now

$$C = 180^\circ - (A + B) = 180^\circ - (45^\circ + 60^\circ) = 75^\circ$$

4.2.3 Solution of a Triangle when Two Sides and the Angle Between Them are Given

Let ABC be the triangle. Suppose that the sides b and c and the angle A are given. From the formula

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

we can find that

$$a = \sqrt{b^2 + c^2 - 2bc \cos A}$$

Therefore, all the three sides are known and hence we can solve the triangle as in Section 4.2.2. We can also use the formulae

$$\tan\left(\frac{B - C}{2}\right) = \frac{b - c}{b + c} \cot \frac{A}{2} \quad (\text{by Theorem 4.8})$$

to get $(B - C)/2$ and hence $B - C$. Also $B + C = 180^\circ - A$. From these two equations we can find B and C . The third side can be found from

$$a = \frac{b \sin A}{\sin B}$$

Example 4.9

Solve the triangle ABC if $a = \sqrt{3}$, $b = 1$ and $C = 30^\circ$.

Now,

Solution: From the formula

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

we get that

$$\begin{aligned} c &= \sqrt{a^2 + b^2 - 2ab \cos C} \\ &= \sqrt{(\sqrt{3})^2 + 1^2 - 2\sqrt{3} \cos 30^\circ} \\ &= \sqrt{3 + 1 - 2\sqrt{3} \cdot \frac{\sqrt{3}}{2}} = 1 \end{aligned}$$

$$\begin{aligned} \cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{1^2 + 1^2 - (\sqrt{3})^2}{2 \times 1 \times 1} = \frac{-1}{2} \end{aligned}$$

Therefore $A = 120^\circ$. Also

$$B = 180^\circ - (A + C) = 180^\circ - (120^\circ + 30^\circ) = 30^\circ$$

4.2.4 Solution of a Triangle when Two Sides and an Angle Opposite to One of Them are Given

Let ABC be a triangle in which the sides a and b and the angle A are given. From the formula

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

we get that

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Therefore

$$\begin{aligned} c^2 - 2bc \cos A + b^2 \cos^2 A &= a^2 - b^2 + b^2 \cos^2 A \\ &= a^2 - b^2 \sin^2 A \end{aligned}$$

which gives

$$\begin{aligned} (c - b \cos A)^2 &= a^2 - b^2 \sin^2 A \\ c - b \cos A &= \pm \sqrt{a^2 - b^2 \sin^2 A} \\ c &= b \cos A \pm \sqrt{a^2 - b^2 \sin^2 A} \end{aligned}$$

This is an equation to find the value of c when a , b and A are given. Here we have to distinguish the following four cases:

Case 1: If $a < b \sin A$, then $a^2 - b^2 \sin^2 A < 0$ and hence $\sqrt{a^2 - b^2 \sin^2 A}$ is not a real number. Therefore the above equation does not determine the value of c .

Case 2: If $a = b \sin A$, then there is only one value for c , namely $b \cos A$, and in this case the triangle is a right-angled triangle, B being the right angle since

$$\frac{\sin B}{b} = \frac{\sin A}{a} = \frac{1}{b}$$

and hence $\sin B = 1$.

Case 3: If $a > b \sin A$, then there are two values for c , that is,

$$c = b \cos A + \sqrt{a^2 - b^2 \sin^2 A}$$

or

$$c = b \cos A - \sqrt{a^2 - b^2 \sin^2 A}$$

However, c must be positive and hence the above two distinct values of c are admissible only when

$$b \cos A - \sqrt{a^2 - b^2 \sin^2 A} > 0$$

That is

$$\sqrt{a^2 - b^2 \sin^2 A} < b \cos A$$

$$a^2 - b^2 \sin^2 A < b^2 \cos^2 A$$

$$a^2 < b^2 \sin^2 A + b^2 \cos^2 A = b^2$$

which implies $a < b$. Thus, when $a > b \sin A$, there is unique triangle if $a \geq b$ and there are two triangles if $b \sin A < a < b$.

Case 4: If A is an obtuse angle, then $\cos A < 0$ and hence

$$b \cos A - \sqrt{a^2 - b^2 \sin^2 A} < 0$$

and therefore the value of c must be $b \cos A + \sqrt{a^2 - b^2 \sin^2 A}$. Moreover this too must be positive, that is,

$$\sqrt{a^2 - b^2 \sin^2 A} > -b \cos A$$

$$a^2 - b^2 \sin^2 A > b^2 \cos^2 A$$

$$a^2 > b^2 \sin^2 A + b^2 \cos^2 A = b^2$$

which implies $a > b$. Thus, when A is obtuse, there is no triangle if $a \leq b$ and there is only one triangle if $a > b$.

Example 4.10

Solve the triangle ABC , given that $a = 100$, $c = 100\sqrt{2}$ and $A = 30^\circ$.

Solution: As in Section 4.2.4, we have

$$b = c \cos A \pm \sqrt{a^2 - c^2 \sin^2 A}$$

In this triangle we have

$$c^2 \sin^2 A = (100\sqrt{2})^2 \left(\frac{1}{2}\right)^2 = 5000 \quad (\because \sin 30^\circ = 1/2)$$

Therefore

$$a^2 = (100)^2 > c^2 \sin^2 A$$

and $\sqrt{a^2 - c^2 \sin^2 A}$ is a real number. Here

$$c \sin A = 100\sqrt{2} = \frac{1}{2} = \frac{100}{\sqrt{2}}$$

Therefore

$$c \sin A < a < c$$

From the discussion made in Case 3 above, there are two triangles with the given elements. First, we have

$$b = c \cos A \pm \sqrt{a^2 - c^2 \sin^2 A}$$

$$= 100\sqrt{2} \times \frac{\sqrt{3}}{2} \pm \sqrt{(100)^2 - (100\sqrt{2})^2 \left(\frac{1}{2}\right)^2}$$

$$= 100 \frac{\sqrt{3}}{\sqrt{2}} \pm \sqrt{5000}$$

$$= 100 \frac{\sqrt{3}}{\sqrt{2}} \pm 50\sqrt{2}$$

$$= 50\sqrt{2}(\sqrt{3} \pm 1)$$

Now

$$\begin{aligned} \cos B &= \frac{c^2 + a^2 - b^2}{2ca} \\ &= \frac{(100\sqrt{2})^2 + (100)^2 - [50\sqrt{2}(\sqrt{3} \pm 1)]^2}{2 \times 100\sqrt{2} \times 100} \\ &= \frac{1 \pm \sqrt{3}}{2\sqrt{2}} \end{aligned}$$

Now two cases arise:

$$(1) \cos B = \frac{1+\sqrt{3}}{2\sqrt{2}} \Rightarrow B = 15^\circ \text{ and } C = 135^\circ$$

$$(2) \cos B = \frac{1-\sqrt{3}}{2\sqrt{2}} \Rightarrow B = 105^\circ \text{ and } C = 45^\circ$$

Thus the two solutions of the triangle ABC are:

$$(1) a_1 = 100, b_1 = 50\sqrt{2}(\sqrt{3}-1), c_1 = 100\sqrt{2}, A_1 = 30^\circ, B_1 = 15^\circ \text{ and } C_1 = 135^\circ$$

$$(2) a_2 = 100, b_2 = 50\sqrt{2}(\sqrt{3}+1), c_2 = 100\sqrt{2}, A_2 = 30^\circ, B_2 = 105^\circ \text{ and } C_2 = 45^\circ$$

Example 4.11

Is a triangle solvable if $a = 5$, $b = 7$ and $\sin A = 3/4$?

Solution: From the sine formula, we have

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

Therefore

$$\sin B = \frac{b}{a} \sin A = \frac{7}{5} \cdot \frac{3}{4} = \frac{21}{20} > 1$$

This is impossible, since $\sin B < 1$ for all real numbers B . Therefore, there is no triangle with the given values for a , b and $\sin A$.

4.2.5 Solution of a Triangle when One Side and Two Angles are Given

Let ABC be a triangle in which side a and angles B and C are given. Then $A = 180^\circ - (B + C)$. Therefore, all the angles are known. Also, from the sine formulae, we have

$$b = \frac{\sin B}{\sin A} a \quad \text{and} \quad c = \frac{\sin C}{\sin A} a$$

4.2.6 Solution of a Triangle when All the Angles are Given

There is no unique solution in this case. We can only say that the ratios of the sides are determined from the sine formulae. That is,

$$\frac{a}{b} = \frac{\sin A}{\sin B}, \quad \frac{b}{c} = \frac{\sin B}{\sin C} \quad \text{and} \quad \frac{c}{a} = \frac{\sin C}{\sin A}$$

Example 4.12

In a triangle ABC , $\cos A = 17/22$ and $\cos C = 1/14$. Find the ratio $a:b:c$.

Solution: We have

$$\cos A = \frac{17}{22} \Rightarrow \sin A = \sqrt{1 - \frac{17^2}{22^2}} = \frac{\sqrt{195}}{22}$$

$$\text{and} \quad \cos C = \frac{1}{14} \Rightarrow \sin C = \sqrt{1 - \frac{1}{14^2}} = \frac{\sqrt{195}}{14}$$

Also,

$$\begin{aligned} \sin B &= \sin[\pi - (A + C)] \\ &= \sin(A + C) \\ &= \sin A \cos C + \cos A \sin C \end{aligned}$$

$$\begin{aligned} &= \frac{\sqrt{195}}{22} \cdot \frac{1}{14} + \frac{17}{22} \cdot \frac{\sqrt{195}}{14} \\ &= \frac{\sqrt{195} \times 18}{22 \times 14} = \frac{\sqrt{195} \times 9}{11 \times 14} \end{aligned}$$

Therefore

$$\begin{aligned} \frac{a}{b} &= \frac{\sin A}{\sin B} = \frac{\sqrt{195}}{22} \times \frac{11 \times 14}{\sqrt{195} \times 9} = \frac{7}{9} \\ \frac{b}{c} &= \frac{\sin B}{\sin C} = \frac{\sqrt{195} \times 9}{11 \times 14} \times \frac{14}{\sqrt{195}} = \frac{9}{11} \end{aligned}$$

Thus $a:b:c = 7:9:11$.

Example 4.13

It is given that the base angles of a triangle are $22\frac{1}{2}^\circ$ and $112\frac{1}{2}^\circ$. Prove that the base of the triangle is twice the height of the triangle.

Solution: Let ABC be the triangle such that $A = 22\frac{1}{2}^\circ$ and $B = 112\frac{1}{2}^\circ$ (Figure 4.15). Then

$$C = 180^\circ - \left(22\frac{1}{2}^\circ + 112\frac{1}{2}^\circ\right) = 45^\circ$$

Also

$$\angle DBC = 180^\circ - 112\frac{1}{2}^\circ = 67\frac{1}{2}^\circ$$

$$\angle BCD = 90^\circ - 67\frac{1}{2}^\circ = 22\frac{1}{2}^\circ$$

where CD is the perpendicular to AB through C .

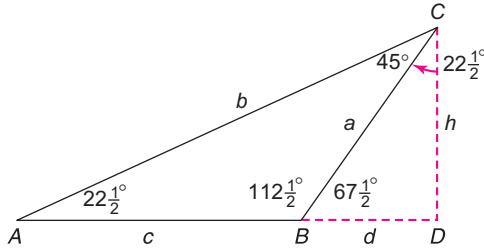


FIGURE 4.15 Example 4.13.

From the triangle ADC ,

$$\tan 22\frac{1}{2}^\circ = \frac{h}{c+d}$$

where $d = BD$. Also, from the triangle BDC ,

$$\tan 22\frac{1}{2}^\circ = \tan \angle BCD = \frac{d}{h}$$

But

$$\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1 \left(1 = \tan 45^\circ = \frac{2 \tan 22\frac{1}{2}^\circ}{1 - \tan^2 22\frac{1}{2}^\circ} \right)$$

Therefore

$$\begin{aligned} \frac{h}{c+d} &= \frac{d}{h} = \sqrt{2} - 1 \\ \Rightarrow d &= h(\sqrt{2} - 1) \end{aligned} \quad (4.2)$$

Substituting the value of d in Eq. (4.2) we get

$$\begin{aligned} \frac{h}{c+h(\sqrt{2}-1)} &= \sqrt{2}-1 \\ h &= c(\sqrt{2}-1) + h(\sqrt{2}-1)^2 \\ c(\sqrt{2}-1) &= h[1 - (\sqrt{2}-1)^2] \\ c &= \frac{1}{\sqrt{2}-1} [1 - (2+1-2\sqrt{2})]h \\ &= \frac{1}{\sqrt{2}-1} \cdot (-2+2\sqrt{2})h \\ &= \frac{2(\sqrt{2}-1)h}{\sqrt{2}-1} = 2h \end{aligned}$$

Thus the base $AB = c = 2$ (height).

Example 4.14

If in a triangle ABC , it is given that the angles are in the ratio $1:2:7$, then what is the ratio of the greatest side to the smallest side?

Solution: Since $A + B + C = 180^\circ$ and $A:B:C = 1:2:7$, we get that $A = 18^\circ$, $B = 36^\circ$ and $C = 126^\circ$. Therefore a is the smallest side and c is the greatest side. Now,

$$\frac{c}{a} = \frac{\sin C}{\sin A} = \frac{\sin 126^\circ}{\sin 18^\circ} = \frac{\sin 54^\circ}{\sin 18^\circ}$$

Recall that

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

and

$$\sin 54^\circ = \cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

Therefore

$$\frac{c}{a} = \frac{\sin 54^\circ}{\sin 18^\circ} = \frac{\sqrt{5}+1}{\sqrt{5}-1}$$

Thus, the required ratio is $\sqrt{5}+1:\sqrt{5}-1$.

4.3 | Properties of a Triangle

In this section we discuss various properties of triangles, mainly related to the circumcircle, incircle and excircle, using the formulae we have derived in the previous sections.

THEOREM 4.14
(AREA OF A TRIANGLE)

The area of a triangle is given by

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} bc \sin A \\ &= \frac{1}{2} ca \sin B \\ &= \frac{1}{2} ab \sin C \\ &= \sqrt{s(s-a)(s-b)(s-c)}\end{aligned}$$

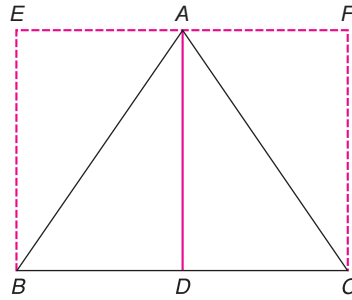


FIGURE 4.16 Theorem 4.14.

PROOF Let ABC be a triangle. Draw a perpendicular to BC through A to meet BC at D . Draw a parallel to BC (say EF) passing through A and to EF draw perpendicular BE and CF (Figure 4.16). Let Δ be the area of the triangle ABC . Then, we have

$$\begin{aligned}\Delta &= \frac{1}{2} \text{ area of the rectangle } BCFE \\ &= \frac{1}{2} BC \cdot CF \\ &= \frac{1}{2} a \cdot AD \\ &= \frac{1}{2} a \cdot (AB \cdot \sin B) \\ &= \frac{1}{2} ac \sin B\end{aligned}$$

Similarly

$$\Delta = \frac{1}{2} bc \sin A = \frac{1}{2} ab \sin C$$

Also, from Theorem 4.7 we have

$$\sin A = \frac{2}{bc} \sqrt{s(s-a)(s-b)(s-c)}$$

Therefore

$$\Delta = \frac{1}{2} bc \sin A = \sqrt{s(s-a)(s-b)(s-c)}$$

Example 4.15

In a triangle ABC , $a = 18$, $b = 24$ and $c = 30$. Find the area of the triangle.

Solution: We have

$$2s = a + b + c = 18 + 24 + 30 = 72$$

Therefore $s = 36$. Now

$$s - a = 18, s - b = 12, s - c = 6$$

Therefore the area of ABC is

$$\begin{aligned} \sqrt{s(s-a)(s-b)(s-c)} &= \sqrt{36 \times 18 \times 12 \times 6} \\ &= 216 \text{ sq. units} \end{aligned}$$

Example 4.16

In a triangle ABC , $B = 45^\circ$, $C = 60^\circ$ and $a = 2(\sqrt{3} + 1)$. Find the area of the triangle.

Solution: Since $A + B + C = 180^\circ$, we get $A = 75^\circ$. From

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

we have

$$\begin{aligned} b &= \frac{a \sin B}{\sin A} = \frac{2(\sqrt{3} + 1) \sin 45^\circ}{\sin 75^\circ} \\ &= \frac{2(\sqrt{3} + 1)(1/\sqrt{2})}{(\sqrt{3} + 1)/2\sqrt{2}} \left(\because \sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} \right) \\ &= 4 \end{aligned}$$

The area of triangle ABC is

$$\begin{aligned} \frac{1}{2} ab \sin C &= \frac{1}{2} 2(\sqrt{3} + 1) \times 4 \times \sin 60^\circ \\ &= 4(\sqrt{3} + 1) \times \frac{\sqrt{3}}{2} \\ &= 2\sqrt{3}(\sqrt{3} + 1) \\ &= 6 + 2\sqrt{3} \text{ sq. units} \end{aligned}$$

DEFINITION 4.1 The circle passing through the three vertices of a triangle is called the *circumcircle* and its centre and radius are called *circumcentre* and the *circumradius*, respectively.

We know that the perpendicular bisectors of the sides of a triangle are concurrent and the point of concurrence is equidistant from the vertices and hence it is the circumcentre.

THEOREM 4.15 Let R be the circumradius of a triangle ABC . Then $a = 2R \sin A$, $b = 2R \sin B$ and $c = 2R \sin C$. That is

$$2R = \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

PROOF Let S be the circumcentre of the triangle ABC . Consider the angle A . We shall distinguish three cases.

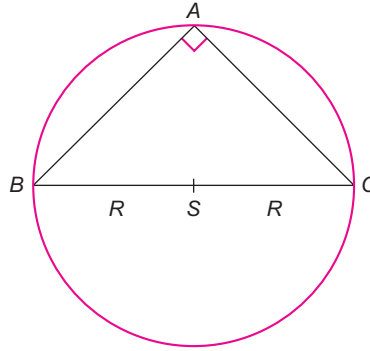


FIGURE 4.17 Case I.

Case I: Suppose that $A = 90^\circ$. Then S is the mid-point of BC and BC is a diameter of the circum-circle (Figure 4.17). Now,

$$a = BC = 2R = 2R \cdot \sin 90^\circ$$

Therefore

$$\frac{a}{\sin A} = 2R$$

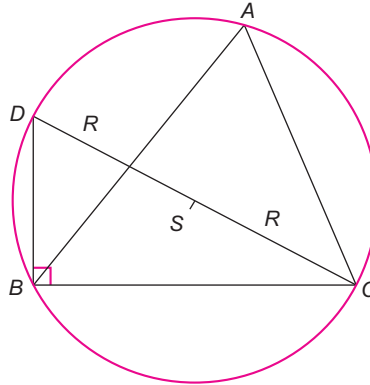


FIGURE 4.18 Case II.

Case II: Suppose that $A < 90^\circ$. Let CD be a diameter of the circumcircle, so that S is its mid-point and $CS = SD = R$. Join BD . Then $\angle DBC = 90^\circ$, since it is an angle in the semicircle (Figure 4.18). Also, $\angle BDC = \angle BAC$ (since they are angles in the same segment made by BC). Now,

$$\sin A = \sin \angle BDC = \frac{BC}{CD} = \frac{a}{2R}$$

and hence $a = 2R \sin A$.

Case III: Suppose that $A > 90^\circ$. Draw diameter CD of the circumcircle through C . Then $CS = SD = R$. Join B and D . Then $\angle DBC = 90^\circ$, since it is an angle made by a semicircle (Figure 4.19).

In the cyclic quadrilateral $BACD$, we have

$$\angle BDC + \angle BAC = 180^\circ$$

Therefore in the right-angled BDC ,

$$\sin \angle BDC = \frac{BC}{CD} = \frac{a}{2R}$$

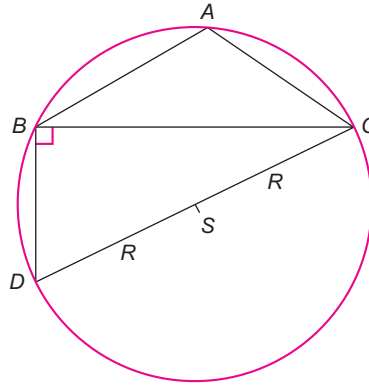


FIGURE 4.19 Case III.

Therefore

$$\sin A = \sin(180^\circ - A) = \sin \angle BDC = \frac{a}{2R}$$

Hence

$$a = 2R \sin A$$

Similarly we can prove that $b = 2R \sin B$ and $c = 2R \sin C$. ■

COROLLARY 4.2 Let Δ be the area of a triangle ABC and R be its circumradius. Then

$$R = \frac{abc}{4\Delta} = \frac{abc}{4\sqrt{s(s-a)(s-b)(s-c)}}$$

PROOF From Theorem 4.7, we have

$$\sin A = \frac{2}{bc} \sqrt{s(s-a)(s-b)(s-c)} = \frac{2\Delta}{bc}$$

Therefore

$$\frac{a}{2R} = \frac{2\Delta}{bc} \quad \text{or} \quad R = \frac{abc}{4\Delta} \quad \blacksquare$$

DEFINITION 4.2 Incircle, Incentre, Inradius The circle that touches the three sides of a triangle internally is called the *incircle* or *inscribed circle* of the triangle. Its centre and radius are called the *incentre* and *inradius*, respectively, and are denoted by I and r , respectively.

It is known from Geometry that the internal bisectors of the angles of a triangle are concurrent and that the point of concurrence is equidistant from the sides and hence there is a circle touching all the three sides internally. Therefore, the point of concurrence of the internal bisectors of the angles is precisely the incentre I and the distance from I to any side is the inradius r . In the following, we express the inradius in terms of the sides of the triangle.

THEOREM 4.16 Let Δ be the area of a triangle ABC . Then the inradius r is equal to Δ/s ; that is,

$$r = \frac{\Delta}{s} = \frac{2\sqrt{s(s-a)(s-b)(s-c)}}{a+b+c}$$

PROOF

Draw internal bisectors of the angles B and C to meet at I . Then I is the incentre. Join A and I . Then AI becomes the bisector of the angle A . Draw perpendiculars ID , IE and IF to the sides BC , CA and AB , respectively (Figure 4.20). Then, since the sides are tangents to the incircle with centre I , we have

$$ID = IE = IF = r, \text{ the inradius}$$

Then

$$\text{Area of triangle } IBC = \frac{1}{2} BC \cdot ID = \frac{1}{2} ar$$

Similarly,

$$\text{Area of triangle } ICA = \frac{1}{2} br$$

and triangle

$$\text{Area of triangle } IAB = \frac{1}{2} cr$$

Hence

$$\text{Area of the triangle } ABC = \text{Sum of the areas of the triangles } IBC, ICA \text{ and } IAB$$

$$= \frac{1}{2} ar + \frac{1}{2} br + \frac{1}{2} cr$$

$$= \frac{a + b + c}{2} \cdot r$$

$$= sr$$

Thus

$$\Delta = sr \quad \text{or} \quad r = \frac{\Delta}{s} = \frac{2\sqrt{s(s-a)(s-b)(s-c)}}{a+b+c}$$

since $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ and $2s = a + b + c$.

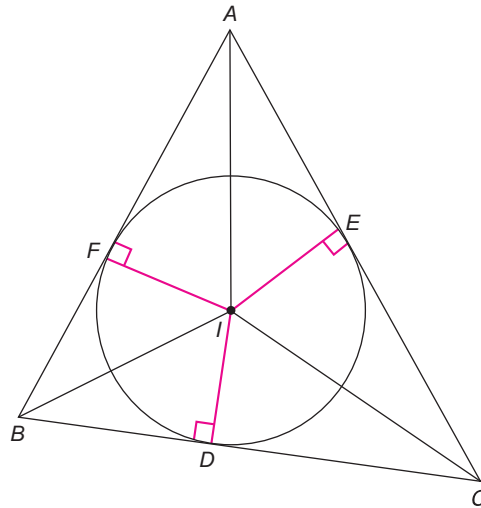


FIGURE 4.20 Theorem 4.16.

THEOREM 4.17 The inradius r of a triangle ABC can be expressed as

$$r = (s - a) \tan \frac{A}{2} = (s - b) \tan \frac{B}{2} = (s - c) \tan \frac{C}{2}$$

PROOF Note that in Figure 4.20 we have $\angle IBD = \angle IBF$ (since BI is the bisector of the angle B) and $\angle IDB = 90^\circ = \angle IFD$. Therefore the two triangles IDB and IFB are equal in all respects and, in particular,

$$BD = BF \quad \text{and} \quad \text{hence } 2BD = BD + BF$$

Similarly

$$AE = AF \quad \text{and} \quad \text{hence } 2AE = AE + AF$$

$$CE = CD \quad \text{and} \quad \text{hence } 2CE = CE + CD$$

By adding these three equations, we get

$$\begin{aligned} 2BD + 2AE + 2CE &= BD + BF + AE + AF + CE + CD \\ &= (BD + CD) + (BF + AF) + (AE + CE) \\ &= BC + AB + AC \\ &= a + b + c = 2s \end{aligned}$$

Therefore

$$\begin{aligned} 2BD + 2(AE + CE) &= 2s \\ 2BD + 2AC &= 2s \\ BD &= s - AC = s - b = BF \end{aligned}$$

Similarly

$$CE = s - c = CD \quad \text{and} \quad AF = s - a = AE$$

Therefore

$$\begin{aligned} r &= ID = BD \tan \angle IBD = BD \tan \frac{B}{2} = (s - b) \tan \frac{B}{2} \\ r &= IE = CE \tan \angle ICE = CE \tan \frac{C}{2} = (s - c) \tan \frac{C}{2} \\ r &= IF = AF \tan \angle IAF = AF \tan \frac{A}{2} = (s - a) \tan \frac{A}{2} \end{aligned}$$

Thus

$$r = (s - a) \tan \frac{A}{2} = (s - b) \tan \frac{B}{2} = (s - c) \tan \frac{C}{2}$$

THEOREM 4.18 Let ABC be a triangle and r its inradius. Then

$$r = a \frac{\sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = b \frac{\sin \frac{C}{2} \sin \frac{A}{2}}{\cos \frac{B}{2}} = c \frac{\sin \frac{A}{2} \sin \frac{B}{2}}{\cos \frac{C}{2}}$$

PROOF Consider Figure 4.20. We have

$$\begin{aligned} a &= BD + CD \\ &= ID \cot \angle IBD + ID \cot \angle ICD \end{aligned}$$

$$\begin{aligned}
 &= r \cot \frac{B}{2} + r \cot \frac{C}{2} \\
 &= r \left(\frac{\cos(B/2)}{\sin(B/2)} \right) + r \left(\frac{\cos(C/2)}{\sin(C/2)} \right)
 \end{aligned}$$

Therefore

$$\begin{aligned}
 a \sin \frac{B}{2} \sin \frac{C}{2} &= r \left(\sin \frac{C}{2} \cos \frac{B}{2} + \cos \frac{C}{2} \sin \frac{B}{2} \right) \\
 &= r \sin \left(\frac{B}{2} + \frac{C}{2} \right) \\
 &= r \sin \left(90^\circ - \frac{A}{2} \right) \quad \left(\text{since } \frac{A}{2} + \frac{B}{2} + \frac{C}{2} = 90^\circ \right) \\
 &= r \cos \frac{A}{2}
 \end{aligned}$$

Hence

$$r = a \frac{\sin(B/2) \sin(C/2)}{\cos(A/2)}$$

Similarly, the other formulae can be proved. ■

COROLLARY 4.3

If R is the circumradius and r is the inradius of a triangle ABC , then

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

PROOF

Since

$$a = 2R \sin A = 4R \sin \frac{A}{2} \cos \frac{A}{2}$$

we have (by Theorem 4.18)

$$\begin{aligned}
 r &= \left(4R \sin \frac{A}{2} \cos \frac{A}{2} \right) \left(\frac{\sin(B/2) \sin(C/2)}{\cos(A/2)} \right) \\
 &= 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}
 \end{aligned}$$
■

Example 4.17

Find the circumradius and inradius of a triangle whose sides are 18 cm, 24 cm and 30 cm.

Solution: Let ABC be the triangle. Let $a = 18$ cm, $b = 24$ cm and $c = 30$ cm. Then

$$s = \frac{a+b+c}{2} = \frac{18+24+30}{2} = 36$$

The area of the triangle is

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = 216 \text{ sq. cm}$$

From Corollary 4.2, the circumradius R is given by

$$R = \frac{abc}{4\Delta} = \frac{18 \times 24 \times 30}{4 \times 216} = 15 \text{ cm}$$

From Theorem 4.16, the inradius r is given by

$$r = \frac{\Delta}{s} = \frac{216}{36} = 6 \text{ cm}$$

DEFINITION 4.3 Excircles, Excentres, Exradius Let ABC be a triangle. The circle that touches BC and the extended sides AB and AC (as shown Figure 4.21) is called the *excircle* or *escribed circle opposite to the vertex A*. Its centre and radius are called the *excentre* or *escentre* and *exradius* or *esradius* opposite to A , respectively. These are denoted by I_1 and r_1 , respectively. Similarly, we can define the escentre and esradius opposite to B and C which are denoted by I_2, r_2 and I_3, r_3 , respectively.

The internal bisector of the angle A and the bisectors of the external angles B and C are known to be concurrent and the point of concurrence is equidistant from the three sides. This point of concurrence is precisely the excentre I_1 and the distance from I_1 to any side is the esradius r_1 . In the following we derive formulae to find the exradius r_1, r_2 and r_3 in terms of the sides of the triangle.

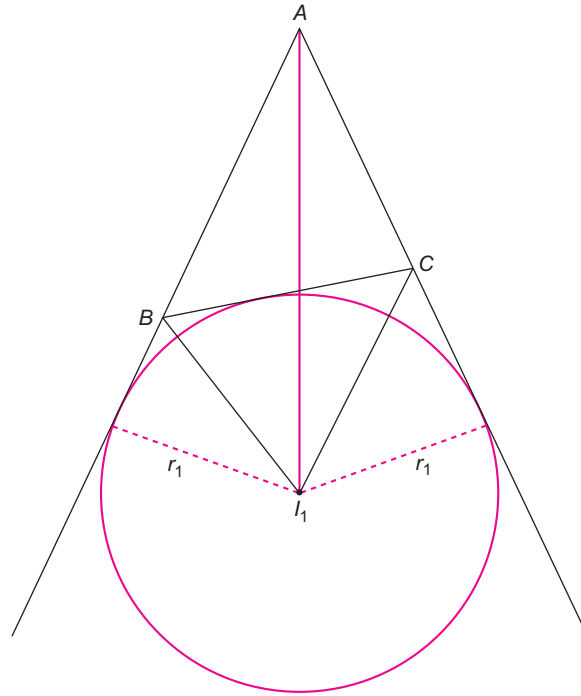


FIGURE 4.21 Figure to depict excircles, excentres, exradius.

THEOREM 4.19 Let ABC be a triangle with area Δ and r_1, r_2 and r_3 be the radii of the escribed circles opposite to A, B and C respectively. Then

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b} \quad \text{and} \quad r_3 = \frac{\Delta}{s-c}$$

PROOF Produce AB and AC to L and M . Bisect the angles CBL and BCM by the lines BI_1 and CI_1 . These meet at I_1 and AI_1 will be the bisector of the angle A . Draw perpendiculars I_1D_1, I_1E_1 and I_1F_1 to the sides BC, AC and AB , respectively.

The two triangles I_1D_1B and I_1F_1B are equal in all respects (since BI_1 is the bisector of the angle $\angle D_1BF_1$ and $\angle BF_1I_1 = 90^\circ = \angle BD_1I_1$). Therefore $I_1F_1 = I_1D_1$. Similarly $I_1E_1 = I_1D_1$. Therefore, $I_1D_1 = I_1E_1 = I_1F_1$. Hence I_1 is the centre of the escribed circle opposite to A and $I_1D_1 = r_1$. Now, the area of the quadrilateral ABI_1C is equal to the sum of the areas of the triangles ABI_1 and ACI_1 and is also equal to the areas of the triangles ABC and BI_1C . Therefore

$$\Delta ABI_1 + \Delta ACI_1 = \Delta ABC + \Delta BI_1C$$

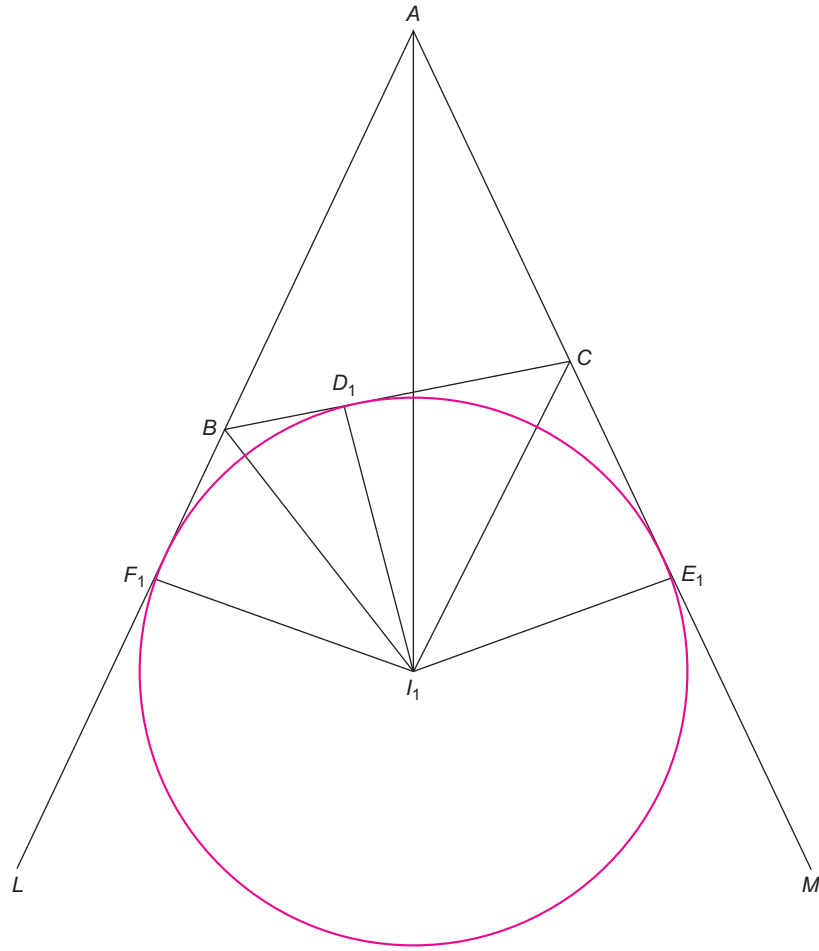


FIGURE 4.22 Theorem 4.19.

where $\Delta AB I_1$ denotes the area of the triangle $AB I_1$, $\Delta AC I_1$, ΔABC and $\Delta BI_1 C$ denote the areas of the triangles $AC I_1$, ABC and $BI_1 C$, respectively. Therefore

$$\frac{1}{2} \cdot I_1 F_1 \cdot AB + \frac{1}{2} I_1 E_1 \cdot CA = \Delta + \frac{1}{2} I_1 D_1 \cdot BC$$

where $\Delta = \Delta ABC$ = the area of ABC . Hence

$$\begin{aligned} \frac{1}{2} r_1 c + \frac{1}{2} r_1 b &= \Delta + \frac{1}{2} r_1 a \\ \Delta &= \frac{r_1(c+b-a)}{2} = r_1 \left(\frac{a+b+c}{2} - a \right) = r_1(s-a) \end{aligned}$$

This implies

$$r_1 = \frac{\Delta}{s-a}$$

Similarly we can prove that

$$r_2 = \frac{\Delta}{s-b} \quad \text{and} \quad r_3 = \frac{\Delta}{s-c}$$

In the following, the exradii r_1 , r_2 and r_3 are expressed in terms of s and the tangents of the half angles of the triangle.

THEOREM 4.20 Let r_1 , r_2 and r_3 be the radii of the escribed circles opposite to A , B and C , respectively, for a triangle ABC . Then

$$r_1 = s \tan \frac{A}{2}, \quad r_2 = s \tan \frac{B}{2} \quad \text{and} \quad r_3 = s \tan \frac{C}{2}$$

PROOF Consider Figure 4.22. Since AE_1 and AF_1 are tangents to the circle from the point A , we get that $AE_1 = AF_1$ and, for a similar reason, we have

$$BF_1 = BD_1 \quad \text{and} \quad CE_1 = CD_1$$

Now,

$$\begin{aligned} 2AE_1 &= AE_1 + AF_1 \\ &= AC + CE_1 + AB + BF_1 \\ &= AC + CD_1 + AB + BD_1 \\ &= AB + AC + CD_1 + BD_1 \\ &= AB + AC + BC \\ &= a + b + c = 2s \end{aligned}$$

Therefore

$$AE_1 = s = AF_1$$

From the right-angled triangle AI_1E_1 , we have

$$\tan \frac{A}{2} = \frac{I_1E_1}{AE_1} = \frac{r_1}{s}$$

Thus

$$r_1 = s \cdot \tan \frac{A}{2}$$

Similarly

$$r_2 = s \tan \frac{B}{2} \quad \text{and} \quad r_3 = s \tan \frac{C}{2}$$

COROLLARY 4.4 Let r_1 , r_2 and r_3 be the radii of the escribed circles opposite to A , B and C , respectively, for a triangle ABC . Then

$$r_1 = (s - c) \cot \frac{B}{2}$$

$$r_2 = (s - a) \cot \frac{C}{2}$$

and

$$r_3 = (s - b) \cot \frac{A}{2}$$

PROOF In Theorem 4.20 we have proved that $s = AF_1$ and hence

$$s - c = AF_1 - AB = BF_1$$

Now, in the right-angled triangle BF_1I_1 , we have

$$\cot \frac{B}{2} = \tan \left(90 - \frac{B}{2} \right) = \frac{I_1F_1}{BF_1} = \frac{r_1}{s-c}$$

Therefore

$$r_1 = (s-c) \cot \frac{B}{2}$$

Similarly

$$r_2 = (s-a) \cot \frac{C}{2} \quad \text{and} \quad r_3 = (s-b) \cot \frac{A}{2}$$

Example 4.18

In a triangle ABC , $a = 13$, $b = 14$ and $c = 15$. Find the area and the exradii are and hence the exradii.

Solution: We have

$$2s = a + b + c = 13 + 14 + 15 = 42$$

Hence

$$s = 21, s-a = 8, s-b = 7 \quad \text{and} \quad s-c = 6$$

Therefore the area is

$$\begin{aligned} \Delta &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{21 \times 8 \times 7 \times 6} = 84 \text{ sq. units} \end{aligned}$$

$$r_1 = \frac{\Delta}{s-a} = \frac{84}{8} = 10.5$$

$$r_2 = \frac{\Delta}{s-b} = \frac{84}{7} = 12$$

$$r_3 = \frac{\Delta}{s-c} = \frac{84}{6} = 14$$



QUICK LOOK 1

In any triangle ABC ,

$$rr_1r_2r_3 = \frac{\Delta}{s} \cdot \frac{\Delta}{s-a} \cdot \frac{\Delta}{s-b} \cdot \frac{\Delta}{s-c} = \frac{\Delta^4}{\Delta^2} = \Delta^2$$

Another formula for the esradius r_1 is obtained in the following in terms of the side a and the angles B and C .

THEOREM 4.21 In any triangle ABC ,

$$r_1 = \frac{a \cos(B/2) \cdot \cos(C/2)}{\cos(A/2)}$$

$$r_2 = \frac{b \cos(C/2) \cdot \cos(A/2)}{\cos(B/2)}$$

and

$$r_3 = \frac{c \cos(A/2) \cdot \cos(B/2)}{\cos(C/2)}$$

PROOF Again consider Figure 4.22. Since I_1C bisects the angle $\angle BCE_1$, we have

$$\angle I_1CD_1 = \frac{1}{2}(180^\circ - C) = 90^\circ - \frac{C}{2}$$

Similarly

$$\angle I_1BD_1 = 90^\circ - \frac{B}{2}$$

Now,

$$\begin{aligned} a &= BC \\ &= BD_1 + D_1C \\ &= I_1D_1 \cdot \cot \angle I_1BD_1 + I_1D_1 \cdot \cot \angle I_1CD_1 \\ &= r_1 \left[\cot \left(90^\circ - \frac{B}{2} \right) + \cot \left(90^\circ - \frac{C}{2} \right) \right] \\ &= r_1 \left(\tan \frac{B}{2} + \tan \frac{C}{2} \right) \\ &= r_1 \left(\frac{\sin(B/2)}{\cos(B/2)} + \frac{\sin(C/2)}{\cos(C/2)} \right) \end{aligned}$$

Therefore

$$\begin{aligned} a \cos \frac{B}{2} \cos \frac{C}{2} &= r_1 \left(\sin \frac{B}{2} \cos \frac{C}{2} + \cos \frac{B}{2} \sin \frac{C}{2} \right) \\ &= r_1 \left[\sin \left(\frac{B}{2} + \frac{C}{2} \right) \right] \\ &= r_1 \sin \left(90^\circ - \frac{A}{2} \right) \\ &= r_1 \cos \frac{A}{2} \end{aligned}$$

Hence

$$r_1 = \frac{a \cos(B/2) \cdot \cos(C/2)}{\cos(A/2)}$$

Similarly,

$$r_2 = \frac{b \cos(C/2) \cdot \cos(A/2)}{\cos(B/2)} \quad \text{and} \quad r_3 = \frac{c \cos(A/2) \cdot \cos(B/2)}{\cos(C/2)}$$

COROLLARY 4.5 In any triangle ABC ,

$$r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$r_2 = 4R \sin \frac{B}{2} \cos \frac{C}{2} \cos \frac{A}{2}$$

and

$$r_3 = 4R \sin \frac{C}{2} \cos \frac{A}{2} \cos \frac{B}{2}$$

PROOF This follows from Theorem 4.21 and from the facts that

$$a = 2R \sin A = 4R \sin \frac{A}{2} \cos \frac{A}{2}$$

$$b = 2R \sin B = 4R \sin \frac{B}{2} \cos \frac{B}{2}$$

and

$$c = 2R \sin C = 4R \sin \frac{C}{2} \cos \frac{C}{2}$$

Example 4.19

In any triangle ABC , prove that

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}$$

Solution: We have

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{s^2}{\Delta^2} + \frac{(s-a)^2}{\Delta^2} + \frac{(s-b)^2}{\Delta^2} + \frac{(s-c)^2}{\Delta^2}$$

$$\begin{aligned} &= \frac{1}{\Delta^2} (s^2 + s^2 + a^2 - 2as + s^2 + b^2 - 2bs + s^2 + c^2 - 2cs) \\ &= \frac{1}{\Delta^2} [4s^2 - 2s(a+b+c) + a^2 + b^2 + c^2] \\ &= \frac{1}{\Delta^2} (4s^2 - 2s \cdot 2s + a^2 + b^2 + c^2) \\ &= \frac{a^2 + b^2 + c^2}{\Delta^2} \end{aligned}$$

Example 4.20

In any triangle ABC , prove that

$$\frac{r_1}{bc} + \frac{r_2}{ca} + \frac{r_3}{ab} = \frac{1}{r} - \frac{1}{2R}$$

Solution: We have

$$\begin{aligned} \frac{r_1}{bc} + \frac{r_2}{ca} + \frac{r_3}{ab} &= \frac{1}{abc} (ar_1 + br_2 + cr_3) \\ &= \frac{1}{abc} \left(a \cdot s \tan \frac{A}{2} + b \cdot s \tan \frac{B}{2} + c \cdot s \tan \frac{C}{2} \right) \\ &= \frac{s}{abc} \left(2R \sin A \tan \frac{A}{2} + 2R \sin B \tan \frac{B}{2} + 2R \sin C \tan \frac{C}{2} \right) \\ &= \frac{s}{abc} \left(4R \sin \frac{A}{2} \cos \frac{A}{2} \tan \frac{A}{2} + 4R \sin \frac{B}{2} \cos \frac{B}{2} \tan \frac{B}{2} \right. \\ &\quad \left. + 4R \sin \frac{C}{2} \cos \frac{C}{2} \tan \frac{C}{2} \right) \end{aligned}$$

$$\begin{aligned} &= \frac{s}{abc} \left(4R \sin^2 \frac{A}{2} + 4R \sin^2 \frac{B}{2} + 4R \sin^2 \frac{C}{2} \right) \\ &= \frac{2Rs}{abc} (1 - \cos A + 1 - \cos B + 1 - \cos C) \\ &= \frac{s}{2\Delta} [3 - (\cos A + \cos B + \cos C)] \quad \left(\because R = \frac{abc}{4\Delta} \right) \\ &= \frac{1}{2r} \left[3 - \left(1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \right] \quad \left(\because r = \frac{\Delta}{s} \right) \\ &= \frac{1}{2r} \left[2 - \frac{4R \sin(A/2) \sin(B/2) \sin(C/2)}{R} \right] \\ &= \frac{1}{2r} \left(2 - \frac{r}{R} \right) = \frac{1}{r} - \frac{1}{2R} \end{aligned}$$

Example 4.21

In a triangle ABC , if $r_1 = 8$, $r_2 = 12$ and $r_3 = 24$, then find the lengths of the sides.

Solution: Let Δ be the area. Then

$$\frac{\Delta}{s-a} = r_1 = 8 \Rightarrow s-a = \frac{\Delta}{8}$$

$$\frac{\Delta}{s-b} = r_2 = 12 \Rightarrow s-b = \frac{\Delta}{12}$$

$$\frac{\Delta}{s-c} = r_3 = 24 \Rightarrow s-c = \frac{\Delta}{24}$$

Therefore

$$3s - (a + b + c) = (s - a) + (s - b) + (s - c)$$

$$= \Delta \left(\frac{1}{8} + \frac{1}{12} + \frac{1}{24} \right) = \frac{\Delta}{4}$$

This gives

$$3s - 2s = \frac{\Delta}{4}$$

$$s = \frac{\Delta}{4}$$

$$r = \frac{\Delta}{s} = 4$$

Since

$$\Delta^2 = rr_1r_2r_3 \quad (\text{see Quick Look 1})$$

$$= 4 \times 8 \times 12 \times 24$$

$$= (12 \times 8)^2$$

Therefore

$$\Delta = 96 \text{ sq. units}$$

Now

$$s = \frac{\Delta}{r} = \frac{96}{4} = 24$$

$$a = s - \frac{\Delta}{r_1} = 24 - \frac{96}{8} = 12$$

$$b = s - \frac{\Delta}{r_2} = 24 - \frac{96}{12} = 16$$

$$c = s - \frac{\Delta}{r_3} = 24 - \frac{96}{24} = 20$$

DEFINITION 4.4 Orthocentre In any triangle, the perpendiculars drawn from the vertices to the respective opposite sides are known to be *concurrent* and the point of concurrence is called the *orthocentre* of the triangle. Also, the triangle formed by the feet of these perpendiculars on the sides is called the *pedal triangle* of the triangle.

Let ABC be a triangle and AK , BL and CM be perpendiculars drawn from A , B and C to BC , CA and AB , respectively. Then the perpendiculars AK , BL and CM are concurrent at a point P . The point P is called the *orthocentre* and the triangle KLM is called the *pedal triangle* of the triangle ABC (Figure 4.23).

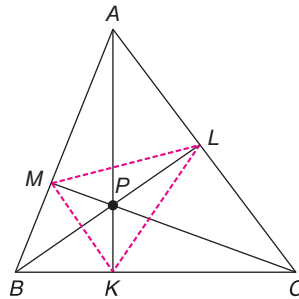


FIGURE 4.23 Orthocenter and pedal triangle.

THEOREM 4.22 In any triangle ABC , the distances of the orthocentre P from the vertices A , B and C are given by

$$AP = 2R \cos A, \quad BP = 2R \cos B \quad \text{and} \quad CP = 2R \cos C$$

where R is the circumradius of the triangle ABC . Also the distances of P from the sides BC , CA and AB are given by

$$PK = 2R \cos B \cos C$$

$$PL = 2R \cos C \cos A$$

$$PM = 2R \cos A \cos B$$

and

PROOF First observe that $\angle KAC = 90^\circ - C$ and hence $\angle BPK = \angle APL = C$ and $\angle PBK = 90^\circ - C$. Now, in the triangle APL ,

$$\cos \angle KAC = \frac{AL}{AP}$$

and hence

$$AP = \frac{AL}{\cos \angle KAC} = \frac{AB \cos A}{\cos(90^\circ - C)} = \frac{C}{\sin C} \cos A = 2R \cos A$$

Similarly

$$BP = 2R \cos B \quad \text{and} \quad CP = 2R \cos C$$

Also

$$\begin{aligned} PK &= KB \tan \angle PBK \\ &= KB \tan(90^\circ - C) \\ &= AB \cos B \cdot \cot C \\ &= AB \cos B \cdot \frac{\cos C}{\sin C} \\ &= \frac{c}{\sin C} \cos B \cos C \\ &= 2R \cos B \cos C \end{aligned}$$

Similarly

$$PL = 2R \cos C \cos A \quad \text{and} \quad PM = 2R \cos A \cos B$$



In the following, we derive formulae for the angles and sides of the pedal triangle of a triangle ABC in terms of the angles and sides of ABC .

THEOREM 4.23

Let ABC be a triangle and K, L and M be the feet of perpendiculars from A, B and C on the sides BC, CA and AB , respectively; that is, KLM is the pedal triangle of the triangle ABC (Figure 4.24). The angles and the sides of the pedal triangle are given by

$$\angle MKL = 180^\circ - 2A, \quad \angle KLM = 180^\circ - 2B, \quad \angle LMK = 180^\circ - 2C$$

$$LM = a \cos A, \quad MK = b \cos B \quad \text{and} \quad KL = c \cos C$$

PROOF

Since $\angle CKP = 90^\circ = \angle CLP$, the points K, C, L and P are concyclic and therefore

$$\angle PKL = \angle PCL = 90^\circ - A$$

(since $\angle CMA = 90^\circ$ in the triangle AMC).

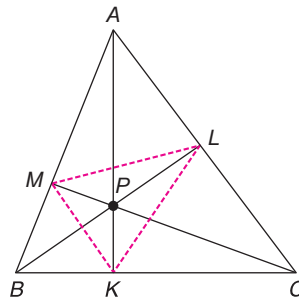


FIGURE 4.24 Theorem 4.23.

Similarly M, B, K and P are concyclic and hence

$$\angle MKP = \angle MBP = 90^\circ - A$$

Now,

$$\angle MKL = \angle MKP + \angle PKL = 90^\circ - A + 90^\circ - A = 180^\circ - 2A$$

Similarly

$$\angle KLM = 180^\circ - 2B \quad \text{and} \quad \angle LMK = 180^\circ - 2C$$

Also, from the triangle MLA , we have

$$\begin{aligned} \frac{LM}{\sin A} &= \frac{AL}{\sin \angle LMA} \\ &= \frac{AB \cos A}{\cos \angle LMP} \\ &= \frac{c \cos A}{\cos \angle LAP} \\ &= \frac{c \cos A}{\cos(90^\circ - C)} \\ &= \frac{c \cos A}{\sin C} \end{aligned}$$

Therefore

$$LM = \frac{c \sin A \cos A}{\sin C} = \left(\frac{c}{\sin C} \right) \sin A \cdot \cos A = a \cos A$$

Similarly, $MK = b \cos B$ and $KL = a \cos C$. ■

Note: In any triangle ABC , the angles of the pedal triangle are the supplements of twice the angles of the triangle. The sides of the pedal triangle are $a \cos A$, $b \cos B$ and $c \cos C$.

DEFINITION 4.5 Excentric Triangle Let ABC be a triangle and I_1, I_2 and I_3 be centres of the escribed circles which are opposite to A, B and C , respectively. Then the triangle $I_1I_2I_3$ is called the *excentric triangle* of ABC .

THEOREM 4.24 Let ABC be a triangle, I its incentre and $I_1I_2I_3$ be the excentric triangle of ABC . Then ABC is the pedal triangle of $I_1I_2I_3$ and I is the orthocentre of $I_1I_2I_3$ (Figure 4.25).

PROOF Let I_1, I_2 and I_3 be the centres of the escribed circles which are opposite to A, B and C , respectively. I is the centre of the incircle of ABC . Then IC bisects the angle $\angle ACB$ and I_1C bisects the angle $\angle BCM$. Therefore

$$\begin{aligned} \angle ICI_1 &= \angle ICB + \angle I_1CB \\ &= \frac{1}{2} \angle ACB + \frac{1}{2} \angle BCM \\ &= \frac{1}{2} (\angle ACB + \angle BCM) \\ &= \frac{1}{2} \times 180^\circ = 90^\circ \end{aligned}$$

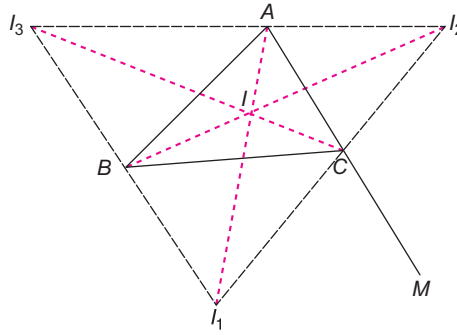


FIGURE 4.25 Theorem 4.24.

Similarly $\angle ICI_2 = 90^\circ$. Therefore

$$\angle I_1CI_2 = \angle ICI_1 + \angle ICI_2 = 90^\circ + 90^\circ = 180^\circ$$

and hence I_1CI_2 is a straight line to which IC is a perpendicular. Also, since IA and I_1A both bisect the angle $\angle BAC$, the three points I_1, I and A are collinear; that is, I_1IA is a straight line. Similarly, I_2IB and I_3IC are straight lines. Thus $I_1I_2I_3$ is a triangle such that A, B and C are the feet of the perpendiculars drawn from I_1, I_2 and I_3 to I_2I_3, I_3I_1 and I_1I_2 , respectively, and further I is the point of concurrence of these perpendiculars. Thus, I is the orthocenter of the triangle $I_1I_2I_3$ and ABC is the pedal triangle of $I_1I_2I_3$. ■

DEFINITION 4.6 Medians and Centroid Let ABC be a triangle and D, E and F be the mid-points of BC, CA and AB , respectively. The lines AD, BE and CF are called the *medians* of the triangle ABC . These three medians are concurrent and the point of concurrence is called the *centroid* of the triangle and is usually denoted by G (Figure 4.26).

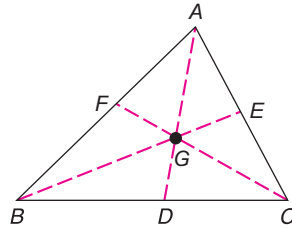


FIGURE 4.26 Medians and centroid.

The concurrency of the medians is proved in Geometry at secondary school level. Also, it is known that

$$AG = \frac{2}{3}AD, \quad BG = \frac{2}{3}BE \quad \text{and} \quad CG = \frac{2}{3}CF$$

This is to say that the centroid G is the point of trisection of each of the medians. In the following, we derive formulae to express the lengths of the medians in terms of the elements of the given triangle.

THEOREM 4.25 Let ABC be a triangle and D, E and F be the mid-points of BC, CA and AB , respectively. Then the lengths of the medians are given by

$$AD = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2} = \frac{1}{2}\sqrt{b^2 + c^2 + 2bc \cos A}$$

$$BE = \frac{1}{2}\sqrt{2c^2 + 2a^2 - b^2} = \frac{1}{2}\sqrt{c^2 + a^2 + 2ca \cos B}$$

and

$$CF = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2} = \frac{1}{2}\sqrt{a^2 + b^2 + 2ab \cos C}$$

PROOF From Theorem 4.2, we have

$$\begin{aligned} AD^2 &= AC^2 + CD^2 - 2AC \cdot CD \cos C \\ &= b^2 + \left(\frac{a}{2}\right)^2 - AC \cdot BC \cos C \\ &= b^2 + \frac{a^2}{4} - ab \cos C \end{aligned}$$

Therefore

$$\begin{aligned} 2AD^2 &= 2b^2 + \frac{a^2}{2} - 2ab \cos C \\ &= b^2 - \frac{a^2}{2} + (b^2 + a^2 - 2ab \cos C) \\ &= b^2 - \frac{a^2}{2} + c^2 \quad (\because c^2 = b^2 + a^2 - 2ab \cos C) \end{aligned}$$

Hence

$$AD^2 = \frac{b^2}{2} - \frac{a^2}{4} + \frac{c^2}{2} = \frac{1}{4}(2b^2 + 2c^2 - a^2)$$

or

$$\begin{aligned} AD &= \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} = \frac{1}{2} \sqrt{b^2 + c^2 + b^2 + c^2 - a^2} \\ &= \frac{1}{2} \sqrt{b^2 + c^2 + 2bc \cos A} \end{aligned}$$

(since $b^2 + c^2 - a^2 = 2bc \cos A$).

Similarly,

$$BE = \frac{1}{2} \sqrt{2c^2 + 2a^2 - b^2} = \frac{1}{2} \sqrt{c^2 + a^2 + 2ca \cos B}$$

and

$$CF = \frac{1}{2} \sqrt{2a^2 + 2b^2 - c^2} = \frac{1}{2} \sqrt{a^2 + b^2 + 2ab \cos C}$$

THEOREM 4.26 Let AD , BE and CF be the medians of a triangle ABC . Then

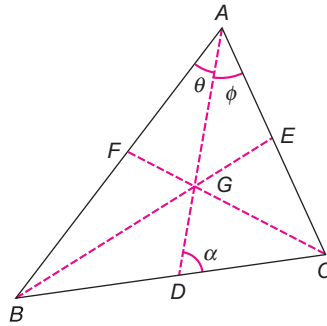


FIGURE 4.27 Theorem 4.26.

$$\sin \angle BAD = \frac{a \sin B}{\sqrt{2b^2 + 2c^2 - a^2}}$$

$$\sin \angle CAD = \frac{a \sin C}{\sqrt{2b^2 + 2c^2 - a^2}}$$

$$\sin \angle ADC = \frac{2b \sin C}{\sqrt{2b^2 + 2c^2 - a^2}}$$

PROOF Put $\theta = \angle BAD$, $\phi = \angle CAD$ and $\alpha = \angle ADC$. See Figure 4.27. Then

$$\frac{\sin \phi}{\sin C} = \frac{DC}{AD} = \frac{a}{2AD}$$

Therefore

$$\sin \phi = \frac{a \sin C}{2AD} = \frac{a \sin C}{\sqrt{2b^2 + 2c^2 - a^2}}$$

Similarly

$$\sin \theta = \frac{a \sin B}{\sqrt{2b^2 + 2c^2 - a^2}}$$

Also

$$\frac{\sin \alpha}{\sin C} = \frac{AC}{AD} = \frac{b}{AD}$$

Therefore

$$\sin \alpha = \frac{2b \sin C}{\sqrt{2b^2 + 2c^2 - a^2}}$$

THEOREM 4.27

In any triangle, the centroid lies on the line joining the circumcentre to the orthocentre and divides it in the ratio 1:2.

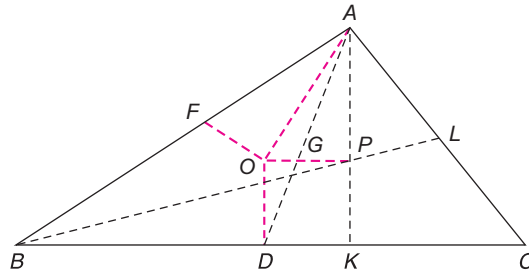


FIGURE 4.28 Theorem 4.27.

PROOF

Let ABC be a triangle and O and P be its circumcentre and orthocentre, respectively. Draw OD and PK perpendicular to BC . Let AD and OP meet at G . The triangles OGD and PGA are similar. See Figure 4.28. From Theorem 4.15, we have $OD = R \cos A$. Also, from Theorem 4.22, $AP = 2R \cos A$.

Since the triangles OGD and PGA are similar, we have

$$\frac{AG}{GD} = \frac{AP}{OD} = 2$$

Therefore, the point G is the centroid of the triangle. Also

$$\frac{OG}{GP} = \frac{OD}{AP} = \frac{1}{2}$$

Therefore, the centroid lies on the line joining the circumcentre to the orthocentre and divides it in the ratio 1:2. ■

DEFINITION 4.7 Nine-point Circle It is known that the feet of the perpendiculars, the mid-points of the sides and the mid-points of the line joining the vertices to the orthocentre are all concyclic. The circle passing through these nine points is called the *nine-point circle*.

Also, it is known that the centre of the nine-point circle lies on the line joining the circumcentre O and the orthocentre P and it bisects OP . Therefore, the circumcentre, the centroid, the centre of the nine-point circle and the orthocentre all lie on a straight line.

THEOREM 4.28 Let ABC be a triangle and R its circumradius. Then the distance between the circumcentre O and the orthocentre P is given by

$$OP = R\sqrt{1 - 8\cos A \cos B \cos C}$$

PROOF Let F be the foot of the perpendicular drawn from O to AB . Then

$$\angle FAO = 90^\circ - \angle FOA = 90^\circ - C$$

and from Figure 4.28

$$\angle LAP = 90^\circ - C$$

Therefore

$$\begin{aligned} \angle PAO &= A - \angle FAO - \angle LAP \\ &= A - 2(90^\circ - C) \\ &= A + 2C - 180^\circ \\ &= A + 2C - (A + B + C) = C - B \end{aligned}$$

Also, $OA = R$ and, by Theorem 4.22, we have

$$PA = 2R \cos A$$

Therefore

$$\begin{aligned} OP^2 &= OA^2 + PA^2 - 2OA \cdot PA \cos \angle PAO \\ &= R^2 + 4R^2 \cos^2 A - 4R^2 \cos A \cos(C - B) \\ &= R^2 + 4R^2 \cos A [\cos A - \cos(C - B)] \\ &= R^2 + 4R^2 \cos A [\cos\{180 - (B + C)\} - \cos(C - B)] \\ &= R^2 + 4R^2 \cos A [-\cos(C + B) - \cos(C - B)] \\ &= R^2 - 4R^2 \cos A [\cos(C + B) + \cos(C - B)] \end{aligned}$$

$$\begin{aligned}
&= R^2 - 4R^2 \cos A \cdot 2 \cos B \cos C \\
&= R^2(1 - 8 \cos A \cos B \cos C)
\end{aligned}$$

Therefore

$$OP = R\sqrt{1 - 8 \cos A \cos B \cos C}$$

THEOREM 4.29

Let ABC be a triangle and R its circumradius. Then the distance between the circumcentre O and the incentre I is given by

$$OI = R\sqrt{1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}} = \sqrt{R(R - 2r)}$$

where r is the inradius.

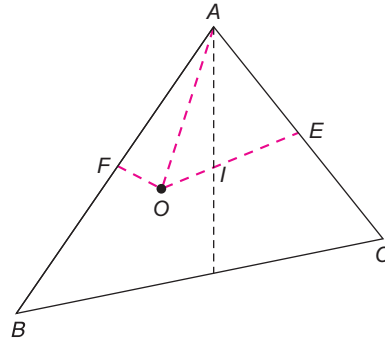


FIGURE 4.29 Theorem 4.29.

PROOF

As above, let OF be perpendicular to AB and IE be perpendicular to AC (Figure 4.29). As in Theorem 4.27, we have $\angle OAF = 90^\circ - C$. Therefore

$$\begin{aligned}
\angle IAO &= \angle FAI - \angle OAF \\
&= \frac{A}{2} - (90^\circ - C) \\
&= \frac{A}{2} + C - \frac{A + B + C}{2} \\
&= \frac{C - B}{2}
\end{aligned}$$

Also

$$IA = \frac{IE}{\sin(A/2)} = \frac{r}{\sin(A/2)} = 4R \sin \frac{B}{2} \sin \frac{C}{2}$$

Therefore

$$\begin{aligned}
OI^2 &= OA^2 + IA^2 - 2OA \cdot IA \cos \angle IAO \\
&= R^2 + \left(4R \sin \frac{B}{2} \sin \frac{C}{2}\right)^2 - 2R \left(4R \sin \frac{B}{2} \sin \frac{C}{2}\right) \cos \frac{C - B}{2} \\
&= R^2 + 16R^2 \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} - 8R^2 \sin \frac{B}{2} \sin \frac{C}{2} \cos \frac{C - B}{2}
\end{aligned}$$

$$\begin{aligned}
&= R^2 \left[1 + 16 \sin^2 \frac{B}{2} \sin^2 \frac{C}{2} - 8 \sin \frac{B}{2} \sin \frac{C}{2} \left(\cos \frac{B}{2} \cos \frac{C}{2} + \sin \frac{B}{2} \sin \frac{C}{2} \right) \right] \\
&= R^2 \left[1 - 8 \sin \frac{B}{2} \sin \frac{C}{2} \left(\cos \frac{B}{2} \cos \frac{C}{2} - \sin \frac{B}{2} \sin \frac{C}{2} \right) \right] \\
&= R^2 \left[1 - 8 \sin \frac{B}{2} \sin \frac{C}{2} \cos \frac{B+C}{2} \right] \\
&= R^2 \left[1 - 8 \sin \frac{B}{2} \sin \frac{C}{2} \sin \frac{A}{2} \right]
\end{aligned}$$

Therefore

$$OI = R \sqrt{1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}}$$

Also

$$OI^2 = R^2 - 2R \cdot 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = R^2 - 2Rr \quad (\text{by Corollary 4.3})$$

Hence

$$OI = \sqrt{R(R - 2r)}$$

QUICK LOOK 2

An argument similar to the above can be used to derive formulae for the distances between the circumcentre O and the excentres I_1, I_2 and I_3 . These formulas are as follows.

$$OI_1 = R \sqrt{1 + 8 \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}} = \sqrt{R(R + 2r_1)}$$

$$OI_2 = R \sqrt{1 + 8 \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}} = \sqrt{R(R + 2r_2)}$$

$$\text{and } OI_3 = R \sqrt{1 + 8 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}} = \sqrt{R(R + 2r_3)}$$

where R, r_1, r_2 and r_3 are, respectively, the circumradius and the exradii.

THEOREM 4.30 Let ABC be a triangle and AD the bisector of the angle A meeting BC at D . Then

$$AD = \frac{2bc}{b+c} \cos \frac{A}{2}$$

and

$$\angle ADC = \frac{A}{2} + B$$

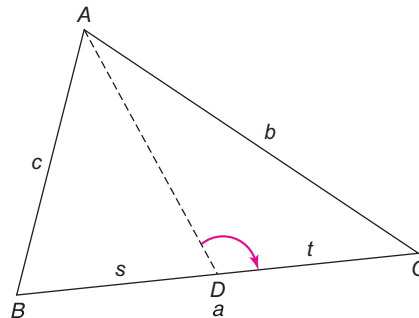


FIGURE 4.30 Theorem 4.30.

PROOF Suppose that D divides BC such that $BD = s$ and $DC = t$ (Figure 4.30). Then, we know from Geometry that

$$\frac{s}{t} = \frac{AB}{AC} = \frac{c}{b}$$

Therefore

$$\frac{s}{c} = \frac{t}{b} = \frac{s+t}{c+b} = \frac{a}{c+b}$$

Hence

$$s = \frac{ac}{b+c} \quad \text{and} \quad t = \frac{ab}{b+c}$$

Also, since the sum of the areas of the triangles ABD and ACD is equal to the area of the triangle ABC , we have

$$\frac{1}{2} AB \cdot AD \sin \frac{A}{2} + \frac{1}{2} AC \cdot AD \sin \frac{A}{2} = \frac{1}{2} AB \cdot AC \sin A$$

Therefore

$$AD \sin \frac{A}{2} \cdot (c+b) = cb \sin A$$

$$AD = \frac{bc}{b+c} \cdot \frac{\sin A}{\sin(A/2)} = \frac{2bc}{b+c} \cos \frac{A}{2}$$

Further,

$$\angle ADC = \angle ABC + \angle BAD = B + \frac{A}{2}$$

Similarly, if BE and CF are the bisectors of the angles B and C respectively, then we have the following (Figure 4.31):

$$BE = \frac{2ca}{c+a} \cos \frac{B}{2} \quad \text{and} \quad \angle BEA = \frac{B}{2} + C$$

$$CF = \frac{2ab}{a+b} \cos \frac{C}{2} \quad \text{and} \quad \angle CFB = \frac{C}{2} + A$$

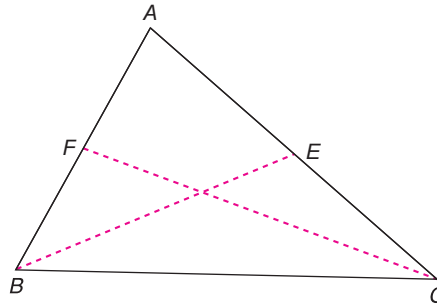


FIGURE 4.31

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. In a triangle ABC ,

$$(a+b)\cos C + (b+c)\cos A + (c+a)\cos B =$$

- (A) $\frac{a+b+c}{4}$ (B) $\frac{a+b+c}{2}$
 (C) $\frac{a+b+c}{3}$ (D) $a+b+c$

Solution:

$$\begin{aligned} & (a+b)\cos C + (b+c)\cos A + (c+a)\cos B \\ &= (a\cos C + c\cos A) + (b\cos C + c\cos B) \\ & \quad + (b\cos A + a\cos B) \\ &= b + a + c \\ &= a + b + c \end{aligned}$$

Answer: (D)2. In a triangle ABC , if $(a+b+c)(b+c-a) = 3bc$, then the angle A is equal to

- (A) 30° (B) 60° (C) 45° (D) 120°

Solution:

$$\begin{aligned} & (a+b+c)(b+c-a) = 3bc \\ \Rightarrow & 2s(2s-a) = 3bc \\ \Rightarrow & \frac{s(s-a)}{bc} = \frac{3}{4} \\ \Rightarrow & \cos^2 \frac{A}{2} = \frac{3}{4} \\ \Rightarrow & \cos \frac{A}{2} = \pm \frac{\sqrt{3}}{2} \\ \Rightarrow & \cos \frac{A}{2} = \frac{\sqrt{3}}{2} \quad \left(\because \frac{A}{2} < 90^\circ \right) \\ \Rightarrow & \frac{A}{2} = 30^\circ \\ \Rightarrow & A = 60^\circ \end{aligned}$$

Answer: (B)3. In a triangle ABC , if

$$\tan \frac{A}{2} = \frac{5}{6} \quad \text{and} \quad \tan \frac{C}{2} = \frac{2}{5}$$

then the sides a, b and c are

- (A) in GP (B) in HP
 (C) in AP (D) equal

Solution: We have

$$\tan \frac{A}{2} \cdot \tan \frac{C}{2} = \frac{5}{6} \cdot \frac{2}{5} = \frac{1}{3}$$

Therefore

$$\begin{aligned} \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \cdot \sqrt{\frac{(s-a)(s-b)}{s(s-c)}} &= \frac{1}{3} \\ \frac{s-b}{s} &= \frac{1}{3} \end{aligned}$$

This gives

$$\begin{aligned} 3s - 3b &= s \\ \Rightarrow 2s &= 3b \end{aligned}$$

Therefore

$$\begin{aligned} a + b + c &= 3b \\ \Rightarrow a + c &= 2b \end{aligned}$$

Answer: (C)4. In a triangle ABC , if $a = 6, b = 5$ and $c = 9$, then angle A is equal to

- (A) $\cos^{-1}\left(\frac{7}{9}\right)$ (B) $\cos^{-1}\left(\frac{5}{9}\right)$
 (C) $\sin^{-1}\left(\frac{7}{9}\right)$ (D) $\cos^{-1}\left(\frac{4}{9}\right)$

Solution: From the cosine formula, we have

$$\begin{aligned} \cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{5^2 + 9^2 - 6^2}{2 \times 5 \times 9} = \frac{7}{9} \end{aligned}$$

Therefore

$$A = \cos^{-1}\left(\frac{7}{9}\right)$$

Answer: (A)5. In a triangle ABC , if $a = 13, b = 14$ and $c = 15$, then circumdiameter of the triangle is

- (A) $\frac{65}{3}$ (B) $\frac{65}{4}$ (C) $\frac{64}{5}$ (D) $\frac{56}{4}$

Solution: We have

$$2s = a + b + c = 13 + 14 + 15 = 42$$

Therefore

$$s = 21, s - a = 8, s - b = 7 \quad \text{and} \quad s - c = 6$$

By Corollary 4.2,

$$\begin{aligned} R &= \frac{abc}{4\sqrt{s(s-a)(s-b)(s-c)}} \\ &= \frac{13 \times 14 \times 15}{4\sqrt{21 \times 8 \times 7 \times 6}} \\ &= \frac{13 \times 14 \times 15}{4 \times 84} \end{aligned}$$

Therefore the circumdiameter is

$$2R = \frac{65}{4}$$

Answer: (B)

6. In a triangle ABC ,

$$b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2} =$$

- (A) $\frac{a+b+c}{3}$ (B) $a+b+c$
 (C) $\frac{a+b+c}{2}$ (D) $\frac{a+b+c}{4}$

Solution: By Theorem 4.5

$$\begin{aligned} b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2} &= b \left(\frac{s(s-c)}{ab} \right) + c \left(\frac{s(s-b)}{ca} \right) \\ &= \frac{s(s-c)}{a} + \frac{s(s-b)}{a} \\ &= \frac{s(2s-b-c)}{a} \\ &= \frac{sa}{a} = s = \frac{a+b+c}{2} \end{aligned}$$

Answer: (C)

7. In $\triangle ABC$, if $A = 60^\circ$ and $B = 30^\circ$, then $a:b:c =$

- (A) $2:1:\sqrt{3}$ (B) $1:\sqrt{3}:2$
 (C) $\sqrt{3}:2:\sqrt{2}$ (D) $\sqrt{3}:1:2$

Solution: Since $A + B + C = 180^\circ$, we have

$$60^\circ + 30^\circ + C = 180^\circ$$

and hence $C = 90^\circ$. From the sine formula, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Therefore

$$\begin{aligned} a:b:c &= \sin A:\sin B:\sin C \\ &= \sin 60^\circ:\sin 30^\circ:\sin 90^\circ \\ &= \frac{\sqrt{3}}{2}:\frac{1}{2}:1 \\ &= \sqrt{3}:1:2 \end{aligned}$$

Answer: (D)

8. In a triangle ABC , if

$$\frac{1}{b+c} + \frac{1}{c+a} = \frac{3}{a+b+c}$$

then $C =$

- (A) 30° (B) 60° (C) 45° (D) 75°

Solution: From the given equation, we have

$$\frac{(c+a)+(b+c)}{(b+c)(c+a)} = \frac{3}{a+b+c}$$

Therefore

$$\begin{aligned} (a+b+2c)(a+b+c) &= 3(b+c)(c+a) \\ a^2 + ab + ac + ab + b^2 + bc + 2ca \\ &+ 2bc + 2c^2 = 3(bc + ab + c^2 + ca) \\ a^2 + b^2 - c^2 &= ab \end{aligned}$$

Hence

$$\begin{aligned} \cos C &= \frac{a^2 + b^2 - c^2}{2ab} = \frac{ab}{2ab} = \frac{1}{2} \\ C &= 60^\circ \end{aligned}$$

Answer: (B)

9. In a triangle ABC ,

$$2 \left(a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2} \right) =$$

- (A) $a+b+c$ (B) $b+c-a$
 (C) $a+b-c$ (D) $a+c-b$

Solution: We have

$$\begin{aligned} 2 \left(a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2} \right) &= a(1 - \cos C) + c(1 - \cos A) \\ &= a - (a \cos C + c \cos A) + c \\ &= a + c - b \end{aligned}$$

Answer: (D)

10. In a triangle ABC , $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) =$
 (A) s (B) 1 (C) 0 (D) $2s$

Solution: We know that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = K, \text{ say}$$

Then

$$\begin{aligned} & a \sin(B - C) + b \sin(C - A) + c \sin(A - B) \\ &= K \sin A \sin(B - C) + K \sin B \sin(C - A) \\ &\quad + K \sin C \sin(A - B) \\ &= K [\sin(B + C) \sin(B - C) + \sin(C + A) \sin(C - A) \\ &\quad + \sin(A + B) \sin(A - B)] \\ &= K [(\sin^2 B - \sin^2 C) + (\sin^2 C - \sin^2 A) \\ &\quad + (\sin^2 A - \sin^2 B)] \\ &= K \times 0 = 0 \end{aligned}$$

Note that K is the diameter of the circumcircle of $\triangle ABC$.

Answer: (C)

11. In a triangle ABC ,

$$c \sin\left(\frac{A - B}{2}\right) =$$

(A) $(a - b) \cos C$ (B) $(a - b) \cos \frac{C}{2}$
 (C) $\frac{a - b}{2} \cdot \cos C$ (D) $\frac{a - b}{2} \cdot \cos \frac{C}{2}$

Solution: Let

$$\frac{a}{\sin A} = K = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Then

$$\begin{aligned} \frac{a - b}{c} &= \frac{K \sin A - K \sin B}{K \sin C} \\ &= \frac{\sin A - \sin B}{\sin C} \\ &= \frac{2 \cos\left(\frac{A + B}{2}\right) \sin\left(\frac{A - B}{2}\right)}{2 \sin \frac{C}{2} \cos \frac{C}{2}} \\ &= \frac{\cos\left(90^\circ - \frac{C}{2}\right) \sin\left(\frac{A - B}{2}\right)}{\sin \frac{C}{2} \cos \frac{C}{2}} \end{aligned}$$

$$= \frac{\sin\left(\frac{A - B}{2}\right)}{\cos \frac{C}{2}}$$

Therefore

$$c \sin\left(\frac{A - B}{2}\right) = (a - b) \cos \frac{C}{2}$$

Answer: (B)

12. If A, B and C are the angles of a triangle, then

$$\frac{\sin(B + C)}{\sin(B - C)} =$$

(A) $\frac{b + c}{b - c}$ (B) $\frac{b^2 + c^2}{b^2 - c^2}$
 (C) $\frac{b^2 - c^2}{a^2}$ (D) $\frac{a^2}{b^2 - c^2}$

Solution:

$$\begin{aligned} \frac{\sin(B + C)}{\sin(B - C)} &= \frac{\sin^2(B + C)}{\sin(B - C) \sin(B + C)} \\ &= \frac{\sin^2(\pi - A)}{\sin^2 B - \sin^2 C} \\ &= \frac{4R^2 \sin^2 A}{4R^2 (\sin^2 B - \sin^2 C)} \\ &= \frac{a^2}{b^2 - c^2} \end{aligned}$$

Answer: (D)

13. If ABC is a triangle with area Δ , then $\cot A + \cot B + \cot C =$

(A) $\frac{a^2 + b^2 + c^2}{2\Delta}$ (B) $\frac{a^2 + b^2 + c^2}{4\Delta}$
 (C) $\frac{a + b + c}{4\Delta}$ (D) $\frac{a^2 + b^2 + c^2}{\Delta}$

Solution:

$$\begin{aligned} & \cot A + \cot B + \cot C \\ &= \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C} \\ &= \frac{b^2 + c^2 - a^2}{2bc \sin A} + \frac{c^2 + a^2 - b^2}{2ca \sin B} + \frac{a^2 + b^2 - c^2}{2ab \sin C} \\ &\quad \text{(by cosine formula)} \end{aligned}$$

$$= \frac{1}{4\Delta} (b^2 + c^2 - a^2 + c^2 + a^2 - b^2 + a^2 + b^2 - c^2)$$

$$\left(\text{since } \Delta = \frac{1}{2} bc \sin \right)$$

$$= \frac{1}{4\Delta} (a^2 + b^2 + c^2)$$

Answer: (B)

14. In a triangle ABC ,

$$(a + b + c) \left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) =$$

$$(A) \quad 2c \cot \frac{C}{2}$$

$$(B) \quad 2c \tan \frac{A}{2}$$

$$(C) \quad (a + b + c) \cot \frac{C}{2}$$

$$(D) \quad c \cot \frac{C}{2}$$

Solution:

$$\begin{aligned} (a + b + c) \left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) &= 2s \left(\frac{\Delta}{s(s-a)} + \frac{\Delta}{s(s-b)} \right) \\ &= 2\Delta \left(\frac{1}{s-a} + \frac{1}{s-b} \right) \\ &= 2\Delta \left(\frac{s-b+s-a}{(s-a)(s-b)} \right) \\ &= 2\Delta \cdot \frac{2s-a-b}{(s-a)(s-b)} \\ &= 2\Delta \cdot \frac{c}{(s-a)(s-b)} \\ &= 2c \cdot \frac{\Delta}{(s-a)(s-b)} \\ &= 2c \cot \frac{C}{2} \end{aligned}$$

Answer: (A)

15. In a triangle ABC , if

$$\frac{\tan A - \tan B}{\tan A + \tan B} = \frac{c-b}{c}$$

then angle A is equal to

$$(A) \quad 30^\circ \quad (B) \quad 45^\circ \quad (C) \quad 60^\circ \quad (D) \quad 75^\circ$$

Solution: The given equation can be written as

$$\frac{\frac{\sin A}{\cos A} - \frac{\sin B}{\cos B}}{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}} = \frac{\sin C - \sin B}{\sin C} \quad \left(\because \frac{\sin B}{b} = \frac{\sin C}{c} \right)$$

$$\frac{\frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B}}{\frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B}} = \frac{\sin C - \sin B}{\sin C}$$

$$\frac{\sin(A-B)}{\sin(A+B)} = \frac{\sin(A+B) - \sin B}{\sin(A+B)}$$

$$[\because C = \pi - (A+B)]$$

$$\sin(A-B) = \sin(A+B) - \sin B$$

$$\sin(A-B) - \sin(A+B) = -\sin B$$

$$\sin(A+B) - \sin(A-B) = \sin B$$

$$2 \cos A \sin B = \sin B$$

$$\cos A = \frac{1}{2} = \cos 60^\circ$$

Therefore $A = 60^\circ$.

Answer: (C)

16. In a triangle ABC , if a^2, b^2, c^2 are in AP, then $\cot A, \cot B, \cot C$ are

(A) in GP (B) in AP (C) in HP (D) equal

Solution: Since a^2, b^2, c^2 are in AP, we have

$$b^2 - a^2 = c^2 - b^2$$

Therefore

$$\sin^2 B - \sin^2 A = \sin^2 C - \sin^2 B$$

$$\sin(B+A) \sin(B-A) = \sin(C+B) \sin(C-B)$$

$$\sin C \sin(B-A) = \sin A \sin(C-B)$$

$$\frac{\sin(A-B)}{\sin A} = \frac{\sin(B-C)}{\sin C}$$

$$\frac{\frac{\sin A \cos B - \cos A \sin B}{\sin A \sin B}}{\frac{\sin A \cos B + \cos A \sin B}{\sin A \sin B}} = \frac{\frac{\sin B \cos C - \cos B \sin C}{\sin B \sin C}}{\frac{\sin B \cos C + \cos B \sin C}{\sin B \sin C}}$$

$$\cot B - \cot A = \cot C - \cot B$$

Therefore $\cot A, \cot B, \cot C$ are in AP.

Answer: (B)

17. In $\triangle ABC$, $a^2 \sin 2B + b^2 \sin 2A$ is equal to

(A) 2Δ (B) 3Δ (C) 4Δ (D) 6Δ

Solution:

$$\begin{aligned} a^2 \sin 2B + b^2 \sin 2A &= 2a^2 \sin B \cos B + 2b^2 \sin A \cos A \\ &= 2a(b \sin A) \cos B + 2b(a \sin B) \cos A \\ &= 2ab \sin(A+B) \end{aligned}$$

$$= 2ab \sin C$$

$$= 4\Delta$$

Answer: (C)

18. In $\triangle ABC$, if $8R^2 = a^2 + b^2 + c^2$, then the triangle is

- (A) isosceles (B) right angled
(C) equilateral (D) scalene

Solution:

$$\begin{aligned} 8R^2 &= 4R^2(\sin^2 A + \sin^2 B + \sin^2 C) \\ 2 &= \frac{1 - \cos 2A}{2} + \frac{1 - \cos 2B}{2} + \frac{1 - \cos 2C}{2} \\ -1 &= \cos 2A + \cos 2B + \cos 2C \\ &= -1 - 4\cos A \cos B \cos C \end{aligned}$$

Hence

$$\cos A \cos B \cos C = 0$$

So one of the angles of the triangle is a right angle.

Answer: (B)

19. In $\triangle ABC$, $a^3 \cos(B - C) + b^3 \cos(C - A) + c^3 \cos(A - B)$ is equal to

- (A) abc (B) $2abc$ (C) 0 (D) $3abc$

Solution:

$$\begin{aligned} a^3 \cos(B - C) &= a^2(2R \sin A) \cos(B - C) \\ &= 2Ra^2 \sin(B + C) \cos(B - C) \\ &= Ra^2(\sin 2B + \sin 2C) \\ &= 2Ra^2(\sin B \cos B + \sin C \cos C) \\ &= 2Ra^2\left(\frac{b}{2R} \cos B + \frac{c}{2R} \cos C\right) \\ &= a^2(b \cos B + c \cos C) \end{aligned}$$

Similarly

$$b^3 \cos(C - A) = b^2(a \cos A + c \cos C)$$

$$\text{and } c^3 \cos(A - B) = c^2(a \cos A + b \cos B)$$

Therefore

$$\begin{aligned} \sum a^3 \cos(B - C) &= ab(a \cos B + b \cos A) + bc(b \cos C + c \cos B) \\ &\quad + ca(c \cos A + a \cos C) \\ &= ab(c) + bc(a) + ca(b) \\ &= 3abc \end{aligned}$$

Answer: (D)

20. In $\triangle ABC$, $a^3 \sin(B - C) + b^3 \sin(C - A) + c^3 \sin(A - B)$ equals

- (A) 0 (B) abc (C) $2abc$ (D) $3abc$

Solution:

$$\begin{aligned} a^3 \sin(B - C) &= a^2(2R \sin A) \sin(B - C) \\ &= 2Ra^2 \sin(B + C) \sin(B - C) \\ &= 2Ra^2(\sin^2 B - \sin^2 C) \\ &= 2Ra^2\left(\frac{b^2}{4R^2} - \frac{c^2}{4R^2}\right) \\ &= \frac{1}{2R}a^2(b^2 - c^2) \end{aligned}$$

Therefore

$$\begin{aligned} \sum a^3 \sin(B - C) &= \frac{1}{2R}[a^2(b^2 - c^2) + b^2(c^2 - a^2) + c^2(a^2 - b^2)] \\ &= \frac{1}{2R}(0) = 0 \end{aligned}$$

Answer: (A)

21. In $\triangle ABC$,

$$(a - b)^2 \cos^2 \frac{C}{2} + (a + b)^2 \sin^2 \frac{C}{2} =$$

- (A) $a^2 + b^2 + c^2$ (B) $abc(a + b + c)$
(C) $(abc)\Delta$ (D) c^2

Solution: We have

$$\begin{aligned} (a - b)^2 \cos^2 \frac{C}{2} + (a + b)^2 \sin^2 \frac{C}{2} &= (a - b)^2 \frac{(1 + \cos C)}{2} + (a + b)^2 \frac{(1 - \cos C)}{2} \\ &= \frac{1}{2}[(a - b)^2 + (a + b)^2] + \frac{1}{2}[(a - b)^2 - (a + b)^2] \cos C \\ &= a^2 + b^2 + \frac{1}{2}(-4ab \cos C) \\ &= a^2 + b^2 - 2ab \cos C \\ &= c^2 \end{aligned}$$

Answer: (D)

22. In $\triangle ABC$, if the angles A , B and C are in the ratio $1:2:3$, then the sides a , b , c are in the ratio

- (A) $1:2:3$ (B) $1:\sqrt{3}:2$
(C) $1:2:\sqrt{3}$ (D) $\sqrt{3}:1:2$

Solution: Let $A = \theta$, $B = 2\theta$ and $C = 3\theta$. Therefore

$$180^\circ = 6\theta$$

$$\Rightarrow \theta = 30^\circ$$

This implies $A = 30^\circ$, $B = 60^\circ$, $C = 90^\circ$. Therefore

$$\begin{aligned} a:b:c &= 2R \sin A : 2R \sin B : 2R \sin C \\ &= \sin 30^\circ : \sin 60^\circ : \sin 90^\circ \\ &= \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 \\ &= 1 : \sqrt{3} : 2 \end{aligned}$$

Answer: (B)

23. In $\triangle ABC$, if $A = 30^\circ$ and the area Δ is $(\sqrt{3}/4)a^2$, then the difference of the angles B and C is

- (A) 75° (B) 60° (C) 90° (D) 45°

Solution:

$$\frac{\sqrt{3}}{4}a^2 = \Delta = \frac{1}{2}ab \sin C = \frac{1}{2}a^2 \frac{\sin B \sin C}{\sin A}$$

Therefore

$$\sin B \sin C = \frac{\sqrt{3}}{4} \quad (\because 2 \sin A = 1)$$

Also

$$\begin{aligned} \frac{\sqrt{3}}{2} &= \cos A \\ &= -\cos(B+C) \\ &= -\cos B \cos C + \sin B \sin C \\ &= -\cos B \cos C + \frac{\sqrt{3}}{4} \end{aligned}$$

Therefore

$$\cos B \cos C = -\frac{\sqrt{3}}{4}$$

Now

$$\begin{aligned} \cos(B-C) &= \cos B \cos C + \sin B \sin C \\ &= -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} = 0 \end{aligned}$$

which gives $|B-C| = 90^\circ$.

Answer: (C)

24. In $\triangle ABC$, if $\cos B + \cos C = 2(1 - \cos A)$, then the order in which the sides a, b, c are in AP is

(A) a, b, c

(B) b, a, c or c, a, b

(C) b, c, a or a, c, b

(D) c, b, a

Solution:

$$\begin{aligned} \cos B + \cos C &= 4 \sin^2 \frac{A}{2} \\ 2 \cos \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right) &= 4 \sin^2 \frac{A}{2} \\ \sin \left(\frac{A}{2} \right) \cos \left(\frac{B-C}{2} \right) &= 2 \sin^2 \frac{A}{2} \\ \cos \left(\frac{B-C}{2} \right) &= 2 \sin \frac{A}{2} \\ 2 \cos \left(\frac{A}{2} \right) \cos \left(\frac{B-C}{2} \right) &= 4 \sin \frac{A}{2} \cos \frac{A}{2} \\ 2 \sin \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right) &= 2 \sin A \\ \sin B + \sin C &= 2 \sin A \\ b + c &= 2a \end{aligned}$$

Answer: (B)

25. In $\triangle ABC$, if AD is the internal bisector of the angle A meeting the side BC in D , then the length of AD is

(A) $\left(\frac{2bc}{b+c} \right) \cos \frac{A}{2}$

(B) $\sqrt{2b^2 + 2c^2 - a^2}$

(C) $\frac{abc}{b+c}$

(D) $\frac{abc}{a+b+c}$

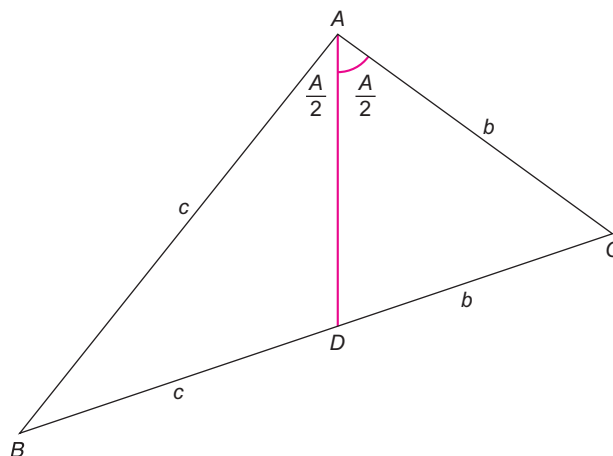


FIGURE 4.32 Single correct choice type question 25.

Solution: See Figure 4.32. It is known that $BD:DC = AB:AC = c:b$. Therefore

$$BD = \frac{ac}{b+c} \quad \text{and} \quad DC = \frac{ab}{b+c}$$

Also

$$\text{Area of } \triangle ABD = \frac{1}{2}(AB)(AD)\sin\frac{A}{2} = \frac{AD[c\sin(A/2)]}{2}$$

$$\text{Area of } \triangle ACD = \frac{AD[b\sin(A/2)]}{2}$$

Now

$$\begin{aligned}\frac{1}{2}bc\sin A &= \text{Area of } \triangle ABC = \triangle ABD + \triangle ACD \\ &= \frac{(AD)\sin(A/2)}{2}(b+c)\end{aligned}$$

Therefore

$$\begin{aligned}bc\left(2\sin\frac{A}{2}\cos\frac{A}{2}\right) &= (AD)(b+c)\sin\frac{A}{2} \\ AD &= \left(\frac{2bc}{b+c}\right)\cos\frac{A}{2}\end{aligned}$$

Answer: (A)

26. In $\triangle ABC$, $BC = 6$ units. BE and CF are medians such that $\angle CBE = 30^\circ$ and $\angle BCF = 60^\circ$. Then the area of $\triangle ABC$ is

- (A) $8\sqrt{3}$ (B) $\frac{9\sqrt{3}}{2}$
(C) $\frac{27\sqrt{3}}{2}$ (D) $9\sqrt{3}$

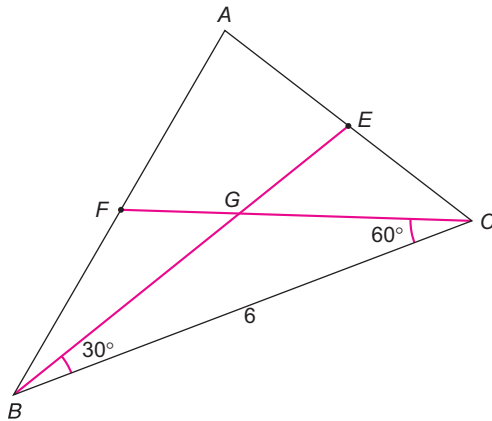


FIGURE 4.33 Single correct choice type question 26.

Solution: See Figure 4.33. It is known that the medians intersect at the centroid G of $\triangle ABC$. By hypothesis $\angle BGC = 90^\circ$. Also, it is known that the area of $\triangle ABC$ is three times of the area of $\triangle BGC$. In $\triangle BGC$,

$$\frac{BG}{\sin 60^\circ} = \frac{CG}{\sin 30^\circ} = \frac{BC}{\sin 90^\circ} = 6$$

Therefore

$$BG = 6\sin 60^\circ = 3\sqrt{3} \quad \text{and} \quad CG = 6\sin 30^\circ = 3$$

Also

$$\begin{aligned}\text{Area of } \triangle ABC &= 3 (\text{Area of } \triangle BGC) \\ &= 3\left(\frac{1}{2}BG \cdot CG\right) \\ &= \frac{3}{2} \times 3\sqrt{3} \times 3 \\ &= \frac{27}{2}\sqrt{3}\end{aligned}$$

Answer: (C)

27. In $\triangle ABC$, if $b = 3$, $c = 5$ and $\cos(B - C) = 7/25$, then the value of “ a ” is

- (A) $\sqrt{10}$ (B) $\sqrt{15}$
(C) 6 (D) $\sqrt{5}$

Solution:

$$\begin{aligned}\frac{7}{25} &= \cos(B - C) = \frac{1 - \tan^2[(B - C)/2]}{1 + \tan^2[(B - C)/2]} \\ 7 + 7\tan^2\left(\frac{B - C}{2}\right) &= 25 - 25\tan^2\left(\frac{B - C}{2}\right) \\ 32\tan^2\left(\frac{B - C}{2}\right) &= 18\end{aligned}$$

Therefore

$$\tan\left(\frac{B - C}{2}\right) = \frac{-3}{4}$$

because $b < c$. Now by Napier’s rule

$$\begin{aligned}\frac{-3}{4} &= \tan\frac{B - C}{2} \\ &= \left(\frac{b - c}{b + c}\right)\cot\frac{A}{2} \\ &= \left(\frac{3 - 5}{3 + 5}\right)\cot\frac{A}{2}\end{aligned}$$

Therefore

$$\tan\frac{A}{2} = \frac{1}{3}$$

Again

$$\cos A = \frac{1 - \tan^2(A/2)}{1 + \tan^2(A/2)} = \frac{1 - (1/9)}{1 + (1/9)} = \frac{4}{5}$$

Therefore

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \quad (\text{cosine formula}) \\ &= 9 + 25 - 2(3)(5)\frac{4}{5} \\ &= 34 - 24 \\ &= 10 \end{aligned}$$

Hence, $a = \sqrt{10}$.

Answer: (A)

- 28.** E and F are interior points on the sides AC and AB , respectively, of $\triangle ABC$. The lines BE and CF intersect in P . If the area of $\triangle BPF$ is 5, the area of the quadrilateral $AEPF$ is 22 and the area of $\triangle CPE$ is 8, then the area of $\triangle BPC$ is equal to

(A) 20 (B) 15 (C) 12 (D) 10

Solution: Let the area of $\triangle APF$ be λ so that the area of $\triangle APE$ is $22 - \lambda$. Suppose the area of $\triangle BPC$ is x . See Figure 4.34.

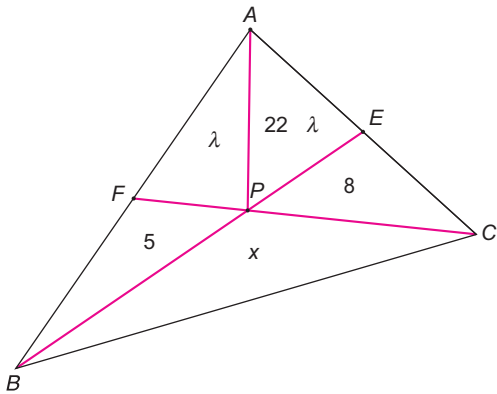


FIGURE 4.34 Single correct choice type question 28.

(A) We have

$$\frac{CP}{PF} = \frac{\text{Area of } \triangle BPC}{\text{Area of } \triangle BPF} = \frac{x}{5}$$

Again,

$$\frac{CP}{PF} = \frac{\text{Area of } \triangle APC}{\text{Area of } \triangle APF} = \frac{30 - \lambda}{\lambda}$$

Therefore

$$\frac{x}{5} = \frac{30 - \lambda}{\lambda} \quad (4.3)$$

(B) In a similar way,

$$\frac{BP}{PE} = \frac{\lambda + 5}{22 - \lambda} \quad \text{and} \quad \frac{BP}{PE} = \frac{x}{8}$$

Therefore

$$\frac{\lambda + 5}{22 - \lambda} = \frac{x}{8} \quad (4.4)$$

From Eqs. (4.3) and (4.4),

$$\frac{5(30 - \lambda)}{\lambda} = x = \frac{8(\lambda + 5)}{22 - \lambda}$$

Therefore

$$\lambda^2 + 100\lambda - 50 \times 22 = 0$$

$$(\lambda + 110)(\lambda - 10) = 0$$

This implies $\lambda = 10$ and $x = 10$. Hence area of $\triangle BPC$ is 10.

Answer: (D)

- 29.** Two circles with radii R and r ($R > r$) touch each other internally at A . ABC is an equilateral triangle with B on one circle and C on the other. Then the length of the side of the equilateral triangle ABC is

$$\begin{aligned} \text{(A)} \quad & \frac{2Rr\sqrt{3}}{\sqrt{R^2 - Rr + r^2}} & \text{(B)} \quad & \frac{Rr\sqrt{3}}{\sqrt{R^2 - Rr + r^2}} \\ \text{(C)} \quad & \frac{Rr\sqrt{3}}{\sqrt{R^2 + Rr + r^2}} & \text{(D)} \quad & \frac{2Rr\sqrt{3}}{\sqrt{R^2 + Rr + r^2}} \end{aligned}$$

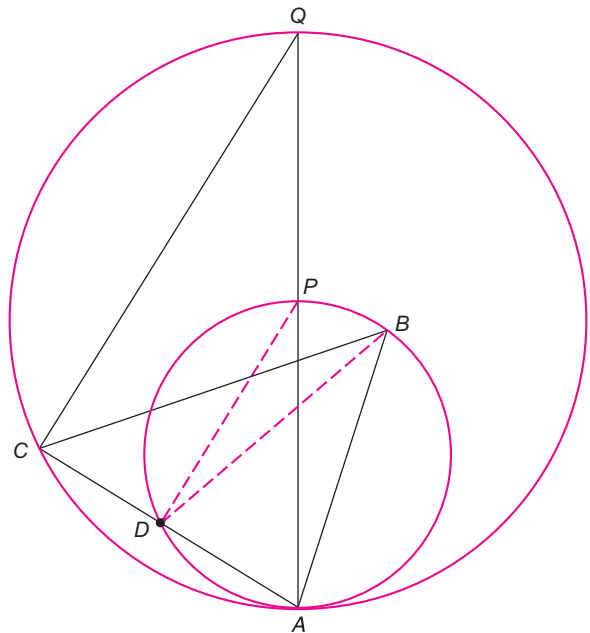


FIGURE 4.35 Single correct choice type question 29.

Solution: Let the diameter AP of the inner circle meet the outer circle in Q . Let x be the length of the side of the equilateral $\triangle ABC$. Suppose the side AC meet the inner circle in D . Join the point C to Q and the point D to P (Figure 4.35).

Therefore

$$\angle ACQ = \angle ADP = 90^\circ$$

Hence $\triangle ACQ$ and $\triangle ADP$ are similar triangles. From $\triangle BDA$,

$$\frac{BD}{\sin 60^\circ} = 2r$$

Therefore

$$BD = \sqrt{3}r$$

Also, from the similarity of triangles

$$\frac{AD}{AC} = \frac{AP}{AQ} = \frac{r}{R}$$

Therefore

$$AD = \frac{rx}{R} \quad (\because AC = x)$$

Using cosine rule for $\triangle ABD$,

$$\begin{aligned} 3r^2 &= (BD)^2 = (AD)^2 + (AB)^2 - 2(AD)(AB)\cos 60^\circ \\ &= \frac{r^2 x^2}{R^2} + x^2 - 2\left(\frac{rx}{R}\right)(x)\left(\frac{1}{2}\right) \\ &= \frac{x^2}{R^2}(R^2 - Rr + r^2) \\ x &= \frac{\sqrt{3}Rr}{\sqrt{R^2 - Rr + r^2}} \end{aligned}$$

Answer: (B)

30. The sides a , b and c of a triangle satisfy the equality $bc^2 = (a+b)(a-b)^2$. If α and β are the angles opposite to a and b , respectively, then

- (A) $\alpha = 3\beta$ (B) $\alpha = 2\beta$
(C) $\beta = 2\alpha$ (D) $\beta = 3\alpha$

Solution: Let $a = BC$, $b = CA$ and $c = AB$. So $\alpha = \angle A$, $\beta = \angle B$. Now by hypothesis

$$bc^2 = (a+b)(a-b)^2$$

Therefore

$$\begin{aligned} \sin B \sin^2 C &= (\sin A + \sin B)(\sin A - \sin B)^2 \\ &= (\sin^2 A - \sin^2 B)(\sin A - \sin B) \\ &= \sin(A+B)\sin(A-B)(\sin A - \sin B) \\ &= \sin C \sin(A-B)(\sin A - \sin B) \end{aligned}$$

$$\sin B \sin C = \sin(A-B)(\sin A - \sin B)$$

$$\sin B \sin(A+B) = \sin A \sin(A-B)$$

$$-\sin B \sin(A-B)$$

$$\sin B[\sin(A+B) + \sin(A-B)] = \sin A \sin(A-B)$$

$$\sin A(\sin 2B) = \sin A \sin(A-B)$$

$$\sin 2B = \sin(A-B)$$

This implies

$$A-B=2B \quad \text{or} \quad A-B=\pi-2B$$

$$\Rightarrow A=3B \quad \text{or} \quad A+B=\pi$$

But $A+B$ cannot be π . Therefore $A=3B$ which means $\alpha=3\beta$.

Answer: (A)

31. In $\triangle ABC$, $\angle ABC = 45^\circ$. D is a point on the segment BC such that $2BD = DC$ and $\angle DAB = 15^\circ$. Then $\angle ACB$ is

- (A) 75° (B) 60° (C) 90° (D) 105°

Solution: Let $\angle DAC = \theta$ so that

$$\angle A = \theta + 15^\circ$$

Note that

$$\angle ADC = 45^\circ + 15^\circ = 60^\circ$$

See Figure 4.36.

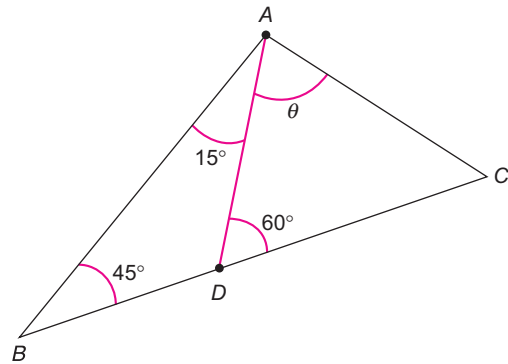


FIGURE 4.36 Single correct choice type question 31.

From $\triangle CAD$,

$$\frac{CD}{\sin \theta} = \frac{AC}{\sin 60^\circ} \quad (4.5)$$

Also from $\triangle ABC$,

$$\frac{BC}{\sin(\theta + 15^\circ)} = \frac{AC}{\sin 45^\circ} \quad (4.6)$$

Therefore from Eqs. (4.5) and (4.6)

$$\frac{(CD)\sin(\theta + 15^\circ)}{(BC)\sin\theta} = \frac{\sin 45^\circ}{\sin 60^\circ}$$

$$\frac{CD}{BC} = \frac{\sin\theta \sin 45^\circ}{\sin(\theta + 15^\circ) \sin 60^\circ}$$

$$\frac{2}{3} = \frac{\sin\theta \sin 45^\circ}{\sin(\theta + 15^\circ) \sin 60^\circ}$$

But

$$\frac{2}{3} = \left(\frac{\sin 45^\circ}{\sin 60^\circ} \right)^2$$

Therefore

$$\left(\frac{\sin 45^\circ}{\sin 60^\circ} \right)^2 = \frac{\sin\theta}{\sin(\theta + 15^\circ)} \cdot \frac{\sin 45^\circ}{\sin 60^\circ}$$

Clearly $\theta = 45^\circ$ is a solution of the above equation. Therefore $\angle A = 60^\circ$, $\angle B = 45^\circ$ and $\angle C = 75^\circ$.

Answer: (A)

32. AB is a diameter of a circle with centre at O and radius r . D is a point on the tangent at B and DC is the other tangent to the circle at C such that $\angle COA = 60^\circ$. Then

- (A) $(DC)\sqrt{3} = r = AC$
 (B) $r\sqrt{2} = DC = (AC)\sqrt{3}$
 (C) $DC = r\sqrt{3} = (AC)\sqrt{3}$
 (D) $(DC)\sqrt{2} = r = AC$

Solution: See Figure 4.37.

$$(AC)^2 = 2r^2 - 2r^2 \cos 60^\circ = r^2 \quad (4.7)$$

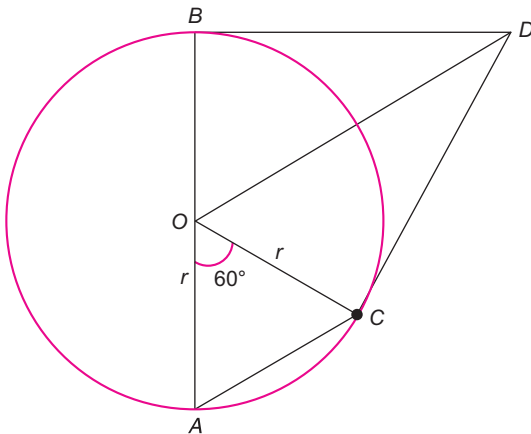


FIGURE 4.37 Single correct choice type question 32.

Also $\angle COD = \angle BOD = 60^\circ$. From $\triangle DOC$,

$$\tan 60^\circ = \frac{DC}{OC}$$

Therefore by Eq. (4.7) we have

$$DC = r\sqrt{3} = (AC)\sqrt{3}$$

Answer: (C)

33. In an isosceles triangle one angle is 120° and the radius of its inscribed circle is $\sqrt{3}$. Then, the area of the triangle is

- (A) $12 + 7\sqrt{3}$ (B) $7 + 12\sqrt{3}$
 (C) $12 - 7\sqrt{3}$ (D) 4π

Solution: See Figure 4.38. In $\triangle ABC$, let $\angle A = 120^\circ$, $AB = AC$. Then

$$\Delta = \frac{1}{2}bc \sin 120^\circ = \frac{1}{2}b^2 \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4} \cdot b^2$$

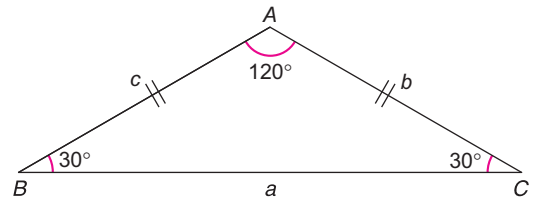


FIGURE 4.38 Single correct choice type question 33.

Again

$$\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} \Rightarrow a = \sqrt{3}b$$

Now

$$2s = a + b + c = a + 2b$$

$$= b(\sqrt{3} + 2)$$

Therefore

$$\frac{\sqrt{3}}{4}b^2 = \Delta = rs = \sqrt{3} \cdot \frac{b(2 + \sqrt{3})}{2}$$

Hence $b = 2(2 + \sqrt{3})$. So

$$\Delta = \frac{\sqrt{3}}{4}b^2 = \frac{\sqrt{3}}{4} \cdot 4(2 + \sqrt{3})^2$$

$$= \sqrt{3}(7 + 4\sqrt{3}) = 12 + 7\sqrt{3}$$

Answer: (A)

34. In $\triangle ABC$, $AB = 3$, $BC = 5$, $AC = 7$ and $\angle ABC = 120^\circ$. The angle bisectors AD and CE of $\angle A$ and $\angle C$ meet in P . Then the length AP is

(A) 6 (B) $\sqrt{6}$ (C) 7 (D) $\sqrt{7}$

Solution: Note that P is the incentre of $\triangle ABC$. Draw PL perpendicular to AB so that PL is the inradius r of $\triangle ABC$ (Figure 4.39).

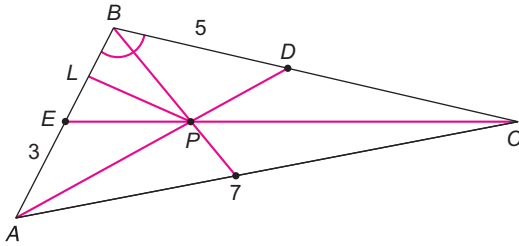


FIGURE 4.39 Single correct choice type question 34.

Now

$$\begin{aligned} rs = \Delta &= \frac{1}{2}(AB)(BC)\sin 120^\circ \\ &= \frac{1}{2}(3)(5)\frac{\sqrt{3}}{2} = \frac{15\sqrt{3}}{4} \end{aligned}$$

Therefore

$$\begin{aligned} r\left(\frac{15}{2}\right) &= \frac{15\sqrt{3}}{4} \\ r &= \frac{\sqrt{3}}{2} \end{aligned} \quad (4.8)$$

From $\triangle BPL$,

$$\tan 60^\circ = \frac{PL}{BL} = \frac{r}{BL} = \frac{\sqrt{3}}{2BL}$$

Therefore

$$\begin{aligned} BL &= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} = \frac{1}{2} \\ AL &= AB - BL = 3 - \frac{1}{2} = \frac{5}{2} \end{aligned}$$

Hence

$$\begin{aligned} AP^2 &= AL^2 + PL^2 = \frac{25}{4} + \frac{3}{4} = 7 \\ \Rightarrow AP &= \sqrt{7} \end{aligned}$$

Alternate Solution:

$$\begin{aligned} \sin \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{bc}} \\ &= \sqrt{\frac{[(15/2)-7][(15/2)-3]}{7 \times 3}} \\ &= \frac{\sqrt{3}}{2\sqrt{7}} \end{aligned}$$

Now

AP = The distance between the vertex A and the incentre P

$$\begin{aligned} &= r \operatorname{cosec} \frac{A}{2} \\ &= \frac{\sqrt{3}}{2} \cdot \frac{2\sqrt{7}}{\sqrt{3}} \quad [\text{By Eq. (4.8)}] \\ &= \sqrt{7} \end{aligned}$$

Answer: (D)

35. In $\triangle ABC$, $\angle C = 90^\circ$ and $AB = 60$. The medians AD and BE intersect at G (centroid). If θ is the acute angle between the medians AD and BE whose tangent value is $1/3$, then the area of $\triangle ABC$ is

(A) 200 (B) 300 (C) 400 (D) 500

Solution: See Figure 4.40. We have

$$\angle BGD = \angle AGE = \theta \quad \text{and} \quad \tan \theta = \frac{1}{3}$$

By hypothesis

$$\begin{aligned} (BC)^2 + (AC)^2 &= (AB)^2 \\ a^2 + b^2 &= 60^2 \end{aligned} \quad (4.9)$$

Consider the quadrilateral $DGEC$ in which

$$\angle DGE + \angle GDC + 90^\circ + \angle CEG = 360^\circ$$

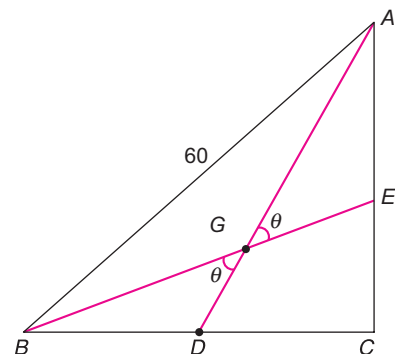


FIGURE 4.40 Single correct choice type question 35.

Therefore

$$(180^\circ - \theta) + \underline{D} + 90^\circ + \underline{E} = 360^\circ$$

where $\underline{D} = \underline{GDC}$ and $\underline{E} = \underline{CEG}$. This gives

$$\underline{D} + \underline{E} = 90^\circ + \theta \quad (4.10)$$

Therefore

$$-3 = -\cot \theta = \tan(90^\circ + \theta) = \tan(D + E) \quad [\text{from Eq. (4.10)}]$$

$$\begin{aligned} &= \frac{\tan D + \tan E}{1 - \tan D \tan E} \\ &= \frac{(AC/DC) + (BC/EC)}{1 - (AC/DC) \cdot (BC/EC)} \\ &= \frac{(2b/a) + (2a/b)}{1 - 4} \end{aligned}$$

Therefore

$$\begin{aligned} 9\left(\frac{ab}{2}\right) &= a^2 + b^2 = 3600 \\ 9\Delta &= 3600 \\ \Delta &= 400 \end{aligned}$$

Answer: (C)

36. In $\triangle ABC$ if the line joining the incentre and centroid is parallel to the side BC , then the sides a , b and c (in certain order) form a/an

(A) AP (B) GP (C) HP (D) AGP

Solution: In $\triangle ABC$, AE is the bisector of $\angle A$, AD is the median, I is the incentre and G is the centroid (Figure 4.41). IG is parallel to BC . This implies

$$\begin{aligned} \frac{AI}{IE} &= \frac{AG}{GD} = \frac{2}{1} \\ \frac{b+c}{a} &= \frac{2}{1} \quad [\because I \text{ divides } AE \text{ in the ratio } (b+c):a] \\ b+c &= 2a \end{aligned}$$

Hence c , a , b are in AP.

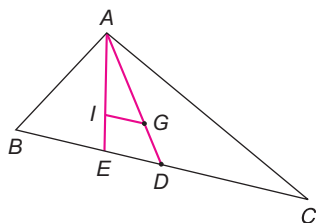


FIGURE 4.41 Single correct choice type question 36.

Answer: (A)

37. AD , BE and CF are medians of $\triangle ABC$. Then $(AD)^2 + (BE)^2 + (CF)^2 = K(a^2 + b^2 + c^2)$ where the value of K is

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

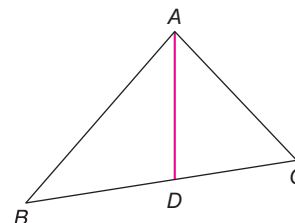


FIGURE 4.42 Single correct choice type question 37.

Solution: D is the mid-point of the side BC (Figure 4.42). Using cosine rule for $\triangle ABD$, we get

$$\begin{aligned} (AD)^2 &= (AB)^2 + (BD)^2 - 2(AB)(BD)\cos B \\ &= c^2 + \frac{a^2}{4} - 2(c)\left(\frac{a}{2}\right)\left(\frac{c^2 + a^2 - b^2}{2ca}\right) \\ &= c^2 + \frac{a^2}{4} - \frac{c^2 + a^2 - b^2}{2} \\ &= \frac{2c^2 + 2b^2 - a^2}{4} \end{aligned}$$

Therefore

$$\begin{aligned} (AD)^2 + (BE)^2 + (CF)^2 &= \frac{1}{4}(4a^2 + 4b^2 + 4c^2 - (a^2 + b^2 + c^2)) \\ &= \frac{3}{4}(a^2 + b^2 + c^2) \end{aligned}$$

Hence $K = 3/4$.

Answer: (A)

38. The sides of a triangle are in AP. Its area is $3/5$ times the area of an equilateral triangle whose perimeter is so as the given triangle. Then the ratio of the sides is
(A) 5:3:7 (B) 3:5:7 (C) 7:4:3 (D) 3:4:7

Solution: Let the sides be $x - d$, x and $x + d$. Therefore the perimeter

$$2s = 3x \quad \text{or} \quad s = \frac{3x}{2}$$

Then

$$\begin{aligned} \Delta &= \sqrt{s(s-x+d)(s-x)(s-x-d)} \\ &= \frac{x}{2} \sqrt{3\left(\frac{x^2}{4} - d^2\right)} \\ &= \frac{x}{4} \sqrt{3(x^2 - 4d^2)} \end{aligned}$$

But the area of the equilateral triangle with the same semi-perimeter $3x/2$ is $(\sqrt{3}/4)x^2$. Therefore

$$\begin{aligned}\Delta &= \frac{3}{5} \left(\frac{\sqrt{3}}{4} x^2 \right) \\ \frac{x^2}{16} \times 3(x^2 - 4d^2) &= \frac{9}{25} \times \frac{3}{16} \times x^4 \\ x^2 - 4d^2 &= \frac{9}{25} \cdot x^2 \\ \frac{16}{25} x^2 &= 4d^2 \\ x &= \frac{5}{2} d\end{aligned}$$

Hence the sides of the triangle are

$$\begin{aligned}x - d &= \frac{5}{2} d - d = \frac{3d}{2} \\ x &= \frac{5d}{2}\end{aligned}$$

and

$$x + d = \frac{5d}{2} + d = \frac{7}{2} d$$

Therefore the ratio of the sides is 3:5:7.

Answer: (B)

39. If p and q are the lengths of the perpendiculars drawn from the vertices A and B of $\triangle ABC$ onto any line passing through the vertex C , then the value of $a^2 p^2 + b^2 q^2 - 2abpq \cos C$ is

- (A) $a^2 b^2 \sin^2 C$ (B) $a^2 b^2 \cos^2 C$
(C) $a^2 b^2 \tan^2 C$ (D) $a^2 b^2 \csc^2 C$

Solution: Let AM and BN be perpendiculars drawn from A and B , respectively, onto a line through the vertex C (Figure 4.43). Then

$$p = AM, q = BN$$

Let $\angle ACM = \alpha$ and $\angle BCN = \beta$. Then

$$\angle C = \pi - (\alpha + \beta)$$

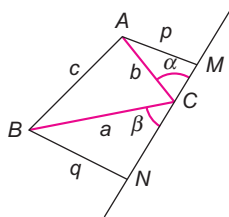


FIGURE 4.43 Single correct choice type question 39.

Now

$$\sin \alpha = \frac{p}{b} \quad \text{and} \quad \sin \beta = \frac{q}{a}$$

Therefore

$$\begin{aligned}\cos \alpha &= \frac{1}{b} \sqrt{b^2 - p^2} \\ \cos \beta &= \frac{1}{a} \sqrt{a^2 - q^2} \\ \sin C &= \sin[\pi - (\alpha + \beta)] \\ &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= \frac{1}{ab} [p \sqrt{a^2 - q^2} + q \sqrt{b^2 - p^2}]\end{aligned}$$

Squaring and cross-multiplying we get

$$a^2 b^2 \sin^2 C = [p \sqrt{a^2 - q^2} + q \sqrt{b^2 - p^2}]^2 \quad (4.11)$$

Now

$$\begin{aligned}a^2 p^2 + b^2 q^2 - 2abpq \cos C &= a^2 p^2 + b^2 q^2 + 2abpq \cos(\alpha + \beta) \\ &= a^2 p^2 + b^2 q^2 + 2abpq [\cos \alpha \cos \beta - \sin \alpha \sin \beta] \\ &= a^2 p^2 + b^2 q^2 + 2abpq \left[\frac{1}{ab} \sqrt{(a^2 - q^2)(b^2 - p^2)} - \frac{pq}{ab} \right] \\ &= a^2 p^2 + b^2 q^2 + 2pq [\sqrt{(a^2 - q^2)(b^2 - p^2)} - pq] \\ &= a^2 p^2 + b^2 q^2 - 2p^2 q^2 + 2pq \sqrt{(a^2 - q^2)(b^2 - p^2)} \\ &= p^2 (a^2 - q^2) + q^2 (b^2 - p^2) + 2pq \sqrt{(a^2 - q^2)(b^2 - p^2)} \\ &= (p \sqrt{a^2 - q^2} + q \sqrt{b^2 - p^2})^2 \quad (4.12)\end{aligned}$$

From Eqs. (4.11) and (4.12) it follows that

$$a^2 p^2 + b^2 q^2 - 2abpq \cos C = a^2 b^2 \sin^2 C$$

Answer: (A)

40. In an acute-angled $\triangle ABC$, the circle described on the altitude AD as diameter cuts the sides AB and AC , respectively, in P and Q . Then the length of the chord PQ is equal to

- (A) $\frac{\Delta}{2R}$ (B) $\frac{\Delta}{2r}$ (C) $\frac{\Delta}{R}$ (D) $\frac{\Delta}{r}$

Solution: See Figure 4.44. From $\triangle APQ$, we have

$$\frac{PQ}{\sin A} = AD$$

Therefore

$$\begin{aligned} PQ &= (AD) \sin A = (b \sin C) \sin A \\ &= b \left(\frac{c}{2R} \right) \sin A \\ &= \left(\frac{1}{2} bc \sin A \right) \frac{1}{R} \\ &= \frac{\Delta}{R} \end{aligned}$$

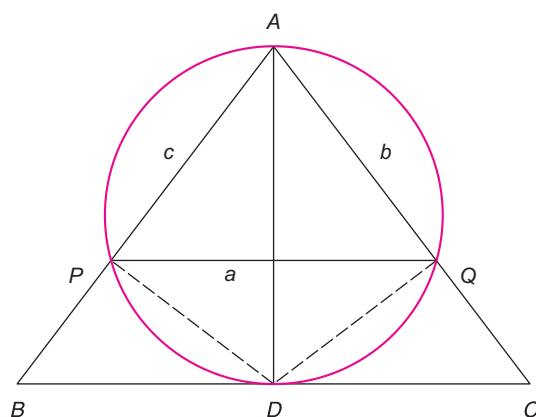


FIGURE 4.44 Single correct choice type question 40.

Answer: (C)

41. $A_1, A_2, \dots, A_n$ are the vertices of a regular polygon of n sides such that

$$\frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$$

Then, the value of n is

- (A) 5 (B) 6 (C) 7 (D) 9

Solution: See Figure 4.45. Let the centre of the polygon be “O”. Therefore

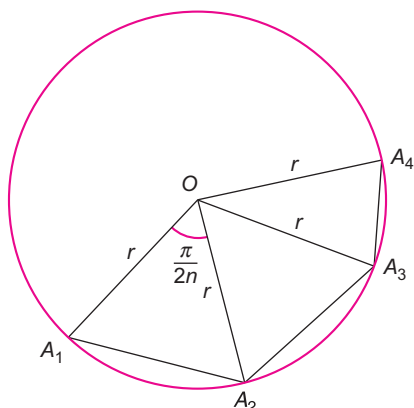


FIGURE 4.45 Single correct choice type question 41.

$$\angle A_1 O A_2 = \frac{\pi}{2n}$$

Let the radius of the circle circumscribing the polygon be r . Therefore

$$\begin{aligned} (A_1 A_2)^2 &= r^2 + r^2 - 2r^2 \cos \left(\frac{\pi}{2n} \right) \\ &= 4r^2 \sin^2 \frac{\pi}{n} \end{aligned}$$

Similarly

$$(A_1 A_3)^2 = 4r^2 \sin^2 \frac{2\pi}{n}$$

and

$$(A_1 A_4)^2 = 4r^2 \sin^2 \frac{3\pi}{n}$$

By hypothesis

$$\frac{1}{\sin(\pi/n)} = \frac{1}{\sin(2\pi/n)} + \frac{1}{\sin(3\pi/n)}$$

Therefore

$$\begin{aligned} \sin \frac{2\pi}{n} \sin \frac{3\pi}{n} &= \sin \frac{\pi}{n} \left(\sin \frac{2\pi}{n} + \sin \frac{3\pi}{n} \right) \\ 2 \cos \frac{\pi}{n} \sin \frac{3\pi}{n} &= \sin \frac{2\pi}{n} + \sin \frac{3\pi}{n} \\ \sin \frac{4\pi}{n} + \sin \frac{2\pi}{n} &= \sin \frac{2\pi}{n} + \sin \frac{3\pi}{n} \\ \sin \frac{4\pi}{n} &= \sin \frac{3\pi}{n} \\ \sin \frac{4\pi}{n} - \sin \frac{3\pi}{n} &= 0 \\ 2 \cos \frac{7\pi}{2n} \sin \frac{\pi}{2n} &= 0 \end{aligned}$$

Now

$$\begin{aligned} \sin \frac{\pi}{2n} \neq 0 &\Rightarrow \cos \frac{7\pi}{2n} = 0 \\ &\Rightarrow \frac{7\pi}{2n} = \frac{\pi}{2} \\ &\Rightarrow n = 7 \end{aligned}$$

Answer: (C)

42. In $\triangle ABC$, angle A is twice angle B . Then $a^2 - b^2$ is equal to

- (A) 2Δ (B) bc (C) ca (D) ab

Solution:

$$\begin{aligned}
 a^2 - b^2 &= 4R^2(\sin^2 A - \sin^2 B) \\
 &= 4R^2 \sin(A+B) \sin(A-B) \\
 &= 4R^2 \sin C \sin B \quad (\because A = 2B) \\
 &= (2R \sin B)(2R \sin C) \\
 &= bc
 \end{aligned}$$

Answer: (B)**43.** In $\triangle ABC$, let

$$\begin{aligned}
 x &= \tan \frac{B-C}{2} \tan \frac{A}{2}, \quad y = \tan \frac{C-A}{2} \tan \frac{B}{2}, \\
 z &= \tan \frac{A-B}{2} \tan \frac{C}{2}
 \end{aligned}$$

Then $x + y + z = K(xyz)$, where the value of K is

- (A) 2 (B) -2 (C) 1 (D) -1

Solution: By Napier's rule

$$\tan \frac{B-C}{2} = \left(\frac{b-c}{b+c} \right) \cot \frac{A}{2}$$

Therefore

$$x = \frac{b-c}{b+c}, \quad y = \frac{c-a}{a+c}, \quad z = \frac{a-b}{a+b}$$

Now

$$x = \frac{b-c}{b+c} \Rightarrow \frac{x+1}{x-1} = \frac{-b}{c}$$

Similarly

$$\frac{y+1}{y-1} = \frac{-c}{a} \quad \text{and} \quad \frac{z+1}{z-1} = \frac{-a}{b}$$

Therefore

$$\left(\frac{x+1}{x-1} \right) \left(\frac{y+1}{y-1} \right) \left(\frac{z+1}{z-1} \right) = -1$$

$$(x+1)(y+1)(z+1) = -(x-1)(y-1)(z-1)$$

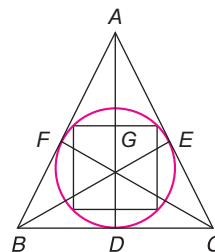
On expanding the products both sides and simplifying we have

$$2xyz = -2(x+y+z)$$

$$x+y+z = -xyz$$

Therefore $K = -1$.**Answer: (D)****44.** A circle is inscribed in an equilateral triangle of side “ a ” units. Then, the area of any square inscribed in the circle is

- (A)
- $\frac{a^2}{4}$
- (B)
- $\frac{a^2}{6}$
- (C)
- $\frac{a^2}{8}$
- (D)
- $\frac{a^2}{12}$

**FIGURE 4.46** Single correct choice type question 44.**Solution:** Let $\triangle ABC$ be equilateral with side lengths “ a ”. Let AD , BE , CF be the medians and hence altitudes and G the centroid (orthocentre) (Figure 4.46). Therefore

$$AD = \frac{a\sqrt{3}}{2} \quad \left(\because \sin 60^\circ = \frac{AD}{a} \right)$$

Now

$$r = \text{Radius of the circle} = GD = \frac{AD}{3} = \frac{a}{2\sqrt{3}}$$

If x is the side of the square, then

$$x\sqrt{2} = 2r = \frac{a}{\sqrt{3}}$$

Therefore

$$\text{Area of the square} = x^2 = \frac{a^2}{6}$$

Answer: (B)**45.** In any $\triangle ABC$, if $\cos A + \cos B + \cos C = 3/2$, then the triangle is

- (A) right angled
-
- (B) right-angled isosceles
-
- (C) isosceles but not equilateral
-
- (D) equilateral

Solution:

$$2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} + 1 - 2 \sin^2 \frac{C}{2} = \frac{3}{2}$$

$$2 \sin \frac{C}{2} \cos \frac{(A-B)}{2} = \frac{1}{2} + 2 \sin^2 \frac{C}{2}$$

$$\begin{aligned}\cos \frac{A-B}{2} &= \frac{1}{4\sin(C/2)} + \sin \frac{C}{2} \\ &= \left(\sqrt{\sin \frac{C}{2}} - \frac{1}{2\sqrt{\sin(C/2)}} \right)^2 + 1 \geq 1\end{aligned}$$

But

$$\cos \frac{A-B}{2} \leq 1$$

Therefore

$$\cos \frac{A-B}{2} = 1 \quad \text{and} \quad \sin \frac{C}{2} = \frac{1}{2}$$

This implies $A = B = C = 60^\circ$.

Answer: (D)

46. In $\triangle ABC$, the minimum value of the sum of the squares of its sides is

- (A) $3\sqrt{3}\Delta$ (B) $2\sqrt{3}\Delta$
(C) $4\sqrt{3}\Delta$ (D) $\sqrt{3}\Delta$

where Δ is the area of $\triangle ABC$.

Solution: By cosine law

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 + b^2 + c^2 = 2(b^2 + c^2) - 2bc \cos A$$

We know that

$$\Delta = \frac{1}{2}bc \sin A$$

Therefore

$$4\Delta\sqrt{3} = (2bc)\sqrt{3} \sin A$$

Hence

$$\begin{aligned}a^2 + b^2 + c^2 - 4\sqrt{3}\Delta &= 2(b^2 + c^2) - 2bc(\cos A + \sqrt{3} \sin A) \\ &= 2(b^2 + c^2) - 4bc \left(\frac{1}{2} \cos A + \frac{\sqrt{3}}{2} \sin A \right) \\ &= 2(b^2 + c^2) - 4bc \cos \left(\frac{\pi}{3} - A \right) \geq 2(b^2 + c^2) - 4bc \\ &= 2(b - c)^2 \geq 0\end{aligned}$$

Equality holds if $b = c$. Similarly

$$a^2 + b^2 + c^2 - 4\sqrt{3}\Delta \geq 2(c - a)^2$$

and equality holds if $c = a$. Hence

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}\Delta$$

and equality holds when $a = b = c$. This means that the minimum value of $a^2 + b^2 + c^2$ is $4\sqrt{3}\Delta$ and this minimum is attained when $\triangle ABC$ is equilateral. Therefore among all triangles, equilateral triangles are those such that the minimum value of $a^2 + b^2 + c^2$ is $4\sqrt{3}\Delta$.

Answer: (C)

47. In an equilateral triangle, 3 coins of radii 1 unit each are kept so that they touch each other and also touch the sides of the triangle. Area of the triangle is

- (A) $4 + 2\sqrt{3}$ (B) $6 + 4\sqrt{3}$
(C) $12 + \frac{7\sqrt{3}}{4}$ (D) $3 + \frac{7\sqrt{3}}{4}$

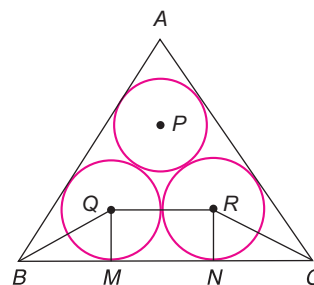


FIGURE 4.47 Single correct choice type question 47.

Solution: Let Q, R be the centres of the circles touching the side BC at M and N , respectively (Figure 4.47). In $\triangle QBM$, $\angle QBM = 30^\circ$. Therefore

$$BM = (QM) \cot 30^\circ = \sqrt{3}$$

$$MN = QR = 2 \quad \text{and} \quad NC = \sqrt{3}$$

Therefore the side BC is equal to

$$BC = BM + MN + NC = 2 + 2\sqrt{3}$$

Hence

$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{\sqrt{3}}{4}(BC)^2 \\ &= \frac{\sqrt{3}}{4}(2 + 2\sqrt{3})^2 \\ &= \frac{\sqrt{3}}{4}(4 + 12 + 8\sqrt{3}) \\ &= 6 + 4\sqrt{3}\end{aligned}$$

Answer: (B)

48. In $\triangle ABC$, let $\angle C = \pi/2$. If r is the inradius and R is the circumradius of the triangle, then $2(r + R)$ is

(A) $a + b - c$ (B) $b + c$ (C) $c + a$ (D) $a + b$

Solution: See Figure 4.48. We have $\angle C = 90^\circ$, so

$$\frac{c}{\sin C} = 2R \Rightarrow R = \frac{c}{2}$$

Also

$$r = (s - c) \tan \frac{C}{2} = s - c$$

Therefore

$$\begin{aligned} 2(r + R) &= 2r + 2R \\ &= 2(s - c) + c \\ &= 2s - c \\ &= (a + b + c) - c \\ &= a + b \end{aligned}$$

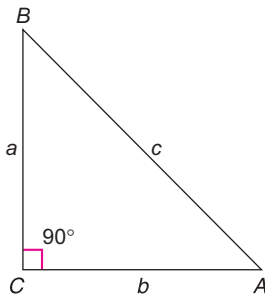


FIGURE 4.48 Single correct choice type question 48.

Answer: (D)

49. The sides of a triangle are in the ratio $1:\sqrt{3}:2$. Then the angles opposite to these sides are in the ratio

(A) 1:3:5 (B) 2:3:4 (C) 1:2:3 (D) 3:1:2

Solution: In $\triangle ABC$, let $BC = a$, $CA = b$, $AB = c$. By hypothesis,

$$a:b:c = 1:\sqrt{3}:2$$

Let $a = K$, $b = \sqrt{3}K$, $c = 2K$. Therefore

$$\begin{aligned} \cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{3 + 4 - 1}{2\sqrt{3}(2)} = \frac{\sqrt{3}}{2} \\ \Rightarrow A &= \frac{\pi}{6} \end{aligned}$$

Also

$$\begin{aligned} \cos B &= \frac{c^2 + a^2 - b^2}{2ca} = \frac{4 + 1 - 3}{2(2)} = \frac{1}{2} \\ \Rightarrow B &= \frac{\pi}{3} \end{aligned}$$

Hence

$$C = \frac{\pi}{2}$$

So $A:B:C = 1:2:3$.

Answer: (C)

50. Let $A_0A_1A_2A_3A_4A_5$ be a regular polygon of six sides inscribed in a circle of unit radius. Then the product of the line segments A_0A_1 , A_0A_2 and A_0A_4 is

(A) $\frac{3}{4}$ (B) 3 (C) $3\sqrt{3}$ (D) $\frac{3\sqrt{3}}{2}$

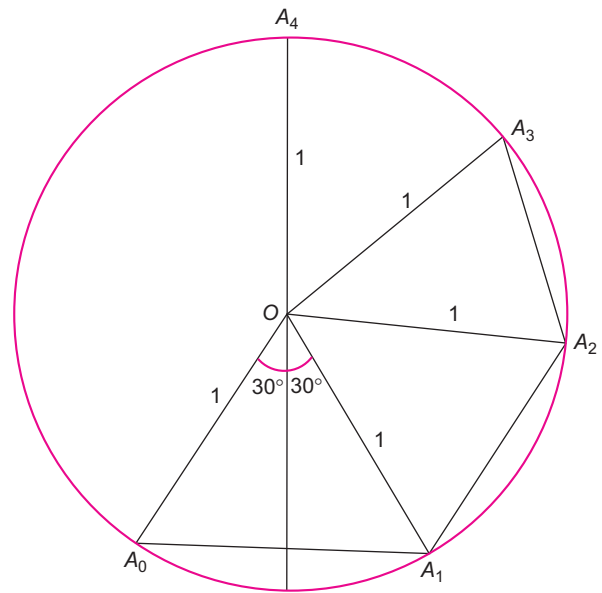


FIGURE 4.49 Single correct choice type question 50.

Solution: See Figure 4.49. Let O be the centre of the hexagon. Therefore

$$\angle A_0OA_1 = 60^\circ, \quad \angle A_0OA_2 = 120^\circ, \quad \angle A_0OA_4 = 120^\circ$$

By cosine formula

$$\begin{aligned} (A_0A_1)^2 &= 1^2 + 1^2 - 2(1)(1)\cos 60^\circ \\ &= 2(1 - \cos 60^\circ) \\ &= 4\sin^2 30^\circ = 1 \\ (A_0A_2)^2 &= 1 + 1 - 2\cos 120^\circ \\ &= 2(1 - \cos 120^\circ) \\ &= 2\left(1 + \frac{1}{2}\right) = 3 \\ (A_0A_4)^2 &= 1 + 1 - 2\cos 120^\circ \\ &= 3 \end{aligned}$$

Therefore

$$(A_0A_1)(A_0A_2)(A_0A_4) = 1 \cdot \sqrt{3} \cdot \sqrt{3} = 3$$

Answer: (B)

51. If the angles of a triangle are in the ratio 4:1:1, then the ratio of the largest side to the perimeter of the triangle is

(A) $\sqrt{3}:(2+\sqrt{3})$ (B) 1:6

(C) $1:2+\sqrt{3}$ (D) 2:3

Solution: Let the angles be $A = 4\theta$, $B = \theta$, $C = \theta$.
Therefore

$$6\theta = 180^\circ \Rightarrow \theta = 30^\circ$$

Hence the angles are $120^\circ, 30^\circ, 30^\circ$. Let $A = 120^\circ$, $B = 30^\circ$, $C = 30^\circ$. Therefore side BC is the largest side. Hence

$$\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ} = 2R$$

which gives

$$a = (2R)\sin 120^\circ = R\sqrt{3}$$

and $b = c = R$

Now

$$2s = \text{Perimeter} = a + b + c = R(\sqrt{3} + 2)$$

So

$$a:2s = \sqrt{3}R:R(2+\sqrt{3}) = \sqrt{3}:(2+\sqrt{3})$$

Answer: (A)

52. In $\triangle ABC$, if

$$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$$

then angle C in degrees is

(A) 60° (B) 45° (C) 30° (D) 90°

Solution:

$$\begin{aligned} \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} &= \frac{bc \cos A + ca \cos B + ab \cos C}{abc} \\ &= \frac{a^2 + b^2}{abc} \quad (\text{given}) \end{aligned}$$

Therefore

$$\left(\frac{b^2 + c^2 - a^2}{2} \right) + \left(\frac{c^2 + a^2 - b^2}{2} \right) + \left(\frac{a^2 + b^2 - c^2}{2} \right) = a^2 + b^2$$

$$a^2 + b^2 + c^2 = 2(a^2 + b^2)$$

$$a^2 + b^2 = c^2$$

$$\angle C = 90^\circ$$

Answer: (D)

53. In $\triangle ABC$, if the median AD is the Geometrical Mean between the sides AB and AC , then $1 + \cos A$ is equal to

(A) $\frac{1}{2} |\cos B - \cos C|$ (B) $\sqrt{2} |\cos B - \cos C|$

(C) $\frac{1}{\sqrt{2}} |\cos B - \cos C|$ (D) $2 |\cos B - \cos C|$

Solution: By hypothesis,

$$(AD)^2 = AB \cdot AC$$

$$(AD)^2 = bc \quad (4.13)$$

But we know that

$$(AD)^2 = \frac{1}{4}(2b^2 + 2c^2 - a^2) \quad (\text{see Theorem 4.25}) \quad (4.14)$$

Therefore from Eqs. (4.13) and (4.14) we get

$$2b^2 + 2c^2 - a^2 = 4bc$$

$$a^2 = 2(b - c)^2$$

$$a = \sqrt{2} |b - c|$$

$$\sin A = \sqrt{2} |\sin B - \sin C|$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = 2\sqrt{2} \left| \cos \frac{B+C}{2} \sin \frac{B-C}{2} \right|$$

$$\cos \frac{A}{2} = \sqrt{2} \left| \sin \frac{B-C}{2} \right|$$

$$2 \cos^2 \frac{A}{2} = \sqrt{2} \left| 2 \cos \frac{A}{2} \sin \frac{B-C}{2} \right|$$

$$= \sqrt{2} \left| 2 \sin \frac{B+C}{2} \sin \frac{B-C}{2} \right|$$

$$= \sqrt{2} |\cos B - \cos C|$$

Therefore

$$1 + \cos A = \sqrt{2} |\cos B - \cos C|$$

Answer: (B)

54. Let ABC be a triangle and $p = \sin A \sin B \sin C$, $q = \cos A \cos B \cos C$. Then $\tan A$, $\tan B$ and $\tan C$ are the roots of the equation

- (A) $qx^3 - px^2 + (1+q)x - p = 0$
 (B) $qx^3 + px^2 - (1+q)x + p = 0$
 (C) $qx^3 - px^2 + (1+q)x + p = 0$
 (D) $qx^3 - px^2 - (1+q)x - p = 0$

Solution:

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C = \frac{p}{q}$$

Now

$$\begin{aligned} \sum \tan B \tan C &= \frac{-\cos(A+B+C) + \cos A \cos B \cos C}{\cos A \cos B \cos C} \\ &= \frac{1+q}{q} \end{aligned}$$

Therefore $\tan A$, $\tan B$, $\tan C$ are roots of the equation

$$\begin{aligned} x^3 - \frac{px^2}{q} + \frac{(1+q)}{q}x - \frac{p}{q} &= 0 \\ qx^3 - px^2 + (1+q)x - p &= 0 \end{aligned}$$

Answer: (A)

55. In $\triangle ABC$, $\angle B = 90^\circ$. A line passing through the vertex A meets the side BC at a point D inside the circumcircle of the triangle ABC . If $\angle DAB = \alpha$, $\angle CAB = \beta$ and the distance between A and the mid-point of DC is d , then the area of the circumcircle is

- (A) $\frac{\pi d^2 \cos^2 \alpha}{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta \cos(\beta - \alpha)}$
 (B) $\frac{2\pi d^2 \cos^2 \alpha}{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta \cos(\beta - \alpha)}$
 (C) $\frac{\pi d^2 \cos^2 \alpha}{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta}$
 (D) $\frac{2\pi d^2 \cos^2 \alpha}{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta}$

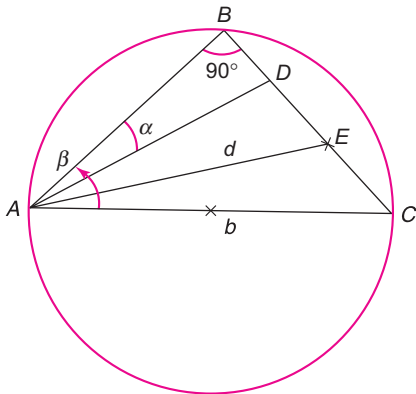


FIGURE 4.50 Single correct choice type question 55.

Solution: See Figure 4.50. Let E be the mid-point of DC so that the distance AE is d . Let R be the circumradius of the $\triangle ABC$. Therefore $b = AC = 2R$. From $\triangle ABC$,

$$\cos \beta = \frac{AB}{AC} = \frac{AB}{2R}$$

Therefore

$$AB = (2R) \cos \beta \quad (4.15)$$

From $\triangle ABD$,

$$\cos \alpha = \frac{AB}{AD} = \frac{2R \cos \beta}{AD}$$

Hence

$$AD = \frac{2R \cos \beta}{\cos \alpha} \quad (4.16)$$

Also

$$\sin \alpha = \frac{BD}{AD}$$

and hence

$$BD = \frac{2R \cos \beta \sin \alpha}{\cos \alpha} \quad (4.17)$$

Now,

$$\begin{aligned} DE &= \frac{1}{2} DC \\ &= \frac{1}{2} (BC - BD) \\ &= \frac{1}{2} \left(2R \sin \beta - \frac{2R \cos \beta \sin \alpha}{\cos \alpha} \right) \\ &= \frac{R \sin(\beta - \alpha)}{\cos \alpha} \end{aligned} \quad (4.18)$$

In $\triangle ACD$, AE being the median,

$$\begin{aligned} 4(AE)^2 &= 2(AC)^2 + 2(AD)^2 - (DC)^2 \\ &= 2(AC)^2 + 2(AD)^2 - 4(DE)^2 \end{aligned}$$

$$(AC)^2 + (AD)^2 = 2(AE)^2 + 2(DE)^2$$

Using Eqs. (4.15)–(4.18) we have

$$4R^2 + \frac{4R^2 \cos^2 \beta}{\cos^2 \alpha} = 2d^2 + \frac{2R^2 \sin^2(\beta - \alpha)}{\cos^2 \alpha}$$

$$R^2 \left[\frac{2 \cos^2 \beta}{\cos^2 \alpha} + 2 - \frac{\sin^2(\beta - \alpha)}{\cos^2 \alpha} \right] = d^2$$

$$R^2 [2 \cos^2 \beta + 2 \cos^2 \alpha - \sin^2(\beta - \alpha)] = d^2 \cos^2 \alpha$$

$$\begin{aligned}
 &R^2[2\cos^2\beta + 2\cos^2\alpha - \sin^2\beta\cos^2\alpha - \sin^2\alpha\cos^2\beta \\
 &\quad + 2\sin\alpha\sin\beta\cos\alpha\cos\beta] = d^2\cos^2\alpha \\
 &R^2[\cos^2\beta + \cos^2\alpha + \cos^2\beta(1 - \sin^2\alpha) + \cos^2\alpha(1 - \sin^2\beta) \\
 &\quad + 2\sin\alpha\sin\beta\cos\alpha\cos\beta] = d^2\cos^2\alpha \\
 &R^2[\cos^2\beta + \cos^2\alpha + 2\cos^2\alpha\cos^2\beta \\
 &\quad + 2\sin\alpha\sin\beta\cos\alpha\cos\beta] = d^2\cos^2\alpha \\
 &R^2[\cos^2\alpha + \cos^2\beta + 2\cos\alpha\cos\beta\cos(\beta - \alpha)] = d^2\cos^2\alpha
 \end{aligned}$$

Therefore area of the circle is

$$\pi R^2 = \frac{\pi d^2 \cos^2 \alpha}{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta \cos(\beta - \alpha)}$$

Answer: (A)

56. In $\triangle ABC$, if the angles A, B, C satisfy $A : B : C = 1 : 2 : 4$, then $(a^2 - b^2)(b^2 - c^2)(c^2 - a^2)$ is equal to

- (A) $a^2 + b^2 + c^2$ (B) $-a^2 b^2 c^2$
 (C) $\frac{1}{2}(a^2 + b^2 + c^2)$ (D) $2a^2 b^2 c^2$

Solution:

$$A + 2A + 4A = \pi \Rightarrow A = \frac{\pi}{7}$$

Therefore

$$B = \frac{2\pi}{7}, C = \frac{4\pi}{7}$$

Now,

$$\begin{aligned}
 &(a^2 - b^2)(b^2 - c^2)(c^2 - a^2) \\
 &= 64R^6(\sin^2 A - \sin^2 B)(\sin^2 B - \sin^2 C)(\sin^2 C - \sin^2 A) \\
 &= 64R^6[\sin(A + B)\sin(A - B)][\sin(B + C)\sin(B - C)] \\
 &\quad [\sin(C + A)\sin(C - A)] \\
 &= 64R^6 \left[\sin \frac{3\pi}{7} \sin \left(\frac{-\pi}{7} \right) \right] \left[\sin \frac{6\pi}{7} \sin \left(\frac{-2\pi}{7} \right) \right] \\
 &\quad \left[\sin \frac{5\pi}{7} \sin \left(\frac{-3\pi}{7} \right) \right] \\
 &= -64R^6 \sin^2 \frac{\pi}{7} \sin^2 \frac{2\pi}{7} \sin^2 \frac{4\pi}{7} \\
 &= - \left(4R^2 \sin^2 \frac{\pi}{7} \right) \left(4R^2 \sin^2 \frac{2\pi}{7} \right) \left(4R^2 \sin^2 \frac{4\pi}{7} \right) \\
 &= -a^2 b^2 c^2
 \end{aligned}$$

Answer: (B)

57. In $\triangle ABC$, if $a^2 + b^2 = (2011)c^2$, then

$$\frac{\cot C}{\cot A + \cot B} =$$

(A) 2010 (B) 1006 (C) 1005 (D) 1005.5

Solution:

$$\begin{aligned}
 \frac{\cot C}{\cot A + \cot B} &= \frac{\cos C \sin A \sin B}{\sin^2 C} \\
 &= \frac{[(a^2 + b^2 - c^2)/2ab](ab/4R^2)}{c^2/4R^2} \\
 &= \frac{a^2 + b^2 - c^2}{2c^2} \\
 &= \frac{(2011)c^2 - c^2}{2c^2} \\
 &= \frac{(2010)c^2}{2c^2} = 1005
 \end{aligned}$$

Answer: (C)

58. In $\triangle ABC$, if the perimeter is 6 times the Arithmetic Mean (AM) of $\sin A, \sin B$ and $\sin C$ and the side BC is of length 1 unit, then the angle A (in degrees) is

- (A) 60° (B) 45° (C) 75° (D) 30°

Solution: Let $a = BC, b = CA$ and $c = AB$. By hypothesis,

$$a + b + c = \frac{6}{3}(\sin A + \sin B + \sin C)$$

Therefore

$$\begin{aligned}
 2R(\sin A + \sin B + \sin C) &= 2(\sin A + \sin B + \sin C) \\
 \Rightarrow R &= 1
 \end{aligned}$$

Now,

$$1 = BC = a = 2R \sin A = 2 \sin A$$

Therefore

$$\sin A = \frac{1}{2} \Rightarrow A = 30^\circ$$

Answer: (D)

59. If $x > 1$, then the largest angle of the triangle whose sides are of lengths $x^2 + x + 1, 2x + 1$ and $x^2 - 1$ is

- (A) 150° (B) 120° (C) 110° (D) 135°

Solution: Let $a = x^2 + x + 1, b = 2x + 1$ and $c = x^2 - 1$. Clearly, a is greater than b and c because $x > 1$ and also sum of any two of a, b, c is greater than the third. Hence they form a triangle and " a " is the greatest side. If α is the angle opposite to the side a , then

$$\begin{aligned}\cos \alpha &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{(2x+1)^2 + (x^2-1)^2 - (x^2+x+1)^2}{2(2x+1)(x^2-1)} \\ &= \frac{-(2x^3+x^2-2x-1)}{2(2x^3+x^2-2x-1)} \\ &= \frac{-1}{2}\end{aligned}$$

Hence $\alpha = 120^\circ$.

Answer: (B)

60. In $\triangle ABC$, if

$$a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$$

then

- (A) $b + c = 2a$ (B) $a + b = 2c$
(C) $ac = b^2$ (D) $a + c = 2b$

Solution: It is given that

$$a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$$

Therefore

$$\begin{aligned}a(1 + \cos C) + c(1 + \cos A) &= 3b \\ a + c + (a \cos C + c \cos A) &= 3b \\ (a + c) + b &= 3b \\ a + c &= 2b\end{aligned}$$

Answer: (D)

61. In $\triangle ABC$, the sides are of lengths 17, 25, 28 units. Then, the length of the greatest altitude is

- (A) $\frac{420}{17}$ (B) 26 (C) 26.4 (D) 27.5

Solution: Let the triangle be ABC where $BC = 17$, $CA = 25$ and $AB = 28$. Suppose AD , BE and CF are the altitudes. Then

$$\begin{aligned}\Delta^2 &= s(s-a)(s-b)(s-c) \\ &= 35 \times 18 \times 10 \times 7 \\ &= 5^2 \times 6^2 \times 7^2 \\ &= (210)^2\end{aligned}$$

So $\Delta = 210$. Now

$$\Delta = \frac{1}{2} AD \cdot BC = \frac{1}{2} BE \cdot CA = \frac{1}{2} CF \cdot AB$$

Therefore

$$\begin{aligned}AD &= \frac{2\Delta}{BC} = \frac{420}{17} \\ BE &= \frac{2\Delta}{CA} = \frac{420}{25} \\ CF &= \frac{2\Delta}{AB} = \frac{420}{28}\end{aligned}$$

So $AD = 420/17$ is the greatest altitude.

Note: The reader can see that, in a triangle with unequal sides, the largest altitude is the altitude drawn from the vertex onto the shortest side.

Answer: (A)

62. In $\triangle ABC$, the sides $a = BC$, $c = AB$ and the angle A are given. If b_1 and b_2 are two values of the side $b = AC$ such that $b_2 = 2b_1$, then the value of $\sin A$ is

- (A) $\frac{\sqrt{9a^2 - c^2}}{2\sqrt{2}c}$ (B) $\frac{\sqrt{9c^2 - a^2}}{2\sqrt{2}a}$
(C) $\frac{\sqrt{9a^2 - c^2}}{2\sqrt{2}a}$ (D) $\frac{\sqrt{9a^2 + c^2}}{2\sqrt{2}c}$

Solution: From cosine rule,

$$b^2 - 2bc \cos A + (c^2 - a^2) = 0$$

for which b_1 and b_2 are roots. Therefore

$$b_1 + b_2 = 2c \cos A \Rightarrow 3b_1 = 2c \cos A \quad (4.19)$$

$$b_1 b_2 = c^2 - a^2 \Rightarrow 2b_1^2 = c^2 - a^2 \quad (4.20)$$

(using $b_2 = 2b_1$). Therefore using Eqs. (4.19) and (4.20) we get

$$\begin{aligned}2\left(\frac{2c \cos A}{3}\right)^2 &= c^2 - a^2 \\ 8c^2 \cos^2 A &= 9(c^2 - a^2) \\ 8c^2(1 - \sin^2 A) &= 9(c^2 - a^2) \\ \sin^2 A &= \frac{9a^2 - c^2}{8c^2} \\ \sin A &= \frac{\sqrt{9a^2 - c^2}}{2\sqrt{2}c}\end{aligned}$$

Answer: (A)

63. The sides BC , CA and AB of $\triangle ABC$ subtend angles α , β and γ at the centre of its circumcircle. Keeping the circle fixed and changing the triangle ABC , then the minimum value of

$$\frac{1}{3} \left[\cos \left(\alpha + \frac{\pi}{2} \right) + \cos \left(\beta + \frac{\pi}{2} \right) + \cos \left(\gamma + \frac{\pi}{2} \right) \right]$$

is

(A) $\frac{-1}{2}$ (B) $\frac{1}{2}$ (C) $\frac{-\sqrt{3}}{2}$ (D) $\frac{-3}{2\sqrt{2}}$

Solution: See Figure 4.51. Note that $\alpha + \beta + \gamma = 2\pi$. Let

$$\begin{aligned} Z &= \frac{1}{3} \left[\cos \left(\alpha + \frac{\pi}{2} \right) + \cos \left(\beta + \frac{\pi}{2} \right) + \cos \left(\gamma + \frac{\pi}{2} \right) \right] \\ &= -\frac{1}{3} (\sin \alpha + \sin \beta + \sin \gamma) \\ &= -\frac{1}{3} \left(4 \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin \frac{\gamma}{2} \right) \\ &= -\frac{1}{3} (4 \sin A \sin B \sin C) \quad (\because \alpha = 2A, \beta = 2B, \gamma = 2C) \end{aligned}$$

In $\triangle ABC$, the maximum value of $\sin A \sin B \sin C$ is $3\sqrt{3}/8$ which will occur when $A = B = C = 60^\circ$. Therefore

$$Z \geq -\frac{1}{3} \left(\frac{3\sqrt{3}}{2} \right) = -\frac{\sqrt{3}}{2}$$

Therefore the minimum value of Z is $-\sqrt{3}/2$.

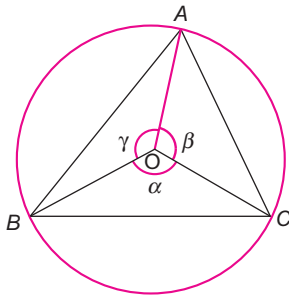


FIGURE 4.51 Single correct choice type question 63.

Answer: (C)

64. In $\triangle ABC$, $\angle A = 90^\circ$ and h is length of the altitude drawn from the vertex A onto the opposite side BC . The side BC is divided into n equal parts (n is odd) such that the segment containing the mid-point D of BC subtends angle α at A . Then $\tan \alpha$ is equal to

(A) $\frac{4nh}{a(n^2 + 1)}$ (B) $\frac{4nh}{a(n^2 - 1)}$
 (C) $\frac{2nh}{a(n^2 + 1)}$ (D) $\frac{2nh}{a(n^2 - 1)}$

Solution: See Figure 4.52. Let $h = AE$ and PQ be the segment containing the mid-point D of the side BC and $\angle PAQ = \alpha$. Since each of BP and QC is divided into $(n-1)/2$ equal parts, we have

$$BP = QC = \left(\frac{n-1}{2} \right) \frac{a}{n}$$

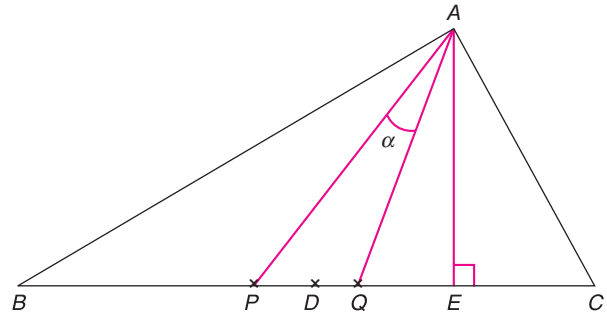


FIGURE 4.52 Single correct choice type question 64.

where a is the length of the side BC . It is clear that $\triangle ABE$ and $\triangle ACE$ are similar triangles. Therefore

$$\begin{aligned} h^2 &= BE \cdot EC = (BP + PE)(QC - QE) \\ &= BP \cdot QC - BP \cdot QE + PE \cdot QC - PE \cdot QE \\ &= BP \cdot QC + BP(PE - QE) - PE \cdot QE \quad (\because BP = QC) \end{aligned}$$

So

$$\begin{aligned} h^2 + PE \cdot QE &= \left(\frac{n-1}{2} \right) \frac{a}{n} \left(\frac{n-1}{2} \right) \frac{a}{n} + BP \cdot PQ \\ &= \frac{(n-1)^2 a^2}{4n^2} + \left(\frac{n-1}{2} \right) \frac{a}{n} \cdot \frac{a}{n} \\ &= \frac{(n^2 - 1)a^2}{4n^2} \end{aligned} \quad (4.21)$$

Let $\angle PAE = \theta_1$ and $\angle QAE = \theta_2$ so that $\theta_1 - \theta_2 = \alpha$. Therefore

$$\begin{aligned} \tan \alpha &= \tan(\theta_1 - \theta_2) \\ &= \frac{\tan \theta_1 - \tan \theta_2}{1 + \tan \theta_1 \tan \theta_2} \\ &= \frac{(PE/h) - (QE/h)}{1 + (PE/h) \cdot (QE/h)} \\ &= \frac{h \cdot PQ}{h^2 + PE \cdot QE} \\ &= \frac{h(a/n)}{[(n^2 - 1)a^2/4n^2]} \quad [\text{from Eq. (4.21)}] \\ &= \frac{(4nh)}{(n^2 - 1)a} \end{aligned}$$

Answer: (B)

65. A is the obtuse angle of $\triangle ABC$. If the median and the altitude from the vertex A divide the angle A into three equal parts, then

$$\left(\sin^3 \frac{A}{3} \right) \left(\cos \frac{A}{3} \right) =$$

- (A) $\frac{3a^3}{16b^2c}$ (B) $\frac{3a^3}{64b^2c}$
 (C) $\frac{9a^3}{128b^2c}$ (D) $\frac{3a^3}{128b^2c}$

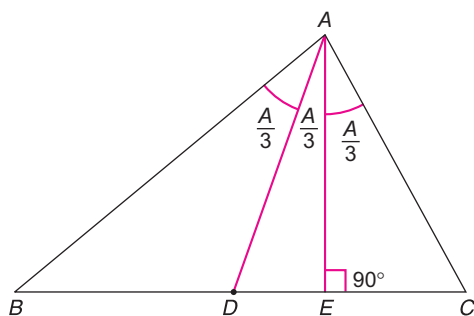


FIGURE 4.53 Single correct choice type question 65.

Solution: See Figure 4.53. In $\triangle ADC$,

$$\angle ADE = 90^\circ - \frac{A}{3} = \angle ACD$$

Therefore

$$AD = AC = b$$

$$DE = EC = \frac{a/2}{2} = \frac{a}{4}$$

Now

$$\begin{aligned} \sin^3 \frac{A}{3} \cos \frac{A}{3} &= \frac{1}{2} \sin^2 \frac{A}{3} \sin \frac{2A}{3} \\ &= \frac{1}{2} \left(\frac{EC}{AC} \right)^2 \left(\frac{BE}{AB} \right) \\ &= \frac{1}{2} \left(\frac{a^2}{16b^2} \right) \left(\frac{3a}{4c} \right) \\ &= \frac{3a^3}{128b^2c} \end{aligned}$$

Answer: (D)

66. In $\triangle ABC$, if the median through A is perpendicular to the side AB , then $\tan A + 2 \tan B$ is equal to

- (A) $\tan C$ (B) $-\tan C$
 (C) 0 (D) $2 \tan C$

Solution: Draw CM perpendicular to BA produced. Since AD is parallel to MC and D is the mid-point of BC , it follows that $MA = AB = x$ (say). Now suppose $AD = y$ so that $MC = 2y$. See Figure 4.54.

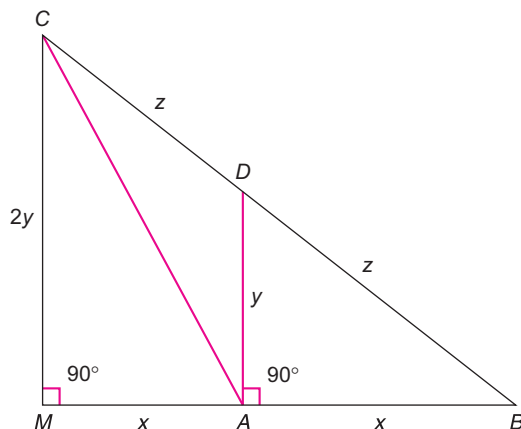


FIGURE 4.54 Single correct choice type question 66.

Let $BD = DC = z$. Then

$$\begin{aligned} \tan A &= \tan(\pi - \angle CAM) \\ &= -\tan \angle CAM \\ &= \frac{-2y}{x} \\ &= -2 \tan B \end{aligned}$$

Therefore

$$\tan A + 2 \tan B = 0$$

Answer: (C)

67. In $\triangle ABC$, if the median through A is perpendicular to the side AB , then the value of $\cos A \cos B$ is equal to

- (A) $\frac{2(b^2 - a^2)}{3ab}$ (B) $\frac{2(c^2 - a^2)}{3ac}$
 (C) $\frac{2(a^2 - b^2)}{3ab}$ (D) $\frac{2(c^2 + a^2)}{3ab}$

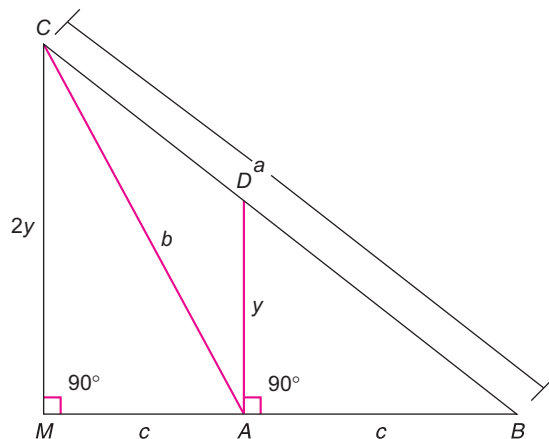


FIGURE 4.55 Single correct choice type question 67.

Solution: See Figure 4.55. Draw CM perpendicular to BA produced. Since AD is parallel to MC and D is the mid-point of BC , it follows that $MA = AB = c$ (say). Now suppose $AD = y$ so that $MC = 2y$. From $\triangle ABD$,

$$\cos B = \frac{AB}{BD} = \frac{c}{a/2} = \frac{2c}{a}$$

Also from $\triangle ACM$,

$$\cos(\pi - A) = \frac{AM}{AC} = \frac{c}{b} \quad (\because MA = AB = c)$$

Therefore

$$\cos A \cos B = -\frac{2c^2}{ab} \quad (4.22)$$

Now

$$(AD)^2 + (AB)^2 = (BD)^2 \Rightarrow y^2 + c^2 = \frac{a^2}{4}$$

and

$$(MC)^2 + (MA)^2 = (AC)^2 \Rightarrow 4y^2 + c^2 = b^2$$

Therefore

$$a^2 - 4c^2 = 4y^2 = b^2 - c^2$$

or

$$c^2 = \frac{a^2 - b^2}{3} \quad (4.23)$$

From Eqs. (4.22) and (4.23), we get

$$\cos A \cos B = \frac{2(b^2 - a^2)}{3ab}$$

Answer: (A)

68. In $\triangle ABC$, AD is the altitude from A onto the side BC . If $b > c$, $\angle C = 23^\circ$ and $AD = abc/(b^2 - c^2)$, then $\angle B$ is equal to

- (A) 83° (B) 113° (C) 123° (D) 75°

Solution: Draw AD perpendicular to CB produced (Figure 4.56).

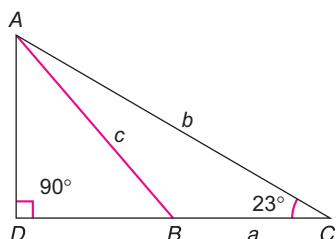


FIGURE 4.56 Single correct choice type question 68.

By hypothesis,

$$\frac{abc}{b^2 - c^2} = AD = c \sin B$$

Therefore

$$\frac{\sin A \sin B}{\sin^2 B - \sin^2 C} = \sin B$$

$$\frac{\sin(B + C)}{\sin(B + C)\sin(B - C)} = 1$$

$$B - C = 90^\circ \quad \text{or} \quad B = 90^\circ + C = 113^\circ$$

Answer: (B)

69. In $\triangle ABC$, if

$$a \tan A + b \tan B = (a + b) \tan \frac{A + B}{2}$$

then the triangle is

- (A) isosceles
(B) isosceles and right angled
(C) right angled
(D) equilateral

Solution: From the hypothesis

$$a \left(\tan A - \tan \frac{A + B}{2} \right) = b \left(\tan \frac{A + B}{2} - \tan B \right)$$

Therefore

$$\begin{aligned} & \frac{a \left[\sin A \cos \left(\frac{A + B}{2} \right) - \cos A \sin \left(\frac{A + B}{2} \right) \right]}{\cos A \cos \left(\frac{A + B}{2} \right)} \\ &= \frac{b \left[\sin \left(\frac{A + B}{2} \right) \cos B - \cos \left(\frac{A + B}{2} \right) \sin B \right]}{\cos \left(\frac{A + B}{2} \right) \cos B} \\ & \frac{a \sin \left(\frac{A - B}{2} \right)}{\cos A} = \frac{b \sin \left(\frac{A - B}{2} \right)}{\cos B} \end{aligned}$$

$$\sin \left(\frac{A - B}{2} \right) (a \cos B - b \cos A) = 0$$

$$\sin \left(\frac{A - B}{2} \right) (\sin A \cos B - \cos A \sin B) = 0$$

$$2 \sin^2 \left(\frac{A - B}{2} \right) \cos \left(\frac{A - B}{2} \right) = 0$$

Now

$$\begin{aligned}\cos \frac{A-B}{2} &\neq 0 \\ \Rightarrow \sin \frac{(A-B)}{2} &= 0 \\ \Rightarrow A &= B\end{aligned}$$

Therefore $\triangle ABC$ is isosceles.

Answer: (A)

70. The sides of a triangle are three consecutive natural numbers and its largest angle is twice its smallest one. Then, the perimeter of the triangle is

- (A) 18 (B) 21 (C) 15 (D) 24

Solution: Let a, b, c be the sides and $a = n, b = n + 1, c = n + 2$ and $\angle C = 2\angle A$. Now $\sin C = \sin 2A = 2 \sin A \cos A$, therefore

$$\begin{aligned}c &= (2a) \frac{b^2 + c^2 - a^2}{2bc} \\ bc^2 &= a(b^2 + c^2 - a^2) \\ (n+1)(n+2)^2 &= n[(n+1)^2 + (n+2)^2 - n^2] \\ &= n[n^2 + 6n + 5] \\ &= n(n+1)(n+5) \\ (n+2)^2 &= n(n+5)\end{aligned}$$

Therefore

$$n^2 + 4n + 4 = n^2 + 5n$$

which gives $n = 4$ and the perimeter $= 4 + 5 + 6 = 15$.

Answer: (C)

71. In $\triangle ABC$, if

$$\tan A \tan B + \tan B \tan C + \tan A \tan C = 9$$

then the sum of the distances of the vertices from the orthocentre is

- (A) $3\sqrt{3}R$ (B) $2\sqrt{3}R$ (C) $\sqrt{3}R$ (D) $3R$

Solution: From the hypothesis, we can see that at least one of $\tan A \tan B, \tan B \tan C$ and $\tan C \tan A$ is not equal to 1. Suppose $\tan A \tan B \neq 1$. Also we know that

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Therefore

$$\tan C = \frac{\tan A + \tan B}{\tan A \tan B - 1} \quad (4.24)$$

Now it is given that

$$\begin{aligned}9 &= \tan A \tan B + \tan C(\tan A + \tan B) \\ &= \tan A \tan B + \frac{(\tan A + \tan B)^2}{\tan A \tan B - 1} \quad [\text{By Eq. (4.24)}]\end{aligned}$$

So

$$\begin{aligned}9(\tan A \tan B - 1) &= \tan A \tan B(\tan A \tan B - 1) \\ &\quad + (\tan A + \tan B)^2 \\ (\tan A + \tan B)^2 - 10 \tan A \tan B + \tan^2 A \tan^2 B + 9 &= 0 \\ (\tan A - \tan B)^2 + (\tan A \tan B - 3)^2 &= 0\end{aligned}$$

This gives

$$\tan A = \tan B \quad \text{and} \quad \tan A \tan B = 3$$

So

$$\tan^2 A = 3$$

and so

$$\tan B = \tan A = \sqrt{3}$$

Therefore $A = B = 60^\circ$ and hence $C = 60^\circ$. This implies $\triangle ABC$ is equilateral.

In an equilateral triangle, orthocentre and circumcentre are same. Therefore, if O is the circumcentre, then

$$OA + OB + OC = 3R$$

Answer: (D)

72. Triangle ABC is equilateral with side length a and P is a point on the side BA produced such that A lies between P and B . Let r_1 be the radius of the circle inscribed in $\triangle PAC$ and r_2 be the radius of the escribed circle of $\triangle PBC$ opposite to the vertex P . Then the value of $r_1 + r_2$ is

- (A) $\frac{\sqrt{3}a}{2}$ (B) $\frac{3a}{2}$
(C) $\frac{3\sqrt{3}a}{2}$ (D) $2\sqrt{3}a$

Solution: See Figure 4.57. Let $\angle APC = \alpha$ and $\angle ACP = \beta$. Therefore

$$\frac{AC}{\sin \alpha} = \frac{BC}{\sin \alpha}$$

which implies that the circumradius is same for $\triangle PAC$ and $\triangle PBC$.

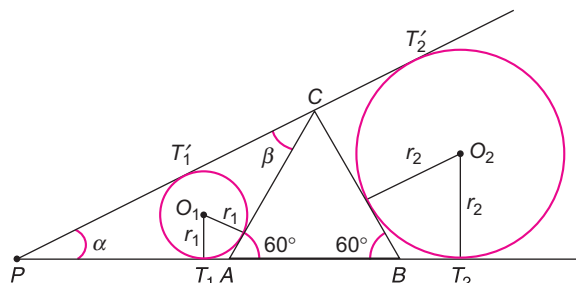


FIGURE 4.57 Single correct choice type question 72.

Let R' be the circumradius of $\triangle PAC$ and $\triangle PBC$. Therefore

$$r_1 = 4R' \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin 60^\circ = 4R' \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \cdot \frac{\sqrt{3}}{2} \quad (4.25)$$

$$r_2 = 4R' \sin \frac{\alpha}{2} \cos \left(\frac{60^\circ + \beta}{2} \right) \frac{\sqrt{3}}{2} \quad (4.26)$$

From Eqs. (4.25) and (4.26),

$$\begin{aligned} r_1 + r_2 &= 2R' \sin \frac{\alpha}{2} \left[\sqrt{3} \sin \frac{\beta}{2} + \sqrt{3} \left(\frac{\sqrt{3}}{2} \cos \frac{\beta}{2} - \frac{1}{2} \sin \frac{\beta}{2} \right) \right] \\ &= \left(2R' \sin \frac{\alpha}{2} \right) \sqrt{3} \left(\sin \frac{\beta}{2} + \frac{\sqrt{3}}{2} \cos \frac{\beta}{2} - \frac{1}{2} \sin \frac{\beta}{2} \right) \\ &= 2R' \sqrt{3} \sin \frac{\alpha}{2} \left(\frac{1}{2} \sin \frac{\beta}{2} + \frac{\sqrt{3}}{2} \cos \frac{\beta}{2} \right) \\ &= 2R' \sqrt{3} \sin \frac{\alpha}{2} \sin \left(60^\circ + \frac{\beta}{2} \right) \\ &= 2R' \sqrt{3} \sin \frac{\alpha}{2} \sin \left(60^\circ + 30^\circ - \frac{\alpha}{2} \right) \quad (\because \beta = 60^\circ - \alpha) \\ &= 2R' \sqrt{3} \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} \\ &= R' \sqrt{3} \sin \alpha \\ &= R' \sqrt{3} \left(\frac{a}{2R'} \right) = \frac{\sqrt{3}}{2} a \end{aligned}$$

Answer: (A)

73. The number of triangles with circumradius R is equal to $25/8$ and whose sides are natural numbers is

- (A) 4 (B) 2
(C) 1 (D) infinite

Solution: In $\triangle ABC$, $a = BC$, $b = CA$ and $c = AB$ are natural numbers and $R = 25/8$ (Figure 4.58). Now

$$a = 2R \sin A = \left(\frac{25}{4} \right) \sin A$$

and a is a positive integer. Therefore $(25/4) \sin A$ must be an integer. Hence

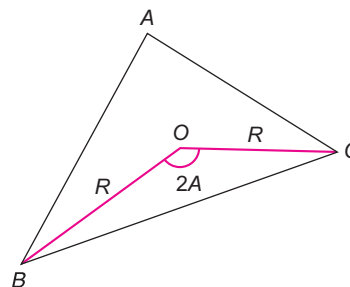


FIGURE 4.58 Single correct choice type question 73.

$$\sin A = \frac{4}{25}, \frac{8}{25}, \frac{12}{25}, \frac{16}{25}, \frac{20}{25} \text{ or } \frac{24}{25}$$

Also, since a, b, c are integers $\cos A, \cos B, \cos C$ are rational numbers where

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Therefore $\sin A, \sin B$ and $\sin C$ belong to the two element set

$$\left\{ \frac{20}{25}, \frac{24}{25} \right\}$$

Therefore the possible triangles whose sides are natural numbers are the triples $(5, 5, 5); (5, 5, 6); (5, 6, 6)$ and $(6, 6, 6)$. Among these triangles, the triangle with sides 5, 5 and 6 satisfies the condition $2R$ is equal to $6\frac{1}{4}$.

Hence the number of such triangles is one.

Answer: (C)

74. Two sides of a triangle are 2 and $\sqrt{3}$ and the included angle is 30° . Then, the inradius of the triangle is

- (A) $\frac{\sqrt{3}-1}{4}$ (B) $\frac{\sqrt{3}+1}{2}$
(C) $\frac{\sqrt{3}-1}{2}$ (D) $\frac{\sqrt{3}+1}{4}$

Solution: In $\triangle ABC$, let $b = 2$, $c = \sqrt{3}$ and $\angle A = 30^\circ$. The third side a is given by

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ &= 4 + 3 - 2(2)(\sqrt{3}) \left(\frac{\sqrt{3}}{2} \right) \\ &= 7 - 6 = 1 \end{aligned}$$

Therefore $a = 1$ and so

$$s = \frac{1}{2}(a + b + c) = \frac{3 + \sqrt{3}}{2}$$

Hence

$$s - a = \frac{\sqrt{3} + 1}{2}$$

Now

$$\begin{aligned} r &= (s - a) \tan \frac{A}{2} = \left(\frac{\sqrt{3} + 1}{2} \right) \tan 15^\circ \\ &= \frac{\sqrt{3} + 1}{2} \times \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = \frac{\sqrt{3} - 1}{2} \end{aligned}$$

Answer: (C)

75. In $\triangle ABC$, let r_1, r_2 and r_3 be the exradii opposite to the vertices A, B and C , respectively. Let

$$x = a(rr_1 + r_2r_3)$$

$$y = b(rr_2 + r_3r_1)$$

$$z = c(rr_3 + r_1r_2)$$

and
$$p = (r_3 - r) \cot \frac{C}{2}$$

Then

(A) $x = y = z = p$

(B) $x = y = z$

(C) $x = y = z = 2p$

(D) $y = z = 2p$

Solution:

$$\begin{aligned} x &= a(rr_1 + r_2r_3) \\ &= a \left(\frac{\Delta^2}{s(s-a)} + \frac{\Delta^2}{(s-b)(s-c)} \right) \\ &= a\Delta^2 \left(\frac{(s-b)(s-c) + s(s-a)}{\Delta^2} \right) \\ &= a[2s^2 - s(a+b+c) + bc] \\ &= abc \end{aligned}$$

Similarly $y = abc, z = abc$. Therefore $x = y = z$. Now

$$\begin{aligned} p &= (r_3 - r) \cot \frac{C}{2} \\ &= \Delta \left(\frac{1}{s-c} - \frac{1}{s} \right) \cot \frac{C}{2} \\ &= \frac{\Delta c}{s(s-a)} \cot \frac{C}{2} \\ &= c \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \cdot \cot \frac{C}{2} \end{aligned}$$

$$\begin{aligned} &= c \tan \frac{C}{2} \cot \frac{C}{2} \\ &= c \end{aligned}$$

Hence $x \neq p$.

Answer: (B)

76. In $\triangle ABC$,

$$\frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} =$$

(A) $\frac{1}{Rr}$

(B) $\frac{2}{Rr}$

(C) $\frac{1}{2Rr}$

(D) $\frac{1}{2(R+r)}$

Solution:

$$\begin{aligned} \frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} &= \frac{a+b+c}{abc} \\ &= \frac{2s}{abc} \\ &= \frac{2(\Delta/r)}{abc} \\ &= \left(\frac{4\Delta}{abc} \right) \frac{1}{2r} \\ &= \frac{1}{2Rr} \end{aligned}$$

Answer: (C)

77. In $\triangle ABC$,

$$\frac{r_1}{(s-b)(s-c)} + \frac{r_2}{(s-c)(s-a)} + \frac{r_3}{(s-a)(s-b)} =$$

(A) $\frac{4}{r}$ (B) $\frac{1}{r}$ (C) $\frac{2}{r}$ (D) $\frac{3}{r}$

Solution:

$$\frac{r_1}{(s-b)(s-c)} = \frac{\Delta}{(s-a)(s-b)(s-c)} = \frac{s}{\Delta}$$

Similarly

$$\frac{r_2}{(s-c)(s-a)} = \frac{s}{\Delta} = \frac{r_3}{(s-a)(s-b)}$$

Therefore

$$\sum \frac{r_i}{(s-b)(s-c)} = \frac{3s}{\Delta} = \frac{3}{r}$$

Answer: (D)

78. In $\triangle ABC$, if $8R^2 = a^2 + b^2 + c^2$, then $\sin^2 A + \sin^2 B + \sin^2 C =$
 (A) 2 (B) 3 (C) 4 (D) 1

Solution:

$$\begin{aligned} 8R^2 &= a^2 + b^2 + c^2 \\ &= 4R^2(\sin^2 A + \sin^2 B + \sin^2 C) \end{aligned}$$

Therefore

$$\sin^2 A + \sin^2 B + \sin^2 C = 2$$

Answer: (A)

79. In $\triangle ABC$, $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$ is equal to

- (A) $\frac{1}{r}$ (B) $\frac{2}{r}$ (C) $\frac{\Delta}{r}$ (D) $\frac{\Delta}{2r}$

Solution:

$$\begin{aligned} \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} &= \frac{s-a}{\Delta} + \frac{s-b}{\Delta} + \frac{s-c}{\Delta} \\ &= \frac{1}{\Delta}(3s - 2s) \\ &= \frac{s}{\Delta} \\ &= \frac{1}{r} \end{aligned}$$

Answer: (A)

80. In $\triangle ABC$, $\angle A = 120^\circ$. Let D, E and F be the points of intersection of the internal bisectors with the opposite sides. Then $\triangle DEF$ is
 (A) right angled
 (B) isosceles
 (C) isosceles and right angled
 (D) equilateral

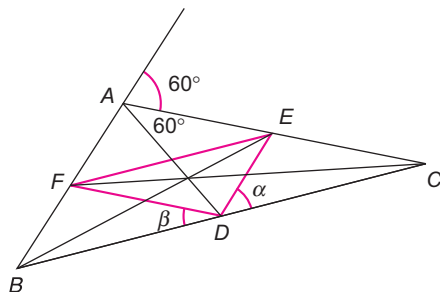


FIGURE 4.59 Single correct choice type question 80.

Solution: For the $\triangle ABD$, AC is the external bisector of $\angle BAD$ and BE is the internal bisector of $\angle DBA$. Therefore E is the excentre to $\triangle ABD$. Hence

$$\angle CDE = \angle ADE = \alpha \text{ (suppose)}$$

Similarly

$$\angle ADF = \angle FDB = \beta \text{ (suppose)}$$

See Figure 4.59. Therefore

$$\begin{aligned} 2\angle EDF &= 2(\alpha + \beta) = 180^\circ \\ \angle EDF &= 90^\circ \end{aligned}$$

Answer: (A)

81. In $\triangle ABC$, if t_1, t_2 and t_3 are the lengths of the tangents drawn from the excentres I_1, I_2 and I_3 of $\triangle ABC$ to the circumcircle of the triangle, then

$$\begin{aligned} \frac{1}{t_1^2} + \frac{1}{t_2^2} + \frac{1}{t_3^2} &= \\ \text{(A) } \frac{1}{Rr} \quad \text{(B) } \frac{2}{Rr} \quad \text{(C) } \frac{2r}{R} \quad \text{(D) } \frac{1}{2Rr} \end{aligned}$$

Solution: Let I_1T_1 be the tangent from the excentre I_1 to the circumcircle and O be the circumcentre (Figure 4.60). It is known that (see Quick Look 2 and Theorem 4.29)

$$OI_1^2 = R^2 + 2Rr_1$$

and

$$OT_1^2 = R^2$$

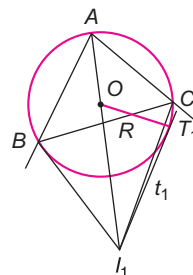


FIGURE 4.60 Single correct choice type question 81.

From the right-angled $\triangle OT_1I_1$

$$\begin{aligned} OI_1^2 &= R^2 + t_1^2 \\ R^2 + 2Rr_1 &= R^2 + t_1^2 \\ t_1^2 &= 2Rr_1 \end{aligned}$$

Similarly

$$t_2^2 = 2Rr_2 \quad \text{and} \quad t_3^2 = 2Rr_3$$

Therefore

$$\begin{aligned} \frac{1}{t_1^2} + \frac{1}{t_2^2} + \frac{1}{t_3^2} &= \frac{1}{2R} \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right) \\ &= \frac{1}{2R} \left(\frac{1}{r} \right) \end{aligned}$$

Answer: (D)

82. In $\triangle ABC$, if $(a-b)(s-c) = (b-c)(s-a)$, then

- (A) $2r_2 = r_1 + r_3$ (B) $r_2^2 = r_1 r_3$
 (C) $r_3 = \frac{2r_1 r_2}{r_1 + r_2}$ (D) $r_2 = \frac{2r_3 r_1}{r_3 + r_1}$

Solution:

$$\begin{aligned} (a-b)(s-c) &= (b-c)(s-a) \\ \Rightarrow \frac{b-a}{s-a} &= \frac{c-b}{s-c} \\ \Rightarrow \frac{b-a}{(s-a)(s-b)} &= \frac{c-b}{(s-b)(s-c)} \\ \Rightarrow \frac{\Delta}{s-b} - \frac{\Delta}{s-a} &= \frac{\Delta}{s-c} - \frac{\Delta}{s-b} \\ \Rightarrow r_2 - r_1 &= r_3 - r_2 \\ \Rightarrow 2r_2 &= r_1 + r_3 \end{aligned}$$

Answer: (A)

83. In $\triangle ABC$, $\angle A = 90^\circ$ and AD is drawn perpendicular to the side BC . If p and q are the radii of the inscribed circles of $\triangle ABD$ and $\triangle ACD$, respectively, and r is the inradius of $\triangle ABC$, then r is equal to

- (A) $2\sqrt{p^2 + q^2}$ (B) $\frac{1}{2}\sqrt{p^2 + q^2}$
 (C) $\sqrt{p^2 + q^2}$ (D) $\frac{1}{2}(p^2 + q^2)$

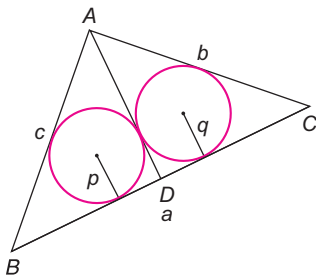


FIGURE 4.61 Single correct choice type question 83.

Solution: See Figure 4.61. We have $a^2 = b^2 + c^2$. By hypothesis, $\triangle ABD$ and $\triangle ACD$ are both similar as well as similar to $\triangle ABC$. Let R and R' be the circumradii of $\triangle ABC$ and $\triangle ABD$, respectively. Therefore

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\text{and } p = 4R' \left(\sin \frac{1}{2} \angle B \right) \left(\sin \frac{1}{2} \angle BAD \right) \left(\sin \frac{1}{2} \angle BDA \right)$$

But the angles of $\triangle ABC$ and $\triangle ABD$ are equal in some order. Therefore

$$\frac{r}{p} = \frac{4R}{4R'} = \frac{2(a/\sin A)}{2(c/\sin D)} = \frac{a}{c}$$

Similarly

$$\frac{r}{q} = \frac{a}{b}$$

Therefore

$$\begin{aligned} r^2(b^2 + c^2) &= a^2(p^2 + q^2) \\ r^2 &= p^2 + q^2 \end{aligned}$$

Note: In general, the inradii of two similar triangles are proportional to their sides.

Answer: (C)

84. In $\triangle ABC$, $r_1 r_2 + r_2 r_3 + r_3 r_1$ is equal to

- (A) s^2 (B) Δ^2 (C) $\frac{r^2}{\Delta^2}$ (D) $\frac{R^2}{\Delta^2}$

Solution:

$$\begin{aligned} r_1 r_2 + r_2 r_3 + r_3 r_1 &= \Delta^2 \left[\frac{1}{(s-a)(s-b)} + \frac{1}{(s-b)(s-c)} + \frac{1}{(s-c)(s-a)} \right] \\ &= s[(s-c) + (s-a) + (s-b)] \\ &= s(3s - 2s) \\ &= s^2 \end{aligned}$$

Answer: (A)

85. Let I be the incentre and r the inradius of $\triangle ABC$. Let D, E, F be the feet of the perpendiculars drawn from I onto the sides BC, CA and AB , respectively. If r_1, r_2 and r_3 are the radii of the circles inscribed in the quadrilaterals $AFIE, BDIF$ and $CEID$, respectively, then

$$\frac{r_1}{r-r_1} + \frac{r_2}{r-r_2} + \frac{r_3}{r-r_3} = \frac{K}{r(r-r_1)(r-r_2)(r-r_3)}$$

where K is equal to

- (A) s^2 (B) Δ^2 (C) $\Delta^2 s^2$ (D) $\Delta^2 r^2$

Solution: Let J_1 be the centre of the circle inscribed in the quadrilateral $AFIE$ touching the side IE in M_1 and the side AE in N_1 . Clearly J_1 lies on the segment AI (Figure 4.62). Clearly $J_1 M_1 E N_1$ is a square, because $J_1 M_1 = J_1 N_1 = r_1$. Also

$$\angle M_1 J_1 I = \angle N_1 J_1 A = 90^\circ - \frac{A}{2}$$

Therefore from $\triangle I M_1 J_1$,

$$\tan \left(90^\circ - \frac{A}{2} \right) = \frac{I M_1}{J_1 M_1} = \frac{IE - M_1 E}{J_1 M_1} = \frac{r - r_1}{r_1}$$

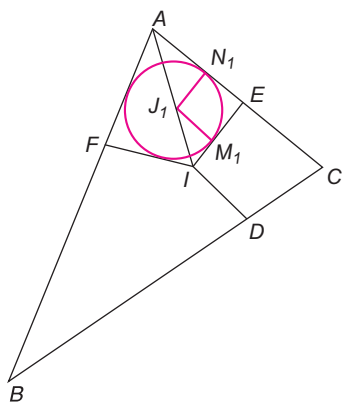


FIGURE 4.62 Single correct choice type question 85.

Therefore

$$\cot \frac{A}{2} = \frac{r - r_1}{r_1}$$

Similarly

$$\cot \frac{B}{2} = \frac{r - r_2}{r_2} \quad \text{and} \quad \cot \frac{C}{2} = \frac{r - r_3}{r_3}$$

In any $\triangle ABC$

$$\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

Therefore

$$\begin{aligned} \frac{r - r_1}{r_1} + \frac{r - r_2}{r_2} + \frac{r - r_3}{r_3} &= \frac{r_1 r_2 r_3}{(r - r_1)(r - r_2)(r - r_3)} \\ &= \frac{r r_1 r_2 r_3}{r(r - r_1)(r - r_2)(r - r_3)} \\ &= \frac{\Delta^2}{r(r - r_1)(r - r_2)(r - r_3)} \end{aligned}$$

Hence K is equal to Δ^2 .

Answer: (B)

86. If r is the inradius and r_1, r_2 and r_3 are exradii of any triangle, then

$$\left(\frac{1}{r} - \frac{1}{r_1} \right) \left(\frac{1}{r} - \frac{1}{r_2} \right) \left(\frac{1}{r} - \frac{1}{r_3} \right) =$$

(A) $\frac{4R}{\Delta}$ (B) $\frac{4R^2}{\Delta}$ (C) $\frac{4R^2}{\Delta^2}$ (D) $\frac{4R}{\Delta^2}$

Solution:

$$\begin{aligned} &\left(\frac{1}{r} - \frac{1}{r_1} \right) \left(\frac{1}{r} - \frac{1}{r_2} \right) \left(\frac{1}{r} - \frac{1}{r_3} \right) \\ &= \left(\frac{s}{\Delta} - \frac{s-a}{\Delta} \right) \left(\frac{s}{\Delta} - \frac{s-b}{\Delta} \right) \left(\frac{s}{\Delta} - \frac{s-c}{\Delta} \right) \\ &= \frac{abc}{\Delta^3} = \frac{4R\Delta}{\Delta^3} = \frac{4R}{\Delta^2} \end{aligned}$$

Answer: (D)

87. In a triangle, if $r_1 = 12$, $r_2 = 18$ and $r_3 = 36$, then r is equal to

- (A) 6 (B) 9 (C) 4 (D) 3

Solution:

$$\begin{aligned} \frac{1}{r} &= \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \\ &= \frac{1}{12} + \frac{1}{18} + \frac{1}{36} \\ &= \frac{6}{36} = \frac{1}{6} \end{aligned}$$

Therefore $r = 6$.

Answer: (A)

88. In $\triangle ABC$,

$$(r_1 + r_2) \cot \left(\frac{A+B}{2} \right) =$$

(A) a (B) b (C) c (D) $a + b + c$

Solution:

$$\begin{aligned} (r_1 + r_2) \cot \left(\frac{A+B}{2} \right) &= \left(\frac{\Delta}{s-a} + \frac{\Delta}{s-b} \right) \tan \frac{C}{2} \\ &= \frac{\Delta c}{(s-a)(s-b)} \left(\tan \frac{C}{2} \right) \\ &= c \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \cdot \tan \frac{C}{2} \\ &= c \cdot \left(\cot \frac{C}{2} \right) \tan \frac{C}{2} \\ &= c \end{aligned}$$

Answer: (C)

89. In $\triangle ABC$, if the median AD makes angle θ with the side BC , then $\sin \theta$ is equal to

- (A) $\frac{2b \sin C}{\sqrt{2b^2 + 2c^2 - a^2}}$ (B) $\frac{b \sin C}{\sqrt{2b^2 + 2c^2 - a^2}}$
 (C) $\frac{2b \sin B}{\sqrt{2b^2 + 2c^2 - a^2}}$ (D) $\frac{b \sin B}{\sqrt{2b^2 + 2c^2 - a^2}}$

Solution: From $\triangle ADC$ (Figure 4.63),

$$\frac{AC}{\sin \theta} = \frac{AD}{\sin C}$$

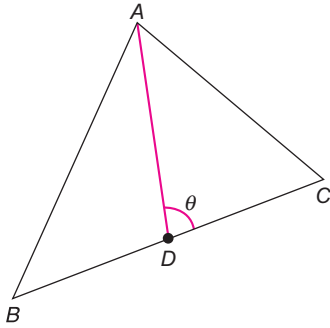


FIGURE 4.63 Single correct choice type question 89.

Therefore

$$\begin{aligned} \sin \theta &= \frac{AC \sin C}{AD} \\ &= \frac{b \sin C}{(1/2)\sqrt{2b^2 + 2c^2 - a^2}} \quad (\text{see Theorem 4.25}) \\ &= \frac{2b \sin C}{\sqrt{2b^2 + 2c^2 - a^2}} \end{aligned}$$

Answer: (A)

90. The internal bisectors of the angles of $\triangle ABC$ meet the circumcircle in D , E and F . If Δ' and Δ are the areas of $\triangle DEF$ and $\triangle ABC$, respectively, then the ratio $\Delta' : \Delta$ is equal to

- (A) $2R:r$ (B) $R:r$
 (C) $(R-r):(R+r)$ (D) $R:2r$

Solution: Since angles in the same segment are equal (Figure 4.64), we have

$$\angle ADE = \angle ABE = \frac{1}{2}\angle B$$

and $\angle ADF = \angle ACF = \frac{1}{2}\angle C$

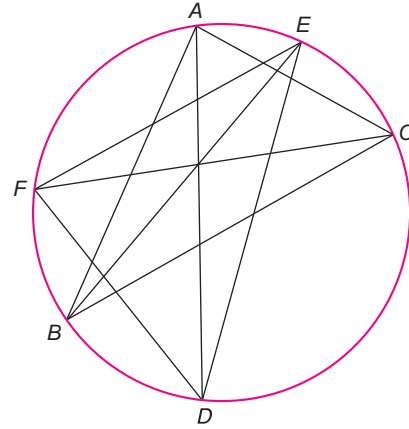


FIGURE 4.64 Single correct choice type question 90.

Therefore

$$\angle EDF = \frac{1}{2}(\angle B + \angle C) = 90^\circ - \frac{1}{2}\angle A$$

Similarly

$$\angle DEF = 90^\circ - \frac{1}{2}\angle B$$

and

$$\angle DFE = 90^\circ - \frac{1}{2}\angle C$$

Using sine rule for $\triangle DEF$ we get

$$\begin{aligned} \frac{EF}{\sin[90^\circ - (A/2)]} &= \frac{FD}{\sin[90^\circ - (B/2)]} \\ &= \frac{DE}{\sin[90^\circ - (C/2)]} = 2R \end{aligned}$$

Therefore

$$EF = 2R \cos \frac{A}{2}, FD = 2R \cos \frac{B}{2}, DE = 2R \cos \frac{C}{2}$$

So

$$\begin{aligned} \Delta &= \frac{1}{2}bc \sin A \\ &= 2R^2 \sin A \sin B \sin C \\ &= 16R^2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \\ \Delta' &= \frac{1}{2}(DE)(DF) \sin \left(90^\circ - \frac{A}{2} \right) \\ &= 2R^2 \sin \left(90^\circ - \frac{C}{2} \right) \sin \left(90^\circ - \frac{B}{2} \right) \sin \left(90^\circ - \frac{A}{2} \right) \\ &= 2R^2 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \end{aligned}$$

Therefore

$$\begin{aligned}\frac{\Delta'}{\Delta} &= \frac{1}{8\sin(A/2)\sin(B/2)\sin(C/2)} \\ &= \frac{R}{2[4R\sin(A/2)\sin(B/2)\sin(C/2)]} \\ &= \frac{R}{2r}\end{aligned}$$

Answer: (D)

91. In $\triangle ABC$, if $r_1 = 8$, $r_2 = 12$, $r_3 = 24$, and $BC = a$, then the value of a is

- (A) 12 (B) 6 (C) 4 (D) 8

Solution: In $\triangle ABC$, it is easy to verify that

$$(r_1 - r)(r_2 + r_3) = a^2$$

and
$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$$

Therefore

$$\begin{aligned}\frac{1}{r} &= \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \\ &= \frac{1}{8} + \frac{1}{12} + \frac{1}{24} \\ &= \frac{1}{4} \\ \Rightarrow r &= 4\end{aligned}$$

So

$$\begin{aligned}a^2 &= (r_1 - r)(r_2 + r_3) \\ &= (8 - 4)(12 + 24) \\ &= 4 \times 36 \\ \Rightarrow a &= 12\end{aligned}$$

Answer: (A)

92. In $\triangle ABC$, if $r_1 = r_2 + r_3 + r$, then angle A must be

- (A) 120° (B) 60° (C) 90° (D) 45°

Solution: Given that $r_1 = r_2 + r_3 + r$. Therefore

$$\begin{aligned}r_1 - r &= r_2 + r_3 \\ 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} - 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \\ &= 4R\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2} + 4R\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2} \\ \sin\frac{A}{2}\cos\frac{B+C}{2} &= \cos\frac{A}{2}\sin\frac{B+C}{2} \\ \sin^2\frac{A}{2} &= \cos^2\frac{A}{2}\end{aligned}$$

This implies

$$\begin{aligned}\frac{A}{2} &= 45^\circ \\ A &= 90^\circ\end{aligned}$$

Answer: (C)

93. In any $\triangle ABC$, $b(r_1 - r) + a(r_2 - r)$ is equal to

- (A) $\frac{abc}{r_3}$ (B) $\frac{a+b+c}{r_3}$ (C) $\frac{cr_3}{r}$ (D) $\frac{c+r}{r_3}$

Solution:

$$\begin{aligned}b(r_1 - r) + c(r_2 - r) &= b\Delta\left(\frac{1}{s-a} - \frac{1}{s}\right) + a\Delta\left(\frac{1}{s-b} - \frac{1}{s}\right) \\ &= \frac{ab\Delta}{s(s-a)} + \frac{ab\Delta}{s(s-b)} \\ &= \frac{ab\Delta}{s}\left(\frac{1}{s-a} + \frac{1}{s-b}\right) \\ &= \frac{ab\Delta}{s}\left(\frac{c}{(s-a)(s-b)}\right) \\ &= \frac{(abc)\Delta(s-c)}{s(s-a)(s-b)(s-c)} \\ &= \frac{(abc)(s-c)}{\Delta} \\ &= \frac{abc}{r_3} \left(\because r_3 = \frac{\Delta}{s-c} \right)\end{aligned}$$

Answer: (A)

Multiple Correct Choice Type Questions

1. In $\triangle ABC$,

- (A) $(2ac)\sin\left(\frac{A-B+C}{2}\right) = c^2 + a^2 - b^2$
(B) $a(b\cos C - c\cos B) = b^2 - c^2$

(C) $(b+c)\cos A + (c+a)\cos B + (a+b)\cos C = \frac{1}{2}(a+b+c)$

(D) If $\angle A = 60^\circ$, then

$$\left(1 + \frac{a}{c} + \frac{b}{c}\right)\left(1 + \frac{c}{b} - \frac{a}{b}\right) = 3$$

Solution:

$$\begin{aligned} \text{(A)} \quad 2ac \sin\left(\frac{A-B+C}{2}\right) &= 2ac \sin(90^\circ - B) \\ &= 2ac \cos B \\ &= c^2 + a^2 - b^2 \end{aligned}$$

Therefore (A) is true.

$$\begin{aligned} \text{(B)} \quad a(b \cos C - c \cos B) &= ab \cos C - ac \cos B \\ &= \frac{a^2 + b^2 - c^2}{2} - \frac{a^2 + c^2 - b^2}{2} \\ &= b^2 - c^2 \end{aligned}$$

Hence (B) is true.

$$\begin{aligned} \text{(C)} \quad (b+c)\cos A + (c+a)\cos B + (a+b)\cos C \\ &= (b \cos A + a \cos B) + (a \cos C + c \cos A) \\ &\quad + (b \cos C + c \cos B) \\ &= c + b + a = 2s \end{aligned}$$

So (C) is not true.

$$\begin{aligned} \text{(D)} \quad \left(1 + \frac{a}{c} + \frac{b}{c}\right)\left(1 + \frac{c}{b} - \frac{a}{b}\right) \\ &= \frac{(c+a+b)(b+c-a)}{bc} \\ &= \frac{(b+c)^2 - a^2}{bc} \\ &= \frac{(b^2 + c^2 + 2bc) - (b^2 + c^2 - 2bc \cos 60^\circ)}{bc} \\ &= \frac{3bc}{bc} \quad \left(\because \cos 60^\circ = \frac{1}{2}\right) \\ &= 3 \end{aligned}$$

Therefore (D) is true.

Answers: (A), (B), (D)

2. In any $\triangle ABC$ which of the following are true?

- (A) If $\angle A = 45^\circ$, $\angle B = 75^\circ$, then $a + c\sqrt{2} = 2b$.
 (B) If the sides a , b and c are in HP, then $\sin^2(A/2)$, $\sin^2(B/2)$ and $\sin^2(C/2)$ are in AP.
 (C) $(a+b+c)\left[\tan \frac{A}{2} + \tan \frac{B}{2}\right] = 2c \cot \frac{C}{2}$
 (D) $bc \cos^2 \frac{A}{2} + ca \cos^2 \frac{B}{2} + ab \cos^2 \frac{C}{2} = \Delta^2$

Solution:

(A) $\angle A = 45^\circ$, $\angle B = 75^\circ \Rightarrow \angle C = 60^\circ$. Now

$$b = a \cos C + c \cos A = \frac{a}{2} + \frac{c}{\sqrt{2}}$$

Therefore

$$a + c\sqrt{2} = 2b$$

(A) is true.

(B) a , b , c are in HP implies

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in AP}$$

Therefore

$$\frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b}$$

$$\frac{a-b}{ab} = \frac{b-c}{bc}$$

$$\frac{\sin A - \sin B}{\sin B \sin A} = \frac{\sin B - \sin C}{\sin C \sin B}$$

$$\frac{2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}}{2 \sin \frac{A}{2} \cos \frac{A}{2}} = \frac{2 \cos \frac{B+C}{2} \sin \frac{B-C}{2}}{2 \sin \frac{C}{2} \cos \frac{C}{2}}$$

$$\frac{\sin \frac{C}{2} \sin \frac{A-B}{2}}{\sin \frac{A}{2} \cos \frac{A}{2}} = \frac{\sin \frac{A}{2} \sin \frac{B-C}{2}}{\sin \frac{C}{2} \cos \frac{C}{2}}$$

$$\sin^2 \frac{C}{2} \cos \frac{C}{2} \sin \frac{A-B}{2} = \sin^2 \frac{A}{2} \cos \frac{A}{2} \sin \frac{B-C}{2}$$

$$\sin^2 \frac{C}{2} \left[\sin \frac{A+B}{2} \cdot \sin \frac{A-B}{2} \right] = \sin^2 \frac{A}{2} \left[\sin \frac{B+C}{2} \sin \frac{B-C}{2} \right]$$

$$\sin^2 \frac{C}{2} \left[\sin^2 \frac{A}{2} - \sin^2 \frac{B}{2} \right] = \sin^2 \frac{A}{2} \left[\sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} \right]$$

$$\sin^2 \frac{B}{2} \left[\sin^2 \frac{C}{2} + \sin^2 \frac{A}{2} \right] = 2 \sin^2 \frac{C}{2} \sin^2 \frac{A}{2}$$

$$\frac{2}{\sin^2(B/2)} = \frac{1}{\sin^2(A/2)} + \frac{1}{\sin^2(C/2)}$$

Therefore $\sin^2(A/2)$, $\sin^2(B/2)$ and $\sin^2(C/2)$ are in HP. Hence (B) is not true.

(C) We have

$$\begin{aligned} &(a+b+c) \left[\tan \frac{A}{2} + \tan \frac{B}{2} \right] \\ &= (a+b+c) \left[\frac{\sin \frac{A}{2}}{\cos \frac{A}{2}} + \frac{\sin \frac{B}{2}}{\cos \frac{B}{2}} \right] \end{aligned}$$

$$\begin{aligned}
 &= \frac{(a+b+c) \sin \frac{A+B}{2}}{\cos \frac{A}{2} \cos \frac{B}{2}} \\
 &= \frac{2R(\sin A + \sin B + \sin C) \cos \frac{C}{2}}{\cos \frac{A}{2} \cos \frac{B}{2}} \\
 &= \frac{2R \left(4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \right) \cos \frac{C}{2}}{\cos \frac{A}{2} \cos \frac{B}{2}} \\
 &= 8R \cos^2 \frac{C}{2} \\
 &= 4 \left(\frac{c}{\sin C} \right) \cos^2 \frac{C}{2} \\
 &= 4 \left(\frac{c}{2 \sin \frac{C}{2} \cos \frac{C}{2}} \right) \cos^2 \frac{C}{2} \\
 &= 2c \cot \frac{C}{2}
 \end{aligned}$$

Therefore (C) is true.

$$\begin{aligned}
 \text{(D)} \quad &bc \cos^2 \frac{A}{2} + ca \cos^2 \frac{B}{2} + ab \cos^2 \frac{C}{2} \\
 &= (bc) \frac{s(s-a)}{bc} + (ca) \frac{s(s-b)}{ca} + (ab) \frac{s(s-c)}{ab} \\
 &= s[3s - a - b - c] = s^2
 \end{aligned}$$

Hence (D) is not true.

Answers: (A), (C)

3. In $\triangle ABC$, which of the following are true?

(A) If the angles are in AP, then $b^2 = c^2 + a^2 - ca$

$$\text{(B)} \quad \frac{\sin(B-C)}{\sin(B+C)} = \frac{b^2 - c^2}{a^2}$$

$$\text{(C)} \quad \sum \frac{b^2 - c^2}{\cos B + \cos C} = 0$$

$$\text{(D)} \quad \frac{1 + \cos(A-B)\cos C}{1 + \cos(A-C)\cos B} = \frac{a^2 + b^2}{a^2 + c^2}$$

Solution:

(A) A, B, C are in AP implies

$$3B = 180^\circ \quad (\because A + C = 2B)$$

$$B = 60^\circ$$

Now

$$\begin{aligned}
 b^2 &= c^2 + a^2 - 2ca \cos B \\
 &= c^2 + a^2 - 2ca \left(\frac{1}{2} \right) \\
 &= c^2 + a^2 - ca
 \end{aligned}$$

Hence (A) is true.

(B) We have

$$\begin{aligned}
 \frac{\sin(B-C)}{\sin(B+C)} &= \frac{\sin(B+C)\sin(B-C)}{\sin^2(B+C)} \\
 &= \frac{\sin^2 B - \sin^2 C}{\sin^2 A} \\
 &= \frac{b^2 - c^2}{a^2}
 \end{aligned}$$

Hence (B) is true.

(C) Consider

$$\begin{aligned}
 \frac{b^2 - c^2}{\cos B + \cos C} &= \frac{4R^2(\sin^2 B - \sin^2 C)}{\cos B + \cos C} \\
 &= \frac{4R^2[\cos^2 C - \cos^2 B]}{\cos B + \cos C} \\
 &= 4R^2(\cos C - \cos B)
 \end{aligned}$$

Therefore

$$\begin{aligned}
 \sum \frac{b^2 - c^2}{\cos B + \cos C} &= 4R^2[\cos C - \cos B + \cos A - \cos C \\
 &\quad + \cos B - \cos A] \\
 &= 4R^2(0) = 0
 \end{aligned}$$

So (C) is true.

(D) We have

$$\begin{aligned}
 \frac{1 + \cos(A-B)\cos C}{1 + \cos(A-C)\cos B} &= \frac{1 - \cos(A-B)\cos(A+B)}{1 - \cos(A-C)\cos(A+C)} \\
 &= \frac{1 - (\cos^2 A - \sin^2 B)}{1 - (\cos^2 A - \sin^2 C)} \\
 &= \frac{\sin^2 A + \sin^2 B}{\sin^2 A + \sin^2 C} \\
 &= \frac{(a^2/4R^2) + (b^2/4R^2)}{(a^2/4R^2) + (c^2/4R^2)} \\
 &= \frac{a^2 + b^2}{a^2 + c^2}
 \end{aligned}$$

Hence (D) is true.

Answers: (A), (B), (C), (D)

4. In $\triangle ABC$, the altitudes AD , BE and CF meet the circumcircle in L , M and N , respectively. Then
- (A) the angles of the triangle LMN are supplements of twice the angles of $\triangle ABC$
 - (B) the angles of the $\triangle LMN$ are complements of the angles of $\triangle ABC$
 - (C) the area of $\triangle LMN$ is $4\Delta \cos A \cos B \cos C$
 - (D) the area of $\triangle LMN$ is $8\Delta \cos A \cos B \cos C$

Solution: See Figure 4.65. We have

$$\begin{aligned} \angle ALM &= \angle ABM \quad (\text{angles in the same segment}) \\ &= 90^\circ - A \end{aligned}$$

$$\text{and } \angle ALN = \angle ACN = 90^\circ - A$$

Hence

$$\angle MLN = (90^\circ - A) + (90^\circ - A) = 180^\circ - 2A$$

Similarly

$$\angle NML = 180^\circ - 2B$$

$$\text{and } \angle MNL = 180^\circ - 2C$$

Therefore (A) is true.

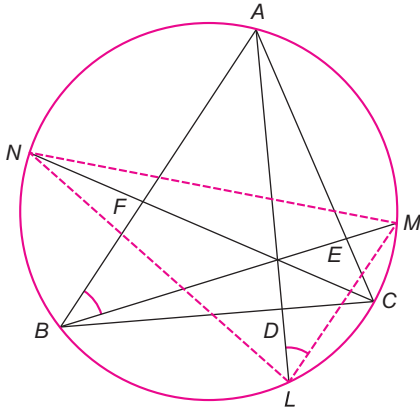


FIGURE 4.65 Multiple correct choice type question 4.

Using sine rule for $\triangle LMN$ we get

$$\frac{MN}{\sin(180^\circ - 2A)} = \frac{LN}{\sin(180^\circ - 2B)} = \frac{ML}{\sin(180^\circ - 2C)} = 2R$$

Therefore

$$MN = 2R \sin 2A, LN = 2R \sin 2B, LM = 2R \sin 2C$$

Hence

$$\text{Area of } \triangle LMN = \frac{1}{2} (LM)(LN) \sin(180^\circ - 2A)$$

(See Theorem 4.14)

$$\begin{aligned} &= \frac{1}{2} (2R \sin 2C)(2R \sin 2B) \sin 2A \\ &= 2R^2 (2 \sin C \cos C)(2 \sin B \cos B)(2 \sin A \cos A) \\ &= 8(2R^2 \sin A \sin B \sin C)(\cos A \cos B \cos C) \\ &= 8\Delta \cos A \cos B \cos C \quad (\because 2R^2 \sin A \sin B \sin C = \Delta) \end{aligned}$$

Hence (D) is true.

Answers: (A), (D)

5. Quadrilateral $ABCD$ is inscribed in a circle with AD as diameter. If $AD = 4$, $AB = BC = 1$, then

$$(A) AC = \sqrt{15}/2$$

$$(B) CD = 7/2$$

$$(C) \text{ area of } \triangle ACD = 7\sqrt{15}/8$$

$$(D) \text{ perimeter of the quadrilateral } ABCD = 19/2$$

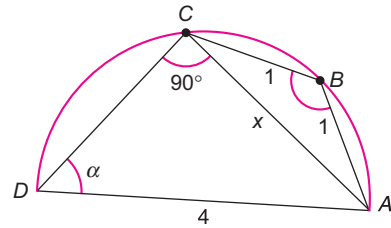


FIGURE 4.66 Multiple correct choice type question 5.

Solution: See Figure 4.66.

- (A) Let $\angle ADC = \alpha$ and $AC = x$. Using cosine formula for $\triangle ABC$,

$$\begin{aligned} x^2 &= 1 + 1^2 - 2(1)(1) \cos(180^\circ - \alpha) \\ &= 2 + 2 \cos \alpha \\ &= 2 + 2 \frac{CD}{4} \\ &= 2 + \frac{1}{2} \sqrt{16 - x^2} \end{aligned}$$

Therefore

$$\begin{aligned} (x^2 - 2)^2 &= \frac{16 - x^2}{4} \\ 4x^4 - 15x^2 &= 0 \\ x^2 &= \frac{15}{4} \\ x &= \frac{\sqrt{15}}{2} \end{aligned}$$

Hence (A) is true.

(B) Now

$$\begin{aligned} CD &= \sqrt{16 - x^2} \\ &= \sqrt{16 - \frac{15}{4}} \\ &= \frac{\sqrt{49}}{2} = \frac{7}{2} \end{aligned}$$

Therefore (B) is true.

(C) The area of $\triangle ACD$ is given by

$$\frac{1}{2} CD \cdot AC = \frac{1}{2} \times \frac{7}{2} \times \frac{\sqrt{15}}{2} = \frac{7\sqrt{15}}{8}$$

Hence (C) is true.

(D) The perimeter of the quadrilateral $ABCD$ is

$$1 + 1 + \frac{7}{2} + 4 = \frac{19}{2}$$

Hence (D) is true.

Answers: (A), (B), (C), (D)

6. Let ABC be an equilateral triangle with side length 20 units. Let P be a point inside the triangle ABC . If PD , PE and PF are drawn perpendiculars to the sides BC , CA and AB , respectively, then

- (A) value of $PD + PE + PF$ depends on the position of P
 (B) value of $PD + PE + PF$ is independent of the position of P
 (C) $PD + PE + PF = 10\sqrt{3}$
 (D) $PD + PE + PF = \frac{PA + PB + PC}{10}$

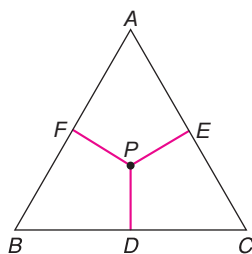


FIGURE 4.67 Multiple correct choice type question 6.

Solution: See Figure 4.67. We have

$$\begin{aligned} \frac{\sqrt{3}}{4} \times 20^2 &= \text{Area of } \triangle ABC \\ &= (\text{Area of } \triangle PBC) + (\text{Area of } \triangle PCA) \\ &\quad + (\text{Area of } \triangle PAB) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2}(BC)(PD) + \frac{1}{2}(PE)(CA) + \frac{1}{2}(AB)(PF) \\ &= \frac{1}{2}(20)(PD + PE + PF) \\ &= 10(PD + PE + PF) \end{aligned}$$

Therefore

$$PD + PE + PF = \frac{\sqrt{3}}{4} \times \frac{20 \times 20}{10} = 10\sqrt{3}$$

Answers: (B), (C)

7. In $\triangle ABC$, $\angle C = 90^\circ$, $BC = 3$, and $AC = 4$. D is a point on the side AB (in between A and B) such that $\angle ACD = 60^\circ$. Then

- (A) $CD = \frac{8\sqrt{3}}{4 + \sqrt{3}}$ (B) $AD = \frac{20}{4 + \sqrt{3}}$
 (C) $BD = \frac{15}{3 + 4\sqrt{3}}$ (D) $CD = \frac{8}{4 + \sqrt{3}}$

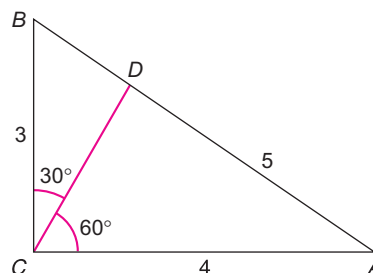


FIGURE 4.68 Multiple correct choice type question 7.

Solution: See Figure 4.68. By Pythagoras theorem, $AB = 5$. From $\triangle ACD$, using sine rule we get

$$\begin{aligned} \frac{AD}{\sin 60^\circ} &= \frac{CD}{\sin A} \\ AD &= \frac{(CD) \sin 60^\circ}{\sin A} \\ &= \frac{5}{3}(CD) \frac{\sqrt{3}}{2} \end{aligned}$$

Similarly from $\triangle BCD$,

$$BD = \frac{5}{4}(CD) \frac{1}{2}$$

Therefore

$$\begin{aligned} 5 &= AB \\ &= AD + BD \end{aligned}$$

$$\begin{aligned}
 &= 5(CD) \left(\frac{1}{2\sqrt{3}} + \frac{1}{8} \right) \\
 &= 5(CD) \frac{(4 + \sqrt{3})}{8\sqrt{3}}
 \end{aligned}$$

Hence

$$CD = \frac{8\sqrt{3}}{4 + \sqrt{3}}$$

Therefore (A) is true. Now

$$\begin{aligned}
 AD &= \frac{5}{3}(CD) \frac{\sqrt{3}}{2} \\
 &= \frac{5}{3} \left(\frac{8\sqrt{3}}{4 + \sqrt{3}} \right) \frac{\sqrt{3}}{2} \\
 &= \frac{20}{4 + \sqrt{3}}
 \end{aligned}$$

Hence (B) is true. Now

$$\begin{aligned}
 BD &= \frac{5}{4}(CD) \frac{1}{2} \\
 &= \frac{5}{4} \left(\frac{8\sqrt{3}}{4 + \sqrt{3}} \right) \frac{1}{2} \\
 &= \frac{5}{4} \left(\frac{24}{3 + 4\sqrt{3}} \right) \frac{1}{2} \\
 &= \frac{15}{3 + 4\sqrt{3}}
 \end{aligned}$$

Hence (C) is true.

Answers: (A), (B), (C)

8. In $\triangle ABC$, $\angle A = 45^\circ$, $BC = \sqrt{8}$, $AB = \sqrt{12}$. Then

- (A) $\angle C = 60^\circ$ or 120°
- (B) $\angle B = 75^\circ$ or 15°
- (C) $AC = 6 + \sqrt{2}$ or $6 - \sqrt{2}$
- (D) $AC = \sqrt{6} + \sqrt{2}$ or $\sqrt{6} - \sqrt{2}$

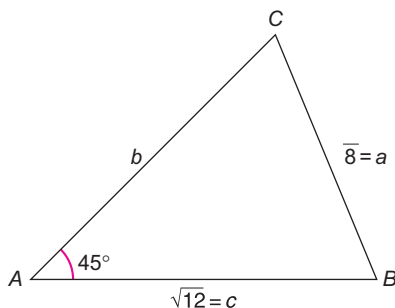


FIGURE 4.69 Multiple correct choice type question 8.

Solution: See Figure 4.69. By sine rule

$$\frac{\sqrt{8}}{\sin 45^\circ} = \frac{\sqrt{12}}{\sin C} = \frac{b}{\sin B}$$

Therefore

$$\begin{aligned}
 \sin C &= \frac{\sqrt{12} \sin 45^\circ}{\sqrt{8}} \\
 &= \frac{\sqrt{12}}{\sqrt{8}} \times \frac{1}{\sqrt{2}} \\
 &= \frac{2\sqrt{3}}{4} \\
 &= \frac{\sqrt{3}}{2}
 \end{aligned}$$

This implies that $\angle C$ is 60° or 120° . Therefore (A) is true. Now

$$\angle C = 60^\circ \Rightarrow \angle B = 180^\circ - (45^\circ + 60^\circ) = 75^\circ$$

$$\angle C = 120^\circ \Rightarrow \angle B = 180^\circ - (45^\circ + 120^\circ) = 15^\circ$$

Hence (B) is true.

Finally depending on the two values of angle B we get

$$\begin{aligned}
 b &= \frac{\sqrt{8}}{\sin 45^\circ} \times \sin B \\
 &= \sqrt{2} \times \sqrt{8} \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right) \quad \text{or} \quad \sqrt{2} \times \sqrt{8} \left(\frac{\sqrt{3} - 1}{2\sqrt{2}} \right) \\
 &= \sqrt{2}(\sqrt{3} + 1) \quad \text{or} \quad \sqrt{2}(\sqrt{3} - 1) \\
 &= \sqrt{6} + \sqrt{2} \quad \text{or} \quad \sqrt{6} - \sqrt{2}
 \end{aligned}$$

Therefore (D) is true.

Answers: (A), (B), (D)

9. In $\triangle ABC$, if

$$a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$$

then

- (A) a, b, c are in HP
- (B) a, b, c are in AP
- (C) $\cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2}$ are in AP
- (D) $\cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2}$ are in HP

Solution: Given that

$$a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$$

Therefore

$$\begin{aligned}\frac{a(1+\cos C)}{2} + \frac{c(1+\cos A)}{2} &= \frac{3b}{2} \\ a+c+(a\cos C+c\cos A) &= 3b \\ a+c+b &= 3b \\ a+c &= 2b\end{aligned}$$

Therefore a, b, c are in AP. Hence (B) is true. Now

$$\begin{aligned}\cot \frac{A}{2} + \cot \frac{C}{2} &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\ &= \frac{\sqrt{s}[s-a+s-c]}{\sqrt{(s-a)(s-b)(s-c)}} \\ &= \frac{\sqrt{s}(2s-2b)}{\sqrt{(s-a)(s-b)(s-c)}} \quad (\because a+c=2b) \\ &= 2\sqrt{\frac{s(s-b)}{(s-c)(s-a)}} \\ &= 2\cot \frac{B}{2}\end{aligned}$$

Therefore $\cot(A/2), \cot(B/2), \cot(C/2)$ are in AP. Therefore (C) is true.

Answers: (B) and (C)

10. In $\triangle ABC$, if

$$\cos A + \cos B = 4\sin^2 \frac{C}{2}$$

then

(A) $a+b=2c$

(B) a, b, c are in HP

(C) $\tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$ are in AP

(D) $\tan \frac{A}{2}, \tan \frac{C}{2}, \tan \frac{B}{2}$ are in HP

Solution: Given that

$$\cos A + \cos B = 4\sin^2 \frac{C}{2}$$

Therefore

$$\begin{aligned}2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) &= 4\sin^2 \frac{C}{2} \\ 2\sin \frac{C}{2}\cos\left(\frac{A-B}{2}\right) &= 4\sin^2 \frac{C}{2}\end{aligned}$$

$$\cos\left(\frac{A-B}{2}\right) = 2\sin \frac{C}{2}$$

$$2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) = \left(2\sin \frac{C}{2}\right)\left(2\cos \frac{C}{2}\right)$$

$$\sin A + \sin B = 2\sin C$$

Therefore $a+b=2c$. Hence (A) is true. Also a, c, b are in AP. Since the sides a, c and b are in AP, from Problem 9 above it follows that $\cot(A/2), \cot(C/2), \cot(B/2)$ are also in AP. Hence $\tan(A/2), \tan(C/2), \tan(B/2)$ are in HP. This implies that (D) is true.

Answers: (A) and (D)

11. In $\triangle ABC$, which of the following are true?

(A) If a^2, b^2, c^2 are in AP, then $\cot A, \cot B, \cot C$ are in AP.

(B) If $\sin^2 A + \sin^2 B + \sin^2 C = 2$, then the triangle is right angled.

(C) If a, b, c are in AP, then $\cot \frac{A}{2} \cot \frac{C}{2}$ is equal to 3.

(D) $b^2 \sin 2C + c^2 \sin 2B = (bc) \sin 2A$

Solution:

(A) Now a^2, b^2, c^2 are in AP. Therefore

$$b^2 - a^2 = c^2 - b^2$$

$$\sin^2 B - \sin^2 A = \sin^2 C - \sin^2 B$$

$$\sin(B+A)\sin(B-A) = \sin(C+B)\sin(C-B)$$

$$\sin C \sin(B-A) = \sin A \sin(C-B)$$

$$\begin{aligned}\sin C(\sin B \cos A - \cos B \sin A) &= \sin A(\sin C \cos B \\ &\quad - \sin B \cos C)\end{aligned}$$

Dividing both sides with $\sin A \sin B \sin C$ we get

$$\frac{\cos A}{\sin A} - \frac{\cos B}{\sin B} = \frac{\cos B}{\sin B} - \frac{\cos C}{\sin C}$$

$$\cot A - \cot B = \cot B - \cot C$$

Therefore $\cot A, \cot B, \cot C$ are in AP. Therefore (A) is true.

(B) Given that $\sin^2 A + \sin^2 B + \sin^2 C = 2$. Therefore

$$\frac{1-\cos 2A}{2} + \frac{1-\cos 2B}{2} + \frac{1-\cos 2C}{2} = 2$$

$$\cos 2A + \cos 2B + \cos 2C = -1$$

$$-1 - 4\cos A \cos B \cos C = -1$$

$$\cos A \cos B \cos C = 0$$

Hence one of the angles must be a right angle and so (B) is true.

(C) By hypothesis $a + c = 2b$. So

$$\begin{aligned}\cot \frac{A}{2} \cot \frac{C}{2} &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \times \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\ &= \frac{s}{s-b} \\ &= \frac{2s}{2s-2b} \\ &= \frac{a+b+c}{a+b+c-2b} \\ &= \frac{3b}{3b-2b} = 3\end{aligned}$$

Hence (C) is true.

(D) We have

$$\begin{aligned}b^2 \sin 2C + c^2 \sin 2B &= 2b^2 \sin C \cos C + 2c^2 \sin B \cos B \\ &= 2b(c \sin B) \cos C + 2c(b \sin C) \cos B \\ &= 2bc(\sin B \cos C + \cos B \sin C) \\ &= 2bc \sin(B+C) \\ &= 2bc \sin A\end{aligned}$$

Hence (D) is not true.

Answers: (A), (B), (C)

12. Let $0 < \alpha, \beta < \pi/2$ satisfying the relation

$$8 \cos \alpha \cos \beta \cos(\alpha + \beta) = -1$$

be two angles of a triangle. Let θ be the third angle of the triangle. Then

(A) $\theta = \frac{7\pi}{12}$

(B) $\theta = \frac{\pi}{3}$

(C) the circumradius $R = a/\sqrt{3}$ where “ a ” is the side opposite to the angle θ

(D) the value of $\cos \alpha + \cos \beta + \cos \theta = 3/2$

Solution: It is given that

$$8 \cos \alpha \cos \beta \cos(\alpha + \beta) = -1$$

Therefore

$$\begin{aligned}4[\cos(\alpha + \beta) + \cos(\alpha - \beta)]\cos(\alpha + \beta) &= -1 \\ 4\cos^2(\alpha + \beta) + 4\cos(\alpha - \beta)\cos(\alpha + \beta) + 1 &= 0 \\ [2\cos(\alpha + \beta) + \cos(\alpha - \beta)]^2 + 1 - \cos^2(\alpha - \beta) &= 0 \\ [2\cos(\alpha + \beta) + \cos(\alpha - \beta)]^2 &= -\sin^2(\alpha - \beta)\end{aligned}$$

This gives

$$2\cos(\alpha + \beta) + \cos(\alpha - \beta) = 0$$

and $\sin(\alpha - \beta) = 0$

Hence

$$\alpha = \beta \quad \text{and} \quad \alpha + \beta = \frac{2\pi}{3} \quad \left(\because 0 < \alpha, \beta < \frac{\pi}{2} \right)$$

which implies

$$\alpha = \beta = \frac{\pi}{3}$$

So

$$\theta = \frac{\pi}{3}$$

Hence (B) is true.

(C) and (D) are also true because the triangle is equilateral.

Answers: (B), (C), (D)

13. In $\triangle ABC$, the internal bisector AD of the angle A is equal to

(A) $\left(\frac{2bc}{b+c} \right) \cos \frac{A}{2}$

(B) $\left(\frac{b+c}{2bc} \right) \cos \frac{C}{2}$

(C) $\frac{b \sin C}{\sin[B + (A/2)]}$

(D) $\frac{b \sin C}{\sin[A + (B/2)]}$

Solution: See Figure 4.70.

$$\text{Area of } \triangle ABD = \frac{1}{2} AB \cdot AD \cdot \sin \frac{A}{2}$$

$$\text{Area of } \triangle ACD = \frac{1}{2} AC \cdot AD \cdot \sin \frac{A}{2}$$

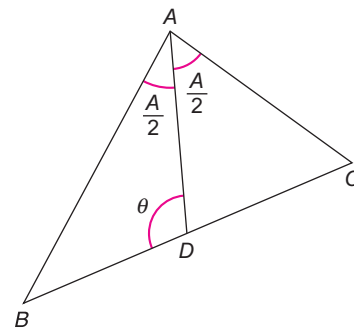


FIGURE 4.70 Multiple correct choice type question 13.

Therefore

$$\begin{aligned}\Delta &= \text{Area of } \triangle ABC = \text{Area of } \triangle ABD + \text{Area of } \triangle ACD \\ &= \frac{1}{2} AD(AB + AC) \sin \frac{A}{2} \\ &= \frac{1}{2} AD(c + b) \sin \frac{A}{2}\end{aligned}$$

This gives

$$\begin{aligned} AD &= \frac{2\Delta}{(b+c)\sin(A/2)} \\ &= \frac{2[(1/2)bc\sin A]}{(b+c)\sin(A/2)} \\ &= \frac{bc[2\sin(A/2)\cos(A/2)]}{(b+c)\sin(A/2)} \\ &= \left(\frac{2bc}{b+c}\right)\cos\frac{A}{2} \end{aligned}$$

So (A) is true.

Again, from $\triangle ACD$

$$\frac{AD}{\sin C} = \frac{AC}{\sin(180^\circ - \theta)}$$

where $\theta = \angle ADB$. Hence

$$AD = \frac{b \sin C}{\sin[B + (A/2)]} \left(\because B + \frac{A}{2} + \theta = 180^\circ \right)$$

Therefore (C) is true.

Answers: (A), (C)

14. In $\triangle ABC$, $\angle A = 90^\circ$, D is the mid-point of BC and m is the length of the altitude AM drawn from A onto BC . It is given that the length of the median AD is geometric mean between the adjacent sides AB and AC . Let R be the circumradius of $\triangle ABC$. Then

- (A) $m = \frac{R}{2}$
 (B) $\angle ADM = 30^\circ$
 (C) the acute angles of $\triangle ABC$ are 15° and 75°
 (D) $\angle C = 60^\circ$ and $\angle B = 30^\circ$

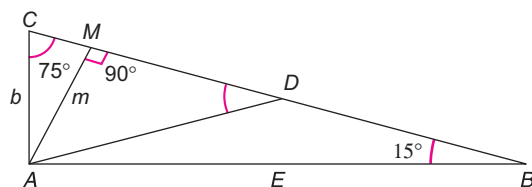


FIGURE 4.71 Multiple correct choice type question 14.

Solution: See Figure 4.71. Since $\angle A = 90^\circ$, BC is the circumdiameter of $\triangle ABC$ and hence $AD = DC = DB = R$. Now

$$\frac{1}{2}bc = \text{Area of } \triangle ABC$$

$$= \frac{1}{2}(m)(a) = \frac{1}{2}m(2R)$$

Therefore

$$\begin{aligned} m &= \frac{bc}{2R} \\ &= \frac{R^2}{2R} \quad (\because bc = AD^2 = R^2) \\ &= \frac{R}{2} \end{aligned}$$

Therefore (A) is true. From $\triangle ADM$,

$$\sin(\angle ADM) = \frac{AM}{AD} = \frac{m}{R} = \frac{1}{2}$$

So $\angle ADM = 30^\circ$. Hence (B) is true. Also

$$\angle ADB = 180^\circ - 30^\circ = 150^\circ$$

and hence

$$\angle DAB = \angle B = 15^\circ \quad (\because AD = DB)$$

This implies

$$\angle C = 75^\circ$$

Also $\angle B = 15^\circ$, $\angle C = 75^\circ$ imply that (C) is true.

Answers: (A), (B), (C)

15. In $\triangle ABC$, let r be the inradius; and r_1, r_2 and r_3 are the three exradii; s is the semiperimeter of the triangle. Then which of the following are true?

- (A) $rr_1 = r_2r_3 \tan^2 \frac{A}{2}$
 (B) $(rr_1) \tan \frac{A}{2} = \text{Area of } \triangle ABC$
 (C) $r_1r_2 + r_2r_3 + r_3r_1 = s^2$
 (D) $(r_3 - r) \cot \frac{C}{2} = c$

Solution: We know that

$$r = \frac{\Delta}{s}, \quad r_1 = \frac{\Delta}{s-a}, \quad r_2 = \frac{\Delta}{s-b} \quad \text{and} \quad r_3 = \frac{\Delta}{s-c}$$

Also

$$\begin{aligned} \cot \frac{C}{2} &= \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\ \tan \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \end{aligned}$$

(A) We have

$$\frac{rr_1}{r_2r_3} = \frac{\frac{\Delta}{s} \times \frac{\Delta}{s-a}}{\frac{\Delta}{s-b} \times \frac{\Delta}{s-c}} = \frac{(s-b)(s-c)}{s(s-a)} = \tan^2 \frac{A}{2}$$

Therefore (A) is true.

(B) We have

$$rr_1 = \frac{\Delta}{s} \times \frac{\Delta}{s-a} = \frac{s(s-a)(s-b)(s-c)}{s(s-a)} = (s-b)(s-c)$$

Therefore

$$\begin{aligned} (rr_1) \cot \frac{A}{2} &= (s-b)(s-c) \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \\ &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \Delta \end{aligned}$$

Hence

$$(rr_1) \tan \frac{A}{2} \neq \Delta$$

(B) is not true.

(C) We have

$$\begin{aligned} r_1r_2 + r_2r_3 + r_3r_1 &= \frac{\Delta^2}{(s-a)(s-b)} + \frac{\Delta^2}{(s-b)(s-c)} \\ &\quad + \frac{\Delta^2}{(s-c)(s-a)} \\ &= \frac{\Delta^2[s-c + s-a + s-b]}{(s-a)(s-b)(s-c)} \\ &= \frac{\Delta^2(3s-2s)}{s(s-a)(s-b)(s-c)} = \frac{\Delta^2 s^2}{\Delta^2} = s^2 \end{aligned}$$

Hence (C) is true.

(D) We have

$$\begin{aligned} (r_3 - r) \cot \frac{C}{2} &= \left(\frac{\Delta}{s-c} - \frac{\Delta}{s} \right) \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\ &= \frac{\Delta(c)}{\sqrt{s(s-a)(s-b)(s-c)}} = \frac{\Delta(c)}{\Delta} = c \end{aligned}$$

So (D) is true.

Answers: (A), (C), (D)

16. In $\triangle ABC$, D , E and F are the feet of the altitudes drawn from the vertices A , B and C , respectively, onto the opposite sides. For $\triangle DEF$, which of the following are true?

(A) $EF = a \cos A$

(B) $\angle EDF = 180^\circ - 2A$

(C) Area of $\triangle DEF = 2\Delta \cos A \cos B \cos C$ where Δ is the area of $\triangle ABC$

(D) The radius of the incircle of $\triangle DEF$ is $2R \cos A \cos B \cos C$

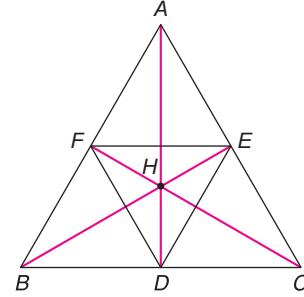


FIGURE 4.72 Multiple correct choice type question 16.

Solution: Let H be the orthocentre of $\triangle ABC$ (Figure 4.72). Then

$$\angle HDC = \angle HEC = 90^\circ$$

which implies $HDCE$ is a cyclic quadrilateral. Therefore

$$\angle HDE = \angle HCE = 90^\circ - A$$

Similarly

$$\angle HDF = \angle HBF = 90^\circ - A$$

Therefore

$$\angle EDF = 180^\circ - 2A$$

Hence (B) is true. Similarly

$$\angle DEF = 180^\circ - 2B \quad \text{and} \quad \angle FED = 180^\circ - 2C$$

Now, from $\triangle AFE$,

$$\frac{EF}{\sin A} = \frac{AE}{\sin(\angle AFE)} \quad (4.27)$$

Now

$$\angle AFE = 90^\circ - \angle EFH = 90^\circ - \angle HAE = C \quad (4.28)$$

Again, from $\triangle ABE$, $\angle ABE = 90^\circ - A$. Therefore

$$\begin{aligned} \cos A &= \sin(90^\circ - A) = \frac{AE}{AB} = \frac{AE}{c} \\ AE &= c \cos A \end{aligned} \quad (4.29)$$

From Eqs. (4.27), (4.28) and (4.29) we get

$$\frac{EF}{\sin A} = \frac{c \cos A}{\sin C} = 2R \cos A$$

Hence

$$EF = 2R \sin A \cos A = a \cos A$$

and so (A) is true. Now

$$\begin{aligned} \text{Area of } \triangle DEF &= \frac{1}{2}(DE)(DF) \sin D \\ &= \frac{1}{2}(c \cos C)(b \cos B) \sin(180^\circ - 2A) \\ &= \frac{1}{2}(bc)(\cos B \cos C)(2 \sin A \cos A) \\ &= (bc \sin A)(\cos A \cos B \cos C) \\ &= 2\Delta \cos A \cos B \cos C \end{aligned}$$

Hence (C) is true. Also

$$\begin{aligned} \text{Inradius of } \triangle DEF &= \frac{\text{Area of } \triangle DEF}{(1/2)(DE + EF + FD)} \\ &= \frac{4\Delta \cos A \cos B \cos C}{a \cos A + b \cos B + c \cos C} \\ &= \frac{4\Delta \cos A \cos B \cos C}{R(\sin 2A + \sin 2B + \sin 2C)} \\ &= \frac{4\Delta \cos A \cos B \cos C}{R(4 \sin A \sin B \sin C)} \\ &= \frac{4\Delta \cos A \cos B \cos C}{R\left(4 \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R}\right)} \\ &= \frac{8R^2(\Delta) \cos A \cos B \cos C}{abc} \\ &= \frac{8R^2(\Delta) \cos A \cos B \cos C}{4R\Delta} \\ &= 2R \cos A \cos B \cos C \end{aligned}$$

Hence (D) is also true.

Answers: (A), (B), (C), (D)

17. For $\triangle ABC$, let $A_1B_1C_1$ be the triangle formed by the excentres of $\triangle ABC$. Let $\triangle A_2B_2C_2$ be formed by the excentres of $\triangle A_1B_1C_1$ and so on. Finally, let $\triangle A_nB_nC_n$ be the n th derived triangle, that is, it is formed by the excentres of $\triangle A_{n-1}B_{n-1}C_{n-1}$. Then

(A) $\angle A_1 = 90^\circ - \frac{A}{2}$

(B) the incentre of $\triangle ABC$ is the orthocentre of $\triangle A_1B_1C_1$

(C) $B_1C_1 = a \operatorname{cosec} \frac{A}{2}$

(D) as $n \rightarrow \infty$, $\triangle A_nB_nC_n$ tends to become equilateral

Solution: See Figure 4.73. Clearly C_1, A, B_1 are collinear and $\triangle ABC$ is the pedal triangle of $\triangle A_1B_1C_1$. Hence the incentre I is the orthocentre of $\triangle A_1B_1C_1$. Therefore (B) is true.

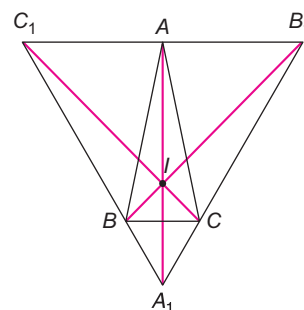


FIGURE 4.73 Multiple correct choice type question 17.

Since $\triangle ABC$ is the pedal triangle of $\triangle A_1B_1C_1$, from Problem 16 (part B) we get $A = 180^\circ - 2A_1$, and therefore

$$A_1 = 90^\circ - \frac{A}{2}$$

Hence (A) is true. Also, we can write

$$A_1 = \frac{\pi}{3} + \left(-\frac{1}{2}\right)\left(A - \frac{\pi}{3}\right)$$

$$\begin{aligned} A_2 &= 90^\circ - \frac{A_1}{2} \\ &= \frac{\pi}{3} + \left(\frac{-1}{2}\right)^2 \left(A - \frac{\pi}{3}\right) \end{aligned}$$

By induction,

$$A_n = \frac{\pi}{3} + \left(\frac{-1}{2}\right)^n \left(A - \frac{\pi}{3}\right)$$

Applying limit we get

$$\lim_{n \rightarrow \infty} A_n = \frac{\pi}{3} \quad \left(\because \lim_{n \rightarrow \infty} \left(-\frac{1}{2}\right)^n = 0 \right)$$

Therefore (D) is true. Again from Problem 16 (part A),

$$\begin{aligned} BC &= (B_1C_1) \cos A_1 \\ &= (B_1C_1) \cos \left(90^\circ - \frac{A}{2}\right) \\ &= (B_1C_1) \sin \frac{A}{2} \end{aligned}$$

Therefore

$$B_1C_1 = a \operatorname{cosec} \frac{A}{2}$$

Hence (C) is true.

Answers: (A), (B), (C), (D)

18. Which of the following statements are true?

- (A) In $\triangle ABC$, if $a:b:c = 4:5:6$, then $R:r$ is $16:7$.
- (B) If $r_1 = 2r_2 = 3r_3$, then $a:b$ is equal to $5:4$.
- (C) In $\triangle ABC$, if $r_1:r_2:r_3 = 4:5:6$, then $\angle A = \angle C$.
- (D) $\triangle ABC$ is uniquely determined, if side a , $\sin A$ and circumradius R are known.

Solution:

(A) Let $a = 4K$, $b = 5K$ and $c = 6K$. Therefore

$$2s = a + b + c = 15K$$

so that

$$s = \frac{15K}{2}$$

Hence

$$\begin{aligned} \Delta^2 &= s(s-a)(s-b)(s-c) \\ &= \frac{15K}{2} \left(\frac{15K}{2} - 4K \right) \left(\frac{15K}{2} - 5K \right) \left(\frac{15K}{2} - 6K \right) \\ &= (15 \times 7 \times 5 \times 3) \frac{K^4}{16} \end{aligned}$$

$$\Delta = \frac{15\sqrt{7}K^2}{4}$$

Now

$$\begin{aligned} r &= \frac{\Delta}{s} = \frac{15\sqrt{7}K^2}{4} \times \frac{2}{15K} = \frac{K\sqrt{7}}{2} \\ R &= \frac{abc}{4\Delta} = \frac{4 \cdot 5 \cdot 6K^3}{15\sqrt{7}K^2} = \frac{8K}{\sqrt{7}} \end{aligned}$$

Therefore

$$R:r = \frac{8}{\sqrt{7}}(K) : \frac{\sqrt{7}}{2}(K) = 16:7$$

which implies that (A) is true.

(B) We have

$$r_1 = \frac{\Delta}{s-a}, \quad r_2 = \frac{\Delta}{s-b}, \quad r_3 = \frac{\Delta}{s-c}$$

Therefore

$$\begin{aligned} \frac{\Delta}{s-a} &= 2 \left(\frac{\Delta}{s-b} \right) = 3 \left(\frac{\Delta}{s-c} \right) \\ \frac{1}{s-a} &= \frac{2}{s-b} = \frac{3}{s-c} = \frac{1+2+3}{3s-2s} = \frac{6}{s} \end{aligned}$$

Equation first with the last one we get

$$s = 6(s-a) \Rightarrow a = \frac{5}{6}s$$

Equation second with the last one we get

$$2s = 6(s-b) \Rightarrow b = \frac{4}{6}s$$

Therefore $a:b = 5:4$. Hence (B) is true.

(C) We have $r_1 = 4k$, $r_2 = 5k$, $r_3 = 6k$. Therefore

$$r_2 - r_1 = r_3 - r_2$$

We have from Corollary 4.5 that

$$r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$r_2 = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}$$

$$\text{and} \quad r_3 = 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

Therefore

$$r_2 - r_1 = r_3 - r_2$$

$$\Rightarrow 4R \cos \frac{C}{2} \sin \left(\frac{A-B}{2} \right) = 4R \cos \frac{A}{2} \sin \left(\frac{B-C}{2} \right)$$

If $A = C$, then from above it follows that

$$\frac{A-B}{2} = \frac{B-C}{2}$$

so that $\triangle ABC$ becomes equilateral. Hence (C) is not true.

- (D) We have that $a/\sin A = 2R$. But, the sides b and c cannot be determined with this hypothesis. Therefore (D) is not true.

Answers: (A), (B)

19. For $\triangle ABC$, which of the following are true?

- (A) If the sides a , b , c and Δ are rational, then a , $\tan(B/2)$, $\tan(C/2)$ are rational.
- (B) If a , $\tan(B/2)$, $\tan(C/2)$ are rational, then a , $\sin A$, $\sin B$ and $\sin C$ are rational.

- (C) If $a, \sin A, \sin B$ and $\sin C$ are rational, then a, b, c and Δ are rational.
- (D) If a, b, c and Δ are rational, then r, r_1, r_2 and r_3 are rational.

Solution:

(A) Suppose a, b, c and Δ are rational. Now

$$s = \frac{a+b+c}{2}$$

and so, s is rational. Now

$$\tan \frac{B}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{(s-b)(s-c)}{\Delta}$$

is rational. Similarly $\tan(C/2)$ is rational. Therefore

$$a, \tan \frac{B}{2}, \tan \frac{C}{2} \text{ are rational}$$

and (A) is true.

(B) Suppose $a, \tan(B/2), \tan(C/2)$ are rational. Therefore

$$\sin B = \frac{2 \tan(B/2)}{1 + \tan^2(B/2)}, \sin C = \frac{2 \tan(C/2)}{1 + \tan^2(C/2)}$$

are rational. Now

$$\begin{aligned} \tan \frac{A}{2} &= \tan \left(90^\circ - \frac{B+C}{2} \right) \\ &= \cot \left(\frac{B+C}{2} \right) \\ &= \frac{1 - \tan(B/2) \tan(C/2)}{\tan(B/2) + \tan(C/2)} \end{aligned}$$

Therefore $\tan(A/2)$ is also rational. Hence

$$\sin A = \frac{2 \tan(A/2)}{1 + \tan^2(A/2)}$$

is rational. So, $a, \sin A, \sin B, \sin C$ are rational and hence (B) is true.

(C) Suppose $a, \sin A, \sin B$ and $\sin C$ are rational. Therefore $2R = a/\sin A$ is rational. Similarly $b = 2R \sin B, c = 2R \sin C$ are rational and hence Δ is also rational. This implies (C) is true.

(D) Suppose a, b, c and Δ are rational. Then s and Δ are rational and hence

$$r = \frac{\Delta}{s}, r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c} \text{ are also rational}$$

Therefore (D) is also true.

Note: In $\triangle ABC$, the following statements are equivalent.

(i) a, b, c and Δ are rational.

(ii) $a, \tan \frac{B}{2}, \tan \frac{C}{2}$ are rational.

(iii) $a, \sin A, \sin B$ and $\sin C$ are rational.

Actually in this problem, we proved that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

Answers: (A), (B), (C), (D)

20. In $\triangle ABC$, if the angles are in AP and $3a^2 = 2b^2$, then

(A) $\angle A = 45^\circ$

(B) $\angle C = 75^\circ$

(C) $\angle C = 45^\circ$

(D) $\angle A = 60^\circ$

Solution: We have

$$A + B + C = 180^\circ$$

and

$$A + C = 2B$$

Solving the two equations we get

$$3B = 180^\circ \Rightarrow B = 60^\circ$$

Therefore

$$\begin{aligned} \frac{b}{\sin 60^\circ} &= \frac{a}{\sin A} \\ \Rightarrow \frac{2b}{\sqrt{3}} &= \frac{a}{\sin A} \\ \Rightarrow a &= \frac{(2 \sin A)b}{\sqrt{3}} \end{aligned}$$

Now

$$3a^2 = 2b^2 \Rightarrow (4 \sin^2 A)b^2 = 2b^2$$

Therefore

$$\begin{aligned} \sin^2 A &= \frac{1}{2} \\ \Rightarrow \sin A &= \frac{1}{\sqrt{2}} \\ \Rightarrow A &= 45^\circ \quad (\because B = 60^\circ \Rightarrow A \neq 135^\circ) \end{aligned}$$

Now, $A = 45^\circ, B = 60^\circ$ implies $C = 75^\circ$.

Answers: (A) and (B)

Matrix-Match Type Questions

1. Match the items of Column I to those of Column II

Column I	Column II
(A) In $\triangle ABC$, if $2a^2 + 4b^2 + c^2 = 4ab + 2ac$ then the value of $8\cos B$ is	(p) 1
(B) In $\triangle ABC$, if $\cos A \cos B + \sin A \sin B \sin C = 1$, then the value of $\cos(A - B)$ equals	(q) 0
(C) Let D be the mid-point of the side BC of $\triangle ABC$. If $\angle DAC = 90^\circ$, then $a^2 - c^2 = mb^2$, where the value of m is	(r) $2/3$
(D) In $\triangle ABC$, if $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then the value of $\cos A \cos B \cos C$ is equal to	(s) 3
	(t) 7

Solution: Consider Figure 4.74.

(A) We have

$$\begin{aligned}
 2a^2 + 4b^2 + c^2 &= 4ab + 2ac \\
 (a^2 + 4b^2 - 4ab) + (a^2 + c^2 - 2ac) &= 0 \\
 (a - 2b)^2 + (a - c)^2 &= 0
 \end{aligned}$$

Therefore

$$a = 2b \quad \text{and} \quad a = c$$

Now

$$\begin{aligned}
 8\cos B &= \frac{8(c^2 + a^2 - b^2)}{2ca} \\
 &= \frac{8(4b^2 + 4b^2 - b^2)}{2(2b)(2b)} = \frac{8 \times 7b^2}{8b^2} = 7
 \end{aligned}$$

Answer: (A) $\rightarrow$ (t)

(B) We have

$$\begin{aligned}
 \cos A \cos B + \sin A \sin B \sin C &= 1 \\
 (\cos A \cos B + \sin A \sin B) &+ (\sin A \sin B \sin C - \sin A \sin B) = 1 \\
 [\sin A \sin B (\sin C - 1)] &= 1 - \cos(A - B)
 \end{aligned}$$

Now $\text{LHS} \leq 0$ and $\text{RHS} \geq 0 \Rightarrow \text{LHS} = \text{RHS} = 0$. Hence

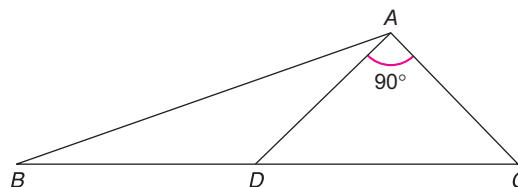
$$\cos(A - B) = 1 \quad \text{and} \quad \sin C = 1$$

Answer: (B) $\rightarrow$ (p)(C) From $\triangle ACD$,

$$\cos C = \frac{AC}{DC} = \frac{2b}{a}$$

Also from $\triangle ABC$,

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

**FIGURE 4.74** Matrix-match type question 1.

Therefore

$$\begin{aligned}
 \frac{2b}{a} &= \frac{a^2 + b^2 - c^2}{2ab} \\
 4b^2 &= a^2 + b^2 - c^2 \\
 3b^2 &= a^2 - c^2
 \end{aligned}$$

Answer: (C) $\rightarrow$ (s)

(D) We have

$$\begin{aligned}
 \cos^2 A + \cos^2 B + \cos^2 C &= 1 \\
 \frac{1 + \cos 2A}{2} + \frac{1 + \cos 2B}{2} + \frac{1 + \cos 2C}{2} &= 1 \\
 \cos 2A + \cos 2B + \cos 2C &= -1 \\
 -1 - 4\cos A \cos B \cos C &= -1 \\
 \cos A \cos B \cos C &= 0 \quad (\because A + B + C = 180^\circ)
 \end{aligned}$$

Answer: (D) $\rightarrow$ (q)

2. Match the items of Column I to those of Column II.

Column I	Column II
(A) In $\triangle ABC$, if $a = 3b$ and $A - B = 90^\circ$, then $\tan B$ equals	(p) 1
(B) AD is a diameter of the circumcircle of $\triangle ABC$, intersecting the side BC internally. Then	(q) $1/3$
(C) In $\triangle ABC$, the altitudes AD , BE and CF meet the circumcircle in P , Q and R respectively. Then	(r) $1/2$
(D) In $\triangle ABC$, if $\sin^2 A + \sin^2 B + \sin^2 C = 2$ and if θ is one of A , B and C , then the value of $\sin \theta$ is equal to	(s) 2

Solution:

(A) From Napier's rule (Theorem 4.8)

$$\begin{aligned}\tan \frac{A-B}{2} &= \frac{a-b}{a+b} \cot \left(\frac{C}{2} \right) \\ 1 &= \left(\frac{3b-b}{3b+b} \right) \cot \frac{C}{2} \\ \cot \frac{C}{2} &= 2\end{aligned}\quad (4.30)$$

Also

$$\begin{aligned}A + B + C &= 180^\circ \Rightarrow (90^\circ + B) + B + C = 180^\circ \\ \Rightarrow B &= 45^\circ - \frac{C}{2}\end{aligned}$$

Therefore

$$\begin{aligned}\tan B &= \tan \left(45^\circ - \frac{C}{2} \right) \\ &= \frac{1 - \tan(C/2)}{1 + \tan(C/2)} \quad [\text{from Eq. (4.30)}] \\ &= \frac{1 - (1/2)}{1 + (1/2)} \\ &= \frac{1}{3}\end{aligned}$$

Answer: (A) → (q)

(B) We have that AD is a diameter (Figure 4.75). This implies that

$$\angle ABD = \angle ACD = 90^\circ$$

Also $\angle ADC = \angle B$ and $\angle ADB = \angle C$. Therefore

$$\angle BDC = B + C$$

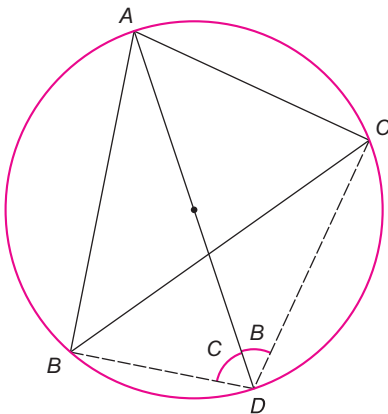


FIGURE 4.75 Matrix-match type question 2 part (B).

From $\triangle ACD$, we have

$$\cot B = \frac{DC}{AC} = \frac{DC}{b}$$

Therefore

$$DC = b \cot B$$

Similarly $BD = c \cot C$. Therefore

$$\begin{aligned}\text{Area of } \triangle BDC &= \frac{1}{2} (BD)(DC) \sin D \\ &= \frac{1}{2} (c \cot C)(b \cot B) \sin(B + C) \\ &= \frac{1}{2} (bc) \cot C \cot B \sin A \\ &= \frac{1}{2} (2R \sin B)(2R \sin C) \cot B \cot C \sin A \\ &= 2R^2 \cos B \cos C \sin A\end{aligned}$$

Therefore

$$\frac{\text{Area of } \triangle BDC}{R^2 \sin A \cos B \cos C} = 2$$

Answer: (B) → (s)

(C) Join BP and CP . Now

$$\angle PBC = \angle PAC = 90^\circ - C$$

$$\angle DPB = C \quad \text{and} \quad \angle DPC = B$$

Therefore

$$\angle BPC = B + C = 180^\circ - A$$

From $\triangle DPC$,

$$\tan B = \frac{DC}{DP}$$

$$DC = (DP) \tan B$$

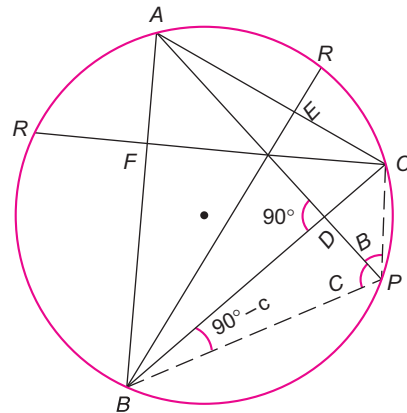


FIGURE 4.76 Matrix-match type question 2 part (C).

Again from $\triangle PBC$,

$$BD = (DP) \tan C$$

$$a = BD + DC = (DP)(\tan B + \tan C)$$

$$\frac{BC}{DP} = \frac{a}{DP} = \tan B + \tan C$$

Similarly

$$\frac{AC}{EQ} = \tan C + \tan A$$

$$\text{and } \frac{AB}{FR} = \tan A + \tan B$$

On adding we get

$$\frac{BC}{DP} + \frac{AC}{EQ} + \frac{AB}{FR} = 2(\tan A + \tan B + \tan C)$$

Therefore $k = 2$.

Answer: (C) $\rightarrow$ (s)

(D) We have

$$\begin{aligned} \sin^2 B + \sin^2 B + \sin^2 C &= 2 \\ \frac{1 - \cos 2A}{2} + \frac{1 - \cos 2B}{2} + \frac{1 - \cos 2C}{2} &= 2 \\ \cos 2A + \cos 2B + \cos 2C &= -1 \\ -1 - 4 \cos A \cos B \cos C &= -1 \\ \cos A \cos B \cos C &= 0 \end{aligned}$$

Therefore $\theta = 90^\circ$ and $\sin \theta = 1$.

Answer: (D) $\rightarrow$ (p)

3. In $\triangle ABC$, it is given that $b = \sqrt{3}$, $c = 1$ and $\angle A = 30^\circ$. Match the items of Column I with those in Column II.

Column I	Column II
(A) $\angle B$ is	(p) 1
(B) $\angle C$ is equal to	(q) 30°
(C) Side 'a' equals	(r) 2
(D) Circumradius R equals	(s) 120°

Solution: We have by Napier's rule that

$$\begin{aligned} \tan \frac{B-C}{2} &= \left(\frac{b-c}{b+c} \right) \cot \frac{A}{2} \\ &= \left(\frac{\sqrt{3}-1}{\sqrt{3}+1} \right) \cot 15^\circ \\ &= 1 \end{aligned}$$

Therefore

$$\frac{B-C}{2} = 45^\circ \quad \text{or} \quad B-C = 90^\circ$$

Also

$$B+C = 180^\circ - A = 180^\circ - 30^\circ = 150^\circ$$

Therefore $B = 120^\circ$, $C = 30^\circ$. Now

$$A = C \Rightarrow a = c = 1$$

$$\text{and } 2R = \frac{a}{\sin A} = \frac{1}{\sin 30^\circ} = 2 \Rightarrow R = 1$$

Answer: (A) $\rightarrow$ (s), (B) $\rightarrow$ (q), (C) $\rightarrow$ (p), (D) $\rightarrow$ (p)

4. The sides of a triangle are 13, 14 and 15. In Column I the inradius and the three exradii of the triangle are given and their values are given in Column II. Match these.

Column I	Column II
(A) r	(p) 14
(B) r_1	(q) $21/2$
(C) r_2	(r) 4
(D) r_3	(s) 12
	(t) $19/2$

Solution: Let $a = 13$, $b = 14$, $c = 15$. We know that $2s = a + b + c$. Therefore

$$2s = 13 + 14 + 15 = 42$$

$$\Rightarrow s = 21$$

From these values we have $s - a = 8$, $s - b = 7$, $s - c = 6$. Now

$$\Delta^2 = s(s-a)(s-b)(s-c) = 21 \times 8 \times 7 \times 6$$

$$\Delta = 7 \times 4 \times 3$$

Therefore

$$r = \frac{\Delta}{s} = \frac{7 \times 4 \times 3}{21} = 4$$

$$r_1 = \frac{\Delta}{s-a} = \frac{7 \times 4 \times 3}{8} = \frac{21}{2}$$

$$r_2 = \frac{\Delta}{s-b} = \frac{7 \times 4 \times 3}{7} = 12$$

$$r_3 = \frac{\Delta}{s-c} = \frac{7 \times 4 \times 3}{6} = 14$$

Answer: (A) $\rightarrow$ (r), (B) $\rightarrow$ (q), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)

5. For $\triangle ABC$, match the items of Column I with those of Column II.

Column I	Column II
(A) $rr_1r_2r_3$ is equal to	(p) $\frac{abc}{2R^2}$
(B) $\frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab}$ equals	(q) Δ^2
(C) $4R \sin A \sin B \sin C$ is equal to	(r) $\frac{1}{2Rr}$
(D) $r^2r_1^2 \cot^2 \frac{A}{2}$ equals	(s) Δ

Solution:

(A) We have

$$\begin{aligned} rr_1r_2r_3 &= \frac{\Delta}{s} \cdot \frac{\Delta}{s-a} \cdot \frac{\Delta}{s-b} \cdot \frac{\Delta}{s-c} \\ &= \frac{\Delta^4}{\Delta^2} = \Delta^2 \end{aligned}$$

Answer: (A) $\rightarrow$ (q)

(B) We have

$$\begin{aligned} \frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} &= \frac{a+b+c}{abc} \\ &= \frac{2s}{abc} = \frac{2s}{4R\Delta} = \frac{1}{2Rr} \end{aligned}$$

Answer: (B) $\rightarrow$ (r)

(C) We have

$$\begin{aligned} 4R \sin A \sin B \sin C &= (4R) \left(\frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} \right) \\ &= \frac{abc}{2R^2} \end{aligned}$$

Answer: (C) $\rightarrow$ (p)

(D) We have

$$\begin{aligned} r^2r_1^2 \cot^2 \frac{A}{2} &= \frac{\Delta^2}{s^2} \cdot \frac{\Delta^2}{(s-a)^2} \cdot \frac{s(s-a)}{(s-b)(s-c)} \\ &= \frac{\Delta^4}{\Delta^2} = \Delta^2 \end{aligned}$$

Answer: (D) $\rightarrow$ (q)

Comprehension-Type Questions

1. **Passage:** In $\triangle ABC$, O , I , H and G respectively denote circumcentre, incentre, orthocentre and centroid. It is known that

$$(OI)^2 = R^2 \left(1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right)$$

$$(OH)^2 = R^2 (1 - 8 \cos A \cos B \cos C)$$

Answer the following questions.

(i) $(OI)^2$ is equal to

- (A) $R^2 + 2Rr$
 (B) $R^2 - 2Rr$
 (C) $R^2 - Rr$
 (D) $R^2 + Rr$

(ii) $(OI)^2$ equals

- (A) $-R^2(2 \cos A + 2 \cos B + 2 \cos C - 3)$
 (B) $R^2(2 \cos A + 2 \cos B + 2 \cos C - 3)$
 (C) $R^2(\cos 2A + \cos 2B + \cos 2C)$
 (D) $-R^2(\cos A + \cos B + \cos C - 2)$

(iii) $(OG)^2$ is equal to

- (A) $R^2 - (a^2 + b^2 + c^2)$
 (B) $R^2 - \frac{1}{3}(a^2 + b^2 + c^2)$

$$(C) R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$$

$$(D) \frac{1}{9}(3R^2 - a^2 - b^2 - c^2)$$

Solution:

(i) We have

$$\begin{aligned} (OI)^2 &= R^2 \left(1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \\ &= R^2 - 2R \left(4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \\ &= R^2 - 2Rr \end{aligned}$$

Answer: (B)

(ii) Again

$$\begin{aligned} OI^2 &= R^2 - 8R^2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\ &= R^2 - 2R^2 \left(4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \\ &= R^2 - 2R^2(\cos A + \cos B + \cos C - 1) \\ &= R^2 - R^2(2 \cos A + 2 \cos B + 2 \cos C - 3 + 1) \\ &= -R^2(2 \cos A + 2 \cos B + 2 \cos C - 3) \end{aligned}$$

Answer: (A)

(iii) By Theorem 4.27

$$OG:GH = 1:2 \Rightarrow OG = \frac{1}{3}(OH)$$

Therefore

$$\begin{aligned} (OG)^2 &= \frac{1}{9}(OH)^2 \\ &= \frac{1}{9}(R^2 - 8R^2 \cos A \cos B \cos C) \quad (\text{Theorem 4.28}) \\ &= \frac{R^2}{9}[1 + 2(\cos 2A + \cos 2B + \cos 2C + 1)] \\ &= \frac{R^2}{9}[3 + 2(1 - 2\sin^2 A + 1 - 2\sin^2 B + 1 - 2\sin^2 C)] \\ &= \frac{R^2}{9}(9 - 4\sin^2 A - 4\sin^2 B - 4\sin^2 C) \\ &= R^2 - \frac{1}{9}(4R^2 \sin^2 A + 4R^2 \sin^2 B + 4R^2 \sin^2 C) \\ &= R^2 - \frac{1}{9}(a^2 + b^2 + c^2) \end{aligned}$$

Answer: (C)

2. Passage: I_1, I_2, I_3 are the three excentres of $\triangle ABC$ opposite to the vertices A, B, C , respectively, and I is its incentre. Answer the following three questions.

(i) II_1 is equal to

- (A) $a \sec \frac{A}{2}$ (B) $a \operatorname{cosec} \frac{A}{2}$
(C) $a \cos \frac{A}{2}$ (D) $2a \sin \frac{A}{2}$

(ii) The length I_2I_3 is equal to

- (A) $a \sec \frac{A}{2}$ (B) $a \operatorname{cosec} \frac{A}{2}$
(C) $2a \operatorname{cosec} \frac{A}{2}$ (D) $2a \sec \frac{A}{2}$

(iii) Area of $\triangle I_1I_2I_3$ is

- (A) $\frac{abc}{r}$ (B) $\frac{abc}{Rr}$
(C) $\frac{abc}{2Rr}$ (D) $\frac{abc}{2r}$

Solution: See Figure 4.77. It is easy to see that

$$\begin{aligned} AI &= r \operatorname{cosec} \frac{A}{2} \\ AI_1 &= r_1 \operatorname{cosec} \frac{A}{2} \end{aligned}$$

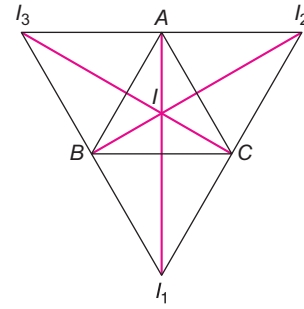


FIGURE 4.77 Comprehension type question 2.

Also, we have proved that (Problem 17 of Multiple Correct Choice Type Questions)

$$\angle I_1 = 90^\circ - \frac{A}{2}$$

$$\angle I_2 = 90^\circ - \frac{B}{2}$$

and

$$\angle I_3 = 90^\circ - \frac{C}{2}$$

(i) We have

$$\begin{aligned} II_1 &= AI_1 - AI = \operatorname{cosec} \frac{A}{2}(r_1 - r) \\ &= \operatorname{cosec} \frac{A}{2} \left(\frac{\Delta}{s-a} - \frac{\Delta}{s} \right) \\ &= \operatorname{cosec} \frac{A}{2} \left(\frac{\Delta a}{s(s-a)} \right) \\ &= a \operatorname{cosec} \frac{A}{2} \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \\ &= a \operatorname{cosec} \frac{A}{2} \tan \frac{A}{2} \\ &= a \sec \frac{A}{2} \end{aligned}$$

Answer: (A)

(ii) Also from Problem 16 (of Multiple Correct Choice Type Questions) we have

$$\begin{aligned} a &= BC = (I_2I_3) \cos \angle I_1 \\ &= I_2I_3 \cos \left(90^\circ - \frac{A}{2} \right) \end{aligned}$$

Therefore

$$I_2I_3 = a \operatorname{cosec} \frac{A}{2}$$

Answer: (B)

(iii) Now

$$I_1 I_2 = c \operatorname{cosec} \frac{C}{2}$$

$$I_1 I_3 = b \operatorname{cosec} \frac{B}{2}$$

and $\angle I_1 = 90^\circ - \frac{A}{2}$

imply that

$$\begin{aligned} \text{Area of } \Delta I_1 I_2 I_3 &= \frac{1}{2} (I_1 I_3) (I_1 I_2) \sin(\angle I_1) \\ &= \frac{1}{2} \left(b \operatorname{cosec} \frac{B}{2} \right) \left(c \operatorname{cosec} \frac{C}{2} \right) \sin \left(90^\circ - \frac{A}{2} \right) \\ &= \frac{1}{2} (bc) \operatorname{cosec} \frac{B}{2} \operatorname{cosec} \frac{C}{2} \cos \frac{A}{2} \\ &= \frac{1}{2} (bc) \operatorname{cosec} \frac{B}{2} \operatorname{cosec} \frac{C}{2} \left(\frac{2 \sin \frac{A}{2} \cos \frac{A}{2}}{2 \sin \frac{A}{2}} \right) \\ &= \frac{1}{2} (bc) \operatorname{cosec} \frac{B}{2} \operatorname{cosec} \frac{C}{2} \left(\frac{\sin A}{2 \sin \frac{A}{2}} \right) \\ &= \frac{1}{2} (bc) \left(\frac{a}{2R} \right) \left(\frac{1}{2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}} \right) \\ &= \frac{abc}{2r} \quad \left(\because r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \end{aligned}$$

Answer: (D)

3. Passage: Using sine and cosine rules and any standard formulae for a triangle, answer the following questions.

(i) $\frac{a \sec A + b \sec B + c \sec C}{\tan A + \tan B + \tan C}$ is equal to

- (A) R (B) $2R$ (C) r (D) $2r$

(ii) If the sides a, b, c are in AP, then the value of

$$\tan \frac{A}{2} + \tan \frac{C}{2} = k \cot \frac{B}{2}$$

where k is equal to

- (A) $\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{2}{3}$

(iii) If $(a+b+c)(b+c-a) = \lambda bc$, then

- (A) $0 < \lambda < 4$ (B) $0 < \lambda \leq 3$
(C) $0 < \lambda \leq 2$ (D) $0 < \lambda \leq 1$

Solution:

(i) We have

$$\begin{aligned} &\frac{a \sec A + b \sec B + c \sec C}{\tan A + \tan B + \tan C} \\ &= \frac{2R \tan A + 2R \tan B + 2R \tan C}{\tan A + \tan B + \tan C} \\ &= 2R \end{aligned}$$

Answer: (B)

(ii) By hypothesis $a + c = 2b$. Also

$$\begin{aligned} \tan \frac{A}{2} + \tan \frac{C}{2} &= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} + \sqrt{\frac{(s-a)(s-b)}{s(s-c)}} \\ &= \frac{\sqrt{s-b}(s-c+s-a)}{\sqrt{s(s-a)(s-c)}} \\ &= \frac{\sqrt{s-b}(3b-2b)}{\sqrt{s(s-a)(s-c)}} \quad (\because a+c=2b) \\ &= \frac{b\sqrt{s(s-b)}}{s\sqrt{(s-a)(s-c)}} \\ &= \frac{b}{s} \cot \frac{B}{2} \\ &= \frac{2b}{3b} \cot \frac{B}{2} \quad (\because 2s = a+b+c = 3b) \\ &= \frac{2}{3} \cot \frac{B}{2} \end{aligned}$$

Answer: (D)

(iii) We have

$$\lambda bc = (b+c)^2 - a^2 = (b^2 + c^2 - a^2) + 2bc$$

Therefore

$$\begin{aligned} bc(\lambda - 2) &= b^2 + c^2 - a^2 \\ \frac{\lambda - 2}{2} &= \frac{b^2 + c^2 - a^2}{2bc} = \cos A \end{aligned}$$

Hence

$$-1 \leq \frac{\lambda - 2}{2} \leq 1$$

But

$$\lambda = \frac{(a+b+c)(b+c-a)}{bc} > 0 \quad (\because b+c > a)$$

Therefore

$$0 < \lambda < 4 \quad (\because \lambda = 4 \Rightarrow \cos A = 1)$$

Answer: (A)

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true

1. Statement I: In $\triangle ABC$, if

$$\frac{a \cos A + b \cos B + c \cos C}{a \sin B + b \sin C + c \sin A} = \frac{a + b + c}{9R}$$

then $\triangle ABC$ is equilateral.

Statement II: If x, y, z are positive, then

$$\frac{x + y + z}{3} \geq \sqrt[3]{xyz}$$

where the equality holds if and only if $x = y = z$.

Solution: Let x_1, y_1 and z_1 be positive real numbers. Now

$$\begin{aligned} x_1^3 + y_1^3 + z_1^3 - 3x_1y_1z_1 &= (x_1 + y_1 + z_1)(x_1^2 + y_1^2 + z_1^2 - x_1y_1 - y_1z_1 - z_1x_1) \\ &= \frac{x_1 + y_1 + z_1}{2} [(x_1 - y_1)^2 + (y_1 - z_1)^2 + (z_1 - x_1)^2] \geq 0 \end{aligned}$$

where equality occurs if and only if $x_1 = y_1 = z_1$. Now put $x = x_1^3, y = y_1^3, z = z_1^3$. Therefore

$$x + y + z \geq 3(xyz)^{1/3}$$

and equality occurs if and only if $x = y = z$. Hence Statement II is true. Now

$$\begin{aligned} \frac{a \cos A + b \cos B + c \cos C}{a \sin B + b \sin C + c \sin A} &= \frac{a + b + c}{9R} \\ \frac{\sin 2A + \sin 2B + \sin 2C}{2 \sum (\sin A \sin B)} &= \frac{2}{9} (\sin A + \sin B + \sin C) \\ \frac{4 \sin A \sin B \sin C}{2 \sum (\sin A \sin B)} &= \frac{2}{9} (\sin A + \sin B + \sin C) \end{aligned}$$

$$9 \sin A \sin B \sin C = (\sin A + \sin B + \sin C)$$

$$\left(\sum (\sin A \sin B) \right)$$

$$9 = (\sin A + \sin B + \sin C)$$

$$(\operatorname{cosec} A + \operatorname{cosec} B + \operatorname{cosec} C)$$

$$\geq 9 \sqrt[3]{\sin A \sin B \sin C}$$

$$\sqrt[3]{\operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C} = 9$$

Therefore equality implies $\sin A = \sin B = \sin C$. Statement I is true.

Answer: (A)

2. Statement I: In $\triangle ABC$, if

$$x = \tan \left(\frac{B - C}{2} \right) \tan \frac{A}{2}$$

$$y = \tan \left(\frac{C - A}{2} \right) \tan \frac{B}{2}$$

$$z = \tan \left(\frac{A - B}{2} \right) \tan \frac{C}{2}$$

then $x + y + z = -xyz$.

Statement II: In $\triangle ABC$,

$$\tan \frac{B - C}{2} = \left(\frac{b - c}{b + c} \right) \cot \frac{A}{2}$$

$$\tan \frac{C - A}{2} = \left(\frac{c - a}{c + a} \right) \cot \frac{B}{2}$$

and $\tan \frac{A - B}{2} = \left(\frac{a - b}{a + b} \right) \cot \frac{C}{2}$

Solution: We have

$$\begin{aligned} \left(\frac{b - c}{b + c} \right) \cot \frac{A}{2} &= \left(\frac{\sin B - \sin C}{\sin B + \sin C} \right) \cot \frac{A}{2} \\ &= \frac{2 \cos \left(\frac{B + C}{2} \right) \sin \left(\frac{B - C}{2} \right)}{2 \sin \left(\frac{B + C}{2} \right) \cos \left(\frac{B - C}{2} \right)} \left(\frac{\cos \frac{A}{2}}{\sin \frac{A}{2}} \right) \\ &= \frac{\sin \left(\frac{B - C}{2} \right)}{\cos \left(\frac{B - C}{2} \right)} \\ &= \tan \left(\frac{B - C}{2} \right) \end{aligned}$$

Therefore Statement II is true. Now

$$x = \tan \frac{B - C}{2} \tan \frac{A}{2} = \frac{b - c}{b + c}$$

Therefore

$$x(b + c) = b - c$$

$$(x+1)c = (1-x)b$$

$$\frac{1+x}{1-x} = \frac{b}{c}$$

Similarly

$$\frac{1+y}{1-y} = \frac{c}{a} \quad \text{and} \quad \frac{1+z}{1-z} = \frac{a}{b}$$

Hence

$$\left(\frac{1+x}{1-x}\right) \cdot \left(\frac{1+y}{1-y}\right) \cdot \left(\frac{1+z}{1-z}\right) = 1$$

$$(1+x)(1+y)(1+z) = (1-x)(1-y)(1-z)$$

Simplifying both sides, we will have

$$2(x+y+z) = -2(xyz)$$

Answer: (A)

3. Statement I: In $\triangle ABC$, the altitudes AD , BE and CF meet the circumcircle in L , M and N , respectively, and Δ is the area of $\triangle ABC$. Then, the area of $\triangle LMN$ is $8\Delta \cos A \cos B \cos C$.

Statement II: $\triangle ABC$ and $\triangle LMN$ have the same circumcircle.

Solution: See Figure 4.78. Statement II is true from the hypothesis.

$$\angle ALM = \angle ABM = 90^\circ - A$$

$$\angle ALN = \angle ACN = 90^\circ - A$$

Therefore

$$\angle MLN = 180^\circ - 2A$$

Similarly,

$$\angle NML = 180^\circ - 2B \quad \text{and} \quad \angle MNL = 180^\circ - 2C$$

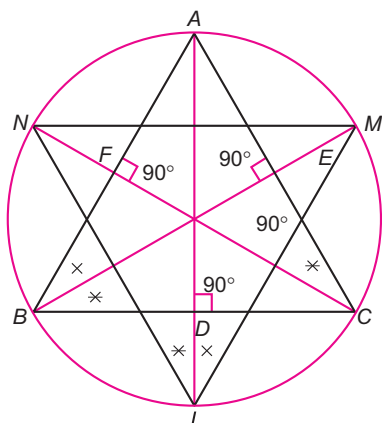


FIGURE 4.78 Assertion–reasoning type question 3.

Let R be the circumradius of $\triangle ABC$. Therefore

$$\frac{MN}{\sin(180^\circ - 2A)} = \frac{NL}{\sin(180^\circ - 2B)} = \frac{LM}{\sin(180^\circ - 2C)} = 2R$$

which gives

$$LM = 2R \sin 2C$$

$$LN = 2R \sin 2B$$

and

$$MN = 2R \sin 2A$$

Now,

$$\begin{aligned} \text{Area of } \triangle LMN &= \frac{1}{2}(LN)(LM)\sin(180^\circ - 2A) \\ &= \frac{1}{2}(2R \sin 2B)(2R \sin 2C) \sin 2A \\ &= 2R^2 (2 \sin B \cos B)(2 \sin C \cos C)(2 \sin A \cos A) \\ &= 16R^2 (\sin A \sin B \sin C)(\cos A \cos B \cos C) \\ &= 16R^2 \left(\frac{a}{2R} \cdot \frac{b}{2R} \sin C\right) (\cos A \cos B \cos C) \\ &= 8 \left(\frac{1}{2} ab \sin C\right) (\cos A \cos B \cos C) \\ &= 8\Delta \cos A \cos B \cos C \end{aligned}$$

Answer: (A)

4. Statement I: In $\triangle ABC$, $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$.

Statement II: For any real θ , $0 \leq \cos^2 \theta \leq 1$.

Solution: Statement II is true because for any real θ , $-1 \leq \cos \theta \leq 1$. Now

$$\begin{aligned} &2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\ &= \left[\cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right] \cos \left(\frac{A+B}{2} \right) \\ &= -\cos^2 \left(\frac{A+B}{2} \right) + \cos \left(\frac{A-B}{2} \right) \cos \left(\frac{A+B}{2} \right) \\ &= -\left[\cos \left(\frac{A+B}{2} \right) - \frac{1}{2} \cos \left(\frac{A-B}{2} \right) \right]^2 + \frac{1}{4} \cos^2 \left(\frac{A-B}{2} \right) \\ &\leq \frac{1}{4} \cos^2 \left(\frac{A-B}{2} \right) \leq \frac{1}{4} \end{aligned}$$

Therefore

$$\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$$

Answer: (A)

5. Statement I: In $\triangle ABC$, $\cot A + \cot B + \cot C = \frac{a^2 + b^2 + c^2}{4\Delta}$.

Statement II: In any $\triangle ABC$,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

and

$$\frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C = \Delta$$

Solution: Statement II is standard formulae. Now

$$\begin{aligned} \cot A + \cot B + \cot C &= \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C} \\ &= \frac{b^2 + c^2 - a^2}{2bc \sin A} + \frac{c^2 + a^2 - b^2}{2ca \sin B} + \frac{a^2 + b^2 - c^2}{2ab \sin C} \\ &= \frac{b^2 + c^2 - a^2}{4\Delta} + \frac{c^2 + a^2 - b^2}{4\Delta} + \frac{a^2 + b^2 - c^2}{4\Delta} \\ &= \frac{a^2 + b^2 + c^2}{4\Delta} \end{aligned}$$

Answer: (A)

6. Statement I: Let G be the centroid of $\triangle ABC$ and $\angle BGC = \alpha$, $\angle CGA = \beta$ and $\angle AGB = \gamma$. Then

$$\cot A + \cot B + \cot C + \cot \alpha + \cot \beta + \cot \gamma = 0$$

Statement II: If AD , BE and CF are the medians of $\triangle ABC$, then

$$2AD = \sqrt{2b^2 + 2c^2 - a^2}$$

$$2BE = \sqrt{2c^2 + 2a^2 - b^2}$$

$$2CF = \sqrt{2a^2 + 2b^2 - c^2}$$

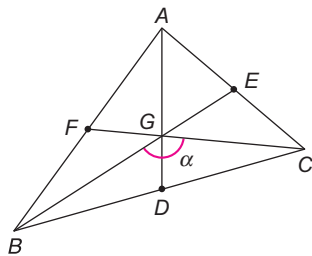


FIGURE 4.79 Assertion–reasoning type question 6.

Solution: From $\triangle ABD$ (Figure 4.79),

$$(AD)^2 = (AB)^2 + (BD)^2 - 2(AB)(BD)\cos B$$

$$= c^2 + \frac{a^2}{4} - 2(c)\left(\frac{a}{2}\right)\left(\frac{c^2 + a^2 - b^2}{2ca}\right)$$

$$= c^2 + \frac{a^2}{4} - \frac{c^2 + a^2 - b^2}{2}$$

$$= \frac{2b^2 + 2c^2 - a^2}{4}$$

$$4(AD)^2 = 2b^2 + 2c^2 - a^2$$

Thus Statement II true. Now

$$BG = \frac{2}{3}(BE)$$

$$\Rightarrow (BG)^2 = \frac{4}{9}(BE)^2 = \frac{1}{9}(2c^2 + 2a^2 - b^2) \quad (4.31)$$

Similarly

$$(CG)^2 = \frac{1}{9}(2a^2 + 2b^2 - c^2) \quad (4.32)$$

In $\triangle BGC$,

$$\cot \alpha = \frac{\cos \alpha}{\sin \alpha}$$

$$= \frac{(BG)^2 + (CG)^2 - (BC)^2}{2(BG)(CG)\sin \alpha}$$

$$= \frac{b^2 + c^2 - 5a^2}{36(\text{Area of } \triangle BGC)} \quad [\text{Eqs. (4.31) and (4.32)}]$$

$$= \frac{b^2 + c^2 - 5a^2}{36(\Delta/3)} \quad [\because \Delta = 3(\text{Area of } \triangle BGC)]$$

$$= \frac{b^2 + c^2 - 5a^2}{12\Delta}$$

Similarly

$$\cot \beta = \frac{c^2 + a^2 - 5b^2}{12\Delta}$$

and

$$\cot \gamma = \frac{a^2 + b^2 - 5c^2}{12\Delta}$$

Therefore

$$\cot \alpha + \cot \beta + \cot \gamma = \frac{-3(a^2 + b^2 + c^2)}{12\Delta}$$

$$= -\frac{(a^2 + b^2 + c^2)}{4\Delta}$$

$$= -(\cot A + \cot B + \cot C)$$

(See Problem 5 of Assertion–Reasoning Type Questions.)

Answer: (A)

Integer Answer Type Questions

1. In a triangle, the lengths of two sides are 4 and $\sqrt{6}$. The angle opposite to the side of length $\sqrt{6}$ is 30° . The number of such triangles is _____.

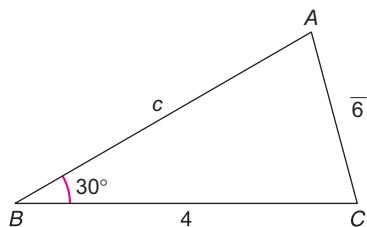


FIGURE 4.80 Integer answer type question 1.

Solution: See Figure 4.80. From $\triangle ABC$,

$$\frac{BC}{\sin A} = \frac{AC}{\sin B}$$

Substituting the values we get

$$\frac{4}{\sin A} = \frac{\sqrt{6}}{\sin 30^\circ}$$

Therefore

$$\sin A = \frac{4 \sin 30^\circ}{\sqrt{6}} = \frac{4}{\sqrt{6}} \times \frac{1}{2} = \sqrt{\frac{2}{3}}$$

$$\Rightarrow A = \sin^{-1} \sqrt{\frac{2}{3}} \quad \text{or} \quad \pi - \sin^{-1} \sqrt{\frac{2}{3}}$$

Hence the number of triangles is 2.

Answer: 2

2. In $\triangle ABC$, it is given that the values of sides $BC = a$, $c = AB$ and the angle A . Then, the third side $AC = b$ has two values b_1 and b_2 such that $b_2 = 2b_1$. Then $3a = c\sqrt{1 + \lambda \sin^2 A}$ where the value of λ is _____.

Solution: By cosine rule

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Therefore

$$b^2 - 2bc \cos A + c^2 - a^2 = 0$$

is a quadratic equation in b whose discriminant is

$$\begin{aligned} 4c^2 \cos^2 A - 4(c^2 - a^2) &= 4[c^2(1 - \sin^2 A) - c^2 + a^2] \\ &= 4(a^2 - c^2 \sin^2 A) \\ &= 4(a^2 - a^2 \sin^2 C) \\ &\quad (\because c \sin A = a \sin C) \\ &= 4a^2(1 - \sin^2 C) \geq 0 \end{aligned}$$

where equality occurs if $C = 90^\circ$ so that b has only one value. Therefore we can suppose that $\sin C \neq 1$. Therefore the equation

$$b^2 - 2bc \cos A + c^2 - a^2 = 0$$

has two distinct roots say b_1 and b_2 . Therefore

$$b_1 + b_2 = 2c \cos A$$

and

$$b_1 b_2 = c^2 - a^2$$

Since $b_2 = 2b_1$ (given) we have

$$3b_1 = 2c \cos A$$

and

$$2b_1^2 = c^2 - a^2$$

Therefore

$$\begin{aligned} c^2 - a^2 &= 2b_1^2 = 2\left(\frac{2}{3}c \cos A\right)^2 \\ 9c^2 - 9a^2 &= 8c^2 \cos^2 A = 8c^2(1 - \sin^2 A) \\ 9a^2 &= c^2(1 + 8 \sin^2 A) \\ 3a &= c\sqrt{1 + 8 \sin^2 A} \end{aligned}$$

Comparing this with the given equation we get $\lambda = 8$.

Answer: 8

3. In $\triangle ABC$, suppose the sides $a = BC$, $b = AC$ and the angle A are known. If B_1, C_1 and B_2, C_2 are the other values of the angles B and C , then

$$\frac{\sin C_1}{\sin B_1} + \frac{\sin C_2}{\sin B_2} = k \cos A$$

where k is equal to _____.

Solution: Since

$$b^2 - 2bc \cos A + c^2 - a^2 = 0 \quad (4.33)$$

we have

$$c^2 - 2c(b \cos A) + b^2 - a^2 = 0$$

Let c_1 and c_2 be values of c satisfying the above quadratic. Therefore

$$c_1 + c_2 = 2b \cos A$$

$$c_1 c_2 = b^2 - a^2$$

But also, if b_1 and b_2 are the two values of b satisfying Eq. (4.33), then we have two angles B_1 and B_2 corresponding to b_1 and b_2 . Therefore

$$\begin{aligned}
\frac{\sin C_1}{\sin B_1} + \frac{\sin C_2}{\sin B_2} &= \frac{\sin C_1}{\sin B_1} + \frac{\sin C_2}{\sin(180^\circ - B_1)} \\
&= \frac{\sin C_1}{\sin B_1} + \frac{\sin C_2}{\sin B_1} \\
&= \frac{\sin C_1 + \sin C_2}{\sin B_1} \\
&= \frac{k(c_1 + c_2)}{kb} \quad (\text{where } k \text{ is a constant}) \\
&= \frac{c_1 + c_2}{b} = \frac{2b \cos A}{b} = 2 \cos A
\end{aligned}$$

Therefore $k = 2$.

Answer: 2

4. In $\triangle ABC$, if the sides a, b, c (in this order) are in AP. Then $\cot(A/2) \cot(C/2)$ is equal to ____.

Solution: By hypothesis $a + c = 2b$. Now

$$\begin{aligned}
2s &= a + b + c = 3b \\
\Rightarrow s &= \frac{3}{2}b
\end{aligned}$$

Also

$$\begin{aligned}
\cot \frac{A}{2} \cot \frac{C}{2} &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \times \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\
&= \frac{s}{s-b} = \frac{2s}{2s-2b} = \frac{3b}{3b-2b} = 3
\end{aligned}$$

Answer: 3

5. The sides of a triangle are x, y and $\sqrt{x^2 + xy + y^2}$. If θ is the greatest angle of the triangle, then the value of $4 \sin^2 \theta$ is ____.

Solution: We have

$$\sqrt{x^2 + xy + y^2} > x, y$$

which implies the greatest angle θ is the angle opposite to the side $\sqrt{x^2 + xy + y^2}$. Therefore by cosine rule

$$x^2 + xy + y^2 = x^2 + y^2 - 2xy \cos \theta$$

Therefore

$$\cos \theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

Hence $4 \sin^2 \theta = 3$.

Answer: 3

6. In $\triangle ABC$, the value of

$$\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} = m + \frac{r}{2R}$$

where m is equal to ____.

Solution: We have

$$\begin{aligned}
&\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} \\
&= \frac{1 + \cos A}{2} + \frac{1 + \cos B}{2} + \frac{1 + \cos C}{2} \\
&= \frac{3}{2} + \frac{1}{2}(\cos A + \cos B + \cos C) \\
&= \frac{3}{2} + \frac{1}{2} \left(1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \\
&= \frac{3}{2} + \frac{1}{2} + 2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\
&= 2 + \frac{4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}}{2R} \\
&= 2 + \frac{r}{2R}
\end{aligned}$$

Comparing with the given equation we get $m = 2$.

Answer: 2

7. In $\triangle ABC$, if r, r_1, r_2 and r_3 are the inradius and exradii, respectively, then $r_1 + r_2 + r_3 - r = nR$ where n is equal to ____.

Solution: According to Corollary 4.3 and Corollary 4.5

$$\begin{aligned}
r &= 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\
r_1 &= 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \\
r_2 &= 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} \\
r_3 &= 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}
\end{aligned}$$

where R is the circumradius. Now,

$$\begin{aligned}
r_1 + r_2 &= 4R \cos \frac{C}{2} \left(\sin \frac{A}{2} \cos \frac{B}{2} + \cos \frac{A}{2} \sin \frac{B}{2} \right) \\
&= 4R \cos \frac{C}{2} \sin \left(\frac{A+B}{2} \right) \\
&= 4R \cos^2 \frac{C}{2}
\end{aligned}$$

$$\begin{aligned}
 r_3 - r &= 4R \sin \frac{C}{2} \left(\cos \frac{A}{2} \cos \frac{B}{2} - \sin \frac{A}{2} \sin \frac{B}{2} \right) \\
 &= 4R \sin \frac{C}{2} \cos \left(\frac{A+B}{2} \right) \\
 &= 4R \sin^2 \frac{C}{2}
 \end{aligned}$$

Therefore

$$r_1 + r_2 + r_3 - r = 4R \left(\cos^2 \frac{C}{2} + \sin^2 \frac{C}{2} \right) = 4R$$

Comparing with the given equation we get $n = 4$.

Answer: 4

8. In any $\triangle ABC$, $\frac{a+b+c}{r_1+r_2+r_3} \left(\frac{a}{r_1} + \frac{b}{r_2} + \frac{c}{r_3} \right)$ is equal to _____.

Solution: We have

$$\begin{aligned}
 \frac{a+b+c}{r_1+r_2+r_3} &= \frac{2s}{\Delta \left[\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} \right]} \\
 &= \frac{2s(s-a)(s-b)(s-c)}{\Delta[(s-b)(s-c) + (s-c)(s-a) + (s-a)(s-b)]}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2\Delta^2}{\Delta[3s^2 - s(b+c+c+a+a+b) + bc+ca+ab]} \\
 &= \frac{2\Delta}{bc+ca+ab-s^2} \quad (4.34)
 \end{aligned}$$

Now

$$\begin{aligned}
 \frac{a}{r_1} + \frac{b}{r_2} + \frac{c}{r_3} &= \frac{a(s-a)}{\Delta} + \frac{b(s-b)}{\Delta} + \frac{c(s-c)}{\Delta} \\
 &= \frac{s(a+b+c) - (a^2+b^2+c^2)}{\Delta} \\
 &= \frac{2s^2 - (a+b+c)^2 + 2(ab+bc+ca)}{\Delta} \\
 &= \frac{2s^2 - 4s^2 + 2(ab+bc+ca)}{\Delta} \\
 &= \frac{2[(ab+bc+ca) - s^2]}{\Delta} \quad (4.35)
 \end{aligned}$$

Therefore, from Eqs. (4.34) and (4.35)

$$\frac{a+b+c}{r_1+r_2+r_3} \left(\frac{a}{r_1} + \frac{b}{r_2} + \frac{c}{r_3} \right) = 4$$

Answer: 4

SUMMARY

Properties of Triangles

Notation: Throughout this summary, the following notation will be used for $\triangle ABC$.

- (i) The lengths of the sides BC , CA and AB will be denoted by a , b , c , respectively and

$$s = \frac{a+b+c}{2}$$

A , B and C represent the vertices as well as the angles as per the context.

- (ii) O denotes the circumcentre, I the incentre I_1 , I_2 , I_3 excentres opposite to the vertices A , B and C , respectively, G the centroid and H the orthocentre.
- (iii) R is the circumradius and r the inradius and r_1 , r_2 , r_3 the exradii of the excircles opposite to the vertices A , B and C , respectively.
- (iv) Δ denotes the area of the triangles.

4.1 Sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

4.2 Cosine formula: $a^2 = b^2 + c^2 - 2bc \cos A$, $b^2 = c^2 + a^2 - 2ca \cos B$ and $c^2 = a^2 + b^2 - 2ab \cos C$.

4.3 Projection formula: $a = b \cos C + c \cos B$, $b = c \cos A + a \cos C$ and $c = a \cos B + b \cos A$.

4.4 Area:

(i) $\frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C = \Delta$

(ii) $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

(iii) $\Delta = \frac{abc}{4R}$

(iv) $\Delta = 2R^2 \sin A \sin B \sin C$

4.5 Half the angles:

(i) $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$

$\sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$

$\sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

$$\begin{aligned}
 \text{(ii)} \quad \cos \frac{A}{2} &= \sqrt{\frac{s(s-a)}{bc}} \\
 \cos \frac{B}{2} &= \sqrt{\frac{s(s-b)}{ca}} \\
 \cos \frac{C}{2} &= \sqrt{\frac{s(s-c)}{ab}} \\
 \text{(iii)} \quad \tan \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \\
 \tan \frac{B}{2} &= \sqrt{\frac{(s-c)(s-a)}{s(s-b)}} \\
 \tan \frac{C}{2} &= \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}
 \end{aligned}$$

4.6 Napiere's rule:

$$\begin{aligned}
 \tan \left(\frac{B-C}{2} \right) &= \left(\frac{b-c}{b+c} \right) \cot \frac{A}{2} \\
 \tan \left(\frac{C-A}{2} \right) &= \left(\frac{c-a}{c+a} \right) \cot \frac{B}{2} \quad \text{and} \\
 \tan \left(\frac{A-B}{2} \right) &= \left(\frac{a-b}{a+b} \right) \cot \frac{C}{2}
 \end{aligned}$$

4.7 Lengths of the medians: Let m_a , m_b and m_c be the lengths of the medians through the vertices A , B and C of $\triangle ABC$. Then

$$\begin{aligned}
 4m_a^2 &= 2b^2 + 2c^2 - a^2 \\
 4m_b^2 &= 2c^2 + 2a^2 - b^2 \\
 4m_c^2 &= 2a^2 + 2b^2 - c^2
 \end{aligned}$$

4.8 Condition for an angle to be obtuse: In $\triangle ABC$, the angle A is obtuse if and only if

$$m_a^2 < \frac{m_b^2 + m_c^2}{5}$$

4.9 Lengths of internal angle bisectors: In $\triangle ABC$, the length of the internal bisector of the angle A is

$$\left(\frac{2bc}{b+c} \right) \cos \frac{A}{2} = \sqrt{bc - \frac{bca^2}{(b+c)^2}}$$

4.10 The area of the triangle formed by the feet of the internal angle bisector of $\triangle ABC$ is

$$\left(\frac{2abc}{(a+b)(b+c)(c+a)} \right) \Delta \quad (\text{For proof see Chapter 6})$$

4.11 Inradius:

$$\begin{aligned}
 \text{(i)} \quad r &= \frac{\Delta}{s} \\
 \text{(ii)} \quad r &= (s-a) \tan \frac{A}{2} \\
 r &= (s-b) \tan \frac{B}{2} \\
 r &= (s-c) \tan \frac{C}{2} \\
 \text{(iii)} \quad r &= 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}
 \end{aligned}$$

4.12 Exradii:

$$\begin{aligned}
 \text{(i)} \quad r_1 &= \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c} \\
 \text{(ii)} \quad r_1 &= s \tan \frac{A}{2}, r_2 = s \tan \frac{B}{2}, r_3 = s \tan \frac{C}{2} \\
 \text{(iii)} \quad r_1 &= 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \\
 r_2 &= 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} \\
 r_3 &= 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} \\
 \text{(iv)} \quad r_1 + r_2 + r_3 - r &= 4R \\
 \text{(v)} \quad \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} &= \frac{1}{r}
 \end{aligned}$$

4.13 If p_1 , p_2 and p_3 are the altitudes drawn from the vertices A , B and C on to the opposite sides, then

$$\begin{aligned}
 \text{(i)} \quad p_1 &= \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c} \\
 \text{(ii)} \quad \frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} &= \frac{1}{r}
 \end{aligned}$$

4.14 Pedal triangle: The triangle whose vertices are the feet of the altitudes drawn from the vertices A , B and C on to the opposite sides is called the pedal triangle of $\triangle ABC$.

4.15 The distances of the orthocentre from the vertices and from the sides: In $\triangle ABC$, D , E and F are the feet of the altitudes from the vertices on to the sides BC , CA and AB , respectively, and H is the orthocentre. Then

$$\begin{aligned}
 \text{(i)} \quad AH &= 2R \cos A, BH = 2R \cos B, CH = 2R \cos C. \\
 \text{(ii)} \quad HD &= 2R \cos B \cos C, HE = 2R \cos C \cos A, HF = 2R \cos A \cos B. \\
 \text{(iii)} \quad H &\text{ divides } \overline{AD} \text{ in the ratio } \cos A : \cos B \cos C \\
 &\text{ divides } \overline{BE} \text{ in the ratio } \cos B : \cos C \cos A \text{ and } \overline{CF} \text{ in} \\
 &\text{ the ratio } \cos C : \cos A \cos B.
 \end{aligned}$$

4.16 The points of intersection of the altitudes with the circumcircle are reflections of the orthocentre with respect to the corresponding sides.

4.17 *The angles and sides of the pedal triangle:* DEF is the pedal triangle of $\triangle ABC$. Then

(i) $\angle D = 180^\circ - 2A$, $\angle E = 180^\circ - 2B$, $\angle F = 180^\circ - 2C$.

(ii) $EF = a \cos A$, $DF = b \cos B$, $DE = c \cos C$.

(iii) (a) The circumradius of $\triangle DEF$ is $\frac{R}{2}$.

(b) The area of $\triangle DEF$ is $2\Delta \cos A \cos B \cos C$.

(c) The inradius of the pedal triangle DEF is $2R \cos A \cos B \cos C$.

4.18 *Excentric triangle:* The triangle formed by the three excentres I_1 , I_2 and I_3 of $\triangle ABC$ is called excentric triangle of $\triangle ABC$.

4.19 *Angles and sides of $\triangle ABC$:*

(i) Note that I_1A , I_2B and I_3C are altitudes of $\triangle I_1I_2I_3$ and hence $\triangle ABC$ is the pedal triangle of $\triangle I_1I_2I_3$ and I is the orthocentre of $\triangle I_1I_2I_3$.

(ii) $\angle I_1 = 90^\circ - \frac{A}{2}$, $\angle I_2 = 90^\circ - \frac{B}{2}$ and $\angle I_3 = 90^\circ - \frac{C}{2}$.

(iii) $I_2I_3 = 4R \cos \frac{A}{2}$, $I_3I_1 = 4R \cos \frac{B}{2}$, $I_1I_2 = 4R \cos \frac{C}{2}$.

(iv) Area of $\triangle I_1I_2I_3$ is $2Rs = \frac{abc}{2r}$.

4.20 In $\triangle ABC$

(i) $AI = r \operatorname{cosec} \frac{A}{2}$

$BI = r \operatorname{cosec} \frac{B}{2}$

$CI = r \operatorname{cosec} \frac{C}{2}$

(ii) $AI_1 = r_1 \operatorname{cosec} \frac{A}{2}$

$BI_2 = r_2 \operatorname{cosec} \frac{B}{2}$

$CI_3 = r_3 \operatorname{cosec} \frac{C}{2}$

4.21 (i) $OH^2 = R^2(1 - 8 \cos A \cos B \cos C)$

(ii) $OI^2 = R^2 \left(1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) = R^2 - 2Rr$

(iii) $OI_1^2 = R^2 \left(1 + 8 \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \right) = R^2 + 2Rr_1$



QUICK LOOK

In $\triangle ABC$, $R \geq 2r$ and the equality holds if $\triangle ABC$ is equilateral.

EXERCISES

Single Correct Choice Type Questions

1. In $\triangle ABC$, $a = 6$, $\angle B = 45^\circ$ and $\angle C = 30^\circ$. AD is drawn perpendicular to the side BC . A circle is described on AD as diameter cuts the sides AB and AC in P and Q , respectively. Then the length of the segment PQ is

(A) $\frac{3\sqrt{3}}{2}$ (B) $\frac{3\sqrt{3}}{4}$ (C) $\frac{3}{\sqrt{2}}$ (D) $2\sqrt{3}$

2. In $\triangle ABC$, the value of

$$(b+c) \tan \frac{A}{2} + (c+a) \tan \frac{B}{2} + (a+b) \tan \frac{C}{2}$$

is

(A) $2(R+r)$ (B) $4(R+r)$
(C) $3(R+r)$ (D) $6(R+r)$

3. In $\triangle ABC$, $a = BC$, $b = CA$, $c = AB$. If a^2 , b^2 , c^2 are in AP (in the given order), then

$$\frac{\sin 3B}{\sin B} =$$

(A) $\left(\frac{a^2 - c^2}{2ac} \right)^2$ (B) $\left(\frac{a^2 + c^2}{2ac} \right)^2$

(C) $\left(\frac{a^2 - c^2}{ac} \right)^2$ (D) $\left(\frac{a^2 + c^2}{ac} \right)^2$

4. In $\triangle ABC$, the median AD is perpendicular to the side AC at A . Then

$$\cos A \cos C = k \left(\frac{c^2 - a^2}{ca} \right)$$

where the value of k is

(A) $\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{2}{3}$

5. In $\triangle ABC$, if $BC = 5$, $CA = 4$ and $\cos(A - B) = 31/32$, then the area of the triangle is

(A) $5\sqrt{7}$ (B) $\frac{15}{4}\sqrt{3}$ (C) $\frac{15}{3}\sqrt{7}$ (D) $\frac{15}{4}\sqrt{7}$

6. In $\triangle ABC$, if $a = 6$, $b = 3$ and $\cos(A - B) = 4/5$, then the third side c is equal to
 (A) $3\sqrt{5}$ (B) 6 (C) $2\sqrt{5}$ (D) $6\sqrt{5}$

7. In $\triangle ABC$, if $\cos A = \lambda \cos B$, then

$$\tan^2\left(\frac{A+B}{2}\right) =$$

- (A) $\left(\frac{a+b}{a-b}\right)\left(\frac{1+\lambda}{1-\lambda}\right)$ (B) $\left(\frac{a+b}{a-b}\right)\left(\frac{1-\lambda}{1+\lambda}\right)$
 (C) $\left(\frac{a-b}{a+b}\right)\left(\frac{1+\lambda}{1-\lambda}\right)$ (D) $\left(\frac{a-b}{a+b}\right)\left(\frac{1-\lambda}{1+\lambda}\right)$

8. In $\triangle ABC$, it is given that

$$\cos A + \cos B + \cos C = \frac{3}{2}$$

If one side of the triangle is of length 4 units, the area of its pedal triangle is

- (A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$ (C) $2\sqrt{3}$ (D) $\sqrt{3}$

(Hint: See Theorem 4.23.)

9. In $\triangle ABC$, $a = 5$, $b = 4$, $c = 3$ and its centroid and circumcentre are, respectively, G and O . Then the magnitude of OG is

- (A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

$$\left(\text{Hint: } OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)\right)$$

10. In $\triangle ABC$, if

$$\cos A \cos B + \sin A \sin B \sin C = 1$$

then

- $\sin^2 A + \sin^2 B + \sin^2 C =$
 (A) $2\sqrt{2}$ (B) $\frac{3}{\sqrt{2}}$ (C) 2 (D) $\frac{3}{2}$

11. In $\triangle ABC$, if

$$\frac{a \cos A + b \cos B + c \cos C}{a \sin B + b \sin C + c \sin A} = \frac{a + b + c}{9R}$$

and I_1 , I_2 and I_3 are the excentres of $\triangle ABC$, then the area of $\triangle I_1 I_2 I_3$ is

- (A) $3R^2$ (B) $3\sqrt{3}R^2$
 (C) $3R^2$ (D) $\left(\frac{3\sqrt{3}}{2}\right)R^2$

(Hint: Area of $\triangle I_1 I_2 I_3$ is $2Rs$.)

12. In a circle of 1 unit radius, AB is a chord whose length is also 1. If C is any point on the major arc of AB , then the maximum value of $(AC)^2 + (BC)^2$ is

- (A) $2 + \sqrt{3}$ (B) $2(2 + \sqrt{3})$
 (C) $2(\sqrt{3} + 1)$ (D) $4(\sqrt{3} + 1)$

(Hint: $\angle BCA = 30^\circ$ and let $\angle BAC = \theta$.)

13. In $\triangle ABC$, if $a = 2$, $b = 3$ and $\sec^2 A = 8/5$, then one of the values of the third side c is

- (A) $\sqrt{10}$ (B) $\sqrt{6}$ (C) $\sqrt{5}$ (D) 6

14. In $\triangle ABC$, if p_1 , p_2 and p_3 are the lengths of the altitudes, then $2\Delta^2$ is equal to

- (A) $Rp_1 p_2 p_3$ (B) $R^2 p_1 p_2 p_3$
 (C) $3R^2 p_1 p_2 p_3$ (D) $\frac{3}{2} R^2 p_1 p_2 p_3$

where R is the circumradius.

15. In $\triangle ABC$, if $\cot A + \cot B + \cot C = \sqrt{3}$, then the sides are in the ratio

- (A) 1:2:3 (B) 1:1: $\sqrt{2}$
 (C) 1: $\sqrt{2}$:1 (D) 1:1:1

16. In $\triangle ABC$, if the distances of the vertices from the incentre are x , y and z and r is its inradius, then r^2 is equal to

- (A) $\frac{x+y+z}{xyz}$ (B) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$
 (C) $\frac{xyz}{4R}$ (D) $\frac{x^2 y^2 z^2}{x+y+z}$

17. In $\triangle ABC$, if the altitude, the angle bisector and the median through A , divide angle A into four equal parts, then angle A is equal to

- (A) 120° (B) 90° (C) 115° (D) 75°

18. In $\triangle ABC$, $r_1 + r_2 + r_3 = r + kR$ where k is

- (A) 2 (B) 4 (C) 6 (D) 8

19. If α , β and γ are the distances of the vertices from its orthocentre, then

$$\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} =$$

- (A) $\frac{a+b+c}{\alpha\beta\gamma}$ (B) $\frac{abc}{\alpha+\beta+\gamma}$
 (C) $\frac{abc}{\alpha\beta\gamma}$ (D) $\frac{a+b+c}{\alpha+\beta+\gamma}$

20. In $\triangle ABC$, if p_1 , p_2 and p_3 are the lengths of the altitudes, then $\Delta(p_1^{-2} + p_2^{-2} + p_3^{-2}) =$

- (A) $\cos A + \cos B + \cos C$
 (B) $\sin A + \sin B + \sin C$
 (C) $\tan A + \tan B + \tan C$
 (D) $\cot A + \cot B + \cot C$

21. If Q is a point on the altitude AD and is lying inside the triangle such that $\angle CBQ = B/3$, then the length AQ is

- (A) $2c \sin \frac{B}{3}$ (B) $2c \sin \frac{C}{3}$
 (C) $2a \sin \frac{B}{3}$ (D) $2b \sin \frac{B}{3}$

22. In $\triangle ABC$, Let m_a , m_b and m_c be the lengths of the medians through the vertices A , B and C , respectively. Then angle A is obtuse if and only if

- (A) $m_a^2 < \frac{m_b^2 + m_c^2}{2}$ (B) $m_a^2 < \frac{m_b^2 + m_c^2}{3}$

- (C) $m_a^2 < \frac{m_b^2 + m_c^2}{4}$ (D) $m_a^2 < \frac{m_b^2 + m_c^2}{5}$

(Hint: Use $4m_a^2 = 2b^2 + 2c^2 - a^2$, etc.)

23. In $\triangle ABC$, if $b^2 + c^2 = (2011)a^2$, then

$$\frac{\cot A}{\cot B + \cot C} =$$

- (A) $\frac{2011}{2}$ (B) 1005 (C) 1006 (D) $\frac{2009}{2}$

24. In $\triangle ABC$, if $a^2 + b^2 + c^2 = ac + bc\sqrt{3}$, then the triangle is

- (A) right angled (B) isosceles
 (C) equilateral (D) obtuse angled

25. In $\triangle ABC$, if $(r_2 - r_1)(r_3 - r_1) = 2r_2r_3$, then the triangle is

- (A) equilateral (B) isosceles
 (C) right angled (D) obtuse angled

Multiple Correct Choice Type Questions

1. In $\triangle ABC$, if $a^2 + b^2 + c^2 = bc + (ac)\sqrt{3}$, then

- (A) $\angle A = 60^\circ$ (B) $\angle B = 90^\circ$
 (C) $\angle C = 90^\circ$ (D) inradius is $\frac{2ab}{c(3 + \sqrt{3})}$

2. In $\triangle ABC$, if

$$(a + b)^2 = ab + c^2$$

and $\sin A + \cos A = \sqrt{3}/2$

then

- (A) $\angle C = 120^\circ$ (B) $\angle A = 15^\circ$
 (C) $\angle B = 45^\circ$ (D) $a:b:c = (\sqrt{3} - 1):2:\sqrt{6}$

3. In any $\triangle ABC$, which of the following statements are true?

- (A) $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$
 (B) $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} \geq \frac{3}{4}$
 (C) $\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} \leq \frac{9}{4}$
 (D) $\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \leq \frac{3\sqrt{3}}{8}$

4. Let ABC be a triangle. Then

- (A) $\sin \frac{A}{2} \leq \frac{a}{b+c}$
 (B) If $\sin A + \sin B + \sin C \leq 1$, then $\min\{B + C, C + A, A + B\} < 30^\circ$

(C) $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$

(D) $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1$

5. If the lengths of the tangents drawn from the vertices A , B and C of $\triangle ABC$ to its incircle are in AP, then

- (A) a, b, c are in AP (B) $\cos A = \frac{4c - 3b}{2c}$
 (C) r_1, r_2, r_3 are in AP (D) r_1, r_2, r_3 are in HP

6. In $\triangle ABC$, D , E and F are the feet of the altitudes drawn from the vertices A , B and C onto the opposite sides. Let R_1 , R_2 and R_3 be the circumradii of $\triangle AEF$, $\triangle BDF$ and $\triangle DCE$, respectively. Then

- (A) $2R_1 = a \cot A$
 (B) $2R_2 = b \cot B$
 (C) $2R_3 = c \cot C$
 (D) The ratio of the perimeter of $\triangle ABC$ to the perimeter of $\triangle DEF$ is $r:R$
 (Hint: See Theorems 4.22 and 4.23.)

7. In $\triangle ABC$, which of the following statements are true?

- (A) $IA \cdot IB \cdot IC = (abc) \tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}$
 (B) $AI_1 \cdot BI_2 \cdot CI_3 = (r_1 r_2 r_3) \operatorname{cosec} \frac{A}{2} \operatorname{cosec} \frac{B}{2} \operatorname{cosec} \frac{C}{2}$
 (C) $II_1 \cdot II_2 \cdot II_3 = 16rR^2$
 (D) $(II_1)^2 + (II_2)^2 + (II_3)^2 = 16R^2$

8. In any $\triangle ABC$

- (A) $(r_1 + r_2) \tan \frac{C}{2} = c$ (B) $(r_3 - r) \cot \frac{C}{2} = c$
 (C) $(r_3 - r) \tan \frac{C}{2} = c$ (D) $r_2 r_3 + r_3 r_1 + r_1 r_2 = \Delta^2$

9. Two sides of a triangle are 12 and $12\sqrt{2}$ and the angle opposite to the shorter side is 30° . Then

- (A) the number of such triangles is two
 (B) the angles of the triangles are $30^\circ, 45^\circ, 105^\circ$ or $30^\circ, 135^\circ, 15^\circ$
 (C) area of the triangle is $\frac{\sqrt{3}+1}{2\sqrt{2}}$ or $\frac{\sqrt{3}-1}{2\sqrt{2}}$
 (D) areas are in the ratio $(\sqrt{3}+1):(\sqrt{3}-1)$

10. In $\triangle ABC$, let

$$\cos \alpha = \frac{a}{b+c}, \cos \beta = \frac{b}{c+a} \quad \text{and} \quad \cos \gamma = \frac{c}{a+b}$$

where each of α, β and γ is positive and less than π . Then

- (A) $\tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2} + \tan^2 \frac{\gamma}{2} = 1$
 (B) $\tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2} = \tan A \tan B \tan C$
 (C) $\tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2} = \tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}$
 (D) $\tan \alpha \tan \beta \tan \gamma = \tan A \tan B \tan C$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with *one or more* statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are (A) $\rightarrow$ (p), (s); (B) $\rightarrow$ (q), (s), (t); (C) $\rightarrow$ (r); (D) $\rightarrow$ (r), (t); that is if the matches are (A) $\rightarrow$ (p) and (s); (B) $\rightarrow$ (q), (s) and (t); (C) $\rightarrow$ (r); and (D) $\rightarrow$ (r), (t) then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

1. In $\triangle ABC$, $\angle A$ is least and its value is 45° . If $\tan A, \tan B$ and $\tan C$ are in AP and the area of the triangle is 27 square units, then match the items of Column I with those of Column II.

Column I	Column II
(A) The side a is of length	(p) $6\sqrt{2}$
(B) The side b is equal to	(q) $3\sqrt{2}$
(C) The side c equals	(r) $\frac{3\sqrt{10}}{2}$
(D) Circumradius of $\triangle ABC$ is	(s) $3\sqrt{5}$
	(t) 9

2. For any $\triangle ABC$, match the items of Column I with those of Column II.

Column I	Column II
(A) $rr_1r_2r_3$ is equal to	(p) $s^3 \tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}$
(B) $\frac{rr_1}{r_2r_3}$ equals	(q) $\tan^2 \frac{A}{2}$
(C) $r_1r_2r_3$ is equal to	(r) Δ^2
(D) $\frac{r_1r_2r_3}{\Delta}(s-a)(s-b)(s-c) =$	(s) $\tan^3 \frac{A}{2}$

3. Match the items of Column I with those of Column II.

Column I	Column II
(A) In $\triangle ABC$, if	(p) $\sqrt{3}$
$\cos A + \cos B + \cos C = \frac{7}{4}$	
then the value of R/r is	(q) $\frac{4}{3}$
(B) In $\triangle ABC$, if $AC = 2AB$, then	
$\cot \frac{A}{2} \cot \frac{B-C}{2} =$	(r) $2\sqrt{3}$
(C) In a right-angled triangle, the hypotenuse is four times the length of the altitude drawn from the opposite vertex and 2θ is the difference of the other two acute angles. Then $\tan^2 2\theta =$	(s) $\frac{2}{3}$
(D) In $\triangle ABC$, if $\angle A = 30^\circ, b = 2, c = \sqrt{3}$ and r is the inradius, then the value of $2r + 1$ is	(t) 3

4. In $\triangle ABC$, it is given that $a = 5$, $b = 3$ and $c = 7$. Match the items of Column I with those of Column II.

Column I	Column II
(A) Area of $\triangle ABC$ is	(p) 5
(B) The circumradius is equal to	(q) $\frac{\sqrt{3}}{2}$
(C) The inradius equals	(r) $\frac{7\sqrt{3}}{3}$
(D) The value of $7 \cos B + 3 \cos C$ is	(s) $\frac{15\sqrt{3}}{4}$

5. In $\triangle ABC$, match the items of Column I with those of Column II.

Column I	Column II
(A) $\cot\left(\frac{A+B}{2}\right)\tan\left(\frac{A-B}{2}\right)$ is	(p) b^2
(B) $\frac{1 - \tan(A/2)\tan(B/2)}{1 + \tan(A/2)\tan(B/2)}$ is equal to	(q) $\frac{a-b}{a+b}$
(C) If the angles A, B, C (in this order) are in AP, then $a^2 + c^2 - ac$ is equal to	(r) $\frac{c}{a+b}$
(D) $\frac{1 + \cos(A-B)\cos C}{1 + \cos(A-C)\cos B}$ equals	(s) $\frac{a^2 + b^2}{a^2 - c^2}$
	(t) $\frac{a^2 + b^2}{a^2 + c^2}$

Comprehension-Type Questions

1. **Passage:** Δ_1, Δ_2 and Δ_3 are the areas of the triangles cut off by the tangents drawn to the incircle of $\triangle ABC$ and drawn parallel to the sides BC, CA and AB , respectively. Answer the following three questions.

(i) $\frac{\Delta_1}{(s-a)^2} = \frac{\Delta_2}{(s-b)^2} = \frac{\Delta_3}{(s-c)^2} = \lambda$, where λ equals

(A) $\frac{\Delta}{s^2}$ (B) $\frac{\Delta^2}{s^2}$ (C) $\frac{\Delta}{s}$ (D) $\frac{\Delta^2}{s}$

- (ii) If R_1 is the circumradius of the triangle whose area is Δ_1 , then R_1 is equal to

(A) $R \cot \frac{B}{2} \cot \frac{C}{2}$ (B) $R \tan \frac{B}{2} \tan \frac{C}{2}$
(C) $\frac{R}{3}$ (D) $\frac{R}{r}$

- (iii) If R_1, R_2 and R_3 are the circumradii of the triangles with areas Δ_1, Δ_2 and Δ_3 then $R_1 + R_2 + R_3$ is

(A) $\frac{2R}{3}$ (B) R (C) $\frac{3R}{2}$ (D) $\frac{4R}{3}$

2. **Passage:** In $\triangle ABC$, let a, b, c be the sides, s the semi-perimeter and Δ the area. Answer the following three questions.

- (i) Minimum value of $a^2 + b^2 + c^2$ is

(A) $3\sqrt{2}\Delta$ (B) $4\sqrt{3}\Delta$ (C) $3\sqrt{3}\Delta$ (D) $2\sqrt{3}\Delta$

- (ii) Minimum value of $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$ is

(A) $\frac{s}{abc}$ (B) $\frac{2s}{abc}$ (C) $\frac{3s}{abc}$ (D) $2\Delta s$

- (iii) Minimum value of

$$\frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a}$$

is

(A) $4s$ (B) $3s$ (C) $2s$ (D) s

3. **Passage:** For any $\triangle ABC$, answer the following three questions.

- (i) If the sides a, b, c are in GP (in the given order), then the three numbers

$$(b^2 - c^2) \left(\frac{\tan B + \tan C}{\tan B - \tan C} \right), (c^2 - a^2) \left(\frac{\tan C + \tan A}{\tan C - \tan A} \right),$$

$$(a^2 - b^2) \left(\frac{\tan A + \tan B}{\tan A - \tan B} \right)$$

are in

(A) AP (B) HP
(C) GP (D) increasing order

- (ii) If $\cos A + 2 \cos B + \cos C = 2$, then

(A) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in HP (B) a, b, c are in GP

(C) a, b, c are in HP (D) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in GP

- (iii) If $\sin A, \sin B$ and $\sin C$ are in AP, then the altitudes of the triangle are in

(A) AP (B) GP (C) HP (D) AGP

4. Passage: In $\triangle ABC$, the incircle touches the sides BC , CA and AB at the points D , E and F , respectively. If the radius of the incircle is 4 and the lengths of BD , CE and AF are consecutive integers, then answer the following three questions.

(i) $\triangle ABC$ is

- (A) equilateral
(B) right-angled

(C) right-angled isosceles

(D) The sides of $\triangle ABC$ are in AP in some order

(ii) Area of $\triangle ABC$ is

- (A) $2\sqrt{21}$ (B) $2\sqrt{17}$ (C) $3\sqrt{11}$ (D) 84

(iii) In some order $\sin A$, $\sin B$ and $\sin C$ are in

- (A) AP (B) GP (C) HP (D) AGP

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
(B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
(C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true

1. Statement I: In $\triangle ABC$, the inradius r is less than $\frac{a^2 + b^2 + c^2}{3(a + b + c)}$

Statement II: Area of $\triangle ABC = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C$

2. Statement I: In an acute-angled $\triangle ABC$, if

$$8(\cos^3 A + \cos^3 B + \cos^3 C - 3\cos A \cos B \cos C) + 36(\cos A \cos B + \cos B \cos C + \cos C \cos A) = 27$$

then $\triangle ABC$ is equilateral.

Statement II: In any $\triangle ABC$,

$$\cos A + \cos B + \cos C \leq \frac{3}{2}$$

where the equality holds if and only if the triangle is equilateral.

3. Statement I: In $\triangle ABC$, $\angle C = 90^\circ$ and the bisector of the angle C meets the side AB internally in D . If $p = AD$ and $q = BD$, then

$$\frac{p}{q} = \frac{1 - \tan[(A - B)/2]}{1 + \tan[(A - B)/2]}$$

Statement II: In a triangle, internal bisector of an angle divides opposite side in the ratio of the other two sides.

4. Statement I: The sides a , b , c of $\triangle ABC$ are, respectively, 18, 24 and 30. Then its inradius is 9.

Statement II: The inradius $r = \Delta/s$, where Δ is the area and s is the semiperimeter of the triangle.

5. Statement I: In $\triangle ABC$, ' I ' and ' O ' denote the incentre and circumcentre, respectively. If $\angle BIO = 90^\circ$, then $a + c = 2b$.

Statement II: In $\triangle ABC$, $(IO)^2 = R^2 - Rr$.

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
0	0	0	0
1	1	1	1
2	●	2	2
3	3	3	3
4	●	4	4
5	5	5	5
6	●	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. In $\triangle ABC$, if $a = 18$, $b = 24$ and $c = 30$, then the value of $\operatorname{cosec} C$ is ____.
2. In $\triangle ABC$, if $a = 25$, $b = 52$ and $c = 63$, then the value of $5 \tan(A/2) + 2 \tan(B/2)$ is equal to ____.
3. In $\triangle ABC$, if $a = 6$, $b = 3$ and $\cos(A - B) = 4/5$, then its area is ____.
4. In $\triangle ABC$, $\frac{(I_2 I_3)^2}{R(r_2 + r_3)}$ is equal to ____.
5. The circumradius of $\triangle ABC$ is ____ times the circumradius of its pedal triangle.
6. In $\triangle ABC$, the distances of the circumcentre from the sides BC , CA and AB are, respectively, p_1 , p_2 and p_3 . Then $abc/p_1 p_2 p_3$ is ____ times $(a/p_1) + (b/p_2) + (c/p_3)$.
7. If the line joining the incentre and the circumcentre of triangle ABC is parallel to the side BC , then the value of $\cos B + \cos C$ is ____.
8. In $\triangle ABC$, the value of
$$\frac{a \cos A + b \cos B + c \cos C}{a + b + c} = \frac{r}{kR}$$
 where k is equal to ____.
9. In $\triangle ABC$, the minimum value of
$$\frac{a}{s-a} + \frac{b}{s-b} + \frac{c}{s-c}$$
 is ____.
10. If H is the orthocentre of $\triangle ABC$ whose circumradius is 4, then the circumradius of $\triangle BHC$ is ____.

ANSWERS

Single Correct Choice Type Questions

1. (C)
2. (B)
3. (A)
4. (D)
5. (D)
6. (A)
7. (B)
8. (D)
9. (A)
10. (C)
11. (B)
12. (B)
13. (A)
14. (A)
15. (D)
16. (C)
17. (B)
18. (B)
19. (C)
20. (D)
21. (A)
22. (D)
23. (B)
24. (A)
25. (C)

Multiple Correct Choice Type Questions

1. (A), (C), (D)
2. (A), (B), (C), (D)
3. (A), (B), (C), (D)
4. (A), (B), (C), (D)
5. (A), (B), (D)
6. (A), (B), (C)
7. (A), (B), (C), (D)
8. (A), (B), (D)
9. (A), (B), (D)
10. (A), (C)

Matrix-Match Type Questions

1. (A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (t), (D) $\rightarrow$ (r)
2. (A) $\rightarrow$ (r), (B) $\rightarrow$ (q), (C) $\rightarrow$ (p), (D) $\rightarrow$ (r)
3. (A) $\rightarrow$ (q), (B) $\rightarrow$ (t), (C) $\rightarrow$ (t), (D) $\rightarrow$ (p)
4. (A) $\rightarrow$ (s), (B) $\rightarrow$ (r), (C) $\rightarrow$ (q), (D) $\rightarrow$ (p)
5. (A) $\rightarrow$ (q), (B) $\rightarrow$ (r), (C) $\rightarrow$ (p), (D) $\rightarrow$ (t)

Comprehension-Type Questions

1. (i) (A); (ii) (B); (iii) (B)
2. (i) (B); (ii) (B); (iii) (C)
3. (i) (C); (ii) (A); (iii) (C)
4. (i) (D); (ii) (D); (iii) (A)

Assertion–Reasoning Type Questions

1. (A)
2. (A)
3. (A)
4. (D)
5. (C)

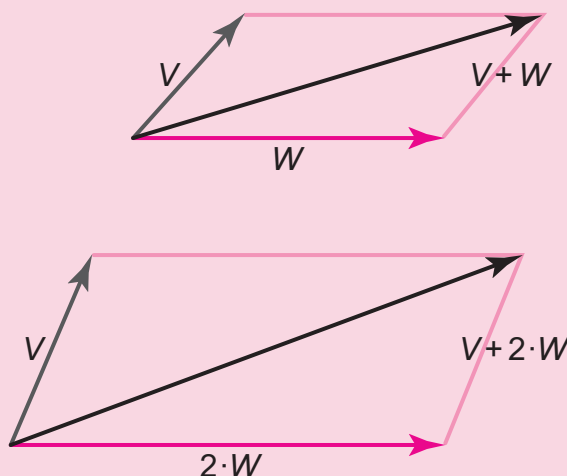
Integer Answer Type Questions

1. 1
2. 2
3. 9
4. 4
5. 2
6. 4
7. 1
8. 1
9. 6
10. 4

Addition and Scalar Multiplication of Vectors

5

Addition and Scalar Multiplication of Vectors



Contents

- 5.1 Definition and Classification of Vectors
- 5.2 Addition of Vectors
- 5.3 Multiplication of a Vector by a Scalar
- 5.4 The Division Formula
- 5.5 Components of a Vector
- 5.6 Vector Equation of a Line and a Plane

Worked-Out Problems
Summary
Exercises
Answers

In the 19th century, the famous mathematical physicist R. Hamilton (1805–1865) introduced the notion of a vector and his studies on vectors brought out their importance. In recent years, the algebra of vectors has become an indispensable tool in the study of Applied Mathematics, Physics and Engineering.

In this chapter, we define the concept of vectors, their addition and two types of multiplication. Also we derive various properties of addition and multiplication of vectors and obtain the vector equations of straight lines and planes.

5.1 | Definition and Classification of Vectors

Physical quantities such as length, area, volume, mass, temperature, density, etc. possess only magnitude and are not related to any definite direction in space. Such quantities are known as *scalar quantities* or simply scalars. On the other hand physical quantities such as displacement, velocity, acceleration, force momentum, couple, electric current, magnetic and gravitational intensities, etc. possess magnitude and are associated with direction in space. Such quantities are known as *vector quantities* or simply vectors. In this section we provide concrete mathematical definitions for these intuitive ideas of scalars and vectors.

The three-dimensional space is simply called the space and the points in the space are associated with ordered triads of real numbers, with respect to a given rectangular coordinate system. We first introduce the concept of a directed line segment.

DEFINITION 5.1 Let A and B be two distinct points in the space. Then the ordered pair (A, B) is denoted by $\overrightarrow{AB}$ and is called the *directed line segment* with initial point A and terminal point B . The straight line $\overline{AB}$ is called the *support* of $\overrightarrow{AB}$.

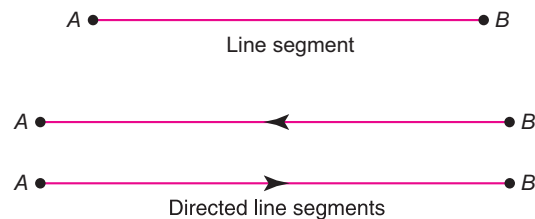


FIGURE 5.1 Directed line segment.

A directed line segment $\overrightarrow{AB}$ is represented by the line segment AB with an arrow mark pointing towards B as shown in Figure 5.1. Note that $\overrightarrow{AB}$ is different from $\overrightarrow{BA}$, though their supports are same.

DEFINITION 5.2 Let $\overrightarrow{AB}$ be a directed line segment. The distance between A and B is called the *magnitude* of $\overrightarrow{AB}$ and is denoted by $|\overrightarrow{AB}|$ or simply AB .

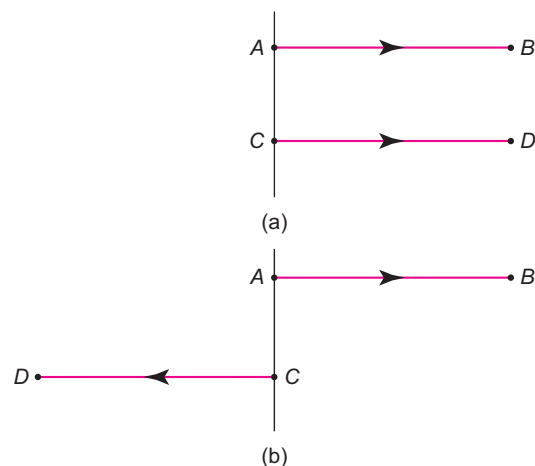


FIGURE 5.2 Directed line segments in (a) same direction; (b) opposite direction.

Note that the line segment and the length of that line segment are denoted in the same way, namely, AB . There is no ambiguity and one should understand as per the context. Two directed line segments $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are said to have

same direction if their supports $\overline{AB}$ and $\overline{CD}$ are parallel and both B and D lie in the same half-plane determined by the line $\overline{AC}$ [see Figure 5.2(a)]. If $\overline{AB}$ and $\overline{CD}$ are parallel but do not have the same direction, then $\overline{AB}$ and $\overline{CD}$ are said to have *opposite direction*. Note that $\overline{AB}$ and $\overline{BA}$ have opposite direction [see Figure 5.2(b)].

DEFINITION 5.3 Equivalence Let DLS be the set of all directed line segments in the space. Two directed line segments are said to be *equivalent* if they have the same magnitude and the same direction.

The relation “being equivalent” is an equivalence relation on DLS (i.e., a binary relation which is reflexive on DLS, symmetric and transitive) and hence partitions DLS into pair-wise disjoint classes (which are called equivalence classes) satisfying the following properties

1. Any two directed line segments in the same equivalence class are equivalent (i.e., they have same magnitude as well as direction).
2. Every directed line segment belongs to exactly one equivalence class.
3. Directed line segments belonging to different equivalence classes are not equivalent.

DEFINITION 5.4 Vector Each equivalence class discussed above is called a *vector* and is usually denoted by $\vec{a}, \vec{b}, \vec{c}$, etc.

If $\overline{AB}$ is a directed line segment in the equivalence class $\vec{a}$, then we simply write $\overline{AB} = \vec{a}$. Note that $\overline{AB} = \overline{CD}$ if and only if $\overline{AB} = \overline{CD}$ and $\overline{AB}$ and $\overline{CD}$ have the same direction.

If $\overline{AB}$ belongs to the vector $\vec{a}$, then we say that $\overline{AB}$ represents the vector $\vec{a}$ and in this case the magnitude of $\vec{a}$ is defined as the magnitude AB of $\overline{AB}$ and is denoted by $|\vec{a}|$. Note that $\overline{AB}$ and $\overline{CD}$ represent the same vector if and only if they have same direction as well as magnitude. Two vectors $\vec{a}$ and $\vec{b}$ are said to have same direction if any $\overline{AB}$ representing $\vec{a}$ and any $\overline{CD}$ representing $\vec{b}$ have same direction.

DEFINITION 5.5 Zero Vector For any two points P and Q in the space, clearly $\overline{PP}$ and $\overline{QQ}$ are equivalent and hence the collection of $\overline{PP}$ where P is a point in the space is a vector (equivalence class of directed line segments), which is called the *zero vector* and is denoted by $\vec{0}$.

DEFINITION 5.6 Support For any non-zero vector $\vec{a}$, a *support* of $\vec{a}$ is defined to be the support of any $\overline{AB}$ representing $\vec{a}$.

The zero vector has neither support nor direction; however, it has magnitude, namely 0, that is, $|\vec{0}| = 0$. Note that

1. Any two supports of a non-zero vector are parallel in the space.
2. Any vector is uniquely determined by its direction and magnitude but not its support.

DEFINITION 5.7 Unit Vector A vector of magnitude one unit is called a *unit vector*. If $\vec{a}$ is a non-zero vector and $\vec{e}$ is a unit vector having the same direction as $\vec{a}$, then $\vec{e}$ is called a *unit vector in the direction of $\vec{a}$* (Figure 5.3).

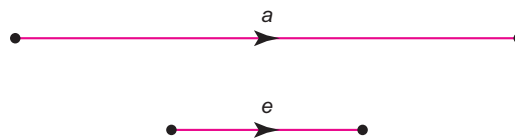


FIGURE 5.3 Unit vector.

DEFINITION 5.8 Collinear, Like and Unlike Vectors Two or more vectors are said to be *collinear* if they have parallel or same supports. Two collinear vectors are said to be *like* or *unlike* vectors according as they have the same direction or opposite direction.

In Figure 5.4(a) $\vec{a}$ and $\vec{b}$ are like vectors and in Figure 5.4(b) $\vec{a}$ and $\vec{b}$ are unlike vector as they are collinear and have opposite direction.

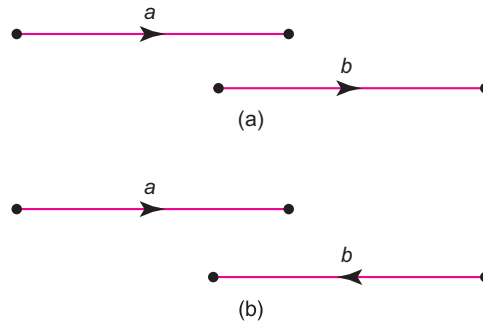


FIGURE 5.4 (a) Like vectors; (b) Unlike vectors.

DEFINITION 5.9 Coplanar and Non-coplanar Vectors A set of vectors is said to be *coplanar* if their supports are in the same plane or parallel to the same plane. Vectors which are not coplanar are called *non-coplanar*.

Note that any two vectors are always coplanar. Further, three vectors $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$ and $\vec{c} = \overrightarrow{OC}$ are coplanar if and only if the points O, A, B and C are all in the same plane.

5.2 | Addition of Vectors

In this section we define the notion of addition of two vectors and derive various properties of vectors involving addition. Before going for the definition of addition, we shall first have the following theorem.

THEOREM 5.1 Let $\vec{a}$ and $\vec{b}$ be two vectors and O, A, B, O_1, A_1 and B_1 be points such that

$$\overrightarrow{OA} = \vec{a} = \overrightarrow{O_1A_1} \quad \text{and} \quad \overrightarrow{AB} = \vec{b} = \overrightarrow{A_1B_1}$$

Then

$$\overrightarrow{OB} = \overrightarrow{O_1B_1}$$

PROOF Since $\overrightarrow{OA} = \overrightarrow{O_1A_1}$ and $\overrightarrow{AB} = \overrightarrow{A_1B_1}$, we have

$$OA = O_1A_1 \quad \text{and} \quad AB = A_1B_1$$

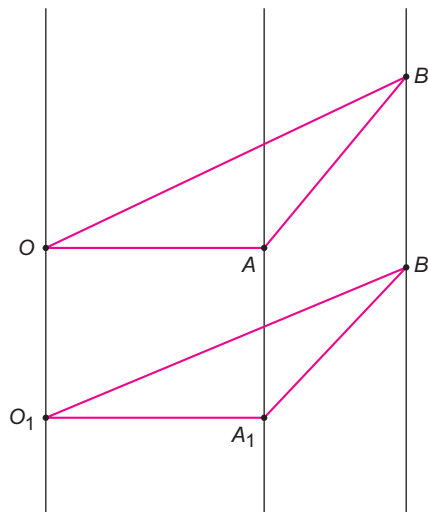


FIGURE 5.5 Theorem 5.1.

Also, the lines OA and O_1A_1 are parallel and so are the lines AB and A_1B_1 . Therefore OO_1A_1A and AA_1B_1B are parallelograms (Figure 5.5) and

$$OO_1 = AA_1 = BB_1$$

Thus OO_1B_1B is a parallelogram, so that $OB = O_1B_1$ and hence

$$\overrightarrow{OB} = \overrightarrow{O_1B_1}$$

Note: $\vec{a}$ and $\vec{b}$ are collinear if and only if O, A, B and O_1, A_1, B_1 are collinear, and, in this case, O, A, B, O_1, A_1 and B_1 lie on the same line.

DEFINITION 5.10 Resultant Vector Let $\vec{a}$ and $\vec{b}$ be vectors and O, A and B be points such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{AB} = \vec{b}$. Then the sum $\vec{a} + \vec{b}$ is defined to be the vector $\overrightarrow{OB}$. $\vec{a} + \vec{b}$ is also called the *resultant vector* of $\vec{a}$ and $\vec{b}$.

DEFINITION 5.11 Triangle Law for Addition The equation

$$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$$

is known as the *triangle law for addition* (Figure 5.6).

DEFINITION 5.12 Parallelogram Law for Addition If $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OC}$ are co-initial vectors, then complete the triangle in Figure 5.6 to form the parallelogram $OACB$ (Figure 5.7) to get that $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{OB} = \vec{a} + \vec{b}$ and, for this reason, the equation

$$\overrightarrow{OA} + \overrightarrow{OC} = \overrightarrow{OB}$$

is known as the *parallelogram law for addition*.

Note: In view of the Theorem 5.1, the sum $\vec{a} + \vec{b}$ does not depend on the initial points or final points of $\vec{a}$ or $\vec{b}$.

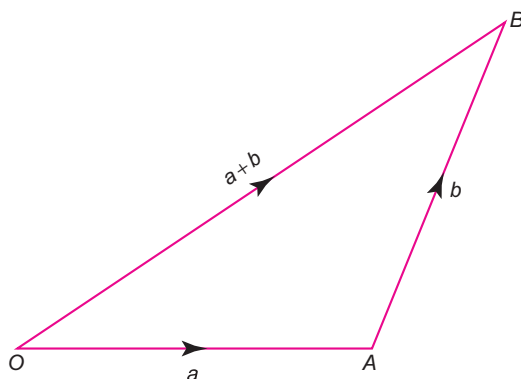


FIGURE 5.6 Triangle law for addition.

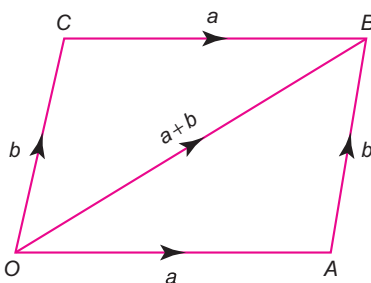


FIGURE 5.7 Parallelogram law for addition.

THEOREM 5.2
(COMMUTATIVE
PROPERTY FOR
ADDITION)

For any vectors $\vec{a}$ and $\vec{b}$,

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

PROOF

Choose points A , B and C such that $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BC} = \vec{b}$ (Figure 5.8).

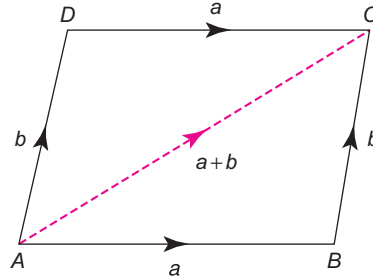


FIGURE 5.8 Theorem 5.2.

Then, by Definition 5.11, we have

$$\vec{a} + \vec{b} = \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC} \quad (5.1)$$

Now, draw parallels to AB and BC through C and A , respectively, and let them intersect at D . Now, $ABCD$ is a parallelogram. Also,

$$\vec{a} = \overrightarrow{AB} = \overrightarrow{DC}$$

and

$$\vec{b} = \overrightarrow{BC} = \overrightarrow{AD}$$

Then, by the triangle law for addition, we have

$$\vec{b} + \vec{a} = \overrightarrow{AD} + \overrightarrow{DC} = \overrightarrow{AC} \quad (5.2)$$

From Eqs. (5.1) and (5.2), it follows that $\vec{a} + \vec{b} = \vec{b} + \vec{a}$. ■

The addition of vectors is defined to be a binary operation, in the sense that given two vectors $\vec{a}$ and $\vec{b}$, we have defined the vector $\vec{a} + \vec{b}$ and proved in Theorem 5.2 that $\vec{a} + \vec{b} = \vec{b} + \vec{a}$. This property is called the *commutativity of addition*.

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three given vectors, the expression $\vec{a} + \vec{b} + \vec{c}$ has no meaning right now, since $+$ is defined only between two vectors. However, $(\vec{a} + \vec{b}) + \vec{c}$ is meaningful, since it is the sum of $\vec{a} + \vec{b}$ and $\vec{c}$. Similarly $\vec{a} + (\vec{b} + \vec{c})$ is meaningful, since it is the sum of $\vec{a}$ and $\vec{b} + \vec{c}$. In the following, we prove that both the expressions $(\vec{a} + \vec{b}) + \vec{c}$ and $\vec{a} + (\vec{b} + \vec{c})$ convey the same meaning and they represent the same vector. This property is called the *associativity of addition*.

THEOREM 5.3
(ASSOCIATIVE
PROPERTY FOR
ADDITION)

For any vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$,

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

PROOF

Choose points A , B , C and D such that $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{BC} = \vec{b}$ and $\overrightarrow{CD} = \vec{c}$. Then, by the triangle law (Figure 5.9),

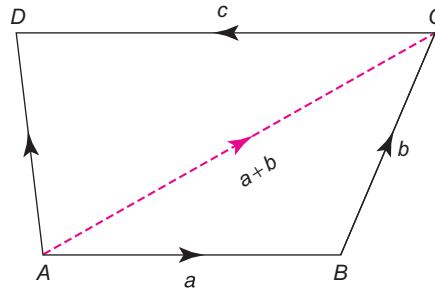


FIGURE 5.9 Theorem 5.3.

$$\begin{aligned}
 (\vec{a} + \vec{b}) + \vec{c} &= (\overrightarrow{AB} + \overrightarrow{BC}) + \overrightarrow{CD} \\
 &= \overrightarrow{AC} + \overrightarrow{CD} \\
 &= \overrightarrow{AD} \\
 &= \overrightarrow{AB} + \overrightarrow{BD} \\
 &= \overrightarrow{AB} + (\overrightarrow{BC} + \overrightarrow{CD}) \\
 &= \vec{a} + (\vec{b} + \vec{c})
 \end{aligned}$$

Note: In view of the above result, we simply write $\vec{a} + \vec{b} + \vec{c}$ for $(\vec{a} + \vec{b}) + \vec{c}$ or $\vec{a} + (\vec{b} + \vec{c})$.

THEOREM 5.4

1. There is a unique vector $\vec{0}$ such that $\vec{a} + \vec{0} = \vec{a}$ for all vectors $\vec{a}$.
2. For each vector $\vec{a}$, there is a unique vector $\vec{b}$ such that $\vec{a} + \vec{b} = \vec{0}$.

PROOF

Recall that for any points A and B , the vectors $\overrightarrow{AA}$ and $\overrightarrow{BB}$ are equal and are denoted by $\vec{0}$.

1. For any vector $\vec{a} = \overrightarrow{AB}$, we have

$$\vec{a} + \vec{0} = \overrightarrow{AB} + \overrightarrow{BB} = \overrightarrow{AB} = \vec{a}$$

Also, if $\vec{z}$ is any vector such that $\vec{a} + \vec{z} = \vec{a}$ for all vectors $\vec{a}$, then

$$\vec{z} = \vec{z} + \vec{0} = \vec{0} + \vec{z} = \vec{0}$$

Thus $\vec{0}$ is the unique vector such that $\vec{a} + \vec{0} = \vec{a}$ for all vectors $\vec{a}$.

2. Let $\vec{a} = \overrightarrow{AB}$ be any given vector. Put $\vec{b} = \overrightarrow{BA}$. Then

$$\vec{a} + \vec{b} = \overrightarrow{AB} + \overrightarrow{BA} = \overrightarrow{AA} = \vec{0}$$

If $\vec{c}$ is any vector such that $\vec{a} + \vec{c} = \vec{0}$, then

$$\begin{aligned}
 \vec{c} &= \vec{c} + \vec{0} \\
 &= \vec{c} + (\vec{a} + \vec{b}) \\
 &= (\vec{c} + \vec{a}) + \vec{b} \\
 &= \vec{0} + \vec{b} = \vec{b}
 \end{aligned}$$

Thus $\vec{b} (= \overrightarrow{BA})$ is the unique vector such that $\vec{a} + \vec{b} = \vec{0}$.

DEFINITION 5.13 The unique vector $\vec{0}$ such that $\vec{a} + \vec{0} = \vec{a}$ for all vectors $\vec{a}$ is called the *additive identity* or *zero vector*.

DEFINITION 5.14 For any vector $\vec{a}$, the unique vector $\vec{b}$ such that $\vec{a} + \vec{b} = \vec{0}$ is called the *additive inverse* of $\vec{a}$ and is denoted by $-\vec{a}$.

Note that, if $\vec{a} = \overrightarrow{AB}$, then $-\vec{a} = \overrightarrow{BA}$.

THEOREM 5.5 The following hold for any vector $\vec{a}, \vec{b}$ and $\vec{c}$.

1. $\vec{a} + \vec{b} = \vec{0} \Leftrightarrow \vec{b} = -\vec{a}$
2. $\vec{a} + \vec{b} = \vec{a} + \vec{c} \Leftrightarrow \vec{b} = \vec{c}$
3. $\vec{a} + \vec{b} = \vec{a} \Leftrightarrow \vec{b} = \vec{0}$
4. $-(-\vec{a}) = \vec{a}$
5. $-(\vec{a} + \vec{b}) = (-\vec{a}) + (-\vec{b})$
6. $\vec{a} + \vec{b} = \vec{c} \Rightarrow \vec{a} = \vec{c} + (-\vec{b})$

PROOF 1. We have $\vec{a} + \vec{b} = \vec{0}$ which implies

$$\begin{aligned}\vec{b} &= \vec{0} + \vec{b} \\ &= [\vec{a} + (-\vec{a})] + \vec{b} \\ &= -\vec{a} + (\vec{a} + \vec{b}) \\ &= -\vec{a} + \vec{0} = -\vec{a}\end{aligned}$$

The other implication is trivial.

2. We have $\vec{a} + \vec{b} = \vec{a} + \vec{c}$ which implies

$$\begin{aligned}-\vec{a} + (\vec{a} + \vec{b}) &= -\vec{a} + (\vec{a} + \vec{c}) \\ (-\vec{a} + \vec{a}) + \vec{b} &= (-\vec{a} + \vec{a}) + \vec{c} \\ \vec{0} + \vec{b} &= \vec{0} + \vec{c} \\ \vec{b} &= \vec{c}\end{aligned}$$

3. We have

$$\begin{aligned}\vec{a} + \vec{b} &= \vec{a} \\ \Rightarrow \vec{a} + \vec{b} &= \vec{a} + \vec{0} \\ \Rightarrow \vec{b} &= \vec{0} \quad [\text{by (2)}]\end{aligned}$$

4. Since $-\vec{a} + \vec{a} = \vec{0}$, it follows from (1) that $\vec{a} = -(-\vec{a})$.

5. Since

$$\begin{aligned}\vec{a} + \vec{b} + (-\vec{a}) + (-\vec{b}) &= [\vec{a} + (-\vec{a})] + [\vec{b} + (-\vec{b})] \\ &= \vec{0} + \vec{0} = \vec{0}\end{aligned}$$

it follows again from (1) that $(-\vec{a}) + (-\vec{b}) = -(\vec{a} + \vec{b})$.

6. If $\vec{a} + \vec{b} = \vec{c}$, then

$$\vec{c} + (-\vec{b}) = \vec{a} + \vec{b} + (-\vec{b}) = \vec{a} + \vec{0} = \vec{a}$$

Note: For any vectors $\vec{a}$ and $\vec{b}$, for simplicity we write $\vec{a} - \vec{b}$ for $\vec{a} + (-\vec{b})$.

DEFINITION 5.15 Let “ O ” be a fixed point. If A is any point in the space, then the vector $\overrightarrow{OA}$ is called the *position vector* of the point A with reference to the origin O .

THEOREM 5.6 Let $\vec{a}$ and $\vec{b}$ be the position vectors of the points A and B respectively with reference to the origin O . Then $\overrightarrow{AB} = \vec{b} - \vec{a}$.

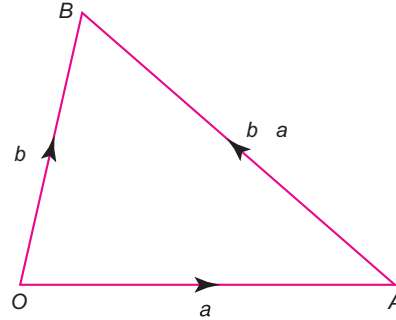


FIGURE 5.10 Theorem 5.6.

PROOF We are given that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$. By the triangle law (Figure 5.10), we have $\overrightarrow{OA} + \overrightarrow{AB} = \vec{b}$. Therefore, by part (6) of Theorem 5.5, we have

$$\overrightarrow{AB} = \vec{b} - \overrightarrow{OA} = \vec{b} - \vec{a}$$

Note: For any three points A, B, C , $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ and hence $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$

THEOREM 5.7 Let $\vec{a}$ and $\vec{b}$ be any vectors. Then

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

Also, $\vec{a}$ and $\vec{b}$ are like vectors if and only if

$$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$$

PROOF Choose points A, B and C such that $\vec{a} = \overrightarrow{AB}$ and $\vec{b} = \overrightarrow{BC}$. We shall distinguish three cases.

Case I: Suppose that $\vec{a}$ and $\vec{b}$ are not collinear. Then A, B and C are not collinear and, in the triangle ABC ,

$$AC < AB + BC$$

(i.e., any side of a triangle is less than the sum of the other two sides, Figure 5.11).

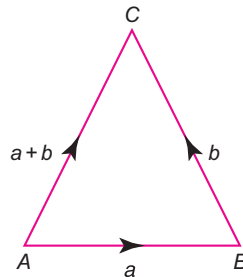


FIGURE 5.11 Case I.

Therefore, we have

$$|\vec{a} + \vec{b}| = |\overrightarrow{AC}| = AC < AB + BC = |\overrightarrow{AB}| + |\overrightarrow{BC}| = |\vec{a}| + |\vec{b}|$$

Case II: Suppose that $\vec{a}$ and $\vec{b}$ are collinear and have opposite directions (Figure 5.12). Then B does not lie between A and C . B is either to the left of A or to the right of C . Then

$$\begin{aligned} |\vec{a} + \vec{b}| &= |\overrightarrow{AB} + \overrightarrow{BC}| \\ &= |\overrightarrow{AC}| \\ &= AC \\ &< AB + BC = |\vec{a}| + |\vec{b}| \end{aligned}$$

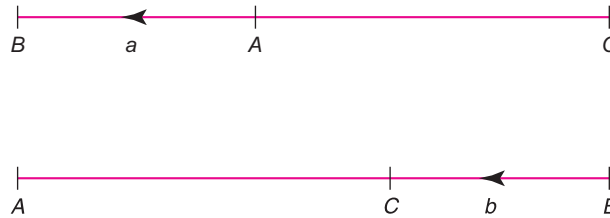


FIGURE 5.12 Case II.

Case III: Suppose that $\vec{a}$ and $\vec{b}$ are collinear and have the same direction (Figure 5.13), that is, $\vec{a}$ and $\vec{b}$ are like vectors. Then B lies between A and C . Then

$$\begin{aligned} |\vec{a} + \vec{b}| &= |\overrightarrow{AB} + \overrightarrow{BC}| \\ &= |\overrightarrow{AC}| \\ &= AC \\ &= AB + BC \\ &= |\vec{a}| + |\vec{b}| \end{aligned}$$

Therefore, if $\vec{a}$ and $\vec{b}$ are not like vectors [Case (I) or (II)], then

$$|\vec{a} + \vec{b}| < |\vec{a}| + |\vec{b}|$$

and, if $\vec{a}$ and $\vec{b}$ are like vectors [Case (III)], then

$$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$$



FIGURE 5.13 Case III.

Note: The inequality $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ is called the *triangular inequality*.

COROLLARY 5.1 For any vectors $\vec{a}$ and $\vec{b}$,

$$||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}|$$

PROOF By the triangular inequality, we have

$$|\vec{a}| = |\vec{a} - \vec{b} + \vec{b}| \leq |\vec{a} - \vec{b}| + |\vec{b}|$$

and hence $|\vec{a}| - |\vec{b}| \leq |\vec{a} - \vec{b}|$. Also,

$$|\vec{b}| = |\vec{a} + \vec{b} - \vec{a}| \leq |\vec{a}| + |\vec{b} - \vec{a}|$$

and hence

$$|\vec{b}| - |\vec{a}| \leq |\vec{b} - \vec{a}| = |-(\vec{a} - \vec{b})| = |\vec{a} - \vec{b}|$$

Thus

$$||\vec{a}| - |\vec{b}|| = \pm(|\vec{a}| - |\vec{b}|) \leq |\vec{a} - \vec{b}|$$

DEFINITION 5.16 For any vector $\vec{a}$ and for any integer n , we define the vector $n\vec{a}$ by

$$n\vec{a} = \begin{cases} \vec{0} & \text{if } n = 0 \\ (n-1)\vec{a} + \vec{a} & \text{if } n > 0 \\ -(-n)\vec{a} & \text{if } n < 0 \end{cases}$$



QUICK LOOK 1

In particular,

$$1\vec{a} = \vec{a}$$

$$2\vec{a} = \vec{a} + \vec{a}$$

$$3\vec{a} = \vec{a} + \vec{a} + \vec{a}$$

$$(-3)\vec{a} = -(3\vec{a}) = -\vec{a} - \vec{a} - \vec{a}$$

$$(-2)\vec{a} = -\vec{a} - \vec{a}$$

$$(-1)\vec{a} = -\vec{a}$$

COROLLARY 5.2 For any integer n and for any vector $\vec{a}$,

$$|n\vec{a}| = |n||\vec{a}|$$

PROOF This can be proved from the equalities

$$|\vec{a} + \vec{a}| = |\vec{a}| + |\vec{a}|$$

and

$$|-\vec{a}| = |\vec{a}|$$

and by using induction on n .

Example

Let $ABCD$ be a parallelogram (Figure 5.14). Then

(1) $\overrightarrow{AB} = \overrightarrow{DC}$

(2) $\overrightarrow{AD} = \overrightarrow{BC}$

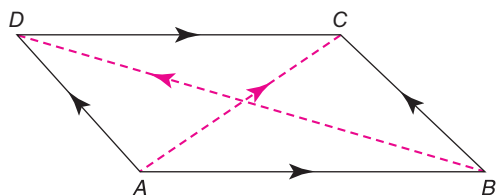


FIGURE 5.14

$$\begin{aligned} \text{(3)} \quad \overrightarrow{AC} + \overrightarrow{BD} &= \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{BC} + \overrightarrow{CD} \\ &= \overrightarrow{AB} + 2\overrightarrow{BC} - \overrightarrow{DC} = 2\overrightarrow{BC} \quad (\text{since } \overrightarrow{DC} = \overrightarrow{AB}) \end{aligned}$$

$$\begin{aligned} \text{(4)} \quad \overrightarrow{AC} - \overrightarrow{BD} &= \overrightarrow{AB} + \overrightarrow{BC} - (\overrightarrow{BC} + \overrightarrow{CD}) \\ &= \overrightarrow{AB} + \overrightarrow{DC} = 2\overrightarrow{AB} \end{aligned}$$

Example

Let $\vec{a} + 2\vec{b}$, $2\vec{a} - \vec{b}$, $\vec{a}$ and $3\vec{a} + \vec{b}$ be the position vectors of the points A , B , C and D , respectively, relative to a point O . Then

$$\vec{OA} = \vec{a} + 2\vec{b}$$

$$\vec{OB} = 2\vec{a} - \vec{b}$$

$$\vec{OC} = \vec{a}$$

$$\vec{OD} = 3\vec{a} + \vec{b}$$

Therefore

$$\begin{aligned}\vec{AC} &= \vec{OC} - \vec{OA} \quad (\text{see Theorem 5.6}) \\ &= \vec{a} - (\vec{a} + 2\vec{b}) = -2\vec{b}\end{aligned}$$

$$\begin{aligned}\vec{DA} &= \vec{OA} - \vec{OD} \\ &= (\vec{a} + 2\vec{b}) - (3\vec{a} + \vec{b}) = -2\vec{a} + \vec{b} \\ \vec{AB} &= \vec{OB} - \vec{OA} \\ &= (2\vec{a} - \vec{b}) - (\vec{a} + 2\vec{b}) = \vec{a} - 3\vec{b} \\ \vec{BC} &= \vec{OC} - \vec{OB} \\ &= \vec{a} - (2\vec{a} - \vec{b}) = -\vec{a} + \vec{b}\end{aligned}$$

Example 5.1

Let ABC be a triangle in which P , Q and R are the mid-points of AB , BC and CA , respectively (Figure 5.15). Then, for any point O , prove that

$$\vec{OA} + \vec{OB} + \vec{OC} = \vec{OP} + \vec{OQ} + \vec{OR}$$

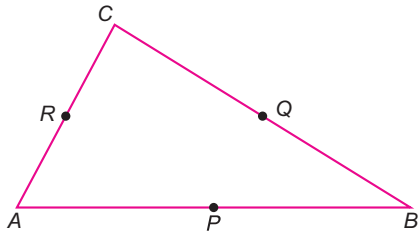


FIGURE 5.15 Example 5.1.

Solution: Since P , Q and R are the mid-points of AB , BC and CA we have

$$\begin{aligned}\vec{OA} + \vec{OB} + \vec{OC} &= \vec{OP} + \vec{PA} + \vec{OQ} + \vec{QB} + \vec{OR} + \vec{RC} \\ &= \vec{OP} + \vec{OQ} + \vec{OR} + \vec{PA} + \vec{RC} + \vec{QB} \\ &= \vec{OP} + \vec{OQ} + \vec{OR} + \frac{1}{2}(\vec{BA} + \vec{AC} + \vec{CB}) \\ &= \vec{OP} + \vec{OQ} + \vec{OR} \\ &\quad (\because \vec{BA} + \vec{AC} + \vec{CB} = \vec{0})\end{aligned}$$

Example 5.2

Let S be the circumcentre and O the orthocentre of a triangle ABC (Figure 5.16). Then prove that

(1) $\vec{OA} + \vec{OB} + \vec{OC} = 2\vec{OS}$

(2) $\vec{SA} + \vec{SB} + \vec{SC} = \vec{SO}$

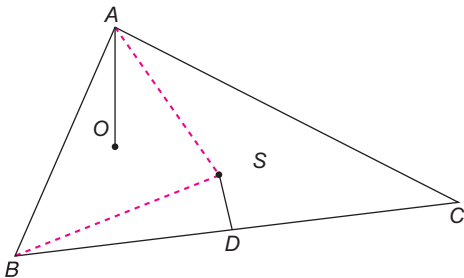


FIGURE 5.16 Example 5.2.

Solution: Let D be the mid-point of BC . Then

$$\vec{DB} + \vec{DC} = -\vec{DC} + \vec{DC} = \vec{0}$$

Also, we know that (Theorem 4.22, Chapter 4)

$$AO = 2R \cos A$$

and from $\triangle BDS$

$$\cos A = \cos \angle BSD = \frac{SD}{R}$$

Hence

$$2\vec{SD} = \vec{AO}$$

(1) Now,

$$\begin{aligned}
 \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} &= \overrightarrow{OA} + (\overrightarrow{OD} + \overrightarrow{DB}) + (\overrightarrow{OD} + \overrightarrow{DC}) \\
 &= \overrightarrow{OA} + 2\overrightarrow{OD} + \overrightarrow{DB} + \overrightarrow{DC} \\
 &= 2\overrightarrow{DS} + 2\overrightarrow{OD} \quad (\because \overrightarrow{DB} + \overrightarrow{DC} = \vec{0}) \\
 &= 2(\overrightarrow{OD} + \overrightarrow{DS}) \\
 &= 2\overrightarrow{OS}
 \end{aligned}$$

(2) Again

$$\begin{aligned}
 \overrightarrow{SA} + \overrightarrow{SB} + \overrightarrow{SC} &= \overrightarrow{SA} + \overrightarrow{SD} + \overrightarrow{DB} + \overrightarrow{SD} + \overrightarrow{DC} \\
 &= \overrightarrow{SA} + 2\overrightarrow{SD} + \overrightarrow{DB} + \overrightarrow{DC} \\
 &= \overrightarrow{SA} + \overrightarrow{AO} + \vec{0} \\
 &= \overrightarrow{SO}
 \end{aligned}$$

(See the Note under “Single Correct Choice Type Question 3” in Worked-Out Problems.)

5.3 | Multiplication of a Vector by a Scalar

Multiplication of a vector by a positive integer is only a repeated addition. In Definition 5.16, we have defined the product $n\vec{a}$ for any integer n and a vector $\vec{a}$. In this section we extend multiplication of vectors by integers to that by any real number.

DEFINITION 5.17 Let r be any real number and $\vec{a}$ any vector.

1. $r\vec{a}$ is defined to be $\vec{0}$ if $r = 0$ or $\vec{a} = \vec{0}$.
2. If $r > 0$ and $\vec{a} \neq \vec{0}$, then $r\vec{a}$ is defined to be the vector whose magnitude is $r|\vec{a}|$ and direction is same as that of $\vec{a}$.
3. If $r < 0$ and $\vec{a} \neq \vec{0}$, then $r\vec{a}$ is defined as $(-r)(-\vec{a})$, which is the vector with the magnitude $(-r)|\vec{a}|$ and the direction same as $-\vec{a}$, or equivalently, the direction is opposite to that of $\vec{a}$.

Note that Definition 5.17 coincides with Definition 5.16 when r is an integer. The following is an immediate consequence of Definition 5.17.

THEOREM 5.8 For any real number r and for any vector $\vec{a}$,

$$(-r)\vec{a} = -(r\vec{a}) = r(-\vec{a})$$

The following is a generalization of the above.

THEOREM 5.9 For any real numbers r and s and for any vector $\vec{a}$,

$$r(s\vec{a}) = (rs)\vec{a} = s(r\vec{a})$$

PROOF The result is trivial if $r = 0$ or $s = 0$ or $\vec{a} = \vec{0}$. Suppose that $r \neq 0$, $s \neq 0$ and $\vec{a} \neq \vec{0}$. We have

$$|r(s\vec{a})| = |r||s\vec{a}| = |r||s||\vec{a}| = |rs||\vec{a}| = |(rs)\vec{a}|$$

Therefore, $r(s\vec{a})$ and $(rs)\vec{a}$ have same magnitude. If $rs > 0$, then ($r > 0$ and $s > 0$) or ($r < 0$ and $s < 0$) and hence both of $r(s\vec{a})$ and $(rs)\vec{a}$ have the same direction as that of $\vec{a}$. If $rs < 0$, then ($r > 0$ and $s < 0$ or $r < 0$ and $s > 0$) both of $r(s\vec{a})$ and $(rs)\vec{a}$ have the same direction which is opposite to that of $\vec{a}$.

Therefore, in all cases, $r(s\vec{a})$ and $(rs)\vec{a}$ have same direction. Thus $r(s\vec{a}) = (rs)\vec{a}$. Since $rs = sr$, we have

$$r(s\vec{a}) = (rs)\vec{a} = (sr)\vec{a} = s(r\vec{a})$$

Note: By taking $s = -1$ in Theorem 5.9, we get Theorem 5.8.

THEOREM 5.10 For any real numbers r and s and for any vector $\vec{a}$,

$$(r + s)\vec{a} = r\vec{a} + s\vec{a}$$

PROOF Without loss of generality, we can assume that $r \neq 0$, $s \neq 0$ and $\vec{a} \neq \vec{0}$. We shall distinguish the following four cases.

Case I: Suppose that $r > 0$ and $s > 0$. Then $r + s > 0$ and hence the directions of $r\vec{a}$, $s\vec{a}$, $r\vec{a} + s\vec{a}$ and $(r + s)\vec{a}$ are all equal to the direction of $\vec{a}$. That is, $(r + s)\vec{a}$ and $r\vec{a} + s\vec{a}$ have the same direction. Also, regarding their magnitudes, we have

$$\begin{aligned} |(r + s)\vec{a}| &= |r + s||\vec{a}| = (r + s)|\vec{a}| \quad (\text{since } r + s > 0) \\ &= r|\vec{a}| + s|\vec{a}| \\ &= |r||\vec{a}| + |s||\vec{a}| \quad (\text{since } r > 0 \text{ and } s > 0) \\ &= |r\vec{a}| + |s\vec{a}| \\ &= |r\vec{a} + s\vec{a}| \quad (\text{by Theorem 5.7}) \end{aligned}$$

Thus $(r + s)\vec{a}$ and $r\vec{a} + s\vec{a}$ have same direction and magnitude and hence they are equal.

Case II: Suppose that $r < 0$ and $s < 0$. Then $-r > 0$ and $-s > 0$. Now we have

$$\begin{aligned} (r + s)\vec{a} &= [-(-r - s)]\vec{a} \\ &= -[(-r - s)\vec{a}] \quad (\text{by Theorem 5.8}) \\ &= -[(-r)\vec{a} + (-s)\vec{a}] \quad [\text{by Case (I) above}] \\ &= -[-(r\vec{a}) - (s\vec{a})] \quad (\text{again by Theorem 5.8}) \\ &= r\vec{a} + s\vec{a} \quad [\text{by part (5) of Theorem 5.5}] \end{aligned}$$

Case III: Suppose that $r > 0$ and $s < 0$.

(A) If $r + s > 0$, then

$$\begin{aligned} r\vec{a} &= [(r + s) + (-s)]\vec{a} \\ &= (r + s)\vec{a} + (-s)\vec{a} \quad [\text{by Case (I) above}] \\ &= (r + s)\vec{a} - s\vec{a} \quad (\text{by Theorem 5.8}) \end{aligned}$$

and hence

$$(r + s)\vec{a} = r\vec{a} + s\vec{a}$$

(B) If $r + s < 0$, then

$$\begin{aligned} s\vec{a} &= [(r + s) + (-r)]\vec{a} \\ &= (r + s)\vec{a} + (-r)\vec{a} \quad [\text{by Case (II) above}] \\ &= (r + s)\vec{a} - r\vec{a} \quad (\text{by Theorem 5.8}) \end{aligned}$$

and hence

$$(r + s)\vec{a} = r\vec{a} + s\vec{a}$$

(C) If $r + s = 0$, then $s = -r$ and

$$\begin{aligned} r\vec{a} + s\vec{a} &= r\vec{a} + (-r)\vec{a} \\ &= r\vec{a} - r\vec{a} \\ &= \vec{0} = (r + s)\vec{a} \end{aligned}$$

Case IV: Suppose that $r < 0$ and $s > 0$. By interchanging r and s and using Case (III), we get that

$$(r + s)\vec{a} = (s + r)\vec{a} = s\vec{a} + r\vec{a} = r\vec{a} + s\vec{a}$$

THEOREM 5.11 For any vectors $\vec{a}$ and $\vec{b}$ and for any real number r ,

$$r(\vec{a} + \vec{b}) = r\vec{a} + r\vec{b}$$

PROOF The result is trivial if $r = 0$ or $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$. Therefore we can suppose that $r \neq 0$, $\vec{a} \neq \vec{0}$ and $\vec{b} \neq \vec{0}$.

Case I: Suppose that $r > 0$. Choose points O, A and B such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{AB} = \vec{b}$ (Figure 5.17). Then

$$\vec{a} + \vec{b} = \overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$$

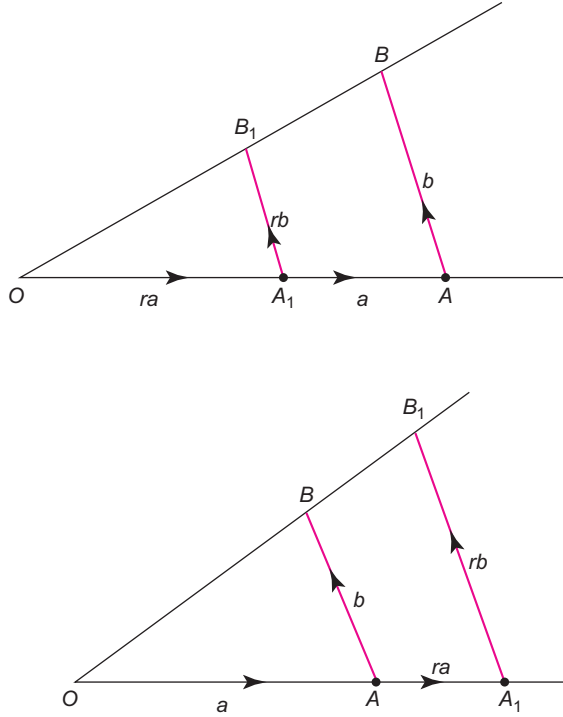


FIGURE 5.17 Theorem 5.11.

Choose a point A_1 on the line OA such that $\overrightarrow{OA_1} = r\vec{a}$ and then choose a point B_1 that $\overrightarrow{A_1B_1} = r\vec{b}$. Then the triangles OAB and OA_1B_1 are similar and B_1 lies on the line OB . Since $r > 0$ and $OB_1 = r \cdot OB$, we get that $\overrightarrow{OB_1} = r \cdot \overrightarrow{OB} = r\vec{b}$. Now

$$\begin{aligned} r(\vec{a} + \vec{b}) &= r \cdot \overrightarrow{OB} \\ &= \overrightarrow{OB_1} \\ &= \overrightarrow{OA_1} + \overrightarrow{A_1B_1} \\ &= r\vec{a} + r\vec{b} \end{aligned}$$

Case II: Suppose that $r < 0$. Then $-r > 0$ and

$$\begin{aligned} r(\vec{a} + \vec{b}) &= [-(-r)] (\vec{a} + \vec{b}) \\ &= -(-r)(\vec{a} + \vec{b}) \quad (\text{by Theorem 5.8}) \\ &= -[(-r)\vec{a} + (-r)\vec{b}] \quad [\text{by Case (I)}] \\ &= -(-r\vec{a} - r\vec{b}) \\ &= r\vec{a} + r\vec{b} \end{aligned}$$



Note: For any vector $\vec{a}$ and non-zero real number r , we shall write $\vec{a}/r$ for $(1/r)\vec{a}$.

DEFINITION 5.18 A vector $\vec{a}$ is said to be *unit vector* if its magnitude is the unit 1; that is $|\vec{a}| = 1$.

If $\vec{a}$ is any non-zero vector, then

$$\left| \frac{\vec{a}}{|\vec{a}|} \right| = \frac{1}{|\vec{a}|} |\vec{a}| = 1$$

and hence $\vec{a}/|\vec{a}|$ is a unit vector for any $\vec{a} \neq \vec{0}$. $\vec{a}/|\vec{a}|$ is called the unit vector in the direction of $\vec{a}$.

For any non-zero vectors $\vec{a}$ and $\vec{b}$, note that $\vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{b} = r\vec{a}$, for some non-zero real number r and, in this case,

$$r = \pm \frac{|\vec{b}|}{|\vec{a}|}$$

depending on whether $\vec{a}$ and $\vec{b}$ are like vectors or opposite vectors. In other words, two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if $\alpha\vec{a} + \beta\vec{b} = \vec{0}$ for some non-zero real numbers α and β .

5.4 | The Division Formula

Given two points A and B whose position vectors with respect to a point O are $\vec{a}$ and $\vec{b}$, we derive a formula for the position vector of any point on the line joining A and B , in terms of the vectors $\vec{a}$ and $\vec{b}$ and the ratio in which the point divides the line segment AB .

THEOREM 5.12 Let $\vec{a}$ and $\vec{b}$ be the position vectors of two points A and B , respectively, with respect to a point O . Then a point C lies on the line $\overline{AB}$ and divides the line segment AB in the ratio $r:s$ if and only if the position vector of C with respect to O is

$$\frac{r\vec{b} + s\vec{a}}{r + s}$$

that is

$$\overrightarrow{OC} = \frac{r \cdot \overrightarrow{OB} + s \cdot \overrightarrow{OA}}{r + s}$$

PROOF Let C be a point on the line $\overline{AB}$, dividing the line segment AB in the ratio $r:s$.

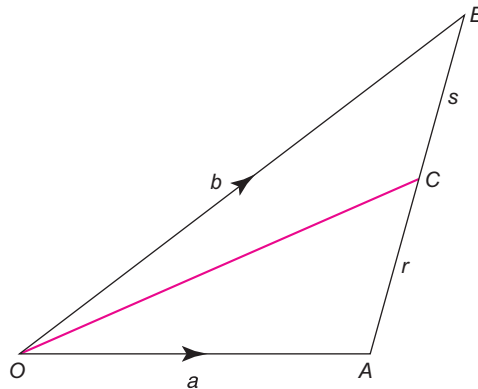


FIGURE 5.18 Case I.

Case I: Suppose that C divides AB internally (Figure 5.18). Then $r \geq 0$, $s \geq 0$ and $r + s > 0$. We have, from Theorem 5.6,

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$

and

$$\overrightarrow{CB} = \overrightarrow{OB} - \overrightarrow{OC}$$

Since C divides AB in the ratio $r:s$, we have

$$AC:CB = r:s$$

that is

$$sAC = rCB$$

Therefore

$$\begin{aligned} s\overrightarrow{AC} &= r\overrightarrow{CB} \\ s(\overrightarrow{OC} - \overrightarrow{OA}) &= r(\overrightarrow{OB} - \overrightarrow{OC}) \\ r\overrightarrow{OB} + s\overrightarrow{OA} &= s\overrightarrow{OC} + r\overrightarrow{OC} = (s+r)\overrightarrow{OC} \\ \overrightarrow{OC} &= \frac{r\overrightarrow{OB} + s\overrightarrow{OA}}{r+s} = \frac{r\vec{b} + s\vec{a}}{r+s} \end{aligned}$$

Case II: Suppose that C divides AB externally (Figure 5.19) in the ratio $r:s$. Then

$$rs < 0 \quad \text{and} \quad \frac{AC}{CB} = \frac{|r|}{|s|}$$

Therefore

$$\begin{aligned} |s|AC &= |r|CB \\ s\overrightarrow{AC} &= r\overrightarrow{CB} \end{aligned}$$

Since $AC \neq CB$, $|r| \neq |s|$ and hence $r + s \neq 0$. Now, from $s\overrightarrow{AC} = r\overrightarrow{CB}$, we have

$$\begin{aligned} s(\overrightarrow{OC} - \overrightarrow{OA}) &= r(\overrightarrow{OB} - \overrightarrow{OC}) \\ s\overrightarrow{OC} + r\overrightarrow{OC} &= r\overrightarrow{OB} + s\overrightarrow{OA} \end{aligned}$$

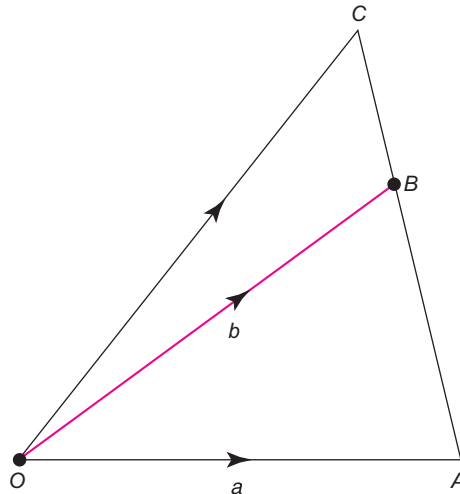


FIGURE 5.19 Case II.

$$(r + s)\overrightarrow{OC} = r\overrightarrow{OB} + s\overrightarrow{OA}$$

$$\overrightarrow{OC} = \frac{r\overrightarrow{OB} + s\overrightarrow{OA}}{r + s} = \frac{r\vec{b} + s\vec{a}}{r + s}$$

Conversely, suppose that C is a point such that

$$\overrightarrow{OC} = \frac{r\vec{b} + s\vec{a}}{r + s}$$

Then

$$(r + s)\overrightarrow{OC} = r\vec{b} + s\vec{a} = r\overrightarrow{OB} + s\overrightarrow{OA}$$

Therefore

$$r(\overrightarrow{OC} - \overrightarrow{OB}) = s(\overrightarrow{OA} - \overrightarrow{OC})$$

$$r\overrightarrow{CB} = s\overrightarrow{CA} \quad \text{and} \quad \frac{CA}{CB} = \frac{r}{s}$$

Therefore $\overrightarrow{CA}$ and $\overrightarrow{CB}$ are parallel vectors with a common initial point C . Hence A , B , C are collinear. Thus, C lies on the line $\overline{AB}$ and C divides AB in the ratio $r:s$. ■

COROLLARY 5.3 Let A and B be points whose position vectors with respect to a point O are $\vec{a}$ and $\vec{b}$. If C is the mid-point of AB , then

$$\overrightarrow{OC} = \frac{\vec{a} + \vec{b}}{2}$$

COROLLARY 5.4 The position vectors of the two points dividing a line segment AB into three equal parts are

$$\frac{2\overrightarrow{OA} + \overrightarrow{OB}}{3} \quad \text{and} \quad \frac{\overrightarrow{OA} + 2\overrightarrow{OB}}{3}$$

where O is the point of reference. These two points are called the *trisecting points* of AB .

THEOREM 5.13 Let A , B , and C be three distinct points whose position vectors with respect to a point O are $\vec{a}$, $\vec{b}$ and $\vec{c}$, respectively. Then A , B and C are collinear if and only if there exist real numbers r , s and t such that atleast one of them is non-zero,

$$r + s + t = 0 \quad \text{and} \quad r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$$

PROOF Suppose that r , s and t are real numbers, not all zero, such that

$$r + s + t = 0 \quad \text{and} \quad r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$$

Suppose that $t \neq 0$. Then

$$t\vec{c} = -r\vec{a} - s\vec{b} \quad \text{and} \quad t = -r - s$$

Therefore

$$\overrightarrow{OC} = \vec{c} = \frac{-r\vec{a} - s\vec{b}}{-r - s} = \frac{r\vec{a} + s\vec{b}}{r + s}$$

and hence, by Theorem 5.12, C lies on the line $\overline{AB}$. Thus A , B and C are collinear.

Conversely, suppose that A , B and C are collinear (Figure 5.20). Without loss of generality, we can assume that B lies in between A and C .

Put $AB = t$ and $BC = r$. Since A , B and C are distinct, t and r are positive real numbers and B divides AC internally in the ratio $t:r$. Hence, by Theorem 5.12, we have

$$\vec{b} = \overrightarrow{OB} = \frac{t\vec{c} + r\vec{a}}{t+r}$$

Now, put $s = -t - r$. Then, clearly $r + s + t = 0$ and $r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$.



FIGURE 5.20 Theorem 5.13.

COROLLARY 5.5

Let O and A be distinct points. Then a point P lies on the line $\overrightarrow{OA}$ if and only if

$$\overrightarrow{OP} = r \cdot \overrightarrow{OA}$$

for some real number r .

Recall that the line joining a vertex of a triangle and the mid-point of the side opposite to it is called a *median* of the triangle.

THEOREM 5.14

The medians of a triangle are concurrent and their point of concurrence divides each median in the ratio 2:1.

PROOF

Let ABC be a triangle and D , E and F be the mid-points of the sides BC , CA and AB , respectively. Fix an arbitrary point O and let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the position vectors of A , B and C , respectively, with respect to O (Figure 5.21). Then, we have

$$\overrightarrow{OD} = \frac{\overrightarrow{OB} + \overrightarrow{OC}}{2} = \frac{\vec{b} + \vec{c}}{2}$$

$$\overrightarrow{OE} = \frac{\overrightarrow{OC} + \overrightarrow{OA}}{2} = \frac{\vec{c} + \vec{a}}{2}$$

and

$$\overrightarrow{OF} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2} = \frac{\vec{a} + \vec{b}}{2}$$

Let G be the point that divides AD internally in the ratio 2:1. Then

$$\overrightarrow{OG} = \frac{2\overrightarrow{OD} + \overrightarrow{OA}}{2+1} = \frac{\vec{b} + \vec{c} + \vec{a}}{3} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

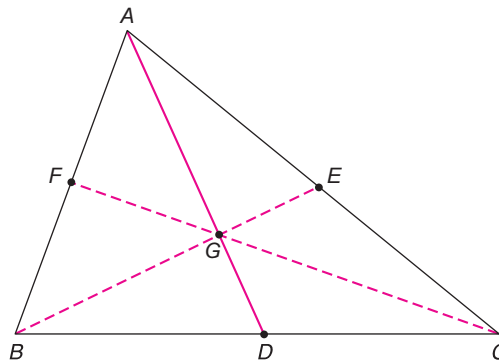


FIGURE 5.21 Theorem 5.14.

Also

$$\begin{aligned}\overrightarrow{OG} &= \frac{\vec{a} + \vec{b} + \vec{c}}{3} \\ &= \frac{2[(\vec{c} + \vec{a})/2] + \vec{b}}{2 + 1} \\ &= \frac{2\overrightarrow{OE} + \overrightarrow{OB}}{2 + 1}\end{aligned}$$

By Theorem 5.12, the point G lies on the median BE and divides it in the ratio 2:1. Again,

$$\begin{aligned}\overrightarrow{OG} &= \frac{\vec{a} + \vec{b} + \vec{c}}{3} \\ &= \frac{2[(\vec{a} + \vec{b})/2] + \vec{c}}{2 + 1} \\ &= \frac{2\overrightarrow{OF} + \overrightarrow{OC}}{2 + 1}\end{aligned}$$

and hence G lies on the median CF and divides it in the ratio 2:1. Therefore, G is a common point of all the three medians. Thus, the medians are concurrent and G is their point of concurrence. Also, G divides each of the medians in the ratio 2:1. ■

THEOREM 5.15

A point P lies on the plane determined by three non-collinear points O , A and B if and only if there exists real numbers r and s such that

$$\overrightarrow{OP} = r\overrightarrow{OA} + s\overrightarrow{OB}$$

and, in this case, the real numbers r and s are unique satisfying this equation.

PROOF

Let O , A and B be three non-collinear points and Π be the plane determined by them. Suppose that P is a point on Π . Then OA , OB and OP are coplanar. If P lies on the line $\overrightarrow{OA}$, then there is a real number r such that

$$\overrightarrow{OP} = r\overrightarrow{OA} = r\overrightarrow{OA} + s\overrightarrow{OB}$$

where $s = 0$. Similarly, if P lies on the line $\overrightarrow{OB}$, then there is a real number s such that

$$\overrightarrow{OP} = s\overrightarrow{OB} = r\overrightarrow{OA} + s\overrightarrow{OB}$$

where $r = 0$. Suppose that P lies on neither $\overrightarrow{OA}$ nor $\overrightarrow{OB}$. Through P draw parallels to the lines $\overrightarrow{OA}$ and $\overrightarrow{OB}$. These parallels intersect $\overrightarrow{OB}$ and $\overrightarrow{OA}$, say at C and D , respectively (since O , A and B are not collinear). Then $ODPC$ is a parallelogram (Figure 5.22). Therefore, there exist real numbers r and s such that

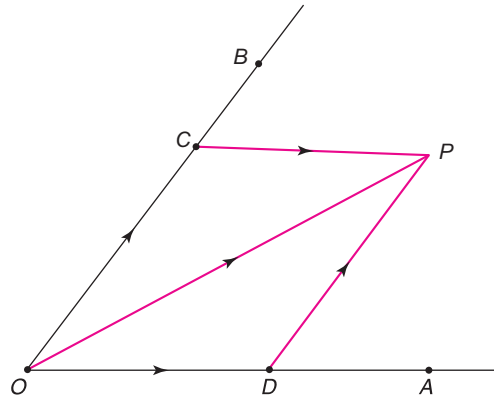
$$\overrightarrow{OD} = r\overrightarrow{OA}$$

and

$$\overrightarrow{OC} = s\overrightarrow{OB}$$

Now, we have

$$\overrightarrow{OP} = \overrightarrow{OD} + \overrightarrow{DP} = \overrightarrow{OD} + \overrightarrow{OC} = r\overrightarrow{OA} + s\overrightarrow{OB}$$

FIGURE 5.22 Parallelogram $ODPC$.

Conversely, suppose that $\overrightarrow{OP} = r\overrightarrow{OA} + s\overrightarrow{OB}$ for some real numbers r and s . If $r = 0$ or $s = 0$, then $\overrightarrow{OP} = s\overrightarrow{OB}$ or $r\overrightarrow{OA}$ and hence P lies on $\overrightarrow{OB}$ or $\overrightarrow{OA}$, so that P lies on the plane Π . Suppose that $r \neq 0$ and $s \neq 0$. Choose points C on $\overrightarrow{OB}$ and D on $\overrightarrow{OA}$ such that $\overrightarrow{OC} = s\overrightarrow{OB}$ and $\overrightarrow{OD} = r\overrightarrow{OA}$. Since O, A and B are non-collinear, it follows that O, D and C are not collinear. Therefore we can form a parallelogram $ODQC$ (Figure 5.23). Here, Q lies on Π and

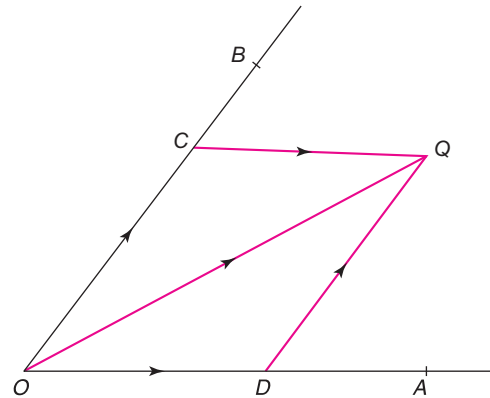
$$\begin{aligned}\overrightarrow{OQ} &= \overrightarrow{OD} + \overrightarrow{DQ} \\ &= \overrightarrow{OD} + \overrightarrow{OC} \\ &= r\overrightarrow{OA} + s\overrightarrow{OB} \\ &= \overrightarrow{OP}\end{aligned}$$

and hence $P = Q$. Thus P lies on the plane Π determined by O, A and B . The uniqueness of r and s can be proved as follows. Suppose that r' and s' are real numbers such that

$$r\overrightarrow{OA} + s\overrightarrow{OB} = \overrightarrow{OP} = r'\overrightarrow{OA} + s'\overrightarrow{OB}$$

Then

$$(r - r')\overrightarrow{OA} + (s - s')\overrightarrow{OB} = \vec{0}$$

FIGURE 5.23 Parallelogram $ODQC$.

If $r \neq r'$, then

$$\overrightarrow{OA} = \frac{s-s'}{r-r'} \cdot \overrightarrow{OB}$$

and hence, by Corollary 5.5, O , A and B are collinear, which is a contradiction to the hypothesis. Therefore $r - r' = 0$ and hence $r = r'$. Similarly, $s = s'$. ■

THEOREM 5.16 A quadrilateral is a parallelogram if and only if its diagonals bisect each other.

PROOF Let $ABCD$ a quadrilateral. Let $\vec{b}$, $\vec{c}$ and $\vec{d}$ be the position vectors of B , C and D with respect to A respectively; that is,

$$\overrightarrow{AB} = \vec{b}, \overrightarrow{AC} = \vec{c} \quad \text{and} \quad \overrightarrow{AD} = \vec{d}$$

Let P be the point of intersection of the diagonals AC and BD (Figure 5.24). Suppose that the diagonals bisect each other. Then,

$$\frac{\overrightarrow{AB} + \overrightarrow{AD}}{2} = \overrightarrow{AP} = \frac{\overrightarrow{AC}}{2}$$

That is

$$\frac{\vec{b} + \vec{d}}{2} = \frac{\vec{c}}{2}$$

Therefore

$$\vec{b} = \vec{c} - \vec{d} \quad \text{and} \quad \vec{d} = \vec{c} - \vec{b}$$

$$\overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{AD} = \overrightarrow{DC} \quad \text{and} \quad \overrightarrow{AD} = \overrightarrow{AC} - \overrightarrow{AB} = \overrightarrow{BC}$$

That is, the opposite sides of the quadrilateral are parallel and equal and hence $ABCD$ is a parallelogram.

Conversely, suppose that $ABCD$ is a parallelogram. Then

$$\vec{b} = \overrightarrow{AB} = \overrightarrow{DC} \quad \text{and} \quad \vec{d} = \overrightarrow{AD} = \overrightarrow{BC}$$

If Q is the mid-point of BD , then

$$\begin{aligned} \overrightarrow{AQ} &= \frac{\overrightarrow{AB} + \overrightarrow{AD}}{2} = \frac{\overrightarrow{AB} + (\overrightarrow{AC} + \overrightarrow{CD})}{2} \\ &= \frac{\overrightarrow{AB} + \overrightarrow{AC} - \overrightarrow{DC}}{2} = \frac{\overrightarrow{AC}}{2} \quad (\because \overrightarrow{AB} = \overrightarrow{DC}) \end{aligned}$$

Therefore Q is the mid-point of AC also. Thus BD and AC bisect each other.

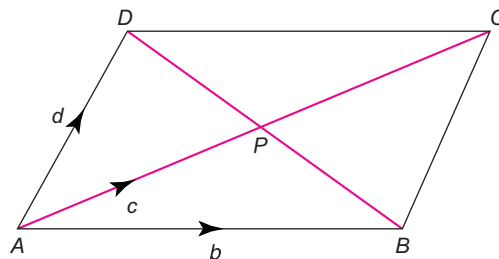


FIGURE 5.24 Theorem 5.16. ■

DEFINITION 5.19 Linear Combination Let $\vec{a}, \vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ be any vectors. $\vec{a}$ is said to be a *linear combination* of $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ if

$$\vec{a} = r_1\vec{a}_1 + r_2\vec{a}_2 + \dots + r_n\vec{a}_n$$

for some real numbers $r_1, r_2, \dots, r_n$.

DEFINITION 5.20 Linearly Dependent and Independent Vectors Non-zero vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are said to be *linearly dependent* if one of them is a linear combination of the others. If they are not linearly dependent, then we say that $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are *linearly independent*.

The following is a very important tool in solving certain dependency problems.

THEOREM 5.17 Let $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ be non-zero vectors. Then $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly independent if and only if, for any real numbers $r_1, r_2, \dots, r_n$,

$$r_1\vec{a}_1 + r_2\vec{a}_2 + \dots + r_n\vec{a}_n = \vec{0} \Rightarrow r_1 = r_2 = \dots = r_n = 0$$

PROOF We present a contra-positive proof. Suppose that $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly dependent. Then one of $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ is a linear combination of the others. Without loss of generality, we can assume that $\vec{a}_1$ is a linear combination of $\vec{a}_2, \dots, \vec{a}_n$. Therefore, there exist real numbers $r_2, r_3, \dots, r_n$ such that

$$\vec{a}_1 = r_2\vec{a}_2 + r_3\vec{a}_3 + \dots + r_n\vec{a}_n$$

Put $r_1 = -1$. Then

$$r_1\vec{a}_1 + r_2\vec{a}_2 + \dots + r_n\vec{a}_n = \vec{0} \quad \text{and} \quad r_1 \neq 0$$

and hence the required condition is not satisfied. Conversely, suppose that the required condition is not satisfied. That is, there exist real numbers $r_1, r_2, \dots, r_n$ such that

$$r_1\vec{a}_1 + r_2\vec{a}_2 + \dots + r_n\vec{a}_n = \vec{0}$$

and not all of $r_1, r_2, \dots, r_n$ are zero. Without loss of generality, we can assume that $r_1 \neq 0$. Then, from the above equation, we get that

$$\vec{a}_1 = \left(\frac{-r_2}{r_1} \right) \vec{a}_2 + \left(\frac{-r_3}{r_1} \right) \vec{a}_3 + \dots + \left(\frac{-r_n}{r_1} \right) \vec{a}_n$$

and hence $\vec{a}_1$ is a linear combination of $\vec{a}_2, \vec{a}_3, \dots, \vec{a}_n$. Thus $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly dependent. ■

THEOREM 5.18 Two non-zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if $\vec{a}, \vec{b}$ are linearly dependent.

PROOF Let O, A and B be points such that $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$. Since $\vec{a}$ and $\vec{b}$ are non-zero, $O \neq A$ and $O \neq B$. Suppose that $\vec{a}$ and $\vec{b}$ are collinear. Then $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are collinear and hence B lies on the line $\overrightarrow{OA}$. Then, by Corollary 5.5, we get that

$$\overrightarrow{OB} = r \cdot \overrightarrow{OA} \quad \text{or} \quad \vec{b} = r\vec{a}$$

for some real number r and hence $\vec{a}, \vec{b}$ are linearly dependent. Conversely, suppose that $\vec{a}, \vec{b}$ are linearly dependent. Then, we can assume, without loss of generality, that

$$\vec{a} = s\vec{b}$$

for some real number s . Since $\overrightarrow{OA} = s \overrightarrow{OB}$, again by Corollary 5.5 we get that A lies on the line $\overrightarrow{OB}$ and hence O, A and B are collinear and therefore $\vec{a}$ and $\vec{b}$ are collinear. ■

COROLLARY 5.6 The following are equivalent to each other for any non-zero vectors $\vec{a}$ and $\vec{b}$.

1. $\vec{a}$ and $\vec{b}$ are non-collinear.
2. $\vec{a}$ and $\vec{b}$ are linearly independent.
3. For any real numbers r and s , $r\vec{a} + s\vec{b} = \vec{0} \Rightarrow r = 0 = s$.

Example 5.3

Let $ABCD$ be a parallelogram and E the mid point of AB . Prove that the point of intersection of DE and AC divide then in the same ratio and find the ratio.

Solution: Let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AD} = \vec{d}$. Then $\overrightarrow{DC} = \vec{b}$ and $\overrightarrow{BC} = \vec{d}$. Now,

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \vec{b} + \vec{d}$$

and
$$\overrightarrow{AE} = \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\vec{b}$$

Let P be the point of intersection of DE and AC (Figure 5.25). Suppose that P divides AC and DE in the ratios $r:1$ and $s:1$, respectively. Then

$$\begin{aligned}\overrightarrow{AP} &= \frac{r \cdot \overrightarrow{AA} + 1 \cdot \overrightarrow{AC}}{r + 1} \\ &= \frac{r \cdot \vec{0} + 1 \cdot (\vec{b} + \vec{d})}{r + 1} = \frac{\vec{b} + \vec{d}}{r + 1}\end{aligned}$$

Also

$$\overrightarrow{AP} = \frac{s \cdot \overrightarrow{AE} + 1 \cdot \overrightarrow{AD}}{s + 1}$$

$$= \frac{s \cdot (1/2)\vec{b} + \vec{d}}{s + 1} = \frac{s\vec{b} + 2\vec{d}}{2(s + 1)}$$

Therefore

$$\frac{\vec{b} + \vec{d}}{r + 1} = \frac{s\vec{b} + 2\vec{d}}{2(s + 1)}$$

Since $\vec{b}$ and $\vec{d}$ are not collinear, by Theorem 5.15, we get

$$\frac{s}{2(s + 1)} = \frac{1}{r + 1} = \frac{2}{2(s + 1)}$$

from which we get that $r = 2 = s$.

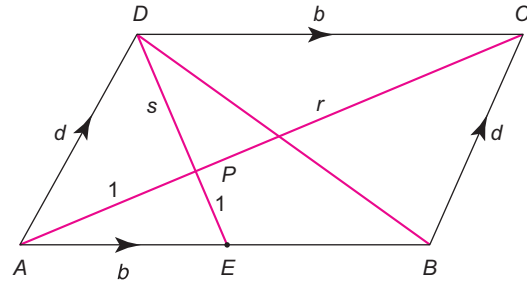


FIGURE 5.25 Example 5.3.

DEFINITION 5.21 Coplanar Vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are said to be coplanar if all of them lie in a single plane. In other words, when $\vec{a}_i = \overrightarrow{OA_i}$, $1 \leq i \leq n$, then $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are said to be coplanar if all the points $O, A_1, A_2, \dots, A_n$ lie on a single plane.

The following is a direct consequence of Theorem 5.15.

THEOREM 5.19 The following are equivalent to each other for any non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$:

1. $\vec{a}, \vec{b}, \vec{c}$ are not coplanar.
2. $\vec{a}, \vec{b}, \vec{c}$ are linearly independent.
3. For any real numbers, r, s and t ,

$$r\vec{a} + s\vec{b} + t\vec{c} = \vec{0} \Rightarrow r = s = t = 0$$

Example 5.4

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-coplanar vectors. Let A , B and C be points whose position vectors with respect to the origin O are $\vec{a} + 2\vec{b} + 3\vec{c}$, $-2\vec{a} + 3\vec{b} + 5\vec{c}$ and $7\vec{a} - \vec{c}$, respectively. Then prove that A , B and C are collinear.

Solution: We are given that

$$\vec{OA} = \vec{a} + 2\vec{b} + 3\vec{c}$$

$$\vec{OB} = -2\vec{a} + 3\vec{b} + 5\vec{c}$$

and $\vec{OC} = 7\vec{a} - \vec{c}$

Then

$$\begin{aligned}\vec{AB} &= \vec{OB} - \vec{OA} \\ &= (-2\vec{a} + 3\vec{b} + 5\vec{c}) - (\vec{a} + 2\vec{b} + 3\vec{c}) \\ &= -3\vec{a} + \vec{b} + 2\vec{c}\end{aligned}$$

and

$$\begin{aligned}\vec{AC} &= \vec{OC} - \vec{OA} \\ &= (7\vec{a} - \vec{c}) - (\vec{a} + 2\vec{b} + 3\vec{c}) \\ &= 6\vec{a} - 2\vec{b} - 4\vec{c} \\ &= -2(-3\vec{a} + \vec{b} + 2\vec{c})\end{aligned}$$

Therefore

$$\vec{AC} = 2\vec{AB}$$

By Corollary 5.6, $\vec{AC}$ and $\vec{AB}$ are collinear and hence A , B and C are collinear.

Example 5.5

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-coplanar vectors. Prove that $\vec{a} + 2\vec{b} - \vec{c}$, $4\vec{a} + \vec{b} + 3\vec{c}$ and $2\vec{a} - 3\vec{b} + 2\vec{c}$ are non-coplanar.

Solution: It is enough if we prove that these are linearly independent. Let r , s and t be any real numbers such that

$$r(\vec{a} + 2\vec{b} - \vec{c}) + s(4\vec{a} + \vec{b} + 3\vec{c}) + t(2\vec{a} - 3\vec{b} + 2\vec{c}) = \vec{0}$$

Then

$$(r + 4s + 2t)\vec{a} + (2r + s - 3t)\vec{b} + (-r + 3s + 2t)\vec{c} = \vec{0}$$

Since $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar, we have, from Theorem 5.19, that

$$\begin{aligned}r + 4s + 2t &= 0 \\ 2r + s - 3t &= 0 \\ -r + 3s + 2t &= 0\end{aligned}$$

Solving these, we get that $r = 0 = s = t$. Therefore the given vectors are non-coplanar.

5.5 | Components of a Vector

Given any three non-coplanar vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, we shall prove in this section that any vector can be expressed as a linear combination of $\vec{a}$, $\vec{b}$ and $\vec{c}$ uniquely which facilitates us in representing any vector by a triad of real numbers. In particular, when the given vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are pairwise perpendicular, we have several advantages of the representation of vectors by triads of real numbers.

THEOREM 5.20

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be any non-coplanar vectors. For any vector $\vec{x}$, there exists a unique triad (r, s, t) of real numbers such that

$$\vec{x} = r\vec{a} + s\vec{b} + t\vec{c}$$

PROOF

Let $\vec{x}$ be a given vector. Choose points O , A , B , C and P such that

$$\vec{OA} = \vec{a}, \vec{OB} = \vec{b}, \vec{OC} = \vec{c} \quad \text{and} \quad \vec{OP} = \vec{x}$$

If $\vec{x} = \vec{0}$, then we can take $r = s = t = 0$ and these are unique since $\vec{a}$, $\vec{b}$ and $\vec{c}$ are linearly independent (Theorem 5.19). Therefore, we can suppose that $\vec{x} \neq \vec{0}$.

Since $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are not coplanar, $\overline{BOC}$, $\overline{COA}$ and $\overline{AOB}$ are three different planes. If P lies on the plane $\overline{AOB}$, then by Theorem 5.15, there exist real numbers r and s such that

$$\overrightarrow{OP} = r\overrightarrow{OA} + s\overrightarrow{OB}$$

Therefore

$$\vec{x} = r\vec{a} + s\vec{b} + 0 \cdot \vec{c}$$

Therefore we may assume that P is not in the plane $\overline{AOB}$. Draw a line through P parallel to OC and let it meet the plane $\overline{AOB}$ at Q (Figure 5.26). Then there exists a real number t such that $\overrightarrow{QP} = t\overrightarrow{OC} = t\vec{c}$. Since Q lies on the plane $\overline{AOB}$, there exist real numbers r and s such that

$$\overrightarrow{OQ} = r\overrightarrow{OA} + s\overrightarrow{OB} \quad (\text{by Theorem 5.15})$$

Now,

$$\overrightarrow{OP} = \overrightarrow{OQ} + \overrightarrow{QP} = r\overrightarrow{OA} + s\overrightarrow{OB} + t\overrightarrow{OC}$$

Therefore

$$\vec{x} = r\vec{a} + s\vec{b} + t\vec{c}$$

Since $\vec{a}$, $\vec{b}$ and $\vec{c}$ are not coplanar, they are linearly independent and hence

$$\begin{aligned} r\vec{a} + s\vec{b} + t\vec{c} &= r_1\vec{a} + s_1\vec{b} + t_1\vec{c} \\ \Rightarrow (r - r_1)\vec{a} + (s - s_1)\vec{b} + (t - t_1)\vec{c} &= \vec{0} \\ \Rightarrow r - r_1 = s - s_1 = t - t_1 &= 0 \end{aligned}$$

This implies $r = r_1$, $s = s_1$ and $t = t_1$. Thus, there exist unique real numbers r , s and t such that

$$\vec{x} = r\vec{a} + s\vec{b} + t\vec{c}$$

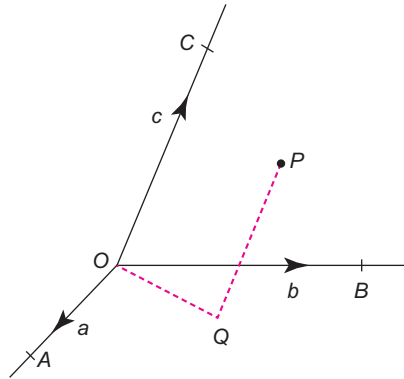


FIGURE 5.26 Theorem 5.20.

COROLLARY 5.7 Any four or more distinct vectors are linearly dependent.

PROOF

Let A be any set of vectors having at least four vectors. Choose distinct vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ in A . If $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly dependent, then so are all the vectors in A . Suppose that $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly independent. Since there are at least four vectors in A , we can choose $\vec{x}$ in A other than $\vec{a}$, $\vec{b}$, $\vec{c}$. Then, by Theorem 5.20, $\vec{x}$ is a linear combination of $\vec{a}$, $\vec{b}$ and $\vec{c}$ and therefore the vectors in A are linearly dependent. ■

Next, we shall define the concept of angle between two vectors. If $\vec{a}$ and $\vec{b}$ are two vectors, we can choose points O, A and B such that $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$. Then the angle between $\vec{a}$ and $\vec{b}$ is defined to be the smaller angle $\angle AOB$ and is denoted by $(\vec{a}, \vec{b})$. However, we have to prove that the angle between $\vec{a}$ and $\vec{b}$ does not depend on the choice of O, A and B . This is proved in the following.

THEOREM 5.21 Let O, A, B, O_1, A_1 and B_1 be points such that

$$\vec{OA} = \vec{O_1A_1} \quad \text{and} \quad \vec{OB} = \vec{O_1B_1}$$

Let $\angle AOB$ and $\angle A_1O_1B_1$ denote the smaller angles between OA and OB and between O_1A_1 and O_1B_1 , respectively. Then

$$\angle AOB = \angle A_1O_1B_1$$

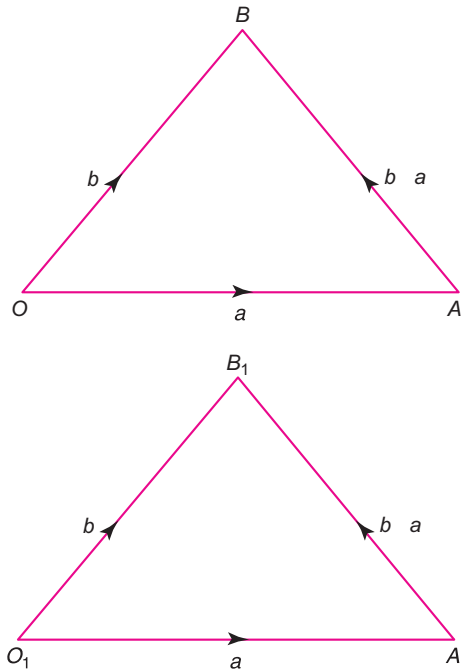


FIGURE 5.27 Theorem 5.21.

PROOF Let $\vec{a} = \vec{OA} = \vec{O_1A_1}$ and $\vec{b} = \vec{OB} = \vec{O_1B_1}$. We shall distinguish the following two cases. First notice that both $\angle AOB$ and $\angle A_1O_1B_1$ are in between 0 and π .

Case I: Suppose that $\vec{a}$ and $\vec{b}$ are parallel. Then O, A and B are collinear and O_1, A_1 and B_1 are also collinear. If $\vec{a}$ and $\vec{b}$ are like vectors, then A and B lie on the same side of O and hence $\angle AOB = 0^\circ$ and similarly $\angle A_1O_1B_1 = 0^\circ$. If $\vec{a}$ and $\vec{b}$ are opposite vectors, then A and B lie on different sides of O on the line $\vec{AOB}$ and hence $\angle AOB = \pi$ and similarly $\angle A_1O_1B_1 = \pi$.

Case II: Suppose that $\vec{a}$ and $\vec{b}$ are not parallel. Then O, A and B are not collinear and O_1, A_1 and B_1 are also not collinear, as shown in Figure 5.27. Since $\vec{OA} = \vec{a} = \vec{O_1A_1}$ and $\vec{OB} = \vec{b} = \vec{O_1B_1}$, we have

$$OA = O_1A_1 \quad \text{and} \quad OB = O_1B_1$$

Also, $\vec{AB} = \vec{b} - \vec{a} = \vec{A_1B_1}$, which implies $AB = A_1B_1$. Therefore, the triangles OAB and $O_1A_1B_1$ are congruent and hence $\angle AOB = \angle A_1O_1B_1$. ■

DEFINITION 5.22 Let $\vec{a}$ and $\vec{b}$ be any vectors and O, A and B be points such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$. Then the measure of the angle AOB which lies between 0 and π is defined as the *angle* between $\vec{a}$ and $\vec{b}$ and is denoted by $(\vec{a}, \vec{b})$. $\vec{a}$ and $\vec{b}$ are said to be *perpendicular vectors* if

$$(\vec{a}, \vec{b}) = \frac{\pi}{2}$$

By Theorem 5.21, $(\vec{a}, \vec{b})$ is independent of the choice of the points O, A and B . Also, $0 \leq (\vec{a}, \vec{b}) \leq \pi$. In Figure 5.28, $\vec{a}$ and $\vec{b}$ are given such that $(\vec{a}, \vec{b}) = 0$ or acute or obtuse or π .

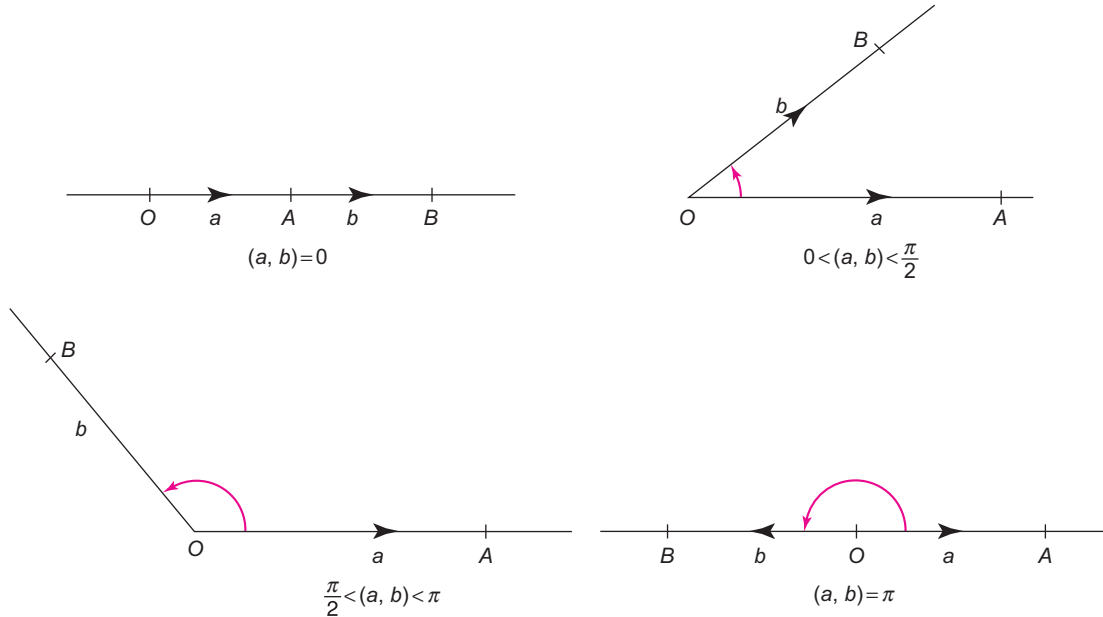


FIGURE 5.28

QUICK LOOK 2

- $(\vec{a}, \vec{b}) = (\vec{b}, \vec{a}) = (-\vec{a}, -\vec{b}) = (-\vec{b}, -\vec{a})$
- $(\vec{a}, -\vec{b}) = \pi - (\vec{a}, \vec{b}) = (-\vec{a}, \vec{b})$
- $(r\vec{a}, s\vec{b}) = (\vec{a}, \vec{b})$ for all positive real numbers r and s
- $(-r\vec{a}, s\vec{b}) = \pi - (\vec{a}, \vec{b}) = (r\vec{a}, -s\vec{b})$ for all $r, s > 0$
- $(\vec{a}, \vec{b}) = 0 \Leftrightarrow \vec{a}$ and $\vec{b}$ are like vectors
- $(\vec{a}, \vec{b}) = \pi \Leftrightarrow \vec{a}$ and $\vec{b}$ are opposite vectors

DEFINITION 5.23 If $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar vectors, then the triad $(\vec{a}, \vec{b}, \vec{c})$ is called a *basis vector triad*. If $(\vec{a}, \vec{b}, \vec{c})$ is a basis vector triad and $\vec{x}$ is any vector, then the unique real numbers r, s and t (as in Theorem 5.20) satisfying

$$\vec{x} = r\vec{a} + s\vec{b} + t\vec{c}$$

are, respectively, called $\vec{a}$ -component, $\vec{b}$ -component and $\vec{c}$ -component of $\vec{x}$ with respect to the triad $(\vec{a}, \vec{b}, \vec{c})$.

Let $(\vec{a}, \vec{b}, \vec{c})$ be a basis vector triad and O, A, B, C be points in the space such that

$$\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b} \quad \text{and} \quad \overrightarrow{OC} = \vec{c}$$

Since $\vec{a}, \vec{b}$ and $\vec{c}$ are linearly independent, the points O, A and B are not collinear (see Theorem 5.18) and hence the angle $(\vec{a}, \vec{b}) < \pi$. Also, in view of Theorem 5.15, $\vec{c}$ does not lie in the plane AOB . The basis vector triad $(\vec{a}, \vec{b}, \vec{c})$ is said

to be a *right-handed system* if, when observed from C , the angle of rotation from OA to OB in anticlockwise direction does not exceed 180° . $(\vec{a}, \vec{b}, \vec{c})$ is said to be a *left-handed system* if it is not a right-handed system.

DEFINITION 5.24 A right-handed system $(\vec{i}, \vec{j}, \vec{k})$ of vectors is said to be *orthogonal* if every pair of these are perpendicular, that is,

$$(\vec{i}, \vec{j}) = (\vec{j}, \vec{k}) = (\vec{k}, \vec{i}) = \frac{\pi}{2}$$

Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed orthogonal triad of unit vectors and O be a point in the space. Let OX , OY and OZ be directed lines along $\vec{i}$, $\vec{j}$ and $\vec{k}$, respectively. Then OX , OY and OZ are pairwise perpendicular and non-coplanar lines. The directed lines OX , OY and OZ are said to be the positive X -axis, the positive Y -axis and the positive Z -axis, respectively. From Figure 5.29, it can be observed that if a right-handed screw with axis along the Z -axis is rotated through an angle 90° from OX to OY , then it advances in the direction of positive Z -axis.

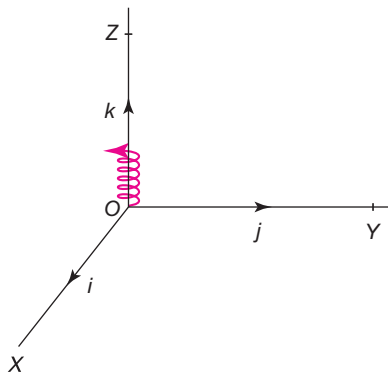


FIGURE 5.29 Right-handed system of orthogonal directed lines.

The triad (OX, OY, OZ) is called a right-handed system of orthogonal directed lines. $\vec{i}$, $\vec{j}$ and $\vec{k}$ are unit vectors along positive X , Y and Z axis, respectively. Hereafter, unless otherwise stated, $(\vec{i}, \vec{j}, \vec{k})$ always denotes a right-handed triad of orthogonal unit vectors.

Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed triad of orthogonal unit vectors. Choose a point O in the three-dimensional space and let A, B, C, A', B' and C' be points in the space such that

$$\overrightarrow{OA} = -\overrightarrow{OA'} = \vec{i}$$

$$\overrightarrow{OB} = -\overrightarrow{OB'} = \vec{j}$$

and

$$\overrightarrow{OC} = -\overrightarrow{OC'} = \vec{k}$$

Then the lines $\overline{AA'}$, $\overline{BB'}$ and $\overline{CC'}$ are pairwise perpendicular. Usually $(\vec{i}, \vec{j}, \vec{k})$ is called an *orthonormal basis* and the lines $\overline{AA'}$, $\overline{BB'}$, $\overline{CC'}$ are called the *coordinate axes* relative to $(\vec{i}, \vec{j}, \vec{k})$. O is called the *origin* and $\overline{AA'}$, $\overline{BB'}$ and $\overline{CC'}$ are, respectively, called the *x-axis*, the *y-axis* and the *z-axis*. $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are called the *positive directions* and $\overrightarrow{OA'}$, $\overrightarrow{OB'}$ and $\overrightarrow{OC'}$ are called the *negative directions* of the coordinate axes.

THEOREM 5.22 Let $(\vec{i}, \vec{j}, \vec{k})$ be an orthonormal basis of vectors. For any vector $\vec{a}$, there exists a unique ordered triad (a_1, a_2, a_3) of real numbers such that

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

and $\vec{a} \mapsto (a_1, a_2, a_3)$ is a one-to-one correspondence between the set V of vectors and the set $\mathbb{R}^3$ of ordered triads of real numbers.

PROOF This follows from the fact that $\vec{i}, \vec{j}, \vec{k}$ are non-coplanar and hence linearly independent and from Theorem 5.20. ■

DEFINITION 5.25 Let $\vec{a}$ be any vector and $(\vec{i}, \vec{j}, \vec{k})$ be an orthonormal basis of vectors. If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, then we denote this by writing $\vec{a} = (a_1, a_2, a_3)$. a_1 , a_2 and a_3 are, respectively, called the x -component (or $\vec{i}$ -component or the first component), y -component (or $\vec{j}$ -component or the second component) and z -component (or $\vec{k}$ -component or the third component) of $\vec{a}$.

THEOREM 5.23 Let $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$ with respect to an orthonormal basis $(\vec{i}, \vec{j}, \vec{k})$ of vectors and r be any real number. Then the following hold:

1. $\vec{a} = \vec{0} \Leftrightarrow a_1 = a_2 = a_3 = 0$
2. $\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$
3. $-\vec{a} = (-a_1, -a_2, -a_3)$
4. $r\vec{a} = (ra_1, ra_2, ra_3)$

PROOF These are consequences of the results that

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

and

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

and Theorems 5.8–5.11. ■

THEOREM 5.24 Let $\vec{a} = (a_1, a_2, a_3)$ with respect to an orthonormal basis $(\vec{i}, \vec{j}, \vec{k})$ of vectors. Then

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

PROOF Let $(\overline{OX}, \overline{OY}, \overline{OZ})$ be an orthogonal right-handed system of directed lines and $\vec{i}, \vec{j}$ and $\vec{k}$ be the unit vectors along $\overline{OX}, \overline{OY}$ and $\overline{OZ}$, respectively. Then $(\vec{i}, \vec{j}, \vec{k})$ is an orthonormal right-handed system of unit vectors. Let P be a point in the space such that

$$\overline{OP} = \vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

Let Q be the foot of the perpendicular from P to the plane ZOX , as shown in Figure 5.30. Draw perpendiculars from Q to the X -axis and Z -axis to meet at A and C , respectively. Then $OCQA$ is a rectangle. We have

$$\begin{aligned} \vec{a} &= \overline{OP} = \overline{OQ} + \overline{QP} \\ &= \overline{OC} + \overline{CQ} + \overline{QP} \\ &= \overline{OC} + \overline{OA} + \overline{QP} \end{aligned}$$

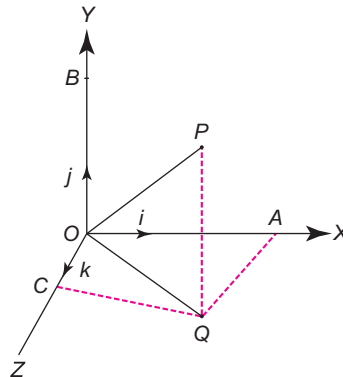


FIGURE 5.30 Theorem 5.24.

Since QP is perpendicular to the plane ZOX , QP is parallel to the Y -axis. Let B be a point on the Y -axis such that $\overline{OB} = \overline{QP}$. Now,

$$\begin{aligned}\vec{a} &= \overline{OC} + \overline{OA} + \overline{QP} \\ &= \overline{OC} + \overline{OA} + \overline{OB} \\ &= \overline{OA} + \overline{OB} + \overline{OC}\end{aligned}$$

Therefore

$$a_1\vec{i} + a_2\vec{j} + a_3\vec{k} = \overline{OA} + \overline{OB} + \overline{OC}$$

Since $\overline{OA}$, $\overline{OB}$ and $\overline{OC}$ are linearly independent, we get that

$$\overline{OA} = a_1\vec{i}, \quad \overline{OB} = a_2\vec{j} \quad \text{and} \quad \overline{OC} = a_3\vec{k}$$

Hence

$$|\overline{OA}| = a_1, \quad |\overline{OB}| = a_2 \quad \text{and} \quad |\overline{OC}| = a_3$$

Now

$$\begin{aligned}|\vec{a}|^2 &= |\overline{OP}|^2 = OP^2 = OQ^2 + QP^2 \quad (\text{since } OQP \text{ is a right angle}) \\ &= OC^2 + CQ^2 + QP^2 \quad (\text{since } OCQ \text{ is a right angle}) \\ &= OC^2 + OA^2 + OB^2 \\ &= a_3^2 + a_1^2 + a_2^2\end{aligned}$$

Thus

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

COROLLARY 5.8

Let $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$ be the position vectors of two points P and Q with respect to an orthonormal basis $(\vec{i}, \vec{j}, \vec{k})$. Then the distance between P and Q is given by

$$PQ = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

PROOF

Let O be the origin and

$$\vec{a} = \overline{OP} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

and

$$\vec{b} = \overline{OQ} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

Then

$$\begin{aligned}\overline{PQ} &= \overline{OQ} - \overline{OP} \\ &= (b_1 - a_1)\vec{i} + (b_2 - a_2)\vec{j} + (b_3 - a_3)\vec{k}\end{aligned}$$

Therefore

$$\begin{aligned}PQ &= |\overline{PQ}| = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2} \\ &= \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}\end{aligned}$$

DEFINITION 5.26

Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed orthonormal vector basis and $\vec{a}$ any vector. Let α , β and γ be the angles made by $\vec{a}$ with $\vec{i}$, $\vec{j}$ and $\vec{k}$, respectively. That is $\alpha = (\vec{a}, \vec{i})$, $\beta = (\vec{a}, \vec{j})$ and $\gamma = (\vec{a}, \vec{k})$. Then $\cos \alpha$, $\cos \beta$, $\cos \gamma$ taken in that order are called the *direction cosines* of $\vec{a}$ with respect to $(\vec{i}, \vec{j}, \vec{k})$ and these are denoted by l , m , n , respectively.

THEOREM 5.25 Let $\vec{0} \neq \vec{a} = (a_1, a_2, a_3)$ with respect to a right-handed orthonormal vector basis $(\vec{i}, \vec{j}, \vec{k})$ and l, m, n be direction cosines of $\vec{a}$. Then

$$l = \frac{a_1}{|\vec{a}|}, m = \frac{a_2}{|\vec{a}|} \quad \text{and} \quad n = \frac{a_3}{|\vec{a}|}$$

and

$$l^2 + m^2 + n^2 = 1$$

PROOF Let O be the origin and OX, OY and OZ be the directed lines along $\vec{i}, \vec{j}$ and $\vec{k}$, respectively (Figure 5.31). Let P be a point such that

$$\overrightarrow{OP} = \vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

Let α, β, γ be the angles made by $\vec{a}$ with $\vec{i}, \vec{j}, \vec{k}$, respectively. Then $\cos \alpha = l, \cos \beta = m$ and $\cos \gamma = n$. From Figure 5.31, we have

$$a_1 = OP \cos \alpha = rl \quad \text{and} \quad \text{hence } l = \frac{a_1}{r}$$

$$a_2 = OP \cos \beta = rm \quad \text{and} \quad \text{hence } m = \frac{a_2}{r}$$

$$a_3 = OP \cos \gamma = rn \quad \text{and} \quad \text{hence } n = \frac{a_3}{r}$$

where $r = OP = |\overrightarrow{OP}| = |\vec{a}|$. Now,

$$\begin{aligned} \vec{a} &= a_1\vec{i} + a_2\vec{j} + a_3\vec{k} \\ &= rli + rmj + rn\vec{k} \end{aligned}$$

Therefore

$$r^2 = |\vec{a}|^2 = (rl)^2 + (rm)^2 + (rn)^2 = r^2(l^2 + m^2 + n^2)$$

Hence

$$l^2 + m^2 + n^2 = 1$$

since $\vec{a} \neq \vec{0}$ and hence $r \neq 0$.

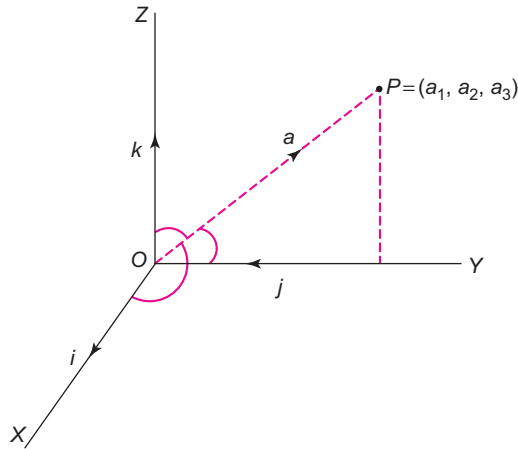


FIGURE 5.31 Theorem 5.25.

DEFINITION 5.27 Let (l, m, n) be direction cosines of a vector $\vec{a}$. Then any triad (r, s, t) of real numbers are said to be *direction ratios* if $(r, s, t) = c(l, m, n)$ for some positive real number c .

Note that, with respect to a given right-handed orthonormal unit vector triad $(\vec{i}, \vec{j}, \vec{k})$, any vector has exactly one triad of direction cosines, whereas it has infinitely many triads of direction ratios, one triad for each positive real number c .

QUICK LOOK 3

Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed orthonormal system of vectors and O the origin. Let P and Q be points such that $\vec{OP} = (a_1, a_2, a_3)$ and $\vec{OQ} = (b_1, b_2, b_3)$.

1. The direction cosines of $\vec{PQ}$ are

$$\left(\frac{b_1 - a_1}{PQ}, \frac{b_2 - a_2}{PQ}, \frac{b_3 - a_3}{PQ} \right)$$

2. The direction ratios of $\vec{PQ}$ are

$$(b_1 - a_1, b_2 - a_2, b_3 - a_3)$$

3. The direction cosines of $\vec{i}, \vec{j}$ and $\vec{k}$ are $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, respectively.

Note: Let us recall that for a 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the determinant is defined as

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

and that, for a 3×3 matrix, the determinant is defined as

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

In the following we obtain a necessary and sufficient condition for three vectors to be linearly independent in terms of the determinant of the matrix of their coordinates with respect to any orthonormal vector triad.

THEOREM 5.26 Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed orthonormal vector triad and $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$ and $\vec{c} = (c_1, c_2, c_3)$ be vectors. Then $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent if and only if

$$\det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = 0$$

PROOF We are given that

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

$$\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

Suppose that $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent. Then, there exist real numbers r, s, t , not all zero, such that

$$r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$$

that is

$$r(a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) + s(b_1\vec{i} + b_2\vec{j} + b_3\vec{k}) + t(c_1\vec{i} + c_2\vec{j} + c_3\vec{k}) = \vec{0}$$

Therefore

$$(ra_1 + sb_1 + tc_1)\vec{i} + (ra_2 + sb_2 + tc_2)\vec{j} + (ra_3 + sb_3 + tc_3)\vec{k} = \vec{0}$$

Since $\vec{i}, \vec{j}, \vec{k}$ are not coplanar and hence linearly independent, it follows that

$$ra_1 + sb_1 + tc_1 = 0 \quad (5.3)$$

$$ra_2 + sb_2 + tc_2 = 0 \quad (5.4)$$

$$ra_3 + sb_3 + tc_3 = 0 \quad (5.5)$$

Since one of r, s and t must be non-zero, we can assume, without loss of generality, that $r \neq 0$. By multiplying Eqs. (5.3)–(5.5) by

$$b_2c_3 - b_3c_2$$

$$b_3c_1 - b_1c_3$$

and

$$b_1c_2 - b_2c_1$$

respectively and by adding all the resultants we get that

$$\det A = 0$$

where

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

Since $r \neq 0$, $\det A = 0$. Now,

$$\det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \det A^T = \det A = 0$$

Conversely, suppose that the determinant is zero. Therefore, the homogeneous system of equations (Vol. 1, Chapter 8)

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

has a nonzero solution. Therefore

$$(a_1x + b_1y + c_1z)\vec{i} + (a_2x + b_2y + c_2z)\vec{j} + (a_3x + b_3y + c_3z)\vec{k} = \vec{0}$$

That is, $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$ where at least one of x, y, z is not zero. Hence $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent. ■

Example 5.6

Let

$$\vec{a} = 3\vec{i} + 2\vec{j} - 4\vec{k}$$

$$\vec{b} = 2\vec{i} - 3\vec{j} + \vec{k}$$

$$\vec{c} = 4\vec{i} - \vec{j} + 2\vec{k}$$

Then find the sum $\vec{a} + \vec{b} + \vec{c}$, its magnitude and direction cosines.

Solution:

$$\vec{a} + \vec{b} + \vec{c} = (3\vec{i} + 2\vec{j} - 4\vec{k}) + (2\vec{i} - 3\vec{j} + \vec{k}) + (4\vec{i} - \vec{j} + 2\vec{k})$$

$$= (3 + 2 + 4)\vec{i} + (2 - 3 - 1)\vec{j} + (-4 + 1 + 2)\vec{k}$$

$$= 9\vec{i} - 2\vec{j} - \vec{k}$$

The magnitude is given by

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{9^2 + (-2)^2 + (-1)^2} = \sqrt{86}$$

The direction cosines of $\vec{a} + \vec{b} + \vec{c}$ are

$$\left(\frac{9}{\sqrt{86}}, \frac{-2}{\sqrt{86}}, \frac{-1}{\sqrt{86}} \right)$$

Example 5.7

Let A and B be points whose position vectors are $2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{i} - 2\vec{j} + 2\vec{k}$, respectively. Find the position vector of the point C that divides AB in the ratio 3:2 internally.

Solution: We are given that

$$\vec{OA} = 2\vec{i} + 3\vec{j} - \vec{k}$$

and

$$\vec{OB} = \vec{i} - 2\vec{j} + 2\vec{k}$$

Then

$$\vec{OC} = \frac{2(2\vec{i} + 3\vec{j} - \vec{k}) + 3(\vec{i} - 2\vec{j} + 2\vec{k})}{2 + 3}$$

$$= \frac{(4 + 3)\vec{i} + (6 - 6)\vec{j} + (-2 + 6)\vec{k}}{5}$$

$$= \frac{7}{5}\vec{i} + \frac{4}{5}\vec{k}$$

Example 5.8

A boat is moving in a river. The velocity of the boat relative to water is represented by $3\vec{i} + 4\vec{j}$ and that of water relative to Earth is $\vec{i} - 3\vec{j}$, where $\vec{i}$ and $\vec{j}$ are unit vectors along North and East, respectively. Find the velocity and direction of the boat relative to the Earth.

Solution: If two particles P and Q are moving with velocities $\vec{a}$ and $\vec{b}$, respectively, then the relative velocity of P with respect to Q is $\vec{a} - \vec{b}$ and that of Q with respect to P is $\vec{b} - \vec{a}$. See Figure 5.32.

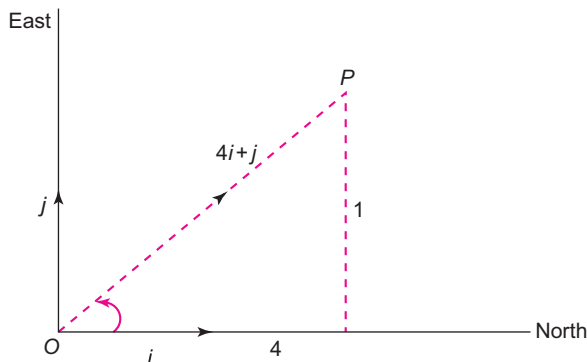


FIGURE 5.32 Example 5.8.

Let us assume that the velocity of Earth is $\vec{0}$.

The velocity of water relative to Earth is $\vec{i} - 3\vec{j}$.

The velocity of the boat relative to water is $3\vec{i} + 4\vec{j}$.

Therefore

$$3\vec{i} + 4\vec{j} = \text{Velocity of boat} - \text{Velocity of water}$$

$$\vec{i} - 3\vec{j} = \text{Velocity of water} - \text{Velocity of Earth}$$

Adding these two, we get

$$4\vec{i} + \vec{j} = \text{Velocity of boat} - \text{Velocity of Earth}$$

$$= \text{Velocity of boat relative to Earth}$$

Therefore, the speed of the boat is

$$|4\vec{i} + \vec{j}| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

and its direction of motion makes an angle

$$\theta = \tan^{-1}\left(\frac{1}{4}\right)$$

due East–North.

Example 5.9

Prove that the points with position vectors $2\vec{i} + \vec{j} + \vec{k}$, $6\vec{i} - \vec{j} + 2\vec{k}$ and $14\vec{i} - 5\vec{j} + 4\vec{k}$ are collinear.

Solution: Let O be the origin and P, Q and R be points such that

$$\overrightarrow{OP} = 2\vec{i} + \vec{j} + \vec{k}$$

$$\overrightarrow{OQ} = 6\vec{i} - \vec{j} + 2\vec{k}$$

and
$$\overrightarrow{OR} = 14\vec{i} - 5\vec{j} + 4\vec{k}$$

We know that P, Q, R are collinear if and only if $\overrightarrow{OP}, \overrightarrow{OQ}, \overrightarrow{OR}$ are linearly dependent. Now, consider

$$\begin{aligned} \det \begin{pmatrix} 2 & 1 & 1 \\ 6 & -1 & 2 \\ 14 & -5 & 4 \end{pmatrix} &= 2[(-1)4 - 2(-5)] - 1(6 \times 4 - 2 \times 14) + 1[6(-5) - (-1)14] \\ &= 2 \times 6 - (-4) + (-16) = 0 \end{aligned}$$

Therefore by Theorem 5.26, P, Q and R are collinear.

Example 5.10

Let P, Q, R and S be points with position vectors $3\vec{i} - 2\vec{j} - \vec{k}$, $2\vec{i} + 3\vec{j} - 4\vec{k}$, $-\vec{i} + \vec{j} + 2\vec{k}$ and $4\vec{i} + 5\vec{j} + r\vec{k}$. Find the value of r so that P, Q, R and S are coplanar.

Solution: Let O be the origin. We are given that

$$\overrightarrow{OP} = 3\vec{i} - 2\vec{j} - \vec{k}$$

$$\overrightarrow{OQ} = 2\vec{i} + 3\vec{j} - 4\vec{k}$$

$$\overrightarrow{OR} = -\vec{i} + \vec{j} + 2\vec{k}$$

$$\overrightarrow{OS} = 4\vec{i} + 5\vec{j} + r\vec{k}$$

Now,

$$\begin{aligned} \overrightarrow{PQ} &= \overrightarrow{OQ} - \overrightarrow{OP} \\ &= (2\vec{i} + 3\vec{j} - 4\vec{k}) - (3\vec{i} - 2\vec{j} - \vec{k}) \\ &= -\vec{i} + 5\vec{j} - 3\vec{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{PR} &= \overrightarrow{OR} - \overrightarrow{OP} \\ &= (-\vec{i} + \vec{j} + 2\vec{k}) - (3\vec{i} - 2\vec{j} - \vec{k}) \\ &= -4\vec{i} + 3\vec{j} + 3\vec{k} \end{aligned}$$

and

$$\begin{aligned} \overrightarrow{PS} &= \overrightarrow{OS} - \overrightarrow{OP} \\ &= (4\vec{i} + 5\vec{j} + r\vec{k}) - (3\vec{i} - 2\vec{j} - \vec{k}) \\ &= \vec{i} + 7\vec{j} + (r+1)\vec{k} \end{aligned}$$

Now P, Q, R, S are coplanar if and only if $\overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{PS}$ are coplanar. Equivalently,

$$\det \begin{pmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & r+1 \end{pmatrix} = 0$$

Now the given determinant equals

$$\begin{aligned} &-1[3(r+1) - 3 \times 7] - 5[-4(r+1) - 3 \times 1] - 3(-4 \times 7 - 3) \\ &= -3r + 18 + 20r + 35 + 93 \\ &= 17r + 146 \end{aligned}$$

Therefore P, Q, R, S are coplanar if and only if $17r + 146 = 0$. That is

$$r = \frac{-146}{17}$$

Example 5.11

Prove that the points $A(1, 1, 1)$, $B(1, 2, 3)$ and $C(2, -1, 1)$ are the vertices of an isosceles triangle and find the direction cosines of $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{CA}$.

Solution: We are given that

$$\overrightarrow{OA} = \vec{i} + \vec{j} + \vec{k}$$

$$\overrightarrow{OB} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$\overrightarrow{OC} = 2\vec{i} - \vec{j} + \vec{k}$$

Therefore

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{j} + 2\vec{k}$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \vec{i} - 3\vec{j} - 2\vec{k}$$

and

$$\overrightarrow{CA} = \overrightarrow{OA} - \overrightarrow{OC} = -\vec{i} + 2\vec{j}$$

Now,

$$AB = |\overrightarrow{AB}| = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}$$

$$BC = |\overline{BC}| = \sqrt{1^2 + (-3)^2 + (-2)^2} = \sqrt{14}$$

$$CA = |\overline{CA}| = \sqrt{(-1)^2 + 2^2 + 0^2} = \sqrt{5}$$

Since $AB = CA$, the triangle ABC is an isosceles triangle.

$$\text{Direction cosines of } \overline{AB} = \left(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

$$\text{Direction cosines of } \overline{BC} = \left(\frac{1}{\sqrt{14}}, \frac{-3}{\sqrt{14}}, \frac{-2}{\sqrt{14}}\right)$$

$$\text{Direction cosines of } \overline{CA} = \left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right)$$

5.6 | Vector Equation of a Line and a Plane

We shall derive vector equation of a straight line and a plane in different forms using the results on collinearity and coplanarity proved earlier.

THEOREM 5.27

Let A be a point with position vector $\vec{a}$ and let $\vec{b}$ be a given vector. Then the vector equation of the straight line parallel to $\vec{b}$ and passing through A is

$$\vec{x} = \vec{a} + r\vec{b}, r \in \mathbb{R}$$

PROOF

Let O be the origin. Then, we are given that $\overline{OA} = \vec{a}$. Let L be the straight line parallel to $\vec{b}$ and passing through A . Let P be arbitrary point on L and $\overline{OP} = \vec{x}$. Since $\overline{AP}$ is parallel to $\vec{b}$, we get that

$$\overline{AP} = r\vec{b}$$

for some real number r . Now,

$$\vec{x} = \overline{OP} = \overline{OA} + \overline{AP} = \vec{a} + r\vec{b}$$

Conversely, let P be a point such that

$$\overline{OP} = \vec{x} = \vec{a} + r\vec{b}$$

where r is a real number. Then

$$\overline{AP} = \overline{OP} - \overline{OA} = (\vec{a} + r\vec{b}) - \vec{a} = r\vec{b}$$

and hence $\overline{AP}$ is parallel to $\vec{b}$, and so AP is parallel to L . Since A lies on L , P also lies on L . Thus P lies on L if and only if

$$\overline{OP} = \vec{a} + r\vec{b}, r \in \mathbb{R}$$

COROLLARY 5.9

The vector equation of the straight line parallel to a given vector $\vec{b}$ and passing through the origin is

$$\vec{x} = r\vec{b}, r \in \mathbb{R}$$

THEOREM 5.28 (CARTESIAN FORM OF A LINE)

Let (a_1, a_2, a_3) be the coordinates of a point A with reference to a right-handed rectangular Cartesian coordinate system $OXYZ$ and let (r_1, r_2, r_3) be a triad of non-zero real numbers. Then the Cartesian equation of the straight line passing through A and with direction ratios (r_1, r_2, r_3) is

$$\frac{x - a_1}{r_1} = \frac{y - a_2}{r_2} = \frac{z - a_3}{r_3} = t, t \in \mathbb{R}$$

PROOF Let $\vec{i}, \vec{j}, \vec{k}$ be the unit vectors along the positive X, Y, Z axes, respectively. Let L be the straight line passing through A and with direction ratios (r_1, r_2, r_3) . The position vector of A is $a_1\vec{i} + a_2\vec{j} + a_3\vec{k} = \vec{a}$, say. L is parallel to the vector

$$\vec{b} = r_1\vec{i} + r_2\vec{j} + r_3\vec{k}$$

Let $P(x, y, z)$ be an arbitrary point in the space. Then

$$P \text{ lies on } L \Leftrightarrow \overrightarrow{OP} = \vec{a} + t\vec{b} \quad \text{for some real number } t$$

$$\Leftrightarrow x\vec{i} + y\vec{j} + z\vec{k} = (a_1 + tr_1)\vec{i} + (a_2 + tr_2)\vec{j} + (a_3 + tr_3)\vec{k}$$

$$\Leftrightarrow x = a_1 + tr_1, y = a_2 + tr_2, z = a_3 + tr_3, t \in \mathbb{R}$$

$$\Leftrightarrow \frac{x - a_1}{r_1} = \frac{y - a_2}{r_2} = \frac{z - a_3}{r_3} = t, t \in \mathbb{R}$$

THEOREM 5.29 Let A and B be two distinct points with position vectors $\vec{a}$ and $\vec{b}$, respectively. The vector equation of the straight line passing through A and B is

$$\vec{x} = (1 - r)\vec{a} + r\vec{b}, r \in \mathbb{R}$$

PROOF Let O be the origin. We are given that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$ and

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{b} - \vec{a} \neq \vec{0}$$

Let L be the straight line passing through A and B . Then L is parallel to the vector $\overrightarrow{AB}$ and passing through A . By Theorem 5.27, the vector equation of L is

$$\begin{aligned} \vec{x} &= \vec{a} + r\overrightarrow{AB} \\ &= \vec{a} + r(\vec{b} - \vec{a}) \end{aligned}$$

that is

$$\vec{x} = (1 - r)\vec{a} + r\vec{b}, r \in \mathbb{R}$$

COROLLARY 5.10 Let $A(a_1, a_2, a_3)$ and $B(b_1, b_2, b_3)$ be two distinct points in the space. Then the Cartesian equation of the straight line passing through A and B is

$$\frac{x - a_1}{b_1 - a_1} = \frac{y - a_2}{b_2 - a_2} = \frac{z - a_3}{b_3 - a_3} = t, t \in \mathbb{R}$$

PROOF Let $OXYZ$ be a right-handed rectangular Cartesian coordinate system and $\vec{i}, \vec{j}$ and $\vec{k}$ be the unit vectors along the positive X, Y and Z axes, respectively. Let L be the straight line passing through A and B . Let $P(x, y, z)$ be an arbitrary point in the space. Then

$$\overrightarrow{OA} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k} = \vec{a}, \quad \text{say}$$

$$\overrightarrow{OB} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k} = \vec{b}, \quad \text{say}$$

and

$$\overrightarrow{OP} = x\vec{i} + y\vec{j} + z\vec{k}$$

Now

$$P \text{ lies on } L \Leftrightarrow x\vec{i} + y\vec{j} + z\vec{k} = \overrightarrow{OP}$$

$$= (1 - r)\vec{a} + r\vec{b}, r \in \mathbb{R} \quad (\text{by Theorem 5.29})$$

$$\Leftrightarrow (x - a_1)\vec{i} + (y - a_2)\vec{j} + (z - a_3)\vec{k} = r[(b_1 - a_1)\vec{i} + (b_2 - a_2)\vec{j} + (b_3 - a_3)\vec{k}]$$

$$\Leftrightarrow x - a_1 = r(b_1 - a_1), y - a_2 = r(b_2 - a_2), z - a_3 = r(b_3 - a_3)$$

$$\Leftrightarrow \frac{x - a_1}{b_1 - a_1} = \frac{y - a_2}{b_2 - a_2} = \frac{z - a_3}{b_3 - a_3} = r, r \in \mathbb{R}$$

Example 5.12

Find the vector equation and the Cartesian equation of the straight line passing through $2\vec{i} + \vec{j} + 3\vec{k}$ and $-4\vec{i} + 3\vec{j} - \vec{k}$.

Solution: From Theorem 5.29, the vector equation of the straight line passing through $\vec{a} = 2\vec{i} + \vec{j} + 3\vec{k}$ and $\vec{b} = -4\vec{i} + 3\vec{j} - \vec{k}$ is

$$\vec{x} = (1 - r)\vec{a} + r\vec{b}, r \in \mathbb{R}$$

That is

$$\vec{x} = (2 - 2r - 4r)\vec{i} + (1 - r + 3r)\vec{j} + (3 - 3r - r)\vec{k}$$

$$\vec{x} = (2 - 6r)\vec{i} + (1 + 2r)\vec{j} + (3 - 4r)\vec{k}, r \in \mathbb{R}$$

Let A and B be the points such that

$$\vec{OA} = 2\vec{i} + \vec{j} + 3\vec{k}$$

and

$$\vec{OB} = -4\vec{i} + 3\vec{j} - \vec{k}$$

The Cartesian coordinates of A and B are $(2, 1, 3)$ and $(-4, 3, -1)$, respectively. From Corollary 5.10, the Cartesian equation of the line joining A and B is

$$\frac{x - 2}{-4 - 2} = \frac{y - 1}{3 - 1} = \frac{z - 3}{-1 - 3} = r, r \in \mathbb{R}$$

$$\frac{x - 2}{-6} = \frac{y - 1}{2} = \frac{z - 3}{-4}$$

$$\frac{x - 2}{3} = y - 1 = \frac{z - 3}{-2}$$

Example 5.13

Does the straight line passing through the points $2\vec{a} + 3\vec{b} - \vec{c}$ and $3\vec{a} + 4\vec{b} - 2\vec{c}$ intersect the straight line passing through the points $\vec{a} - 2\vec{b} + 3\vec{c}$ and $\vec{a} - 6\vec{b} + 6\vec{c}$. If so, find the point of intersection. (Here $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors.)

Solution: Let L_1 and L_2 be the first and second straight lines given. Then the equations of L_1 and L_2 are, respectively,

$$\begin{aligned} \vec{x} &= (1 - r)(2\vec{a} + 3\vec{b} - \vec{c}) + r(3\vec{a} + 4\vec{b} - 2\vec{c}) \\ &= (2 + r)\vec{a} + (3 + r)\vec{b} - (1 + r)\vec{c}, r \in \mathbb{R} \end{aligned} \quad (5.6)$$

$$\text{and } \vec{x} = (1 - s)(\vec{a} - 2\vec{b} + 3\vec{c}) + s(\vec{a} - 6\vec{b} + 6\vec{c})$$

$$= \vec{a} + (-2 - 4s)\vec{b} + (3 + 3s)\vec{c}, r \in \mathbb{R} \quad (5.7)$$

For L_1 and L_2 to have a common point, we have to find r and s such that the corresponding coefficients in Eqs. (5.6) and (5.7) of L_1 and L_2 are equal, that is

$$2 + r = 1,$$

$$3 + r = -2 - 4s$$

and

$$-1 - r = 3 + 3s$$

These give $r = -1$ and $s = -1$. Also, the point of intersection is $\vec{a} + 2\vec{b}$.

Example 5.14

Using vector methods prove that

$$\frac{x}{a} + \frac{y}{b} = 1$$

is the equation of a straight line in intercept form (where the coordinate axes may be perpendicular or oblique).

Solution: Let $\vec{OX}, \vec{OY}, \vec{OZ}$ be any non-coplanar vectors. Let $\vec{i}$ and $\vec{j}$ be the unit vectors along $\vec{OX}$ and $\vec{OY}$,

respectively. Let A and B be points on OX and OY , respectively, and

$$\vec{OA} = a\vec{i} \quad \text{and} \quad \vec{OB} = b\vec{j}$$

Then

$$\vec{i} = \frac{\vec{OA}}{a} \quad \text{and} \quad \vec{j} = \frac{\vec{OB}}{b}$$

Let $P(x, y)$ be an arbitrary point on the straight line $\overline{AB}$. Choose a point Q on $\overline{OA}$ such that QP is parallel to OB (Figure 5.33). Then

$$\overline{OQ} = x\vec{i} = \frac{x}{a}\overline{OA}$$

and

$$\overline{QP} = y\vec{j} = \frac{y}{b}\overline{OB}$$

Also

$$\overline{OP} = \overline{OQ} + \overline{QP} = \frac{x}{a}\overline{OA} + \frac{y}{b}\overline{OB}$$

Since P is a point on the line $\overline{AB}$,

$$\overline{OP} = (1-r)\overline{OA} + r\overline{OB}$$

for some $r \in \mathbb{R}$. Thus

$$\frac{x}{a}\overline{OA} + \frac{y}{b}\overline{OB} = (1-r)\overline{OA} + r\overline{OB}$$

Since $\overline{OA}$ and $\overline{OB}$ are not collinear and hence linearly independent, we get that

$$\frac{x}{a} = 1-r \quad \text{and} \quad \frac{y}{b} = r$$

From these two, we get that

$$\frac{x}{a} + \frac{y}{b} = 1$$

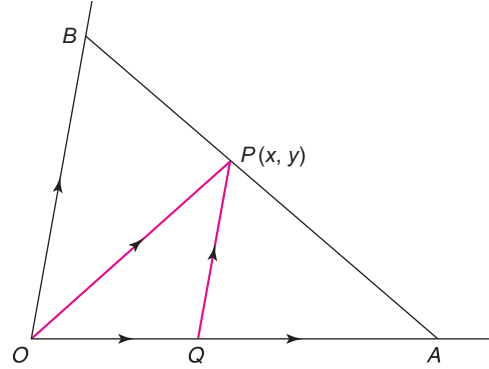


FIGURE 5.33 Example 5.14.

THEOREM 5.30

Let A be a point in the space with position vector $\vec{a}$ with respect to the origin O . Let $\vec{b}$ and $\vec{c}$ be non-collinear vectors. Then the vector equation of the plane through the point A and parallel to $\vec{b}$ and $\vec{c}$ is

$$\vec{x} = \vec{a} + r\vec{b} + s\vec{c}, \quad r \text{ and } s \in \mathbb{R}$$

PROOF

We are given that $\overline{OA} = \vec{a}$. Choose points B and C such that $\overline{AB} = \vec{b}$ and $\overline{AC} = \vec{c}$. The points A , B and C are not collinear, since $\overline{AB}$ and $\overline{AC}$ are not collinear (Figure 5.34). Therefore, there exists a unique plane passing through A , B and C . Let that plane be Π . Then Π is the plane passing through the point A and parallel to the vectors $\vec{b}$ and $\vec{c}$.

Let P be any point with position vector $\vec{x}$ with respect to the origin O ; that is,

$$\overline{OP} = \vec{x}$$

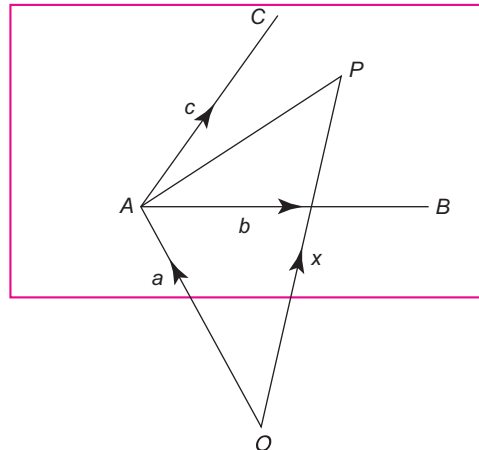


FIGURE 5.34 Theorem 5.30.

Since P lies on the plane Π , then by Theorem 5.15, there exist real numbers r and s such that

$$\overrightarrow{AP} = r\overrightarrow{AB} + s\overrightarrow{AC} = r\vec{b} + s\vec{c}$$

and hence

$$\vec{x} = \overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = \vec{a} + r\vec{b} + s\vec{c}$$

Conversely, if P is a point and r and s are real number such that $\overrightarrow{OP} = \vec{a} + r\vec{b} + s\vec{c}$, then

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \overrightarrow{OP} - \vec{a} = r\vec{b} + s\vec{c}$$

and therefore, again by Theorem 5.15, P lies on the plane determined by A, B and C . Thus P lies on Π if and only if $\overrightarrow{OP} = \vec{a} + r\vec{b} + s\vec{c}$ for some real numbers r and s . ■

COROLLARY 5.11

The vector equation of the plane passing through origin and parallel to the vectors $\vec{b}$ and $\vec{c}$ is

$$\vec{x} = r\vec{b} + s\vec{c}, \quad r, s \in \mathbb{R}$$

THEOREM 5.31

Let A, B and C be three non-collinear points with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$, respectively, relative to the origin O . Then the vector equation of the plane passing through A, B and C is

$$\vec{x} = (1 - r - s)\vec{a} + r\vec{b} + s\vec{c}, \quad r \text{ and } s \in \mathbb{R}$$

PROOF

We are given that $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$. Let Π be the plane passing through A, B and C . Then Π is parallel to $\overrightarrow{AB}$ and $\overrightarrow{AC}$ and is passing through A (Figure 5.35). Therefore, by Theorem 5.30, the vector equation of the plane Π is

$$\begin{aligned} \vec{x} &= \vec{a} + r\overrightarrow{AB} + s\overrightarrow{AC}, \quad r, s \in \mathbb{R} \\ &= \vec{a} + r(\overrightarrow{OB} - \overrightarrow{OA}) + s(\overrightarrow{OC} - \overrightarrow{OA}) \\ &= \vec{a} + r(\vec{b} - \vec{a}) + s(\vec{c} - \vec{a}) \\ &= (1 - r - s)\vec{a} + r\vec{b} + s\vec{c}, \quad r \text{ and } s \in \mathbb{R} \end{aligned}$$

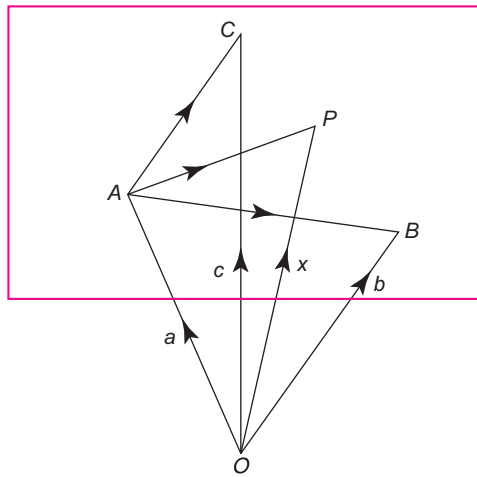


FIGURE 5.35 Theorem 5.31. ■

COROLLARY 5.12 Let A, B and C be non-collinear points with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$, respectively, relative to the origin O . Then the vector equation of the plane passing through A and B parallel to $\vec{c}$ is

$$\vec{x} = (1-r)\vec{a} + r\vec{b} + s\vec{c}, r \text{ and } s \in \mathbb{R}$$

PROOF We have $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$. Also,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{b} - \vec{a}$$

Let Π be the plane passing through A and B and parallel to $\vec{c}$. Then Π is the plane passing through A and parallel to $\overrightarrow{AB}$ and $\vec{c}$. By the Theorem 5.30 the vector equation of Π is

$$\begin{aligned}\vec{x} &= \vec{a} + r\overrightarrow{AB} + s\vec{c}, r, s \in \mathbb{R} \\ &= \vec{a} + r(\vec{b} - \vec{a}) + s\vec{c}\end{aligned}$$

that is

$$\vec{x} = (1-r)\vec{a} + r\vec{b} + s\vec{c}, r \text{ and } s \in \mathbb{R}$$

Example 5.15

Find the vector equation of the plane passing through the points $(1, -2, 5)$, $(0, -5, -1)$ and $(-3, 5, 0)$.

Solution: Let $A = (1, -2, 5)$, $B = (0, -5, -1)$ and $C = (-3, 5, 0)$. Let the position vectors of A, B and C relative to the origin be $\vec{a}, \vec{b}$ and $\vec{c}$, respectively. Then

$$\overrightarrow{OA} = \vec{a} = \vec{i} - 2\vec{j} + 5\vec{k}$$

$$\overrightarrow{OB} = \vec{b} = -5\vec{j} - \vec{k}$$

and

$$\overrightarrow{OC} = \vec{c} = -3\vec{i} + 5\vec{j}$$

By Theorem 5.31, the vector equation of the plane passing through A, B and C is

$$\begin{aligned}\vec{x} &= (1-r-s)\vec{a} + r\vec{b} + s\vec{c}, r, s \in \mathbb{R} \\ &= (1-r-s)(\vec{i} - 2\vec{j} + 5\vec{k}) + r(-5\vec{j} - \vec{k}) + s(-3\vec{i} + 5\vec{j})\end{aligned}$$

that is

$$\vec{x} = (1-r-4s)\vec{i} + (-2-3r+7s)\vec{j} + (5-6r-5s)\vec{k}$$

THEOREM 5.32

Four points A, B, C and D with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$, respectively, are coplanar if and only if there exist real numbers r, s, t and u , not all zero, such that

$$r + s + t + u = 0$$

and

$$r\vec{a} + s\vec{b} + t\vec{c} + u\vec{d} = \vec{0}$$

PROOF

Suppose that there exist real numbers r, s, t and u , not all zero, such that

$$r + s + t + u = 0$$

and

$$r\vec{a} + s\vec{b} + t\vec{c} + u\vec{d} = \vec{0}$$

Without loss of generality, we can assume that $r \neq 0$. Then

$$s + t + u = -r \neq 0$$

and

$$-(s + t + u)\vec{a} + s\vec{b} + t\vec{c} + u\vec{d} = \vec{0}$$

Therefore

$$s(\vec{b} - \vec{a}) + t(\vec{c} - \vec{a}) + u(\vec{d} - \vec{a}) = \vec{0}$$

and hence

$$s\overrightarrow{AB} + t\overrightarrow{AC} + u\overrightarrow{AD} = \vec{0}$$

and one of s, t, u is not zero (since $r + s + t + u = 0$ and $r \neq 0$). Therefore, $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar vectors and hence the points A, B, C and D are coplanar.

Conversely, suppose that the points A, B, C and D are coplanar. Then the vectors $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar and hence there exist real numbers α and β such that

$$\overrightarrow{AD} = \alpha \overrightarrow{AB} + \beta \overrightarrow{AC}$$

that is

$$\vec{d} - \vec{a} = \alpha(\vec{b} - \vec{a}) + \beta(\vec{c} - \vec{a})$$

Therefore

$$(1 - \alpha - \beta)\vec{a} + \alpha\vec{b} + \beta\vec{c} + (-1)\vec{d} = \vec{0}$$

By taking $r = 1 - \alpha - \beta, s = \alpha, t = \beta$ and $u = -1$, we get that

$$r + s + t + u = 0$$

and

$$r\vec{a} + s\vec{b} + t\vec{c} + u\vec{d} = \vec{0}$$



Finally we conclude this chapter by obtaining the equation of angle bisectors of the angle between two lines.

THEOREM 5.33

Equation of the angle bisectors of the lines whose equations are $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{a} + s\vec{c}$ is

$$\vec{r} = \vec{a} + \lambda \left[\frac{\pm \vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right]$$

where t, s and λ are scalar parameters.

PROOF

The given lines are intersecting in the point A with position vector $\vec{a}$ and parallel to the vectors $\vec{b}$ and $\vec{c}$ respectively. Let P be a point with position vector $\vec{r}$ on the bisector of angle LOM (see Figure 5.36). Through P draw a line parallel to $\vec{c}$ meeting the line AL in Q . Therefore

$$\angle QAP = \angle APQ \Rightarrow AQ = PQ$$

Hence

$$\overrightarrow{AQ} = \lambda \left(\frac{\vec{b}}{|\vec{b}|} \right) \quad \text{and} \quad \overrightarrow{QP} = \lambda \left(\frac{\vec{c}}{|\vec{c}|} \right)$$

Therefore

$$\begin{aligned} \overrightarrow{AP} &= \overrightarrow{AQ} + \overrightarrow{QP} \\ \Rightarrow \vec{r} - \vec{a} &= \lambda \left(\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right) \\ \Rightarrow \vec{r} &= \vec{a} + \lambda \left(\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right) \end{aligned}$$

Similarly, the other angle bisector can be seen as

$$\vec{r} = \vec{a} + \lambda \left(\frac{-\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right)$$

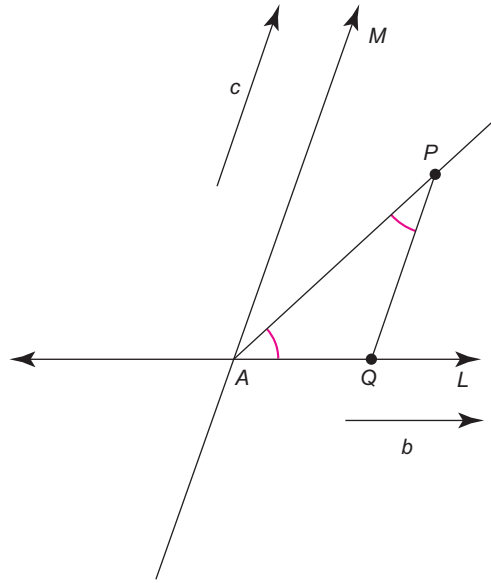


FIGURE 5.36 Theorem 5.33.

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. Let $\vec{a}$ and $\vec{b}$ be non-collinear vectors. If the vectors $(\lambda - 1)\vec{a} + 2\vec{b}$ and $3\vec{a} + \lambda\vec{b}$ are collinear vectors, then the value of λ is
- (A) 2 or 3 (B) -2 or 3
(C) -2 or -3 (D) 2 or -3

Solution: Since the vectors $(\lambda - 1)\vec{a} + 2\vec{b}$ and $3\vec{a} + \lambda\vec{b}$ are collinear vectors, there exists scalar x such that

$$\begin{aligned} 3\vec{a} + \lambda\vec{b} &= x[(\lambda - 1)\vec{a} + 2\vec{b}] \\ &= x(\lambda - 1)\vec{a} + 2x\vec{b} \end{aligned} \quad (5.8)$$

Since $\vec{a}$ and $\vec{b}$ are non-collinear vectors, the corresponding scalar coefficients on both sides of Eq. (5.8) must be equal. Therefore

$$x(\lambda - 1) = 3 \quad \text{and} \quad 2x = \lambda$$

On solving the equations for x we get

$$\frac{\lambda}{2}(\lambda - 1) = 3$$

and hence $\lambda = -2$ or 3.

Answer: (B)

2. If $ABCDE$ is a pentagon (see Figure 5.37), then $\vec{AB} + \vec{AE} + \vec{BC} + \vec{DC} + \vec{ED} + \vec{AC}$ is equal to
- (A) $6\vec{AC}$ (B) $5\vec{AC}$ (C) $4\vec{AC}$ (D) $3\vec{AC}$

Solution: We have

$$\begin{aligned} \vec{AB} + \vec{AE} + \vec{BC} + \vec{DC} + \vec{ED} + \vec{AC} \\ &= (\vec{AB} + \vec{BC}) + (\vec{AE} + \vec{ED} + \vec{DC}) + \vec{AC} \\ &= \vec{AC} + \vec{AC} + \vec{AC} \\ &= 3\vec{AC} \end{aligned}$$

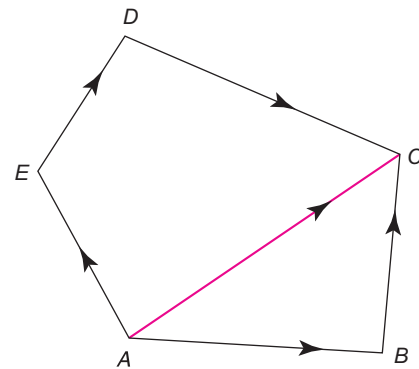


FIGURE 5.37 Single correct choice type question 2.

Answer: (D)

3. In $\triangle ABC$, let O and H denote the circumcentre and orthocentre, respectively. Then $\vec{OA} + \vec{OB} + \vec{OC} =$
- (A) $\vec{OH}$ (B) $2\vec{OH}$ (C) $\frac{2}{3}\vec{OH}$ (D) $3\vec{OH}$

Solution: In Chapter 4 we have proved that

$$AO = 2R \cos A$$

and

$$OD = R \cos B$$

where D is the mid-point of the side BC (Figure 5.38).

Now

$$\begin{aligned}\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} &= \overrightarrow{OA} + 2\overrightarrow{OD} \\ &= \overrightarrow{OA} + \overrightarrow{AH} \\ &= \overrightarrow{AH}\end{aligned}$$

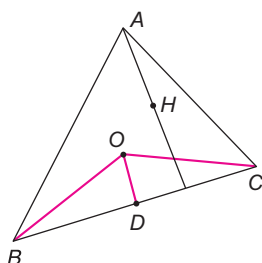


FIGURE 5.38 Single correct choice type question 3.

Answer: (A)

Note: With circumcentre as origin of reference, the position vector of the orthocentre is sum of the position vectors of the vertices of the triangle. Equivalently, in the Argand's plane (complex numbers), if the circumcentre of a triangle is the origin of the coordinate axes, then the complex number representing the orthocentre is the sum of the complex numbers representing the vertices.

4. $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar vectors. If $\vec{a} + \vec{b} + \vec{c} = x\vec{d}$ and $\vec{b} + \vec{c} + \vec{d} = y\vec{a}$, where x , y are scalars, then

- (A) $x = y = 1$
 (B) $x = y = -1$
 (C) $x = 1, y = -1$
 (D) $\vec{a} + \vec{b} + \vec{c} + \vec{d} = (x + y)(\vec{a} + \vec{d})$

Solution: From the given equations we get

$$\vec{a} + \vec{b} + \vec{c} = x\vec{d} = x(y\vec{a} - \vec{b} - \vec{c})$$

Equating the corresponding coefficients we get

$$xy = 1, x = -1, y = -1$$

Answer: (B)

5. D , E and F are the mid-points of the sides AB , AC and BC , respectively. Then the vector $\overrightarrow{BE} + \overrightarrow{AF}$ equals

- (A) $\frac{3}{2}\overrightarrow{BF}$ (B) $\frac{1}{2}\overrightarrow{BF}$ (C) $2\overrightarrow{BF}$ (D) $\overrightarrow{DC}$

Solution: Take A as origin (Figure 5.39) and let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AC} = \vec{c}$. Therefore

$$\overrightarrow{AD} = \frac{1}{2}\vec{b}$$

$$\overrightarrow{AE} = \frac{1}{2}\vec{c}$$

and

$$\overrightarrow{AF} = \frac{\vec{b} + \vec{c}}{2}$$

Now

$$\begin{aligned}\overrightarrow{BE} + \overrightarrow{AF} &= (\overrightarrow{BA} + \overrightarrow{AE}) + \overrightarrow{AF} \\ &= \left(-\vec{b} + \frac{\vec{c}}{2}\right) + \frac{\vec{b} + \vec{c}}{2} \\ &= \vec{c} - \frac{\vec{b}}{2} \\ &= \overrightarrow{DC}\end{aligned}$$

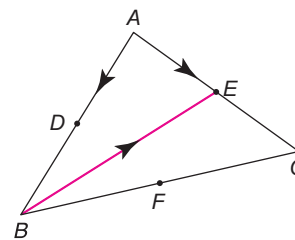


FIGURE 5.39 Single correct choice type question 5.

Answer: (D)

6. If $\vec{a} = (m, -2, 5)$ and $\vec{b} = (1, n, -3)$ are collinear vectors, then

- (A) $m = \frac{5}{3}, n = \frac{-6}{5}$ (B) $m = \frac{-5}{3}, n = \frac{-6}{5}$
 (C) $m = \frac{-5}{3}, n = \frac{6}{5}$ (D) $m = \frac{5}{3}, n = \frac{6}{5}$

Solution: Since the vectors $\vec{a}$ and $\vec{b}$ are collinear vectors, there exists a scalar x such that $\vec{b} = x\vec{a}$. Therefore

$$(1, n, -3) = x(m, -2, 5)$$

Hence

$$mx = 1, -2x = n \quad \text{and} \quad 5x = -3$$

Solving these we get

$$x = \frac{-3}{5}, m = \frac{-5}{3}, n = \frac{6}{5}$$

Answer: (C)

7. Points D , E and F divide the sides BC , CA and AB internally in the ratio $1:4$, $3:2$ and $3:7$. If the point P divides the side AB in the ratio $1:3$, then $\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF}$ is equal to

- (A) $\frac{2}{3}\overline{CP}$ (B) $\frac{3}{5}\overline{CP}$ (C) $\frac{4}{5}\overline{CP}$ (D) $\frac{2}{5}\overline{CP}$

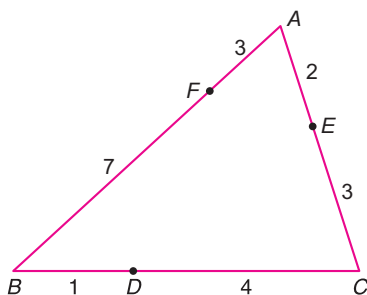


FIGURE 5.40 Single correct choice type question 7.

Solution: Take A as origin and let $\overline{AB} = \vec{b}$, $\overline{AC} = \vec{c}$. Therefore

$$\overline{AD} = \frac{\vec{c} + 4\vec{b}}{5}$$

$$\overline{AE} = \frac{2}{5}\vec{c}$$

$$\overline{AF} = \frac{3}{10}\vec{b}$$

$$\overline{AP} = \frac{\vec{b}}{4}$$

Now

$$\begin{aligned}\overline{AD} + \overline{BE} + \overline{CF} &= \frac{\vec{c} + 4\vec{b}}{5} + \left(\frac{2}{5}\vec{c} - \vec{b}\right) + \left(\frac{3}{10}\vec{b} - \vec{c}\right) \\ &= \frac{1}{10}(-4\vec{c} + \vec{b}) \\ &= \frac{1}{10}(\vec{b} - 4\vec{c})\end{aligned}$$

But

$$\begin{aligned}\overline{CP} &= \frac{\vec{b}}{4} - \vec{c} = \frac{1}{4}(\vec{b} - 4\vec{c}) \\ &= \frac{10}{4} \left(\frac{\vec{b} - 4\vec{c}}{10} \right) \\ &= \frac{5}{2}(\overline{AD} + \overline{BE} + \overline{CF})\end{aligned}$$

Therefore

$$\overline{AD} + \overline{BE} + \overline{CF} = \frac{2}{5}\overline{CP}$$

Answer: (D)

8. Let $\vec{a} = \vec{i} + \vec{j} - \vec{k}$, $\vec{b} = 5\vec{i} - 3\vec{j} - 3\vec{k}$ and $\vec{c} = 3\vec{i} - \vec{j} + 2\vec{k}$. If $\vec{r}$ is a vector in the direction of $\vec{c}$ with magnitude $|\vec{a} + \vec{b}|$, then $\vec{r}$ is equal to

- (A) $3\vec{c}$ (B) $2\vec{c}$ (C) $\frac{3}{2}\vec{c}$ (D) $\frac{2}{3}\vec{c}$

Solution: We have

$$\begin{aligned}\vec{a} + \vec{b} &= 6\vec{i} - 2\vec{j} - 4\vec{k} \\ \Rightarrow |\vec{a} + \vec{b}| &= \sqrt{6^2 + 2^2 + 4^2} = \sqrt{56} = 2\sqrt{14}\end{aligned}$$

Also

$$\text{Unit vector in the direction of } \vec{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{\vec{c}}{\sqrt{14}}$$

Therefore

$$\vec{r} = |\vec{a} + \vec{b}| \left(\frac{\vec{c}}{|\vec{c}|} \right) = 2\sqrt{14} \left(\frac{\vec{c}}{\sqrt{14}} \right) = 2\vec{c}$$

Answer: (B)

9. The position vectors of three points are $2\vec{a} - \vec{b} + 3\vec{c}$, $\vec{a} - 2\vec{b} + m\vec{c}$ and $n\vec{a} - 5\vec{b}$ where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and m, n are scalars. The three points are collinear if and only if

- (A) $m = -2, n = \frac{9}{4}$ (B) $m = \frac{9}{4}, n = -2$
(C) $m = 2, n = \frac{9}{4}$ (D) $m = -\frac{9}{4}, n = 2$

Solution: Let the points be P, Q and R , respectively. Then P, Q, R are collinear if and only if $\overline{PQ} = x\overline{QR}$ for some scalar x . This implies and is implied by

$$\begin{aligned}-\vec{a} - \vec{b} + (m - 3)\vec{c} &= x[(n - 1)\vec{a} - 3\vec{b} - m\vec{c}] \\ \Leftrightarrow x(n - 1) &= -1, -3x = -1 \quad \text{and} \quad m - 3 = -xm \\ \Leftrightarrow x &= \frac{1}{3}, m = \frac{9}{4} \quad \text{and} \quad n = -2\end{aligned}$$

Answer: (B)

10. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar vectors. Suppose A, B, C and D are four points with position vectors $-\vec{a} + 4\vec{b} - 3\vec{c}$, $3\vec{a} + 2\vec{b} - 5\vec{c}$, $-3\vec{a} + 8\vec{b} - 5\vec{c}$ and $-3\vec{a} + 2\vec{b} + \vec{c}$. If $\overline{AB} = x\overline{AC} + y\overline{AD}$, then

- (A) $x = 1, y = 1$ (B) $x = 1, y = -1$
(C) $x = -1, y = 1$ (D) $x = -1, y = -1$

Solution: We have

$$\begin{aligned}\overline{AB} &= x\overline{AC} + y\overline{AD} \\ \Rightarrow 4\vec{a} - 2\vec{b} - 2\vec{c} &= x(-2\vec{a} + 4\vec{b} - 2\vec{c}) + y(-2\vec{a} - 2\vec{b} + 4\vec{c})\end{aligned}$$

Equating the corresponding coefficients of both sides we get

$$-2x - 2y = +4 \quad (5.9)$$

$$4x - 2y = -2 \quad (5.10)$$

$$-2x + 4y = -2 \quad (5.11)$$

Solving Eqs. (5.9) and (5.10), we get that $x = -1$ and $y = -1$ which also satisfy Eq. (5.11).

Answer: (D)

11. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors which are the position vectors of three non-collinear points A, B and C , respectively. Points D and E divide the sides BC and BA in the ratios $5:2$ and $2:1$. Let the lines AD and CE meet in P . If O is the origin of reference, then $4\vec{a} + 2\vec{b} + 5\vec{c} = \lambda(\vec{OP})$, where λ equals

(A) 11 (B) 7 (C) 3 (D) 4

Solution: By hypothesis,

$$\vec{OD} = \frac{2\vec{b} + 5\vec{c}}{7} \quad \text{and} \quad \vec{OE} = \frac{2\vec{a} + \vec{b}}{3}$$

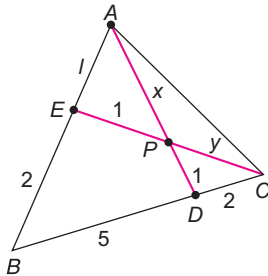


FIGURE 5.41 Single correct choice type question 11.

Suppose $AP:PD = x:1$ and $CP:PE = y:1$. Therefore

$$\frac{x[(2\vec{b} + 5\vec{c})/7] + \vec{a}}{x+1} = \vec{OP} = \frac{y[(2\vec{a} + \vec{b})/3] + \vec{c}}{y+1}$$

Since $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, in the above equation the corresponding coefficients on both sides must be equal. So

$$\frac{1}{x+1} = \frac{2y}{3(y+1)} \quad (5.12)$$

$$\frac{2x}{7(x+1)} = \frac{y}{3(y+1)} \quad (5.13)$$

$$\frac{5x}{7(x+1)} = \frac{1}{y+1} \quad (5.14)$$

From Eqs. (5.12) and (5.13) we get

$$x = \frac{7}{4} \quad \text{and} \quad y = \frac{6}{5}$$

whose values also satisfy Eq. (5.14). Therefore

$$\vec{OP} = \frac{4\vec{a} + 2\vec{b} + 5\vec{c}}{11}$$

$$\lambda = 11$$

Answer: (A)

12. In $\triangle ABC$, E is the mid-point of AB and D is a point on the side BC such that $BD:DC = 2:1$. The lines AD and CE intersect in Q . Then the ratio $AQ:QD$ is
(A) 2:1 (B) 2:3 (C) 3:1 (D) 3:2

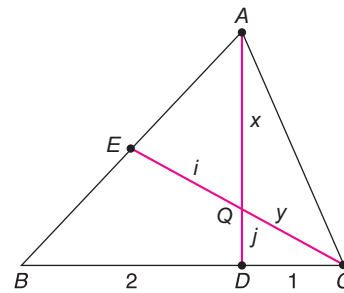


FIGURE 5.42 Single correct choice type question 12.

Solution: See Figure 5.42. Taking A as origin, let $\vec{AB} = \vec{b}$ and $\vec{AC} = \vec{c}$. Therefore

$$\vec{AE} = \frac{1}{2}\vec{b}$$

and

$$\vec{AD} = \frac{\vec{b} + 2\vec{c}}{3}$$

Let $AQ:QD = x:1$ and $CQ:QE = y:1$. Therefore

$$\frac{x[(\vec{b} + 2\vec{c})/3] + 1(\vec{0})}{x+1} = \frac{y(\vec{b}/2) + \vec{c}}{y+1}$$

Since $\vec{b}$ and $\vec{c}$ are non-collinear vectors, we have

$$\frac{x}{3(x+1)} = \frac{y}{2(y+1)} \quad (5.15)$$

and

$$\frac{2x}{3(x+1)} = \frac{1}{y+1} \quad (5.16)$$

Therefore from Eqs. (5.15) and (5.16) we have

$$2\left(\frac{y}{2(y+1)}\right) = \frac{1}{y+1}$$

Hence $y = 1$ and $x = 3$.

Answer: (C)

13. P, Q and R are the mid-points of the sides BC, CA and AB of $\triangle ABC$. If O is any point in the space (or in the plane of $\triangle ABC$), then

$$\vec{OA} + \vec{OB} + \vec{OC} = \lambda(\vec{OP} + \vec{OQ} + \vec{OR})$$

where the value of λ is

- (A) 2 (B) 3 (C) $\frac{2}{3}$ (D) 1

Solution: Since P , Q and R are mid-points of BC , CA and AB , we have

$$\overrightarrow{OP} = \frac{\overrightarrow{OB} + \overrightarrow{OC}}{2}$$

$$\overrightarrow{OQ} = \frac{\overrightarrow{OC} + \overrightarrow{OA}}{2}$$

and
$$\overrightarrow{OR} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2}$$

Therefore

$$2(\overrightarrow{OP} + \overrightarrow{OQ} + \overrightarrow{OR}) = 2(\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC})$$

Hence $\lambda = 1$.

Answer: (D)

14. If the points

$$(2 - \alpha)\vec{i} + 2\vec{j} + 2\vec{k}$$

$$2\vec{i} + (2 - \beta)\vec{j} + 2\vec{k}$$

$$2\vec{i} + 2\vec{j} + (2 - \gamma)\vec{k}$$

and
$$\vec{i} + \vec{j} + \vec{k}$$

are coplanar, and $\alpha\beta\gamma \neq 0$, then

(A) $\alpha + \beta + \gamma = 1$

(B) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 1$

(C) $\frac{1}{1-\alpha} + \frac{1}{1-\beta} + \frac{1}{1-\gamma} = 1$

(D) $\alpha + \beta + \gamma = 0$

Solution: Let the given points be respectively A , B , C and D . Therefore, the four points A , B , C and D are coplanar if and only if the three vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar. Hence the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ are linearly dependent (see Theorem 5.19). Therefore

$$\begin{vmatrix} \alpha & -\beta & 0 \\ \alpha & 0 & -\gamma \\ \alpha - 1 & -1 & -1 \end{vmatrix} = 0 \quad (\text{See Theorem 5.26})$$

$$\Rightarrow \alpha(0 - \gamma) + \beta[-\alpha + \gamma(\alpha - 1)] = 0$$

$$\Rightarrow -\alpha\gamma - \alpha\beta + \alpha\beta\gamma - \beta\gamma = 0$$

$$\Rightarrow \beta\gamma + \gamma\alpha + \alpha\beta = \alpha\beta\gamma$$

$$\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 1$$

Answer: (B)

15. The perimeter of the triangle with vertices $(3, 1, 5)$, $(-1, -1, 9)$ and $(0, -5, 1)$ is

- (A) 27 (B) $3\sqrt{31}$ (C) 15 (D) $15 + \sqrt{61}$

Solution: Let the given points be A , B and C respectively. Then

$$\overrightarrow{AB} = (-4, -2, 4)$$

$$\overrightarrow{BC} = (1, -4, -8)$$

$$\overrightarrow{CA} = (3, 6, 4)$$

Therefore

$$|\overrightarrow{AB}| = \sqrt{4^2 + 2^2 + 4^2} = 6$$

$$|\overrightarrow{BC}| = \sqrt{1^2 + 4^2 + 8^2} = 9$$

$$|\overrightarrow{CA}| = \sqrt{3^2 + 6^2 + 4^2} = \sqrt{61}$$

Answer: (D)

16. The position vectors of the vertices A , B and C of a triangle are respectively (a, b, c) , (b, c, a) and (c, a, b) . Then $\triangle ABC$ is

- (A) right angled
(B) right-angled isosceles
(C) equilateral
(D) obtuse angled

Solution: The sides are represented by the vectors

$$\overrightarrow{AB} = (b - a)\vec{i} + (c - b)\vec{j} + (a - c)\vec{k}$$

$$\overrightarrow{BC} = (c - b)\vec{i} + (a - c)\vec{j} + (b - a)\vec{k}$$

and $\overrightarrow{CA} = (a - c)\vec{i} + (b - a)\vec{j} + (c - b)\vec{k}$

Therefore

$$|\overrightarrow{AB}| = |\overrightarrow{BC}| = |\overrightarrow{CA}| = \sqrt{(a - b)^2 + (b - c)^2 + (c - a)^2}$$

Hence $\triangle ABC$ is equilateral.

Answer: (C)

17. $ABCD$ is a quadrilateral. M and N are mid-points of the sides AB and CD , respectively, and the diagonals AC and BD intersect in 'O'. If the points M , N and O are collinear, then $ABCD$ is a

- (A) trapezium (not necessarily parallelogram, rectangle, square or rhombus)
(B) parallelogram (not necessarily rectangle, square or rhombus)
(C) rectangle (not necessarily square or rhombus)
(D) rhombus (not necessarily square)

Solution: See Figure 5.43. Take 'O' as origin. Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \lambda\vec{a}$ and $\overrightarrow{OD} = \mu\vec{b}$ where λ and μ are scalars. Clearly $\vec{a}$ and $\vec{b}$ are non-collinear vectors. Therefore,

$$\overrightarrow{OM} = \frac{\vec{a} + \vec{b}}{2}$$

$$\overrightarrow{ON} = \frac{\lambda\vec{a} + \mu\vec{b}}{2}$$

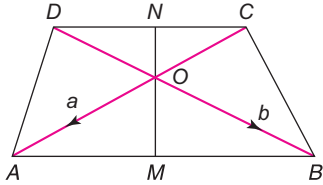


FIGURE 5.43 Single correct choice type question 17.

Since M, O and N are collinear, there exists scalar K such that

$$\overrightarrow{ON} = K\overrightarrow{OM}$$

That is

$$\frac{\lambda\vec{a} + \mu\vec{b}}{2} = K \frac{(\vec{a} + \vec{b})}{2}$$

Therefore

$$\lambda = K = \mu$$

and hence

$$\overrightarrow{DC} = \lambda\vec{a} - \mu\vec{b} = K(\vec{a} - \vec{b}) = K\overrightarrow{BA}$$

So, the sides AB and CD are parallel and ABCD is a trapezium.

Answer: (A)

18. $\vec{a}$ and $\vec{b}$ are non-collinear vectors and

$$\vec{\alpha} = x\vec{a} + 2y\vec{b}$$

$$\vec{\beta} = -2y\vec{a} + 3x\vec{b}$$

and

$$\vec{\gamma} = 4\vec{a} - 2\vec{b}$$

If $\vec{\gamma} = 2\vec{\alpha} - \vec{\beta}$, then

$$(A) \ x = \frac{10}{7}, y = \frac{4}{7} \quad (B) \ x = \frac{4}{7}, y = \frac{6}{7}$$

$$(C) \ x = \frac{8}{7}, y = \frac{2}{7} \quad (D) \ x = 2, y = 3$$

Solution: By hypothesis,

$$\begin{aligned} 4\vec{a} - 2\vec{b} &= \vec{\gamma} \\ &= 2\vec{\alpha} - \vec{\beta} \end{aligned}$$

$$= 2(x\vec{a} + 2y\vec{b}) - (-2y\vec{a} + 3x\vec{b})$$

$$= (2x + 2y)\vec{a} + (4y - 3x)\vec{b}$$

Therefore

$$2x + 2y = 4 \quad (5.17)$$

$$4y - 3x = -2 \quad (5.18)$$

Solving Eqs. (5.17) and (5.18), we get that

$$x = \frac{10}{7}, y = \frac{4}{7}$$

Answer: (A)

19. ABCD is a parallelogram and point P divides AD in the ratio 3:1 internally. The line BP meets the diagonal AC in Q. Then the ratio AQ:QC is equal to
(A) 2:3 (B) 3:2 (C) 4:3 (D) 3:4

Solution: Take A as origin (Figure 5.44) and let

$$\overrightarrow{AB} = \vec{b} \quad \text{and} \quad \overrightarrow{AD} = \vec{d}$$

so that

$$\overrightarrow{AC} = \vec{b} + \vec{d}$$

Therefore

$$\overrightarrow{AP} = \frac{3}{4}\vec{d}$$

Suppose AQ:QC = μ :1 and BQ:QP = λ :1. then

$$\begin{aligned} \frac{\lambda[(3/4)\vec{d}] + \vec{b}}{\lambda + 1} &= \frac{\mu(\vec{b} + \vec{d})}{\mu + 1} \\ \frac{1}{\lambda + 1} &= \frac{\mu}{\mu + 1} = \frac{3\lambda}{4(\lambda + 1)} \end{aligned}$$

This gives

$$\lambda = \frac{4}{3} \quad \text{and} \quad \frac{3}{7} = \frac{\mu}{\mu + 1} \Rightarrow \mu = \frac{3}{4}$$

Hence AQ:QC = 3:4.

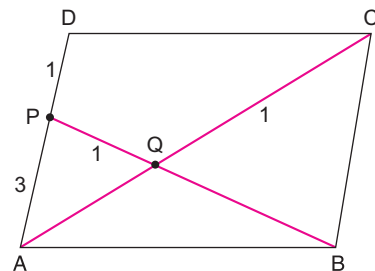


FIGURE 5.44 Single correct choice type question 19.

20. If none of the scalars x, y, z is equal to 1 and the vectors $x\vec{i} + \vec{j} + \vec{k}$, $\vec{i} + y\vec{j} + \vec{k}$ and $\vec{i} + \vec{j} + z\vec{k}$ are coplanar, then

$$\frac{1}{1-x} + \frac{1}{1-y} + \frac{1}{1-z} =$$

(A) 0 (B) 1 (C) -1 (D) xyz

Solution: According to Theorems 5.19 and 5.26, we have

$$\begin{vmatrix} x & 1 & 1 \\ 1 & y & 1 \\ 1 & 1 & z \end{vmatrix} = 0$$

The row operations $R_1 - R_2$ and $R_2 - R_3$ give

$$\begin{vmatrix} x-1 & 1-y & 0 \\ 0 & y-1 & 1-z \\ 1 & 1 & z \end{vmatrix} = 0$$

$$\Rightarrow (x-1)[z(y-1) - (1-z)] - (1-y)[0 - (1-z)] = 0$$

$$\Rightarrow z(x-1)(y-1) - (x-1)(1-z) + (1-y)(1-z) = 0$$

$$\Rightarrow z(1-x)(1-y) + (1-x)(1-z) + (1-y)(1-z) = 0$$

$$\Rightarrow \frac{z}{1-z} + \frac{1}{1-y} + \frac{1}{1-x} = 0$$

$$\Rightarrow \left(\frac{z}{1-z} + 1 \right) + \frac{1}{1-y} + \frac{1}{1-x} = 1$$

$$\Rightarrow \frac{1}{1-z} + \frac{1}{1-y} + \frac{1}{1-x} = 1$$

Answer: (B)

21. In the two-dimensional plane, if a vector $\vec{r}$ of magnitude 4 makes an angle 30° with the positive direction of x -axis, then $\vec{r}$ is equal to

(A) $4\sqrt{3}\vec{i} + \vec{j}$ (B) $2\vec{i} + \sqrt{3}\vec{j}$
(C) $2\vec{i} + 2\sqrt{3}\vec{j}$ (D) $2\sqrt{3}\vec{i} + 2\vec{j}$

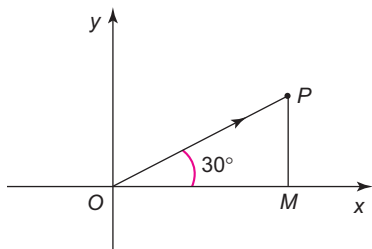


FIGURE 5.45 Single correct choice type question 21.

Solution: Let $\vec{OP} = \vec{r}$ where O is the origin (Figure 5.45). Therefore its components along x -axis and y -axis are $(|\vec{OP}|\cos 30^\circ)\vec{i}$ and $(|\vec{OP}|\sin 30^\circ)\vec{j}$, respectively.

Hence

$$\vec{r} = 4\left(\frac{\sqrt{3}}{2}\right)\vec{i} + 4\left(\frac{1}{2}\right)\vec{j}$$

$$= 2\sqrt{3}\vec{i} + 2\vec{j}$$

Answer: (D)

22. If the vectors $a\vec{i} + a\vec{j} + c\vec{k}$, $\vec{i} + \vec{k}$ and $c\vec{i} + c\vec{j} + b\vec{k}$ are coplanar, then

(A) $a + c = 2b$ (B) $\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$
(C) $c^2 = ab$ (D) $b^2 = ca$

Solution: Since the vectors are coplanar, we have

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

$$\Rightarrow a(0 - c) - a(b - c) + c(c - 0) = 0$$

$$\Rightarrow -ac - ab + ac + c^2 = 0$$

$$\Rightarrow c^2 = ab$$

Answer: (C)

23. In $\triangle ABC$, D is the mid-point of BC and E is the mid-point of AD . BE is produced to meet the side AC in F . Then BF is equal to

(A) $4EF$ (B) $3EF$ (C) $2EF$ (D) $6EF$

Solution: See Figure 5.46. Take A as origin, let $\vec{AB} = \vec{b}$ and $\vec{AC} = \vec{c}$. Then

$$\vec{AD} = \frac{\vec{b} + \vec{c}}{2}$$

and

$$\vec{AE} = \frac{\vec{b} + \vec{c}}{4}$$

According to Theorem 5.29, the equations of the lines BE and AC are, respectively,

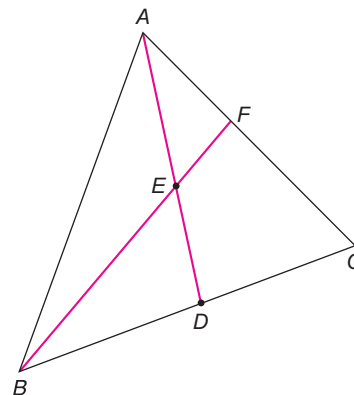


FIGURE 5.46 Single correct choice type question 23.

$$\vec{r} = (1-x)\vec{b} + x\frac{(\vec{b} + \vec{c})}{4}$$

and $\vec{r} = y\vec{c}$

where x, y are scalars. The two lines BE and AC intersect in F . Therefore, the position vector F must satisfy both equations. Hence equating the corresponding coefficients we get

$$1-x+\frac{x}{4}=0 \quad \text{and} \quad \frac{x}{4}=y$$

This gives

$$x = \frac{4}{3} \quad \text{and} \quad y = \frac{1}{3}$$

Therefore

$$\overrightarrow{AF} = \frac{1}{3}\vec{c}$$

Now

$$\overrightarrow{EF} = \frac{1}{3}\vec{c} - \frac{\vec{b} + \vec{c}}{4} = \frac{1}{12}(\vec{c} - 3\vec{b})$$

$$\begin{aligned} \overrightarrow{BF} &= \frac{1}{3}\vec{c} - \vec{b} \\ &= \frac{\vec{c} - 3\vec{b}}{3} \\ &= \frac{4(\vec{c} - 3\vec{b})}{12} \\ &= 4\overrightarrow{EF} \end{aligned}$$

Answer: (A)

24. If the vectors $\vec{a} + \vec{b} - \lambda\vec{c}$, $3\vec{a} - 2\vec{b} + 4\vec{c}$, $3\vec{a} - 7\vec{b} + 14\vec{c}$ are linearly dependent ($\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar) then the value of λ is equal to

- (A) 14 (B) 2 (C) 3 (D) 4

Solution: Since the vectors are linearly dependent, by Theorem 5.26,

$$\begin{vmatrix} 1 & 1 & -\lambda \\ 3 & -2 & 4 \\ 3 & -7 & 14 \end{vmatrix} = 0$$

$$\Rightarrow 1(42 - 28) - \lambda(42 - 12) - \lambda(-21 + 6) = 0$$

$$\Rightarrow -30 + 15\lambda = 0$$

$$\Rightarrow \lambda = 2$$

Answer: (B)

25. The line joining the points $\vec{i} - 2\vec{j} + \vec{k}$ and $-2\vec{j} + 3\vec{k}$ meets the plane passing through the points origin

(i.e., $\vec{O}$) and the two points $4\vec{j}$, $2\vec{i} + \vec{k}$ at the point whose position vector is

- (A) $\frac{1}{5}(6\vec{i} - 10\vec{j} + 3\vec{k})$ (B) $\frac{1}{5}(6\vec{i} + 10\vec{j} + 3\vec{k})$
(C) $\frac{1}{5}(6\vec{i} + 10\vec{j} - 3\vec{k})$ (D) $\frac{1}{5}(6\vec{i} - 10\vec{j} - 3\vec{k})$

Solution: By Theorem 5.31, the equation of the plane passing through $\vec{O}$, $4\vec{j}$ and $2\vec{i} + \vec{k}$ is

$$\vec{r} = x(4\vec{j}) + y(2\vec{i} + \vec{k}) \quad (5.19)$$

Again, by Theorem 5.29, the equation of the line passing through the points $\vec{i} - 2\vec{j} + \vec{k}$ and $-2\vec{j} + 3\vec{k}$ is

$$\vec{r} = (1-z)(\vec{i} - 2\vec{j} + \vec{k}) + z(-2\vec{j} + 3\vec{k}) \quad (5.20)$$

The common point of the plane and the line satisfies both Eqs. (5.19) and (5.20). Therefore, by equating the corresponding coefficient of Eqs. (5.19) and (5.20), we have

$$2y = (1-z) \quad (5.21)$$

$$4x = -2(1-z) - 2z \quad (5.22)$$

$$y = 1 - z + 3z \quad (5.23)$$

Solving Eqs. (5.21)–(5.23), we get

$$x = \frac{-1}{2}, y = \frac{3}{5} \quad \text{and} \quad z = \frac{-1}{5}$$

Substituting these values in Eq. (5.19), we have

$$\begin{aligned} \vec{r} &= \frac{-1}{2}(4\vec{j}) + \frac{3}{5}(2\vec{i} + \vec{k}) \\ &= \frac{1}{5}(6\vec{i} - 10\vec{j} + 3\vec{k}) \end{aligned}$$

Answer: (A)

26. If the vectors

$$\vec{a} = \vec{i} + \vec{j} + \vec{k}$$

$$\vec{b} = 4\vec{i} + 3\vec{j} + 4\vec{k}$$

and $\vec{c} = \vec{i} + \alpha\vec{j} + \beta\vec{k}$

are linearly dependent and $|\vec{c}| = \sqrt{3}$, then

- (A) $\alpha = \pm 1, \beta = 1$ (B) $\alpha = 1, \beta = -1$
(C) $\alpha = 1, \beta = \pm 1$ (D) $\alpha = -1, \beta = \pm 1$

Solution: Since the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly dependent, by Theorem 5.26, we have

$$\begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0$$

$$\begin{aligned}\Rightarrow (3\beta - 4\alpha) - (4\beta - 4) + (4\alpha - 3) &= 0 \\ \Rightarrow -\beta + 1 &= 0 \\ \Rightarrow \beta &= 1\end{aligned}$$

Now $|\vec{c}| = \sqrt{3}$ implies

$$\begin{aligned}1^2 + \alpha^2 + \beta^2 &= 3 \\ \Rightarrow \alpha^2 &= 1 \\ \Rightarrow \alpha &= \pm 1\end{aligned}$$

Answer: (A)

27. If $3\vec{i} + 2\vec{j} - 5\vec{k} = x(2\vec{i} - \vec{j} + \vec{k}) + y(\vec{i} + 3\vec{j} - 2\vec{k}) + z(-2\vec{i} + \vec{j} - 3\vec{k})$ then

- (A) $x = 1, y = 2, z = 3$ (B) $x = 2, y = 3, z = 1$
(C) $x = 3, y = 1, z = 2$ (D) $x = 1, y = 3, z = 2$

Solution: Equating the coefficients of $\vec{i}, \vec{j}$ and $\vec{k}$ on both sides we get

$$2x + y - 2z = 3 \quad (5.24)$$

$$-x + 3y + z = 2 \quad (5.25)$$

$$x - 2y - 3z = -5 \quad (5.26)$$

Solving these equations, we have $x = 3, y = 1$ and $z = 2$.

Answer: (C)

28. The position vectors of the vertices A, B and C are, respectively, $\vec{i} - \vec{j} - 3\vec{k}, 2\vec{i} + \vec{j} - 2\vec{k}$ and $-5\vec{i} + 2\vec{j} - 6\vec{k}$. The internal bisector of angle A meets the side BC in D . Then the length AD is

- (A) $\frac{1}{4}$ (B) $\frac{11}{2}$ (C) $\frac{15}{2}$ (D) $\frac{3\sqrt{10}}{4}$

Solution: It is known that D divides the segment BC internally in the ratio of $AB:AC$. Now

$$\overrightarrow{AB} = \vec{i} + 2\vec{j} + \vec{k}$$

$$\overrightarrow{AC} = -6\vec{i} + 3\vec{j} - 3\vec{k}$$

and hence $AB = \sqrt{6}$ and $AC = \sqrt{54} = 3\sqrt{6}$. Therefore

$$AB:AC = 1:3$$

Now

$$BD:DC = AB:AC = 1:3$$

$$\Rightarrow \overrightarrow{AD} = \frac{3}{4}(-\vec{i} + 3\vec{j})$$

$$\Rightarrow AD = \frac{3}{4}\sqrt{10}$$

Answer: (D)

29. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar vectors. Let L and Π be a line and a plane whose equations are

$$\vec{r} = \vec{a} + 2\vec{b} + t(\vec{a} - \vec{c})$$

$$\text{and } \vec{r} = x(\vec{a} + 2\vec{b}) + y(2\vec{b} + \vec{c})$$

respectively, where t, x, y are scalars. Then

- (A) the line L meets the plane Π in the point $\vec{a} - \vec{b} + \vec{c}$
(B) the line L meets the plane Π in the point $2\vec{a} + 2\vec{b} - \vec{c}$
(C) the line L and the plane Π have infinitely many common points
(D) the line L does not intersect the plane Π

Solution: Equating the corresponding coefficient of the line and the plane, we have

$$x = 1 + t$$

$$x + y = 1$$

and

$$y = -t$$

Therefore, the equation of the line L is

$$\begin{aligned}\vec{r} &= \vec{a} + 2\vec{b} - y(\vec{a} - \vec{c}) \\ &= (x + y)(\vec{a} + 2\vec{b}) - y(\vec{a} - \vec{c}) \\ &= x(\vec{a} + 2\vec{b}) + y(2\vec{b} + \vec{c})\end{aligned}$$

All points of the line L lie in the plane Π .

Answer: (C)

30. If

$$\begin{vmatrix} a & a^2 & a^3 + 1 \\ b & b^2 & b^3 + 1 \\ c & c^2 & c^3 + 1 \end{vmatrix} = 0$$

and the vectors $\vec{A} = (1, a, a^2), \vec{B} = (1, b, b^2)$ and $\vec{C} = (1, c, c^2)$ are non-coplanar, then abc is equal to

- (A) 1 (B) -1 (C) 2 (D) -2

Solution: Let

$$\begin{aligned}\Delta &= \begin{vmatrix} a & a^2 & a^3 + 1 \\ b & b^2 & b^3 + 1 \\ c & c^2 & c^3 + 1 \end{vmatrix} \\ &= \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} + \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}\end{aligned}$$

$$\begin{aligned}
 &= abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} + \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \\
 &= (abc + 1) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}
 \end{aligned}$$

Since $\vec{A}, \vec{B}, \vec{C}$ are non-coplanar, they are linearly independent and hence

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \neq 0$$

(by Theorem 5.26). Therefore $\Delta = 0 \Rightarrow abc + 1 = 0$.

Answer: (B)

Multiple Correct Choice Type Questions

1. Consider the cube $OAFBDCEP$ (see Figure 5.47) Then

- (A) $\vec{OA} + \vec{OB} + \vec{OC} = \vec{OP}$
 (B) $\vec{OE} + \vec{OF} + \vec{OD} = 2\vec{OP}$
 (C) $\vec{OP} + \vec{CF} + \vec{BE} + \vec{DA} = 4\vec{OA}$
 (D) $\vec{OA} + \vec{OC} + \vec{OE} = 3\vec{OP}$

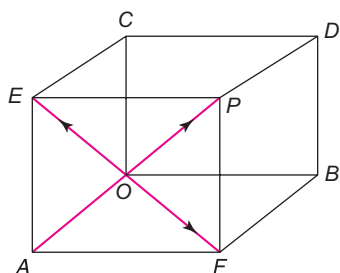


FIGURE 5.47 Multiple correct choice type question 1.

Solution: We can suppose, the cube is unit cube and $\vec{OA} = \vec{i}, \vec{OB} = \vec{j}, \vec{OC} = \vec{k}$. Therefore

(A) We have

$$\vec{OA} + \vec{OB} + \vec{OC} = \vec{i} + \vec{j} + \vec{k} = \vec{OP}$$

Therefore (A) is true.

(B) We have

$$\begin{aligned}
 \vec{OE} + \vec{OF} + \vec{OD} &= (\vec{i} + \vec{k}) + (\vec{i} + \vec{j}) + (\vec{j} + \vec{k}) \\
 &= 2(\vec{i} + \vec{j} + \vec{k}) \\
 &= 2\vec{OP}
 \end{aligned}$$

Hence (B) is true.

(C) We have

$$\begin{aligned}
 \vec{OP} + \vec{CF} + \vec{BE} + \vec{DA} &= (\vec{i} + \vec{j} + \vec{k}) + (-\vec{k} + \vec{i} + \vec{j}) \\
 &\quad + (-\vec{j} + \vec{i} + \vec{k}) + (-\vec{j} - \vec{k} + \vec{i}) \\
 &= 4\vec{i} = 4\vec{OA}
 \end{aligned}$$

(C) is true.

(D) We have

$$\begin{aligned}
 \vec{OA} + \vec{OC} + \vec{OE} &= \vec{i} + \vec{k} + (\vec{i} + \vec{k}) \\
 &= 2(\vec{i} + \vec{k})
 \end{aligned}$$

Hence (D) is not true.

Answers: (A), (B), (C)

2. Let $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors. Then

- (A) the vectors $\vec{a} - 2\vec{b} + 3\vec{c}, -2\vec{a} + 3\vec{b} - 4\vec{c}, \vec{a} - 3\vec{b} + 5\vec{c}$ are coplanar.
 (B) the vectors given in (A) are linearly independent.
 (C) if $\vec{a} - 2\vec{b} + 3\vec{c} = x(-2\vec{a} + 3\vec{b} - 4\vec{c}) + y(\vec{a} - 3\vec{b} + 5\vec{c})$ then $x = -1/3$ and $y = 1/3$.
 (D) $\vec{a} - 2\vec{b} + 3\vec{c}$ cannot be written as a linear combination of the vectors $-2\vec{a} + 3\vec{b} - 4\vec{c}$ and $\vec{a} - 3\vec{b} + 5\vec{c}$.

Solution:

(A) We have

$$\begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 1(15 - 12) + 2(-10 + 4) + 3(6 - 3) = 3 - 12 + 9 = 0$$

Therefore the vectors $\vec{a} - 2\vec{b} + 3\vec{c}, -2\vec{a} + 3\vec{b} - 4\vec{c}$ and $\vec{a} - 3\vec{b} + 5\vec{c}$ are linearly dependent and hence they are coplanar. This implies (A) is true and (B) is not true.

(C) It is true because by equating the corresponding coefficients on both sides we have

$$\begin{aligned}
 -2x + y &= 1 \\
 3x - 3y &= -2
 \end{aligned}$$

and

$$-4x + 5y = 3$$

On solving these equations, we get that

$$x = \frac{-1}{3}, y = \frac{1}{3}$$

Answers: (A), (C)

3. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors. Consider the four points A, B, C and D whose position vectors are, respectively, $-\vec{a} + 4\vec{b} - 3\vec{c}, 3\vec{a} + 2\vec{b} - 5\vec{c}, 3\vec{a} + 8\vec{b} - 5\vec{c}$ and $-3\vec{a} + 2\vec{b} + \vec{c}$. Then

- (A) A, B, C and D are non-coplanar
 (B) A, B, C and D are coplanar
 (C) $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ are linearly dependent
 (D) $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ are linearly independent

Solution: Points A, B, C and D are coplanar if and only if the vectors $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar and hence $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are linearly dependent (Theorem 5.19). We have

$$\overrightarrow{AB} = 4\vec{a} - 2\vec{b} - 2\vec{c}$$

$$\overrightarrow{AC} = 4\vec{a} + 4\vec{b} - 2\vec{c}$$

$$\overrightarrow{AD} = -2\vec{a} - 2\vec{b} + 4\vec{c}$$

Now

$$\begin{vmatrix} 4 & -2 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 4 \end{vmatrix} = 8 \begin{vmatrix} 2 & -1 & -1 \\ 2 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= 8[2(4 - 1) + 1(4 - 1) - 1(-2 + 2)]$$

$$= 8(9) \neq 0$$

Hence $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ are not linearly dependent. Hence they are independent and non-coplanar.

Answers: (A), (D)

4. In $\triangle ABC$, D and E are points on BC and AC , respectively, such that $BD = 3 DC$ and $AE = 4 EC$. The lines AD and BE intersect in P . Then

- (A) $AP:PD = 16:3$
 (B) $AP:PD = 15:3$
 (C) $BP:PE = 15:3$
 (D) $BP:PE = 15:4$

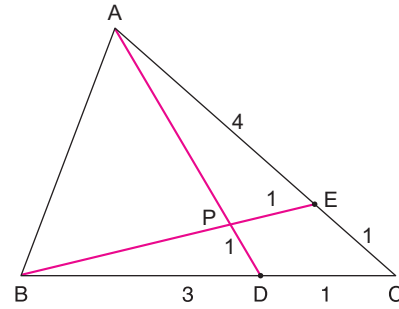


FIGURE 5.48 Multiple correct choice type question 4.

Solution: Take A as origin. Let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AC} = \vec{c}$. Therefore by hypothesis,

$$\overrightarrow{AD} = \frac{\vec{b} + 3\vec{c}}{4} \quad \text{and} \quad \overrightarrow{AE} = \frac{4\vec{c}}{5}$$

Suppose $BP:PE = \lambda:1$ and $AP:PD = \mu:1$. Then

$$\frac{\vec{b} + \lambda(4\vec{c}/5)}{\lambda + 1} = \frac{\mu(\vec{b} + 3\vec{c})}{4(\mu + 1)}$$

Equating the coefficients of $\vec{b}$ and $\vec{c}$ on both sides we get

$$\frac{1}{\lambda + 1} = \frac{\mu}{4(\mu + 1)} \quad (5.27)$$

$$\text{and} \quad \frac{4\lambda}{5(\lambda + 1)} = \frac{3\mu}{4(\mu + 1)} \quad (5.28)$$

Therefore

$$\frac{4\lambda}{5(\lambda + 1)} = \frac{3}{\lambda + 1}$$

$$4\lambda = 15 \quad \text{or} \quad \lambda = \frac{15}{4}$$

Hence $BP:PE = 15:4$ and so (D) is true.

Substituting the value of $\lambda = 15/4$ in Eq. (5.27) we get

$$\frac{4}{19} = \frac{\mu}{4(\mu + 1)}$$

$$16\mu + 16 = 19\mu$$

$$\mu = \frac{16}{3}$$

Therefore $AP:PD = 16:3$ and so (A) is true.

Answers: (A), (D)

Matrix-Match Type Questions

1. The position vectors of the points A, B, C and D are, respectively, $\vec{a}, \vec{b}, 2\vec{a} + 3\vec{b}$ and $\vec{a} - 2\vec{b}$. Match the items of Column I with those of Column II.

Column I	Column II
(A) $\overrightarrow{AC}$ is	(p) $3\vec{b} - \vec{a}$
(B) $\overrightarrow{DB}$ is	(q) $2\vec{a} + 2\vec{b}$
(C) $\overrightarrow{BC}$ is	(r) $\vec{a} + 3\vec{b}$
(D) $\overrightarrow{AC} + \overrightarrow{DB} + \overrightarrow{BC}$ equals	(s) $2\vec{a} + 8\vec{b}$

Solution:

$$\overrightarrow{AC} = (2\vec{a} + 3\vec{b}) - \vec{a} = \vec{a} + 3\vec{b}$$

$$\overrightarrow{DB} = \vec{b} - (\vec{a} - 2\vec{b}) = 3\vec{b} - \vec{a}$$

$$\overrightarrow{BC} = (2\vec{a} + 3\vec{b}) - \vec{b} = 2\vec{a} + 2\vec{b}$$

$$\overrightarrow{AC} + \overrightarrow{DB} + \overrightarrow{BC} = 2\vec{a} + 8\vec{b}$$

Answer: (A) $\rightarrow$ (r), (B) $\rightarrow$ (p), (C) $\rightarrow$ (q), (D) $\rightarrow$ (s)

2. $ABCD$ is a quadrilateral and $ADEF$ is a parallelogram (see Figure 5.49) with $\overrightarrow{AF} = \overrightarrow{BC}$. If $\overrightarrow{AB} = \vec{b}$, $\overrightarrow{AC} = \vec{c}$ and $\overrightarrow{AD} = \vec{d}$, match the items of Column I with those of Column II.

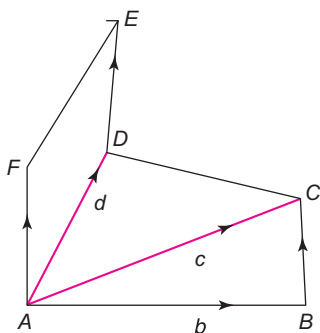


FIGURE 5.49 Matrix-match type question 2.

Column I	Column II
(A) $\overrightarrow{DE}$ is equal to	(p) $\vec{c} - \vec{b}$
(B) $\overrightarrow{BC}$ equals	(q) $\frac{\vec{c} + \vec{d}}{2}$
(C) If Y is the mid-point of DC then $\overrightarrow{AY}$ is equal to	(r) $\frac{\vec{c} + \vec{d} - \vec{b}}{2}$
(D) If X is the mid-point of AB and Y is the mid-point of DC then $\overrightarrow{XY}$ is	(s) $\vec{c} - \vec{d}$

Solution:

(A) and (B) $\overrightarrow{DE} = \overrightarrow{AF} = \overrightarrow{BC} = \vec{c} - \vec{b}$

(C) $\overrightarrow{AY} = \frac{\overrightarrow{AC} + \overrightarrow{AD}}{2} = \frac{\vec{c} + \vec{d}}{2}$

(D) $\overrightarrow{AX} = \frac{1}{2}\vec{b}$ and $\overrightarrow{AY} = \frac{\vec{c} + \vec{d}}{2} \Rightarrow \overrightarrow{XY} = \overrightarrow{AY} - \overrightarrow{AX} = \frac{\vec{c} + \vec{d} - \vec{b}}{2}$

Answer: (A) $\rightarrow$ (p), (B) $\rightarrow$ (p), (C) $\rightarrow$ (q), (D) $\rightarrow$ (r)

3. Let $\vec{i}, \vec{j}, \vec{k}$ be unit vectors along positive x -, y - and z -axes, respectively, with 'O' being the origin. Match the items of Column I with those of Column II.

Column I	Column II
(A) The points $\vec{i} + \vec{j} + \vec{k}, \vec{i} + 2\vec{j} + 3\vec{k}$ and $2\vec{i} - \vec{j} + \vec{k}$ form	(p) right angled
(B) The vectors $\vec{i} + 2\vec{j} + 3\vec{k}, 2\vec{i} + \vec{j} + 3\vec{k}$ and $-3\vec{i} - 3\vec{j} - 6\vec{k}$ form	(q) scalene
(C) The vectors $-\vec{i} - \vec{j} - 4\vec{k}, -3\vec{i} + 3\vec{j}$ and $4\vec{i} - 2\vec{j} + 4\vec{k}$ form	(r) isosceles
(D) The points $7\vec{j} + 10\vec{k}, -\vec{i} + 6\vec{j} + 6\vec{k}$ and $-4\vec{i} + 9\vec{j} + 6\vec{k}$ constitute the vertices of a triangle which is	(s) equilateral

Solution:

- (A) Let the given points be A, B and C , respectively. Therefore

$$\overrightarrow{AB} = \vec{j} + 2\vec{k}$$

$$\overrightarrow{BC} = \vec{i} - 3\vec{j} - 2\vec{k}$$

$$\overrightarrow{CA} = -\vec{i} + 2\vec{j}$$

Now

$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0} \Rightarrow \text{the points } A, B, C \text{ form a triangle}$$

$$\text{Also } |\overrightarrow{AB}| = |\overrightarrow{CA}| = \sqrt{1^2 + 2^2} \Rightarrow \Delta ABC \text{ is isosceles.}$$

Answer: (A) $\rightarrow$ (r)

- (B) Let $\vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}, \vec{b} = 2\vec{i} + \vec{j} + 3\vec{k}$ and $\vec{c} = -3\vec{i} - 3\vec{j} - 6\vec{k}$. Now

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

implies that $\vec{a}, \vec{b}, \vec{c}$ form a triangle. Also

$$|\vec{a}| = |\vec{b}| = \sqrt{14}$$

Therefore $\vec{a}, \vec{b}, \vec{c}$ form an isosceles triangle.

Answer: (B) → (r)

(C) Let

$$\vec{a} = -\vec{i} - \vec{j} - 4\vec{k}$$

$$\vec{b} = -3\vec{i} + 3\vec{j}$$

and

$$\vec{c} = 4\vec{i} - 2\vec{j} + 4\vec{k}$$

Now

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

implies that $\vec{a}, \vec{b}, \vec{c}$ form a triangle. Also

$$|\vec{a}|^2 + |\vec{b}|^2 = 18 + 18 = 36 = |\vec{c}|^2$$

Therefore $\vec{a}, \vec{b}$ and $\vec{c}$ form a right-angled isosceles triangle.

Answer: (C) → (p), (r)

(D) Let the given points be A, B and C , respectively, so that

$$\vec{AB} = -\vec{i} - \vec{j} - 4\vec{k}$$

$$\vec{BC} = -3\vec{i} + 3\vec{j}$$

and

$$\vec{CA} = 4\vec{i} - 2\vec{j} + 4\vec{k}$$

Now $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0} \Rightarrow A, B, C$ form a triangle. Also

$$|\vec{AB}|^2 = 18 = |\vec{BC}|^2 = 18$$

and

$$|\vec{CA}|^2 = 36$$

implies that $\triangle ABC$ is right-angled isosceles.

Answer: (D) → (p), (r)

4. In the Cartesian plane, a man starts at origin and walks a distance of 3 units in the North-East direction and reaches a point P . From P , he walks a distance of 4 units in the North-West direction to reach a point Q . Construct the parallelogram $OPQR$ with PO and PQ as adjacent sides. Let M be the mid-point of PQ . Match the items of Column I with those of Column II.

Column I	Column II
(A) The position vector of P is	(p) $\frac{3}{\sqrt{2}}(\vec{i} + \vec{j})$
(B) The position vector of R is	(q) $\frac{1}{\sqrt{2}}(\vec{i} + 5\vec{j})$
(C) The position vector of M is	(r) $2\sqrt{2}(-\vec{i} + \vec{j})$
(D) If the line OM meets the diagonal PR in the point T , then $\vec{OT}$ equals	(s) $\frac{\sqrt{2}}{3}(\vec{i} + 5\vec{j})$

Solution: Let $\vec{i}$ and $\vec{j}$ be the unit vectors along $\vec{Ox}$ and $\vec{Oy}$, respectively (Figure 5.50). Now $OP = 3$ and $\angle xOP = 45^\circ$ implies

$$\begin{aligned}\vec{OP} &= (3\cos 45^\circ)\vec{i} + (3\sin 45^\circ)\vec{j} \\ &= \frac{3}{\sqrt{2}}(\vec{i} + \vec{j})\end{aligned}$$

Answer: (A) → (p)

Again, $\angle xOR = 135^\circ$ and $OR = 4$ implies that

$$\begin{aligned}\vec{OR} &= \frac{4}{\sqrt{2}}(-\vec{i} + \vec{j}) \\ &= 2\sqrt{2}(-\vec{i} + \vec{j})\end{aligned}$$

Answer: (B) → (r)

The position vector of Q is given by

$$\vec{OP} + \vec{PQ} = \vec{OP} + \vec{OR} = \frac{1}{\sqrt{2}}(-\vec{i} + 7\vec{j})$$

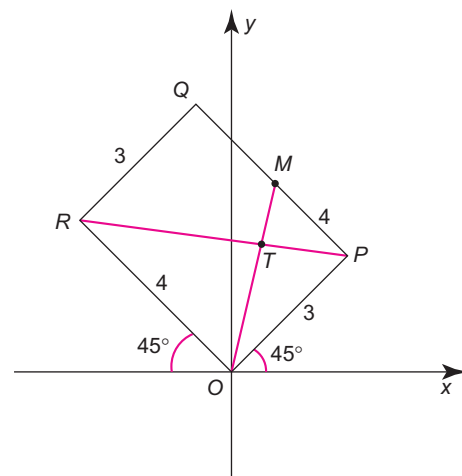


FIGURE 5.50 Matrix-match type question 4.

Therefore

$$\begin{aligned}\overline{OM} &= \frac{(3/\sqrt{2})(\vec{i} + \vec{j}) + (1/\sqrt{2})(-\vec{i} + 7\vec{j})}{2} \\ &= \frac{2\vec{i} + 10\vec{j}}{2\sqrt{2}} = \frac{\vec{i} + 5\vec{j}}{\sqrt{2}}\end{aligned}$$

Answer: (C) → (q)

Now, by Example 5.4,

$$PT:TR = 1:2$$

Therefore

$$\begin{aligned}\overline{OT} &= \frac{1(\overline{OR}) + 2(\overline{OP})}{3} \\ &= \frac{(4/\sqrt{2})(-\vec{i} + \vec{j}) + 2[(3/\sqrt{2})(\vec{i} + \vec{j})]}{3} \\ &= \frac{\sqrt{2}}{3}(\vec{i} + 5\vec{j})\end{aligned}$$

Answer: (D) → (s)

Comprehension-Type Questions

1. Passage: If $\vec{a}$ and $\vec{b}$ are the position vectors of two points A and B respectively, then the position vector of the point P which divides the segment $\overline{AB}$ in the ratio $m:n(m+n \neq 0)$ is

$$\frac{n\vec{a} + m\vec{b}}{m+n}$$

Answer the following three questions.

- (i) In $\triangle ABC$, D is the mid-point of AB and E is a point on the side BC such that $BE:EC = 1:2$. The lines AE and CD meet in P . Then, the ratio $AP:PE =$
(A) 2:3 (B) 3:2 (C) 3:4 (D) 4:3
- (ii) In the Question (i), the ratio $CP:PD$ is
(A) 4:1 (B) 1:4 (C) 3:1 (D) 3:1
- (iii) In $\triangle ABC$, a line is drawn parallel to the side BC meeting the sides AB and AC in M and N respectively. The lines BN and CM intersect in E . Then the line AE divides the segment $\overline{BC}$ in the ratio
(A) 1:2 (B) 3:1
(C) 1:1 (D) 2:3 externally

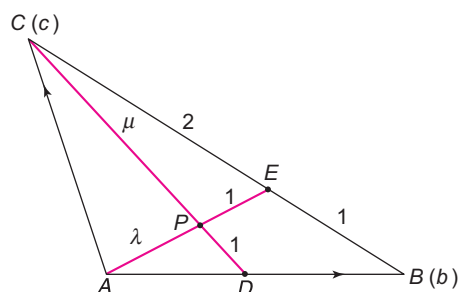


FIGURE 5.51 Comprehension-type question 1 part (i).

Solution: (i) and (ii): Take A as origin and let $\overline{AB} = \vec{b}$ and $\overline{AC} = \vec{c}$. Therefore

$$\overline{AD} = \frac{1}{2}\vec{b}$$

and $\overline{AE} = \frac{2\vec{b} + \vec{c}}{3}$

Suppose $AP:PE = \lambda:1$ and $CP:PD = \mu:1$. Therefore

$$\frac{\lambda[(2\vec{b} + \vec{c})/3]}{\lambda + 1} = \overline{AP} = \frac{\mu[(1/2)\vec{b}] + \vec{c}}{\mu + 1}$$

Since $\vec{b}$ and $\vec{c}$ are non-collinear vectors, the corresponding coefficients of $\vec{b}$ and $\vec{c}$ on both sides are equal. Therefore

$$\frac{2\lambda}{3(\lambda + 1)} = \frac{\mu}{2(\mu + 1)} \quad (5.29)$$

and $\frac{\lambda}{3(\lambda + 1)} = \frac{1}{(\mu + 1)} \quad (5.30)$

Solving Eqs. (5.29) and (5.30) for λ and μ we get

$$\mu = 4 \quad \text{and} \quad \lambda = \frac{3}{2}$$

Hence $AP:PE = 3:2$ and $CP:PD = 4:1$.

Answer: (i) → (B); (ii) → (A)

- (iii) Take A as origin and let $\overline{AB} = \vec{b}$, $\overline{AC} = \vec{c}$ (see Figure 5.52). Suppose the line AE meets the side BC in D .

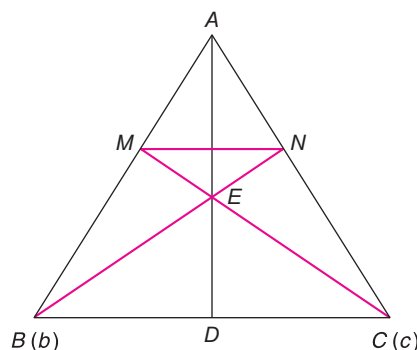


FIGURE 5.52 Comprehension-type question 1 part (iii).

Since MN is parallel to BC , it follows that

$$\frac{AM}{AB} = \frac{AN}{AC} = \lambda \quad (\text{suppose})$$

so that $\overrightarrow{AM} = \lambda \vec{b}$ and $\overrightarrow{AN} = \lambda \vec{c}$ ($\lambda > 0$). Therefore from Theorem 5.29, the equation of the line BN is

$$\vec{r} = (1-t)\vec{b} + t(\lambda\vec{c}) \quad (5.31)$$

and the equation of the line CM is

$$\vec{r} = (1-s)\vec{c} + s(\lambda\vec{b}) \quad (5.32)$$

From Eqs. (5.31) and (5.32)

$$1-t = s\lambda \quad \text{and} \quad 1-s = t\lambda$$

so that

$$t = \frac{1}{1+\lambda} = s$$

There the position vector of E is

$$\frac{\lambda}{\lambda+1}(\vec{b} + \vec{c})$$

Now, the equation of the line AE is

$$\vec{r} = x \frac{\lambda}{\lambda+1}(\vec{b} + \vec{c}) \quad (5.33)$$

and the equation of the line BC is

$$\vec{r} = (1-y)\vec{b} + y\vec{c} \quad (5.34)$$

Since D is a common point of the lines AE and BC , from Eqs. (5.33) and (5.34), we have

$$1-y = \frac{\lambda x}{\lambda+1} = y$$

which implies that $y = 1/2$. Hence the position vector of D is $(\vec{b} + \vec{c})/2$. Therefore the ratio $BD:DC = 1:1$.

Answer: (iii) → (C)

2. Passage: If $\vec{i}, \vec{j}$ and $\vec{k}$ are unit vectors along positive directions of the coordinate axes, then every vector $\vec{r}$ can be represented as $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ uniquely. Answer the following three questions.

- (i) If $x(\vec{i} + \vec{j} + 3\vec{k}) + y(3\vec{i} - 3\vec{j} + \vec{k}) + z(-4\vec{i} + 5\vec{j}) = \lambda(x\vec{i} + y\vec{j} + z\vec{k})$ where at least one of x, y, z is not zero, then the number of distinct values of λ is
(A) 1 (B) 2 (C) 3 (D) 0

- (ii) If $x(\vec{j} + \vec{k}) + y(-\vec{j} + 2\vec{k}) + z\vec{k} = \vec{i}$, then
(A) $x = 0, y = 1, z = -1$
(B) $x = -1, y = 1, z = -1$

$$(C) x = -1, y = -1, z = -1$$

$$(D) \text{ no real values of } x, y, z \text{ exist}$$

- (iii) If $x(2\vec{i} - \vec{j} + \vec{k}) + y(\vec{i} + 2\vec{j} - 3\vec{k}) + z(3\vec{i} + a\vec{j} + 5\vec{k}) = \vec{0}$ where x, y, z are scalars such that $(x, y, z) \neq (0, 0, 0)$ then the value of a is
(A) 3 (B) -3 (C) -4 (D) 4

Solution:

- (i) Equating corresponding coefficients of both sides we get

$$x + 3y - 4z = \lambda x$$

$$x - 3y + 5z = \lambda y$$

$$3x + y = \lambda z$$

Therefore

$$(1-\lambda)x + 3y - 4z = 0$$

$$x - (3+\lambda)y + 5z = 0$$

$$3x + y - \lambda z = 0$$

The above is a homogeneous system of equations and hence it has non-zero solution, if

$$\begin{vmatrix} 1-\lambda & 3 & -4 \\ 1 & -3-\lambda & 5 \\ 3 & 1 & -\lambda \end{vmatrix} = 0 \quad (\text{See Volume 1, Chapter 8})$$

In expanding the determinant, we have

$$\lambda(\lambda+1)^2 = 0$$

Therefore $\lambda = 0, -1$. If $\lambda = 0$, such scalars x, y, z with $(x, y, z) \neq (0, 0, 0)$ do not exist. Hence $\lambda = -1$.

Answer: (A)

- (ii) From the given equation

$$-\vec{i} + (x-y)\vec{j} + (x+2y+z)\vec{k} = \vec{0}$$

Therefore

$$-1 = 0$$

$$x = y$$

$$\text{and} \quad x + 2y + z = 0$$

which is impossible. Therefore no real values of x, y, z exist.

Answer: (D)

- (iii) The given relation implies

$$2x + y + 3z = 0$$

$$-x + 2y + az = 0$$

$$\text{and} \quad x - 3y + 5z = 0$$

Therefore

$$\begin{vmatrix} 2 & 1 & 3 \\ -1 & 2 & a \\ 1 & -3 & 5 \end{vmatrix} = 0 \quad (\text{Volume 1, Chapter 8})$$

$$\Rightarrow 2(10 + 3a) - 1(-5 - a) + 3(3 - 2) = 0$$

$$\Rightarrow 7a + 28 = 0$$

$$\Rightarrow a = -4$$

Answer: (C)

- 3. Passage:** Let $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{a} + t\vec{c}$ be lines passing through the point $\vec{a}$ and parallel to the vectors $\vec{b}$ and $\vec{c}$, respectively. Then the equations

$$\vec{r} = \vec{a} + t \left(\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right)$$

and
$$\vec{r} = \vec{a} + t \left(\frac{-\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right)$$

are the angle bisectors of the two lines. Answer the following three questions.

- (i) If $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then the vector $\vec{a} + \vec{b}$ bisects the angle between $\vec{a}$ and $\vec{b}$, if

(A) $|\vec{a}| = |\vec{b}|$

(B) angle between $\vec{a}$ and $\vec{b}$ is 0 or π

(C) $\vec{a}, \vec{b}$ form adjacent sides of a parallelogram

(D) $\vec{a}, \vec{b}$ form adjacent sides of a rectangle

- (ii) The vector $-\vec{i} + \vec{j} - \vec{k}$ bisects angle between the vectors $\vec{a}$ and $3\vec{i} + 4\vec{j}$. Then, unit vector along $\vec{a}$ is

(A) $\frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$

(B) $\frac{2\vec{i} - \vec{j} + 2\vec{k}}{3}$

(C) $\frac{1}{15}(-11\vec{i} + 10\vec{j} - 2\vec{k})$

(D) $\frac{1}{15}(11\vec{i} + 10\vec{j} + 2\vec{k})$

- (iii) The position vectors of the points A and B are, respectively, $\vec{i} + 3\vec{j} - 2\vec{k}$ and $3\vec{i} + \vec{j} - 2\vec{k}$. If O is the origin and the internal bisector of $\angle AOB$ meets the line AB in P , then $\vec{OP}$ is equal to

(A) $\vec{i} + \vec{j} - \vec{k}$

(B) $2(\vec{i} + \vec{j} - \vec{k})$

(C) $\frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$

(D) $\frac{2}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$

Solution:

- (i) $\vec{a} + \vec{b}$ is a diagonal of the parallelogram for which $\vec{a}$ and $\vec{b}$ are two adjacent sides. A diagonal of

a parallelogram bisects the angle between two adjacent sides if either it is a rhombus or a square. In any case the two adjacent sides must be of equal lengths. Hence $\vec{a} + \vec{b}$ bisects angle between $\vec{a}$ and $\vec{b}$ if $|\vec{a}| = |\vec{b}|$.

Answer: (A)

- (ii) Let

$$\frac{\vec{a}}{|\vec{a}|} = x\vec{i} + y\vec{j} + z\vec{k}$$

so that $x^2 + y^2 + z^2 = 1$. Let $\vec{b} = 3\vec{i} + 4\vec{j}$. Therefore by Theorem 5.33, equation of the bisector of the angle between $\vec{a}$ and $\vec{b}$ is

$$\begin{aligned} \vec{r} &= t \left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} \right) \\ &= t \left(x\vec{i} + y\vec{j} + z\vec{k} + \frac{3\vec{i} + 4\vec{j}}{5} \right) \end{aligned}$$

Therefore

$$-\vec{i} + \vec{j} - \vec{k} = t \left(x\vec{i} + y\vec{j} + z\vec{k} + \frac{3\vec{i} + 4\vec{j}}{5} \right)$$

for some scalar t . Hence

$$t \left(x + \frac{3}{5} \right) = -1 \Rightarrow x = -\frac{1}{t} - \frac{3}{5}$$

$$t \left(y + \frac{4}{5} \right) = 1 \Rightarrow y = \frac{1}{t} - \frac{4}{5}$$

$$tz = -1 \Rightarrow z = -\frac{1}{t}$$

Now

$$x^2 + y^2 + z^2 = 1$$

$$\Rightarrow (3t + 5)^2 + (5 - 4t)^2 + 25 = 25t^2$$

$$\Rightarrow t = \frac{15}{2}$$

Therefore

$$x = \frac{-11}{15}, y = \frac{10}{15} \quad \text{and} \quad z = \frac{-2}{15}$$

Answer: (C)

- (iii) $|\vec{OA}| = |\vec{OB}| = \sqrt{14} \Rightarrow \triangle AOB$ is an isosceles triangle. Hence P is the mid-point of AB . Therefore

$$\begin{aligned} \vec{OP} &= \frac{1}{2}(\vec{OA} + \vec{OB}) \\ &= 2(\vec{i} + \vec{j} - \vec{k}) \end{aligned}$$

Answer: (B)

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true

1. Statement I: Let $ABCD$ be a trapezium in which AB and DC are parallel sides. The diagonals AC and BD intersect in O . If M and N are the mid-points of AB and CD , then M, O, N are collinear.

Statement II: If $\vec{a}$ and $\vec{b}$ are non-collinear vectors (i.e., not parallel), then every vector $\vec{r}$ in the plane of $\vec{a}$ and $\vec{b}$ can be expressed as $\vec{r} = x\vec{a} + y\vec{b}$ where x and y are scalars uniquely.

Solution: Statement II is Theorem 5.15. Take the point O as origin and let $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$ so that $\vec{OC} = x\vec{a}$ and $\vec{OD} = y\vec{b}$, where x and y are scalars (Figure 5.53).

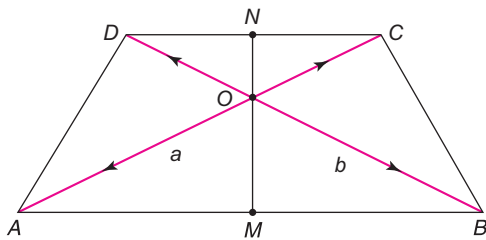


FIGURE 5.53 Assertion–reasoning type question 1.

Since M and N are mid-points of AB and CD respectively

$$\vec{OM} = \frac{\vec{a} + \vec{b}}{2}$$

and

$$\vec{ON} = \frac{x\vec{a} + y\vec{b}}{2}$$

Since $\vec{DC}$ is parallel to $\vec{AB}$ we have

$$x\vec{a} - y\vec{b} = \lambda(\vec{b} - \vec{a})$$

Again, since $\vec{a}$ and $\vec{b}$ are non-collinear, by Statement II

$$x = -\lambda = y$$

Therefore

$$\vec{ON} = -\lambda \frac{(\vec{a} + \vec{b})}{2} = -\lambda \vec{OM}$$

Therefore the points M, O, N are collinear.

Answer: (A)

2. Statement I: If the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar, then the points with position vectors $\vec{a} - 2\vec{b} + 3\vec{c}$, $2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$ are collinear.

Statement II: Three points P, Q, R are collinear if and only if $\vec{PQ} = \lambda \vec{QR}$ for some scalar λ .

Solution: Statement II is true according to Definition 5.17. Let A, B and C be the three points in Statement I so that

$$\vec{AB} = \vec{a} + 5\vec{b} - 7\vec{c}$$

and

$$\begin{aligned} \vec{BC} &= -2\vec{a} - 10\vec{b} + 14\vec{c} \\ &= -2\vec{AB} \end{aligned}$$

Therefore A, B, C are collinear points.

Answer: (A)

3. Statement I: Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors. Then the vectors $3\vec{a} - 7\vec{b} - 4\vec{c}$, $3\vec{a} - 2\vec{b} + \vec{c}$ and $\vec{a} + \vec{b} + 2\vec{c}$ are linearly dependent.

Statement II: Three vectors are linearly dependent if and only if one of them is a linear combination of the other two vectors.

Solution: Proof of Statement II: Suppose $\vec{a}, \vec{b}$ and $\vec{c}$ are linearly dependent. Therefore there exist scalars x, y, z not all zero such that

$$x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$$

If $z \neq 0$, then

$$\vec{c} = \left(\frac{-x}{z}\right)\vec{a} + \left(\frac{-y}{z}\right)\vec{b}$$

so that $\vec{c}$ is a linear combination of $\vec{a}$ and $\vec{b}$. Conversely suppose if $x\vec{a} + y\vec{b} = \vec{c}$, then

$$x\vec{a} + y\vec{b} + (-1)\vec{c} = \vec{0}$$

and $-1 \neq 0$. Hence $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent. Hence Statement II is true.

Let the vectors in Statement I be $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ respectively. Let $\vec{\gamma} = x\vec{\alpha} + y\vec{\beta}$. That is,

$$\vec{a} + \vec{b} + 2\vec{c} = x(3\vec{a} - 7\vec{b} - 4\vec{c}) + y(3\vec{a} - 2\vec{b} + \vec{c})$$

On equating the corresponding coefficients both sides we get

$$3x + 3y = 1 \quad (5.35)$$

$$-7x - 2y = 1 \quad (5.36)$$

$$-4x + y = 2 \quad (5.37)$$

Solving Eqs. (5.35) and (5.37), we have

$$x = -\frac{1}{3} \quad \text{and} \quad y = \frac{2}{3}$$

which also satisfy Eq. (5.36). Therefore

$$\vec{\gamma} = -\frac{1}{3}\vec{\alpha} + \frac{2}{3}\vec{\beta}$$

Hence by Statement II, $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are linearly dependent.

Answer: (A)

Integer Answer Type Questions

1. If the vectors $-2\vec{i} + 3\vec{j} + y\vec{k}$ and $x\vec{i} - 6\vec{j} + 2\vec{k}$ are collinear vectors, then the value of $x - y$ is ____.

Solution: Since the vectors are collinear, there exists scalar λ such that

$$-2\vec{i} + 3\vec{j} + y\vec{k} = \lambda(x\vec{i} - 6\vec{j} + 2\vec{k})$$

Therefore $\lambda x = -2$, $-6\lambda = 3$ and $2\lambda = y$. Now $\lambda = -1/2$ implies that $x = 4$, $y = -1$. Therefore $x - y = 5$.

Answer: 5

2. A, B, C are three points whose position vectors are, respectively, $\vec{a} - 2\vec{b} + 3\vec{c}$, $2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$ where $\vec{a}, \vec{b}$ are non-coplanar vectors. If $\overline{AC} = m\overline{AB}$, then the value of $-m$ is ____.

Solution: We have

$$\begin{aligned} \overline{AC} &= m\overline{AB} \\ \Rightarrow -\vec{a} - 5\vec{b} + 7\vec{c} &= m(\vec{a} + 5\vec{b} - 7\vec{c}) \\ &= -m(-\vec{a} - 5\vec{b} + 7\vec{c}) \\ \Rightarrow -m &= 1 \end{aligned}$$

Answer: 1

3. If the vectors $3\vec{i} + 3\vec{j} + \sqrt{3}\vec{k}$, $\vec{i} + \vec{k}$ and $\sqrt{3}\vec{i} + \sqrt{3}\vec{j} + x\vec{k}$ are coplanar, then the value of x is ____.

Solution: Since the given vectors are coplanar, they are linearly dependent (Theorem 5.19). Hence

$$\begin{vmatrix} 3 & 3 & \sqrt{3} \\ 1 & 0 & 1 \\ \sqrt{3} & \sqrt{3} & \lambda \end{vmatrix} = 0 \quad (\text{Theorem 5.26})$$

Therefore

$$3(0 - \sqrt{3}) - 3(\lambda - \sqrt{3}) + \sqrt{3}(\sqrt{3} - 0) = 0$$

That is

$$-3\sqrt{3} - 3\lambda + 3\sqrt{3} + 3 = 0$$

Therefore $\lambda = 1$.

Answer: 1

4. Let $\vec{a} = \vec{i} + \lambda\vec{j} + \vec{k}$ and $\vec{b} = \vec{i} + \vec{j} + \vec{k}$. If $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$ then the value of λ is ____.

Solution: Since

$$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$$

by Theorem 5.7 the vectors $\vec{a}$ and $\vec{b}$ are like vectors. Therefore there exists positive scalar x such that

$$\vec{a} = x\vec{b}$$

On equating the corresponding coefficients of both sides we get $x = 1$, $x = \lambda$. Therefore $\lambda = 1$.

Answer: 1

5. Let A, B, C be the vertices of a triangle whose position vectors are, respectively, $\vec{a}, \vec{b}$ and $\vec{c}$. M is the mid-point of the side AB . If T is a point on the segment MC whose position vector is

$$\frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

then the ratio $CT:TM = \lambda:1$ where λ is equal to ____.

Solution: Clearly T is the centroid of $\triangle ABC$. Hence $CT:TM = 2:1$.

Answer: 2

SUMMARY

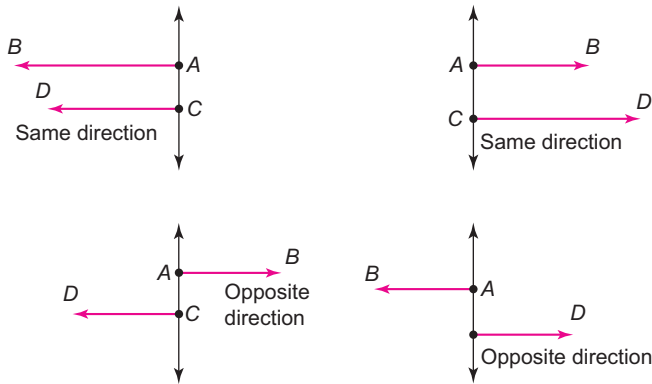
5.1 Directed line segment: If A and B are distinct points, then the ordered pair (A, B) is called directed line segment and is denoted by $\overrightarrow{AB}$.

5.2 Support of a directed line segment: For the directed line segment $\overrightarrow{AB}$, the straight line AB is called support.

QUICK LOOK

Many directed line segments may have the same support.

5.3 Same direction or opposite direction: Let $\overrightarrow{AB}$ and $\overrightarrow{CD}$ be two directed line segments. Consider the line $\overleftrightarrow{AC}$. If the supports of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel and both B and D lie in the same half plane determined by the line $\overleftrightarrow{AC}$, then $\overrightarrow{AB}$ and $\overrightarrow{CD}$ said to have the same direction. If B and D lie on opposite half planes of $\overleftrightarrow{AC}$, then $\overrightarrow{AB}$ and $\overrightarrow{CD}$ said to have opposite directions.



QUICK LOOK

$\overrightarrow{AB}$ and $\overrightarrow{BA}$ are of opposite direction.

5.4 Magnitude or length: The distance AB is called magnitude of the directed line segment $\overrightarrow{AB}$ and it is denoted by $|\overrightarrow{AB}|$. Note that $|\overrightarrow{AB}|$ is equal to $|\overrightarrow{BA}|$.

5.5 Equivalent directed line segments: Two directed line segments $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are said to be equivalent if they have same direction and same magnitude.

5.6 Theorem: Let S be the set of all directed line segments. Define relation R on S as $R = \{(\overrightarrow{AB}, \overrightarrow{CD}) | \overrightarrow{AB} \text{ and } \overrightarrow{CD} \text{ are equivalent}\}$

Then R is an equivalence relation in S , which divides S into equivalence classes having the following three properties.

- (1) Every directed line segment must belong to exactly one equivalence classes.
- (2) Any two directed line segments belonging to the same equivalence class are equivalent (i.e., same direction and same magnitude).
- (3) Directed line segments belonging to different equivalence classes are *not* equivalent.

Note: For the above theorem see Chapter 1 of Volume 1.

5.7 Vector: Each equivalence class discussed in the above theorem is called a *vector*. Usually we denote vectors by $\vec{a}, \vec{b}, \vec{c}, \vec{r}, \vec{p}$, etc.

QUICK LOOK

- (i) Vector $\vec{a}$ means it is an equivalence class of all equivalent directed line segments.
- (ii) Vector $\vec{a}$ can be represented by any directed line segment $\overrightarrow{AB}$ belonging to the equivalence class $\vec{a}$. For example, if $\overrightarrow{AB}$ belongs to $\vec{a}$, then we write $\vec{a} = \overrightarrow{AB}$. The magnitude of $\vec{a}$ is defined to be $|\overrightarrow{AB}|$.
- (iii) Two directed line segments $\overrightarrow{AB}$ and $\overrightarrow{CD}$ represent the same vector if and only if they have the same direction and same magnitude.

5.8 Zero vector or null vector: The collection of all directed line segments $\overrightarrow{PP}$, where P is any point from a vector is called zero vector and is denoted by $\vec{0}$.

QUICK LOOK

Zero vector has no direction and has zero magnitude. We can consider $\vec{0}$ in any direction.

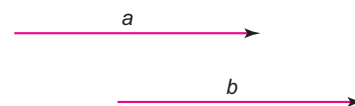
5.9 Support of $\vec{a}$: The support of any directed line segment $\overrightarrow{AB}$ belonging to $\vec{a}$ is considered to be the support of $\vec{a}$.

5.10 Unit vector: The equivalence class of all equivalent directed line segments of magnitude 1 is called unit vector.

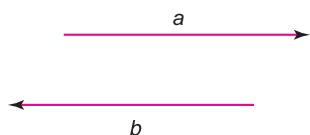
5.11 Parallel or collinear vectors: Two vectors $\vec{a}$ and $\vec{b}$ are said to be collinear vectors (parallel vectors) if any support of $\vec{a}$ is parallel to any support of $\vec{b}$.

5.12 Like and unlike vectors: Two collinear vectors are said to be like or unlike vectors according as they have the same direction or opposite directions.

The vectors $\vec{a}$ and $\vec{b}$ in the figure below are like vectors.



Also $\vec{a}$ and $\vec{b}$ in the following figure are unlike vectors.



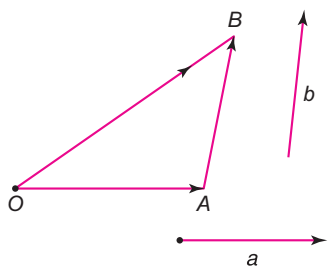
5.13 Unit vector in the direction of $\vec{a}$: If a non-zero vector $\vec{a}$ and a unit vector $\vec{e}$ are like vectors, then $\vec{e}$ is said to be unit vector in the direction of $\vec{a}$.

5.14 Coplanar and non-coplanar vectors: A set of vectors (or collection of vectors) are said to be coplanar vectors if their supports are in the same plane or parallel to the same plane, Vectors which are *not coplanar* are called non-coplanar.

QUICK LOOK

- (i) Any two vectors are coplanar because if $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$, then the lines $\overrightarrow{OA}$ and $\overrightarrow{OB}$ determine a plane.
- (ii) If $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$ and $\vec{c} = \overrightarrow{OC}$, then $\vec{a}$, $\vec{b}$, $\vec{c}$ are coplanar if and only if the four points O , A , B and C are coplanar. Hence coplanarity or non-coplanarity arises only when there are more than two vectors.

5.15 Vector addition: If $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{AB}$, then $\vec{a} + \vec{b}$ is defined as the vector $\overrightarrow{OB}$. $\vec{a} + (-\vec{b})$ is denoted by $\vec{a} - \vec{b}$.



5.16 Theorem: Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors. Then

- (i) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (Commutative law)
- (ii) $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ (Associative law)
- (iii) $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$ for all vectors $\vec{a}$
- (iv) $\vec{a} + (-\vec{a}) = -\vec{a} + \vec{a} = \vec{0}$ ($-\vec{a}$ is called additive inverse of $\vec{a}$)
- (v) $\vec{a} + \vec{x} = \vec{b} \Leftrightarrow \vec{x} = \vec{b} - \vec{a}$

5.17 Theorem: Let $\vec{a}$ and $\vec{b}$ be two vectors. Then

- (i) $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ and equality holds if and only if $\vec{a}$ and $\vec{b}$ are like vectors.
- (ii) $||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}|$ and equality holds if and only if $\vec{a}$ and $\vec{b}$ are like vectors and $|\vec{a}| \geq |\vec{b}|$.

5.18 Multiplication of a vector with a scalar: Let $\vec{a}$ be any vector and λ any scalar (i.e., real number). Then

- (i) $\lambda\vec{a}$ is equal to $\vec{0}$ if either $\lambda = 0$ or $\vec{a} = \vec{0}$.
- (ii) If $\lambda > 0$ and $\vec{a} \neq \vec{0}$, then $\lambda\vec{a}$ is defined to be the vector in the direction of $\vec{a}$ with magnitude $\lambda|\vec{a}|$.
- (iii) If $\lambda < 0$ and $\vec{a} \neq \vec{0}$, then $\lambda\vec{a}$ is defined to be the vector in the opposite direction of $\vec{a}$ with magnitude $(-\lambda)|\vec{a}|$.

QUICK LOOK

- (i) $|\lambda\vec{a}| = |\lambda||\vec{a}|$
- (ii) $\vec{a}$ and $\lambda\vec{a}$ are collinear vectors.

5.19 Theorem: Let $\vec{a}$, $\vec{b}$ be vectors and m , n be scalars. Then the following hold.

- (i) $m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$
- (ii) $m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$
- (iii) $(m + n)\vec{a} = m\vec{a} + n\vec{a}$
- (iv) $m(-\vec{a}) = (-m)\vec{a} = -(m\vec{a})$
- (v) $1\vec{a} = \vec{a}$

QUICK LOOK

If $\vec{a} \neq \vec{0}$, then $\vec{a}/|\vec{a}|$ is a unit vector in the direction of $\vec{a}$ and $-(\vec{a}/|\vec{a}|)$ is a unit vector in the opposite direction of $\vec{a}$.

5.20 Position vector of a point: Let 'O' be a fixed point in the space, called origin of reference. If A is any point in the space, then the vector $\overrightarrow{OA}$ is called the position vector of the point A with reference to the origin.

QUICK LOOK

If O is the origin and $\vec{a}$ is any vector, then there exists point A in the space such that $\overrightarrow{OA} = \vec{a}$.

5.21 Theorem: If $\vec{a}$ and $\vec{b}$ are the position vectors of the points A and B , respectively, then $\vec{AB} = \vec{b} - \vec{a}$.

5.22 Division formula (Section formula): Let $\vec{a}$ and $\vec{b}$ be the position vectors of two points A and B , respectively. Then the position vector of the point which divides $\vec{AB}$ in the ratio $m:n$ is

$$\frac{n\vec{a} + m\vec{b}}{m + n}$$

The same formula is valid for both internal and external division. The position vector of the mid-point is

$$\frac{\vec{a} + \vec{b}}{2}$$



5.23 Results:

(i) the position vector of the centroid G of $\triangle ABC$ is

$$\frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are, respectively, the position vectors of A , B and C .

(ii) The segments joining the vertices A , B , C and D of a tetrahedron $ABCD$ to the centroids of the opposite faces are concurrent. This point is denoted by G and is called the centroid or centroid of the tetrahedron. G divides the segment joining the vertex to the centroid of opposite face in the ratio 3:1 and its position vector is

$$\frac{1}{4}(\vec{a} + \vec{b} + \vec{c} + \vec{d})$$

where $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are position vectors of A , B , C and D , respectively.

(iii) If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are the position vectors of A , B and C , respectively, of $\triangle ABC$ with respect to circumcentre as origin, then the position vector of the orthocentre of $\triangle ABC$ is $\vec{a} + \vec{b} + \vec{c}$.

5.24 Theorem: Three points with position vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are collinear if and only if there exist scalars x , y , and z not all zero such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$ and $x + y + z = 0$.

5.25 Plane representation: If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors, then every vector $\vec{r}$ in the plane determined by $\vec{a}$ and $\vec{b}$ can be represented as

$\vec{r} = x\vec{a} + y\vec{b}$ in one and only one way. In particular, if $\vec{i}$ and $\vec{j}$ are unit vectors along positive directions of x and y axes, respectively, and $\vec{r}$ is any vector in the plane, then $\vec{r} = x\vec{i} + y\vec{j}$, where (x, y) are the coordinates of the point P such that $\vec{OP} = \vec{r}$.

QUICK LOOK

In the Cartesian plane, point $P(x, y)$ is identified with the vector $\vec{OP} = x\vec{i} + y\vec{j}$ where ' O ' is the origin of the coordinate axes.

5.26 Space representation: Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{r}$ be any vector. Then there exists unique triad (x, y, z) of scalars such that $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$. In particular, let $\vec{i}$, $\vec{j}$ and $\vec{k}$ be unit vectors along the positive directions of the x , y and z axes respectively. Let $\vec{r}$ be any vector. Then there exist unique triad (x, y, z) of scalars such that $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and (x, y, z) are coordinates of the point P such that $\vec{r} = \vec{OP}$. Hence in the space, we identify vector $x\vec{i} + y\vec{j} + z\vec{k}$ as the point (x, y, z) .

5.27 Linear combination: If $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots, \vec{a}_n$ are vectors and $x_1, x_2, \dots, x_n$ are scalars, then the vector $x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n$ is called a linear combination of $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$.

QUICK LOOK

- (i) If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors then every vector in the plane of $\vec{a}$ and $\vec{b}$ can be written as a linear combination of $\vec{a}$ and $\vec{b}$ uniquely.
- (ii) Every vector can be uniquely represented as a linear combination of three non-coplanar vectors.

5.28 Linear independence and dependence: Vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are said to be linearly dependent, if there exist scalars $x_1, x_2, \dots, x_n$, not all zero (that is at least one $x_i \neq 0$) such that $x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n = \vec{0}$. Vectors which are not linearly dependent are called linearly independent.

QUICK LOOK

Vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly independent if and only if

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n = \vec{0} \\ \Rightarrow x_1 = x_2 = \dots = x_n = 0$$

5.29 Theorem: Vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly dependent if and only if one of them is a linear combination of the remaining.

- 5.30** (i) Two vectors are non-collinear if and only if they are linearly independent.
 (ii) Three vectors are non-coplanar if and only if they are linearly independent.



QUICK LOOK

- (i) Two vectors are collinear if and only if they are linearly dependent.
 (ii) Three vectors are coplanar if and only if they are dependent.
 (iii) In the three-dimensional space the maximum number of linearly independent vectors (the vectors may change) is three. That is, more than three vectors are linearly dependent.
 (iv) If a finite set of vectors contains zero vector then they are linearly dependent.

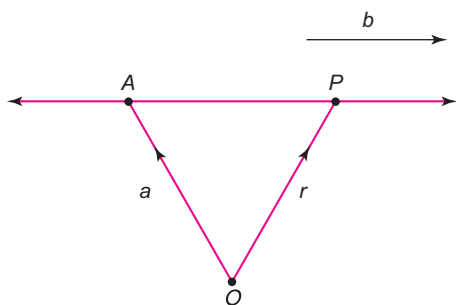
5.31 Theorem (Important): Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ and $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$. Then $\vec{a}, \vec{b}, \vec{c}$ are linear dependent (hence coplanar) if and only if the

$$\det \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

Note: The above condition also holds good even if $\vec{a}, \vec{b}, \vec{c}$ are represented as linear combination of any three non-coplanar vectors, that is by taking some $\vec{l}, \vec{m}$ and $\vec{n}$ in place of $\vec{i}, \vec{j}$ and $\vec{k}$, respectively.

5.32 Parametric vector equation of a line:

- (i) The equation of the line passing through point whose position vector is $\vec{a}$ and parallel to a vector $\vec{b}$ is $\vec{r} = \vec{a} + t\vec{b}$, $t \in \mathbb{R}$.
 (ii) Vector equation of the line passing through two points $\vec{a}$ and $\vec{b}$ is $\vec{r} = (1-t)\vec{a} + t\vec{b}$, $t \in \mathbb{R}$.



5.33 Plane equations:

- (i) The vector equation of a plane passing through a point whose position vector is $\vec{a}$ and parallel to two vectors $\vec{b}$ and $\vec{c}$ is

$$\vec{r} = \vec{a} + t\vec{b} + s\vec{c}, t \text{ and } s \in \mathbb{R}$$

- (ii) Equation of the plane passing through two points $\vec{a}$ and $\vec{b}$ and parallel to a vector $\vec{c}$ is

$$\vec{r} = (1-t)\vec{a} + t\vec{b} + s\vec{c}, t \text{ and } s \in \mathbb{R}$$

- (iii) Equation of the plane passing through three points $\vec{a}, \vec{b}, \vec{c}$ is

$$\vec{r} = (1-t-s)\vec{a} + t\vec{b} + s\vec{c}, t \text{ and } s \in \mathbb{R}$$

5.34 Theorem: Three points A, B and C with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$, respectively, are collinear if and only if there exist scalars x, y, z not all zero such that

$$x + y + z = 0$$

$$\text{and } x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$$

5.35 Theorem: Four points A, B, C and D with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar if and only if there exist scalars x, y, z and u not all zero such that

$$x + y + z + u = 0$$

$$\text{and } x\vec{a} + y\vec{b} + z\vec{c} + u\vec{d} = \vec{0}$$

5.36 Angle between two vectors: Let $\vec{a} = \vec{OA}$ and $\vec{b} = \vec{OB}$. Then the measure of $\angle AOB$ which lies between 0 and π is defined to be the angle between $\vec{a}$ and $\vec{b}$ and is denoted by $(\vec{a}, \vec{b})$.



QUICK LOOK

$0 \leq (\vec{a}, \vec{b}) \leq \pi$. The left equality holds if and only if $\vec{a}, \vec{b}$ are like vectors and right equality holds if and only if $\vec{a}, \vec{b}$ are unlike vectors.

5.37 Note: Let $\vec{a}$ and $\vec{b}$ be two vectors. Then

$$(1) (\vec{a}, \vec{b}) = (\vec{b}, \vec{a}) = (-\vec{a}, -\vec{b}) = (-\vec{b}, -\vec{a})$$

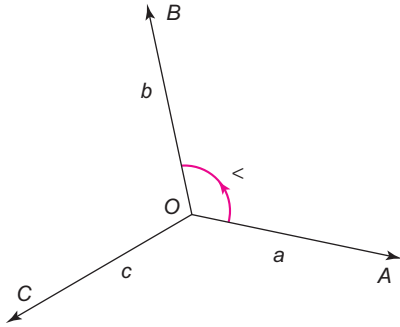
$$(2) (\vec{a}, -\vec{b}) = \pi - (\vec{a}, \vec{b}) = (-\vec{a}, \vec{b})$$

$$(3) (\vec{a}, \vec{b}) = (x\vec{a}, y\vec{b}) \text{ for all positive } x \text{ and } y$$

$$(4) (-x\vec{a}, y\vec{b}) = \pi - (x\vec{a}, y\vec{b}) = (x\vec{c}, -y\vec{b}) \text{ for all positive } x \text{ and } y$$

5.38 If $(\vec{a}, \vec{b}) = 90^\circ$, then $\vec{a}$ and $\vec{b}$ are called perpendicular or orthogonal vectors.

5.39 Right- and left-handed systems: Let $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$, $\vec{c} = \overrightarrow{OC}$. When observed from the point C, if the angle of rotation of $\overrightarrow{OA}$ towards $\overrightarrow{OB}$ in anticlockwise sense does not exceed π , then $(\vec{a}, \vec{b}, \vec{c})$ is called right-handed system. System which is not right handed is called left-handed system.



Note:

- (1) In a system $(\vec{a}, \vec{b}, \vec{c})$ if two vectors are interchanged then the system will change.
- (2) In a system if a vector is replaced by its additive inverse, the system will change.
- (3) If the vectors are cyclically permuted, the system does not change.

5.40 Components of a vector: Let $(\vec{a}, \vec{b}, \vec{c})$ be a right-handed system of non-coplanar vectors. If $\vec{r}$ is any vector then $\vec{r}$ can be expressed as $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$, and x, y, z are called components of $\vec{r}$ w.r.t. the system $(\vec{a}, \vec{b}, \vec{c})$. In particular, if $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ then x, y, z are x -component, y -component and z -component of $\vec{r}$, respectively.



QUICK LOOK

- (i) If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, then $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.
- (ii) Let $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$ so that

$$\overrightarrow{OP} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$$

$$\text{and } \overrightarrow{OQ} = x_2\vec{i} + y_2\vec{j} + z_2\vec{k}$$

Hence

$$\overrightarrow{PQ} = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} + (z_2 - z_1)\vec{k}$$

$$\text{and } |\overrightarrow{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

5.41 Direction cosines (DCS) and direction ratios (DRS) of a vector: Let $\overrightarrow{X'OX}$, $\overrightarrow{Y'OY}$ and $\overrightarrow{Z'OZ}$ be mutually perpendicular lines called coordinate axes. Let $\vec{i}, \vec{j}, \vec{k}$ be unit vectors in the directions of $\overrightarrow{OX}$, $\overrightarrow{OY}$ and $\overrightarrow{OZ}$ such that $(\vec{i}, \vec{j}, \vec{k})$ is a right-handed system. Let $\vec{a}$ be any vector, and let $\alpha = (\vec{a}, \vec{i})$, $\beta = (\vec{a}, \vec{j})$ and $\gamma = (\vec{a}, \vec{k})$ (i.e., the angles made by $\vec{a}$ with $\vec{i}, \vec{j}$ and $\vec{k}$ respectively). Then the ordered triad $(\cos \alpha, \cos \beta, \cos \gamma)$ is called direction cosines of $\vec{a}$.

5.42 If l, m, n are DCS of $\vec{a}$ w.r.t. $(\vec{i}, \vec{j}, \vec{k})$, then $l^2 + m^2 + n^2 = 1$.



QUICK LOOK

- (i) If (l, m, n) are DCS of $\vec{a}$, then $l\vec{i} + m\vec{j} + n\vec{k}$ is a unit vector parallel to $\vec{a}$. In a sense unit vector parallel to $\vec{a}$ given the DCS of $\vec{a}$.
- (ii) If (l, m, n) are DCS of a vector $\vec{a}$ and $\lambda \neq 0$, then $(\lambda l, \lambda m, \lambda n)$ are called DRS of $\vec{a}$.

EXERCISES

Single Correct Choice Type Questions

1. In a regular hexagon $ABCDEF$, vector $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF}$ is equal to
(A) $2\overrightarrow{AD}$ (B) $3\overrightarrow{AD}$ (C) $4\overrightarrow{AD}$ (D) $6\overrightarrow{AD}$
2. $ABCD$ is a quadrilateral and not a parallelogram, described in this order. Let P, Q, R and S be the mid-points of the sides AB, BC, CD and AD , respectively. Then $PQRS$ is a
(A) parallelogram (B) rectangle
(C) square (D) rhombus
3. Let A and B be points whose position vectors are, respectively, $\vec{i} + 3\vec{j} - 2\vec{k}$ and $3\vec{i} + \vec{j} - 2\vec{k}$. The bisector of the angle AOB (' O ' is the origin) meets the segment AB internally in C . Then the vector $\overrightarrow{OC}$ is equal to
(A) $\vec{i} + \vec{j} + \vec{k}$ (B) $2(\vec{i} + \vec{j} - \vec{k})$
(C) $\vec{i} + \vec{j} - \vec{k}$ (D) $4(\vec{i} + \vec{j} + \vec{k})$
4. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors. If
 $\vec{a} + 3\vec{b} + 4\vec{c} = x(\vec{a} + 5\vec{b} - 2\vec{c}) + y(\vec{a} - 2\vec{b} + 3\vec{c})$
 $+ z(6\vec{a} + 14\vec{b} + 4\vec{c})$

then

- (A) $x = -3, y = 2, z = 1$ (B) $x = 3, y = -2, z = 1$
 (C) $x = -3, y = -2, z = 1$ (D) $x = 3, y = 2, z = 1$

5. $\vec{a}$ is a vector in the Cartesian plane (in which $\vec{OX}$ and $\vec{OY}$ are coordinate axes) has components $2p$ and 1 . The axes are rotated in the same plane about origin, through the angle 45° in the counterclockwise sense. The components $\vec{a}$ with respect to the new system of axes are $2p + 1$ and 1 . Then the value of p is

- (A) 0 (B) -1 or $\frac{1}{3}$ (C) $-\frac{1}{3}$ or 1 (D) 1 or -1

6. Let the points A, B and C be represented by the vectors $\vec{i} + \vec{j}, -4\vec{i} + 2\vec{j}$ and $2\vec{i} + 3\vec{j}$, respectively, in the Cartesian plane. Then the length of the median through B of $\triangle ABC$ is

- (A) $\frac{5}{2}$ (B) $\frac{2}{5}$ (C) $\frac{11}{3}$ (D) $\frac{11}{2}$

7. $ABCD$ is a tetrahedron and O is a point in side the tetrahedron. The lines AO, BO, CO and DO meet the opposite faces in P, Q, R and S , respectively. Then

- (A) $\frac{AP}{OP} + \frac{BQ}{OQ} + \frac{CR}{OR} + \frac{DS}{OS} = 1$
 (B) $\frac{OP}{AP} + \frac{OQ}{BQ} + \frac{OR}{CR} + \frac{OS}{DS} = 1$
 (C) $\frac{AO}{OP} + \frac{BO}{OQ} + \frac{CO}{OR} + \frac{DO}{OS} = 1$
 (D) $\frac{OP}{OA} + \frac{OQ}{OB} + \frac{OR}{OC} + \frac{OS}{OD} = 1$

8. The line passing through the point $2\vec{a} + 3\vec{b}$ and parallel to the vector $\vec{c}$ cuts the plane

$$\vec{r} = \vec{a} - \vec{b} + t(\vec{a} + \vec{b} - \vec{c}) + s(\vec{a} + \vec{c} - \vec{b})$$

where t and s are scalars in the point whose position vector is

- (A) $2\vec{a} + 3\vec{b} + 4\vec{c}$ (B) $2\vec{a} - 3\vec{b} + 4\vec{c}$
 (C) $2\vec{a} - 3\vec{b} - 4\vec{c}$ (D) $2\vec{a} + 3\vec{b} - 4\vec{c}$

9. P is the mid-point of the side AD of the parallelogram $ABCD$. The line BP meets the diagonal AC in Q and the line CD in R . Then $RQ:QB$ is equal to
- (A) 2:1 (B) 1:2 (C) 3:1 (D) 1:3

10. The three points $\vec{i} - 2\vec{j} + 3\vec{k}, 2\vec{i} + 3\vec{j} - 4\vec{k}$ and $-7\vec{j} + 10\vec{k}$
- (A) are collinear
 (B) form the vertices of a right-angled triangle

- (C) form an isosceles triangle
 (D) non-collinear

11. $OXYZ$ is a rectangular coordinate system. Keeping Z -axis fixed, the X - and Y -axes are rotated in their plane through angle 45° in the anti-clockwise sense. If the components of a vector with respect new system are $2\sqrt{2}, 3\sqrt{2}$ and 4 , then the components of the same vector with respect to the original systems are

- (A) $5, -1, 4$ (B) $-1, 5, 4$
 (C) $-1, -5, 4\sqrt{2}$ (D) $-1, 5, 4\sqrt{2}$

12. In $\triangle ABC$, $\vec{AB} = 2\vec{i} + 4\vec{j} + 4\vec{k}$ and $\vec{AC} = 2\vec{i} + 2\vec{j} + \vec{k}$. Then the length of the median through A is

- (A) $\frac{\sqrt{77}}{2}$ (B) 5 (C) $5\sqrt{2}$ (D) $10\sqrt{2}$

13. $\vec{a}, \vec{b}, \vec{c}$ are non-zero vectors such that they are pairwise non-collinear. It is given that $\vec{a} + 2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 3\vec{c}$ is collinear with $\vec{a}$. Then the vector $\vec{a} + 2\vec{b} + 3\vec{c}$ is

- (A) equal to $\vec{0}$ (B) parallel to $\vec{b}$
 (C) parallel to $\vec{a}$ (D) parallel to $\vec{c}$

14. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors and $\vec{x} = 2\vec{a} + 3\vec{b} - \vec{c}$, $\vec{y} = \vec{a} - 2\vec{b} + 2\vec{c}$ and $\vec{z} = -2\vec{a} + \vec{b} - 2\vec{c}$. Then the vector $3\vec{a} - \vec{b} + 2\vec{c}$ in terms of $\vec{x}, \vec{y}$ and $\vec{z}$ is

- (A) $5\vec{x} + 2\vec{y} + 3\vec{z}$ (B) $3\vec{x} + 5\vec{y} + 2\vec{z}$
 (C) $2\vec{x} + 5\vec{y} + 3\vec{z}$ (D) $5\vec{x} + 3\vec{y} + 2\vec{z}$

15. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{p} = 5\vec{a} + 6\vec{b} + 7\vec{c}, \vec{q} = 7\vec{a} - 8\vec{b} + 9\vec{c}$ and $\vec{r} = 3\vec{a} + 20\vec{b} + 5\vec{c}$. If $\vec{r} = x\vec{p} + y\vec{q}$ where x and y are scalars, then

- (A) $x = 1, y = 1$ (B) $x + y = 1$
 (C) $x = \frac{3}{2}, y = 1$ (D) $x = 2, y = -1$

16. A, B, C and D are four points whose position vectors are, respectively, $4\vec{i} + 5\vec{j} + \vec{k}, -\vec{i} - \vec{k}, 5\vec{i} + 9\vec{j} + 4\vec{k}$ and $-\vec{i} + \vec{k}$. If $\vec{AB} = x\vec{AC} + y\vec{AD}$, then

- (A) $x = \frac{-2}{3}, y = \frac{2}{3}$
 (B) $x = \frac{2}{3}, y = \frac{-2}{3}$
 (C) $x = \frac{3}{2}, y = \frac{2}{3}$

- (D) real values of x and y do not exist

17. Let A and B be two points whose position vectors are $5\vec{i}$ and $5\vec{j}$, respectively. A point P divides the line joining A and B in the ratio $AP:PB = \lambda:1$. If the position vector of P has magnitude less than or equal to $\sqrt{37}$, then
- (A) $\frac{-1}{6} \leq \lambda \leq \frac{1}{6}$ (B) $\lambda \leq -6$ or $\lambda \geq \frac{-1}{6}$
 (C) $\lambda \geq 0$ (D) $0 < \lambda < 1$
18. In $\triangle ABC$, E is the mid-point of the median AD . The line BE meets the side AC in F . Then AC equals
- (A) $3 AF$ (B) $2 AF$
 (C) $3 CF$ (D) $\frac{3}{2} CF$
19. $ABCD$ is a quadrilateral. P, Q, R and S are the mid-points of the sides AB, BC, CD and AD , respectively. The lines PR and QS intersect in E . If ' O ' is any point other than any point mentioned above, then the vector $\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD}$ is equal to
- (A) $8\overrightarrow{OE}$ (B) $6\overrightarrow{OE}$
 (C) $4\overrightarrow{OE}$ (D) $3\overrightarrow{OE}$
20. $ABCDE$ is pentagon in which AB, CD are parallel and AE, BC are parallel. The lines BE and AD intersect in X . Then the ratio $AX:XD$ is equal to
- (A) 2:5 (B) 5:2 (C) 2:3 (D) 3:2

ANSWERS

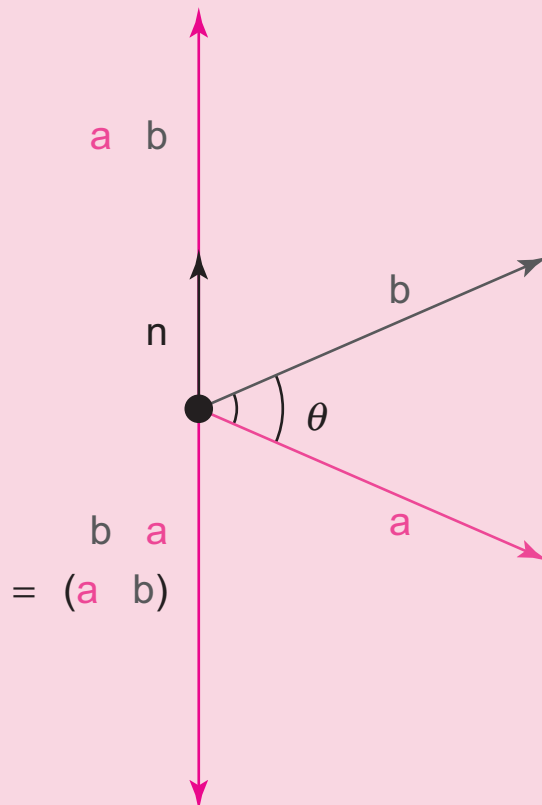
Single Correct Choice Type Questions

- | | |
|---------|---------|
| 1. (B) | 11. (B) |
| 2. (A) | 12. (A) |
| 3. (B) | 13. (A) |
| 4. (C) | 14. (C) |
| 5. (C) | 15. (D) |
| 6. (D) | 16. (D) |
| 7. (B) | 17. (B) |
| 8. (D) | 18. (A) |
| 9. (A) | 19. (C) |
| 10. (A) | 20. (A) |

Multiplication of Vectors

6

Multiplication of Vectors



Contents

- 6.1 *Scalar or Dot Product*
- 6.2 *Vector Equations of a Plane and a Sphere*
- 6.3 *Vector or Cross Product*
- 6.4 *Vector Areas*
- 6.5 *Scalar Triple Product*
- 6.6 *Vector Triple Product and Multiple Products*

Worked-Out Problems
Summary
Exercises
Answers

Vector multiplication may concern any of the following articles:

- **Dot product:** Also known as the “scalar product,” an operation which takes two vectors and returns a **scalar** quantity.
- **Cross product:** Also known as the “vector product,” a binary operation on two vectors that results in another **vector**.
- **Triple products:** Products involving three vectors.
- **Multiple cross products:** Products involving more than three vectors.

In the previous chapter, we have discussed the operations of addition of vectors and multiplication of vectors by scalars. In this chapter, we consider two more important operations on vectors, namely, scalar product (or dot product) and vector product (or cross product) and discuss certain applications and properties of these. These will be used to derive vector equations of a plane in several forms.

6.1 | Scalar or Dot Product

In this section, we introduce the concept of scalar product or dot product of two vectors and present a geometrical interpretation which will be used in deriving the vector equation of a plane in the normal form and the orthogonal projection of a vector on another vector. Let us begin with the following.

DEFINITION 6.1 Let $\vec{a}$ and $\vec{b}$ be vectors. Then we define

$$\vec{a} \times \vec{b} = 0 \quad \text{if } \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0}$$

$$\text{and} \quad \vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\cos\theta \quad \text{if } \vec{a} \neq \vec{0} \text{ and } \vec{b} \neq \vec{0}$$

where θ is the angle between the vectors $\vec{a}$ and $\vec{b}$. $\vec{a} \times \vec{b}$ is called the *dot product* of $\vec{a}$ and $\vec{b}$, just because a dot $\times$ is used here. Since $\vec{a} \times \vec{b}$ is defined as a scalar, it is also called the *scalar product* of $\vec{a}$ and $\vec{b}$.

For any two vectors $\vec{a}, \vec{b}$ let us recall that the angle between $\vec{a}$ and $\vec{b}$ is denoted by $(\vec{a}, \vec{b})$ (see Definition 5.22, Chapter 5) and we note that

$$(\vec{a}, \vec{b}) = \pi - (\vec{a}, -\vec{b}) = (\vec{a}, \vec{b})$$

$$\text{and} \quad (\vec{a}, \vec{b}) = (\vec{b}, \vec{a})$$

In this notation, we have

$$\begin{aligned} \vec{a} \times \vec{b} &= |\vec{a}||\vec{b}|\cos(\vec{a}, \vec{b}) \\ &= |\vec{b}||\vec{a}|\cos(\vec{b}, \vec{a}) \\ &= \vec{b} \times \vec{a} \end{aligned}$$

for any non-zero vectors $\vec{a}$ and $\vec{b}$. If either of $\vec{a}$ or $\vec{b}$ is a zero vector, then clearly by Definition 6.1 we have $\vec{a} \times \vec{b} = 0 = \vec{b} \times \vec{a}$ and therefore, for any two vectors $\vec{a}$ and $\vec{b}$,

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$$

That is, the scalar product operation is commutative. However, we cannot talk on the associativity of this operation, since $\vec{a} \times \vec{b}$ is scalar and hence $(\vec{a} \times \vec{b}) \times \vec{c}$ is not defined whereas $(\vec{a} \times \vec{b}) \vec{c}$ has meaning according to the concept of multiplication of a vector with a scalar.

The following are easy verifications and listed for the sake of ready reference on the angle $(\vec{a}, \vec{b})$.



QUICK LOOK 1

The following hold for any non-zero vectors $\vec{a}$ and $\vec{b}$.

$$1. \quad 0 < (\vec{a}, \vec{b}) < 180^\circ$$

$$2. \quad 0 < (\vec{a}, \vec{b}) < 90^\circ \quad \cos(\vec{a}, \vec{b}) > 0 \quad \vec{a} \times \vec{b} > 0$$

$$3. \quad (\vec{a}, \vec{b}) = 90^\circ \quad \vec{a} \times \vec{b} = 0$$

$$4. \quad 90^\circ < (\vec{a}, \vec{b}) < 180^\circ \quad \cos(\vec{a}, \vec{b}) < 0 \quad \vec{a} \times \vec{b} < 0$$

Before going for the various properties of the scalar product of two vectors, we will first give a geometrical interpretation.

DEFINITION 6.2 Let $\vec{a}$ and $\vec{b}$ be non-zero vectors and O, A and B be points in the space such that $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$. Let Q be the foot of the perpendicular from B to the line OA . (Figure 6.1). Then

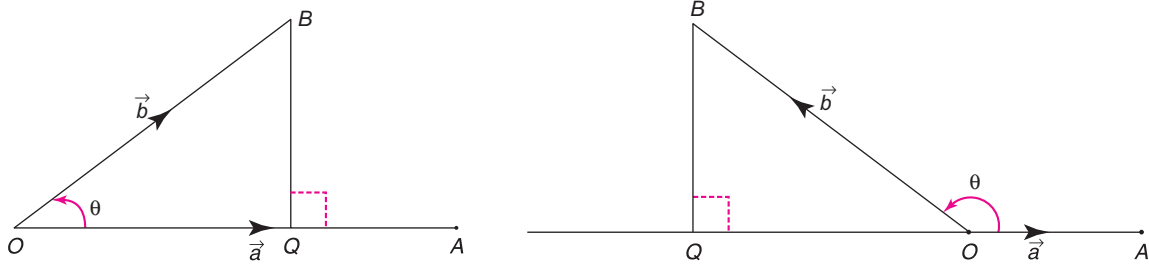


FIGURE 6.1 Definition 6.2.

1. The *scalar component* (or, simply, the *component* or the *projection*) of $\vec{b}$ along $\vec{a}$ is defined as OQ if $(\vec{a}, \vec{b}) \leq 90^\circ$ and as $-OQ$ if $(\vec{a}, \vec{b}) > 90^\circ$.
2. The *vector component* (or the *orthogonal projection*) of $\vec{b}$ on $\vec{a}$ is defined as the vector $\overrightarrow{OQ}$.

QUICK LOOK 2

1. The component of $\vec{b}$ on $\vec{a}$ is a scalar and the orthogonal projection of $\vec{b}$ on $\vec{a}$ is a vector.
2. If $\vec{e}$ is a unit vector and $\vec{b}$ is any vector, then $(\vec{b} \cdot \vec{e}) \vec{e}$ is the orthogonal projection of $\vec{b}$ on $\vec{e}$ and, for any non-zero vector $\vec{a}$,
3. Let A, B, C and D be coplanar points, $\vec{a} = \overrightarrow{AB}$ and $\vec{b} = \overrightarrow{CD}$. If the perpendiculars from C and D to the line AB meet AB at P and Q , respectively, then $\overrightarrow{PQ}$ is the orthogonal projection of $\vec{b}$ on $\vec{a}$. See Figure 6.3.

$$\left(\vec{b} \cdot \frac{\vec{a}}{|\vec{a}|} \right) \frac{\vec{a}}{|\vec{a}|}$$

is the orthogonal projection of $\vec{b}$ on $\vec{a}$. See Figure 6.2.

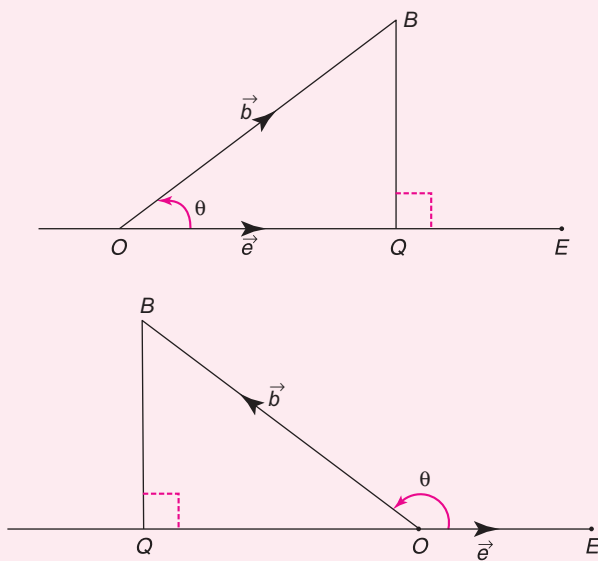


FIGURE 6.2

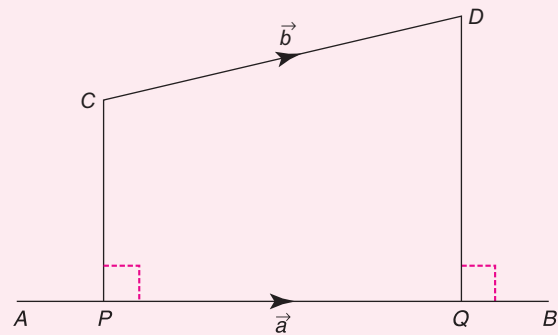


FIGURE 6.3

4. If $\vec{b}$ is perpendicular to $\vec{a}$, then the orthogonal projection of $\vec{b}$ on $\vec{a}$ is the zero vector.
5. If $\vec{b}$ is parallel to $\vec{a}$, then the orthogonal projection of $\vec{b}$ on $\vec{a}$ is $\vec{b}$.

THEOREM 6.1 Let $\vec{a}$ and $\vec{b}$ be non-zero non-parallel vectors. Then

1. the projection (or the scalar component) of $\vec{b}$ on $\vec{a}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$.
2. the orthogonal projection (or the vector component) of $\vec{b}$ on $\vec{a}$ is $\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$.

PROOF Choose points O, A and B such that $\vec{a} = \overrightarrow{OA}$ and $\vec{b} = \overrightarrow{OB}$. Let $\angle AOB = \theta$. Draw perpendicular BP to meet $\overrightarrow{OA}$ at P . Then OP or $-OP$ is the projection of $\vec{b}$ on $\vec{a}$, according as $\theta < 90^\circ$ or $\theta > 90^\circ$ as shown in Figure 6.4. Then, if $\theta < 90^\circ$,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = |\vec{a}| |\overrightarrow{OB}| \cos \theta = |\vec{a}| OP$$

and hence

$$OP = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

If $\theta > 90^\circ$, then

$$\begin{aligned} \vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos \theta \\ &= |\vec{a}| |\vec{b}| (\cos(180^\circ - \theta)) \\ &= |\vec{a}| OB \cos(180^\circ - \theta) \\ &= |\vec{a}| OP \end{aligned}$$

and hence

$$OP = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

Thus $(\vec{a} \cdot \vec{b})/|\vec{a}|$ is the projection of $\vec{b}$ on $\vec{a}$.

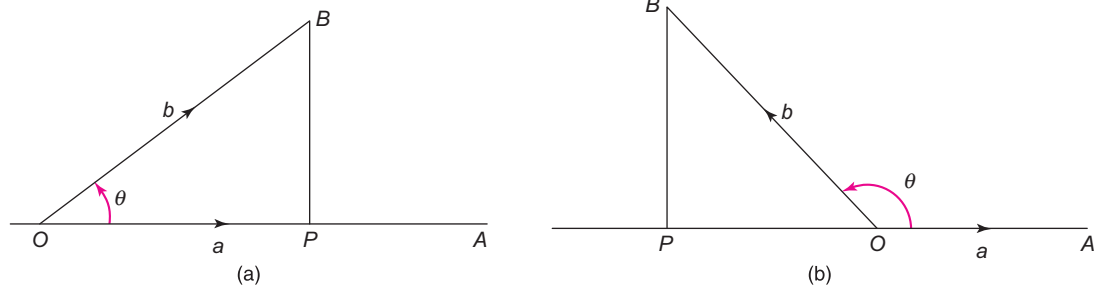


FIGURE 6.4 Theorem 6.1.

In the above, if $\vec{e}$ is the unit vector, in the direction of $\vec{a}$, that is,

$$\vec{e} = \frac{\vec{a}}{|\vec{a}|}$$

then

$$\overrightarrow{OP} = (\vec{b} \cdot \vec{e}) \vec{e} = \vec{b} \cdot \frac{\vec{a}}{|\vec{a}|} \cdot \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$$

 QUICK LOOK 3

From Figure 6.4(b), we have

$$\overrightarrow{PB} = \overrightarrow{OB} - \overrightarrow{OP} = \vec{b} - \frac{(\vec{b} \cdot \vec{a})\vec{a}}{|\vec{a}|^2}$$

and therefore

$$\vec{b} - \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$$

is the orthogonal projection of $\vec{b}$ in the direction perpendicular to $\vec{a}$ in the AOB plane.

THEOREM 6.2 The following hold for any vectors $\vec{a}$ and $\vec{b}$ and scalars α and β .

1. $(\alpha\vec{a}) \cdot \vec{b} = \alpha(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (\alpha\vec{b})$
2. $(\alpha\vec{a}) \cdot (\beta\vec{b}) = \alpha\beta(\vec{a} \cdot \vec{b})$
3. $(-\vec{a}) \cdot \vec{b} = -(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (-\vec{b})$
4. $(-\vec{a}) \cdot (-\vec{b}) = \vec{a} \cdot \vec{b}$

PROOF

These are trivial if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\alpha = 0$ or $\beta = 0$. Therefore, we can assume that $\vec{a}$ and $\vec{b}$ are non-zero vectors and α and β are non-zero scalars. Recall that $(\vec{a}, \vec{b})$ denotes the angle between $\vec{a}$ and $\vec{b}$ and that

$$(\vec{a}, \vec{b}) = \pi - (\vec{a}, -\vec{b}) = (\vec{a}, \vec{b})$$

If $\alpha > 0$, then

$$\begin{aligned} (\alpha\vec{a}) \cdot \vec{b} &= |\alpha\vec{a}| |\vec{b}| \cos(\alpha\vec{a}, \vec{b}) \\ &= \alpha |\vec{a}| |\vec{b}| \cos(\vec{a}, \vec{b}) \\ &= \alpha(\vec{a} \cdot \vec{b}) \end{aligned}$$

If $\alpha < 0$, then

$$\begin{aligned} (\alpha\vec{a}) \cdot \vec{b} &= |\alpha\vec{a}| |\vec{b}| \cos(\alpha\vec{a}, \vec{b}) \\ &= |\alpha| |\vec{a}| |\vec{b}| \cos[\pi - (\vec{a}, \vec{b})] \\ &= |\alpha| |\vec{a}| |\vec{b}| \cos[\pi - (\vec{a}, \vec{b})] \quad (\text{since } \alpha < 0) \\ &= (-\alpha) |\vec{a}| |\vec{b}| [-\cos(\vec{a}, \vec{b})] \\ &= \alpha |\vec{a}| |\vec{b}| \cos(\vec{a}, \vec{b}) \\ &= \alpha(\vec{a} \cdot \vec{b}) \end{aligned}$$

This proves (1). To prove (2), consider

$$\begin{aligned} (\alpha\vec{a}) \cdot (\beta\vec{b}) &= \alpha[\vec{a} \cdot (\beta\vec{b})] \\ &= \alpha[(\beta\vec{b}) \cdot \vec{a}] \\ &= \alpha\beta(\vec{b} \cdot \vec{a}) \\ &= \alpha\beta(\vec{a} \cdot \vec{b}) \end{aligned}$$

(3) and (4) follow from (1) by taking $\alpha = -1$ and/or $\beta = -1$. ■

QUICK LOOK 4

1. For any vectors $\vec{a}$ and $\vec{b}$,

$$\cos(\vec{a}, \vec{b}) = 0 \quad (\vec{a}, \vec{b}) = 90^\circ$$

Therefore

$$\vec{a} \times \vec{b} = 0 \quad \vec{a} = \vec{0} \quad \text{or} \quad \vec{b} = \vec{0} \quad \text{or} \quad \vec{a} \parallel \vec{b}$$

2. If $\vec{a}$ and $\vec{b}$ are like vectors, then $(\vec{a}, \vec{b}) = 0^\circ$ and hence $\cos(\vec{a}, \vec{b}) = 1$, so that $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|$.

3. If $\vec{a}$ and $\vec{b}$ are unlike vectors, then $(\vec{a}, \vec{b}) = 180^\circ$ and hence $\cos(\vec{a}, \vec{b}) = -1$, so that $\vec{a} \times \vec{b} = -|\vec{a}||\vec{b}|$.

4. $\vec{a} \times \vec{a} = |\vec{a}||\vec{a}| = |\vec{a}|^2$. Here afterwards, $\vec{a}^2$ means $\vec{a} \times \vec{a}$.

THEOREM 6.3 The following hold for any vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

$$1. \vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

$$2. (\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$$

PROOF

We prove this in the case when $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar vectors and skip the general case. Let us suppose that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-zero coplanar vectors and O , A , B and C are coplanar points such that

$$\vec{a} = \overrightarrow{OA}, \vec{b} = \overrightarrow{OB} \quad \text{and} \quad \vec{c} = \overrightarrow{BC}$$

Then

$$\vec{b} + \vec{c} = \overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OC}$$

Draw perpendiculars BP and CQ from B and C to the line $\overrightarrow{OA}$ to meet at P and Q respectively. Let $\vec{e}$ be the unit vector in the direction of $\vec{a}$; that is,

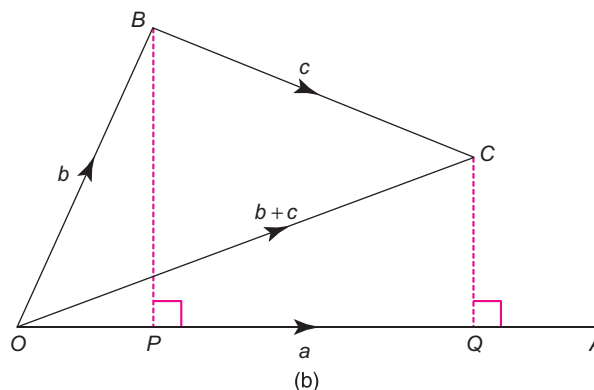
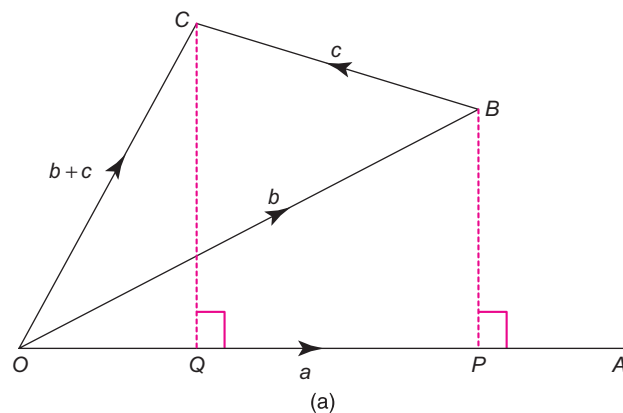


FIGURE 6.5 Theorem 6.3.

$$\vec{e} = \frac{1}{|\vec{a}|} \vec{a}$$

Now, we have that

$\overline{OP}$ = the orthogonal projection of $\vec{b}$ on $\vec{a}$

$\overline{PQ}$ = the orthogonal projection of $\vec{c}$ on $\vec{a}$

and

$\overline{OQ}$ = the orthogonal projection of $\vec{b} + \vec{c}$ on $\vec{a}$

and therefore, by Quick Look 2 [part (2)], we have

$$\overline{OP} = (\vec{b} \times \vec{e}) \vec{e}$$

$$\overline{PQ} = (\vec{c} \times \vec{e}) \vec{e}$$

and

$$\overline{OQ} = [(\vec{b} + \vec{c}) \times \vec{e}] \vec{e}$$

Now

$$\begin{aligned} [(\vec{b} + \vec{c}) \times \vec{e}] \vec{e} &= \overline{OQ} \\ &= \overline{OP} + \overline{PQ} \\ &= (\vec{b} \times \vec{e}) \vec{e} + (\vec{c} \times \vec{e}) \vec{e} \\ &= [(\vec{b} \times \vec{e}) + (\vec{c} \times \vec{e})] \vec{e} \end{aligned}$$

Since $\vec{e}$ is a unit vector, it follows that

$$(\vec{b} + \vec{c}) \times \vec{e} = \vec{b} \times \vec{e} + \vec{c} \times \vec{e}$$

Now,

$$\begin{aligned} \vec{a} \times (\vec{b} + \vec{c}) &= (\vec{b} + \vec{c}) \times \vec{a} \\ &= (\vec{b} + \vec{c}) \times (|\vec{a}| \vec{e}) \\ &= |\vec{a}| [(\vec{b} + \vec{c}) \times \vec{e}] \\ &= |\vec{a}| (\vec{b} \times \vec{e} + \vec{c} \times \vec{e}) \\ &= |\vec{a}| (\vec{b} \times \vec{e}) + |\vec{a}| (\vec{c} \times \vec{e}) \\ &= \vec{b} \times (|\vec{a}| \vec{e}) + \vec{c} \times (|\vec{a}| \vec{e}) \\ &= \vec{b} \times \vec{a} + \vec{c} \times \vec{a} \\ &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c} \end{aligned}$$

Also,

$$\begin{aligned} (\vec{a} + \vec{b}) \times \vec{c} &= \vec{c} \times (\vec{a} + \vec{b}) \\ &= \vec{c} \times \vec{a} + \vec{c} \times \vec{b} \\ &= \vec{a} \times \vec{c} + \vec{b} \times \vec{c} \end{aligned}$$

COROLLARY 6.1

For any vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n, \vec{b}_1, \vec{b}_2, \dots, \vec{b}_m$,

$$\sum_{i=1}^n \vec{a}_i \times \sum_{j=1}^m \vec{b}_j = \sum_{i,j} (\vec{a}_i \times \vec{b}_j)$$

COROLLARY 6.2 For any vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$,

$$\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

and

$$(\vec{a} + \vec{b}) \times \vec{c} = (\vec{a} \times \vec{c}) + (\vec{b} \times \vec{c})$$

COROLLARY 6.3 For any non-zero vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$,

$$\vec{a} \times \vec{c} = \vec{b} \times \vec{c} \quad \vec{a} = \vec{b}$$

or

$\vec{a} - \vec{b}$ is perpendicular to $\vec{c}$

PROOF We have

$$\vec{a} \times \vec{c} = \vec{b} \times \vec{c}$$

$$(\vec{a} - \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} - \vec{b} \times \vec{c} = \vec{0}$$

$$\vec{a} - \vec{b} = \vec{0} \text{ or } \vec{a} - \vec{b} \text{ is perpendicular to } \vec{c}$$

The following are direct consequences of the above and of the facts that $|\vec{a}|^2 = \vec{a} \cdot \vec{a}$ and $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$ for any vectors $\vec{a}$ and $\vec{b}$.

COROLLARY 6.4 The following hold for any vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

1. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b}$
2. $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$
3. $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$
4. $|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$

THEOREM 6.4 The following hold for any vectors $\vec{a}$ and $\vec{b}$.

1. $\vec{a} \times \vec{a} = \vec{0}$
2. $\vec{a} \times \vec{a} = \vec{0} \quad \vec{a} = \vec{0}$
3. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$
4. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \quad \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0} \text{ or } \vec{a} \text{ and } \vec{b} \text{ are parallel}$
5. $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$
6. $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}| \quad \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0} \text{ or } \vec{a} \text{ and } \vec{b} \text{ are like vectors}$
7. $||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}| \leq |\vec{a}| + |\vec{b}|$

PROOF (1) and (2) follow from the fact that

$$\vec{a} \times \vec{a} = |\vec{a}|^2 \sin 0 = \vec{0}$$

3. We have

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$$

$$= |\vec{a}||\vec{b}| |\sin \theta|$$

$$|\vec{a}||\vec{b}|$$

since $|\sin \theta| \leq 1$.

4. If $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then

$$|\vec{a} \times \vec{b}| = 0 = |\vec{a}| |\vec{b}|$$

Now, for any non-zero vectors $\vec{a}$ and $\vec{b}$,

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \quad |\cos \theta| = 1 \quad \theta = 0 \quad \text{or} \quad \pi$$

where θ is the angle between $\vec{a}$ and $\vec{b}$.

5. We have

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} \\ &= |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}| |\vec{b}| \cos \theta \\ &= (|\vec{a}| + |\vec{b}|)^2 \end{aligned}$$

and hence $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$.

6. If θ is the angle between $\vec{a}$ and $\vec{b}$, then

$$\begin{aligned} |\vec{a} + \vec{b}| &= |\vec{a}| + |\vec{b}| \\ \vec{a} \times \vec{b} &= |\vec{a}| |\vec{b}| \sin \theta \\ |\vec{a}| |\vec{b}| (\cos \theta - 1) &= 0 \\ \vec{a} = \vec{0} \quad \text{or} \quad \vec{b} = \vec{0} \quad \text{or} \quad \cos \theta &= 1 \\ \vec{a} = \vec{0} \quad \text{or} \quad \vec{b} = \vec{0} \quad \text{or} \quad \theta &= 0^\circ \\ \vec{a} = \vec{0} \quad \text{or} \quad \vec{b} = \vec{0} \quad \text{or} \quad \vec{a} \text{ and } \vec{b} &\text{ are like vectors} \end{aligned}$$

7. Consider

$$|\vec{a}| = |\vec{a} - \vec{b} + \vec{b}| \quad |\vec{a} - \vec{b}| + |\vec{b}|$$

and hence

$$|\vec{a}| - |\vec{b}| \leq |\vec{a} - \vec{b}|$$

Similarly,

$$|\vec{b}| - |\vec{a}| \leq |\vec{b} - \vec{a}| = |\vec{a} - \vec{b}|$$

Therefore

$$||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}| \leq |\vec{a}| + |\vec{b}|$$



Next, we express the dot product of given two vectors in terms of the components on the unit vectors along the coordinate axes with respect to a given rectangular coordinate system $OXYZ$. If $\vec{i}, \vec{j}$ and $\vec{k}$ be unit vectors along $\overrightarrow{OX}, \overrightarrow{OY}$ and $\overrightarrow{OZ}$, respectively, then, since $\cos 0^\circ = 1$ and $\cos 90^\circ = 0$, we have

$$\vec{i} \cdot \vec{i} = 1 = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k}$$

and

$$\vec{i} \cdot \vec{j} = 0 = \vec{j} \cdot \vec{k} = \vec{k} \cdot \vec{i}$$

Recall that $(\vec{i}, \vec{j}, \vec{k})$ is called a *triad* of orthogonal unit vectors.

THEOREM 6.5 Let $\vec{a}$ be a vector and $(\vec{i}, \vec{j}, \vec{k})$ be a triad of orthogonal unit vectors. Then

$$\vec{a} = (\vec{a} \times \vec{i})\vec{i} + (\vec{a} \times \vec{j})\vec{j} + (\vec{a} \times \vec{k})\vec{k}$$

PROOF We have learnt earlier that there exists a triad (a_1, a_2, a_3) of real numbers such that

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

Now,

$$\begin{aligned}\vec{a} \times \vec{i} &= (a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) \times \vec{i} \\ &= a_1(\vec{i} \times \vec{i}) + a_2(\vec{j} \times \vec{i}) + a_3(\vec{k} \times \vec{i}) \\ &= a_2\vec{k} - a_3\vec{j}\end{aligned}$$

and, similarly $\vec{a} \times \vec{j} = a_3\vec{i} - a_1\vec{k}$ and $\vec{a} \times \vec{k} = a_1\vec{j} - a_2\vec{i}$. Thus

$$\vec{a} = (\vec{a} \times \vec{i})\vec{i} + (\vec{a} \times \vec{j})\vec{j} + (\vec{a} \times \vec{k})\vec{k}$$

THEOREM 6.6 Let $(\vec{i}, \vec{j}, \vec{k})$ be a triad of orthogonal unit vectors and

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

and

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

Then the dot product of $\vec{a}$ and $\vec{b}$ is given by

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

and in particular,

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

PROOF From Theorem 6.5, we get

$$\begin{aligned}\vec{a} \cdot \vec{b} &= \vec{a} \cdot (b_1\vec{i} + b_2\vec{j} + b_3\vec{k}) \\ &= b_1(\vec{a} \cdot \vec{i}) + b_2(\vec{a} \cdot \vec{j}) + b_3(\vec{a} \cdot \vec{k}) \\ &= b_1a_1 + b_2a_2 + b_3a_3 \\ &= a_1b_1 + a_2b_2 + a_3b_3\end{aligned}$$

Also, we have

$$|\vec{a}|^2 = \vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2$$

and hence

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

COROLLARY 6.5 For any triads of real numbers (a_1, a_2, a_3) and (b_1, b_2, b_3) ,

(CAUCHY-SCHWARTZ INEQUALITY)

$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

and the equality holds if and only if a_1, a_2, a_3 are proportional to b_1, b_2, b_3 .

PROOF This follows from Theorem 6.6 and Theorem 6.4 [parts (3) and (4)] by taking $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$.

THEOREM 6.7 Let $(\vec{i}, \vec{j}, \vec{k})$ be a triad of orthogonal unit vectors and

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

and

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

be two non-zero vectors. Then the angle between $\vec{a}$ and $\vec{b}$ is given by

$$\cos^{-1} \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \cos^{-1} \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

PROOF If θ is angle between $\vec{a}$ and $\vec{b}$, then

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$$

and hence

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

so that

$$\theta = \cos^{-1} \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

COROLLARY 6.6 Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ be two non-zero vectors. Then

1. $\vec{a}$ and $\vec{b}$ are perpendicular to each other if and only if $a_1b_1 + a_2b_2 + a_3b_3 = 0$.
2. $\vec{a}$ and $\vec{b}$ are parallel if and only if there exists a scalar t such that $a_1 = tb_1$, $a_2 = tb_2$ and $a_3 = tb_3$.

Example 6.1

Find the dot product of the vectors $6\vec{i} + 2\vec{j} + 3\vec{k}$ and $2\vec{i} - 9\vec{j} + 6\vec{k}$ and the angle between them.

where $\theta = (\vec{a}, \vec{b})$, the angle between $\vec{a}$ and $\vec{b}$. Now,

$$\vec{a} \cdot \vec{b} = 6 \cdot 2 + 2 \cdot (-9) + 3 \cdot 6 = 12$$

Solution: Let

$$\vec{a} = 6\vec{i} + 2\vec{j} + 3\vec{k}$$

and

$$\vec{b} = 2\vec{i} - 9\vec{j} + 6\vec{k}$$

Then

$$|\vec{a}| = \sqrt{6^2 + 2^2 + 3^2} = \sqrt{36 + 4 + 9} = 7$$

and

$$|\vec{b}| = \sqrt{2^2 + (-9)^2 + 6^2} = \sqrt{4 + 81 + 36} = 11$$

We know that

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$$

Therefore

$$\begin{aligned} \theta &= \cos^{-1} \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \\ &= \cos^{-1} \frac{12}{7 \cdot 11} \\ &= \cos^{-1} \frac{12}{77} \end{aligned}$$

Example 6.2

Let $\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{b} = 3\vec{i} - \vec{j} + 2\vec{k}$. Prove that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular to each other.

$$\begin{aligned} &= [1^2 + 2^2 + (-3)^2] - [3^2 + (-1)^2 + 2^2] \\ &= 0 \end{aligned}$$

Solution: We have, by Corollary 6.4 [part (3)], that

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$$

Therefore, $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular to each other.

Example 6.3

Find the angles of the triangle whose position vectors of the vertices are $\vec{i} + 2\vec{j} - 5\vec{k}$, $2\vec{i} + 2\vec{j} + \vec{k}$ and $2\vec{i} + \vec{j} - \vec{k}$.

Solution: Let ABC be the given triangle and

$$\vec{a} = \overrightarrow{OA} = \vec{i} + 2\vec{j} - 5\vec{k}$$

$$\vec{b} = \overrightarrow{OB} = 2\vec{i} + 2\vec{j} + \vec{k}$$

and

$$\vec{c} = \overrightarrow{OC} = 2\vec{i} + \vec{j} - \vec{k}$$

Then

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 3\vec{i} + 6\vec{k}$$

and

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \vec{j} + 4\vec{k}$$

Now $\angle A$ is the angle between $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Therefore

$$\begin{aligned} \angle A &= \cos^{-1} \frac{|\overrightarrow{AB} \times \overrightarrow{AC}|}{|\overrightarrow{AB}| |\overrightarrow{AC}|} \\ &= \cos^{-1} \frac{(-3)(1+0) - (1)(6) - 4}{\sqrt{(-3)^2 + 0^2 + 6^2} \sqrt{1^2 + (4)^2}} \end{aligned}$$

This implies

$$\angle A = \cos^{-1} \frac{7}{3\sqrt{10}}$$

Similarly, we can find that $\angle B$, the angle between $\overrightarrow{BA}$ and $\overrightarrow{BC}$ is given by

$$\begin{aligned} \angle B &= \cos^{-1} \frac{|\overrightarrow{BA} \times \overrightarrow{BC}|}{|\overrightarrow{BA}| |\overrightarrow{BC}|} \\ &= \cos^{-1} \frac{8}{\sqrt{105}} \end{aligned}$$

and

$$\begin{aligned} \angle C &= \cos^{-1} \frac{|\overrightarrow{CA} \times \overrightarrow{CB}|}{|\overrightarrow{CA}| |\overrightarrow{CB}|} \\ &= \cos^{-1} \frac{1}{\sqrt{42}} \end{aligned}$$

In the following we prove certain important geometrical results using the dot product of vectors and its properties.

THEOREM 6.8

Let A , B and C represent the angles of a triangle ABC and a , b and c represent the lengths of the sides opposite to them, respectively. Then

1. $a^2 = b^2 + c^2 - 2bc \cos A$ (Law of cosines)
2. $a = b \cos C + c \cos B$ (Projection formula)

PROOF

See Figure 6.6. Let $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{AC} = \vec{b}$ and $\overrightarrow{BA} = \vec{c}$. Then

$$|\vec{a}| = a, |\vec{b}| = b \quad \text{and} \quad |\vec{c}| = c$$

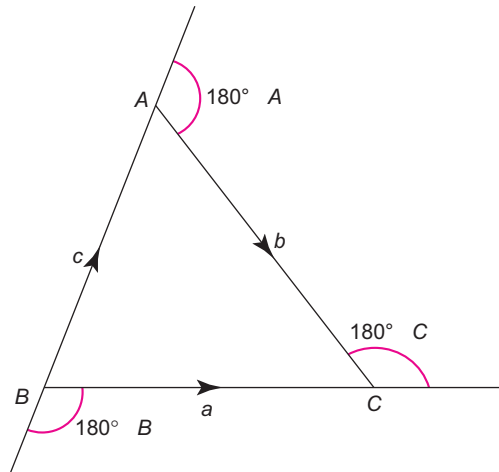


FIGURE 6.6 Theorem 6.8.

1. We have

$$(\vec{c}, \vec{b}) = 180^\circ - A$$

where $(\vec{c}, \vec{b})$ is the angle between $\vec{c}$ and $\vec{b}$. Since $\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC}$, we have $\vec{a} = \vec{c} + \vec{b}$. Therefore

$$\begin{aligned} a^2 &= |\vec{a}|^2 = |\vec{c} + \vec{b}|^2 \\ &= |\vec{c}|^2 + |\vec{b}|^2 + 2\vec{c} \cdot \vec{b} \\ &= c^2 + b^2 + 2|\vec{c}||\vec{b}|\cos(\vec{c}, \vec{b}) \\ &= c^2 + b^2 + 2cb\cos(\pi - A) \\ &= b^2 + c^2 - 2bc\cos A \end{aligned}$$

2. Consider

$$\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC} = -(\overrightarrow{CA} + \overrightarrow{AB})$$

We have

$$\begin{aligned} a^2 &= |\vec{a}|^2 \\ &= |\overrightarrow{BC}|^2 \\ &= \overrightarrow{BC} \cdot \overrightarrow{BC} \\ &= (\overrightarrow{CA} + \overrightarrow{AB}) \cdot \overrightarrow{BC} \\ &= \overrightarrow{CA} \cdot \overrightarrow{BC} + \overrightarrow{AB} \cdot \overrightarrow{BC} \\ &= |\overrightarrow{CA}||\overrightarrow{BC}|\cos(\overrightarrow{CA}, \overrightarrow{BC}) + |\overrightarrow{AB}||\overrightarrow{BC}|\cos(\overrightarrow{AB}, \overrightarrow{BC}) \\ &= ba\cos(\pi - C) + ca\cos(\pi - B) \\ &= ab\cos C + ac\cos B \end{aligned}$$

Therefore

$$a^2 = a(b\cos C + c\cos B)$$

or

$$a = b\cos C + c\cos B$$



 **Try it out** We can prove that other laws of cosines and projection formulae, namely,

1. $b^2 = c^2 + a^2 - 2ca\cos B$
2. $c^2 = a^2 + b^2 - 2ab\cos C$
3. $b = c\cos A + a\cos C$
4. $c = a\cos B + b\cos A$

THEOREM 6.9

The altitudes of a triangle are concurrent.

PROOF

Consider a triangle ABC and draw perpendiculars from A and B to BC and CA to meet them at D and E , respectively. Suppose that O is the point of intersection of AD and BE . Join C and O and produce it to meet AB at F . With reference to O , let the position vectors of A , B and C be $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively. From Figure 6.7 we have the following:

$$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -\vec{b} + \vec{c}$$

$$\overrightarrow{CA} = \overrightarrow{CO} + \overrightarrow{OA} = -\vec{c} + \vec{a}$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\vec{a} + \vec{b}$$

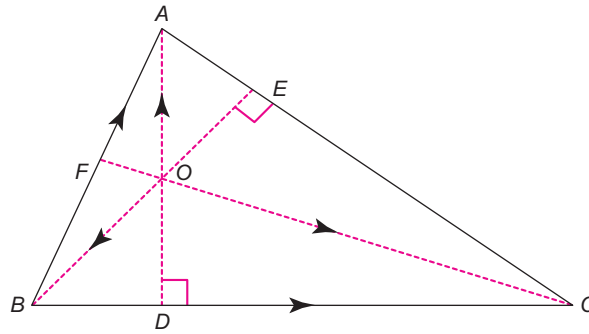


FIGURE 6.7 Theorem 6.9.

Since $AD \perp BC$, $\overrightarrow{OA} \times \overrightarrow{BC} = 0$ and hence $\vec{a} \times (\vec{c} - \vec{b}) = 0$ so that

$$\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$$

Similarly, from $BE \perp AC$, we get that

$$\vec{b} \times \vec{a} = \vec{b} \times \vec{c}$$

From these two, we have

$$\vec{a} \times \vec{c} = \vec{b} \times \vec{c}$$

and hence $(\vec{b} - \vec{a}) \times \vec{c} = 0$. Therefore $CF \perp AB$. Thus the altitudes of the triangle ABC are concurrent. ■

THEOREM 6.10

The perpendicular bisectors of the sides of a triangle are concurrent.

PROOF

Let ABC be a triangle and D , E and F be the mid-points of the sides BC , CA and AB , respectively. Let the perpendicular bisectors drawn to the BC and CA at D and E meet at O . Join O and F . Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be the position vectors of A , B , C , respectively, with respect to O . Now (see Figure 6.8), we have

$$\vec{a} = \overrightarrow{OA}, \vec{b} = \overrightarrow{OB}, \vec{c} = \overrightarrow{OC}$$

Hence

$$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -\vec{b} + \vec{c}$$

$$\overrightarrow{CA} = \overrightarrow{CO} + \overrightarrow{OA} = -\vec{c} + \vec{a}$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\vec{a} + \vec{b}$$

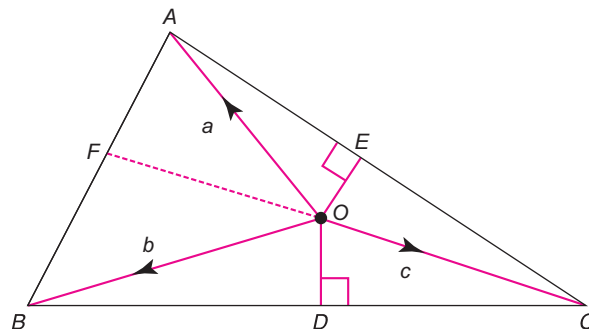


FIGURE 6.8 Theorem 6.10.

Also

$$\overrightarrow{OD} = \frac{1}{2}(\vec{b} + \vec{c})$$

$$\overrightarrow{OE} = \frac{1}{2}(\vec{c} + \vec{a})$$

$$\overrightarrow{OF} = \frac{1}{2}(\vec{a} + \vec{b})$$

Since $OD \perp BC$,

$$\frac{1}{2}(\vec{b} + \vec{c}) \times (\vec{b} + \vec{c}) = 0$$

and hence $|\vec{c}|^2 = |\vec{b}|^2$. Similarly, from $OE \perp CA$, we get $|\vec{a}|^2 = |\vec{c}|^2$. Therefore $|\vec{a}|^2 = |\vec{b}|^2$ and hence

$$\begin{aligned} \overrightarrow{OF} \times \overrightarrow{AB} &= \frac{1}{2}(\vec{a} + \vec{b}) \times (\vec{a} + \vec{b}) \\ &= \frac{1}{2}(|\vec{b}|^2 - |\vec{a}|^2) = 0 \end{aligned}$$

Thus OF is perpendicular to AB , so that OF becomes the perpendicular bisector of AB . Thus, the perpendicular bisectors of the sides of a triangle are concurrent. ■

THEOREM 6.11

Let α and β be any non-negative real numbers such that $\alpha + \beta \leq \pi$. Then

1. $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
2. $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

PROOF

Let $\vec{i}$ and $\vec{j}$ be the unit vectors along the rectangular coordinate axes OX and OY , respectively. Choose a point A in the XY plane such that $\overrightarrow{OA} = \vec{a}$ is a unit vector making an angle α with the positive X -axis. Also choose a point B in the XY plane such that $\overrightarrow{OB} = \vec{b}$ is a unit vector making an angle β with the positive X -axis. Draw perpendiculars AP and BQ to OX (Figure 6.9). Then we have

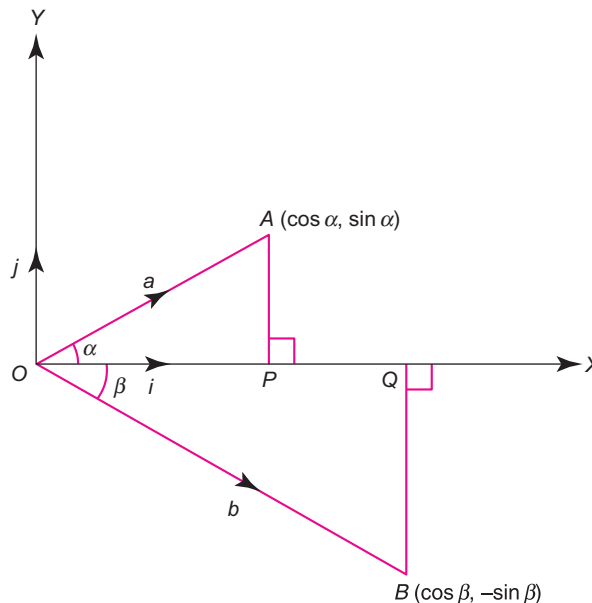


FIGURE 6.9 Theorem 6.11.

$$\begin{aligned}
 \vec{a} &= \overrightarrow{OA} \\
 &= \overrightarrow{OP} + \overrightarrow{PA} \\
 &= (OP)\vec{i} + (PA)\vec{j} \\
 &= \cos \alpha \vec{i} + \sin \alpha \vec{j} \\
 \vec{b} &= \overrightarrow{OB} \\
 &= \overrightarrow{OQ} + \overrightarrow{QB} \\
 &= (OQ)\vec{i} + (QB)\vec{j} \\
 &= \cos \beta \vec{i} + \sin \beta \vec{j}
 \end{aligned}$$

Therefore, we get that

$$\begin{aligned}
 \cos(\alpha + \beta) &= |\vec{a}| |\vec{b}| \cos(\vec{a}, \vec{b}) \quad (\text{since } |\vec{a}| = 1 = |\vec{b}|) \\
 &= \vec{a} \cdot \vec{b} \\
 &= (\cos \alpha \vec{i} + \sin \alpha \vec{j}) \cdot (\cos \beta \vec{i} + \sin \beta \vec{j}) \\
 &= \cos \alpha \cos \beta + \sin \alpha \sin \beta
 \end{aligned}$$

Similar technique yields

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

THEOREM 6.12

The angle in a semicircle is a right angle.

PROOF

Consider a circle with AOB as a diameter and O as the centre. Let P be an arbitrary point on the semicircle, as shown in Figure 6.10. With reference to O as the origin, let the position vectors of A and P be $\vec{a}$ and $\vec{p}$, respectively. Then

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \vec{p} - \vec{a}$$

and

$$\overrightarrow{BP} = \overrightarrow{OP} - \overrightarrow{OB} = \vec{p} - (-\vec{a}) = \vec{p} + \vec{a}$$

Now,

$$\begin{aligned}
 \overrightarrow{AP} \cdot \overrightarrow{BP} &= (\vec{p} - \vec{a}) \cdot (\vec{p} + \vec{a}) \\
 &= |\vec{p}|^2 - |\vec{a}|^2 \\
 &= r^2 - r^2 = 0
 \end{aligned}$$

where r is the radius of the circle. Thus AP is perpendicular to BP and hence $\angle APB = 90^\circ$.

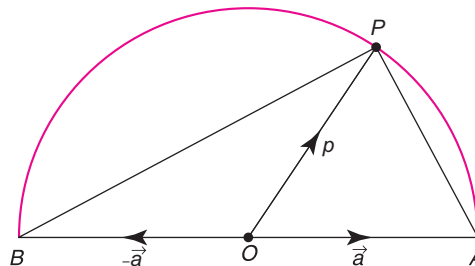


FIGURE 6.10 Theorem 6.12.

Example 6.4

In a triangle ABC , let D be the midpoint of BC . Then prove that

$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

Solution: See Figure 6.11. Consider A as the origin. Let $\vec{b}$ and $\vec{c}$ be the position vectors of B and C respectively. Then

$$\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC} = -\vec{b} + \vec{c}$$

$$\text{and } \overrightarrow{BD} = \frac{1}{2}(\vec{c} - \vec{b})$$

Also,

$$\begin{aligned} \overrightarrow{AD} &= \overrightarrow{AB} + \overrightarrow{BD} \\ &= \vec{b} + \frac{1}{2}(\vec{c} - \vec{b}) \\ &= \frac{1}{2}(\vec{b} + \vec{c}) \end{aligned}$$

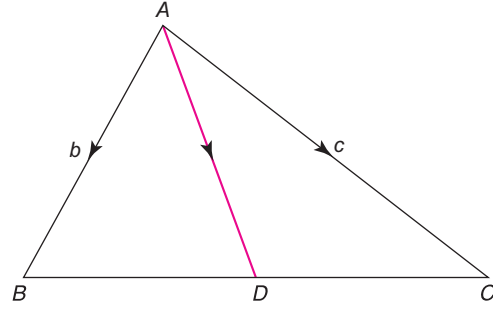


FIGURE 6.11 Example 6.4.

Now consider

$$\begin{aligned} 2(AD^2 + BD^2) &= 2(|\overrightarrow{AD}|^2 + |\overrightarrow{BD}|^2) \\ &= 2 \left(\frac{1}{4} |\vec{b} + \vec{c}|^2 + \frac{1}{4} |\vec{c} - \vec{b}|^2 \right) \end{aligned}$$

By Corollary 6.4, we have

$$\begin{aligned} 2(AD^2 + BD^2) &= \frac{1}{2} (|\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b} \cdot \vec{c} + |\vec{c}|^2 + |\vec{b}|^2 - 2\vec{b} \cdot \vec{c}) \\ &= |\vec{b}|^2 + |\vec{c}|^2 = AB^2 + AC^2 \end{aligned}$$

Example 6.5

Prove that the smaller angle between any two diagonals of a cube is

$$\cos^{-1} \frac{1}{3}$$

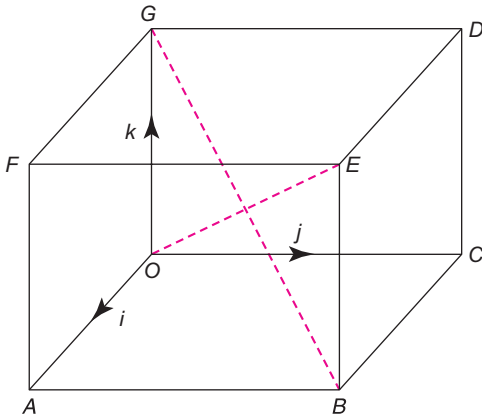


FIGURE 6.12 Example 6.5.

Solution: Let $OABCDEFG$ be a unit cube as shown in Figure 6.12. Keeping O as origin, let $\overrightarrow{OA} = \vec{i}$, $\overrightarrow{OB} = \vec{j}$ and $\overrightarrow{OG} = \vec{k}$. Then $\vec{i}, \vec{j}, \vec{k}$ form an orthogonal unit vector triad. Consider the diagonals OE and BG . From Figure 6.12, we have

$$\begin{aligned} \overrightarrow{OE} &= \overrightarrow{OB} + \overrightarrow{BE} \\ &= \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BE} \\ &= \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{CG} \quad (\text{since } \overrightarrow{AB} = \overrightarrow{OC}, \overrightarrow{BE} = \overrightarrow{CG}) \\ &= \vec{i} + \vec{j} + \vec{k} \end{aligned}$$

and

$$\begin{aligned} \overrightarrow{GB} &= \overrightarrow{GO} + \overrightarrow{OB} \\ &= -\vec{k} + \vec{j} \\ &= \vec{j} - \vec{k} \end{aligned}$$

Let θ be the smaller angle between the diagonals OE and GB . Then

$$\begin{aligned} \cos \theta &= \frac{|\overrightarrow{OE} \cdot \overrightarrow{GB}|}{|\overrightarrow{OE}| |\overrightarrow{GB}|} \\ &= \frac{(1)(0) + (1)(1) + (1)(-1)}{\sqrt{1^2 + 1^2 + 1^2} \sqrt{1^2 + 1^2}} \\ &= \frac{1}{3} \end{aligned}$$

Thus

$$\theta = \cos^{-1} \frac{1}{3}$$

6.2 | Vector Equations of a Plane and a Sphere

Using vector addition and multiplication of a vector by a scalar we have derived vector equations of a straight line and a plane in the previous chapter. In this chapter, using dot product of vectors, we derive vector equation of a plane at a given distance from the origin and perpendicular to a given vector.

THEOREM 6.13 The equation of the plane, whose unit normal drawn away from the origin is $\vec{n}$ and whose distance from the origin is a , is given by

$$\vec{r} \cdot \vec{n} = a$$

This is called the *equation of a plane in normal form*.

PROOF See Figure 6.13. Let R be an arbitrary point in the required plane and the position vector of R with respect to the origin O be $\vec{r}$. Let a perpendicular be drawn from the origin to the plane to meet the plane at N . Then $\overline{RN}$ is perpendicular to $\overline{ON}$. Since the distance from the origin to the required plane is given to be a , we have

$$a = ON = OR \cos(\vec{r}, \vec{n}) = OR |\vec{n}| \cos(\vec{r}, \vec{n})$$

since $\vec{n}$ is a unit normal. Therefore

$$a = \vec{r} \cdot \vec{n}$$

On the other hand, if P is any point such that

$$\overline{OP} \cdot \vec{n} = a$$

then

$$\begin{aligned} \overline{NP} \cdot \vec{n} &= (\overline{OP} - \overline{ON}) \cdot \vec{n} \\ &= \overline{OP} \cdot \vec{n} - \overline{ON} \cdot \vec{n} \\ &= a - a\vec{n} \cdot \vec{n} \\ &= a - a = 0 \end{aligned}$$

and hence PN is perpendicular to ON . Since the required plane is perpendicular to ON and passes through N , it follows that P lies in the required plane. Thus $\vec{r} \cdot \vec{n} = a$ is the equation of the required plane.

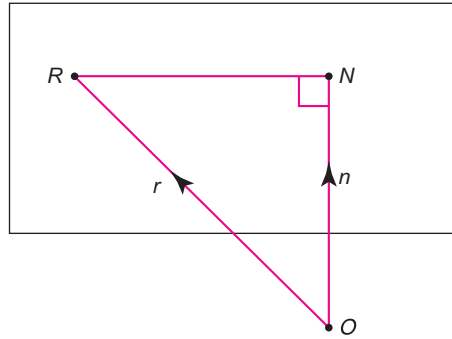


FIGURE 6.13 Theorem 6.13.

COROLLARY 6.7 Equation of the plane passing through the origin and perpendicular to the unit vector $\vec{n}$ is $\vec{r} \cdot \vec{n} = 0$.

PROOF The proof follows from Theorem 6.13 and the requirement that the plane passes through the origin and hence the distance from the origin to the plane is zero.

QUICK LOOK 5

If $R = (x, y, z)$ with respect to an orthogonal system of coordinate axes passing through the origin O and the direction cosines of the normal to the plane are l, m, n , respectively, then, with the usual notation,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

and

$$\vec{n} = l\vec{i} + m\vec{j} + n\vec{k}$$

and hence

$$\vec{r} \cdot \vec{n} = xl + ym + zn$$

Therefore the equation of the plane in Theorem 6.13 reduces to

$$xl + ym + zn = a$$

This is the Cartesian equation of the plane in normal form. Also, the Cartesian equation of the plane passing through the origin and having l, m, n as the direction cosines of its normal is

$$xl + ym + zn = 0$$

Next we derive the vector equation of a sphere. First recall that if C is a fixed point in the space and a is a fixed non-negative real number, then the set of all points P in the space such that $CP = a$ is called the sphere with centre at C and radius a .

THEOREM 6.14

Let $\vec{c}$ be the position vector of the centre C of a sphere S with radius a . Then the vector equation of S is given by

$$|\vec{r}|^2 - 2\vec{r} \cdot \vec{c} + |\vec{c}|^2 = a^2$$

PROOF

See Figure 6.14. Let P be an arbitrary point in the space and $\vec{r}$ be the position vector of P with respect to the origin O . The position vector of the centre C is given as $\vec{c}$. Then we have

$$\overrightarrow{OP} = \vec{r}$$

$$\overrightarrow{OC} = \vec{c}$$

and

$$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \vec{r} - \vec{c}$$

Now

$$P \text{ is a point on the sphere } S \quad CP = a$$

$$CP^2 = a^2$$

$$|\vec{r} - \vec{c}|^2 = a^2$$

$$|\vec{r}|^2 + |\vec{c}|^2 - 2\vec{r} \cdot \vec{c} = a^2$$

Thus the vector equation of the sphere with radius a and centre at C with a position vector $\vec{c}$ is

$$|\vec{r}|^2 - 2\vec{r} \cdot \vec{c} + |\vec{c}|^2 = a^2$$

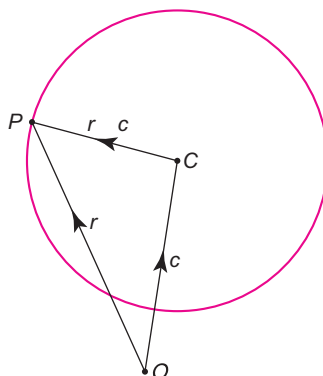


FIGURE 6.14 Theorem 6.14.

COROLLARY 6.8 If the origin O lies on the sphere S whose radius is a and centre at C with position vector $\vec{c}$, then the vector equation of the sphere S is given by

$$|\vec{r}|^2 = 2\vec{r} \cdot \vec{c}$$

PROOF The general equation of a sphere with radius a and centre at C (with position vector $\vec{c}$) is given by

$$|\vec{r}|^2 - 2\vec{r} \cdot \vec{c} + |\vec{c}|^2 = a^2 \quad (6.1)$$

If the origin lies on the sphere, then

$$|\vec{0}|^2 - 2\vec{0} \cdot \vec{c} + |\vec{c}|^2 = a^2$$

and hence $|\vec{c}|^2 = a^2$. Substituting this in Eq. (6.1), we get the equation of the sphere as

$$|\vec{r}|^2 = 2\vec{r} \cdot \vec{c}$$

COROLLARY 6.9 Equation of the sphere with radius a and centre at the origin is given by

$$|\vec{r}|^2 = a^2$$

PROOF Substituting $\vec{c} = \vec{0}$ in Eq. (6.1), we get the equation as $|\vec{r}|^2 = a^2$.

QUICK LOOK 6

Let us consider the Cartesian form of the sphere with radius a and centre at $C = (c_1, c_2, c_3)$. Then

$$\vec{c} = \overrightarrow{OC} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

Let $P = (x, y, z)$ be an arbitrary point in the space and

$$\vec{r} = \overrightarrow{OP} = x\vec{i} + y\vec{j} + z\vec{k}$$

Then P lies on the sphere S if and only if $CP = a^2$; that is,

$$(x - c_1)^2 + (y - c_2)^2 + (z - c_3)^2 = a^2$$

This is the Cartesian form of the equation of the sphere with radius a and centre at $C = (c_1, c_2, c_3)$.

If the origin lies on the sphere, then

$$c_1^2 + c_2^2 + c_3^2 = a^2$$

and the equation of the sphere reduces to

$$x^2 + y^2 + z^2 = 2(c_1x + c_2y + c_3z)$$

If origin is the centre of the sphere, then equation of sphere reduces to

$$x^2 + y^2 + z^2 = a^2$$

Try it out The following is a generalization of Theorem 6.12:

Any diameter of a sphere S subtends a right angle at any point on S other than extremities of the diameter.

THEOREM 6.15 Let AB be a diameter of a sphere S and P be a point on S . Then

$$\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$$

PROOF Let a be the radius and C the centre of the sphere S . Then

$$\overrightarrow{AC} = \overrightarrow{CB} = \overrightarrow{BC}$$

$$\overrightarrow{AP} = \overrightarrow{AC} + \overrightarrow{CP}$$

$$\overrightarrow{BP} = \overrightarrow{BC} + \overrightarrow{CP}$$

Hence

$$\begin{aligned}
 \overrightarrow{AP} \times \overrightarrow{BP} &= (\overrightarrow{AC} + \overrightarrow{CP}) \times (\overrightarrow{BC} + \overrightarrow{CP}) \\
 &= (\overrightarrow{CP} + \overrightarrow{AC}) \times (\overrightarrow{CP} - \overrightarrow{AC}) \\
 &= |\overrightarrow{CP}|^2 - |\overrightarrow{AC}|^2 \\
 &= a^2 - a^2 = 0
 \end{aligned}$$

THEOREM 6.16

Let A and B be any distinct points in the space and $a = (1/2)AB$. Let C be the mid-point of AB . Then the sphere S with radius a and centre at C is given by

$$S = \{P : \overrightarrow{AP} \times \overrightarrow{BP} = 0\}$$

PROOF

By Theorem 6.15,

$$S = \{P : \overrightarrow{AP} \times \overrightarrow{BP} = 0\}$$

On the other hand, let P be any point in the space such that $\overrightarrow{AP} \times \overrightarrow{BP} = 0$. Then

$$\overrightarrow{CP} = \overrightarrow{CA} + \overrightarrow{AP}$$

and

$$\overrightarrow{CP} = \overrightarrow{CB} + \overrightarrow{BP} = -\overrightarrow{CA} + \overrightarrow{BP}$$

Therefore

$$\begin{aligned}
 CP^2 &= \overrightarrow{CP} \times \overrightarrow{CP} \\
 &= (\overrightarrow{CA} + \overrightarrow{AP}) \times (\overrightarrow{CB} + \overrightarrow{BP}) \\
 &= (\overrightarrow{AP} + \overrightarrow{CA}) \times (\overrightarrow{BP} - \overrightarrow{CA}) \\
 &= \overrightarrow{AP} \times \overrightarrow{BP} - \overrightarrow{AP} \times \overrightarrow{CA} + \overrightarrow{CA} \times \overrightarrow{BP} - \overrightarrow{CA} \times \overrightarrow{CA} \\
 &= 0 + \overrightarrow{CA} \times (\overrightarrow{BP} - \overrightarrow{AP}) - a^2 \\
 &= \overrightarrow{CA} \times \overrightarrow{BA} - a^2 \\
 &= \overrightarrow{CA} \times (2\overrightarrow{CA}) - a^2 \\
 &= 2a^2 - a^2 = a^2
 \end{aligned}$$

and hence P lies on the sphere. Thus

$$S = \{P : \overrightarrow{AP} \times \overrightarrow{BP} = 0\}$$

COROLLARY 6.10

Let A and B be two given points with position vectors $\vec{a}$ and $\vec{b}$, respectively. Then the equation of the sphere with AB as a diameter is

$$(\vec{r} - \vec{a}) \times (\vec{r} - \vec{b}) = 0$$

PROOF

If P is a point with position vector $\vec{r}$, then $\overrightarrow{AP} = \vec{r} - \vec{a}$ and $\overrightarrow{BP} = \vec{r} - \vec{b}$. Therefore, by Theorem 6.16, P lies on the sphere with AB as a diameter if and only if $(\vec{r} - \vec{a}) \times (\vec{r} - \vec{b}) = 0$. ■

DEFINITION 6.3

Let π_1 and π_2 be two planes and $\vec{n}_1$ and $\vec{n}_2$ be unit normals to π_1 and π_2 , respectively. Then the angle between $\vec{n}_1$ and $\vec{n}_2$ is called the angle between π_1 and π_2 .

 QUICK LOOK 7

Since a given plane has two different unit normals (one is opposite to the other), angle between a given pair of planes is not unique. In general, there are two angles between a given pair of planes. If θ is one of them, then

$\pi - \theta$ is the other. If $\vec{n}_1$ and $\vec{n}_2$ are unit normals to the planes π_1 and π_2 , respectively, then $\cos^{-1}(\vec{n}_1 \cdot \vec{n}_2)$ is an angle between the planes π_1 and π_2 .

THEOREM 6.17

Let A be a point in the space with position vector $\vec{a}$ and $\vec{n}$ be any non-zero vector. Then the vector equation of the plane passing through A and perpendicular to $\vec{n}$ is

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

PROOF

If P is a point in the space with position vector $\vec{r}$, then P lies on the required plane if and only if $\overrightarrow{AP}$ is perpendicular to $\vec{n}$; that is, $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$. ■

Example 6.6

Find the equation of the plane passing through $(-2, 1, 3)$ and perpendicular to $3\vec{i} + \vec{j} + 5\vec{k}$.

Solution: Let O be the origin and $A = (-2, 1, 3)$ be the given point. Then the position vector of A with respect to the origin is

$$\overrightarrow{OA} = -2\vec{i} + \vec{j} + 3\vec{k}$$

If $P = (x, y, z)$ is any point in the space, then the position vector of P is

$$\overrightarrow{OP} = x\vec{i} + y\vec{j} + z\vec{k} = \vec{r}, \text{ say}$$

Then P lies on the required plane if and only if

$$\overrightarrow{AP} \text{ is perpendicular to } 3\vec{i} + \vec{j} + 5\vec{k}$$

$$(\overrightarrow{OP} - \overrightarrow{OA}) \cdot (3\vec{i} + \vec{j} + 5\vec{k}) = 0$$

$$[(x+2)\vec{i} + (y-1)\vec{j} + (z-3)\vec{k}] \cdot (3\vec{i} + \vec{j} + 5\vec{k}) = 0$$

$$(x+2)3 + (y-1) + (z-3)5 = 0$$

$$3x + y + 5z = 10$$

This is the equation of the required plane.

Example 6.7

Find the angles between the planes $2x - 3y - 6z = 5$ and $6x + 2y - 9z = 4$.

Solution: Let

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{a} = 2\vec{i} - 3\vec{j} - 6\vec{k}$$

and $\vec{b} = 6\vec{i} + 2\vec{j} - 9\vec{k}$

Then vector equations of the given planes are

$$\vec{r} \cdot \vec{a} = 5 \quad \text{and} \quad \vec{r} \cdot \vec{b} = 4$$

Here $\vec{a}/|\vec{a}|$ and $\vec{b}/|\vec{b}|$ are unit vectors perpendicular to the given planes. If θ is the angle between these planes, then

$$\cos \theta = \frac{1}{|\vec{a}|} \vec{a} \cdot \frac{1}{|\vec{b}|} \vec{b} = \frac{1}{|\vec{a}||\vec{b}|} (\vec{a} \cdot \vec{b})$$

$$= \frac{2(-6) + (-3)(2) + (-6)(-9)}{\sqrt{2^2 + (-3)^2 + (-6)^2} \sqrt{6^2 + 2^2 + (-9)^2}}$$

$$= \frac{60}{77}$$

Therefore

$$\theta = \cos^{-1} \frac{60}{77}$$

The other angle is

$$\pi - \cos^{-1} \frac{60}{77}$$

Example 6.8

If $|\vec{r}|^2 - 2\vec{r} \cdot (3\vec{i} + 4\vec{j} + 6\vec{k}) + 4 = 0$ represents the equation of a sphere, then find the centre and radius of the sphere.

Solution: The equation of a sphere whose centre is the point with position vector $\vec{c}$ and radius a is

$$(\vec{r} - \vec{c}) \cdot (\vec{r} - \vec{c}) = a^2$$

Simplifying we get

$$\begin{aligned} \vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{c} + \vec{c} \cdot \vec{c} &= a^2 \\ |\vec{r}|^2 - 2\vec{r} \cdot \vec{c} &= a^2 - |\vec{c}|^2 \end{aligned} \quad (6.2)$$

The given equation is

$$|\vec{r}|^2 - 2\vec{r} \cdot (3\vec{i} + 4\vec{j} + 6\vec{k}) + 4 = 0 \quad (6.3)$$

Therefore comparing Eqs. (6.2) and (6.3) we get

$$\vec{c} = 3\vec{i} + 4\vec{j} + 6\vec{k}$$

and

$$a^2 = |\vec{c}|^2 - 4$$

Thus the centre is $(3, 4, 6)$ and the radius is

$$\sqrt{|\vec{c}|^2 - 4} = \sqrt{3^2 + 4^2 + 6^2 - 4} = 2\sqrt{15}$$

6.3 | Vector or Cross Product

We introduce the concept of vector product or cross product of two vectors and study its various properties and applications in finding vector areas and volumes of certain geometrical figures.

DEFINITION 6.4 Let $\vec{a}$ and $\vec{b}$ be any vectors. We define

$$\vec{a} \times \vec{b} = \begin{cases} \vec{0} & \text{if } \vec{a} \text{ and } \vec{b} \text{ are collinear or } \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0} \\ |\vec{a}||\vec{b}|\sin\theta\vec{n} & \text{otherwise} \end{cases}$$

where $\theta = (\vec{a}, \vec{b})$ is the angle between $\vec{a}$ and $\vec{b}$ and $\vec{n}$ is the unit vector orthogonal to both $\vec{a}$ and $\vec{b}$ such that $(\vec{a}, \vec{b}, \vec{n})$ form a triad of right-handed vectors. $\vec{a} \times \vec{b}$ is called the *vector product* (since it is a vector) or *cross product* (since $\times$ is used in between $\vec{a}$ and $\vec{b}$ in contrast with the dot product $\vec{a} \cdot \vec{b}$) of $\vec{a}$ and $\vec{b}$. Figure 6.15 shows the vector product.

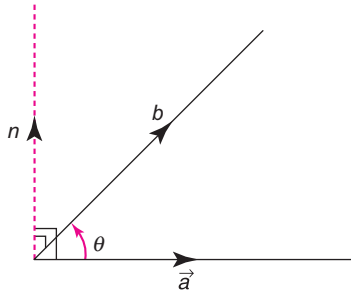


FIGURE 6.15 Definition 6.4.

Note that if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a}$ and $\vec{b}$ are obviously collinear and hence $\vec{a} \times \vec{b} = \vec{0}$. Conversely, if $\vec{a} \times \vec{b} = \vec{0}$, then $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a}$ and $\vec{b}$ are non-zero collinear (parallel) vectors.

THEOREM 6.18 For any vectors $\vec{a}$ and $\vec{b}$

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

PROOF We can assume that $\vec{a}$ and $\vec{b}$ are non-zero non-collinear vectors (otherwise $\vec{a} \times \vec{b} = \vec{0}$ and the result is trivial). Then $0 < (\vec{a}, \vec{b}) < \pi$ and hence $0 < \sin(\vec{a}, \vec{b}) \leq 1$ so that

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin(\vec{a}, \vec{b}) = |\vec{a}||\vec{b}|\sin\theta$$

since $\vec{n}$ is a unit vector. ■

We have proved earlier that the dot product is commutative in the sense that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ for all vectors $\vec{a}$ and $\vec{b}$. The vector product is not commutative, that is, $\vec{a} \times \vec{b}$ may not be equal to $\vec{b} \times \vec{a}$. However, we have the following.

THEOREM 6.19 For any vectors $\vec{a}$ and $\vec{b}$,

$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

That is, $\vec{a} \times \vec{b}$ and $\vec{b} \times \vec{a}$ are opposite to each other.

PROOF If one of $\vec{a}$ and $\vec{b}$ is the zero vector or if $\vec{a}$ and $\vec{b}$ are collinear, then

$$\vec{a} \times \vec{b} = \vec{0} = -\vec{0} = -(\vec{b} \times \vec{a})$$

Therefore, we can assume that $\vec{a}$ and $\vec{b}$ are non-zero and non-collinear vectors. Let θ be the angle between $\vec{a}$ and $\vec{b}$ and $\vec{n}$ be the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $(\vec{a}, \vec{b}, \vec{n})$ is a right-handed system. Then, by Definition 6.4, we have

$$\vec{a} \times \vec{b} = (|\vec{a}||\vec{b}|\sin\theta)\vec{n}$$

Since $(\vec{b}, \vec{a}, -\vec{n})$ is a right-handed system, we have

$$\begin{aligned}\vec{b} \times \vec{a} &= (|\vec{b}||\vec{a}|\sin\theta)(-\vec{n}) \\ &= (|\vec{a}||\vec{b}|\sin\theta)(-\vec{n}) \\ &= -(\vec{a} \times \vec{b})\end{aligned}$$

COROLLARY 6.11 We have

$$|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{a}| = |\vec{a}||\vec{b}|\sin\theta$$

where θ is the angle between $\vec{a}$ and $\vec{b}$.

THEOREM 6.20 The following hold for any vectors $\vec{a}$ and $\vec{b}$ and scalars l and m :

1. $(-\vec{a}) \times \vec{b} = \vec{b} \times \vec{a} = -(\vec{a} \times \vec{b})$
2. $(\vec{a}) \times (-\vec{b}) = -\vec{a} \times \vec{b}$
3. $(l\vec{a}) \times \vec{b} = l(\vec{a} \times \vec{b}) = \vec{a} \times (l\vec{b})$
4. $(l\vec{a}) \times (m\vec{b}) = lm(\vec{a} \times \vec{b})$

PROOF All these equalities hold good trivially if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a}$ and $\vec{b}$ are collinear or $l = 0$ or $m = 0$. Therefore we can assume that $\vec{a}$ and $\vec{b}$ are non-zero non-collinear vectors and l and m are non-zero scalars. Let θ be the angle between $\vec{a}$ and $\vec{b}$ and $\vec{n}$ be the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $(\vec{a}, \vec{b}, \vec{n})$ is a right-handed system.

1. The vector triad $(-\vec{a}, \vec{b}, \vec{n})$ becomes a left-handed system while $(\vec{a}, \vec{b}, -\vec{n})$ becomes a right-handed system. Also, the angle between $-\vec{a}$ and $\vec{b}$ is $\pi - \theta$. Therefore

$$\begin{aligned}(-\vec{a}) \times \vec{b} &= [|\vec{a}||\vec{b}|\sin(\pi - \theta)](-\vec{n}) \\ &= (|\vec{a}||\vec{b}|\sin\theta)(-\vec{n}) \\ &= -(\vec{a} \times \vec{b}) \\ &= \vec{b} \times \vec{a}\end{aligned}$$

and

$$\begin{aligned}\vec{a} \times (-\vec{b}) &= -[(-\vec{b}) \times \vec{a}] \\ &= -(\vec{a} \times \vec{b}) \quad (\text{by the above equality})\end{aligned}$$

2. We have

$$(-\vec{a}) \times (-\vec{b}) = -[\vec{a} \times (-\vec{b})] = -[-(\vec{a} \times \vec{b})] = \vec{a} \times \vec{b}$$

3. Let us first assume that $l > 0$. Then the angle between $l\vec{a}$ and $\vec{b}$ is θ and $|l\vec{a}| = l|\vec{a}|$. Therefore since $(l\vec{a}, \vec{b}, \vec{n})$ is a right-handed system

$$\begin{aligned}(l\vec{a}) \times \vec{b} &= (|l\vec{a}||\vec{b}| \sin \theta) \vec{n}, \\ &= (l|\vec{a}||\vec{b}| \sin \theta) \vec{n} \\ &= l(\vec{a} \times \vec{b})\end{aligned}$$

If $l < 0$, then $-l > 0$ and, by (1) above, we have

$$\begin{aligned}(l\vec{a}) \times \vec{b} &= -[(-l)\vec{a} \times \vec{b}] \\ &= -(-l)(\vec{a} \times \vec{b}) \\ &= l(\vec{a} \times \vec{b})\end{aligned}$$

Also,

$$\vec{a} \times (l\vec{b}) = -(l\vec{b}) \times \vec{a} = -l(\vec{b} \times \vec{a}) = l(-\vec{b} \times \vec{a}) = l(\vec{a} \times \vec{b})$$

4. We have

$$(l\vec{a}) \times (m\vec{b}) = l[\vec{a} \times (m\vec{b})] = lm(\vec{a} \times \vec{b})$$

THEOREM 6.21 For any vectors $\vec{a}$ and $\vec{b}$,

$$\vec{a} \times (\vec{a} + \vec{b}) = \vec{a} \times \vec{b}$$

PROOF

This equality trivially holds if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$. Therefore, we can assume that $\vec{a}$ and $\vec{b}$ are non-zero vectors. Let O be the origin and choose points A and B such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$. Complete the parallelogram $OABC$. Let D be the foot of the perpendicular from C to the line OA (Figure 6.16). Now we have

$$\overrightarrow{OC} = \vec{a} + \vec{b}$$

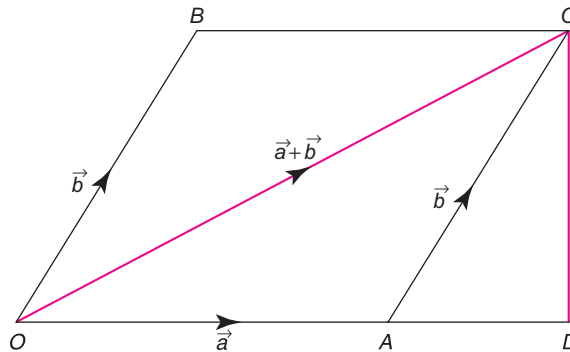


FIGURE 6.16 Theorem 6.21.

Clearly, $\vec{a} \cdot (\vec{a} + \vec{b})$ and $\vec{a} \cdot \vec{b}$ have the same direction. Also, regarding the magnitudes, we have

$$\begin{aligned} |\vec{a} \cdot (\vec{a} + \vec{b})| &= |\vec{a}| |\vec{a} + \vec{b}| \sin \angle COD \\ &= |\vec{a}| OC \sin \angle COD = |\vec{a}| CD \\ &= |\vec{a}| AC \sin \angle DAC \\ &= |\vec{a}| |\vec{b}| \sin \angle AOB = |\vec{a} \cdot \vec{b}| \end{aligned}$$

Thus $\vec{a} \cdot (\vec{a} + \vec{b}) = \vec{a} \cdot \vec{b}$. ■

COROLLARY 6.12 For any vectors $\vec{a}$ and $\vec{b}$ and any scalar λ ,

$$\vec{a} \cdot (\lambda \vec{a} + \vec{b}) = \vec{a} \cdot \vec{b}$$

PROOF If $\lambda = 0$, this is trivial. Suppose that $\lambda \neq 0$. Then

$$\begin{aligned} \vec{a} \cdot (\lambda \vec{a} + \vec{b}) &= \vec{a} \cdot \left(\lambda \vec{a} + \frac{1}{\lambda} \vec{b} \right) \\ &= \lambda \vec{a} \cdot \vec{a} + \frac{1}{\lambda} \vec{a} \cdot \vec{b} \\ &= \lambda \vec{a} \cdot \frac{1}{\lambda} \vec{b} \quad (\text{by Theorem 6.21}) \\ &= \vec{a} \cdot \vec{b} \end{aligned}$$
■

Note: The proof of the following theorem is bit lengthy. The reader may skip the proof in first reading and assume its validity.

THEOREM 6.22
(DISTRIBUTIVE LAW)

For any vectors $\vec{a}, \vec{b}$ and $\vec{c}$,

1. $\vec{a} \cdot (\vec{b} + \vec{c}) = (\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{c})$
2. $(\vec{a} + \vec{b}) \cdot \vec{c} = (\vec{a} \cdot \vec{c}) + (\vec{b} \cdot \vec{c})$

PROOF (2) is a simple consequence of (1), because

$$\begin{aligned} (\vec{a} + \vec{b}) \cdot \vec{c} &= [\vec{c} \cdot (\vec{a} + \vec{b})] \\ &= [(\vec{c} \cdot \vec{a}) + (\vec{c} \cdot \vec{b})], \quad \text{by (1)} \\ &= \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} \\ &= \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} \end{aligned}$$

1. This is trivial if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{c} = \vec{0}$. Therefore we can assume that $\vec{a}, \vec{b}$ and $\vec{c}$ are non-zero vectors. We shall distinguish the following cases.

Case I: Suppose that $\vec{b}$ and $\vec{c}$ are linearly dependent. In this case there exists a scalar, α such that $\vec{b} = \alpha \vec{c}$ and hence

$$\begin{aligned} \vec{a} \cdot (\vec{b} + \vec{c}) &= \vec{a} \cdot (\alpha \vec{c} + \vec{c}) \\ &= \vec{a} \cdot [(\alpha + 1)\vec{c}] \end{aligned}$$

$$\begin{aligned}
&= (\alpha + 1)(\vec{a} \times \vec{c}) \\
&= \alpha(\vec{a} \times \vec{c}) + (\vec{a} \times \vec{c}) \\
&= [\vec{a} \times (\alpha\vec{c})] + (\vec{a} \times \vec{c}) \\
&= (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})
\end{aligned}$$

Case II: Suppose that $\vec{b}$ and $\vec{c}$ are linearly independent and $\vec{a} \times (\vec{b} + \vec{c}) = \vec{0}$. Then $\vec{a}$ and $\vec{b} + \vec{c}$ are non-zero parallel vectors and hence $\vec{b} + \vec{c} = \lambda\vec{a}$ for some scalar λ . Now, we have

$$\begin{aligned}
\vec{a} \times \vec{c} &= \vec{a} \times (\lambda\vec{a} - \vec{b}) \\
&= \frac{1}{\lambda}[\lambda\vec{a} \times (\lambda\vec{a} - \vec{b})], \text{ by Theorem 6.20 [part (3)]} \\
&= \frac{1}{\lambda}[\lambda\vec{a} \times (-\vec{b})], \text{ by Theorem 6.21} \\
&= \vec{a} \times (-\vec{b}), \text{ by Theorem 6.20 [part (3)]} \\
&= -\vec{a} \times \vec{b}
\end{aligned}$$

Therefore

$$(\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c}) = \vec{0} = \vec{a} \times (\vec{b} + \vec{c})$$

Case III: Suppose that $\vec{b}$ and $\vec{c}$ are linearly independent and $\vec{a}$ is perpendicular to both $\vec{b}$ and $\vec{c}$. Let O be the origin and B and C be points such that $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$. Then O, B, C are not collinear (since $\vec{b}$ and $\vec{c}$ are linearly independent, Figure 6.17). Choose a point D such that $OBDC$ is a parallelogram. Then $\overrightarrow{OD} = \vec{b} + \vec{c}$. Now, choose points B' and C' such that $\overrightarrow{OB'} = \vec{a} \times \vec{b}$ and $\overrightarrow{OC'} = \vec{a} \times \vec{c}$. Then B' and C' lie in the OBC plane, since $\vec{a}$ is perpendicular to both $\vec{b}$ and $\vec{c}$. Also,

$$OB' = |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| = |\vec{a}|(OB)$$

and

$$OC' = |\vec{a} \times \vec{c}| = |\vec{a}||\vec{c}| = |\vec{a}|(OC)$$

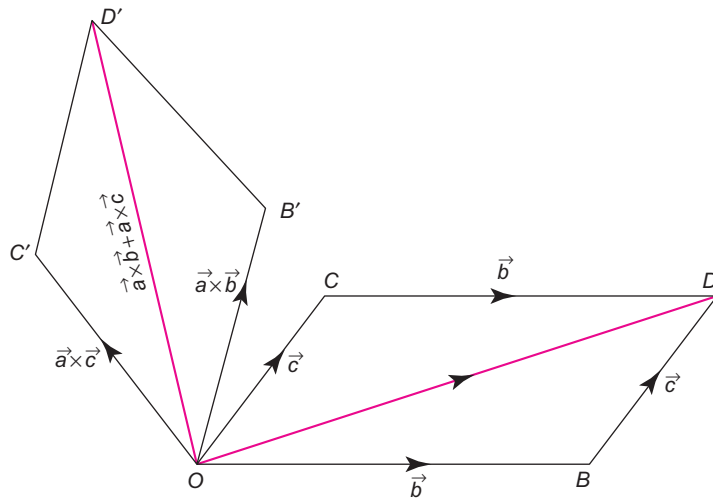


FIGURE 6.17 Theorem 6.22 [part (1), Case III].

Next, choose a point D such that $OBDC$ is a parallelogram, as shown in Figure 6.17. Then we have, by the parallelogram law,

$$\overrightarrow{OD} = (\vec{a} - \vec{b}) + (\vec{a} - \vec{c}) \quad (6.4)$$

Let L be a straight line passing through O and perpendicular to the OBC plane. Since $(\vec{a}, \vec{b}, \vec{a} - \vec{b})$ and $(\vec{a}, \vec{c}, \vec{a} - \vec{c})$ are right-handed orthogonal systems, a rotation about the line L through the angle $\angle BOC$, so that the new direction of $\overrightarrow{OB}$ coincides with the old direction of $\overrightarrow{OC}$, brings $\overrightarrow{OB}$ along the old direction of $\overrightarrow{OC}$. Therefore

$$\angle BOC = \angle BOC$$

$$\text{and} \quad \angle OBD = \pi - \angle BOC = \pi - \angle BOC = \angle OBD$$

Since

$$\frac{OB}{OD} = |\vec{a}| = \frac{BD}{OD}$$

it follows that the triangles $OB D$ and OBD are similar. Hence

$$\frac{OD}{OD} = |\vec{a}|$$

and

$$\angle BOD = \angle BOD$$

Also, since $(\vec{a}, \vec{b}, \vec{a} - \vec{b})$ and $(\vec{a}, \vec{c}, \vec{a} - \vec{c})$ are right-handed orthogonal systems, a rotation about the line L through 90° , so that the new direction of $\overrightarrow{OB}$ coincides with the old direction of $\overrightarrow{OB}$, brings $\overrightarrow{OC}$ along the old direction of $\overrightarrow{OC}$. Therefore the same rotation brings $\overrightarrow{OD}$ along the old direction of $\overrightarrow{OD}$ and hence it follows that $(\vec{a}, \overrightarrow{OD}, \overrightarrow{OD})$ is a right-handed orthogonal system. Therefore $\overrightarrow{OD}$ and $\vec{a} - (\vec{b} + \vec{c})$ have the same direction. Now

$$OD = |\vec{a}|(OD) = |\vec{a}||\vec{b} + \vec{c}| = |\vec{a} - (\vec{b} + \vec{c})|$$

Thus $\overrightarrow{OD}$ and $\vec{a} - (\vec{b} + \vec{c})$ have the same direction and magnitude, so that

$$\vec{a} - (\vec{b} + \vec{c}) = \overrightarrow{OD} = (\vec{a} - \vec{b}) + (\vec{a} - \vec{c}) \quad [\text{by Eq. (6.4)}]$$

Case IV: Suppose that $\vec{b}$ and $\vec{c}$ are linearly independent, $\vec{a} - (\vec{b} + \vec{c}) = \vec{0}$ and $\vec{a}$ is not perpendicular to $\vec{b}$ or $\vec{c}$. If $\vec{b} = \lambda \vec{a}$ for some non-zero scalar λ , then

$$\begin{aligned} \vec{a} - (\vec{b} + \vec{c}) &= \vec{a} - (\lambda \vec{a} + \vec{c}) \\ &= \vec{a} - \vec{c} \quad (\text{by Corollary 6.12}) \\ &= \lambda \vec{0} + (\vec{a} - \vec{c}) \\ &= \lambda(\vec{a} - \vec{a}) + (\vec{a} - \vec{c}) \\ &= (\vec{a} - \lambda \vec{a}) + (\vec{a} - \vec{c}) \\ &= (\vec{a} - \vec{b}) + (\vec{a} - \vec{c}) \end{aligned}$$

Similarly, we are through if $\vec{c} = \lambda \vec{a}$ for some scalar $\lambda \neq 0$. Therefore, we can assume that $\vec{a}$ and $\vec{b}$ are linearly independent and that $\vec{a}$ and $\vec{c}$ are linearly independent. Let O denote the origin. Choose points A , B and C such that

$$\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b} \quad \text{and} \quad \overrightarrow{OC} = \vec{c}$$

Then the points O , A and B as well as the points O , A and C are non-collinear. Choose a point D in the OAB plane such that OD is perpendicular to OA and $(\vec{a}, \vec{OD}, \vec{a} \times \vec{b})$ is a right-handed system (see Figure 6.18).

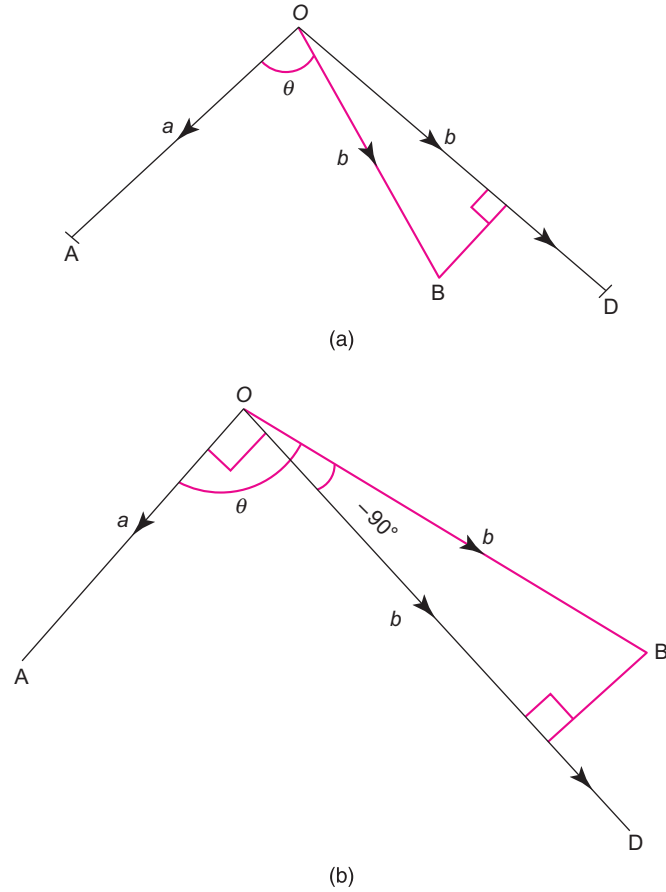


FIGURE 6.18 Theorem 6.22 [part (1), case (iv)].

Let $\vec{b}'$ be the orthogonal projection of $\vec{b}$ on $\vec{OD}$. Then $\vec{a} \times \vec{b}' = \vec{a} \times \vec{b}$ and $\vec{b}' \times \vec{b}$ is parallel to $\vec{a}$. Similarly we can choose a vector $\vec{c}$ such that $\vec{a}$ is perpendicular to $\vec{c}$, $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$ and $\vec{c} \times \vec{c}$ is parallel to $\vec{a}$. If $\vec{b}'$ and $\vec{c}$ are linearly dependent, then as in case (i),

$$\begin{aligned}\vec{a} \times (\vec{b}' + \vec{c}) &= \vec{a} \times (\vec{b}' + \vec{c}) \\ &= (\vec{a} \times \vec{b}') + (\vec{a} \times \vec{c}) \\ &= (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})\end{aligned}$$

Now, suppose that $\vec{b}'$ and $\vec{c}$ are linearly independent. Since $\vec{a}$ is perpendicular to both $\vec{b}'$ and $\vec{c}$, it follows from case (iii) that

$$\vec{a} \times (\vec{b}' + \vec{c}) = (\vec{a} \times \vec{b}') + (\vec{a} \times \vec{c})$$

Since both $\vec{b}' \times \vec{b}$ and $\vec{c} \times \vec{c}$ are parallel to $\vec{a}$, it follows that $\vec{b}' + \vec{c} \times (\vec{b}' + \vec{c})$ is parallel to $\vec{a}$ and hence

$$\vec{a} \times [(\vec{b}' + \vec{c}) \times (\vec{b}' + \vec{c})] = \vec{0}$$

and therefore, from case (ii),

$$\vec{a} \times (\vec{b}' + \vec{c}) \times \vec{a} \times (\vec{b}' + \vec{c}) = \vec{a} \times [(\vec{b}' + \vec{c}) \times (\vec{b}' + \vec{c})] = \vec{0}$$

and hence

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

But we have

$$\begin{aligned}\vec{a} \cdot (\vec{b} + \vec{c}) &= (\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{c}) \\ &= (\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{c})\end{aligned}$$

Thus, in any case, $\vec{a} \cdot (\vec{b} + \vec{c}) = (\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{c})$. ■

Note: The above proof is apparently involved and complicated. We offer an alternate proof later (see Theorem 6.45) which is elegant and simple. The strength of the above proof is that it is direct from the definition of the vector product.

Next, we derive a simple formula for the vector product of two vectors in terms of their components in the directions of an orthogonal right-handed system of unit vectors. First, let us have the following.

THEOREM 6.23 Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed orthogonal unit vector triad. Then

1. $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k} = 1$
2. $\vec{i} \cdot \vec{j} = \vec{j} \cdot \vec{k} = \vec{k} \cdot \vec{i} = 0$
3. $\vec{j} \cdot \vec{i} = -\vec{i} \cdot \vec{j}$, $\vec{k} \cdot \vec{j} = -\vec{j} \cdot \vec{k}$ and $\vec{i} \cdot \vec{k} = -\vec{k} \cdot \vec{i}$

PROOF

1. Follows from the fact that $\vec{a} \cdot \vec{a} = |\vec{a}|^2$ for any vector $\vec{a}$.
2. Since $(\vec{i}, \vec{j}, \vec{k})$ is a right-handed system and $\vec{k}$ is the unit vector orthogonal to both $\vec{i}$ and $\vec{j}$, we have

$$\vec{i} \times \vec{j} = |\vec{i}| |\vec{j}| \sin \frac{\pi}{2} \vec{k} = \vec{k}$$

Similarly $\vec{j} \times \vec{k} = \vec{i}$ and $\vec{k} \times \vec{i} = \vec{j}$.

3. Follows from (2) and the fact that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ for any vectors $\vec{a}$ and $\vec{b}$. ■

THEOREM 6.24 Let $(\vec{i}, \vec{j}, \vec{k})$ be a right-handed orthogonal system of unit vectors and $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$. Then

$$\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2)\vec{i} + (a_3b_1 - a_1b_3)\vec{j} + (a_1b_2 - a_2b_1)\vec{k}$$

PROOF

We have

$$\begin{aligned}\vec{a} \times \vec{b} &= (a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) \times (b_1\vec{i} + b_2\vec{j} + b_3\vec{k}) \\ &= a_1b_1(\vec{i} \times \vec{i}) + a_1b_2(\vec{i} \times \vec{j}) + a_1b_3(\vec{i} \times \vec{k}) \\ &\quad + a_2b_1(\vec{j} \times \vec{i}) + a_2b_2(\vec{j} \times \vec{j}) + a_2b_3(\vec{j} \times \vec{k}) \quad (\text{by Theorem 6.22}) \\ &\quad + a_3b_1(\vec{k} \times \vec{i}) + a_3b_2(\vec{k} \times \vec{j}) + a_3b_3(\vec{k} \times \vec{k}) \\ &= a_1b_2\vec{k} - a_1b_3\vec{j} - a_2b_1\vec{k} + a_2b_3\vec{i} + a_3b_1\vec{j} - a_3b_2\vec{i} \quad (\text{by Theorem 6.23}) \\ &= (a_2b_3 - a_3b_2)\vec{i} + (a_3b_1 - a_1b_3)\vec{j} + (a_1b_2 - a_2b_1)\vec{k}\end{aligned}$$
■

Note: The above expression for $\vec{a} \times \vec{b}$ can be given in the form of a determinant as

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

THEOREM 6.25

Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ be non-zero vectors where $(\vec{i}, \vec{j}, \vec{k})$ is a right-handed system of orthogonal unit vectors. Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then

$$\sin \theta = \sqrt{\frac{(a_2b_3 - a_3b_2)^2}{(a_1^2)(b_1^2)}}$$

and

$$\cos \theta = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2} \sqrt{b_1^2}}$$

PROOF

Since $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \vec{n}$, we have

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} = \frac{\sqrt{(a_2b_3 - a_3b_2)^2}}{\sqrt{a_1^2} \sqrt{b_1^2}}$$

and hence

$$\begin{aligned} \cos \theta &= \sqrt{1 - \sin^2 \theta} \\ &= \frac{\sqrt{(a_1^2)(b_1^2) - (a_2b_3 - a_3b_2)^2}}{a_1^2 \times b_1^2} \\ &= \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}} \end{aligned}$$

Note: In the above, the value of $\sin \theta$ will not give us the value of θ , since the sine function is not an injection on the interval $[0, \pi]$. However the cosine function is an injection on the interval $[0, \pi]$ and hence θ is determined uniquely by the value of $\cos \theta$. For these reasons, the formula for $\cos \theta$ is more useful in determining the angle between two vectors.

THEOREM 6.26

For any vectors $\vec{a}$ and $\vec{b}$,

$$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} \\ \vec{a} \times \vec{b} & \vec{b} \times \vec{b} \end{vmatrix}$$

PROOF

These equalities hold trivially if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a}$ and $\vec{b}$ are collinear. Therefore we can assume that $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$ and $\vec{a}$ and $\vec{b}$ are not collinear (parallel). Then

$$\vec{a} \times \vec{b} = (|\vec{a}||\vec{b}| \sin \theta) \vec{n}$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ and $\vec{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $(\vec{a}, \vec{b}, \vec{n})$ is a right-handed system. Now, we have

$$\begin{aligned} |\vec{a} \times \vec{b}|^2 &= |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta \quad (\text{since } |\vec{n}| = 1) \\ &= |\vec{a}|^2 |\vec{b}|^2 (1 - \cos^2 \theta) \\ &= |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta \\ &= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 \\ &= (\vec{a} \times \vec{a}) \cdot (\vec{b} \times \vec{b}) - (\vec{a} \cdot \vec{b})(\vec{a} \cdot \vec{b}) \\ &= \text{the determinant} \begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} \\ \vec{a} \times \vec{b} & \vec{b} \times \vec{b} \end{vmatrix} \end{aligned}$$

THEOREM 6.27 Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-parallel vectors. Then a unit vector $\vec{n}$ perpendicular to both $\vec{a}$ and $\vec{b}$ is given by

$$\vec{n} = \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \pm \frac{(a_2b_3 - a_3b_2)\vec{i}}{\sqrt{(a_2b_3 - a_3b_2)^2}}$$

where $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$.

PROOF This is a direct consequence of Definition 6.4 and Theorem 6.24. ■

Note: The signs + or - depend on whether $(\vec{a}, \vec{b}, \vec{n})$ is a right-handed system or a left-handed system.

In the following we derive a formula for the cross product of two vectors when they are expressed as linear combinations of three given non-coplanar vectors. This generalizes Theorem 6.24.

THEOREM 6.28 Let $(\vec{a}, \vec{b}, \vec{c})$ be a non-coplanar vector triad and

$$\vec{\alpha} = r_1\vec{a} + r_2\vec{b} + r_3\vec{c}$$

and

$$\vec{\beta} = s_1\vec{a} + s_2\vec{b} + s_3\vec{c}$$

Then

$$\vec{\alpha} \times \vec{\beta} = \begin{vmatrix} \vec{b} & \vec{c} & \vec{c} & \vec{a} & \vec{a} & \vec{b} \\ r_1 & & r_2 & & r_3 & \\ s_1 & & s_2 & & s_3 & \end{vmatrix}$$

PROOF Using the distributive law of cross product over addition (Theorem 6.22), we have

$$\begin{aligned} \vec{\alpha} \times \vec{\beta} &= (r_1\vec{a} + r_2\vec{b} + r_3\vec{c}) \times (s_1\vec{a} + s_2\vec{b} + s_3\vec{c}) \\ &= r_1s_2(\vec{a} \times \vec{b}) + r_1s_3(\vec{a} \times \vec{c}) + r_2s_1(\vec{b} \times \vec{a}) + r_2s_3(\vec{b} \times \vec{c}) \\ &\quad + r_3s_1(\vec{c} \times \vec{a}) + r_3s_2(\vec{c} \times \vec{b}) \\ &= (r_2s_3 - r_3s_2)(\vec{b} \times \vec{c}) + (r_1s_3 - r_3s_1)(\vec{c} \times \vec{a}) + (r_1s_2 - r_2s_1)(\vec{a} \times \vec{b}) \\ &= \begin{vmatrix} \vec{b} & \vec{c} & \vec{c} & \vec{a} & \vec{a} & \vec{b} \\ r_1 & & r_2 & & r_3 & \\ s_1 & & s_2 & & s_3 & \end{vmatrix} \end{aligned}$$

Try it out If $(\vec{a}, \vec{b}, \vec{c})$ is an orthogonal right-handed system of unit vectors, then Theorem 6.28 implies Theorem 6.24, since $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$ and $\vec{c} \times \vec{a} = \vec{b}$. Prove it.

Example 6.9

Let $\vec{a} = 2\vec{i} - 3\vec{j} + 5\vec{k}$ and $\vec{b} = \vec{i} + 4\vec{j} + 2\vec{k}$. Then find the cross product $\vec{a} \times \vec{b}$ and unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 5 \\ 1 & 4 & 2 \end{vmatrix}$$

Solution: Recall the formula

Substituting the values we get

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 5 \\ 1 & 4 & 2 \end{vmatrix} \\ &= (6 - 20)\vec{i} - (4 + 5)\vec{j} + (8 - 3)\vec{k} \\ &= -14\vec{i} - 9\vec{j} + 5\vec{k}\end{aligned}$$

Example 6.10

Let ABC be a triangle, $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$ and $\overrightarrow{AB} = \vec{c}$. Then prove that

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

Solution: First observe that

$$\vec{a} + \vec{b} + \vec{c} = \overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB} = \overrightarrow{BA} + \overrightarrow{AB} = \vec{0}$$

The unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$ are

$$\pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \pm \frac{1}{\sqrt{782}} (-14\vec{i} - 9\vec{j} + 5\vec{k})$$

and hence $\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$. Now, by Theorem 6.21, we have

$$\vec{a} \times \vec{b} = \vec{a} \times (\vec{a} + \vec{b}) = \vec{a} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0} + \vec{a} \times \vec{b} = \vec{a} \times \vec{b}$$

and

$$\vec{b} \times \vec{c} = \vec{b} \times (\vec{b} + \vec{c}) = \vec{b} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0} + \vec{b} \times \vec{c} = \vec{b} \times \vec{c}$$

Therefore

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

Example 6.11

If $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$ and $\vec{b} = 3\vec{i} + 4\vec{j} - \vec{k}$, find angle between $\vec{a}$ and $\vec{b}$.

Solution: If θ is the required angle, then

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

Now

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 1 \\ 3 & 4 & -1 \end{vmatrix} = -3\vec{i} + 5\vec{j} + 11\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(-3)^2 + 5^2 + 11^2} = \sqrt{155}$$

$$|\vec{a}| = \sqrt{2^2 + 1^2 + 1^2} = \sqrt{6}$$

$$|\vec{b}| = \sqrt{3^2 + 4^2 + (-1)^2} = \sqrt{26}$$

Therefore

$$\sin \theta = \frac{\sqrt{155}}{\sqrt{6}\sqrt{26}} = \frac{\sqrt{155}}{\sqrt{156}}$$

Also,

$$\vec{a} \cdot \vec{b} = 2 \times 3 + 1 \times 4 + 1 \times (-1) = 1$$

Therefore

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{1}{\sqrt{156}}$$

Hence

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{156}} \right)$$

Note that for finding the angle between $\vec{a}$ and $\vec{b}$, use of the dot product is more suitable.

Example 6.12

Let $\vec{a} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} + \vec{j}$. Let $\vec{c}$ be a vector such that $|\vec{a} \times \vec{c}| = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a}$ and $\vec{c}$ is 30° . Then find the value of $|\vec{a} \times \vec{b} \times \vec{c}|$.

Solution: We have

$$|\vec{a}| = \sqrt{2^2 + 1^2 + (-2)^2} = 3$$

and

$$|\vec{b}| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

Now, $2\sqrt{2} = |\vec{c} - \vec{a}|$ implies that $|\vec{c} - \vec{a}|^2 = 8$. Therefore

$$|\vec{c}|^2 + |\vec{a}|^2 - 2(\vec{c} \cdot \vec{a}) = 8$$

$$|\vec{c}|^2 + 9 - 2|\vec{c}| = 8$$

$$(|\vec{c}| - 1)^2 = 0$$

$$|\vec{c}| = 1$$

and

$$|\vec{a} - \vec{b}| = 3$$

Therefore

$$\begin{aligned} |(\vec{a} - \vec{b}) \cdot \vec{c}| &= |\vec{a} - \vec{b}| |\vec{c}| \sin 30^\circ \\ &= 3(1) \frac{1}{2} = \frac{3}{2} \end{aligned}$$

Also

$$\vec{a} - \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 2 \\ 1 & 1 & 0 \end{vmatrix} = 2\vec{i} - 2\vec{j} + \vec{k}$$

6.4 | Vector Areas

The orientation of a closed curve is usually denoted by an ordered triad of non-collinear points on the curve as indicated in Figure 6.19. Consider the plane region D bounded by a closed curve described in the anticlockwise sense. It is indicated by the ordered triad (P_1, P_2, P_3) of points on it (see figure (a) below). In the figure (b), the region D is bounded by a curve described in the clockwise sense and it is indicated by the ordered triad (P_1, P_3, P_2) . While the area of the region D is independent of the orientation of the bounding curve, the vector area depends on the orientation of the bounding curve, as we see from the definition given below.

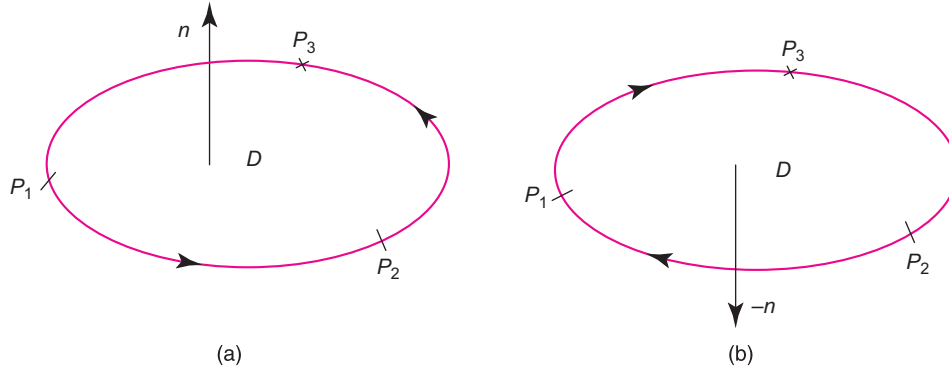


FIGURE 6.19 Orientation of closed curve.

DEFINITION 6.5 Let D be the plane region bounded by an oriented closed curve C whose orientation is indicated by an ordered triad (Q_1, Q_2, Q_3) of non-collinear points on C . The vector area of D is defined as $A \vec{n}$, where A is the area of the region D and $\vec{n}$ is the unit vector perpendicular to D such that $(\vec{Q_1Q_2}, \vec{Q_1Q_3}, \vec{n})$ is a right-handed system. In Figure 6.19(a), $(\vec{P_1P_2}, \vec{P_1P_3}, \vec{n})$ is a right-handed system and in this case $A \vec{n}$ is the vector area of the region D . In Figure 6.19(b), $(\vec{P_1P_3}, \vec{P_1P_2}, -\vec{n})$ is a right-handed system and hence, in this case, the vector area of the region D is $A(-\vec{n})$.

QUICK LOOK 8

1. For any plane region, there will be two vector areas and they are opposite to each other.
2. The magnitude of the vector area of the plane region is the usual area of the region and its direction is perpendicular to the plane of the region.

THEOREM 6.29 The vector area of a triangle ABC is

$$\frac{1}{2}(\overrightarrow{AB} \times \overrightarrow{AC})$$

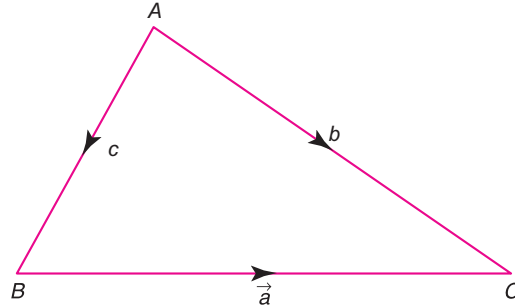


FIGURE 6.20 Theorem 6.29.

PROOF Consider a triangle ABC (Figure 6.20) and let its area be A . Let $\vec{n}$ be the unit vector perpendicular to the plane of the triangle ABC such that $(\overrightarrow{AB}, \overrightarrow{AC}, \vec{n})$ forms a right-handed system. Then the vector area of the triangle is $A\vec{n}$. But, we have

$$= \text{Area of the triangle } ABC$$

$$= \frac{1}{2}bc \sin A$$

$$= \frac{1}{2}|\overrightarrow{AC}||\overrightarrow{AB}| \sin A$$

Therefore the vector area of the triangle ABC is

$$\frac{1}{2}(\overrightarrow{AB} \times \overrightarrow{AC})$$

■

COROLLARY 6.13 The vector area of a parallelogram (Figure 6.21) with adjacent sides $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$.

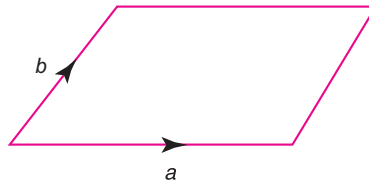


FIGURE 6.21 Corollary 6.13.

COROLLARY 6.14 Let ABC be a triangle and $\vec{a}, \vec{b}$ and $\vec{c}$ be the position vectors of A, B and C (with respect to the origin O), respectively. Then the vector area of the triangle ABC is

$$\frac{1}{2}[(\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) + (\vec{a} \times \vec{b})]$$

PROOF The vector area of the triangle ABC is given by

$$\frac{1}{2}(\overrightarrow{AB} \times \overrightarrow{AC}) = \frac{1}{2}[(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})]$$

$$= \frac{1}{2}[(\vec{b} \times \vec{c}) + \{\vec{b} \times (-\vec{a})\} + \{(-\vec{a}) \times \vec{c}\} + \{(-\vec{a}) \times (-\vec{a})\}]$$

$$\begin{aligned}
 &= \frac{1}{2}[\vec{b} \quad \vec{c} \quad \vec{b} \quad \vec{a} \quad \vec{a} \quad \vec{c} + \vec{0}] \\
 &= \frac{1}{2}[(\vec{b} \quad \vec{c}) + (\vec{c} \quad \vec{a}) + (\vec{a} \quad \vec{b})]
 \end{aligned}$$

Note that three points A, B, C are collinear if and only if the vector area of the triangle ABC is $\vec{0}$. ■

THEOREM 6.30 The vector area of a parallelogram $ABCD$ is

$$\frac{1}{2}(\overrightarrow{AC} \quad \overrightarrow{BD})$$

PROOF Consider a parallelogram $ABCD$ (Figure 6.22). Let O be the point of intersection of the diagonals AC and BD . Let $\overrightarrow{AC} = \vec{a}$ and $\overrightarrow{BD} = \vec{b}$. Then, we have

$$\overrightarrow{AO} = \overrightarrow{OC} = \frac{\vec{a}}{2}$$

and

$$\overrightarrow{BO} = \overrightarrow{OD} = \frac{\vec{b}}{2}$$

Now

$$\begin{aligned}
 \text{Vector area of the parallelogram } ABCD &= \text{Vector area of } OAB \\
 &\quad + \text{Vector area of } OBC \\
 &\quad + \text{Vector area of } OCD \\
 &\quad + \text{Vector area of } ODA
 \end{aligned}$$

Hence vector area of the parallelogram $ABCD$ is given by

$$\begin{aligned}
 &\frac{1}{2}[(\overrightarrow{OA} \quad \overrightarrow{OB}) + (\overrightarrow{OB} \quad \overrightarrow{OC}) + (\overrightarrow{OC} \quad \overrightarrow{OD}) + (\overrightarrow{OD} \quad \overrightarrow{OA})] \\
 &= \frac{1}{2} \left[\frac{\vec{a}}{2} \div \frac{\vec{b}}{2} + \frac{\vec{b}}{2} \div \frac{\vec{a}}{2} + \frac{\vec{a}}{2} \div \frac{\vec{b}}{2} + \frac{\vec{b}}{2} \div \frac{\vec{a}}{2} \right] \\
 &= \frac{1}{8}[(\vec{a} \quad \vec{b}) \quad (\vec{b} \quad \vec{a}) + (\vec{a} \quad \vec{b}) \quad (\vec{b} \quad \vec{a})] \\
 &= \frac{1}{8}[(\vec{a} \quad \vec{b}) + (\vec{a} \quad \vec{b}) + (\vec{a} \quad \vec{b}) + (\vec{a} \quad \vec{b})] \\
 &= \frac{1}{2}(\vec{a} \quad \vec{b}) \\
 &= \frac{1}{2}(\overrightarrow{AC} \quad \overrightarrow{BD})
 \end{aligned}$$

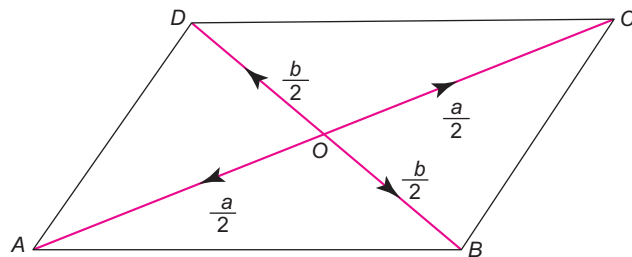


FIGURE 6.22 Theorem 6.30. ■

THEOREM 6.31

Let A, B and C denote the internal angles of a triangle ABC (Figure 6.23) and a, b, c denote the lengths of the sides opposite to these angles respectively. Let

$$s = \frac{1}{2}(a + b + c)$$

Then the following hold:

$$1. \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

(Note that this is called the **sine rule** for triangles.)

2. The area of the triangle is given by

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

PROOF

1. From Theorem 6.29, the vector area of the triangle is

$$\begin{aligned} \frac{1}{2}(\overrightarrow{AB} \times \overrightarrow{AC}) &= \frac{1}{2}(\overrightarrow{BC} \times \overrightarrow{BA}) \\ &= \frac{1}{2}(\overrightarrow{CA} \times \overrightarrow{CB}) \end{aligned}$$

Therefore

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = |\overrightarrow{BC} \times \overrightarrow{BA}| = |\overrightarrow{CA} \times \overrightarrow{CB}|$$

That is,

$$cb \sin A = ac \sin B = ba \sin C$$

Hence

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

2. Let $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$ and $\overrightarrow{AB} = \vec{c}$. Then $|\vec{a}| = a$, $|\vec{b}| = b$ and $|\vec{c}| = c$. We have

$$\overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{BA} = -\overrightarrow{AB}$$

Therefore

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

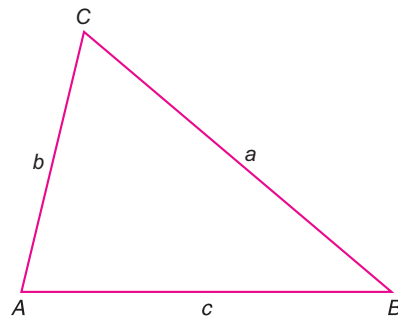


FIGURE 6.23 Theorem 6.31.

Hence

$$\begin{aligned}
 (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) &= (-\vec{c}) \cdot (-\vec{c}) = c^2 \\
 a^2 + b^2 + 2(\vec{a} \cdot \vec{b}) &= c^2 \\
 \vec{a} \cdot \vec{b} &= \frac{1}{2}[c^2 - (a^2 + b^2)] \quad (6.5)
 \end{aligned}$$

Now

$$\Delta = \frac{1}{2} |\overrightarrow{CA} \times \overrightarrow{CB}|$$

Therefore

$$\begin{aligned}
 4\Delta^2 &= |\vec{b} \times \vec{a}|^2 \\
 &= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 \\
 &= (|\vec{a}| |\vec{b}| + \vec{a} \cdot \vec{b}) (|\vec{a}| |\vec{b}| - \vec{a} \cdot \vec{b}) \\
 &= \left(ab + \frac{1}{2}(c^2 - a^2 - b^2) \right) \left(ab - \frac{1}{2}(c^2 - a^2 - b^2) \right), \quad [\text{by Eq. (6.5)}] \\
 &= \frac{1}{4} [c^2 - (a - b)^2] [(a + b)^2 - c^2] \\
 &= \frac{1}{4} [c - (a - b)] (c + a - b) (a + b - c) (a + b + c) \\
 &= \frac{1}{4} (a + b + c) (a + b - c) (b + c - a) (c + a - b) \\
 &= \frac{1}{4} 2s \cdot 2(s - c) \cdot 2(s - a) \cdot 2(s - b) \\
 &= 4s(s - a)(s - b)(s - c)
 \end{aligned}$$

So

$$\Delta = \sqrt{s(s - a)(s - b)(s - c)}$$

THEOREM 6.32

For any θ and $\phi \in [0, \pi]$,

$$\sin(\theta - \phi) = \sin \theta \cos \phi - \cos \theta \sin \phi$$

PROOF

Fix a right-handed rectangular Cartesian coordinate system $OXYZ$ and $\vec{i}, \vec{j}$ and $\vec{k}$ be the unit vectors along $\overrightarrow{OX}, \overrightarrow{OY}$ and $\overrightarrow{OZ}$, respectively (Figure 6.24). The required equation is trivial if $\theta = \phi$ or one of them is zero and the other is π . Therefore, we can assume that $\theta \neq \phi$ and $\{\theta, \phi\} \neq \{0, \pi\}$. Choose points P and Q in the XY -plane such that $OP = OQ = 1$, OP makes an angle ϕ and OQ makes an angle θ with the positive direction of the X -axis. Then the points O, P and Q are non-collinear, since $\theta \neq \phi \in [0, \pi]$ and $\{\theta, \phi\} \neq \{0, \pi\}$. Now, we have

$$\overrightarrow{OP} = (\cos \phi) \vec{i} + (\sin \phi) \vec{j}$$

and

$$\overrightarrow{OQ} = (\cos \theta) \vec{i} + (\sin \theta) \vec{j}$$

Therefore

$$\overrightarrow{OP} \times \overrightarrow{OQ} = (\sin \theta \cos \phi - \cos \theta \sin \phi) \vec{k}$$

But

$$\begin{aligned}\overline{OP} \cdot \overline{OQ} &= (OP)(OQ) \sin(\overline{OP}, \overline{OQ}) \vec{n} \\ &= \sin(\overline{OP}, \overline{OQ}) \vec{n}\end{aligned}$$

where $\vec{n}$ is the unit vector orthogonal to the OPQ plane ($= OXY$ plane) such that $(\overline{OP}, \overline{OQ}, \vec{n})$ is a right-handed system. Also,

$$(\overline{OP}, \overline{OQ}) = \theta - \phi \quad \text{and} \quad \vec{n} = \vec{k} \quad \text{if } \theta > \phi$$

and

$$(\overline{OP}, \overline{OQ}) = \phi - \theta \quad \text{and} \quad \vec{n} = -\vec{k} \quad \text{if } \phi > \theta$$

In either case, we have

$$\overline{OP} \cdot \overline{OQ} = \sin(\theta - \phi) \vec{k}$$

and hence

$$\sin(\theta - \phi) = \sin \theta \cos \phi - \cos \theta \sin \phi$$

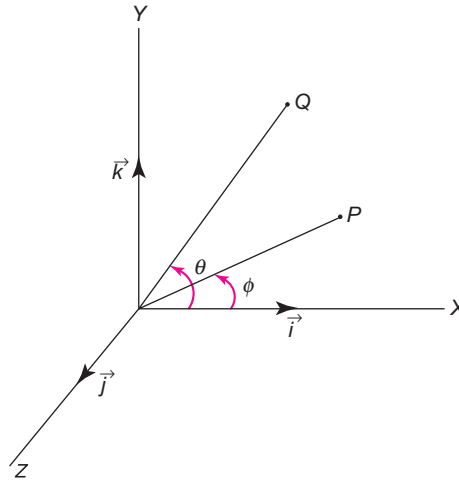


FIGURE 6.24 Theorem 6.32.

QUICK LOOK 9

Though we have proved the above theorem only for $\theta - \phi$, as we have learnt it from trigonometry, and $\phi \in [0, \pi]$, this holds good for all real numbers θ and ϕ .

We conclude this section with few examples in which we make use of the results proved in this section and earlier sections.

Example 6.13

Find the area of the parallelogram having $2\vec{i} - 3\vec{j}$ and $3\vec{i} + \vec{k}$ as adjacent sides.

Solution: The vector area of the given parallelogram is given by

$$(2\vec{i} - 3\vec{j}) \times (3\vec{i} + \vec{k}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 0 \\ 3 & 0 & 1 \end{vmatrix}$$

$$\begin{aligned}&= (3 - 0)\vec{i} - (2 - 0)\vec{j} + (0 + 9)\vec{k} \\ &= 3\vec{i} - 2\vec{j} + 9\vec{k}\end{aligned}$$

Therefore the area of the parallelogram is

$$|3\vec{i} - 2\vec{j} + 9\vec{k}| = \sqrt{3^2 + 2^2 + 9^2} = \sqrt{94} \text{ sq. units}$$

Example 6.14

Find the vector area of the triangle having vertices $(1, 2, 3)$, $(2, 5, 1)$ and $(1, 1, 2)$. Also find its magnitude.

Solution: Let $A = (1, 2, 3)$, $B = (2, 5, 1)$ and $C = (1, 1, 2)$. Then the position vectors of A , B and C with respect to the origin are

$$\vec{OA} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$\vec{OB} = 2\vec{i} + 5\vec{j} + \vec{k}$$

and

$$\vec{OC} = \vec{i} + \vec{j} + 2\vec{k}$$

Then

$$\begin{aligned}\vec{AB} &= \vec{OB} - \vec{OA} \\ &= (2\vec{i} + 5\vec{j} + \vec{k}) - (\vec{i} + 2\vec{j} + 3\vec{k}) \\ &= \vec{i} + 3\vec{j} - 2\vec{k}\end{aligned}$$

and

$$\begin{aligned}\vec{AC} &= (\vec{i} + \vec{j} + 2\vec{k}) - (\vec{i} + 2\vec{j} + 3\vec{k}) \\ &= -\vec{j} - \vec{k}\end{aligned}$$

Therefore

$$\begin{aligned}\text{Vector area of } \triangle ABC &= \frac{1}{2}(\vec{AB} \times \vec{AC}) \\ &= \frac{1}{2} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -2 \\ 0 & -1 & -1 \end{vmatrix} \\ &= \frac{1}{2}(-7\vec{i} + 9\vec{j} + 5\vec{k})\end{aligned}$$

The magnitude of the vector area of $\triangle ABC$ is

$$\frac{1}{2}\sqrt{49 + 81 + 25} = \frac{1}{2}\sqrt{155} \text{ sq. units}$$

Example 6.15

For any vector $\vec{a}$, prove that

$$|\vec{a} \times \vec{i}|^2 + |\vec{a} \times \vec{j}|^2 + |\vec{a} \times \vec{k}|^2 = 2|\vec{a}|^2$$

Solution: Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$. Then we have

$$\vec{a} \times \vec{i} = \begin{vmatrix} \vec{j} & \vec{k} \\ a_2 & a_3 \end{vmatrix} = a_3\vec{j} - a_2\vec{k}$$

Therefore

$$|\vec{a} \times \vec{i}|^2 = a_2^2 + a_3^2$$

Similarly

$$|\vec{a} \times \vec{j}|^2 = a_1^2 + a_3^2$$

and

$$|\vec{a} \times \vec{k}|^2 = a_1^2 + a_2^2$$

Hence

$$\begin{aligned}|\vec{a} \times \vec{i}|^2 + |\vec{a} \times \vec{j}|^2 + |\vec{a} \times \vec{k}|^2 &= (a_2^2 + a_3^2) + (a_1^2 + a_3^2) + (a_1^2 + a_2^2) \\ &= 2(a_1^2 + a_2^2 + a_3^2) \\ &= 2|\vec{a}|^2\end{aligned}$$

6.5 | Scalar Triple Product

We shall introduce the concept of scalar triple product and discuss certain properties and geometrical interpretation of this.

DEFINITION 6.6 The scalar triple product of three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is defined as $(\vec{a} \times \vec{b}) \cdot \vec{c}$ and is denoted by $[\vec{a} \vec{b} \vec{c}]$. This is also called the box product or simply box $[\vec{a} \vec{b} \vec{c}]$.

**QUICK LOOK 10**

- $[\vec{a} \vec{b} \vec{c}]$ is a scalar, since it is the dot (scalar) product of the vectors $\vec{a} \times \vec{b}$ and $\vec{c}$.
- $[\vec{a} \vec{b} \vec{c}] = 0$ if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{c} = \vec{0}$, or $\vec{a}$, $\vec{b}$ or $\vec{c}$ are collinear, or $\vec{c}$ is perpendicular to $\vec{a} \times \vec{b}$.
- If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-zero and non-coplanar vectors, then $[\vec{a} \vec{b} \vec{c}] \neq 0$.

For non-zero and non-coplanar vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, the scalar triple product $[\vec{a} \ \vec{b} \ \vec{c}]$ can be interpreted geometrically as the volume of the parallelepiped with $\vec{a}$, $\vec{b}$ and $\vec{c}$ as coterminal edges. This is proved in the following.

THEOREM 6.33

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-zero non-coplanar vectors. Let O , A , B and C be points such that $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and $\vec{OC} = \vec{c}$. Let V be the volume of the parallelepiped with $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ as coterminal edges. Then

1. $(\vec{a} \ \vec{b}) \times \vec{c} = V$ if $(\vec{a}, \vec{b}, \vec{c})$ is a right-handed system.
2. $(\vec{a} \ \vec{b}) \times \vec{c} = -V$ if $(\vec{a}, \vec{b}, \vec{c})$ is a left-handed system.

PROOF

Since $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar, the points O , A , B and C are non-coplanar. Complete the parallelepiped $OADBFCGE$ having $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ as coterminal edges (Figure 6.25).

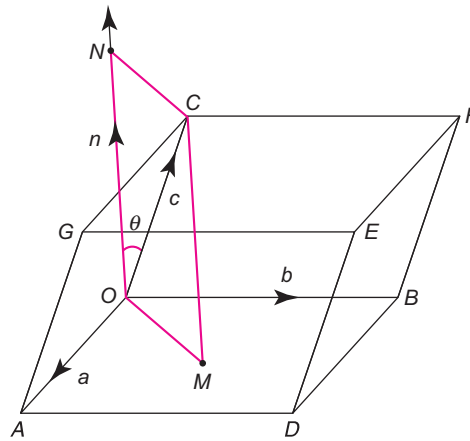


FIGURE 6.25 Theorem 6.33.

1. Suppose that $(\vec{a}, \vec{b}, \vec{c})$ is a right-handed system. Draw the perpendicular CM to the plane determined by $\vec{OA}$ and $\vec{OB}$ and N be the foot of the perpendicular to the support of $\vec{a} \ \vec{b}$ from C . Let $\vec{n}$ be the unit vector in the direction of $\vec{a} \ \vec{b}$. Then $(\vec{a}, \vec{b}, \vec{n})$ is a right-handed system. Let θ be the angle between $\vec{a} \ \vec{b}$ and $\vec{c}$; that is, $\angle CON = \theta$. Then

$$\begin{aligned}
 V &= (\text{Area of the base parallelogram } OADB) \ (\text{Height of the vertex } C \text{ from the base}) \\
 &= |\vec{a} \ \vec{b}| (CM) \\
 &= |\vec{a} \ \vec{b}| (ON) \\
 &= |\vec{a} \ \vec{b}| (OC) \cos \theta \\
 &= |\vec{a} \ \vec{b}| |\vec{c}| \cos \theta \\
 &= (\vec{a} \ \vec{b}) \times \vec{c} \\
 &= [\vec{a} \ \vec{b} \ \vec{c}]
 \end{aligned}$$

2. Suppose that $(\vec{a}, \vec{b}, \vec{c})$ is a left-handed system. Then $(\vec{a}, \vec{b}, -\vec{c})$ is a right-handed system and hence

$$V = (\vec{a} \ \vec{b}) \times (-\vec{c}) = -[(\vec{a} \ \vec{b}) \times \vec{c}]$$

Therefore

$$(\vec{a} \ \vec{b}) \times \vec{c} = V$$

THEOREM 6.34 Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear. Then

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0 \quad \vec{a}, \vec{b} \text{ and } \vec{c} \text{ are coplanar}$$

PROOF Suppose that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar. Since $\vec{a} \times \vec{b}$ is perpendicular to the plane determined by $\vec{a}$ and $\vec{b}$, it follows that $\vec{a} \times \vec{b}$ is perpendicular to $\vec{c}$ and hence

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = 0$$

Conversely, suppose that $(\vec{a} \times \vec{b}) \cdot \vec{c} = 0$. Then $\vec{a} \times \vec{b}$ is perpendicular to $\vec{c}$. But, by the definition of $\vec{a} \times \vec{b}$, we have that $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$. Therefore $\vec{a} \times \vec{b}$ is perpendicular to $\vec{a}$, $\vec{b}$ and $\vec{c}$. Thus $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar. ■

COROLLARY 6.15 Four distinct points A , B , C and D are coplanar if and only if $[\overrightarrow{AB} \ \overrightarrow{AC} \ \overrightarrow{AD}] = 0$.

PROOF This follows from Theorem 6.34 and the fact that A , B , C and D are coplanar if and only if $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar. ■

THEOREM 6.35 Let $OXYZ$ be a rectangular Cartesian coordinate system and $\vec{i}$, $\vec{j}$ and $\vec{k}$ be unit vectors along the positive axes $\overrightarrow{OX}$, $\overrightarrow{OY}$ and $\overrightarrow{OZ}$, respectively. Let

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

$$\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

be any three vectors. Then the scalar triple product $[\vec{a} \ \vec{b} \ \vec{c}]$ is precisely given by

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

PROOF From Theorem 6.24, we have

$$\begin{aligned} \vec{a} \times \vec{b} &= (a_2b_3 - a_3b_2)\vec{i} + (a_3b_1 - a_1b_3)\vec{j} + (a_1b_2 - a_2b_1)\vec{k} \\ &= (a_2b_3 - a_3b_2)\vec{i} - (a_1b_3 - a_3b_1)\vec{j} + (a_1b_2 - a_2b_1)\vec{k} \end{aligned}$$

Therefore

$$\begin{aligned} [\vec{a} \ \vec{b} \ \vec{c}] &= (\vec{a} \times \vec{b}) \cdot \vec{c} \\ &= (a_2b_3 - a_3b_2)c_1 - (a_1b_3 - a_3b_1)c_2 + (a_1b_2 - a_2b_1)c_3 \\ &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \end{aligned}$$

COROLLARY 6.16 Any vectors

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

$$\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

are coplanar if and only if the following determinant is zero:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

THEOREM 6.36 For any vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$,

$$[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$$

that is

$$(\vec{a} \ \vec{b}) \times \vec{c} = (\vec{b} \ \vec{c}) \times \vec{a} = (\vec{c} \ \vec{a}) \times \vec{b}$$

PROOF If one of $\vec{a}$, $\vec{b}$ and $\vec{c}$ is $\vec{0}$ or if any two of them are collinear, then the equality trivially holds. Therefore, we can assume that all of them are non-zero and no-two of them are collinear. Also, we can assume that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar. If $(\vec{a}, \vec{b}, \vec{c})$ is a right-handed system, then so are $(\vec{b}, \vec{c}, \vec{a})$ and $(\vec{c}, \vec{a}, \vec{b})$ and hence, by the Theorem 6.35

$$(\vec{a} \ \vec{b}) \times \vec{c} = (\vec{b} \ \vec{c}) \times \vec{a} = (\vec{c} \ \vec{a}) \times \vec{b} = \text{Volume of the parallelepiped} = V$$

If $(\vec{a}, \vec{b}, \vec{c})$ is a left-handed system, then so are $(\vec{b}, \vec{c}, \vec{a})$ and $(\vec{c}, \vec{a}, \vec{b})$ and hence

$$(\vec{a} \ \vec{b}) \times \vec{c} = -V = (\vec{b} \ \vec{c}) \times \vec{a} = (\vec{c} \ \vec{a}) \times \vec{b}$$

THEOREM 6.37 For any three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$,

$$(\vec{a} \ \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \ \vec{c})$$

(That is, in a scalar triple product, the operations dot and cross can be interchanged.)

PROOF From Theorem 6.36,

$$(\vec{a} \ \vec{b}) \times \vec{c} = (\vec{b} \ \vec{c}) \times \vec{a} = \vec{a} \times (\vec{b} \ \vec{c})$$

since dot product is commutative.

COROLLARY 6.17 Let $\vec{\alpha}$, $\vec{\beta}$ and $\vec{\gamma}$ be three non-coplanar vectors and

$$\vec{a} = a_1 \vec{\alpha} + a_2 \vec{\beta} + a_3 \vec{\gamma}$$

$$\vec{b} = b_1 \vec{\alpha} + b_2 \vec{\beta} + b_3 \vec{\gamma}$$

$$\vec{c} = c_1 \vec{\alpha} + c_2 \vec{\beta} + c_3 \vec{\gamma}$$

be any vectors. Then $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar if and only if

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

PROOF By Theorem 6.28, we have

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{\beta} & \vec{\gamma} & \vec{\gamma} & \vec{\alpha} & \vec{\alpha} & \vec{\beta} \\ a_1 & & a_2 & & a_3 & \\ b_1 & & b_2 & & b_3 & \end{vmatrix} \\ &= (a_2b_3 - a_3b_2)(\vec{\beta} \times \vec{\gamma}) - (a_1b_3 - a_3b_1)(\vec{\gamma} \times \vec{\alpha}) + (a_1b_2 - a_2b_1)(\vec{\alpha} \times \vec{\beta}) \end{aligned}$$

Therefore

$$\begin{aligned} [\vec{a} \times \vec{b} \times \vec{c}] &= (\vec{a} \times \vec{b}) \times \vec{c} \\ &= (a_2b_3 - a_3b_2)(\vec{\beta} \times \vec{\gamma}) \times c_1\vec{\alpha} - (a_1b_3 - a_3b_1)(\vec{\gamma} \times \vec{\alpha}) \times c_2\vec{\beta} \\ &\quad + (a_1b_2 - a_2b_1)(\vec{\alpha} \times \vec{\beta}) \times c_3\vec{\gamma} \quad [\text{since } (\vec{\beta} \times \vec{\gamma}) \times \vec{\beta} = 0 = (\vec{\beta} \times \vec{\gamma}) \times \vec{\gamma}, \text{ etc.}] \\ &= [(a_2b_3 - a_3b_2)c_1 - (a_1b_3 - a_3b_1)c_2 + (a_1b_2 - a_2b_1)c_3][\vec{\alpha} \times \vec{\beta} \times \vec{\gamma}] \end{aligned}$$

since, $(\vec{\beta} \times \vec{\gamma}) \times \vec{\alpha} = (\vec{\gamma} \times \vec{\alpha}) \times \vec{\beta} = (\vec{\alpha} \times \vec{\beta}) \times \vec{\gamma} = [\vec{\alpha} \times \vec{\beta} \times \vec{\gamma}]$ (see Theorems 6.36 and 6.37). Therefore

$$[\vec{a} \times \vec{b} \times \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} [\vec{\alpha} \times \vec{\beta} \times \vec{\gamma}] \quad (6.6)$$

Since $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are coplanar, $[\vec{\alpha} \times \vec{\beta} \times \vec{\gamma}] = 0$. Therefore

$$\vec{a}, \vec{b} \text{ and } \vec{c} \text{ are coplanar} \quad [\vec{a} \times \vec{b} \times \vec{c}] = 0$$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

QUICK LOOK 11

1. During the course of the proof of Corollary 6.17, we have proved the following result.

If $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ are non-coplanar vectors and

$$\vec{a} = a_1\vec{\alpha} + a_2\vec{\beta} + a_3\vec{\gamma}$$

$$\vec{b} = b_1\vec{\alpha} + b_2\vec{\beta} + b_3\vec{\gamma}$$

$$\vec{c} = c_1\vec{\alpha} + c_2\vec{\beta} + c_3\vec{\gamma}$$

Then

$$[\vec{a} \times \vec{b} \times \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} [\vec{\alpha} \times \vec{\beta} \times \vec{\gamma}]$$

which is same as Eq. (6.6).

2. Using the property that determinant of product of two square matrices is equal to the product of the determinants of the matrices, the following result can be easily proved:

If $\vec{l}, \vec{m}, \vec{n}, \vec{a}, \vec{b}, \vec{c}$ are two sets of vectors each of which is expressed in terms of $\vec{i}, \vec{j}$ and $\vec{k}$ then

$$(i) [\vec{l} \times \vec{m} \times \vec{n}](\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{l} & \vec{l} \times \vec{a} & \vec{l} \times \vec{b} \\ \vec{m} & \vec{m} \times \vec{a} & \vec{m} \times \vec{b} \\ \vec{n} & \vec{n} \times \vec{a} & \vec{n} \times \vec{b} \end{vmatrix}$$

$$(ii) [\vec{l} \times \vec{m} \times \vec{n}][\vec{a} \times \vec{b} \times \vec{c}] = \begin{vmatrix} \vec{l} \times \vec{a} & \vec{l} \times \vec{b} & \vec{l} \times \vec{c} \\ \vec{m} \times \vec{a} & \vec{m} \times \vec{b} & \vec{m} \times \vec{c} \\ \vec{n} \times \vec{a} & \vec{n} \times \vec{b} & \vec{n} \times \vec{c} \end{vmatrix}$$

THEOREM 6.38 The volume of a tetrahedron with $\vec{a}, \vec{b}$ and $\vec{c}$ as coterminus edges is

$$\frac{1}{6} |[\vec{a} \times \vec{b} \times \vec{c}]|$$

PROOF Let $OABC$ be a tetrahedron and $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$. Let V be the volume of the tetrahedron $OABC$ (Figure 6.26). Then the volume is given by

$$V = \frac{1}{3}(\text{Area of the base } \triangle OAB)(\text{Length of the perpendicular from } C \text{ to the base } \triangle OAB)$$

Draw perpendicular CN from C to the triangle OAB and perpendicular CM from C to the supporting line of $\vec{a} \times \vec{b}$.

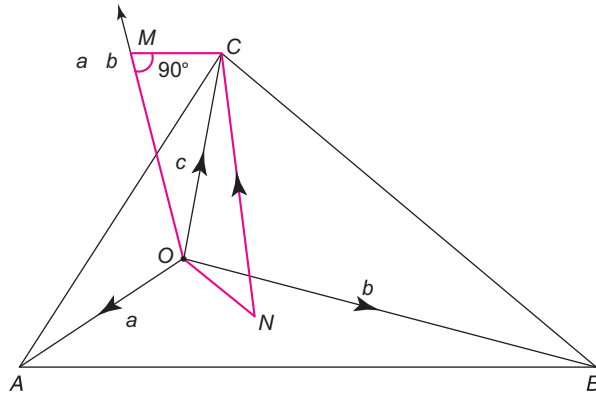


FIGURE 6.26 Theorem 6.38.

Then

$$\begin{aligned} CN &= OM \\ &= \text{Length of the projection of } \vec{c} \text{ on } \vec{a} \times \vec{b} \\ &= \frac{|(\vec{a} \times \vec{b}) \cdot \vec{c}|}{|\vec{a} \times \vec{b}|} \\ &= \frac{|[\vec{a} \ \vec{b} \ \vec{c}]|}{|\vec{a} \times \vec{b}|} \end{aligned}$$

Also,

$$\text{Area of } \triangle OAB = \frac{1}{2} |\vec{a} \times \vec{b}|$$

Therefore

$$V = \frac{1}{3} \left(\frac{1}{2} |\vec{a} \times \vec{b}| \right) \left(\frac{|[\vec{a} \ \vec{b} \ \vec{c}]|}{|\vec{a} \times \vec{b}|} \right) = \frac{|[\vec{a} \ \vec{b} \ \vec{c}]|}{6}$$

■

COROLLARY 6.18 The volume of a tetrahedron whose vertices are A, B, C and D is

$$\frac{1}{6} (|[\overrightarrow{AB} \ \overrightarrow{AC} \ \overrightarrow{AD}]|)$$

PROOF Since $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coterminal edges of the tetrahedron $ABCD$, it follows from Theorem 6.38 that the volume of the tetrahedron is

$$\frac{1}{6} (|[\overrightarrow{AB} \ \overrightarrow{AC} \ \overrightarrow{AD}]|)$$

■

Example 6.16

Consider the vectors $\vec{a} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{b} = \vec{i} - 3\vec{j} - 5\vec{k}$ and $\vec{c} = 3\vec{i} - 4\vec{j} - 4\vec{k}$. Prove that they are coplanar.

Solution: For the given vectors we have

$$[\vec{a} \vec{b} \vec{c}] = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$= \begin{vmatrix} 2 & -1 & 1 \\ 1 & -3 & -5 \\ 3 & -4 & -4 \end{vmatrix}$$

$$= 2(12 - 20) + 1(-4 + 15) + 1(-4 + 9)$$

$$= 0$$

Hence $\vec{a}, \vec{b}, \vec{c}$ are coplanar.

Example 6.17

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar vectors and let $\vec{\alpha} = 3\vec{a} + 2\vec{b} - 5\vec{c}$, $\vec{\beta} = -3\vec{a} + 8\vec{b} - 5\vec{c}$, $\vec{\gamma} = -3\vec{a} + 2\vec{b} + \vec{c}$ and $\vec{\delta} = -\vec{a} + 4\vec{b} - 3\vec{c}$ be four points. Prove that $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ and $\vec{\delta}$ are coplanar.

Solution: Let A, B, C and D be four points and O be the origin such that $\vec{OA} = \vec{\alpha}$, $\vec{OB} = \vec{\beta}$, $\vec{OC} = \vec{\gamma}$ and $\vec{OD} = \vec{\delta}$. Then

$$\vec{AB} = \vec{\beta} - \vec{\alpha}$$

$$= (-3\vec{a} + 8\vec{b} - 5\vec{c}) - (3\vec{a} + 2\vec{b} - 5\vec{c})$$

$$= -6\vec{a} + 6\vec{b}$$

$$\vec{AC} = \vec{\gamma} - \vec{\alpha} = -6\vec{a} + 6\vec{c}$$

$$\vec{AD} = \vec{\delta} - \vec{\alpha} = -4\vec{a} + 2\vec{b} + 2\vec{c}$$

Therefore

$$[\vec{AB} \vec{AC} \vec{AD}] = \begin{vmatrix} -6 & 6 & 0 \\ -6 & 0 & 6 \\ -4 & 2 & 2 \end{vmatrix} [\vec{a} \vec{b} \vec{c}] \quad (\text{see Quick Look 11})$$

$$= [-6(-12) - 6(-12 + 24)][\vec{a} \vec{b} \vec{c}] = 0$$

Therefore $\vec{AB}, \vec{AC}$ and $\vec{AD}$ are coplanar. Hence the points A, B, C and D are coplanar. So $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ and $\vec{\delta}$ are coplanar.

Example 6.18

Find the volume of the parallelepiped with $2\vec{i} - 3\vec{j}$, $\vec{i} + \vec{j} - \vec{k}$ and $3\vec{i} - \vec{k}$ as coterminal edges.

$$= 2(-1) + 3[-1 - (-3)] + 0(3)$$

$$= 4$$

Solution: Let $\vec{a} = 2\vec{i} - 3\vec{j}$, $\vec{b} = \vec{i} + \vec{j} - \vec{k}$ and $\vec{c} = 3\vec{i} - \vec{k}$. Then

Therefore the volume of the parallelepiped is

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix}$$

$$|[\vec{a} \vec{b} \vec{c}]| = 4$$

We have discussed in the previous chapter about the vector equation of a plane in various forms. In the following, we obtain vector equation of a plane using dot, cross and scalar triple products. Also, we introduce the concept of skew lines and derive a formula for the shortest distance between two skew lines. In the following, we fix a point O as the origin of reference and adopt the notation that $A(\vec{a})$ is a point, that is, $\vec{a}$ is the position vector of a point A with respect to the origin O . Let us first prove the following.

THEOREM 6.39

Let $A(\vec{a})$ be a point and $\vec{b}$ and $\vec{c}$ be two non-collinear vectors. Then the vector equation of the plane passing through A and parallel to the vectors $\vec{b}$ and $\vec{c}$ is

$$[\vec{r} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]$$

PROOF Let $P(\vec{r})$ be any point in the required plane. We may assume that $A \neq P$. Then

P lies in the plane Vectors $\overrightarrow{AP}$, $\vec{b}$ and $\vec{c}$ are coplanar

$$[\overrightarrow{AP} \vec{b} \vec{c}] = 0$$

$$\overrightarrow{AP} \times (\vec{b} \times \vec{c}) = 0$$

$$(\vec{r} - \vec{a}) \times (\vec{b} \times \vec{c}) = 0$$

$$\vec{r} \times (\vec{b} \times \vec{c}) = \vec{a} \times (\vec{b} \times \vec{c})$$

$$[\vec{r} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]$$

Thus the equation of the plane passing through A and parallel to $\vec{b}$ and $\vec{c}$ is

$$[\vec{r} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]$$

THEOREM 6.40 Let $A(\vec{a})$ and $B(\vec{b})$ be two given distinct points and $\vec{c}$ a vector. Then the vector equation of the plane passing through A and B and parallel to $\vec{c}$ is

$$[\vec{r} \vec{b} \vec{c}] + [\vec{r} \vec{c} \vec{a}] = [\vec{a} \vec{b} \vec{c}]$$

PROOF Consider an arbitrary point $P(\vec{r}) \neq A$. Then

$P(\vec{r})$ lies in the required plane Vector $\overrightarrow{AP} \perp \overrightarrow{AB}$ is perpendicular to the plane

$\overrightarrow{AP} \perp \overrightarrow{AB}$ is perpendicular to $\vec{c}$

$$(\overrightarrow{AP} \perp \overrightarrow{AB}) \times \vec{c} = 0$$

$$\overrightarrow{AP} \times (\overrightarrow{AB} \times \vec{c}) = 0$$

$$(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) \times \vec{c} = 0$$

$$(\vec{r} - \vec{a}) \times (\vec{b} \times \vec{c} + \vec{c} \times \vec{a}) = 0$$

$$\vec{r} \times (\vec{b} \times \vec{c}) + \vec{r} \times (\vec{c} \times \vec{a}) = \vec{a} \times (\vec{b} \times \vec{c})$$

$$[\vec{r} \vec{b} \vec{c}] + [\vec{r} \vec{c} \vec{a}] = [\vec{a} \vec{b} \vec{c}]$$

Thus the vector equation of the required plane is

$$[\vec{r} \vec{b} \vec{c}] + [\vec{r} \vec{c} \vec{a}] = [\vec{a} \vec{b} \vec{c}]$$

THEOREM 6.41 Let $A(\vec{a})$, $B(\vec{b})$ and $C(\vec{c})$ be three given non-collinear points. Then the vector equation of the plane passing through A , B and C is

$$[\vec{r} \vec{b} \vec{c}] + [\vec{r} \vec{c} \vec{a}] + [\vec{r} \vec{a} \vec{b}] = [\vec{a} \vec{b} \vec{c}]$$

PROOF Let π be the plane passing through A , B and C and $P(\vec{r})$ be an arbitrary point. Then

$P(\vec{r})$ lies on π A, B, C and P are coplanar

$\overrightarrow{PA}$, $\overrightarrow{PB}$ and $\overrightarrow{PC}$ are coplanar

$\vec{a} - \vec{r}$, $\vec{b} - \vec{r}$ and $\vec{c} - \vec{r}$ are coplanar

$$[\vec{a} \quad \vec{r} \quad \vec{b} \quad \vec{r} \quad \vec{c} \quad \vec{r}] = 0$$

$$(\vec{a} \quad \vec{r}) \times (\vec{b} \quad \vec{r}) \cdot (\vec{c} \quad \vec{r}) = 0$$

$$\vec{r} \times (\vec{b} \quad \vec{c}) + (\vec{c} \quad \vec{a}) + (\vec{a} \quad \vec{b}) = \vec{a} \times (\vec{b} \quad \vec{c})$$

$$[\vec{r} \quad \vec{b} \quad \vec{c}] + [\vec{r} \quad \vec{c} \quad \vec{a}] + [\vec{r} \quad \vec{a} \quad \vec{b}] = [\vec{a} \quad \vec{b} \quad \vec{c}]$$

Thus the vector equation of the plane is

$$[\vec{r} \quad \vec{b} \quad \vec{c}] + [\vec{r} \quad \vec{c} \quad \vec{a}] + [\vec{r} \quad \vec{a} \quad \vec{b}] = [\vec{a} \quad \vec{b} \quad \vec{c}]$$

THEOREM 6.42 The vector equation of the plane containing the line $\vec{r} = \vec{a} + t\vec{b}$, $t \in \mathbb{R}$ and perpendicular to the plane $\vec{r} \times \vec{c} = q$ is $[\vec{r} \quad \vec{b} \quad \vec{c}] = [\vec{a} \quad \vec{b} \quad \vec{c}]$.

PROOF Recall that the vector $\vec{c}$ is normal to the plane $\vec{r} \times \vec{c} = q$. Since the required plane contains the line $\vec{r} = \vec{a} + t\vec{b}$, $t \in \mathbb{R}$, it should pass through the point $\vec{a}$ and is parallel to the vectors $\vec{b}$ and $\vec{c}$. Thus, by Theorem 6.39, the equation of the required plane is $[\vec{r} \quad \vec{b} \quad \vec{c}] = [\vec{a} \quad \vec{b} \quad \vec{c}]$.

DEFINITION 6.7 Skew Lines Two lines in the space are said to be *skew lines* if there is no plane containing both the lines.

Note that if two lines intersect, then there is a plane containing both of them and hence they are not skew lines. Therefore, two skew lines have no common point. However, they will have a common perpendicular. In the following we prove that any two skew lines have a common perpendicular and derive a formula to find its length.

THEOREM 6.43 Let L_1 and L_2 be two skew lines. Then there exists unique pair (P, Q) of points with P on L_1 and Q on L_2 such that PQ is perpendicular to both L_1 and L_2 .

PROOF Let $A(\vec{a})$ and $C(\vec{c})$ be points on L_1 and L_2 , respectively, and let $\vec{b}$ and $\vec{d}$ be parallel to L_1 and L_2 , respectively (Figure 6.27). Then the equations of L_1 and L_2 are

$$\vec{r} = \vec{a} + t\vec{b}$$

and

$$\vec{r} = \vec{c} + s\vec{d}$$

where t and s are scalars. Observe that L_1 and L_2 are not parallel if and only if $|\vec{b} \times \vec{d}| \neq |\vec{b}||\vec{d}|$. Consider the following simultaneous equations:

$$(\vec{b} \times \vec{d})s - |\vec{b}|^2 t = (\vec{a} - \vec{c}) \times \vec{b} \quad (6.7)$$

$$|\vec{d}|^2 s - |\vec{b} \times \vec{d}| t = (\vec{a} - \vec{c}) \times \vec{d} \quad (6.8)$$

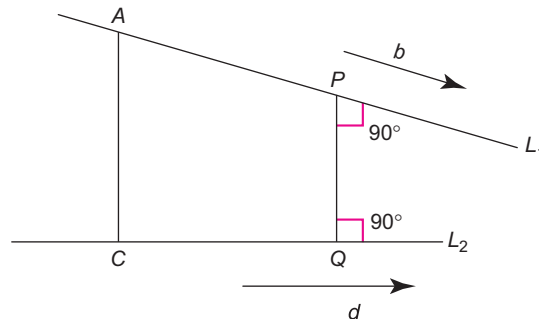


FIGURE 6.27 Theorem 6.43.

Since $|\vec{b}||\vec{d}| = |\vec{b} \times \vec{d}|$, Eqs. (6.7) and (6.8) have unique common solution, say (s_1, t_1) . Let P and Q be the points on L_1 and L_2 corresponding to t_1 and s_1 , respectively. Now, we have

$$\begin{aligned}\vec{b} \times \overrightarrow{PQ} &= \vec{b} \times (\vec{c} + s_1 \vec{d} - (\vec{a} + t_1 \vec{b})) \\ &= \vec{b} \times (\vec{c} - \vec{a}) + (s_1 \vec{d} - t_1 \vec{b}) \times \vec{b} \\ &= \vec{b} \times (\vec{c} - \vec{a}) + s_1 \vec{b} \times \vec{d} - t_1 \vec{b} \times \vec{b} \\ &= 0 \quad [\text{from Eq. (6.7)}]\end{aligned}$$

Therefore, $\overrightarrow{PQ}$ is perpendicular to $\vec{b}$ and hence to L_1 (since $\vec{b}$ is parallel to L_1). Similarly, using Eq. (6.8), we can prove that $\overrightarrow{PQ}$ is perpendicular to L_2 .

Next, we will prove that P and Q are unique. If P and Q are points on L_1 and L_2 , respectively, such that $\overrightarrow{PQ}$ is perpendicular to both L_1 and L_2 , then the corresponding scalars s and t satisfy Eqs. (6.7) and (6.8). By the uniqueness of (s_1, t_1) it follows that $P = P$ and $Q = Q$. Thus P and Q are unique on L_1 and L_2 , respectively, such that $\overrightarrow{PQ}$ is perpendicular to both L_1 and L_2 . ■

Note that, if P and Q are points on L_1 and L_2 respectively, then $\overrightarrow{PQ} \perp \vec{b}$ and $\overrightarrow{PQ} \perp \vec{d}$. Therefore $\overrightarrow{PQ}$ is the *shortest distance* between L_1 and L_2 .

THEOREM 6.44

The shortest distance between two skew lines whose equations are $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{c} + s\vec{d}$, $s, t \in \mathbb{R}$, is

$$\frac{|(\vec{a} - \vec{c}) \times (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}$$

PROOF

By Theorem 6.43, there are unique points P and Q on L_1 and L_2 , respectively, such that $\overrightarrow{PQ}$ is perpendicular to both L_1 and L_2 . Then $\overrightarrow{PQ}$ is parallel to $\vec{b} \times \vec{d}$. Choose points A and C such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OC} = \vec{c}$, where O is the origin. Now, we have

$$\begin{aligned}PQ &= \text{Magnitude of the projection vector of } \overrightarrow{AC} \text{ on } \overrightarrow{PQ} \\ &= \text{Magnitude of the projection vector of } \overrightarrow{AC} \text{ on } \vec{b} \times \vec{d} \\ &= \frac{|(\vec{c} - \vec{a}) \times (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|} \quad (\text{see Theorem 6.1}) \\ &= \frac{|(\vec{a} - \vec{c}) \times (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}\end{aligned}$$

Note that PQ is the shortest distance between L_1 and L_2 . ■

Example 6.19

Let $A = (2, 3, -1)$, $B = (4, 5, 2)$ and $C = (3, 6, 5)$. Find the equation of the plane passing through A , B and C .

Solution: Let O be the origin. Then, we are given that

$$\begin{aligned}\vec{a} &= \overrightarrow{OA} = 2\vec{i} + 3\vec{j} - \vec{k} \\ \vec{b} &= \overrightarrow{OB} = 4\vec{i} + 5\vec{j} + 2\vec{k} \\ \vec{c} &= \overrightarrow{OC} = 3\vec{i} + 6\vec{j} + 5\vec{k}\end{aligned}$$

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ be the position vector (with respect to the origin O) of an arbitrary point P in the space.

Then, by Theorem 6.41, the equation of the plane passing through A , B and C is

$$[\vec{r} \ \vec{b} \ \vec{c}] + [\vec{r} \ \vec{c} \ \vec{a}] + [\vec{r} \ \vec{a} \ \vec{b}] = [\vec{a} \ \vec{b} \ \vec{c}]$$

$$\begin{vmatrix} x & y & z \\ 4 & 5 & 2 \\ 3 & 6 & 5 \end{vmatrix} + \begin{vmatrix} x & y & z \\ 3 & 6 & 5 \\ 2 & 3 & -1 \end{vmatrix} + \begin{vmatrix} x & y & z \\ 2 & 3 & -1 \\ 4 & 5 & 2 \end{vmatrix} = \begin{vmatrix} 2 & 3 & -1 \\ 4 & 5 & 2 \\ 3 & 6 & 5 \end{vmatrix}$$

Alternate Solution: Problems of this type can be solved directly as follows without referring to Theorem 6.41.

$P(\vec{r})$ lies on the plane passing through A, B and C and

$\overrightarrow{AP}, \overrightarrow{AB}$ and $\overrightarrow{AC}$ are coplanar

$$[\overrightarrow{AP} \overrightarrow{AB} \overrightarrow{AC}] = 0$$

We have

$$\overrightarrow{AP} = \vec{r} \quad \vec{a} = (x-2)\vec{i} + (y-3)\vec{j} + (z+1)\vec{k}$$

$$\overrightarrow{AB} = \vec{b} \quad \vec{a} = 2\vec{i} + 2\vec{j} + 3\vec{k}$$

$$\overrightarrow{AC} = \vec{c} \quad \vec{a} = \vec{i} + 3\vec{j} + 6\vec{k}$$

$$\begin{aligned} [\overrightarrow{AP} \overrightarrow{AB} \overrightarrow{AC}] &= \begin{vmatrix} x-2 & y-3 & z+1 \\ 2 & 2 & 3 \\ 1 & 3 & 6 \end{vmatrix} \\ &= (x-2)(12-9) - (y-3)(12-3) \\ &\quad + (z+1)(6-2) \\ &= 3x - 9y + 4z + 25 \end{aligned}$$

Thus, the equation of the plane passing through A, B and C is

$$3x - 9y + 4z + 25 = 0$$

Example 6.20

Find the shortest distance between the following skew lines: Then

$$L_1: \vec{r} = (\vec{i} - 2\vec{j} - \vec{k}) + t(4\vec{i} - 3\vec{k})$$

$$L_2: \vec{r} = (\vec{i} + 2\vec{j} - \vec{k}) + s(2\vec{i} - 4\vec{j} - 5\vec{k})$$

$$\vec{a} - \vec{c} = 4\vec{j}$$

$$\vec{b} - \vec{d} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & 0 & 3 \\ 2 & 4 & 5 \end{vmatrix} = 12\vec{i} + 14\vec{j} - 16\vec{k}$$

Solution: Let

$$\vec{a} = \vec{i} - 2\vec{j} - \vec{k}$$

$$\vec{b} = 4\vec{i} - 3\vec{k}$$

$$\vec{c} = \vec{i} + 2\vec{j} - \vec{k}$$

$$\vec{d} = 2\vec{i} - 4\vec{j} - 5\vec{k}$$

From Theorem 6.44, the shortest distance between L_1 and L_2 is

$$\begin{aligned} \frac{|(\vec{a} - \vec{c}) \times (\vec{b} - \vec{d})|}{|\vec{b} - \vec{d}|} &= \frac{|(4\vec{j}) \times (12\vec{i} + 14\vec{j} - 16\vec{k})|}{|12\vec{i} + 14\vec{j} - 16\vec{k}|} \\ &= \frac{|(-4)14|}{\sqrt{(12)^2 + (14)^2 + (-16)^2}} = \frac{56}{\sqrt{596}} = \frac{28}{\sqrt{149}} \end{aligned}$$

As per note under Theorem 6.22, the following is an alternate proof of Theorem 6.22.

THEOREM 6.45 For any three vectors $\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

PROOF Let $\vec{r}$ be any vector. Since the dot product is distributive

$$\begin{aligned} \vec{r} \times [\vec{a} \cdot (\vec{b} + \vec{c}) - (\vec{a} \cdot \vec{b}) - (\vec{a} \cdot \vec{c})] &= \vec{r} \times [\vec{a} \cdot (\vec{b} + \vec{c})] - \vec{r} \times [\vec{a} \cdot \vec{b}] - \vec{r} \times [\vec{a} \cdot \vec{c}] \\ &= (\vec{r} \times \vec{a}) \cdot (\vec{b} + \vec{c}) - (\vec{r} \times \vec{a}) \cdot \vec{b} - (\vec{r} \times \vec{a}) \cdot \vec{c} \quad (\because \text{dot and cross can be interchanged}) \\ &= (\vec{r} \times \vec{a}) \cdot (\vec{b} + \vec{c}) - (\vec{r} \times \vec{a}) \cdot \vec{b} - (\vec{r} \times \vec{a}) \cdot \vec{c} \quad (\because \text{dot and cross can be interchanged}) \\ &= (\vec{r} \times \vec{a}) \cdot [\vec{b} + \vec{c} - \vec{b} - \vec{c}] \quad (\because \text{dot product is distributive}) \\ &= 0 \end{aligned}$$

This is true for any vector $\vec{r}$. Therefore

$$\vec{a} \cdot (\vec{b} + \vec{c}) - (\vec{a} \cdot \vec{b}) - (\vec{a} \cdot \vec{c}) = 0$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

6.6 | Vector Triple Product

The concept of scalar triple product of three vectors is introduced in the previous section and using this we have derived equations of a plane in different forms. In this section we introduce the notion of the vector triple product of three vectors and discuss certain properties.

DEFINITION 6.8 For any three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, $(\vec{a} \times \vec{b}) \times \vec{c}$ is called the *vector triple product* of $\vec{a}$, $\vec{b}$ and $\vec{c}$.

In respect of scalar triple products, we have proved that

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$$

Such type of associativity does not hold for vector triple products. For, consider the following example.

Example 6.21

Let $\vec{a} = \vec{i} + \vec{j} + 2\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$ and $\vec{c} = 2\vec{i} + \vec{j} + \vec{k}$. Prove that associatively does not hold. That is prove that $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$

Solution: For the given vectors we have

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 3\vec{i} + \vec{j} + \vec{k}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = 5\vec{j} - 5\vec{k}$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{vmatrix} = \vec{i} + \vec{j} - 3\vec{k}$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 2 \\ 1 & 1 & 3 \end{vmatrix} = 5\vec{i} + 5\vec{j}$$

Therefore $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$.

THEOREM 6.46 The following hold for any vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

1. $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
2. $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

PROOF

1. If $\vec{a}$ and $\vec{b}$ are collinear, say $\vec{a} = t\vec{b}$, $t \in \mathbb{R}$, then

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{0} \times \vec{c} = \vec{0}$$

and

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (t\vec{b} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})(t\vec{b}) = \vec{0}$$

and hence (1) holds. Also, if $\vec{c}$ is parallel to $\vec{a} \times \vec{b}$, then $\vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$ so that $\vec{a} \cdot \vec{c} = 0 = \vec{b} \cdot \vec{c}$ and therefore

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{0} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

Therefore, we can assume that $\vec{a}$ and $\vec{b}$ are non-collinear and $\vec{c}$ is not parallel to $\vec{a} \times \vec{b}$.

Fix the origin O and select points A and B such that $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$. Let $\vec{i}$ be the unit vector in the direction of $\vec{OA}$ and let $\vec{j}$ be the unit vector perpendicular to $\vec{i}$ in the OAB plane. Let $\vec{k}$ be the unit vector perpendicular to the OAB plane such that $(\vec{i}, \vec{j}, \vec{k})$ is a right-handed system of orthogonal unit vectors. Note that the plane OAB is same as the XY plane. Now, we have

$$\vec{a} = a_1\vec{i}$$

$$\vec{b} = b_1\vec{i} + b_2\vec{j}$$

$$\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

Therefore

$$\begin{aligned}
 (\vec{a} \times \vec{b}) \times \vec{c} &= (a_1 b_2 \vec{k}) \times (c_1 \vec{i} + c_2 \vec{j} + c_3 \vec{k}) \\
 &= a_1 b_2 c_1 \vec{j} - a_1 b_2 c_2 \vec{i} \\
 &= a_1 c_1 (b_1 \vec{i} + b_2 \vec{j}) - (b_1 c_1 + b_2 c_2) a_1 \vec{i} \\
 &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a}
 \end{aligned}$$

2. We have

$$\begin{aligned}
 \vec{a} \times (\vec{b} \times \vec{c}) &= -[(\vec{b} \times \vec{c}) \times \vec{a}] \\
 &= -[(\vec{b} \cdot \vec{a}) \vec{c} - (\vec{c} \cdot \vec{a}) \vec{b}] \\
 &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}
 \end{aligned}$$

COROLLARY 6.19 Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be vectors such that $\vec{a}$ and $\vec{b}$ are non-collinear and $\vec{b}$ is perpendicular to neither $\vec{a}$ nor $\vec{c}$. Then

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

if and only if the vectors $\vec{a}$ and $\vec{c}$ are collinear.

PROOF Suppose that

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

Then

$$(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a} = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

Therefore

$$(\vec{b} \cdot \vec{c}) \vec{a} = (\vec{a} \cdot \vec{b}) \vec{c}$$

Since $\vec{b}$ is not perpendicular to $\vec{c}$ and to $\vec{a}$, we get that $\vec{b} \cdot \vec{c}$ and $\vec{a} \cdot \vec{b}$ are non-zero scalars. Therefore $\vec{a}$ and $\vec{c}$ are collinear.

Conversely, suppose that $\vec{a}$ and $\vec{c}$ are collinear. Then $\vec{c} = \lambda \vec{a}$ for some scalar λ . Now, consider

$$\begin{aligned}
 (\vec{a} \times \vec{b}) \times \vec{c} - \vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \times \vec{b}) \times (\lambda \vec{a}) - \vec{a} \times (\vec{b} \times \lambda \vec{a}) \\
 &= \lambda [(\vec{a} \times \vec{b}) \times \vec{a} - \{\vec{a} \times (\vec{b} \times \vec{a})\}] \\
 &= \lambda [(\vec{a} \cdot \vec{a}) \vec{b} - (\vec{b} \cdot \vec{a}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b} + (\vec{b} \cdot \vec{a}) \vec{a}] \\
 &= \vec{0}
 \end{aligned}$$

Therefore

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

THEOREM 6.47 Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be any vectors such that $\vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{c}$. Then

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

PROOF We have $\vec{a} \cdot \vec{b} = 0 = \vec{c} \cdot \vec{b}$. Now,

$$\begin{aligned}
 (\vec{a} \times \vec{b}) \times \vec{c} &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a} = (\vec{a} \cdot \vec{c}) \vec{b} \\
 \vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = (\vec{a} \cdot \vec{c}) \vec{b}
 \end{aligned}$$

Therefore

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

In the following we derive formulae for scalar and vector products of four vectors and apply there to get a representation of vectors in terms of three non-coplanar vectors.

THEOREM 6.48 The following hold for any four vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$.

$$1. (\vec{a} \ \vec{b}) \times (\vec{c} \ \vec{d}) = \begin{vmatrix} \vec{a} \times \vec{c} & \vec{a} \times \vec{d} \\ \vec{b} \times \vec{c} & \vec{b} \times \vec{d} \end{vmatrix} = (\vec{a} \times \vec{c})(\vec{b} \times \vec{d}) - (\vec{a} \times \vec{d})(\vec{b} \times \vec{c})$$

and, in particular,

$$|\vec{a} \ \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \times \vec{b})^2$$

$$2. (\vec{a} \ \vec{b}) \cdot (\vec{c} \ \vec{d}) = [\vec{a} \ \vec{c} \ \vec{d}] \vec{b} - [\vec{b} \ \vec{c} \ \vec{d}] \vec{a} = [\vec{a} \ \vec{b} \ \vec{d}] \vec{c} - [\vec{a} \ \vec{b} \ \vec{c}] \vec{d}$$

PROOF 1. We have

$$\begin{aligned} (\vec{a} \ \vec{b}) \times (\vec{c} \ \vec{d}) &= \vec{a} \times [\vec{b} \cdot (\vec{c} \ \vec{d})] \quad (\text{by Theorem 6.37}) \\ &= \vec{a} \times [(\vec{b} \times \vec{d}) \cdot \vec{c} - (\vec{b} \times \vec{c}) \cdot \vec{d}] \quad [\text{by Theorem 6.46, part (2)}] \\ &= (\vec{b} \times \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \times \vec{c})(\vec{a} \times \vec{d}) \\ &= \begin{vmatrix} \vec{a} \times \vec{c} & \vec{a} \times \vec{d} \\ \vec{b} \times \vec{c} & \vec{b} \times \vec{d} \end{vmatrix} \end{aligned}$$

and

$$\begin{aligned} |\vec{a} \ \vec{b}|^2 &= (\vec{a} \ \vec{b}) \times (\vec{a} \ \vec{b}) \\ &= (\vec{b} \times \vec{b})(\vec{a} \times \vec{a}) - (\vec{b} \times \vec{a})(\vec{a} \times \vec{b}) \\ &= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \times \vec{b})^2 \end{aligned}$$

2. We have

$$\begin{aligned} (\vec{a} \ \vec{b}) \cdot (\vec{c} \ \vec{d}) &= [\vec{a} \times \vec{c} \ \vec{d}] \vec{b} - [\vec{b} \times \vec{c} \ \vec{d}] \vec{a} \quad [\text{by Theorem 6.46, part (1)}] \\ &= [\vec{a} \ \vec{c} \ \vec{d}] \vec{b} - [\vec{b} \ \vec{c} \ \vec{d}] \vec{a} \\ (\vec{a} \ \vec{b}) \cdot (\vec{c} \ \vec{d}) &= [(\vec{a} \ \vec{b}) \times \vec{d}] \cdot \vec{c} - [(\vec{a} \ \vec{b}) \times \vec{c}] \cdot \vec{d} \quad [\text{by Theorem 6.46, part (2)}] \\ &= [\vec{a} \times \vec{b} \ \vec{d}] \cdot \vec{c} - [\vec{a} \times \vec{b} \ \vec{c}] \cdot \vec{d} \\ &= [\vec{a} \ \vec{b} \ \vec{d}] \vec{c} - [\vec{a} \ \vec{b} \ \vec{c}] \vec{d} \end{aligned}$$

If $\vec{a}, \vec{b}$ and $\vec{c}$ are any non-coplanar vectors, then we have proved earlier in Chapter 5 (Theorem 5.20) that any vector $\vec{r}$ can be uniquely expressed as $x\vec{a} + y\vec{b} + z\vec{c}$ for some triad (x, y, z) of real numbers. We express these components x, y and z in terms of scalar triple products in the following.

THEOREM 6.49 Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar vectors. Then any vector $\vec{r}$ can be expressed as

$$\vec{r} = \frac{[\vec{b} \ \vec{c} \ \vec{r}]}{[\vec{a} \ \vec{b} \ \vec{c}]} \vec{a} + \frac{[\vec{c} \ \vec{a} \ \vec{r}]}{[\vec{a} \ \vec{b} \ \vec{c}]} \vec{b} + \frac{[\vec{a} \ \vec{b} \ \vec{r}]}{[\vec{a} \ \vec{b} \ \vec{c}]} \vec{c}$$

PROOF By replacing $\vec{d}$ with $\vec{r}$ in part (2) of Theorem 6.48, we get that

$$[\vec{a} \ \vec{c} \ \vec{r}] \vec{b} - [\vec{b} \ \vec{c} \ \vec{r}] \vec{a} = [\vec{a} \ \vec{b} \ \vec{r}] \vec{c} - [\vec{a} \ \vec{b} \ \vec{c}] \vec{r}$$

Therefore

$$[\vec{a} \ \vec{b} \ \vec{c}] \vec{r} = [\vec{b} \ \vec{c} \ \vec{r}] \vec{a} + [\vec{c} \ \vec{a} \ \vec{r}] \vec{b} + [\vec{a} \ \vec{b} \ \vec{r}] \vec{c}$$

Since $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar, $[\vec{a} \vec{b} \vec{c}] \neq 0$ and therefore we get that

$$\vec{r} = \frac{[\vec{b} \vec{c} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]} \vec{a} + \frac{[\vec{c} \vec{a} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]} \vec{b} + \frac{[\vec{a} \vec{b} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]} \vec{c}$$

Note: In the above, if $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, then

$$x = \frac{[\vec{b} \vec{c} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]}, y = \frac{[\vec{c} \vec{a} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]} \text{ and } z = \frac{[\vec{a} \vec{b} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]}$$

THEOREM 6.50

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar vectors. Then $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$ are also non-coplanar and any vector $\vec{d}$ can be expressed in terms of $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$ as

$$\vec{d} = \frac{\vec{a} \times \vec{d}}{[\vec{a} \vec{b} \vec{c}]} (\vec{b} \times \vec{c}) + \frac{\vec{b} \times \vec{d}}{[\vec{a} \vec{b} \vec{c}]} (\vec{c} \times \vec{a}) + \frac{\vec{c} \times \vec{d}}{[\vec{a} \vec{b} \vec{c}]} (\vec{a} \times \vec{b})$$

PROOF

We have

$$[\vec{b} \times \vec{c} \quad \vec{c} \times \vec{a} \quad \vec{a} \times \vec{b}] = [\vec{a} \vec{b} \vec{c}]^2 \neq 0$$

since $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar. Therefore, $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$ are also non-coplanar. Let

$$\vec{d} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$$

Since $\vec{a} \times (\vec{c} \times \vec{a}) = 0 = \vec{a} \times (\vec{a} \times \vec{b})$, we have

$$\begin{aligned} \vec{a} \times \vec{d} &= x[\vec{a} \times (\vec{b} \times \vec{c})] \\ &= x[\vec{a} \vec{b} \vec{c}] \end{aligned}$$

Similarly,

$$\vec{b} \times \vec{d} = y[\vec{b} \times (\vec{c} \times \vec{a})] = y[\vec{a} \vec{b} \vec{c}]$$

and

$$\vec{c} \times \vec{d} = z[\vec{c} \times (\vec{a} \times \vec{b})] = z[\vec{a} \vec{b} \vec{c}]$$

Therefore

$$\vec{d} = \frac{\vec{a} \times \vec{d}}{[\vec{a} \vec{b} \vec{c}]} (\vec{b} \times \vec{c}) + \frac{\vec{b} \times \vec{d}}{[\vec{a} \vec{b} \vec{c}]} (\vec{c} \times \vec{a}) + \frac{\vec{c} \times \vec{d}}{[\vec{a} \vec{b} \vec{c}]} (\vec{a} \times \vec{b})$$

Example 6.22

Let $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 2\vec{i} - \vec{j} + 3\vec{k}$, $\vec{c} = \vec{i} - \vec{j}$ and $\vec{d} = 6\vec{i} + 2\vec{j} + 3\vec{k}$. Express $\vec{d}$ in terms of $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$.

Therefore $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar and hence so are $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$. We have

Solution: First note that

$$[\vec{b} \times \vec{c} \quad \vec{c} \times \vec{a} \quad \vec{a} \times \vec{b}] = [\vec{a} \vec{b} \vec{c}]^2$$

$$\begin{aligned} &= \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{vmatrix} \\ &= 5^2 = 25 \end{aligned}$$

$$[\vec{a} \vec{b} \vec{c}] = 5$$

$$\vec{a} \times \vec{d} = (1 \cdot 6) + (1 \cdot 2) + (1 \cdot 3) = 11$$

$$\vec{b} \times \vec{d} = (2 \cdot 6) + (1 \cdot 2) + (3 \cdot 3) = 19$$

$$\vec{c} \times \vec{d} = (1 \cdot 6) + (1 \cdot 2) = 4$$

From Theorem 6.50, we have

$$\vec{d} = \frac{11}{5}(\vec{b} \times \vec{c}) + \frac{19}{5}(\vec{c} \times \vec{a}) + \frac{4}{5}(\vec{a} \times \vec{b})$$

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. Let $\vec{a} = 5\vec{i} + \vec{j} + 8\vec{k}$ and $\vec{b} = \vec{i} + \vec{j} + \lambda\vec{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are orthogonal to each other then $|\lambda|$ is equal to

(A) $\sqrt{80}$ (B) $\sqrt{90}$ (C) $\sqrt{88}$ (D) $\sqrt{99}$

Solution: By hypothesis

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$$

Therefore

$$|\vec{a}|^2 - |\vec{b}|^2 = 0 \quad 90 = 2 + \lambda^2$$

This implies

$$|\lambda| = \sqrt{88}$$

Answer: (C)

2. If $\vec{a}$ is a unit vector and $\vec{x}$ is any vector such that

$$(\vec{x} + \vec{a}) \cdot (\vec{x} - \vec{a}) = 48$$

then $|\vec{x}|$ is equal to

(A) 4 (B) 5 (C) 6 (D) 7

Solution: We have

$$\begin{aligned} 48 &= (\vec{x} + \vec{a}) \cdot (\vec{x} - \vec{a}) \\ &= |\vec{x}|^2 - |\vec{a}|^2 \\ &= |\vec{x}|^2 - 1 \end{aligned}$$

Therefore

$$|\vec{x}| = 7$$

Answer: (D)

3. If θ is the angle between the vectors $\vec{i} + 3\vec{j} + 7\vec{k}$ and $\vec{i} - 3\vec{j} + 7\vec{k}$, then $\cos \theta$ is equal to

(A) $\frac{41}{59}$ (B) $\frac{40}{59}$ (C) $\frac{42}{59}$ (D) $\frac{39}{59}$

Solution: Let

$$\vec{a} = \vec{i} + 3\vec{j} + 7\vec{k}$$

and

$$\vec{b} = \vec{i} - 3\vec{j} + 7\vec{k}$$

By Theorem 6.7,

$$\cos \theta = \frac{1 \cdot 9 + 49}{\sqrt{59}\sqrt{59}} = \frac{41}{59}$$

Answer: (A)

4. The vectors $\lambda x\vec{i} - 6\vec{j} + 3\vec{k}$ and $x\vec{i} + 2\vec{j} + 2\lambda x\vec{k}$ make an obtuse angle with each other for all real x . Then λ belongs to the interval

(A) $\frac{4}{3}, 0$ (B) $\frac{4}{3}, 0$
(C) $\frac{4}{3}, 0$ (D) $0, \frac{4}{3}$

Solution: By hypothesis,

$$(\lambda x\vec{i} - 6\vec{j} + 3\vec{k}) \cdot (x\vec{i} + 2\vec{j} + 2\lambda x\vec{k}) < 0$$

for all real x . Therefore

$$\lambda x^2 + 6\lambda x - 12 < 0$$

for all real x . This implies

$$36\lambda^2 + 48\lambda < 0$$

$$3\lambda^2 + 4\lambda < 0$$

$$\frac{4}{3} < \lambda < 0$$

When $\lambda = 0$, then clearly

$$(-6\vec{j} + 3\vec{k}) \cdot (x\vec{i} + 2\vec{j}) = -12 < 0$$

Therefore

$$\frac{4}{3} < \lambda < 0$$

Answer: (B)

5. Let $\vec{a}$ and $\vec{b}$ two vectors such that $|\vec{a}| = 2\sqrt{2}$ and $|\vec{b}| = 3$ and contain angle 45° . A parallelogram is constructed with $\vec{a} - 3\vec{b}$ and $5\vec{a} + 2\vec{b}$ as adjacent sides. Then the length of the larger diagonal is

(A) $\sqrt{393}$ (B) $\sqrt{493}$ (C) $\sqrt{593}$ (D) $\sqrt{693}$

Solution: The diagonal vectors are

$$(5\vec{a} + 2\vec{b}) + (\vec{a} - 3\vec{b}) = 6\vec{a} - \vec{b}$$

and

$$(5\vec{a} + 2\vec{b}) - (\vec{a} - 3\vec{b}) = 4\vec{a} + 5\vec{b}$$

Now

$$|6\vec{a} - \vec{b}|^2 = 36|\vec{a}|^2 - 12\vec{a} \cdot \vec{b} + |\vec{b}|^2 \quad (\text{see Corollary 6.4})$$

$$= 36(8) - 12(2\sqrt{2})(3) \frac{1}{\sqrt{2}} + 9$$

$$= 288 - 72 + 9 = 225$$

Therefore

$$|6\vec{a} - \vec{b}| = 15$$

Also

$$\begin{aligned} |4\vec{a} + 5\vec{b}|^2 &= 16|\vec{a}|^2 + 40(\vec{a} \cdot \vec{b}) + 25|\vec{b}|^2 \\ &= 16(8) + 40(2\sqrt{2})(3)\frac{1}{\sqrt{2}} + 25(9) \\ &= 128 + 240 + 225 \\ &= 593 \end{aligned}$$

Therefore

$$|4\vec{a} + 5\vec{b}| = \sqrt{593}$$

Hence this is the larger diagonal.

Answer: (C)

6. In the two-dimensional analytical plane having rectangular coordinate system, A and B are two points on the curve $y = 2^{x+2}$ such that $\vec{OA} \cdot \vec{i} = 1$ and $\vec{OB} \cdot \vec{i} = 2$, where $\vec{i}$ is the unit vector in the positive direction of the X -axis. Then $|\vec{OA} - \vec{OB}|$ is equal to

- (A) 10 (B) 12 (C) 8 (D) 6

Solution: Let $A = (x_1, y_1)$ and $B = (x_2, y_2)$. Therefore

$$y_1 = 2^{x_1+2}$$

and

$$y_2 = 2^{x_2+2}$$

Now

$$\vec{OA} \cdot \vec{i} = 1 \quad x_1 = 1$$

and

$$\vec{OB} \cdot \vec{i} = 2 \quad x_2 = 2$$

Therefore $y_1 = 2$ and $y_2 = 16$. Hence

$$\vec{OA} = (1, 2) = \vec{i} + 2\vec{j}$$

and

$$\vec{OB} = (2, 16) = 2\vec{i} + 16\vec{j}$$

Therefore

$$\begin{aligned} 4\vec{OA} - \vec{OB} &= (4\vec{i} + 8\vec{j}) - (2\vec{i} + 16\vec{j}) \\ &= 2\vec{i} - 8\vec{j} \end{aligned}$$

Hence

$$|4\vec{OA} - \vec{OB}| = \sqrt{(2)^2 + (-8)^2} = 10$$

Answer: (A)

7. Let $\vec{a} = \vec{i} + \vec{j}$ and $\vec{b} = \vec{j} + \vec{k}$ and the angle between $\vec{a}$ and $\vec{b}$ be θ . If $\vec{c} = x\vec{i} + y\vec{j} + z\vec{k}$ such that $y = 0$, $|\vec{c}| = \sqrt{2}$ and makes angle θ with $\vec{a}$ and $\vec{b}$, then $\vec{c}$ is equal to

- (A) $\frac{1}{3}(\vec{i} + \vec{j} + 4\vec{k})$ (B) $\frac{1}{3}(4\vec{i} + \vec{j} + \vec{k})$
(C) $\frac{1}{3}(\vec{i} - 4\vec{j} + \vec{k})$ (D) $\frac{1}{3}(\vec{i} + 2\vec{j} + \vec{k})$

Solution: We have $|\vec{c}| = \sqrt{2}$ which implies

$$x^2 + y^2 + z^2 = 1 \quad (6.9)$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{1}{2} \quad (6.10)$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{c}}{|\vec{a}||\vec{c}|} = \frac{x + y}{2} \quad (6.11)$$

Also

$$\cos \theta = \frac{y + z}{2} \quad (6.12)$$

From Eqs. (6.10), (6.11) and (6.12),

$$x + y = 1$$

and

$$y + z = 1$$

Substituting these values in Eq. (6.9) we get

$$x^2 + (1 - x)^2 + x^2 = 2$$

Therefore

$$x = 1, \frac{1}{3}$$

Now $x = 1$ $y = 0$. But by hypothesis $y = 0$. Therefore

$$x = \frac{1}{3}, y = \frac{4}{3}, z = \frac{1}{3}$$

and so

$$\vec{c} = \frac{1}{3}(\vec{i} - 4\vec{j} + \vec{k})$$

Answer: (C)

8. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors such that $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{c}| = \sqrt{24}$ and sum of any two vectors is orthogonal to the third vector, then $|\vec{a} + \vec{b} + \vec{c}|$ is equal to

- (A) 7 (B) $5\sqrt{2}$ (C) $7\sqrt{2}$ (D) $6\sqrt{2}$

Solution: By hypothesis

$$(\vec{a} + \vec{b}) \cdot \vec{c} = 0$$

$$(\vec{b} + \vec{c}) \cdot \vec{a} = 0$$

$$(\vec{c} + \vec{a}) \cdot \vec{b} = 0$$

Therefore

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$$

Now

$$\begin{aligned} |\vec{a} + \vec{b} + \vec{c}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\ &= 9 + 16 + 24 + 2(0) \\ &= 49 \end{aligned}$$

Hence $|\vec{a} + \vec{b} + \vec{c}| = 7$.

Answer: (A)

9. If $|\vec{a}| = |\vec{b}| = |\vec{a} + \vec{b}| = 1$, then $|\vec{a} - \vec{b}|$ equals

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\sqrt{3}$

Solution: By hypothesis

$$1 = |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b}) = 1 + 1 + 2\cos\theta$$

where θ is the angle between $\vec{a}$ and $\vec{b}$. Therefore

$$\cos\theta = \frac{1}{2}$$

Hence

$$\begin{aligned} |\vec{a} - \vec{b}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 - 2(\vec{a} \cdot \vec{b}) \\ &= 1 + 1 - 2(1)(1) = \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

Answer: (D)

10. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors having same magnitude, then the vector $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to each of $\vec{a}$, $\vec{b}$ and $\vec{c}$ at an angle

- (A) $\cos^{-1} \frac{1}{3}$ (B) $\cos^{-1} \frac{1}{\sqrt{3}}$
(C) $\cos^{-1} \frac{2}{3}$ (D) $\cos^{-1} \frac{1}{2}$

Solution: We have

$$\begin{aligned} |\vec{a} + \vec{b} + \vec{c}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 \quad (\because \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0) \\ &= 3K^2 \end{aligned}$$

where $K = |\vec{a}| = |\vec{b}| = |\vec{c}|$. Therefore

$$|\vec{a} + \vec{b} + \vec{c}| = K\sqrt{3}$$

Now, let α , β and γ be the angles of inclination of $\vec{a} + \vec{b} + \vec{c}$ with $\vec{a}$, $\vec{b}$, $\vec{c}$, respectively. Therefore

$$\cos\alpha = \frac{\vec{a} \cdot (\vec{a} + \vec{b} + \vec{c})}{|\vec{a}| |\vec{a} + \vec{b} + \vec{c}|} = \frac{K^2}{K(K\sqrt{3})} = \frac{1}{\sqrt{3}}$$

Answer: (B)

11. Let $\vec{a} = \vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$ and $\vec{c}$ be a unit vector in the plane determined by $\vec{a}$ and $\vec{b}$. If $\vec{c}$ is perpendicular to the vector $\vec{i} + \vec{j} + \vec{k}$ and makes an obtuse angle with $\vec{a}$, then $\vec{c}$ is

- (A) $\frac{\vec{j} - \vec{k}}{\sqrt{2}}$ (B) $\frac{\vec{j} + \vec{k}}{\sqrt{2}}$
(C) $\frac{\vec{i} - \vec{k}}{\sqrt{2}}$ (D) $\frac{\vec{i} + \vec{k}}{\sqrt{2}}$

Solution: Since $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$ (notice that $\vec{a}$ and $\vec{b}$ are not collinear vectors), let $\vec{c} = x\vec{a} + y\vec{b}$ where x and y are scalars. Now

$$|\vec{c}| = 1 \quad \sqrt{(x+y)^2 + (x+2y)^2 + (2x+y)^2} = 1 \quad (6.13)$$

Also $\vec{c}$ is perpendicular to $\vec{i} + \vec{j} + \vec{k}$, we have

$$\begin{aligned} \vec{c} \cdot (\vec{i} + \vec{j} + \vec{k}) &= 0 \\ (x+y) + (x+2y) + (2x+y) &= 0 \\ 4x + 4y &= 0 \\ x &= -y \end{aligned}$$

Substituting the value of $y = -x$ in Eq. (6.13), we have

$$\sqrt{2}x = \pm 1$$

so that

$$x = \pm \frac{1}{\sqrt{2}}$$

Therefore

$$x = \frac{1}{\sqrt{2}} \quad \vec{c} = \frac{\vec{j}}{\sqrt{2}} + \frac{\vec{k}}{\sqrt{2}}$$

and

$$x = -\frac{1}{\sqrt{2}} \quad \vec{c} = -\frac{\vec{j}}{\sqrt{2}} - \frac{\vec{k}}{\sqrt{2}}$$

But $\vec{c}$ makes obtuse angle with $\vec{a}$. This means

$$\vec{c} = -\frac{\vec{j}}{\sqrt{2}} - \frac{\vec{k}}{\sqrt{2}}$$

because

$$\vec{a} \cdot \vec{c} = 0 + \frac{1}{\sqrt{2}} - \frac{2}{\sqrt{2}} = -\frac{1}{\sqrt{2}} < 0$$

Answer: (A)

12. Consider the following two statements:

S_1 : Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be unit vectors such that $\vec{a}$ is perpendicular to both $\vec{b}$ and $\vec{c}$ and further the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$. Then $\vec{a} = \pm 2(\vec{b} \times \vec{c})$

S_2 : The points with position vectors $\vec{a} + \vec{b}$, $\vec{a} - \vec{b}$ and $\vec{a} + \lambda\vec{b}$ are collinear for all real values of λ where $\vec{a}$ and $\vec{b}$ are non-collinear vectors.

Then,

- (A) Both S_1 and S_2 are true
 (B) S_1 is true, but S_2 is false
 (C) S_1 is false and S_2 is true
 (D) Both S_1 and S_2 are false

Solution: S_1 : $\vec{a}$ is perpendicular to both $\vec{b}$ and $\vec{c}$. This implies

$$\vec{a} = \lambda(\vec{b} \times \vec{c})$$

for some scalar λ . Therefore

$$\begin{aligned} 1 &= |\vec{a}|^2 = \lambda^2 |\vec{b} \times \vec{c}|^2 \\ &= \lambda^2 [|\vec{b}|^2 |\vec{c}|^2 - (\vec{b} \cdot \vec{c})^2] \quad (\text{see Theorem 6.26}) \\ &= \lambda^2 \left(1 - \cos^2 \frac{\pi}{6} \right) \\ &= \lambda^2 \left(1 - \frac{3}{4} \right) \\ &= \frac{\lambda^2}{4} \end{aligned}$$

Therefore $\lambda = \pm 2$ and so

$$\vec{a} = \pm 2(\vec{b} \times \vec{c})$$

Hence S_1 is true. Let A , B and C be the points whose position vectors are $\vec{a} + \vec{b}$, $\vec{a} - \vec{b}$ and $\vec{a} + \lambda\vec{b}$, respectively. Now, A , B and C are collinear if and only if $\vec{AB} = x\vec{BC}$ for some scalar x . That is,

$$(\vec{a} - \vec{b}) - (\vec{a} + \vec{b}) = x[\vec{a} + \lambda\vec{b} - (\vec{a} - \vec{b})]$$

That is,

$$-2\vec{b} = x(\lambda + 1)\vec{b}$$

Therefore $x(\lambda + 1) = -2$ for all real λ and for some real x . This is not possible when $\lambda = -1$, in which case, $C = B$. Hence S_2 is also true.

Answer: (A)

- 13.** The number of distinct real values of λ for which the vectors $-\lambda^2\vec{i} + \vec{j} + \vec{k}$, $\vec{i} - \lambda^2\vec{j} + \vec{k}$ and $\vec{i} + \vec{j} - \lambda^2\vec{k}$ are coplanar is

- (A) 0 (B) 1 (C) 2 (D) 3

Solution: The given vectors are coplanar if and only if

$$\begin{vmatrix} -\lambda^2 & 1 & 1 \\ 1 & -\lambda^2 & 1 \\ 1 & 1 & -\lambda^2 \end{vmatrix} = 0 \quad (\text{see Corollary 6.16})$$

That is

$$\lambda^6 - 3\lambda^2 - 2 = 0$$

$$(\lambda^2 + 1)^2 (\lambda^2 - 2) = 0 \quad (\because \pm 1 \text{ and } \pm i \text{ are repeated roots})$$

This gives

$$\lambda = \pm\sqrt{2}$$

Answer: (C)

- 14.** Let $\vec{a} = x\vec{i} - \vec{j} + \vec{k}$ (x is a scalar) and $\vec{b} = 2\vec{i} - \vec{j} + 5\vec{k}$. If the scalar projection of $\vec{a}$ on $\vec{b}$ is $1/\sqrt{30}$, then x is equal to

- (A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{-5}{3}$ (D) $\frac{-5}{2}$

Solution: By hypothesis

$$\begin{aligned} \frac{1}{\sqrt{30}} &= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \\ &= \frac{2x + 6}{\sqrt{30}} \end{aligned}$$

Therefore

$$x = \frac{-5}{2}$$

Answer: (D)

- 15.** If $|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 1$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ is

- (A) 12 (B) -12 (C) -13 (D) 13

Solution: We have

$$\begin{aligned} \vec{a} + \vec{b} + \vec{c} = \vec{0} &\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = 0 \\ &\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2\sum \vec{a} \cdot \vec{b} = 0 \\ &\Rightarrow 9 + 16 + 1 + 2\sum \vec{a} \cdot \vec{b} = 0 \\ &\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -13 \end{aligned}$$

Answer: (C)

- 16.** Let $\vec{a} = 2\vec{i} + 3\vec{j} + \vec{k}$, $\vec{b} = 4\vec{i} + \vec{j}$ and $\vec{c} = \vec{i} + 3\vec{j} + 2\vec{k}$. If $\vec{d}$ is such that $\vec{d} \cdot \vec{a} = 9$, $\vec{d} \cdot \vec{b} = 7$ and $\vec{d} \cdot \vec{c} = 6$, then $\vec{d}$ is equal to

- (A) $\vec{i} + 3\vec{j} - 2\vec{k}$ (B) $\vec{i} - 3\vec{j} + 2\vec{k}$
 (C) $-\vec{i} + 3\vec{j} - 2\vec{k}$ (D) $\vec{i} + 3\vec{j} + 2\vec{k}$

Solution: Let $\vec{d} = x\vec{i} + y\vec{j} + z\vec{k}$.

$$\vec{d} \cdot \vec{a} = 9 \Rightarrow 2x + 3y + z = 9 \quad (6.14)$$

$$\vec{d} \cdot \vec{b} = 7 \Rightarrow 4x + y = 7 \quad (6.15)$$

$$\vec{d} \cdot \vec{c} = 6 \Rightarrow x + 3y + 2z = 6 \quad (6.16)$$

Solving Eqs. (6.14), (6.15) and (6.16) we have $x = 1$, $y = 3$ and $z = 2$. Therefore

$$\vec{d} = \vec{i} + 3\vec{j} - 2\vec{k}$$

Answer: (A)

17. The points $A(2, -1, 1)$, $B(1, -3, -5)$ and $C(3, -4, -4)$ form the vertices of

- (A) an equilateral triangle
- (B) isosceles but not equilateral
- (C) right angled
- (D) right angled isosceles

Solution: We have

$$\overrightarrow{AB} = \vec{i} - 2\vec{j} - 6\vec{k}$$

$$\overrightarrow{CA} = \vec{i} + 3\vec{j} + 5\vec{k}$$

$$\overrightarrow{BC} = 2\vec{i} - \vec{j} + \vec{k}$$

$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$$

Therefore A, B, C form a triangle and

$$\overrightarrow{BC} \times \overrightarrow{CA} = 2\vec{i} - 3\vec{j} + 5\vec{k}$$

Therefore $\angle C = 90^\circ$.

Answer: (C)

18. If the vectors $2\vec{i} + a\vec{j} + 4\vec{k}$ and $3\vec{i} + 5\vec{j} - 3\vec{k}$ are at right angles, then the value of a is

- (A) $\frac{18}{5}$
- (B) 1
- (C) $\frac{1}{5}$
- (D) 1

Solution: We have

$$(2\vec{i} + a\vec{j} + 4\vec{k}) \cdot (3\vec{i} + 5\vec{j} - 3\vec{k}) = 0$$

$$6 + 5a - 12 = 0$$

$$a = \frac{18}{5}$$

Answer: (A)

19. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors, then the value of

$$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$$

is less than or equal to

- (A) 8
- (B) 12
- (C) 9
- (D) 6

Solution: We have

$$|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$$

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0$$

$$\frac{3}{2} (\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0 \quad (6.17)$$

Now

$$\begin{aligned} |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 &= 2(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\ &= 6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 6 - 2 \left(\frac{3}{2} \right) = 3 \quad [\text{by Eq. (6.17)}] \end{aligned}$$

Answer: (C)

20. The angle between the two vectors $\vec{a} = 2\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{b} = 6\vec{i} - 3\vec{j} + 2\vec{k}$ is

- (A) $\cos^{-1} \frac{1}{3}$
- (B) $\cos^{-1} \frac{1}{2}$
- (C) $\cos^{-1} \frac{5\sqrt{7}}{21}$
- (D) $\cos^{-1} \frac{4}{21}$

Solution: We have

$$\vec{a} \cdot \vec{b} = 12 - 6 - 2 = 4$$

$$|\vec{a}||\vec{b}| = \sqrt{36 + 9 + 4} \sqrt{4 + 4 + 1} = 21$$

Therefore the angle between the vectors, by Theorem 6.7 is

$$\cos^{-1} \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \cos^{-1} \frac{4}{21}$$

Answer: (D)

21. If the scalar product of the vector $\vec{i} + \vec{j} + \vec{k}$ with the unit vector in the direction of the resultant of the vectors $2\vec{i} + 4\vec{j} - 5\vec{k}$ and $\lambda\vec{i} + 2\vec{j} + 3\vec{k}$ is unity, then λ equals

- (A) 5
- (B) 2
- (C) 1
- (D) -1

Solution: Let

$$\vec{r} = (2\vec{i} + 4\vec{j} - 5\vec{k}) + (\lambda\vec{i} + 2\vec{j} + 3\vec{k})$$

$$= (2 + \lambda)\vec{i} + 6\vec{j} - 2\vec{k}$$

$$\vec{e} = \text{unit vector in the direction of } \vec{r}$$

$$= \frac{(2 + \lambda)\vec{i} + 6\vec{j} - 2\vec{k}}{\sqrt{(2 + \lambda)^2 + 36 + 4}}$$

Therefore

$$\vec{e} \cdot (\vec{i} + \vec{j} + \vec{k}) = 1$$

$$(2 + \lambda) + 6 - 2 = \sqrt{(2 + \lambda)^2 + 40}$$

$$(\lambda + 6)^2 = (2 + \lambda)^2 + 40$$

$$12\lambda + 36 = 4\lambda + 44$$

$$8\lambda = 8$$

$$\lambda = 1$$

Answer: (C)

22. Let $\vec{a} = 2\vec{i} + 3\vec{j} + 2\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$. The vector component of $\vec{a}$ perpendicular to the direction of $\vec{b}$ is

(A) $\frac{1}{3}(\vec{i} + \vec{j} + \vec{k})$ (B) $\frac{1}{3}(\vec{i} - \vec{j} + \vec{k})$
 (C) $\frac{1}{3}(\vec{i} + \vec{j} - \vec{k})$ (D) $\frac{1}{3}(-\vec{i} + \vec{j} + \vec{k})$

Solution: Vector component of $\vec{a}$ perpendicular to the direction of $\vec{b}$ is given by (see Quick Look 3)

$$\vec{a} - \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b} = (2\vec{i} + 3\vec{j} + 2\vec{k}) - \frac{5}{3}(\vec{i} + 2\vec{j} + \vec{k})$$

$$= \frac{1}{3}(\vec{i} - \vec{j} + \vec{k})$$

Answer: (B)

23. If $\vec{a}$ is collinear with $\vec{b} = 3\vec{i} + 6\vec{j} + 6\vec{k}$ and $\vec{a} \cdot \vec{b} = 27$, then $\vec{a}$ is

(A) $3(\vec{i} + \vec{j} + \vec{k})$ (B) $\vec{i} + 2\vec{j} + 2\vec{k}$
 (C) $2\vec{i} + 2\vec{j} + 2\vec{k}$ (D) $\vec{i} + 3\vec{j} + 3\vec{k}$

Solution: $\vec{a}$ is collinear with $\vec{b}$. This implies

$$\vec{a} = \lambda \vec{b}$$

for some scalar λ . Now

$$\begin{aligned}\vec{a} \cdot \vec{b} &= 27 \\ \lambda(\vec{b} \cdot \vec{b}) &= 27 \\ \lambda(9 + 36 + 36) &= 27 \\ \lambda &= \frac{1}{3}\end{aligned}$$

Therefore

$$\vec{a} = \frac{1}{3}\vec{b} = \vec{i} + 2\vec{j} + 2\vec{k}$$

Answer: (B)

24. Let $\vec{a}$ and $\vec{b}$ be unit vectors. If the vectors $\vec{x} = \vec{a} + 2\vec{b}$ and $\vec{y} = 5\vec{a} - 4\vec{b}$ are orthogonal vectors, then the angle between $\vec{a}$ and $\vec{b}$ is

(A) 60° (B) 30° (C) 45° (D) 75°

Solution: By hypothesis

$$\begin{aligned}\vec{x} \cdot \vec{y} &= 0 \\ 5|\vec{a}|^2 - 8|\vec{b}|^2 + 6\vec{a} \cdot \vec{b} &= 0\end{aligned}$$

Assuming θ is the angle between $\vec{a}$ and $\vec{b}$ we get

$$5 - 8 + 6\cos\theta = 0$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

Answer: (A)

25. Angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$. If $|\vec{b}| = 2|\vec{a}|$ and the vectors $\vec{a} + x\vec{b}$ and $\vec{a} - \vec{b}$ are at right angles, then x is equal to

(A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{5}$ (D) $\frac{1}{3}$

Solution: We have

$$(\vec{a} + x\vec{b}) \cdot (\vec{a} - \vec{b}) = 0$$

$$|\vec{a}|^2 - x|\vec{b}|^2 + (x - 1)(\vec{a} \cdot \vec{b}) = 0$$

$$|\vec{a}|^2 - 4x|\vec{a}|^2 + (x - 1)2|\vec{a}|^2 \cos \frac{2\pi}{3} = 0$$

$$1 - 4x + (x - 1) = 0$$

$$x = \frac{2}{5}$$

Answer: (A)

26. A plane is at a distance of 5 units from the origin and perpendicular to the vector $2\vec{i} + \vec{j} + 2\vec{k}$. Then its equation is

(A) $\vec{r} \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = 15$

(B) $\vec{r} \cdot (2\vec{i} + \vec{j} + \vec{k}) = 5$

(C) $\vec{r} \cdot (\vec{i} + 2\vec{j} + 2\vec{k}) = 15$

(D) $\vec{r} \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = 5$

Solution: Unit normal to the given plane is

$$\vec{n} = \frac{2\vec{i} + \vec{j} + 2\vec{k}}{3}$$

By Theorem 6.13, the plane equation

$$\vec{r} \cdot \frac{(2\vec{i} + \vec{j} + 2\vec{k})}{3} = 5$$

That is

$$\vec{r} \cdot (2\vec{i} + \vec{j} + 2\vec{k}) = 15$$

Answer: (A)

27. M is the foot of the perpendicular drawn from the point $A(2, 4, 3)$ onto the line joining the points $P(1, 2, 4)$ and $Q(3, 4, 5)$. Then the vector representing the point M is

- (A) $\frac{1}{9}(19\vec{i} + 28\vec{j} + 40\vec{k})$
 (B) $\frac{1}{9}(28\vec{i} + 9\vec{j} + 41\vec{k})$
 (C) $\frac{1}{9}(21\vec{i} + 28\vec{j} + 19\vec{k})$
 (D) $\frac{1}{9}(19\vec{i} + 28\vec{j} + 41\vec{k})$

Solution: By Theorem 5.29 (Chapter 5), the equation of the line PQ is

$$\vec{r} = (1-t)(\vec{i} + 2\vec{j} + 4\vec{k}) + t(3\vec{i} + 4\vec{j} + 5\vec{k})$$

That is

$$\vec{r} = (1+2t)\vec{i} + (2+2t)\vec{j} + (4+t)\vec{k} \quad (6.18)$$

Therefore, the position vector of M is given by Eq. (6.18). So

$$\overrightarrow{AM} = (-1+2t)\vec{i} + (-2+2t)\vec{j} + (1+t)\vec{k}$$

Since $\overrightarrow{AM}$ is perpendicular to $\overrightarrow{PQ} = 2\vec{i} + 2\vec{j} + \vec{k}$, we have

$$\overrightarrow{AM} \cdot \overrightarrow{PQ} = 0$$

Therefore

$$\begin{aligned} 2(-1+2t) + 2(-2+2t) + (1+t) &= 0 \\ -5 + 9t &= 0 \\ t &= \frac{5}{9} \end{aligned}$$

Substituting the value $t = 5/9$ in Eq. (6.18), we have

$$M = \frac{1}{9}(19\vec{i} + 28\vec{j} + 41\vec{k})$$

Answer: (D)

28. Let p be real and $|p| \geq 2$. If A , B and C are variable angles such that

$$\sqrt{p^2 - 4} \tan A + p \tan B + \sqrt{p^2 + 4} \tan C = 6p$$

then the minimum value of $\tan^2 A + \tan^2 B + \tan^2 C$ is

- (A) 8 (B) 12 (C) 18 (D) 6

Solution: Consider the vectors

$$\vec{a} = \sqrt{p^2 - 4}\vec{i} + p\vec{j} + \sqrt{p^2 + 4}\vec{k}$$

and $\vec{b} = (\tan A)\vec{i} + (\tan B)\vec{j} + (\tan C)\vec{k}$

so that

$$\vec{a} \cdot \vec{b} = \sqrt{p^2 - 4} \tan A + p \tan B + \sqrt{p^2 + 4} \tan C$$

Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then

$$\begin{aligned} 36p^2 &= (\vec{a} \cdot \vec{b})^2 \\ &= |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta \leq |\vec{a}|^2 |\vec{b}|^2 \\ &= (p^2 - 4 + p^2 + p^2 + 4)(\tan^2 A + \tan^2 B + \tan^2 C) \\ &= 3p^2(\tan^2 A + \tan^2 B + \tan^2 C) \end{aligned}$$

Hence

$$\tan^2 A + \tan^2 B + \tan^2 C \geq 12$$

and is equal to 12 when θ is either 0 or π .

Answer: (B)

29. In $\triangle ABC$ (Figure 6.28), $AB = BC = 8$ and $AC = 12$. P is a point on the side AB such that $AP:PB = 1:3$. Then the angle between the vectors $\overrightarrow{CP}$ and $\overrightarrow{CA}$ is

- (A) $\cos^{-1}\left(\frac{3\sqrt{5}}{8}\right)$ (B) $\cos^{-1}\left(\frac{3\sqrt{3}}{8}\right)$
 (C) $\cos^{-1}\left(\frac{3\sqrt{7}}{8}\right)$ (D) $\cos^{-1}\left(\frac{3\sqrt{2}}{8}\right)$

Solution: Take A as origin and let $\overrightarrow{AB} = \vec{\alpha}$ and $\overrightarrow{AC} = \vec{\beta}$ so that

$$|\vec{\alpha}| = 8 = |\vec{\beta} - \vec{\alpha}|$$

and

$$|\vec{\beta}| = 12$$

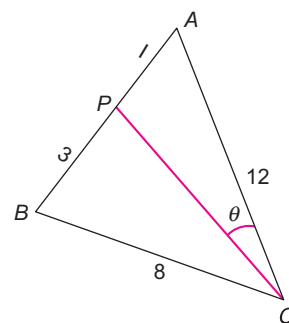


FIGURE 6.28 Single correct choice type question 29.

Now $AB = BC = 8$ implies

$$\begin{aligned} |\vec{\alpha}| &= |\vec{\beta} - \vec{\alpha}| \\ \Rightarrow |\vec{\alpha}|^2 &= |\vec{\beta}|^2 + |\vec{\alpha}|^2 - 2\vec{\alpha} \cdot \vec{\beta} \\ \Rightarrow 2\vec{\alpha} \cdot \vec{\beta} &= |\vec{\beta}|^2 \\ \Rightarrow 2|\vec{\alpha}||\vec{\beta}|\cos A &= |\vec{\beta}|^2 \\ \Rightarrow 2(8)(12)\cos A &= 144 \\ \Rightarrow \cos A &= \frac{3}{4} \end{aligned} \quad (6.19)$$

Now

$$\overrightarrow{AP} = \frac{\vec{\alpha}}{4}$$

and $\overrightarrow{CP} = \frac{\vec{\alpha}}{4} \quad \vec{\beta} = \frac{1}{4}(\vec{\alpha} - 4\vec{\beta})$

Again

$$\begin{aligned} |\overrightarrow{CP}|^2 &= \frac{1}{16} |\vec{\alpha} - 4\vec{\beta}|^2 \\ &= \frac{1}{16} [8^2 + 16 - 12^2 - 8(\vec{\alpha} \times \vec{\beta})] \\ &= \frac{1}{16} [64 + 16 - 144 - 8(8)(12)\frac{3}{4}] \quad \because \cos A = \frac{3}{4} \\ &= 4 + 144 - 36 \\ &= 112 \end{aligned}$$

Therefore

$$|\overrightarrow{CP}| = \sqrt{112} = 4\sqrt{7}$$

Let θ be the angle between $\overrightarrow{CP}$ and $\overrightarrow{CA}$. Then

$$\begin{aligned} \cos \theta &= \frac{\overrightarrow{CP} \cdot \overrightarrow{CA}}{|\overrightarrow{CP}| |\overrightarrow{CA}|} \\ &= \frac{(1/4)(\vec{\alpha} - 4\vec{\beta}) \cdot (\vec{\alpha} - \vec{\beta})}{4\sqrt{7} \cdot 12} \\ &= \frac{4|\vec{\beta}|^2 - \vec{\alpha} \cdot \vec{\beta}}{16\sqrt{7} \cdot 12} \\ &= \frac{4 - 144 - 8(12)(3/4)}{16\sqrt{7} \cdot 12} \\ &= \frac{144 - 18}{4\sqrt{7} \cdot 12} \\ &= \frac{126}{4\sqrt{7} \cdot 12} \\ &= \frac{21}{8\sqrt{7}} \\ &= \frac{3\sqrt{7}}{8} \end{aligned}$$

Therefore

$$\theta = \cos^{-1} \frac{3\sqrt{7}}{8}$$

Answer: (C)

30. Let $\overrightarrow{OA} = \vec{i} + 2\vec{j} + 2\vec{k}$. In the plane of $\overrightarrow{OA}$ and $\vec{i}$, rotate $\overrightarrow{OA}$ through 90° about the origin O such that

the new position of $\overrightarrow{OA}$ makes an acute angle with the positive X -axis. Then the new position of $\overrightarrow{OA}$ is

(A) $\frac{1}{\sqrt{2}}(4\vec{i} - 2\vec{j} - 2\vec{k})$

(B) $\frac{1}{\sqrt{2}}(4\vec{i} + 2\vec{j} + 2\vec{k})$

(C) $2\vec{j} - 2\vec{k}$

(D) $6\vec{i} - 3\vec{k}$

Solution: Let the new position of $\overrightarrow{OA}$ be $\vec{r}$. Since $\vec{r}$ is coplanar with $\overrightarrow{OA}$ and $\vec{i}$, let

$$\begin{aligned} \vec{r} &= x(\overrightarrow{OA}) + y\vec{i} \\ &= (x + y)\vec{i} + 2x\vec{j} + 2x\vec{k} \end{aligned}$$

Since $\vec{r}$ is perpendicular to $\overrightarrow{OA}$, we have

$$\begin{aligned} \vec{r} \cdot \overrightarrow{OA} &= 0 \\ (x + y) + 4x + 4x &= 0 \\ y &= -9x \end{aligned} \quad (6.20)$$

Also

$$\begin{aligned} |\vec{r}| &= 3 \quad (x + y)^2 + 4x^2 + 4x^2 = 9 \\ 64x^2 + 8x^2 &= 9 \quad [\text{by Eq. (6.20)}] \\ x &= \pm \frac{1}{2\sqrt{2}} \end{aligned}$$

so that

$$y = \mp \frac{9}{2\sqrt{2}}$$

Now

$$\vec{r} = \frac{1}{\sqrt{2}}(4\vec{i} - \vec{j} - \vec{k})$$

or

$$\vec{r} = \frac{1}{\sqrt{2}}(4\vec{i} + \vec{j} + \vec{k})$$

But $\vec{r}$ makes acute angle with the positive X -axis. Hence

$$\vec{r} = \frac{1}{\sqrt{2}}(4\vec{i} - \vec{j} - \vec{k})$$

Answer: (A)

31. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are vectors such that $\vec{a} + \vec{b} + 2\vec{c} = \vec{0}$ and $|\vec{a}| = 1$, $|\vec{b}| = 4$ and $|\vec{c}| = 2$, then the value of $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is

(A) $\frac{13}{2}$ (B) $\frac{13}{2}$ (C) $\frac{17}{2}$ (D) $\frac{17}{2}$

Solution: By hypothesis

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

Therefore

$$\begin{aligned} 4 &= |\vec{c}|^2 = |\vec{a} + \vec{b} + \vec{c}|^2 \\ &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b}) \\ &= 1 + 16 + 4 + 2(\vec{a} \cdot \vec{b}) \end{aligned}$$

Therefore

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{17}{2}$$

Answer: (D)

32. If $\vec{a}$ and $\vec{b}$ are unit vectors such that $|\vec{a} + \vec{b}| = \sqrt{3}$, then $(3\vec{a} - 4\vec{b}) \times (2\vec{a} + 5\vec{b})$ is equal to

- (A) $\frac{15}{2}$ (B) $\frac{15}{2}$ (C) $\frac{21}{2}$ (D) $\frac{21}{2}$

Solution: We have

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= 3 \\ 1 + 1 + 2(\vec{a} \cdot \vec{b}) &= 3 \\ \vec{a} \cdot \vec{b} &= \frac{1}{2} \end{aligned}$$

Now

$$\begin{aligned} (3\vec{a} - 4\vec{b}) \times (2\vec{a} + 5\vec{b}) &= 6|\vec{a}|^2 - 20|\vec{b}|^2 + 7(\vec{a} \times \vec{b}) \\ &= 6 - 20 + \frac{7}{2} \\ &= -\frac{21}{2} \end{aligned}$$

Answer: (C)

33. In quadrilateral $ABCD$, if

$$|\vec{AB}|^2 + |\vec{CD}|^2 = |\vec{BC}|^2 + |\vec{AD}|^2$$

then the angle between the diagonals AC and BD is

- (A) 60° (B) 75° (C) 90° (D) 120°

Solution: Take A as origin and let $\vec{AB} = \vec{b}$, $\vec{AC} = \vec{c}$ and $\vec{AD} = \vec{d}$. Then

$$\begin{aligned} |\vec{AB}|^2 + |\vec{CD}|^2 &= |\vec{BC}|^2 + |\vec{AD}|^2 \\ |\vec{b}|^2 + |\vec{d} - \vec{c}|^2 &= (\vec{c} - \vec{b})^2 + |\vec{d}|^2 \\ |\vec{b}|^2 + |\vec{d}|^2 + |\vec{c}|^2 - 2(\vec{d} \cdot \vec{c}) &= |\vec{c}|^2 + |\vec{b}|^2 - 2(\vec{b} \cdot \vec{c}) + |\vec{d}|^2 \\ \vec{d} \cdot \vec{c} &= \vec{b} \cdot \vec{c} \end{aligned}$$

$$(\vec{d} - \vec{b}) \cdot \vec{c} = 0$$

$$\vec{BD} \cdot \vec{AC} = 0$$

Therefore angle between AC and BD is 90° .

Answer: (C)

34. In $\triangle ABC$ (Figure 6.29), if $BC = a$, $CA = b$ and $AB = c$ and the internal bisector of the angle A meets the side BC in L , then $|\vec{AL}|^2$ is equal to

- (A) $bc \frac{bca^2}{(b+c)^2}$ (B) $bc \frac{abc}{(b+c)^2}$
(C) $a^2bc + \frac{ab}{(b+c)^2}$ (D) $a^2bc - \frac{bc}{(b+c)^2}$

Solution: It is known that $BL:LC = c:b$. Take A as origin, let $\vec{AB} = \vec{\alpha}$ and $\vec{AC} = \vec{\beta}$ so that $|\vec{\alpha}| = c$, $|\vec{\beta}| = b$ and the angle between $\vec{\alpha}$ and $\vec{\beta}$ is A .

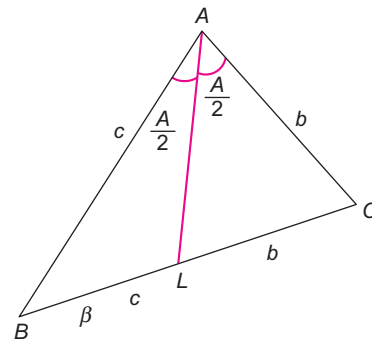


FIGURE 6.29 Single correct choice type question 34.

Now $BL:LC = c:b$. This means that the position vector $\vec{AL}$ of L is given by

$$\vec{AL} = \frac{b\vec{\alpha} + c\vec{\beta}}{b+c}$$

Therefore

$$\begin{aligned} |\vec{AL}|^2 &= \frac{1}{(b+c)^2} [b^2|\vec{\alpha}|^2 + c^2|\vec{\beta}|^2 + 2bc(\vec{\alpha} \cdot \vec{\beta})] \\ &= \frac{1}{(b+c)^2} [b^2c^2 + b^2c^2 + 2b^2c^2 \cos A] \\ &= \frac{1}{(b+c)^2} [2b^2c^2 + 2b^2c^2 \frac{(b^2 + c^2 - a^2)}{2bc}] \\ &= \frac{2b^2c^2}{(b+c)^2} \frac{2bc + b^2 + c^2 - a^2}{2bc} \\ &= \frac{bc}{(b+c)^2} [(b+c)^2 - a^2] \\ &= bc \frac{bca^2}{(b+c)^2} \end{aligned}$$

Answer: (A)

35. $\vec{a}, \vec{b}, \vec{c}$ are unit vectors. If $\vec{r}$ is a vector such that $|\vec{r}| = 2$, $\vec{a} + \vec{b} + \vec{c} = \vec{r}$, $\vec{a} \cdot \vec{r} = 1$, $\vec{b} \cdot \vec{r} = 3/2$, then the angle between $\vec{c}$ and $\vec{r}$ is

- (A) $\cos^{-1} \frac{1}{4}$ (B) $\cos^{-1} \frac{3}{4}$
 (C) $\cos^{-1} \frac{1}{2\sqrt{2}}$ (D) $\cos^{-1} \frac{1}{3}$

Solution: We have

$$(\vec{a} + \vec{b} + \vec{c}) \cdot \vec{r} = \vec{r} \cdot \vec{r} = 4$$

$$\vec{a} \cdot \vec{r} + \vec{b} \cdot \vec{r} + \vec{c} \cdot \vec{r} = 4$$

$$1 + \frac{3}{2} + 2\cos\theta = 4$$

where θ is the angle between $\vec{c}$ and $\vec{r}$. Therefore

$$2\cos\theta = \frac{3}{2}$$

$$\theta = \cos^{-1} \frac{3}{4}$$

Answer: (B)

36. In $\triangle ABC$ (Figure 6.30), if “O” is the circumcentre and R is the circumradius and G is the centroid, then $|\overrightarrow{OG}|^2$ equals

- (A) $R \frac{1}{9}(a^2 + b^2 + c^2)$
 (B) $R^2 \frac{1}{9}(a^2 + b^2 + c^2)$
 (C) $R^2 \frac{1}{6}(a^2 + b^2 + c^2)$
 (D) $R^2 \frac{1}{3}(a^2 + b^2 + c^2)$

Solution: Take “O” as the origin. Let $\overrightarrow{OA} = \vec{\alpha}$, $\overrightarrow{OB} = \vec{\beta}$ and $\overrightarrow{OC} = \vec{\gamma}$. Therefore $(\vec{\beta}, \vec{\gamma}) = 2A$, $(\vec{\gamma}, \vec{\alpha}) = 2B$ and $(\vec{\alpha}, \vec{\beta}) = 2C$.

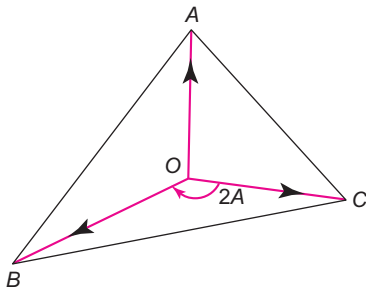


FIGURE 6.30 Single correct choice type question 36.

It is known that

$$\overrightarrow{OG} = \frac{\vec{\alpha} + \vec{\beta} + \vec{\gamma}}{3} \quad (\text{see Corollary 5.4})$$

Therefore

$$\begin{aligned} 9|\overrightarrow{OG}|^2 &= |\vec{\alpha} + \vec{\beta} + \vec{\gamma}|^2 \\ &= |\vec{\alpha}|^2 + |\vec{\beta}|^2 + |\vec{\gamma}|^2 + 2(\vec{\beta} \cdot \vec{\gamma}) \\ &= 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C) \\ &= 3R^2 + 2R^2[1 - 2\sin^2 A + 1 - 2\sin^2 B + 1 - 2\sin^2 C] \\ &= 9R^2 - 4R^2 \sin^2 A - 4R^2 \sin^2 B - 4R^2 \sin^2 C \\ &= 9R^2 - a^2 - b^2 - c^2 \quad \because \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \end{aligned}$$

Hence

$$|\overrightarrow{OG}|^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$$

Answer: (B)

37. Quadrilateral $ABCD$ is inscribed in a circle of radius R (Figure 6.31). If $(AB)^2 + (CD)^2 = 4R^2$ then the angle between the diagonals AC and BD is

- (A) 120° (B) 90° (C) 75° (D) 60°

Solution: Let “O” be the centre of the circle. Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \vec{c}$ and $\overrightarrow{OD} = \vec{d}$ so that

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{d}| = R$$

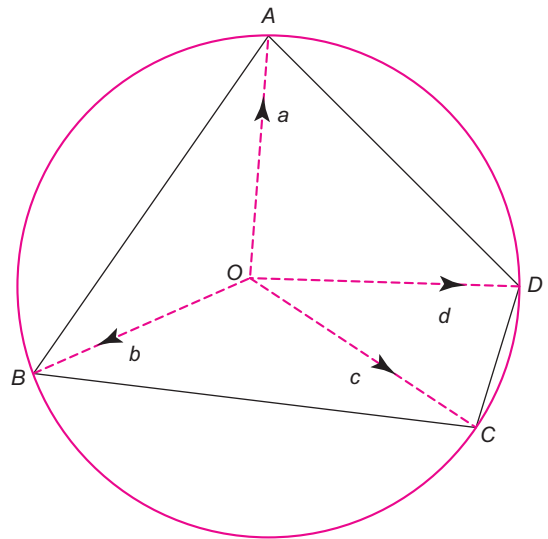


FIGURE 6.31 Single correct choice type question 37.

By hypothesis,

$$\begin{aligned} |\overline{AB}|^2 + |\overline{CD}|^2 &= 4R^2 \\ \Rightarrow |\vec{b} - \vec{a}|^2 + |\vec{d} - \vec{c}|^2 &= 4R^2 \\ \Rightarrow -2(\vec{a} \cdot \vec{b}) - 2(\vec{c} \cdot \vec{d}) &= 0 \end{aligned} \quad (6.21)$$

$$\begin{aligned} \Rightarrow -2R^2[\cos \angle AOB + \cos \angle COD] &= 0 \\ \Rightarrow \cos \angle AOB + \cos \angle COD &= 0 \\ \Rightarrow \angle AOB + \angle COD &= \pi \end{aligned} \quad (6.22)$$

Therefore

$$\angle BOC + \angle AOD = \pi$$

Now

$$\begin{aligned} \overline{AC} \cdot \overline{BD} &= (\vec{c} - \vec{a}) \cdot (\vec{d} - \vec{b}) \\ &= (\vec{c} \cdot \vec{d}) - (\vec{b} \cdot \vec{c}) - (\vec{a} \cdot \vec{d}) + (\vec{a} \cdot \vec{b}) \\ &= -(\vec{b} \cdot \vec{c} + \vec{a} \cdot \vec{d}) \\ &\quad (\text{because from Eq. (6.21) } \vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{d} = 0) \\ &= -R^2[\cos \angle BOC + \cos \angle AOD] \quad [\text{by Eq. (6.22)}] \\ &= -R^2(0) \\ &= 0 \end{aligned}$$

Therefore AC and BD are at right angles.

38. “ O ” is the circumcentre and H is the orthocentre of a $\triangle ABC$. If P is any arbitrary point, then the vector $\overline{PA} + \overline{PB} + \overline{PC} - \overline{PH}$ is equal to

- (A) $\overline{PO}$ (B) $2\overline{PO}$ (C) $\overline{OP}$ (D) $2\overline{OP}$

Solution: With respect to the circumcentre ‘ O ’ as origin, let $\overline{OA} = \vec{a}$, $\overline{OB} = \vec{b}$ and $\overline{OC} = \vec{c}$. Therefore by Example 5.3 (Chapter 5)

$$\overline{OH} = \vec{a} + \vec{b} + \vec{c}$$

Let $\overline{OP} = \vec{p}$. Therefore

$$\begin{aligned} \overline{PA} + \overline{PB} + \overline{PC} - \overline{PH} &= (\vec{a} - \vec{p}) + (\vec{b} - \vec{p}) + (\vec{c} - \vec{p}) \\ &\quad - (\vec{a} + \vec{b} + \vec{c} - \vec{p}) \\ &= -2\vec{p} \\ &= 2\overline{PO} \end{aligned}$$

Answer: (B)

39. In $\triangle ABC$ (Figure 6.32), angle B is a right angle. If the medians AD and BE are perpendicular to each other, then angle C is

- (A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

- (C) $\tan^{-1}(\sqrt{2})$ (D) $\tan^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

Solution: With respect to origin B , let $\overline{BA} = \vec{a}$ and $\overline{BC} = \vec{c}$ so that $\vec{c} \cdot \vec{a} = 0$.

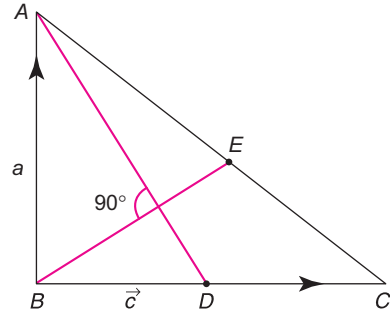


FIGURE 6.32 Single correct choice type question 39.

Now,

$$\overline{BD} = \frac{\vec{c}}{2}$$

and

$$\overline{BE} = \frac{\vec{c} + \vec{a}}{2}$$

BE and AD are perpendicular to each other. This implies

$$\begin{aligned} \left(\frac{\vec{c} + \vec{a}}{2}\right) \cdot \left(\frac{\vec{c}}{2} - \vec{a}\right) &= 0 \\ \Rightarrow (\vec{c} + \vec{a}) \cdot (\vec{c} - 2\vec{a}) &= 0 \\ \Rightarrow |\vec{c}|^2 - 2|\vec{a}|^2 - \vec{c} \cdot \vec{a} &= 0 \\ \Rightarrow |\vec{c}|^2 - 2|\vec{a}|^2 &= 0 \quad (\because \vec{c} \cdot \vec{a} = 0) \\ \Rightarrow \frac{|\vec{a}|^2}{|\vec{c}|^2} &= \frac{1}{2} \\ \Rightarrow \frac{(AB)^2}{(BC)^2} &= \frac{1}{2} \\ \Rightarrow \frac{AB}{BC} &= \frac{1}{\sqrt{2}} \\ \Rightarrow \tan C &= \frac{1}{\sqrt{2}} \\ \Rightarrow C &= \tan^{-1}\left(\frac{1}{\sqrt{2}}\right) \end{aligned}$$

Answer: (A)

40. In a $\triangle ABC$, if $a^2 + b^2 = 2c^2$ and m_a, m_b and m_c are the lengths of the medians through the vertices A, B and C , respectively then $am_b + bm_a$ is equal to

- (A) $c m_c$ (B) $2c m_c$ (C) $3c m_c$ (D) $\sqrt{3}c m_c$

Solution: In Chapter 4, we have proved (see Theorem 4.25) that

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

$$m_b = \frac{1}{2}\sqrt{2c^2 + 2a^2 - b^2}$$

$$m_c = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

Therefore

$$\begin{aligned} a m_b + b m_a &= \frac{a}{2}\sqrt{2c^2 + 2a^2 - b^2} + \frac{b}{2}\sqrt{2b^2 + 2c^2 - a^2} \\ &= \frac{a}{2}\sqrt{3a^2} + \frac{b}{2}\sqrt{3b^2} \quad (\because a^2 + b^2 = 2c^2) \\ &= \frac{\sqrt{3}}{2}(a^2 + b^2) \\ &= \frac{\sqrt{3}}{2}(2c^2) \\ &= \sqrt{3}c^2 \\ &= 2c \left(\frac{\sqrt{2a^2 + 2b^2 - c^2}}{2} \right) \\ &= 2c m_c \end{aligned}$$

Answer: (B)

41. In $\triangle ABC$ (Figure 6.33), the median CM is perpendicular to the angle bisector AL of angle A . If $|\overline{CM}| : |\overline{AL}| = m : 1$, then

$$(A) \quad m^2 = \frac{9}{4} \left(\frac{1 + \cos A}{1 - \cos A} \right) \quad (B) \quad m^2 = \frac{9}{4} \left(\frac{1 - \cos A}{1 + \cos A} \right)$$

$$(C) \quad m^2 = \frac{4}{9} \left(\frac{1 - \cos A}{1 + \cos A} \right) \quad (D) \quad m^2 = \frac{4}{9} \left(\frac{1 + \cos A}{1 - \cos A} \right)$$

Solution: Taking “A” as origin, let $\overline{AB} = \vec{\alpha}$, $\overline{AC} = \vec{\beta}$ and so that $|\vec{\alpha}| = c$, $|\vec{\beta}| = b$ and $(\vec{\alpha}, \vec{\beta}) = A$. Therefore

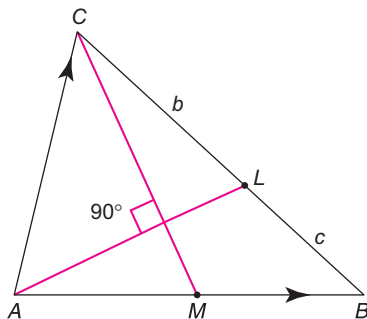


FIGURE 6.33 Single correct choice type question 41.

$$\overline{AM} = \frac{1}{2}\vec{\alpha}$$

$$\overline{AL} = \frac{b\vec{\alpha} + c\vec{\beta}}{b + c} \quad (\because BL : LC = c : b)$$

$$\overline{CM} = \frac{\vec{\alpha}}{2} - \vec{\beta}$$

By hypothesis,

$$\begin{aligned} \overline{AL} \cdot \overline{CM} &= 0 \\ \Rightarrow \left(\frac{b\vec{\alpha} + c\vec{\beta}}{2} \right) \cdot \left(\frac{\vec{\alpha} - 2\vec{\beta}}{2} \right) &= 0 \\ \Rightarrow b|\vec{\alpha}|^2 - 2c|\vec{\beta}|^2 + (c - 2b)(\vec{\alpha} \cdot \vec{\beta}) &= 0 \\ \Rightarrow bc^2 - 2cb^2 + bc(c - 2b)\cos A &= 0 \\ \Rightarrow c - 2b + (c - 2b)\cos A &= 0 \\ \Rightarrow (c - 2b)(1 + \cos A) &= 0 \\ \Rightarrow c &= 2b \end{aligned} \quad (6.23)$$

Again

$$\begin{aligned} |\overline{CM}| : |\overline{AL}| &= m : 1 \\ \Rightarrow |\overline{CM}|^2 &= m^2 |\overline{AL}|^2 \\ \Rightarrow \frac{|\vec{\alpha} - 2\vec{\beta}|^2}{4} &= m^2 \frac{|b\vec{\alpha} + c\vec{\beta}|^2}{(b + c)^2} \end{aligned}$$

Therefore

$$\begin{aligned} \frac{|\vec{\alpha} - 2\vec{\beta}|^2}{4} &= \frac{m^2}{9} |\vec{\alpha} + 2\vec{\beta}|^2 \quad [\text{By Eq. (6.23)}] \\ 9[|\vec{\alpha}|^2 + 4|\vec{\beta}|^2 - 4(\vec{\alpha} \cdot \vec{\beta})] &= 4m^2[|\vec{\alpha}|^2 + 4|\vec{\beta}|^2 + 4(\vec{\alpha} \cdot \vec{\beta})] \\ 9[c^2 + 4b^2 - 4bc\cos A] &= 4m^2[c^2 + 4b^2 + 4bc\cos A] \\ 9[4b^2 + 4b^2 - 8b^2\cos A] &= 4m^2[4b^2 + 4b^2 + 8b^2\cos A] \\ & \quad [\text{by Eq. (6.23)}] \end{aligned}$$

$$9(1 - \cos A) = 4m^2(1 + \cos A)$$

$$m^2 = \frac{9}{4} \left(\frac{1 - \cos A}{1 + \cos A} \right)$$

Answer: (B)

42. Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors of same magnitude. If the vectors $\vec{a} + 3\vec{b}$ and $5\vec{a} + 3\vec{b}$ are at right angles to each other, then the angle between $\vec{a}$ and $\vec{b}$ is

$$(A) \quad \cos^{-1}\left(\frac{7}{9}\right) \quad (B) \quad \pi - \cos^{-1}\left(\frac{7}{9}\right)$$

$$(C) \quad \cos^{-1}\left(\frac{1}{3}\right) \quad (D) \quad \pi - \cos^{-1}\left(\frac{1}{3}\right)$$

Solution: It is given that $|\vec{a}| = |\vec{b}|$. Let $\theta = (\vec{a}, \vec{b})$. Now

$$(\vec{a} + 3\vec{b}) \times (5\vec{a} + 3\vec{b}) = 0$$

$$5|\vec{a}|^2 + 9|\vec{b}|^2 + 18(\vec{a} \times \vec{b}) = 0$$

$$5|\vec{a}|^2 + 9|\vec{b}|^2 + 18|\vec{a}||\vec{b}|\cos\theta = 0$$

$$5 + 9 + 18\cos\theta = 0 \quad (\because |\vec{a}| = |\vec{b}|)$$

$$\cos\theta = -\frac{7}{9}$$

$$\theta = \pi - \cos^{-1} \frac{7}{9}$$

Answer: (B)

43. a, b, c are positive reals and are the l th, m th and n th terms of a geometric progression, respectively. If

$$\vec{A} = (\log a)\vec{i} + (\log b)\vec{j} + (\log c)\vec{k}$$

$$\text{and } \vec{B} = (m - n)\vec{i} + (n - l)\vec{j} + (l - m)\vec{k}$$

then the angle between $\vec{A}$ and $\vec{B}$ is

(A) 60° (B) 45° (C) 30° (D) 90°

Solution: Let α be the first term and x be the common ratio of the GP. Then

$$a = \alpha x^{l-1}$$

$$b = \alpha x^{m-1}$$

$$c = \alpha x^{n-1}$$

Therefore

$$\log a = (l-1)\log x + \log \alpha$$

$$\log b = (m-1)\log x + \log \alpha$$

$$\log c = (n-1)\log x + \log \alpha$$

Now

$$\begin{aligned} \vec{A} \times \vec{B} &= (m-n)\log a + (n-l)\log b + (l-m)\log c \\ &= (\log x)[(m-n)(l-1) + (n-l)(m-1) + (l-m)(n-1)] \\ &\quad + (\log \alpha)[(m-n) + (n-l) + (l-m)] \\ &= (\log x)(0) + (\log \alpha)(0) = 0 \end{aligned}$$

Answer: (D)

44. $ABCD$ is a trapezium (Figure 6.34) in which the sides BC and AD are parallel and the adjacent sides AB and AD are at right angles and AC, BD are at right angles. If $BC:AD = n:1$, then the ratio of the diagonals $AC:BD$ is equal to

(A) $(n+1):1$ (B) $n:1$ (C) $\sqrt{n}:1$ (D) $n^2:1$

Solution: Take A as origin and let $\vec{AB} = \vec{b}$ and $\vec{AD} = \vec{d}$ so that $\vec{BC} = n\vec{d}$, $\vec{b} \times \vec{d} = 0$. Now AC and BD are at right angles. This implies

$$\vec{AC} \times \vec{BD} = 0$$

$$(\vec{b} + n\vec{d}) \times (\vec{d} - \vec{b}) = 0$$

$$\vec{b} \times \vec{d} - |\vec{b}|^2 + n|\vec{d}|^2 - n(\vec{b} \times \vec{d}) = 0$$

$$n|\vec{d}|^2 - |\vec{b}|^2 = 0 \quad (\because \vec{b} \times \vec{d} = 0)$$

$$n|\vec{d}|^2 = |\vec{b}|^2 \quad (6.24)$$

Now,

$$\begin{aligned} \frac{|\vec{AC}|^2}{|\vec{BD}|^2} &= \frac{|\vec{b} + n\vec{d}|^2}{|\vec{d} - \vec{b}|^2} \\ &= \frac{|\vec{b}|^2 + n^2|\vec{d}|^2 + 2n(\vec{b} \times \vec{d})}{|\vec{d}|^2 + |\vec{b}|^2 - \vec{b} \times \vec{d}} \\ &= \frac{|\vec{b}|^2 + n^2|\vec{d}|^2}{|\vec{d}|^2 + |\vec{b}|^2} \quad (\because \vec{b} \times \vec{d} = 0) \\ &= \frac{n|\vec{d}|^2 + n^2|\vec{d}|^2}{|\vec{d}|^2 + n|\vec{d}|^2} \quad [\text{by Eq. (6.24)}] \\ &= n \frac{(|\vec{d}|^2 + n|\vec{d}|^2)}{|\vec{d}|^2 + n|\vec{d}|^2} \\ &= n \end{aligned}$$

Therefore

$$AC:BD = \sqrt{n}:1$$

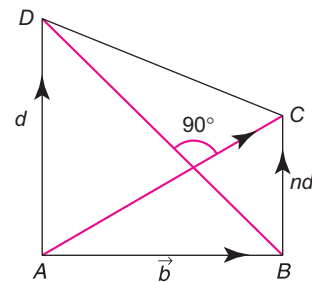


FIGURE 6.34 Single correct choice type question 44.

Answer: (C)

45. If $|\vec{a}| = 13$, $|\vec{b}| = 5$ and $\vec{a} \times \vec{b} = 30$, then $|\vec{a} - \vec{b}|$ is equal to

(A) 30 (B) $\frac{65}{23}\sqrt{493}$
(C) $\frac{30}{23}\sqrt{493}$ (D) $\frac{65}{13}\sqrt{133}$

Solution: We have

$$\begin{aligned} 30 &= \vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\cos\theta \\ &= (65)\cos\theta \end{aligned}$$

where $\theta = (\vec{a}, \vec{b})$. Therefore

$$\cos\theta = \frac{30}{65} = \frac{6}{13}$$

$$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{36}{169}} = \frac{\sqrt{133}}{13}$$

So

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta = \frac{65 \times \sqrt{133}}{13}$$

Answer: (D)

46. For any vector $\vec{a}$,

$$\begin{aligned} |\vec{a} \cdot \vec{i}|^2 + |\vec{a} \cdot \vec{j}|^2 + |\vec{a} \cdot \vec{k}|^2 &= \\ \text{(A) } |\vec{a}|^2 &\quad \text{(B) } 2|\vec{a}|^2 \quad \text{(C) } \frac{1}{2}|\vec{a}|^2 \quad \text{(D) } 3|\vec{a}|^2 \end{aligned}$$

Solution: Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$. Then

$$\vec{a} \cdot \vec{i} = a_1\vec{k} + a_3\vec{j}$$

$$\vec{a} \cdot \vec{j} = a_1\vec{k} - a_3\vec{i}$$

$$\vec{a} \cdot \vec{k} = a_1\vec{j} + a_2\vec{i}$$

Therefore

$$\begin{aligned} |\vec{a} \cdot \vec{i}|^2 + |\vec{a} \cdot \vec{j}|^2 + |\vec{a} \cdot \vec{k}|^2 &= \\ &= (a_2^2 + a_3^2) + (a_1^2 + a_3^2) + (a_1^2 + a_2^2) \\ &= 2(a_1^2 + a_2^2 + a_3^2) \\ &= 2|\vec{a}|^2 \end{aligned}$$

Answer: (B)

47. $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the vertices A, B and C of $\triangle ABC$, respectively. Then the vector perpendicular to the plane of $\triangle ABC$ is

- (A) $\vec{b} \times \vec{c}$
 (B) $\vec{a} + \vec{b} + \vec{c}$
 (C) $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}$
 (D) $(\vec{a} \times \vec{b}) \cdot (\vec{a} + \vec{b} + \vec{c})$

Solution: We have $\vec{AB} = \vec{b} - \vec{a}$, $\vec{BC} = \vec{c} - \vec{b}$, $\vec{AC} = \vec{c} - \vec{a}$. The vector perpendicular to the plane of the triangle ABC is

$$\begin{aligned} \vec{AB} \times \vec{AC} &= (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) \\ &= \vec{b} \times \vec{c} - \vec{b} \times \vec{a} + \vec{c} \times \vec{a} - \vec{a} \times \vec{a} \\ &= \vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b} \end{aligned}$$

Answer: (C)

48. $\vec{a}$ and $\vec{b}$ are non-collinear vectors such that $|\vec{a}| = 2\sqrt{2}$ and $|\vec{b}| = 3$ and the angle between $(\vec{a}, \vec{b})$ is $\pi/4$. If $5\vec{a} + 2\vec{b}$ and $\vec{a} - 3\vec{b}$ are adjacent sides of a parallelogram, then its area in square units is

- (A) 72 (B) 92 (C) 102 (D) 112

Solution: The area of a the parallelogram with $\vec{\alpha}$ and $\vec{\beta}$ as adjacent sides is (see Corollary 6.13)

$$|\vec{\alpha} \times \vec{\beta}|$$

Therefore area of the parallelogram whose adjacent sides are $5\vec{a} + 2\vec{b}$ and $\vec{a} - 3\vec{b}$ is

$$\begin{aligned} |(5\vec{a} + 2\vec{b}) \times (\vec{a} - 3\vec{b})| &= |17(\vec{a} \times \vec{b})| \\ &= 17|\vec{a} \times \vec{b}| \\ &= 17|\vec{a}||\vec{b}|\sin\frac{\pi}{4} \\ &= 17(2\sqrt{2})(3) \frac{1}{\sqrt{2}} \\ &= 102 \text{ sq. units} \end{aligned}$$

Answer: (C)

49. The vector $\vec{r}$ is perpendicular to the vectors $\vec{a} = 3\vec{i} + 2\vec{j} + 2\vec{k}$ and $\vec{b} = 18\vec{i} - 22\vec{j} - 5\vec{k}$ and makes an obtuse angle with $\vec{i}$. If $|\vec{r}| = 14$, then $\vec{r}$ equals

- (A) $4\vec{i} - 6\vec{j} + 12\vec{k}$ (B) $4\vec{i} + 6\vec{j} + 6\vec{k}$
 (C) $4\vec{i} + 6\vec{j} - 6\vec{k}$ (D) $6\vec{i} - 4\vec{j} - 12\vec{k}$

Solution: $\vec{r}$ is parallel to $\vec{a} \times \vec{b}$. This implies

$$\begin{aligned} \vec{r} &= \lambda(\vec{a} \times \vec{b}) \\ &= \lambda \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & 2 \\ 18 & -22 & -5 \end{vmatrix} \\ &= (\lambda)17(2\vec{i} + 3\vec{j} - 6\vec{k}) \end{aligned}$$

Now $\vec{r}$ makes obtuse angle with $\vec{i}$. This means $\lambda < 0$.
Hence

$$|\vec{r}| = 14 \quad 14^2 = 17^2(\lambda^2)(49)$$

$$\lambda^2 = \frac{4}{17^2}$$

$$\lambda = \frac{2}{17} \quad (\because \lambda < 0)$$

$$\vec{r} = 2(2\vec{i} + 3\vec{j} - 6\vec{k})$$

Answer: (A)

50. $A(1, y_1)$ and $B(x_2, 11)$ are two points on a plane curve $y = x^2 - 2x + 3$ with $x_2 > 0$. Then the area of the parallelogram with $\overrightarrow{OA}$ and $\overrightarrow{OB}$ ("O" is the origin) as adjacent sides is

- (A) 3 (B) 9 (C) 6 (D) 4

Solution: We have

$$y_1 = 1 - 2 + 3 = 2$$

and

$$11 = x_2^2 - 2x_2 + 3$$

$$x_2^2 - 2x_2 - 8 = 0$$

$$(x_2 - 4)(x_2 + 2) = 0$$

$$x_2 = 4, -2$$

Therefore $B = (4, 11)$ and $A = (1, 2)$. Hence

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = (\vec{i} + 2\vec{j}) \cdot (4\vec{i} + 11\vec{j})$$

$$= 11\vec{k} - 8\vec{k} = 3\vec{k}$$

So

$$\text{Area} = |\overrightarrow{OA} \cdot \overrightarrow{OB}| = 3$$

Answer: (A)

51. The area of the parallelogram whose diagonals are represented by the vectors $\vec{a} = 3\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} - 3\vec{j} + 4\vec{k}$ is

- (A) $4\sqrt{3}$ (B) $5\sqrt{3}$ (C) $6\sqrt{3}$ (D) $3\sqrt{3}$

Solution: The area of the parallelogram $ABCD$ in terms of its diagonal vectors is (see Theorem 6.30)

$$\frac{1}{2} |\overrightarrow{AC} \cdot \overrightarrow{BD}|$$

Now

$$\begin{aligned} \vec{a} \cdot \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 2 \\ 1 & 3 & 4 \end{vmatrix} \\ &= 2\vec{i} - 14\vec{k} - 10\vec{k} \\ &= 2(\vec{i} + 7\vec{k} + 5\vec{k}) \end{aligned}$$

Therefore

$$\begin{aligned} \text{Area} &= \frac{1}{2} |\vec{a} \cdot \vec{b}| \\ &= \sqrt{1^2 + 7^2 + 5^2} \\ &= \sqrt{75} = 5\sqrt{3} \end{aligned}$$

Answer: (B)

52. Let $\vec{a} = 3\vec{i} + 2\vec{j} + 2\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$. Then a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is

- (A) $\frac{1}{3}(2\vec{i} + 2\vec{j} + \vec{k})$ (B) $\frac{1}{3}(2\vec{i} + 2\vec{j} - \vec{k})$
(C) $\frac{1}{3}(2\vec{i} - 2\vec{j} + \vec{k})$ (D) $\frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$

Solution: We have

$$\vec{a} + \vec{b} = 4\vec{i} + 4\vec{j}$$

$$\vec{a} - \vec{b} = 2\vec{i} + 4\vec{k}$$

Then

$$\begin{aligned} (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & 4 & 0 \\ 2 & 0 & 4 \end{vmatrix} \\ &= 16\vec{i} - 16\vec{j} - 8\vec{k} \\ &= 8(2\vec{i} - 2\vec{j} - \vec{k}) \end{aligned}$$

Unit vector perpendicular to both $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ is

$$\pm \frac{1}{3}(2\vec{i} - 2\vec{j} - \vec{k})$$

One of them is

$$\frac{1}{3}(2\vec{i} + 2\vec{j} + \vec{k})$$

Answer: (A)

53. The perpendicular distance of a vertex of a unit cube from a diagonal not passing through it is

(A) $\frac{2}{3}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

Solution: Let $\vec{OA} = (1, 0, 0)$, $\vec{OB} = (0, 1, 0)$ and $\vec{OC} = (0, 0, 1)$ be three coterminous edges of the unit cube (see Figure 6.35). Clearly the vector $\vec{OR} = \vec{i} + \vec{j} + \vec{k}$. OR is a diagonal not passing through A .

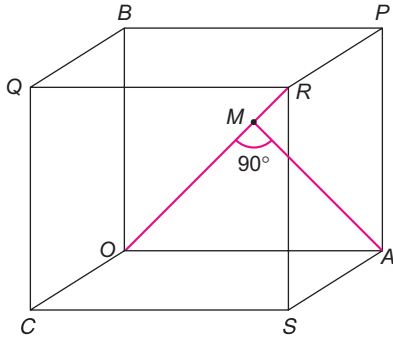


FIGURE 6.35 Single correct choice type question 53.

It is known that the area of the triangle whose vertices have the position vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is (see Corollary 6.14)

$$\frac{1}{2} |\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}|$$

Therefore the area of $\triangle OAR$ is

$$\frac{1}{2} |\vec{i} \times (\vec{i} + \vec{j} + \vec{k}) + (\vec{i} + \vec{j} + \vec{k}) \times \vec{0} + \vec{0} \times \vec{i}| = \frac{1}{2} |\vec{k} - \vec{j}| = \frac{1}{\sqrt{2}}$$

But,

$$\text{Area of } \triangle OAR = \frac{1}{2} |\vec{OR}| \cdot AM$$

where AM is the perpendicular distance of the vertex A from the side OR given by

$$\frac{1}{2} \sqrt{3} \cdot (AM)$$

Therefore

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2} AM$$

$$AM = \frac{\sqrt{2}}{\sqrt{3}} = \sqrt{\frac{2}{3}}$$

Answer: (B)

54. The area of the triangle whose vertices are $\vec{i} + \vec{j} + \vec{k}$, $\vec{i} - \vec{j} + \vec{k}$ and $\vec{k}$ is

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

Solution: Let A , B and C be represented by $\vec{i} + \vec{j} + \vec{k}$, $\vec{i} - \vec{j} + \vec{k}$ and $\vec{k}$, respectively. Then

$$\vec{AB} = -2\vec{j}$$

$$\vec{AC} = -\vec{i} - \vec{k}$$

Now, the area of the $\triangle ABC$ (see Theorem 6.29) is

$$\frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} |2\vec{j} \times (\vec{i} + \vec{k})|$$

$$= |-\vec{k} + \vec{i}| = \sqrt{2}$$

Answer: (B)

55. Let $\vec{a} = 2\vec{i} + \vec{j} - \vec{k}$, $\vec{b} = -\vec{i} + 2\vec{j} - 4\vec{k}$ and $\vec{c} = \vec{i} + \vec{j} + \vec{k}$. Then $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ is equal to

(A) 26 (B) 24 (C) -24 (D) -26

Solution: We have (see Theorem 6.48)

$$(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 6 & 2 \\ 4 & -3 \end{vmatrix}$$

$$= -26$$

Answer: (D)

56. Let $\vec{\alpha}$, $\vec{\beta}$ and $\vec{\gamma}$ be non-coplanar vectors. If $x\vec{\alpha} + y\vec{\beta} + z\vec{\gamma}$ is equal to $a(\vec{\alpha} + \vec{\gamma}) + b(-\vec{\alpha} + \vec{\gamma}) + c(\vec{\alpha} + \vec{\beta} + \vec{\gamma})$, then a is equal to

(A) $-\frac{x}{2} + y - \frac{z}{2}$ (B) $-\frac{x}{2} + \frac{z}{2}$

(C) $\frac{x}{2} - y + \frac{z}{2}$ (D) $x - \frac{y}{2} + \frac{z}{2}$

Solution: We have

$$x\vec{\alpha} + y\vec{\beta} + z\vec{\gamma} = (a - b + c)\vec{\alpha} + c\vec{\beta} + (a + b + c)\vec{\gamma}$$

Therefore

$$a - b + c = x$$

$$y = c$$

$$a + b + c = z$$

Now

$$a - b = -y + x$$

and

$$a + b = z - y$$

Hence

$$a = \frac{x - 2y + z}{2} = \frac{x}{2} - y + \frac{z}{2}$$

Answer: (C)

57. Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = 10\vec{a} + 2\vec{b}$ and $\overrightarrow{OC} = \vec{b}$ where O , A and C are non-collinear points. Let p denote the area of the quadrilateral $OABC$ and q denote the area of the parallelogram with $\overrightarrow{OA}$ and $\overrightarrow{OC}$ as adjacent sides. If $p = \lambda q$, then λ equals

(A) 6 (B) 8 (C) 10 (D) 9

Solution: $q = \text{area of the parallelogram} = |\vec{a} \times \vec{b}|$.

$$\begin{aligned} p &= \text{Area of the Quadrilateral } OABC = \frac{1}{2} |\overrightarrow{OB} \times \overrightarrow{AC}| \\ &= \frac{1}{2} |(10\vec{a} + 2\vec{b}) \times (\vec{b} - \vec{a})| \\ &= \frac{1}{2} |10(\vec{a} \times \vec{b}) - 2(\vec{b} \times \vec{a})| \\ &= \frac{12}{2} |\vec{a} \times \vec{b}| = 6q \end{aligned}$$

Answer: (A)

58. Let $\vec{a} = \vec{i}$, $\vec{b} = x\vec{i} + \vec{j} + (1-x)\vec{k}$ and $\vec{c} = y\vec{i} + x\vec{j} + (1+x-y)\vec{k}$. Then $[\vec{a} \vec{b} \vec{c}]$ depends on

(A) only x (B) only y
(C) neither x nor y (D) both x and y

Solution: Using Theorem 6.35 we know that

$$\begin{aligned} [\vec{a} \vec{b} \vec{c}] &= \begin{vmatrix} 1 & 0 & 1 \\ x & 1 & 1+x \\ y & x & 1+x-y \end{vmatrix} \\ &= (1+x-y) \cdot x(1-x) - (x^2-y) \\ &= 1+x-y-x+x^2-x^2+y \\ &= 1 \end{aligned}$$

Answer: (C)

59. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct?

(A) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$
(B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$
(C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} = \vec{0}$
(D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

Solution: We have $\vec{a} + \vec{b} = -\vec{c}$. Therefore

$$\begin{aligned} (\vec{a} + \vec{b}) \times \vec{c} &= \vec{0} \\ \vec{b} \times \vec{c} &= -\vec{a} \times \vec{c} = \vec{c} \times \vec{a} \end{aligned}$$

Again

$$\begin{aligned} (\vec{b} + \vec{c}) \times \vec{a} &= \vec{0} \\ \vec{c} \times \vec{a} &= -\vec{b} \times \vec{a} = \vec{a} \times \vec{b} \end{aligned}$$

Therefore

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$$

Answer: (B)

60. A non-zero vector $\vec{a}$ is parallel to the line of intersection of the planes determined by the vectors $\vec{i}, \vec{i} + \vec{j}$ and the plane determined by the vectors $\vec{i} - \vec{j}, \vec{i} + \vec{k}$. The angle between $\vec{a}$ and the vector $\vec{i} - 2\vec{j} + 2\vec{k}$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

Solution: $\vec{n}_1 = \vec{i} \times (\vec{i} + \vec{j}) = \vec{k}$ is normal to the first plane $\vec{n}_2 = (\vec{i} - \vec{j}) \times (\vec{i} + \vec{k}) = \vec{j} + \vec{k}$ is normal to second plane. Since $\vec{a}$ is parallel to the line of intersection of the planes, it is perpendicular to both $\vec{n}_1$ and $\vec{n}_2$ and hence $\vec{a}$ is parallel to the vector

$$\vec{n}_1 \times \vec{n}_2 = \vec{k} \times (\vec{j} + \vec{k}) = \vec{i} - \vec{j}$$

The angle between $\vec{a}$ and $\vec{i} - 2\vec{j} + 2\vec{k}$ is same as the angle between $\vec{n}_1 \times \vec{n}_2 = \vec{i} - \vec{j}$ and $\vec{i} - 2\vec{j} + 2\vec{k}$. If θ is the angle between these two vectors, then

$$\cos \theta = \frac{(\vec{i} - \vec{j}) \cdot (\vec{i} - 2\vec{j} + 2\vec{k})}{|\vec{i} - \vec{j}| |\vec{i} - 2\vec{j} + 2\vec{k}|} = \frac{1+2}{\sqrt{2}(3)} = \frac{1}{\sqrt{2}}$$

Therefore

$$\theta = \frac{\pi}{4}$$

Answer: (A)

61. The perpendicular distance of the point $\vec{i} + \vec{j} + \vec{k}$ from the line joining the points $3\vec{i} + 4\vec{j} - \vec{k}$ and $2\vec{i} + \vec{j} - \vec{k}$ is

(A) $\frac{3}{\sqrt{10}}$ (B) $\frac{\sqrt{10}}{2}$ (C) $\frac{7}{\sqrt{10}}$ (D) $2\sqrt{10}$

Solution: Let A and B be the points $3\vec{i} + 4\vec{j} - \vec{k}$ and $2\vec{i} + \vec{j} - \vec{k}$, respectively, and P the point $\vec{i} + \vec{j} + \vec{k}$. Draw AM perpendicular to the line AB (Figure 6.36). Then

$$\begin{aligned} \overrightarrow{PA} &= 2\vec{i} + 3\vec{j} - 2\vec{k} \\ \overrightarrow{PB} &= \vec{i} - 2\vec{k} \end{aligned}$$

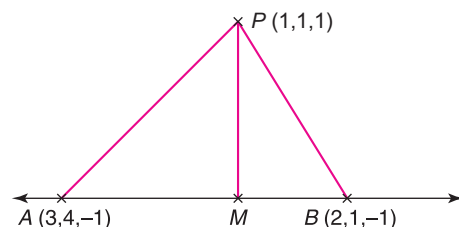


FIGURE 6.36 Single correct choice type question 61.

Now

$$\overrightarrow{PA} \quad \overrightarrow{PB} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 2 \\ 1 & 0 & 2 \end{vmatrix} = 6\vec{i} + 2\vec{j} - 3\vec{k}$$

and $\overrightarrow{AB} = \vec{i} - 3\vec{j}$

Now (see Theorem 6.29)

$$\begin{aligned} \frac{1}{2}(AB)(PM) &= \text{Area of } \triangle PAB = \frac{1}{2}|\overrightarrow{PA} \times \overrightarrow{PB}| \\ &= \frac{1}{2}\sqrt{(6)^2 + 2^2 + (-3)^2} \\ &= \frac{7}{2} \end{aligned}$$

Hence

$$PM = \frac{7}{AB} = \frac{7}{\sqrt{(1)^2 + (-3)^2}} = \frac{7}{\sqrt{10}}$$

Answer: (C)

62. If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the vertices A, B and C of a triangle, then, the angle between the vectors $\vec{b} - \vec{c}$ and $\vec{c} - \vec{a}$ is

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$
(C) $\cos^{-1} \frac{1}{3}$ (D) $\cos^{-1} \frac{1}{\sqrt{3}}$

Solution: It is known that $\overrightarrow{AB} \cdot \overrightarrow{AC}$ is perpendicular to the plane of $\triangle ABC$. But

$$\begin{aligned} \overrightarrow{AB} \cdot \overrightarrow{AC} &= (\vec{b} - \vec{a}) \cdot (\vec{c} - \vec{a}) \\ &= \vec{b} \cdot \vec{c} - \vec{b} \cdot \vec{a} - \vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{a} \\ &= \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{a} \end{aligned}$$

Answer: (A)

63. Let $\vec{a}$ be the position vector of a point and $\vec{b}$ be a non-zero vector. Then, the locus of the point P whose position vector $\vec{r}$ satisfies the relation $\vec{r} - \vec{b} = \vec{a} - \vec{b}$ is

- (A) a straight line
(B) a line passing through $\vec{a}$ and perpendicular to the line $\vec{r} = \vec{a} + t\vec{b}, t \in \mathbb{R}$
(C) a line making an angle of 60° with the line $\vec{r} = \vec{a} + t\vec{b}$
(D) is a circle with centre at the point $\vec{a}$.

Solution: We have

$$\begin{aligned} \vec{r} - \vec{b} &= \vec{a} - \vec{b} \\ (\vec{r} - \vec{a}) - \vec{b} &= 0 \end{aligned}$$

$$\vec{r} - \vec{a} = t\vec{b}, t \in \mathbb{R}$$

$$\vec{r} = \vec{a} + t\vec{b}, t \in \mathbb{R}$$

Answer: (A)

64. Let p be the area of a quadrilateral $ABCD$ and q the area of a parallelogram inscribed in the quadrilateral with its sides drawn parallel to the diagonals of the quadrilateral. Then

- (A) $p = 2q$ (B) $p > 3q$
(C) $p = 3q$ (D) $p < 2q$

Solution: Take A as origin and let $\overrightarrow{AB} = \vec{b}$, $\overrightarrow{AC} = \vec{c}$ and $\overrightarrow{AD} = \vec{d}$. Let $PQRS$ be the parallelogram inscribed in the quadrilateral with P, Q, R and S on the segments AB, BC, CD and DA , respectively (Figure 6.37). Let

$$AP:PB = CQ:QB = \lambda:1$$

Therefore

$$AS:SD = CR:RD = \lambda:1$$

Hence

$$\overrightarrow{AP} = \frac{\lambda\vec{b}}{\lambda+1}$$

$$\overrightarrow{AQ} = \frac{\lambda\vec{b} + \vec{c}}{\lambda+1}$$

$$\overrightarrow{AS} = \frac{\lambda\vec{d}}{\lambda+1}$$

$$\overrightarrow{PQ} = \frac{\vec{c}}{\lambda+1}$$

$$\overrightarrow{PS} = \frac{\lambda}{\lambda+1}(\vec{d} - \vec{b}) = \frac{\lambda}{\lambda+1}(\overrightarrow{BD})$$

Now $q = \text{Area of the parallelogram } PQRS$ is given by

$$\begin{aligned} |\overrightarrow{PQ} \times \overrightarrow{PS}| &= \frac{\lambda}{(\lambda+1)^2} |\overrightarrow{AC} \times \overrightarrow{BD}| \\ &= \frac{2\lambda}{(\lambda+1)^2} (\text{Area of } ABCD) \\ &= \frac{2\lambda}{(\lambda+1)^2} (p) \end{aligned}$$

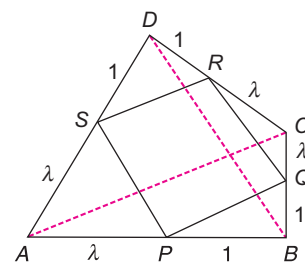


FIGURE 6.37 Single correct choice type question 64.

Therefore

$$p = \frac{(\lambda + 1)^2}{2\lambda}(q) = \left[\left(\frac{\lambda}{2} + \frac{1}{2\lambda} \right) + 1 \right] (q) \geq 2q$$

Equality holds when $\lambda = 1$, that is, P, Q, R and S are the mid-points of the sides.

Answer: (A)

65. In the rhombus $ABCD$, $\angle A = 60^\circ$. K is any point on the segment AC . L and M are points on the segments AB and BC , respectively, such that $KLBM$ is a parallelogram. Then the angles of $\triangle LMD$ are

- (A) $90^\circ, 45^\circ, 45^\circ$ (B) $60^\circ, 60^\circ, 60^\circ$
(C) $90^\circ, 60^\circ, 30^\circ$ (D) $75^\circ, 30^\circ, 75^\circ$

Solution: Take A as origin. Let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AD} = \vec{d}$ so that $\overrightarrow{AC} = \vec{b} + \vec{d}$, $|\vec{b}| = |\vec{d}| = a$ (suppose) and $(\vec{b}, \vec{d}) = 60^\circ$. Let

$$\overrightarrow{AK} = \lambda \overrightarrow{AC} = \lambda(\vec{b} + \vec{d})$$

and $0 < \lambda < 1$. $\overrightarrow{LK}$ is parallel to BC implies that

$$\frac{AL}{AB} = \frac{AK}{AC} = \lambda \quad (\text{suppose})$$

Therefore

$$\overrightarrow{AL} = \lambda \vec{b}$$

Equation of the line KM is

$$\vec{r} = \lambda(\vec{b} + \vec{d}) + t\vec{b} \quad (6.25)$$

where $t \in \mathbb{R}$ (see Theorem 5.27, Chapter 5). Equation of the line BC is

$$\vec{r} = \vec{b} + s\vec{d} \quad (6.26)$$

Since M is the point of intersection of the lines KM and BC , on equating the coefficients of Eqs. (6.25) and (6.26) we have

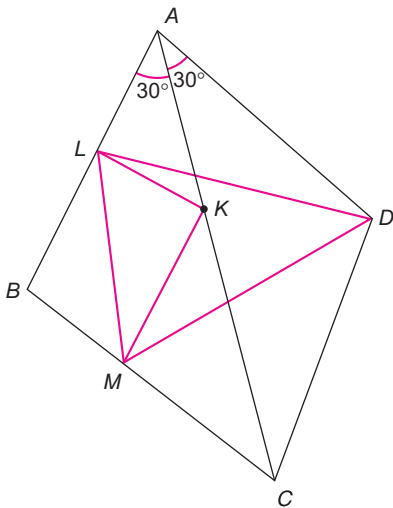


FIGURE 6.38 Single correct choice type question 65.

$$\lambda + t = 1 \quad \text{and} \quad \lambda = s$$

Therefore the position vectors of M, L and K are, respectively, $\vec{b} + \lambda\vec{d}$, $\lambda\vec{b}$ and $\lambda(\vec{b} + \vec{d})$. Hence

$$\begin{aligned} |\overrightarrow{LM}|^2 &= |\vec{b} + \lambda\vec{d} - \lambda\vec{b}|^2 \\ &= |(1 - \lambda)\vec{b} + \lambda\vec{d}|^2 \\ &= (1 - \lambda)^2 |\vec{b}|^2 + \lambda^2 |\vec{d}|^2 + 2\lambda(1 - \lambda)(\vec{b} \cdot \vec{d}) \\ &= a^2[(1 - \lambda)^2 + \lambda^2 + \lambda(1 - \lambda)] \quad \left(\because \vec{b} \cdot \vec{d} = a^2 \cos 60^\circ = \frac{a^2}{2} \right) \\ &= a^2(\lambda^2 - \lambda + 1) \end{aligned}$$

Note that $\lambda^2 - \lambda + 1 > 0$. Now

$$\begin{aligned} |\overrightarrow{MD}|^2 &= |\vec{b} + \lambda\vec{d} - \vec{d}|^2 \\ &= |\vec{b} + (\lambda - 1)\vec{d}|^2 = |\vec{b}|^2 + (\lambda - 1)^2 |\vec{d}|^2 + 2(\lambda - 1)(\vec{b} \cdot \vec{d}) \\ &= a^2[1 + (\lambda - 1)^2 + \lambda - 1] \\ &= a^2(\lambda^2 - \lambda + 1) \\ |\overrightarrow{LD}|^2 &= |\lambda\vec{b} - \vec{d}|^2 \\ &= \lambda^2 |\vec{b}|^2 - 2\lambda(\vec{b} \cdot \vec{d}) + |\vec{d}|^2 \\ &= a^2(\lambda^2 - \lambda + 1) \end{aligned}$$

Therefore

$$LM = DL = DM$$

and so $\triangle LMD$ is equilateral.

Answer: (B)

66. P is a point on the circumcircle of $\triangle ABC$ other than the vertices A, B and C . H is the orthocentre, M is the mid-point of PH and D is the mid-point of BC . Then the angle between the vectors $\overrightarrow{DM}$ and $\overrightarrow{AP}$ is
(A) 75° (B) 60° (C) 120° (D) 90°

Solution: Take the circumcentre " O " of the triangle as origin and let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$. See Figure 6.39.

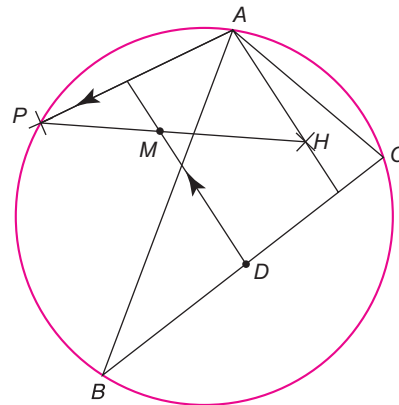


FIGURE 6.39 Single correct choice type question 66.

Therefore by Example 5.3 (Chapter 5)

$$\overrightarrow{OH} = \vec{a} + \vec{b} + \vec{c}$$

Let $\overrightarrow{OP} = \vec{p}$ so that

$$\overrightarrow{OM} = \frac{\vec{p} + \vec{a} + \vec{b} + \vec{c}}{2}$$

and

$$\begin{aligned}\overrightarrow{DM} &= \frac{\vec{p} + \vec{a} + \vec{b} + \vec{c}}{2} - \frac{\vec{b} + \vec{c}}{2} \\ &= \frac{\vec{p} + \vec{a}}{2}\end{aligned}$$

Now

$$\begin{aligned}\overrightarrow{DM} \cdot \overrightarrow{AP} &= \left(\frac{\vec{p} + \vec{a}}{2} \right) \cdot (\vec{p} - \vec{a}) \\ &= \frac{1}{2} (|\vec{p}|^2 - |\vec{a}|^2) \\ &= \frac{1}{2} (0) \quad (\because |\vec{p}| = |\vec{a}| = R) \\ &= 0\end{aligned}$$

Therefore $\overrightarrow{DM}$ and $\overrightarrow{AP}$ are at right angles.

Answer: (D)

67. $ABCD$ is a parallelogram. Through the vertex A , an arbitrary line is drawn to meet the diagonal BD in P , the side BC in Q and the side DC produced in R . Then AP is

- (A) AM between PR and PQ
- (B) GM between PR and PQ
- (C) HM between PR and PQ
- (D) $AR = PQ \cdot PR$

Solution: Take A as origin and let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AD} = \vec{d}$. Suppose $BP:PD = \lambda:1$ so that

$$\overrightarrow{AP} = \frac{\vec{b} + \lambda \vec{d}}{\lambda + 1}$$

Therefore the equation of the line AP is (see Theorem 5.29)

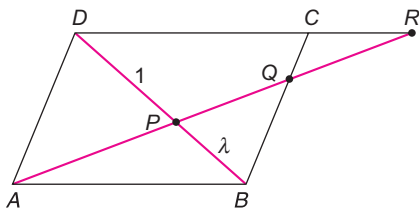


FIGURE 6.40 Single correct choice type question 67.

$$\vec{r} = t \frac{(\lambda \vec{d} + \vec{b})}{\lambda + 1} \quad (6.27)$$

Equation of the line BC is (see Theorem 5.27)

$$\vec{r} = \vec{b} + s\vec{d}, \quad s \in \mathbb{R} \quad (6.28)$$

Equation of the line DC is (see Theorem 5.27)

$$\vec{r} = \vec{d} + u\vec{b}, \quad u \in \mathbb{R} \quad (6.29)$$

From Eqs. (6.27) and (6.28), we have

$$\frac{t}{\lambda + 1} = 1 \quad \text{and} \quad \frac{t\lambda}{\lambda + 1} = s$$

and hence $s = \lambda$, $t = \lambda + 1$. Therefore

$$\overrightarrow{AQ} = \vec{b} + \lambda \vec{d} \quad (6.30)$$

Also from Eqs. (6.27) and (6.29), we have

$$\frac{t\lambda}{\lambda + 1} = 1 \quad \text{and} \quad \frac{t}{\lambda + 1} = u$$

Therefore

$$\lambda u = 1 \quad \text{or} \quad u = \frac{1}{\lambda}$$

Hence

$$\overrightarrow{AR} = \vec{d} + \frac{1}{\lambda} \vec{b} \quad (6.31)$$

Therefore

$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{AQ} - \overrightarrow{AP} \\ &= (\vec{b} + \lambda \vec{d}) - \left(\frac{\vec{b} + \lambda \vec{d}}{\lambda + 1} \right) \\ &= \frac{\lambda}{\lambda + 1} (\vec{b} + \lambda \vec{d}) \\ &= \lambda (\overrightarrow{AP})\end{aligned}$$

This gives

$$\frac{PQ}{AP} = \lambda \quad (6.32)$$

Also

$$\begin{aligned}\overrightarrow{PR} &= \overrightarrow{AR} - \overrightarrow{AP} \\ &= \left(\vec{d} + \frac{1}{\lambda} \vec{b} \right) - \frac{\vec{b} + \lambda \vec{d}}{\lambda + 1} \quad [\text{see Eq. (6.31)}] \\ &= (\vec{b} + \lambda \vec{d}) \left(\frac{1}{\lambda} - \frac{1}{\lambda + 1} \right)\end{aligned}$$

$$\begin{aligned}
 &= \frac{\vec{b} + \lambda \vec{d}}{\lambda(\lambda + 1)} \\
 &= \frac{1}{\lambda}(\overline{AP})
 \end{aligned}$$

Hence

$$\frac{AP}{PR} = \lambda \quad (6.33)$$

Equations (6.32) and (6.33) give

$$\begin{aligned}
 \frac{PQ}{AP} &= \frac{AP}{PR} \\
 (AP)^2 &= PQ \times PR
 \end{aligned}$$

Answer: (B)

68. On the sides BC , CA and AB of $\triangle ABC$, points D , E and F are taken, respectively, such that $BD:DC = CE:EA = AF:FB = \lambda:1$. Then the ratio of the area of $\triangle DEF$ to the area of $\triangle ABC$ is

- (A) $(\lambda^2 - \lambda + 1):(\lambda + 1)^2$
 (B) $(\lambda^2 - \lambda + 1):(\lambda + 1)$
 (C) $(\lambda^2 + \lambda + 1):(\lambda + 1)^2$
 (D) $(\lambda^2 + \lambda + 1):(\lambda + 1)$

Solution: Take A as origin and let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AC} = \vec{c}$. Therefore

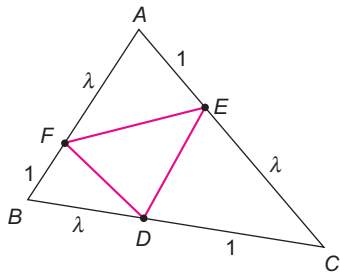


FIGURE 6.41 Single correct choice type question 68.

$$\overrightarrow{AD} = \frac{\vec{b} + \lambda \vec{c}}{\lambda + 1}$$

$$\overrightarrow{AE} = \frac{\vec{c}}{\lambda + 1}$$

$$\overrightarrow{AF} = \frac{\lambda \vec{b}}{\lambda + 1}$$

$$\overrightarrow{DE} = \overrightarrow{AE} - \overrightarrow{AD} = \frac{\vec{b} + (1 - \lambda)\vec{c}}{\lambda + 1}$$

$$\overrightarrow{DF} = \frac{(\lambda - 1)\vec{b} - \lambda \vec{c}}{\lambda + 1}$$

Therefore (using Theorem 6.29)

$$\begin{aligned}
 \text{Area of } \triangle DEF &= \frac{1}{2} |\overrightarrow{DE} \times \overrightarrow{DF}| \\
 &= \frac{1}{2} \left| \frac{\vec{b} + (1 - \lambda)\vec{c}}{\lambda + 1} \times \frac{(\lambda - 1)\vec{b} - \lambda \vec{c}}{\lambda + 1} \right| \\
 &= \frac{1}{2(\lambda + 1)^2} |\lambda(\vec{b} \times \vec{c}) + (\lambda - 1)^2(\vec{b} \times \vec{c})| \\
 &= \frac{\lambda^2 - \lambda + 1}{2(\lambda + 1)^2} |\vec{b} \times \vec{c}| \\
 &= \frac{\lambda^2 - \lambda + 1}{(\lambda + 1)^2} (\text{Area of } \triangle ABC) \\
 \therefore \frac{1}{2} |\vec{b} \times \vec{c}| &= \text{Area } \triangle ABC
 \end{aligned}$$

Answer: (A)

69. A vector of magnitude $\sqrt{2}$ and coplanar with the vectors $3\vec{i} - \vec{j} + \vec{k}$ and $\vec{i} + \vec{j} - 2\vec{k}$ which is perpendicular to the vector $2\vec{i} + 2\vec{j} + \vec{k}$ is one of the following vectors. Which one is it?

- (A) $\frac{\sqrt{2}}{3}(\vec{i} + \vec{j} + \vec{k})$ (B) $\frac{1}{5}(3\vec{i} - 5\vec{j} + 4\vec{k})$
 (C) $\frac{1}{5}(3\vec{i} + 5\vec{j} + 4\vec{k})$ (D) $\frac{2}{\sqrt{6}}(\vec{i} - \vec{j} + \vec{k})$

Solution: Let $\vec{r}$ be the required vector. Since $\vec{r}$ is coplanar with the vectors $\vec{b} = 3\vec{i} - \vec{j} + \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} - 2\vec{k}$ (observe that $\vec{b}$ and $\vec{c}$ are non-collinear), let

$$\begin{aligned}
 \vec{r} &= x\vec{b} + y\vec{c} \\
 &= (3x + y)\vec{i} + (y - x)\vec{j} + (x + 2y)\vec{k} \quad (6.34)
 \end{aligned}$$

$$|\vec{r}| = \sqrt{2} \quad (3x + y)^2 + (y - x)^2 + (x + 2y)^2 = 2 \quad (6.35)$$

where x and y are scalars. Now $\vec{r}$ is perpendicular to $2\vec{i} + 2\vec{j} + \vec{k}$ implies

$$2(3x + y) + 2(y - x) + x + 2y = 0$$

$$3x + 2y = 0$$

$$y = -\frac{3x}{2}$$

Putting this value of $y = -3x/2$ in Eq. (6.35) we get

$$3x^2 + \frac{3x^2}{2} + (x - 3x)^2 = 2$$

$$\frac{9x^2}{4} + \frac{25x^2}{4} + 4x^2 = 2$$

$$50x^2 = 8$$

$$x = \pm \frac{2\sqrt{2}}{5\sqrt{2}} = \pm \frac{2}{5}$$

Now

$$x = \frac{2}{5} \Rightarrow y = -\frac{3x}{2} = -\frac{3}{5}$$

$$x = -\frac{2}{5} \Rightarrow y = \frac{3}{5}$$

In tabular form we have

$$\begin{array}{c|c} x & y = -\frac{3x}{2} \\ \hline \frac{2}{5} & -\frac{3}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{array}$$

Answer: (B)

- 70.** The position vectors of the points A, B, C and D are, respectively, $3\vec{i} - 2\vec{j} - \vec{k}$, $2\vec{i} + 3\vec{j} - 4\vec{k}$, $-\vec{i} + \vec{j} + 2\vec{k}$ and $4\vec{i} + 5\vec{j} + \lambda\vec{k}$. If all the four points lie in a plane, then the value of λ is

(A) $-\frac{146}{17}$ (B) $\frac{146}{7}$ (C) $\frac{98}{17}$ (D) $-\frac{98}{17}$

Solution: We have

$$\overrightarrow{AB} = -\vec{i} + 5\vec{j} - 3\vec{k}$$

$$\overrightarrow{AC} = -4\vec{i} + 3\vec{j} + 3\vec{k}$$

$$\overrightarrow{AD} = \vec{i} + 7\vec{j} + (\lambda + 1)\vec{k}$$

Now

A, B, C and D are coplanar

$\Leftrightarrow \overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar

$$\Leftrightarrow \begin{vmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & (\lambda + 1) \end{vmatrix} = 0 \quad (\text{see Corollary 6.16})$$

$$\Leftrightarrow 17\lambda + 146 = 0$$

Answer: (A)

- 71.** The value of a so that the volume of the parallelopiped formed by the vectors $\vec{i} + a\vec{j} - \vec{k}$, $\vec{j} + a\vec{k}$, $\vec{i} + \vec{j} + \vec{k}$ becomes minimum is

(A) $\sqrt{3}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{7}{4}$

Solution: We have

$$V = \text{Volume of the parallelopiped} = |\det A|$$

(see Theorem 6.33 and Theorem 6.35) where A is the matrix

$$\begin{bmatrix} 1 & a & -1 \\ 0 & 1 & a \\ 1 & 1 & 1 \end{bmatrix}$$

Now

$$\det A = 1 - a + a^2 + 1$$

$$= a^2 - a + 2$$

$$= \left(a - \frac{1}{2}\right)^2 + \frac{7}{4} \geq \frac{7}{4}$$

Therefore

$$V = \det A = \left(a - \frac{1}{2}\right)^2 + \frac{7}{4} \geq \frac{7}{4}$$

V is equal to $7/4$ when $a = 1/2$. Hence V is minimum when $a = 1/2$ and the minimum value of V is $7/4$.

Answer: (B)

- 72.** If the volume of the parallelopiped whose set of coterminous edges are represented by the vectors $-12\vec{i} + \lambda\vec{k}$, $3\vec{j} - \vec{k}$ and $2\vec{i} + \vec{j} - 15\vec{k}$ is 546 cubic units, then a value of λ is

(A) -3 (B) 3 (C) -179 (D) 178

Solution: We have

$$V = \text{Volume}$$

$$= \text{Absolute value of } \begin{vmatrix} -12 & 0 & \lambda \\ 0 & 3 & -1 \\ 2 & 1 & -15 \end{vmatrix}$$

$$= |12 \times 44 - 6\lambda|$$

Therefore

$$88 - \lambda = \pm 91$$

Now

$$88 - \lambda = 91 \Rightarrow \lambda = -3$$

and

$$88 - \lambda = -91 \Rightarrow \lambda = 179$$

Answer: (A)

- 73.** Unit vector orthogonal to $\vec{a} = 3\vec{i} + 2\vec{j} + 6\vec{k}$ and coplanar with $\vec{b} = 2\vec{i} + \vec{j} + \vec{k}$ and $\vec{c} = \vec{i} - \vec{j} + \vec{k}$ is

(A) $\frac{6\vec{i} - 5\vec{k}}{\sqrt{61}}$

(B) $\frac{3\vec{j} - \vec{k}}{\sqrt{10}}$

(C) $\frac{2\vec{i} - 5\vec{j}}{\sqrt{29}}$

(D) $\frac{2\vec{i} + \vec{j} - 2\vec{k}}{3}$

Solution: By the definition of cross product, the vector $\vec{a} \times (\vec{b} \times \vec{c})$ is coplanar with $\vec{b}$ and $\vec{c}$ and is orthogonal to $\vec{a}$. Now (using Theorem 6.46) we have

$$\begin{aligned}\vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \\ &= 7\vec{b} - 14\vec{c} = 7(3\vec{j} - \vec{k})\end{aligned}$$

Therefore the required unit vector is

$$\frac{7(3\vec{j} - \vec{k})}{7\sqrt{9+1}} = \frac{3\vec{j} - \vec{k}}{\sqrt{10}}$$

Answer: (B)

74. If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{b}$ is a vector such that $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \vec{j} - \vec{k}$, then $\vec{b}$ is

- (A) $\vec{i} - \vec{j} + \vec{k}$ (B) $2\vec{j} - \vec{k}$ (C) $\vec{i}$ (D) $2\vec{i}$

Solution: We have

$$(\vec{a} \times \vec{b}) \times \vec{a} = (\vec{j} - \vec{k}) \times (\vec{i} + \vec{j} + \vec{k})$$

Therefore (using Theorem 6.46)

$$\begin{aligned}(\vec{a} \cdot \vec{a})\vec{b} - (\vec{b} \cdot \vec{a})\vec{a} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} \\ 3\vec{b} - \vec{a} &= 2\vec{i} - \vec{j} - \vec{k} \quad (\because \vec{a} \cdot \vec{b} = 1) \\ 3\vec{b} &= (\vec{i} + \vec{j} + \vec{k}) + (2\vec{i} - \vec{j} - \vec{k}) \\ \vec{b} &= \frac{1}{3}(3\vec{i}) = \vec{i}\end{aligned}$$

Answer: (C)

75. Let $\vec{a} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} + \vec{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30° , then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to

- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

Solution: We have

$$\begin{aligned}8 &= |\vec{c} - \vec{a}|^2 \\ &= |\vec{c}|^2 - 2(\vec{c} \cdot \vec{a}) + |\vec{a}|^2 \\ &= |\vec{c}|^2 - 2|\vec{c}| + 9\end{aligned}$$

Therefore

$$(|c| - 1)^2 = 0 \quad \text{or} \quad |\vec{c}| = 1$$

Now

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 2\vec{i} - 2\vec{j} + \vec{k}$$

Hence

$$|\vec{a} \times \vec{b}| = 3$$

So

$$\begin{aligned}|(\vec{a} \times \vec{b}) \times \vec{c}| &= |\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ \\ &= 3(1)\left(\frac{1}{2}\right) = \frac{3}{2}\end{aligned}$$

Answer: (B)

76. The vectors $\vec{a} = (\sec^2 \alpha)\vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + (\sec^2 \beta)\vec{j} + \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} + (\sec^2 \gamma)\vec{k}$ are coplanar only if

- (A) α, β, γ are distinct such that $\alpha + \beta + \gamma = \pi$
(B) $\alpha = \beta = \gamma = n\pi, n \in \mathbb{R}$
(C) $\alpha = \beta = \gamma = \frac{\pi}{4}$
(D) no real values exist

Solution: The vector are coplanar only if

$$\begin{vmatrix} \sec^2 \alpha & 1 & 1 \\ 1 & \sec^2 \beta & 1 \\ 1 & 1 & \sec^2 \gamma \end{vmatrix} = 0$$

Replacing $\sec^2 \alpha$, $\sec^2 \beta$ and $\sec^2 \gamma$ with $1 + \tan^2 \alpha$, $1 + \tan^2 \beta$ and $1 + \tan^2 \gamma$ and simplifying we have

$$\sum (\tan^2 \beta \tan^2 \gamma) + \tan^2 \alpha \tan^2 \beta \tan^2 \gamma = 0$$

which is possible only when

$$\tan^2 \alpha = \tan^2 \beta = \tan^2 \gamma = 0$$

Answer: (B)

77. Let $\vec{a}, \vec{b}$ be unit vectors. If $\vec{c}$ is a vector such that $\vec{b} = \vec{c} + \vec{c} \times \vec{a}$ and if

$$|(\vec{a} \times \vec{b}) \cdot \vec{c}| = \frac{1}{2}$$

then the angle between $\vec{a}$ and $\vec{b}$ is

- (A) 60° (B) 30° (C) 45° (D) 90°

Solution: Let θ be the angle between $\vec{a}$ and $\vec{c}$. Now

$$\begin{aligned}|(\vec{a} \times \vec{b}) \cdot \vec{c}| &= |\vec{b} \cdot (\vec{c} \times \vec{a})| = |(\vec{c} + \vec{c} \times \vec{a}) \cdot (\vec{c} \times \vec{a})| \\ &= |\vec{c} \times \vec{a}|^2 = |\vec{c}|^2 \sin^2 \theta\end{aligned}\quad (6.36)$$

Also

$$|\vec{b}|^2 = |\vec{c} + \vec{c} \times \vec{a}|^2 = |\vec{c}|^2 + |\vec{c} \times \vec{a}|^2 = |\vec{c}|^2 + |\vec{c}|^2 \sin^2 \theta$$

Therefore

$$1 = |\vec{b}|^2 = (1 + \sin^2 \theta)|\vec{c}|^2 \quad (\because |\vec{b}| = 1)$$

So

$$|\vec{c}|^2 = \frac{1}{1 + \sin^2 \theta} \quad (6.37)$$

Therefore from Eqs. (6.36) and (6.37) we get

$$|(\vec{a} \times \vec{b}) \cdot \vec{c}| = \frac{\sin^2 \theta}{1 + \sin^2 \theta} = \frac{1}{2}$$

if and only if $\theta = 90^\circ$. Now

$$\begin{aligned} \vec{b} \times \vec{a} &= \vec{c} \times \vec{a} + (\vec{c} \times \vec{a}) \times \vec{a} \\ &= 0 + 0 \quad (\because \vec{c} \times \vec{a} = 0) \end{aligned}$$

Therefore $\vec{a}$ and $\vec{b}$ are perpendicular to each other.

Answer: (D)

78. $\vec{a}$, $\vec{b}$ and $\vec{c}$ are vectors of magnitudes 1, 1 and 2, respectively. If $\vec{a} \cdot (\vec{a} \times \vec{c}) + \vec{b} \cdot \vec{c} = 0$, then acute angle between $\vec{a}$ and $\vec{c}$ is

- (A) 90° (B) 60° (C) 45° (D) 30°

Solution: We have (using Theorem 6.46)

$$\vec{b} \cdot \vec{a} \cdot (\vec{a} \times \vec{c}) = (\vec{a} \times \vec{c}) \cdot \vec{a} \cdot (\vec{a} \times \vec{c})$$

Therefore

$$\begin{aligned} 1 &= |\vec{b}|^2 = (\vec{a} \times \vec{c})^2 |\vec{a}|^2 + |\vec{c}|^2 - 2(\vec{a} \times \vec{c})^2 \quad (\because |\vec{a}| = 1 = |\vec{b}|) \\ &= (\vec{a} \times \vec{c})^2 + 4 - 2(\vec{a} \times \vec{c})^2 \\ &= 4 - (\vec{a} \times \vec{c})^2 \\ &= 4 - 4 \cos^2 \theta \end{aligned}$$

where $\theta = (\vec{a}, \vec{c})$. Therefore

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

Hence θ is acute $\theta = 30^\circ$.

Answer: (D)

79. Let $A = (1, 2, 1)$, $B = (2, 1, 2)$, $C = (0, 4, 4)$, $D = (2, 2, 2)$ and $E = (4, 1, 2)$ be five points. The line AB meets the plane CDE at the points

- (A) $(3, 1, 3)$ (B) $(3, 0, 3)$
(C) $(1, 1, 1)$ (D) $(2, 0, 2)$

Solution: Equation of the line AB is (using Theorem 5.29)

$$\begin{aligned} \vec{r} &= (1 - t)(2\vec{i} + \vec{j} + 2\vec{k}) + t(\vec{i} + 2\vec{j} + \vec{k}), t \in \mathbb{R} \\ \vec{r} &= (2 - t)\vec{i} + (1 + t)\vec{j} + (2 - t)\vec{k} \end{aligned} \quad (6.38)$$

For the plane CDE , $\vec{CD} \times \vec{CE}$ is a normal. So

$$\begin{aligned} \vec{CD} \times \vec{CE} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 2 & 2 \\ 4 & 5 & 2 \end{vmatrix} \\ &= 6\vec{i} - 4\vec{j} + 2\vec{k} \end{aligned}$$

Therefore $3\vec{i} - 2\vec{j} + \vec{k}$ is also normal to the plane CDE . The plane passes through the point C . By Theorem 6.17, the equation of the plane is

$$[\vec{r} \cdot (3\vec{i} - 2\vec{j} + \vec{k})] \cdot (3\vec{i} - 2\vec{j} + \vec{k}) = 0$$

where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ is any point in the plane CDE . That is

$$\begin{aligned} [x\vec{i} + (y + 4)\vec{j} + (z - 4)\vec{k}] \cdot (3\vec{i} - 2\vec{j} + \vec{k}) &= 0 \\ 3x - 2(y + 4) + z - 4 &= 0 \\ 3x - 2y + z - 12 &= 0 \end{aligned} \quad (6.39)$$

From Eqs. (6.38) and (6.39), we have

$$\begin{aligned} 3(2 - t) - 2(1 + t) + (2 - t) - 12 &= 0 \\ 6 - 3t - 2 - 2t + 2 - t - 12 &= 0 \\ 6t - 6 &= 0 \\ t &= 1 \end{aligned}$$

Substituting the value of $t = 1$ in Eq. (6.38), we have

$$\vec{r} = 3\vec{i} + 3\vec{k}$$

Therefore $(3, 0, 3)$ is the point where the line AB meets the plane CDE .

Answer: (B)

80. Let $\vec{a} = \vec{i}$, $\vec{b} = \vec{j}$, $\vec{c} = \vec{k}$. $\vec{d}$ is a unit vector such that $\vec{a} \times \vec{d} = 0 = [\vec{b} \times \vec{c}]$, then $\vec{d}$ equals

- (A) $\pm \frac{(\vec{i} + \vec{j} - 2\vec{k})}{\sqrt{6}}$ (B) $\pm \frac{(\vec{i} - \vec{j} - 2\vec{k})}{\sqrt{6}}$
(C) $\pm \frac{(2\vec{i} + \vec{j} - \vec{k})}{\sqrt{6}}$ (D) $\pm \frac{(\vec{i} + 2\vec{j} - \vec{k})}{\sqrt{6}}$

Solution: Let $\vec{d} = x\vec{i} + y\vec{j} + z\vec{k}$ and $|\vec{d}| = 1$.

$$\vec{a} \times \vec{d} = 0 \quad x - y = 0 \quad (6.40)$$

$$\begin{aligned} [\vec{b} \times \vec{c}] \cdot \vec{d} &= 0 \quad \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ x & y & z \end{vmatrix} = 0 \\ x + y + z &= 0 \end{aligned} \quad (6.41)$$

$$|\vec{d}| = 1 \quad x^2 + y^2 + z^2 = 1 \quad (6.42)$$

From Eqs. (6.40) and (6.41) we get

$$y = x \quad \text{and} \quad z = 2x$$

Substituting these values in Eq. (6.42), we have

$$6x^2 = 1 \quad \text{or} \quad x = \pm \frac{1}{\sqrt{6}}$$

Therefore

$$\vec{d} = \pm \frac{1}{\sqrt{6}}(\vec{i} + \vec{j} - 2\vec{k})$$

Answer: (A)

81. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-coplanar vectors and $2\vec{a} + 3\vec{b} - \vec{c}$, $\vec{a} - 2\vec{b} + 3\vec{c}$, $3\vec{a} + 4\vec{b} - 2\vec{c}$ and $\vec{a} - 6\vec{b} + 6\vec{c}$ be position vectors of the points A, B, C and D , respectively. Then the scalar triple product $[\vec{AB} \vec{AC} \vec{AD}]$ is equal to

- (A) $9[\vec{a} \vec{b} \vec{c}]$ (B) $6[\vec{a} \vec{b} \vec{c}]$
(C) $4[\vec{a} \vec{b} \vec{c}]$ (D) 0

Solution: We have

$$\vec{AB} = \vec{a} - 5\vec{b} + 4\vec{c}$$

$$\vec{AC} = \vec{a} + \vec{b} - \vec{c}$$

$$\vec{AD} = \vec{a} - 9\vec{b} + 7\vec{c}$$

Therefore (using Quick Look 11)

$$\begin{aligned} [\vec{AB} \vec{AC} \vec{AD}] &= \begin{vmatrix} 1 & -5 & 4 \\ 1 & 1 & -1 \\ 1 & -9 & 7 \end{vmatrix} [\vec{a} \vec{b} \vec{c}] \\ &= (2 + 30 - 32)[\vec{a} \vec{b} \vec{c}] \\ &= 0 \end{aligned}$$

Answer: (D)



QUICK LOOK

The points A, B, C and D are coplanar.

82. In Figure 6.42, $\vec{AB} = 3\vec{i} - \vec{j}$, $\vec{AC} = 2\vec{i} + 3\vec{j}$ and $\vec{DE} = 4\vec{i} - 2\vec{j}$. Then the area of the shaded region in square units is

- (A) 5 (B) 6 (C) 7 (D) 8

Solution: We have

$$\begin{aligned} \vec{BC} &= \vec{BA} + \vec{AC} \\ &= (-3\vec{i} + \vec{j}) + (2\vec{i} + 3\vec{j}) \\ &= -\vec{i} + 4\vec{j} \end{aligned}$$

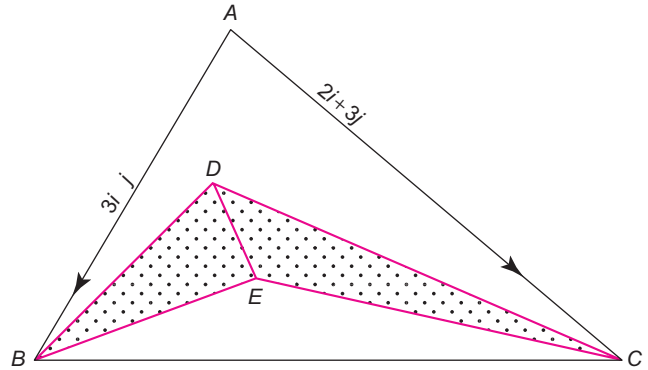


FIGURE 6.42 Single correct choice type question 82.

Vector area of the shaded region is given by

$$\begin{aligned} \frac{1}{2} \vec{ED} \times \vec{EB} + \frac{1}{2} \vec{EC} \times \vec{ED} &= \frac{1}{2} [\vec{ED} \times \vec{EB} + \vec{ED} \times \vec{EC}] \\ &= \frac{1}{2} [\vec{ED} \times (\vec{EB} + \vec{EC})] \\ &= \frac{1}{2} (\vec{ED} \times \vec{CB}) \\ &= \frac{1}{2} [(4\vec{i} - 2\vec{j}) \times (\vec{i} - 4\vec{j})] \\ &= \frac{1}{2} (16\vec{k} - 2\vec{k}) \\ &= 7\vec{k} \end{aligned}$$

Therefore

$$\text{Area} = |\text{Vector area}| = 7$$

Answer: (C)

83. The angle between two unit vectors $\vec{a}$ and $\vec{b}$ is $\cos^{-1}(1/4)$. If $\vec{c}$ is a vector such that $|\vec{c}| = 4$ and $\vec{c} - 2\vec{b} = \lambda\vec{a}$, then value of λ is

- (A) 3, 4 (B) 3, 4 (C) 3, 4 (D) 3, 4

Solution: We have

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \cos^{-1} \frac{1}{4} = \frac{1}{4} \quad (6.43)$$

$$\vec{c} = \lambda\vec{a} + 2\vec{b}$$

$$16 = |\vec{c}|^2$$

$$= |\lambda\vec{a} + 2\vec{b}|^2$$

$$= \lambda^2 |\vec{a}|^2 + 4\lambda(\vec{a} \cdot \vec{b}) + 4|\vec{b}|^2$$

$$= \lambda^2 + 4\lambda \frac{1}{4} + 4 \quad [\text{by Eq. (6.43)}]$$

Therefore

$$\lambda^2 + \lambda - 12 = 0$$

$$(\lambda + 4)(\lambda - 3) = 0$$

$$\lambda = 3 \quad \text{or} \quad -4$$

Answer: (A)

- 84.** In $\triangle ABC$ (Figure 6.43), if $(BC)^2 + (AC)^2 = 5(AB)^2$, then the angle between the medians AD and BE is
 (A) 60° (B) 90° (C) 120° (D) 45°

Solution: Take A as origin. Let $\vec{AB} = \vec{\alpha}$ and $\vec{AC} = \vec{\beta}$ so that $|\vec{\alpha}| = c$, $|\vec{\beta}| = b$ and $(\vec{\alpha}, \vec{\beta}) = A$. Therefore

$$\vec{AD} = \frac{\vec{\alpha} + \vec{\beta}}{2}$$

and

$$\vec{AE} = \frac{1}{2}\vec{\beta}$$

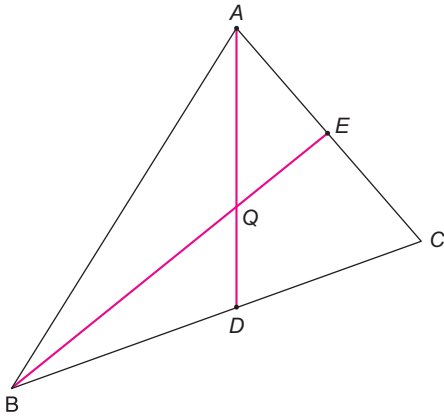


FIGURE 6.43 Single correct choice type question 84.

Now

$$\begin{aligned} \vec{AD} \times \vec{BE} &= \frac{\vec{\alpha} + \vec{\beta}}{2} \times \left[\frac{(\vec{\beta}/2) - \vec{\alpha}}{2} \right] \\ &= \frac{1}{8} (|\vec{\beta}|^2 - 2|\vec{\alpha}|^2 - \vec{\alpha} \times \vec{\beta}) \end{aligned} \quad (6.44)$$

But by hypothesis

$$\begin{aligned} |\vec{\beta}|^2 - |\vec{\alpha}|^2 + |\vec{\beta}|^2 &= 5|\vec{\alpha}|^2 \\ 2|\vec{\beta}|^2 - 2(\vec{\alpha} \times \vec{\beta}) &= 4|\vec{\alpha}|^2 \\ |\vec{\beta}|^2 - (\vec{\alpha} \times \vec{\beta}) - 2|\vec{\alpha}|^2 &= 0 \end{aligned} \quad (6.45)$$

From Eqs. (6.44) and (6.45), it follows that $\vec{AD} \times \vec{BE} = 0$. Therefore the medians AD and BE are at right angles.

Answer: (B)

- 85.** In $\triangle ABC$, $AB = BC$ and the median AD is perpendicular to the bisector of the angle C (Figure 6.44). Then $\angle ACB$ is

(A) 60°

(B) 45°

(C) $\cos^{-1} \frac{1}{3}$

(D) $\cos^{-1} \frac{1}{4}$

Solution: Take A as origin. Let $\vec{AB} = \vec{\alpha}$ and $\vec{AC} = \vec{\beta}$ so that $|\vec{\alpha}| = c$, $|\vec{\beta}| = b$ and $(\vec{\alpha}, \vec{\beta}) = A$. Let CE be the bisector of $\angle ACB$ so that $AE:EB = b:a$. Therefore

$$\vec{AD} = \frac{\vec{\alpha} + \vec{\beta}}{2}$$

and

$$\vec{AE} = \frac{b\vec{\alpha}}{a+b}$$

By hypothesis,

$$AB = BC$$

$$|\vec{AB}| = |\vec{BC}|$$

$$|\vec{\alpha}|^2 = |\vec{\alpha} - \vec{\beta}|^2$$

$$|\vec{\beta}|^2 - 2(\vec{\alpha} \times \vec{\beta}) = 0$$

$$|\vec{\beta}|^2 - 2|\vec{\alpha}||\vec{\beta}|\cos A = 0$$

$$b = 2c \cos A \quad (6.46)$$

Now

$$\vec{AD} \times \vec{CE} = 0 \quad \frac{\vec{\alpha} + \vec{\beta}}{2} \times \frac{b\vec{\alpha}}{a+b} - \vec{\beta} = 0$$

$$(\vec{\alpha} + \vec{\beta}) \times (b\vec{\alpha} - (a+b)\vec{\beta}) = 0$$

$$b|\vec{\alpha}|^2 - (a+b)(\vec{\alpha} \times \vec{\beta}) + b(\vec{\alpha} \times \vec{\beta}) - (a+b)|\vec{\beta}|^2 = 0$$

$$b|\vec{\alpha}|^2 - a(\vec{\alpha} \times \vec{\beta}) - (a+b)|\vec{\beta}|^2 = 0$$

$$bc^2 - abc \cos A - (a+b)b^2 = 0$$

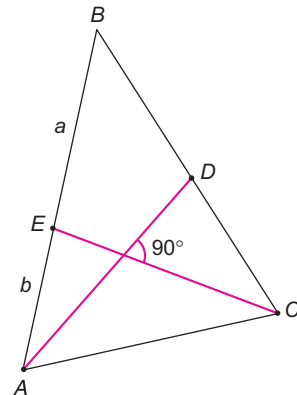


FIGURE 6.44 Single correct choice type question 85.

$$ba^2 - ba^2 \cos A - (a+b)b^2 = 0 \quad (\because AB = BC \quad c = a)$$

$$a^2 - a^2 \cos A - ab - b^2 = 0$$

$$a^2 - a^2 \cos A - a(2a \cos A) - 4a^2 \cos^2 A = 0$$

[from Eq. (6.46) and $c = a$]

$$1 - 3 \cos A - 4 \cos^2 A = 0$$

$$4 \cos^2 A + 3 \cos A - 1 = 0$$

$$(4 \cos A - 1)(\cos A + 1) = 0$$

$$\cos A = \frac{1}{4}$$

But

$$\angle A = \angle ACB \quad (\because AB = BC)$$

Therefore

$$\cos C = \frac{1}{4} \quad \text{or} \quad \angle ACB = \cos^{-1} \frac{1}{4}$$

Answer: (D)

86. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors and $\vec{r}$ be a non-zero vector such that

$$\vec{r} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$$

where x, y, z are scalars and $[\vec{a} \vec{b} \vec{c}](x + y + z) = 1$.
Then $\vec{r} \times (\vec{a} + \vec{b} + \vec{c})$ is equal to

$$(A) \ 3 \quad (B) \ 2 \quad (C) \ 1 \quad (D) \ \frac{1}{3}$$

Solution: We have

$$\vec{r} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$$

Therefore

$$\vec{r} \times \vec{a} = x[\vec{a} \vec{b} \vec{c}]$$

$$\vec{r} \times \vec{b} = y[\vec{a} \vec{b} \vec{c}]$$

$$\vec{r} \times \vec{c} = z[\vec{a} \vec{b} \vec{c}]$$

So

$$\vec{r} \times \vec{a} + \vec{r} \times \vec{b} + \vec{r} \times \vec{c} = (x + y + z)[\vec{a} \vec{b} \vec{c}] = 1$$

This implies

$$\vec{r} \times (\vec{a} + \vec{b} + \vec{c}) = 1$$

Answer: (C)

87. If $\vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular vectors of magnitudes 1, 2 and 2, respectively, then the value of $||[\vec{a} \vec{b} \vec{c}]||$ is

$$(A) \ 4 \quad (B) \ 5 \quad (C) \ \sqrt{5} \quad (D) \ 2$$

Solution: Since $\vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular vectors, the parallelepiped formed by them is a rectangular box. Therefore

$$||[\vec{a} \vec{b} \vec{c}]|| = ||\vec{a}|| ||\vec{b}|| ||\vec{c}|| = 4$$

Answer: (A)

88. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar unit vectors such that

$$\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{\sqrt{2}}(\vec{b} + \vec{c})$$

and $\vec{b}, \vec{c}$ are non-collinear vectors, then the angle between $\vec{a}$ and $\vec{b}$ is

$$(A) \ \frac{\pi}{4} \quad (B) \ \frac{3\pi}{4} \quad (C) \ \frac{\pi}{3} \quad (D) \ \frac{2\pi}{3}$$

Solution: By Theorem 6.46

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \times \vec{b}) \vec{c}$$

Therefore

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \times \vec{b}) \vec{c} = \frac{1}{\sqrt{2}}(\vec{b} + \vec{c})$$

This implies (using Theorem 5.15)

$$\vec{a} \times \vec{b} = \frac{1}{\sqrt{2}} \quad \text{and} \quad \vec{a} \times \vec{b} = \frac{1}{\sqrt{2}} \vec{c}$$

Therefore

$$\frac{1}{\sqrt{2}} = \vec{a} \times \vec{b} = ||\vec{a}|| ||\vec{b}|| \cos \theta$$

where $\theta = (\vec{a}, \vec{b})$. Hence

$$\frac{1}{\sqrt{2}} = \cos \theta \quad \theta = \frac{3\pi}{4}$$

Answer: (B)

89. If $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$ and $\vec{a}, \vec{c}$ are not collinear, then

$$(A) \ (\vec{a} \times \vec{c}) \times \vec{b} = \vec{0} \quad (B) \ (\vec{a} \times \vec{c}) \times \vec{b} = 0$$

$$(C) \ \vec{a} \times \vec{b} = \vec{a} \times \vec{c} \quad (D) \ \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{a} \times \vec{b}$$

Solution: We have

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$$

This implies

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \times \vec{b}) \times \vec{c} \quad (\text{see Theorem 6.46})$$

which further gives

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{b} \times \vec{c}) \times \vec{a}$$

But $\vec{a}$ and $\vec{c}$ are non-collinear vectors. Therefore

$$\vec{a} \times \vec{b} = 0 = \vec{b} \times \vec{c}$$

Hence $\vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{c}$ and so $\vec{b}$ is parallel to $\vec{a} \times \vec{c}$. Therefore

$$(\vec{a} \times \vec{c}) \cdot \vec{b} = 0$$

Answer: (A)

90. In $\triangle ABC$ (Figure 6.45), if $|\overline{AC}| = 3$ and $|\overline{BC}| = 4$ and the medians AD and BE are at right angles, then the area of $\triangle ABC$ is

- (A) 12 (B) $\sqrt{7}$ (C) 7 (D) $\sqrt{11}$

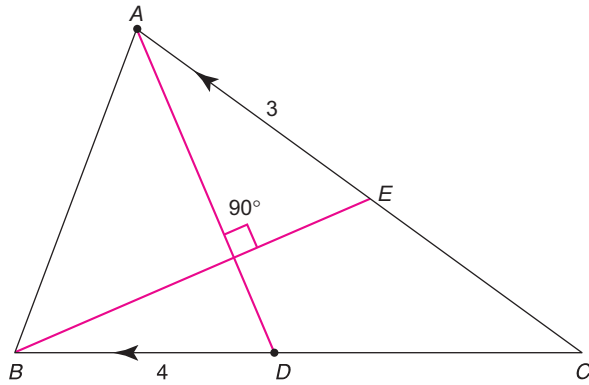


FIGURE 6.45 Single correct choice type question 90.

Solution: Take C as origin and let $\overrightarrow{CA} = \vec{a}$ and $\overrightarrow{CB} = \vec{b}$ so that $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $(\vec{a}, \vec{b}) = \angle C$. Now

$$\overrightarrow{AD} \times \overrightarrow{BE} = 0$$

$$\frac{\vec{b}}{2} \times \frac{\vec{a}}{2} = 0$$

$$(\vec{b} \times 2\vec{a}) \times (\vec{a} \times 2\vec{b}) = 0$$

$$2|\vec{a}|^2 - 2|\vec{b}|^2 + 5(\vec{a} \times \vec{b}) = 0$$

$$2(9) - 2(16) + 5(3)(4)\cos C = 0$$

$$60 \cos C = 50$$

$$\cos C = \frac{5}{6}$$

Therefore

$$\sin C = \sqrt{1 - \frac{25}{36}} = \frac{\sqrt{11}}{6}$$

Now area of $\triangle ABC$

$$= \frac{1}{2} |\overrightarrow{CA} \times \overrightarrow{CB}| = \frac{1}{2} |\vec{a}| |\vec{b}| \sin C$$

$$= \frac{1}{2} (3)(4) \frac{\sqrt{11}}{6} = \sqrt{11}$$

Answer: (D)

91. If the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar vectors, then the value of the determinant

$$\begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \times \vec{c} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \times \vec{c} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \times \vec{c} \end{vmatrix} =$$

- (A) 0 (B) 2 (C) 2 (D) 1

Solution: In Quick Look 11, part (2), take $\vec{l} = \vec{a}$, $\vec{m} = \vec{b}$ and $\vec{n} = \vec{c}$. Then

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \times \vec{c} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \times \vec{c} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \times \vec{c} \end{vmatrix}$$

But $\vec{a}$, $\vec{b}$, $\vec{c}$ are coplanar. This implies

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0$$

Direct Solution:

Let

$$\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$$

$$\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$$

$$\vec{c} = c_1 \vec{i} + c_2 \vec{j} + c_3 \vec{k}$$

Therefore

$$\begin{aligned} [\vec{a} \ \vec{b} \ \vec{c}]^2 &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \\ &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad [\because \det A = \det(A^T)] \\ &= \begin{vmatrix} a_1^2 & a_1 b_1 & a_1 c_1 \\ b_1 a_1 & b_1^2 & b_1 c_1 \\ c_1 a_1 & c_1 b_1 & c_1^2 \end{vmatrix} \\ &= \begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \times \vec{c} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \times \vec{c} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \times \vec{c} \end{vmatrix} \end{aligned}$$

Answer: (A)

92. If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors, then the value of $[\vec{b} \times \vec{c} \quad \vec{c} \times \vec{a} \quad \vec{a} \times \vec{b}]$ is
 (A) 2 (B) 1 (C) 0 (D) $|\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}|$

Solution: We have

$$\begin{aligned} [\vec{b} \times \vec{c} \quad \vec{c} \times \vec{a} \quad \vec{a} \times \vec{b}] &= ((\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})) \cdot (\vec{a} \times \vec{b}) \\ &= ([\vec{b} \vec{c} \vec{a}] \vec{c} - [\vec{c} \vec{c} \vec{a}] \vec{b}) \cdot (\vec{a} \times \vec{b}) \\ &= [\vec{b} \vec{c} \vec{a}] [\vec{c} \vec{a} \vec{b}] - 0 \\ &= [\vec{a} \vec{b} \vec{c}]^2 \quad (\because [\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]) \\ &= 0 \quad (\because \vec{a}, \vec{b}, \vec{c} \text{ are coplanar}) \end{aligned}$$

For deriving this we have used Theorem 6.48 part (2) and Theorem 6.36.

Answer: (C)

93. In the $\triangle ABC$ (Figure 6.46), points K and L are taken respectively on the segments AB and BC such that $AK:KB = 1:2$ and $BL:LC = 1:2$. Let P be the point of intersection of the lines AL and CK . If the area of the $\triangle BCP$ is 2 sq. units, then the area of the $\triangle ABC$ in sq. units is

- (A) $\frac{7}{4}$ (B) $\frac{7}{2}$ (C) 7 (D) $\sqrt{7}$

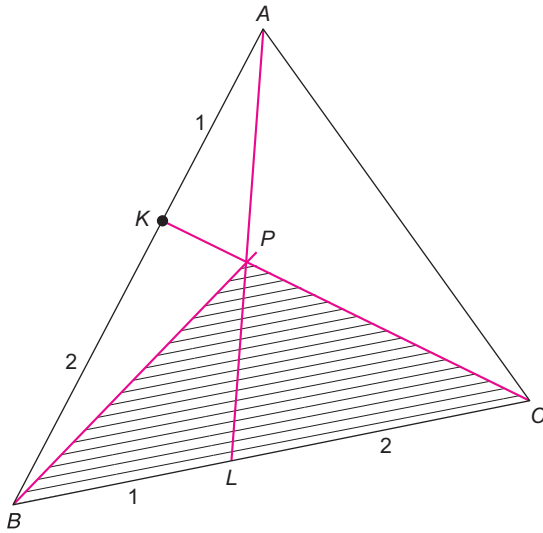


FIGURE 6.46 Single correct choice type question 93.

Solution: Take A as origin and let $\overrightarrow{AB} = \vec{\alpha}$, $\overrightarrow{AC} = \vec{\beta}$, so that the angle between $\vec{\alpha}$ and $\vec{\beta}$ is A . Note

$$AK:KB = 1:2 \Rightarrow \overrightarrow{AK} = \frac{1}{3}\vec{\alpha}$$

$$BL:LC = 1:2 \Rightarrow \overrightarrow{AL} = \frac{2\vec{\alpha} + \vec{\beta}}{3}$$

Equation of the line AL is

$$\vec{r} = t \frac{(2\vec{\alpha} + \vec{\beta})}{3}, t \in \mathbb{R} \quad (6.47)$$

Equation of the line CK is

$$\vec{r} = \frac{s}{3}\vec{\alpha} + (1-s)\vec{\beta} \quad (6.48)$$

[For Eqs. (6.47) and (6.48) see Theorem 5.30.] Equating the corresponding coefficients of $\vec{\alpha}$ and $\vec{\beta}$ in Eqs. (6.47) and (6.48) we have

$$\frac{2t}{3} = \frac{s}{3} \quad \text{and} \quad \frac{t}{3} = 1 - s$$

Therefore

$$\begin{aligned} \frac{s}{6} &= \frac{t}{3} = 1 - s \\ \Rightarrow s &= \frac{6}{7} \quad \text{and} \quad t = \frac{3}{7} \end{aligned}$$

Therefore position vector $\overrightarrow{AP}$ of P is

$$\frac{1}{7}(2\vec{\alpha} + \vec{\beta})$$

Hence

$$\overrightarrow{PB} = \vec{\alpha} - \frac{1}{7}(2\vec{\alpha} + \vec{\beta}) = \frac{1}{7}(5\vec{\alpha} - \vec{\beta})$$

and

$$\overrightarrow{PC} = \vec{\beta} - \frac{1}{7}(2\vec{\alpha} + \vec{\beta}) = \frac{1}{7}(6\vec{\beta} - 2\vec{\alpha}) = \frac{2}{7}(3\vec{\beta} - \vec{\alpha})$$

Now

$$\begin{aligned} 2 &= \text{Area of } \triangle ABC \\ &= \frac{1}{2} |\overrightarrow{PB} \times \overrightarrow{PC}| \\ &= \frac{1}{2} \times \frac{2}{49} |(5\vec{\alpha} - \vec{\beta}) \times (3\vec{\beta} - \vec{\alpha})| \\ &= \frac{1}{49} |15\vec{\alpha} \times \vec{\beta} - \vec{\alpha} \times \vec{\beta}| = \frac{2}{7} |\vec{\alpha} \times \vec{\beta}| \end{aligned}$$

Therefore $|\vec{\alpha} \times \vec{\beta}| = 7$. Now

$$\text{Area of } \triangle ABC = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2} |\vec{\alpha} \times \vec{\beta}| = \frac{7}{2}$$

Answer: (B)

94. Let $\vec{a} = 3\vec{i} - 6\vec{j} - \vec{k}$, $\vec{b} = \vec{i} + 4\vec{j} - 3\vec{k}$ and $\vec{c} = 3\vec{i} - 4\vec{j} - 12\vec{k}$ then the projection of $\vec{a} \times \vec{b}$ on $\vec{c}$ in vector form is

$$(A) \frac{14\vec{c}}{13} \quad (B) \frac{14\vec{c}}{13} \quad (C) \frac{12\vec{c}}{13} \quad (D) \frac{15\vec{c}}{13}$$

Solution: Projection vector of $\vec{\beta}$ on $\vec{\alpha}$ is [see Theorem 6.1 part (2)]

$$\frac{\vec{\beta} \cdot \vec{\alpha}}{|\vec{\alpha}|^2} \vec{\alpha}$$

Therefore vector projection of $\vec{a} - \vec{b}$ on $\vec{c}$ is

$$\begin{aligned} \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} (\vec{c}) &= \frac{1}{169} \begin{vmatrix} 3 & 6 & 1 \\ 1 & 4 & 3 \\ 3 & 4 & 12 \end{vmatrix} \vec{c} \\ &= \frac{1}{169} [3(48 - 12) + 6(12 + 9) - 1(4 - 12)] \vec{c} \\ &= \frac{1}{169} (180 - 18 + 16) \vec{c} \\ &= \frac{14\vec{c}}{13} \end{aligned}$$

Answer: (B)

95. If $|\vec{a}| = \sqrt{6}$ and $|\vec{b}| = \sqrt{5}$, then $[(\vec{a} - \vec{b}) \cdot \vec{b}] \cdot \vec{b}$ is equal to

$$(A) 5(\vec{b} - \vec{a}) \quad (B) 5(\vec{a} - \vec{b}) \\ (C) 6(\vec{a} - \vec{b}) \quad (D) 6(\vec{b} - \vec{a})$$

Solution: We have

$$\begin{aligned} [(\vec{a} - \vec{b}) \cdot \vec{b}] \cdot \vec{b} &= [(\vec{a} - \vec{b}) \cdot \vec{b}] \vec{b} = (\vec{b} \cdot \vec{b})(\vec{a} - \vec{b}) \\ &= 0(\vec{b}) - 5(\vec{a} - \vec{b}) \\ &= 5(\vec{b} - \vec{a}) \end{aligned}$$

Answer: (A)

96. Let the angle between $\vec{a}$ and $\vec{b}$ be 120° and $|\vec{a}| = 1$, $|\vec{b}| = 2$. Then

$$|(\vec{a} + 3\vec{b}) - (3\vec{a} - \vec{b})|^2 =$$

$$(A) 300 \quad (B) 225 \quad (C) 275 \quad (D) 325$$

Solution: We have

$$\begin{aligned} |(\vec{a} + 3\vec{b}) - (3\vec{a} - \vec{b})|^2 &= |(\vec{a} - \vec{b}) + 9(\vec{b} - \vec{a})|^2 \\ &= 10^2 |\vec{a} - \vec{b}|^2 \\ &= 100 |\vec{a}|^2 |\vec{b}|^2 \sin^2 120^\circ \\ &= 100 (1)(4) \left(\frac{3}{4}\right)^2 \\ &= 300 \end{aligned}$$

Answer: (A)

97. If $(\vec{a} - \vec{b}) \cdot \vec{c} = 5\vec{a} + 4\vec{b}$ and $\vec{a} \cdot \vec{b} = 3$, where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to

$$(A) 4\vec{b} - 3\vec{c} \quad (B) 3\vec{b} - 5\vec{c} \\ (C) 3\vec{c} - 4\vec{b} \quad (D) 5\vec{b} - 3\vec{c}$$

Solution: We have

$$5\vec{a} + 4\vec{b} = (\vec{a} - \vec{b}) \cdot \vec{c} = (\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{b} \times \vec{c}) \cdot \vec{a}$$

Therefore

$$\vec{a} \cdot \vec{c} = 4 \quad \text{and} \quad \vec{b} \cdot \vec{c} = 5$$

Now

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= (\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{a} \times \vec{b}) \cdot \vec{c} \\ &= 4\vec{b} - 3\vec{c} \end{aligned}$$

Answer: (A)

98. Let $\vec{a} = \vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b} = 2\vec{i} - \vec{k}$. The point of intersection of the lines represented by the equations $\vec{r} - \vec{a} = \vec{b} - \vec{a}$ and $\vec{r} - \vec{b} = \vec{a} - \vec{b}$ is

$$(A) 3\vec{i} - \vec{j} + \vec{k} \quad (B) 3\vec{i} + \vec{j} + \vec{k} \\ (C) \vec{i} - \vec{j} - \vec{k} \quad (D) \vec{i} + \vec{j} + \vec{k}$$

Solution: We have

$$\begin{aligned} \vec{r} - \vec{a} &= \vec{b} - \vec{a} \\ (\vec{r} - \vec{b}) - \vec{a} &= \vec{b} - \vec{a} \\ \vec{r} - \vec{b} &\text{ is collinear with } \vec{a} \\ \vec{r} - \vec{b} &= t\vec{a}, t \in \mathbb{R} \\ \vec{r} &= \vec{b} + t\vec{a} \end{aligned} \tag{6.49}$$

Similarly

$$\vec{r} - \vec{b} = \vec{a} - \vec{b} \quad \vec{r} = \vec{a} + s\vec{b} \tag{6.50}$$

where $s \in \mathbb{R}$. The two lines given by Eqs. (6.49) and (6.50) intersect if $t = s = 1$. Therefore the point of intersection is

$$\vec{a} + \vec{b} = 3\vec{i} + \vec{j} + \vec{k}$$

Answer: (B)

99. $\vec{a} \cdot [(\vec{b} + \vec{c}) - (\vec{a} + \vec{b} + \vec{c})]$ is equal to

$$(A) 0 \quad (B) 2[\vec{a} \cdot \vec{b} \cdot \vec{c}] \\ (C) [\vec{a} \cdot \vec{b} \cdot \vec{c}] \quad (D) [\vec{a} \cdot \vec{b} \cdot \vec{c}]$$

Solution: We have

$$\begin{aligned} \vec{a} \cdot [(\vec{b} + \vec{c}) - (\vec{a} + \vec{b} + \vec{c})] &= \vec{a} \cdot [(\vec{b} + \vec{c}) - \vec{a}] \\ &= \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{c} \\ &= [\vec{a} \cdot \vec{b} \cdot \vec{a}] + [\vec{a} \cdot \vec{c} \cdot \vec{a}] \\ &= 0 + 0 = 0 \end{aligned}$$

Answer: (A)

100. $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors. Then

$$(\vec{a} + \vec{b} + \vec{c}) \times [(\vec{a} + \vec{b}) \times (\vec{a} + \vec{c})] =$$

- (A) 0 (B) $[\vec{a} \vec{b} \vec{c}]$
(C) $2[\vec{a} \vec{b} \vec{c}]$ (D) $[\vec{a} \vec{b} \vec{c}]$

Solution: Given scalar is

$$(\vec{a} + \vec{b} + \vec{c}) \times [\vec{c} \times \vec{a} - \vec{a} \times \vec{b} + \vec{b} \times \vec{c}] = [\vec{a} \vec{b} \vec{c}] - [\vec{b} \vec{c} \vec{a}] + [\vec{c} \vec{a} \vec{b}]$$

$$= [\vec{a} \vec{b} \vec{c}]$$

Answer: (D)

101. Let $\vec{a} = x\vec{i} + y\vec{j} + z\vec{k}$ be a vector of magnitude $2\sqrt{3}$ and makes an obtuse angle with y -axis. If $\vec{a}$ makes equal angles with the vectors $\vec{b} = y\vec{i} - 2z\vec{j} + 3x\vec{k}$ and $\vec{c} = 2z\vec{i} + 3x\vec{j} - y\vec{k}$ and is perpendicular to $\vec{d} = \vec{i} + \vec{j} + 2\vec{k}$, then $\vec{a}$ is equal to

- (A) $2\vec{i} - 2\vec{j} - 2\vec{k}$ (B) $2\vec{i} + 2\vec{j} - 2\vec{k}$
(C) $2\vec{i} - 2\vec{j} + 2\vec{k}$ (D) $2\vec{i} + 2\vec{j} + 2\vec{k}$

Solution: We have $|\vec{a}| = 2\sqrt{3}$. Therefore

$$x^2 + y^2 + z^2 = 12 \quad (6.51)$$

Since $|\vec{b}| = |\vec{c}|$ and $\vec{a}$ makes equal angles with $\vec{b}$ and $\vec{c}$, we have

$$\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$$

Therefore

$$xy - 2yz + 3zx = 2zx + 3xy - yz$$

$$2xy + yz - zx = 0 \quad (6.52)$$

Also

$$\vec{a} \times \vec{d} = 0 \quad x - y + 2z = 0 \quad (6.53)$$

Also $\vec{a}$ makes an obtuse angle with y -axis. This implies

$$y < 0 \quad (6.54)$$

From Eqs. (6.52) and (6.53) we have

$$2x(x + 2z) + (x + 2z)z - zx = 0$$

$$2x^2 + 4zx + 2z^2 = 0$$

$$(z + x)^2 = 0$$

$$z = -x$$

From Eq. (6.53) we have $y = -x$. Therefore, from Eq. (6.51)

$$x^2 + x^2 + x^2 = 12 \quad x^2 = 4$$

$$x = \pm 2$$

Since from Eq. (6.54) $y < 0$, we have $x = 2$, $y = -2$ and $z = -2$. Therefore

$$\vec{a} = 2\vec{i} - 2\vec{j} - 2\vec{k}$$

Answer: (A)

102. If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of equal magnitudes and $\vec{x}$ is a vector satisfying the equation

$$\vec{a} [(\vec{x} \times \vec{b})\vec{a}] + \vec{b} [(\vec{x} \times \vec{c})\vec{b}] + \vec{c} [(\vec{x} \times \vec{a})\vec{c}] = \vec{0}$$

then $\vec{x}$ equals

- (A) $\frac{1}{2}(\vec{a} + \vec{b} - 2\vec{c})$ (B) $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$
(C) $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$ (D) $\frac{1}{3}(2\vec{a} + \vec{b} - \vec{c})$

Solution: We have

$$\vec{a} [(\vec{x} \times \vec{b})\vec{a}] = (\vec{a} \times \vec{x})(\vec{x} \times \vec{b}) - \vec{a} \times (\vec{x} \times \vec{b})\vec{a}$$

$$= |\vec{a}|^2 (\vec{x} \times \vec{b}) - (\vec{a} \times \vec{x} \times \vec{a})\vec{b} \quad (6.55a)$$

Similarly

$$\vec{b} [(\vec{x} \times \vec{c})\vec{b}] = |\vec{b}|^2 (\vec{x} \times \vec{c}) - (\vec{b} \times \vec{x} \times \vec{b})\vec{c} \quad (6.54b)$$

$$\vec{c} [(\vec{x} \times \vec{a})\vec{c}] = |\vec{c}|^2 (\vec{x} \times \vec{a}) - (\vec{c} \times \vec{x} \times \vec{c})\vec{a} \quad (6.55c)$$

Let $|\vec{a}| = |\vec{b}| = |\vec{c}| = \lambda$. Since $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = 0$, by adding Eqs. (6.55a), (6.55b) and (6.55c), we have

$$3\lambda^2 \vec{x} - \lambda^2(\vec{a} + \vec{b} + \vec{c}) - [(\vec{a} \times \vec{x})\vec{a} + (\vec{b} \times \vec{x})\vec{b} + (\vec{c} \times \vec{x})\vec{c}] = 0 \quad (6.56)$$

Suppose

$$\vec{x} = a_1\vec{a} + b_1\vec{b} + c_1\vec{c}$$

(This is possible, since $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar.) Therefore

$$\vec{a} \times \vec{x} = a_1 |\vec{a}|^2 = \lambda^2 a_1$$

$$\vec{b} \times \vec{x} = \lambda^2 b_1$$

$$\vec{c} \times \vec{x} = \lambda^2 c_1$$

Substituting these values in Eq. (6.56) we have

$$3\lambda^2 \vec{x} - \lambda^2(\vec{a} + \vec{b} + \vec{c}) - \lambda^2 \vec{x} = 0$$

$$2\vec{x} = \vec{a} + \vec{b} + \vec{c}$$

$$\vec{x} = \frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$$

Note: Since $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of same magnitude, one can take $\vec{a} = p\vec{i}, \vec{b} = p\vec{j}$ and $\vec{c} = p\vec{k}$ and proceed.

Answer: (B)

103. Let the vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$. Let P_1 and P_2 be the planes determined by the pairs of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively. Then the angle between P_1 and P_2 is

- (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Solution: $(\vec{a} \ \vec{b}) \ (\vec{c} \ \vec{d}) = 0$ $\vec{a} \ \vec{b}$ and $\vec{c} \ \vec{d}$ are parallel. But $\vec{a} \ \vec{b}$ and $\vec{c} \ \vec{d}$ are normals to the planes P_1 and P_2 , respectively. Also, since $(\vec{a}, \vec{b}, \vec{a} \ \vec{b})$ and $(\vec{c}, \vec{d}, \vec{c} \ \vec{d})$ are right-handed systems, it follows that $\vec{a} \ \vec{b}$ and $\vec{c} \ \vec{d}$ are like vectors. Since the angle between P_1 and P_2 is the angle between their normals (see Definition 6.3) it follows that the angle between P_1 and P_2 is 0.

Answer: (A)

104. A plane P_1 is parallel to the vectors $\vec{a} = 2\vec{j} + 3\vec{k}$ and $\vec{b} = 4\vec{j} - 3\vec{k}$ and a plane P_2 is parallel to the vectors $\vec{c} = \vec{j} - \vec{k}$ and $\vec{d} = 3\vec{i} + 3\vec{k}$. Vector $\vec{A}$ is parallel to the line of intersection of P_1 and P_2 . Then the angle θ between $\vec{A}$ and the vector $2\vec{i} + \vec{j} - 2\vec{k}$, $0 < \theta < \pi/2$ is

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

Solution: Normal to $P_1 = \vec{a} \ \vec{b} = 18\vec{i}$ and normal to $P_2 = \vec{c} \ \vec{d} = 3\vec{i} - 3\vec{j} - 3\vec{k}$. By hypothesis, $\vec{A}$ is perpendicular to both $\vec{a} \ \vec{b}$ and $\vec{c} \ \vec{d}$ and hence we can take

$$\vec{A} = (\vec{a} \ \vec{b}) \ (\vec{c} \ \vec{d}) = 54(\vec{j} + \vec{k})$$

Therefore

$$\cos \theta = \frac{54(\vec{j} + \vec{k}) \cdot (2\vec{i} + \vec{j} - 2\vec{k})}{54\sqrt{2} \cdot 3} = \frac{54 \cdot 3}{54\sqrt{2} \cdot 3} = \frac{1}{\sqrt{2}}$$

and so

$$\theta = \frac{\pi}{4}$$

Note: If $\vec{A} = (\vec{c} \ \vec{d}) \ (\vec{a} \ \vec{b})$, then $\theta = \frac{3\pi}{4}$.

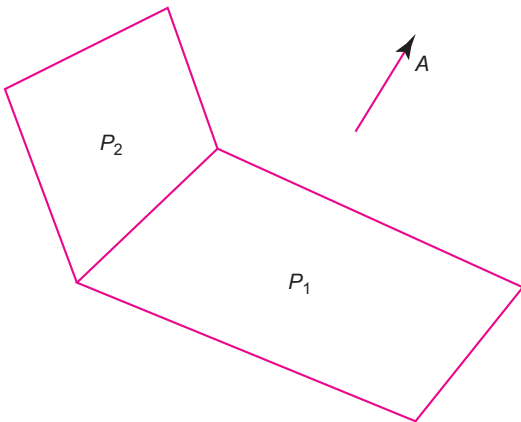


FIGURE 6.47 Single correct choice type question 104.

Answer: (B)

105. In $\triangle ABC$ (Figure 6.48), D is the mid-point of AC and E is a point on the side BC such that $BE:EC = 2:1$. If AE and BD intersect in P , then the ratio $AP:PE$ is

- (A) 2:1 (B) 2:3 (C) 3:2 (D) 1:2

Solution: Take A as origin, let $\vec{AB} = \vec{b}$ and $\vec{AC} = \vec{c}$. Therefore

$$\vec{AD} = \frac{1}{2}\vec{c}$$

and

$$\vec{AE} = \frac{\vec{b} + 2\vec{c}}{3}$$

Suppose the ratio $AP:PE = \lambda:1$ and the ratio $BP:PD = \mu:1$. Then

$$\frac{\vec{b} + \mu(\vec{c}/2)}{\mu + 1} = \frac{\lambda(\vec{b} + 2\vec{c})}{3(\lambda + 1)}$$

$\vec{b}$ and $\vec{c}$ are non-collinear vectors. This implies

$$\frac{1}{\mu + 1} = \frac{\lambda}{3(\lambda + 1)}$$

and

$$\frac{\mu}{2(\mu + 1)} = \frac{2\lambda}{3(\lambda + 1)}$$

Solving these we get that

$$\mu = 4 \quad \text{and} \quad \lambda = \frac{3}{2}$$

Therefore

$$AP:PE = 3:2$$

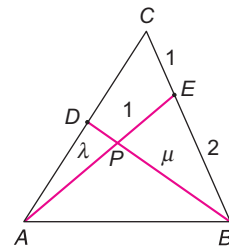


FIGURE 6.48 Single correct choice type question 105.

Answer: (C)

106. If $\vec{a}, \vec{b}, \vec{c}$ are unit coplanar vectors, then the scalar triple product $[2\vec{a} \ \vec{b} \ 2\vec{c}] \ [\vec{c} \ 2\vec{c} \ \vec{a}]$ equals

- (A) $\sqrt{2}$ (B) 1 (C) $\sqrt{3}$ (D) 0

Solution: The given scalar triple product is

$$\begin{aligned} &= (2\vec{a} \ \vec{b}) \cdot (2\vec{b} \ \vec{c}) \ (\vec{c} \ \vec{a}) \\ &= (2\vec{a} \ \vec{b}) \cdot [4(\vec{b} \ \vec{c}) + 2(\vec{a} \ \vec{b}) + (\vec{c} \ \vec{a})] \\ &= 8[\vec{a} \ \vec{b} \ \vec{c}] \ [\vec{b} \ \vec{c} \ \vec{a}] \\ &= 7[\vec{a} \ \vec{b} \ \vec{c}] \\ &= 7(0) = 0 \end{aligned}$$

Answer: (D)

107. Let $\vec{a} = 2\vec{i} + 3\vec{j} + 4\vec{k}$, $\vec{b} = \vec{i} + 5\vec{j} + 2\vec{k}$ and $\vec{c} = 3\vec{i} + 15\vec{j} + 6\vec{k}$. Then

$$\begin{vmatrix} \vec{a} \times \vec{b} & \vec{a} \times \vec{c} & \vec{b} \times \vec{c} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{c} & \vec{c} \times \vec{a} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{a} \times \vec{b} \end{vmatrix} =$$

- (A) 0 (B) 8 (C) 4 (D) 6

Solution: Under Quick Look 11, part (2), put $\vec{l} = \vec{a}$, $\vec{m} = \vec{b}$ and $\vec{n} = \vec{c}$. Then the given determinant value is $[\vec{a} \vec{b} \vec{c}]^2$. But by hypothesis $\vec{c} = 3\vec{b}$, so that $[\vec{a} \vec{b} \vec{c}]$ is zero.

Answer: (A)

108. Let A, B, C and D be any four points in the space. Then the minimum value of $(AC)^2 + (BD)^2 + (AD)^2 + (BC)^2$ is

- (A) $(AD)^2 + (BC)^2$
(B) $(AB)^2 + (CD)^2$
(C) $(AC)^2 + (BD)^2$
(D) no minimum value

Solution: Take A as origin and let $\vec{AB} = \vec{b}$, $\vec{AC} = \vec{c}$ and $\vec{AD} = \vec{d}$. Then

$$\begin{aligned} & (AC)^2 + (BD)^2 + (AD)^2 + (BC)^2 \\ &= |\vec{c}|^2 + |\vec{d} - \vec{b}|^2 + |\vec{d}|^2 + |\vec{c} - \vec{b}|^2 \\ &= |\vec{c}|^2 + |\vec{b}|^2 + |\vec{d}|^2 - 2(\vec{b} \times \vec{d}) + |\vec{d}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - 2(\vec{b} \times \vec{c}) \\ &= |\vec{b} - \vec{c}|^2 + |\vec{d}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + |\vec{d}|^2 - 2(\vec{c} \times \vec{d}) \\ &= |\vec{b} - \vec{c}|^2 + |\vec{d}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + |\vec{d}|^2 - 2(\vec{b} \times \vec{c}) \end{aligned}$$

where the equality occurs if and only if $\vec{b} = \vec{c} + \vec{d}$ which is equivalent to $\vec{b} - \vec{c} = \vec{d}$ or $\vec{CB} = \vec{AD}$. Hence the minimum value is

$$|\vec{b}|^2 + |\vec{c} - \vec{d}|^2 = (AB)^2 + (CD)^2$$

Answer: (B)

109. Let $\vec{u}, \vec{v}$ and $\vec{w}$ be non-coplanar unit vectors and the angles $(\vec{u}, \vec{v}) = \alpha$, $(\vec{v}, \vec{w}) = \beta$ and $(\vec{w}, \vec{u}) = \gamma$. Suppose $\vec{a}, \vec{b}$ and $\vec{c}$ be the unit vectors along the bisectors of α, β and γ respectively. Then the value of the scalar triple product

$$[\vec{a} \vec{b} \vec{c}] = \frac{1}{K} [\vec{u} \vec{v} \vec{w}]^2 \sec^2 \frac{\alpha}{2} \sec^2 \frac{\beta}{2} \sec^2 \frac{\gamma}{2}$$

where the value of K is

- (A) 2 (B) 4 (C) 8 (D) 16

Solution: As per Theorem 5.33

$$\vec{a} = \frac{\vec{u} + \vec{v}}{|\vec{u} + \vec{v}|}$$

$$\vec{b} = \frac{\vec{v} + \vec{w}}{|\vec{v} + \vec{w}|}$$

$$\vec{c} = \frac{\vec{w} + \vec{u}}{|\vec{w} + \vec{u}|}$$

Now

$$\begin{aligned} [\vec{a} \vec{b} \vec{c}] &= (\vec{a} \times \vec{b}) \cdot (\vec{c}) \\ &= (\vec{a} \times \vec{b}) \cdot \left[\frac{\vec{v} + \vec{w}}{|\vec{v} + \vec{w}|} \right] \\ &= \frac{1}{|\vec{v} + \vec{w}|} (\vec{a} \times \vec{b}) \cdot (\vec{v} + \vec{w}) \\ &= \frac{1}{|\vec{v} + \vec{w}|} [\vec{a} \vec{b} \vec{v} + \vec{a} \vec{b} \vec{w}] \end{aligned}$$

Substituting the values of $\vec{a}, \vec{b}$ and $\vec{c}$ we have

$$\begin{aligned} [\vec{a} \vec{b} \vec{c}] &= \frac{1}{|\vec{u} + \vec{v}| |\vec{v} + \vec{w}| |\vec{w} + \vec{u}|} [\vec{u} + \vec{v} \cdot \vec{v} + \vec{w} \cdot \vec{w} + \vec{w} + \vec{u}] \\ &= \frac{1}{4 \cos^2 \frac{\alpha}{2} \cdot 4 \cos^2 \frac{\beta}{2} \cdot 4 \cos^2 \frac{\gamma}{2}} (2[\vec{u} \vec{v} \vec{w}])^2 \\ &= \frac{1}{16 \cos^2 \frac{\alpha}{2} \cos^2 \frac{\beta}{2} \cos^2 \frac{\gamma}{2}} [\vec{u} \vec{v} \vec{w}]^2 \end{aligned}$$

Answer: (D)

110. In a Quadrilateral $ABCD$, if $(AB)^2 + (CD)^2 = (BC)^2 + (AD)^2$ then the angle between the diagonals AC and BD is

- (A) 120° (B) 90° (C) 60° (D) 45°

Solution: Take A as origin and let $\vec{AB} = \vec{b}$, $\vec{AC} = \vec{c}$ and $\vec{AD} = \vec{d}$. Now

$$(AB)^2 + (CD)^2 = (BC)^2 + (AD)^2$$

$$|\vec{b}|^2 + |\vec{d} - \vec{c}|^2 = |\vec{c} - \vec{b}|^2 + |\vec{d}|^2$$

$$\vec{c} \times \vec{d} = \vec{c} \times \vec{b} \quad (6.57)$$

Again

$$\begin{aligned} \vec{AC} \times \vec{BD} &= \vec{c} \times (\vec{d} - \vec{b}) \\ &= \vec{c} \times \vec{d} - \vec{c} \times \vec{b} \\ &= 0 \quad [\text{by Eq. (6.57)}] \end{aligned}$$

Therefore AC and BD are at right angles.

Answer: (B)

Multiple Correct Choice Type Questions

1. Let $\vec{a} = 2\vec{i} + \vec{j}$, $\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$ and $\vec{c} = 3\vec{i} - \vec{j} + \vec{k}$.

Then

(A) the volume of the parallelepiped whose coterminus edges are $\vec{a}, \vec{b}, \vec{c}$ is 5 cubic units.

(B) $|\vec{a} \cdot (\vec{b} \times \vec{c})| = 5\sqrt{5}$

(C) area of the face with $\vec{a}, \vec{b}$ as adjacent sides is $\sqrt{83}$

(D) $[\vec{a} \cdot \vec{b} \cdot \vec{c}] = 25$

Solution:

(A) Volume of the parallelepiped $= |[\vec{a} \vec{b} \vec{c}]|$ where

$$\begin{aligned} [\vec{a} \vec{b} \vec{c}] &= \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix} \\ &= 2(2+3) - 1(1+9) + 1(-1+6) \\ &= 5 \end{aligned}$$

Therefore

$$|[\vec{a} \vec{b} \vec{c}]| = 5$$

Hence (A) is correct.

(B) We have

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= (\vec{a} \times \vec{c}) \cdot \vec{b} = (\vec{a} \times \vec{b}) \cdot \vec{c} \\ &= \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix} = 5 \\ &= 8(\vec{i} + 2\vec{j} + 3\vec{k}) \cdot (3\vec{i} - \vec{j} + \vec{k}) \\ &= 5\vec{i} \cdot 15\vec{j} \cdot 25\vec{k} \\ &= 5(\vec{i} + 3\vec{j} + 5\vec{k}) \end{aligned}$$

Therefore

$$\begin{aligned} |\vec{a} \cdot (\vec{b} \times \vec{c})| &= 5\sqrt{1+9+25} \\ &= 5\sqrt{35} \end{aligned}$$

Hence (B) is not correct.

(C) Area of the parallelogram with $\vec{a}$ and $\vec{b}$ as adjacent sides $= |\vec{a} \times \vec{b}|$. Now

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 5\vec{i} - 7\vec{j} + 3\vec{k}$$

Therefore

$$|\vec{a} \times \vec{b}| = \sqrt{25+49+9} = \sqrt{83}$$

Hence (C) is correct.

(D) We have

$$[\vec{a} \cdot \vec{b} \cdot \vec{c}] = [\vec{a} \vec{b} \vec{c}] = 5 = 25$$

So (D) is correct.

Answers: (A), (C), (D)

2. Let λ be a real scalar and

$$\vec{a} = \lambda\vec{i} + 2\lambda\vec{j} - 3\lambda\vec{k}$$

$$\vec{b} = (2\lambda + 1)\vec{i} + (2\lambda + 3)\vec{j} + (\lambda + 1)\vec{k}$$

$$\vec{c} = (3\lambda + 5)\vec{i} + (\lambda + 5)\vec{j} + (\lambda + 2)\vec{k}$$

Then $\vec{a}, \vec{b}, \vec{c}$ are coplanar

(A) for exactly one real values of λ

(B) for exactly two real values of λ

(C) for exactly three real values of λ

(D) for no non-zero real value of λ

Solution: The vectors are coplanar if and only if

$$\begin{vmatrix} \lambda & 2\lambda & 3\lambda \\ 2\lambda + 1 & 2\lambda + 3 & \lambda + 1 \\ 3\lambda + 5 & \lambda + 5 & \lambda + 2 \end{vmatrix} = 0$$

The operations $C_2 - 2C_1$ and $C_3 - 3C_1$ imply

$$\begin{aligned} &= \begin{vmatrix} \lambda & 0 & 0 \\ 2\lambda + 1 & 2\lambda + 1 & 7\lambda + 4 \\ 3\lambda + 5 & 5\lambda & 10\lambda + 17 \end{vmatrix} \\ &= \lambda[(2\lambda + 1)(10\lambda + 17) + 5(\lambda + 1)(7\lambda + 4)] \\ &= \lambda[20\lambda^2 + 24\lambda + 17 + 35\lambda^2 + 55\lambda + 20] \\ &= \lambda(15\lambda^2 + 31\lambda + 37) \end{aligned}$$

Clearly $\lambda = 0$ is a root and the vectors are coplanar. In this case $\vec{a} = \vec{0}$ which is coplanar with any two vectors. Now

$$15\lambda^2 + 31\lambda + 37 = 0$$

for all $\lambda \neq 0$ because its discriminant is negative. Therefore (A) and (D) are correct.

Answers: (A), (D)

3. Let $\vec{a} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{b}$ is a vector such that $\vec{a} \times \vec{b} = \vec{i}$ and $\vec{a} \cdot \vec{b} = \vec{j} \cdot \vec{k}$. Then

(A) $\vec{b} = 2\vec{i}$

(B) $\vec{b} = \vec{i}$

(C) $(\vec{a} \times \vec{b}) \cdot \vec{b} = (\vec{j} + \vec{k})$

(D) $(\vec{a} \times \vec{b}) \cdot \vec{b} = \vec{k} - \vec{j}$

Solution: We have

$$\begin{aligned}\vec{j} - \vec{k} &= \vec{a} \times \vec{b} \Rightarrow \vec{a} \times (\vec{j} - \vec{k}) = \vec{a} \times (\vec{a} \times \vec{b}) \\ &\Rightarrow (\vec{i} + \vec{j} + \vec{k}) \times (\vec{j} - \vec{k}) = (\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} \\ &\Rightarrow -2\vec{i} + \vec{j} + \vec{k} = \vec{a} - 3\vec{b} \\ &\Rightarrow \vec{b} = \vec{i}\end{aligned}$$

Hence (B) is correct.

Also

$$\begin{aligned}(\vec{a} \times \vec{b}) \times \vec{b} &= (\vec{a} \times \vec{i}) \times \vec{i} \\ &= (\vec{a} \cdot \vec{i})\vec{i} - (\vec{i} \cdot \vec{i})\vec{a} \\ &= \vec{i} - \vec{a} = -\vec{j} - \vec{k}\end{aligned}$$

So (C) is correct.

Answers: (B), (C)

4. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the position vectors of three non-collinear points A , B and C , respectively, and $\vec{p} = \vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}$. Then

(A) $\vec{p}$ is perpendicular to the plane of $\triangle ABC$

(B) $|\vec{p}|$ is twice the area of $\triangle ABC$

(C) $[\vec{b} \times \vec{c} \quad \vec{c} \times \vec{a} \quad \vec{a} \times \vec{b}] = 2[\vec{a} \quad \vec{b} \quad \vec{c}]$

(D) $\vec{a} \cdot \vec{p} + \vec{b} \cdot \vec{p} + \vec{c} \cdot \vec{p} = 3[\vec{a} \quad \vec{b} \quad \vec{c}]$

Solution: We have $\vec{AB} = \vec{b} - \vec{a}$, $\vec{AC} = \vec{c} - \vec{a}$. Then

$$\begin{aligned}\vec{AB} \times \vec{AC} &= (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) \\ &= \vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} \\ &= \vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}\end{aligned}$$

But $\vec{AB} \times \vec{AC}$ is perpendicular to the plane of $\triangle ABC$. Therefore $\vec{p}$ is perpendicular the plane of $\triangle ABC$. Hence (A) is correct.

Also $(1/2)|\vec{AB} \times \vec{AC}|$ is the area of $\triangle ABC$. So (B) is correct. Again

$$[\vec{b} \times \vec{c} \quad \vec{c} \times \vec{a} \quad \vec{a} \times \vec{b}] = [\vec{a} \quad \vec{b} \quad \vec{c}]^2$$

This implies (C) is not correct. Finally

$$\vec{a} \cdot \vec{p} + \vec{b} \cdot \vec{p} + \vec{c} \cdot \vec{p} = [\vec{a} \quad \vec{b} \quad \vec{c}] + [\vec{b} \quad \vec{c} \quad \vec{a}] + [\vec{c} \quad \vec{a} \quad \vec{b}] = 3[\vec{a} \quad \vec{b} \quad \vec{c}]$$

This means (D) is correct.

Answers: (A), (B), (D)

5. Let $\vec{a}$ and $\vec{b}$ be unit vectors and θ be the angle between $\vec{a}$ and $\vec{b}$. Let $\vec{x} = \vec{a} + \vec{b}$ and $\vec{y} = \vec{a} - \vec{b}$. Then

(A) $\vec{x} \cdot \vec{y} = 1$

(B) $\vec{x} \cdot \vec{y} = 0$

(C) $|\vec{x} \times \vec{y}| = 2 \sin \theta$

(D) $|\vec{x} \times \vec{y}|$, two times the area of the parallelogram with $\vec{a}$ and $\vec{b}$ as adjacent sides

Solution: We have

$$\begin{aligned}\vec{x} \cdot \vec{y} &= (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) \\ &= |\vec{a}|^2 - |\vec{b}|^2 \\ &= 1 - 1 = 0\end{aligned}$$

So (B) is correct. Also

$$\begin{aligned}\vec{x} \times \vec{y} &= (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) \\ &= -2(\vec{a} \times \vec{b})\end{aligned}$$

Therefore

$$\begin{aligned}|\vec{x} \times \vec{y}| &= 2|\vec{a} \times \vec{b}| \\ &= 2|\vec{a}||\vec{b}|\sin \theta \\ &= 2 \sin \theta\end{aligned}$$

So (C) is correct. Also

$$|\vec{x} \times \vec{y}| = 2|\vec{a} \times \vec{b}|$$

This implies (D) is correct.

Answers: (B), (C), (D)

6. $ABCD$ is a regular tetrahedron. P and Q are the mid-points of the edges AC and AB , respectively, G is the centroid of the face BCD and θ is the angle between the vectors $\vec{PG}$ and $\vec{DG}$. Then

(A) the angle between $\vec{AB}$ and $\vec{CD}$ is 90°

(B) the angle θ is $\pi - \cos^{-1}\left(\frac{5}{6\sqrt{3}}\right)$

(C) θ is $\cos^{-1}\left(\frac{5}{6\sqrt{3}}\right)$

(D) angle between $\vec{AB}$ and $\vec{CD}$ is 120°

Solution: Take D as the origin and let $\vec{DA} = \vec{a}$, $\vec{DB} = \vec{b}$ and $\vec{DC} = \vec{c}$. See Figure 6.49. Then

$$\begin{aligned}\vec{DP} &= \frac{(\vec{a} + \vec{c})}{2} \\ \vec{DQ} &= \frac{\vec{a} + \vec{b}}{2} \\ \vec{DG} &= \frac{\vec{b} + \vec{c}}{3}\end{aligned}$$

Since $ABCD$ is regular all the faces are equilateral triangles with equal edges and hence the angles of each

triangular face are 60° . In a tetrahedron $ABCD$, the pairs of edges (AD, BC) , (CD, AB) and (BD, AC) are called opposite pairs of edges. Now

$$\begin{aligned}\overrightarrow{AD} \cdot \overrightarrow{BC} &= (-\vec{a}) \cdot (\vec{c} - \vec{b}) = \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} \\ &= |\vec{a}| |\vec{b}| \cos 60^\circ \\ &= -|\vec{a}| |\vec{c}| \cos 60^\circ = 0\end{aligned}$$

because $|\vec{a}| = |\vec{b}| = |\vec{c}|$.

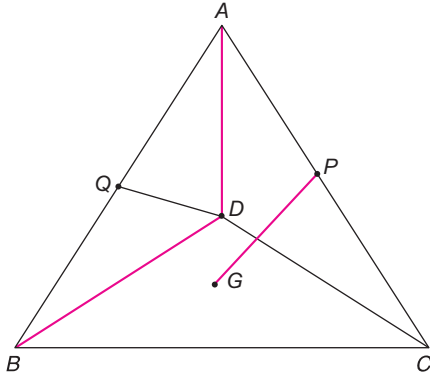


FIGURE 6.49 Multiple correct choice type question 6.

Therefore AD and BC are at right angles. Hence opposite pairs of edges are at right angles. (A) is correct. Now

$$\begin{aligned}\overrightarrow{PG} &= \frac{1}{3}(\vec{b} + \vec{c}) - \frac{1}{2}(\vec{a} + \vec{c}) \\ &= \frac{1}{6}(2\vec{b} - \vec{c} - 3\vec{a}) = -\frac{1}{6}(3\vec{a} - 2\vec{b} + \vec{c})\end{aligned}$$

$$\overrightarrow{DQ} = \frac{1}{2}(\vec{a} + \vec{b})$$

$$\begin{aligned}36|\overrightarrow{PG}|^2 &= 9|\vec{a}|^2 + 4|\vec{b}|^2 + |\vec{c}|^2 - 12(\vec{a} \cdot \vec{b}) - 4(\vec{b} \cdot \vec{c}) + 6(\vec{c} \cdot \vec{a}) \\ &= 9|\vec{a}|^2 + 4|\vec{b}|^2 + |\vec{c}|^2 - 12|\vec{a}| |\vec{b}| \cos 60^\circ \\ &\quad - 4|\vec{b}| |\vec{c}| \cos 60^\circ + 6|\vec{c}| |\vec{a}| \cos 60^\circ\end{aligned}$$

It is known that $|\vec{a}| = |\vec{b}| = |\vec{c}| = K$ (suppose). Therefore

$$36|\overrightarrow{PG}|^2 = K^2(14 - 6 - 2 + 3) = 9K^2 \quad (6.58)$$

Again

$$\begin{aligned}4|\overrightarrow{DQ}|^2 &= |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}| |\vec{b}| \cos 60^\circ \\ &= 3K^2\end{aligned} \quad (6.59)$$

From Eqs. (6.58) and (6.59) we have

$$|\overrightarrow{PG}| = \frac{K}{2} \quad \text{and} \quad |\overrightarrow{DQ}| = \frac{\sqrt{3}K}{2} \quad (6.60)$$

Now

$$\overrightarrow{PG} \cdot \overrightarrow{DQ} = -\frac{1}{6}(3\vec{a} - 2\vec{b} + \vec{c}) \cdot \frac{1}{2}(\vec{a} + \vec{b})$$

$$\begin{aligned}&= -\frac{1}{12} \left[3K^2 + \frac{3}{2}K^2 - K^2 - 2K^2 + \frac{K^2}{2} + \frac{K^2}{2} \right] \\ &= -\frac{5K^2}{24}\end{aligned} \quad (6.61)$$

Therefore, using Eqs. (6.60) and (6.61) we have

$$\begin{aligned}\cos \theta &= \frac{\overrightarrow{PG} \cdot \overrightarrow{DQ}}{|\overrightarrow{PG}| |\overrightarrow{DQ}|} = -\frac{5K^2}{24} \times \frac{4}{\sqrt{3}K^2} \\ &= -\frac{5}{6\sqrt{3}}\end{aligned}$$

Hence

$$\begin{aligned}\theta &= \cos^{-1} \left(-\frac{5}{6\sqrt{3}} \right) \\ &= \pi - \cos^{-1} \left(\frac{5}{6\sqrt{3}} \right)\end{aligned}$$

(B) is correct.

Answers: (A), (B)

7. A, B, C and D are fixed points in space or in plane and M is any arbitrary point in the space. Which of the following true?

- (A) $\overrightarrow{MA} \cdot \overrightarrow{MC} = \overrightarrow{MB} \cdot \overrightarrow{MD}$ if $ABCD$ is a rectangle.
- (B) $|\overrightarrow{MA}|^2 + |\overrightarrow{MC}|^2 = |\overrightarrow{MB}|^2 + |\overrightarrow{MD}|^2$ if $ABCD$ is rectangle.
- (C) If $\overrightarrow{AM} \cdot \overrightarrow{CM} \neq \overrightarrow{BM} \cdot \overrightarrow{DM}$ for all positions of M , then $ABCD$ is a parallelogram without being a rectangle.
- (D) If $\overrightarrow{AM} \cdot \overrightarrow{CM} \neq \overrightarrow{BM} \cdot \overrightarrow{DM}$ for all positions of M , then $ABCD$ is a trapezium which is not a parallelogram.

Solution: Take A as origin and let $\overrightarrow{AB} = \vec{b}$, $\overrightarrow{AD} = \vec{d}$ and $\overrightarrow{AM} = \vec{m}$. See Figure 6.50. Suppose $ABCD$ is a rectangle so that $\vec{b} \cdot \vec{d} = 0$ and $\overrightarrow{AC} = \vec{b} + \vec{d}$. Now

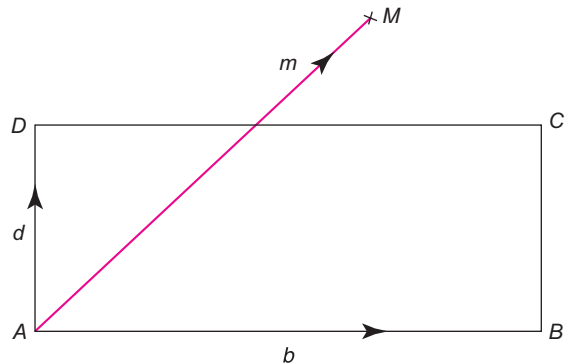


FIGURE 6.50 Multiple correct choice type question 7.

$$\begin{aligned}\overline{MA} \times \overline{MC} &= (\vec{m}) \times (\vec{b} + \vec{d} - \vec{m}) \\ &= \vec{m} \times \vec{b} - \vec{m} \times \vec{d} + |\vec{m}|^2\end{aligned}\quad (6.62)$$

$$\begin{aligned}\overline{MB} \times \overline{MD} &= (\vec{b} - \vec{m}) \times (\vec{d} - \vec{m}) \\ &= \vec{b} \times \vec{d} - \vec{b} \times \vec{m} - \vec{m} \times \vec{d} + |\vec{m}|^2 \\ &= \vec{m} \times \vec{b} - \vec{m} \times \vec{d} + |\vec{m}|^2 \quad (\because \vec{b} \times \vec{d} = 0)\end{aligned}\quad (6.63)$$

Therefore Eqs. (6.62) and (6.63) imply

$$\overline{MA} \times \overline{MC} = \overline{MB} \times \overline{MD}$$

Hence (A) is correct.

Again

$$\begin{aligned}|\overline{MA}|^2 + |\overline{MC}|^2 &= |\vec{m}|^2 + |\vec{b} + \vec{d} - \vec{m}|^2 \\ &= 2|\vec{m}|^2 + |\vec{b}|^2 + |\vec{d}|^2 - 2\vec{m} \times (\vec{b} + \vec{d}) \quad (\because \vec{b} \times \vec{d} = 0)\end{aligned}\quad (6.64)$$

Also

$$\begin{aligned}|\overline{MB}|^2 + |\overline{MD}|^2 &= |\vec{b} - \vec{m}|^2 + |\vec{d} - \vec{m}|^2 \\ &= 2|\vec{m}|^2 + |\vec{b}|^2 + |\vec{d}|^2 - 2\vec{m} \times (\vec{b} + \vec{d})\end{aligned}\quad (6.65)$$

Equations (6.64) and (6.65) imply

$$|\overline{MA}|^2 + |\overline{MC}|^2 = |\overline{MB}|^2 + |\overline{MD}|^2$$

Hence (B) is correct.

Suppose $\overline{AM} \times \overline{CM} = \overline{BM} \times \overline{DM}$ for all points M . Then

$$\begin{aligned}\vec{m} \times (\vec{m} - \vec{c}) &= (\vec{m} - \vec{b}) \times (\vec{m} - \vec{d}) \\ \vec{m} \times (\vec{d} + \vec{b} - \vec{c}) &= (\vec{b} \times \vec{d})\end{aligned}\quad (6.66)$$

Let $\vec{\alpha} = \vec{b} + \vec{d} - \vec{c}$. We prove that $\vec{\alpha} = \vec{0}$ so that $\vec{b} - \vec{c} = \vec{d}$ in which case $\overline{BC} = \overline{AD}$ and hence $ABCD$ is a parallelogram. Suppose $\vec{\alpha} \neq \vec{0}$. Then the vector

$$\vec{m} = \frac{\vec{b} \times \vec{d}}{|\vec{\alpha}|^2} \vec{\alpha}$$

satisfies the equation $\vec{m} \times \vec{\alpha} = \vec{b} \times \vec{d}$ which is a contradiction to Eq. (6.66). Therefore $\vec{\alpha} = \vec{0}$. Now $\overline{AM} \times \overline{CM} = \overline{BM} \times \overline{DM}$ assures that $ABCD$ is not a rectangle because of (A).

Answers: (A), (B), (C)

8. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-coplanar vectors such that $(\vec{a}, \vec{b}, \vec{c})$ is a right-handed system. Let

$$\begin{aligned}\vec{\alpha} &= \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]} \\ \vec{\beta} &= \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}\end{aligned}$$

$$\vec{\gamma} = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}$$

and $x > 0$. Then

- (A) $x[\vec{a} \vec{b} \vec{c}] + \frac{[\vec{\alpha} \vec{\beta} \vec{\gamma}]}{x}$ has least value 2
 (B) $[\vec{\alpha} \vec{\beta} \vec{\gamma}]$ is a positive scalar
 (C) $\vec{a} \times \vec{\alpha} + \vec{b} \times \vec{\beta} + \vec{c} \times \vec{\gamma} = 3$
 (D) $\vec{b} \times \vec{\alpha} + \vec{c} \times \vec{\beta} + \vec{a} \times \vec{\gamma} = 3$

Solution: We have

$$\begin{aligned}[\vec{\alpha} \vec{\beta} \vec{\gamma}] &= \frac{1}{[\vec{a} \vec{b} \vec{c}]^3} [\vec{b} \times \vec{c} \cdot \vec{a} \times \vec{a} \times \vec{b}] \\ &= \frac{[\vec{a} \vec{b} \vec{c}]^2}{[\vec{a} \vec{b} \vec{c}]^3} = \frac{1}{[\vec{a} \vec{b} \vec{c}]} = 0\end{aligned}$$

Also $(\vec{a}, \vec{b}, \vec{c})$ is a right-handed system $(\vec{\alpha}, \vec{\beta}, \vec{\gamma})$ is also right-handed system.

$$[\vec{\alpha} \vec{\beta} \vec{\gamma}][\vec{a} \vec{b} \vec{c}] = 1$$

and $[\vec{a} \vec{b} \vec{c}] > 0$ $[\vec{\alpha} \vec{\beta} \vec{\gamma}] > 0$

Hence (B) is correct.

Now AM - GM implies

$$x[\vec{a} \vec{b} \vec{c}] + \frac{[\vec{\alpha} \vec{\beta} \vec{\gamma}]}{x} \geq 2\sqrt{x[\vec{a} \vec{b} \vec{c}]\frac{[\vec{\alpha} \vec{\beta} \vec{\gamma}]}{x}} = 2$$

and hence least value is 2. So (A) is correct.

Now

$$\vec{a} \times \vec{\alpha} + \vec{b} \times \vec{\beta} + \vec{c} \times \vec{\gamma} = \frac{[\vec{a} \vec{b} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]} + \frac{[\vec{b} \vec{c} \vec{a}]}{[\vec{a} \vec{b} \vec{c}]} + \frac{[\vec{c} \vec{a} \vec{b}]}{[\vec{a} \vec{b} \vec{c}]} = 1 + 1 + 1 = 3$$

So (C) is correct.

Also, $\vec{b} \times \vec{\alpha} = 0$, $\vec{c} \times \vec{\beta} = 0$ and $\vec{a} \times \vec{\gamma} = 0$, therefore (D) is not correct.

Answers: (A), (B), (C)

9. $\vec{b}$ and $\vec{c}$ are non-collinear vectors. If $\vec{a}$ is a vector such that $|\vec{c}|^2 \vec{a} = \vec{c}$ and

$$\vec{a} \cdot (\vec{b} \times \vec{c}) + (\vec{a} \times \vec{b}) \cdot \vec{b} = (4 - 2x \sin y)\vec{b} + (x^2 - 1)\vec{c}$$

then

- (A) $x = 1$
 (B) $y = (4n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$
 (C) $y = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$
 (D) $x = 1$

Solution: By hypothesis

$$(\vec{a} \times \vec{c})\vec{b} - (\vec{a} \times \vec{b})\vec{c} + (\vec{a} \times \vec{b})\vec{b} = (4 - 2x - \sin y)\vec{b} + (x^2 - 1)\vec{c}$$

and $|\vec{c}|^2 \vec{a} = \vec{c}$

Now

$$|\vec{c}|^2 \vec{a} = \vec{c}$$

$$|\vec{c}|^2 (\vec{a} \times \vec{c}) = |\vec{c}|^2$$

$$\vec{a} \times \vec{c} = 1$$

Therefore

$$(1 + \vec{a} \times \vec{b})\vec{b} - (\vec{a} \times \vec{b})\vec{c} = (4 - 2x - \sin y)\vec{b} + (x^2 - 1)\vec{c}$$

This gives

$$1 + \vec{a} \times \vec{b} = 4 - 2x - \sin y$$

and $(\vec{a} \times \vec{b}) = x^2 - 1$

Solving we get

$$1 + 1 - x^2 = 4 - 2x - \sin y$$

$$x^2 - 2x + 2 = \sin y$$

$$(x - 1)^2 + 1 = \sin y$$

So $1 - \sin y - 1 - x - 1 = 0$ and $\sin y = 1$. Therefore

$$x = 1$$

and

$$y = (4n + 1)\frac{\pi}{2}$$

Answers: (B), (D)

10. Let $ABCDEF$ be a regular hexagon (Figure 6.51) such that $\vec{AD} = x\vec{BC}$ and $\vec{CF} = y\vec{AB}$. Then

(A) $xy = 4$ (B) $\vec{AF} \times \vec{AB} + \frac{1}{2}\vec{BC}^2 = 0$

(C) $xy = 4$ (D) $\vec{AB} \times \vec{AF} + \frac{1}{3}\vec{BC}^2 = 0$

Solution: Let “O” be the centre of the hexagon. Then

$$x\vec{BC} = \vec{AD} = 2\vec{AO} = 2\vec{BC}$$

$$x = 2$$

$$y\vec{AB} = \vec{CF} = 2\vec{CO} = 2\vec{AB}$$

$$y = 2$$

Hence

$$xy = 4$$

(A) is correct. Now

$$\vec{AB} \times \vec{AF} = |\vec{AB}|^2 \cos 120^\circ = \frac{1}{2}(\vec{BC})^2$$

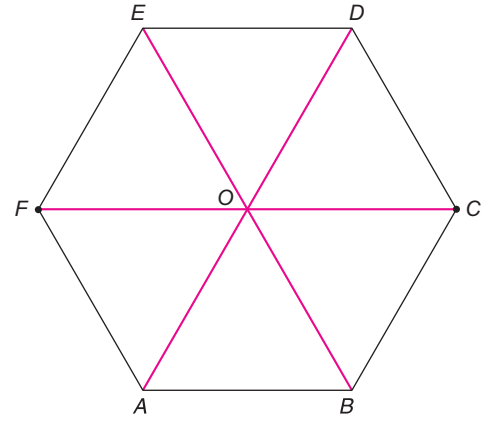


FIGURE 6.51 Multiple correct choice type question 10.

(B) is correct.

Answers: (A), (B)

11. Let $\vec{a}$ and $\vec{b}$ be two non-collinear unit vectors. If $\vec{\alpha} = \vec{a} - (\vec{a} \times \vec{b})\vec{b}$ and $\vec{\beta} = \vec{a} - \vec{b}$, then $|\vec{\beta}|$ is

(A) $|\vec{\alpha}|$ (B) $|\vec{\alpha}| + |\vec{\alpha} \times \vec{a}|$

(C) $|\vec{\alpha}| + |\vec{\alpha} \times \vec{b}|$ (D) $|\vec{\alpha}| + \vec{\alpha} \times (\vec{a} + \vec{b})$

Solution: Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \cos \theta = \cos \theta$$

Now

$$\begin{aligned} |\vec{\alpha}|^2 &= |\vec{a} - (\vec{a} \times \vec{b})\vec{b}|^2 \\ &= |\vec{a}|^2 + (\vec{a} \times \vec{b})^2 |\vec{b}|^2 - 2(\vec{a} \times \vec{b}) \\ &= 1 + \cos^2 \theta - 2\cos^2 \theta \\ &= \sin^2 \theta \\ &= (|\vec{a}||\vec{b}| \sin \theta)^2 \quad (\because |\vec{a}| = |\vec{b}| = 1) \\ &= |\vec{a} - \vec{b}|^2 \\ &= |\vec{\beta}|^2 \end{aligned}$$

Therefore

$$|\vec{\beta}| = |\vec{\alpha}|$$

Hence (A) is correct. Now

$$\begin{aligned} \vec{\alpha} \times \vec{b} &= [\vec{a} - (\vec{a} \times \vec{b})\vec{b}] \times \vec{b} \\ &= \vec{a} \times \vec{b} - (\vec{a} \times \vec{b})(\vec{b} \times \vec{b}) \\ &= \vec{a} \times \vec{b} - \vec{a} \times \vec{b} \quad (\because \vec{b} \times \vec{b} = 0) \\ &= 0 \end{aligned}$$

Since

$$|\vec{\alpha}| + |\vec{\alpha} \cdot \vec{b}| = |\vec{\alpha}| + 0 = |\vec{\alpha}| = |\vec{\beta}|$$

Therefore (C) is correct.

Answers: (A), (C)

12. The vector $(1/3)(2\vec{i} - 2\vec{j} + \vec{k})$ is

- (A) a unit vector
- (B) makes an angle $\pi/3$ with the vector $2\vec{i} - 4\vec{j} + 3\vec{k}$
- (C) parallel to the vector $-\vec{i} + \vec{j} - \frac{1}{2}\vec{k}$
- (D) perpendicular to the vector $3\vec{i} + 2\vec{j} - 2\vec{k}$

Solution: Let

$$\vec{a} = \frac{1}{3}(2\vec{i} - 2\vec{j} + \vec{k})$$

so that

$$|\vec{a}|^2 = \frac{1}{9}(2^2 + 2^2 + 1) = 1$$

Therefore

$$|\vec{a}| = 1$$

(A) is correct.

Let $\vec{b} = 2\vec{i} - 4\vec{j} + 3\vec{k}$. Then

$$\vec{a} \cdot \vec{b} = \frac{1}{3}(4 + 8 + 3) = 5$$

Let $\theta = (\vec{a}, \vec{b})$. Then

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{5}{1 \cdot \sqrt{4 + 16 + 9}} = \frac{5}{\sqrt{29}}$$

Hence (B) is not correct.

Now

$$-\vec{i} + \vec{j} - \frac{1}{2}\vec{k} = -\frac{1}{2}(2\vec{i} - 2\vec{j} + \vec{k}) = -\frac{3}{2}\vec{a}$$

Therefore $\vec{a}$ is parallel $\vec{b}$. Hence (C) is correct. Also

$$\vec{a} \cdot (3\vec{i} + 2\vec{j} - 2\vec{k}) = \frac{1}{3}(6 - 4 - 2) = 0$$

Therefore (D) is correct.

Answers: (A), (C), (D)

13. For three vectors $\vec{a}, \vec{b}, \vec{c}$, $\vec{b} \cdot (\vec{a} \times \vec{c})$ is

- (A) not equal to $\vec{a} \cdot (\vec{b} \times \vec{c})$
- (B) not equal to $(\vec{b} \times \vec{c}) \cdot \vec{a}$

(C) not equal to $(\vec{a} \times \vec{b}) \cdot \vec{c}$

(D) not equal to $\vec{c} \cdot (\vec{b} \times \vec{a})$

Solution: It is known that

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

Therefore

$$\vec{b} \cdot (\vec{a} \times \vec{c}) = \vec{c} \cdot (\vec{b} \times \vec{a})$$

Therefore (D) is not correct.

Answers: (A), (B), (C)

14. For any two vectors $\vec{a}$ and $\vec{b}$ which of the following are true?

- (A) $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$
- (B) $|1 - \vec{a} \cdot \vec{b}|^2 + |\vec{a} + \vec{b} + \vec{a} \times \vec{b}|^2 = (1 + |\vec{a}|^2)(1 + |\vec{b}|^2)$
- (C) $(\vec{a} \times \vec{b}) \times \vec{a} = |\vec{a}|^2 \vec{b} + (\vec{a} \cdot \vec{b})\vec{a}$
- (D) $\vec{a} \times (\vec{b} \times \vec{a}) = |\vec{a}|^2 \vec{b} - (\vec{a} \cdot \vec{b})\vec{a}$

Solution:

(A) We have

$$|\vec{a} \times \vec{b}|^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} \end{vmatrix} \quad (\text{Theorem 6.48})$$

$$= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$$

Therefore

$$|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$$

So (A) is correct.

(B) We have

$$\begin{aligned} & |1 - \vec{a} \cdot \vec{b}|^2 + |\vec{a} + \vec{b} + \vec{a} \times \vec{b}|^2 \\ &= 1 - 2(\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{b})^2 + |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b}) + |\vec{a} \times \vec{b}|^2 \\ & \quad [\because \vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{a} \times \vec{b}) = 0] \\ &= 1 + |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta + |\vec{a}|^2 + |\vec{b}|^2 + |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta \\ &= 1 + |\vec{a}|^2 + |\vec{b}|^2 + |\vec{a}|^2 |\vec{b}|^2 (\cos^2 \theta + \sin^2 \theta) \\ &= (1 + |\vec{a}|^2)(1 + |\vec{b}|^2) \end{aligned}$$

This implies (B) is correct.

(C) From Theorem 6.46 we have

$$(\vec{a} \times \vec{b}) \times \vec{a} = (\vec{a} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}$$

Hence (C) is not correct.

(D) Now

$$\vec{a} \times (\vec{b} \times \vec{a}) = (\vec{a} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}$$

Therefore (D) is correct.

Answers: (A), (B), (D)

15. Let $\overrightarrow{AB} = 3\vec{i} + 4\vec{k}$ and $\overrightarrow{AC} = 5\vec{i} - 2\vec{j} + 4\vec{k}$. Then
- (A) length of the median through A of $\triangle ABC$ is $\sqrt{33}$
 - (B) the length of the altitude from A to the base BC of $\triangle ABC$ is $\sqrt{\frac{41}{2}}$
 - (C) the area of $\triangle ABC$ is $\sqrt{41}$ sq. units
 - (D) the area of $\triangle ABC$ is 10 sq. units

Solution:

(A) Suppose AD is the median. Then

$$\begin{aligned}\overrightarrow{AD} &= \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2} \\ &= \frac{8\vec{i} - 2\vec{j} + 8\vec{k}}{2} \\ &= 4\vec{i} - \vec{j} + 4\vec{k}\end{aligned}$$

So

$$|\overrightarrow{AD}| = \sqrt{4^2 + 1^2 + 4^2} = \sqrt{33}$$

Hence (A) is correct.

(B) We have

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC} \Rightarrow \overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB} = 2\vec{i} - 2\vec{j}$$

Suppose AM is the altitude from A to BC (Figure 6.52). Then

$$\begin{aligned}\sin B &= \frac{AM}{AB} \\ \Rightarrow AM &= (AB)\sin B \\ &= \frac{(BC)(AB)\sin B}{BC} \\ &= \frac{|\overrightarrow{BC} \times \overrightarrow{AB}|}{|\overrightarrow{BC}|}\end{aligned}\quad (6.67)$$

Now

$$\begin{aligned}\overrightarrow{BC} \times \overrightarrow{AB} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 0 \\ 3 & 0 & 4 \end{vmatrix} \\ &= -8\vec{i} - 8\vec{j} + 6\vec{k}\end{aligned}$$

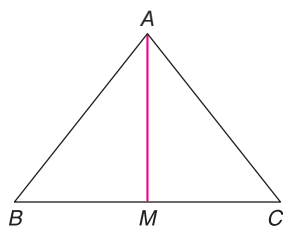


FIGURE 6.52 Multiple correct choice type question 15.

Therefore

$$|\overrightarrow{BC} \times \overrightarrow{AB}| = \sqrt{64 + 64 + 36} = \sqrt{164}$$

Also

$$|\overrightarrow{BC}| = \sqrt{8}$$

Now by Eq. (6.67)

$$\begin{aligned}AM &= \frac{\sqrt{164}}{\sqrt{8}} \\ &= \sqrt{\frac{41}{2}}\end{aligned}$$

Hence (B) is correct.

(C) We have

$$\begin{aligned}\overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 0 & 4 \\ 5 & -2 & 4 \end{vmatrix} \\ &= 8\vec{i} + 8\vec{j} + 6\vec{k}\end{aligned}$$

Therefore by Theorem 6.29,

$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{1}{2} |\overrightarrow{BC} \times \overrightarrow{AB}| \\ &= \frac{1}{2} \sqrt{164} = \sqrt{41}\end{aligned}$$

This means (C) is correct.

Answers: (A), (B), (C)

16. If the lines

$$\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-K}$$

and

$$\frac{x-1}{K} = \frac{y-4}{2} = \frac{z-5}{1}$$

are coplanar, then K may be

- (A) 0 (B) 3 (C) -2 (D) -3

Solution: Let the given lines be L_1 and L_2 , respectively. L_1 is passing through the point $(2, 3, 4)$ and having direction ratios $1, 1$ and $-K$. L_2 is passing through the point $(1, 4, 5)$ and having direction ratios $K, 2, 1$ (see Theorem 5.28, Chapter 5). Let $A = (2, 3, 4)$ and $B = (1, 4, 5)$. Also $\vec{a} = (1, 1, -K)$ and $\vec{b} = (K, 2, 1)$. The lines are coplanar if and only if vectors $\overrightarrow{AB}, \vec{a}$ and $\vec{b}$ are coplanar. Therefore by Corollary 6.16

$$\begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -K \\ K & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow (1 + 2K) + (1 + K^2) - (2 - K) = 0$$

$$K^2 + 3K = 0$$

$$K = 0 \quad \text{or} \quad 3$$

Answers: (A), (D)

17. Consider the lines

$$L_1: \frac{x}{3} = \frac{y}{1} = \frac{z}{1}$$

$$L_2: \frac{x+3}{3} = \frac{y+7}{2} = \frac{z}{4}. \text{ Then}$$

- (A) L_1 and L_2 are coplanar
 (B) L_1 and L_2 are not coplanar and shortest distance between them is $2\sqrt{30}$
 (C) L_1 and L_2 are skew lines
 (D) the shortest distance between L_1 and L_2 is $3\sqrt{30}$

Solution: The line L_1 is passing through the point $A(3, 8, 3)$ and having direction ratios 3, 1, 1. The line L_2 is passing through the point $B(-3, -7, 6)$ and having direction ratios -3, 2, 4. L_1 and L_2 are coplanar if and only if $\overrightarrow{AB}, \vec{a} = (3, 1, 1)$ and $\vec{b} = (-3, 2, 4)$ are coplanar. Therefore by Corollary 6.16, this would happen if and only if

$$\begin{vmatrix} 6 & 15 & 3 \\ 3 & 1 & 1 \\ 3 & 2 & 4 \end{vmatrix} = 0$$

Now

$$\begin{vmatrix} 6 & 15 & 3 \\ 3 & 1 & 1 \\ 3 & 2 & 4 \end{vmatrix} = 3[2(4-2) + 5(12+3) + (6-3)] \\ = 3[12 + 75 + 3] = 0$$

Therefore L_1 and L_2 are not coplanar and hence they are skew lines. Therefore (C) is correct.

The shortest distance between L_1 and L_2 is (using Theorem 6.44)

$$\frac{|\overrightarrow{AB} \times (\vec{a} - \vec{b})|}{|\vec{a} - \vec{b}|} \quad (6.68)$$

Now

$$\overrightarrow{AB} = (-6, -15, 3) = 3(-2, -5, 1) \\ \vec{a} - \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 1 \\ 3 & 2 & 4 \end{vmatrix} = 6\vec{i} - 15\vec{j} + 3\vec{k}$$

Therefore

$$\overrightarrow{AB} \times (\vec{a} - \vec{b}) = 3(12 + 75 + 3) = 270 \\ |\vec{a} - \vec{b}| = 3\sqrt{4 + 25 + 1} = 3\sqrt{30}$$

Hence by Eq. (6.68)

$$\text{Shortest distance} = \frac{270}{3\sqrt{30}} \\ = 3\sqrt{30}$$

So (D) is correct.

Answers: (C), (D)

18. Let $\vec{a}$ be a vector parallel to the line of intersection of the planes P_1 and P_2 . Plane P_1 is parallel to the vectors $2\vec{j} + 3\vec{k}$ and $4\vec{j} - 3\vec{k}$. Plane P_2 is parallel to the vectors $\vec{j} - \vec{k}$ and $3\vec{i} + 3\vec{j}$. Then the angle between the vector $\vec{a}$ and a given vector $2\vec{i} + \vec{j} - 2\vec{k}$ is

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{2\pi}{3}$

Solution: We have

$$\vec{n}_1 = \text{normal to } P_1 = (2\vec{j} + 3\vec{k}) \times (4\vec{j} - 3\vec{k}) = 18\vec{i}$$

$$\vec{n}_2 = \text{normal to } P_2 = (\vec{j} - \vec{k}) \times (3\vec{i} + 3\vec{j}) = 3(\vec{i} - \vec{j} - \vec{k})$$

Now $\vec{a}$ is parallel to $\vec{n}_1 \times \vec{n}_2$ angle between $\vec{a}$ and $2\vec{i} + \vec{j} - 2\vec{k}$ is same as angle between $\vec{n}_1 \times \vec{n}_2$ and $2\vec{i} + \vec{j} - 2\vec{k}$. Then

$$\vec{n}_1 \times \vec{n}_2 = 54 \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & * \\ 1 & 1 & 1 \end{vmatrix} = 54(\vec{j} + \vec{k})$$

Let θ be the angle between $\vec{n}_1 \times \vec{n}_2$ and $2\vec{i} + \vec{j} - 2\vec{k}$. Therefore

$$\cos \theta = \frac{54(1-2)}{54\sqrt{2}(3)} = \frac{1}{\sqrt{2}}$$

This implies

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

Angle depends on the direction of $\vec{a}$ which is parallel to $\vec{n}_1 \times \vec{n}_2$. If we take $\vec{a} = \vec{n}_1 \times \vec{n}_2$, then

$$\theta = \frac{\pi}{4}$$

Answers: (B), (C)

19. $\vec{x}, \vec{y}$ and $\vec{z}$ are vectors of equal magnitudes $\sqrt{2}$ and each is inclined to the others at an angle of 60° . If $\vec{a} = \vec{x} \times \vec{y}$, $\vec{b} = \vec{y} \times \vec{z}$ and $\vec{c} = \vec{z} \times \vec{x}$, then

- (A) $\vec{x} \times \vec{y} = \vec{y} \times \vec{z} = \vec{z} \times \vec{x} = 1$
 (B) $\vec{x} = \vec{a} - \vec{c}$
 (C) $\vec{y} = \vec{b} - \vec{c}$
 (D) $\vec{z} = \vec{b} + (\vec{a} - \vec{c})$

Solution: We have

$$\vec{x} \times \vec{y} = |\vec{x}| |\vec{y}| \cos 60^\circ = \sqrt{2} \cdot \sqrt{2} \cdot \frac{1}{2} = 1$$

Therefore (A) is correct.

$$\vec{a} = \vec{x} \times (\vec{y} \times \vec{z}) = (\vec{x} \times \vec{y}) \times \vec{z} = \vec{y} \times \vec{z} \quad (6.69)$$

$$\vec{b} = \vec{y} \times (\vec{z} \times \vec{x}) = (\vec{y} \times \vec{x}) \times \vec{z} = \vec{z} \times \vec{x} \quad (6.70)$$

Now by Eq. (6.69)

$$\begin{aligned} \vec{a} \cdot \vec{c} &= (\vec{y} \times \vec{z}) \cdot (\vec{x} \times \vec{y}) \\ &= \vec{y} \cdot (\vec{x} \times \vec{y}) \times \vec{z} \cdot (\vec{x} \times \vec{y}) \\ &= (\vec{y} \times \vec{x}) \cdot \vec{z} \cdot (\vec{y} \times \vec{x}) \cdot \vec{z} \\ &= 2\vec{x} \cdot \vec{y} \cdot \vec{x} \cdot \vec{y} = \vec{x} \cdot \vec{x} \cdot \vec{y} \cdot \vec{y} = 1 \end{aligned}$$

Hence (B) is correct. Again

$$\begin{aligned} \vec{b} \cdot \vec{c} &= (\vec{z} \times \vec{x}) \cdot (\vec{x} \times \vec{y}) \\ &= \vec{z} \cdot (\vec{x} \times \vec{y}) \times \vec{x} \cdot (\vec{x} \times \vec{y}) \\ &= (\vec{z} \times \vec{x}) \cdot \vec{x} \cdot (\vec{z} \times \vec{x}) \cdot \vec{y} \\ &= \vec{x} \cdot \vec{y} \cdot \vec{x} \cdot \vec{y} = 1 \end{aligned}$$

Hence (C) is correct. From Eq. (6.70),

$$\vec{z} = \vec{b} + \vec{x} = \vec{b} + \vec{a} \cdot \vec{c}$$

So (D) is correct.

Answers: (A), (B), (C), (D)

20. Let $P(3, 2, 6)$ be a point in the space and Q is a point on the line $\vec{r} = (\vec{i} + \vec{j} + 2\vec{k}) + \lambda(3\vec{i} + \vec{j} + 5\vec{k})$. Then

- (A) the value of λ for which $\overrightarrow{PQ}$ is perpendicular to the vector $\vec{i} + 4\vec{j} + 3\vec{k}$ is $1/4$
- (B) the value of λ for which $\overrightarrow{PQ}$ has magnitude $\sqrt{29}$ is 0 or $34/35$
- (C) the value of λ for which $\overrightarrow{PQ}$ is equal to $\vec{i} + 4\vec{j} + 9\vec{k}$ is 1
- (D) the value of λ for which $\overrightarrow{PQ}$ makes angle 90° with X -axis is $3/2$

Solution: We have

$$\begin{aligned} Q &= (1 + 3\lambda)\vec{i} + (\lambda + 1)\vec{j} + (2 + 5\lambda)\vec{k} \\ P &= 3\vec{i} + 2\vec{j} + 6\vec{k} \end{aligned}$$

Therefore

$$\overrightarrow{PQ} = (2 - 3\lambda)\vec{i} + (\lambda - 3)\vec{j} + (5\lambda - 4)\vec{k}$$

Now $\overrightarrow{PQ}$ is perpendicular to $\vec{i} + 4\vec{j} + 3\vec{k}$. This implies

$$\begin{aligned} (2 - 3\lambda)(1) + (\lambda - 3)(4) + (5\lambda - 4)(3) &= 0 \\ 8\lambda &= 2 \quad \lambda = \frac{1}{4} \end{aligned}$$

Hence (A) is correct. Now

$$\begin{aligned} 29 &= |\overrightarrow{PQ}|^2 = (2 - 3\lambda)^2 + (\lambda - 3)^2 + (5\lambda - 4)^2 \\ 35\lambda^2 - 34\lambda &= 0 \\ \lambda &= 0 \quad \text{or} \quad \frac{34}{35} \end{aligned}$$

Hence (B) is correct. Again

$$\begin{aligned} \overrightarrow{PQ} &= \vec{i} + 4\vec{j} + 9\vec{k} \\ 2 - 3\lambda &= 1, \lambda - 3 = 4 \quad \text{and} \quad 5\lambda - 4 = 9 \end{aligned}$$

Hence $\lambda = 1$ and (C) is correct. $\overrightarrow{PQ}$ is perpendicular to X -axis means

$$\begin{aligned} \overrightarrow{PQ} \cdot \vec{x} &= 0 \\ 2 - 3\lambda &= 0 \\ \lambda &= \frac{2}{3} \end{aligned}$$

So (D) is not correct.

Answers: (A), (B), (C)

21. $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are unit vectors such that

$$\begin{aligned} (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) &= 1 \\ \text{and} \quad \vec{a} \cdot \vec{c} &= \frac{1}{2} \end{aligned}$$

Then

- (A) $\vec{a}, \vec{b}, \vec{c}$ are coplanar
- (B) $\vec{b}, \vec{c}, \vec{d}$ are coplanar
- (C) $\vec{b}, \vec{d}$ are non-parallel
- (D) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel

Solution: $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ the angle between $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$ is 0. Therefore $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$ are parallel vectors. So

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 0$$

which implies

$$[\vec{a} \times \vec{b} \times \vec{c}] \cdot \vec{d} = 0 \quad (6.71)$$

$$\text{and} \quad [\vec{a} \times \vec{c} \times \vec{d}] \cdot \vec{b} = 0 \quad (6.72)$$

If $\vec{a}, \vec{b}, \vec{c}$ are not coplanar, then from Eq. (6.71) it follows that $\vec{c}$ and $\vec{d}$ are parallel so that $\vec{c} \times \vec{d} = \vec{0}$ which is not possible because $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 1$. Therefore $\vec{a}, \vec{b}, \vec{c}$ are coplanar. Similarly from Eq. (6.72) $\vec{b}, \vec{c}, \vec{d}$ are coplanar. Hence (A) and (B) are correct.

Again

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 1 \quad |\vec{a} \times \vec{b}| |\vec{c} \times \vec{d}| \cos \alpha = 1$$

where α is the angle between $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$. So $\alpha = 0$. This means

$$|\vec{a} \times \vec{b}| |\vec{c} \times \vec{d}| = 1$$

$$(|\vec{a}| |\vec{b}| \sin \beta) (|\vec{c}| |\vec{d}| \sin \gamma) = 1$$

where $\beta = (\vec{a}, \vec{b})$ and $\gamma = (\vec{c}, \vec{d})$. Hence

$$\sin \beta \sin \gamma = 1$$

$$\beta = \gamma = \frac{\pi}{2}$$

So $\vec{a}$ and $\vec{b}$ are at right angles and $\vec{c}$ and $\vec{d}$ are at right angles. Now suppose $\vec{b}$ and $\vec{d}$ are parallel. Since $\vec{a}$ is perpendicular to $\vec{b}$, $\vec{a}$ is perpendicular to $\vec{d}$. Now

$$\begin{aligned} 1 &= (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{a} \times [\vec{b} \times (\vec{c} \times \vec{d})] \\ &= \vec{a} \times [(\vec{b} \times \vec{c}) \times \vec{d}] \\ &= (\vec{a} \times \vec{c}) (\vec{b} \times \vec{d}) - (\vec{a} \times \vec{d}) (\vec{b} \times \vec{c}) \\ &= (\vec{a} \times \vec{c}) (\vec{b} \times \vec{d}) \quad (\because \vec{a} \times \vec{d} = 0) \\ &= \frac{1}{2} (\vec{b} \times \vec{d}) \end{aligned}$$

Hence $\vec{b} \times \vec{d} = 2$ which is not possible because $\vec{b}$ and $\vec{d}$ are unit vectors. So $\vec{b}$ and $\vec{d}$ cannot be parallel. Hence (C) is correct.

Answers: (A), (B), (C)

22. Let $\vec{a}$ and $\vec{b}$ be perpendicular unit vectors and $\vec{c}$ be a unit vector equally inclined to both $\vec{a}$ and $\vec{b}$ at an angle θ . If $\vec{c} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, then

(A) $x = y = \cos \theta$

(B) $x = \cos \theta, y = \sin \theta$

(C) $x = \cos \theta, z = \cos \theta$

(D) $x = y = \cos \theta, z = \cos \theta$

Solution: By hypothesis $\vec{a} \times \vec{b} = 0, \vec{a} \times \vec{c} = \cos \theta = \vec{b} \times \vec{c}$. Now

$$\begin{aligned} \vec{a} \times \vec{c} &= x(\vec{a} \times \vec{a}) + y(\vec{a} \times \vec{b}) + z[\vec{a} \times (\vec{a} \times \vec{b})] \\ &= x + 0 + 0 \end{aligned}$$

Therefore

$$x = \vec{a} \times \vec{c} = \cos \theta$$

Similarly

$$y = \vec{b} \times \vec{c} = \cos \theta$$

Again

$$\begin{aligned} 1 &= |\vec{c}|^2 \\ &= x^2 |\vec{a}|^2 + y^2 |\vec{b}|^2 + z^2 |\vec{a} \times \vec{b}|^2 + 2xy(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) \\ &\quad + 2zx[\vec{a} \times (\vec{a} \times \vec{b})] + 2yz[\vec{b} \times (\vec{a} \times \vec{b})] \\ &= x^2 + y^2 + z^2 |\vec{a}|^2 |\vec{b}|^2 \sin^2 90^\circ \\ &= \cos^2 \theta + \cos^2 \theta + z^2 \end{aligned}$$

Therefore

$$z^2 = \cos 2\theta$$

Answers: (A), (C)

23. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be unit vectors such that $\vec{a}$ is perpendicular to both $\vec{b}$ and $\vec{c}$. If the angle between $\vec{b}$ and $\vec{c}$ is $\pi/6$, then $\vec{a} = \lambda(\vec{b} \times \vec{c})$ where λ may be

(A) 4 (B) 4 (C) 2 (D) 2

Solution: We have

$$\begin{aligned} 1 &= |\lambda| |\vec{b} \times \vec{c}| \\ &= |\lambda| |\vec{b}| |\vec{c}| \sin \frac{\pi}{6} \\ &= |\lambda| \frac{1}{2} \end{aligned}$$

Therefore

$$\lambda = \pm 2$$

Answers: (C), (D)

24. The position vectors of two points A and C are $9\vec{i} + \vec{j} + 2\vec{k}$ and $7\vec{i} - 2\vec{j} + 7\vec{k}$, respectively (Figure 6.53). The point of intersection of the vectors $\vec{AB} = 4\vec{i} + \vec{j} + 3\vec{k}$ and $\vec{CD} = 2\vec{i} - \vec{j} + 2\vec{k}$ is P. $\vec{PQ}$ is perpendicular to both $\vec{AB}$ and $\vec{CD}$ and $|\vec{PQ}| = 15$. Then the position vectors of the point Q is

(A) (4, 11, 11)

(B) (6, 9, 9)

(C) (-4, 11, 11)

(D) (-6, 9, 9)

Solution: Equation of the lines AB and CD are, respectively (see Theorem 5.27, Chapter 5)

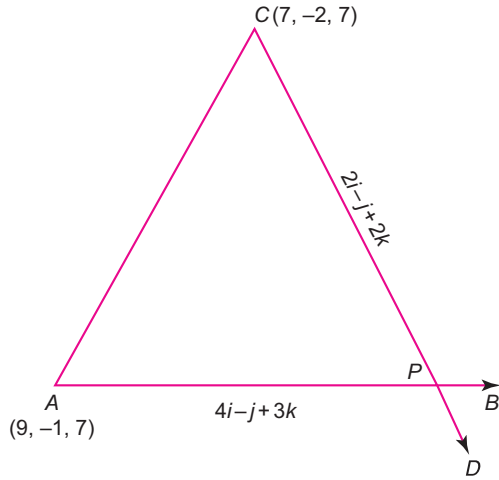


FIGURE 6.53 Multiple choice type question 24.

$$\vec{r} = (9\vec{i} - \vec{j} + 7\vec{k}) + t(4\vec{i} - \vec{j} + 3\vec{k}), t \in \mathbb{R}$$

$$\text{and } \vec{r} = (7\vec{i} - 2\vec{j} + 7\vec{k}) + s(2\vec{i} - \vec{j} + 2\vec{k}), s \in \mathbb{R}$$

Equating the corresponding coefficients in the above equations we get

$$9 + 4t = 7 + 2s$$

$$\text{or } 4t - 2s = -2 \quad (6.73)$$

$$-1 - t = -2 - s$$

$$\text{or } t - s = 1 \quad (6.74)$$

$$7 + 3t = 7 + 2s$$

$$\text{or } 3t - 2s = 0 \quad (6.75)$$

Now $t = -2$ and $s = -3$ satisfy Eqs. (6.73), (6.74) and (6.75). Therefore

$$P = (9\vec{i} - \vec{j} + 7\vec{k}) - 2(4\vec{i} - \vec{j} + 3\vec{k}) = \vec{i} + \vec{j} + \vec{k}$$

Now

$$\begin{aligned} \overrightarrow{AB} \times \overrightarrow{CD} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & -1 & 3 \\ 2 & -1 & 2 \end{vmatrix} \\ &= \vec{i} - 2\vec{j} - 2\vec{k} \end{aligned}$$

Since $\overrightarrow{PQ}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{CD}$, let

$$\overrightarrow{PQ} = \lambda(\overrightarrow{AB} \times \overrightarrow{CD}) = \lambda(\vec{i} - 2\vec{j} - 2\vec{k}), \lambda \in \mathbb{R}$$

Then

$$\begin{aligned} |\overrightarrow{PQ}| &= 15 \\ \Rightarrow 3|\lambda| &= 15 \\ \Rightarrow \lambda &= \pm 5 \end{aligned}$$

Case 1: $\lambda = 5$. This implies

$$\overrightarrow{PQ} = 5\vec{i} - 10\vec{j} - 10\vec{k}$$

Hence

$$\begin{aligned} \overrightarrow{OQ} &= \overrightarrow{PQ} + \overrightarrow{OP} \\ &= (5\vec{i} - 10\vec{j} - 10\vec{k}) + (\vec{i} + \vec{j} + \vec{k}) \\ &= 6\vec{i} - 9\vec{j} - 9\vec{k} \end{aligned}$$

So (B) is correct.

Case 2: $\lambda = -5$. This implies

$$\overrightarrow{PQ} = -5\vec{i} + 10\vec{j} + 10\vec{k}$$

Therefore

$$\begin{aligned} \overrightarrow{OQ} &= \overrightarrow{PQ} + \overrightarrow{OP} = (-5\vec{i} + 10\vec{j} + 10\vec{k}) + (\vec{i} + \vec{j} + \vec{k}) \\ &= -4\vec{i} + 11\vec{j} + 11\vec{k} \end{aligned}$$

So (C) is correct.

Answers: (B), (C)

25. In $\triangle ABC$, let I and O be the incentre and circumcentre, respectively. Then

(A) $a\overrightarrow{AI} + b\overrightarrow{BI} + c\overrightarrow{CI} = \vec{0}$ where a, b and c are the sides lengths of BC, CA and AB , respectively

(B) $\overrightarrow{OI} = \frac{1}{2s}(a\overrightarrow{OA} + b\overrightarrow{OB} + c\overrightarrow{OC})$ where $2s = a + b + c$

(C) if H is the orthocentre, then $\overrightarrow{OH} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$

(D) $\overrightarrow{OG} = \frac{1}{3}(\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC})$

Solution: Let AD be the internal bisector of $\angle A$ meeting the side BC in D . Then

$$BD : DC = c : b$$

Take the circumcentre " O " as origin and let $\overrightarrow{OA} = \vec{\alpha}$, $\overrightarrow{OB} = \vec{\beta}$ and $\overrightarrow{OC} = \vec{\gamma}$. Then

$$\overrightarrow{OD} = \frac{b\vec{\beta} + c\vec{\gamma}}{b + c}$$

Now I divides AD in the ratio $(b + c) : a$. So

$$\begin{aligned} \overrightarrow{OI} &= \frac{a\vec{\alpha} + (b + c)\left(\frac{b\vec{\beta} + c\vec{\gamma}}{b + c}\right)}{a + (b + c)} \\ &= \frac{1}{2s}(a\vec{\alpha} + b\vec{\beta} + c\vec{\gamma}) \end{aligned}$$

Hence (B) is correct. Also

$$\begin{aligned}\overline{AI} &= \overline{OI} - \overline{OA} = \frac{1}{2s}(a\vec{\alpha} + b\vec{\beta} + c\vec{\gamma}) - \vec{\alpha} \\ &= \frac{1}{2s}(b\vec{\beta} + c\vec{\gamma} - b\vec{\alpha} - c\vec{\alpha})\end{aligned}$$

Similarly

$$\begin{aligned}\overline{BI} &= \frac{1}{2s}(a\vec{\alpha} + c\vec{\gamma} - a\vec{\beta} - c\vec{\beta}) \\ \overline{CI} &= \frac{1}{2s}(a\vec{\alpha} + b\vec{\beta} - a\vec{\gamma} - b\vec{\gamma})\end{aligned}$$

Therefore

$$\begin{aligned}a\overline{AI} + b\overline{BI} + c\overline{CI} &= \frac{1}{2s}[ab(\vec{\beta} - \vec{\alpha}) - ab(\vec{\beta} - \vec{\alpha}) + \dots] \\ &= \frac{1}{2s}(\vec{0}) = \vec{0}\end{aligned}$$

So (A) is correct. For (C) see Example 5.3, Chapter 5, and (D) follows from the fact the centroid G divides the median through A in the ratio 2:1 reckoning from A .

Answers: (A), (B), (C), (D)

Matrix-Match Type Questions

1. $\vec{e}_1, \vec{e}_2$ and $\vec{e}_3$ are unit vectors such that

$$\begin{aligned}\vec{e}_1 + \vec{e}_2 + \vec{e}_3 &= \vec{a} \\ \vec{e}_1 - (\vec{e}_2 - \vec{e}_3) &= \vec{b} \\ (\vec{e}_1 - \vec{e}_2) - \vec{e}_3 &= \vec{c} \\ \vec{a} \times \vec{e}_1 &= \frac{3}{2} \\ \vec{a} \times \vec{e}_2 &= \frac{7}{4} \\ |\vec{a}| &= 2\end{aligned}$$

Match the items of Column I to those of Column II.

Column I	Column II
(A) $\vec{e}_1 \times \vec{e}_2$ is	(p) $\frac{2}{3}$
(B) $\vec{e}_2 \times \vec{e}_3$ is	(q) $\frac{1}{4}$
(C) $\vec{e}_3 \times \vec{e}_1$ is	(r) $\frac{1}{2}$
	(s) 0
(D) $\vec{c} = \lambda \vec{e}_2$ where λ is	(t) $\frac{3}{4}$

Solution: We have

$$\vec{e}_1 - (\vec{e}_2 - \vec{e}_3) = \vec{b} \quad (\vec{e}_1 \times \vec{e}_3)\vec{e}_2 - (\vec{e}_1 \times \vec{e}_2)\vec{e}_3 = \vec{b} \quad (6.76)$$

and

$$(\vec{e}_1 - \vec{e}_2) - \vec{e}_3 = \vec{c} \quad (\vec{e}_1 \times \vec{e}_3)\vec{e}_2 - (\vec{e}_2 \times \vec{e}_3)\vec{e}_1 = \vec{c} \quad (6.77)$$

Now

$$\begin{aligned}\vec{e}_1 + \vec{e}_2 + \vec{e}_3 &= \vec{a} \quad 1 + 1 + 1 + 2(\vec{e}_1 \times \vec{e}_2) + 2(\vec{e}_2 \times \vec{e}_3) \\ &+ 2(\vec{e}_3 \times \vec{e}_1) = |\vec{a}|^2 = 4\end{aligned}$$

Therefore

$$\vec{e}_1 \times \vec{e}_2 + \vec{e}_2 \times \vec{e}_3 + \vec{e}_3 \times \vec{e}_1 = \frac{1}{2} \quad (6.78)$$

Again

$$\begin{aligned}\vec{a} &= \vec{e}_1 + \vec{e}_2 + \vec{e}_3 \\ \vec{a} \times \vec{e}_1 &= \frac{3}{2} \\ \vec{a} \times \vec{e}_2 &= \frac{7}{4}\end{aligned}$$

These imply

$$\frac{3}{2} = \vec{a} \times \vec{e}_1 = 1 + \vec{e}_1 \times \vec{e}_2 + \vec{e}_1 \times \vec{e}_3$$

and

$$\frac{7}{4} = \vec{a} \times \vec{e}_2 = \vec{e}_1 \times \vec{e}_2 + 1 + \vec{e}_2 \times \vec{e}_3$$

Therefore

$$\vec{e}_1 \times \vec{e}_2 + \vec{e}_1 \times \vec{e}_3 = \frac{1}{2}$$

and

$$\vec{e}_1 \times \vec{e}_2 + \vec{e}_2 \times \vec{e}_3 = \frac{3}{4}$$

Using these values in Eq. (6.78) we have

$$\vec{e}_2 \times \vec{e}_3 = 0$$

$$\vec{e}_3 \times \vec{e}_1 = \frac{1}{4}$$

$$\vec{e}_1 \times \vec{e}_2 = \frac{3}{4}$$

Using the values of $\vec{e}_1 \times \vec{e}_2 = 3/4$, $\vec{e}_2 \times \vec{e}_3 = 0$ and $\vec{e}_3 \times \vec{e}_1 = (1/4)$ in Eqs. (6.76) and (6.77) we have

$$\frac{1}{4}\vec{e}_2 - \frac{3}{4}\vec{e}_3 = \vec{b}$$

and $\vec{c} = \frac{1}{4}\vec{e}_2$

Answer: (A) → (t), (B) → (s), (C) → (q), (D) → q

2. Match the items of Column I to those of Column II.

Column I	Column II
(A) If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{a} \times \vec{b} = 1$ and $\vec{a} \cdot \vec{b} = \vec{j} \cdot \vec{k}$ then $ \vec{b} $ is	(p) 0
(B) If $\vec{a} = \vec{i} - 3\vec{j} + \vec{k}$, $\vec{b} = 2\vec{i} + \lambda\vec{j} + \vec{k}$ and $\vec{c} = 3\vec{i} + \vec{j} - 2\vec{k}$ are coplanar, then λ equals	(q) 1
(C) If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors, then $[\vec{a} \ \vec{b} \ \vec{b} + \vec{c} \ \vec{c} + \vec{a}]$ is equal to	(r) $\frac{1}{4}$
(D) $\vec{a}, \vec{b}$ are unit vector which include angle $\pi/6$. If $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, then $[\vec{a} \ \vec{b} \ \vec{c}]^2$ equals	(s) 4

Solution:

(A) We have

$$\begin{aligned} (\vec{a} \cdot \vec{b}) \vec{a} &= (\vec{j} \cdot \vec{k}) (\vec{i} + \vec{j} + \vec{k}) \\ (\vec{a} \times \vec{a}) \vec{b} &= (\vec{b} \times \vec{a}) \vec{a} = 2\vec{i} - \vec{j} - \vec{k} \\ 3\vec{b} - \vec{a} &= 2\vec{i} - \vec{j} - \vec{k} \end{aligned}$$

Therefore

$$3\vec{b} = 3\vec{i} \quad \text{or} \quad \vec{b} = \vec{i}$$

so that $|\vec{b}| = 1$.

Answer: (A) → (q)

(B) $\vec{a}, \vec{b}, \vec{c}$ are coplanar. This implies

$$\begin{vmatrix} 1 & 3 & 1 \\ 2 & \lambda & 1 \\ 3 & 1 & 2 \end{vmatrix} = 0$$

$$2\lambda - 1 + 3(4 - 3) + (2 - 3\lambda) = 0$$

$$5\lambda - 20 = 0$$

$$\lambda = 4$$

Answer: (B) → (s)

(C) We have from Quick Look 11 that

$$[\vec{a} \ \vec{b} \ \vec{b} + \vec{c} \ \vec{c} + \vec{a}] = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} [\vec{a} \ \vec{b} \ \vec{c}] = 0$$

Answer: (C) → (p)

(D) We have

$$|\vec{a} \ \vec{b}| = \sin \frac{\pi}{6} = \frac{1}{2}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = (\vec{a} \cdot \vec{b}) \times \vec{c} = |\vec{a} \ \vec{b}| |\vec{c}| \sin 90^\circ = \frac{1}{2}$$

Therefore

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \frac{1}{4}$$

Answer: (D) → (r)

3. Match the items of Column I to those of Column II.

Column I	Column II
(A) Let $\vec{a} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} - 2\vec{k}$. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ has magnitude $\sqrt{2/3}$ is	(p) $2\vec{i} - \vec{j} + 5\vec{k}$
(B) Let $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - \vec{k}$. Then a vector coplanar with $\vec{a}$ and $\vec{b}$ and having magnitude $2\sqrt{5}$ and perpendicular to $\vec{a}$ is	(q) $\sqrt{5}(\vec{i} + \vec{j} - 2\vec{k})$
(C) Let $\vec{a} = \vec{i} - \vec{j}$, $\vec{b} = \vec{j} - \vec{k}$, $\vec{c} = \vec{k} - \vec{i}$. If $\vec{d}$ is a vector of magnitude $\sqrt{30}$ such that $\vec{a} \times \vec{d} = 0 = [\vec{b} \ \vec{c} \ \vec{d}]$, then $\vec{d}$ may be	(r) $2\vec{i} + 3\vec{j} - 3\vec{k}$
(D) $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{c} = \vec{j} - \vec{k}$. If $\vec{b}$ is a vector such that $\vec{a} \times \vec{b} = 3$ and $\vec{a} \cdot \vec{b} = \vec{c}$ is	(s) $\sqrt{10}(\vec{j} + \vec{k})$
	(t) $\frac{1}{3}(5\vec{i} + 2\vec{j} + 2\vec{j})$

Solution:

(A) Neither $\vec{b}$ nor $\vec{c}$ is a required vector. Let $\vec{d}$ be the required vector and

$$\vec{d} = \vec{b} + x\vec{c} = (1+x)\vec{i} + (2+x)\vec{j} - (1+2x)\vec{k}$$

Therefore

$$\begin{aligned}\frac{|\vec{d} \times \vec{a}|}{|\vec{a}|} &= \sqrt{\frac{2}{3}} \\ (\vec{d} \times \vec{a})^2 &= 4 \\ [2(1+x) \quad (2+x) \quad (1+2x)]^2 &= 4 \\ x+1 &= \pm 2 \\ x &= 1, -3\end{aligned}$$

Answer: (A) → (p), (r)

(B) Let $\vec{c}$ be required vector and

$$\vec{c} = x\vec{a} + y\vec{b} = (2x+y)\vec{i} + (x+2y)\vec{j} + (x-y)\vec{k}$$

Then

$$\begin{aligned}\vec{c} \times \vec{a} &= 0 \quad 2(2x+y) + 1(x+2y) + (x-y) = 0 \\ 6x + 3y &= 0 \\ 2x + y &= 0\end{aligned} \quad (6.79)$$

Also

$$\begin{aligned}|\vec{c}| &= 2\sqrt{5} \quad (2x+y)^2 + (x+2y)^2 + (x-y)^2 = 20 \\ 9x^2 + 9y^2 &= 20 \quad [\text{by Eq. (6.79)}] \\ x &= \pm \frac{\sqrt{10}}{3}\end{aligned}$$

Now $x = \sqrt{10}/3$ and $y = -2\sqrt{10}/3$ imply

$$\vec{c} = \sqrt{10}(\vec{j} + \vec{k})$$

Answer: (B) → (s)

(C) Let $\vec{d} = x\vec{i} + y\vec{j} + z\vec{k}$. Therefore

$$\vec{d} \times \vec{a} = 0 \quad x - y = 0 \quad (6.80)$$

$$\begin{aligned}[\vec{b} \quad \vec{c} \quad \vec{d}] &= 0 \quad \begin{vmatrix} x & y & z \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 0 \\ x + y + z &= 0\end{aligned} \quad (6.81)$$

From Eqs. (6.80) and (6.81) we have

$$y = x \quad \text{and} \quad z = -2x \quad (6.82)$$

Now

$$\begin{aligned}|\vec{d}| &= 30 \quad x^2 + y^2 + z^2 = 30 \\ 6x^2 &= 30 \quad [\text{by Eq. (6.82)}] \\ x &= \pm \sqrt{5}\end{aligned}$$

Also

$$x = \sqrt{5} \quad \vec{d} = \sqrt{5}\vec{i} + \sqrt{5}\vec{j} - 2\sqrt{5}\vec{k} = \sqrt{5}(\vec{i} + \vec{j} - 2\vec{k})$$

Answer: (C) → (q)

(D) We have

$$\begin{aligned}\vec{a} &= \vec{i} + \vec{j} + \vec{k} \\ \vec{c} &= \vec{j} - \vec{k} \\ \vec{a} \times \vec{b} &= 3 \\ \vec{a} \cdot \vec{b} &= \vec{c}\end{aligned}$$

Now

$$\begin{aligned}\vec{a} \cdot \vec{b} &= \vec{c} = \vec{j} - \vec{k} \\ (\vec{a} \cdot \vec{b}) \cdot \vec{a} &= (\vec{j} - \vec{k}) \cdot (\vec{i} + \vec{j} + \vec{k}) \\ (\vec{a} \times \vec{b}) \cdot \vec{b} &= (\vec{b} \times \vec{a}) \cdot \vec{a} = 2\vec{i} \cdot \vec{j} - \vec{k} \\ 3\vec{b} \cdot 3\vec{a} &= 2\vec{i} \cdot \vec{j} - \vec{k} \\ 3\vec{b} \cdot 3(\vec{i} + \vec{j} + \vec{k}) &= 2\vec{i} \cdot \vec{j} - \vec{k} \\ \vec{b} &= \frac{1}{3}(5\vec{i} + 2\vec{j} + 2\vec{k})\end{aligned}$$

Answer: (D) → (t)

4. Let $\vec{a} = 2\vec{i} + \vec{j}$, $\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$, $\vec{c} = -3\vec{i} - \vec{j} + \vec{k}$ and suppose

$$\begin{aligned}\vec{l} &= \frac{\vec{b} \times \vec{c}}{[\vec{a} \quad \vec{b} \quad \vec{c}]} \\ \vec{m} &= \frac{\vec{c} \times \vec{a}}{[\vec{a} \quad \vec{b} \quad \vec{c}]} \\ \vec{n} &= \frac{\vec{a} \times \vec{b}}{[\vec{a} \quad \vec{b} \quad \vec{c}]}\end{aligned}$$

Let $\vec{d}$ be $\vec{i} + \vec{j} + \vec{k}$. If $\vec{d} = x\vec{l} + y\vec{m} + z\vec{n}$, then match the items of Column I to those of Column II.

Column I	Column II
(A) The value of $[\vec{a} \quad \vec{b} \quad \vec{c}]$ equals	(p) 5
(B) Value of x is	(q) 2
(C) Value of y equals	(r) 1
(D) z is equal to	(s) 1
	(t) 2

Solution: We have

$$\vec{d} = x\vec{l} + y\vec{m} + z\vec{n} = x \frac{(\vec{b} \times \vec{c})}{[\vec{a} \quad \vec{b} \quad \vec{c}]} + y \frac{(\vec{c} \times \vec{a})}{[\vec{a} \quad \vec{b} \quad \vec{c}]} + z \frac{(\vec{a} \times \vec{b})}{[\vec{a} \quad \vec{b} \quad \vec{c}]}$$

Now

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix}$$

$$= 2(2+3) - 1(1+9) - 1(-1+6)$$

$$= 5$$

Also

$$2 = \vec{d} \times \vec{a} = x \frac{[\vec{a} \ \vec{b} \ \vec{c}]}{[\vec{a} \ \vec{b} \ \vec{c}]} = x$$

$$x = 2$$

$$1 = \vec{d} \times \vec{b} = y \frac{[\vec{a} \ \vec{b} \ \vec{c}]}{[\vec{a} \ \vec{b} \ \vec{c}]} = y$$

$$y = 1$$

$$1 = \vec{d} \times \vec{c} = z$$

$$z = 1$$

Answer: (A) → (p), (B) → (q), (C) → (r), (D) → (s)

5. Match the items of Column I to those of Column II.

Column I	Column II
(A) If $\vec{a}$ and $\vec{b}$ are non-zero vectors and $ \vec{a} + \vec{b} = \vec{a} + \vec{b} $, then the angle between $\vec{a}$ and $\vec{b}$ is	(p) 0
(B) If $\vec{a}$ and $\vec{b}$ non-zero vectors such that $ \vec{a} + \vec{b} = \vec{a} - \vec{b} $, then angle θ between them is	(q) $\cos^{-1} \frac{1}{3}$
(C) In a regular tetrahedron, the angle between the opposite pair of edges is	(r) $\frac{\pi}{2}$
(D) In a cube, the angle between any two diagonals is	(s) $\frac{\pi}{3}$
	(t) $\cos^{-1} \frac{1}{\sqrt{3}}$

Comprehension-Type Questions

1. **Passage:** If $\vec{a}$ and $\vec{b}$ are non-zero vectors and θ is the angle between $\vec{a}$ and $\vec{b}$, then

$$\theta = \cos^{-1} \frac{\vec{a} \times \vec{b}}{|\vec{a}| |\vec{b}|}$$

Answer the following questions.

(i) If $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that each is perpendicular to sum of the other two and $|\vec{a}| = 2, |\vec{b}| = 3$ and $|\vec{c}| = 6$. Then the angle between $\vec{a}$ and $\vec{a} + \vec{b} + \vec{c}$ is

(A) $\cos^{-1} \frac{2}{7}$ (B) $\frac{\pi}{4}$

(C) $\cos^{-1} \frac{1}{7}$ (D) $\frac{\pi}{2}$

Solution:

(A) Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then

$$|\vec{a} + \vec{b}|^2 = (|\vec{a}| + |\vec{b}|)^2 \quad 2|\vec{a}||\vec{b}|\cos\theta = 2|\vec{a}||\vec{b}|$$

$$\theta = 0$$

Answer: (A) → (p)

(B) We have

$$|\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2 \quad 2(\vec{a} \times \vec{b}) = -2(\vec{a} \times \vec{b})$$

$$\vec{a} \times \vec{b} = 0$$

Answer: (B) → (r)

(C) Let $ABCD$ be a regular tetrahedron. Therefore, all the faces are equal triangles of equal edges. Take A as origin and let $\vec{AB} = \vec{b}, \vec{AC} = \vec{c}$ and $\vec{AD} = \vec{d}$. (AB, CD) is one pair of opposite edges and

$$\vec{AB} \times \vec{CD} = \vec{b} \times (\vec{d} - \vec{c})$$

$$= \vec{b} \times \vec{d} - \vec{b} \times \vec{c}$$

$$= |\vec{b}||\vec{d}|\cos 60^\circ - |\vec{c}||\vec{d}|\cos 60^\circ$$

$$= 0 \quad (\because |\vec{b}| = |\vec{c}| = |\vec{d}|)$$

Therefore angle between AB or CD is $\pi/2$.

Answer: (C) → (r)

(D) Let $\vec{OA} = \vec{a}, \vec{OB} = \vec{b}$ and $\vec{OC} = \vec{c}$ such that $|\vec{a}| = |\vec{b}| = |\vec{c}|$. Consider the cube with $\vec{a}, \vec{b}, \vec{c}$ as coterminal edges (student is advised to draw the diagram). We can see that $\vec{a} + \vec{b} + \vec{c}$ and $\vec{a} + \vec{b} - \vec{c}$ are a pair of diagonals. If θ is the angle between them, then

$$\cos\theta = \frac{\vec{a} \times \vec{a} + \vec{b} \times \vec{b} - \vec{c} \times \vec{c}}{|\vec{a} + \vec{b} + \vec{c}||\vec{a} + \vec{b} - \vec{c}|} = \frac{1}{\sqrt{3} \sqrt{3}} = \frac{1}{3}$$

Answer: (D) → (q)

(ii) In the above problem, the angle between $\vec{a} - \vec{b}$ and $\vec{a} + \vec{b} + \vec{c}$ is

(A) $\cos^{-1} \frac{3}{7}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) $\cos^{-1} \frac{6}{7}$

(iii) If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular unit vectors, then the angle between the vectors $\vec{b} - \vec{c} + \vec{c} - \vec{a} + \vec{a} - \vec{b}$ and $\vec{a} + \vec{b} + \vec{c}$ is

(A) $\cos^{-1} \frac{1}{7}$ (B) $\cos^{-1}(1)$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Solution:

- (i) By hypothesis $\vec{a} \cdot (\vec{b} + \vec{c}) = 0$, $\vec{b} \cdot (\vec{c} + \vec{a}) = 0$ and $\vec{c} \cdot (\vec{a} + \vec{b}) = 0$. Therefore

$$\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0$$

$$\vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{a} = 0$$

$$\vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} = 0$$

Adding all the three equations, we have

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$$

and hence

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$$

Therefore

$$[\vec{a} \vec{b} \vec{c}] = |\vec{a}| |\vec{b}| |\vec{c}| = 2 \times 3 \times 6 = 36$$

Now

$$\begin{aligned} |\vec{a} + \vec{b} + \vec{c}|^2 &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2 \sum (\vec{b} \cdot \vec{c}) \\ &= 4 + 9 + 36 + 0 = 49 \end{aligned}$$

So

$$|\vec{a} + \vec{b} + \vec{c}| = 7$$

Now, angle between $\vec{a} + \vec{b} + \vec{c}$ and $\vec{a}$ is

$$\begin{aligned} \cos^{-1} \left(\frac{(\vec{a} + \vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a} + \vec{b} + \vec{c}| |\vec{a}|} \right) &= \cos^{-1} \left(\frac{\vec{a} \cdot \vec{a}}{7 \times 2} \right) \\ &= \cos^{-1} \left(\frac{4}{7 \times 2} \right) \\ &= \cos^{-1} \left(\frac{2}{7} \right) \end{aligned}$$

Answer: (A)

- (ii) We have

$$|\vec{a} \times \vec{b}| = 2 \times 3 = 6$$

Now since $\vec{a}, \vec{b}, \vec{c}$ are at right angles,

$$\begin{aligned} (\vec{a} \times \vec{b}) \cdot (\vec{a} + \vec{b} + \vec{c}) &= [\vec{a} \vec{b} \vec{c}] = |\vec{a}| |\vec{b}| |\vec{c}| \\ &= 2 \times 3 \times 6 = 36 \end{aligned}$$

Therefore angle between $\vec{a} \times \vec{b}$ and $\vec{a} + \vec{b} + \vec{c}$ is

$$\cos^{-1} \left(\frac{(\vec{a} \times \vec{b}) \cdot (\vec{a} + \vec{b} + \vec{c})}{|\vec{a} \times \vec{b}| |\vec{a} + \vec{b} + \vec{c}|} \right) = \cos^{-1} \left(\frac{36}{6 \times 7} \right) = \cos^{-1} \left(\frac{6}{7} \right)$$

Answer: (D)

- (iii) We have

$$(\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 3[\vec{a} \vec{b} \vec{c}] = 3$$

Now

$$\begin{aligned} |\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}|^2 &= |\vec{b} \times \vec{c}|^2 + |\vec{c} \times \vec{a}|^2 \\ &\quad + |\vec{a} \times \vec{b}|^2 + 2 \sum (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) \\ &= 1 + 1 + 1 + 2 \sum (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) \end{aligned} \quad (6.83)$$

By Theorem 6.48

$$\begin{aligned} (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) &= \begin{vmatrix} \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{a} \\ \vec{c} \cdot \vec{c} & \vec{c} \cdot \vec{a} \end{vmatrix} \\ &= \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = 0 \end{aligned}$$

From Eq. (6.83)

$$|\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}| = \sqrt{3}$$

Also

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}$$

Therefore angle between $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}$ and $\vec{a} + \vec{b} + \vec{c}$ is

$$\cos^{-1} \left(\frac{3}{\sqrt{3} \times \sqrt{3}} \right) = \cos^{-1}(1) = 0$$

Note: You can take $\vec{a}, \vec{b}$ and $\vec{c}$ as $\vec{i}, \vec{j}, \vec{k}$.

Answer: (B)

- 2. Passage:** If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and $\vec{r}$ is any vector, then there exist unique set of scalars $\{x, y, z\}$ such that $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$. Also three vectors $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar if and only if $[\vec{a} \vec{b} \vec{c}] \neq 0$. Let

$$\vec{a} = \vec{i} + \vec{j} - \vec{k}$$

$$\vec{b} = -\vec{i} + 2\vec{j} + 2\vec{k}$$

$$\vec{c} = -\vec{i} + 2\vec{j} - \vec{k}$$

Answer the following three questions.

- (i) If $\vec{d} = 2\vec{i} + 3\vec{j} + \vec{k}$ and $\vec{d} = x\vec{a} + y\vec{b} + z\vec{c}$, then

(A) $x = -\frac{7}{9}, y = \frac{11}{9}, z = -\frac{8}{9}$

(B) $x = \frac{7}{9}, y = \frac{11}{9}, z = -\frac{8}{9}$

(C) $x = \frac{7}{3}, y = \frac{11}{9}, z = -\frac{8}{9}$

(D) $x = \frac{11}{3}, y = -\frac{7}{3}, z = -\frac{8}{9}$

(ii) If the same $\vec{d}$ as in (i) is equal to $x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$, then

(A) $x = -\frac{4}{9}, y = -\frac{6}{9}, z = -\frac{1}{3}$

(B) $x = \frac{2}{9}, y = -\frac{6}{9}, z = -\frac{1}{9}$

(C) $x = -\frac{2}{9}, y = \frac{6}{9}, z = -\frac{1}{9}$

(D) $x = \frac{2}{9}, y = \frac{6}{9}, z = -\frac{1}{9}$

(iii) If the same

$$\vec{d} = x \left(\frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]} \right) + y \left(\frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]} \right) + z \left(\frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]} \right)$$

then

(A) $x = -2, y = -6, z = -1$

(B) $x = -2, y = 6, z = 1$

(C) $x = 4, y = 6, z = 3$

(D) $x = 2, y = 3, z = 1$

Solution:

(i) We have

$$\begin{aligned} 2\vec{i} + 3\vec{j} + \vec{k} &= x\vec{a} + y\vec{b} + z\vec{c} \\ &= (x - y - z)\vec{i} + (x + 2y + 2z)\vec{j} + (-x + 2y - z)\vec{k} \end{aligned}$$

Therefore

$$x - y - z = 2 \quad (6.84)$$

$$x + 2y + 2z = 3 \quad (6.85)$$

$$-x + 2y - z = 1 \quad (6.86)$$

Solving Eqs. (6.84), (6.85) and (6.86) we get that

$$x = \frac{7}{3}, y = \frac{11}{9}, z = -\frac{8}{9}$$

Answer: (C)

(ii) We have

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 1 & 1 & -1 \\ -1 & 2 & 2 \\ -1 & 2 & -1 \end{vmatrix} = -9$$

$$\vec{d} = 2\vec{i} + 3\vec{j} + \vec{k} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$$

$$4 = \vec{d} \cdot \vec{a} = x[\vec{a} \vec{b} \vec{c}] = x(-9) \Rightarrow x = -\frac{4}{9}$$

$$6 = \vec{d} \cdot \vec{b} = y[\vec{b} \vec{c} \vec{a}] = y(-9) \Rightarrow y = -\frac{2}{3}$$

$$3 = \vec{d} \cdot \vec{c} = z[\vec{a} \vec{b} \vec{c}] = z(-9) \Rightarrow z = -\frac{1}{3}$$

Answer: (A)

(iii) We have

$$2\vec{i} + 3\vec{j} + \vec{k} = \vec{d} = x \left(\frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]} \right) + y \left(\frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]} \right) + z \left(\frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]} \right)$$

Therefore

$$4 = \vec{d} \cdot \vec{a} = x \frac{[\vec{a} \vec{b} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]} = x$$

$$6 = \vec{d} \cdot \vec{b} = y$$

$$3 = \vec{d} \cdot \vec{c} = z$$

Answer: (C)

3. Passage: The equation of a straight line passing through the points with position vectors $\vec{a}$ and $\vec{b}$ is $\vec{r} = (1-t)\vec{a} + t\vec{b}, t \in \mathbb{R}$. It is given that in $\triangle ABC$, $\angle A = 90^\circ$ and the vertices B and C lie on the line joining the points $(-1, 3, 2)$ and $(1, 1, 3)$ and $\angle ABC = 30^\circ$. If $A = (-1, 2, 3)$ (see Figure 6.54), then answer the following three questions.

(i) The vertex B may be

(A) $\left(\frac{2\sqrt{3}-1}{3}, \frac{7-2\sqrt{3}}{3}, \frac{7+\sqrt{3}}{3} \right)$

(B) $\left(\frac{2-\sqrt{3}}{3}, \frac{7-2\sqrt{3}}{3}, \frac{7+\sqrt{3}}{3} \right)$

(C) $\left(\frac{2\sqrt{3}+1}{3}, \frac{7-2\sqrt{3}}{3}, \frac{7+\sqrt{3}}{3} \right)$

(D) $\left(\frac{2\sqrt{3}-1}{3}, \frac{7+\sqrt{3}}{3}, \frac{7-2\sqrt{3}}{3} \right)$

(ii) The vertex C may be

(A) $\left(\frac{-3-2\sqrt{3}}{9}, \frac{2\sqrt{3}-3}{9}, \frac{21-\sqrt{3}}{9} \right)$

(B) $\left(\frac{2\sqrt{3}-3}{9}, \frac{21-2\sqrt{3}}{9}, \frac{21+\sqrt{3}}{9} \right)$

(C) $\left(\frac{2\sqrt{3}-3}{9}, \frac{-3-2\sqrt{3}}{9}, \frac{21+\sqrt{3}}{9} \right)$

(D) $\left(\frac{-3-2\sqrt{3}}{9}, \frac{21+\sqrt{3}}{9}, \frac{21-\sqrt{3}}{9} \right)$

(iii) If G is the centroid of the triangle, then $\vec{GA} + \vec{GB} + \vec{GC}$ equals

(A) $\vec{AB} + \vec{BC} + \vec{CA}$ (B) $\vec{AB} \times \vec{AC}$

(C) $\vec{BC} \times \vec{BA}$ (D) $\vec{CA} \times \vec{CB}$

Solution: Equation of the line BC is

$$\begin{aligned}\vec{r} &= (1-t)(-1, 3, 2) + t(1, 1, 3) \\ &= (2t-1, 3-2t, 2+t)\end{aligned}$$

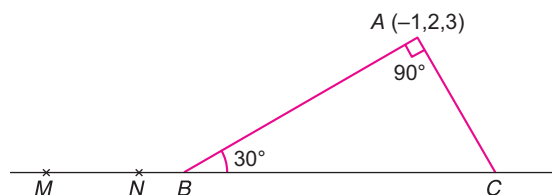


FIGURE 6.54 Comprehension-type question 3.

Therefore

$$\begin{aligned}\overrightarrow{BA} &= (-2t, 2t-1, 1-t) \\ \overrightarrow{MN} &= (2, -2, 1)\end{aligned}$$

Now

$$\begin{aligned}\frac{\sqrt{3}}{2} &= \cos 30^\circ = \frac{|-4t-4t+2+1-t|}{3\sqrt{4t^2+4t^2+1-4t+1+t^2-2t}} \\ &= \frac{|9t-3|}{3\sqrt{9t^2-6t+2}}\end{aligned}$$

Therefore

$$\begin{aligned}3(9t^2-6t+2) &= 4(3t-1)^2 \\ 9t^2-6t-2 &= 0 \\ t &= \frac{6 \pm 6\sqrt{3}}{18} = \frac{1 \pm \sqrt{3}}{3}\end{aligned}$$

Let $\overrightarrow{CA} = (-2t_1, 2t_1-1, 1-t_1)$ and $\overrightarrow{BA} = (-2t, 2t-1, 1-t)$. Then

$$\begin{aligned}\overrightarrow{BA} \cdot \overrightarrow{CA} &= 0 \Rightarrow 9tt_1 - 3t - 3t_1 + 2 = 0 \\ &\Rightarrow 3(3t-1)t_1 = 3t-2 \\ \Rightarrow t_1 &= \frac{3t-2}{3(3t-1)} = \frac{1}{3} \left[1 - \frac{1}{3t-1} \right]\end{aligned}$$

Therefore

$$\begin{aligned}t &= \frac{1+\sqrt{3}}{3} \Rightarrow t_1 = \frac{\sqrt{3}-1}{3\sqrt{3}} \\ t &= \frac{1-\sqrt{3}}{3} \Rightarrow t_1 = \frac{\sqrt{3}+1}{3\sqrt{3}}\end{aligned}$$

So

$$B = \left(\frac{2\sqrt{3}-1}{3}, \frac{7-2\sqrt{3}}{3}, \frac{7+\sqrt{3}}{3} \right)$$

$$\text{and } C = \left(\frac{-2-\sqrt{3}}{3\sqrt{3}}, \frac{2+7\sqrt{3}}{3\sqrt{3}}, \frac{7\sqrt{3}-1}{3\sqrt{3}} \right)$$

OR

$$B = \left(\frac{-1-2\sqrt{3}}{3}, \frac{7+2\sqrt{3}}{3}, \frac{7-\sqrt{3}}{3} \right)$$

$$\text{and } C = \left(\frac{2-\sqrt{3}}{3\sqrt{3}}, \frac{-2+7\sqrt{3}}{3\sqrt{3}}, \frac{7\sqrt{3}+1}{3\sqrt{3}} \right)$$

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the position vectors of the vertices A , B and C respectively. Then

$$\begin{aligned}G &= \frac{\vec{a} + \vec{b} + \vec{c}}{3} \\ \overrightarrow{GA} &= \vec{a} - \frac{\vec{a} + \vec{b} + \vec{c}}{3} = \frac{2\vec{a} - \vec{b} - \vec{c}}{3}\end{aligned}$$

Similarly

$$\begin{aligned}\overrightarrow{GB} &= \frac{2\vec{b} - \vec{c} - \vec{a}}{3} \\ \overrightarrow{GC} &= \frac{2\vec{c} - \vec{a} - \vec{b}}{3}\end{aligned}$$

and

Hence

$$\overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC} = \vec{0} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}$$

Answer: (i) $\rightarrow$ (A), (ii) $\rightarrow$ (B), (iii) $\rightarrow$ (A)

4. Passage: Two non-zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if $\vec{b} = \lambda\vec{a}$, $\lambda \in \mathbb{R}$. Non-zero vectors $\vec{a}$ and $\vec{b}$ are at right angles if and only if $\vec{a} \cdot \vec{b} = 0$. If $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then every vector $\vec{r}$ in the plane determined by $\vec{a}$ and $\vec{b}$ can be written as $\vec{r} = x\vec{a} + y\vec{b}$ in one and only one way. Answer the following questions.

(i) If $\vec{b} = x\vec{i} + y\vec{j} + z\vec{k}$ such that $|\vec{b}| = 10$, $xyz < 0$ and is collinear with the vector $\vec{a} = 2\sqrt{2}\vec{i} - \vec{j} + 4\vec{k}$ then $\vec{b}$ is

- (A) $(-4\sqrt{2}, 2, 8)$ (B) $(4\sqrt{2}, -2, 8)$
(C) $(-4\sqrt{2}, -2, -8)$ (D) $(4\sqrt{2}, 2, -8)$

(ii) Let $\vec{a} = (2, 3, -1)$ and $\vec{b} = (1, -2, 3)$. $\vec{c}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$ and satisfies $\vec{c} \cdot \vec{d} = -6$ where $\vec{d} = 2\vec{i} - \vec{j} + \vec{k}$. Then $\vec{c}$ is equal to

- (A) $(-3, 3, 3)$ (B) $(2, 2, -2)$
(C) $(2, 1, -1)$ (D) $(0, 1, 1)$

(iii) Let $\vec{a} = (-1, 1, 1)$ and $\vec{b} = (2, 0, 1)$. If $\vec{c}$ is vector coplanar with the vectors $\vec{a}$ and $\vec{b}$, is perpendicular to $\vec{b}$ and satisfies the condition $\vec{a} \cdot \vec{c} = 7$, then $\vec{c}$ equals

- (A) $\left(-\frac{3}{2}, \frac{5}{2}, 3\right)$ (B) $(1, 0, -2)$

(C) $(-3, 0, 6)$ (D) $(\frac{3}{2}, 0, 3)$

Solution:

(i) Let

$$x\vec{i} + y\vec{j} + z\vec{k} = \vec{b} = \lambda\vec{a} = \lambda(2\sqrt{2}, 1, 4)$$

Therefore

$$x = 2\lambda\sqrt{2}, y = \lambda, z = 4\lambda$$

Now

$$|\vec{b}| = 10 \quad 8\lambda^2 + \lambda^2 + 16\lambda^2 = 100$$

$$\lambda = \pm 2$$

So

$$\lambda = 2 \quad x = 4\sqrt{2}, y = 2, z = 8$$

$$\lambda = -2 \quad x = -4\sqrt{2}, y = -2, z = -8$$

We have

$$xyz < 0 \quad x = 4\sqrt{2}, y = 2, z = 8$$

Answer: (B)

(ii) $\vec{a} = (2, 3, 1)$ and $\vec{b} = (1, 2, 3)$. Then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 7\vec{i} - 7\vec{j} + 7\vec{k} = 7(\vec{i} - \vec{j} + \vec{k})$$

Since $\vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$, let

$$\vec{c} = \lambda(\vec{a} \times \vec{b}) = 7\lambda(\vec{i} - \vec{j} + \vec{k})$$

then

$$\vec{c} \cdot \vec{d} = 6 \quad 7\lambda(\vec{i} - \vec{j} + \vec{k}) \cdot (2\vec{i} - \vec{j} + \vec{k}) = 6$$

$$7\lambda(2 + 1 + 1) = 6$$

$$\lambda = \frac{3}{7}$$

Therefore

$$\vec{c} = 7\lambda(\vec{i} - \vec{j} + \vec{k}) = 3(\vec{i} - \vec{j} + \vec{k}) = (3, -3, 3)$$

Answer: (A)

(iii) $\vec{a} = (1, 1, 1)$ and $\vec{b} = (2, 0, 1)$. Let

$$\vec{c} = x\vec{a} + y\vec{b}$$

$$= (x + 2y, x, x + y) \quad (6.87)$$

Now

$\vec{c}$ is perpendicular to $\vec{b}$ $\vec{c} \cdot \vec{b} = 0$

$$2x + 4y + x + y = 0$$

$$5y - x = 0 \quad (6.88)$$

Again

$$\vec{a} \cdot \vec{c} = 7 \quad x + 2y + x + x + y = 7$$

$$3x - y = 7 \quad (6.89)$$

Solving Eqs. (6.88) and (6.89) we have

$$x = \frac{5}{2}, y = \frac{1}{2}$$

Substituting these values in Eq. (9.87) we get

$$\vec{c} = \frac{5}{2}\vec{a} + \frac{1}{2}\vec{b} = \frac{5}{2}(1, 1, 1) + \frac{1}{2}(2, 0, 1) = \left(\frac{7}{2}, \frac{3}{2}, \frac{3}{2}\right)$$

Answer: (A)

5. Passage: For any three vectors $\vec{a}, \vec{b}, \vec{c}$, the following hold good:

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{a} \times \vec{c}) \cdot \vec{b} = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{c}) \cdot \vec{b} = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

Let $\vec{a}$ and $\vec{b}$ be mutually perpendicular unit vectors. Answer the following three questions.

(i) If $\vec{d}$ is any vector such that $\vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{d}$, then $(\vec{d} \times \vec{b}) \cdot \vec{b}$ equal

(A) $\vec{a}$ (B) $\vec{d}$ (C) $\vec{d}$ (D) $2\vec{b}$

(ii) If $\vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{d}$, then $\vec{d}$ equals

(A) $\frac{(\vec{a} \times \vec{b}) \cdot \vec{a}}{2}$ (B) $\frac{\vec{a} + (\vec{a} \times \vec{b})}{2}$

(C) $\vec{a} \times \vec{b}$ (D) $\frac{\vec{a} - (\vec{a} \times \vec{b})}{2}$

(iii) If $\vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{d}$ which of the following statements is false?

(A) Vectors $\vec{a}, \vec{d}$ and $\vec{d} \times \vec{b}$ are linearly dependent

(B) $[\vec{d} \times \vec{a} \times \vec{d} \times \vec{b}] = 0$

(C) $\vec{d} \cdot \vec{b} = 0$ and $|\vec{d}| = \frac{1}{\sqrt{2}}$

(D) $[\vec{d} \times \vec{a} \times \vec{d} \times \vec{b}] = 0$

Solution: By hypothesis $\vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{d}$ and $\vec{a} \times \vec{b} = 0$.
Therefore

$$0 = (\vec{d} \cdot \vec{b}) \times \vec{b} = \vec{a} \times \vec{b} \quad \vec{d} \times \vec{b} = 0 \quad \vec{d} \times \vec{b} = 0$$

$$\vec{d} \times \vec{b} = 0$$

Now

$$\vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{d} \quad (\vec{d} \cdot \vec{b}) \cdot \vec{b} = \vec{a} \cdot \vec{b} \quad \vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{b} \quad (6.90)$$

$$\vec{d} = \vec{a} \quad \vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b} \quad (\vec{a} \cdot \vec{d})$$

$$(\because \vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{d})$$

$$\vec{d} = \frac{\vec{a} \cdot (\vec{a} \cdot \vec{b})}{2}$$

$$(\vec{d} \cdot \vec{b}) \cdot \vec{b} = \vec{a} \cdot \vec{b} \quad \vec{d} \cdot \vec{b} = \vec{a} \cdot \vec{b}$$

$$= \vec{a} \cdot \vec{b} \quad \vec{d} \cdot (\vec{a} \cdot \vec{b}) \text{ [by Eq. (6.90)]}$$

$$= \vec{d}$$

Also

$$\vec{d} = \frac{\vec{a} \cdot (\vec{a} \cdot \vec{b})}{2}$$

Again

$$[\vec{d} \vec{a} \vec{d} \vec{b}] = (\vec{d} \cdot \vec{a}) (\vec{d} \cdot \vec{b})$$

$$= \left| \begin{array}{cc} \vec{d} \times \vec{a} & \vec{d} \times \vec{b} \\ \vec{a} \times \vec{d} & \vec{a} \times \vec{b} \end{array} \right|$$

$$= \left| \begin{array}{cc} \vec{d} \times \vec{a} & 0 \\ \vec{a} \times \vec{d} & 0 \end{array} \right| = 0$$

Therefore (B) is false.

Answer: (i) $\rightarrow$ (b), (ii) $\rightarrow$ (D), (iii) $\rightarrow$ (B)

Assertion-Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
(B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
(C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true

1. Statement I: If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, then $\vec{b} \cdot \vec{c}, \vec{c} \cdot \vec{a}$ and $\vec{a} \cdot \vec{b}$ are also non-coplanar.

Statement II: Vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar if and only if $[\vec{a} \vec{b} \vec{c}] = 0$.

Solution: According to Theorem 6.34 Statement II is correct. Now

$$[\vec{b} \vec{c} \vec{c} \vec{a} \vec{a} \vec{b}] = [\vec{a} \vec{b} \vec{c}]^2 = 0$$

because by Statement I $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar.

Answer: (A)

2. Statement I: If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors, then

$$\left| \begin{array}{ccc} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \end{array} \right| = 0$$

Statement II: $[\vec{a} \vec{b} \vec{c}] (\vec{a} \cdot \vec{b}) = \left| \begin{array}{ccc} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \end{array} \right|$

Solution: Let

$$\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$$

$$\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$$

$$\vec{c} = c_1 \vec{i} + c_2 \vec{j} + c_3 \vec{k}$$

Then

$$[\vec{a} \vec{b} \vec{c}] (\vec{a} \cdot \vec{b}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

If A and B are two matrices of 3 $\times$ 3 order, then we know that $\det(AB) = (\det A)(\det B)$. Therefore

$$[\vec{a} \vec{b} \vec{c}] (\vec{a} \cdot \vec{b}) = \left| \begin{array}{ccc} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \end{array} \right|$$

$\vec{a}, \vec{b}, \vec{c}$ are coplanar $[\vec{a} \vec{b} \vec{c}] = 0$. So

$$[\vec{a} \vec{b} \vec{c}] (\vec{a} \cdot \vec{b}) = 0$$

Answer: (A)

3. Statement I: The shortest distance between the lines

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$$

and $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$

is $2\sqrt{19}$.

Statement II: The shortest distance between two skew lines $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{c} + s\vec{d}$ is

$$\frac{|(\vec{a} - \vec{c}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}$$

Solution: According to Theorem 6.44, Statement II is correct.

In the given lines $\vec{a} = (3, 5, 7)$, $\vec{b} = (1, -2, 1)$, $\vec{c} = (-1, -1, -1)$ and $\vec{d} = (7, -6, 1)$. Now

$$\vec{a} - \vec{c} = (4, 6, 8)$$

and $\vec{b} \times \vec{d} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix} = 4\vec{i} + 6\vec{j} + 8\vec{k}$

Therefore the shortest distance is

$$\begin{aligned} \frac{|(\vec{a} - \vec{c}) \cdot (4\vec{i} + 6\vec{j} + 8\vec{k})|}{2\sqrt{4 + 9 + 16}} &= \frac{|16 + 36 + 64|}{2\sqrt{29}} \\ &= \frac{58}{\sqrt{29}} = 2\sqrt{29} \end{aligned}$$

Hence Statement I is false.

Answer: (D)

4. Statement I: In any $\triangle ABC$, $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$.

Statement II: If O is the circum centre of $\triangle ABC$, then $|\vec{OA} + \vec{OB} + \vec{OC}| \geq 0$.

Solution: Statement II is clearly true. It is known that the angles $(\vec{OB}, \vec{OC}) = 2A$, $(\vec{OC}, \vec{OA}) = 2B$ and $(\vec{OA}, \vec{OB}) = 2C$. Now

$$\begin{aligned} 0 \leq |\vec{OA} + \vec{OB} + \vec{OC}|^2 &= |\vec{OA}|^2 + |\vec{OB}|^2 + |\vec{OC}|^2 + 2\sum (\vec{OB} \cdot \vec{OC}) \\ &= 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C) \end{aligned}$$

where $OA = OB = OC = R$. Therefore

$$0 \leq 3 + 2(\cos 2A + \cos 2B + \cos 2C)$$

Hence

$$-\frac{3}{2} \leq \cos 2A + \cos 2B + \cos 2C$$

Answer: (A)

5. $\vec{PQ}, \vec{QR}, \vec{RS}, \vec{ST}, \vec{TU}$ and $\vec{UP}$ represent the sides of a regular hexagon (Figure 6.55).

Statement I: $\vec{PQ} \times (\vec{RS} + \vec{ST}) \neq \vec{0}$

Statement II: $\vec{PQ} \times \vec{RS} = \vec{0}$ and $\vec{PQ} \times \vec{ST} \neq \vec{0}$.

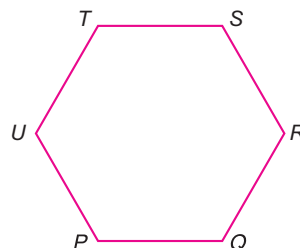


FIGURE 6.55 Assertion–reasoning type question 5.

Solution: We have

$$\vec{PQ} \times (\vec{RS} + \vec{ST}) = \vec{PQ} \times \vec{RT} \neq \vec{0}$$

because $\vec{PQ}$ and $\vec{RT}$ are not parallel. Statement I is true. Now

$$\vec{PQ} \times \vec{RS} \neq \vec{0}$$

because they are not parallel vectors. Also $\vec{PQ} \times \vec{ST} = \vec{0}$ because $\vec{PQ}$ and $\vec{ST}$ are parallel. Hence Statement II is false.

Answer: (C)

6. Consider the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Statement I: Let $\vec{n}_1 = (3, -6, -2)$ and $\vec{n}_2 = (2, 1, -2)$ be normals to the planes. Then $\vec{n}_1 \times \vec{n}_2$ is parallel to the line of intersection of the planes.

Statement II: The vector $14\vec{i} + 2\vec{j} + 15\vec{k}$ is parallel to the line of intersection of the planes.

Solution: We have

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = 14\vec{i} + 2\vec{j} + 15\vec{k}$$

Answer: (B)

Integer Answer Type Questions

1. Let $\vec{a} = (8, -10, 13)$. If $\vec{b} = (x, y, z)$ is a vector of magnitude $\sqrt{37}$ and collinear with $\vec{a}$ and making an acute angle with Z-axis, then the integral part of $x + y + z$ is _____.

Solution: We have

$$\vec{b} = \lambda \vec{a} \Rightarrow x = 8\lambda, y = -10\lambda, z = 13\lambda$$

Now

$$37 = |\vec{b}|^2 = x^2 + y^2 + z^2 = (333)\lambda^2$$

Therefore

$$\lambda^2 = \frac{1}{9} \quad \text{or} \quad \lambda = \pm \frac{1}{3}$$

But it makes acute angle with Z-axis. This implies

$$\lambda = \frac{1}{3}$$

Therefore

$$x = \frac{8}{3}, y = -\frac{10}{3} \quad \text{and} \quad z = \frac{13}{3}$$

So

$$x + y + z = \frac{11}{3} = 3.66 \dots$$

$$[x + y + z] = 3$$

Answer: 3

2. Let $\vec{a} = \vec{i} - \vec{j} + 3\vec{k}$, $\vec{b} = 3\vec{i} - 5\vec{j} + 6\vec{k}$. Then integer part of the magnitude of the projection of $2\vec{a} - \vec{b}$ on to the vector $\vec{a} + \vec{b}$ is _____.

Solution:

$$\vec{a} + \vec{b} = 4\vec{i} - 6\vec{j} + 9\vec{k}$$

$$2\vec{a} - \vec{b} = -\vec{i} + 3\vec{j}$$

$$(\vec{a} + \vec{b}) \cdot (2\vec{a} - \vec{b}) = -4 - 18 = -22$$

$$|\vec{a} + \vec{b}| = \sqrt{133}$$

According to Quick Look 2, the magnitude of the projection of $2\vec{a} - \vec{b}$ on $(\vec{a} + \vec{b})$ is

$$\frac{|(\vec{a} + \vec{b}) \cdot (2\vec{a} - \vec{b})|}{|\vec{a} + \vec{b}|} = \frac{22}{\sqrt{133}}$$

whose integer part is 1.

Answer: 1

3. $\vec{OA} = \vec{a}$, $\vec{OB} = 10\vec{a} + 2\vec{b}$ and $\vec{OC} = \vec{b}$ where O , A and C are non-collinear points. Let p be the area of the quadrilateral $OABC$ and q be the area of the parallelogram with OA and OC as adjacent sides. The $p = Kq$ where K is _____.

Solution: See single correct choice type question 57.

Answer: 6

4. $P(1, -1, 2)$, $Q(2, 0, -1)$ and $R(0, 2, 1)$ are three points, then $[\vec{QR} \times \vec{RP}, \vec{RP} \times \vec{PQ}, \vec{PQ} \times \vec{QR}]$ is equal to _____.

Solution: We have

$$\vec{PQ} = \vec{i} + \vec{j} - 3\vec{k}$$

$$\vec{QR} = -2\vec{i} + 2\vec{j} + 2\vec{k}$$

$$\vec{RP} = \vec{i} - 3\vec{j} + \vec{k}$$

Now

$$[\vec{PQ}, \vec{QR}, \vec{RP}] = \begin{vmatrix} 1 & 1 & -3 \\ -2 & 2 & 2 \\ 1 & -3 & 1 \end{vmatrix} = 0$$

This is also evident from $\vec{PQ} + \vec{QR} + \vec{RP} = \vec{0}$. Now

$$[\vec{QR} \times \vec{RP}, \vec{RP} \times \vec{PQ}, \vec{PQ} \times \vec{QR}] = [\vec{PQ}, \vec{QR}, \vec{RP}]^2 = 0$$

Answer: 0

5. In a quadrilateral $ABCD$, $\vec{AB} = \vec{a}$, $\vec{AD} = \vec{d}$ and $\vec{AC} = m\vec{a} + n\vec{d}$ where m and n are positive integers. If the area of $ABCD$ is $5|\vec{a} \times \vec{d}|$, then the possible number of pairs (m, n) is _____.

Solution:

$$\begin{aligned} \text{Area of } ABCD &= \frac{1}{2} |\vec{AC} \times \vec{BD}| \\ &= \frac{1}{2} |(m\vec{a} + n\vec{d}) \times (\vec{d} - \vec{a})| \\ &= \frac{1}{2} |(m+n)(\vec{a} \times \vec{d})| \\ &= \frac{m+n}{2} (\vec{a} \times \vec{d}) \\ &= 5|\vec{a} \times \vec{d}| \end{aligned}$$

This implies

$$\frac{m+n}{2} = 5$$

Therefore $m+n=10$. Hence m takes values from 1 to 9 so that n takes values from 9 to 1. Hence number of pairs = 9.

Answer: 9

6. If $(\vec{b} \ \vec{c}) \ (\vec{c} \ \vec{a}) = 3\vec{c}$, then the value of $[\vec{b} \ \vec{c} \ \vec{a}]$ is ____.

Solution: We have

$$3\vec{c} = (\vec{b} \ \vec{c}) \ (\vec{c} \ \vec{a}) = [\vec{b} \ \vec{c} \ \vec{a}] \vec{c} = 0$$

Therefore

$$[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = 3$$

$$[\vec{b} \ \vec{c} \ \vec{a}] = [\vec{a} \ \vec{b} \ \vec{c}] = 3^2$$

Answer: 9

7. If the point of intersection of the lines

$$\frac{x}{1} = \frac{y}{2} = \frac{z+3}{3}$$

$$\text{and} \quad \frac{x}{2} = \frac{y}{2} = \frac{z}{4}$$

is (x, y, z) , then $y+x$ is ____.

Solution: Let L_1 and L_2 be the given lines, respectively. P is a point on L_1 and Q is a point on L_2 . Therefore

$$P = (t, 2+2t, 3+3t)$$

$$\text{and} \quad Q = (2+2s, 6+2s, 3+4s)$$

where $t, s \in \mathbb{R}$. Now $P=Q$ implies

$$t = 2+2s \quad \text{or} \quad t-2s = 2 \quad (6.91)$$

$$2+2t = 6+2s \quad \text{or} \quad 2t-2s = 4 \quad (6.92)$$

$$\text{and} \quad 3+3t = 3+4s \quad \text{or} \quad 3t-4s = 6 \quad (6.93)$$

From Eqs. (6.91) and (6.92) $t=+2, s=0$ which also satisfy Eq. (6.93). Therefore the point of intersection is

$$(2, 6, 3) = (x, y, z)$$

So

$$y+x=8$$

Answer: 8

8. If $A_1, A_2, \dots, A_8$ are vertices of a regular octagon, then

$$\sum_{j=1}^7 (\vec{OA}_j \ \vec{OA}_{j+1}) = K(\vec{OA}_1 \ \vec{OA}_2)$$

where the value of K is ____.

Solution: Each side subtends an angle of $2\pi/8 = \pi/4$ at the centre of the octagon. Let “ O ” be the centre of the octagon and r the radius of the circumcircle of the octagon. Therefore

$$\vec{OA}_1 \ \vec{OA}_2 = r^2 \sin \frac{\pi}{4} \vec{n}$$

where $\vec{n}$ is the vector perpendicular to the plane of the polygon such that from the side of $\vec{n}$, the points $A_1, A_2, A_3, \dots, A_8$ are in counterclockwise sense. Hence

$$\vec{OA}_2 \ \vec{OA}_3 = r^2 \sin \frac{\pi}{4} \vec{n}$$

$$\vec{OA}_3 \ \vec{OA}_4 = r^2 \sin \frac{\pi}{4} \vec{n}, \dots \text{etc.}$$

Therefore

$$\begin{aligned} \sum_{j=1}^7 (\vec{OA}_j \ \vec{OA}_{j+1}) &= 7 r^2 \sin \frac{\pi}{4} \vec{n} \\ &= 7(\vec{OA}_1 \ \vec{OA}_2) \end{aligned}$$

Answer: 7

9. $\vec{OA} = 2\vec{i} - 2\vec{j}, \vec{OB} = \vec{i} + \vec{j} - \vec{k}$ and $\vec{OC} = 3\vec{i} - \vec{k}$ are edges of a parallelepiped. Then the volume of the parallelepiped in cubic units is ____.

Solution:

$$\begin{aligned} [\vec{OA} \ \vec{OB} \ \vec{OC}] &= \begin{vmatrix} 2 & 2 & 0 \\ 1 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix} \\ &= 2(1+0) + 2(1+3) \\ &= 2+4 \\ &= 2 \end{aligned}$$

$$\text{Volume of } OABC = |[\vec{OA} \ \vec{OB} \ \vec{OC}]| = 2$$

Answer: 2

10. Let $\vec{OA} = \vec{a}, \vec{OB} = \vec{b}$ and $\vec{OC} = \vec{c}$. Let V_1 be the volume of the parallelepiped with edges $\vec{a}, \vec{b}$ and $\vec{c}$ and V_2 be the volume of the parallelepiped with coterminal edges $\vec{a} + \vec{b}, \vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$. Then $V_2 = KV_1$ where K equals ____.

Solution: We have

$$V_1 = |[\vec{a} \vec{b} \vec{c}]|$$

$$V_2 = |[\vec{a} + \vec{b} \vec{b} + \vec{c} \vec{c} + \vec{a}]| = |2[\vec{a} \vec{b} \vec{c}]| = 2V_1$$

Answer: 2

- 11.** Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors and $A = 2\vec{a} + 3\vec{b} - \vec{c}$, $B = \vec{a} - 2\vec{b} + 3\vec{c}$, $C = 3\vec{a} + 4\vec{b} - 2\vec{c}$ and $D = \vec{a} - 6\vec{b} + 6\vec{c}$ be four points. Then $[\vec{AB} \vec{AC} \vec{AD}]$ is equal to _____.

Solution: We have

$$\vec{AB} = \vec{a} - 5\vec{b} + 4\vec{c}$$

$$\vec{AC} = 2\vec{a} + 7\vec{b} - 3\vec{c}$$

$$\vec{AD} = \vec{a} - 9\vec{b} + 7\vec{c}$$

Therefore using Quick Look 11 [part (1)] we have

$$\begin{aligned} [\vec{AB} \vec{AC} \vec{AD}] &= \begin{vmatrix} 1 & -5 & 4 \\ 2 & 7 & -3 \\ 1 & -9 & 7 \end{vmatrix} [\vec{a} \vec{b} \vec{c}] \\ &= (2 + 30 - 32)[\vec{a} \vec{b} \vec{c}] = 0 \end{aligned}$$

Answer: 0

- 12.** Let $ABCD$ be a parallelogram (Figure 6.56) in which AB is parallel to CD and AD is parallel to BC . K is a point on the side AD such that $AK = (1/5)(AD)$. L is a point on the diagonal AC such that $AL = (1/6)(AC)$. Then the area of the parallelogram for which $\vec{LB}$ and $\vec{LK}$ are adjacent sides is _____.

Solution: Take A as origin and let $\vec{AB} = \vec{b}$ and $\vec{AD} = \vec{d}$ so that $\vec{AC} = \vec{b} + \vec{d}$. By hypothesis

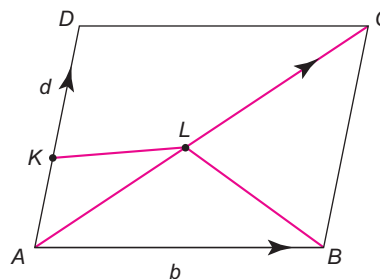


FIGURE 6.56 Integer type question 12.

$$\vec{AK} = \frac{1}{5}\vec{d}$$

$$\vec{AL} = \frac{1}{6}(\vec{b} + \vec{d})$$

$$\vec{KL} = \frac{1}{6}(\vec{b} + \vec{d}) - \frac{1}{5}\vec{d}$$

$$= \frac{1}{30}(5\vec{b} - \vec{d})$$

$$\vec{BL} = \frac{1}{6}(\vec{b} + \vec{d}) - \vec{b}$$

$$= \frac{1}{6}(\vec{d} - 5\vec{b})$$

$$= -\frac{1}{6}(5\vec{b} - \vec{d})$$

$$= -\frac{1}{30}(5\vec{b} - \vec{d})$$

$$= -\frac{1}{30}\vec{KL}$$

Therefore the vectors $\vec{KL}$ and $\vec{BL}$ are collinear vectors. Hence the area of the parallelogram for which $\vec{KL}$ and $\vec{BL}$ are adjacent sides is zero.

Answer: 0

SUMMARY

- 6.1 Scalar product (or dot product):** Let $\vec{a}$ and $\vec{b}$ two vectors. Then we define $\vec{a} \cdot \vec{b} = 0$ if either of them is zero vector. If $\vec{a} \neq \vec{0}, \vec{b} \neq \vec{0}$ and $\theta = (\vec{a}, \vec{b})$ is the angle between $\vec{a}$ and $\vec{b}$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$.



QUICK LOOK

- (i) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$. But $(\vec{a} \cdot \vec{b}) \cdot \vec{c}$ has no meaning.
- (ii) $0 < (\vec{a}, \vec{b}) < 90^\circ \Rightarrow \cos(\vec{a}, \vec{b}) > 0 \Rightarrow \vec{a} \cdot \vec{b} > 0$.
- (iii) $(\vec{a}, \vec{b}) = 90^\circ \Rightarrow \vec{a} \cdot \vec{b} = 0$, when $\vec{a} \perp \vec{b}$.
- (iv) $(\vec{a}, \vec{b}) > 90^\circ \Rightarrow \vec{a} \cdot \vec{b} < 0$.

- 6.2 Scalar and vector components:** Let $\vec{a}$ and $\vec{b}$ be non-zero vectors and O, A and B be points in the space

such that $\vec{a} = \vec{OA}$ and $\vec{b} = \vec{OB}$. Let Q be the foot of the perpendicular from B onto the support $\vec{OA}$ of $\vec{a}$. Then OQ is called the scalar component (or simply the component or the projection) of $\vec{b}$ on $\vec{a}$ if $(\vec{a}, \vec{b}) < 90^\circ$, and $-OQ$ if $(\vec{a}, \vec{b}) > 90^\circ$. $\vec{OQ}$ is called the vector component (or the orthogonal projection) of $\vec{b}$ on $\vec{a}$.

- 6.3 Orthogonal projection:** The orthogonal projection of $\vec{b}$ on $\vec{a}$ is

$$\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} = \vec{b} \cdot \frac{\vec{a}}{|\vec{a}|} \frac{\vec{a}}{|\vec{a}|} = (\vec{b} \cdot \vec{e}) \vec{e}$$

where $\vec{e} = \vec{a}/|\vec{a}|$ is the unit vector in the direction of $\vec{a}$.

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- (i) The orthogonal projection of $\vec{b}$ on $\lambda\vec{a}$ ($\lambda \neq 0$) is same as the orthogonal projection of $\vec{b}$ on $\vec{a}$.
- (ii) The magnitude of the orthogonal projection of $\vec{b}$ on $\vec{a}$ is $\frac{|\vec{b} \cdot \vec{a}|}{|\vec{a}|}$.

6.4 Notation: $\vec{a} \cdot \vec{a}$ is denoted by $\vec{a}^2$ and note that $\vec{a}^2 = |\vec{a}|^2$.

6.5 For any two vectors $\vec{a}, \vec{b}$:

- (i) $(\vec{a} + \vec{b})^2 = \vec{a}^2 + 2(\vec{a} \cdot \vec{b}) + \vec{b}^2$
- (ii) $(\vec{a} - \vec{b})^2 = \vec{a}^2 - 2(\vec{a} \cdot \vec{b}) + \vec{b}^2$

For three vectors $\vec{a}, \vec{b}, \vec{c}$,

$$(\vec{a} + \vec{b} + \vec{c})^2 = \vec{a}^2 + \vec{b}^2 + \vec{c}^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

6.6 Magnitude of sum and difference: If $\vec{a}$ and $\vec{b}$ are two vectors, then

- (i) $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$. Equality holds either of $\vec{a}, \vec{b}$ is a zero vector or $\vec{a}, \vec{b}$ are like vectors.
- (ii) $||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}| \leq |\vec{a}| + |\vec{b}|$. Equality holds if and only if either $\vec{a}, \vec{b}$ is $\vec{0}$ or $\vec{a}, \vec{b}$ are like vectors.

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$|\vec{a} \cdot \vec{b}| = |\vec{a}||\vec{b}|$. Equality holds either of $\vec{a}, \vec{b}$ is $\vec{0}$ or they are collinear vectors.

6.7 Formulae:

- (i) $\vec{a} = (\vec{a} \cdot \vec{i})\vec{i} + (\vec{a} \cdot \vec{j})\vec{j} + (\vec{a} \cdot \vec{k})\vec{k}$
- (ii) If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ then $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$ and in particular $|\vec{a}|^2 = \vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2$

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If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, then $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.

6.8 Expression for the angle between two vectors: Let $\vec{a}$ and $\vec{b}$ be non-zero vectors and $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ and $\theta = (\vec{a}, \vec{b})$. Then

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{a_1\vec{i} + a_2\vec{j} + a_3\vec{k}}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

6.9 Cauchy-Schwartz inequality: For any triads of real numbers (a_1, a_2, a_3) and (b_1, b_2, b_3) ,

$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

and the equality holds if and only if a_1, a_2, a_3 are proportional to b_1, b_2, b_3 . In vector language,

$$|\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2$$

where the equality holds if and only if $\vec{a}$ and $\vec{b}$ are collinear vectors.

6.10 Vector equation of a plane (Normal form): The equation of the plane, whose unit normal drawn from the origin is $\vec{n}$ and whose distance from the origin is a , is given by

$$\vec{r} \cdot \vec{n} = a$$

In particular, if the plane passes through origin, its equation is $\vec{r} \cdot \vec{n} = 0$.

6.11 Equation of the plane in Cartesian form: If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $\vec{n} = l\vec{i} + m\vec{j} + n\vec{k}$, then the equation of the plane is $lx + my + nz = a$.

6.12 Cross-product: Let $\vec{a}$ and $\vec{b}$ be any vectors. We define

$$\vec{a} \times \vec{b} = \begin{cases} \vec{0} & \text{if either of } \vec{a}, \vec{b} \text{ is } \vec{0} \text{ or } \vec{a}, \vec{b} \text{ are collinear} \\ |\vec{a}||\vec{b}| \sin \theta \vec{n} & \text{otherwise} \end{cases}$$

where $\theta = (\vec{a}, \vec{b})$ and $\vec{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $(\vec{a}, \vec{b}, \vec{n})$ is a triad of right-handed system. $\vec{a} \times \vec{b}$ is called vector product or cross-product of $\vec{a}$ and $\vec{b}$.

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- (i) $\vec{a} \times \vec{b} = \vec{0}$ either of $\vec{a}, \vec{b}$ is $\vec{0}$ or $\vec{a}, \vec{b}$ are collinear vectors.
- (ii) $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|$ equality holds if and only if $\vec{a}, \vec{b}$ are orthogonal to each other.
- (iii) $(\vec{a}, \vec{b}, \vec{n})$ is a right-handed system $(\vec{b}, \vec{a}, -\vec{n})$ is a right-handed system so that

$$\vec{b} \times \vec{a} = |\vec{b}||\vec{a}| \sin \theta (-\vec{n}) = -(\vec{a} \times \vec{b})$$

6.13 About $(\vec{i}, \vec{j}, \vec{k})$: Let $(\vec{i}, \vec{j}, \vec{k})$ be a triad of mutually perpendicular unit vectors forming right-handed system. Then

- (i) $\vec{i} \times \vec{i} = \vec{j} \times \vec{j} = \vec{k} \times \vec{k} = \vec{0}$.
- (ii) $\vec{i} \times \vec{j} = \vec{k}, \vec{j} \times \vec{k} = \vec{i}, \vec{k} \times \vec{i} = \vec{j}$.

6.14 Formulae:

- (i) If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, then

$$\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2)\vec{i} - (a_1b_3 - a_3b_1)\vec{j} + (a_1b_2 - a_2b_1)\vec{k}$$

which is denoted by

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

and can be expanded as in the case of usual determinant expansion.

- (ii) If $\vec{a} = a_1\vec{\alpha} + a_2\vec{\beta} + a_3\vec{\gamma}$ and $\vec{b} = b_1\vec{\alpha} + b_2\vec{\beta} + b_3\vec{\gamma}$ where $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are non-coplanar vectors, then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{\beta} & \vec{\alpha} & \vec{\gamma} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

- 6.15** If θ is the angle between two non-zero vectors $\vec{a}$ and $\vec{b}$ then

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} = \frac{\sqrt{(a_1b_2 - a_2b_1)^2}}{\sqrt{a_1^2} \sqrt{b_1^2}}$$

where $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$

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- (i) $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.
(ii) Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ and hence perpendicular to the plane determined by $\vec{a}$ and $\vec{b}$ is

$$\pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

- 6.16 Important advise:** To find the angle between two vectors, always use dot product because the cross-product gives $\sin \theta$ and $\sin \theta > 0$ for $0 < \theta < 180^\circ$.

- 6.17 Vector area:** Let D be a plane region bounded by closed curve ζ . On ζ mark three points P_1, P_2 and P_3 . Let $\vec{n}$ be unit vector perpendicular to the plane of the region D such that from the side of $\vec{n}$, the points P_1, P_2 and P_3 be in counterclockwise sense. If $\vec{A}$ is the area of the plane region D , then $\vec{A}\vec{n}$ is called the vector area of the region D . From the side of $\vec{n}$ if

P_1, P_2 and P_3 are clockwise, then $(-\vec{n})$ is the vector area of the region D .

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- (i) Vector area of the region $D = (\pm \vec{n})$ where $\vec{n}$ is unit normal to the plane region D and A is its area.
(ii) $|\text{Vector area}| = \text{Area}$.

6.18 Vector area of a quadrilateral and a parallelogram:

- (i) $\vec{a} \times \vec{b}$ is the vector area of the parallelogram with $\vec{a}$ and $\vec{b}$ as adjacent sides and $(1/2)(\vec{a} \times \vec{b})$ is the area of the triangle for which $\vec{a}$ and $\vec{b}$ are two of the sides.
(ii) The vector area of a quadrilateral $ABCD$ in terms of its diagonal vectors is $(1/2)(\vec{AC} \times \vec{BD})$ which is also valid, if $ABCD$ is a parallelogram.

- 6.19** (i) The vector area of $\triangle ABC$ is

$$\frac{1}{2}\vec{AB} \times \vec{AC} = \frac{1}{2}\vec{CB} \times \vec{BA} = \frac{1}{2}\vec{CA} \times \vec{CB}$$

- (ii) If $\vec{a}, \vec{b}$ and $\vec{c}$ are the vertices of a triangle, then its vector area is

$$\frac{1}{2}(\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b})$$

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In both 6.18 and 6.19 the corresponding modulus of vector area will give area.

- 6.20 Formula:** If $\vec{a}$ is any vector, then

$$|\vec{a} \times \vec{i}|^2 + |\vec{a} \times \vec{j}|^2 + |\vec{a} \times \vec{k}|^2 = 2|\vec{a}|^2$$

- 6.21 Scalar triple product:** If $\vec{a}, \vec{b}, \vec{c}$ are three vectors, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is called the scalar triple product of $\vec{a}, \vec{b}$ and $\vec{c}$ and is denoted by $[\vec{a} \vec{b} \vec{c}]$.

- 6.22 Note:**

- (i) $\vec{a}, \vec{b}, \vec{c}$ are coplanar if and only if $[\vec{a} \vec{b} \vec{c}] = 0$.
(ii) Four points A, B, C and D are coplanar if the vectors $\vec{AB}, \vec{AC}$ and $\vec{AD}$ are coplanar $[\vec{AB} \vec{AC} \vec{AD}] = 0$.

- 6.23 Geometrical interpretation of $[\vec{a} \vec{b} \vec{c}]$:** Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors and V be the volume of the parallelepiped with $\vec{a}, \vec{b}, \vec{c}$ as coterminal edges.

$[\vec{a}, \vec{b}, \vec{c}] = V$ or $-V$ according as $(\vec{a}, \vec{b}, \vec{c})$ is a right-handed system or left-handed system. In fact $||[\vec{a}, \vec{b}, \vec{c}]|| = V$. The volume of the tetrahedron whose coterminus edges are $\vec{a}, \vec{b}$ and $\vec{c}$

$$\frac{1}{6} ||[\vec{a}, \vec{b}, \vec{c}]||$$

6.24 Interchange of cross and dot: Since the triads $(\vec{a}, \vec{b}, \vec{c})$, $(\vec{b}, \vec{c}, \vec{a})$ and $(\vec{c}, \vec{a}, \vec{b})$ are all either right-handed systems or all left-handed systems, it follows that

$$[\vec{a}, \vec{b}, \vec{c}] = [\vec{b}, \vec{c}, \vec{a}] = [\vec{c}, \vec{a}, \vec{b}] = +V \text{ or } -V$$

Hence

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a}, \vec{b}, \vec{c}] = [\vec{b}, \vec{c}, \vec{a}] = (\vec{b} \times \vec{c}) \cdot \vec{a} = \vec{a} \cdot (\vec{b} \times \vec{c})$$

since dot is commutative.

6.25 Formula for $[\vec{a}, \vec{b}, \vec{c}]$:

(i) Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ and $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$ then

$$[\vec{a}, \vec{b}, \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(ii) If $\vec{a} = a_1\vec{l} + a_2\vec{m} + a_3\vec{n}$, $\vec{b} = b_1\vec{l} + b_2\vec{m} + b_3\vec{n}$ and $\vec{c} = c_1\vec{l} + c_2\vec{m} + c_3\vec{n}$ where $\vec{l}, \vec{m}$ and $\vec{n}$ are non-coplanar vectors, then

$$[\vec{a}, \vec{b}, \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} [\vec{l}, \vec{m}, \vec{n}]$$

6.26 Important note: If $\vec{a}, \vec{b}, \vec{c}$ are expressed in $\vec{i}, \vec{j}$ and $\vec{k}$ or any three non-coplanar vectors $\vec{l}, \vec{m}$ and $\vec{n}$, the necessary and sufficient condition for $\vec{a}, \vec{b}, \vec{c}$ to be coplanar is

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

6.27 Two important formulae: Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$, $\vec{l} = l_1\vec{i} + l_2\vec{j} + l_3\vec{k}$, $\vec{m} = m_1\vec{i} + m_2\vec{j} + m_3\vec{k}$ and $\vec{n} = n_1\vec{i} + n_2\vec{j} + n_3\vec{k}$ then

$$(i) (a) [\vec{l}, \vec{m}, \vec{n}](\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{l} \times \vec{a} & \vec{l} \times \vec{b} & \vec{l} \\ \vec{m} \times \vec{a} & \vec{m} \times \vec{b} & \vec{m} \\ \vec{n} \times \vec{a} & \vec{n} \times \vec{b} & \vec{n} \end{vmatrix}$$

$$(b) [\vec{a}, \vec{b}, \vec{c}](\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \end{vmatrix}$$

$$(ii) [\vec{l}, \vec{m}, \vec{n}][\vec{a}, \vec{b}, \vec{c}] = \begin{vmatrix} \vec{l} \times \vec{a} & \vec{l} \times \vec{b} & \vec{l} \times \vec{c} \\ \vec{m} \times \vec{a} & \vec{m} \times \vec{b} & \vec{m} \times \vec{c} \\ \vec{n} \times \vec{a} & \vec{n} \times \vec{b} & \vec{n} \times \vec{c} \end{vmatrix}$$

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In 6.27 (i) (b), if $\vec{a}, \vec{b}, \vec{c}$ are coplanar, then

$$\begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \end{vmatrix} = \vec{0}$$

In (ii) if either of the triad $(\vec{l}, \vec{m}, \vec{n})$ or $(\vec{a}, \vec{b}, \vec{c})$ is a set of coplanar vectors, then

$$\begin{vmatrix} \vec{l} \times \vec{a} & \vec{l} \times \vec{b} & \vec{l} \times \vec{c} \\ \vec{m} \times \vec{a} & \vec{m} \times \vec{b} & \vec{m} \times \vec{c} \\ \vec{n} \times \vec{a} & \vec{n} \times \vec{b} & \vec{n} \times \vec{c} \end{vmatrix} = 0$$

6.28 Volume of parallelopiped and tetrahedron: Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$. Then

(i) Volume of the parallelopiped with $\vec{a}, \vec{b}, \vec{c}$ as coterminus edges is the absolute value of the determinant

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(ii) The volume of the tetrahedron with $\vec{a}, \vec{b}$ and $\vec{c}$ as coterminus edges is the absolute value of

$$\frac{1}{6} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(iii) Volume of the prism is half of the volume of the parallelopiped.

6.29 Vector product of three vectors: For any three vectors $\vec{a}, \vec{b}$ and $\vec{c}$

$$(i) (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \times \vec{c}) \times \vec{b} = (\vec{b} \times \vec{c}) \times \vec{a}$$

$$(ii) \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{c}) \times \vec{b} = (\vec{a} \times \vec{b}) \times \vec{c}$$

6.30 Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be vectors such that $\vec{a}$ and $\vec{b}$ are non-collinear and $\vec{b}$ is perpendicular to neither $\vec{a}$ nor $\vec{c}$. Then

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$$

if and only if the vectors $\vec{a}$ and $\vec{c}$ are collinear vectors.

6.31 Vector product and dot product of four vectors:

Let $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ be four vectors. Then

$$(i) (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \vec{c} \vec{d}] \vec{b} - [\vec{b} \vec{c} \vec{d}] \vec{a}$$

and also

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \vec{b} \vec{d}] \vec{c} - [\vec{a} \vec{b} \vec{c}] \vec{d}$$

From this we have

$$[\vec{a} \vec{b} \vec{c}] \vec{d} = [\vec{b} \vec{c} \vec{d}] \vec{a} + [\vec{c} \vec{a} \vec{d}] \vec{b} + [\vec{a} \vec{b} \vec{d}] \vec{c}$$

and hence if $[\vec{a} \vec{b} \vec{c}] \neq 0$ (i.e. $\vec{a}, \vec{b}, \vec{c}$ are non coplanar) then

$$\vec{d} = \frac{[\vec{b} \vec{c} \vec{d}]}{[\vec{a} \vec{b} \vec{c}]} \vec{a} + \frac{[\vec{c} \vec{a} \vec{d}]}{[\vec{a} \vec{b} \vec{c}]} \vec{b} + \frac{[\vec{a} \vec{b} \vec{d}]}{[\vec{a} \vec{b} \vec{c}]} \vec{c}$$

Therefore as per the space representation theorem (Theorem 5.20, Chapter 5), if $\vec{d} = x\vec{a} + y\vec{b} + z\vec{c}$, then

$$x = \frac{[\vec{b} \vec{c} \vec{d}]}{[\vec{a} \vec{b} \vec{c}]}, \quad y = \frac{[\vec{c} \vec{a} \vec{d}]}{[\vec{a} \vec{b} \vec{c}]} \quad \text{and} \quad z = \frac{[\vec{a} \vec{b} \vec{d}]}{[\vec{a} \vec{b} \vec{c}]}$$

$$(ii) (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix}$$

6.32 For any three vectors $\vec{a}, \vec{b}, \vec{c}$:

$$(i) [\vec{a} + \vec{b} \vec{b} + \vec{c} \vec{c} + \vec{a}] = 2[\vec{a} \vec{b} \vec{c}]$$

$$(ii) [\vec{b} \times \vec{c} \vec{c} \times \vec{a} \vec{a} \times \vec{b}] = [\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

6.33 Reciprocal systems: Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors. Define

$$\vec{a'} = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}$$

$$\vec{b'} = \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}$$

$$\vec{c'} = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}$$

Then $(\vec{a'}, \vec{b'}, \vec{c'})$ is called reciprocal system of $(\vec{a}, \vec{b}, \vec{c})$. In these systems the pairs $\vec{a}, \vec{a'}; \vec{b}, \vec{b'}; \vec{c}, \vec{c'}$

are called corresponding vectors and the other pairs are called non-corresponding vectors.

6.34 Properties of reciprocal systems:

$$(i) \vec{a} \cdot \vec{a'} = \vec{b} \cdot \vec{b'} = \vec{c} \cdot \vec{c'} = 1 \quad \text{and} \quad \vec{a} \cdot \vec{b'} = \vec{a} \cdot \vec{c'} = \vec{b} \cdot \vec{a'} = \vec{b} \cdot \vec{c'} = \vec{c} \cdot \vec{a'} = \vec{c} \cdot \vec{b'} = 0$$

$$(ii) [\vec{a} \vec{b} \vec{c}][\vec{a'} \vec{b'} \vec{c'}] = 1$$

$$(iii) \text{ If } (\vec{a'}, \vec{b'}, \vec{c'}) \text{ is the reciprocal system of } (\vec{a}, \vec{b}, \vec{c}) \text{ then } \vec{a''} = \vec{a}, \vec{b''} = \vec{b} \text{ and } \vec{c''} = \vec{c}.$$

Remark: Without using the name reciprocal systems, questions have been asked in the past JEE examinations.

Note: If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, then $\vec{a'}, \vec{b'}, \vec{c'}$ (defined above) are also non-coplanar and hence if $\vec{r}$ is any vector, such that $\vec{r} = x\vec{a'} + y\vec{b'} + z\vec{c'}$, then $x = \vec{r} \cdot \vec{a}, y = \vec{r} \cdot \vec{b}$ and $z = \vec{r} \cdot \vec{c}$.

6.35 Vector equation of a plane using a scalar triple product:

(i) The equation of the plane passing through a point $A(\vec{a})$ and parallel to two non-collinear vectors $\vec{b}$ and $\vec{c}$ is

$$[\vec{r} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]$$

(ii) The equation of the plane passing through two points $A(\vec{a}), B(\vec{b})$ and parallel to a vector $\vec{c}$ is

$$[\vec{r} \vec{b} \vec{c}] + [\vec{r} \vec{c} \vec{a}] = [\vec{a} \vec{b} \vec{c}]$$

(iii) The equation of the plane passing three non-collinear points $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$ is

$$[\vec{r} \vec{b} \vec{c}] + [\vec{r} \vec{c} \vec{a}] + [\vec{r} \vec{a} \vec{b}] = [\vec{a} \vec{b} \vec{c}]$$

(iv) The equation of the plane containing the line $\vec{r} = \vec{a} + t\vec{b}$ and perpendicular plane $\vec{r} \cdot \vec{c} = q$ is

$$[\vec{r} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]$$

Finally we conclude the summary with the concept of skew lines and shortest distance between two skew lines.

6.36 Skew lines: Any two non-coplanar lines are called skew lines. That is two lines are said to be skew lines, if there is no plane containing both.

6.37 Shortest distance between two skew lines: Two lines $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{c} + s\vec{d}$ where $s, t \in \mathbb{R}$ are skew lines if and only if $[\vec{a} - \vec{c} \vec{b} \vec{d}] \neq 0$. If these lines are skew, then the shortest distance between them is

$$\frac{|(\vec{a} - \vec{c}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}$$

EXERCISES

Single Correct Choice Type Questions

- If $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}$ and $\vec{b} = 2\vec{i} - \vec{j}$, then the angle between the vectors $\vec{a} - \vec{b}$ and $\vec{a} + \vec{b}$ is
 - $\cos^{-1}\left(\frac{1}{13}\right)$
 - $\cos^{-1}\left(\frac{1}{11}\right)$
 - $\frac{\pi}{3}$
 - $\frac{\pi}{4}$
- If the angle between $\vec{a}$ and $\vec{b}$ is 120° , $|\vec{a}| = 3$ and $|\vec{b}| = 4$, then the scalar product of $3\vec{a} - 2\vec{b}$ and $\vec{a} + 2\vec{b}$ is
 - 61
 - 41
 - 41
 - 61
- Let $\vec{a} = \vec{i} + \vec{j}$, $\vec{b} = \vec{i} - \vec{j}$. Suppose $\vec{x}$ and $\vec{y}$ are two vectors satisfying the two equations $2\vec{x} + \vec{y} = \vec{a}$ and $\vec{x} + 2\vec{y} = \vec{b}$. If θ is the angle between the two vectors $\vec{x}$ and $\vec{y}$, then $\cos \theta$ is equal to
 - 1
 - 0
 - $\frac{1}{\sqrt{2}}$
 - 1
- Let $\vec{a} = 6\vec{i} - 8\vec{j} - (15/2)\vec{k}$. If $\vec{b}$ is collinear with $\vec{a}$, making obtuse angle with z -axis and $|\vec{b}| = 50$, then $\vec{b}$ is equal to
 - $-24\vec{i} + 32\vec{j} + 30\vec{k}$
 - $24\vec{i} + 32\vec{j} - 30\vec{k}$
 - $24\vec{i} - 32\vec{j} - 30\vec{k}$
 - $24\vec{i} - 32\vec{j} - 15\vec{k}$
- $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors such that no two are collinear. If $\vec{a} + \vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + \vec{c}$ is collinear with $\vec{a}$, then $\vec{a} + \vec{b} + \vec{c}$ is equal to
 - $\vec{0}$
 - $2(\vec{a} + \vec{c})$
 - $2\vec{b}$
 - $3\vec{b}$
- If $A(3, 2, -3)$, $B(5, 1, -1)$ and $C(1, -2, 1)$ are the vertices of a triangle, then the external angle at the vertex A is
 - $\pi - \cos^{-1}\left(\frac{2}{9}\right)$
 - $\pi - \cos^{-1}\left(\frac{4}{9}\right)$
 - $\cos^{-1}\left(\frac{4}{9}\right)$
 - $\cos^{-1}\left(\frac{2}{9}\right)$
- In a trapezium $ABCD$, $\vec{BC} = \lambda \vec{AD}$. If the vector $\vec{x} = \vec{AC} + \vec{BD}$ is collinear with $\vec{AD}$ and $\vec{x} = \mu \vec{AD}$, μ equals
 - $\lambda - 1$
 - $1 - \lambda$
 - $2 + \lambda$
 - $\lambda + 1$
- In a parallelogram $ABCD$, it is given that $|\vec{AB}| = a$, $|\vec{AD}| = b$ and $|\vec{BD}| = c$, then the scalar product of $\vec{AC}$ and $\vec{AD}$ is equal to
 - $\frac{a^2 + 3b^2 - c^2}{2}$
 - $\frac{3a^2 + b^2 - c^2}{2}$
 - $\frac{a^2 + 2b^2 + c^2}{2}$
 - $\frac{c^2 + b^2 - a^2}{2}$
- $ABCD$ is a parallelogram in which $\vec{AB} = \vec{p}$ and $\vec{AD} = \vec{q}$. DM is drawn perpendicular to AB . Then $\vec{DM}$ is equal to
 - $\vec{q} - \frac{(\vec{p} \cdot \vec{q})\vec{p}}{|\vec{p}|^2}$
 - $\frac{(\vec{p} \cdot \vec{q})\vec{p}}{|\vec{p}|^2} - \vec{q}$
 - $\frac{(\vec{q} \cdot \vec{p})\vec{q}}{|\vec{p}|} - \vec{p}$
 - $\vec{q} - \frac{(\vec{q} \cdot \vec{p})\vec{p}}{|\vec{p}|}$
- In $\triangle ABC$, $\angle C = 90^\circ$ and $|\vec{AB}| = c$. Then the sum $\vec{AB} \cdot \vec{AC} + \vec{BC} \cdot \vec{BA} + \vec{CA} \cdot \vec{CB}$ equals
 - 0
 - $2c^2$
 - $3c^2$
 - c^2
- $A(1, y_1)$ and $B(x_2, 1)$ are points on the curve $y = x^2 - 4x + 5$ in the rectangular Cartesian coordinate system. If " O " is the origin of the axes, then the scalar product of $\vec{OA}$ and $\vec{OB}$ is
 - 4
 - 2
 - 6
 - 8
- In a quadrilateral $ABCD$, angle A is equal to 120° and the diagonal AC is the bisector of the angle A . If $|\vec{AC}| = (1/5)|\vec{AB}| = (1/3)|\vec{AD}|$ and θ is the angle between the vectors $\vec{BA}$ and $\vec{CD}$, then $\cos \theta$ is equal to
 - $\frac{1}{\sqrt{7}}$
 - $\frac{2}{\sqrt{7}}$
 - $\frac{3}{\sqrt{17}}$
 - $\frac{4}{\sqrt{17}}$
- If $\vec{a}$ and $\vec{b}$ are non-collinear vectors and

$$[(x-2)\vec{a} + \vec{b}] \times [(2x+1)\vec{a} - \vec{b}] = \vec{0}$$
 then the scalar x is equal to
 - $\frac{1}{2}$
 - $-\frac{1}{2}$
 - $-\frac{1}{3}$
 - $\frac{1}{3}$
- $ABCD$ is a rhombus having sides of one unit length. If the angle BAD is 30° , then $|\vec{AC}|$ is equal to
 - $\sqrt{2 - \sqrt{3}}$
 - $\sqrt{2 + \sqrt{3}}$
 - $\sqrt{1 + \sqrt{3}}$
 - $\sqrt{\sqrt{3} - 1}$
- The vectors $\vec{a}$ and $\vec{b}$ are non-zero vectors of equal magnitudes. If the vectors $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are orthogonal to each other, then the angle between $\vec{a}$ and $\vec{b}$ is
 - $\frac{\pi}{6}$
 - $\frac{\pi}{4}$
 - $\frac{\pi}{3}$
 - $\frac{2\pi}{3}$

16. In a triangle, the vertices are at the points $A(3, 2)$, $B(5, 1)$ and $C(1, 2)$. Then the angle at the vertex A is
 (A) $\cos^{-1} \frac{4}{9}$ (B) $\cos^{-1} \frac{2}{9}$
 (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$
17. The vector $\vec{a} + 3\vec{b}$ is perpendicular to the vector $7\vec{a} - 5\vec{b}$ and the vector $\vec{a} - 4\vec{b}$ is perpendicular to the vector $7\vec{a} - 2\vec{b}$. Then the angle between $\vec{a}$ and $\vec{b}$ is
 (A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$
18. $\vec{a}$ and $\vec{b}$ are two vectors which include angle 120° . If $\vec{a} + \lambda\vec{b}$ is orthogonal to the vectors $\vec{a} - \vec{b}$ and $|\vec{a}| = (1/2)|\vec{b}|$, then the value of λ is
 (A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{1}{5}$
19. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 1$ and $|\vec{c}| = 4$, then the value of $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is equal to
 (A) 12 (B) 12 (C) 13 (D) 13
20. The angle between the vectors $(\vec{a} \times \vec{c})\vec{b}$, $(\vec{a} \times \vec{b})\vec{c}$ and $\vec{a}$ is
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$
21. $ABCD$ is a quadrilateral and R is the radius of the circle circumscribing the quadrilateral. If $|\vec{AB}|^2 + |\vec{CD}|^2 = 4R^2$, then the angle between the diagonals AC and BD is
 (A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{12}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{3}$
22. In $\triangle ABC$, a line is drawn parallel to the side AB meeting AC in E and BC in F . If $AF = BE$, then $\triangle ABC$ is
 (A) right angled
 (B) right-angled isosceles
 (C) isosceles
 (D) equilateral
23. The area of $\triangle ABC$ is 6 square units. K is a point on the side AB such that $AK:KB = 2:3$. L is a point on the side AC such that $AL:LC = 5:3$. P is the point of intersection of the lines BL and CK inside the triangle. If the distance of P from the side of AB is $3/2$ units, then the length of the side AB is equal to
 (A) 6 (B) 5 (C) 4 (D) 3
24. OAB is a triangle right angled at the vertex O . Squares $OALM$ and $OBPQ$ on the sides OA and OB , respectively, are constructed externally. The lines AP and BL meet in H . Then the angle between $\vec{OH}$ and $\vec{AB}$ is
 (A) 90° (B) 60° (C) 45° (D) 75°
25. $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a}$ is perpendicular to both $\vec{b}$ and $\vec{c}$. If the angle between $\vec{b}$ and $\vec{c}$ is $\pi/3$, then the value of $|\vec{a} + \vec{b} + \vec{c}|$ is
 (A) 2 (B) 3 (C) $\frac{5}{2}$ (D) $\frac{7}{2}$
26. Let $\vec{i} + 2\vec{j} + 3\vec{k}$ and $2\vec{i} - \vec{j} - \vec{k}$ be the position vectors of the points A and B , respectively. Then the projection of $\vec{AB}$ on the vector $\vec{i} + \vec{j} + \vec{k}$ is
 (A) $2(\vec{i} + \vec{j} + \vec{k})$ (B) $\frac{7}{3}(\vec{i} + \vec{j} + \vec{k})$
 (C) $\frac{7}{3}(\vec{i} + \vec{j} + \vec{k})$ (D) $2(\vec{i} + \vec{j} + \vec{k})$
27. $\vec{AB} = 3\vec{i} - 2\vec{j} + 2\vec{k}$ and $\vec{BC} = \vec{i} - 2\vec{k}$ represent the adjacent sides of a parallelogram. Then the acute angle between the diagonals is
 (A) $\frac{\pi}{6}$ (B) $\cos^{-1} \sqrt{\frac{3}{10}}$
 (C) $\cos^{-1} \sqrt{\frac{3}{5}}$ (D) $\cos^{-1} \sqrt{\frac{2}{5}}$
28. Let π be a plane and P a point not in π . Q is the foot of the perpendicular drawn from P onto π . QM is drawn perpendicular to a line L in π . Then the angle between the lines PM and L is
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$
 (C) $\cos^{-1} \frac{3}{5}$ (D) $\cos^{-1} \frac{\sqrt{2}}{3}$
 (This result is called "Theorem of three perpendiculars" in solid geometry.)
29. Let $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} - 5\vec{k}$, $\vec{c} = x\vec{i} + 2\vec{j} + 3\vec{k}$. If the scalar product of $\vec{a}$ with a unit vector parallel to $\vec{b} + \vec{c}$ is unity, then the value of x is equal to
 (A) 5 (B) 1 (C) 3 (D) 1
30. Let $\vec{a} = 3\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} - 3\vec{j} + 4\vec{k}$. Then the area of the parallelogram with $\vec{a}$ and $\vec{b}$ as diagonal vectors is
 (A) $3\sqrt{5}$ (B) 3 (C) $5\sqrt{3}$ (D) 5

31. Let $A(2, 3, 5)$, $B(1, 3, 2)$ and $C(x, 5, y)$ be the vertices of $\triangle ABC$. If the median $\overline{AD}$ is equally inclined to the coordinate axes, then

(A) $x = 10, y = 7$ (B) $x = 7, y = 10$
(C) $x = 7, y = 10$ (D) $x = 10, y = 7$

32. Let $\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$, $\vec{c} = 3\vec{i} - \vec{j} + \vec{k}$ and $\vec{x} = \vec{a} + \vec{b} + \vec{c}$, $\vec{y} = \vec{a} - \vec{b} + \vec{c}$ and $\vec{z} = \vec{a} + \vec{b} - \vec{c}$. Then the volume of the parallelepiped with $\vec{x}$, $\vec{y}$ and $\vec{z}$ as coterminal edges is

(A) 24 (B) 6 (C) 28 (D) 7

(Hint: Use part (2) of Quick Look 11.)

33. Let $\vec{a} = \lambda(\vec{i} + 2\vec{j} - 3\vec{k})$, $\vec{b} = (2\lambda + 1)\vec{i} + (2\lambda + 3)\vec{j} + (\lambda + 1)\vec{k}$, $\vec{c} = (3\lambda + 5)\vec{i} - \vec{b} + \vec{c} + (\lambda + 5)\vec{j} + (\lambda + 2)\vec{k}$. Then $\vec{a}$, $\vec{b}$, $\vec{c}$ are

(A) coplanar for all real values of λ
(B) non-coplanar for all real values of λ
(C) non-coplanar for all non-zero real values of λ
(D) coplanar for all non-zero real values of λ

34. $\vec{b}$ and $\vec{c}$ are non-collinear vectors. $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 4$ and

$$\vec{a} \cdot (\vec{b} - \vec{c}) = (p^2 - 2p + 6)\vec{b} + (\sin q)\vec{c}$$

where $0 < q < \pi$. Then the point (p, q) is

(A) $(1, \pi)$ (B) $\left(\frac{\pi}{2}, 1\right)$ (C) $(1, 1)$ (D) $\left(1, \frac{\pi}{2}\right)$

35. Let L be the line of intersection of the planes $\vec{r} \times (3\vec{i} + 3\vec{j} + 2\vec{k}) = 0$ and $\vec{r} \times (\vec{i} + 2\vec{j} + 3\vec{k}) = 0$, then the angle between the line L and the unit vector along the positive X -axis is

(A) $\cos^{-1} \frac{5}{\sqrt{83}}$ (B) $\cos^{-1} \frac{8}{\sqrt{83}}$
(C) $\pi - \cos^{-1} \frac{7}{\sqrt{83}}$ (D) $\cos^{-1} \frac{7}{\sqrt{83}}$

36. Let $\vec{a} = \vec{i} + \vec{j} - \vec{k}$ and $\vec{b} = \vec{i} + 3\vec{k}$. If $\vec{c}$ is a unit vector such that $[\vec{a} \vec{b} \vec{c}]$ is maximum, then $\vec{c}$ can be

(A) $\frac{1}{\sqrt{26}}(3\vec{i} + 4\vec{j} + \vec{k})$ (B) $\frac{1}{\sqrt{26}}(3\vec{i} - 4\vec{j} - \vec{k})$
(C) $\frac{1}{\sqrt{26}}(4\vec{i} + 3\vec{j} - \vec{k})$ (D) $\frac{1}{\sqrt{26}}(\vec{i} - 3\vec{j} + 4\vec{k})$

37. Let $\vec{p} = p_1\vec{i} + p_2\vec{j} + p_3\vec{k}$ where p_1, p_2 and p_3 are rational numbers and $|\vec{p}|$ is also rational. If $\vec{p}$ makes an angle $\pi/4$ with the vector $\vec{a} = \sqrt{2}\vec{i} + 3\sqrt{2}\vec{j} + 4\vec{k}$, then $\vec{p}$ lies in

(A) XY -plane
(B) ZX -plane
(C) YZ -plane
(D) along the bisector of the angle between $\vec{k}$ and $\vec{a}$.

38. $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and

$$\vec{p} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$$

where x, y, z are scalars. If $\vec{p}$ is perpendicular to the vector $\vec{a} + \vec{b} + \vec{c}$, then

(A) $xy + yz + zx > 0$ (B) $x^3 + y^3 + z^3 = 3xyz$
(C) $x = y = z$ (D) $x^2 + y^2 + z^2 = xyz$

39. If $\vec{a}$ and $\vec{b}$ are orthogonal vectors, then $[(\vec{a} \times \vec{b}) \cdot \vec{a}] \cdot \vec{a}$ is equal to

(A) $|\vec{a}|^4 \vec{b}$ (B) $|\vec{a}|^4 \vec{b}$
(C) $|\vec{a}|^2 |\vec{b}|^2 \vec{a}$ (D) $|\vec{b}|^4 \vec{a}$

40. If $\vec{a}$ and $\vec{b}$ are non-collinear unit vectors, then $|(\vec{a} \times \vec{b}) \cdot (\vec{a} + \vec{b})|$ is equal to

(A) $2\sqrt{1 - (\vec{a} \times \vec{b})^2}$ (B) $\sqrt{2 - (\vec{a} \times \vec{b})^2}$
(C) $\sqrt{4 - (\vec{a} \times \vec{b})^2}$ (D) $\sqrt{4 - 2(\vec{a} \times \vec{b})^2}$

41. Let $\vec{b}, \vec{c}, \vec{d}$ be three non-coplanar vectors. Then for any vector $\vec{a}$, the vector $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) + (\vec{a} \times \vec{c}) \cdot (\vec{d} \times \vec{b}) + (\vec{a} \times \vec{d}) \cdot (\vec{b} \times \vec{c})$ is equal to

(A) $2[\vec{b} \vec{c} \vec{d}]\vec{a}$ (B) $2[\vec{c} \vec{b} \vec{d}]\vec{a}$
(C) $\vec{0}$ (D) $[\vec{a} \vec{b} \vec{c}]\vec{c}$

42. If the volume of tetrahedron having coterminal edges $12\vec{i} + \lambda\vec{k}, 3\vec{j} - \vec{k}$ and $2\vec{i} + \vec{j} - 15\vec{k}$ is 91, then one of the values of λ is

(A) 6 (B) 5 (C) 4 (D) 3

43. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar vectors and

$$\vec{a} = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}, \vec{b} = \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}, \vec{c} = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}$$

Then $[\vec{a} \vec{b} \vec{c}][\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}]$ equals

(A) $\vec{0}$ (B) $\vec{a} + \vec{b} + \vec{c}$
(C) $[\vec{a} \vec{b} \vec{c}](\vec{a} + \vec{b} + \vec{c})$ (D) $3(\vec{a} + \vec{b} + \vec{c})$

44. On the sides AB and BC of $\triangle ABC$, points P and Q are taken, respectively, such that $AP:PB = 2:3$ and $CQ:QB = 3:2$. The lines AQ and CP intersect in R .

If the area of BRC is one square unit, then the area of ABC is

- (A) $\frac{9}{19}$ (B) $\frac{7}{4}$ (C) $\frac{17}{4}$ (D) $\frac{19}{9}$

45. Let $\vec{a} = \vec{i} + \vec{j}$ and $\vec{b} = \vec{j} + \vec{k}$. $\vec{r}$ is a vector such that $\vec{r} \cdot \vec{a} = \vec{b} \cdot \vec{a}$ and $\vec{r} \cdot \vec{b} = \vec{a} \cdot \vec{b}$. Then $\vec{r}$ is equal to

- (A) $\vec{i} + \vec{j} - 2\vec{k}$ (B) $\vec{i} - \vec{j} + 2\vec{k}$
(C) $\vec{i} - \vec{j} + \vec{k}$ (D) $\vec{i} + 2\vec{j} + \vec{k}$

Multiple Correct Choice Type Questions

1. Let $A(1, 3, 7)$, $B(2, 1, 0)$ and $C(0, 1, 5)$ be three points. Then,

- (A) $|\overline{AB}| = \sqrt{74}$
(B) $(2\overline{AB} - \overline{CB}) \times (2\overline{BC} + \overline{BA}) = 19$
(C) $|\overline{BC}| = \sqrt{33}$
(D) $(\overline{AB} - \overline{CB}) \times (\overline{BC} + \overline{AB}) = 19$

2. If the vectors $\vec{a} = (3, x, 1)$ and $\vec{b} = (2, 1, y)$ are orthogonal vectors and $|\vec{a}| = |\vec{b}|$, then

- (A) $x = \frac{31}{12}$ (B) $y = \frac{41}{12}$
(C) $x = \frac{31}{12}$ (D) $y = \frac{41}{12}$

3. Let $\vec{a}, \vec{b}$ be two vectors such that $|\vec{a}| = 2\sqrt{2}$, $|\vec{b}| = 3$ and the angle between $\vec{a}$ and $\vec{b}$ is $\pi/4$. Then, the lengths of the diagonals constructed with $\vec{p} = 5\vec{a} + 2\vec{b}$ and $\vec{q} = \vec{a} - 3\vec{b}$ as adjacent sides of a parallelogram and its area are

- (A) 15 (B) $\sqrt{693}$
(C) $\sqrt{593}$ (D) 102

4. Consider the parallelepiped with $\overline{OA} = \vec{i} + \vec{j} + \vec{k}$, $\overline{OB} = 2\vec{i} + 4\vec{j} - \vec{k}$ and $\overline{OC} = \vec{i} + \vec{j} + 3\vec{k}$ as edges. Then

- (A) $[\overline{OA} \overline{OB} \overline{OC}] = 4$
(B) $[\overline{OA} \overline{OB} \overline{OC}] = 4$
(C) the length of the altitude CM drawn from the corner C onto the base of parallelogram having $\overline{OA}$ and $\overline{OB}$ as adjacent sides is $4/\sqrt{38}$

- (D) $CM = \frac{2}{3}$

5. Let $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} - \vec{j} + \vec{k}$, $\vec{c} = \vec{i} + \vec{j} - \vec{k}$ and $\vec{d} = \vec{i} + \vec{j} + \vec{k}$. Then

- (A) $(\vec{a} - \vec{b}) \cdot (\vec{c} - \vec{d}) = 8\vec{j}$
(B) $(\vec{a} - \vec{c}) \cdot (\vec{b} - \vec{d}) = 8\vec{k}$
(C) $(\vec{a} - \vec{b}) \times (\vec{c} - \vec{d}) = 0$
(D) $(\vec{a} - \vec{c}) \times (\vec{b} - \vec{d}) = 0$

6. Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be four vectors. Then

- (A) $[(\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{c})] \times \vec{d} = [\vec{a} \cdot \vec{b} \cdot \vec{c}] (\vec{d} \times \vec{a})$
(B) $[(\vec{a} - \vec{b}) \cdot (\vec{b} - \vec{c})] \times \vec{d} = [\vec{a} \cdot \vec{b} \cdot \vec{c}] (\vec{b} \times \vec{d})$
(C) $[(\vec{a} - \vec{c}) \cdot (\vec{b} - \vec{d})] \times \vec{d} = [\vec{a} \cdot \vec{b} \cdot \vec{c}] (\vec{b} \times \vec{d}) + [\vec{a} \cdot \vec{b} \cdot \vec{c}] (\vec{c} \times \vec{d})$
(D) $[(\vec{a} - \vec{d}) \cdot (\vec{c} - \vec{d})] \times \vec{d} = [\vec{a} \cdot \vec{c} \cdot \vec{d}] (\vec{d} \times \vec{d})$

7. Consider the four points $A(0, 1, 2)$, $B(3, 0, 1)$, $C(4, 3, 6)$ and $D(2, 3, 2)$. Then,

- (A) area of ABC is $3\sqrt{10}$
(B) the altitude drawn from A on to the side BC has magnitude $6/7\sqrt{14}$
(C) volume of the tetrahedron $ABCD$ is 6
(D) the shortest distance between the lines AB and CD is $\sqrt{6}$

8. Consider the line L whose equation is

$$\frac{x}{2} - \frac{1}{3} = \frac{y+1}{3} = \frac{z+10}{3}$$

Which of the following are correct?

- (A) The direction cosines of L are $(2, 3, 8)$.
(B) The foot of the perpendicular drawn from the point $P(1, 0, 0)$ onto the line is $(3, 4, 2)$.
(C) The image of the point $P(1, 0, 0)$ in the line L is $(5, 8, 4)$.
(D) The distance of the point $P(1, 0, 0)$ from the line L is $2\sqrt{6}$.

9. Let a, b, c be non-zero real numbers such that $a + b = 0$ and $c = 2b$. Then,

- (A) the vectors $\vec{p} = a\vec{i} + b\vec{j} + c\vec{k}$ and $\vec{r} = \vec{i} + \vec{j} + 2\vec{k}$ are collinear vectors
(B) the number of vectors $\vec{p} = a\vec{i} + b\vec{j} + c\vec{k}$ such that $|\vec{p}|$ is equal to $\sqrt{6}$ is infinite
(C) the magnitude of $\vec{p} = a\vec{i} + b\vec{j} + c\vec{k}$ is $2\sqrt{6}$ if $2a + 3b + 6c = 26$
(D) the vectors $\vec{p} = a\vec{i} + b\vec{j} + c\vec{k}$ and $\vec{q} = 4\vec{i} + 2\vec{j} + \vec{k}$ are perpendicular to each other

10. Let A , B and C be the points $(1, 1, 1)$, $(1, -1, 1)$ and $(-1, 3, 5)$, respectively. Then,
- (A) equation of the plane ABC is $3x - z - 2 = 0$
- (B) the vector $3\vec{i} - \vec{k}$ is normal to the plane ABC

- (C) foot of the perpendicular drawn from the point $(1, 1, 2)$ is $\frac{3}{10}, 1, \frac{19}{10}$
- (D) image of $(1, 1, 2)$ in the plane ABC is $(1, 1, 3)$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with *one or more* statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are (A) (p), (s); (B) (q), (s), (t); (C) (r); (D) (r), (t); that is if the matches are (A) (p) and (s); (B) (q), (s) and (t); (C) (r); and (D) (r), (t), then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

1. Match the items of Column I with those of Column II.

Column I	Column II
(A) In the rectangular Cartesian coordinate plane A and B are points on the curve $xy = 6$ such that $\vec{OA} \cdot \vec{AB} = 2$ and $\vec{OB} \cdot \vec{AB} = 3$ where O is the origin of the coordinate axes. Then $ 2\vec{OA} + 3\vec{OB} $ is equal to	(p) 1
(B) $\vec{a} = (2, 3, -5)$, $\vec{b} = (3, 0, 1)$ and $\vec{c} = (4, -3, 2)$. If $\vec{d} = 3\vec{a} + \vec{b} - \vec{c}$, then $\frac{1}{\sqrt{17}} \vec{d} $ is	(q) 1
(C) $A(-3, -2, 0)$, $B(3, -3, 1)$ and $C(5, 0, 2)$ are three consecutive vertices of a parallelogram. Then the sum of the coordinates of its fourth vertex is	(r) 5
(D) If $\vec{a} = (x, -2, 1)$ and $\vec{b} = (x, x, -3)$ are orthogonal vectors, then possible values of x are	(s) 3
	(t) 3

2. Match the items of Column I with those of Column II.

Column I	Column II
(A) If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $ \vec{a} = 3$, $ \vec{b} = 5$ and $ \vec{c} = 7$ then the angle between $\vec{a}$ and $\vec{b}$ is	(p) $\frac{\pi}{2}$
(B) If $\vec{a}$ and $\vec{b}$ are unit vectors such that $ \vec{a} - \vec{b} $ is also unit vector, then the angle between $\vec{a}$ and $\vec{b}$ is	(q) $\frac{\pi}{4}$
(C) In a regular tetrahedron (i.e., all edges are equal) the angle between opposite pair of edges is	(r) $\frac{\pi}{6}$
(D) If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors such that $ \vec{a} + \vec{b} = \vec{a} - \vec{b} $, then the angle between $\vec{a}$ and $\vec{b}$ is	(s) $\frac{\pi}{3}$

3. Match the items of Column I with those of Column II.

Column I	Column II
(A) The line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+3}{4}$ meets the plane $2x + 4y - z - 1 = 0$ in the point (a, b, c) , then the value of $a + b + c$ is	(p) 2
(B) The square of the reciprocal of the distance of the point $(1, -2, 3)$ from the plane $x - y + z - 5 = 0$ is	(q) 3
(C) The distance of the line $\frac{x-1}{3} = \frac{y+2}{2} = \frac{z-1}{2}$ from the plane $2x + 2y - z = 0$ is	(r) 7
(D) Let V be the volume of a tetrahedron σ and V' be the volume of the tetrahedron formed by the centroids of the faces of σ , then $V'/9V$ is equal to	(s) 1

4. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $[\vec{a} \ \vec{b} \ \vec{c}] = 2$. Match the items of Column I with those of Column II.

Column I	Column II
(A) The value of $[\vec{a} + \vec{b} \ \vec{b} + \vec{c} \ \vec{c} + \vec{a}]$ is	(p) 2
(B) The value of $[\vec{b} \times \vec{c} \ \vec{c} \times \vec{a} \ \vec{a} \times \vec{b}]$ equals	(q) 6
(C) $[(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})] \cdot \vec{c}$ is equal to	(r) 4
(D) $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ equals	(s) 8
	(t) 12

Comprehension-Type Questions

1. **Passage:** Let $\vec{a} = \vec{i} - \vec{j}$, $\vec{b} = \vec{j} - \vec{k}$ and $\vec{c} = \vec{k} - \vec{i}$. Answer the following three questions.

- (i) If $\vec{a} \cdot \vec{d} = 0 = [\vec{b} \ \vec{c} \ \vec{d}]$ and $|\vec{d}| = 1$, then $\vec{d}$ equals

(A) $\pm \frac{1}{\sqrt{6}}(\vec{i} + \vec{j} - 2\vec{k})$ (B) $\pm \frac{1}{\sqrt{3}}(\vec{i} + \vec{j} - \vec{k})$

(C) $\pm \frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$ (D) $\pm \vec{k}$

- (ii) $(\vec{a} \times \vec{b}) \times (\vec{a} \times \vec{c})$ is equal to

(A) $\vec{i} + \vec{j} + \vec{k}$ (B) $-2\vec{i} + 2\vec{j}$

(C) $2\vec{i} + \vec{k}$ (D) $\vec{0}$

- (iii) $[\vec{b} \times \vec{c} \ \vec{c} \times \vec{a} \ \vec{a} \times \vec{b}]$ is equal to

(A) 0 (B) 1 (C) 2 (D) 4

2. **Passage:** Let $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}$, $\vec{b} = \vec{i} - \vec{j} + \vec{k}$, $\vec{c} = \vec{i} + \vec{j} - \vec{k}$. Answer the following questions.

- (i) A vector coplanar with $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $1/\sqrt{3}$ is

(A) $4\vec{i} - \vec{j} + 4\vec{k}$ (B) $3\vec{i} + \vec{j} - 3\vec{k}$

(C) $2\vec{i} + \vec{j} - 2\vec{k}$ (D) $2\vec{i} + \vec{j} + 2\vec{k}$

- (ii) $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \end{vmatrix}$ equals

(A) $9(\vec{i} - \vec{k})$ (B) $9(\vec{i} - \vec{j})$

(C) $18(\vec{i} - \vec{k})$ (D) $18(2\vec{j} - \vec{k})$

(Hint: Use part (2) Quick Look 11.)

- (iii) The area of the parallelogram for which $\vec{a} + \vec{b}$ and $\vec{b} + \vec{c}$ are adjacent sides is

(A) $2\sqrt{2}$ (B) 2 (C) $3\sqrt{2}$ (D) $4\sqrt{2}$

3. **Passage:** If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors and $\vec{d} = x\vec{a} + y\vec{b} + z\vec{c}$, then

$$x = \frac{[\vec{b} \ \vec{c} \ \vec{d}]}{[\vec{a} \ \vec{b} \ \vec{c}]}, \quad y = \frac{[\vec{c} \ \vec{a} \ \vec{d}]}{[\vec{a} \ \vec{b} \ \vec{c}]}, \quad z = \frac{[\vec{a} \ \vec{b} \ \vec{d}]}{[\vec{a} \ \vec{b} \ \vec{c}]}$$

Now let $\vec{a} = 2\vec{i} - \vec{j} + \vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} - 2\vec{k}$. Answer the following three questions.

- (i) If $\vec{i} + \vec{j} + \vec{k} = x\vec{a} + y\vec{b} + z\vec{c}$, then

(A) $x = \frac{3}{8}, y = \frac{9}{8}, z = -\frac{7}{8}$

(B) $x = -\frac{3}{8}, y = \frac{9}{8}, z = -\frac{7}{8}$

(C) $x = \frac{3}{8}, y = \frac{9}{8}, z = \frac{7}{8}$

(D) $x = -\frac{3}{8}, y = -\frac{9}{8}, z = \frac{7}{8}$

- (ii) If $\vec{i} - \vec{j} + \vec{k} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$, then

(A) $x = -\frac{1}{2}, y = -\frac{1}{4}, z = \frac{1}{4}$

(B) $x = \frac{1}{2}, y = \frac{1}{4}, z = \frac{2}{8}$

(C) $x = -\frac{1}{2}, y = \frac{1}{4}, z = \frac{1}{4}$

(D) $x = -\frac{1}{8}, y = -\frac{1}{8}, z = \frac{2}{8}$

- (iii) $\vec{i} + \vec{j} - \vec{k} = x \frac{(\vec{b} \times \vec{c})}{[\vec{a} \ \vec{b} \ \vec{c}]} + y \frac{(\vec{c} \times \vec{a})}{[\vec{a} \ \vec{b} \ \vec{c}]} + z \frac{(\vec{a} \times \vec{b})}{[\vec{a} \ \vec{b} \ \vec{c}]}$, then

(A) $x = 0, y = 2, z = -2$ (B) $x = 0, y = 4, z = 4$

(C) $x = 0, y = -4, z = -2$ (D) $x = 0, y = -4, z = 4$

4. Passage: Let $\vec{a}$ and $\vec{b}$ be two non-zero non-collinear vectors. Then the projection vector of $\vec{b}$ on $\vec{a}$ is $\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$ and the vector component of $\vec{b}$ perpendicular to the direction of $\vec{a}$ is $\vec{b} - \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$. Answer the following three questions.

- (i) Let $\vec{a}, \vec{b}, \vec{c}$ be non-zero and non-coplanar vectors and

$$\vec{b}_1 = \vec{b} - \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$$

$$\vec{b}_2 = \vec{b} + \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$$

$$\vec{c}_1 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} + \frac{\vec{b} \cdot \vec{c}}{|\vec{c}|^2} \vec{b}_1$$

$$\vec{c}_2 = \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} + \frac{\vec{b} \cdot \vec{c}}{|\vec{b}_1|^2} \vec{b}_1 + \vec{c}$$

$$\vec{c}_3 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{c}|^2} \vec{a} + \frac{\vec{b} \cdot \vec{c}}{|\vec{c}|^2} \vec{b}_1$$

$$\vec{c}_4 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{c}|^2} \vec{a} = \frac{\vec{b} \cdot \vec{c}}{|\vec{b}|^2} \vec{b}_1$$

Then which of the following triads are orthogonal?

(A) $(\vec{a}, \vec{b}_1, \vec{c}_3)$

(B) $(\vec{a}, \vec{b}_1, \vec{c}_2)$

(C) $(\vec{a}, \vec{b}_1, \vec{c}_1)$

(D) $(\vec{a}, \vec{b}_2, \vec{c}_2)$

- (ii) Let $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}$, $\vec{b} = \vec{i} - \vec{j} + \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} - \vec{k}$. Then the projection vector of $\vec{c}$ on $\vec{a} - \vec{b}$ is

(A) $\sqrt{2}(\vec{i} - \vec{k})$

(B) $3\sqrt{2}(\vec{i} + 2\vec{k})$

(C) $\frac{2\sqrt{2}}{3}(\vec{i} - \vec{k})$

(D) $\vec{i} - \vec{k}$

- (iii) The component of $\vec{a}$ perpendicular to the direction of $(\vec{b} \times \vec{c})$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are same as in (ii) is

(A) $2\vec{i} + \vec{j} - \vec{k}$

(B) $\frac{1}{2}(2\vec{i} + \vec{j} - \vec{k})$

(C) $\vec{j} - \vec{k}$

(D) $\frac{1}{2}(\vec{j} - \vec{k})$

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
 (B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true

1. Statement I: Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, $\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$ where a_r, b_r, c_r ($r = 1, 2, 3$) are non-negative real numbers and $(a_r + b_r + c_r) = L$. If V is the volume of the parallelepiped with $\vec{a}, \vec{b}$ and $\vec{c}$ as coterminal edges, then $V \leq \frac{L^3}{3\sqrt{3}}$.

Statement II: The Arithmetic mean of non-negative real numbers is greater than or equal to their Geometric mean.

2. Statement I: A, B, C and D are four points such that for any point P in the space $\overrightarrow{AP} \times \overrightarrow{CP} = \overrightarrow{BP} \times \overrightarrow{DP}$. Then $ABCD$ is a parallelogram which cannot be a rectangle.

Statement II: If the points A, B, C and D are the vertices of a rectangle, then for any point Q , $\overrightarrow{QA} \times \overrightarrow{QC} = \overrightarrow{QB} \times \overrightarrow{QD}$ and $|\overrightarrow{QA}|^2 + |\overrightarrow{QC}|^2 = |\overrightarrow{QB}|^2 + |\overrightarrow{QD}|^2$.

(Hint: See multiple correct choice type question 7 in Worked-Out Problems.)

3. Statement I: The angle at the vertex A of $\triangle ABC$ is $\pi/6$, H is the orthocentre and M is the mid-point of BC . If T is a point on the line HM produced such that $HM = MT$, then $|\overrightarrow{AT}| = 2|\overrightarrow{BC}|$.

Statement II: In $\triangle ABC$, if O is the circumcentre and H is the orthocentre, then $\overrightarrow{OH} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$.

4. Statement I: The sum of the squares of the sides of any quadrilateral equals the sum of the squares

of its diagonals together with four times the square of the distance between the middle points of the diagonals.

Statement II: If $\vec{a}$ and $\vec{b}$ are two vectors, then $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 - 2(\vec{a} \times \vec{b}) + |\vec{b}|^2$.

5. Statement I: If the incident ray on a surface is along unit vector $\vec{v}$, the reflected ray along the unit vector $\vec{w}$ and the normal is along the unit ray $\vec{a}$ outward, then $\vec{w} = \vec{v} - 2(\vec{a} \times \vec{v})\vec{a}$.

Statement II: The incident vector, the reflected vector and the normal vector at the point of incidence are

coplanar and the angle of incidence is equal to the angle of reflection. See Figure 6.57.

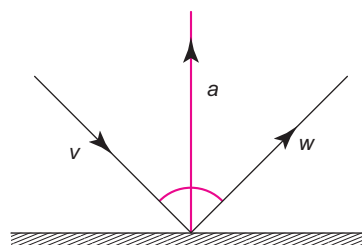


FIGURE 6.57 Assertion–reasoning type question 5.

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
0	0	0	0
1	1	1	1
2	●	2	2
3	3	3	3
4	●	4	4
5	5	5	5
6	●	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. The volume of a parallelepiped with $\vec{a}$, $\vec{b}$, $\vec{c}$ as coterminus edges is 3. Then the volume of the parallelepiped with $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$ as coterminus edges is _____.

2. The volume of a parallelepiped with coterminus edges $\vec{a}$, $\vec{b}$, $\vec{c}$ is 3. Then the value of the determinant

$$\begin{vmatrix} \vec{a} \times \vec{a} & \vec{a} \times \vec{b} & \vec{a} \times \vec{c} \\ \vec{b} \times \vec{a} & \vec{b} \times \vec{b} & \vec{b} \times \vec{c} \\ \vec{c} \times \vec{a} & \vec{c} \times \vec{b} & \vec{c} \times \vec{c} \end{vmatrix}$$

is _____.

3. If $\vec{e}_1$, $\vec{e}_2$, $\vec{e}_3$ are mutually perpendicular unit vectors forming a right-handed system, then the value of $[\vec{e}_1 + \vec{e}_2 + \vec{e}_3, \vec{e}_1 + \vec{e}_2, \vec{e}_2 + \vec{e}_3]$ is _____.

4. If $|\vec{a}| = 2$ and $\vec{a} \times \vec{b} = 0$, then $\vec{a} \cdot [\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b}))]$ is $n\vec{b}$ where $\sqrt{n}$ is _____.

5. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non-coplanar vectors and $\vec{p} = \vec{a} + 2\vec{b} + 3\vec{c}$, $\vec{q} = 2\vec{a} + 3\vec{b} + 4\vec{c}$ and $\vec{r} = \vec{a} + 3\vec{b} + 5\vec{c}$. Then $[\vec{p}, \vec{q}, \vec{r}]$ equals _____.

6. A (1, 1, 1), B (1, -1, 1) and C (0, 0, 1) are the vertices of a triangle. Then the area of $\triangle ABC$ in square units is _____.

7. M and N are the mid-points of the sides AC and AB, respectively, of $\triangle ABC$. Through the vertex C, a line is drawn parallel to the side AB meeting the line BM in P. Then the area of $\triangle ABC$ is equal to λ times the area of $\triangle PMN$ where λ equals _____.

8. D, E and F are points on the sides BC, CA and AB, respectively, such that $BD:DC = CE:EA = AF:FB = 2:1$. Then m times the area of $\triangle DEF$ is equal to n times the area of $\triangle ABC$ where m/n is equal to _____.

9. If V is the volume of the tetrahedron whose vertices are (2, -1, 3), (4, 1, 3), (3, 2, -1) and (1, 4, 2), then the integer part of V is _____.

10. If the vectors $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 4\vec{i} + 3\vec{j} + 4\vec{k}$, $\vec{c} = \vec{i} + \alpha\vec{j} + \beta\vec{k}$ are coplanar and $|\vec{c}| = \sqrt{3}$, then $\alpha^2 + \beta$ is _____.

ANSWERS**Single Correct Choice Type Questions**

- | | |
|---------|---------|
| 1. (B) | 24. (A) |
| 2. (D) | 25. (A) |
| 3. (A) | 26. (D) |
| 4. (C) | 27. (B) |
| 5. (A) | 28. (A) |
| 6. (B) | 29. (D) |
| 7. (D) | 30. (C) |
| 8. (A) | 31. (C) |
| 9. (B) | 32. (C) |
| 10. (D) | 33. (C) |
| 11. (A) | 34. (D) |
| 12. (B) | 35. (A) |
| 13. (D) | 36. (B) |
| 14. (B) | 37. (A) |
| 15. (C) | 38. (B) |
| 16. (A) | 39. (A) |
| 17. (B) | 40. (A) |
| 18. (C) | 41. (B) |
| 19. (D) | 42. (D) |
| 20. (B) | 43. (B) |
| 21. (A) | 44. (D) |
| 22. (C) | 45. (D) |
| 23. (C) | |

Multiple Correct Choice Type Questions

- | | |
|-----------------------|-------------------|
| 1. (A), (B), (C) | 6. (A), (B), (D) |
| 2. (C), (D) | 7. (A), (C), (D) |
| 3. (A), (C), (D) | 8. (B), (C), (D) |
| 4. (B), (C) | 9. (A), (D) |
| 5. (A), (B), (C), (D) | 10. (A), (B), (C) |

Matrix-Match Type Questions

- | | |
|--|---------------------------------------|
| 1. (A) (r), (B) (r), (C) (p), (D) (q), (t) | 3. (A) (q), (B) (q), (C) (s), (D) (q) |
| 2. (A) (s), (B) (s), (C) (p), (D) (p) | 4. (A) (r), (B) (r), (C) (r), (D) (r) |

Comprehension-Type Questions

- | | |
|---------------------------------|---------------------------------|
| 1. (i) (A); (ii) (D); (iii) (A) | 3. (i) (A); (ii) (C); (iii) (B) |
| 2. (i) (D); (ii) (C); (iii) (A) | 4. (i) (B); (ii) (D); (iii) (B) |

Assertion–Reasoning Type Questions

- | | |
|--------|--------|
| 1. (A) | 4. (A) |
| 2. (A) | 5. (A) |
| 3. (A) | |

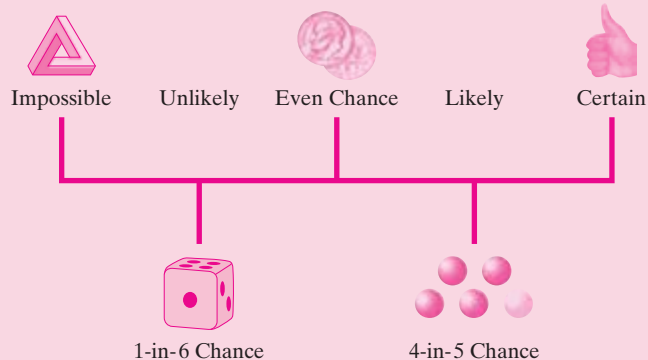
Integer Answer Type Questions

- | | |
|------|-------|
| 1. 6 | 6. 1 |
| 2. 9 | 7. 4 |
| 3. 1 | 8. 3 |
| 4. 4 | 9. 7 |
| 5. 0 | 10. 2 |

Probability

7

Probability



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Sometimes 'Human being' vices lead to great inventions. One such, gambling disease of some 17th century French people, is the main root for the origin of Probability. In modern day science, statistics and probability are treated as almost exact sciences. Probability is vastly applicable in pharmaceuticals, veterinary science and defence fields.

Probability is a measure of uncertainty and deals with the phenomenon of chance or randomness. We quite often make unknowingly some decisions about probability. For example, when one leaves his house in the morning on a cloudy day he may decide to take an umbrella with him, even if it is not raining, because he thinks that it may rain later in the day.

The theory of probability is developed to explain such kind of decisions mathematically. Infact, the theory of probability has its origins in gambling and games of chance. It is known that a French gambler Chevalier de Mere approached a French mathematician Blaise Pascal for a solution of a problem concerning gambling. Pascal gave a solution and subsequently he corresponded with another French mathematician Pierre de Farnat and established the foundations of the theory of probability. Later Laplace, a French scientist, made first attempt towards giving some mathematical rigour to the subject of probability. The present day theory of probability is credited to Andrei Nikolaevich Kolmogorov, a 20th century Russian probabilist, who laid the set theoretic foundations of probability.

7.1 | Random Experiments and Events

In this section we define certain important elementary technical terms related to probability and discuss some examples concerning these. Let us begin with the following.

DEFINITION 7.1 Random Experiment A random experiment is an experiment in which

1. The experiment can be repeated any number of times under identical conditions.
2. All possible outcomes of the experiment are known in advance.
3. The actual outcome in a particular experiment is not known in advance.

Examples

(1) Rolling of an Unbiased Die

Rolling of an unbiased die is a random experiment in which all the possible outcomes are 1, 2, 3, 4, 5 and 6, if we denote the six faces of the die with the numbers 1, 2, 3, 4, 5 and 6. The faces of the die may also contain dots in numbers 1, 2, 3, 4, 5 and 6. In any case, we identify the faces of a die with the numbers 1, 2, 3, 4, 5 and 6. The actual outcome in a particular experiment (rolling of the die) is the number that appears on the uppermost face of the die and this is not known in advance.

This experiment can be performed any number of times under identical conditions.

(2) Tossing of an Unbiased Coin

Tossing of an unbiased coin is a random experiment in which there are only two possible outcomes, namely, Head (H) and Tail (T). In a particular experiment (tossing of the coin), the outcome is not known in advance. This experiment can also be performed any number of times under identical conditions and therefore this is a random experiment.

“Tossing of an unbiased coin till tail appears” is also a random experiment. However, experiments such as “measuring the acceleration due to gravity using a compound pendulum” is not a random experiment, since the experiment cannot be repeated under identical conditions and the possible outcomes are not known in advance.

Throughout this chapter, by a coin or die we always mean an unbiased coin (or a fair coin) or unbiased die (or a fair die) unless otherwise mentioned.

DEFINITION 7.2 Some of the important definitions are as follows:

1. A set of possible outcomes of an experiment is called an *event*.
2. Two or more events are said to be *exhaustive* if the performance of the experiment always results in the occurrence of atleast one of them.
3. Two or more events are said to be *mutually exclusive* if the occurrence of one of the events prevents the occurrence of any of the remaining events.
4. Two or more events are said to be *equally likely* (or *equiprobable*) if there is no reason to expect one of them in preference to others.

Example

In rolling of a die, consider the events:

E_1 : occurrence of an even number (that is, an even number appears on the upper most face)

and E_2 : occurrence of an odd number.

Then E_1 and E_2 are exhaustive, mutually exclusive and equiprobable.

Example

If a coin is tossed, “the occurrence of Head” and “the occurrence of Tail” are exhaustive, mutually exclusive and equiprobable.

Example 7.1

Consider the experiment of “throwing a pair of dice and finding the sum of the two numbers that show up (on the upper most faces).” The possible outcomes (events) are numbers 2 to 12 (i.e., $a + b$ where $1 \leq a \leq 6$ and $1 \leq b \leq 6$). These events are exhaustive and mutually exclusive, but not equiprobable. Why?

Solution: These sums are not equally likely, as we can observe from the following table.

The sum 12 occurs only once, the sum 10 appears thrice and hence “the sum 12 occurs” and “the sum 10 occurs” are not equiprobable (or equally likely).

		Die-1					
Die-2	+	1	2	3	4	5	6
	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Example 7.2

In rolling of a die, consider the events

E_1 : occurrence of an even number

E_2 : occurrence of a prime number

and E_3 : occurrence of 1

Then E_1 , E_2 and E_3 are exhaustive, but not mutually exclusive. Why?

Solution: This is because 2 occurs in both E_1 and E_2 as it is both even and prime.

7.2 | Classical Definition of Probability

In the following we give a classical definition of probability and discuss certain examples.

DEFINITION 7.3 Suppose that in a random experiment there are n exhaustive, mutually exclusive and equiprobable possible outcomes. If m of them are favourable to the happening of an event E , then the *probability of occurrence* of E (or simply the *probability* of E) is defined as m/n and is denoted by $P(E)$. That is,

$$P(E) = \frac{m}{n}$$

Clearly $0 \leq P(E) \leq 1$.

Note that since the number of outcomes not favourable to an event E is $n - m$, the probability of non-occurrence of the event E is $(n - m)/n$ and is denoted by $P(E^c)$. That is,

$$P(E^c) = \frac{n - m}{n} = 1 - \frac{m}{n} = P(E)$$

and hence

$$P(E) + P(E^c) = 1$$

Example 7.3

Suppose that an integer is picked from among 1 to 20 (both inclusive). What is the probability of picking a prime?

Solution: There are 20 outcomes of the experiment of picking an integer. The primes between 1 and 20 are

2, 3, 5, 7, 11, 13, 17 and 19 and these are 8 in number. Therefore, 8 are favourable to the event of picking a prime and hence the probability of picking a prime is

$$\frac{8}{20} = \frac{2}{5}$$

Example 7.4

Suppose that 8 dice are thrown. Find the probability that none of the dice shows 3 (on the upper most face).

Solution: Any outcome of “throwing 8 dice” can be expressed as an 8-tuple of integers from 1 to 6 and hence the total number of possible outcomes is 6^8 .

An outcome that none of the dice shows 3 can be expressed as 8-tuple of integers from the 5-element set $\{1, 2, 4, 5, 6\}$ and there are 5^8 such outcomes. Thus, the probability that none of the dice shows 3 is

$$\frac{5^8}{6^8} = \left(\frac{5}{6}\right)^8$$

Example 7.5

Suppose that a bag contains 6 red, 5 black and 4 blue balls. Find the probability that three balls drawn simultaneously are one blue, one black and one red.

Solution: The total number of balls is

$$6 + 5 + 4 = 15$$

Out of these 15 balls, 3 balls can be drawn in ${}^{15}C_3$ ways. Therefore, 3 balls can be drawn simultaneously in

$$\frac{15 \cdot 14 \cdot 13}{1 \cdot 2 \cdot 3} = 455 \text{ ways}$$

Drawing one blue, one black and one red ball simultaneously can be expressed as a triple (a_1, a_2, a_3) , where $1 \leq a_1 \leq 4$, $1 \leq a_2 \leq 5$ and $1 \leq a_3 \leq 6$. The number of such tuples is

$$4 \times 5 \times 6 = 120$$

Thus, the probability that 3 balls drawn simultaneously are one blue, one black and one red is

$$\frac{120}{455} = \frac{24}{91}$$

Example 7.6

A number x is drawn arbitrarily from the set of integers from 1 to 100, both inclusive. Find the probability that

$$x > 29 - \frac{100}{x}$$

Solution: When we draw an arbitrary number x from the set $\{1, 2, \dots, 100\}$ the total number of outcomes is

100. Let A be the event that the number x has the property

$$x > 29 - \frac{100}{x}$$

Now

$$x > 29 - \frac{100}{x}$$

$$\Leftrightarrow x + \frac{100}{x} > 29$$

$$\Leftrightarrow x^2 - 29x + 100 > 0$$

$$\Leftrightarrow (x - 25)(x - 4) > 0$$

$$\Leftrightarrow x > 25 \quad \text{or} \quad x < 4$$

$$\Leftrightarrow x \in \{1, 2, 3\} \cup \{26, 27, 28, \dots, 100\}$$

Therefore the number of cases favourable to A is 78.
Thus the required probability is

$$P(A) = \frac{78}{100} = 0.78$$

Example 7.7

Out of 30 consecutive integers two are drawn at random. Find the probability that their sum is odd.

Solution: The total number of ways of choosing 2 out of 30 integers is

$${}^{30}C_2 = 435$$

Out of these 30 integers, 15 are odd and 15 are even (since the given integers are consecutive). For the sum

of the chosen two integers to be odd, one must be odd and the other must be even. Therefore, the number of cases favourable is

$$15 \times 15 = 225$$

Thus, the required probability is equal to

$$\frac{225}{435} = \frac{15}{29}$$

7.2.1 Limitations of the Classical Definition

1. If the outcomes of a random experiment are not equally likely, the probability of an event in such an experiment cannot be found. For example, the probability of a student passing an examination is not $1/2$ as the outcomes of success and failure in the examination are not equally likely.
2. If the random experiment contains infinitely many outcomes, then the classical definition cannot be applied for the probability of an event in such an experiment. For example, tossing a coin until Head appears is a random experiment in which there are infinite number of outcomes, namely all finite sequences of H's and T's (Heads and Tails). The classical definition gives no answer for finding the probability that the Head never appears before 100th tossing.

We have noticed certain limitations and deficiencies in the classical definition of probability. In order to overcome one of these limitations, we now give the following definition which is known as the *statistical* (or *empirical* or *relative frequency*) *definition of probability*.

DEFINITION 7.4 Let E be an event of a random experiment. Let the experiment be repeated n times out of which E occurs m_n times. Then the ratio

$$r_n = \frac{m_n}{n}$$

is called the n th *relative frequency* of the event E . Clearly $0 \leq m_n \leq n$ and hence $0 \leq r_n \leq 1$. If $\{r_n\}$ is a convergent sequence, then its limit is defined to be the *probability* of E and is denoted by $P(E)$. That is,

$$P(E) = \lim_{n \rightarrow \infty} r_n = \lim_{n \rightarrow \infty} \frac{m_n}{n}$$

It can be easily seen that $0 \leq P(E) \leq 1$, since $0 \leq r_n \leq 1$ for all $n \in \mathbb{Z}^+$.

One can easily observe the following deficiencies in the above statistical definition of probability

1. Repeating a random experiment infinitely many times is impossible.
2. The sequence of relative frequencies must be assumed to be convergent, which may not be true all the time.

7.3 | Axiomatic Approach to Probability

We have noticed certain shortcomings and deficiencies both in the classical definition (Definition 7.3) and the statistical definition (Definition 7.4). In order to overcome these shortcomings and deficiencies, a Russian probabilist A.N. Kolmogorov has proposed an axiomatic approach to probability. Before going to this, first let us have certain basic definitions.

DEFINITION 7.5

1. An outcome of a random experiment is called an *elementary event* or *simple event*.
2. The set of all outcomes of a random experiment is called the *sample space* associated with the experiment. In other words, when ξ is a random experiment and S is the sample space associated with ξ , then every element of S is an outcome of ξ and any performance of ξ results in an outcome that corresponds to exactly one element of S .

DEFINITION 7.6 Some important terms are:

Let S be the sample space associated with a random experiment ξ .

1. The elements of S are called the *sample points* or *elementary events* of ξ .
2. A subset E of S is called an *event*. That is, a set of elementary events is called an event.
3. An event E is said to *happen* or *occurs* if an outcome of the experiment ξ belongs to E .
4. Otherwise, we say that E *does not happen* or E *does not occur*.
5. The empty set ϕ and the whole set S , being subsets of S , are called *impossible event* and *certain* (or *sure*) *event*, respectively.
6. The complement of the event E is the event given by $S-E$ and is denoted by E^c or $\bar{E}$. Such event called the *complementary event* of E . Usually E^c is denoted by $\bar{E}$.

Example

Let ξ be the experiment of rolling a die. Then the sample space of ξ is the set $S = \{1, 2, 3, 4, 5, 6\}$, where the integers denote the faces of the die. “Occurrence of a positive integer ≤ 6 on the uppermost face of the die” is a sure

event, “occurrence of a positive even integer ≤ 6 on the uppermost face of the die” is an event and “occurrence of a prime ≥ 6 on the uppermost face of the die” is an impossible event.

DEFINITION 7.7

1. A set S is said to be *countably infinite* if it is bijective with the set $\mathbb{Z}^+$ of positive integers and, in this case we can write $S = \{s_1, s_2, \dots, s_n, \dots\}$.
2. A set S is said to be *finite* if S is empty or S is bijective with the set $\{1, 2, \dots, n\}$ for some $n \in \mathbb{Z}^+$ and in this case we write $S = \{s_1, s_2, \dots, s_n\}$ and say that S is an *n-element set* or the number of elements in S is n .
3. A set S is said to be *atmost countable* if S is either finite or countably infinite. For example $\mathbb{Z}^+, \mathbb{Z}$ are countably infinite sets.

Note that the sample space S of a random experiment ξ may be finite or infinite. Throughout our discussions in this chapter, we take the sample space S of an experiment to be atmost countable; that is, either finite or countably infinite. For example, the sample space in Examples (1) and (2) below Definition 7.1 are finite and that of the experiment “Tossing a coin till tail appears” is countably infinite.

DEFINITION 7.8 Let $E_1, E_2, \dots, E_n$ be some events of a random experiment ξ . That is, $E_1, E_2, \dots, E_n$ are subsets of the sample space S of ξ .

1. The events $E_1, E_2, \dots, E_n$ are said to be *mutually exclusive* if $E_i \cap E_j = \phi$ for all $i \neq j$; that is, if the happening of an event E_i prevents the happening of any other event $E_j, j \neq i$.
2. The events $E_1, E_2, \dots, E_n$ are said to be *equiprobable* or *equally likely* if there is no reason to expect one of them to happen in preference to others.

3. The events $E_1, E_2, \dots, E_n$ are said to be exhaustive if $E_1 \cup E_2 \cup \dots \cup E_n = S$; that is, if the performance of the experiment always results in the occurrence of at least one of E_i 's.

Note that for any random experiment ξ whose sample space is S , the set of all possible events of ξ is $\wp(S)$, the power set of S (the set of all subjects of S). In the following we introduce Kolmogorov's axiomatic approach to the theory of probability.

DEFINITION 7.9 Let ξ be a random experiment and S its sample space and suppose that S is finite. Then a function $P: \wp(S) \rightarrow \mathbb{R}$ is called a *probability function* if it satisfies the following conditions.

1. **Positive axiom:** $P(E) \geq 0$ for all $E \in \wp(S)$
2. **Completeness axiom:** $\wp(S) = 1$
3. **Union axiom:** If E_1 and $E_2 \in \wp(S)$ and $E_1 \cap E_2 = \phi$, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

In this case, for any $E \in \wp(S)$, $P(E)$ is called the *probability of the event E* . If $E = \{s\}$, then we write $P(s)$ for $P(\{s\})$ for simplicity. In the case when S is countably infinite, we have to replace (3) above by (3') given below.

3'. If $\{E_n\}$ is a sequence of mutually exclusive events, then

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

provided this infinite sum exist.

Example 7.8

Let $S = \{1, 2, 3, 4, 5, 6\}$ be the sample space of a random experiment. Define $P(E) = \sum_{s \in E} P(s)$ for any $E \subseteq S$, where

$$\begin{aligned} P(1) &= 0.1 \\ P(2) &= 0.2 = P(3) \\ P(4) &= 0.4 \\ P(5) &= 0.05 = P(6) \end{aligned}$$

Show that P is a probability function.

Solution: We have

(1) Positive axiom:

$$P(E) = \sum_{s \in E} P(s) > 0$$

since $P(s) > 0$ for all $s \in S$.

(2) Completeness axiom:

$$\begin{aligned} P(S) &= \sum_{s \in S} P(s) = P(1) + P(2) + P(3) + P(4) + P(5) + P(6) \\ &= 0.1 + 0.2 + 0.2 + 0.4 + 0.05 + 0.05 \\ &= 1 \end{aligned}$$

(3) Union axiom: For any E_1 and $E_2 \in \wp(S)$ with $E_1 \cap E_2 = \phi$, we have

$$P(E_1 \cup E_2) = \sum_{s \in E_1 \cup E_2} P(s) = \sum_{s \in E_1} P(s) + \sum_{s \in E_2} P(s) = P(E_1) + P(E_2)$$

since $s \in E_1 \cup E_2$ if and only if $s \in E_1$ or $s \in E_2$ but not both.

Thus P is a probability function.

Try it out Let $S = \{H, T\}$ be the sample space of a random experiment ξ and define

$$P(H) = \frac{1}{2} = P(T)$$

$$P(\phi) = 0$$

$$P(S) = 1$$

and

Then verify that P is a probability function.

In the following we prove certain important elementary properties of probability functions.

THEOREM 7.1

The following hold good for any probability function P on $\wp(S)$, where S is the sample space of a random experiment ξ .

1. $P(\phi) = 0$
2. If E^c is the complementary event of E , then

$$P(E^c) = 1 - P(E)$$

3. $0 \leq P(E) \leq 1$ for all $E \in \wp(S)$
4. If $E_1 \subseteq E_2 \subseteq S$, then $P(E_2 - E_1) = P(E_2) - P(E_1)$
5. If $E_1 \subseteq E_2 \subseteq S$, then $P(E_1) \leq P(E_2)$
6. If $E_1, E_2, \dots, E_n$ are mutually exclusive events, then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

PROOF

All these are derived from the axioms (1), (2) and (3) of Definition 7.9.

1. Since $S \cap \phi = \phi$, we have, from Axiom (3) of Definition 7.9 that

$$P(S) = P(S \cup \phi) = P(S) + P(\phi)$$

and hence $P(\phi) = 0$.

2. Since $E \cap E^c = \phi$ and $E \cup E^c = S$, we have

$$1 = P(S) = P(E \cup E^c) = P(E) + P(E^c)$$

and hence $P(E^c) = 1 - P(E)$

3. Since $1 - P(E) = P(E^c) \geq 0$ [by Axiom (1), Definition 7.9], we get that

$$0 \leq P(E) \leq 1$$

4. Let $E_1 \subseteq E_2 \subseteq S$, Then

$$E_2 = E_1 \cup (E_2 - E_1)$$

and

$$E_1 \cap (E_2 - E_1) = \phi$$

Therefore, we have

$$P(E_2) = P(E_1) + P(E_2 - E_1)$$

and hence

$$P(E_2 - E_1) = P(E_2) - P(E_1)$$

5. If $E_1 \subseteq E_2 \subseteq S$, then

$$P(E_2) - P(E_1) = P(E_2 - E_1) \geq 0$$

[by Axiom (1), Definition 7.9] and hence

$$P(E_1) \leq P(E_2)$$

6. This follows by induction on n and by using Axiom (3), Definition 7.9. ■

Note: For any event E , $P(E):P(\bar{E})$ (ratio) is called odds in favour of E and the ratio $P(\bar{E}):P(E)$ is odds against E .

THEOREM 7.2
(ADDITION
THEOREM ON
PROBABILITY)

Let A , B and C be three events of a random experiment ξ and P is a probability function on $\wp(S)$, where S is the sample space of ξ . Then the following hold.

1. $P(A \cup B) + P(A \cap B) = P(A) + P(B)$
2. $P(A - B) = P(A) - P(A \cap B)$
3. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

PROOF

1. From the Venn diagram in Figure 7.1, we have

$$A \cup B = [A - (A \cap B)] \cup (A \cap B) \cup [B - (A \cap B)]$$

Now $A - (A \cap B)$, $A \cap B$ and $B - (A \cap B)$ are mutually exclusive and hence

$$\begin{aligned} P(A \cup B) &= P[A - (A \cap B)] + P(A \cap B) + P[B - (A \cap B)] \\ &= P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B) \quad [\text{by (4), Theorem 7.1}] \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

Thus

$$P(A \cup B) + P(A \cap B) = P(A) + P(B)$$

2. Since $A - B = A - (A \cap B)$ and $A \cap B \subseteq A$, we have

$$P(A - B) = P[A - (A \cap B)] = P(A) - P(A \cap B)$$

3. We make use of (1) repeatedly. Consider

$$\begin{aligned} P(A \cup B \cup C) &= P(A \cup B) + P(C) - P[(A \cup B) \cap C] \\ &= P(A) + P(B) - P(A \cap B) + P(C) - P[(A \cap C) \cup (B \cap C)] \\ &= P(A) + P(B) + P(C) - P(A \cap B) - [P(A \cap C) + P(B \cap C) - P(A \cap C \cap B \cap C)] \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \end{aligned}$$

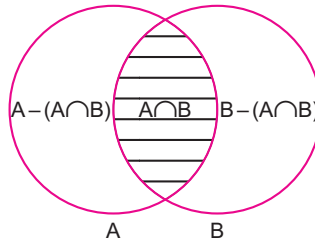


FIGURE 7.1 Theorem 7.2.

Note that a probability function P on $\wp(S)$, where S is given to be finite, is completely determined by $P(s)$, $s \in S$. If $P(s)$ is a non-negative real number for each s in a finite set S such that

$$\sum_{s \in S} P(s) = 1$$

then the function $P: \wp(S) \rightarrow \mathbb{R}$, defined by

$$P(A) = \sum_{s \in A} P(s)$$

for any $A \subseteq S$, is a probability function.

In Table 7.1 we give set theoretic descriptions of various events, which will be useful in solving problems on probability later.

Table 7.1 Set theoretic descriptions of various events

S. No.	Event	Set theoretic description
1.	Events A or B to occur	$A \cup B$
2.	Events A and B to occur	$A \cap B$
3.	A occurs and B does not occur	$A - B$ or $A \cap B^c$ or $A \cap \bar{B}$
4.	Neither A nor B occurs	$A^c \cap B^c$ or $\bar{A} \cap \bar{B}$
5.	Exactly one of the events A and B occurs	$(A \cap \bar{B}) \cup (\bar{A} \cap B)$ or $(A \cap B^c) \cup (A^c \cap B)$ or $(A \cup B) - (A \cap B)$
6.	Not more than one of the events A or B occurs	$(A \cap B^c) \cup (A^c \cap B) \cup (A^c \cap B^c)$ or $(A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (\bar{A} \cap \bar{B})$
7.	The event B occurs whenever A occurs	$A \subseteq B$

Example 7.9

A pack of cards means a pack containing 52 cards, out of which 26 are with red figures and another 26 are with black figures. These 52 cards are divided into 4 sets, namely Hearts (in red colour), Diamonds (in red colour), Spades (in black) and Clubs (in black). Each set consists of 13 cards labelled as

A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K

where A is Ace, J is Jack, Q is Queen, K is King. Find the probability of drawing an ace or a spade from a well-shuffled pack of 52 cards.

Solution: Let E_1 and E_2 be the events of drawing an ace and drawing a spade, respectively. Then we have to find $P(E_1 \cup E_2)$. Note that E_1 and E_2 are not mutually exclusive, since there is a spade in aces. In fact, $E_1 \cap E_2$ has exactly one member. Note that

$$n(E_1) = 4, n(E_2) = 13 \quad \text{or} \quad n(E_1 \cap E_2) = 1$$

Therefore

$$P(E_1) = \frac{4}{52}, P(E_2) = \frac{13}{52} \quad \text{and} \quad P(E_1 \cap E_2) = \frac{1}{52}$$

So

$$\begin{aligned} P(E_1 \cup E_2) &= P(E_1) + P(E_2) - P(E_1 \cap E_2) \\ &= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} \\ &= \frac{16}{52} = \frac{4}{13} \end{aligned}$$

Thus $4/13$ is the probability of drawing an ace or a spade from the pack.

Example 7.10

A , B and C are three newspapers from a city. 25% of the population reads A , 20% reads B , 15% reads C , 16% reads both A and B , 10% reads both B and C , 8% reads both A and C and 4% reads all the three. Find the percentage of the population who read atleast one of A , B and C .

Solution: We are given that

$$\begin{aligned} P(A) &= \frac{25}{100}, P(B) = \frac{20}{100}, P(C) = \frac{15}{100} \\ P(A \cap B) &= \frac{16}{100}, P(B \cap C) = \frac{10}{100}, P(C \cap A) = \frac{8}{100} \end{aligned}$$

and

$$P(A \cap B \cap C) = \frac{4}{100}$$

We have to find $P(A \cup B \cup C)$. We can use the formula

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) \\ &\quad - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \\ &= \frac{1}{100}(25 + 20 + 15 - 16 - 10 - 8 + 4) \\ &= \frac{30}{100} \end{aligned}$$

Thus 30% of the people read atleast one of the newspapers.

Example 7.11

If two numbers are selected randomly from 30 consecutive natural numbers, find the probability that the sum of the two numbers is

- (a) even.
(b) odd.

Solution: Two numbers can be selected from 30 in ${}^{30}C_2 = 435$ ways. Since the given 30 numbers are consecutive, there are 15 even and 15 odd numbers among them.

- (a) For $a + b$ to be even, it is necessary and sufficient that either both a and b are even or both odd. There

are ${}^{15}C_2 + {}^{15}C_2$ such ways. Therefore the probability that the sum of the two numbers selected is even is

$$\frac{{}^{15}C_2 + {}^{15}C_2}{{}^{30}C_2} = \frac{15 \times 14}{435} = \frac{14}{29}$$

- (b) The event “ $a + b$ is odd” is complementary to the event “ $a + b$ is even” and therefore, the probability that the sum of the two numbers selected odd is

$$1 - \frac{14}{29} = \frac{15}{29}$$

7.4 | Independent and Dependent Events

An icecream vendor sells more icecreams during a sunny day than other days. The probability of getting more profit depends on the weather conditions that the day is sunny and hot. Situations like these lead to the following, in which we assume that there is a probability function P on the set of events of a random experiment.

DEFINITION 7.10 Let ξ be a random experiment and A and B be two events of ξ . Then the event “occurrence of B after the occurrence of A ” is called a *conditional event* and is denoted by B/A . Similarly, the event “occurrence of A after the occurrence of B ” is denoted by A/B .

DEFINITION 7.11 Let A and B be two events of a random experiment ξ . The conditional probability of occurrence of A after the occurrence of B is defined by

$$\frac{P(A \cap B)}{P(B)}$$

where $P(B) > 0$ and is denoted by $P(A/B)$. That is,

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \quad \text{when } P(B) > 0$$

Similarly,

$$P(B/A) = \frac{P(B \cap A)}{P(A)}, \quad \text{when } P(A) > 0$$

**QUICK LOOK 1**

$$P(A/B) = \frac{\text{Number of favourable cases to } A \text{ among } B}{\text{Number of favourable cases to } B}$$

Example 7.12

A pair of fair dice is thrown. Find the probability that either of the dice shows 2 if the sum is 6.

Solution: The sample space of the experiment “throwing a pair of fair dice” consists of $36 (= 6 \times 6)$ ordered pairs (a, b) , where a and b can be any integers

from 1 to 6. Let A be the event “2 appears on either of the dice” and B be the event “sum is 6”. We want to find $P(A/B)$. Note that

$$A = \{(2, b) | 1 \leq b \leq 6\} \cup \{(a, 2) | 1 \leq a \leq 6\}$$

$$\text{and } B = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

Also,

$$A \cap B = \{(2, 4), (4, 2)\}$$

Therefore

$$P(B) = \frac{5}{36} \quad \text{and} \quad P(A \cap B) = \frac{2}{36}$$

So

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{2/36}{5/36} = \frac{2}{5}$$

Example 7.13

In a class, 30% of the students failed in Physics, 25% failed in Mathematics and 15% failed in both Physics and Mathematics. If a student is selected at random failed in Mathematics, find the probability that he failed in Physics also.

Solution: Let A be the event “failed in Physics” and B be the event “failed in Mathematics”. We want to find $P(A/B)$. It is given that

$$P(A) = \frac{30}{100} \quad \text{and} \quad P(B) = \frac{25}{100}$$

Also,

$$P(A \cap B) = \frac{15}{100}$$

Therefore

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{15/100}{25/100} = \frac{15}{25} = \frac{3}{5}$$

THEOREM 7.3 (MULTIPLICATION THEOREM ON CONDITIONAL PROBABILITY)

PROOF

Let A and B be two events of a random experiment such that $P(A) > 0$ and $P(B) > 0$. Then

$$P(A \cap B) = P(A)P(B/A) = P(B)P(A/B)$$

This is an immediate consequence of the definition of the conditional probabilities $P(A/B)$ and $P(B/A)$. This can be extended to any finite number of events $E_1, E_2, \dots, E_n$, by using induction on n . ■

COROLLARY 7.1

Let $E_1, E_2, \dots, E_n$ be n events of a random experiment such that

$$P\left(\bigcap_{i=1}^{n-1} E_i\right) > 0$$

Then

$$P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1)P(E_2/E_1)P(E_3/(E_1 \cap E_2)) \cdots P\left(E_n/\left(\bigcap_{i=1}^{n-1} E_i\right)\right)$$

Example 7.14

A bag contains 20 identical balls of which 8 are black and 12 are blue. Three balls are taken out at random from the bag one after the other without replacement. Find the probability that all the three balls drawn are blue.

Solution: The probability that the first ball drawn is blue is $12/20$, since there are 12 blue balls among 20 balls in the bag. If the first ball is blue, then the probability that the second ball drawn is blue is $11/19$, since 11 of the remaining 19 are blue. Similarly, if the first two balls drawn are blue,

then the probability that the third ball drawn is blue is $10/18$. Therefore, by Corollary 7.1, the required probability is

$$\frac{12}{20} \cdot \frac{11}{19} \cdot \frac{10}{18} = \frac{11}{57}$$

Note: If the drawn ball is replaced every time, then the probability is

$$\left(\frac{12}{20}\right)^3 = \left(\frac{3}{5}\right)^3$$

Example 7.15

A jar contains 10 white balls and 6 blue balls, all are of equal size. Two balls are drawn without replacement. Find the probability that the second ball is white if it is known that the first is white.

Solution: Let E_1 be the event “the first ball drawn is white” and E_2 be the event “the second ball drawn is white again. Then,

$$P(E_1) = \frac{10}{16}$$

since 10 out of 10 + 6 balls are white. But, after one ball is chosen, there remain 9 white balls and 6 blue balls. Therefore the required probability is

$$\begin{aligned} P(E_2/E_1) &= \frac{P(E_1 \cap E_2)}{P(E_1)} \\ &= \frac{10}{16} \cdot \frac{9}{15} = \frac{9}{15} = \frac{3}{5} \end{aligned}$$

THEOREM 7.4

Let S be the sample space of a random experiment ξ and P be a probability function on $\wp(S)$, the set of all events of ξ . Let $A \in \wp(S)$ such that $P(A) > 0$. Define $P_A: \wp(S) \rightarrow \mathbb{R}$ by

$$P_A(E) = P(E/A)$$

for any $E \in \wp(S)$. Then P_A is also a probability function.

PROOF

Recall that

$$P_A(E) = P(E/A) = \frac{P(E \cap A)}{P(A)}$$

Since $P(A) > 0$ and $P(E \cap A) \geq 0$, $P_A(E) \geq 0$ for all $E \in \wp(S)$. Also,

$$P_A(S) = \frac{P(S \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

since $A \subseteq S$. Further, let E_1 and E_2 be two mutually exclusive events (i.e., E_1 and $E_2 \subseteq S$ and $E_1 \cap E_2 = \phi$). Then,

$$\begin{aligned} P_A(E_1 \cup E_2) &= \frac{P[(E_1 \cup E_2) \cap A]}{P(A)} \\ &= \frac{P[(E_1 \cap A) \cup (E_2 \cap A)]}{P(A)} \\ &= \frac{P(E_1 \cap A) + P(E_2 \cap A)}{P(A)} \quad (\text{since } E_1 \cap A \cap E_2 \cap A = \phi) \\ &= P_A(E_1) + P_A(E_2) \end{aligned}$$

Thus, P_A is a probability function on $\wp(S)$. ■

DEFINITION 7.12

Let ξ be a random experiment and S its sample space. Let P be a probability function on $\wp(S)$. Two events A and B are said to be *independent of each other* if the occurrence of one of them does not influence the occurrence of the other. That is, B is independent of A if

$$P(B) = P(B/A)$$

If A and B are not independent, then we say that they are *dependent*.

THEOREM 7.5

The following are equivalent to each other for any two events A and B of a random experiment.

1. A and B are independent of each other.
2. $P(B) = P(B/A)$

3. $P(A) = P(A/B)$
 4. $P(A \cap B) = P(A) P(B)$

PROOF(1) $\Leftrightarrow$ (2) follows from Definition 7.12.(2) $\Leftrightarrow$ (3) $\Leftrightarrow$ (4) follow from the multiplication theorem (Theorem 7.3), where we have

$$P(A \cap B) = P(A)P(B/A) = P(B)P(A/B)$$

Example 7.16

Let A and B be independent events with $P(A) = 0.6$ and $P(B) = 0.2$. Find the following.

- (1) $P(A \cap B)$
 (2) $P(A/B)$
 (3) $P(B/A)$
 (4) $P(A \cup B)$

Solution: From Theorem 7.5, we have(1) $P(A \cap B)$ is given by

$$P(A \cap B) = P(A)P(B) = (0.6)(0.2) = 0.12 = \frac{3}{25}$$

(2) $P(A/B)$ is given by

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.12}{0.2} = \frac{3}{5}$$

(3) $P(B/A)$ is given by

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{0.12}{0.6} = \frac{1}{5}$$

(4) $P(A \cup B)$ is given by

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.6 + 0.2 - 0.12 = 0.68 = \frac{17}{25} \end{aligned}$$

Example 7.17

A bag contains 20 balls out of which 10 balls are white and others are black. Two balls are drawn from the bag at random with replacement. Let A be the event “first ball drawn is white” and B be the event “second ball drawn is white”. Verify whether A and B are independent.

Solution: It is given that the first ball drawn is replaced before drawing the second ball. Therefore, there are 20×20 ways to draw the two balls, of which 10×20 have the property that the first ball is white. Therefore,

$$P(A) = \frac{10 \times 20}{20 \times 20} = \frac{1}{2}$$

Similarly

$$P(B) = \frac{1}{2}$$

Further, there are 10×10 ways to draw the balls so that both the first and the second balls are white. Hence

$$P(A \cap B) = \frac{10 \times 10}{20 \times 20} = \frac{1}{4}$$

Now,

$$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B)$$

Therefore, by Theorem 7.5, A and B are independent of each other.**Example 7.18**

A bag B_1 contains 3 white balls and 4 black balls and another bag B_2 contains 2 black balls and 4 white balls. A bag is drawn at random and a ball is drawn at random from it. Find the probability that the ball drawn is white.

Solution: Let E_1 and E_2 be the events of choosing B_1 and B_2 , respectively. Then,

$$P(E_1) = \frac{1}{2} = P(E_2)$$

Let A be the event that the ball chosen is white. Then,

$$P(A/E_1) = \frac{3}{7} \quad \text{and} \quad P(A/E_2) = \frac{4}{6} = \frac{2}{3}$$

Since E_1 and E_2 are exhaustive and mutually exclusive, it follows that

$$A = (A \cap E_1) \cup (A \cap E_2)$$

and

$$(A \cap E_1) \cap (A \cap E_2) = \phi$$

Therefore

$$\begin{aligned}
 P(A) &= P(A \cap E_1) + P(A \cap E_2) \\
 &= P(E_1)P(A/E_1) + P(E_2)P(A/E_2)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \cdot \frac{3}{7} + \frac{1}{2} \cdot \frac{2}{3} \\
 &= \frac{3}{14} + \frac{1}{3} = \frac{23}{42}
 \end{aligned}$$

THEOREM 7.6
(TOTAL
PROBABILITY)

Let $E_1, E_2, E_3, \dots, E_n$ be mutually exclusive and exhaustive events with non-zero probabilities of a random experiment ξ . Then for any event A connected with ξ ,

$$P(A) = \sum_{i=1}^n P(E_i)P(A/E_i)$$

PROOF

We have

$$\begin{aligned}
 A &= A \cap S \\
 &= A \cap (E_1 \cup E_2 \cup \dots \cup E_n) \\
 &= (A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)
 \end{aligned}$$

Therefore

$$P(A) = \sum_{i=1}^n P(A \cap E_i) = \sum_{i=1}^n P(E_i)P(A/E_i)$$

THEOREM 7.7
(BAYES'
THEOREM)

Let $E_1, E_2, \dots, E_n$ be exhaustive and mutually exclusive events of a random experiment ξ and S be the sample space of ξ with a probability function P on $\wp(S)$ such that $P(E_i) > 0$ for all $1 \leq i \leq n$. Then for any event A of the random experiment ξ ,

$$P(E_j/A) = \frac{P(E_j)P(A/E_j)}{\sum_{i=1}^n P(E_i)P(A/E_i)}$$

for any $1 \leq j \leq n$.

PROOF

By hypothesis $E_1, E_2, \dots, E_n$ are exhaustive and mutually exclusive and hence

$$E_1 \cup E_2 \cup \dots \cup E_n = S$$

and

$$E_i \cap E_j = \phi \quad \text{for } i \neq j$$

Also, it is given that $P(E_i) > 0$ for all $1 \leq i \leq n$. For any event A (i.e., $A \subseteq S$), we have

$$A = A \cap S = A \cap \left(\bigcup_{i=1}^n E_i \right) = \bigcup_{i=1}^n (A \cap E_i)$$

Since $A \cap E_i \cap A \cap E_j = A \cap E_i \cap E_j = \phi$ for $i \neq j$ we have

$$\begin{aligned}
 P(A) &= \sum_{i=1}^n P(A \cap E_i) \\
 &= \sum_{i=1}^n P(E_i)P(A/E_i) \quad (\text{by Theorem 7.3})
 \end{aligned}$$

Thus, by Theorem 7.3 again, for any $1 \leq i \leq n$, we have

$$\begin{aligned}
 P(E_j/A) &= \frac{P(E_j \cap A)}{P(A)} \\
 &= \frac{P(E_j \cap A)}{\sum_{i=1}^n P(E_i)P(A/E_i)}
 \end{aligned}$$

Example 7.19

Three bags, B_1 , B_2 and B_3 contain balls as given in Table 7.2.

Table 7.2 Example 7.19

	Red	White	Black
B_1	2	2	1
B_2	4	3	2
B_3	2	4	3

A die is thrown. B_1 is chosen if either 1 or 2 turns up, B_2 is chosen if either 3 or 4 turns up and B_3 is chosen if either 4 or 5 turns up. Having chosen a bag in this way, a ball is chosen at random from this bag. If the ball chosen is of red colour, find the probability that it comes from bag B_2 ?

Solution: Let E_i be the event of choosing the bag B_i , for $i = 1, 2, 3$. Let R be the event of choosing a red ball. Then

$$P(E_i) = \frac{1}{3} \quad \text{for } i = 1, 2, 3$$

Having chosen the bag B_i , the probability of choosing a red ball is $P(R/E_i)$ and is given by

$$P(R/E_1) = \frac{2}{5}, P(R/E_2) = \frac{4}{9} \quad \text{and} \quad P(R/E_3) = \frac{2}{9}$$

We want to find the probability $P(E_2/R)$. By Bayes' theorem, we get that

$$\begin{aligned} P(E_2/R) &= \frac{P(E_2)P(R/E_2)}{P(E_1)P(R/E_1) + P(E_2)P(R/E_2) + P(E_3)P(R/E_3)} \\ &= \frac{\frac{1}{3} \times \frac{4}{9}}{\left(\frac{1}{3} \times \frac{2}{5}\right) + \left(\frac{1}{3} \times \frac{4}{9}\right) + \left(\frac{1}{3} \times \frac{2}{9}\right)} \\ &= \frac{\frac{4}{27}}{\left(\frac{18 + 20 + 10}{3 \times 5 \times 9}\right)} \\ &= \frac{4}{27} \times \frac{3 \times 5 \times 9}{48} = \frac{5}{12} \end{aligned}$$

7.5 | Random Variables and Probability Distributions

In this section we define the concepts of random variables and probability distributions and discuss certain important properties of these. First we define a random variable which quantifies the events of a sample space of a random experiment.

DEFINITION 7.13 Let S be the sample space of a random experiment. Then any real-valued function defined on S is called a *random variable* on S . That is, a random variable of S is simply a function $\vartheta: S \rightarrow \mathbb{R}$.

Examples

- (1) Let S be the sample space of the experiment “rolling a fair die”. That is, $S = \{1, 2, 3, 4, 5, 6\}$. Define $\vartheta: S \rightarrow \mathbb{R}$ by

$$\vartheta(s) = \begin{cases} 1 & \text{if } s \text{ is even} \\ 0 & \text{if } s \text{ is odd} \end{cases}$$

That is, $\vartheta(1) = \vartheta(3) = \vartheta(5) = 0$ and $\vartheta(2) = \vartheta(4) = \vartheta(6) = 1$. Then ϑ is a random variable on S .

- (2) Let S be the sample space of the experiment “tossing a fair coin”. That is, $S = \{H, T\}$, where H = Head and T = Tail. Define $\vartheta: S \rightarrow \mathbb{R}$ by

$$\vartheta(H) = 1 \quad \text{and} \quad \vartheta(T) = 0$$

Then ϑ is a random variable on S .

- (3) Let S be the sample space of the experiment “tossing three fair coins simultaneously”. Then

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Define $\vartheta: S \rightarrow \mathbb{R}$ by

$$\vartheta(s) = \text{The number of heads in } s$$

For example, $\vartheta(HHH) = 3$, $\vartheta(HHT) = 2$, $\vartheta(THT) = 1$, $\vartheta(TTT) = 0$. Then ϑ is a random variable on S .

For any random experiment ξ and its sample space S , we prove in the following that any probability function on $\wp(S)$ and a random variable on S give rise to a probability function on $P(\mathbb{R})$.

THEOREM 7.8

Let S be the sample space of a random experiment. Let $P: \wp(S) \rightarrow \mathbb{R}$ be a probability function and $\vartheta: S \rightarrow \mathbb{R}$ be a random variable. Define $P': \wp(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$P'(T) = P(\vartheta^{-1}(T)) \quad \text{for any } T \subseteq \mathbb{R},$$

where $\vartheta^{-1}(T) = \{s \in S \mid \vartheta(s) \in T\}$. Then P' is a probability function on $\wp(\mathbb{R})$ and is called the *probability function induced by P and ϑ* .

PROOF

Since $P(E) \geq 0$ for all $E \subseteq S$, we have

$$P'(T) = P(\vartheta^{-1}(T)) \geq 0 \quad \text{for all } T \subseteq \mathbb{R}$$

since $\vartheta^{-1}(T) \subseteq S$. Also,

$$P'(R) = P(\vartheta^{-1}(\mathbb{R})) = P(S) = 1$$

If A and $B \subseteq R$ such that $A \cap B = \emptyset$, then

$$\vartheta^{-1}(A) \cap \vartheta^{-1}(B) = \vartheta^{-1}(A \cap B) = \vartheta^{-1}(\emptyset) = \emptyset$$

and

$$\vartheta^{-1}(A \cup B) = \vartheta^{-1}(A) \cup \vartheta^{-1}(B)$$

and therefore

$$\begin{aligned} P'(A \cup B) &= P[\vartheta^{-1}(A \cup B)] \\ &= P[\vartheta^{-1}(A) \cup \vartheta^{-1}(B)] \\ &= P[\vartheta^{-1}(A)] + P[\vartheta^{-1}(B)] \\ &= P'(A) + P'(B) \end{aligned}$$

Thus P' is a probability function on $\wp(\mathbb{R})$. ■

Example

Let S be the sample space of the experiment “rolling a fair die” and P be the usual probability function on $\wp(S)$ determined by

$$P(s) = \frac{1}{6} \quad \text{for any } s \in S$$

Note that S is the 6-element set $\{1, 2, 3, 4, 5, 6\}$.

Let $\vartheta: S \rightarrow \mathbb{R}$ be the random variable given by $\vartheta(1) = \vartheta(3) = \vartheta(5) = 0$ and $\vartheta(2) = \vartheta(4) = \vartheta(6) = 1$. Then, for any $T \subseteq \mathbb{R}$,

$$\vartheta^{-1}(0) = \{1, 3, 5\}$$

$$\vartheta^{-1}(1) = \{2, 4, 6\}$$

$$\vartheta^{-1}(T) = \begin{cases} \emptyset & \text{if } 0 \notin T \text{ and } 1 \notin T \\ \{1, 3, 5\} & \text{if } 0 \in T \text{ and } 1 \notin T \\ \{2, 4, 6\} & \text{if } 0 \notin T \text{ and } 1 \in T \\ S & \text{if } 0 \in T \text{ and } 1 \in T \end{cases}$$

Also, since

$$P(\{1, 3, 5\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$$\text{and} \quad P(\{2, 4, 6\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

we have

$$P'(T) = P(\vartheta^{-1}(T)) = \begin{cases} 0 & \text{if } 0, 1 \notin T \\ \frac{1}{2} & \text{if } 0, 1 \in T \text{ but not both} \\ 1 & \text{if } 0, 1 \in T \end{cases}$$

It is a convention to use the letter X to denote a random variable and as such a random variable is simply a function X from the sample space S into the real number system. Here afterwards, we use X to denote a random variable.

DEFINITION 7.14 Let S be the sample space of a random experiment and $X: S \rightarrow \mathbb{R}$ be a random variable. Then, to each real number r , the event

$$X^{-1}((-\infty, r]) = \{s \in S \mid X(s) \leq r\}$$

is denoted by $(X \leq r)$. That is, for each $r \in \mathbb{R}$,

$$(X \leq r) = \{s \in S \mid X(s) \leq r\}$$

DEFINITION 7.15 For any random variable X , the function $F: \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$F(r) = P((X \leq r)) \quad \text{for any } r \in \mathbb{R},$$

is called the *probability distribution function of X* .

THEOREM 7.9 Let X be a random variable and F the probability distribution of X . Then the following hold.

1. $0 \leq F(r) \leq 1$ for all $r \in \mathbb{R}$
2. $r \leq s \Rightarrow F(r) \leq F(s)$ for any r and $s \in \mathbb{R}$
3. $\lim_{r \rightarrow \infty} F(r) = 1$ and $\lim_{r \rightarrow -\infty} F(r) = 0$
4. $F(r+0) = \lim_{\substack{s \rightarrow r \\ s > r}} F(s) = F(r)$

PROOF

For any real number, we have

$$F(r) = P((X \leq r))$$

where

$$(X \leq r) = \{s \in S \mid X(s) \leq r\}$$

Since P is a probability function, we have $0 \leq P(E) \leq 1$ for all $E \subseteq S$ and therefore

$$0 \leq F(r) \leq 1$$

Also, if r and $s \in \mathbb{R}$, then

$$\begin{aligned} r \leq s &\Rightarrow (-\infty, r] \subseteq (-\infty, s] \\ &\Rightarrow X^{-1}((-\infty, r]) \subseteq X^{-1}((-\infty, s]) \\ &\Rightarrow (X \leq r) \subseteq (X \leq s) \\ &\Rightarrow P((X \leq r)) \leq P((X \leq s)) \\ &\Rightarrow F(r) \leq F(s) \end{aligned}$$

The remaining results follow from the fact that F is an increasing function and

$$\text{lub}_{r \in \mathbb{R}} F(r) = 1, \text{glb}_{r \in \mathbb{R}} F(r) = 0$$

and, for any $r \in \mathbb{R}$,

$$(-\infty, r] = \bigcap_{r < s \in \mathbb{R}} (-\infty, s]$$

DEFINITION 7.16 A random variable whose range is at most countable (i.e., finite or countably infinite) is called a *discrete random variable*. A random variable which is not discrete is called a *continuous random variable*.

The random variables given in all the above examples are discrete. Though there are certain continuous random variables, we confine our discussion to the discrete random variables only. In the following we introduce the notions of mean, variance, standard deviation and probability distribution of discrete random variables.

DEFINITION 7.17 Let X be a discrete random variable on the sample space S of a random experiment and let the range of X be $\{x_1, x_2, \dots, x_n, \dots\}$. Let P be a probability function on $\wp(S)$. For each $n \in \mathbb{Z}^+$, let $P(X = x_n)$ be defined by

$$P(X = x_n) = P(X^{-1}\{x_n\})$$

1. If $\sum_n x_n P(X = x_n)$ is finite, then it is called the *mean* of X and is denoted by μ_X , or simply μ when there is no ambiguity about X . That is,

$$\mu = \sum_n x_n P(X = x_n)$$

2. If $\sum_n (x_n - \mu)^2 P(X = x_n)$ is finite, then it is called the *variance* of X and is denoted by v_X or simply v .
3. If v is the variance of X , then $\sqrt{v}$ is called the *standard deviation* of X and is denoted by σ_X , or simply σ . Therefore $\sqrt{v} = \sigma$ or $\sigma^2 = v$.
4. The system of numbers

$$X: x_1 \ x_2 \ x_3 \ \cdots \ x_n \ \cdots$$

$$P(X = x_i): p_1 \ p_2 \ p_3 \ \cdots \ p_n \ \cdots$$

is called the *probability distribution* of X .

(Note that $p_i = P(X = x_i) = P(X^{-1}\{x_i\})$.)

QUICK LOOK 2

From the points given in Definition 7.17, we have

$$\begin{aligned} \sigma^2 &= \sum_n (x_n - \mu)^2 p_n \\ &= \sum_n x_n^2 p_n - 2\mu \sum_n x_n p_n + \mu^2 \sum_n p_n \end{aligned}$$

$$\begin{aligned} &= \sum_n x_n^2 p_n - 2\mu^2 + \mu^2 \quad (\text{since } \sum_n x_n p_n = \mu \text{ and } \sum_n p_n = 1) \\ &= \sum_n x_n^2 p_n - \mu^2 \end{aligned}$$

Therefore

$$\sigma^2 + \mu^2 = \sum_n x_n^2 p_n$$

Example 7.20

The probability distribution of a random variable X is given in Table 7.3.

Table 7.3 Example 7.20

$X = x$	1	2	3	4	5	6	7
$P(X = x)$	0.25	0.50	0.75	1.0	1.25	1.50	1.75

Find the mean, variance and standard deviation of X .

Solution: The mean is given by

$$\begin{aligned} \mu &= \sum_{r=1}^7 x_r P(X = x_r) \\ &= 1 \times (0.25) + 2(0.50) + 3(0.75) + 4(1.0) \\ &\quad + 5(1.25) + 6(1.50) + 7(1.75) \end{aligned}$$

$$\begin{aligned} &= 0.25 + 1.00 + 2.25 + 4 + 6.25 + 9 + 12.25 \\ &= 35 \end{aligned}$$

The variance is given by

$$\begin{aligned} v &= \sum_{r=1}^7 (x_r - \mu)^2 P(X = x_r) \\ &= (-34)^2(0.25) + (-33)^2(0.5) + (-32)^2(0.75) + (-31)^2(1.00) \\ &\quad + (-30)^2(1.25) + (-29)^2(1.5) + (-28)^2(1.75) \\ &= 289 + 544.5 + 768 + 961 + 1125 + 1261.5 + 1372 \\ &= 6321 \end{aligned}$$

The standard deviation is

$$\sigma = \sqrt{v} = \sqrt{6321} = 7\sqrt{129}$$

Example 7.21

Let X be a discrete random variable whose probability distribution is given by

$$P(x = K) = \frac{(K+1)a}{2^K}$$

for $K = 0, 1, 2, 3, 4$ and 5 . Find the value of a .

Solution: Since

$$\sum_{K=0}^5 P(x = K) = 1$$

we have

$$\sum_{K=0}^5 \frac{(K+1)a}{2^K} = 1$$

Therefore

$$a \sum_{K=0}^5 \frac{K+1}{2^K} = 1$$

$$a \left(1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} + \frac{6}{2^5} \right) = 1$$

$$a \left(\frac{32 + 32 + 24 + 16 + 10 + 6}{32} \right) = 1$$

$$a = \frac{4}{15}$$

7.6 | Theoretical Discrete Distribution

In this section, we discuss certain probability distributions in which the variables are distributed according to some definite probability law that can be expressed mathematically. In particular, binomial distributions and Poisson distributions are of special importance.

A random variable X which takes just two values 0 and 1 with the probabilities p and q , respectively, is of particular interest. In this case, note that

$$P(X = 0) = p, \quad P(X = 1) = q \quad \text{and} \quad p + q = 1$$

Quite often we come across such random variables. For example, in coin tossing experiment, we can define $P(H) = p$ and $P(T) = 1 - p$, for some $0 < p < 1$ and the random variable X can be defined by $X(H) = 1$ and $X(T) = 0$, so that

$$P(X = 1) = p \quad \text{and} \quad P(X = 0) = 1 - p$$

Each repetition of such an experiment is called a *trial*.

7.6.1 Binomial Distribution

Let E and E^c be two complementary events in a random experiment with probabilities p and q , respectively. Let us call the occurrence of the event E as a *success* and the occurrence of the event E^c as a *failure* in a trial. If the experiment is repeated n times and E_k is the event having k successes in these trials, then one of these cases is “ E occurs in the first k trials, E^c occurs in the remaining $n - k$ trials”. The required probability for such an event is $p^k q^{n-k}$.

However the number of such cases is

$${}^nC_k = \frac{n!}{(n-k)!k!}$$

Thus

$$P(E_k) = {}^nC_k p^k q^{n-k}$$

Let X denote the number of successes in these n trials. Then X is a random variable with range $\{0, 1, 2, \dots, n\}$. Further

$$P(X = k) = {}^nC_k p^k q^{n-k} = {}^nC_k p^k (1-p)^{n-k} \quad (\text{since } p + q = 1)$$

The distribution of X is summarized in Table 7.4. This distribution shown in Table 7.4 is called a *binomial distribution*. Here n and p are called the *parameters* of X .

Table 7.4 Binomial distribution

$X = k$	0	1	2	...	r	...	n
$P(X = k)$	${}^nC_0 p^0 q^{n-0}$	${}^nC_1 p^1 q^{n-1}$	${}^nC_2 p^2 q^{n-2}$	...	${}^nC_r p^r q^{n-r}$	...	${}^nC_n p^n q^0$

DEFINITION 7.18 A random variable X is said to follow *binomial distribution*, or simply, it is a *binomial variate* with parameters n and p if

$$P(X = k) = {}^nC_k p^k q^{n-k} \quad \text{for } k = 0, 1, 2, \dots, n$$

and this is described by writing $X \sim B(n, p)$.

Note that $P(X = k)$ are nothing but the terms of the binomial expansion of $(p + q)^n$ and this justifies the name ‘binomial distribution’. Also, recall that $q = 1 - p$.

The following can be easily proved by using the results in Chapter 7, Binomial Theorem, Vol. 1.

THEOREM 7.10 If $X \sim B(n, p)$, then the mean μ and the variance σ^2 of X are equal to np and npq , respectively.

Notice that a binomial distribution takes place under the following experimental conditions.

1. Each trial results in two mutually exclusive outcomes, termed as success and failure.
2. The number n of trials is finite.
3. The trials are independent of each other.
4. The probability p of success is constant for each trial.

7.6.2 Poisson Distribution

Another important theoretical discrete distribution is the Poisson distribution which is indeed a limiting case of the binomial distribution. A random variable X is said to follow *Poisson distribution with parameter λ* and X is called a *Poisson variate* if

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for } k = 0, 1, 2, \dots$$

where $\lambda > 0$ is a constant. This is called the *Poisson law*. Table 7.5 shows the Poisson distribution.

Table 7.5 Poisson distribution

$X = k$	0	1	2	3	...	k	...
$P(X = k)$	$e^{-\lambda}$	$\frac{\lambda}{1!} e^{-\lambda}$	$\frac{\lambda^2}{2!} e^{-\lambda}$	$\frac{\lambda^3}{3!} e^{-\lambda}$	...	$\frac{\lambda^k}{k!} e^{-\lambda}$	...

The following is an important result and we skip the proof of this.

THEOREM 7.11 Let X be a Poisson random variate with parameter λ . Then the mean and variance of X are both equal to λ .

Note that the Poisson distribution can be used under the following experimental conditions.

1. Each trial results in two mutually exclusive outcomes, termed as success and failure.
2. The number n of trials is sufficiently large.
3. The probability p of success is very small.
4. The trials are independent of each other.

 QUICK LOOK 3

The Poisson distribution can be used, for instance, in problems like the following:

1. The number of defective items in a packing
2. The number of deaths from non-epidemic diseases such as heart attack or cancer or snake bites

3. The number of vehicles passing a given spot per minute during the peak hours of a day in a given city.

Example 7.22

Eight coins are tossed simultaneously. Find the probability of getting atleast six Heads, using the binomial distribution.

Solution: Let

$$p = \text{the probability of getting a head} = \frac{1}{2}$$

and $q = \text{the probability of getting a tail} = \frac{1}{2}$

The probability of getting r heads in a random toss is

$$P(X = r) = {}^8C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{8-r} = {}^8C_r \left(\frac{1}{2}\right)^8$$

for $r = 0, 1, 2, \dots, 8$. Therefore, the probability of getting atleast 6 heads is

$$\begin{aligned} P(X \geq 6) &= P(X = 6) + P(X = 7) + P(X = 8) \\ &= {}^8C_6 \left(\frac{1}{2}\right)^8 + {}^8C_7 \left(\frac{1}{2}\right)^8 + {}^8C_8 \left(\frac{1}{2}\right)^8 \\ &= \frac{7}{64} + \frac{1}{32} + \frac{1}{256} = \frac{37}{256} \end{aligned}$$

Example 7.23

In a book of 750 pages, there are 500 typographical errors. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.

Solution: The average member of errors per page in the book is

$$\lambda = \frac{500}{750} = \frac{2}{3}$$

The probability of r errors per page is

$$P(X = r) = \frac{\lambda^r}{r!} e^{-\lambda} = \left(\frac{2}{3}\right)^r \frac{1}{r!} e^{-2/3}$$

Therefore

$$P(X = 0) = e^{-2/3}$$

The required probability that a random sample of 5 pages will contain no error is

$$(P(X = 0))^5 = (e^{-2/3})^5 = e^{-10/3}$$

Example 7.24

A Poisson variate X satisfies $P(X = 1) = P(X = 2)$. Find $P(X = 6)$.

Solution: Recall that

$$P(X = r) = \frac{\lambda^r}{r!} e^{-\lambda}, \lambda > 0$$

for $r = 0, 1, 2, \dots$. It is given that $P(X = 1) = P(X = 2)$. Therefore

$$\frac{\lambda}{1!} e^{-\lambda} = \frac{\lambda^2}{2!} e^{-\lambda}$$

$$(2\lambda - \lambda^2)e^{-\lambda} = 0$$

$$\lambda(2 - \lambda) = 0$$

Since $\lambda > 0$, we should have $\lambda = 2$. Therefore

$$P(X = 6) = \frac{\lambda^6}{6!} e^{-\lambda} = \frac{64}{720} e^{-2} = \left(\frac{4}{45}\right) e^{-2}$$

WORKED-OUT PROBLEMS

Single Correct Choice Type Questions

1. A , B and C are three athletes running in a race. If the probability of A winning is twice as likely to win as probability of B and that of B is as likely to win as of C , then the probability of A 's win is

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$

Solution: By hypothesis $P(A) = 2P(B)$ and $P(B) = 2P(C)$. Since

$$P(A) + P(B) + P(C) = 1$$

we have

$$4P(C) + 2P(C) + P(C) = 1$$

Therefore

$$7P(C) = 1 \quad \text{or} \quad P(C) = \frac{1}{7}$$

So

$$P(A) = 4P(C) = \frac{4}{7}$$

Answer: (D)

2. In Problem (1), the probability that one of B or C will win is

(A) $\frac{2}{7}$ (B) $\frac{3}{7}$ (C) $\frac{4}{7}$ (D) $\frac{5}{7}$

Solution: The events that either of B or C will win are mutually exclusive events so that $P(B \cap C) = 0$. Hence

$$\begin{aligned} P(B \cup C) &= P(B) + P(C) \\ &= \frac{2}{7} + \frac{1}{7} = \frac{3}{7} \end{aligned}$$

Answer: (B)

3. Let $S = \{x_1, x_2, x_3, x_4\}$ be a four-element sample space. Which of the following functions defines a probability function on S ?

(A) $P(x_1) = \frac{1}{4}, P(x_2) = \frac{1}{8}, P(x_3) = \frac{1}{2}, P(x_4) = \frac{1}{8}$

(B) $P(x_1) = \frac{1}{3}, P(x_2) = \frac{1}{5}, P(x_3) = \frac{1}{4}, P(x_4) = \frac{1}{2}$

(C) $P(x_1) = \frac{1}{2}, P(x_2) = -\frac{1}{3}, P(x_3) = \frac{1}{4}, P(x_4) = \frac{1}{2}$

(D) $P(x_1) = 0, P(x_2) = \frac{1}{2}, P(x_3) = \frac{1}{3}, P(x_4) = \frac{2}{3}$

Solution: Since probability of an event is a non-negative real number and sum of all the probabilities is equal to 1, the correct answer is (A).

Answer: (A)

4. A die is so weighed such that the probability of a number appearing when tossed is proportional to the number on the face. Then, the probability of a prime numbered face to appear is

(A) $\frac{4}{7}$ (B) $\frac{5}{21}$ (C) $\frac{10}{21}$ (D) $\frac{11}{21}$

Solution: Let p be the constant of proportionality so that

$$P(K) = Kp \text{ for } K = 1, 2, 3, 4, 5, 6$$

Sum of the probabilities = 1. This implies

$$21p = 1 \Rightarrow p = \frac{1}{21}$$

Let E be the event of a prime number face. That is $E = \{2, 3, 5\}$. Then

$$P(E) = 2p + 3p + 5p = \frac{10}{21}$$

Answer: (C)

5. A and B are two events of a random experiment. If $P(A \cup B) = 7/8$, $P(A \cap B) = 1/4$ and $P(\bar{A}) = 5/8$, then $P(A \cap \bar{B})$ is equal to

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{3}{8}$

Solution: We have

$$\begin{aligned} P(A \cap \bar{B}) &= P(A - B) = P(A) - P(A \cap B) \\ &= \left(1 - \frac{5}{8}\right) - \frac{1}{4} \\ &= \frac{3}{8} - \frac{1}{4} = \frac{1}{8} \end{aligned}$$

Answer: (A)

6. Two dice of different colours are thrown at a time. The probability that the sum of the faces appeared is either 7 or 11 is

(A) $\frac{7}{36}$ (B) $\frac{4}{9}$ (C) $\frac{2}{3}$ (D) $\frac{5}{9}$

Solution: The total number of possible outcomes = $6 \times 6 = 36$. Let A = Event of the sum is 7 and B = event of the sum is 11. Therefore

$$A = \{(1, 6), (2, 5), (3, 4), (6, 1), (5, 2), (4, 3)\}$$

$$B = \{(5, 6), (6, 5)\}$$

It is observed that $A \cap B = \phi$. Hence

$$P(A \cup B) = P(A) + P(B) = \frac{6}{36} + \frac{2}{36} = \frac{4}{9}$$

Answer: (B)

7. From a well-shuffled pack of 52 cards, four cards are selected at random. The probability of drawing exactly 2 spades and exactly 2 aces is

$$\begin{array}{ll} \text{(A)} \frac{1494}{270725} & \text{(B)} \frac{1594}{270725} \\ \text{(C)} \frac{1296}{270725} & \text{(D)} \frac{1396}{270725} \end{array}$$

Solution: Total number of outcomes = ${}^{52}C_4 = 270725$
 E = Event of exactly two spade cards and exists two aces.
 A = Event of 1 spade ace, 1 non-spade ace and 1 spade card and 1 non-spade card
 B = Event of 2 non-spade aces and 2 spade cards.

$$P(A) = \frac{1({}^3C_1)({}^{12}C_1)({}^{36}C_1)}{{}^{52}C_4} = \frac{3 \times 12 \times 36}{270725} = \frac{1296}{270725}$$

$$P(B) = \frac{{}^3C_2 \times {}^{12}C_2}{{}^{52}C_4} = \frac{3 \times 66}{270725} = \frac{198}{270725}$$

Now $E = A \cup B$ and $A \cap B = \phi$. This implies

$$\begin{aligned} P(E) &= P(A \cup B) \\ &= P(A) + P(B) = \frac{1494}{270725} \end{aligned}$$

Answer: (A)

8. Six married couples (a couple means wife and husband) are standing in a room. Four people are chosen at random. The probability that the selection contains exactly one couple is

$$\text{(A)} \frac{1}{11} \quad \text{(B)} \frac{1}{33} \quad \text{(C)} \frac{16}{33} \quad \text{(D)} \frac{17}{33}$$

Solution: From 12 people, 4 can be selected in ${}^{12}C_4 = 495$ ways.
 E = Event of the selection contains exactly one couple.
 Therefore the selection must contain one couple and the rest 2 must be non-couple. This can be done in

$${}^6C_1 \times 2({}^5C_1 \times {}^4C_1) = 6 \times 40 = 240 \text{ ways}$$

Therefore

$$P(E) = \frac{240}{495} = \frac{16}{33}$$

Answer: (C)

9. There are three boxes B_1 , B_2 and B_3 . B_1 contains 1 white and 2 black balls. B_2 contains 3 white, 1 black ball and B_3 contain 2 white and 3 black balls. One ball is taken from each box. Among the balls drawn, the probability that there are two black and 1 white ball is

$$\text{(A)} \frac{7}{12} \quad \text{(B)} \frac{23}{60} \quad \text{(C)} \frac{37}{60} \quad \text{(D)} \frac{5}{12}$$

Solution: Let E be the event that the draw contains 2 black and one white ball. Let w_i drawing white ball from B_i and b_j be the event of drawing a black ball from the box B_j ($i = 1, 2, 3$, and $j = 1, 2, 3$). Therefore

$$E = (b_1 \cap b_2 \cap w_3) \cup (b_1 \cap w_2 \cap b_3) \cup (w_1 \cap b_2 \cap b_3)$$

Each of the events in the union is mutually exclusive and independent. Hence

$$\begin{aligned} P(E) &= P(b_1 \cap b_2 \cap w_3) + P(b_1 \cap w_2 \cap b_3) \\ &\quad + P(w_1 \cap b_2 \cap b_3) \\ &= P(b_1)P(b_2)P(w_3) + P(b_1)P(w_2)P(b_3) \\ &\quad + P(w_1)P(b_2)P(b_3) \\ &= \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{2}{5} + \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{3}{5} + \frac{1}{3} \cdot \frac{1}{4} \cdot \frac{3}{5} \\ &= \frac{25}{60} = \frac{5}{12} \end{aligned}$$

Answer: (D)

10. A class contains 20 boys and 20 girls of which half the boys and half the girls have cat eyes. If one student is selected from the class, the probability that either the student is a boy or has cat eyes is

$$\text{(A)} \frac{1}{2} \quad \text{(B)} \frac{3}{4} \quad \text{(C)} \frac{3}{8} \quad \text{(D)} \frac{2}{3}$$

Solution: Let A be the event of a boy and B the event of having cat eyes. So

$$P(A) = \frac{20}{40} = \frac{1}{2}$$

and

$$P(B) = \frac{20}{40} = \frac{1}{2}$$

Now

$$P(A \cap B) = \frac{10}{40} = \frac{1}{4}$$

Therefore

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &\quad - P(A \cap B) \quad (\text{see Theorem 7.2}) \\ &= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4} \end{aligned}$$

Answer: (B)

11. If a positive divisor of 10^{49} is selected at random, then the probability that it is an integer multiple of 10^{38} is

(A) $\frac{9}{625}$ (B) $\frac{18}{625}$ (C) $\frac{36}{625}$ (D) $\frac{9}{2500}$

Solution: $10^{49} = 2^{49} \cdot 5^{49} \Rightarrow$ there are $50 \times 50 = 2500$ positive divisions of 10^{49} . Now $10^{38} = 2^{38} \cdot 5^{38} \Rightarrow$ any integer multiple of 10^{38} must be of the form $2^a \cdot 5^b$ where $38 \leq a, b \leq 49$. That is, a and b will have 12 choices each. Hence required probability is

$$\frac{12 \times 12}{2500} = \frac{36}{625}$$

Answer: (C)

12. Let A and B be two events such that $P(A) = 0.3$, $P(B) = 0.6$ and $P(B/A) = 0.5$. Then $P(\bar{A}/\bar{B})$ equals

(A) $\frac{3}{4}$ (B) $\frac{5}{8}$ (C) $\frac{9}{40}$ (D) $\frac{1}{4}$

Solution: We have by Theorem 7.3

$$P(A \cap B) = P(A)P(B/A) = (0.3)(0.5) = 0.15$$

Now

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.3 + 0.6 - 0.15 \\ &= 0.75 \end{aligned}$$

Also

$$\begin{aligned} P(\bar{A}/\bar{B}) &= \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} \\ &= \frac{P(\overline{A \cup B})}{P(\bar{B})} \\ &= \frac{1 - P(A \cup B)}{1 - P(B)} \\ &= \frac{1 - 0.75}{1 - 0.6} \\ &= \frac{0.25}{0.4} = \frac{250}{400} = \frac{5}{8} \end{aligned}$$

Answer: (B)

13. A positive integer is selected at random from the first 200 natural numbers. The probability that it is divisible by 4 or 5 is

(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) $\frac{3}{5}$ (D) $\frac{3}{10}$

Solution: Any integer divisible by 4 or 5 must be divisible by 20.

Number of multiples of 4 lying between 1 and 200 (including 200) is 50.

Number of multiples of 5 is 20.

Number of multiples of 20 is 10.

Therefore required probability is

$$\frac{50 + 20 - 10}{200} = \frac{60}{200} = \frac{3}{10}$$

Answer: (D)

14. Out of 5 men and 6 women a committee of 5 is to be formed. If the selection is at random, the probability that the committee consists of at least 3 women is

(A) $\frac{141}{462}$ (B) $\frac{143}{462}$ (C) $\frac{131}{462}$ (D) $\frac{1}{4}$

Solution: Let

E : Event that committee consists of at least 3 women.

The favourable cases to E are:

Women	Men	No. of selections
3	2	${}^6C_3 \times {}^5C_2$
4	1	${}^6C_4 \times {}^5C_1$
5	0	${}^6C_5 \times {}^5C_0$

Therefore

$$\begin{aligned} P(E) &= \frac{{}^6C_3 \times {}^5C_2 + {}^6C_4 \times {}^5C_1 + {}^6C_5 \times {}^5C_0}{{}^{11}C_5} \\ &= \frac{6 \times 10 + 15 \times 5 + 6 \times 1}{462} \\ &= \frac{141}{462} \end{aligned}$$

Answer: (A)

15. Each of two bags A and B contain n cards numbered 1 to n . One card from each of A and B is drawn. The probability that the card drawn from A bears number smaller than the number on the card drawn from B is

(A) $\frac{n+1}{2n}$ (B) $\frac{n-1}{2n}$ (C) $\frac{n+1}{n}$ (D) $\frac{n-1}{n}$

Solution: Let

E_1 = Number drawn from A is smaller than that drawn from B .

E_2 = Number drawn from A is greater than that drawn from B .

E_3 = Both numbers are equal.

Note that

$$P(E_1) = P(E_2)$$

and

$$P(E_3) = \frac{n}{n^2} = \frac{1}{n}$$

Therefore

$$\begin{aligned} 2P(E_1) &= 1 - P(E_3) = 1 - \frac{1}{n} \\ \Rightarrow P(E_1) &= \frac{n-1}{2n} \end{aligned}$$

Answer: (B)

16. A, B, C are three events such that $P(A) = 0.3$, $P(B) = 0.4$, $P(C) = 0.8$, $P(A \cap B) = 0.08$, $P(C \cap A) = 0.28$, $P(A \cap B \cap C) = 0.09$, and $P(A \cup B \cup C) \geq 0.75$. Then $P(B \cap C)$ belongs to the interval:

- (A) $[0.11, 0.23]$ (B) $[0.23, 0.48]$
(C) $[0.13, 0.25]$ (D) $[0.15, 0.20]$

Solution: We have

$$\begin{aligned} P(A \cup B \cup C) &= \left(\sum P(A) \right) - P(A \cap B) - P(B \cap C) \\ &\quad - P(C \cap A) + P(A \cap B \cap C) \\ &= (0.3 + 0.4 + 0.8) - 0.08 - P(B \cap C) \\ &\quad - 0.28 + 0.09 \end{aligned} \quad (7.1)$$

Therefore

$$\begin{aligned} P(B \cap C) &= 1.23 - P(A \cup B \cup C) \geq 1.23 - 1 \\ &= 0.23 \quad [\because P(A \cup B \cup C) \leq 1] \\ 0.23 &\leq P(B \cap C) \end{aligned} \quad (7.2)$$

Also

$$\begin{aligned} P(B \cap C) &\leq 1.23 - (0.75) = 0.48 \\ [\because 0.75 &\leq P(A \cup B \cup C)] \end{aligned}$$

So

$$P(B \cap C) \leq 0.48 \quad (7.3)$$

Equations (7.2) and (7.3) imply

$$0.23 \leq P(B \cap C) \leq 0.48$$

Answer: (B)

17. Let E, F and G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P((\bar{E} \cap \bar{F})/G)$ is equal to

- (A) $P(\bar{E}) + P(\bar{F})$ (B) $P(\bar{E}) - P(\bar{F})$
(C) $P(\bar{E}) - P(F)$ (D) $P(E) - P(\bar{F})$

Solution: Note that

$$P(A - B) = P(A - A \cap B) \quad (\because A - B = A - A \cap B)$$

and also

$$P(A - B) = P(A \cap \bar{B}) \quad (\because A - B = A \cap \bar{B})$$

Therefore

$$\begin{aligned} P(\bar{E} \cap \bar{F} \cap G) &= P(G - (\overline{E \cap F})) \\ &= P(G - (E \cup F)) \\ &= P(G - G \cap (E \cup F)) \\ &= P(G) - P((G \cap E) \cup (G \cap F)) \\ &\quad \text{(See Theorem 7.2)} \\ &= P(G) - P(G \cap E) - P(G \cap F) \\ &\quad + P(G \cap E \cap F) \quad \text{(By Theorem 7.2)} \\ &= P(G) - P(G)P(E) - P(G)P(F) + 0 \\ &\quad (\because P(E \cap F \cap G) = 0 \text{ and } E, F \\ &\quad \text{and } G \text{ are pairwise independent}) \\ &= P(G)(1 - P(E) - P(F)) \\ &= P(G)(P(\bar{E}) - P(F)) \end{aligned}$$

Hence

$$\begin{aligned} P((\bar{E} \cap \bar{F})/G) &= \frac{P(\bar{E} \cap \bar{F} \cap G)}{P(G)} \\ &= \frac{P(G)[P(\bar{E}) - P(F)]}{P(G)} = P(\bar{E}) - P(F) \end{aligned}$$

Answer: (C)

18. Six students are to be selected for a quiz competition from 10 aspirants. The probability that two particular students are excluded is

- (A) $\frac{2}{15}$ (B) $\frac{1}{3}$ (C) $\frac{1}{5}$ (D) $\frac{2}{3}$

Solution: Total number of selections of 6 from 10 = ${}^{10}C_6$. Since two particular students are to be excluded, we have to select 6 from 8 only. This can be done in 8C_6 ways. The required probability is

$$\begin{aligned} \frac{{}^8C_6}{{}^{10}C_6} &= \frac{|8|}{|6 \times 2|} \times \frac{|6 \times 4|}{|10|} \\ &= \frac{3 \cdot 4}{9 \cdot 10} = \frac{2}{15} \end{aligned}$$

Answer: (A)

19. $\bar{E}$ and $\bar{F}$ are complementary events of the events E and F . If $0 < P(F) < 1$, then

- (A) $P(E/F) + P(\bar{E}/F) < 1$
 (B) $P(E/F) + P(E/\bar{F}) = 1$
 (C) $P(\bar{E}/F) + P(E/\bar{F}) = 1$
 (D) $P(E/\bar{F}) + P(\bar{E}/\bar{F}) = 1$

Solution:

(A) We have

$$\begin{aligned} P(E/F) + P(\bar{E}/F) &= \frac{P(E \cap F)}{P(F)} + \frac{P(\bar{E} \cap F)}{P(F)} \\ &= \frac{P(F \cap (E \cup \bar{E}))}{P(F)} \\ &= \frac{P(F)}{P(F)} = 1 \end{aligned}$$

Thus, (A) is not correct.

(B) We have

$$\begin{aligned} P(E/F) + P(E/\bar{F}) &= \frac{P(E \cap F)}{P(F)} + \frac{P(E \cap \bar{F})}{P(\bar{F})} \\ &= \frac{P(E \cap F)P(\bar{F}) + P(E \cap \bar{F})P(F)}{P(F)P(\bar{F})} \\ &= \frac{P(E \cap F)(1 - P(F)) + P(E \cap \bar{F})P(F)}{P(F)P(\bar{F})} \\ &= \frac{[P(E \cap \bar{F}) - P(E \cap F)]P(F) + P(E \cap F)}{P(F)P(\bar{F})} \neq 1 \end{aligned}$$

Hence (B) is not correct.

(C) We have

$$\begin{aligned} P(\bar{E}/F) + P(E/\bar{F}) &= \frac{P(\bar{E} \cap F)}{P(F)} + \frac{P(E \cap \bar{F})}{P(\bar{F})} \\ &= \frac{P(\bar{E} \cap F)P(\bar{F}) + P(E \cap \bar{F})P(F)}{P(F)P(\bar{F})} \\ &= \frac{P(\bar{E} \cap F)(1 - P(F)) + P(E \cap \bar{F})P(F)}{P(F)P(\bar{F})} \\ &= \frac{P(F)[P(E \cap \bar{F}) - P(\bar{E} \cap F)] + P(\bar{E} \cap F)}{P(F)P(\bar{F})} \neq 1 \end{aligned}$$

So (C) is not correct.

(D) We have

$$\begin{aligned} P(E/\bar{F}) + P(\bar{E}/\bar{F}) &= \frac{P(E \cap \bar{F})}{P(\bar{F})} + \frac{P(\bar{E} \cap \bar{F})}{P(\bar{F})} \\ &= \frac{P(E \cap \bar{F}) + P(\bar{E} \cap \bar{F})}{P(\bar{F})} \\ &= \frac{P(\bar{F} \cap (E \cup \bar{E}))}{P(\bar{F})} \\ &= \frac{P(\bar{F})}{P(\bar{F})} = 1 \end{aligned}$$

Therefore (D) is correct.

Answer: (D)

20. If $P(B) = 3/4$, $P(A \cap B \cap \bar{C}) = 1/3$ and $P(\bar{A} \cap B \cap \bar{C}) = 1/3$, then $P(B \cap C)$ is

- (A) $\frac{1}{12}$ (B) $\frac{1}{6}$ (C) $\frac{1}{15}$ (D) $\frac{1}{9}$

Solution: From Figure 7.2 we have

$$B = (B \cap C) \cup (A \cap B \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C})$$

Therefore

$$\begin{aligned} P(B \cap C) &= P(B) - P(A \cap B \cap \bar{C}) - P(\bar{A} \cap B \cap \bar{C}) \\ &= \frac{3}{4} - \left(\frac{1}{3} + \frac{1}{3} \right) \\ &= \frac{3}{4} - \frac{2}{3} = \frac{1}{12} \end{aligned}$$

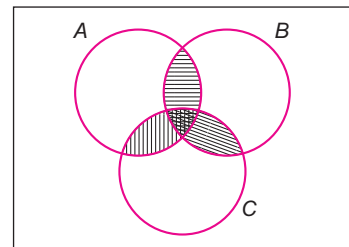


FIGURE 7.2 Single correct choice type question 20.

Answer: (A)

21. If three distinct natural numbers are chosen randomly from the first 100 natural numbers, then the probability that all three of them are divisible by both 2 and 3 is

- (A) $\frac{4}{25}$ (B) $\frac{4}{35}$ (C) $\frac{4}{33}$ (D) $\frac{4}{1155}$

Solution: A number is divisible by 6, only when it is divisible by both 2 and 3. Therefore the number of multiples of 6 (<100) is 16 (they are 6, 12, 18, ..., 96).

From 100 numbers three are selected in $^{100}C_3$ ways.
 From 16 numbers 3 are selected in $^{16}C_3$ ways.
 Therefore probability is

$$\begin{aligned}\frac{{}^{16}C_3}{{}^{100}C_3} &= \frac{|16|}{|3|13} \times \frac{|3|197}{|100|} \\ &= \frac{16 \times 15 \times 14}{100 \times 99 \times 98} = \frac{4}{1155}\end{aligned}$$

Alternate Solution

Since the three numbers are distinct, we can select the three one after other without replacement. Therefore the probability is

$$\frac{16 \times 15 \times 14}{100 \times 99 \times 98} = \frac{4}{1155}$$

Answer: (D)

- 22.** Let A , B and C be mutually independent events. Consider the following two statements.
 Statement I: A and $B \cup C$ are independent.
 Statement II: A and $B \cap C$ are independent.
 Then,
 (A) both I and II are true
 (B) only I is true
 (C) only II is true
 (D) both I and II are false

Solution: We know that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Now

$$\begin{aligned}P(A \cap (B \cup C)) &= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \\ &= P(A)P(B) + P(A)P(C) - P(A)P(B)P(C) \\ &= P(A)[P(B) + P(C) - P(B)P(C)] \\ &= P(A)[P(B) + P(C) - P(B \cap C)] \\ &= P(A)P(B \cup C)\end{aligned}$$

Therefore Statement I is true. Again

$$\begin{aligned}P(A \cap (B \cap C)) &= P(A)P(B)P(C) \\ &= P(A)P(B \cap C)\end{aligned}$$

Hence A and $B \cap C$ are independent. So Statement II is also true.

Answer: (A)

- 23.** A natural number x is selected at random from the first 100 natural numbers. The probability that $x + (100/x) > 50$ is

- (A) $\frac{11}{20}$ (B) $\frac{9}{20}$ (C) $\frac{13}{20}$ (D) $\frac{3}{4}$

Solution: We have

$$\begin{aligned}x + \frac{100}{x} &> 50 \\ \Leftrightarrow (x - 25)^2 &> 525 \\ \Leftrightarrow |x - 25| &> 22\end{aligned}$$

Hence $x < 3$ or $x > 47$. So the number of favourable cases to x is $2 + 53 = 55$. The required probability is

$$\frac{55}{100} = \frac{11}{20}$$

Answer: (A)

- 24.** The first 12 letters of English alphabet are written in a row at random. The probability that there are exactly four letters in between A and B is

- (A) $\frac{5}{66}$ (B) $\frac{1}{22}$ (C) $\frac{7}{66}$ (D) $\frac{1}{11}$

Solution: A and B can be arranged in $^{12}P_2 = 11 \times 12$ ways. Since we want 4 letters in between A and B , the order of the four letters appearing has no importance. A and B can take the following places.

Place for A	Place for B
1	6
2	7
3	8
4	9
5	10
6	11
7	12

A and B can be interchanged. Therefore required probability is

$$\frac{14}{11 \times 12} = \frac{7}{66}$$

Alternate Solution

12 letters can be arranged in $|12|$ ways. In between A and B , four letters can be arranged in $^{10}P_4$ ways. Now 6 letters for whom A and B are at extreme positions together with 6 other letters can be arranged in $|7|$ ways.

Further A and B can be interchanged. Therefore the required probability is

$$\frac{2 \times {}^{10}P_4 \times \underline{7}}{\underline{12}} = \frac{2 \times \underline{10} \times \underline{7}}{\underline{6} \times \underline{12}} \\ = \frac{2 \times 7}{11 \times 12} = \frac{7}{66}$$

Answer: (C)

25. If the letters of the word UNIVERSITY are arranged in a random order, the probability that all the vowels come together is

- (A) $\frac{1}{15}$ (B) $\frac{2}{15}$ (C) $\frac{1}{5}$ (D) $\frac{1}{30}$

Solution: The word UNIVERSITY consists of 10 letters out of which the vowels are e, i, i, u . The 10 letters can be arranged in $\underline{10}/\underline{2}$ ways (since there two identical letters i). Since all the vowels are to be together, treat all the four written as a single object. The $6 + 1 = 7$ objects can be permuted in

$$\underline{7} \left(\frac{\underline{4}}{\underline{2}} \right) \text{ ways}$$

(since 4 vowels can be arranged among themselves in $\underline{4}/\underline{2}$ ways). Therefore required probability is

$$\frac{7(\underline{4}/\underline{2})}{\underline{10}/\underline{2}} = \frac{24}{8 \times 9 \times 10} = \frac{1}{3 \times 10} = \frac{1}{30}$$

Answer: (D)

26. A cubical die has four blank faces, one face marked 2 and another marked 3. If the die is thrown 5 times, the probability that the sum is 12 is

- (A) $\frac{5}{1296}$ (B) $\frac{5}{1294}$ (C) $\frac{5}{2596}$ (D) $\frac{5}{2592}$

Solution: Total number of outcomes is

$$6 \times 6 \times 6 \times 6 \times 6 = 6^5$$

Let

E = sum of the faces is 12.

Favourable cases to E : 2 three times, 3 two times or 3 four times. This is possible in

$$\frac{\underline{5}}{\underline{2} \times \underline{3}} + \frac{\underline{5}}{\underline{4}} \text{ ways}$$

Therefore

$$P(E) = \frac{10 + 5}{6^5} = \frac{15}{6^4 \times 6} = \frac{5}{2 \times 6^4} = \frac{5}{2592}$$

Answer: (D)

27. Each of the letters A, B, C, D, E and F are, respectively, written on six cards (one letter on one card only) and they are well-shuffled and then the top four cards are turned face up without changing their order. Then the probability that they form the word "DEAF" is

- (A) $\frac{1}{120}$ (B) $\frac{1}{360}$ (C) 0.03 (D) 0.13

Solution: Total number of arrangements by cards = ${}^6P_4 = 360$. In only one way, the letters form the word DEAF. Therefore

$$\text{Probability} = \frac{1}{360}$$

Answer: (B)

28. A boy remembers all but the last digit of his girlfriend's mobile number. He randomly chooses a digit from 0 to 9 (including 0 and 9). If he attempts two times, the probability that he reaches her at least once is

- (A) 0.2 (B) 0.3 (C) 0.02 (D) 0.03

Solution: Let

E = Event that the boy makes correct dial at least once.

$\bar{E}$ = Event of failing in both attempts.

In the first attempt, the probability of failing is $9/10$. In the second attempt, the probability of failing is $8/9$. Therefore

$$P(\bar{E}) = \frac{9}{10} \times \frac{8}{9} = \frac{4}{5}$$

$$P(E) = 1 - P(\bar{E}) = 1 - \frac{4}{5} = \frac{1}{5} = 0.2$$

Answer: (A)

29. The probability of the birth dates of all 6 persons to fall in only two different months is

- (A) $\frac{341}{12^6}$ (B) $\frac{341}{12^5}$ (C) $\frac{341}{12^4}$ (D) $\frac{541}{12^6}$

Solution: Since the birth date of any person can fall in anyone of the 12 months, the number of total outcomes is 12^6 . Let

E : Event that the birth dates of all 6 fall in two different months.

Number of possibilities of 6 persons' birth dates to fall in two different months (say February and March) is $2^6 - 2$ (this 2 corresponds to either all their birth dates fall in February or all fall in March). The number of ways that we can select 2 months out of 12 is ${}^{12}C_2$. Therefore

$$P(E) = \frac{{}^{12}C_2(2^6 - 2)}{12^6} = \frac{66 \times 62}{12^6} = \frac{11 \times 31}{12^5} = \frac{341}{12^5}$$

Answer: (B)

30. A is one of the six horses entered for a race and one of the two jockeys B and C ride it. If B rides A , then all the six horses are equally likely to win. If C rides A , then chances of A 's win will be trebled. Then, the odds in favour of A 's win is

- (A) 2:1 (B) 3:2 (C) 1:2 (D) 2:3

Solution: Let

A = Event of A 's win.

B = Event of B riding A .

C = Event of C riding A .

Then

$$P(A/B) = \frac{1}{6}$$

and

$$P(A/C) = 3\left(\frac{1}{6}\right) = \frac{1}{2}$$

Now

$$A = A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Since B and C are mutually exclusive we have

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap C) \\ &= P(B)P(A/B) + P(C)P(A/C) \\ &= \frac{1}{2} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} = \frac{1+3}{12} = \frac{1}{3} \end{aligned}$$

Therefore

$$P(\bar{A}) = 1 - \frac{1}{3} = \frac{2}{3}$$

Hence odds in favour (see Note under Theorem 7.1) is $P(E):P(\bar{E}) = 1:2$.

Answer: (C)

31. An urn contains m white balls and n black balls. A ball is drawn at random and put back into the urn along with k additional balls of the same colour as that of the drawn ball. A ball is again taken out at random. The probability that it is white is

- (A) $\frac{m}{m+n}$ (B) $\frac{n}{m+n}$
(C) $\frac{m+n+k}{mn+k}$ (D) $\frac{mn}{mn+k}$

Solution: Let

W_1 = Event of drawing a white ball in the first draw.

B_1 = Event of drawing black ball in the first draw.

W_2 = Drawing white ball in the second draw.

Now

$$W_2 = W_2 \cap (W_1 \cup B_1) = (W_2 \cap W_1) \cup (W_2 \cap B_1)$$

So

$$P(W_2) = P(W_2 \cap W_1) + P(W_2 \cap B_1) (\because B_1 \cap W_1 = \phi)$$

Therefore by Theorem 7.3

$$\begin{aligned} P(W_2) &= P(W_1)P(W_2/W_1) + P(B_1)P(W_2/B_1) \\ &= \frac{m}{m+n} \cdot \frac{m+k}{m+n+k} + \frac{n}{m+n} \cdot \frac{m}{m+n+k} \\ &= \frac{m(m+k) + mn}{(m+n)(m+n+k)} \\ &= \frac{m(m+k+n)}{(m+n)(m+n+k)} = \frac{m}{m+n} \end{aligned}$$

Answer: (C)

32. Let A, B and C be three events such that $p = P(\text{exactly one of } A \text{ or } B) = P(\text{exactly one of } B \text{ or } C) = P(\text{exactly one of } C \text{ or } A)$ and $P(A, B, C \text{ simultaneously}) = p^2$ where $0 < p < 1/2$. Then $P(\text{at least one of } A, B \text{ or } C)$ is equal to

- (A) $\frac{3p+2p^2}{2}$ (B) $\frac{2p+3p^2}{2}$
(C) $\frac{2p+3p^2}{4}$ (D) $\frac{3p+2p^2}{4}$

Solution: Exactly one of A or B means $(A - B) \cup (B - A)$. By hypothesis

$$P((A - B) \cup (B - A)) = p$$

$$P((B - C) \cup (C - B)) = p$$

$$P((C - A) \cup (A - C)) = p$$

Therefore

$$\begin{aligned} p &= P((A - B) \cup (B - A)) = P(A - B) + P(B - A) \\ &= [P(A) - P(A \cap B)] + [P(B) - P(B \cap A)] \end{aligned}$$

So

$$P(A) + P(B) - 2P(A \cap B) = p \quad (7.4)$$

Similarly

$$P(B) + P(C) - 2P(B \cap C) = p \quad (7.5)$$

$$\text{and} \quad P(C) + P(A) - P(C \cap A) = p \quad (7.6)$$

Adding Eqs. (7.4)–(7.6), we have

$$\begin{aligned} 2[P(A) + P(B) + P(C) - P(A \cap B) \\ - P(B \cap C) - P(C \cap A)] = 3p \end{aligned} \quad (7.7)$$

Now $P(\text{atleast one of } A, B \text{ or } C)$ is given by [see part (3), Theorem 7.2]

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) \\ &\quad - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \\ &= \frac{3p}{2} + p^2 \text{ [Eq. (7.7) and } P(A \cap B \cap C) = p^2] \\ &= \frac{3p + 2p^2}{2} \end{aligned}$$

Answer: (A)

33. A and B are two events with positive probabilities. Consider the following two statements.

Statement I: $P(\bar{A}/B) = 1 - P(A/B)$.

Statement II: $P(\bar{A} \cup \bar{B}) = 1 - P(A)P(B/A)$.

Then

(A) both I and II are true (B) I and II are false

(C) only I is true (D) only II is true

Solution:

(1) We have

$$\begin{aligned} P(\bar{A}/B) &= \frac{P(\bar{A} \cap B)}{P(B)} \\ &= \frac{P(B) - P(A \cap B)}{P(B)} \\ &= \frac{P(B) - P(A \cap B)}{P(B)} \\ &= 1 - \frac{P(A \cap B)}{P(B)} \\ &= 1 - P(A/B) \end{aligned}$$

Hence Statement I is true.

(2) By De'Morgans law we have

$$\begin{aligned} P(\bar{A} \cup \bar{B}) &= P(\overline{A \cap B}) \\ &= 1 - P(A \cap B) \\ &= 1 - P(A)P(B/A) \text{ (Theorem 7.3)} \end{aligned}$$

Therefore Statement II is also true. Hence both I and II are true.

Answer: (A)

34. Purse A contains 9 coins of 50 paise denomination and a one rupee coin. Another purse B contains 10 coins of 50 paise denomination. 9 coins are selected at random from A and transferred to B . Again 9 coins are selected at random from B and transferred to A . The probability that the rupee coin is still in purse A is

(A) $\frac{11}{19}$ (B) $\frac{10}{19}$ (C) $\frac{9}{19}$ (D) $\frac{7}{19}$

Solution: Let

E_1 = Event of the rupee coin transferred from A to B and coming to A .

E_2 = Event of rupee coin does not get transferred from A .

E = Event of the rupee coin is still in A .

Clearly $E = E_1 \cup E_2$ and E_1, E_2 are mutually exclusive. Therefore

$$\begin{aligned} P(E) &= P(E_1) + P(E_2) \\ &= \frac{{}^9C_8 \times {}^1C_1}{{}^{10}C_9} \times \frac{{}^{18}C_8 \times {}^1C_1}{{}^{19}C_9} + \frac{{}^9C_9}{{}^{10}C_9} \\ &= \frac{9}{10} \times \frac{18}{8 \times 19} \times \frac{9 \times 10}{19} + \frac{1}{10} \\ &= \frac{9}{10} \times \frac{9}{19} + \frac{1}{10} \\ &= \frac{81 + 19}{10 \times 19} = \frac{10}{19} \end{aligned}$$

Answer: (B)

35. A set S contains n elements. A subset P of S is selected at random and after inspecting the elements, they are put back into S and then a subset Q is selected at random. The probability that P and Q are disjoint subsets is equal to

(A) $\left(\frac{2}{3}\right)^n$ (B) $\left(\frac{1}{4}\right)^n$ (C) $\left(\frac{1}{3}\right)^n$ (D) $\left(\frac{3}{4}\right)^n$

Solution: Since a set containing n has 2^n subsets, the total number of outcomes $= 2^n \times 2^n = 4^n$. Suppose P contains " r " elements. Then, Q will have 2^{n-r} choices because $P \cap Q = \phi$. For each selection of an r -element set P , there correspond 2^{n-r} choices for Q . Therefore the total number of choices for P and Q with $P \cap Q$ being empty set is (since P can be selected in nC_r ways)

$$\begin{aligned} \sum_{r=0}^n {}^nC_r 2^{n-r} &= {}^nC_0 2^n + {}^nC_1 2^{n-1} + {}^nC_2 2^{n-2} + \dots + {}^nC_n 2^0 \\ &= (2 + 1)^n = 3^n \end{aligned}$$

Therefore the required probability is

$$\frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n$$

Answer: (D)

36. Out of $2n + 1$ consecutive natural numbers, if three numbers are selected at random, the probability that they are in Arithmetic progression is

(A) $\frac{3n}{4n^2 - 1}$ (B) $\frac{3n}{4n^2 + 1}$
(C) $\frac{2n}{4n^2 - 1}$ (D) $\frac{2n}{4n^2 + 1}$

Solution: Three numbers out of $2n + 1$ numbers can be selected in

$${}^{(2n+1)}C_3 = \frac{n(4n^2 - 1)}{3} \text{ ways}$$

Let a, b, c be three numbers from the given $2n + 1$ numbers that are in AP. We need not distinguish between a, b, c and c, b, a .

$$a, b, c \text{ are in AP} \Leftrightarrow a + c = 2b$$

Therefore either both a and c are even or both are odd, because b is fixed as per the relation

$$a + c = 2b$$

Now $2n + 1$ consecutive numbers contain n even and $n + 1$ odd or $n + 1$ even and n odd numbers. Therefore the total number of choices for a and c is

$${}^nC_2 + {}^{(n+1)}C_2 = n^2$$

Hence required probability is

$$\frac{n^2}{[n(4n^2 - 1)]/3} = \frac{3n}{4n^2 - 1}$$

Answer: (A)

37. Two numbers are selected randomly from the set $S = \{1, 2, 3, 4, 5, 6\}$ one by one without replacement. The probability that minimum of the numbers is less than 4 is

- (A) $\frac{1}{15}$ (B) $\frac{14}{15}$ (C) $\frac{1}{5}$ (D) $\frac{4}{5}$

Solution: Two numbers can be selected one after other without replacement in $6 \cdot 5 = 30$ ways. Let $E =$ The minimum of the two is less than 4.

The favourable cases to E are

- (1, 2) (1, 3), (1, 4), (1, 5), (1, 6)
 (2, 1), (2, 3), (2, 4), (2, 5), (2, 6)
 (3, 1), (3, 2), (3, 4), (3, 5), (3, 6)
 (4, 1), (4, 2), (4, 3)
 (5, 1), (5, 2), (5, 3)
 (6, 1), (6, 2), (6, 3)

Therefore number of favourable cases to E is 24. The required probability is

$$P(E) = \frac{24}{30} = \frac{4}{5}$$

Alternate Solution

$\bar{E}$ is the event that both numbers must be different and among 4, 5, 6. The number of favourable cases to $\bar{E}$ is $3 \cdot 2 = 6$. Therefore

$$P(\bar{E}) = \frac{6}{30} = \frac{1}{5}$$

Hence

$$P(E) = 1 - P(\bar{E}) = 1 - \frac{1}{5} = \frac{4}{5}$$

Answer: (D)

Try it out Find the fault in the following argument. One number can be selected from $\{1, 2, 3, 4\}$ and the other from $S = \{1, 2, 3, 4, 5, 6\}$. Therefore total number of favourable cases to E is $4 \times 6 = 24$ and hence

$$P(E) = \frac{24}{30} = \frac{4}{5}$$

38. Two numbers are selected at random from the naturals 1, 2, 3, ..., 100 and are multiplied together. The probability that the product thus obtained is divisible by 3 (upto two places of decimals) is

- (A) 0.65 (B) 0.55 (C) 0.45 (D) 0.35

Solution: Two numbers can be selected from 100 in ${}^{100}C_2 = 99 \times 50$ ways. A product of two positive integers is divisible if and only if at least one of them is divisible by 3 (since 3 is a prime).

Let E be the event that the product is divisible by 3. Let a and b be the selected numbers. Therefore 3 divides ab if and only if either 3 divides a or 3 divides b . Among the numbers 1, 2, 3, ..., 100, there are 33 multiples of 3.

Now $\bar{E} =$ Event of none of a and b is divisible by 3 so that there ${}^{67}C_2$ favourable cases to $\bar{E}$. Therefore

$$\begin{aligned} P(\bar{E}) &= \frac{{}^{67}C_2}{{}^{100}C_2} \\ &= \frac{67 \times 33}{99 \times 50} \\ &= \frac{67}{150} \end{aligned}$$

Hence

$$P(E) = 1 - P(\bar{E}) = \frac{83}{150} = 0.55$$

Answer: (B)

39. There are four machines and it is known that exactly two of them are faulty. They are tested one by one, in a random order till both the faulty machines are identified. The probability that only two tests are needed is

- (A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Solution: The procedure ends in first two tests if either both are faulty or both are good. Therefore the probability is

$$\frac{2}{4} \times \frac{1}{3} + \frac{2}{4} \times \frac{1}{3} = \frac{1}{3}$$

Answer: (A)

40. Let $A = \{1, 2, 3, 4\}$ and $B = \{x_1, x_2\}$ be two sets. If a function from A to B is selected at random, the probability that it is a surjection is

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{5}$ (D) $\frac{7}{8}$

Solution: Total number of functions from A to B is $2^4 = 16$, out of which 2 are constant functions and the remaining 14 are surjection. Hence the required probability is

$$\frac{14}{16} = \frac{7}{8}$$

Answer: (D)

41. A pair of dice is rolled together till the sum of the faces is either 5 or 7. The probability that 5 comes before 7 is

- (A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$

Solution: Let A be the event that the sum is 5 and B the event that the sum is 7. $\bar{A}$ and $\bar{B}$ are, respectively, their complements. Therefore

$$A = A \cup (\bar{A} \cap \bar{B} \cap A) \cup (\bar{A} \cap \bar{B} \cap \bar{A} \cap \bar{B} \cap A) \cup \dots$$

For A , the favourable cases are

$$\{(1, 4), (4, 1), (2, 3), (3, 2)\}$$

For B , the favourable cases are

$$\{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$$

Therefore

$$P(A) = \frac{4}{36} = \frac{1}{9} \quad \text{and} \quad P(B) = \frac{6}{36} = \frac{1}{6}$$

So

$$P(\bar{A}) = 1 - \frac{1}{9} = \frac{8}{9}$$

$$P(\bar{B}) = 1 - \frac{1}{6} = \frac{5}{6}$$

$$P(A \cup B) = P(A) + P(B) = \frac{1}{9} + \frac{1}{6} = \frac{5}{18}$$

and hence

$$P(\bar{A} \cap \bar{B}) = 1 - \frac{5}{18} = \frac{13}{18}$$

Now

$$\begin{aligned} P(A) &= P(A) + P(\bar{A} \cap \bar{B} \cap A) \\ &\quad + P(\bar{A} \cap \bar{B} \cap \bar{A} \cap \bar{B} \cap A) + \dots + \infty \\ &= P(A) + P((\bar{A} \cup \bar{B}) \cap A) \\ &\quad + P((\bar{A} \cup \bar{B}) \cap (\bar{A} \cup \bar{B}) \cap A) + \dots + \infty \\ &= P(A) + P(\bar{A} \cup \bar{B}) \cdot P(A) + (P(\bar{A} \cup \bar{B}))^2 + \dots + \infty \\ &= \frac{1}{9} + \frac{13}{18} \cdot \frac{1}{9} + \left(\frac{13}{18}\right)^2 \cdot \frac{1}{9} + \dots + \infty \\ &= \frac{1}{9} \times \left[1 + \frac{13}{18} + \left(\frac{13}{18}\right)^2 + \dots + \infty \right] \\ &= \frac{1}{9} \times \frac{1}{1 - (13/18)} \\ &= \frac{18}{9 \times 5} = \frac{2}{5} \end{aligned}$$

Answer: (B)

42. Three faces of a fair die are yellow, two faces are green and one face is blue. If the die is tossed three times, the probability that the colours yellow, green and blue appear in the first, second and third toss, respectively, is

- (A) $\frac{1}{36}$ (B) $\frac{1}{12}$ (C) $\frac{11}{12}$ (D) $\frac{35}{36}$

Solution: The probability of yellow in the first toss is $3/6$, the probability of green in the second toss is $2/6$ and the probability of blue in the third toss is $1/6$. As these events are independent, the required probability is

$$\frac{3}{6} \times \frac{2}{6} \times \frac{1}{6} = \frac{1}{36}$$

Answer: (A)

43. A boy whose hobby is tossing a fair coin is to score one point for every tail and 2 points for every head. The boy goes on tossing the coin, till his score reaches n or exceeds n where $n \geq 2$. If p_n is the probability that his score attains exactly n , then p_n is equal to

- (A) $p_{n-1} + p_{n-2}$ (B) $\frac{1}{2} p_{n-1} + p_{n-2}$
(C) $\frac{2}{3} + \frac{(-1)^n}{2^n}$ (D) $\frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$

Solution: As usual H denotes head and T denotes tail so that

$$P(H) = \frac{1}{2} = P(T)$$

Let E_n denote the event that the score is n . One can easily see that

$$E_n = (E_{n-2} \cap H) \cup (E_{n-1} \cap T)$$

Therefore

$$\begin{aligned} p_n &= P(E_n) \\ &= P(E_{n-2} \cap H) + P(E_{n-1} \cap T) \\ &= P(E_{n-2})P(H) + P(E_{n-1})P(T) \\ &= \frac{1}{2}(p_{n-2} + p_{n-1}) \end{aligned}$$

Now

$$\begin{aligned} p_n + \frac{1}{2}p_{n-1} &= p_{n-1} + \frac{1}{2}p_{n-2} \\ &= p_{n-2} + \frac{1}{2}p_{n-3} \\ &= p_{n-3} + \frac{1}{2}p_{n-4} \\ &\dots\dots\dots \\ &\dots\dots\dots \\ &= p_2 + \frac{1}{2}p_1 \end{aligned}$$

Since

$$p_1 = P(T) = \frac{1}{2}$$

and

$$\begin{aligned} p_2 &= P((T \cap T) \cup H) \\ &= P(T)P(T) + P(H) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} = \frac{3}{4} \end{aligned}$$

we have

$$p_n + \frac{1}{2}p_{n-1} = p_2 + \frac{1}{2}p_1 = \frac{3}{4} + \frac{1}{4} = 1$$

Therefore

$$\begin{aligned} p_n - \frac{2}{3} &= \frac{1}{3} - \frac{1}{2}p_{n-1} \\ &= -\frac{1}{2}\left(p_{n-1} - \frac{2}{3}\right) \end{aligned}$$

$$\begin{aligned} &= \left(-\frac{1}{2}\right)^2 \left(p_{n-2} - \frac{2}{3}\right) \\ &= \left(-\frac{1}{2}\right)^3 \left(p_{n-3} - \frac{2}{3}\right) \\ &\dots \\ &= \left(-\frac{1}{2}\right)^{n-1} \left(p_1 - \frac{2}{3}\right) \\ &= \left(-\frac{1}{2}\right)^{n-1} \left(\frac{1}{2} - \frac{2}{3}\right) \end{aligned}$$

Hence

$$\begin{aligned} p_n &= \frac{2}{3} + \left(-\frac{1}{2}\right)^{n-1} \left(-\frac{1}{6}\right) \\ &= \frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2}\right)^n \\ &= \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n} \end{aligned}$$

Answer: (D)

44. In Problem 43, the probability that his score is exactly 5 is

- (A) $\frac{21}{32}$ (B) $\frac{21}{64}$ (C) $\frac{65}{96}$ (D) $\frac{31}{96}$

Solution: In the formula,

$$p_n = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

put $n = 5$ so that

$$p_5 = \frac{63}{3 \cdot 32} = \frac{21}{32}$$

Answer: (A)

45. A pair of fair dice is rolled. The probability that the sum is 9 given that 5 appeared on the first die is

- (A) $\frac{1}{9}$ (B) $\frac{8}{9}$ (C) $\frac{1}{6}$ (D) $\frac{5}{6}$

Solution: Let

E_1 = Event of 5 occurs on the first die.

E_2 = Sum of the faces is 9.

Favourable cases to E_1 are

(5, 1), (5, 2), (5, 3), (5, 4), (5, 5) and (5, 6)

Favourable cases to E_2 are

(5, 4), (4, 5), (3, 6) and (6, 3)

Therefore

$$E_1 \cap E_2 = \{(5, 4)\}$$

Hence

$$\begin{aligned} P(E_2/E_1) &= \frac{P(E_1 \cap E_2)}{P(E_1)} \\ &= \left(\frac{1}{36}\right) \div \left(\frac{6}{36}\right) = \frac{1}{6} \end{aligned}$$

Answer: (C)

46. Two numbers x and y are selected at random from the set of the first $3n$ natural numbers ($n \geq 1$) one after other without replacing the first drawn number. The probability that $x^2 - y^2$ is divisible by 3 is

- (A) $\frac{4n-1}{3(3n-1)}$ (B) $\frac{2n-3}{3(3n-1)}$
(C) $\frac{2n+3}{3(3n-1)}$ (D) $\frac{5n-3}{3(3n-1)}$

Solution: Fermat's theorem states that, if p is a prime and x is any natural number, then either p divides x or $x^{p-1} - 1$ is divisible by p . Hence, if x is a positive integer, then either 3 divides x or $x^2 - 1$ is divisible by 3. Similarly, either 3 divides y or $y^2 - 1$ is divisible by 3. Observe that $x^2 - y^2 = (x^2 - 1) - (y^2 - 1)$ is divisible by 3 if both x and y are not multiples of 3.

Among 1, 2, 3 ..., $3n$, there are n multiples of 3. Among the multiples of 3, we can select two one after other without replacement in $n(n-1)$ ways. Out of the remaining $2n$ numbers, we can select two one after other without replacement $2n(2n-1)$ ways. Hence the required probability is

$$\frac{n(n-1) + 2n(2n-1)}{3n(3n-1)} = \frac{5n-3}{3(3n-1)}$$

Answer: (D)

47. The coefficients b and c of the quadratic expression $x^2 + bx + c$ will be determined by throwing a fair die. If one such quadratic expression is selected at random, the probability that the equation $x^2 + bx + c = 0$ has real roots is equal to

- (A) $\frac{13}{36}$ (B) $\frac{17}{36}$ (C) $\frac{19}{36}$ (D) $\frac{7}{12}$

Solution: Total number of quadratic expressions formed is 6×6 . The equation $x^2 + bx + c = 0$ has real roots if $b^2 \geq 4c$. Table 7.6 gives you the number of values taken by b and c such that $b^2 \geq 4c$.

Table 7.6 Single correct choice type question 47

"c" value	4c value	b value ($b^2 \geq 4c$)	No. of values of b
1	4	2, 3, 4, 5, 6	5
2	8	3, 4, 5, 6	4
3	12	4, 5, 6	3
4	16	4, 5, 6	3
5	20	5, 6	2
6	24	5, 6	2
			Total 19

Therefore required probability is $19/36$.

Answer: (C)

48. In the game of cricket, an over means 6 balls. A batsman can score 0 or 1 or 2 or 3 or 4 or 6 runs per ball. In the last over of the match, a team needs 30 runs to win. The probability to win the match is

- (A) $\frac{70}{6^6}$ (B) $\frac{51}{6^6}$ (C) $\frac{91}{6^6}$ (D) $\frac{71}{6^6}$

Solution: There are four possibilities to win.

- (i) five sixes and a duck (means zero)
(ii) four sixes, a 2 and a 4
(iii) four sixes and two 3's
(iv) three sixes and three fours

Table 7.7 gives the arrangements.

Table 7.7 Single correct choice type question 48

Case	No. of arrangements
(i) Five sixes and a 0	$\frac{6}{5} - 1 = 5$ ($\because$ 666660 cannot occur)
(ii) Four sixes, a 2 and a 4	$\frac{6}{4} = 30$
(iii) Four sixes and two 3's	$\frac{6}{4 \times 2} = 15$
(iv) Three sixes and three fours	$\frac{6}{3 \times 3} = 20$
Total 70	

Therefore the required probability is

$$\frac{70}{6^6}$$

(Since for each ball the batsman can score 0, 1, 2, 3, 4 or 6, the number of total cases is 6^6 .)

Answer: (A)

Try it out Six-digit numbers are formed randomly using the digits 0, 1, 2, 3, 4 and 6 with repetitions. If a number is selected at random from these, what is the probability that the sum of the digits is 30 with 0 not in the units place?
(This question is another form of Question 48.)

49. The number of throws with a single fair die required for a person such that his chance of getting the face 6 is $1/2$ is

(A) 6 (B) 5 (C) 4 (D) 3

Solution: Let x be the number of chances required.

$$E = \text{Event of throwing 6 so that } P(E) = \frac{1}{6}$$

Therefore

$$P(\bar{E}) = 1 - \frac{1}{6} = \frac{5}{6}$$

Probability of not throwing 6 in all the x chances = $\left(\frac{5}{6}\right)^x$

Therefore

$$\text{Probability of throwing 6 at least once} = 1 - \left(\frac{5}{6}\right)^x = \frac{1}{2}$$

This implies

$$x = \frac{\log(1/2)}{\log(5/6)} = 3.8 \text{ approximately}$$

Therefore the number of chances required is 4.

Answer: (C)

50. A person is assigned three works A , B and C . The probabilities of the person finishing the works A , B and C are, respectively, p , q and $1/2$. The person will get full payment only if he either completes atleast the works A and B or A and C . If the probability of his getting full payment is $1/2$, then

(A) $p + q = pq$ (B) $p + pq = 1$
(C) $p + pq = \frac{2}{3}$ (D) $p - pq = \frac{1}{2}$

Solution: Let A , B and C denote the events of the person finishing the works A , B and C , respectively. By hypothesis

$$P(A) = p, P(B) = q \text{ and } P(C) = \frac{1}{2}$$

Let E be the event of the person getting full payment. Therefore

$$E = (A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (A \cap B \cap C)$$

Now

$$\begin{aligned} \frac{1}{2} &= P(E) \\ &= P(A)P(B)P(\bar{C}) + P(A)P(\bar{B})P(C) + P(A)P(B)P(C) \\ &= pq\left(\frac{1}{2}\right) + p(1-q)\left(\frac{1}{2}\right) + pq\left(\frac{1}{2}\right) \end{aligned}$$

Therefore

$$\begin{aligned} pq + p(1-q) + pq &= 1 \\ p + pq &= 1 \end{aligned}$$

Answer: (B)

51. Four friends put their car keys on a table. When they leave the place, they picked up their keys at random. The probability that no person picks his own key is

(A) $\frac{3}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{4}$

Solution: This problem is nothing but derangement problem (See Theorem 6.20 and Try it out, page 295, Vol. 1). The number of derangements of 4 distinct elements is

$$4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 12 - 4 + 1 = 9$$

Therefore the required probability is

$$\frac{9}{4!} = \frac{9}{24} = \frac{3}{8}$$

Answer: (A)

52. An unbiased coin is tossed. If the result is head, then a pair of fair dice is rolled and the number obtained by adding the numbers on the faces is noted. If the result is a tail, then a card is picked from a well-shuffled pack of 11 cards numbered 2, 3, 4, ..., 12 and the number is noted. The probability that the noted number is either 7 or 8 is

(A) $\frac{183}{792}$ (B) $\frac{193}{792}$ (C) $\frac{173}{792}$ (D) $\frac{153}{792}$

Solution: Let H and T be the events of head and tail, respectively. E is the event of the noted number is 7 or 8. Then

$$E = E \cap (H \cup T) = (E \cap H) \cup (E \cap T)$$

and $E \cap H$, $E \cap T$ are mutually exclusive. Therefore

$$\begin{aligned} P(E) &= P(E \cap H) + P(E \cap T) \\ &= P(H)P(E/H) + P(T)P(E/T) \quad (\text{Theorem 7.3})(7.8) \end{aligned}$$

When a pair of dice is rolled, the favourable cases to the event of the sum is 7 or 8 are

$$(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), \\ (4, 3), (4, 4), (5, 3), (3, 5), (2, 6), (6, 2)$$

Therefore

$$P(E/H) = \frac{11}{36} \quad \text{and} \quad P(E/T) = \frac{2}{11}$$

Therefore from Eq. (7.8),

$$P(E) = \frac{1}{2} \times \frac{11}{36} + \frac{1}{2} \times \frac{2}{11} \\ = \frac{121 + 72}{72 \times 11} = \frac{193}{792}$$

Answer: (B)

53. The odds in favour of a book reviewed by three independent critics are, respectively, 5:2, 4:3 and 3:4. The probability that majority of the critics give favourable remark is

$$(A) \frac{210}{343} \quad (B) \frac{209}{343} \quad (C) \frac{211}{343} \quad (D) \frac{205}{343}$$

Solution: Let E_1 , E_2 and E_3 be the events of the critics giving favourable remarks. Then

$$P(E_1) = \frac{5}{7}, P(E_2) = \frac{4}{7} \quad \text{and} \quad P(E_3) = \frac{3}{7}$$

Let E be the event that majority reviewed favourably. Therefore

$$E = (E_1 \cap E_2 \cap \bar{E}_3) \cup (\bar{E}_1 \cap E_2 \cap E_3) \\ \cup (E_1 \cap \bar{E}_2 \cap E_3) \cup (E_1 \cap E_2 \cap E_3)$$

Hence

$$P(E) = P(E_1)P(E_2)P(\bar{E}_3) + P(\bar{E}_1)P(E_2)P(E_3) \\ + P(E_1)P(\bar{E}_2)P(E_3) + P(E_1)P(E_2)P(E_3) \\ = \frac{5}{7} \times \frac{4}{7} \times \left(1 - \frac{3}{7}\right) + \left(1 - \frac{5}{7}\right) \times \frac{4}{7} \times \frac{3}{7} + \frac{5}{7} \\ \times \left(1 - \frac{4}{7}\right) \times \frac{3}{7} + \frac{5}{7} \times \frac{4}{7} \times \frac{3}{7} \\ = \frac{5}{7} \times \frac{4}{7} \times \frac{4}{7} + \frac{2}{7} \times \frac{4}{7} \times \frac{3}{7} + \frac{5}{7} \times \frac{3}{7} \times \frac{3}{7} + \frac{5}{7} \times \frac{4}{7} \times \frac{3}{7} \\ = \frac{80 + 24 + 45 + 60}{7 \times 7 \times 7} \\ = \frac{209}{343}$$

Answer: (B)

54. The probability that India winning a hockey match against Pakistan is $1/2$. In a 5-match series, India surely wins the third match is

$$(A) \frac{1}{8} \quad (B) \frac{1}{4} \quad (C) \frac{2}{3} \quad (D) \frac{1}{2}$$

Solution: India winning the third match is independent of the results of the first two matches. Hence, the probability that India surely winning the third match is $1/2$.

Answer: (D)

55. The probability of India winning a test match against England is $1/2$. In a five match series, the probability that India registers its second win in the third test is (you can assume the independence from match to match)

$$(A) \frac{1}{8} \quad (B) \frac{1}{2} \quad (C) \frac{1}{4} \quad (D) \frac{2}{3}$$

Solution: India has to win one of the first two tests and a win in the third test. Therefore required probability is

$$P(W \cap L \cap W) + P(L \cap W \cap W) = P(W)P(L)P(W) \\ + P(L)P(W)P(W) \\ = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \\ = \frac{1}{4}$$

where W denotes win and L denotes loss.

Answer: (C)

56. A person has to go through three successive tests. The probability of his passing first test is p . If he fails in one of the tests, then the probability his passing next test is $p/2$, otherwise it remains the same. For selection, the person must pass atleast two tests. The probability that the person to be selected is

$$(A) p^2 + p^3 \quad (B) 2p^2 - p^3 \\ (C) p^2 - 2p^3 \quad (D) p^2 + \frac{1}{2}p^3$$

Solution: Let E_i ($i = 1, 2, 3$) be the event of the person passing the i th test and E is the event that he is selected. Then

$$E = (E_1 \cap E_2) \cup (E_1 \cap \bar{E}_2 \cap E_3) \cup (\bar{E}_1 \cap E_2 \cap E_3)$$

Therefore

$$P(E) = P(E_1)P(E_2/E_1) + P(E_1)P(\bar{E}_2/E_1)P(E_3/\bar{E}_2) \\ + P(\bar{E}_1)P(E_2/E_1)P(E_3/E_2) \\ = p \cdot p + p(1-p)\frac{p}{2} + (1-p)\frac{p}{2} \cdot p \\ = 2p^2 - p^3$$

Answer: (B)

57. A and B are independent events and C is the event in which exactly one of A or B occurs. Then

- (A) $P(C) \leq P(A \cup B)P(\bar{A} \cap \bar{B})$
 (B) $P(C) \geq P(A \cup B)P(\bar{A} \cap \bar{B})$
 (C) $P(C) = P(A \cup B)P(\bar{A} \cap \bar{B})$
 (D) $P(C) = P(A \cup B)P(A \cap B)$

Solution: We have

$$C = (A - B) \cup (B - A) = (A \cap \bar{B}) \cup (\bar{A} \cap B)$$

Since A and B are independent, the pairs $(A, \bar{B})$, $(\bar{A}, B)$ and $(\bar{A}, \bar{B})$ are also pairs of independent events. Further $A \cap \bar{B}$ and $\bar{A} \cap B$ are mutually exclusive. Now, let $P(A) = x$ and $P(B) = y$. Therefore

$$\begin{aligned} P(C) &= P(A \cap \bar{B}) + P(\bar{A} \cap B) \\ &= P(A)P(\bar{B}) + P(\bar{A})P(B) \\ &= x(1 - y) + (1 - x)y \end{aligned} \quad (7.9)$$

Also

$$\begin{aligned} P(A \cup B)P(\bar{A} \cap \bar{B}) &= [P(A) + P(B) - P(A \cap B)]P(\bar{A})P(\bar{B}) \\ &= (x + y - xy)(1 - x)(1 - y) \\ &= (x + y)(1 - x)(1 - y) - xy(1 - x)(1 - y) \\ &\leq (x + y)(1 - x)(1 - y) \quad [\because xy(1 - x)(1 - y) \geq 0] \\ &= x(1 - x)(1 - y) + y(1 - x)(1 - y) \\ &= x(1 - y) - x^2(1 - y) + y(1 - x) - y^2(1 - x) \\ &= x(1 - y) + y(1 - x) - [x^2(1 - y) + y^2(1 - x)] \\ &\leq x(1 - y) + y(1 - x) = P(C) \quad [\text{by Eq. (7.9)}] \end{aligned}$$

Answer: (B)

58. A factory A produces 10% of defective screws and another factory B produces 20% of defective screws. A bag contains 4 screws of factory A and 5 screws of factory B . If two screws are drawn at random from the bag, then the probability that atleast one screw is defective (correct to three places of decimals) is

- (A) 0.168 (B) 0.158 (C) 0.165 (D) 0.155

Solution: Let E be the event that atleast one screw is defective. Then E can be regarded as the union of the following events.

E_1 = one defective from A and one non-defective from B .

E_2 = one defective from B and one non-defective from A .

E_3 = one defective from A and one defective from B .

E_4 = both defective from A .

E_5 = both defective from B .

It is given that

$$P(\text{defective screw of } A) = \frac{1}{10}$$

and $P(\text{defective screw of } B) = \frac{1}{5}$

Therefore

$$P(E) = P(E_1) + P(E_2) + P(E_3) + P(E_4) + P(E_5) \quad (7.10)$$

Now

$$P(E_1) = \frac{{}^4C_1 \times {}^5C_1}{{}^9C_2} \times \frac{1}{10} \times \frac{4}{5} = \frac{2}{45}$$

$$P(E_2) = \frac{{}^5C_1 \times {}^4C_1}{{}^9C_2} \times \frac{1}{5} \times \frac{9}{10} = \frac{1}{10}$$

$$P(E_3) = \frac{{}^5C_1 \times {}^4C_1}{{}^9C_2} \times \frac{1}{10} \times \frac{1}{5} = \frac{1}{90}$$

$$P(E_4) = \frac{{}^4C_2}{{}^9C_2} \times \left(\frac{1}{10}\right)^2 = \frac{1}{600}$$

$$P(E_5) = \frac{{}^5C_2}{{}^9C_2} \times \left(\frac{1}{5}\right)^2 = \frac{1}{90}$$

Substituting these values in Eq. (7.10), we have

$$P(E) = \frac{303}{1800} = 0.168 \text{ (approximately)}$$

Answer: (A)

59. If a circle is selected at random touching all the sides of a triangle, then the probability that it touches two sides externally and one side internally is

- (A) $\frac{1}{4}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Solution: It is known that there are four circles touching all the three sides of a triangle out of which one is the incircle and the other three are ex-circles. Therefore the required probability is $3/4$.

Answer: (B)

60. Three groups A , B and C are competing for the positions on the Board of Directors of a company. The probabilities of their winning are 0.5, 0.3 and 0.2, respectively. If the group A wins, the probability of introducing a new product is 0.7 and other corresponding probabilities for groups B and C are, respectively, 0.6 and 0.5. The probability that new product will be introduced is

- (A) 0.43 (B) 0.53 (C) 0.63 (D) 0.73

Solution: Let A, B and C be the events winning of A, B and C , respectively. E is the introduction of new product. Then

$$E = E \cap (A \cup B \cup C) = (E \cap A) \cup (E \cap B) \cup (E \cap C)$$

Therefore

$$\begin{aligned} P(E) &= P(E \cap A) + P(E \cap B) + P(E \cap C) \\ &= P(A)P(E/A) + P(B)P(E/B) + P(C)P(E/C) \\ &= \frac{5}{10} \times \frac{7}{10} + \frac{3}{10} \times \frac{6}{10} + \frac{2}{10} \times \frac{5}{10} \\ &= \frac{63}{100} = 0.63 \end{aligned}$$

Answer: (C)

- 61.** A box contains m green balls and n yellow balls. It is given that the probability of drawing 2 yellow balls from the bag is 5 times the probability of drawing 2 green balls. Also, the probability of drawing 1 ball of each colour is equal to 6 times the probability of drawing 2 green balls. Then the pair (m, n) is equal to
(A) $(6, 3)$ (B) $(3, 5)$ (C) $(4, 6)$ (D) $(3, 6)$

Solution: Let G_K and Y_K be the events of drawing K green balls and K yellow balls. Therefore

$$P(Y_2) = {}^nC_2 \div {}^{(m+n)}C_2 = \frac{n(n-1)}{(m+n)(m+n-1)}$$

$$P(G_2) = {}^mC_2 \div {}^{(m+n)}C_2 = \frac{m(m-1)}{(m+n)(m+n-1)}$$

$$P(G_1 \cap Y_1) = ({}^mC_1 \times {}^nC_1) \div {}^{(m+n)}C_2 = \frac{2mn}{(m+n)(m+n-1)}$$

Now by hypothesis

$$P(Y_2) = 5P(G_2) \Rightarrow n(n-1) = 5m(m-1) \quad (7.11)$$

$$\text{and } P(G_1 \cap Y_1) = 6P(G_2) \Rightarrow 2mn = 6m(m-1) \quad (7.12)$$

From Eq. (7.12), we have $n = 3(m-1)$. Put this value of n in Eq. (7.11) so that

$$3(m-1)(3m-4) = 5m(m-1)$$

$$4m^2 - 16m + 12 = 0$$

$$(m-1)(m-3) = 0$$

$$m = 1, 3$$

Now

m	$n = 3m - 3$
1	0
3	6

Since $n \neq 0$, the pair $(m, n) = (3, 6)$.

Answer: (D)

- 62.** Bag A contains 5 red and 7 white balls and bag B contains 3 red and 12 white balls. One of the bags is selected at random and one ball is drawn from it. The probability that the drawn ball is red is

(A) $\frac{37}{120}$ (B) $\frac{83}{120}$ (C) $\frac{63}{120}$ (D) $\frac{17}{120}$

Solution: Let A and B denote the events of selecting bags A and B , respectively, and R denote drawing a red ball. Then

$$R = (A \cup B) \cap R = (A \cap R) \cup (B \cap R)$$

Therefore

$$\begin{aligned} P(R) &= P(A \cap R) + P(B \cap R) \\ &= P(A)P(R/A) + P(B)P(R/B) \\ &= \frac{1}{2} \times \frac{5}{12} + \frac{1}{2} \times \frac{3}{15} \\ &= \frac{1}{2} \left(\frac{5}{12} + \frac{3}{15} \right) \\ &= \frac{25 + 12}{2 \times 60} = \frac{37}{120} \end{aligned}$$

Answer: (A)

- 63.** On each evening a boy either watches DOORDARSHAN channel or TEN SPORTS. The probability that he watches TEN SPORTS is $4/5$. If he watches DOORDARSHAN, there is a chance of $3/4$ that he will fall asleep, while it is $1/4$ when he watches TEN SPORTS. On one day, the boy is found to be asleep. The probability that the boy watched DOORDARSHAN is

(A) $\frac{5}{7}$ (B) $\frac{2}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$

Solution: Let E_1 and E_2 be the events of the boy watching DOORDARSHAN and TEN SPORTS, respectively. It is given that

$$P(E_1) = \frac{1}{5} \quad \text{and} \quad P(E_2) = \frac{4}{5}$$

Let E be the event of the boy falls asleep. Again by hypothesis

$$P(E/E_1) = \frac{3}{4} \quad \text{and} \quad P(E/E_2) = \frac{1}{4}$$

Now

$$E = E \cap (E_1 \cup E_2) = (E \cap E_1) \cup (E \cap E_2)$$

so that

$$P(E) = P(E_1)P(E/E_1) + P(E_2)P(E/E_2)$$

By Bayes' theorem (Theorem 7.7)

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2)}$$

$$= \frac{(1/5) \times (3/4)}{(1/5) \times (3/4) + (4/5) \times (1/4)} = \frac{3}{7}$$

Answer: (C)

- 64.** The chance that doctor A will diagnose disease X correctly is 60%. The chance that a patient of doctor A dies after correct treatment is 75% while it is 80% after wrong diagnosis. A patient of doctor A having disease X dies. The probability that his disease is correctly diagnosed is

(A) $\frac{8}{17}$ (B) $\frac{9}{17}$ (C) $\frac{11}{17}$ (D) $\frac{6}{17}$

Solution: Let A denote the event of correct diagnosis and E the event of patient's death. It is given that

$$P(A) = \frac{60}{100}, P(\bar{A}) = \frac{40}{100},$$

$$P(E/A) = \frac{75}{100} \quad \text{and} \quad P(E/\bar{A}) = \frac{80}{100}$$

By Bayes' theorem (Theorem 7.7)

$$P(A/E) = \frac{P(A)P(E/A)}{P(A)P(E/A) + P(\bar{A})P(E/\bar{A})}$$

$$= \frac{\frac{60}{100} \times \frac{75}{100}}{\frac{60}{100} \times \frac{75}{100} + \frac{40}{100} \times \frac{80}{100}}$$

$$= \frac{\frac{3}{5} \times \frac{3}{4}}{\frac{3}{5} \times \frac{3}{4} + \frac{2}{5} \times \frac{4}{5}}$$

$$= \frac{9}{17}$$

Answer: (B)

- 65.** A person has two coins in his pocket of which one is a fair coin and the other has heads on both the sides (i.e., two headed coin). He selects one of the coins at random and tosses. If head appears, then he will toss the other coin otherwise he tosses the same coin. The probability that head appears in the second toss is

(A) $\frac{5}{8}$ (B) $\frac{3}{8}$ (C) $\frac{7}{8}$ (D) $\frac{1}{8}$

Solution: Let

H_1 : First toss head.

T_1 : First toss tail.

H_2 : Second toss head.

E_1 : Selecting fair coin.

E_2 : Selecting two headed coin.

Then

$$H_2 = (E_1 \cap H_1 \cap H_2) \cup (E_1 \cap T_1 \cap H_2) \cup (E_2 \cap H_1 \cap H_2)$$

Therefore

$$P(H_2) = P(E_1 \cap H_1)P(H_2/(E_1 \cap H_1)) + P(E_1 \cap T_1)$$

$$P(H_2/(E_1 \cap H_1)) + P(E_2 \cap H_1)P(H_2/(E_2 \cap H_1))$$

$$= P(E_1)P(H_1/E_1)P(H_2/(E_1 \cap H_1)) + P(E_1)P(T_1/E_1)$$

$$P(H_2/(E_1 \cap T_1)) + P(E_2)P(H_1/E_2)$$

$$P(H_2/(E_2 \cap H_1)) \quad (\text{Corollary 7.1})$$

$$= \frac{1}{2} \times \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 1 \times \frac{1}{2}$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{4} = \frac{5}{8}$$

Answer: (A)

- 66.** In a test, an examinee either guesses or copies or knows the answer for a multiple choice question having FOUR choices of which exactly one is correct. The probability that he makes a guess is $1/3$ and the probability for copying is $1/6$. The probability that his answer is correct, given that he copied it is $1/8$. The probability that he knew the answer, given that his answer is correct is

(A) $\frac{5}{29}$ (B) $\frac{9}{29}$ (C) $\frac{24}{29}$ (D) $\frac{20}{29}$

Solution: Let the events be defined as

E_1 : Guessing

E_2 : Copying

E_3 : Knowing

E : Correct answer

By hypothesis,

$$P(E_1) = \frac{1}{3}, P(E_2) = \frac{1}{6}, P(E_3) = 1 - \frac{1}{3} - \frac{1}{6} = \frac{1}{2}$$

$$P(E/E_1) = \frac{1}{4} \quad (\text{out of four choices only one is correct})$$

$$P(E/E_2) = \frac{1}{8}$$

$$P(E/E_3) = 1$$

Therefore by Bayes' theorem,

$$P(E_3/E) = \frac{P(E_3)P(E/E_3)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2) + P(E_3)P(E/E_3)}$$

$$\begin{aligned}
 &= \frac{\frac{1}{2} \times 1}{\frac{1}{3} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{8} + \frac{1}{2} \times 1} \\
 &= \frac{24}{29}
 \end{aligned}$$

Answer: (C)

- 67.** A letter is to come from either LONDON or CLIFTON. The postal mark on the letter legibly shows consecutive letters "ON". The probability that the letter has come from LONDON is

(A) $\frac{12}{17}$ (B) $\frac{13}{17}$ (C) $\frac{5}{17}$ (D) $\frac{4}{17}$

Solution: Let the events be defined as
 E_1 : Letter coming from LONDON.
 E_2 : Letter coming from CLIFTON.
 E : Two consecutive letters ON.

The word LONDON contains 5 types of consecutive letters (LO, ON, ND, DO, ON) of which there are two ON's. The word CLIFTON contains 6 types of consecutive letters (CL, LI, IF, FT, TO, ON) of which there is one "ON". Now

$$P(E_1) = \frac{1}{2} = P(E_2)$$

$$P(E/E_1) = \frac{2}{5}$$

and $P(E/E_2) = \frac{1}{6}$

By Bayes' theorem,

$$P(E_1/E) = \frac{\frac{1}{2} \times \frac{2}{5}}{\frac{1}{2} \times \frac{2}{5} + \frac{1}{2} \times \frac{1}{6}} = \frac{12}{17}$$

Answer: (A)

- 68.** Three persons Mr. Iyyengar, Dr. Singh and Prof. Mukherjee are competing for the post of the principal of a degree college exclusively meant for boys. Their chances are, respectively, 0.5, 0.3 and 0.2. If Mr. Iyyengar is selected, he will introduce co-education with probability 0.5 while the probabilities are 0.7 and 0.6 with regard to Dr. Singh and Prof. Mukherjee, respectively. Co-education is introduced in the college. The probability that Dr. Singh is selected as principal is

(A) $\frac{31}{58}$ (B) $\frac{21}{58}$ (C) $\frac{27}{58}$ (D) $\frac{37}{58}$

Solution: Let the events be

E_1 : Event of Mr. Iyyengar's selection

E_2 : Event of Dr. Singh's selection

E_3 : Event of Prof. Mukherjee's selection.

E : Event of introduction of co-education.

By hypothesis,

$$P(E_1) = 0.5, P(E_2) = 0.3, P(E_3) = 0.2$$

and $P(E/E_1) = 0.5, P(E/E_2) = 0.7, P(E/E_3) = 0.6$

By Bayes' theorem,

$$\begin{aligned}
 P(E_2/E) &= \frac{P(E_2)P(E/E_2)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2) + P(E_3)P(E/E_3)} \\
 &= \frac{(0.3)(0.7)}{(0.5)(0.5) + (0.3)(0.7) + (0.2)(0.6)} \\
 &= \frac{0.21}{0.25 + 0.21 + 0.12} = \frac{0.21}{0.58} = \frac{21}{58}
 \end{aligned}$$

Answer: (B)

- 69.** Bag A contains 3 white and 2 black balls. Bag B contains 2 white and 2 black balls. One ball is drawn at random from A and transferred to B. One ball is selected at random from B and is found to be white. The probability that the transferred ball is white is

(A) $\frac{8}{13}$ (B) $\frac{5}{13}$ (C) $\frac{4}{13}$ (D) $\frac{9}{13}$

Solution: Let E_1 and E_2 denote the events of the transferred ball being white and black, respectively. W denotes the drawn ball from B is white. By hypothesis,

$$P(E_1) = \frac{{}^3C_1}{{}^5C_1} = \frac{3}{5}$$

$$P(E_2) = \frac{{}^2C_1}{{}^5C_1} = \frac{2}{5}$$

$$P(W/E_1) = \frac{{}^3C_1}{{}^5C_1} = \frac{3}{5}$$

$$P(W/E_2) = \frac{{}^2C_1}{{}^5C_1} = \frac{2}{5}$$

By Bayes' theorem,

$$\begin{aligned}
 P(E_1/W) &= \frac{P(E_1)P(W/E_1)}{P(E_1)P(W/E_1) + P(E_2)P(W/E_2)} \\
 &= \frac{\frac{3}{5} \times \frac{3}{5}}{\frac{3}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{2}{5}} = \frac{9}{13}
 \end{aligned}$$

Answer: (D)

70. A bag contains 4 black, 2 white and 6 red balls. Another bag contains 3 black and 5 white balls. An unbiased die is thrown. If either 1 or 2 appears, a ball is chosen from the first bag, otherwise a ball from the second bag. If the drawn ball is black then the probability that 2 appeared on the die is

(A) $\frac{2}{13}$ (B) $\frac{11}{13}$ (C) $\frac{6}{13}$ (D) $\frac{7}{13}$

Solution: Let the events be

E_1 : Event of 1 appearing.

E_2 : Event of 2 appearing.

E_3 : $\overline{E_1 \cup E_2} = \overline{E_1} \cap \overline{E_2}$.

B : Event of drawing a black ball.

Now

$$P(E_1) = \frac{1}{6}, P(E_2) = \frac{1}{6}, P(E_3) = \frac{4}{6}$$

and $P(B/E_1) = \frac{4}{12}$

$$P(B/E_2) = \frac{4}{12}$$

$$P(B/E_3) = \frac{3}{8}$$

By Bayes' theorem

$$\begin{aligned} P(E_2/B) &= \frac{P(E_2)P(B/E_2)}{P(E_1)P(B/E_1) + P(E_2)P(B/E_2) + P(E_3)P(B/E_3)} \\ &= \frac{\frac{1}{6} \times \frac{4}{12}}{\frac{1}{6} \times \frac{4}{12} + \frac{1}{6} \times \frac{4}{12} + \frac{4}{6} \times \frac{3}{8}} \\ &= \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{3} + \frac{3}{2}} = \frac{1}{3} \times \frac{6}{13} = \frac{2}{13} \end{aligned}$$

Answer: (A)

71. Bag A contains 5 white and 3 black balls. Bag B is empty. Four balls are taken at random from A and transferred to empty bag B. From B, a ball is drawn at random and is found to be black. Then, the probability that among the transferred balls three are black and one is white is

(A) $\frac{1}{8}$ (B) $\frac{7}{8}$ (C) $\frac{6}{7}$ (D) $\frac{1}{7}$

Solution: Let B_j be the number of black balls transferred ($j = 0, 1, 2, 3$). B is the event of drawing a black ball. Therefore

$$P(B_0) = \frac{{}^5C_4}{{}^8C_4} = \frac{5}{70}$$

$$P(B_1) = \frac{{}^3C_1 \times {}^5C_3}{{}^8C_4} = \frac{30}{70}$$

$$P(B_2) = \frac{{}^3C_2 \times {}^5C_2}{{}^8C_4} = \frac{30}{70}$$

$$P(B_3) = \frac{{}^3C_3 \times {}^5C_1}{{}^8C_4} = \frac{5}{70}$$

Also

$$P(B/B_0) = 0 \quad (\because \text{no black ball is transferred})$$

$$P(B/B_1) = \frac{1}{4}$$

$$P(B/B_2) = \frac{2}{4}$$

$$P(B/B_3) = \frac{3}{4}$$

Therefore by Bayes' theorem,

$$\begin{aligned} P(B_3/B) &= \frac{P(B_3)P(B/B_3)}{\sum_{i=0}^3 P(B_i)P(B/B_i)} \\ &= \frac{\frac{5}{70} \times \frac{3}{4}}{\frac{5}{70} \times 0 + \frac{30}{70} \times \frac{1}{4} + \frac{30}{70} \times \frac{2}{4} + \frac{5}{70} \times \frac{3}{4}} \\ &= \frac{15}{30 + 60 + 15} = \frac{15}{105} = \frac{1}{7} \end{aligned}$$

Answer: (D)

72. A person goes to office either by car, scooter, bus or train whose probabilities are, respectively, $1/7$, $3/7$, $2/7$ and $1/7$. The probability that he reaches office on time, if he takes car, scooter bus or train is $7/9$, $8/9$, $5/9$ and $8/9$, respectively. Given that he reached office in time, the probability that he travelled by car is

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{6}{7}$ (D) $\frac{5}{7}$

Solution: Let E_1 , E_2 , E_3 and E_4 denote the events of the person travelling by car, scooter, bus or train,

respectively. Let E denote the event of his reaching office in time. By hypothesis

$$P(E_1) = \frac{1}{7}, P(E_2) = \frac{3}{7}, P(E_3) = \frac{2}{7}, P(E_4) = \frac{1}{7}$$

$$P(E/E_1) = \frac{7}{9}, P(E/E_2) = \frac{8}{9}$$

$$P(E/E_3) = \frac{5}{9}, P(E/E_4) = \frac{8}{9}$$

By Bayes' theorem

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{\sum_{j=1}^4 P(E_j)P(E/E_j)}$$

$$= \frac{\frac{1}{7} \times \frac{7}{9}}{\frac{1}{7} \times \frac{7}{9} + \frac{3}{7} \times \frac{8}{9} + \frac{2}{7} \times \frac{5}{9} + \frac{1}{7} \times \frac{8}{9}}$$

$$= \frac{7}{49} = \frac{1}{7}$$

Answer: (A)

73. A natural number a is selected from the first 100 natural numbers. The probability that

$$a = \left[\frac{a}{2} \right] + \left[\frac{a}{3} \right] + \left[\frac{a}{5} \right]$$

where for any real x , $[x]$ denotes the integral part of x , is

(A) 0.4 (B) 0.3 (C) 0.2 (D) 0.1

Proof: The natural number “ a ” can be selected from the 100 numbers in 100 ways. Let

$$a = 30n + r$$

where $0 \leq r < 30$. Therefore

$$15n + 10n + 6n + \left[\frac{r}{2} \right] + \left[\frac{r}{3} \right] + \left[\frac{r}{5} \right] = 30n + r$$

$$n = r - \left[\frac{r}{2} \right] - \left[\frac{r}{3} \right] - \left[\frac{r}{5} \right]$$

Now, for each value of r , there is exactly one value of n and hence one value of a . Therefore the number of values of “ a ” satisfying the given conditions is 30. The required probability is

$$\frac{30}{100} = 0.3$$

Answer: (B)

74. If a natural number n is selected from the set $\{4, 5, 6, \dots, 23\}$, then the probability that

$$x_1 x_2 x_3 x_4 + x_2 x_3 x_4 x_5 + x_3 x_4 x_5 x_6 + \dots + x_n x_1 x_2 x_3 = 0$$

where each of x_j is either 1 or -1 is

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

Solution: Out of 20 numbers, one is selected in 20 ways. Let E be the event that

$$x_1 x_2 x_3 x_4 + x_2 x_3 x_4 x_5 + \dots + x_n x_1 x_2 x_3 = 0$$

where each $x_j = 1$ or -1 ($j = 1, 2, \dots, n$). Let

$$y_1 = x_1 x_2 x_3 x_4, y_2 = x_2 x_3 x_4 x_5, y_n = x_n x_1 x_2 x_3$$

so that

$$y_1 + y_2 + y_3 + \dots + y_n = 0$$

and each of $y_j = 1$ or -1 ($\because x_j = \pm 1$). Therefore n must be even say $n = 2m$. Among y_j , m are $+1$ and m are -1 . Therefore

$$y_1 y_2 y_3 \dots y_n = (-1)^m (1)^m = (-1)^m \quad (7.13)$$

But

$$y_1 y_2 y_3 \dots y_n = x_1^4 x_2^4 x_3^4 \dots x_n^4 = 1 \quad (\because x_j = \pm 1)$$

From Eq. (7.13), $(-1)^m = 1$ and $m = 2K$. So

$$n = 2m = 4K$$

Therefore n is a multiple of 4. Among the numbers 4, 5, 6, ..., 23 there are 5 multiples of 4. Hence

$$P(4) = \frac{5}{20} = \frac{1}{4}$$

Answer: (C)

75. “ A ” is an officer of a company and B is his subordinate. B noticed that his officer A is happy on 60% of his calls. B also noticed that if A is happy then A accepts his request with probability 0.4, and when A is not happy he accepts B 's request with probability 0.1. On one day B calls on A for a request and A accepts it. The probability that A is happy is

(A) $\frac{1}{7}$ (B) $\frac{6}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$

Solution: Let H be the event that A is happy when B calls on him and $\bar{H}$ the event that A is not happy. Let E be the event that A accepts the request of B . By hypothesis,

$$P(H) = \frac{60}{100} = \frac{3}{5}$$

$$P(\bar{H}) = \frac{2}{5}$$

$$P(E/H) = \frac{4}{10}$$

$$P(E/\bar{H}) = \frac{1}{10}$$

By Bayes' theorem,

$$\begin{aligned} P(H/E) &= \frac{P(H)P(E/H)}{P(H)P(E/H) + P(\bar{H})P(E/\bar{H})} \\ &= \frac{\frac{3}{5} \times \frac{2}{5}}{\frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{1}{10}} \\ &= \frac{6}{25} \times \frac{50}{14} = \frac{6}{7} \end{aligned}$$

Answer: (B)

76. A bag contains 4 balls of unknown colours. A ball is drawn at random from it and is found to be white. The probability that all the balls in the bag are white is

- (A) $\frac{4}{5}$ (B) $\frac{1}{5}$ (C) $\frac{3}{5}$ (D) $\frac{2}{5}$

Solution: Let W_j ($j = 1, 2, 3, 4$) denote 1, 2, 3 and 4 white balls are in the bag. Let W be the ball drawn is white. Then

$$P(W_1) = P(W_2) = P(W_3) = P(W_4) = \frac{1}{4}$$

$$P(W/W_1) = \frac{1}{4}$$

$$P(W/W_2) = \frac{2}{4}$$

$$P(W/W_3) = \frac{3}{4}$$

$$P(W/W_4) = 1$$

Therefore by Bayes' theorem

$$\begin{aligned} P(W_4/W) &= \frac{P(W_4)P(W/W_4)}{\sum_{j=1}^4 P(W_j)P(W/W_j)} \\ &= \frac{\frac{1}{4} \times 1}{\frac{1}{4} \left(\frac{1}{4} + \frac{2}{4} + \frac{3}{4} + \frac{4}{4} \right)} \\ &= \frac{4}{10} = \frac{2}{5} \end{aligned}$$

Answer: (D)

77. Let S be the set of all matrices of order 3×3 whose elements are 1, 2, 3, 0, -1, -2, -3. If a matrix is selected at random from S , the probability that it is skew-symmetric is

- (A) $\frac{1}{7^6}$ (B) $\frac{2}{7^6}$ (C) $\frac{1}{7^3}$ (D) $\frac{2}{7^3}$

Solution: Let $A \in S$. Each element of A can be among the given 7 elements. Hence the total number of matrices in S is 7^9 .

Let E be the event that A is skew-symmetric. For a skew-symmetric matrix, the principal diagonal elements must be zeros. Therefore, place 0 in the principal diagonal elements positions. When once the elements above the principal diagonal are fixed, then automatically, the positions of the elements below the principal diagonal will be fixed ($\because a_{ji} = -a_{ij}$). Hence number of favourable cases to E is 7^3 . Therefore,

$$P(E) = \frac{7^3}{7^9} = \frac{1}{7^6}$$

Answer: (A)

78. Let a and b be two consecutive integers selected from the first 20 natural numbers. The probability that $\sqrt{a^2 + b^2 + a^2b^2}$ is an odd positive integer is

- (A) $\frac{9}{19}$ (B) $\frac{10}{19}$ (C) $\frac{13}{19}$ (D) 1

Solution: With the first 20 natural numbers, there are 19 pairs of consecutive numbers. Let $a = n$, $b = n + 1$ so that $n = 1, 2, 3, \dots, 19$. Now

$$\begin{aligned} a^2 + b^2 + a^2b^2 &= n^2 + (n+1)^2 + n^2(n+1)^2 \\ &= n^4 + 2n^3 + 3n^2 + 2n + 1 \\ &= (n^2 + n + 1)^2 \end{aligned}$$

Therefore

$$\sqrt{a^2 + b^2 + a^2b^2} = n^2 + n + 1 = n(n+1) + 1$$

which is always odd because $n(n+1)$ is always even. So

$$\text{Probability} = 1$$

Answer: (D)

79. x, y, z are three numbers selected at random from the first 15 natural numbers and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$. The probability that $\vec{r} \cdot \vec{a} \leq 12$ where $\vec{a} = \vec{i} + \vec{j} + \vec{k}$ is

- (A) $\frac{44}{91}$ (B) $\frac{37}{91}$ (C) $\frac{46}{91}$ (D) $\frac{47}{91}$

Solution: The number of ways of selecting x, y, z is ${}^{15}C_3 = 455$. Now

$$\vec{r} \cdot \vec{a} \leq 12 \Leftrightarrow x + y + z \leq 12 \quad (7.14)$$

The number of possible integral solutions satisfying the inequality Eq. (7.14) is (Theorem 6.15, Vol. 1)

$$\begin{aligned}
 \sum_{n=3}^{12} {}^{(n-1)}C_2 &= {}^2C_2 + {}^3C_2 + {}^4C_2 + \cdots + {}^{11}C_2 \\
 &= ({}^3C_3 + {}^3C_2) + ({}^4C_4 + {}^4C_3 + \cdots + {}^{11}C_2) \\
 &\quad (\because {}^3C_3 = 1 = {}^2C_2) \\
 &= ({}^4C_3 + {}^4C_2) + ({}^5C_4 + {}^5C_3 + \cdots + {}^{11}C_2) \\
 &= ({}^5C_3 + {}^5C_2) + \cdots + {}^{11}C_2 \\
 &\quad \dots\dots\dots \\
 &\quad \dots\dots\dots \\
 &= ({}^{11}C_3 + {}^{11}C_2) \\
 &= {}^{12}C_3 = {}^{12}C_9 = 220
 \end{aligned}$$

The required probability is

$$\frac{220}{455} = \frac{44}{91}$$

Answer: (A)

80. Using the vertices of a cube, triangles are formed. If a triangle is selected at random from these, the probability that it is an equilateral triangle is

(A) $\frac{2}{7}$ (B) $\frac{3}{8}$ (C) $\frac{1}{7}$ (D) $\frac{5}{7}$

Solution: A cube has 8 vertices out of which 3 can be elected in ${}^8C_3 = 56$ ways. Consider Figure 7.3 in which the vertices of the cube $A_1, B_1, C_1, D_1; A_2, B_2, C_2$ and D_2 . If we consider the vertex C_1 , then $C_1D_2B_2$ is an equilateral triangle because the sides are the diagonals of 3 faces. Again $C_2A_2B_1$ is equilateral. For each vertex, there corresponds an equilateral triangle so that there are 8 equilateral triangles. Hence the probability is

$$\frac{8}{56} = \frac{1}{7}$$

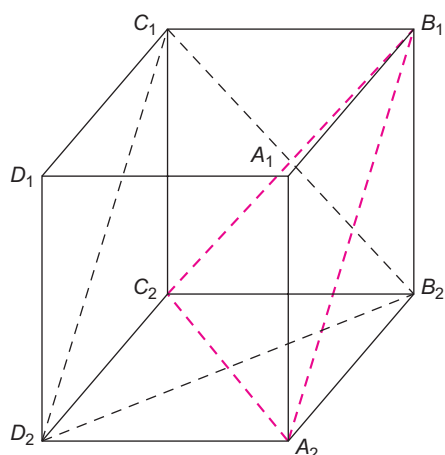


FIGURE 7.3 Single correct choice type question 80.

Answer: (C)

81. An urn contains 5 red balls and 3 blue balls. A ball is selected at random and discarded, but 2 more balls of the other colour are added into the bag. A second ball is selected at random. Then the probability that this ball is red is

(A) $\frac{31}{72}$ (B) $\frac{13}{36}$ (C) $\frac{41}{72}$ (D) $\frac{23}{36}$

Solution: Let R_1 and B_1 denote the events that first drawn ball is red and blue, respectively. R_2 is the event of the second drawn ball being red. Then

$$R_2 = R_2 \cap (R_1 \cup B_1) = (R_2 \cap R_1) \cup (R_2 \cap B_1)$$

Therefore

$$P(R) = P(R_1)P(R_2/R_1) + P(B_1)P(R_2/B_1)$$

$$\begin{aligned}
 &= \frac{5}{8} \times \frac{4}{9} + \frac{3}{8} \times \frac{7}{9} \\
 &= \frac{20 + 21}{72} = \frac{41}{72}
 \end{aligned}$$

Answer: (C)

82. Let $S = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$. A subset of S is said to be nice if it is a four-element subset such that the four elements have no common divisor except 1. If a four-element subset of S is chosen at random, the probability that it is nice is

(A) $\frac{121}{126}$ (B) $\frac{41}{42}$ (C) $\frac{1}{42}$ (D) $\frac{5}{126}$

Solution: The number of four-element subsets of $S = {}^9C_4 = 126$. Let E be the event of nice subset. Any four numbers selected from the numbers 2, 4, 6, 8, 10 have common divisor greater than 1. Four elements from 2, 4, 6, 8 and 10 can be selected in ${}^5C_4 = 5$ ways. Therefore

$$P(\bar{E}) = \frac{5}{126} \Rightarrow P(E) = 1 - P(\bar{E}) = \frac{121}{126}$$

Answer: (A)

83. An urn contains 6 white balls and 4 black balls. Another urn contains 4 white and 6 black balls. A ball is drawn from first urn and transferred to second urn and again ball is transferred from second to first. Now, a ball is drawn from the first urn. The probability that it is white is

(A) $\frac{181}{275}$ (B) $\frac{81}{275}$ (C) $\frac{221}{275}$ (D) $\frac{123}{275}$

Solution: Let

W_1 : Event of transfer of white ball from the first urn.

B_1 : Event of transfer of black ball from first urn.

W_2 : Event of white ball from second to first.

B_2 : Event of black ball from second to first.

W : Event of drawing white ball after the trans-
actions.

Therefore

$$\begin{aligned} W &= W \cap [(W_1 \cup B_1) \cap W_2 \cup (W_1 \cup B_1) \cap B_2] \\ &= W \cap (W_1 \cap W_2 \cup B_1 \cap W_2 \cup W_1 \cap B_2 \cup B_1 \cap B_2) \\ &= (W_1 \cap W_2 \cap W) \cup [(B_1 \cap W_2) \cap W] \\ &\quad \cup (W_1 \cap B_2 \cap W) \cup (B_1 \cap B_2 \cap W) \end{aligned}$$

Hence

$$\begin{aligned} P(W) &= \frac{6}{10} \times \frac{5}{11} \times \frac{6}{10} + \frac{4}{10} \times \frac{4}{11} \times \frac{7}{10} + \frac{6}{10} \times \frac{6}{11} \times \frac{5}{10} \\ &\quad + \frac{4}{10} \times \frac{7}{11} \times \frac{6}{10} \end{aligned}$$

(since all the events are independent).

Answer: (A)

- 84.** In a city 60% are males and 40% are females. Suppose 50% of males and 30% of females have colour blindness. One is selected at random and is found to be colour blind. The probability that the selected person is male is

(A) $\frac{10}{19}$ (B) $\frac{12}{87}$ (C) $\frac{9}{19}$ (D) $\frac{5}{7}$

Solution: Let

E_1 = Event of selecting male.

E_2 = Event of selecting female.

B = Event of colour blindness.

Then

$$\begin{aligned} P(E_1) &= \frac{60}{100} = \frac{3}{5} \\ P(E_2) &= \frac{40}{100} = \frac{2}{5} \\ P(B/E_1) &= \frac{30}{60} = \frac{1}{2} \\ P(B/E_2) &= \frac{12}{40} = \frac{3}{10} \end{aligned}$$

By Bayes' theorem

$$\begin{aligned} P(E_1/B) &= \frac{P(E_1)P(B/E_1)}{P(E_1)P(B/E_1) + P(E_2)P(B/E_2)} \\ &= \frac{\frac{3}{5} \times \frac{1}{2}}{\frac{3}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{3}{10}} \end{aligned}$$

$$\begin{aligned} &= \frac{3}{10} \times \frac{50}{15+6} \\ &= \frac{15}{21} = \frac{5}{7} \end{aligned}$$

Answer: (D)

- 85.** Box A contains 2 white, 1 black and 3 red balls. Box B contains 3 white, 2 black and 4 red balls. Box C contains 4 white, 3 black and 2 red balls. A die is rolled. If 1, 2 or 3 appears, then a ball from box A will be drawn. If 4 or 5 appear, then a ball from box B will be drawn, otherwise a ball from box C will be drawn. A ball is chosen from one of the boxes and it is found to be red. The probability that is from box B is

(A) $\frac{12}{47}$ (B) $\frac{16}{47}$ (C) $\frac{11}{47}$ (D) $\frac{13}{47}$

Solution: Let A , B and C denote the selections of the box A , B and C , respectively, and R denote drawing red ball. By hypothesis

$$P(A) = \frac{3}{6}, \quad P(B) = \frac{2}{6} \quad \text{and} \quad P(C) = \frac{1}{6}$$

Also

$$P(R/A) = \frac{3}{6}, \quad P(R/B) = \frac{4}{9} \quad \text{and} \quad P(R/C) = \frac{2}{9}$$

Therefore by Bayes' theorem

$$\begin{aligned} P(B/R) &= \frac{P(B)P(R/B)}{P(A)P(R/A) + P(B)P(R/B) + P(C)P(R/C)} \\ &= \frac{\frac{2}{6} \times \frac{4}{9}}{\frac{3}{6} \times \frac{3}{6} + \frac{2}{6} \times \frac{4}{9} + \frac{1}{6} \times \frac{2}{9}} \\ &= \frac{\frac{4}{27}}{\frac{1}{4} + \frac{4}{27} + \frac{1}{27}} \\ &= \frac{4}{27} \times \frac{108}{27+20} = \frac{16}{47} \end{aligned}$$

Answer: (B)

- 86.** Let X be a random variable whose range is $\{-1, 0, 1\}$. If mean μ of X is 0.1, $P(X=0) = 0.3$, then $P(X=1)$ is
- (A) 0.2 (B) 0.4 (C) 0.32 (D) 0.3

Solution: By Definition 7.17

$$\mu = \sum x_k P(X = x_k)$$

Therefore

$$0.1 = (-1)P(X = -1) + (0)P(X = 0) + (1)P(X = 1)$$

This implies

$$0.1 = -u + v \quad (7.15)$$

where $u = P(X = -1)$ and $v = P(X = 1)$. Again sum of the probabilities of $X = 1$. This implies

$$\begin{aligned} u + 0.3 + v &= 1 \\ \Rightarrow u + v &= 0.7 \end{aligned} \quad (7.16)$$

From Eqs. (7.15) and (7.16), $u = 0.3$, $v = 0.4$. Therefore

$$P(X = 1) = 0.4$$

Answer: (B)

87. The range of random variable X is $\{1, 2, 3\}$ and $P(X = 1) = 3\lambda^3$, $P(X = 2) = 4\lambda - 10\lambda^2$, $P(X = 3) = 5\lambda - 1$ where λ is constant. Then $P(2 \leq X \leq 3)$ is equal to

- (A) $\frac{8}{9}$ (B) $\frac{2}{3}$ (C) $\frac{4}{9}$ (D) $\frac{1}{3}$

Solution: We have

$$\begin{aligned} \sum_{K=1}^3 P(X = K) &= 1 \\ \Rightarrow 3\lambda^3 - 10\lambda^2 + 9\lambda - 1 &= 1 \\ \Rightarrow 3\lambda^3 - 10\lambda^2 + 9\lambda - 2 &= 0 \\ \Rightarrow (\lambda - 1)(3\lambda - 1)(\lambda - 2) &= 0 \end{aligned}$$

λ cannot be 1 and 2. Therefore $\lambda = 1/3$. Now

$$\begin{aligned} P(2 \leq X \leq 3) &= P(X = 2) + P(X = 3) \\ &= 4\left(\frac{1}{3}\right) - 10\left(\frac{1}{9}\right) + \frac{5}{3} - 1 \\ &= -\frac{10}{9} + 2 = \frac{8}{9} \end{aligned}$$

Answer: (A)

88. The range of a random variable is the set of all natural numbers and λ is a positive constant. If

$$P(X = k) = \frac{\lambda^k}{k} \quad (k = 1, 2, 3, \dots)$$

then λ equals

- (A) $\log_e 2$ (B) $\log_e 3$ (C) $\frac{1}{2}$ (D) $\frac{1}{2} \log_e 3$

Solution: We have

$$\sum_{K=1}^{\infty} P(X = K) = 1 \Rightarrow \frac{\lambda}{1} + \frac{\lambda^2}{2} + \frac{\lambda^3}{3} + \dots + \infty = 1$$

It is known that

$$e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \infty$$

where x is real or complex. Therefore

$$\begin{aligned} e^\lambda - 1 &= 1 \\ \Rightarrow e^\lambda &= 2 \\ \Rightarrow \lambda &= \log_e 2 \end{aligned}$$

Answer: (A)

89. The distribution of a random variable X whose range is $\{1, 2, 3, 4\}$ is given in Table 7.8.

Table 7.8 Single correct choice type question 89

X	1	2	3	4
$P(X = x)$	K	$2K$	$3K$	$4K$

Then, the mean and variance of X are, respectively

- (A) 2, 3 (B) 3, 1 (C) 3, 2 (D) 2, 4

Solution: We have

$$\begin{aligned} \sum_{x=1}^4 P(X = x) &= 1 \\ \Rightarrow 10K &= 1 \\ \Rightarrow K &= \frac{1}{10} \end{aligned}$$

By Definition 7.17, mean is

$$\begin{aligned} \mu &= \sum xP(X = x) \\ &= 1(K) + 2(2K) + 3(3K) + 4(4K) \\ &= 30K = \frac{30}{10} = 3 \end{aligned}$$

Also, if σ^2 is the variance, then by Quick Look 2

$$\begin{aligned} \sigma^2 + \mu^2 &= \sum x^2 P(X = x) \\ &= 1^2(K) + 2^2(2K) + 3^2(3K) + 4^2(4K) \\ &= 100K = \frac{100}{10} = 10 \end{aligned}$$

Therefore

$$\sigma^2 = 10 - 3^2 = 1$$

Answer: (B)

90. Let X be a binomial variate with parameters n and p . If the mean is 20 and variance is 15, then p is equal to

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

Solution: Mean $\mu = np$ and variance $\sigma^2 = npq$ (see Theorem 7.10). Therefore

$$np = \mu = 20$$

and $15 = npq$

Solving we get

$$15 = 20q$$

$$\Rightarrow q = \frac{3}{4}$$

$$\Rightarrow p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$$

Answer: (B)

91. X is a binomial variate with mean 10 and variance 5. Then the number of trials is equal to

- (A) 10 (B) 15 (C) 20 (D) 16

Solution: Let n be the number of trial and p the probability of a success. Then by Theorem 7.10

$$10 = np, 5 = npq$$

Hence

$$5 = 10q \Rightarrow q = \frac{1}{2}$$

and $p = 1 - q = \frac{1}{2}$

So

$$10 = np = n \left(\frac{1}{2} \right) \\ \Rightarrow n = 20$$

Answer: (C)

92. If X is a Poisson variate such that $P(X = 1) = p(X = 2)$, then the parameter m is equal to

- (A) 2 (B) 3 (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

Solution: By hypothesis (Section 7.6.2)

$$P(X = k) = e^{-m} \frac{m^k}{k!}$$

By hypothesis

$$\frac{e^{-m} m}{1!} = \frac{e^{-m} m^2}{2!}$$

Therefore

$$m = 2$$

Answer: (A)

93. X is a Poisson variate such that

$$P(X = 2) = \frac{2}{3} P(X = 1)$$

then $P(X = 3)$ is

- (A) $\frac{34}{81} e^{-3/4}$ (B) $\frac{36}{81} e^{-4/3}$
(C) $\frac{32}{81} e^{-4/3}$ (D) $\frac{36}{81} e^{-3/4}$

Solution: Let m be the parameter of the Poisson distribution. Then

$$P(X = k) = \frac{e^{-m} m^k}{k!}$$

Hence

$$P(X = 2) = \frac{2}{3} P(X = 1)$$

$$\Rightarrow \frac{e^{-m} \cdot m^2}{2!} = \frac{2}{3} e^{-m} \cdot m$$

$$\Rightarrow m = \frac{4}{3}$$

and so

$$P(X = 3) = \frac{e^{-4/3} (4/3)^3}{3!} = \frac{64}{27 \times 6} e^{-4/3} = \frac{32}{81} e^{-4/3}$$

Answer: (C)

94. The probability that a candidate securing admission in IIT through entrance test is $1/10$. Seven candidates are selected at random from a centre. The probability that two will get admission in IIT through entrance test is

- (A) $20(0.1)^2 (0.9)^5$ (B) $15(0.1)^2 (0.9)^5$
(C) $21(0.1)^2 (0.9)^5$ (D) $2(0.1)^2 (0.9)^5$

Solution: Let

$$p = \text{Probability of success} = \frac{1}{10} = 0.1$$

$$q = \text{Probability of failure} = 1 - 0.1 = 0.9$$

Therefore

$$P(X = 2) = \text{Probability of 2 success and 5 failures} \\ = {}^7C_2 (0.1)^2 (0.9)^5 \quad (\text{Definition 7.18}) \\ = 21(0.1)^2 (0.9)^5$$

Answer: (C)

95. A person has three coins A , B and C in his pocket out of which A is a fair coin. The probability of B showing head is $2/3$ and that of C is $1/3$. He selected

one of the coins at random and tossed it three times and observed 2 heads and 1 tail. The probability that the selected coin is A is

- (A) $\frac{7}{25}$ (B) $\frac{18}{25}$ (C) $\frac{9}{25}$ (D) $\frac{16}{25}$

Solution: Let A , B and C denote the events of selecting the coins A , B and C , respectively, so that

$$P(A) = P(B) = P(C) = \frac{1}{3}$$

Let E be the event of getting 2 heads and 1 tail. Hence by the binomial distribution we have

$$P(E/A) = {}^3C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right) = \frac{3}{8}$$

$$P(E/B) = {}^3C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right) = \frac{4}{9}$$

$$P(E/C) = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = \frac{2}{9}$$

Therefore by Bayes' theorem

$$\begin{aligned} P(A/E) &= \frac{P(A)P(E/A)}{P(A)P(E/A) + P(B)P(E/B) + P(C)P(E/C)} \\ &= \frac{\frac{1}{3} \times \frac{3}{8}}{\frac{1}{3} \times \frac{3}{8} + \frac{1}{3} \times \frac{4}{9} + \frac{1}{3} \times \frac{2}{9}} \\ &= \frac{\frac{3}{8}}{\frac{3}{8} + \frac{4}{9} + \frac{2}{9}} \\ &= \frac{27}{75} = \frac{9}{25} \end{aligned}$$

Answer: (C)

96. A man takes a step forward with probability 0.4 and a step backward with probability 0.6. After 11 steps, the probability that he is one step away from the starting point is

- (A) $462 \times (0.24)^5$ (B) $462 \times (0.25)^5$
(C) $362 \times (0.24)^5$ (D) $362 \times (0.25)^5$

Solution: One step away means, six forward and five backward steps or six backward and five forward steps. Therefore

$$\begin{aligned} P(\text{one step away}) &= {}^{11}C_6 (0.4)^6 (0.6)^5 + {}^{11}C_6 (0.6)^6 (0.4)^5 \\ &= {}^{11}C_6 (0.4)^5 (0.6)^5 (0.4 + 0.6) \\ &= 462 \times (0.24)^5 \end{aligned}$$

Answer: (A)

97. The probability of a coin showing head is p . 100 coins are tossed at a time. If the probability of 50 coins showing heads is same as 51 coins showing heads, then the value of p is

- (A) $\frac{49}{100}$ (B) $\frac{51}{101}$ (C) $\frac{1}{2}$ (D) $\frac{49}{101}$

Solution: By hypothesis,

$${}^{100}C_{50} p^{50} q^{50} = {}^{100}C_{51} p^{51} q^{49}$$

where $q = 1 - p$. Therefore

$$\begin{aligned} \frac{{}^{100}C_{50}}{{}^{100}C_{51}} (q) &= \frac{{}^{100}C_{51}}{{}^{100}C_{50}} (p) \\ \frac{q}{50} &= \frac{p}{51} \\ 51q &= 50p \\ 51(1 - p) &= 50p \\ p &= \frac{51}{101} \end{aligned}$$

Answer: (B)

98. In an experiment, the odds in favour of success are 2:1. In six trials, the probability of at least five success is

- (A) $\frac{240}{729}$ (B) $\frac{256}{729}$ (C) $\frac{496}{729}$ (D) $\frac{396}{729}$

Solution: Let X denote the number of successes. Then X is a binomial variate with

$$\begin{aligned} p(\text{probability of success}) &= \frac{2}{3} \\ q &= 1 - p = \frac{1}{3} \end{aligned}$$

Then

$$\begin{aligned} P(\text{at least five successes}) &= P(X = 5) + P(X = 6) \\ &= {}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) + {}^6C_6 \left(\frac{2}{3}\right)^6 \left(\frac{1}{3}\right)^0 \\ &= \frac{6 \times 32}{729} + \frac{64}{729} = \frac{256}{729} \end{aligned}$$

Answer: (B)

99. A book writer writes a good book with probability $1/2$. If it is a good book, the probability that it will be published is $2/3$, otherwise it is $1/4$. If he writes 2 books, the probability that at least one book will be published is

- (A) $\frac{407}{576}$ (B) $\frac{411}{576}$ (C) $\frac{405}{576}$ (D) $\frac{307}{576}$

Solution: Let

G = Event of good book

G' = Event of not a good book

E = Event of publication

Then

$$E = (G \cup G') \cap E = (G \cap E) \cup (G' \cap E)$$

Now

$$P(E/G) = \frac{2}{3}$$

$$P(E/G') = \frac{1}{4}$$

$$P(G) = \frac{1}{2} = P(G')$$

Therefore

$$P(E) = \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{1}{4} = \frac{11}{24}$$

Further, X denotes the number of books published. Then

$$P(\text{at least one book will be published}) = P(X=1) + P(X=2)$$

$$= {}^2C_1 \left(\frac{11}{24} \right) \left(\frac{13}{24} \right) + {}^2C_2 \left(\frac{11}{24} \right)^2 \left(\frac{13}{24} \right)^0$$

$$= 2 \times \frac{11}{24} \times \frac{13}{24} + \left(\frac{11}{24} \right)^2$$

$$= \frac{407}{576}$$

Answer: (A)

100. A fair coin is tossed 6 times. The probability of getting at least four heads is

(A) $\frac{21}{32}$ (B) $\frac{19}{32}$ (C) $\frac{13}{32}$ (D) $\frac{11}{32}$

Solution: With a fair coin,

Probability of getting head = Probability of getting tail

$$= \frac{1}{2}$$

Let X be the number of heads. Now

$$P(\text{getting at least four heads}) = P(X=4) + P(X=5)$$

$$+ P(X=6)$$

$$= {}^6C_4 \left(\frac{1}{2} \right)^4 \left(\frac{1}{2} \right)^2 + {}^6C_5 \left(\frac{1}{2} \right)^5 \left(\frac{1}{2} \right) + {}^6C_6 \left(\frac{1}{2} \right)^6 \left(\frac{1}{2} \right)^0$$

$$\begin{aligned} &= \frac{15}{64} + \frac{6}{64} + \frac{1}{64} \\ &= \frac{22}{64} = \frac{11}{32} \end{aligned}$$

Answer: (D)

101. If on average 1 ship in every 10 is wrecked, then out of 5 ships expected to reach the shore, the probability that 4 at least will arrive safely is

(A) 0.9 (B) 0.905 (C) 0.891 (D) 0.99

Solution: Let p be the probability of safe arrival and q be the probability of wreck. By hypothesis

$$q = \frac{1}{10} \quad \text{and} \quad p = 1 - \frac{1}{10} = \frac{9}{10}$$

Let X denote the number of ships arriving safely. Now,

$$P(\text{at least four safe arrivals}) = P(X=4) + P(X=5)$$

$$= {}^5C_4 \left(\frac{9}{10} \right)^4 \left(\frac{1}{10} \right)$$

$$+ {}^5C_5 \left(\frac{9}{10} \right)^5 \left(\frac{1}{10} \right)^0$$

$$= 5 \times \frac{9^4}{10^5} + \frac{9^5}{10^5}$$

$$= \frac{5 \times 6561}{10^5} + \frac{59049}{10^5}$$

$$= \frac{91854}{10 \times 10^4}$$

$$= \frac{45927}{50000}$$

$$= 0.905$$

Answer: (B)

102. An electrical bulb manufacturing company supplies bulbs to a firm. It is known that 1 bulb out of every 10 is defective. If 8 bulbs of the company are selected at random, the probability that the selection contains 5 good and 3 defective bulbs is

(A) $\frac{56 \times 9^5}{10^8}$

(B) $\frac{44 \times 9^5}{10^8}$

(C) $\frac{46 \times 9^5}{10^8}$

(D) $\frac{54 \times 9^5}{10^8}$

Solution: We have

$$p = \text{Probability for a good bulb} = \frac{9}{10}$$

$$q = \text{Probability for a defective bulb} = \frac{1}{10}$$

Let X denote the number of good bulbs out of 8. Then

$$P(X=5) = {}^8C_5 \left(\frac{9}{10}\right)^5 \left(\frac{1}{10}\right)^3$$

$$= \frac{56 \times 9^5}{10^8}$$

Answer: (A)

- 103.** A random variable X has Poisson distribution such that $P(X=1) = P(X=2)$. If μ and σ^2 are the mean and variance of the distribution, then

$$\frac{\mu}{\sigma^2} =$$

- (A) 2 (B) $\frac{1}{2}$ (C) 1 (D) $\sqrt{2}$

Solution: Let λ be the parameter of X . By hypothesis

$$P(X=1) = P(X=2)$$

$$\Rightarrow \frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\Rightarrow \lambda = 2$$

For a Poisson distribution, mean and variance are equal (see Theorem 7.11)

$$\frac{\lambda}{\sigma^2} = \frac{2}{2} = 1$$

Answer: (C)

- 104.** A random variable X has Poisson distribution with parameter 2. Then $P(X \geq 3)$ is

- (A) $1 + \frac{5}{e^2}$ (B) $1 - \frac{5}{e^2}$

- (C) $1 + \frac{3}{e^2}$ (D) $1 - \frac{3}{e^2}$

Solution: We know that

$$\sum_{k=0}^{\infty} P(X=k) = 1$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{e^{-2} 2^k}{k!} = 1$$

Therefore

$$\sum P(X \geq 3) = 1 - e^{-2} \left(1 + \frac{2}{1!} + \frac{2^2}{2!}\right)$$

$$= 1 - \frac{5}{e^2}$$

Answer: (B)

- 105.** If the mean of a Poisson distribution is $1/2$ then the ratio of $P(X=3)$ is to $P(X=2)$ is

- (A) 1:6 (B) 1:8 (C) 1:4 (D) 1:2

Solution: For a Poisson distribution, the mean is equal to the parameter. Hence the parameter is $1/2$. Now,

$$P(X=3) = \frac{e^{-1/2} (1/2)^3}{3!}$$

$$P(X=2) = \frac{e^{-1/2} (1/2)^2}{2!}$$

Therefore

$$P(X=3):P(X=2) = \frac{1}{48} : \frac{1}{8}$$

$$= 1:6$$

Answer: (A)

Multiple Correct Choice Type Questions

- 1.** A and B are two events of a random experiment such that $P(A) > 0$ and $P(B) \neq 1$. Then $P(\bar{A}/\bar{B})$ equals

- (A) $1 - P(A/B)$ (B) $1 - P(A/\bar{B})$
 (C) $1 - P(\bar{A}/B)$ (D) $\frac{1 - P(A \cup B)}{P(\bar{B})}$

Solution: We have

$$P(\bar{A}/\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})}$$

$$= \frac{P(\overline{A \cup B})}{P(\bar{B})}$$

$$= \frac{1 - P(A \cup B)}{P(\bar{B})}$$

Therefore (D) is correct. Also,

$$P(A/\bar{B}) + P(\bar{A}/\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} + \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})}$$

$$= \frac{P((A \cup \bar{A}) \cap \bar{B})}{P(\bar{B})}$$

$$= \frac{P(\bar{B})}{P(\bar{B})} = 1$$

This implies

$$P(\bar{A}/\bar{B}) = 1 - P(A/\bar{B})$$

So (B) is correct.

Answers: (B), (D)

2. If M and N are two events, then the probability of exactly one of them occurs is

- (A) $P(M) + P(N) - 2P(M \cap N)$
- (B) $P(M) + P(N) - P(M \cap N)$
- (C) $P(\bar{M}) + P(\bar{N}) - 2P(\bar{M} \cap \bar{N})$
- (D) $P(\bar{M} \cap N) + P(M \cap \bar{N})$

Solution: We have

$$\begin{aligned} P(\text{exactly one of } M \text{ and } N \text{ occurs}) &= P(M \Delta N) \\ &= P((M - N) \cup (N - M)) \\ &= P(M - N) + P(N - M) \\ &= P(M) - P(M \cap N) + P(N) - P(M \cap N) \\ &\quad [\text{part (2), Theorem 7.2}] \\ &= P(M) + P(N) - 2P(M \cap N) \end{aligned}$$

Hence (A) is correct. Also,

$$\begin{aligned} P(M \cap \bar{N}) + P(\bar{M} \cap N) &= P(M - N) + P(N - M) \\ &= P(M \Delta N) \end{aligned}$$

So (D) is correct. Again

$$\begin{aligned} P(\bar{M}) + P(\bar{N}) - 2P(\bar{M} \cap \bar{N}) &= 1 - P(M) + 1 - P(N) - 2P(\overline{M \cup N}) \\ &= 2 - P(M) - P(N) - 2[1 - P(M \cup N)] \\ &= -P(M) - P(N) + 2[P(M) + P(N) - P(M \cap N)] \\ &= P(M) + P(N) - 2P(M \cap N) \\ &= P(M - N) + P(N - M) \end{aligned}$$

Hence (C) is correct.

Answers: (A), (C), (D)

3. If E and F are independent events such that $0 < P(E)$, $P(F) < 1$, then

- (A) E and F are mutually exclusive
- (B) E and $\bar{F}$ are mutually independent
- (C) $\bar{E}$ and $\bar{F}$ are mutually independent
- (D) $P(E/F) + P(\bar{E}/F) = 1$

Solution: Mutually exclusive events means one event prevents the other. Hence they are dependent events. Now E and F are independent events implies that

$$\begin{aligned} P(E/F) &= P(E) \\ \Rightarrow P(E \cap F) &= P(E)P(F) \end{aligned} \quad (7.17)$$

Now

$$\begin{aligned} P(E/\bar{F}) &= \frac{P(E \cap \bar{F})}{P(\bar{F})} \\ &= \frac{P(E - F)}{P(\bar{F})} \\ &= \frac{P(E) - P(E \cap F)}{P(\bar{F})} \\ &= \frac{P(E) - P(E)P(F)}{P(\bar{F})} \quad [\text{From Eq. (7.17)}] \\ &= \frac{P(E)(1 - P(F))}{P(\bar{F})} \\ &= \frac{P(E)P(\bar{F})}{P(F)} = P(E) \end{aligned}$$

Hence (B) is correct.

(C) follows from (B) and so (C) is correct. Now,

$$\begin{aligned} P(E/F) + P(\bar{E}/F) &= \frac{P(E \cap F)}{P(F)} + \frac{P(\bar{E} \cap F)}{P(F)} \\ &= \frac{P((E \cup \bar{E}) \cap F)}{P(F)} \\ &= \frac{P(F)}{P(F)} = 1 \end{aligned}$$

So (D) is correct.

Answers: (B), (C), (D)

4. A bag contains 9 coins of which 3 are fair, 2 are two headed and 4 are two tailed coins. Define the following events,

- A: Drawing a fair coin
- B: Drawing two headed coin
- C: Drawing coin having tails on both sides
- E: Drawn coin showing tail

Then,

$$\begin{aligned} \text{(A) } P(A \cap E) &= \frac{1}{6} & \text{(B) } P(C \cap E) &= \frac{4}{9} \\ \text{(C) } P(E) &= \frac{1}{3} & \text{(D) } P(C/E) &= \frac{8}{11} \end{aligned}$$

Solution: We have

$$P(A) = \frac{3}{9} = \frac{1}{3}$$

$$P(E/A) = \frac{1}{2}$$

Therefore

$$P(E \cap A) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

So (A) is correct. Now

$$P(C) = \frac{4}{9} \quad \text{and} \quad P(E/C) = 1$$

Therefore

$$P(C \cap E) = \frac{4}{9} \times 1 = \frac{4}{9}$$

So (B) is correct. Now the event E is given by

$$E = (A \cup C) \cap E = (A \cap E) \cup (C \cap E)$$

Therefore

$$P(E) = P(A \cap E) + P(C \cap E)$$

$$= \frac{1}{6} + \frac{4}{9} = \frac{11}{18}$$

Hence (C) is not correct. Finally,

$$P(C/E) = \frac{P(C \cap E)}{P(E)}$$

$$= \frac{4}{9} \div \frac{11}{18} = \frac{8}{11}$$

So (D) is correct.

Answers: (A), (B), (D)

5. A, B are two events with positive probabilities. If $P(A) = 0.4$, $P(B/A) = 0.9$, $P(\bar{B}/\bar{A}) = 0.6$, then

- (A) $P(A \cup B) = 0.64$ (B) $P(A \cap B) = 0.36$
 (C) $P(A/B) = 0.6$ (D) $P(A/\bar{B}) = 0.2$

Solution: We have

$$0.6 = P(\bar{B}/\bar{A})$$

$$= \frac{P(\bar{B} \cap \bar{A})}{P(\bar{A})}$$

$$= \frac{P(\overline{A \cup B})}{P(\bar{A})}$$

$$= \frac{1 - P(A \cup B)}{1 - P(A)}$$

$$= \frac{1 - P(A \cup B)}{1 - (0.4)}$$

$$= \frac{1 - P(A \cup B)}{0.6}$$

Therefore

$$P(A \cup B) = 1 - (0.6)(0.6)$$

$$= 1 - 0.36 = 0.64$$

So (A) is correct. Now

$$P(A \cap B) = P(A)P(B/A)$$

$$= (0.4)(0.9) = 0.36$$

Hence (B) is correct. Again

$$0.64 = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$= 0.4 + P(B) - (0.36)$$

Therefore

$$P(B) = 1 - (0.4) = 0.6$$

Given $P(B)$ we have

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{0.36}{0.6} = 0.6$$

So (C) is correct. Again

$$P(A/\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})}$$

$$= \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

$$= \frac{0.4 - (0.36)}{1 - (0.61)}$$

$$= \frac{0.04}{0.4} = 0.1$$

Here we have used $A = (A \cap \bar{B}) \cup (A \cap B)$. This implies (D) is not correct.

Answers: (A), (B), (C)

6. Two fair dice are thrown. Let A be the event of both faces are even numbers and B the event of sum of the faces is 10. Then

$$(A) P(A) = \frac{1}{4}$$

$$(B) P(B) = \frac{1}{18}$$

$$(C) P(A \cap B) = \frac{1}{18}$$

$$(D) P(\text{exactly one of } A \text{ and } B) = \frac{2}{9}$$

Solution: For the event A , the favourable cases are

(2, 2), (2, 4), (2, 6),
(4, 2), (4, 4), (4, 6),
(6, 2), (6, 4), (6, 6)

Therefore

$$P(A) = \frac{9}{36} = \frac{1}{4}$$

Hence (A) is correct. For the event B , the favourable cases are (4, 6), (5, 5) and (6, 4). Therefore

$$P(B) = \frac{3}{36} = \frac{1}{12}$$

So (B) is not correct. Now

$$A \cap B = \{(4, 6), (6, 4)\}$$

This implies

$$P(A \cap B) = \frac{2}{36} = \frac{1}{18}$$

So (C) is correct. Finally

$$P(\text{exactly one of } A \text{ and } B \text{ is correct})$$

$$= P((A - B) \cup (B - A))$$

$$= P(A - B) + P(B - A)$$

$$= \frac{7}{36} + \frac{1}{36} = \frac{8}{36} = \frac{2}{9}$$

So (D) is correct.

Answers: (A), (C), (D)

7. A natural number is selected at random from the first 100 natural numbers. Let A , B and C denote the events of selection of even number, a multiple of 3 and a multiple of 5, respectively. Then

$$(A) P(A \cap B) = \frac{4}{25}$$

$$(B) P(B \cap C) = \frac{3}{50}$$

$$(C) P(C \cap A) = \frac{1}{10}$$

$$(D) P(A \cup B \cup C) = \frac{37}{50}$$

Solution: We have

Number of even numbers ≤ 100 is equal to 50.

Number of multiples of 3 ≤ 100 is 33.

Number of multiples of 5 ≤ 100 is 20.

Number of common multiples of 2 and 3 is 16.

Number of common multiples of 3 and 5 is 6.

Number of common multiples of 2 and 5 is 10.

Number of common multiples of 2, 3 and 5 is 3.

Now,

$$P(A) = \frac{50}{100}, P(B) = \frac{33}{100}, P(C) = \frac{20}{100}$$

$$P(A \cap B) = \frac{16}{100}, P(B \cap C) = \frac{6}{100}, P(C \cap A) = \frac{10}{100}$$

Also

$$P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C)$$

$$- P(C \cap A) + P(A \cap B \cap C)$$

$$= \frac{50}{100} + \frac{33}{100} + \frac{20}{100} - \frac{16}{100} - \frac{6}{100} - \frac{10}{100} + \frac{3}{100}$$

$$= \frac{106 - 32}{100} = \frac{74}{100} = \frac{37}{50}$$

Hence all (A), (B), (C) and (D) are correct.

Answers: (A), (B), (C), (D)

8. A and B are two among 10 students from which 3 students are to be selected at random to form a committee. Then,

$$(A) \text{ Probability that } A \text{ belongs to the committee is } \frac{3}{10}.$$

$$(B) \text{ Probability that } A \text{ and } B \text{ belong to the committee is } \frac{1}{15}.$$

$$(C) \text{ Probability that either } A \text{ or } B \text{ belong to the committee is } \frac{8}{15}.$$

$$(D) \text{ Probability that exactly one of } A \text{ and } B \text{ belongs to the committee is } \frac{7}{15}.$$

Solution: Let A and B denote the events “ A belongs to the committee” and “ B belongs to the committee,” respectively. Then

$$P(A) = \frac{{}^9C_2}{{}^{10}C_3} = \frac{36}{120} = \frac{3}{10}$$

$$P(B) = \frac{3}{10}$$

$$P(A \cap B) = \frac{{}^8C_1}{{}^{10}C_3} = \frac{8}{120} = \frac{1}{15}$$

Now

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{3}{10} + \frac{3}{10} - \frac{1}{15} = \frac{16}{30} = \frac{8}{15} \end{aligned}$$

So

$$\begin{aligned} P(\text{exactly one of } A \text{ and } B) &= P((A - B) \cup (B - A)) \\ &= P(A - B) + P(B - A) \\ &= \frac{{}^8C_2}{{}^{10}C_3} + \frac{{}^8C_2}{{}^{10}C_3} \\ &= \frac{28}{120} + \frac{28}{120} \\ &= \frac{56}{120} \\ &= \frac{7}{15} \end{aligned}$$

Hence all (A), (B), (C) and (D) are correct.

Answers: (A), (B), (C), (D)

9. A coin is so weighted such that the probability of it showing H (Head) is $2/3$ and that of T (Tail) is $1/3$ when it is tossed. If head appears, then a number from the first 9 naturals is selected at random, otherwise a number from 1, 2, 3, 4, 5 will be selected. Let E be the event of getting an even number. Then,

(A) $P(E/H) = \frac{4}{9}$

(B) $P(E/T) = \frac{2}{5}$

(C) $P(E) = \frac{58}{135}$

(D) $P(H/E) = \frac{20}{29}$

Solution: Since there are 4 even numbers among 1 to 9, it follows that

$$P(E/H) = \frac{4}{9}$$

Again, since there are 2 even numbers among 1 to 5, it follows that

$$P(E/T) = \frac{2}{5}$$

Therefore (A) and (B) are correct. Now,

$$E = (H \cup T) \cap E = (H \cap E) \cup (T \cap E)$$

This implies

$$\begin{aligned} P(E) &= P(H \cap E) + P(T \cap E) \\ &= P(H)(E/H) + P(T)P(E/T) \quad (\text{Theorem 7.3}) \\ &= \frac{2}{3} \times \frac{4}{9} + \frac{1}{3} \times \frac{2}{5} = \frac{58}{135} \end{aligned}$$

So (C) is correct. By Bayes' theorem,

$$\begin{aligned} P(H/E) &= \frac{P(H)P(E/H)}{P(H)P(E/H) + P(T)P(E/T)} \\ &= \frac{\frac{2}{3} \times \frac{4}{9}}{\frac{2}{3} \times \frac{4}{9} + \frac{1}{3} \times \frac{2}{5}} \\ &= \frac{8}{27} \times \frac{135}{58} = \frac{20}{29} \end{aligned}$$

Hence (D) is correct.

Answers: (A), (B), (C), (D)

10. G_1 and G_2 are two groups of students. G_1 consists of 4 boys and 3 girls. G_2 consists of 3 boys and 3 girls. Let p_1 be the probability of arranging the boys and girls of G_1 alternately. p_1' is the probability of arranging the boys and girls of G_1 such that all the 4 boys are together and all the 3 girls together. p_2 is the probability of arranging boys and girls of G_2 alternately. p_2' is the probability of arranging boys and girls of G_2 such that no two boys and no of two girls are separated. Then,

(A) $p_1 = \frac{1}{35}$

(B) $p_1' = \frac{2}{35}$

(C) $p_2 = \frac{1}{10}$

(D) $p_2' = \frac{1}{20}$

Solution: Since number of boys is greater than number of girls, they can be arranged alternately only when the

first place is occupied by a boy (example BG BG BG B where B stands for boy and G for girl). Therefore

$$P_1 = \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{1} = \frac{1}{35}$$

Therefore (A) is correct. Since boys are together and girls are together (example $b_1, b_2, b_3, b_4, g_1, g_2, g_3$), we consider all the 4 boys as a single object and all the 3 girls as a single object. Two objects can be arranged in $|2|$ ways. In each of these arrangements boys can be interchanged among themselves in $|4|$ ways and girls can be interchanged among themselves in $|3|$ ways. Therefore the number of arrangements of boys and girls satisfying the condition is $|2| \times |4| \times |3|$. So

$$P_1' = \frac{|2| \times |4| \times |3|}{|7|} = \frac{2 \times 6}{5 \times 6 \times 7} = \frac{2}{35}$$

Hence (B) is correct. Number of arrangements of boys and girls alternately (example: bgbgbg or gbgbgb) = $2 \times |3| \times |3|$. Therefore

$$P_2 = \frac{2 \times |3| \times |3|}{|6|} = \frac{72}{720} = \frac{1}{10}$$

Hence (C) is correct. Finally

$$P_2' = \frac{2 \times |3| \times |3|}{|6|} = \frac{1}{10}$$

(Example: $b_1 b_2 b_3 g_1 g_2 g_3$ or $g_1 g_2 g_3 b_1 b_2 b_3$.) Therefore (D) is not correct.

Answers: (A), (B), (C)

11. Let A and B be any two events in a sample space. Then

(A) When $P(B) \neq 0$,

$$P(A/B) \geq \frac{P(A) + P(B) - 1}{P(B)}$$

is always true

(B) $P(A \cap \bar{B}) = P(A) - P(A \cap B)$ is always true

(C) $P(A \cup B) = 1 - P(\bar{A})P(\bar{B})$ if A and B are independent

(D) $P(A \cup B) = 1 - P(\bar{A})P(\bar{B})$ if A and B are disjoint.

Solution: We have

$$\begin{aligned} P(A/B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A) + P(B) - P(A \cup B)}{P(B)} \end{aligned}$$

$$\geq \frac{P(A) + P(B) - 1}{P(B)} \quad [\because P(A \cup B) \leq 1]$$

Hence (A) is correct. Again

$$\begin{aligned} A &= (A - B) \cup (A \cap B) \\ \Rightarrow P(A) &= P(A - B) + P(A \cap B) \\ \Rightarrow P(A - B) &= P(A) - P(A \cap B) \\ \Rightarrow P(A \cap \bar{B}) &= P(A) - P(A \cap B) \end{aligned}$$

Hence (B) is correct. Now if A and B are independent, then so are $\bar{A}$ and $\bar{B}$ and hence

$$P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$$

Therefore

$$\begin{aligned} P(A \cup B) &= 1 - P(\overline{A \cup B}) \\ &= 1 - P(\bar{A} \cap \bar{B}) \\ &= 1 - P(\bar{A})P(\bar{B}) \end{aligned}$$

Consider the following example which shows $P(A \cup B) \neq 1 - P(\bar{A})P(\bar{B})$ when $A \cap B = \phi$.

Let S be the sample space obtained when a fair die is rolled so that $S = \{1, 2, 3, 4, 5, 6\}$. Let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ so that $A \cap B = \phi$ and $P(A \cup B) = P(S) = 1$. But

$$P(\bar{A}) = P(B) = \frac{1}{2} = P(A) = P(\bar{B})$$

Therefore

$$\begin{aligned} 1 - P(\bar{A})P(\bar{B}) &= 1 - \frac{1}{2} \times \frac{1}{2} \\ &= \frac{3}{4} \\ &\neq P(A \cup B) \end{aligned}$$

Hence (D) is not correct.

Answers: (A), (B), (C)

12. A and B are two random events such that $0 < P(A) < 1$ and $0 < P(B) < 1$. Then which of the following are true?

(A) $P(A/B) > P(A) \Rightarrow P(B/A) > P(B)$

(B) $P(B/A) + P(B/\bar{A}) = 1$

(C) $P(A/B) = P(\bar{A}/B)$ only when $A \cap B = \phi$

(D) $P(B/A) = P(B/\bar{A})$ only when A and B are independent

Solution: Suppose

$$\begin{aligned} P(A/B) &> P(A) \\ \Rightarrow \frac{P(A \cap B)}{P(B)} &> P(A) \\ \Rightarrow \frac{P(A \cap B)}{P(A)} &> P(B) \quad [\because 0 < P(A), P(B)] \\ \Rightarrow P(B/A) &> P(B) \end{aligned}$$

So (A) is correct. We now show that (B) is false. Consider the following example: Let $S = \{1, 2, 3, 4, 5, 6\}$ which is obtained by rolling a fair die. Let $A = \{2, 6\}$ and $B = \{2, 3, 5\}$. Then

$$\begin{aligned} P(A/B) &= \frac{1}{3} \\ P(A/\bar{B}) &= \frac{1}{3} \end{aligned}$$

so that

$$P(A/B) + P(A/\bar{B}) = \frac{2}{3} \neq 1$$

(C) is also false for which consider the same sample space: $S = \{1, 2, 3, 4, 5, 6\}$ and let $A = \{2, 4, 6\}$ and $B = \{2, 3, 5\}$. Then

$$\begin{aligned} P(\bar{A}/B) &= \frac{2}{3} \\ P(A/\bar{B}) &= \frac{2}{3} \end{aligned}$$

but $A \cap B \neq \phi$. Finally

$$\begin{aligned} P(B/A) &= P(B/\bar{A}) \\ \Leftrightarrow \frac{P(A \cap B)}{P(A)} &= \frac{P(B \cap \bar{A})}{P(\bar{A})} \\ \Leftrightarrow [1 - P(A)]P(A \cap B) &= P(A)P(B \cap \bar{A}) \\ \Leftrightarrow [1 - P(A)]P(A \cap B) &= P(A)[P(B) - P(A \cap B)] \\ \Leftrightarrow P(A \cap B) &= P(A)P(B) \\ \Leftrightarrow A \text{ and } B &\text{ are independent events} \end{aligned}$$

Hence (D) is correct.

Answers: (A), (D)

Matrix-Match Type Questions

1. A man will live 10 more years for which the probability is $1/3$ and the probability that his wife will live 10 more years is $1/4$. Match the items of Column I with those of Column II.

Column I	Column II
(A) Probability that both wife and husband will live 10 more years is	(p) $\frac{1}{2}$
(B) Probability that at least one of them will live 10 more years is	(q) $\frac{3}{4}$
(C) Probability that neither will be alive in 10 more years is	(r) $\frac{1}{4}$
(D) Probability that only husband will live 10 more years is	(s) $\frac{1}{12}$
	(t) $\frac{2}{3}$

Solution: Let H be the event that husband will live 10 more years and W be the event that wife will live 10 more years.

(A) We have that since H and W are independent

$$\begin{aligned} P(H \cap W) &= P(H)P(W) \\ &= \frac{1}{3} \times \frac{1}{4} = \frac{1}{12} \end{aligned}$$

Answer: (A) $\rightarrow$ (s)

(B) We have

$$\begin{aligned} P(H \cup W) &= P(H) + P(W) - P(H \cap W) \\ &= \frac{1}{3} + \frac{1}{4} - \frac{1}{12} \\ &= \frac{6}{12} = \frac{1}{2} \end{aligned}$$

Answer: (B) $\rightarrow$ (p)

(C) We have

$$\begin{aligned} P(\bar{H} \cap \bar{W}) &= P(\overline{H \cup W}) \\ &= 1 - P(H \cup W) \\ &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Answer: (C) $\rightarrow$ (p)

(D) We have that since $H, \bar{W}$ are independent

$$\begin{aligned} P(H \cap \bar{W}) &= P(H)P(\bar{W}) \\ &= \frac{1}{3} \times \left(1 - \frac{1}{4}\right) = \frac{1}{4} \end{aligned}$$

Answer: (D) $\rightarrow$ (r)

2. In a class, there are 10 boys and 5 girls. Three students are selected one after other without allowing the earlier selected student to join the class. Match the items of Column I with the items of Column II.

Column I	Column II
(A) The probability that the first two are boys and the third a girl is	(p) $\frac{9}{13}$
(B) The probability that first and third are boys and the second a girl is	(q) $\frac{3}{7}$
(C) The probability that first and third are of same sex and the second is of opposite sex is	(r) $\frac{5}{21}$
(D) If the first and third are of same sex and the second is of opposite sex, then probability that the second student is girl is	(s) $\frac{15}{91}$

Solution: Let b and g denote boy and girl, respectively.

- (A) The probability that the first two are boys and the third a girl is

$$P(bbg) = \frac{10}{15} \times \frac{9}{14} \times \frac{5}{13} = \frac{15}{91}$$

Answer: (A) $\rightarrow$ (s)

- (B) The probability that first and third are boys and the second a girl is

$$P(bgb) = \frac{10}{15} \times \frac{5}{14} \times \frac{9}{13} = \frac{15}{91}$$

Answer: (B) $\rightarrow$ (s)

- (C) The probability that first and third are of same sex and the second is of opposite sex is

$$\begin{aligned} P((bgb) \cup (gbg)) &= P(bgb) + P(gbg) \\ &= \frac{15}{91} + \frac{5}{15} \times \frac{10}{14} \times \frac{4}{13} \\ &= \frac{15}{91} + \frac{20}{3 \times 91} \\ &= \frac{65}{3 \times 91} = \frac{5}{21} \end{aligned}$$

Answer: (C) $\rightarrow$ (r)

- (D) Let $E_1 = (bgb) \cup (gbg)$ and $E_2 = bgb$. Then

$$\begin{aligned} P(E_2/E_1) &= \frac{P(E_2)}{P(E_1)} \\ &= \frac{15}{91} \times \frac{21}{5} = \frac{9}{13} \end{aligned}$$

Answer: (D) $\rightarrow$ (p)

3. A and B are two events with $P(A) = 1/3$, $P(B) = 1/4$ and $P(A \cup B) = 1/2$. Match the items of Column I with those of Column II.

Column I	Column II
(A) $P(A/B)$ is equal to	(p) $\frac{1}{3}$
(B) $P(B/A)$ equals	(q) $\frac{1}{4}$
(C) The value of $P(A \cap \bar{B})$ is	(r) $\frac{1}{2}$
(D) $P(A/\bar{B})$ is equal to	(s) $\frac{2}{3}$

Solution: We have

$$\begin{aligned} \frac{1}{2} &= P(A \cup B) \\ &= P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{3} + \frac{1}{4} - P(A \cap B) \end{aligned}$$

Therefore

$$\begin{aligned} P(A \cap B) &= \frac{1}{3} + \frac{1}{4} - \frac{1}{2} \\ &= \frac{4+3-6}{12} = \frac{1}{12} \end{aligned}$$

- (A) We have

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{12} \div \frac{1}{4} = \frac{1}{3}$$

Answer: (A) $\rightarrow$ (p)

- (B) We have

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{1}{12} \div \frac{1}{3} = \frac{1}{4}$$

Answer: (B) $\rightarrow$ (q)

- (C) We have

$$\begin{aligned} P(A \cap \bar{B}) &= P(A - B) \\ &= P(A) - P(A \cap B) \\ &= \frac{1}{3} - \frac{1}{12} \\ &= \frac{3}{12} = \frac{1}{4} \end{aligned}$$

Answer: (C) $\rightarrow$ (q)

(D) We have

$$P(A/\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})}$$

$$= \frac{1/4}{1 - (1/4)} = \frac{1}{3}$$

Answer: (D) → (p)

4. Bag X contains 5 red, 3 white and 8 black balls. Bag Y contains 3 red and 5 white balls. A fair die is tossed. If 2 or 5 appears a ball from bag Y is chosen, otherwise a ball from bag X is chosen. Match the items of Column I with those of Column II. Here R , W and B denote drawing red ball, white ball and black ball, respectively.

Column I	Column II
(A) $P(R/X) =$	(p) $\frac{5}{8}$
(B) $P(R/Y) =$	(q) $\frac{3}{16}$
(C) $P(W/X) =$	(r) $\frac{3}{8}$
(D) $P(X/R) =$	(s) $\frac{5}{16}$

Solution: We have

$$P(X) = \frac{4}{6} = \frac{2}{3}$$

$$P(Y) = \frac{2}{6} = \frac{1}{3}$$

(A) The required probability is

$$P(R/X) = \frac{5}{16}$$

Answer: (A) → (s)

(B) The required probability is

$$P(R/Y) = \frac{3}{18}$$

Answer: (B) → (r)

(C) The required probability is

$$P(W/X) = \frac{3}{16}$$

Answer: (C) → (q)

(D) By Bayes' theorem,

$$P(X/R) = \frac{P(X)P(R/X)}{P(X)P(R/X) + P(Y)P(R/Y)}$$

$$= \frac{\frac{2}{3} \times \frac{5}{16}}{\frac{2}{3} \times \frac{5}{16} + \frac{1}{3} \times \frac{3}{8}}$$

$$= \frac{10}{10 + 6} = \frac{5}{8}$$

Answer: (D) → (p)

5. A , B and C are three students of Mathematics. Each is given two problems in probability. Their chances of solving the problems are, respectively, $1/2$, $1/3$ and $1/6$. They try independently. Let A_1 , A_2 , B_1 , B_2 , C_1 , C_2 denote the events of their solving the first and second problems. Match the items of Column I with those of Column II.

Column I	Column II
(A) $P(A_1 \cap B_2)$	(p) $\frac{1}{4}$
(B) $P(C_1 \cap A_2)$	(q) $\frac{1}{6}$
(C) $P(B_1 \cap C_2)$	(r) $\frac{1}{12}$
	(s) $\frac{1}{18}$
(D) $P(A_1 \cap A_2)$	(t) $\frac{1}{36}$

Solution:

(A) The required probability is

$$P(A_1 \cap B_2) = P(A_1)P(B_2)$$

$$= \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Answer: (A) → (q)

(B) The required probability is

$$P(C_1 \cap A_2) = P(C_1)P(A_2)$$

$$= \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

Answer: (B) → (r)

(C) The required probability is

$$P(B_1 \cap C_2) = P(B_1)P(C_2)$$

$$= \frac{1}{3} \times \frac{1}{6} = \frac{1}{18}$$

Answer: (C) → (s)

(E) The required probability is

$$P(A_1 \cap A_2) = P(A_1)P(A_2) \\ = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Answer: (D) → (p)

6. The probability that a person can hit a target is $3/4$. He tries 5 times. Let X denote binomial variate with parameters $n = 5$ and $p = 3/4$, $q = 1/4$. Match the items of Column I with those of Column II.

Column I	Column II
(A) $P(X \leq 2)$	(p) $\frac{459}{512}$
(B) $P(X \geq 3)$	(q) $\frac{15}{16}$
(C) Mean of X	(r) $\frac{53}{512}$
(D) Variance of X	(s) $\frac{15}{4}$

Solution: By hypothesis

$$P(X = k) = {}^5C_k p^k q^{5-k} \\ = {}^5C_k \left(\frac{3}{4}\right)^k \left(\frac{1}{4}\right)^{5-k}$$

for $k = 0, 1, 2, 3, 4, 5$.

(A) The required probability is

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= {}^5C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^5 + {}^5C_1 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^4 + {}^5C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^3 \\ = \left(\frac{1}{4}\right)^5 + \frac{5 \times 3}{4^5} + \frac{10 \times 9}{4^5} \\ = \frac{106}{4^5} \\ = \frac{53}{512}$$

Answer: (A) → (r)

(B) The required probability is

$$P(X \geq 3) = 1 - P(X \leq 2) \\ = 1 - \frac{106}{4^5} \\ = \frac{1024 - 106}{1024} \\ = \frac{918}{1024} \\ = \frac{459}{512}$$

Answer: (B) → (p)

(C) Mean is given by

$$np = 5 \times \frac{3}{4} = \frac{15}{4}$$

Answer: (C) → (s)

(D) Variance is given by

$$npq = 5 \times \frac{3}{4} \times \frac{1}{4} = \frac{15}{16}$$

Answer: (D) → (q)

Comprehension-Type Questions

1. **Passage:** We consider a two-player game in which there is always a winner. A , B and C are three players.

If A plays B , then the probability of A beating B is $\frac{2}{3}$.

If B plays C , then the probability of B beating C is $\frac{2}{3}$.

If C plays A , then the probability of C beating A is $\frac{2}{3}$.

Answer the following questions:

- (i) When A plays B , the winner plays C . Then, the probability that A is the final winner is

(A) $\frac{2}{9}$ (B) $\frac{7}{9}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

- (ii) When B plays C , the winner plays A . Then, the probability that A is the final winner is

(A) $\frac{2}{9}$ (B) $\frac{5}{9}$ (C) $\frac{7}{9}$ (D) $\frac{4}{9}$

- (iii) When C plays A , the winner plays B . Then, the probability that A is the final winner is

(A) $\frac{5}{9}$ (B) $\frac{4}{9}$ (C) $\frac{2}{9}$ (D) $\frac{7}{9}$

Solution:

- (i) First, A must beat B and then A must beat C . The probability that A is the final winner is

$$\frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$$

Answer: (A)

- (ii) Whether B beats C or C beats B , the winner must lose to A . Therefore

$$\begin{aligned} & \text{Probability } (A \text{ is final winner}) \\ &= P(B \text{ beats } C \text{ and loses to } A) \\ & \quad + P(C \text{ beats } B \text{ and loses to } A) \\ &= \frac{2}{3} \times \frac{2}{3} + \frac{1}{3} \times \frac{1}{3} = \frac{5}{9} \end{aligned}$$

Answer: (B)

- (iii) First, A must beat C and then A must beat B . The probability that A is the final winner is

$$\frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$$

Answer: (C)

2. Passage: A and B are independent events with

$$P(A) = \frac{1}{2} \quad \text{and} \quad P(A \cup B) = \frac{2}{3}$$

Answer the following questions.

- (i) $P(B)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{4}$

- (ii) $P(A/B)$ equals

(A) $\frac{3}{4}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

- (iii) $P(\bar{B}/A)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

Solution:

- (i) We have

$$\begin{aligned} \frac{2}{3} &= P(A \cup B) \\ &= P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{2} + P(B) - \frac{1}{2}P(B) \\ &= \frac{1}{2} + \frac{1}{2}P(B) \end{aligned}$$

Therefore

$$\frac{4}{3} = 1 + P(B) \Rightarrow P(B) = \frac{1}{3}$$

Answer: (B)

- (ii) The required probability is

$$\begin{aligned} P(A/B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A)P(B)}{P(B)} \\ &= P(A) = \frac{1}{2} \end{aligned}$$

Answer: (D)

- (iii) A and B are independent. This implies A and $\bar{B}$ are also independent. So

$$\begin{aligned} P(\bar{B}/A) &= \frac{P(\bar{B} \cap A)}{P(A)} \\ &= \frac{P(\bar{B})P(A)}{P(A)} \\ &= P(\bar{B}) \\ &= 1 - P(B) \\ &= 1 - \frac{1}{3} = \frac{2}{3} \end{aligned}$$

Answer: (B)

- 3. Passage:** A box contains 3 coins out of which two are fair coins and the third is a two headed coin. A coin is selected at random and tossed. If head appears, the same coin is tossed again. If tail appears, another is selected from the remaining two coins and tossed. Answer the following questions.

- (i) The probability that head appears twice is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

- (ii) If the same is tossed twice, then the probability that it is a two headed coin is

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

- (iii) The probability that tail appears twice is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{12}$ (D) $\frac{1}{4}$

Solution: Let E_1 be the event of selecting fair coin and E_2 be the event of selecting two headed coin. H_j denotes head and T_j denotes tail where $j = 1, 2$.

(i) We have

$$H_1 \cap H_2 = (E_1 \cap H_1 \cap H_2) \cup (E_2 \cap H_1 \cap H_2)$$

Then

$$\begin{aligned} P(H_1 \cap H_2) &= P(E_1 \cap H_1 \cap H_2) + P(E_2 \cap H_1 \cap H_2) \\ &= \frac{2}{3} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{3} \times 1 \times 1 = \frac{1}{2} \end{aligned}$$

Answer: (A)

(ii) Let E be the event of the same coin being tossed twice. Then

$$E = (E_1 \cap H) \cup E_2$$

Therefore

$$\begin{aligned} P(E) &= P(E_1 \cap H) + P(E_2) \\ &= \frac{2}{3} \times \frac{1}{2} + \frac{1}{3} = \frac{2}{3} \end{aligned}$$

Now

$$\begin{aligned} P(E_2/E) &= \frac{P(E_2)P(H/E_2)}{P(E_1)P(H/E_1) + P(E_2)} \\ &= \frac{\frac{1}{3} \times 1}{\frac{2}{3} \times \frac{1}{2} + \frac{1}{3}} = \frac{1}{2} \end{aligned}$$

Answer: (B)

(iii) T = Tail appearing twice. This implies selection of fair coin first, getting tail and then selection of fair coin and again getting tail. Hence

$$\begin{aligned} P(T) &= P(E_1 \cap T_1) \times P(E_1 \cap T_2) \\ &= \left(\frac{2}{3} \times \frac{1}{2}\right) \times \left(\frac{1}{2} \times \frac{1}{2}\right) \\ &= \frac{1}{12} \end{aligned}$$

Answer: (C)

4. Passage: A and B are two tennis players participating in a singles match as opponents. As per the rules of the lawn tennis, the one who wins atleast six games with at least two games more than his opponent wins the set. They serve alternately from game to game (i.e., for each game). A starts to serve after winning the toss. It is given that the probability of A winning his service game is $2/3$ and that of losing it is $1/3$. Answer the following questions:

(i) The probability of A winning the first set at six-LOVE (6-0) is

(A) $\left(\frac{2}{3}\right)^6$ (B) $\left(\frac{1}{3}\right)^6$ (C) $\frac{2^3}{3^6}$ (D) $\frac{1}{2}$

(ii) The probability of A winning first set 6-1 is

(A) $\frac{40}{3^7}$ (B) $\frac{40}{3^6}$ (C) $\frac{20}{3^7}$ (D) $\frac{32}{3^7}$

(iii) The probability of B winning the first set 6-1 is

(A) $\frac{20}{3^6}$ (B) $\frac{20}{3^7}$ (C) $\frac{40}{3^6}$ (D) $\frac{4}{3^6}$

Solution:

(i) “ A ” wins the set at 6-0 means A has to win all his 3 service games and B has to lose all his 3 service games. Therefore the probability is

$$\left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^3 = \frac{2^3}{3^6}$$

Answer: (C)

(ii) Probability of A winning by 6-1 is

$${}^3C_1 \left(\frac{2}{3}\right) \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 + {}^3C_1 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^3 = \frac{40}{3^6}$$

Answer: (B)

(iii) Probability of B winning the first set by 6-1 is

$${}^3C_1 \left(\frac{2}{3}\right) \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^3 + {}^3C_1 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^4 = \frac{20}{3^6}$$

Answer: (A)

5. Passage: A fair die is tossed repeatedly until face six appears. Let X denote the number of tosses required. Answer the following questions:

(i) The probability that X is equal to 3 is

(A) $\frac{25}{216}$ (B) $\frac{25}{36}$ (C) $\frac{5}{36}$ (D) $\frac{125}{216}$

(ii) The probability that $X \geq 3$ is

(A) $\frac{125}{216}$ (B) $\frac{25}{36}$ (C) $\frac{5}{36}$ (D) $\frac{25}{216}$

(iii) The conditional probability that $X \geq 6$ given $X > 3$ equals

(A) $\frac{125}{216}$ (B) $\frac{25}{216}$ (C) $\frac{5}{36}$ (D) $\frac{25}{36}$

Solution: Let p be the probability of getting six (success) in a single throw of a die so that

$$p = \frac{1}{6}$$

and the probability of failure

$$q = 1 - p = \frac{5}{6}$$

(i) The required probability is

$$P(X=3) = q^2 p = \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) = \frac{25}{216}$$

Answer: (A)

(ii) The required probability is

$$\begin{aligned} P(X \geq 3) &= q^2 p + q^3 p + q^4 p + \cdots + \infty \\ &= q^2 p(1 + q + q^2 + \cdots + \infty) \\ &= \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) \left(\frac{1}{1 - (5/6)}\right) = \frac{25}{36} \end{aligned}$$

Answer: (B)

(iii) The required probability is ($\because X > 3$)

$$\begin{aligned} P((X \geq 6)/(X > 3)) &= 1 - (P(X=4) + P(X=5)) \\ &= 1 - (p + pq) \\ &= 1 - p(1 + q) \\ &= 1 - \frac{1}{6} \left(1 + \frac{5}{6}\right) \\ &= 1 - \frac{11}{36} = \frac{25}{36} \end{aligned}$$

Answer: (D)

6. Passage: Urns A, B, C, respectively, contain 2 white and 3 black balls, 1 white and 4 black balls, 4 white and 1 black balls. The probabilities of choosing the urns are, respectively, $2/5$, $2/5$, and $1/5$. One of the urns is chosen at random and a ball is drawn from it. Answer the following questions:

(i) Probability that the drawn ball is white is

(A) $\frac{12}{25}$ (B) $\frac{13}{25}$ (C) $\frac{2}{5}$ (D) $\frac{3}{5}$

(ii) Probability that the drawn ball is black is

(A) $\frac{13}{25}$ (B) $\frac{12}{25}$ (C) $\frac{2}{5}$ (D) $\frac{3}{5}$

(iii) If the drawn ball is white, then the probability that it is from urn C is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{12}{25}$ (D) $\frac{13}{25}$

Solution:

(i) Let W denote that drawn ball is white. Then

$$\begin{aligned} W &= (A \cup B \cup C) \cap W \\ &= (A \cap W) \cup (B \cap C) \cup (C \cap A) \end{aligned}$$

Therefore

$$\begin{aligned} P(W) &= P(A \cap W) + P(B \cap W) + P(C \cap W) \\ &= P(A)P(W/A) + P(B)P(W/B) + P(C)P(W/C) \\ &= \frac{2}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{1}{5} + \frac{1}{5} \times \frac{4}{5} \\ &= \frac{10}{25} = \frac{2}{5} \end{aligned}$$

Answer: (C)

(ii) Let E denote that drawn ball is black. Then

$$\begin{aligned} P(E) &= P(A)P(E/A) + P(B)P(E/B) + P(C)P(E/C) \\ &= \frac{2}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{4}{5} + \frac{1}{5} \times \frac{1}{5} \\ &= \frac{15}{25} = \frac{3}{5} \end{aligned}$$

Answer: (D)

(iii) By Bayes' theorem we have

$$\begin{aligned} P(C/W) &= \frac{P(C)P(W/C)}{P(A)P(W/A) + P(B)P(W/B) + P(C)P(W/C)} \\ &= \frac{\frac{1}{5} \times \frac{4}{5}}{\frac{2}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{1}{5} + \frac{1}{5} \times \frac{4}{5}} \\ &= \frac{4}{10} = \frac{2}{5} \end{aligned}$$

Answer: (A)

Assertion-Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
(B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I

- (C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true

1. Statement I: Let A and B be two events and

$$A \Delta B = (A - B) \cup (B - A)$$

Then

$$|P(A) - P(B)| \leq P(A \Delta B)$$

Statement II: $P(A) - P(B)$ is not greater than $P(A - B)$.

Solution: We have

$$P(A) - P(B) \leq P(A \cup B) - P(B) = P(A - B)$$

Statement II is true. Similarly

$$P(B) - P(A) \leq P(A \cup B) - P(A) = P(B - A)$$

Therefore

$$|P(A) - P(B)| \leq P(A - B) + P(B - A) = P(A \Delta B)$$

Statement I is correct.

Answer: (A)

2. Statement I: A is an event with non-zero probability. If A is independent of itself, then $P(A) = 1$.

Statement II: Two independent events cannot be mutually exclusive.

Solution: A is independent of itself. This implies

$$\begin{aligned} P(A) &= P(A \cap A) = P(A)P(A) \\ \Rightarrow P(A) &= 1 \quad (\because P(A) \neq 0) \end{aligned}$$

Statement I is true. If E and F are independent, then

$$P(E \cap F) = P(E)P(F) \neq 0$$

but if E and F are mutually exclusive, then $P(E \cap F) = 0$.

Answer: (B)

3. Consider the system of equations $ax + by = 0$ and $cx + dy = 0$, where a, b, c, d belong to the set $\{0, 1\}$.

Statement I: The probability that the system has unique solution is $3/8$.

Statement II: The probability the system has a solution is 1.

Solution: Statement II is correct, because $x = 0, y = 0$ is always a solution. The system has unique solution if and only if the matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is non-singular. Since each of a, b, c and d is either 0 or 1 it follows that the number of matrices is 2^4 out of which the matrices

$$\begin{aligned} &\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \\ &\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \end{aligned}$$

are non-singular matrices. Therefore the probability is

$$\frac{6}{16} = \frac{3}{8}$$

Answer: (B)

4. Statement I: If $P(A \cup B) = 0.65$ and $P(A \cap B) = 0.15$, then

$$P(\bar{A}) + P(\bar{B}) = 1.2$$

Statement II: For any two events E and F ,

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Solution: Statement II is the addition theorem for probabilities, and hence statement II is correct. Now

$$\begin{aligned} P(\bar{A}) + P(\bar{B}) &= 1 - P(A) + 1 - P(B) \\ &= 2 - (P(A) + P(B)) \\ &= 2 - (P(A \cup B) + P(A \cap B)) \\ &= 2 - (0.65 + 0.15) \\ &= 1.2 \end{aligned}$$

Answer: (A)

5. Statement I: A and B are two events such that

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{3} \quad \text{and} \quad P(A \cap B) = \frac{1}{6}$$

Then

$$P(\text{exactly one of } A \text{ and } B) = \frac{1}{2}$$

Statement II: For any two events E and F

$$P(\text{exactly one of } E \text{ and } F) = P(E) + P(F) - 2P(E \cap F)$$

Solution: We have

$$\begin{aligned} P(\text{exactly one of } E \text{ and } F) &= P((E - F) \cup (F - E)) \\ &= P(E - F) + P(F - E) \\ &= P(E) - P(E \cap F) + P(F) - P(F \cap E) \\ &\quad (\because E - F = E - E \cap F) \\ &= P(E) + P(F) - 2P(E \cap F) \end{aligned}$$

Therefore Statement II is correct. Now

$$\begin{aligned} P(\text{exactly one of } A \text{ and } B) &= P(A) + P(B) - 2P(A \cap B) \\ &= \frac{1}{2} + \frac{1}{3} - 2\left(\frac{1}{6}\right) \\ &= \frac{1}{2} \end{aligned}$$

Answer: (A)

6. A and B are two equally strong players in tennis. Consider the two events
 E_1 : A beats B exactly in 3 games out of 4.
 E_2 : A beats B exactly 5 games out of 8.

Statement I: E_1 is more possible than E_2 .

Statement II: In a binomial distribution $B(n, p)$ where n is the number of trials and p is the probability of success in a single trial and $q = 1 - p$, the probability k success $= {}^nC_k p^k q^{n-k}$.

Solution: Statement II correct (see Section 7.6.1). Now

$$\begin{aligned} P(E_1) &= {}^4C_3 \cdot p^3 q \\ &= 4 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right) \left(\because p = q = \frac{1}{2}\right) \end{aligned}$$

$$= \frac{4}{16} = \frac{1}{4}$$

$$\begin{aligned} P(E_2) &= {}^8C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^3 \\ &= \frac{56}{2^8} = \frac{7}{32} \end{aligned}$$

Therefore

$$P(E_1) > P(E_2)$$

Answer: (A)

Integer Answer Type Questions

1. Let n be a positive integer and $A = \{1, 2, 3, \dots, n\}$. Let S be the set of all mappings $f: A \rightarrow A$. The probability that a randomly chosen element of S to be injective mapping is given to be $3/32$. Then the value of n is _____.

Solution: Number of mappings from A to A is n^n , that is S contains n^n elements. It is known that the number of injections from A to A is $\lfloor n \rfloor$ (see Theorem 6.18, page 293, Vol. 1). Therefore by hypothesis

$$\frac{\lfloor n \rfloor}{n^n} = \frac{3}{32} = \frac{\lfloor 4 \rfloor}{4^4}$$

So

$$n = 4$$

Answer: 4

2. 5 pairs of socks are in a cupboard of which 4 socks are drawn randomly. The probability that the drawn sample contains exactly one pair is m/n , where m and n are natural numbers having no common factor except unity. Then $n - m$ is _____.

Solution: 4 socks out of 10 can be selected in ${}^{10}C_4 = 210$ ways. Let E be the event that the draw contains only one pair. Out of 5 pairs 1 pair can be selected in ${}^5C_1 = 5$ ways. From the remaining 8 socks 2 can be selected in ${}^8C_2 = 28$ ways. But these 28 ways contain 4 pairs also. Therefore number of ways of selecting exactly one pair is

$$5 \times (28 - 4) = 5 \times 24$$

Hence

$$P(E) = \frac{5 \times 24}{210} = \frac{4}{7} = \frac{m}{n}$$

This gives

$$n - m = 7 - 4 = 3$$

Answer: 3

3. A bag contains n balls ($n > 3$) of different colours. If the bag has 3 white balls and the probability of drawing 2 white balls from the bag is $3/10$, the value of n is _____.

Solution: By hypothesis

$$\frac{{}^3C_2}{{}^nC_2} = \frac{3}{10}$$

$$\frac{3 \times 2}{n(n-1)} = \frac{3}{10}$$

$$n^2 - n - 20 = 0$$

$$(n-5)(n+4) = 0$$

$$n = 5$$

Answer: 5

4. A sportsman's chance of shooting an animal at a distance $r > a$ ("a" is constant) is given to be a^2/r^2 . He fires at $r = 2a$ and if he misses, then again fires at $r = 3a$. He repeats the same process at $r = 4a, 5a$ and $6a$. When he misses at $r = 6a$, the animal escapes into the jungle. If the odds against the sportsman are $p:q$, then $q - p$ is _____.

Solution: The sportsman's chance of missing when $r = ja$ is

$$1 - \frac{a^2}{j^2 a^2} = 1 - \frac{1}{j^2} \quad (j = 2, 3, 4, 5, 6)$$

The animal escapes when the sportsman misses in all the five shots. Therefore the probability of animal escaping to jungle is

$$\begin{aligned}\prod_{j=2}^6 \left(1 - \frac{1}{j^2}\right) &= \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{2^3}\right) \left(1 - \frac{1}{2^4}\right) \left(1 - \frac{1}{2^5}\right) \left(1 - \frac{1}{2^6}\right) \\ &= \left[\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right)\right] \\ &\quad \times \left[\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{4}\right) \left(1 + \frac{1}{5}\right) \left(1 + \frac{1}{6}\right)\right] \\ &= \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}\right) \left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \frac{7}{6}\right) \\ &= \frac{1}{6} \times \frac{7}{2} = \frac{7}{12} = \frac{p}{q}\end{aligned}$$

Therefore

$$q - p = 12 - 7 = 5$$

Answer: 5

5. Boxes B_1 , B_2 , B_3 contain different coloured balls as given in Table 7.9. The probabilities of selecting boxes are, respectively, $1/6$, $1/2$ and $1/3$. One of the boxes is chosen at random and a ball is drawn from it. If the probability of the drawn ball is black is $23/90$ then the value of n is equal to ____.

Table 7.9 Integer answer type question 5 (n is a positive integer)

	White	Black	Red
B_1	2	n	2
B_2	3	2	4
B_3	4	3	2

Solution: Let B be denote the event of drawing a black ball. Then

$$\begin{aligned}B &= (B_1 \cup B_2 \cup B_3) \cap B \\ &= (B_1 \cap B) \cup (B_2 \cap B) \cup (B_3 \cap B)\end{aligned}$$

Therefore

$$\begin{aligned}P(B) &= P(B_1)P(B/B_1) + P(B_2)P(B/B_2) + P(B_3)P(B/B_3) \\ &= \frac{1}{6} \times \frac{n}{n+4} + \frac{1}{2} \times \frac{2}{9} + \frac{1}{3} \times \frac{3}{9} \\ &= \frac{n}{6(n+4)} + \frac{2}{9} \\ &= \frac{3n+4(n+4)}{18(n+4)}\end{aligned}$$

By hypothesis

$$P(B) = \frac{23}{90}$$

Therefore

$$\frac{3n+4(n+4)}{18(n+4)} = \frac{23}{90}$$

$$35n + 80 = 23(n+4)$$

$$12n = 12$$

$$n = 1$$

Answer: 1

6. A and B are two independent events whose probabilities are, respectively, $1/n$ and $1/(n+1)$. If the probability of $A \cap B$ is $1/12$, then n equals ____.

Solution: A and B are independent events. This implies

$$P(A \cap B) = P(A)P(B)$$

Therefore

$$\begin{aligned}\frac{1}{12} &= P(A)P(B) \\ &= \frac{1}{n(n+1)}\end{aligned}$$

which gives

$$n = 3$$

Answer: 3

7. A number x is selected from the set of first 9 natural numbers (i.e., $x = 1, 2, 3, \dots, 9$). If the probability that $f(f(x)) = x$ where

$$f(x) = x^2 - 3x + 3$$

is $m/9$, then m is equal to ____.

Solution: Clearly all the solutions of $f(x) = x$ are also solutions of $f(f(x)) = x$. First, we solve $f(x) = x$.

$$\begin{aligned}f(x) = x &\Rightarrow x^2 - 3x + 3 = x \\ &\Rightarrow x^2 - 4x + 3 = 0 \\ &\Rightarrow (x-1)(x-3) = 0 \\ &\Rightarrow x = 1, 3\end{aligned}$$

Therefore $x = 1, 3$ are also solutions of $f(f(x)) = x$. We want to seek if there are any more solutions of $f(f(x)) = x$ other than 1 and 3.

$$\begin{aligned}f(f(x)) = x &\Rightarrow f(x^2 - 3x + 3) = x \\ &\Rightarrow (x^2 - 3x + 3)^2 - 3(x^2 - 3x + 3) + 3 = 0\end{aligned}$$

$$\begin{aligned} \Rightarrow x^4 - 6x^3 + 12x^2 - 9x + 3 &= 0 \\ \Rightarrow (x^2 - 4x + 3)(x^2 - 2x + 1) &= 0 \\ \Rightarrow (x - 1)(x - 3)(x - 1)^2 &= 0 \\ \Rightarrow x &= 1, 3 \end{aligned}$$

In this case we have no additional solutions. Therefore the probability that x satisfies equation $f(f(x)) = x$ is $2/9$. Therefore $m = 2$.

Answer: 2

8. A fair coin is tossed n times. Let X denote the number of heads appeared. If $P(X = 4)$, $P(X = 5)$ and $P(X = 6)$ are in AP, then the smallest values of n is _____.

Solution: Since the coin is fair,

$$P(H) = P(T) = \frac{1}{2}$$

By binomial distribution,

$$\begin{aligned} P(X = K) &= {}^nC_K \left(\frac{1}{2}\right)^{n-K} \left(\frac{1}{2}\right)^K \\ &= {}^nC_K \left(\frac{1}{2}\right)^n \end{aligned}$$

By hypothesis

$$2P(X = 5) = P(X = 4) + P(X = 6)$$

Therefore

$$\begin{aligned} 2({}^nC_5) &= {}^nC_4 + {}^nC_6 \\ n^2 - 21n + 98 &= 0 \\ n &= 7, 14 \end{aligned}$$

Therefore $n = 7$ (smaller value).

Answer: 7

9. $P_1, P_2, \dots, P_8$ are equally strong players (i.e., for each of them the probability of win or lose is $1/2$) are participating in tennis singles tournament. If the probability of P_1 losing to eventual winner of the tournament is m/n then $n - m$ is equal to _____.

Solution: Let X be the eventual winner. P_1 may lose to X in I, II and III rounds.

$$\begin{aligned} P(P_1 \text{ to lose in I}) &= P(P_1 \text{ pairing with } X \text{ and losing}) \\ &= \frac{1}{4} \times \frac{1}{2} \end{aligned}$$

since

$$P(P_1 \text{ pairing with } X) = \frac{1}{4}$$

as there are four pairs.

$$P(P_1 \text{ to lose in II}) = P(P_1 \text{ wins in I}) P(P_1 \text{ pairing with } X \text{ in II})$$

$$P(P_1 \text{ losing})$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{8}$$

$$P(P_1 \text{ to lose in III}) = P(P_1 \text{ wining I and II}) P(P_1 \text{ losing})$$

$$= \left(\frac{1}{2} \times \frac{1}{2}\right) \times \frac{1}{2}$$

$$= \frac{1}{8}$$

Therefore probability of P_1 losing to X is

$$\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

Hence

$$n - m = 8 - 3 = 5$$

Answer: 5

10. The odds against an event A is 2:3 and odds in favour of another event B is 1:2. If A and B are independent and $P(A \cup B) = m/n$, then $|m - n|$ is _____. Here m and n do not have proper common divisor.

Solution: We have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{5} + \frac{1}{3} - \frac{3}{5} \times \frac{1}{3}$$

$$= \frac{9 + 5 - 3}{15}$$

$$= \frac{11}{15}$$

Therefore $m = 11$ and $n = 15$. So $|m - n| = 4$.

Answer: 4

SUMMARY

7.1 Random experiment: An experiment is called random experiment if the following conditions are satisfied.

- (1) The experiment can be repeated any number of times under similar conditions.
- (2) All possible outcomes of the experiment are known in advance
- (3) The actual outcome in a particular experiment cannot be exactly predicted.

7.2 Sample space and event: The set of all possible outcomes of a random experiment is called *sample space* of the experiment and any subset of the sample space is called an *event*.

7.3 Exhaustive events: Two or more events are called *exhaustive events* if the performance of the experiment results in the occurrence of at least one of these events.

7.4 Mutually exclusive events: Two or more events are said to be *mutually exclusive* if the occurrence of one of the events prevents the occurrence of any one of the other events.

7.5 Equally likely events: Two or more events are said to be *equally likely* (or *equiprobable*) if there is no reason to expect one of them in preference to the others.

7.6 Probability (classical definition): Suppose in a random experiment there are n exhaustive, mutually exclusive, equally likely outcomes. If m of them are favourable to an event E , then the probability $P(E)$ of E is defined as

$$P(E) = \frac{m}{n}$$



QUICK LOOK

$$0 \leq P(E) \leq 1 \quad (\because 0 \leq m \leq n)$$

7.7 Complementary event: If E is an event, then non-occurrence of E is called *complementary event* of E and is denoted by $\bar{E}$ (or E^c).



QUICK LOOK

$$P(E^c) = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(E)$$

7.8 Usual probability: The classical probability is also called usual probability.

7.9 Sample points and sample space: Any possible outcome of a random experiment is called a sample point and the set of all sample points is called the sample space of the random experiment. An elementary event means a sample point. Generally sample space is denoted by S .

7.10 Finite sample space: A set A is called finite if either A is an empty set or it is bijective with the set $\{1, 2, 3, \dots, n\}$ for some positive integer n . If a sample space is finite, then it is called a finite sample space.

7.11 Countably infinite sample space: A set A is called countably infinite set if it is bijective with the set $\mathbb{Z}^+$ of all positive integers. If the sample space of a random experiment is countably infinite set, then the sample space is called countably infinite. For example, tossing a fair coin till head appears has a countably infinite sample space.

7.12 Definition: Here afterwards events mean subsets of the sample space. If A and B are two events, then $A \cup B$ means at least one of A or B and $A \cap B$ means both A and B . Impossible event is denoted by empty set ϕ and a certain event means the entire sample space.

7.13 Various events in set theoretical form:

- (1) Events $E_1, E_2, \dots, E_n$ are said to be *mutually exclusive* if $E_i \cap E_j = \phi$ for $i \neq j$.
- (2) $E_1, E_2, \dots, E_n$ are called *exhaustive events* if $E_1 \cup E_2 \cup \dots \cup E_n = S$ (sample space).
- (3) *Mutually exclusive and exhaustive events* means $E_1 \cup E_2 \cup \dots \cup E_n = S$ and $E_i \cap E_j = \phi$ for $i \neq j$.
- (4) $E_1, E_2, \dots, E_n$ are called *equally likely* (or *equiprobable*) if there is no reason to expect one of them in preference to the others.

7.14 Probability function (Axiomatic): Let S be the sample space connected with a random experiment and $\wp(S)$ is the power set of S (i.e., the set of all subsets of S). Then a function $P: \wp(S) \rightarrow \mathbb{R}$ is called a *probability function on the sample S* , if it satisfies the following conditions:

- (1) $P(E) \geq 0$ for all $E \in \wp(S)$
- (2) $P(S) = 1$
- (3) If E_1 and E_2 belong to $\wp(S)$ and $E_1 \cap E_2 = \phi$, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

If P is a probability function on S , then for any $E \in \wp(S)$, $P(E)$ is called the probability of the event E .

If $E = \{s\}$, then we write $P(s)$ instead of $P(\{s\})$.

If S is a countably infinite space then (3) will be replaced by the following:

(3') If $\{E_n\}$ is a sequence of mutually exclusive events, then

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

provided the infinite sum exists.

QUICK LOOK

One can verify that the usual probability (classical definition) is also a probability function according to modern probability definition given above in 7.14.

7.15 Theorem: The following hold good for any probability function P defined on a sample space S .

- (1) $P(\phi) = 0$
- (2) $P(\bar{E}) = 1 - P(E)$
- (3) $0 \leq P(E) \leq 1$ for all $E \in \wp(S)$
- (4) $P(E_1 - E_2) = P(E_1) - P(E_1 \cap E_2)$ and in particular $P(E_1 - E_2) = P(E_1) - P(E_2)$ whenever $E_2 \subset E_1$ and hence $E_2 \subset E_1 \Rightarrow P(E_1) \geq P(E_2)$.
- (5) If $E_1, E_2, \dots, E_n$ are mutually exclusive events, then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

7.16 Definition (Odds in favour, odds against): If E is an event, $\bar{E}$ is its complementary event and P is a probability function, then the ratio $P(E):P(\bar{E})$ is called odds in favour of E and the ratio $P(\bar{E}):P(E)$ is called odds against E .

7.17 Addition theorem: If A and B are two events of a random experiment, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

and in particular, if $A \cap B = \phi$, then

$$P(A \cup B) = P(A) + P(B)$$

QUICK LOOK

If A, B and C are any three events, then

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(B \cap C) \\ &\quad - P(C \cap A) + P(A \cap B \cap C) \end{aligned}$$

7.18 Conditional probability: Let A and B be two events and P be a probability function with $P(B) > 0$, then we define $P(A \cap B)/P(B)$ as the conditional probability of the occurrence of event A after the occurrence of the event B and it is denoted by $P(A/B)$. That is

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

QUICK LOOK

If the sample space S is finite, then

$$P(A/B) = \frac{\text{Number of favourable cases to both } A \text{ and } B}{\text{Number of favourable cases to } B}$$

7.19 Multiplication theorem: Let A and B be two events of a random experiment with positive probabilities, then

$$\begin{aligned} P(A)P(B/A) &= P(A \cap B) \\ &= P(B)P(A/B) \end{aligned}$$

7.20 Corollary: Let $E_1, E_2, \dots, E_n$ be n events with positive probabilities and $P\left(\bigcap_{i=1}^{n-1} E_i\right) > 0$, then

$$\begin{aligned} P(E_1 \cap E_2 \cap \dots \cap E_n) &= P(E_1)P(E_2/E_1)P(E_3/(E_1 \cap E_2)) \\ &\quad \dots P(E_n/(\bigcap_{i=1}^{n-1} E_i)) \end{aligned}$$

7.21 Definition (independent events): Two events A and B are said to be independent of each other if the occurrence of one does not effect the occurrence of the other. That is, B is independent of A , if $P(B/A) = P(B)$.

7.22 Theorem: Two events A and B are independent if and only if $P(A \cap B) = P(A)P(B)$.

7.23 Theorem (total probability): Let $E_1, E_2, \dots, E_n$ be mutually exclusive and exhaustive events with positive probabilities. Then for any event E ,

$$P(E) = \sum_{j=1}^n P(E_j)P(E/E_j)$$

7.24 Bayes' theorem: Let $E_1, E_2, \dots, E_n$ be mutually exclusive and exhaustive events with positive probabilities. If E is an event then

$$P(E_k/E) = \frac{P(E_k)P(E/E_k)}{\sum_{j=1}^n P(E_j)P(E/E_j)}$$

7.25 Definition (random variable): If S is a sample space and $\mathbb{R}$ is the real number set then any function from S to $\mathbb{R}$ is called a random variable on S . Generally, random variable is denoted by the capital letters X, Y, Z , etc.

7.26 Probability function induced by a random variable: Let S be a sample space, P a probability function on S and X a random variable on S . Define $P_X: \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}$ by $P_X(T) = P(X^{-1}(T))$ for $T \in \mathcal{P}(\mathbb{R})$. Then P_X is a probability function on $\mathbb{R}$ and is called the probability function induced by X on $\mathbb{R}$.

7.27 Probability distribution: Let S be a sample space with probability function P and X a random variable. Then the function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(r) = P(X \leq r), r \in \mathbb{R}$$

is called the probability distribution function of the random variable X .

7.28 Theorem: Let X be a random variable on a sample space S and F the probability distribution function of X . Then the following hold.

- (1) $0 \leq F(r) \leq 1$ for all $r \in \mathbb{R}$.
- (2) $r \leq s \Rightarrow F(r) \leq F(s)$ for any r and s of $\mathbb{R}$.
- (3) $\lim_{r \rightarrow -\infty} F(r) = 0$ and $\lim_{r \rightarrow \infty} F(r) = 1$.
- (4) $F(r+0) = \lim_{s \rightarrow r, s > r} F(s) = F(r)$.

7.29 Definition (discrete random variable): A random variable X is called discrete if its range is finite or countably infinite.

7.30 Definition (mean and variance): Let X be a random variable on a sample space S and suppose the range of $X = \{x_1, x_2, \dots, x_n, \dots\}$. Let P be a probability function on S . Let $P(X = x_n) = P(X^{-1}\{x_n\})$. Then

- (1) If $\sum_n x_n P(X = x_n)$ is finite, then it is called the mean of X and is denoted by μ_X or simply μ when there is no confusion about X . That is

$$\mu = \sum_n x_n P(X = x_n)$$

- (2) If $\sum (x_n - \mu)^2 P(X = x_n)$ is finite, then it is called variance, denoted by σ^2 and $|\sigma|$ is called standard deviation of X .
- (3) Table 7.10 is called the probability distribution table of X where $p_i = P(X = x_i) = P(X^{-1}(\{x_i\}))$.

Table 7.10

X	x_1	x_2	$x_3 \dots$	$x_n \dots$
$(X = x_n)$	p_1	p_2	$p_3 \dots$	$p_n \dots$

7.31 Formula: If μ and σ^2 are, respectively, the mean and variance of a random variable X whose range is $\{x_1, x_2, x_3, \dots, x_n, \dots\}$ and $P(X = x_i) = p_i$, then

$$\sigma^2 + \mu^2 = \sum_n x_n^2 p_n$$

7.32 Table 7.11 enables for quick calculation of mean and variance.

Table 7.11

$X = x$	$P(X = x)$	$xP(X = x)$	$x^2P(X = x)$
x_1	$p_1 = P(X = x_1)$	$x_1 p_1$	$x_1^2 p_1$
x_2	$p_2 = P(X = x_2)$	$x_2 p_2$	$x_2^2 p_2$
x_3	$p_3 = P(X = x_3)$	$x_3 p_3$	$x_3^2 p_3$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	$p_n = P(X = x_n)$	$x_n p_n$	$x_n^2 p_n$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
		Total = μ	Total = $\sigma^2 + \mu^2$

7.33 Bernoulli trial: A random experiment in which we consider an event called success and its complementary called failure is called Bernoulli trial.

7.34 Binomial variate: Let n be the number of independent Bernoulli trials in which the probability for a success is p and that of failure is $q = 1 - p$. If X denotes the number of successes, then

$$P(X = k) = {}^n C_k p^k q^{n-k}$$

$P(X = k)$ is called the probability of k successes and $n - k$ failures. n and p are called parameters of the binomial variate X and we write $X \sim B(n, p)$.

7.35 Theorem (mean and variance of $X \sim B(n, p)$):

If $X \sim B(n, p)$, then the mean of X is np and the variance is npq .



QUICK LOOK

The mean of a binomial variate is greater than or equal to its variance.

Note: If a set of n Bernoulli trials are repeated N times, then the expected number of sets which contain r success and $n - r$ failures is $N({}^n C_r p^r q^{n-r})$.

7.36 Poisson distribution: Let X be a random variable whose range is $\{0, 1, 2, 3, \dots, n, \dots\}$ and λ a positive number. We define

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

then X is called a Poisson variate and λ is called the parameter of X .

7.37 Theorem: The mean and variance of a Poisson variate X with parameter λ are equal and equal to λ .

Note: In Vol. 3 (Calculus) we will prove that the Poisson distribution is a limiting case of binomial distribution.

EXERCISES

Single Correct Choice Type Questions

- A bag contains 8 white, 6 black, 2 red and 4 green balls. Two balls are picked at random at a time. The probability that they are of different colours is
(A) $\frac{7}{19}$ (B) $\frac{9}{19}$ (C) $\frac{14}{19}$ (D) $\frac{12}{19}$
- An urn contains 6 white, 4 red and 10 green marbles. Three are drawn one after other without replacement. The probability that they are of different colours is
(A) $\frac{2}{9}$ (B) $\frac{3}{19}$ (C) $\frac{6}{19}$ (D) $\frac{2}{57}$
- A bag contains 4 black and 7 white balls. Two balls are drawn randomly. The probability that at least one of them is black is
(A) 0.71 (B) 0.618 (C) 0.728 (D) 0.628
- Three fair coins are tossed at a time. The probability of getting at least one head is
(A) $\frac{7}{8}$ (B) $\frac{1}{8}$ (C) $\frac{3}{8}$ (D) $\frac{4}{8}$
- There are 9 books of which one is very popular. The books are arranged in a row at random. The probability that the popular book is always at the middle place is
(A) $\frac{2}{9}$ (B) $\frac{1}{9}$ (C) $\frac{(4!)^2}{9!}$ (D) $\frac{2(4!)^2}{9!}$
- The letters of the word PECULIAR are arranged at random. The probability that all the vowels are together is
(A) $\frac{3}{7}$ (B) $\frac{1}{7}$ (C) $\frac{1}{14}$ (D) $\frac{3}{14}$
- A and B are two students among six students. If all the six students are arranged in a row, the probability that B immediately succeeds A is
(A) $\frac{1}{5}$ (B) $\frac{1}{6}$ (C) $\frac{2}{5}$ (D) $\frac{2}{3}$
- An ellipse of eccentricity $2\sqrt{2}/3$ is inscribed in a circle. A point inside the circle is selected at random. The probability that the point lies outside the ellipse is
(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$
- Two subsets A and B are chosen at random from a set containing 6 elements one after other with replacement. The probability that A and B contain equal number of elements is
(A) $\frac{131}{1021}$ (B) $\frac{231}{1021}$ (C) $\frac{331}{1021}$ (D) $\frac{101}{1021}$
- A lot contains 12 cell phones out of which 4 are defective. If two cell phones are drawn from the lot at random, the probability that one is defective and the other is non-defective is
(A) $\frac{16}{33}$ (B) $\frac{15}{33}$ (C) $\frac{13}{33}$ (D) $\frac{17}{33}$
- On the real line (i.e., x -axis) points p and q are selected at random such that $-2 \leq q \leq 3$ and $0 \leq p \leq 3$. The distance between p and q is greater than 3 with probability
(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$
- A and B are two events such that
$$P(A) = \frac{3}{8}, P(B) = \frac{1}{2} \text{ and } P(A \cap B) = \frac{1}{4}$$

Then $P(A \cap \bar{B})$ is equal to
(A) $\frac{1}{12}$ (B) $\frac{1}{3}$ (C) $\frac{1}{16}$ (D) $\frac{1}{4}$
- A bag contains 4 red, 3 white, 2 blue and 1 green marbles. Four marbles are drawn from it at random.

The probability that the draw contains one each from the given colours is

- (A) $\frac{1}{7}$ (B) $\frac{6}{7}$ (C) $\frac{4}{35}$ (D) $\frac{31}{35}$

14. A gardener plants 3 apple trees, 4 neem trees and 5 mango trees in a row at random. The probability that no two mango trees are next to one another is

- (A) $\frac{7}{99}$ (B) $\frac{1}{9}$ (C) $\frac{8}{99}$ (D) $\frac{1}{11}$

15. The first 11 letters of the English alphabet are arranged in all possible ways at random. The probability that there are exactly 4 letters in between A and B is

- (A) $\frac{12}{55}$ (B) $\frac{14}{55}$ (C) $\frac{7}{55}$ (D) $\frac{6}{55}$

16. Two positive real numbers x and y such that $x + y = 12$ are selected at random. The probability that xy is greater than or equal to $3/4$ times their greatest product is

- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

17. From 15 consecutive positive integers, 3 are selected at random. The probability that they are in AP is

- (A) $\frac{3}{65}$ (B) $\frac{7}{65}$ (C) $\frac{9}{65}$ (D) $\frac{11}{65}$

18. Five numbers are selected at random from the first 50 natural numbers and arranged in the increasing order $x_1 < x_2 < x_3 < x_4 < x_5$. The probability that x_3 is 30 is

- (A) $\frac{{}^{29}C_2 \times {}^{20}C_2}{{}^{50}C_5}$ (B) $\frac{{}^{39}C_2 \times {}^{19}C_2}{{}^{50}C_5}$
(C) $\frac{{}^{30}C_2 \times {}^{20}C_2}{{}^{50}C_5}$ (D) $\frac{55}{1029}$

19. Two different coloured dice are rolled. A is the event that the sum of the faces appeared is odd and B is the event that one of the faces turned up shows face 1. Then $P(A \cap \bar{B})$ is

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

20. Two dice are thrown at a time. The probability that the sum is greater than 8 is

- (A) $\frac{1}{6}$ (B) $\frac{1}{9}$ (C) $\frac{1}{3}$ (D) $\frac{5}{18}$

21. A problem in combinatorics is given to three students whose probabilities of solving it are, respectively,

$1/2$, $3/4$ and $1/4$. They try the problem independently. The probability that majority of them solving the problem is

- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

22. A lot contains 15 cameras out of which 4 are defective. Cameras are drawn one after other to identify the defective pieces. The process continues till all the defective pieces are identified. The probability that the procedure ends at the ninth testing is

- (A) $\frac{11}{195}$ (B) $\frac{13}{195}$ (C) $\frac{8}{195}$ (D) $\frac{9}{195}$

23. The probability that man aged x years to die in five years is $1/3$. Out of persons P_1, P_2, P_3, P_4 and P_5 , each aged x years, P_1 will die in 5 years and he is the first person to die in 5 years is

- (A) $\frac{211}{1215}$ (B) $\frac{311}{1315}$ (C) $\frac{111}{1115}$ (D) $\frac{1}{5}$

24. The probabilities of three mutually exclusive events are

$$\frac{1+3p}{3}, \frac{1-p}{4}, \frac{1-2p}{2}$$

Then

- (A) $0 < p \leq \frac{1}{3}$ (B) $\frac{1}{3} \leq p \leq \frac{1}{2}$
(C) $\frac{1}{2} < p < 1$ (D) $\frac{1}{5} \leq p \leq \frac{1}{4}$

25. Two events A and B are such that $P(A) = 0.7$, $P(B) = 0.4$ and $P(A \cap \bar{B}) = 0.5$. Then $P(B/(A \cup \bar{B}))$ is equal to

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$

26. A fair die is rolled four times. Out of the four face values obtained, the probability that the minimum face value is greater than or equal to 2 and the maximum value is less than or equal to 5 is

- (A) $\frac{16}{81}$ (B) $\frac{1}{81}$ (C) $\frac{8}{81}$ (D) $\frac{65}{81}$

27. A purse contains four 50 paise coins and three Rs. 1 coins. Another purse contains six 50 paise coins and two Rs. 1 coins. One of the purses is selected random and a coin is drawn from it. The probability that it is a 50 paise coin is

- (A) $\frac{35}{56}$ (B) $\frac{25}{56}$ (C) $\frac{37}{56}$ (D) $\frac{27}{56}$

28. If each of a , b and c takes values from the set $\{1, 2, 3, 4, 5, 6\}$, then the probability that the equation $ax^2 + bx + c = 0$ has real roots is equal to
 (A) $\frac{53}{216}$ (B) $\frac{43}{216}$ (C) $\frac{23}{216}$ (D) $\frac{63}{216}$
29. In a regular hexagon $ABCDEF$, if three vertices are selected at random, then the probability that they form the vertices of an equilateral triangle is
 (A) $\frac{1}{5}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{10}$
30. From the set $\{1, 2, 3, \dots, 15\}$, seven numbers are selected at random one after other with replacement. The probability that the largest of these is 9 is
 (A) $\left(\frac{8}{15}\right)^7$ (B) $\left(\frac{9}{10}\right)^7$ (C) $\left(\frac{7}{10}\right)^7$ (D) $\left(\frac{3}{5}\right)^7$
31. Two cards are drawn at random from a deck of 52 playing cards one after other, replacing the earlier drawn card. The probability that the first card is diamond and the second card is Queen is
 (A) $\frac{3}{13}$ (B) $\frac{4}{13}$ (C) $\frac{5}{52}$ (D) $\frac{1}{52}$
32. Let S be a four-element set. If a two-element subset of S is selected at random, the probability that it belongs to a partition class of S is
 (A) $\frac{2}{9}$ (B) $\frac{1}{5}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$
 (Hint: See Definition 1.28 and Single Correct Choice Type Question 13, Chapter 1, Vol. 1, page 60.)
33. Let $w \neq 1$ be a cube root of unity. A fair die is thrown three times. If r_1 , r_2 and r_3 are the numbers that appeared on the faces, the probability that $w^{r_1} + w^{r_2} + w^{r_3} = 0$ is
 (A) $\frac{1}{18}$ (B) $\frac{1}{9}$ (C) $\frac{2}{9}$ (D) $\frac{1}{36}$
34. "A" is one of the six horses entered for a race and it is to ridden by one of the two jockeys B and C . If B rides A , then all the horses are equally likely to win. If C rides A , the chances of A 's win will be trebled, then, the odds in favour of A is
 (A) 1:2 (B) 2:1 (C) 1:3 (D) 3:1
35. A and B are two persons. A speaks truth 2 out of 3 times, while B speaks truth 3 out of 4 times. They are independent witnesses in an accident case which is in a police court. The probability that they have given identical statements is
 (A) $\frac{5}{12}$ (B) $\frac{1}{3}$ (C) $\frac{7}{12}$ (D) $\frac{2}{3}$
36. There are five pairs of socks in a cupboard. If 4 socks are drawn at random, the probability that the draw contains exactly one matching is
 (A) $\frac{8}{21}$ (B) $\frac{2}{7}$ (C) $\frac{5}{7}$ (D) $\frac{13}{21}$
37. In a bolt-producing factory, machines A , B and C give 25%, 35% and 40% of the total output, respectively. It is also known that 5%, 4% and 2% of them are defective. If a bolt of the factory is selected at random then the probability that it is defective is
 (A) $\frac{69}{2000}$ (B) $\frac{63}{2000}$ (C) $\frac{73}{2000}$ (D) $\frac{83}{2000}$
38. A and B are two politicians who settle disputes between parties independently. The probabilities of their settling disputes amicably are, respectively, $\frac{4}{7}$ and $\frac{7}{15}$. A firm has engaged them for a settlement of dispute between the firm and the employees. The probability that neither of them settle the dispute is
 (A) $\frac{73}{105}$ (B) $\frac{32}{105}$ (C) $\frac{8}{35}$ (D) $\frac{16}{35}$
39. S is a five-element set. A subset P of A is selected at random. After inspecting the elements of P , the elements are restored to S . Again a subset Q of S is selected at random. The probability that P and Q form a partition of S (i.e., $P \cup Q = S$ and $P \cap Q = \phi$) is
 (A) $\frac{1}{2}$ (B) $\frac{1}{16}$ (C) $\frac{1}{8}$ (D) $\frac{1}{32}$
40. A set X has 10 elements. Subset A of X is selected at random and then the elements of A are returned to X . Again a subset B of X is chosen at random. The probability that B is disjoint with A is
 (A) $\frac{1}{2^{10}}$ (B) $\frac{1}{2^5}$ (C) $\left(\frac{3}{4}\right)^{10}$ (D) $\left(\frac{3}{4}\right)^5$
41. Three groups of workers contain 3 women, 1 male; 2 women and 2 male and 3 males and 1 woman. One worker from each group is selected for a work at random. The probability that the selection constitutes 2 males and 1 lady is
 (A) $\frac{9}{32}$ (B) $\frac{3}{32}$ (C) $\frac{1}{32}$ (D) $\frac{13}{32}$

42. Two persons A and B each tosses a fair coin thrice. The probability that they get same number of heads is
 (A) $\frac{5}{16}$ (B) $\frac{11}{16}$ (C) $\frac{7}{16}$ (D) $\frac{9}{16}$
43. If A and B are two independent events in a random experiment then $P(A \cup B) + P(\bar{A} \cap \bar{B})$ is equal to
 (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{2}{3}$
44. Three bags have the following combination of balls.
 Bag I: 7 black, 3 white
 Bag II: 4 black, 6 white
 Bag III: 2 black, 8 white
 The probabilities of selecting bags are, respectively, $\frac{1}{5}$, $\frac{3}{5}$ and $\frac{1}{5}$. One of the bags is chosen at random and two balls are drawn from it, one after other without replacement. The probability that both balls are black is
 (A) $\frac{4}{45}$ (B) $\frac{1}{9}$ (C) $\frac{41}{45}$ (D) $\frac{8}{45}$
 (Hint: Total probability.)
45. Five men out of 100 and 25 women out of 1000 have colour blindness. If a colour blind person is selected at random, the probability that the person is a male is
 (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
 (Hint: Use Bayes' theorem.)
46. A fair coin is tossed $(m + n)$ times where $m > n$. The probability for m consecutive heads is
 (A) $\frac{n+2}{2^{m+1}}$ (B) $\frac{n+1}{2^{m+1}}$ (C) $\frac{m+2}{2^{n+1}}$ (D) $\frac{m+1}{2^{n+1}}$
47. Three identical bags contain the following composition of balls.
 Bag I: 2 white, 1 black
 Bag II: 1 white, 2 black
 Bag III: 2 white, 2 black
 The probabilities of selecting the bags are equal. One of the bags is selected and a ball is drawn from it. The drawn ball is found to be black and it is kept outside. Then the probability of a black ball again is
 (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{2}{5}$ (D) $\frac{3}{5}$
48. A fair die is rolled. The probability that first 1 appears at even number of trials is
 (A) $\frac{6}{11}$ (B) $\frac{1}{6}$ (C) $\frac{5}{36}$ (D) $\frac{5}{11}$
49. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes B should have such that A and B are independent is
 (A) 2, 4 or 3 (B) 3, 6 or 9
 (C) 4 or 8 (D) 5 or 10
50. If the mean and variance of a Binomial variate X are, respectively, 2 and 1, then $P(X > 1)$ is
 (A) $\frac{11}{16}$ (B) $\frac{5}{16}$ (C) $\frac{7}{16}$ (D) $\frac{9}{16}$

Multiple Correct Choice Type Questions

1. If A and B are independent events, then
 (A) A and $\bar{B}$ are independent
 (B) $\bar{A}$ and B are independent
 (C) $\bar{A}$ and $\bar{B}$ are independent
 (D) A and B are exclusive events
2. Two different coloured fair dice are thrown at a time. Then
 (A) (Sum of the faces is odd) = $\frac{1}{2}$
 (B) $P(\text{At least one face is odd}) = \frac{11}{36}$
 (C) $P(\text{Sum of the faces is } 5) = \frac{1}{9}$
 (D) $P(\text{Neither of the faces is } 1) = \frac{25}{36}$
3. Let P be a probability function on a sample space S and B a fixed non-empty subset of S . For any subset A of S , define

$$P_B(A) = P(A/B)$$

 Then
 (A) $P_B(S) = 1$
 (B) $P_B(A \cap C) = \frac{P(A \cap B \cap C)}{P(B)}$
 (C) $P_B(A \cup C) = P_B(A) + P_B(C) - P_B(A \cap C)$
 (D) $P_B(A \cap C) = P_B(A) P_B(C)$, whenever A and C are independent.
4. Let A , B , C be mutually exclusive and exhaustive events of a random experiment with sample space S .

Suppose P is a probability function on S . Which of the following are not correct?

(A) $P(A) = 0.24, P(B) = 0.4, P(C) = 0.2$

(B) $P(A) = \frac{1}{4}, P(B) = \frac{1}{4}, P(C) = \frac{1}{2}$

(C) $P(A) = 0.1, P(B) = 0.2, P(C) = 0.6$

(D) $P(A) = 0.6, P(A \cap \bar{B}) = 0.5$

5. Let A and B be two events of a random experiment. Suppose $P(A) = p_1, P(B) = p_2$ and $P(A \cup B) = p_3$. Then

(A) $P(\bar{A} \cup B) = 1 - p_1 + p_3$

(B) $P(\bar{A} \cap \bar{B}) = 1 - p_1 - p_2 + p_3$

(C) $P(A \cup (\bar{A} \cap B)) = p_1 + p_2 - p_3$

(D) $P(\overline{A \cup B}) = p_1 + p_2 - p_3$

6. Two fair dice are thrown at a time. Let A, B, C be the following events:

$A = \{(x, y) \mid x \text{ is odd}\}$

$B = \{(x, y) \mid y \text{ is odd}\}$

$C = \{(x, y) \mid x + y \text{ is odd}\}$

Then

(A) $P(A \cap B) = P(A) P(B)$

(B) $P(B \cap C) = P(B) P(C)$

(C) $P(C \cap A) = P(C) P(A)$

(D) $P(A \cap B \cap C) = P(A) P(B) P(C)$

7. The probabilities of the events $A \cap B, A, B$ and $A \cup B$ of a random experiment are in AP in the given order, with the common different as $P(A)$. Assume that $P(A) > 0$. Then A and B are

(A) mutually exclusive

(B) independent

(C) exhaustive and dependent

(D) such that one is twice as likely as the other

8. The letters of the word PROBABILITY are arranged in a row at random. Let E_1 be the event that the two I's are together and E_2 is the event that two B's are together. Then

(A) $P(E_1 \cap E_2) = \frac{2}{55}$

(B) $P(E_1 \cup E_2) = \frac{18}{55}$

(C) $P(E_1/E_2) = \frac{1}{5}$

(D) $P(E_2/E_1) = \frac{2}{5}$

9. The probabilities of a 12th standard student passing the subjects Maths, Physics and Chemistry are, respectively, m, p and c , the probability of passing at least one subject is $3/4$, the probability of passing at least two subjects is $1/2$ and the probability of passing exactly two subjects is $2/5$. Which of the following relations are true?

(A) $p + m + c = \frac{19}{20}$

(B) $p + m + c = \frac{27}{20}$

(C) $pmc = \frac{1}{10}$

(D) $pmc = \frac{1}{4}$

10. A and B are independent events of a random experiment. If $P(A \cap B) = 1/6$ and $P(\bar{A} \cap \bar{B}) = 1/3$. Then

(A) $P(A) = \frac{1}{2}$

(B) $P(A) = \frac{1}{3}$

(C) $P(B) = \frac{1}{3}$

(D) $P(B) = \frac{1}{2}$

Matrix-Match Type Questions

In each of the following questions, statements are given in two columns, which have to be matched. The statements in *Column I* are labeled as (A), (B), (C) and (D), while those in *Column II* are labeled as (p), (q), (r), (s) and (t). Any given statement in *Column I* can have correct matching with *one or more* statements in *Column II*. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

Example: If the correct matches are $(A) \rightarrow (p), (s); (B) \rightarrow (q), (s), (t); (C) \rightarrow (r); (D) \rightarrow (r), (t)$; that is if the matches are $(A) \rightarrow (p)$ and $(s); (B) \rightarrow (q), (s)$ and $(t); (C) \rightarrow (r)$; and $(D) \rightarrow (r), (t)$, then the correct darkening of bubbles will look as follows:

	p	q	r	s	t
A	☒	☐	☐	☒	☐
B	☐	☒	☐	☒	☒
C	☐	☐	☒	☐	☐
D	☐	☐	☒	☐	☒

1. A, B and C are three events of a random experiment and P is a probability function. If $P(A) = 0.6, P(B) = 0.4, P(C) = 0.5, P(A \cup B) = 0.8, P(A \cap C) = 0.3, P(A \cap B \cap C) = 0.2$ and $P(A \cup B \cup C) = 0.85$, then match the items of Column I with those of Column II.

Column I	Column II
(A) $P(A \cap B)$ is	(p) 0.35
(B) $P(B \cap C)$ is	(q) 0.8
(C) $P(\bar{A}/B)$ is equal to	(r) 0.2
(D) $P(\bar{A} \cup \bar{B})$ equals	(s) 0.3
	(t) $\frac{2}{3}$

2. A, B, C are three events such that

$$P(B) = \frac{3}{4}, \quad P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$$

$$\text{and} \quad P(A \cap B \cap \bar{C}) = \frac{1}{3}$$

Match items of Column I with those of Column II.

Column I	Column II
(A) $P(B \cap \bar{C})$ is	(p) $\frac{1}{12}$
(B) $P(B \cap C)$ is	(q) $\frac{4}{9}$
(C) $P((\bar{A} \cap \bar{C})/B)$ is	(r) $\frac{1}{3}$
(D) $P((A \cap \bar{C})/B)$ is	(s) $\frac{2}{3}$

3. A and B are two independent events. It is given that $P(A \cap B) = 1/12$ and $P(\bar{A} \cap \bar{B}) = 1/2$. Match the items of Column I with those of Column II.

Column I	Column II
(A) $P(A)$ is equal to	(p) $\frac{1}{3}$
(B) $P(B)$ is equal to	(q) $\frac{1}{6}$
(C) $P(A \cap \bar{B})$ is	(r) $\frac{1}{2}$
(D) $P(\bar{A} \cap B)$ is	(s) $\frac{1}{4}$

4. Match the items of Column I with those of Column II.

Column I	Column II
(A) If two fair dice are rolled, then the probability of sum of the faces is 7 is	(p) $\frac{1}{4}$
(B) If a card is picked from a deck of 52 playing cards, then the probability of getting a red card is	(q) $\frac{1}{2}$
(C) $P(x)$ is a polynomial satisfying the relation $P(x) + P(2x) = 5x^2 - 18$ for all real x . Now, each coefficient a, b and c of the quadratic expression $ax^2 + bx + c$ is one of the roots of the equation $P(x) = 0$. The probability that $ax^2 + bx + c = 0$ has real roots is	(r) $\frac{3}{8}$
(D) In class X of a school 75% are boys and 25% are girls. Probability of boy getting first class is $1/3$ while girl getting first class is $1/2$. If one candidate is selected at random, the probability of the candidate getting first class is	(s) $\frac{1}{6}$

5. The letters of the word MATHEMATICS are written in a row at random. Match the items of Column I with those of Column II.

Column I	Column II
(A) Probability of the two M's together is	(p) $\frac{2}{11}$
(B) The probability that two M's and two A's are together is	(q) $\frac{4}{55}$
(C) The probability that the vowels are together is	(r) $\frac{6}{55}$
(D) The probability of beginning with M and end with M is	(s) $\frac{1}{55}$

Comprehension-Type Questions

1. **Passage:** A and B are two bags. A contains 4 fair coins and 3 counterfeit coins while B has 5 fair and 7 counterfeit coins. The counterfeit coins have tail on both

sides. Two coins are transferred from bag A to bag B and then a coin is taken from the bag B and tossed. Answer the following three questions.

- (i) If both coins transferred from A to B are counterfeit coins, then probability of getting a tail is

(A) $\frac{3}{4}$ (B) $\frac{5}{28}$ (C) $\frac{1}{2}$ (D) $\frac{23}{28}$

- (ii) If tail appears, then the probability of both coins transferred from bag A are counterfeit is

(A) $\frac{1}{7}$ (B) $\frac{23}{28}$ (C) $\frac{23}{153}$ (D) $\frac{23}{66}$

- (iii) If head appears, then the probability of both coins transferred from bag A are fair is

(A) $\frac{14}{43}$ (B) $\frac{21}{43}$ (C) $\frac{25}{43}$ (D) $\frac{14}{19}$

- 2. Passage:** In a tennis tournament, there are 12 players $S_1, S_2, \dots, S_{12}$ and they are divided into 6 pairs at random. From each pair, the winner will be decided on the basis of the game played between the two players of the pair. Assume that all players are of equal strength so that the probability of any player beating any other player is $1/2$. Answer the following questions.

- (i) The probability that S_1 and S_2 are not playing against other is

(A) $\frac{1}{2}$ (B) $\frac{10}{11}$ (C) $\frac{1}{11}$ (D) $\frac{5}{22}$

- (ii) The probability that exactly one of S_1 and S_2 is among the losers is

(A) $\frac{1}{2}$ (B) $\frac{5}{22}$ (C) $\frac{6}{11}$ (D) $\frac{5}{11}$

- (iii) The probability that both S_1 and S_2 are among the winners is

(A) $\frac{1}{4}$ (B) $\frac{5}{11}$ (C) $\frac{6}{11}$ (D) $\frac{5}{12}$

- 3. Passage:** A box contains 6 coins out of which at least one is biased. Let $E_K (1 \leq K \leq 6)$ denote the event that exactly K out of 6 coins are biased. Also let $P(E_K)$ be directly proportional to $K(K+1)$. Answer the following questions.

- (i) The proportionality constant is equal to

(A) $\frac{1}{74}$ (B) $\frac{1}{148}$ (C) $\frac{1}{112}$ (D) $\frac{1}{504}$

- (ii) If E is the event of selecting a biased coin out of 6 coins, then $P(E_1/E)$ is

(A) $\frac{1}{266}$ (B) $\frac{1}{132}$ (C) $\frac{1}{133}$ (D) $\frac{2}{133}$

- (iii) If the coin selected is found to be biased, then $P(E_6/E)$ is equal to

(A) $\frac{2}{19}$ (B) $\frac{3}{38}$ (C) $\frac{5}{38}$ (D) $\frac{3}{19}$

- 4. Passage:** There are n urns numbered from 1 to n . The K th urn contains K white balls and $(n+1-K)$ black balls. Let E_K denote the event of selecting the K th urn at random and let W denote the event that the ball drawn from the selected urn is white. Let $P(A)$ denote the probability of an event A . Answer the following questions.

- (i) If $P(E_K)$ is proportional to K for $K = 1, 2, 3, \dots, n$, then $\lim_{n \rightarrow \infty} P(W)$ is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

- (ii) If $P(E_K) = C$, a constant for all $K = 1, 2, 3, \dots, n$, then $P(E_n/W)$ is equal to

(A) $\frac{2}{n+1}$ (B) $\frac{1}{n+1}$ (C) $\frac{n}{n+1}$ (D) $\frac{1}{2}$

- (iii) If $P(E_K) = 1/n$ for all $K = 1, 2, 3, \dots, n$ and E denote the event of choosing an even numbered urn, then $P(W/E)$ is equal to

(A) $\frac{1}{2n}$ (B) $\frac{n+4}{2(n+1)}$
(C) $\frac{n+2}{2(n+1)}$ (D) $\frac{n+1}{2n}$

Assertion–Reasoning Type Questions

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statements I and II are true and Statement II is a correct explanation for Statement I
(B) Both Statements I and II are true but Statement II is not a correct explanation for Statement I
(C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true

- 1.** A fair coin is tossed 3 times. Let A be the event that the first toss gives head; B is the event that the second toss gives head and C be the event that there are exactly two consecutive heads or exactly two consecutive tails.

Statement I: A, B, C are independent events

Statement II: A, B, C are pairwise independent.

2. Suppose two persons A and B have $n + 1$ and n coins. All the coins are fair. They toss their coins.

Statement I: The probability that A gets more heads than B is $1/2$.

Statement II: The probability that A gets more heads than B is the same as A gets more tails than B .

3. In throwing a pair of fair dice, let A_1 be the event that the first die turns up odd number, A_2 be the event that the second die turns up odd number and A_3 be the event that the sum of the faces turn up is odd.

Statement I: The events A_1, A_2, A_3 are independent.

Statement II: A_1, A_2, A_3 are pairwise independent.

4. Let $E_1, E_2, \dots, E_n$ be mutually exclusive and exhaustive events of a random experiment with $P(E_i) > 0$ for $i = 1, 2, 3, \dots, n$. Let E be any event connected to the experiment with $0 < P(E) < 1$.

Statement I: $P(E_i/E) > P(E_i)P(E/E_i)$

Statement II: $\sum_{i=1}^n P(E_i) = 1$

5. A random number selector can only select one of the numbers from the set $\{1, 2, 3, \dots, 9\}$ and he will make these selections with equal probabilities.

Statement I: After n trials ($n > 1$), the probability that the product of these selected numbers is divisible by 5 is

$$1 - \left(\frac{8}{9}\right)^n - \left(\frac{5}{9}\right)^n + \left(\frac{4}{9}\right)^n$$

Statement II: For the product to be divisible by 10, the selections must contain at least one 5 and one even number.

Integer Answer Type Questions

The answer to each of the questions in this section is a non-negative integer. The appropriate bubbles below the respective question numbers have to be darkened. For example, as shown in the figure, if the correct answer to the question number Y is 246, then the bubbles under Y labeled as 2, 4, 6 are to be darkened.

X	Y	Z	W
0	0	0	0
1	1	1	1
2	●	2	2
3	3	3	3
4	●	4	4
5	5	5	5
6	●	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. Each of two bags contain both black and white balls. The total number of balls in both bags is 25. One ball is taken from each bag. The probability of both balls are black is $27/50$. Then the probability of both balls are white is $p/50$ where p is ____.
2. 10 girls and 4 boys are to sit at a round table. If the arrangement is random, the probability of no two boys sit next to each other is $a/(b!c!)$. Then $(b + c) - a$ is equal to ____.

3. Let A and B be two sets. A contains 4 elements while B contains 3. The probability of choosing a surjective map from A to B is m/n (m/n is in lowest terms). Then the value of $n - m$ is ____.
(Hint: See Theorem 6.19, Vol. 1.)

4. A number from the set $\{1, 2, 3, \dots, 15\}$ is selected at random. The probability that it is a proper divisor of 30 is a/b (in lowest terms). Then $a + b$ is equal to ____.
5. Let m/n , in lowest terms, be the probability that a randomly selected divisor of 10^{19} is an integer multiple of 10^{10} . Then $m + n$ is ____.
6. A box contains 3 gold coins and 4 silver coins. Coins are drawn one by one without replacement until all the gold coins are drawn. If a/b (in lowest terms) is the probability that the number of draws required is more than 4, then $b - a$ value is ____.
7. If the integers m and n are chosen from the set $\{1, 2, 3, \dots, 100\}$, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 is $1/K$ where K is equal to ____.
8. The probability of a bomb hitting a bridge is $1/3$ and two direct hits are enough to destroy the bridge. The least number of bombs required so that the probability of the bridge getting destroyed is greater than 0.5 is ____.

ANSWERS

Single Correct Choice Type Questions

- | | |
|---------|---------|
| 1. (C) | 26. (A) |
| 2. (D) | 27. (C) |
| 3. (B) | 28. (B) |
| 4. (A) | 29. (D) |
| 5. (B) | 30. (D) |
| 6. (C) | 31. (D) |
| 7. (B) | 32. (A) |
| 8. (A) | 33. (C) |
| 9. (B) | 34. (A) |
| 10. (A) | 35. (C) |
| 11. (B) | 36. (D) |
| 12. (A) | 37. (A) |
| 13. (C) | 38. (C) |
| 14. (A) | 39. (D) |
| 15. (D) | 40. (C) |
| 16. (B) | 41. (D) |
| 17. (B) | 42. (A) |
| 18. (A) | 43. (C) |
| 19. (A) | 44. (D) |
| 20. (D) | 45. (B) |
| 21. (B) | 46. (A) |
| 22. (C) | 47. (A) |
| 23. (A) | 48. (D) |
| 24. (B) | 49. (D) |
| 25. (C) | 50. (A) |

Multiple Correct Choice Type Questions

- | | |
|-----------------------|------------------------|
| 1. (A), (B), (C) | 6. (A), (B), (C) |
| 2. (A), (B), (C), (D) | 7. (A), (D) |
| 3. (A), (B), (C), (D) | 8. (A), (B), (C) |
| 4. (A), (C), (D) | 9. (B), (C) |
| 5. (A), (B), (C) | 10. (A), (B), (C), (D) |

Matrix-Match Type Questions

- | | |
|--|---|
| 1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (p); (C) $\rightarrow$ (t); (D) $\rightarrow$ (q) | 4. (A) $\rightarrow$ (s); (B) $\rightarrow$ (q); (C) $\rightarrow$ (p); (D) $\rightarrow$ (r) |
| 2. (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (q) | 5. (A) $\rightarrow$ (p); (B) $\rightarrow$ (q); (C) $\rightarrow$ (r); (D) $\rightarrow$ (s) |
| 3. (A) $\rightarrow$ (p), (s); (B) $\rightarrow$ (p), (s); (C) $\rightarrow$ (q), (s);
(D) $\rightarrow$ (q), (s) | |

Comprehension Type Questions

- | | |
|---------------------------------|---------------------------------|
| 1. (i) (D); (ii) (C); (iii) (A) | 3. (i) (C); (ii) (A); (iii) (B) |
| 2. (i) (B); (ii) (C); (iii) (D) | 4. (i) (B); (ii) (A); (iii) (C) |

Assertion-Reasoning Type Questions

- | | |
|--------|--------|
| 1. (B) | 4. (D) |
| 2. (A) | 5. (A) |
| 3. (D) | |

Integer Answer Type Questions

1. 2
2. 9
3. 5
4. 3

5. 5
6. 4
7. 4
8. 5

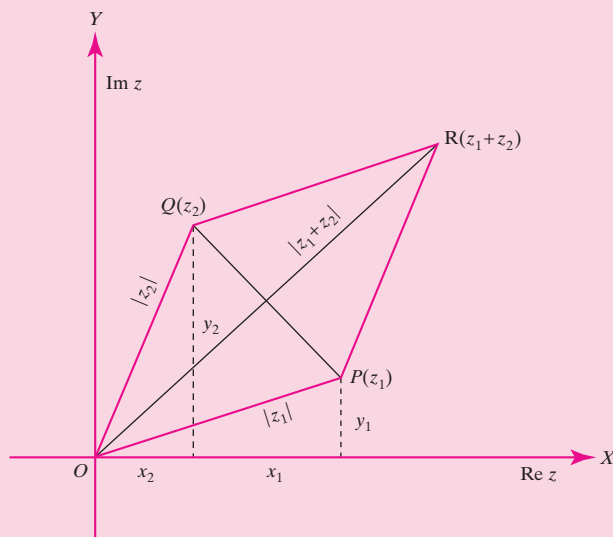
Inequalities

8

Inequalities



Euclid



Contents

8.1 Introduction

Worked-Out Problems
Exercises

Euclid, also known as **Euclid of Alexandria**, was a Greek mathematician, often referred to as the “Father of Geometry.” He was active in Alexandria during the reign of Ptolemy I (323–283 BC). His *Elements* is one of the most influential works in the history of mathematics, serving as the main textbook for teaching mathematics (especially geometry) from the time of its publication until the late 19th or early 20th century. The origin of inequalities stemmed from the famous theorem of Euclid: *The sum of two sides of a triangle is greater than the third side.*

8.1 | Introduction

The origin of inequalities stemmed from the famous theorem of EUCLID, namely, the sum of two sides of a triangle is greater than the third side. Almost all geometric inequalities, in some way or other, are based on this result. In this chapter, we deal with geometric inequalities as well as algebraic inequalities, especially more with Arithmetic–Geometric Mean inequality. Let us begin with formal definitions of Arithmetic Mean (AM), Geometric Mean (GM) and Harmonic Mean (HM) of real numbers.

DEFINITION 8.1 (i) Let $a_1, a_2, \dots, a_n$ be n positive real numbers. Then the number

$$\frac{a_1 + a_2 + \dots + a_n}{n}$$

is called Arithmetic Mean or simply AM of $a_1, a_2, \dots, a_n$.

(ii) $(a_1, a_2, a_3, \dots, a_n)^{1/n}$ is called Geometric Mean (GM).

(iii) $\frac{n}{(1/a_1) + (1/a_2) + \dots + (1/a_n)}$ is called Harmonic Mean (HM).

Note that $n \geq 2$.

THEOREM 8.1 If $a_1, a_2, \dots, a_n$ are n positive real numbers ($n \geq 2$), then

$$\frac{1}{n}(a_1 + a_2 + \dots + a_n) \geq (a_1 a_2 \dots a_n)^{1/n}$$

Equality holds if and only if $a_1 = a_2 = \dots = a_n$.

PROOF We prove the theorem by the principle of mathematical induction. If a_1, a_2 are two positive real numbers, then clearly

$$\frac{a_1 + a_2}{2} \geq \sqrt{a_1 a_2}$$

because $(\sqrt{a_1} - \sqrt{a_2})^2 \geq 0$ and equality occurs if and only if $a_1 = a_2$. Assume the validity of the theorem for $(n-1)$ positive real numbers and suppose $a_1, a_2, \dots, a_n$ be n positive real numbers not all equal.

Without loss of generality, we may suppose a_1 is the greatest and a_n is the least. Let G_n be $(a_1 a_2 \dots a_n)^{1/n}$ so that using the assumption for the numbers $a_2, a_3, \dots, a_{n-1}, a_1 a_n / G_n$, we have

$$\frac{1}{n-1} \left(a_2 + a_3 + \dots + a_{n-1} + \frac{a_1 a_n}{G_n} \right) \geq \left[(a_2 a_3 \dots a_{n-1}) \frac{a_1 a_n}{G_n} \right]^{1/(n-1)}$$

That is

$$a_2 + a_3 + \dots + a_{n-1} + \frac{a_1 a_n}{G_n} \geq (n-1)G_n$$

Therefore

$$\begin{aligned} nG_n &\leq \left(a_2 + a_3 + \dots + a_{n-1} + \frac{a_1 a_n}{G_n} \right) + G_n \\ &= [nA_n - (a_1 + a_n)] + \frac{a_1 a_n}{G_n} + G_n \\ &= nA_n + \frac{[G_n^2 - (a_1 + a_n)G_n + a_1 a_n]}{G_n} \end{aligned}$$

$$\begin{aligned}
 &= nA_n + \frac{(G_n - a_1)(G_n - a_n)}{G_n} \\
 &< nA_n \quad (\because a_1 > G_n > a_n)
 \end{aligned}$$

So

$$G_n < A_n$$

Hence, the theorem follows from induction. ■

COROLLARY 8.1

If $a_1, a_2, \dots, a_n$ are positive and $a_1 + a_2 + \dots + a_n = s$ (constant), then the product $p = a_1 a_2 \dots a_n$ is greatest when

$$a_1 = a_2 = \dots = a_n = \frac{s}{n}$$

COROLLARY 8.2

If $a_1, a_2, \dots, a_n$ are positive such that their product $a_1 a_2 \dots a_n = p$ (constant), then their sum $a_1 + a_2 + \dots + a_n$ is least when

$$a_1 = a_2 = \dots = a_n = p^{1/n}$$

THEOREM 8.2 (CAUCHY'S INEQUALITY)

Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be two sets of real numbers. Then

$$(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2) \times (b_1^2 + b_2^2 + \dots + b_n^2)$$

Equality occurs if and only if $a_1, a_2, \dots, a_n$ are proportional to $b_1, b_2, \dots, b_n$.

PROOF

We proved this inequality when $n = 3$ in Chapter 6 (see Corollary 6.5). Let

$$A = \sum_{i=1}^n a_i^2, B = \sum_{i=1}^n a_i b_i \quad \text{and} \quad C = \sum_{i=1}^n b_i^2$$

For any real x , we have

$$0 \leq \sum_{i=1}^n (a_i + x b_i)^2 = A + 2Bx + Cx^2$$

This implies

$$\begin{aligned}
 4B^2 - 4AC &\leq 0 \\
 \Rightarrow B^2 - AC &\leq 0 \\
 \Rightarrow B^2 &\leq AC
 \end{aligned}$$

and equality holds if and only if $a_i + x b_i = 0$ for $i = 1, 2, \dots, n$. ■

WORKED-OUT PROBLEMS

Caution: When AM–GM inequality is to be applied, be sure that the numbers are positive.

1. If a and b are positive real numbers such that $a + b = 1$, then prove that **Proof:** By Theorem 8.1

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq \frac{25}{2}$$

$$\frac{a+b}{2} \geq \sqrt{ab} \Rightarrow ab \leq \frac{1}{4} \quad (8.1)$$

Now

$$\begin{aligned}
 \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 &= (a^2 + b^2) + 4 + \left(\frac{1}{a^2} + \frac{1}{b^2}\right) \\
 &= (a + b)^2 - 2ab + 4 + \left(\frac{1}{a^2} + \frac{1}{b^2}\right) \\
 &\geq 1 - 2ab + 4 + \frac{2}{ab} \\
 &\geq 1 - 2\left(\frac{1}{4}\right) + 4 + 2(4) \quad [\text{By Eq. (8.1)}] \\
 &= 12 + \frac{1}{2}
 \end{aligned}$$

Equality holds if

$$a = b = \frac{1}{2}$$

2. If $x_1, x_2, \dots, x_n$ are positive real numbers, then show that

$$(x_1 + x_2 + \dots + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right) \geq n^2$$

Equality occurs if and only if $x_1 = x_2 = \dots = x_n$.

Proof: By AM–GM inequality we have

$$x_1 + x_2 + \dots + x_n \geq n(x_1 x_2 \dots x_n)^{1/n} \quad (8.2)$$

and

$$\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \geq n \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \dots \frac{1}{x_n} \right)^{1/n} \quad (8.3)$$

Multiplying Eqs. (8.2) and (8.3), we have

$$(x_1 + x_2 + \dots + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right) \geq n^2$$

3. In $\triangle ABC$, if

$$\cos A + \cos B + \cos C = \frac{3}{2}$$

then prove that the triangle is equilateral.

Proof: We have

$$\cos A + \cos B + \cos C = \frac{3}{2}$$

Therefore

$$2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} + 1 - 2 \sin^2 \frac{C}{2} = \frac{3}{2}$$

$$2 \sin \frac{C}{2} \cos \frac{A-B}{2} - 2 \sin^2 \frac{C}{2} = \frac{1}{2}$$

$$\cos \frac{A-B}{2} = \sin \frac{C}{2} + \frac{1}{4 \sin \frac{C}{2}}$$

$$= \left(\sqrt{\sin \frac{C}{2}} - \frac{1}{2\sqrt{\sin(C/2)}} \right)^2 + 1$$

$$\geq 1$$

Hence we get

$$1 \leq \cos \frac{A-B}{2} \leq 1$$

which implies that

$$\cos \frac{A-B}{2} = 1$$

so that $A = B$. Similarly $B = C$. Therefore $\triangle ABC$ is equilateral.

4. Assuming that $e^x \geq 1 + x$ for all real x (this result can be proved using differential calculus), prove that AM of positive real numbers is greater than or equal to their GM.

Proof: In $e^x \geq 1 + x$ replace x on both sides with $x - 1$. We get

$$e^{x-1} \geq x \quad (8.4)$$

Let $x_1, x_2, \dots, x_n$ be n positive real numbers. Now, replace x with $x_1/\bar{x}, x_2/\bar{x}, \dots, x_n/\bar{x}$ in Eq. (8.4) where

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Now multiply all of these so that

$$e^{(x_1/\bar{x})-1} \cdot e^{(x_2/\bar{x})-1} \dots e^{(x_n/\bar{x})-1} \geq \frac{x_1}{\bar{x}} \cdot \frac{x_2}{\bar{x}} \dots \frac{x_n}{\bar{x}}$$

Therefore

$$\exp \left(\frac{x_1 + x_2 + \dots + x_n}{\bar{x}} - n \right) \geq \frac{(x_1 x_2 \dots x_n)}{(\bar{x})^n}$$

$$\exp \left(\frac{n\bar{x}}{\bar{x}} - n \right) \geq \frac{x_1 x_2 \dots x_n}{(\bar{x})^n}$$

$$1 = e^0 \geq \frac{x_1 x_2 \dots x_n}{(\bar{x})^n}$$

$$\bar{x} \geq (x_1 x_2 \dots x_n)^{1/n}$$

5. In $\triangle ABC$, if $\tan A \tan B + \tan B \tan C + \tan C \tan A = 9$, then prove that $\triangle ABC$ is equilateral.

Proof: First we show that the angles A, B, C are acute. Suppose one of the angles say A is obtuse so that $\tan A$ is negative. Therefore B and C are acute, and $B + C$ is also acute. Also

$$\tan(B + C) = -\tan A > 0$$

Therefore

$$\frac{\tan B + \tan C}{1 - \tan B \tan C} = -\tan A > 0$$

and $\tan B \tan C \neq 1$

so that $\tan B \tan C < 1$. Now

$$\begin{aligned} 9 &= \sum \tan A \tan B \\ &= \tan A(\tan B + \tan C) + \tan B \tan C \end{aligned}$$

which implies that

$$\begin{aligned} 0 &> \tan A(\tan B + \tan C) \\ &= 9 - \tan B \tan C \\ &> 0 \end{aligned}$$

which is a contradiction. Therefore A cannot be obtuse and hence A, B, C are acute. This means $\tan A, \tan B$ and $\tan C$ are positive. Again

$$\begin{aligned} 9 &= \tan A \tan B + \tan B \tan C + \tan C \tan A \\ &= \frac{\tan A \tan B \tan C}{\tan C} + \frac{\tan A \tan B \tan C}{\tan A} \\ &\quad + \frac{\tan C \tan B \tan A}{\tan B} \\ &= \frac{\tan A + \tan B + \tan C}{\tan C} + \frac{\tan A + \tan B + \tan C}{\tan A} \\ &\quad + \frac{\tan A + \tan B + \tan C}{\tan B} \end{aligned}$$

because

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Therefore

$$\begin{aligned} 6 &= \frac{\tan A}{\tan C} + \frac{\tan B}{\tan C} + \frac{\tan B}{\tan A} + \frac{\tan C}{\tan A} + \frac{\tan A}{\tan B} + \frac{\tan C}{\tan B} \\ &\geq 6 \left[\frac{\tan A}{\tan C} \cdot \frac{\tan B}{\tan C} \cdot \frac{\tan B}{\tan A} \cdot \frac{\tan C}{\tan A} \cdot \frac{\tan A}{\tan B} \cdot \frac{\tan C}{\tan B} \right]^{1/6} \\ &\quad \text{(Theorem 8.1)} \\ &= 6 \end{aligned}$$

Because equality is holding,

$$\frac{\tan A}{\tan B} = \frac{\tan B}{\tan C} = \frac{\tan C}{\tan A}$$

which implies that $\tan A = \tan B = \tan C$. Hence the triangle is equilateral.

6. In $\triangle ABC$, if

$$\frac{a \cos A + b \cos B + c \cos C}{a \sin B + b \sin C + c \sin A} = \frac{a + b + c}{9R}$$

where R is the circumradius of $\triangle ABC$, prove that the triangle is equilateral.

Proof: In Chapter 4, we proved that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \quad \text{(Theorem 4.1)}$$

Therefore $a = 2R \sin A$, $b = 2R \sin B$ and $c = 2R \sin C$. According to the hypothesis

$$\begin{aligned} \frac{\sin 2A + \sin 2B + \sin 2C}{2 \sum (\sin A \sin B)} &= \frac{2}{9} (\sin A + \sin B + \sin C) \\ \frac{4 \sin A \sin B \sin C}{2 \sum (\sin A \sin B)} &= \frac{2}{9} (\sum \sin A) \end{aligned}$$

$$9 \sin A \sin B \sin C = (\sin A + \sin B + \sin C) (\sum \sin A \sin B)$$

$$9 = (\sin A + \sin B + \sin C)$$

$$\left(\frac{1}{\sin A} + \frac{1}{\sin B} + \frac{1}{\sin C} \right)$$

$$\geq 9 \quad \text{(by Problem 2)}$$

Equality implies that $\sin A = \sin B = \sin C$. Therefore $\triangle ABC$ is equilateral.

7. Let $a_1, a_2, \dots, a_n; b_1, b_2, \dots, b_n$ be two sets of positive real numbers such that

$$a_1 + a_2 + \dots + a_n = b_1 + b_2 + \dots + b_n$$

Then prove that

$$\sum_{i=1}^n \left(\frac{a_i^2}{a_i + b_i} \right) \geq \frac{1}{2} (a_1 + a_2 + \dots + a_n)$$

$$\text{and} \quad \sum_{i=1}^n \left(\frac{b_i^2}{a_i + b_i} \right) \geq \frac{1}{2} (b_1 + b_2 + \dots + b_n)$$

Proof: We have

$$\frac{a_i^2}{a_i + b_i} = \frac{a_i^2 + a_i b_i - a_i b_i}{a_i + b_i} = a_i - \frac{a_i b_i}{a_i + b_i} \quad (8.5)$$

Again,

$$(a_i + b_i) \left(\frac{1}{a_i} + \frac{1}{b_i} \right) \geq 2^2 \quad \text{(Problem 2)}$$

which implies that

$$\begin{aligned}\frac{a_i + b_i}{4} &\geq \frac{1}{(1/a_i) + (1/b_i)} \\ &= \frac{a_i b_i}{a_i + b_i}\end{aligned}\quad (8.6)$$

From Eq. (8.5)

$$\begin{aligned}\frac{a_i^2}{a_i + b_i} &= a_i - \frac{a_i b_i}{a_i + b_i} \\ &\geq a_i - \frac{a_i + b_i}{4} \quad [\text{from Eq. (8.6)}]\end{aligned}$$

Therefore

$$\frac{a_i^2}{a_i + b_i} \geq \frac{3a_i - b_i}{4}$$

So

$$\begin{aligned}\sum_{i=1}^n \frac{a_i^2}{a_i + b_i} &\geq \frac{1}{4}[3(a_1 + a_2 + \cdots + a_n) - (b_1 + b_2 + \cdots + b_n)] \\ &= \frac{1}{4}[3(a_1 + a_2 + \cdots + a_n) - (a_1 + a_2 + \cdots + a_n)] \\ &= \frac{1}{2}(a_1 + a_2 + \cdots + a_n)\end{aligned}$$

8. If $x_1, x_2, x_3, \dots, x_n$ are positive real numbers such that $x_1 + x_2 + \cdots + x_n = s$ (constant), then prove that

$$(1 + x_1)(1 + x_2) \cdots (1 + x_n) \leq 1 + \frac{s_1}{1} + \frac{s_2}{2} + \cdots + \frac{s_n}{n}$$

Proof: Note that

$${}^nC_r = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}$$

so that

$$\frac{{}^nC_r}{n^r} \leq \frac{1}{r!} \quad (8.7)$$

Now by Theorem 8.1, we have

$$\begin{aligned}(1 + x_1)(1 + x_2) \cdots (1 + x_n) &\leq \left[\frac{(1 + x_1) + (1 + x_2) + \cdots + (1 + x_n)}{n} \right]^n \\ &= \left(1 + \frac{s}{n} \right)^n \\ &= 1 + {}^nC_1 \left(\frac{s}{n} \right) + {}^nC_2 \left(\frac{s}{n} \right)^2 + \cdots + {}^nC_n \left(\frac{s}{n} \right)^n \\ &\leq 1 + \frac{s}{1} + \frac{s^2}{2} + \cdots + \frac{s^n}{n} \quad [\text{By Eq. (8.7)}]\end{aligned}$$

9. If x, y, z are positive real numbers and $x + y + z = 3$, then prove that

$$\sqrt{x} + \sqrt{y} + \sqrt{z} \geq xy + yz + zx$$

Proof: We have

$$2\sqrt{x} - x(y + z) = 2\sqrt{x} - x(3 - x) \quad (8.8)$$

Now

$$\begin{aligned}[2\sqrt{x} - x(3 - x)][2\sqrt{x} + x(3 - x)] &= 4x - x^2(3 - x)^2 \\ &= x[4 - x(9 - 6x + x^2)] \\ &= x(-x^3 + 6x^2 - 9x + 4) \\ &= x(x - 1)(-x^2 + 5x - 4) \\ &= (x - 1)(x - 1)^2(4 - x) \\ &\geq 0 \quad (\because 0 < x < 3)\end{aligned}$$

Hence from Eq. (8.8), we have

$$2\sqrt{x} - x(y + z) \geq 0$$

Similarly

$$2\sqrt{y} - y(z + x) \geq 0$$

and

$$2\sqrt{z} - z(x + y) \geq 0$$

Therefore

$$\begin{aligned}2(\sqrt{x} + \sqrt{y} + \sqrt{z}) - 2(xy + yz + zx) &\geq 0 \\ \sqrt{x} + \sqrt{y} + \sqrt{z} &\geq xy + yz + zx\end{aligned}$$

10. In $\triangle ABC$, if

$$\begin{aligned}(\tan A + 2 \tan B + 3 \tan C)^2 \\ \geq 14(\tan^2 A + \tan^2 B + \tan^2 C)\end{aligned}$$

then show that $\angle B + \angle C$ is $3\pi/4$.

Proof: In Theorem 8.2 (Cauchy's Inequality), take $a_1 = 1$, $a_2 = 2$, $a_3 = 3$ and $b_1 = \tan A$, $b_2 = \tan B$, $b_3 = \tan C$. Then

$$\begin{aligned}(\tan A + 2 \tan B + 3 \tan C)^2 &\leq (1^2 + 2^2 + 3^2) \\ &\quad (\tan^2 A + \tan^2 B + \tan^2 C) \\ (\tan A + 2 \tan B + 3 \tan C)^2 &\leq 14(\tan^2 A + \tan^2 B + \tan^2 C) \\ &\leq (\tan A + 2 \tan B + 3 \tan C)^2 \\ &\quad (\text{by hypothesis})\end{aligned}$$

Therefore

$$(\tan A + 2 \tan B + 3 \tan C)^2 = 14(\tan^2 A + \tan^2 B + \tan^2 C)$$

But it is known that equality occurs if and only if $a_1 : a_2 : a_3 = b_1 : b_2 : b_3$. So

$$\frac{\tan A}{1} = \frac{\tan B}{2} = \frac{\tan C}{3}$$

Hence

$$A = \frac{\pi}{4} \quad \text{and} \quad |B| + |C| = \frac{3\pi}{4}$$

11. In Figure 8.1 P is a point inside $\triangle ABC$. Let d_1, d_2, d_3 be the distances of P from the sides BC, CA and AB , respectively. R and r are, respectively, the circumradius and inradius of $\triangle ABC$. Prove that

- (i) The greatest value of $(\sqrt{d_1} + \sqrt{d_2} + \sqrt{d_3})^2$ is $\frac{a^2 + b^2 + c^2}{2R}$.
- (ii) The least value of $\frac{a}{d_1} + \frac{b}{d_2} + \frac{c}{d_3}$ is $\frac{a+b+c}{r}$.
- (iii) $\frac{a}{d_1} + \frac{b}{d_2} + \frac{c}{d_3}$ is least when P is the incentre of $\triangle ABC$.

Proof: Let $a_1 = \sqrt{ad_1}, a_2 = \sqrt{bd_2}$ and $a_3 = \sqrt{cd_3}$ and $b_1 = 1/\sqrt{a}, b_2 = 1/\sqrt{b}, b_3 = 1/\sqrt{c}$.

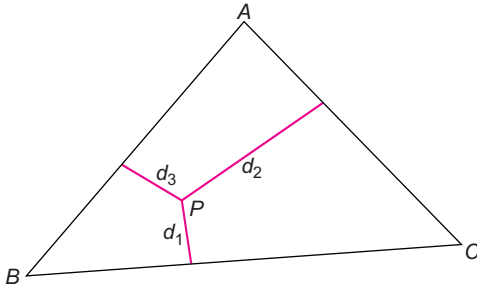


FIGURE 8.1 Worked-out problem 11.

(i) Using Cauchy inequality (Theorem 8.2) we have

$$(\sqrt{d_1} + \sqrt{d_2} + \sqrt{d_3})^2 \leq (ad_1 + bd_2 + cd_3) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \quad (8.9)$$

But

$$ad_1 + bd_2 + cd_3 = 2\Delta PBC + 2\Delta PCA + 2\Delta PAB = 2\Delta.$$

From Eq. (8.9),

$$\begin{aligned} (\sqrt{d_1} + \sqrt{d_2} + \sqrt{d_3})^2 &\leq 2\Delta \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \\ &= \frac{2\Delta}{abc} (bc + ca + ab) \\ &\leq \frac{2\Delta}{abc} (a^2 + b^2 + c^2) = \frac{a^2 + b^2 + c^2}{2R} \end{aligned}$$

Therefore the least value of $(\sqrt{d_1} + \sqrt{d_2} + \sqrt{d_3})^2$ is

$$\frac{a^2 + b^2 + c^2}{2R}$$

(ii) We have

$$\begin{aligned} (ad_1 + bd_2 + cd_3) \left(\frac{a}{d_1} + \frac{b}{d_2} + \frac{c}{d_3} \right) &= (a^2 + b^2 + c^2) \\ &\quad + \sum ab \left(\frac{d_1}{d_2} + \frac{d_2}{d_1} \right) \end{aligned}$$

and

$$\begin{aligned} (ad_1 + bd_2 + cd_3) \times \left(\frac{a}{d_1} + \frac{b}{d_2} + \frac{c}{d_3} \right) &\geq (a^2 + b^2 + c^2) + \sum 2ab \left(\because \frac{d_1}{d_2} + \frac{d_2}{d_1} \geq 2 \right) \\ &= (a + b + c)^2 \end{aligned}$$

Therefore

$$\begin{aligned} \frac{a}{d_1} + \frac{b}{d_2} + \frac{c}{d_3} &\geq \frac{(a + b + c)^2}{2\Delta} \\ &= \frac{2s}{2\Delta} (a + b + c) = \frac{a + b + c}{r} \end{aligned}$$

(iii) is obvious.

Note: The above problem may be asked under comprehension.

12. If x, y, z are positive real numbers and $x + y + z = 1$, then prove that the minimum value of

$$\frac{4}{x} + \frac{9}{y} + \frac{16}{z}$$

is 81.

Proof: Since $x + y + z = 1$ we have

$$\begin{aligned} \frac{4}{x} + \frac{9}{y} + \frac{16}{z} &= (x + y + z) \left(\frac{4}{x} + \frac{9}{y} + \frac{16}{z} \right) \\ &= (4 + 9 + 16) + \left(\frac{4z}{x} + \frac{16x}{z} \right) + \left(\frac{9z}{y} + \frac{16y}{z} \right) + \left(\frac{4y}{x} + \frac{9x}{y} \right) \\ &\geq 29 + 2\sqrt{64} + 2\sqrt{144} + 2\sqrt{36} = 81 \end{aligned}$$

Equality holds if and only if

$$x = \frac{2}{9}, y = \frac{3}{9}, z = \frac{4}{9}$$

13. If a, b, c are positive real numbers such that $a + b + c = 1$, then prove that

$$a^2 + b^2 + c^2 \geq 4(ab + bc + ca) - 1$$

where the equality holds if and only if $a = b = c = 1/3$.

Proof: We have

$$\begin{aligned}
 a^2 + b^2 + c^2 &= (a + b + c)^2 - 2\sum ab \\
 &= 1 - 2\sum ab \geq 4(ab + bc + ca) - 1 \\
 \Leftrightarrow 2 &\geq 6\sum ab \\
 \Leftrightarrow 1 &\geq 3\sum ab \\
 \Leftrightarrow (a + b + c)^2 &\geq 3\sum ab \\
 \Leftrightarrow a^2 + b^2 + c^2 - ab - bc - ca &\geq 0 \\
 \Leftrightarrow \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2] &\geq 0
 \end{aligned}$$

which is true and equality occurs if and only if $a = b = c = 1/3$.

14. If a, b, c are real numbers such that $0 < a, b, c < 1$ and $a + b + c = 2$, then prove that

$$\frac{abc}{(1-a)(1-b)(1-c)} \geq 8$$

Equality occurs when $a = b = c = 2/3$.

Proof: We have

$$\begin{aligned}
 a &= \frac{(a + b - c) + (a - b + c)}{2} \\
 &\geq \sqrt{(a + b - c)(a - b + c)} \\
 b &= \frac{(b + c - a) + (b - c + a)}{2} \\
 &\geq \sqrt{(b + c - a)(b - c + a)} \\
 c &= \frac{(c + a - b) + (c - a + b)}{2} \\
 &\geq \sqrt{(c + a - b)(c - a + b)}
 \end{aligned}$$

Therefore

$$\begin{aligned}
 abc &\geq (b + c - a)(c + a - b)(a + b - c) \\
 &= (2 - 2a)(2 - 2b)(2 - 2c)
 \end{aligned}$$

So

$$\frac{abc}{(1-a)(1-b)(1-c)} \geq 8$$

15. If each of x, y, z is greater than unity and

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 2$$

then prove that

$$\sqrt{x + y + z} \geq \sqrt{x - 1} + \sqrt{y - 1} + \sqrt{z - 1}$$

(Maths Olympiads IRAN, 1998–99)

Proof: In Cauchy's inequality (Theorem 8.2) for $n = 3$, take

$$\begin{aligned}
 a_1 &= \sqrt{x}, a_2 = \sqrt{y}, a_3 = \sqrt{z} \\
 b_1 &= \sqrt{\frac{x-1}{x}}, b_2 = \sqrt{\frac{y-1}{y}}, b_3 = \sqrt{\frac{z-1}{z}}
 \end{aligned}$$

Therefore

$$(a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

This implies

$$\begin{aligned}
 (\sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1})^2 &\leq (x + y + z) \\
 \left(\frac{x-1}{x} + \frac{y-1}{y} + \frac{z-1}{z} \right) &\leq 1 \quad (8.10)
 \end{aligned}$$

But

$$\begin{aligned}
 \frac{x-1}{x} + \frac{y-1}{y} + \frac{z-1}{z} &= 3 - \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) \\
 &= 3 - 2 = 1
 \end{aligned}$$

So by Eq. (8.10) we have

$$\sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1} \leq \sqrt{x + y + z}$$

16. If x, y, z are positive real numbers such that $x^3 y^2 z^4 = 7$, then show that

$$2x + 5y + 3z \geq 9 \left(\frac{525}{2^7} \right)^{1/9}$$

Proof: Using AM $\geq$ GM for the nine numbers

$$\frac{2x}{3}, \frac{2x}{3}, \frac{2x}{3}, \frac{5y}{2}, \frac{5y}{2}, \frac{5y}{2}, \frac{3z}{4}, \frac{3z}{4}, \frac{3z}{4}$$

we have

$$\frac{2x + 5y + 3z}{9} \geq \left(\frac{2x}{3} \cdot \frac{2x}{3} \cdot \frac{2x}{3} \cdot \frac{5y}{2} \cdot \frac{5y}{2} \cdot \frac{5y}{2} \cdot \frac{3z}{4} \cdot \frac{3z}{4} \cdot \frac{3z}{4} \right)^{1/9}$$

Therefore

$$\begin{aligned}
 2x + 5y + 3z &\geq 9 \left[\left(\frac{2}{3} \right)^3 \left(\frac{5}{2} \right)^3 \left(\frac{3}{4} \right)^3 x^3 y^2 z^4 \right]^{1/9} \\
 &= 9 \left[\left(\frac{2}{3} \right)^3 \left(\frac{5}{2} \right)^3 \left(\frac{3}{4} \right)^3 (7) \right]^{1/9} = 9 \left(\frac{525}{2^7} \right)^{1/9}
 \end{aligned}$$

17. If a, b, c are positive real numbers such that $abc = 1$, then prove that

$$\frac{ab}{a^5 + b^5 + ab} + \frac{bc}{b^5 + c^5 + bc} + \frac{ca}{c^5 + a^5 + ca} \leq 1$$

Equality holds when $a = b = c = 1$.

Proof: We have

$$\begin{aligned} a^5 + b^5 &= (a+b)[a^4 - a^3b + a^2b^2 - ab^3 + b^4] \\ &= (a+b)[a^3(a-b) - b^3(a-b) + a^2b^2] \\ &= (a+b)[(a^3 - b^3)(a-b) + a^2b^2] \\ &= (a+b)[(a-b)^2(a^2 + ab + b^2) + a^2b^2] \\ &\geq (a+b)a^2b^2 \end{aligned}$$

Therefore

$$\begin{aligned} \frac{1}{a^5 + b^5 + ab} &\leq \frac{1}{(a+b)a^2b^2 + ab} \\ \frac{ab}{a^5 + b^5 + ab} &\leq \frac{ab}{(a+b)a^2b^2 + ab} \\ &= \frac{1}{ab(a+b) + 1} \\ &= \frac{abc}{ab(a+b) + abc} \quad (\because abc = 1) \\ &= \frac{c}{a+b+c} \end{aligned}$$

So

$$\frac{ab}{a^5 + b^5 + ab} \leq \frac{c}{a+b+c}$$

Similarly

$$\begin{aligned} \frac{bc}{b^5 + c^5 + bc} &\leq \frac{a}{a+b+c} \\ \frac{ca}{c^5 + a^5 + ca} &\leq \frac{b}{a+b+c} \end{aligned}$$

Hence

$$\frac{ab}{a^5 + b^5 + ab} + \frac{bc}{b^5 + c^5 + bc} + \frac{ca}{c^5 + a^5 + ca} \leq \frac{a+b+c}{a+b+c} = 1$$

18. If $0 < a < 1$ and $0 < b < 1$, then prove that $\sqrt{ab} + \sqrt{(1-a)(1-b)} \leq 1$.

Proof: Put $a = \sin^2 \theta$ and $b = \sin^2 \phi$. Then

$$\begin{aligned} \sqrt{ab} + \sqrt{(1-a)(1-b)} &= \sin \theta \sin \phi + \cos \theta \cos \phi \\ &= \cos(\theta - \phi) \\ &\leq 1 \end{aligned}$$

Note that equality holds when $a = b$.

19. If a, b, c are positive real numbers, then prove that

$$a + b + c \leq \frac{b^2 + c^2}{b+c} + \frac{c^2 + a^2}{c+a} + \frac{a^2 + b^2}{a+b} < 2(a+b+c)$$

where the preceding equality holds if $a = b = c$.

Proof: We have

$$\begin{aligned} b^2 + c^2 &\geq 2bc \\ \Rightarrow 2(b^2 + c^2) &\geq 2bc + b^2 + c^2 = (b+c)^2 \\ \Rightarrow \frac{b^2 + c^2}{b+c} &\geq \frac{b+c}{2} \end{aligned}$$

Therefore

$$\begin{aligned} \sum \left(\frac{b^2 + c^2}{b+c} \right) &\geq \frac{b+c}{2} + \frac{c+a}{2} + \frac{a+b}{2} \\ &= a+b+c \end{aligned}$$

So

$$a + b + c \leq \sum \frac{b^2 + c^2}{b+c}$$

Also

$$\begin{aligned} b^2 + c^2 &< (b+c)^2 \\ \Rightarrow \frac{b^2 + c^2}{b+c} &< b+c \end{aligned}$$

Therefore

$$\begin{aligned} \sum \frac{b^2 + c^2}{b+c} &< (b+c) + (c+a) + (a+b) \\ &= 2(a+b+c) \end{aligned}$$

20. If a, b, c are positive real numbers such that $a + b + c = 1$, then prove that

$$(1+a)(1+b)(1+c) \geq 8(1-a)(1-b)(1-c)$$

and equality occurs if $a = b = c = 1/3$.

Proof: We have

$$\begin{aligned} 1+a &= 1 + (1-b-c) \\ &= (1-b) + (1-c) \\ &\geq 2\sqrt{(1-b)(1-c)} \end{aligned}$$

Similarly

$$\begin{aligned} 1+b &\geq 2\sqrt{(1-c)(1-a)} \\ 1+c &\geq 2\sqrt{(1-a)(1-b)} \end{aligned}$$

Therefore

$$(1+a)(1+b)(1+c) \geq 8(1-a)(1-b)(1-c)$$

21. Prove that

$$\frac{1}{2\sqrt{n+1}} < \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n} < \frac{1}{\sqrt{2n+1}}$$

Proof: Let

$$a_n = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n}$$

Therefore

$$a_n < \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n+1} \quad \left(\because \frac{1}{2} < \frac{2}{3}, \frac{3}{4} < \frac{4}{5} \cdots \right)$$

So

$$\begin{aligned} a_n^2 &< \left(\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n} \right) \left(\frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n+1} \right) \\ &= \frac{1}{2n+1} \end{aligned}$$

which gives

$$a_n < \frac{1}{\sqrt{2n+1}}$$

Again,

$$\begin{aligned} (2n+1)a_n &= \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{7}{6} \cdots \left(\frac{2n+1}{2n} \right) \\ &> \frac{4}{3} \cdot \frac{6}{5} \cdot \frac{8}{7} \cdots \left(\frac{2n+2}{2n+1} \right) \end{aligned}$$

So

$$\begin{aligned} (2n+1)^2 a_n^2 &> \frac{2n+2}{2} = n+1 \\ \Rightarrow a_n^2 &> \frac{n+1}{(2n+1)^2} > \frac{1}{4(n+1)} \\ \Rightarrow a_n &> \frac{1}{2\sqrt{n+1}} \end{aligned}$$

22. If a and b are two given positive real numbers and x, y are variable real numbers satisfying the relation $x^2 + y^2 = 1$, then show that the minimum value of

$$\frac{ax^2 + by^2}{\sqrt{a^2x^2 + b^2y^2}}$$

is $2\sqrt{ab}/(a+b)$.

Proof: We have

$$\begin{aligned} (a+b)(ax^2 + by^2) &= a^2x^2 + b^2y^2 + ab(x^2 + y^2) \\ &= (a^2x^2 + b^2y^2) + ab \\ &\geq 2\sqrt{(a^2x^2 + b^2y^2)ab} \end{aligned}$$

Therefore

$$\frac{ax^2 + by^2}{\sqrt{a^2x^2 + b^2y^2}} \geq \frac{2\sqrt{ab}}{a+b}$$

23. If $a_1, a_2, a_3; b_1, b_2, b_3$ are positive real numbers, then prove that

$$(a_1b_1 + a_2b_2 + a_3b_3) \left(\frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3} \right) \geq (a_1 + a_2 + a_3)^2$$

and equality holds if and only if $b_1 = b_2 = b_3$.

Proof: We have

$$\begin{aligned} &(a_1b_1 + a_2b_2 + a_3b_3) \left(\frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3} \right) \\ &= a_1^2 + a_2^2 + a_3^2 + \left(\frac{a_1b_1a_2}{b_2} + \frac{a_2b_2a_1}{b_1} \right) + \left(\frac{a_2a_3b_2}{b_3} + \frac{a_2a_3b_3}{b_2} \right) \\ &\quad + \left(\frac{a_3a_1b_1}{b_3} + \frac{a_3a_1b_3}{b_1} \right) \\ &\geq a_1^2 + a_2^2 + a_3^2 + 2a_1a_2 + 2a_2a_3 + 2a_3a_1 \quad (\because \text{AM} \geq \text{GM}) \end{aligned}$$

and equality occurs if and only if $b_1 = b_2 = b_3$.

24. If a, b, c are positive integers such that $a + b + c = n$, then prove that

$$(a^a b^b c^c)^{1/n} + (a^b b^c c^a)^{1/n} + (a^c b^a c^b)^{1/n} \leq n$$

Equality holds when $a = b = c$ and n is a multiple of 3.

Proof:

Step 1: Consider a number of a 's, b number of b 's and c number of c 's and use $\text{AM} \geq \text{GM}$. Then

$$\begin{aligned} \frac{a^2 + b^2 + c^2}{a+b+c} &\geq (a^a b^b c^c)^{1/(a+b+c)} \\ \frac{a^2 + b^2 + c^2}{n} &\geq (a^a b^b c^c)^{1/n} \end{aligned}$$

Step 2: Consider b number of a 's, c number of b 's and a number of c 's and use $\text{AM} \geq \text{GM}$. Then

$$\frac{ab + bc + ca}{n} \geq (a^b b^c c^a)^{1/n}$$

Step 3: Consider c number of a 's, a number of b 's and b number of c 's and use $AM \geq GM$. Then

$$\frac{ac + ab + bc}{n} \geq (a^c b^a c^b)^{1/n}$$

Adding all the results in steps 1 to 3, we have

$$\begin{aligned} \frac{1}{n}(a^2 + b^2 + c^2 + 2ab + 2bc + 2ca) \\ \geq (a^a b^b c^c)^{1/n} + (a^b b^c c^a)^{1/n} + (a^c b^a c^b)^{1/n} \end{aligned}$$

that is

$$\frac{(a + b + c)^2}{n} \geq (a^a b^b c^c)^{1/n} + (a^b b^c c^a)^{1/n} + (a^c b^a c^b)^{1/n}$$

Result follows because $a + b + c = n$.

25. If a, b, c are positive real numbers, then prove that

$$[(1+a)(1+b)(1+c)]^7 > 7^7 a^4 b^4 c^4$$

(IIT-JEE, 2004)

Proof: We have

$$(1+a)(1+b)(1+c) = 1 + a + b + c + ab + bc + ca + abc$$

This implies

$$(1+a)(1+b)(1+c) > a + b + c + ab + bc + ca + abc \quad (8.11)$$

Using $AM \geq GM$, we have

$$\begin{aligned} \frac{1}{7}(a + b + c + ab + bc + ca + abc) \\ \geq [(abc)(ab \cdot bc \cdot ca)abc]^{1/7} \\ = (a^4 b^4 c^4)^{1/7} \end{aligned}$$

Therefore from Eq. (8.11) we have

$$(1+a)(1+b)(1+c) > 7(a^4 b^4 c^4)^{1/7}$$

So

$$[(1+a)(1+b)(1+c)]^7 > 7^7 a^4 b^4 c^4$$

26. For any positive integer n , prove that

$$\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$$

(IIT-JEE, 2000)

Proof: We have

$$\begin{aligned} n &< \sqrt{n(n+1)} \\ \Rightarrow 2n &< 2\sqrt{n(n+1)} \end{aligned}$$

$$\Rightarrow 4n+1 < 2n+1 + 2\sqrt{n(n+1)}$$

$$= (\sqrt{n} + \sqrt{n+1})^2$$

Therefore

$$\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} \quad (8.12)$$

Again

$$2\sqrt{n(n+1)} < 2n+1$$

$$\Rightarrow n + n + 1 + 2\sqrt{n(n+1)} < 4n+2$$

$$\Rightarrow (\sqrt{n} + \sqrt{n+1})^2 < 4n+2$$

$$\Rightarrow \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2} \quad (8.13)$$

From Eqs. (8.12) and (8.13) we have

$$\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$$

27. If $a_1, a_2, \dots, a_n$ are positive real numbers whose product is a fixed number c , then the minimum value of $a_1 + a_2 + \dots + a_{n-1} + 2a_n$ is

- (A) $n(2c)^{1/n}$ (B) $(n+1)c^{1/n}$
(C) $(2n)c^{1/n}$ (D) $(n+1)(2c)^{1/n}$

(IIT-JEE, 2002)

Solution: Using $AM \geq GM$ for the n numbers $a_1, a_2, \dots, a_{n-1}, 2a_n$, we have

$$\begin{aligned} \frac{a_1 + a_2 + \dots + a_{n-1} + 2a_n}{n} &\geq (a_1 a_2 a_3 \dots a_{n-1} (2a_n))^{1/n} \\ &= (2c)^{1/n} \end{aligned}$$

Therefore

$$a_1 + a_2 + \dots + a_{n-1} + 2a_n \geq n(2c)^{1/n}$$

Answer: (A)

28. If a, b, c and d are positive real numbers such that $a + b + c + d = 2$, then $M = (a + b)(c + d)$ satisfies the relation

- (A) $0 < M \leq 1$ (B) $1 < M \leq 2$
(C) $2 < M \leq 3$ (D) $3 \leq M \leq 4$

(IIT-JEE, 2000)

Solution: Using $AM \geq GM$ for the two numbers $(a + b)$ and $(c + d)$ we have

$$\frac{(a + b) + (c + d)}{2} \geq [(a + b)(c + d)]^{1/2}$$

Therefore

$$\frac{2}{2} \geq M^{1/2}$$

and M cannot be zero because a, b, c, d, e are positive. Hence $0 < M \leq 1$ and equality occurs if and only if $a + b = c + d = 1$.

Answer: (A)

29. The product of n positive numbers is unity, then their sum is

- (A) a positive integer
- (B) divisible by n
- (C) equal to $n + \frac{1}{n}$
- (D) never less than n

(IIT-JEE, 1991)

Solution: Let $a_1, a_2, \dots, a_n$ be positive such that the product $a_1 a_2 a_3 \dots a_n = 1$. Using AM $\geq$ GM we have

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n} = 1$$

This implies

$$a_1 + a_2 + \dots + a_n \geq n$$

Answer: (D)

30. If a, b, c are positive real numbers, then prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

and equality occurs when $a = b = c$.

Proof: We have

$$\frac{(b+c) + (c+a) + (a+b)}{3} \geq [(b+c)(c+a)(a+b)]^{1/3}$$

Therefore

$$\frac{2}{3}(a+b+c) \geq [(b+c)(c+a)(a+b)]^{1/3} \quad (8.14)$$

Again

$$\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \geq \left(\frac{1}{b+c} \cdot \frac{1}{c+a} \cdot \frac{1}{a+b} \right)^{1/3} \quad (8.15)$$

Multiplying Eqs. (8.14) and (8.15), we have

$$(a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right) \geq \frac{9}{2}$$

$$\left(\frac{a}{b+c} + 1 \right) + \left(\frac{b}{c+a} + 1 \right) + \left(\frac{c}{a+b} + 1 \right) \geq \frac{9}{2}$$

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{9}{2} - 3 = \frac{3}{2}$$

31. If a, b, c are positive real numbers, then prove that

$$\frac{2}{b+c} + \frac{2}{c+a} + \frac{2}{a+b} \geq \frac{9}{a+b+c}$$

(Ireland Olympiads, 1998–99)

Proof: As in Problem 30 we have

$$\frac{2}{9}(a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right) \geq 1$$

Therefore

$$\frac{2}{b+c} + \frac{2}{c+a} + \frac{2}{a+b} \geq \frac{9}{a+b+c}$$

32. If a, b, c are the sides of a triangle, then show that

$$1 \leq \frac{a^2 + b^2 + c^2}{ab + bc + ca} < 2$$

and left side equality holds when the triangle is equilateral.

Proof: We have

$$\begin{aligned} a^2 + b^2 + c^2 - ab - bc - ca \\ = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0 \end{aligned}$$

Therefore

$$\frac{a^2 + b^2 + c^2}{ab + bc + ca} \geq 1$$

Also, since a, b, c are the sides of a triangle, we have $|b-c| < a$, $|c-a| < b$ and $|a-b| < c$. Therefore

$$\begin{aligned} (a-b)^2 + (b-c)^2 + (c-a)^2 &< a^2 + b^2 + c^2 \\ 2(a^2 + b^2 + c^2) - 2(ab + bc + ca) &< a^2 + b^2 + c^2 \\ a^2 + b^2 + c^2 &< 2(ab + bc + ca) \\ \frac{a^2 + b^2 + c^2}{ab + bc + ca} &< 2 \end{aligned}$$

33. If a, b, c are sides of a triangle ABC , then prove that

- (i) $(a+b+c)^3 \geq 27(b+c-a)(c+a-b)(a+b-c)$
- (ii) $abc \geq (b+c-a)(c+a-b)(a+b-c)$
- (iii) $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$

In all the three relations equality occurs when $\triangle ABC$ is equilateral.

Proof: (i) In any triangle, since sum of any two sides is greater than the third side, we have

$$b + c - a > 0$$

$$c + a - b > 0$$

$$a + b - c > 0$$

Put

$$x = b + c - a$$

$$y = c + a - b$$

and

$$z = a + b - c$$

so that x, y and z are positive. Also $x + y + z = a + b + c$. Now $AM \geq GM$ implies

$$\frac{x + y + z}{3} \geq \sqrt[3]{xyz}$$

Therefore

$$(x + y + z)^3 \geq 27xyz$$

that is

$$(a + b + c)^3 \geq 27(b + c - a)(c + a - b)(a + b - c)$$

(ii) Again

$$\frac{y + z}{2} \geq \sqrt{yz}$$

$$\frac{z + x}{2} \geq \sqrt{zx}$$

$$\frac{x + y}{2} \geq \sqrt{xy}$$

This implies

$$\left(\frac{y + z}{2}\right)\left(\frac{z + x}{2}\right)\left(\frac{x + y}{2}\right) \geq xyz$$

that is

$$abc \geq (b + c - a)(c + a - b)(a + b - c)$$

(iii) We have

$$\sin \frac{A}{2} = \sqrt{\frac{(s - b)(s - c)}{bc}}$$

$$\sin \frac{B}{2} = \sqrt{\frac{(s - c)(s - a)}{ca}}$$

$$\sin \frac{C}{2} = \sqrt{\frac{(s - a)(s - b)}{ab}}$$

(See Theorem 4.4 of Chapter 4.) Therefore

$$\begin{aligned} \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} &= \frac{(s - a)(s - b)(s - c)}{abc} \\ &= \frac{(2s - 2a)(2s - 2b)(2s - 2c)}{8abc} \\ &= \frac{(b + c - a)(c + a - b)(a + b - c)}{8abc} \\ &\leq \frac{1}{8} \quad [\text{from (ii)}] \end{aligned}$$

34. In $\triangle ABC$, prove that

$$\text{Area } \triangle \leq \frac{1}{4} \sqrt{(a + b + c)abc}$$

(IIT-JEE, 2001)

Proof: In part (ii) of Problem 33, we proved that

$$(b + c - a)(c + a - b)(a + b - c) \leq abc$$

Therefore

$$(s - a)(s - b)(s - c) \leq abc$$

$$(s - a)(s - b)(s - c) \leq \frac{abc}{8}$$

$$s(s - a)(s - b)(s - c) \leq \left(\frac{a + b + c}{2}\right)\left(\frac{abc}{8}\right)$$

$$\Delta^2 \leq \frac{1}{16}(a + b + c)abc$$

$$\Delta \leq \frac{1}{4} \sqrt{(a + b + c)abc}$$

35. In $\triangle ABC$, prove that

$$1 < \cos A + \cos B + \cos C \leq \frac{3}{2}$$

Proof: We have

$$\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} > 1$$

Again

$$\cos A + \cos B + \cos C$$

$$= 1 - 2 \sin^2 \frac{A}{2} + 2 \cos \left(\frac{B + C}{2}\right) \cos \left(\frac{B - C}{2}\right)$$

$$\begin{aligned}
&= 1 - 2\sin^2 \frac{A}{2} + 2\sin \frac{A}{2} \cos \left(\frac{B-C}{2} \right) \\
&\leq 1 - 2\sin^2 \frac{A}{2} + 2\sin \frac{A}{2} \left(\because \cos \left(\frac{B-C}{2} \right) \leq 1 \right) \\
&= 1 - 2 \left(\sin \frac{A}{2} - \frac{1}{2} \right)^2 + \frac{1}{2} \leq \frac{3}{2}
\end{aligned}$$

Note that

$$\cos A + \cos B + \cos C = \frac{3}{2}$$

if and only if $\triangle ABC$ is equilateral.

36. In $\triangle ABC$, prove that

$$\cos A \cos B \cos C \leq \frac{1}{8}$$

Equality occurs if and only if it is an equilateral triangles.

Proof: If $\triangle ABC$ is right-angled or obtuse, the inequality is clear. Suppose $\triangle ABC$ is acute angled. By AM-GM inequality,

$$\frac{\cos A + \cos B + \cos C}{3} \geq (\cos A \cos B \cos C)^{1/3}$$

Therefore

$$\begin{aligned}
\cos A \cos B \cos C &\leq \left(\frac{\cos A + \cos B + \cos C}{3} \right)^3 \\
&\leq \left(\frac{3/2}{3} \right)^3 = \frac{1}{8}
\end{aligned}$$

37. In $\triangle ABC$, prove that

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

Equality holds if and only if the triangle is equilateral.

Proof: First, we prove that

$$(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$$

where x, y, z are positive real numbers. Consider

$$\begin{aligned}
(x + y + z)^2 &= x^2 + y^2 + z^2 + 2xy + 2yz + 2zx \\
&\leq (x^2 + y^2 + z^2) + (x^2 + y^2) + (y^2 + z^2) \\
&\quad + (z^2 + x^2) \\
&= 3(x^2 + y^2 + z^2)
\end{aligned}$$

We know that $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$ and a, b, c are positive. Therefore

$$\begin{aligned}
&(a + b + c)^2 \leq 3(a^2 + b^2 + c^2) \\
4R^2(\sin A + \sin B + \sin C)^2 &\leq 12R^2(\sin^2 A + \sin^2 B + \sin^2 C) \\
(\sin A + \sin B + \sin C)^2 &\leq 3 \left(\frac{1 - \cos 2A}{2} + \frac{1 - \cos 2B}{2} + \frac{1 - \cos 2C}{2} \right) \\
&= \frac{3}{2} [3 - (\cos 2A + \cos 2B + \cos 2C)] \\
&= \frac{3}{2} [3 - (-1 - 4 \cos A \cos B \cos C)] \\
&= \frac{3}{2} (4 + 4 \cos A \cos B \cos C) \\
&= 6(1 + \cos A \cos B \cos C) \\
&\leq 6 \left(1 + \frac{1}{8} \right) \quad (\text{By Problem 36}) \\
&= \frac{27}{4}
\end{aligned}$$

Therefore

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

38. In $\triangle ABC$, prove that

$$\operatorname{cosec} A + \operatorname{cosec} B + \operatorname{cosec} C \geq \frac{9}{4} \left(\sec \frac{A}{2} \sec \frac{B}{2} \sec \frac{C}{2} \right)$$

Proof: We know from Problem 2 that

$$(\sin A + \sin B + \sin C) \left(\frac{1}{\sin A} + \frac{1}{\sin B} + \frac{1}{\sin C} \right) \geq 3^2$$

Therefore

$$\begin{aligned}
&\left(4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \right) (\operatorname{cosec} A + \operatorname{cosec} B + \operatorname{cosec} C) \geq 9 \\
\operatorname{cosec} A + \operatorname{cosec} B + \operatorname{cosec} C &\geq \frac{9}{4} \left(\sec \frac{A}{2} \sec \frac{B}{2} \sec \frac{C}{2} \right)
\end{aligned}$$

39. In $\triangle ABC$, prove that

$$\begin{aligned}
\cos A \cos B \cos C &\leq \frac{1}{24} [\cos^2(A - B) + \cos^2(B - C) \\
&\quad + \cos^2(C - A)]
\end{aligned}$$

Proof: If x and y are any two real numbers, then we have

$$\begin{aligned}(x+y)^2 - 4xy &= (x-y)^2 \geq 0 \\ \Rightarrow (x+y)^2 &\geq 4xy\end{aligned}$$

Now take $x = \cos A$ and $y = 2\cos B \cos C$. Then

$$\begin{aligned}(\cos A + 2\cos B \cos C)^2 &\geq 8\cos A \cos B \cos C \\ [-\cos(B+C) + 2\cos B \cos C]^2 &\geq 8\cos A \cos B \cos C \\ (\cos B \cos C + \sin B \sin C)^2 &\geq 8\cos A \cos B \cos C \\ \cos^2(B-C) &\geq 8\cos A \cos B \cos C\end{aligned}$$

Similarly,

$$\begin{aligned}\cos^2(C-A) &\geq 8\cos A \cos B \cos C \\ \cos^2(A-B) &\geq 8\cos A \cos B \cos C\end{aligned}$$

Therefore

$$\begin{aligned}\cos^2(B-C) + \cos^2(C-A) + \cos^2(A-B) \\ \geq 24\cos A \cos B \cos C\end{aligned}$$

and hence the result.

40. In $\triangle ABC$, prove that

$$a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \leq 3abc$$

Proof: We have

$$\begin{aligned}a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \\ = a(b^2 + c^2 - a^2) + b(c^2 + a^2 - b^2) + c(a^2 + b^2 - c^2) \\ = 2abc \cos A + 2abc \cos B + 2abc \cos C \\ = 2abc(\cos A + \cos B + \cos C) \\ \leq 2abc\left(\frac{3}{2}\right)\end{aligned}$$

The last step has been derived using Problem 35.

41. In $\triangle ABC$, prove that

$$\frac{3}{2} \leq \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < 2$$

Proof: In Problem 30, we have proved that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

for any positive real numbers a, b and c . Now we prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < 2$$

Consider

$$\begin{aligned}b+c &> a \\ \Rightarrow 2(b+c) &> a+b+c \\ \Rightarrow b+c &> \frac{a+b+c}{2} \\ \Rightarrow \frac{1}{b+c} &< \frac{2}{a+b+c} \\ \Rightarrow \frac{a}{b+c} &< \frac{2a}{a+b+c}\end{aligned}$$

Therefore

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < \frac{2a+2b+2c}{a+b+c} = 2$$

42. In $\triangle ABC$, prove that

$$3(bc+ca+ab) \leq (a+b+c)^2 < 4(bc+ca+ab)$$

Proof: We have

$$a^2 + b^2 \geq 2ab$$

$$b^2 + c^2 \geq 2bc$$

and

$$c^2 + a^2 \geq 2ca$$

Now

$$2(a^2 + b^2 + c^2) \geq 2(bc + ca + ab)$$

Therefore

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

Adding $2(ab + bc + ca)$ to both sides we have

$$(a+b+c)^2 \geq 3(ab+bc+ca)$$

Also

$$|a-b| < c$$

$$|b-c| < a$$

$$|c-a| < b$$

This implies

$$(a-b)^2 + (b-c)^2 + (c-a)^2 < a^2 + b^2 + c^2$$

So

$$a^2 + b^2 + c^2 < 2ab + 2bc + 2ca$$

Adding to both sides $2ab + 2bc + 2ca$, we have

$$(a+b+c)^2 < 4(ab+bc+ca)$$

43. In $\triangle ABC$, prove that

$$\frac{a+b+c}{abc} \leq \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

Proof: We have

$$\begin{aligned} b^2c^2 + c^2a^2 &\geq 2\sqrt{b^2c^2 \cdot c^2a^2} = 2abc^2 \\ c^2a^2 + a^2b^2 &\geq 2bca^2 \\ a^2b^2 + b^2c^2 &\geq 2cab^2 \end{aligned}$$

Therefore

$$\begin{aligned} 2(b^2c^2 + c^2a^2 + a^2b^2) &\geq 2abc(a+b+c) \\ b^2c^2 + c^2a^2 + a^2b^2 &\geq abc(a+b+c) \end{aligned}$$

Dividing both sides with $a^2b^2c^2$, we have

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{a+b+c}{abc}$$

which is the required one.

44. In $\triangle ABC$, prove that

$$\begin{aligned} 48(s-a)(s-b)(s-c) &\leq bc(b+c) \\ &\quad + ca(c+a) + ab(a+b) \end{aligned}$$

where $2s = a + b + c$.

Proof: We have

$$\begin{aligned} bc(b+c) + ca(c+a) + ab(a+b) \\ = b(c^2 + a^2) + c(a^2 + b^2) + a(b^2 + c^2) \\ \geq b(2ca) + c(2ab) + a(2bc) \end{aligned}$$

Therefore

$$bc(b+c) + ca(c+a) + ab(a+b) \geq 6abc \quad (8.16)$$

But by part (ii) of Problem 33

$$\begin{aligned} abc &\geq (b+c-a)(c+a-b)(a+b+c) \\ &= 8(s-a)(s-b)(s-c) \end{aligned} \quad (8.17)$$

Therefore from Eqs. (8.16) and (8.17)

$$\begin{aligned} 48(s-a)(s-b)(s-c) &\leq 6abc \leq bc(b+c) \\ &\quad + ca(c+a) + ab(a+b). \end{aligned}$$

45. In $\triangle ABC$, prove that

$$4\Delta \leq \min\{b^2 + c^2, c^2 + a^2, a^2 + b^2\}$$

Proof: We have

$$4\Delta = 2bc \sin A \leq 2bc \leq b^2 + c^2$$

Similarly

$$4\Delta \leq c^2 + a^2$$

$$4\Delta \leq a^2 + b^2$$

Therefore

$$4\Delta \leq \min\{b^2 + c^2, c^2 + a^2, a^2 + b^2\}$$

46. In $\triangle ABC$, prove that

$$b^2c^2 + c^2a^2 + a^2b^2 \geq 16\Delta^2$$

Proof: We have

$$\begin{aligned} 16\Delta^2 &= (2s)2(s-a)2(s-b)2(s-c) \\ &= (a+b+c)(b+c-a)(c+a-b)(a+b-c) \\ &= [(b+c)^2 - a^2][a^2 - (b-c)^2] \\ &= a^2[(b+c)^2 + (b-c)^2] - a^4 - (b^2 - c^2)^2 \\ &= 2(a^2b^2 + b^2c^2 + c^2a^2) - a^4 - b^4 - c^4 \\ &= a^2b^2 + b^2c^2 + c^2a^2 - \frac{1}{2}[(a^2 - b^2)^2 + (b^2 - c^2)^2 \\ &\quad + (c^2 - a^2)^2] \\ &\leq a^2b^2 + b^2c^2 + c^2a^2 \end{aligned}$$

47. In $\triangle ABC$, prove that

$$bc + ca + ab \geq (4\sqrt{3})\Delta$$

Proof: We have

$$\Delta = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C$$

This implies

$$\begin{aligned} bc + ca + ab &= 2\Delta \left(\frac{1}{\sin A} + \frac{1}{\sin B} + \frac{1}{\sin C} \right) \\ &\geq (2\Delta)(3) \left[\frac{1}{\sin A} \cdot \frac{1}{\sin B} \cdot \frac{1}{\sin C} \right]^{1/3} \\ &\geq (2\Delta)(3) \left(\frac{8}{3\sqrt{3}} \right)^{1/3} \\ &\quad \left(\because \sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8} \right) \\ &= (2\Delta)(3) \left(\frac{2}{\sqrt{3}} \right) \\ &= 4\sqrt{3}\Delta \end{aligned}$$

48. In $\triangle ABC$, prove that

$$\begin{aligned} & \frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} \\ & \geq \max \left\{ \frac{9}{a+b+c}, \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right\} \end{aligned}$$

Proof: Let $x = b + c - a$, $y = c + a - b$ and $z = a + b - c$ so that x, y, z are positive and

$$x + y + z = a + b + c$$

By Problem 2,

$$(x + y + z) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) \geq 3^2$$

This implies

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{x + y + z}$$

Therefore

$$\frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} \geq \frac{9}{a+b+c} \quad (8.18)$$

Again,

$$\begin{aligned} \frac{1}{b+c-a} + \frac{1}{c+a-b} &= \frac{2c}{c^2 - (a-b)^2} \\ &\geq \frac{2c}{c^2} = \frac{2}{c} \end{aligned}$$

Similarly

$$\begin{aligned} \frac{1}{c+a-b} + \frac{1}{a+b-c} &\geq \frac{2}{a} \\ \frac{1}{a+b-c} + \frac{1}{b+c-a} &\geq \frac{2}{b} \end{aligned}$$

Adding all the three in equalities we have

$$\frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} \geq \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \quad (8.19)$$

From Eqs. (8.18) and (8.19), the result follows.

49. For $\triangle ABC$, prove that there exist positive real numbers x, y, z such that $y + z = a$, $z + x = b$ and $x + y = c$ and hence prove that

$$\frac{R}{r} \geq \frac{b}{c} + \frac{c}{a}$$

where r is the inradius and R is the circumradius.

Proof: Let $x = s - a$, $y = s - b$, $z = s - c$ where $2s = a + b + c$, so that

$$y + z = a$$

$$z + x = b$$

$$x + y = c$$

Now

$$\begin{aligned} \frac{R}{r} &= \left(\frac{abc}{4\Delta} \right) \left(\frac{s}{\Delta} \right) \\ &= \frac{abc}{4(s-a)(s-b)(s-c)} \\ &= \frac{(y+z)(z+x)(x+y)}{4xyz} \\ &= \frac{y(z+x)(x+y) + z(z+x)(x+y)}{4xyz} \\ &= \frac{(z+x)(x+y)}{4zx} + \frac{(z+x)(x+y)}{4xy} \quad (8.20) \end{aligned}$$

Now

$$4zx \leq (z+x)^2$$

and

$$4xy \leq (x+y)^2$$

This implies

$$\frac{1}{4zx} \geq \frac{1}{(z+x)^2}$$

and

$$\frac{1}{4xy} \geq \frac{1}{(x+y)^2}$$

Therefore using Eq. (8.20) we have

$$\begin{aligned} \frac{R}{r} &\geq \frac{(z+x)(x+y)}{(z+x)^2} + \frac{(z+x)(x+y)}{(x+y)^2} \\ \frac{R}{r} &\geq \frac{x+y}{z+x} + \frac{z+x}{x+y} = \frac{c}{b} + \frac{b}{c} \end{aligned}$$

50. If a, b, c are positive real numbers, then prove that

$$\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} \leq \frac{1}{2} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

(Ireland Olympiads, 1998–99)

Proof: We have

$$\frac{1}{a} + \frac{1}{b} \geq \frac{2}{\sqrt{ab}}$$

$$\frac{1}{b} + \frac{1}{c} \geq \frac{2}{\sqrt{bc}}$$

$$\frac{1}{c} + \frac{1}{a} \geq \frac{2}{\sqrt{ca}}$$

which imply that

$$\begin{aligned} 2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) &\geq 2\left(\frac{1}{\sqrt{bc}} + \frac{1}{\sqrt{ca}} + \frac{1}{\sqrt{ab}}\right) \\ \frac{1}{a} + \frac{1}{b} + \frac{1}{c} &\geq \frac{1}{\sqrt{bc}} + \frac{1}{\sqrt{ca}} + \frac{1}{\sqrt{ab}} \\ &\geq \frac{2}{b+c} + \frac{2}{c+a} + \frac{2}{a+b} \end{aligned}$$

because

$$\begin{aligned} \frac{a+b}{2} &\geq \sqrt{ab} \\ \frac{b+c}{2} &\geq \sqrt{bc} \\ \frac{c+a}{2} &\geq \sqrt{ca} \end{aligned}$$

51. If a, b, c are the sides of a triangle and $a + b + c = 2$, then

$$a^2 + b^2 + c^2 + 2abc < 2$$

Proof: We have

$$4 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2\sum ab$$

Therefore

$$\begin{aligned} a^2 + b^2 + c^2 + 2abc &= 4 - 2\sum ab + 2abc \\ a^2 + b^2 + c^2 + 2abc &< 2 \\ \Leftrightarrow 4 - 2\sum ab + 2abc &< 2 \\ \Leftrightarrow 2 - \sum ab + abc &< 1 \\ \Leftrightarrow 1 - \sum ab + abc &< 0 \end{aligned} \quad (8.21)$$

Now

$$2s = a + b + c = 2 \Rightarrow s = 1$$

Therefore

$$\begin{aligned} \Delta^2 &= s(s-a)(s-b)(s-c) \\ &= (1-a)(1-b)(1-c) \\ &= 1 + ab + bc + ca - (a+b+c) - abc \\ &= 1 + ab + bc + ca - 2 - abc \\ &= ab + bc + ca - 1 - abc \end{aligned}$$

So

$$\begin{aligned} ab + bc + ca - 1 - abc &= \Delta^2 > 0 \\ \Rightarrow 1 - ab - bc - ca + abc &< 0 \end{aligned}$$

This means that Eq. (8.21) is true. Therefore

$$a^2 + b^2 + c^2 + 2abc < 2$$

52. In $\triangle ABC$, prove that $a^2 + b^2 + c^2 \geq 4\sqrt{3}\Delta$.

Proof: We know that

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Therefore

$$\begin{aligned} a^2 + b^2 + c^2 &= 2(b^2 + c^2) - 2bc \cos A \\ a^2 + b^2 + c^2 - 4\sqrt{3}\Delta &= 2(b^2 + c^2) - 2bc \cos A - 2\sqrt{3} \sin A \\ &= 2(b^2 + c^2) - 2bc(\cos A + \sqrt{3} \sin A) \\ &= 2(b^2 + c^2) - 4bc \cos\left(A - \frac{\pi}{3}\right) \\ &\geq 2(b^2 + c^2) - 4bc \left(\because \cos\left(A - \frac{\pi}{3}\right) \leq 1\right) \\ &= 2(b-c)^2 \geq 0 \end{aligned}$$

So

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}\Delta$$

Equality holds, if the triangle is equilateral.

Note: In Problem 47, it is proved that $bc + ca + ab \geq 4\sqrt{3}\Delta$ and hence Problem 52 follows from Problem 47, because $a^2 + b^2 + c^2 \geq ab + bc + ca$. In both cases equality holds if the triangle is equilateral.

53. In $\triangle ABC$, if p_1, p_2 and p_3 are the lengths of the altitudes from the vertices A, B and C on to the opposite sides respectively, then prove that

$$\frac{1}{p_1 - 2r} + \frac{1}{p_2 - 2r} + \frac{1}{p_3 - 2r} \geq \frac{3}{r}$$

where r is the in radius of $\triangle ABC$.

Proof: It is known that

$$\Delta = \frac{1}{2}(BC)p_1 = \frac{ap_1}{2}$$

Therefore

$$\frac{1}{p_1} = \frac{a}{2\Delta}$$

From this we get

$$\begin{aligned}\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} &= \frac{a}{2\Delta} + \frac{b}{2\Delta} + \frac{c}{2\Delta} \\ &= \frac{s}{\Delta} = \frac{1}{r}\end{aligned}$$

Now

$$\begin{aligned}-\frac{2r}{p_1} - \frac{2r}{p_2} - \frac{2r}{p_3} &= -2 \\ \Rightarrow \left(1 - \frac{2r}{p_1}\right) + \left(1 - \frac{2r}{p_2}\right) + \left(1 - \frac{2r}{p_3}\right) &= 3 - 2 = 1\end{aligned}$$

Therefore

$$\frac{p_1 - 2r}{p_1} + \frac{p_2 - 2r}{p_2} + \frac{p_3 - 2r}{p_3} = 1 \quad (8.22)$$

Also

$$\begin{aligned}p_1 - 2r &= \frac{2\Delta}{a} - \frac{2\Delta}{s} \\ &= 2\Delta \left(\frac{1}{a} - \frac{1}{s} \right) \\ &= \Delta \left(\frac{2}{a} - \frac{2}{s} \right) \\ &= \frac{\Delta(b + c - a)}{as} > 0\end{aligned}$$

By Problem 2, we have

$$\begin{aligned}\left(\frac{p_1 - 2r}{p_1} + \frac{p_2 - 2r}{p_2} + \frac{p_3 - 2r}{p_3} \right) \\ \left(\frac{p_1}{p_1 - 2r} + \frac{p_2}{p_2 - 2r} + \frac{p_3}{p_3 - 2r} \right) &\geq 3^2\end{aligned}$$

So from Eq. (8.22), we have

$$\begin{aligned}\frac{p_1}{p_1 - 2r} + \frac{p_2}{p_2 - 2r} + \frac{p_3}{p_3 - 2r} &\geq 9 \\ \left(\frac{p_1}{p_1 - 2r} - 1 \right) + \left(\frac{p_2}{p_2 - 2r} - 1 \right) + \left(\frac{p_3}{p_3 - 2r} - 1 \right) &\geq 6 \\ \frac{2r}{p_1 - 2r} + \frac{2r}{p_2 - 2r} + \frac{2r}{p_3 - 2r} &\geq 6 \\ \frac{1}{p_1 - 2r} + \frac{1}{p_2 - 2r} + \frac{1}{p_3 - 2r} &\geq \frac{3}{r}\end{aligned}$$

54. In $\triangle ABC$, if p_1, p_2, p_3 are the lengths of the altitudes drawn from the vertices onto the opposite sides and

r_1, r_2, r_3 are the ex-radii opposite to the vertices A, B and C , respectively, then prove that

$$\frac{r_1}{p_1} + \frac{r_2}{p_2} + \frac{r_3}{p_3} \geq 3$$

Proof: We have

$$\Delta = \frac{1}{2}ap_1 \Rightarrow p_1 = \frac{2\Delta}{a}$$

Now

$$\frac{1}{r_2} + \frac{1}{r_3} = \frac{s-b}{\Delta} + \frac{s-c}{\Delta} = \frac{a}{\Delta} = \frac{2}{p_1}$$

Therefore

$$\frac{r_1}{r_2} + \frac{r_1}{r_3} = \frac{2r_1}{p_1}$$

Similarly

$$\begin{aligned}\frac{r_2}{r_3} + \frac{r_2}{r_1} &= \frac{2r_2}{p_2} \\ \frac{r_3}{r_1} + \frac{r_3}{r_2} &= \frac{2r_3}{p_3}\end{aligned}$$

Adding all the three equalities, we have

$$\begin{aligned}2 \left(\frac{r_1}{p_1} + \frac{r_2}{p_2} + \frac{r_3}{p_3} \right) &= \frac{r_1}{r_2} + \frac{r_1}{r_3} + \frac{r_2}{r_3} + \frac{r_2}{r_1} + \frac{r_3}{r_1} + \frac{r_3}{r_2} \\ &\geq 6\end{aligned}$$

(using AM $\geq$ GM for the six numbers on the RHS.)

Therefore

$$\frac{r_1}{p_1} + \frac{r_2}{p_2} + \frac{r_3}{p_3} \geq 3$$

55. In $\triangle ABC$, prove that

$$\sum \left(\frac{\cos^2 B + \cos^2 C}{\cos B + \cos C} \right) \geq 1 + \frac{r}{R}$$

Proof: First, we prove that, if x, y and z are real numbers such that the sum of any two is positive, then

$$\frac{x^2 + y^2}{x + y} + \frac{y^2 + z^2}{y + z} + \frac{z^2 + x^2}{z + x} \geq x + y + z$$

We have

$$\begin{aligned}x^2 + y^2 &\geq 2xy \Rightarrow 2(x^2 + y^2) \geq (x + y)^2 \\ &\Rightarrow \frac{x^2 + y^2}{x + y} \geq \frac{x + y}{2}\end{aligned}$$

Therefore

$$\begin{aligned}\sum \frac{x^2 + y^2}{x + y} &\geq \frac{x + y}{2} + \frac{y + z}{2} + \frac{z + x}{2} \\ &= x + y + z\end{aligned}$$

To prove the given problem, take $x = \cos A$, $y = \cos B$ and $z = \cos C$ so that

$$\begin{aligned}x + y &= \cos A + \cos B \\ &= 2 \cos \frac{A + B}{2} \cos \left(\frac{A - B}{2} \right) \\ &> 0\end{aligned}$$

Therefore $x + y$, $y + z$ and $z + x$ are positive. So

$$\begin{aligned}\sum \left(\frac{x^2 + y^2}{x + y} \right) &\geq x + y + z \\ \Rightarrow \sum \left(\frac{\cos^2 A + \cos^2 B}{\cos A + \cos B} \right) &\geq \cos A + \cos B + \cos C \\ &= 1 + \frac{r}{R}\end{aligned}$$

56. In $\triangle ABC$, prove that

$$\sqrt{\sin A} + \sqrt{\sin B} + \sqrt{\sin C} \leq 3 \left(\frac{3}{4} \right)^{1/4}$$

Proof: Let x, y, z be positive real numbers. Then

$$\begin{aligned}(x + y + z)^2 &= x^2 + y^2 + z^2 + 2xy + 2yz + 2zx \\ &\leq (x^2 + y^2 + z^2) + (x^2 + y^2) + (y^2 + z^2) \\ &\quad + (z^2 + x^2) \\ &= 3(x^2 + y^2 + z^2)\end{aligned}$$

This implies

$$x + y + z \leq \sqrt{3(x^2 + y^2 + z^2)}$$

Put $x = \sqrt{\sin A}$, $y = \sqrt{\sin B}$, $z = \sqrt{\sin C}$. Therefore

$$\begin{aligned}\sqrt{\sin A} + \sqrt{\sin B} + \sqrt{\sin C} &\leq \sqrt{3(\sin A + \sin B + \sin C)} \\ &\leq \sqrt{3 \left(\frac{3\sqrt{3}}{2} \right)} \quad (\text{see Problem 37}) \\ &= 3 \left(\frac{3}{4} \right)^{1/4}\end{aligned}$$

57. Consider Figure 8.2. P is a point interior to $\triangle ABC$. The lines AP , BP and CP meet the opposite sides in D , E and F respectively. Prove that

$$\frac{AP}{PD} + \frac{BP}{PE} + \frac{CP}{PF} \geq 6$$

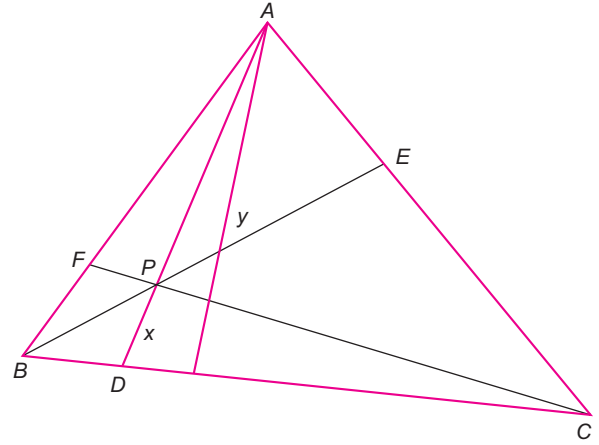


FIGURE 8.2 Worked-out problem 57.

Proof: Let Δ_1 = area of $\triangle PBC$, Δ_2 = area of $\triangle PCA$, Δ_3 = area of $\triangle PAB$ and Δ = area of $\triangle ABC$. Let the length of the altitude drawn from the vertex A on to BC be y and the length of the altitude drawn from P onto BC be x . Now since x and y are parallel we have

$$\frac{\Delta}{\Delta_1} = \frac{y}{x} = \frac{AD}{PD}$$

Therefore

$$\begin{aligned}\frac{\Delta_1 + \Delta_2 + \Delta_3}{\Delta_1} &= \frac{AD}{PD} \\ \frac{\Delta_2}{\Delta_1} + \frac{\Delta_3}{\Delta_1} &= \frac{AD}{PD} - 1 = \frac{AP}{PD}\end{aligned}$$

Similarly

$$\begin{aligned}\frac{\Delta_3}{\Delta_2} + \frac{\Delta_1}{\Delta_2} &= \frac{BP}{PE} \\ \frac{\Delta_1}{\Delta_3} + \frac{\Delta_2}{\Delta_3} &= \frac{CP}{PF}\end{aligned}$$

Adding all we have

$$\begin{aligned}\frac{AP}{PD} + \frac{BP}{PE} + \frac{CP}{PF} &= \sum \left(\frac{\Delta_2}{\Delta_1} + \frac{\Delta_3}{\Delta_1} \right) \\ &\geq 6 \left[\frac{\Delta_2}{\Delta_1} \cdot \frac{\Delta_3}{\Delta_1} \cdot \frac{\Delta_2}{\Delta_3} \cdot \frac{\Delta_1}{\Delta_3} \cdot \frac{\Delta_3}{\Delta_2} \cdot \frac{\Delta_1}{\Delta_2} \right]^{1/6} \\ &= 6\end{aligned}$$

58. In $\triangle ABC$, prove that

$$\sum \left(\frac{\sqrt{\sin A}}{\sqrt{\sin B} + \sqrt{\sin C} - \sqrt{\sin A}} \right) \geq 3$$

Proof: Since $a = 2R \sin A$, $b = 2R \sin B$ and $c = 2R \sin C$, we have to prove that

$$\sum \left(\frac{\sqrt{a}}{\sqrt{b} + \sqrt{c} - \sqrt{a}} \right) \geq 3$$

Let

$$x = \sqrt{b} + \sqrt{c} - \sqrt{a}$$

$$y = \sqrt{c} + \sqrt{a} - \sqrt{b}$$

$$z = \sqrt{a} + \sqrt{b} - \sqrt{c}$$

Since

$$\begin{aligned} (\sqrt{b} + \sqrt{c})^2 &= b + c + 2\sqrt{bc} \\ &> b + c > a \end{aligned}$$

This implies

$$\sqrt{b} + \sqrt{c} - \sqrt{a} > 0$$

So x, y, z are positive. Now $y + z = 2\sqrt{a}$, $z + x = 2\sqrt{b}$ and $x + y = 2\sqrt{c}$ imply that

$$\frac{\sqrt{a}}{\sqrt{b} + \sqrt{c} - \sqrt{a}} = \frac{y + z}{2x}$$

$$\frac{\sqrt{b}}{\sqrt{c} + \sqrt{a} - \sqrt{b}} = \frac{z + x}{2y}$$

$$\frac{\sqrt{c}}{\sqrt{a} + \sqrt{b} - \sqrt{c}} = \frac{x + y}{2z}$$

Therefore

$$2 \sum \left(\frac{\sqrt{a}}{\sqrt{b} + \sqrt{c} - \sqrt{a}} \right) \geq 6 \left(\frac{z}{y} \cdot \frac{x}{y} \cdot \frac{y}{x} \cdot \frac{z}{x} \cdot \frac{x}{z} \cdot \frac{y}{z} \right)^{1/6}$$

$$\sum \left(\frac{\sqrt{a}}{\sqrt{b} + \sqrt{c} - \sqrt{a}} \right) \geq 3$$

59. If x, y, z are distinct positive real numbers and

$$x + \frac{1}{y} = y + \frac{1}{z} = z + \frac{1}{x}$$

then prove that

$$x + y + z > 3 \quad \text{and} \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} > 3$$

Proof: By hypothesis,

$$\frac{xy + 1}{y} = \frac{yz + 1}{z} = \frac{zx + 1}{x}$$

Therefore

$$xyz + z = y^2 z + y$$

$$xyz + x = z^2 x + z$$

$$x^2 y + x = xyz + y$$

Now

$$xyz + z = y^2 z + y$$

$$\Rightarrow yz(x - y) = y - z$$

$$\Rightarrow xyz(x - y) = x(y - z) \quad (8.23)$$

Similarly

$$xyz(y - z) = y(z - x) \quad (8.24)$$

$$xyz(z - x) = z(x - y) \quad (8.25)$$

On multiplying Eqs. (8.23), (8.24) and (8.25) and cancelling $xyz(x - y)(y - z)(z - x)$ we have $(xyz)^2 = 1$ which implies that $xyz = 1$. Therefore

$$x + y + z > 3(xyz)^{1/3} = 3$$

and

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} > 3$$

Here equality cannot occur, because x, y, z are distinct.

60. If a, b and c are positive rational numbers such that the sum of any two is greater than the third, then prove that

$$\left(1 + \frac{b-c}{a} \right)^a \left(1 + \frac{c-a}{b} \right)^b \left(1 + \frac{a-b}{c} \right)^c \leq 1$$

Equality occurs if and only if $a = b = c$.

Proof: We can suppose, $a = x/m, b = y/m, c = z/m$ where x, y, z and m are positive integers. Also note that $b + c - a, c + a - b$ and $a + b - c$ are positive and

$$x + y + z = m(a + b + c)$$

Consider

$$1 + \frac{b-c}{a} \quad (x \text{ times})$$

$$1 + \frac{c-a}{b} \quad (y \text{ times})$$

$$1 + \frac{a-b}{c} \quad (z \text{ times})$$

and use $AM \geq GM$. Therefore

$$\begin{aligned} & \frac{x\left(1 + \frac{b-c}{a}\right) + y\left(1 + \frac{c-a}{b}\right) + z\left(1 + \frac{a-b}{c}\right)}{x+y+z} \\ & \geq \left[\left(1 + \frac{b-c}{a}\right)^x \left(1 + \frac{c-a}{b}\right)^y \left(1 + \frac{a-b}{c}\right)^z \right]^{1/(x+y+z)} \\ & \frac{(x+y+z) + m[(b-c) + (c-a) + (a-b)]}{x+y+z} \\ & \geq \left[\left(1 + \frac{b-c}{a}\right)^x \left(1 + \frac{c-a}{b}\right)^y \left(1 + \frac{a-b}{c}\right)^z \right]^{1/m(a+b+c)} \\ & 1 \geq \left[\left(1 + \frac{b-c}{a}\right)^{x/m} \left(1 + \frac{c-a}{b}\right)^{y/m} \left(1 + \frac{a-b}{c}\right)^{z/m} \right]^{1/(a+b+c)} \\ & 1 \geq \left[\left(1 + \frac{b-c}{a}\right)^a \left(1 + \frac{c-a}{b}\right)^b \left(1 + \frac{a-b}{c}\right)^c \right]^{1/(a+b+c)} \end{aligned}$$

Therefore

$$\left(1 + \frac{b-c}{a}\right)^a \left(1 + \frac{c-a}{b}\right)^b \left(1 + \frac{a-b}{c}\right)^c \leq (1)^{a+b+c} = 1$$

- 61.** If x and y are positive real numbers such that $x + y = 1$, then show that

$$\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right) \geq 9$$

Equality occurs when $x = y = 1/2$.

Proof: We have

$$\frac{x+y}{2} \geq \sqrt{xy} \Rightarrow \frac{1}{2} \geq \sqrt{xy} \quad (8.26)$$

By Eq. (8.26) we have

$$\begin{aligned} \left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right) &= 1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{xy} \\ &\geq 1 + \frac{2}{\sqrt{xy}} + \frac{1}{xy} \\ &\geq 1 + 2(2) + 4 \end{aligned}$$

Therefore

$$\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right) \geq 9$$

- 62.** In $\triangle ABC$, prove that

$$\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3$$

Equality holds when the triangle is equilateral.

Proof: Using $AM \geq GM$ we get

$$\begin{aligned} & \frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \\ & \geq 3 \left[\frac{a}{c+a-b} \cdot \frac{b}{a+b-c} \cdot \frac{c}{b+c-a} \right]^{1/3} \end{aligned}$$

But, from Problem 33, part (ii), we have that

$$abc \geq (b+c-a)(c+a-b)(a+b-c)$$

Therefore

$$\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3$$

EXERCISES

Subjective Questions Only

- 1.** If a, b, c are positive real numbers and $a + b + c = 1$, then prove that

$$8 \leq \left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \leq \frac{8}{27abc}$$

Equality holds when $a = b = c = 1/3$.

- 2.** If a, b, c are positive real numbers, then prove that

$$\begin{aligned} 2(a^3 + b^3 + c^3) &\geq ab(a+b) + bc(b+c) + ca(c+a) \\ &\geq 6abc \end{aligned}$$

- 3.** If $n^4 < 10^n$ for a positive integer $n \geq 2$, then show that $(n+1)^4 < 10^{n+1}$.

- 4.** If x, y are positive such that $3x + 4y = 5$, then show that

$$x^2 y^3 \leq \frac{3}{16}$$

and equality holds if and only if $x = 2/3, y = 3/4$.

- 5.** If x, y, z are positive such that $x + y + z = 18$, then prove that the maximum value of $x^2 y^3 z^4$ is $4^2 6^3 8^4$.

6. If a, b, c, e, f, g are positive, then show that

$$\left(\frac{a}{e} + \frac{b}{f} + \frac{c}{g}\right)\left(\frac{e}{a} + \frac{f}{b} + \frac{g}{c}\right) \geq 9$$

7. If a, b, c are positive, then prove that

$$(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

When does the equality occur?

8. If a, b are positive, then prove that $\sqrt{2}$ lies between a/b and $(a + 2b)/(a + b)$.

9. If $0 < x \leq \pi/2$, then show that the minimum value of

$$\frac{9x^2 \sin^2 x + 4}{x \sin x}$$

is 12.

10. In $\triangle ABC$, prove that the inradius

$$r < \frac{a^2 + b^2 + c^2}{3(a + b + c)}$$

11. If a, b, c are the sides of $\triangle ABC$, then prove that

$$|a^2(b - c) + b^2(c - a) + c^2(a - b)| < abc$$

(Hint: Factorize the expression and use that the difference of two sides is less than the third side.)

12. If $\theta_1, \theta_2, \dots, \theta_n$ are real, then show that

$$\left(\sum_{i=1}^n \cos \theta_i\right)^2 + \left(\sum_{i=1}^n \sin \theta_i\right)^2 \leq n^2$$

13. If $a_1, a_2, \dots, a_n$ are positive real numbers such that $a_1 + a_2 + \dots + a_n = s$ then prove that

$$\frac{s}{s - a_1} + \frac{s}{s - a_2} + \frac{s}{s - a_3} + \dots + \frac{s}{s - a_n} \geq \frac{n^2}{n - 1}$$

where $n \geq 2$

14. If A, B, C are independent events with positive probabilities of a random experiment, then prove that

$$P(A) + P(B) + P(C) \geq 3(P(A \cap B \cap C))^{1/3}$$

When will the equality hold?

15. Let A, B, C be mutually exclusive and exhaustive events with non-zero probabilities of a random experiment, and E is any event of the experiment with $P(E) > 0$, then prove that

$$P(E) \geq 3(P(A)P(B)P(C))^{1/3} \times [P(E/A)P(E/B)P(E/C)]^{1/3}$$

16. If $a_1, a_2, \dots, a_n$ are positive real numbers such that each $a_i < 1$ and $a_1 + a_2 + \dots + a_n = s_n \neq 1$, then prove that

$$(1 + a_1)(1 + a_2) \cdots (1 + a_n) < \frac{1}{1 - s_n}$$

17. If $n > 1$ integer, then prove that $\lfloor n \rfloor > n^{n/2}$.

18. If a, b, c are positive real numbers, then prove that

$$(a + b + c)(bc + ca + ab) \geq 9abc$$

19. If $n > 1$ integer, then prove that $n^n > 1 \cdot 3 \cdot 5 \cdots (2n - 1)$.

- *20. In many books, one finds the following problem: If $a_1, a_2, \dots, a_n$ are positive and $(n - 1)s = a_1 + a_2 + \dots + a_n$, then show that the product

$$a_1 \cdot a_2 \cdot a_3 \cdots a_n > (n - 1)^n (s - a_1)(s - a_2) \cdots (s - a_n)$$

This result is wrong, if we consider $a_1 = 1, a_2 = 2$ and $a_3 = a_4 = 6$. Find the fallacy in the argument (or proof) given by the authors.

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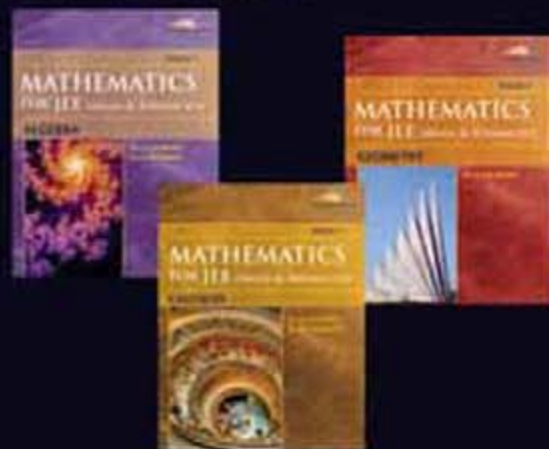
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